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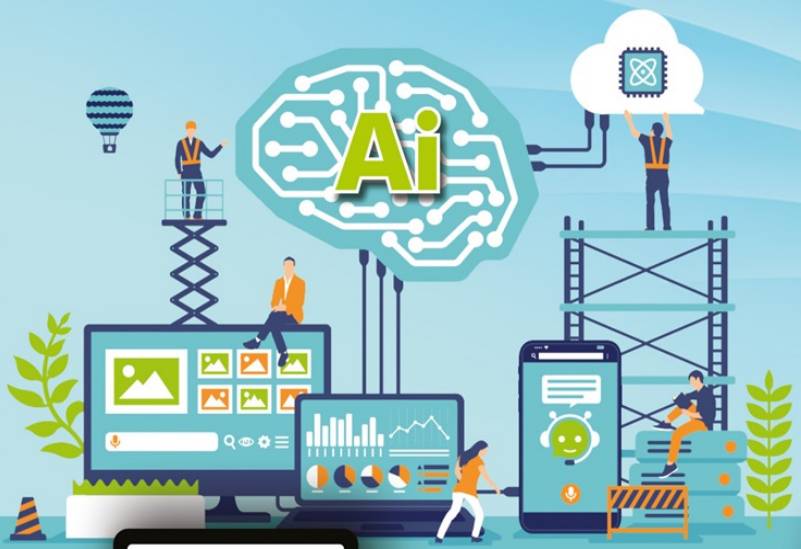


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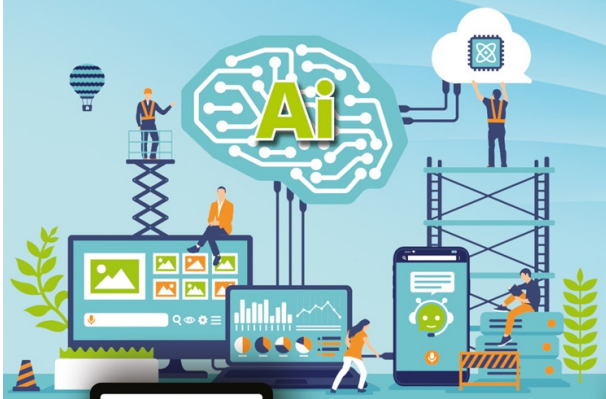


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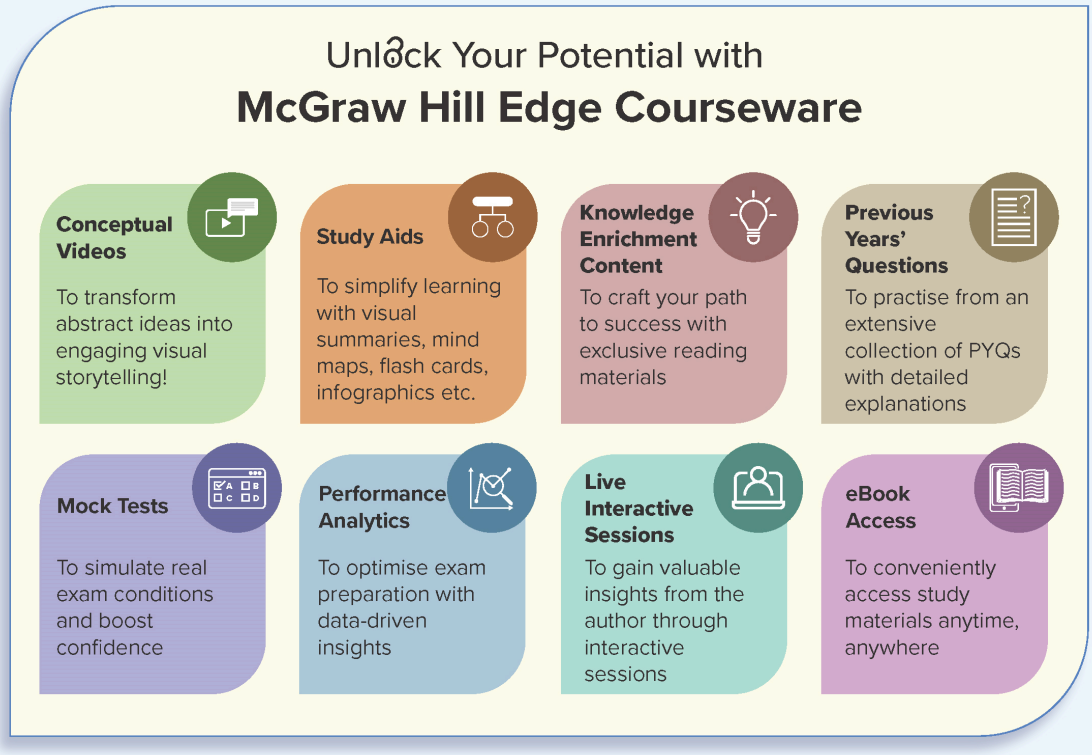


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## Dedicated to

*My Parents*

***Shri Om Prakash Agrahari***

*and*

***Late Smt. Meena Agrahari***

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# ABOUT THE AUTHOR

**Dr. Ravi P. Agrahari**, an **ex-scientist** and **Ph.D. from IIT Delhi**, is a recipient of Junior Research Fellowship and Senior Research Fellowship from CSIR, Government of India. Dr. Agrahari is presently working as an independent consultant in the field of Environment, Renewable and Sustainable Energy and Biogas issues at the **Bag Energy Research Society, Varanasi**. He is an expert in the field of energy and has published various papers in many reputed international journals. He has attended many international conferences in Sweden and France. Dr Agrahari is well known for his expertise in '**Environment Science**' and '**Science and Technology**' and has authored several books on the same for UPSC examinations. He is also popular as an '**Anthropology Optional Teacher**' among UPSC aspirants. As a teacher of UPSC aspirants for over two decades, Dr. Agrahari has been associated with over 50 reputed coaching institutes in more than 15 states in India. He is also associated with various NGOs and Civil Societies for the welfare of weaker section and minority students.

He also runs an online institute on '**Ravi Agrahari Classes App**' which provides sustainable and quality education to the students, who face difficulties in joining institutes in metropolitan cities.

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To answer subject queries of aspirants, he can be reached at a telegram group **https://t.me/drravipagrahari/Dr.RaviP.Agrahari(Official)**.

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# PREFACE

It is with immense pleasure that I present this edition of the book with all new updates and new and improved infographics. This compilation represents the culmination of over two decades of teaching experience. The topics have been meticulously selected, bearing in mind the dynamic nature and latest trends of the UPSC examinations. This interdisciplinary approach ensures comprehensive coverage of major government initiatives and international relations, which are key focus areas for examiners in the Science and Technology segment.

Recognising the evolving patterns of the UPSC Examination, this edition includes infographics on all important topics to facilitate easier retention of information. We have updated the content to include recent developments in Science and Technology, both nationally and internationally, along with related fields where necessary. Furthermore, a question bank featuring previous years' UPSC examinations has been incorporated, enhancing the book's utility as a significant resource for aspirants. All this is further complemented with McGraw Hill Edge, an exclusive digital platform with high-quality learning resources. On the McGraw Hill Edge platform, you can access mock tests, practice tests, PYQ's of Prelims examination with answers and explanations, model questions and answers for Mains, 2025 Mains answers (after the 2025 Mains) and videos explaining key concepts and how to prepare for the

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exam along with webinars on important and upcoming topics from exams point of view to help you excel in the examination.

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The creation of such a comprehensive book would not have been possible without the unwavering support of my family and friends, and the invaluable input and suggestions from my students. I extend my heartfelt thanks to *Vision IAS* a monthly magazine that provided useful material for updating this book, and to Parmeet Singh Jaggi for his assistance in simplifying complex concepts.

A special note of gratitude goes to **Gaurav Kumar Sharma (gauravkumarsharma@icloud.com)**, one of my dedicated students, who has designed the exceptional infographics in this book. His artistic vision and keen design sense have transformed intricate information into visually compelling and accessible representations. Gaurav's meticulous attention to detail and deep understanding of the subject matter are evident in every infographic, enhancing both the aesthetic appeal and educational value of the book.

I would also like to thank Dr. Ankita Kar, Dr. Poonam Saini, Ipsita Beborta, Ruchi Jain and Milan Singh for their constant support during the compilation process. I am deeply grateful to my mentor, **Prof. G. N. Tiwari (IIT Delhi)**, for his continuous encouragement. My heartfelt thanks go to my wife Babeeta Agrahari, Col. Vinod Kumar, Radha Rani Goyal, Shailesh Agrahari, Madhu Agrahari, Namrata Agrahari, Hemant Goyal, Stuti Agrahari, Shani Agrahari and Ankit Agrahari for their patience and support. I am especially thankful to my sons, Aksh and Advait (Adi), for their assistance in editing this edition.

I extend my sincere appreciation to Ms. Kajal for her diligent efforts throughout the writing process. I also thank my students Poonam Phoolwani, Aayushi Mukherjee, Raj Aishwarya Shree, Prashant Jha, Satata Sarkar, Saila Suman Bhuyan, Payal Mahapatra, Peace, Mangte, Pradip, Ajeesh and Zama Jeesan for their help in completing the book on time. Special thanks to Tanweer Ahmad, Nikhil Wadhera, Rahul Bhatadia, Aastha Khanna, Suhaib Ali, Dharmender Sharma and the entire team at McGraw Hill for their relentless encouragement to bring this book to fruition.

Lastly, I am grateful to Atul Photostat (in Mukherjee Nagar and Old Rajender Nagar) and the entire team of '**Ravi Agrahari Classes**' **online platform**, for keeping me informed about readers' requirements. To further

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support this endeavor, I have created a Telegram group: **https://t.me/drravipagrahari/Dr.RaviP.Agrahari(Official)**

This book is a testament to the collaborative effort of many individuals, and I hope it serves as a valuable resource for all UPSC aspirants.

**RAVI P. AGRAHARI**

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# PAPER ANALYSIS

## Analysis of Paper-I (General Science) (2011–2025)

| SUBJECT | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2 |
|---|---|---|---|---|---|---|---|---|
| Current events of National & International importance | | | | | | | | |
| History of India and Indian National Movement | | | | | | | | |
| General Sciences | 21 | 11 | 15 | 20 | 19 | 18 | 15 | |
| India and World Geography | | | | | | | | |

## Prelims Exam Analysis of General Sciences (year wise number of question)(2011–2025)

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| SUBJECT | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 201 |
|---|---|---|---|---|---|---|---|---|
| Biology | 0 | 1 | 6 | 8 | 3 | 0 | 2 | 0 |
| Physics | 0 | 0 | 4 | 0 | 1 | 0 | 0 | 0 |
| Chemistry | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| Science & Technology | 21 | 10 | 5 | 11 | 15 | 18 | 13 | 17 |
| General Sciences | 21 | 11 | 15 | 20 | 19 | 18 | 15 | 17 |
| **Grand Total** | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 10 |

## Analysis of UPSC Preliminary Examination Questions Asked in Previous Years (2011–2025) for Science & Technology

| Units | Topic | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 |
|---|---|---|---|---|---|---|---|
| I | Science & Technology in India | | | | | | |
| II | Energy | 5 | 2 | 2 | 4 | 4 | 6 |
| III | Space Technology | 3 | | | 1 | 1 | 2 |
| IV | Communicaton & Information Technology | 3 | | | 1 | 1 | 4 |
| V | Defence Technology | | | | 1 | | 2 |
| VI | Artificial Intelligence, | | 1 | | | 1 | |

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# PART A
# SCIENCE AND TECHNOLOGY IN INDIA

## IN THIS PART...


**UNIT – I** Science and Technology in India
**UNIT – II** Energy
**UNIT – III** Space Technology
**UNIT – IV** Communication and Information Technology
**UNIT – V** Defence Technology
**UNIT – VI** Artificial Intelligence, Robotics and Nanotechnology
**UNIT – VII** Intellectual Property Rights (IPR) and Related Rights
**UNIT – VIII** Universe and Particle Physics
**UNIT – IX** Biotechnology
**UNIT – X** Diseases and Various Immunisation Programmes
**UNIT – XI** Miscellaneous Topics


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**UNIT-I** Science and Technology in India

# CHAPTER 1
# DEVELOPMENT OF SCIENCE AND TECHNOLOGY IN THE PAST, PRESENT AND FUTURE

## IN THIS CHAPTER...


**1.1 Science in Everyday Life**

**1.2 How is Science and Technology Affecting Human Life?**

**1.3 Indigenisation of Technology and Developing New Technology**

**1.4 Development of Science and Technology in Ancient India**


Pythagoras theorem is found in Budhayan's **Shulba Sutra.**

In **Brahma Sputa Siddantika**, there is the introduction of negative numbers and operations on zero.

The first arithmetic textbook, is **Ganit Sara Sangraha**, which was written by Mahaviracharya in 850 A.D.

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**1.5 Development of Science and Technology in Medieval India**

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During Medieval times, the two outstanding works in Mathematics were **Ganitasara** by Sridhara which was about numbers, multiplication, cubes, division, square roots, mensuration among others, and **Lilavati** written by Bhaskara.

**1.6 Development of Science and Technology in Modern India**
- 1.6.1 First Phase (1947–1967)
- 1.6.2 Second Phase (1967–1987)
- 1.6.3 Third Phase (1987 to Present)

**1.7 Indian Scientists and Their Contributions**
- 1.7.1 Srinivasa Ramanujan Aiyangar (1887–1920)
- 1.7.2 Chandrasekhara V. Raman
- 1.7.3 Jagdish Chandra Bose
- 1.7.4 Homi Jahangir Bhabha
- 1.7.5 Vikram Ambalal Sarabhai
- 1.7.6 A.P.J. Abdul Kalam
- 1.7.7 Krishnaswamy Kasturirangan
- 1.7.8 Anant Mashelkar
- 1.7.9 Tessy Thomas
- 1.7.10 Udupi Ramachandra Rao
- 1.7.11 Venkatraman Ramakrishnan
- 1.7.12 Verghese Kurien (1921–2012)
- 1.7.13 Satyendra Nath Bose
- 1.7.14 Subrahmanyan Chandrasekhar (1910–1995)
- 1.7.15 Har Gobind Khorana
- 1.7.16 M. Visvesvarya
- 1.7.17 Raja Ramana

**1.8 Rational Use of Science and Technology: Key to Development of Rural India**

**1.9 Lacunas in Science and Technology Research in India**

India spends approx. 0.7% of its GDP on R&D, but is still behind many developing and developed nations. The world average is around 1.8%.

**1.10 Recent Schemes and Programmes in News**

**1.11 Atal Innovation Mission (AIM)**

Atal Innovation Mission (AIM) including Self-Employment and Talent Utilisation (SETU) is Government of India's endeavour to promote a culture of innovation and entrepreneurship.

- 1.11.1 Core Functions of AIM

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1.11.2 Impact of Atal Innovation Mission
1.11.3 Atal Tinkering Laboratories (ATLs)
1.11.4 Atal Community Innovation Centre (ACIC) Programme
**1.12 Atal Innovation Mission (AIM) 2.0**

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Science is defined as any systematic and empirical activity that enhances the physical world knowledge, which produces the same result under different conditions. Technology can be defined as that activity which makes a productive use of this knowledge.

## 1.1 Science in Everyday Life

In today's world, science and technology plays a very crucial role in our daily lives, impacting everything we do. It empowers us to understand and adapt to constant changes. Advances in medicine, communication and transportation have transformed our lives significantly. Science has revolutionised our use of electricity, enabling communication via telephone lines, entertainment through television, and illumination. The introduction of high-yielding crop varieties in the 1940s, coupled with new fertilisers and pesticides, sparked the Green Revolution, revolutionising agriculture and increasing food production. Breakthroughs like **Edward Jenner's vaccine for smallpox in the 1700s**, the discovery of germs causing diseases in the 1800s, and Alexander Fleming's first antibiotic, penicillin in the 1920s have revolutionised medicine, eradicating diseases and improving health worldwide. Scientific knowledge influences public policies and personal decisions across various domains, from energy and health to transportation and conservation, shaping our modern way of life.

## 1.2 How is Science and Technology Affecting Human Life?

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Science is a base of methodical knowledge, which includes a chain of steps to get the possible outcome. Science is generally related to theories and it also gives special focus on analysis and brings many advantages to mankind, while technology is concerned with the applied science and includes processes, wherein the study of a particular science is undertaken with the help of tools and knowledge.

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## Advantages

- Science and technology makes human **life simpler** through various innovations in every sector.
- With the invention of **Artificial Intelligence**, people are making **lesser mistakes**. The use of technology has many benefits, it **saves time** and effort and also provides **quality outcomes and life with high production.**
- In medical science, technology brought a revolution through **discovery of various medical or surgical equipments, medicines and procedures** which are helping in **extending the life span of human beings** and saving many lives.
- Science and technology also make life easy to bring a **revolution in communication** system. Communicating through telephones, cellular phones and emails with the help of computers and the internet brought a revolution in communication system. Satellites-based communication system is also very useful for human beings.
- The events broadcast across the globe have eased with the help of **TVs and satellite radios.**
- The **software industry, being one of the world's most progressive industries, has evolved as a result of computers.** Everyone, including the teenagers, is using these latest gadgets to stay entertained.

## Disadvantages

With such advancements in the field of latest technology, it also brings out many disadvantages.

- It negatively **affects the environment.** The **advanced transportation** results in air pollution.

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- With the adoption of automatic machineries in the factories for improving production with accuracy, **people are losing their jobs.**
- These **machines may seem inexpensive** and good to use but **require lots of maintenances, which ultimately make them expensive.**
- Our healthiness is also being affected by technology, thereby **causing eye disease, enhanced body weight and cancer due to specific radiation technology.**
- People are becoming lazy and dependent due to technology.
- Now-a-days, children prefer to play computer games in place of outdoor games, which are not good for their physical growth.
- People are also **facing privacy issues like email hacking, cell phone signals interceptions,** etc.
- There is a huge increase in the number of **cyber-crimes, fraud** and **cheating cases** due to the advancement of technology.
- Above all, due to the latest technology, there is **increase in the issues of internal security and terrorist threats.**
- If the usage of these advancement and inventions takes place in a right way, we can benefit from them. On the other hand, human beings will face the disadvantages due to technology, if their intentions are not - positive.

## 1.3 Indigenisation of Technology and Developing New Technology

Indigenisation is an action or process to bring anything under the control, dominance or influence of the people those are native to an area. It is a substitution of imported products which are manufactured within the country.

'Make in India' scheme was launched in 2014, to improve India's manufacturing. This scheme was to support the indigenisation of technologies in various sectors like electronics, ICT, defence, space, transport, etc.

It leads to the utilisation of its own resources by the nation to develop itself. Due to use of own natural resources, it will be cheaper and easy to

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harness. It is not only a technological advancement, but also a huge morale booster for any nation. Science has always made work easier and even more efficient over the period of time.

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### Advantages of Indigenisation of Technology

There are various advantages of indigenisation of technology:

- Firstly, it has a huge impact on psychological and socioe-conomical development.
- It will open a new vista for various jobs in manufacturing and other sectors.
- It will make the product very cost effective, and much cheaper for the same quality of product that we import from other nations.
- It will improve scope of research and development sector in India. Hence, India will lead more projects in the name of innovation.

### Limitations of Indigenisation of Technologies

- Indigenous technology may be cheaper but inferior to other technologies, available in the global market.
- Indigenous technology may lead to higher wastage and increase production costs which will reduce the demand for a product.
- Since indigenous technology is tested in only one country, there are chances that it may not work if environment or other factors change.

### Development of New Technologies

**Since its independence, India has been working immensely towards becoming a self-sustained nation and has integrated indigenisation of technology in various sectors of economy which was completed in five phases to strengthen science and technological advancements.**

Phase I: It was focused towards building of relevant infrastructure.
Phase II: It included reorientation of various facilities.
Phase III: It promoted the indigenous technology.
Phase IV: It involved in moving towards the economic liberalisation.
Phase V: It involved development of science and technology towards becoming an economically liberalised nation.

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**There are six main sectors which promotes indigenisation and development of new technology in India:**

**Agriculture:** It always has a major share in our GDP.

**Automobile:** It is a leading employment provider in India.

**Defence:** The prime sector for maintaining security of India.

**Information and Communication Technology:** Fastest growing and the most needed sector in India.

**Healthcare and Biotechnology:** An emerging field and the future of science and technology.

**Space:** Symbolic significance in the field of science and technology.

## 1.4 Development of Science and Technology in Ancient India

- A good knowledge of measurement and geometry is showed in the town planning of Harappa. The **Shulba Sutras** is supposed to be the origination of Indian Mathematics.
- From the Indus Valley excavation points, bronze and copper artefacts, and glazed potteries were found, which shows their knowledge of highly modified metallurgy.
- There was awareness in the Vedic people regarding the fermenting of grains and fruits, tanning leather, as well as dyeing process.
- The first century A.D. also witnessed mass production of metals such as iron, copper, silver, gold and also alloys such as bronze and brass.
- Indian contribution to ship building can be witnessed at Lothal, a site in Gujarat, which houses the remains of dockyard, further providing proofs of trade flourishing through sea in those days.
- **Varahamihira** was among the nine gems in the court of Vikramaditya. The king gave him the title of 'Varaha' because of his accurate predictions. His **Brihat-Samhita** is considered a pioneer work in the field of Astronomy. He observed that the moon revolves around the earth and similarly, the earth revolves around the sun.

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- Baudhayan was the first person to calculate the value of pie; Pythagoras theorem is found in Budhayan's **Shulba Sutra.**
- The summary of Mathematics was mentioned in Aryabhatta's book, known as **Aryabhatiya.** He explained zero as a symbol as well as a concept, which helped him in understanding the accurate distance between the earth the moon. He also mentioned that **'Earth is round and rotates on its own axis'**. He also correctly stated that reflected sunlight is the reason behind the lustre of the moon and the planets.
- Brahmagupta introduced multiplication methods in which there is the usage of place value in the exact way as it is used these days. In his famous work, **Brahma Sputa Siddantika,** there is the introduction of negative numbers and operations on zero.
- Bhaskaracharya is renowned for his book **Siddanta Shiromani,** which has four sections: **Beejaganit (Algebra), Lilavati (Arithmetic), Giahaganit (Planets Mathematics) and Goladhyaya (Sphere).** He had also introduced a method to solve algebraic equations called as Chakrawat Method or the Cyclic Method.
- The first arithmetic textbook, is **Ganit Sara Sangraha,** which was written by Mahaviracharya in 850 A.D.
- Kanad, a sixth-century scientist, propounded atomic theory, which is very much similar to the present-day theory of atom. As per the theory, material world, which is formed of kanas, is invisible and indestructible.
- Varahamihira played an important role in the fields of Geology, Hydrology and Ecology. He had also stated that the presence of termites and plants could very well be the indicators of the presence of underground water. In his book, **Brihat Samhita**, he tried to correlate the natural process like undersea activities, planetary motions and unusual cloud formation, which are believed to be the causes of earthquake.
- Ayurveda is the oldest branch of medical science, which was developed in ancient times. The oldest known medical book of the world is **Atreya Samhita.**
- **Charak is known as the Father of Susruta and Ayurvedic medicine** and is also known as the **Father of Surgery.** In Atharva Veda, for the

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first time, diseases, their cure and medicines were mentioned for fever, cough, consumption, diarrhoea, dropsy, sores, leprosy, etc.

- In India, there were emergence of medicine and learning centres at places such as Takshila and Varanasi.
- **Charaksamhita** by Charak and **Sushrutsamhita** by Sushruta are the two vital texts in this field. In the court of Kanishka, Charak was the Raj Vaidya (royal doctor). He is credited as the first person who talked about digestion, metabolism and immunity. He also knew the concepts of Genetics. He greatly contributed in Rhinoplasty (plastic surgery) as well as in ophthalmic surgery (removal of cataracts). Surprisingly, modern surgeons use techniques similar to that of Sushruta during plastic surgeries.
- Yoga originated during ancient time as an allied Ayurveda science, for curing the body at the physical and mental level without any medicine. The Sanskrit word **Yoktra (yoking of the mind to the inner self after detaching it from the outer subjects of senses)** is used for the derivation of the term **Yoga**. Hathyoga is physical and mental yoga, whereas Rajayoga is mental yoga. The purpose of yoga is the achievement of mental, physical, emotional and spiritual balance through bondage liberation and self-realisation. Patanjali is the person behind this great science. In his book Yoga Sutras, he mentioned **'Aum'** as a cosmic sound, spoken of as the symbol of God. Among his other works, Patanjali is known for his book on medicine and his work on Panini's grammar, known as **Mahabhasaya.**

## 1.5 Development of Science and Technology in Medieval India

- In the medieval period (eleventh to eighteenth centuries), India's science and technology was influenced by dual scenarios: First, by the earlier ancient tradition and second, by Islamic and European traditions.
- People were invited from Arabia, Persia and also from Central Asia to **teach in the *madarsas*.** Arithmetic, Mensuration, Geometry, Public Administration, Astronomy, Accountancy and Agriculture were some

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subjects included in primary education. Although numerous efforts were made by the rulers of that time to reform the education system, science did not make much development.

- Akbar was focussed on producing different domestic animal breeds, including elephants and horses. The **Tuzuk-i-Jahangiri** is Jahangir's book having weeding and hybridisation records of his experiments and observation.
- **Lilavati Kammdipika, Suddhantadipika and Lilavati Vyakhya** are the books written by Gangadhar in which he confers trigonometric terms such as sine, cosine, tangent and cotangent and their rules.
- Akbar made Mathematics as a study subject during his regime.
- During the early medieval period, there were two outstanding works in Mathematics, one was **Ganitasara** by Sridhara which was about numbers, multiplication, cubes, division, square roots, mensuration among others, and **Lilavati** written by Bhaskara.
- During Tipu Sultan's regime, a paper-making factory was established in Mysore, that produced a gold-surfaced paper. The Mughals were familiar with the technique of producing gunpowder with the use of sulphur, saltpetre and charcoal in variable ratios for its usage in different guns. The casting of cannons was mentioned in **Tuzuk-i-Baburi.**
- **Ain-i-Akbari** gives a description of the 'Regulations of the Perfume Office of Akbar'. The attar of roses was among the most famous perfumes.
- An astronomical observation post was established by Firoz Shah Tughlaq. An astronomical instrument named as **Yantraja** was developed by Mehendra Suri, who was the court astronomer of Firoz Shah.
- Five astronomical observatories were developed by Sawai Jai Singh, the Maharaja of Jaipur, in five different cities, i.e., Delhi, Ujjain, Varanasi, Mathura and Jaipur.
- **Sarangdhara Samhita** recommended opium as a medicine.
- A new medicine system was introduced in India by Muslims during the eleventh century, i.e., Unani medicine system. Majinye Diyae by Hakim Diya Muhammad is a compilation of Arabic, Persian and Ayurvedic medical knowledge.

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- The field of agriculture witnessed some important changes like introduction of trees, new crops and horticultural plants. The major crops during that time were wheat, rice, barley, millets, pulses, oilseeds, sugarcane, indigo and cotton. New and improved horticultural methods were adopted.

## 1.6 Development of Science and Technology in Modern India

### 1.6.1 First Phase (1947–1967)

Pt. Jawaharlal Nehru, the first Prime Minister of India, initiated various reforms to promote higher education in the field of science and technology in India. **The first IIT (Indian Institute of Technology) was established in 1951 at Kharagpur in West Bengal by Maulana Abul Kalam Azad.** Thereafter, many other IITs were shortly established at Bombay, Madras, Kanpur and Delhi during 1950s and 1960s.

Various achievements between 1947 and 1962 were food grain production rose through 34 million metric tonnes, installed capacity of power generation increased through 79 million kilowatts and an overall increase of industrial production to 94% during these periods. **Radio broadcasting in India was started in 1927. It was named 'All India Radio' in 1937.** Since 1957, it has been called '**Akashvani**'. The Tata Group established Tata Consultancy Services in 1968, which was among the largest software producers in the 1960s.

During 1960s, India's relation with Soviet Union helped the Indian Space Research Organisation (ISRO) to build up the Indian space programme and advanced nuclear programme. **Operation 'Smiling Buddha' is the name of the first nuclear test that was conducted on 18 May 1974 at Pokhran in Rajasthan.**

The Indian space programme strives for self- reliance in the usage of space technology in support of national development. **It includes the launching of the first Indian space satellite Aryabhatta in 1975,** followed by **Bhaskara I** and **Bhaskara II** with Soviet Union assistance, the

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**Apple satellite on the European Arianne rocket** and the **Rohini satellite on India's own SLV-3 rocket.** Subsequently, **INSAT I-B was launched in the year 1983**, which catered to television, radio, meteorological and telecommunication services. The **Council for Scientific and Industrial Research (CSIR)** and **Defence Research and Development Organisation (DRDO),** the two government organisations, are involved in an extensive range of research in the field of science and technology for both defence and civil purpose. India's space programme started with the Soviet Union's financial support, which assisted the Indian Space Research Organisation in the achievement of its aims such as the establishment of **Thumba Equatorial Rocket Launching Station** for remote sensing satellites launching, and also in development of Aryabhatta – India's first satellite, while also sending space astronauts.

### 1.6.2 Second Phase (1967–1987)

America supported India to begin first nuclear power generation in **Tarapur Atomic Power Station and Canada's CANDU reactors.** The **Green Revolution** was brought in India during 1960s, which led to the production of high-yield food varieties and use of chemical fertilisers in India. The farmers belonging to wheat-growing areas (Haryana, Punjab and western Uttar Pradesh) were most benefited. The application of modern science and technology supported the agriculture to flourish rapidly. The efforts of **Indian Council for Agricultural Research (ICAR)** have played a major role in the cultivation of high-yielding food crops and post-harvest technology**.** This made India self-dependent in maintaining the food buffer stock for the people.

### 1.6.3 Third Phase (1987 to Present)

In 1986, the **Department for Biotechnology** Research was established under the aegis of Ministry of Science and Technology. Both the Indian government and the private sector could invest in the field of Biotechnology for medical and agricultural applications. **Bangalore is known as the technological capital of India**. Hyderabad, Bangalore and Chennai are

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considered as the golden triangle of IT and technology, which constitute the vertebrae of R&D, Science and Technology and Indian Manufacturing. Under the Prime Minister's charge, in the year 1981, the Department of Ocean Development was established to harmonise and direct the country's activities in ocean research**.** It works in biological, chemical, physical and geological field besides Meteorology and Geophysical Oceanography.

**India first entered the space mission with the successful launch of Chandrayaan-I on 22 October 2008.** The Moon Impact probe from Chandrayaan-I was released on 18 November 2008, from a height of 100 kilometres. During its 25-minute descent, Chandra's Altitudinal - Composition Explorer (CHACE) recorded evidence of water. **On 5 November 2013, the Indian Space Research Organisation (ISRO) launched a second space mission of India, i.e., Mars Orbiter Mission,** which is also known as Mangalyaan. It is the first interplanetary mission of India, making ISRO the fourth space agency to reach Mars after Soviet Space Programme, NASA and European Space Agency. India is the first Asian nation to reach the orbit of Mars and also the first nation to get the success in its first effort. India also launched Chandrayaan-II mission, on 22nd July 2019. **The Thirty Meter Telescope (TMT)** is an anticipated astronomical observatory having an extremely large telescope, which is to be constructed on the summit of Mauna Kea in the United States, near Hawaii, and further going to be planned for ultraviolet to mid-infrared observations (0.31 to 28 μm wavelengths). There is government support from several nations that spend on R&D, such as **Japan**, **United States**, **Canada**, **China** and **India** for the development of this telescope. Indian participation in the project was approved by the Union Cabinet in 2014. The project has faced considerable challenges. Mauna Kea, an inactive volcano, and by some measures the tallest mountain in the world, hosts multiple telescopes. However, the project has faced local oppositions on the grounds of religious and cultural sentiments. Though permits for constructing the TMT were availed, the Supreme Court of Hawaii invalidated them in 2015. Permissions were restored in 2018, though construction is yet to begin because of continued local opposition.

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## 1.7 Indian Scientists and Their Contributions

### 1.7.1 Srinivasa Ramanujan Aiyangar (1887–1920)

- Srinivasa **Ramanujan** worked during the British Rule in India.
- He was an **Indian mathematician** and made substantial contributions to **mathematical analysis, number theory, infinite series and continued fractions,** including solutions to mathematical problems then considered unsolvable.
- He was one of **the first Indian to be elected a Fellow of Trinity College, Cambridge and the youngest Fellows of the Royal Society.**

### 1.7.2 Chandrasekhara V. Raman (1888–1970)

- He was the first Asian to be conferred with the Nobel Prize in Physics in the year 1930.
- Various experiments were conducted and also a research paper on molecular scattering of light was prepared by him.
- The debate on the nature of light, whether it is waves or particles, was ended by the **'Raman Effect'**, which proved that light is made up of particles known as photons.

### 1.7.3 Jagdish Chandra Bose (1858–1937)

- J.C. Bose was knighted for his paper in 1917 on 'The Electromagnetic Radiation and Polarisation of Electric Ray'**.**
- Dr. Bose invented **Crescograph**, which is used for the recording of even the millionth part of a millimetre of plant growth and its - movement.
- He has demonstrated through cresograph that the sap's upward movement in plants is the living cells activity.

### 1.7.4 Homi Jahangir Bhabha (1909–1966)

- Bhabha is also known as the **Father of Indian Nuclear Science.**

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- He suggested Sir Dorab Ji Tata for establishing an institution to lay the country's foundation as a world nuclear power, which led to the establishment of Tata Institute of Fundamental Research (TIFR).
- The country's first atomic research centre, which is Bhabha Atomic Research Centre (BARC), was set up at Trombay, under his expert guidance.
- **The country's first atomic reactor – Apsara –** was established during his period.
- He became the first chairman of Atomic Energy Commission, which was established in the year 1948.
- He was a recipient of Padma Bhushan by the Government of India. Homi Bhabha Prize is given for the special contribution in **'Theoretical Physics'.**

### 1.7.5 Vikram Ambalal Sarabhai (1919–1971)

- Under the expert guidance of Dr. Vikram Ambalal Sarabhai, **India launched its first satellite, Aryabhatta.**
- He established industries like Sarabhai Geigy Ltd., Sarabhai Glass, Sarabhai Chemicals, Sarabhai Merck Ltd. and many others.
- His efforts saved crores of Indian rupees by initiating the mission of antibiotics and penicillin production and military hardware manufacturing in India.
- The credit for establishing various reputed institutes in India goes to Dr. Vikram Ambalal Sarabhai.
- One such prominent example is Indian Institutes of Management (IIMs).
- Dr. Sarabhai also served as the chairman of the Indian National Commission for Space Research (INCOSPAR) and the Atomic Energy Commission.
- He was the director during the establishment of the Thumba Equatorial Rocket Launching Station (TERLS).
- Sarabhai was conferred with the Padma Bhushan and the Padma Vibhushan.
- He had also taken initiatives for spreading education with the help of satellite communication to the villages.

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### 1.7.6 A.P.J. Abdul Kalam (1931–2015)

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- **Dr. A.P.J. Abdul Kalam** was born on 15 October 1931 in the island town of Rameshwaram in Tamil Nadu.
- He was the eleventh President of India.
- He was honoured with the highest civilian award of India, Bharat Ratna, in the year 1997 for his contributions in the science and engineering field.
- Dr. Kalam had served in Indian Space Research Organisation from the year 1963 to 1982. Dr. Kalam developed the **Satellite Launch Vehicle (SLV 3)** at Vikram Sarabhai Space Centre, which placed the satellite - Rohini into orbit.
- During his tenure as the Director of Defence Research Development Organisation (DRDO), he was entrusted with the responsibility of **Integrated Guided Missile Development Programme (IGMDP)** and subsequently he developed five defence services projects, namely, Prithvi, Trishul, Akash, Nag and Agni.

In his life, **A.P.J. Kalam has written many books**: *Target 3 Billion, Developments in Fluid Mechanics and Space Technology, India 2020: A vision for the New Millennium, Envisioning an Empowered Nation, A Manifesto for Change: A Sequel to India, You Are Born to Blossom, Reignited: Scientific Pathways to a Brighter Future, Transcendence: My Spiritual Experiences with Pramukh Swamiji, My Journey: Transforming Dreams into Actions, Indomitable Spirit, Ignited Minds: Unleashing the Power Within India, The Luminous Sparks, Mission India, Inspiring Thoughts, Forget Your Future: Candid, Forthright, Inspiring, Turning Points: A Journey Through Challenges.*

### 1.7.7 Krishnaswamy Kasturirangan(Born in October 1940)

- He headed ISRO from 1994 to 2003. Earlier, he also served as the Director of ISRO Satellite Centre and under his supervision, new generation spacecraft – the Indian National Satellite (INSAT-2) and the Indian Remote Sensing Satellites (IRS-1A and 1B), besides several scientific satellites, were developed.
- He served as the **Project Director for India's first two experimental earth observation satellites, Bhaskara-I and II.**
- He also led the country's initiative for entering into an era of the planetary exploration through extensive studies which led to the

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conceptualisation of Chandrayaan-I.

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- He was honoured with the Padma Shri (1982), Padma Bhushan (1992) and Padma Vibhushan (2000).

### 1.7.8 Raghunath Anant Mashelkar(Born in January 1943)

- He is also known by his other name Ramesh Mashelkar. He served as the Director General of the Council of Scientific and Industrial Research (CSIR).
- After independence, he played a significant role in shaping India's policies of science and technology. He served as a member of the Scientific Advisory Council to the Prime Minister while also leading the successive governments.
- Under the chairmanship of Mashelkar, a technical expert group on **patent laws was established in 2005**, so as to find out whether the amendments made in Indian patent law were TRIPS compliant or not. The committee concluded unanimously that the amendments were not TRIPS compliant.
- He was awarded the Padma Vibhushan in 2014.

### 1.7.9 Tessy Thomas (Born in April 1963)

- Tessy Thomas headed the programme on the Agni-IV missile of Defence Research and Development Organisation (DRDO), which was successfully tested in 2011. Thomas is also recognised as **India's 'Missile Woman' (Agni Putri).**
- In 2009, the Advanced Systems Laboratory in Hyderabad appointed her as the Project Director for 5,000 km range Agni-IV.
- Her outstanding contribution in the field of missile technology for making India self-reliant has made her win the Lal Bahadur Shastri National Award.

### 1.7.10 Udupi Ramachandra Rao (1932–2017)

- U.R. Rao was the former chairman of Indian Space Research Organisation.
- He was conferred with the Padma Bhushan in 1976.

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- He made a great contribution for the establishment of Indian satellite technology in 1972.
- Under his supervision, many satellites were designed and launched with an aim to provide communication, remote sensing and meteorological services. These include India's first satellite '**Aryabhatta**' in 1975, Bhaskara, APPLE, Rohini, INSAT-1 and INSAT-2 series of multipurpose satellites and the IRS-1A and IRS-1B remote sensing - satellites.

### 1.7.11 Venkatraman Ramakrishnan(Born in April 1952)

- He is an Indian-American and British structural biologist of Indian origin.
- He won the Nobel Prize in 2009 in Chemistry, **'for studies of the structure and function of the ribosome'**.

### 1.7.12 Verghese Kurien (1921–2012)

- Verghese Kurien is **Father of the White Revolution** in India.
- He is the person behind **Operation Flood**, which was the largest agricultural dairy development programme in the world and India is the top ranked milk producer in the world.
- **India's largest food brand Amul Cooperative was established with his help.**
- He also made India self-sufficient in the production of edible oil, taking on a powerful, entrenched and violently resistant oil supplying lobby.
- He is one of the greatest supporters of the **cooperative movement** in the world.
- His work helped in the upliftment of millions of people out of poverty in India as well as outside.

### 1.7.13 Satyendra Nath Bose (1894–1974)

- He was an Indian physicist and mathematician, who is best known for his work with Albert Einstein on the **'Bose–Einstein Statistics'** that has application in information retrieval in present time.
- He worked on boson also called God particles, which revolutionised the modern Physics.

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- The name Boson was given after his name and Bose is known as the **'Father of God Particle'.**

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### 1.7.14 Subrahmanyan Chandrasekhar(1910–1995)

- He was an astrophysicist. He discovered and concluded that massive stars can collapse under their own gravity to reach enormous or even infinite densities. Now, these collapsed stars are referred as **neutron stars** and **black holes.**
- Small and dense stars go on to become white dwarfs while larger stars, after a supernova, can become neutron stars or black holes.
- He was awarded with the Noble prize in Physics in 1983 for 'theoretical studies of the physical process of importance to the structure and evolution of stars', sharing the prize with William Fowler.
- NASA's **Chandra X-ray observatory** is named after him.

### 1.7.15 Har Gobind Khorana (1922–2011)

- **Har Gobind Khorana** was an **Indo-American biochemist.**
- He received **Physiology** or **Medicine Nobel Prize in 1968** with two other colleagues for the study of nucleotides in nucleic acids, and analyse genetic code of the cell those are responsible for the control of protein synthesis.
- Khorana also received the **Louisa Gross Horwitz Prize** from Columbia University in the same year, with Nirenberg.
- Khorana served as faculties of three universities in North America.
- He also received the **National Medal of Science in 1987.**

### 1.7.16 M. Visvesvaraya (1861–1962)

- He was an **Indian civil engineer and statesman** and the 19th Diwan of Mysore, who served from 1912 to 1919.
- He received India's highest honour, **Bharat Ratna, in 1955.**
- He was knighted as a **Knight Commander of the British Indian Empire (KCIE)** by King George V for his contributions to public good.
- **India, Sri Lanka and Tanzania** are celebrating **15 September as Engineer's Day in his memory.**

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- He was **Chief Engineer to work for flood protection system for city of Hyderabad** and also involved in the **construction of Krishna Raja Sagara dam** in North-West suburb of Mysore.

### 1.7.17 Raja Ramanna (1925–2004)

- **Raja Ramanna** was an **Indian physicist** and has played a very important role in earlier stages of **India's nuclear programme.**
- In the beginning Ramana had worked under Homi Jehangir Bhabha, and later became the director of this programme in 1967.
- Ramanna had **expanded and supervised many scientific research on nuclear weapons.** He was the first directing officer of the small team of scientists who **supervised and carried out the test of the nuclear device,** under the codename ***Smiling Buddha,*** in **1974.**
- Ramanna was associated **for more than four decades with India's nuclear programme and also directed it.** He also initiated industrial defence programmes for the Indian Armed Forces.
- He received **Padma Vibhushan**, in the honour of his services to **build India's nuclear programme.**

## 1.8 Rational Use of Science and Technology: Key to Development of Rural India

Approximately 70% of the country's population resides in rural areas, making rural development pivotal for national progress. To achieve this, it is crucial to prioritise conserving natural resources and adopting sustainable technologies. Many rural households still rely on fossil fuels for cooking, exposing them to harmful smoke. To address this, the government launched an initiative to provide LPG connections to the underprivileged. Additionally, the introduction of innovative solutions like the **'Solar Dome',** which was launched in 2016, will help to bring light to rural homes without electricity. These solar-powered devices store sunlight to provide illumination for 8–10 hours, benefiting both adults and school children who

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often study under inadequate lighting. In December 2022, the **first solar dome was opened for public viewing in Bengal's Silicon Valley, Eco Park, New Town, Kolkata.** The dome is 28.950 metre high and has a 45-metre diameter. It has 2000 active solar panels that continuously generate solar power that can produce 332 KW of solar power at high noon.

**First solar dome in Bengal's Silicon Valley**

In construction, advancements in low-cost, eco-friendly materials and techniques, such as the **'Fluorogypsum Plaster'** and portable C-brick machine, offer affordable housing solutions, especially in rural areas. Moreover, research institutions like Structural Engineering Research Centre (SERC) and Central Building Research Institute (CBRI) are pioneering earthquake-resistant structures and sustainable building technologies.

In healthcare, despite the establishment of Primary Health Centres (PHCs) and Community Health Centres (CHCs), rural areas face shortages of medical staff and modern facilities. Improved infrastructure and technology integration could vastly enhance healthcare accessibility.

Science and technology also plays a crucial role in agriculture through systems like the **Agriculture Information System (AIS)** and educational platforms like the **DD Kisan Channel** and online universities. Initiatives like Vigyan Prasar's outreach programmes and the National Informatics Centre's District Information System further facilitate rural development by

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disseminating essential information and promoting education and skill development.

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## 1.9 Lacunas in Science and Technology Research in India

We have transitioned from environmental determinism, where nature's forces dominated, to possibilism, where technology humanises nature. Despite its drawbacks, technology, when leveraged effectively, can alleviate human suffering by tapping into nature's hidden potential through quality research.

Unfortunately, the Indian government has not prioritised bolstering the country's research capabilities. Several factors contribute to this apathy:

- Many Indian universities lack a positive stance towards research, failing to inspire students or provide adequate support.
- Government expenditure on research in India is mediocre, with only about 0.8% of GDP allocated despite persistent calls for increased funding.
- Limited resources and research opportunities primarily benefit prestigious institutions, leaving regional universities struggling and students under undue pressure.

**R&D FACTUAL DATA**

India spends approx. 0.7% of its GDP on R&D, but is still behind many developing and developed nations. Brazil, Russia, China and South Africa spend around 1.2%, 1.1%, above 2% and 0.8%, respectively. The world average is around 1.8%. The developed countries such as the United States, Sweden and Switzerland spend about 2.9%, 3.2% and 3.4%, respectively. Israel spends 4.5% of its GDP on R&D, the highest in the world.

The share of Central government's spending on R&D is nearly 45.1% of total expenditure, while others like private sector industries having contributed about 38.1%, followed by state governments, i.e., up to 7.4%.

There is hardly any increase in participation of women in R&D, in spite of so many provisions and schemes announced for them.

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Although the share of India in global research is low, it is trending positively year after year. And the number of researchers in the recent decade had also gone up.

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Note: **In the Interim Budget 2024–25, a corpus of ₹1 lakh crore was announced to promote technological innovation in sunrise sectors. This year the government's moto is '*Jai Jawan, Jai Kisan, Jai Vigyan and Jai Anusandhan*'.**

- Governance issues, including poor leadership, political interference in funding allocation, and bureaucratic dominance in research decisions, hinder progress.
- Corporations prioritise research for immediate financial gains, neglecting fundamental scientific exploration, which discourages students from pursuing scientific careers.
- A lack of research culture necessitates a pedagogical shift to promote scientific inquiry among youth. The government must establish institutions and frameworks to cultivate a research-centric ethos.

Addressing these challenges requires a concerted effort to prioritise research, allocate adequate funding, and foster a culture of scientific exploration and innovation.

## 1.10 Recent Schemes and Programmes in News

### *1. Anusandhan National Research Foundation (ANRF)*

- It was established with the Anusandhan National Research Foundation (ANRF) Act, 2023. ANRF acts as an apex body to provide high-level strategic direction of scientific research in the country as per the **recommendations of the National Education Policy (NEP), 2020.**
- It facilitates **collaborations among industry, academia, and government departments.**

### *2. Skill India Programme*

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- Council of Scientific and Industrial Research (CSIR) – National Physical Laboratory (CSIR-NPL) has **played a role as the custodian of Indian Standard Time (IST), generated using an atomic time** in standardising **measurements of gas and airborne particles for monitoring atmospheric pollution.**
- It provides training under the "Skill India" programme.

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### 3. One Week – One Lab Campaign

- It aims to create awareness about technologies and services offered by CSIR-NPL.
- It provides solutions to societal problems.
- It develops scientific temperament among students through interactions with schools. ***180 schools in Delhi-NCR have been exposed to NPL labs for different activities***, and more schools will be included in future interactions.

### 4. Science and Heritage Research Initiative (SHRI)

- It explores the **soundproofing qualities** of ***traditional Pattamadai mats*** made up from ***korai grass and cotton threads*** and investigates their potential use in noise reduction in classrooms and recording studios.
- It aims to increase demand for this traditional art from Tirunelveli, Tamil Nadu.

### 5. National Deep Tech Startup Policy (NDTSP)

In **July 2023**, a ***draft of the National Deep Tech Startup Policy (NDTSP)*** was released for public comment, aiming to foster innovation, economic growth, and societal development while laying the foundation for India's emerging Deep Tech Startup ecosystem.

#### What is Deep Tech?

- Deep tech refers to **those startups whose business model is based on high-tech innovation in engineering, or significant scientific advances and have developed intellectual property that promises a huge impact** but is yet to be realised.

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- These are **mainly based** on artificial intelligence or machine learning, or new or emerging technologies like blockchain, computer imaging, and VR.
- Examples: **AI used to predict natural disasters.**

***Key Pillars of NDTSP***

- Ensuring the Security of India's Economic Future
- Facilitating a Seamless Transition to a Knowledge-Driven Economy
- Bolstering National Capability and Sovereignty through the Atma Nirbhar Bharat Imperative
- Promoting Ethical Innovation

***Key Foundational Elements of the Policy***

- Funding and Innovation
- Talent Development
- Access to Advanced Infrastructure and Technology
- Public Procurement and Market Opportunities
- Intellectual Property (IP) Protection

## 1.11 Atal Innovation Mission (AIM)

- Atal Innovation Mission (AIM) is mandated to create an umbrella structure to oversee innovation ecosystem of the country and revolutionise the innovation ecosystem – touching upon the entire innovation life cycle through various programmes.

### 1.11.1 Core Functions of AIM

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The Atal Innovation Mission shall have two core functions:

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1. Entrepreneurship promotion through ***Self-Employment and Talent Utilisation*****,** wherein innovators would be supported and mentored to become successful entrepreneurs.

**STUDENT ENTREPRENEURSHIP PROGRAMME**

**Atal Innovation Mission (AIM), NITI Aayog**, in collaboration with **Dell Technologies** has launched **Student Entrepreneurship Programme 2.0 (SEP 2.0)** for **young innovators of Atal Tinkering Labs (ATLs)**.

**About SEP 2.0 and its significance:**

- It will allow student innovators to work closely with Dell volunteers.
- They will receive **mentor support; prototyping and testing support; intellectual property registration and patenting of ideas, processes, and products**; manufacturing support; as well as the launch support of the product in the market.

**About Student Entrepreneurship Programme:**

- SEP 1.0 began in January 2019.

2. ***Innovation Promotion:*** to provide a platform where innovative ideas are generated.
   - ❑ Atal Innovation Mission (AIM) including **Self-Employment and Talent Utilization (SETU)** is Government of India's endeavour to promote a culture of innovation and entrepreneurship.
   - ❑ Its **objective** is to serve as a platform for promotion of world-class Innovation Hubs, Grand Challenges, Startup businesses and other self-employment activities, particularly in technology driven areas.

## 1.11.2 Impact of Atal Innovation Mission

- The Mission has undertaken many bold and forward-looking initiatives such as **Atal Tinkering Labs (ATL) and Atal Incubation Centres (AIC),** which have received great traction.
- Many ministries/departments of Government of India have initiated innovation related activities with the help and technical support of AIM.

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- Under the ATL programme, more than 10,000 schools were established till 2020.
- More than 100 Atal Incubation Centres (AICs) are likely to be established around the country, supporting at least 50–60 startups each, over the first five years.
- More than 100 innovators/startups are expected to receive some support for productising their innovations.
- Each incubator is expected to foster 50–60 technology-driven innovative startups every four years.

### 1.11.3 Atal Tinkering Laboratories (ATLs)

With a vision to '**Cultivate one million children in India as Neoteric Innovators'**, **Atal Innovation Mission is establishing 'Atal Tinkering Laboratories' (ATLs) in schools across India**.

**Objective:** The objective of this scheme is to foster curiosity, creativity and imagination in young minds; and inculcate skills such as design mindset, computational thinking, adaptive learning, physical computing, etc.

**Financial Support:** AIM will provide grant-in-aid that includes a one-time establishment cost of ₹10 lakh and operational expenses of ₹10 lakh for a maximum period of 5 years to each ATL.

**Eligibility:** Schools (minimum Grade VI–X) managed by Government, local body or private trusts/society can set up ATL.

#### *Need for Such Labs*

- As the world grapples with evolving technologies, a new set of skills have gained popular acceptance and has come to be in high demand. For India, to contribute significantly during this age of rapid technological advancement, there is an urgent need to empower our youth with these 'skills of the future'.
- Equipped with modern technologies to help navigate and impart crucial skills in the age of the Fourth Industrial Revolution, the ATLs are at the vanguard of promoting the scientific temper and an entrepreneurial spirit in children today.

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### 1.11.4 Atal Community Innovation Centre (ACIC) Programme

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Atal Innovation Mission (AIM) launches **Atal Community Innovation Centre** (ACIC) Programme. ACIC programme aims at **spurring community Innovation in underserved and unserved areas of the country.** ACIC can serve as the bridge between the knowledge base existing in communities and the advanced technical ecosystem prevalent in the market base, addressing the needs of society.

#### *Features of the Programme*

- ACIC is a new initiative of Atal Innovation Mission **to support community innovation drive in the country.**
- The programme is directed to encourage the spirit of innovation through solution-driven design thinking to serve the society.
- It will focus on underserved/unserved regions of the country, which at present lack a vibrant start-up and innovation ecosystem.
- **ACIC will be established either in PPP mode or with support of PSUs and other agencies.**
- The maximum grant-in-aid support form AIM will be up to 2.5 crores subject following compliance to ACIC guidelines.

## 1.12 Atal Innovation Mission (AIM) 2.0

#### *Overview*

- The **Union Cabinet** has approved **AIM 2.0 (25 November 2024)** with an expanded scope and a budget of **₹2,750 crore.**
- It will be implemented by **NITI Aayog** till 31 **March 2028.**
- It aims to enhance **India's innovation and entrepreneurship ecosystem** and improve **global competitiveness.**
- **AIM 1.0** (launched in **2016**) focused on building innovation infrastructure.
- AIM 2.0 is a **step towards Viksit Bharat** that **aims to expand, strengthen and deepen India's already vibrant innovation and entrepreneurship ecosystem.**

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- With India at **rank 39 on the Global Innovation Index** and **home to world's third-largest start-up ecosystem**, the **next phase of AIM 2.0 is expected to further enhance India's global competitiveness.**
- The continuation of AIM will **directly contribute to creating better jobs, innovative products and high-impact services across sectors.**
- Whereas AIM 1.0 involved implementing programs that **built new innovation infrastructure** to strengthen India's then nascent ecosystem, AIM 2.0 involves **piloting new initiatives designed to fill gaps** in the ecosystem and scaling successes through central and state **governments, industry, academia and community**.

AIM 2.0 is designed to **strengthen India's innovation and entrepreneurship ecosystem in the following three ways:**

(a) by increasing input (i.e., ushering more innovators and entrepreneurs)

(b) by improving the success rate or 'throughput' (i.e., helping more startups succeed)

(c) by improving the quality of 'output' (i.e., producing better jobs, products and services)

**Two programs target increasing input to the ecosystem:**

The **Language Inclusive Program of Innovation (LIPI) aims** to build innovation and entrepreneurship ecosystems in India's 22 scheduled langes for lowering entry barrier confronting innovators, entrepreneurs and investors who do not speak English.

**Thirty Vernacular Innovation Centres** will be established in existing incubators. **These centres aim to foster innovation and entrepreneurship in India by making these fields more accessible to people who do not primarily speak English.**

The **Frontier Program aims to create customised templates** for the innovation and entrepreneurship ecosystems of Jammu and Kashmir (J&K), Ladakh, the North Eastern states (NE), Aspirational Districts and Blocks where 15% of India's citizens live. About 2500 new ATLs will be created for template development.

Both Atal Tinkering Labs (ATLs) and Atal Incubation Centres (AICs) are initiatives under the Atal Innovation Mission (AIM), **a flagship**

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**program of NITI Aayog, Government of India.**

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**ATLs** are innovation spaces in schools that provide students with hands-on experience with various technologies and tools.

**AICs**, on the other hand, are incubation centres that support the growth and development of startups and entrepreneurs.

**Atal Tinkering Labs (ATLs)**

**Purpose:** The purpose of ATLs is to promote a culture of creativity and innovation among students.

**Location:** ATLs are **established in schools across India**, including government, government-aided and girls/co-ed schools.

**Facilities:** ATLs provide access to tools and technologies like 3D printers, robotics, miniaturized electronics and do-it-yourself (DIY) kits.

**Impact:** ATLs aim to cultivate a problem-solving and innovative mindset in students.

**Atal Incubation Centres (AICs):**

**Purpose:** The purpose of AICs is to support the creation and nurturing of innovative startups and enterprises.

**Location:** AICs are established in universities, institutions and corporate sectors.

**Facilities:** AICs provide a range of resources and support, including access to mentoring, business planning, seed capital, industry partnerships and training. (Seed capital is the initial amount of money an entrepreneur uses to start a business.)

**Impact:** AICs aim to foster world-class innovation and create a supportive ecosystem for startups.

**Key Initiatives Under AIM**

1. **Existing Programs**:
    1. Atal Tinkering Labs (ATLs)
    2. Atal Incubation Centres (AICs)
    3. Atal New India Challenges

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4. Atal Grand Challenges
5. Mentor India, etc.

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2. **New Initiatives Under AIM 2.0:**
    1. **Piloting new programs** to address gaps in the ecosystem.
    2. **Scaling successful models** through government, industry and academia.

**Improving Output Quality (Better Jobs, Products and Services)**

**Industrial Accelerator Program:**

- **10 industry-led accelerators** in **critical sectors** under the **PPP (Public – Private Partnership) model**.
- **Atal Sectoral Innovation Launchpads (ASILs)**:

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- Establishes **iDEX-like platforms** in **central ministries** for startup integration.
- **10 launchpads** to be developed across key ministries.

**iDEX: The Innovations for Defence Excellence (iDEX) scheme**

Launched by the Hon'ble Prime Minister in 2018, iDEX aims to foster innovation and technology development in defence and aerospace by engaging innovators and entrepreneurs to deliver technologically advanced solutions for modernizing Indian Military.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. Atal Innovation Mission is set up under the:** **[UPSC(Pre) – 2019]**

(a) Department of Science and Technology

(b) Ministry of Employment

(c) NITI Aayog

(d) Ministry of Skill Development and Entrepreneurship

**2. Match List I (Indian Scientists/Scholars) with List II (Area of Work) and select the correct answer using the code given below the lists:** **[UPSC(Pre) – 2006]**

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| | List I (Indian Scientists/Scholars) | List II(Area of Work) |
|---|---|---|
| A. | C.N.R. Rao | 1. Telecommunication |
| B. | Jagdish Bhagwati | 2. Physics |
| C. | G.N. Ramachandran | 3. Economics |
| D. | Ashok Jhunjhunwala | 4. Solid-state Chemistry and Material Science |

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 4 | 3 | 2 | 1 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 4 | 1 | 2 | 3 |
| (d) | 2 | 3 | 4 | 1 |

**Answer Key**

| | |
|---|---|
| 1. (c) | 2. (a) |

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# CHAPTER 2
# DIFFERENT ORGANISATIONS, INSTITUTIONS AND DEPARTMENTS FOR THE DEVELOPMENT OF SCIENCE AND TECHNOLOGY

## IN THIS CHAPTER...


**2.1 Role of Science and Technology in the Developing World in the Twenty-First Century**

**2.2 Ministries and Departments for Science and Technology Under Central Government**

- 2.2.1 Department of Science and Technology (DST)
- 2.2.2 Department of Scientific and Industrial Research (DSIR)
- 2.2.3 Department of Atomic Energy (DAE)
- 2.2.4 Department of Space (DoS)
- 2.2.5 Department of Biotechnology (DBT)
- 2.2.6 Indian Council of Medical Research (ICMR) – Strategic Plan and Agenda 2030


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ICMR is working closely with the National AIDS Control Organisation and is

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committed to the global call to ‘End AIDS’ by 2030.

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**2.3 Science and Technology Policies in India**
2.3.1 Science, Technology and Innovation Policy, 2020
**2.4 National AYUSH Mission (AYUSH: Ayurveda, Yoga, Siddha, Unani and Homeopathy)**
2.4.1 Objective
2.4.2 Components of the Mission
2.4.3 Institutional Mechanism
2.4.4 Resource Allocation
2.4.5 Expected Effect of the Mission

The New Education Policy (NEP) approved by the Union Cabinet on July 29th, 2020 emphasises on schools to have more experimentbased classes and hand-on learning.

**2.5 Stem Education:Strengthening Policies for Future Skilled Workforce**
2.5.1 Stem Education andChallenges
2.5.2 Operation of the Scheme
**2.6 Indian Science Congress Association**

Celebrated on 28th February, the theme for National Science Day 2024 was ‘Indigenous Technologies for Viksit Bharat’.

**2.7 National Science Day**
**2.8 Some AutonomousInstitutions Engaged inBasic and AppliedSciences Research**

## 2.1 Role of Science and Technology in the Developing World in the Twenty-First Century

Advancements in science and technology are reshaping how people live, communicate, and do business. They are also driving economic growth. To keep up with this tech-driven world, developing countries need to invest in educating their youth and constantly update the skills of their workforce.

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New sectors like telecommunications, biotechnology, and nanotechnology are emerging, changing the way businesses operate. Technology can also improve healthcare and education in poorer nations.

However, many developing countries face challenges like a lack of skilled workers and capital. This contributes to the income gap between nations. Technologies like computing can boost productivity and improve services like education and healthcare, empowering individuals.

Technological innovations like nuclear batteries and solar power are making energy more accessible and affordable. But managing these revolutions brings challenges, such as ethical concerns and high costs, which can worsen inequality.

Investment climate and incentive structures are crucial for guiding resource distribution and encouraging research. Successful countries invest in research and development to drive innovation and growth.

In this rapidly changing world, countries and companies that can adapt quickly will benefit the most from technological advancements, leading to reduced inequality and poverty.

## 2.2 Ministries and Departments for Science and Technology under Central Government

### 2.2.1 Department of Science and Technology (DST)

In May 1971, the Department of Science and Technology (DST) was established to promote advancements in science and technology in India. It serves as the main body for organising and coordinating such activities nationwide. Here are some of its key responsibilities:

1. Formulating policies related to science and technology.
2. Addressing issues concerning the Scientific Advisory Committee of the Cabinet (SACC).
3. Focusing on emerging areas of science and technology.
4. Coordinating and integrating various scientific research institutions and bodies, and providing grants.
5. Overseeing matters related to:

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- ❑ Science and Engineering Research Board
- ❑ Technology Development Board
- ❑ National Council for Science and Technology Communication
- ❑ National Science and Technology Entrepreneurship Development Board
- ❑ International science and technology cooperation
- ❑ Autonomous institutions such as Institute of Astrophysics and Institute of Geomagnetism under DST
- ❑ Promotion of National Spatial Data Infrastructure
- ❑ Professional science academies
- ❑ Survey of India and National Atlas and Thematic Mapping Organisation
- ❑ Promotion of Geographic Information System (GIS)
- ❑ National Innovation Foundation, Ahmedabad.

6. Managing and coordinating information systems for science and technology.
7. Facilitating inter-agency and inter-departmental coordination for science and technology missions.
8. Promoting the application of science and technology for national development and security.
9. Enhancing institutional capacity building through setting up new institutions and infrastructure development.
10. Promoting science and technology for grassroots development at various levels.
11. Applying science and technology for the benefit of marginalised groups such as women and disadvantaged sections of society.

### 2.2.1.1 Programmes Under DST

***Inspire/Inspiring Creativity and Innovation***

The Department of Science and Technology (DST), Government of India, introduced the 'INSPIRE Programme' to spark interest in science among students and promote innovative thinking. Unlike competitive exams, INSPIRE relies on existing educational systems to identify talent and

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collaborates with various stakeholders to nurture a scientific mindset among youth. For more details, visit the website (https://online-inspire.gov.in/).

The programme covers the age group in the range of 10–32 years, and its components comprise the following:

1. Scheme for Early Attraction of Talent (SEATS)
2. Scholarship for Higher Education (SHE)
3. Assured Opportunity for Research Careers (AORC)

### *INSPIRE Scheme*

The five sub-components of the scheme are:

I. **INSPIRE Award:** This is for the age group of 10–15 years.
II. **INSPIRE Internship:** For the age group of 16–17 years, at a science camp, with a chance to interact with global leaders in the field of science.
III. **INSPIRE Scholarship for Higher Education (SHE):** For the 17–22 year age group, ₹80,000 per year for continuing education at Bachelor of Science and Master of Science levels.
IV. **INSPIRE Fellowship:** For doctoral research (For the age group of 22–27 years).
V. **INSPIRE Faculty:** For assuring career opportunity (For the age group of 27–32 years).

The scheme's first component, INSPIRE awards, is centrally implemented with the help of states/UTs while the scheme's other components are implemented centrally with the help of DST, through the academic/research institutes and universities concerned, etc.

### *Scheme for Early Attraction of Talent (SEAT)*

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The scheme provides a platform to millions of students to portray their creativity where they can experiment and innovate. Under this scheme, students in the age group of 10–15 years are provided with financial assistance to experience the joy and innovation in science. Young minds are also presented with INSPIRE Awards and are exposed to INSPIRE Internships, summer/winter camps.

- **INSPIRE Awards:** It motivates Class VI–X (aged 10–15) students nationwide to innovate, selecting 2 lakh students annually. Each receives ₹5,000, covering project costs and display expenses. Principals nominate students for district, state, and national competitions. All VI–X schools, public or private, are eligible, fostering widespread participation regardless of funding source.

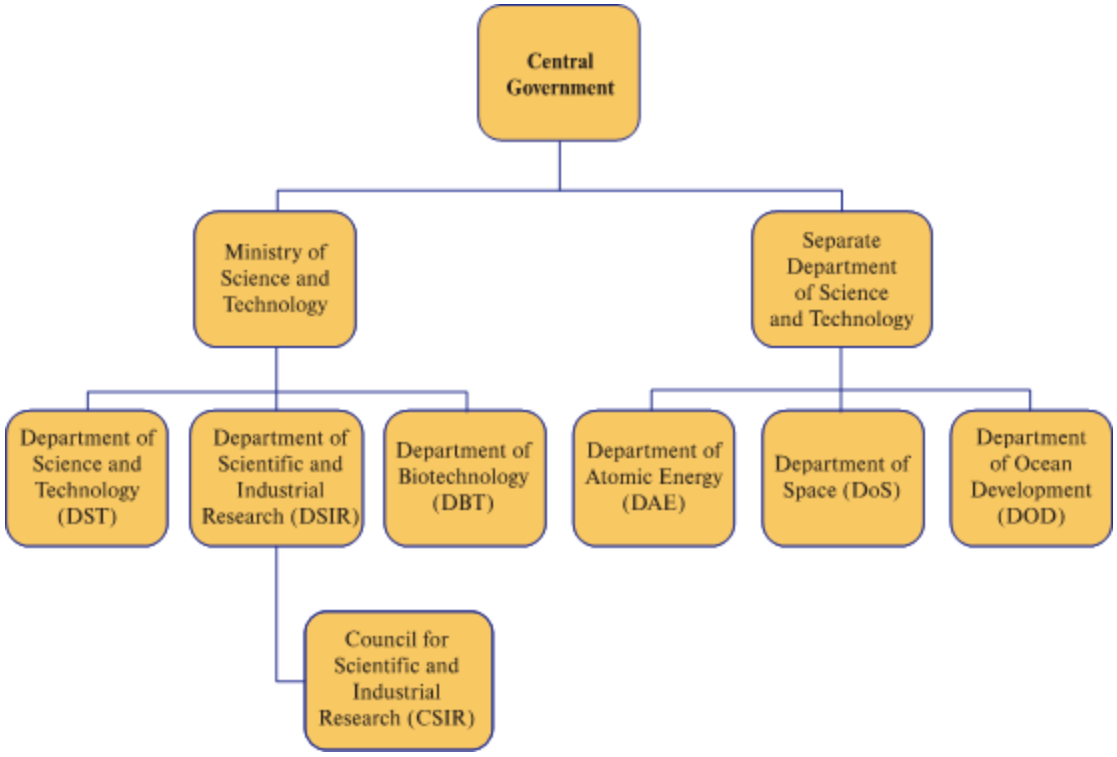


**Figure 2.1** The Science and Technology departments functioning under the Central Government

- **INSPIRE Internship:** The Scheme for Early Attraction of Talent hosts science camps to expose top 1% Class X science students to eminent national and international scientists, including Nobel Laureates. About 50,000 students attend annually, nurturing their curiosity and fostering a new generation of science enthusiasts. These camps are becoming

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influential in promoting science icons amidst a culture dominated by entertainment figures.

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### *Scholarship for Higher Education (SHE)*

The Scholarship for Higher Education is aimed towards driving young students towards science – intensive programmes through providing scholarships and mentoring. Scholarship is a boon for those who are unable to continue their education. The scheme offers ₹10,000 scholarships annually, with each valued at ₹80,000, to young minds in the age group of 17–22 years, for opting natural and basic sciences in Bachelors and Masters level.

### *Assured Opportunity for Research Careers (AORC)*

This programme is for assuring career opportunities to those interested in a career in the R&D sector. It consists of:

- **INSPIRE Fellowships:** This is focussed on students of the age group 22–27 years, who want to pursue a doctoral degree in basic and applied sciences, including engineering, medicine, etc. The fellowships are provided to the first ranker in basic and applied sciences as well as INSPIRE scholars, who secure 65% marks in aggregate at PG level.
- **INSPIRE Faculty Scheme:** This helps young researchers who are in the age group of 27–32 years to develop an independent scientific profile and helps them to emerge as science and technology leaders. Every year, this scheme provides assured career opportunities to 1000 post-doctoral researchers by offering both contractual as well as tenure track positions for five years in both basic and applied sciences.

### *INSPIRE Programme Summary*

## INSPIRE AWARDS – MANAK

The INSPIRE Award programme is renamed as INSPIRE AWARD – MANAK (Million Minds Augmenting National Aspiration and Knowledge) and is executed by DST, in coordination with the National Innovation Foundation (NIF), India, to align it with the action plan for 'Start-Up India' initiative launched by the Honourable Prime Minister of India. The main aim of the programme is to encourage the culture of creative and innovative

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thinking among young minds to address the societal needs or problems through science and technology.

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### 2.2.2 Department of Scientific and Industrial Research (DSIR)

DSIR is a division of the Ministry of Science and Technology with the authorisation for the promotion, development and utilisation of indigenous technologies. It was established in 1986.

- It is tasked with the implementation of the planned scheme 'Technology Promotion, Development and Utilisation (TPDU) Programme' and also the coordination of two autonomous bodies' activities, i.e., Council of Scientific and Industrial Research (CSIR) and Consultancy Development Centre (CDC) and two public-sector undertakings, i.e., National Research Development Corporation (NRDC) and Central Electronics Limited (CEL).

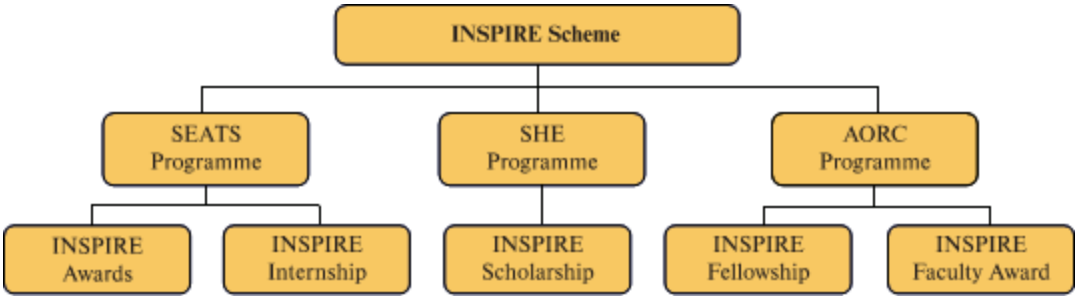


**Figure 2.2** INSPIRE Scheme

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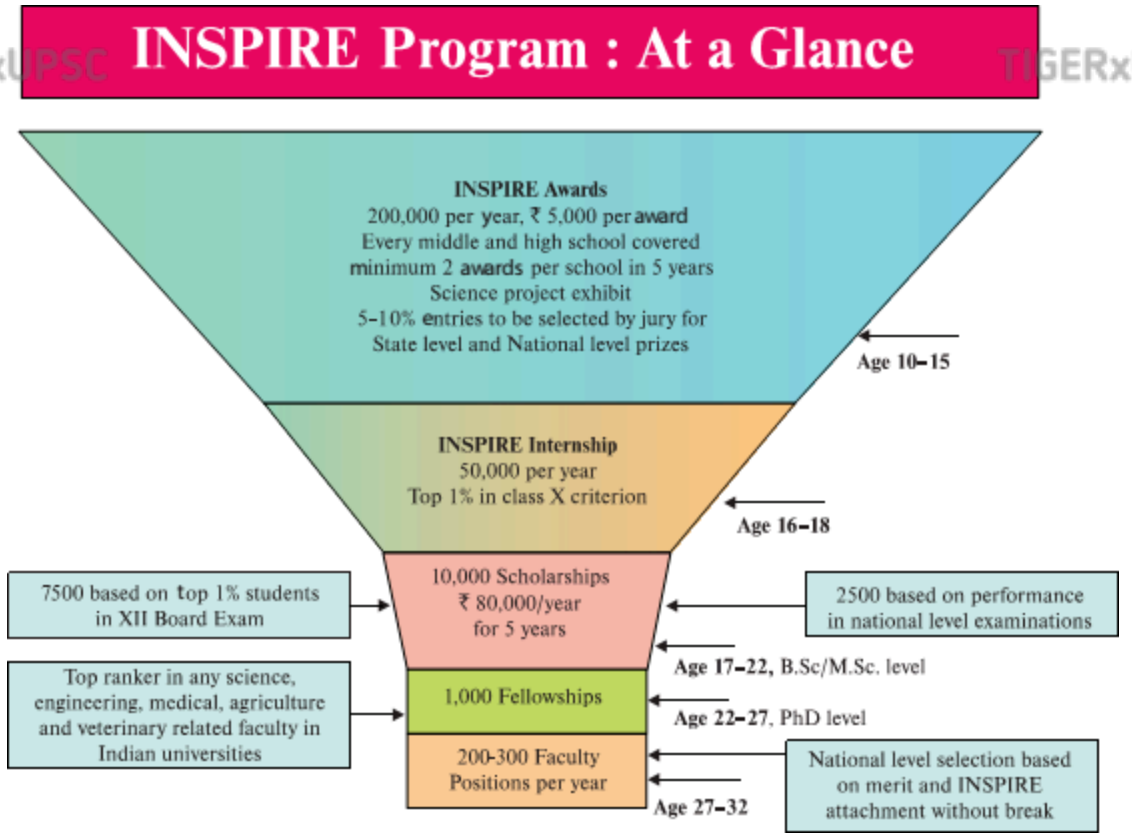


**Figure 2.3** INSPIRE Programme Summary

- Consultancy Development Centre (CDC) is an independent institution working under the Department of Scientific and Industrial Research, Ministry of Science and Technology and the Government of India.

**Objective** To promote and develop consultancy.

### 2.2.2.1 Council of Scientific and Industrial Research (CSIR)

It is an autonomous, non-profit, publicity funded R&D organisation under the Department of Scientific and Industrial Research, Ministry of Science and Technology. It was established in 1942.

**Objective** To maximise the economic, environmental and societal benefits of Indian people through scientific and industrial R&D.

**CSIR INITIATIVES – ADDRESSING EMERGING NEEDS**

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- Leveraging Science, Technology and Innovation (STI) for meeting the needs of the country and economic value creation.
- Enhancing Industry Partnerships and Last Mile Linkages: with the objective to accelerate technology deployment and commercialisation at large scale.
- Enhancing S&T-based entrepreneurship and job creation to build a structured large scale Skill Development Initiative that meets the demands of the country using the state-of-the-art infrastructure and human resources of CSIR.
- Progressive Human Resources Policies to attract and retain the best talents.

The Council of Scientific and Industrial Research (CSIR) is the major organisation under DSIR. It has 40 institutes and around 100 field stations that are dedicated to research and development in well-defined areas. Some other programmes are providing support in R&D by industry, technological self-reliance, schemes to enhance efficacy of transfer of technology and a National Information System for Science and Technology (NISSAT).

### 2.2.3 Department of Atomic Energy (DAE)

The department was set up on 03 August 1954 under the direct control of the then Prime Minister, through a Presidential Order. Through a resolution, the Atomic Energy Commission was constituted and the Secretary to the Government of India in the Department of Atomic Energy is declared as the ex-officio Chairman of the Atomic Energy Commission. DAE is concerned with nuclear power technology development and its uses in the medicine, agriculture, basic research and industrial fields.

### 2.2.4 Department of Space (DoS)

The main objective of Department of Space (DoS) is the promotion of space science and technology development, towards the achievement of becoming self-reliant and advance space applications so as to facilitate the overall nation's development.

The objectives of Department of Space are:

- To achieve the goal of total self-reliance through R&D in satellite and launch-vehicle technology.
- In order to fulfil the telecommunications and broadcasting needs of the country. It aims to provide a national space infrastructure.

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- To provide essential satellite services for weather forecasting, monitoring, etc.
- To provide satellite imagery, which is used for the natural resources survey, management of natural disasters, public goods services and monitoring of environment in the country.
- To undertake proof of concept demonstration of space applications.

### 2.2.5 Department of Biotechnology (DBT)

The Indian government established the National Biotechnology Board (NBTB) in the year 1982. Its main functions include the identification of priority sectors and development of long-term perspective for Biotechnology in India. It was also responsible for fostering programmes and strengthening indigenous capabilities in this newly emerging discipline.

A separate **Department of Biotechnology (DBT)** was established in February 1986 for the following functions:
Popularise Biotechnology for large-scale use;
Promote R&D and manufacturing in Biology;
Enhance industry and university interaction;

- Identify and establish R&D centres for promoting technology in this field;
- Popularise Human Resource Development programmes;
- Act as a nodal agency for some specific international collaboration;
- Improve bio-safety guidelines.

**MANY A FIRST – CSIR LEADING AND PAVING THE WAY**

- First to develop baby milk powder from bufflo milk with excellent digestibility.
- First to develop baby milk powder from bufflo milk with excellent digestibility.
- First to develop baby milk powder from bufflo milk with excellent digestibility.
- First ever flwering of tissue-cultured bamboo with the discovery of one of the smallest protein molecules – seminal plasmin.
- First to extract poly-metallic nodules from the ocean bed.
- First to build an all-composite small aircraft (Hansa) and 5-seater multipurpose aircraft CNM-5.
- First of its kind broad spectrum reflctance spectroscopy based confocal microscope using super continuum light source.

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- First in world bio-processing technology for leather, leading to paradigm shift in leather processing.
- First hand-held battery operated micro PCR – patented in over 100 countries.
- First of its kind portfolio of Streptokinase – a vital, life save injectable drug for cardiac problems.
- First of its kind DNA-based Macro Chip for detection of eye infection – covering diffrent organisms in one test within seven to eight hours, as compared to the conventional 3–7 days.

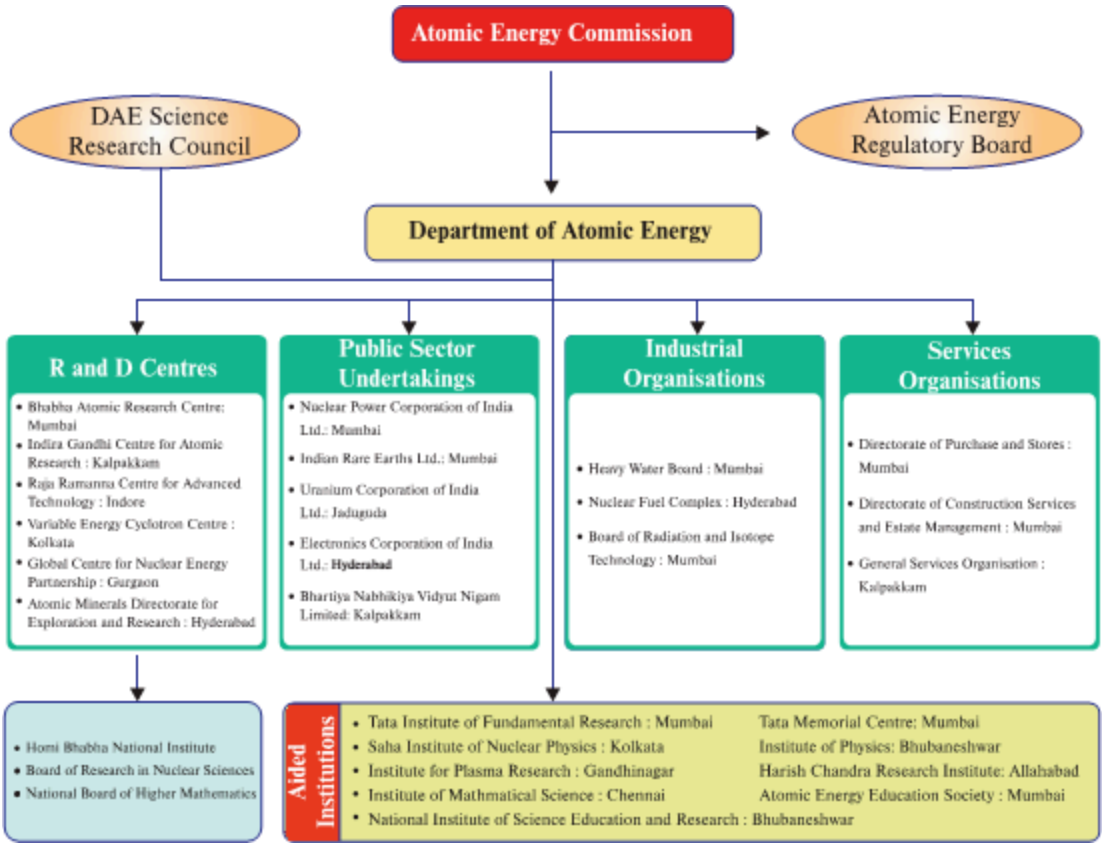


**Figure 2.4** The Organisational Structure of the Atomic Energy Commission

## 2.2.6 Indian Council of Medical Research(ICMR) – Strategic Plan and Agenda 2030

In 2017, ICMR developed its Strategic Plan and Vision 2030 document. It is conceptualised on five primary pillars:

1. Strengthen health research capacity in the country.
2. Organise data systems and research platforms.
3. Leverage traditional medicine.

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4. Enable evidence to policy translation.
5. Strengthen programme implementation through research.

- ICMR's research priorities and projects are well-aligned and harmonised with the National Health Policy (NHP) 2017. In recent years, ICMR has also focussed on diseases targeted for elimination (Kala-azar, Filariasis, Leprosy, Malaria, HIV/AIDS and Tuberculosis), child and adolescent health, universal immunisation, communicable diseases.
- ICMR has initiated a mission mode project on Tuberculosis, namely **'India TB Research Consortium (ITRC)'** to boost TB elimination efforts, which has been sanctioned ₹139 crore from the government of India.
- With a target of Tuberculosis elimination by 2025, another ambitious pilot project is focussing on active case finding in five states for bridging the gap in the last mile delivery of services amongst tribal populations using 35 mobile TB diagnostic vans under the TIE-TB project.
- ICMR is also working closely with the National AIDS Control Organisation and is committed to the global call to **'End AIDS' by 2030.**

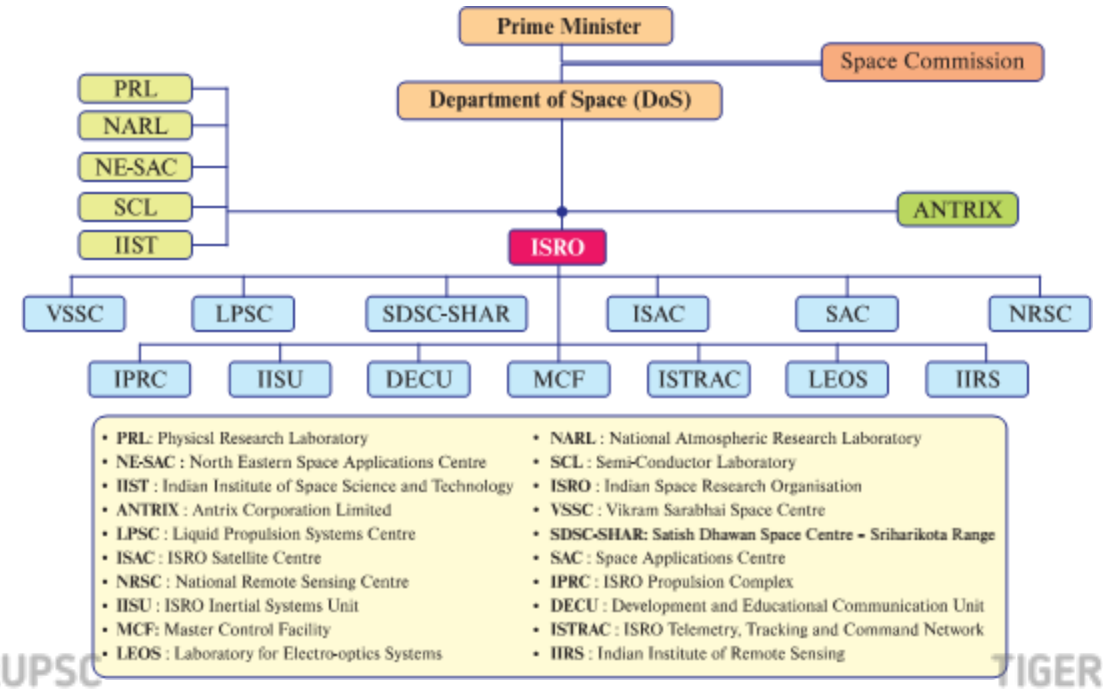

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Figure 2.5 Space Organisation in India

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## 2.3 Science and Technology Policies in India

The various science and technology policies in India are listed as follows:

| | |
|---|---|
| 1958 | First scientific policy resolution of India. |
| 1983 | The main goal of the second policy was to attain technological c |
| 2003 | This third policy wanted to –<br>(i) Bring science and technology together.<br>(ii) Bring higher investment into R&D for addressing national pro |
| 2013 | This fourth policy was framed by the Department of Science an |
| 2020 | The fifth policy has been formulated with an aim to bring aroun Technology, and Innovation. |

Note: The decade of 2010 to 2020 was declared as a decade of innovation by the then Prime Minister, Manmohan Singh. There was a need to synergise science, technology and innovation to position India amongst the top five global scientific powers. With this objective, the fourth policy, the STIP 2013 was implemented.

### 2.3.1 Science, Technology and Innovation Policy, 2020

The formulation process of Science, Technology and Innovation Policy, 2020 (STIP 2020) will be facilitated jointly by the Office of the Principal Scientific Adviser to the Government of India and the Department of Science and Technology (DST).

A participative model with four interconnected tracks has been envisioned to formulate the STIP 2020. The details on various tracks and the entire process is given below.

**Track I** aims to create a repository of public voices that will act as a guiding force for the drafting process.

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**Track II** consultations comprise of 21 expert-driven thematic collectives to feed evidence-based recommendations into the policy drafting process.

**Track III** brings together Ministries and States in extensive engagement through nominated nodal officers.

**Track IV** is the binding force that draws upon apex level multi-stakeholder engagement at the national and global levels. Inputs from these wide-ranging deliberations will finally lead to STIP 2020.

## 2.4 National AYUSH Mission (AYUSH: Ayurveda, Yoga, Siddha, Unani and Homeopathy)

The Government of India has launched National AYUSH Mission (NAM) during 12th Plan, under Ministry of Health and Family Welfare.

The word AYUSH is derived from Sanskrit, meaning 'life'. AYUSH stands for the Department of Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homeopathy.

### 2.4.1 Objective

- Its main objective is to give universal access to AYUSH health care across the nation.
- Other objectives include strengthening of institutional capacity at state level, supporting cultivation of medicinal plants by adopting good agricultural practice.

### 2.4.2 Components of the Mission

#### 2.4.2.1 Mandatory Components

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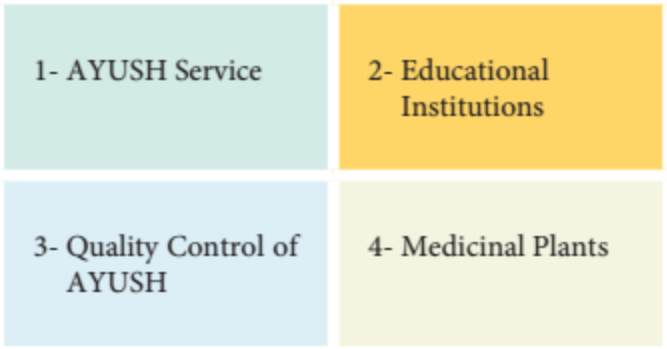


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#### 2.4.2.2 Flexible Components

1. AYUSH wellness centre including Yoga and Naturopathy
2. Tele-medicine
3. Sports medicine through AYUSH
4. Innovation of AYUSH through PPP model
5. Component of interest subsidy for private AYUSH educational instructions
6. Research and development in areas related to medicinal plants
7. Voluntary certification scheme
8. Crop insurance for medicinal plants

### 2.4.3 Institutional Mechanism

At national level, it is administered by National AYUSH Mission directorate, which consists of nine members.

The mission will be assisted by a National AYUSH Mission Appraisal Committee, which will be responsible for reviewing the State Annual Action plan (SAAP).

At state level, it is administered and executed by the State AYUSH Mission Society.

### 2.4.4 Resource Allocation

- Programme Management Unit (PMU) will be provided for strengthening the infrastructure of the mission and assistance will be given by both the Centre and the State. PMU consists of management and technical professionals.

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- For AYUSH services, grant-in-aid would be provided for special category states [North-Eastern States, and three hilly States of HP, J&K (now UTs) and Uttarakhand] in the ratio of 90:10 (90% by the Centre and 10% by the State). For other state/ UTs, it will be 75:25.
- For medicinal plants, special category states are 100% financed by central government whereas it will be shared in 90:10 ratio in other states.

### 2.4.5 Expected Effect of the Mission

1. It will improve AYUSH education through upgraded AYUSH educational institutions.
2. Universal access to AYUSH hospitals and availability of drugs.
3. Encourage the traditional knowledge of AYUSH.

## 2.5 Stem Education: Strengthening Policies for Future Skilled Workforce

- There are certain lacunae in the education system in the country due to which it is becoming difficult to produce skilled workforce.
- Though the Indian school curriculum is strong with respect to basic science and mathematics content, but not good enough neither in case of engineering nor technology.
- In today's highly competitive world, STEM or **Science, Technology, Engineering and Mathematics education** is in high demand with industries.
- The Indian education system has become the largest and most complex in the world, with about 26 crore children enrolled in classes I to XII.
- The National Policy on Education 2016 during the 12th Five-Year Plan recognised education as the most important tool for social, economic and cultural transformation and emphasised on innovation, critical thinking and skill development.
- The New Education Policy (NEP) approved by the Union Cabinet on July 29th, 2020 emphasises on schools to have more experiment-based classes and hand-on learning. It also recognises that evidence-based

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learning, scientific temper and coding skills should be imparted in the students.

- Introduction of coding from class 6 onwards is a very welcome step which will strengthen STEM learning. Artificial Intelligence (AI), Block Programming will be incorporated in school education that will lay a solid foundation for key skills required for 21st century careers.

### 2.5.1 Stem Education and Challenges

- The Indian education system is facing more challenges as compared to other developed nations due to its cultural diversity and huge population. Despite the Right to Free and Compulsory Education (RTE) Act, 2009 poor infrastructure, lack of quality teachers, overhaul of the science curriculum and the failure to identify early talent and provide them with specialised education are some key points that need to be taken care of to improve STEM education in India.
- There are large numbers of dropouts from the science stream at the high school level. The parallel business of tuition/coaching classes is a clear index of the lack of credibility of the school system.

### 2.5.2 Operation of the Scheme

- The primary scheme will be operated in conjunction with National Apprenticeship Promotion Scheme (NAPS) which provides for placing of apprentices up to 10% of the total work force in every business/industry.
- The scheme will be implemented by the Sector Skill Councils (SSCs), initially the Banking Finance Insurance Services (BFSI), Retail, Health care, Telecom, Logistics, Media, Management services, ITeS and Apparel. More sectors would be added over time with emerging apprenticeship demand and curriculum adjustments.

## 2.6 Indian Science Congress Association

The Indian Science Congress Association was ***started in the year 1914 in Kolkata and has a membership of more than 30,000 scientists.*** It owes its

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origin to the foresight and initiative of ***two British chemists, namely, Professor J.L. Simonsen and Professor P.S. MacMahon***. They thought that scientific research in India might be stimulated if an annual meeting of research workers somewhat on the lines of the British Association for the Advancement of Science could be arranged.

***Objectives***

1. To advance and promote the cause of science in India.
2. To hold an annual congress at a suitable place in India.
3. To publish such proceedings, journals, transactions and other publications as may be considered desirable.
4. To secure and manage funds and endowments for the promotion of science including the rights of disposing of or selling all or any portion of the properties of the association.
5. To do and perform any or all other acts, matters and things as are conductive to, or incidental to, or necessary for, the above objects.

**109th Indian Science Congress** recently held at **Lovely Professional University, Phagwara, Punjab (second time after 2019) focussed emphasises the crucial role of global cooperation and innovation in addressing the challenges of the modern world.**

***Theme:*** The Global Perspective on Science and Technology for a Sustainable Future.

***Indian Science Congress*** is organised by ***the Indian Science Congress Association*** every year in the first week of January.

## 2.7 National Science Day

**28th February is celebrated as National Science Day (NSD)** in India. NSD is celebrated to commemorate discovery of the **'Raman Effect',** which led to Sir C.V. Raman winning the Noble Prize. The **first National Science Day** was celebrated on February 28, 1987.

**Theme** The theme for National Science Day 2024 was Indigenous Technologies for Viksit Bharat.

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**RAMAN EFFECT**

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- A phenomenon in spectroscopy was discovered by the eminent physicist Sir Chandrasekhara Venkata Raman in 1928.
- Raman Effct is **a change in the wavelength of light that occurs when a light beam is deflcted by molecules.**
- When a beam of light traverses a dust-free, transparent sample of a chemical compound, a small fraction of the light emerges in directions other than that of the incident (incoming) beam.
- Most of this scattered light is of unchanged wavelength. A small part, however, has wavelengths diffrent from that of the incident light; its presence is a result of the Raman Effct.

## 2.8 Some Autonomous Institutions Engaged in Basic and Applied Sciences Research

1. **Bose Institute, Kolkata:** Devoted to fundamental and applied research in physical and life sciences.
2. **Agharkar Research Institute, Pune:** Research in basic and applied aspects in the field of biological sciences.
3. **Sree Ghitra Tirunal Institute for Medical Science and Technology, Thiruvananthapuram:** For developing biomedical engineering and technology or providing and also demonstrating high standards of patient care in advanced medical specialities.
4. **Indian Association for the Cultivation of Sciences, Kolkata:** It studies solid state physics, material sciences and chemistry, including biological chemistry, polymer science, theoretical physics, spectroscopy, energy research, etc.
5. **Indian Institute of Astrophysics, Bengaluru:** Research in emerging as well as interdisciplinary areas of astrophysics and heavenly bodies and development of instruments used in astrophysical studies.
6. **Jawaharlal Nehru Centre for Advanced Scientific Research, Bengaluru:** Research in frontier areas of chemistry, physics of materials, animal behaviour, genetics, and geodynamics condensed matter theory, etc.

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7. **Raman Research Institute, Bengaluru:** Research in basic sciences such as astronomy, astrophysics, liquid crystals, etc.
8. **S.N. Bose National Centre for Basic Sciences, Kolkata:** Promotes advanced studies in selected branches of basic sciences.
9. **Birbal Shahni Institute of Paleobotany, Lucknow:** Research in the area of paleobotany and its relevance in modern context.
10. **Indian Institute of Geomagnetism, Mumbai:** Observing and understanding the physical processes in the innermost part of the earth.
11. **Wadia Institute of Himalayan Geology, Dehradun:** Basic research in areas of biostratigraphy, sedimentology, tectonics, petrology and geochemistry and environmental geology to understand the geodynamics of the Himalayan region.
12. **International Advanced Research Centre for Powder Metallurgy and New Materials, Hyderabad:** It involves in the development of high-performance materials and processes.
13. **Technology Information Forecasting and Assessment Council, New Delhi**
14. **Vigyan Prasar, Noida (U.P.)**
15. **National Accreditation Board for Testing and Calibration Laboratories, New Delhi**
16. **Centre for Liquid Crystal Research, Bengaluru**
17. **Aryabhatta Research Institute of Advanced Studies (ARIES), Nainital**
18. **Institute of Advanced Study in Science and Technology, Guwahati**
19. **National Innovation Foundation, Ahmedabad**

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**UNIT-II** Energy

# CHAPTER 3
# STRUCTURE OF ENERGY DEVELOPMENT IN INDIA: RENEWABLE AND NON-RENEWABLE ENERGY

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3.7.1 Fuel Cell/Hydrogen Fuel Cell
3.7.2 Microbial Fuel Cells (MFCs)


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3.7.3 Lithium Ion (Li-ion) Battery

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India has the **seventh largest resource of lithium globally**, though it will take time to convert this into reserves. Approximately 5.9 million tons of lithium resources have been identified in the rugged Salal-Haimana region of the Reasi district in the union territory of Jammu and Kashmir.

3.7.4 Sodium Ion Battery
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3.8.3 Sustainable Aviation Fuel (SAF) and Synthetic Petrol

SAF is a type of fuel **produced from biomass** sources such as used cooking oil, waste animal fat, manure and sewage, and various types of organic waste, including maize, foodgrains, rice, and waste wood.

Synthetic petrol, also known as **synfuel**, is a liquid or gaseous fuel derived from syngas, which is **a mixture of hydrogen and carbon monoxide**.

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## 3.1 Energy

Over the last few years, India has achieved substantial synergy and momentum with the help of parallel integration of the administrative departments of coal, power and exhaustible energy. The national commitment goals set at **COP26 (2021) in Glasgow (Scotland) Convention by Hon'ble Prime Minister Sri Narendra Modi**, could possibly be achieved *via* dynamic policies and a vision aiming at sustainable development.

## 3.2 Power Ministry

Indian government's **Ministry of Power** is empowered to supervise and assure adequate production of electricity which can sustain infrastructure development, on top of propagation and delivery. The ministry serves as a channel for communication among the various resources of electricity from central and state governments and also rural electrification projects are undertaken by it.

The power ministry got identity as a ministry, on 2 July 1992. The sectionalised parts of the ministry include the **Power Ministry**, the **Coal**

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**Ministry** and the **Ministry of New and Renewable Energy (MNRE).** The ministry launched the smart grid project in 2012.

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## 3.3 Coal Ministry

The coal and lignite reserves including explorations, production, distribution, supply and even the pricing are under the surveillance of the Ministry of Coal through the government's **Coal India Limited** and its subsidiaries along with **Neyveli Lignite Corporation.**

## 3.4 Ministry of New and Renewable Energy (MNRE)

Department of Non-conventional Energy Sources (DNES) was established in 1982 and became the Ministry of Non-traditional Energy Sources in 1992. The Ministry was re-christened as the **Ministry of New and Renewable Energy (MNRE)** in **October 2006**. Due to the fact that the concern related to country's energy security is growing, the role of new and renewable energy resources is assuming increasing significance recently.

### COAL CONTROLLER'S ORGANISATION (CCO)

Ministry of Coal has one subordinate body 'the Coal Controller's Organisation' (CCO) which **collects and maintains coal production data** of all public and private sector coal mining in our country on a monthly basis. Its headquarters is at **Kolkata** with various field offices at Dhanbad, Ranchi, Bilaspur, Nagpur, Sambalpur, Kothagudem and Asansol.

**Functions and Responsibilities of CCO**

- It does **inspection** of collieries to ensure the quality with respect to grade or size of coal.
- It works as the **appellate authority** in case of any dispute that arises between consumers and owner in respect of grade and size of coal.
- It also **regulates disposal** of coal stock.
- It **provides grant** for opening or re-opening permission of any coal mine.

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- It does the **assessment and collection of excise duty** levied on all raw coal and its dispatch.
- It also **submits monthly coal data** to central and various state government's ministries.

## 3.5 Indian Renewable Energy Development Agency (IREDA)

India established Renewable Energy Development Agency (IREDA) in 1987 as a **public-sector non-banking company** under MNRE with the primary goal of **providing loans** for new and renewable sources of energy, thereby **promoting project viability**. Additionally, to promote participation of new entrepreneurs, IREDA has started entrepreneur development programme, providing training and assistance to new entrepreneurs in this field. The major objectives of IREDA are:

1. **Financial support** to certain projects and schemes related to energy and electricity production with the help of renewable and latest sources, enhancing efficiency of energy.
2. **Betterment in efficiency** through periodical improvement systems, processes and resources, and subsequently assuring consumer satisfaction in addition to the share of IREDA in the sector of renewable energy through innovation.

## 3.6 Non-Renewable and Renewable Sources of Energy

Renewable sources of energy are the energy sources, which are naturally refilled on a human timescale, and can be used many times due to its inexhaustible property. Examples include sunlight, wind, rain, tides, waves and geothermal heat, etc. Non-renewable source of energy is a resource that does not revive itself at a sufficient rate, being exhaustible in nature. Most

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of the fossil fuels like coal, petroleum and natural gas are examples of non-renewable sources. While non-renewable resources are considered to be polluting agents due to the release of carbon dioxide, nitrogen oxides, sulphur oxides, methane and carbon monoxide; renewable sources are considered to be clean sources of energy, as no pollutant is emitted upon harvesting these resources. Over all, grid installed capacity from all sources are given in **Table 3.1**.

### STUDY OF FUEL ON THE BASIS OF CALORIFIC VALUE

Calorific value of a fuel is the amount of heat evolved, in kilojoules, when one gram of the fuel is burned completely to form carbon dioxide and water. Its unit is kilojoule per gram (kJ/g). Higher calorific value of fuel means better quality of fuel. Hydrogen has the highest calorific value among all fuels.

| Fuel | Calorific Value (kJ/g) |
|---|---|
| Coal | 30 |
| Petrol | 48 |
| LPG | 50 |
| CNG | 55 |
| Hydrogen | 150 |

### THE PETROLEUM AND NATURAL GAS REGULATORY BOARD ACT, 2006 (PNGRB ACT, 2006)

PNGRB was constituted in 2006. This Act provides Petroleum and Natural Gas Regulatory Board to **regulate the refining, processing, storage, transportation, distribution, marketing and sale of petroleum, petroleum products and natural gas excluding production** of crude oil and natural gas so as to protect the interests of consumers and entities engaged in specified activities relating to petroleum, petroleum products and natural gas and to ensure uninterrupted and adequate supply of petroleum, petroleum products and natural gas in all parts of the country and to promote competitive markets and for matters connected therewith or incidental

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thereto. This clearly indicates that it ensures a competitive market for gas. An Appellate Tribunal established under section 110 of the Electricity Act, 2003 (36 of 2003) shall be the Appellate Tribunal for PNGRB.

**Table 3.1** Grid-connected, Total Including Non-renewable and Renewable Sources

| Total Installed Capacity (As on 31 May, 2025) | | |
|---|---|---|
| **(A) Installed Generation Capacity (Sector-Wise)** | | |
| **Sector** | **MW** | **% of Share in Total** |
| Central Sector | 1,06,963 | 22.5% |
| State Sector | 1,10,417 | 23.2% |
| Private Sector | 2,58,210 | 53.3% |
| Total | 4,75,590 | **100%** |
| **(B) Installed Generation Capacity (Fuel-Wise)** | | |
| **Category** | **Installed Generation Capacity (MW)** | **% of Share in Total** |
| **(i) Fossil Fuel** | | |
| Coal | 2,12,718 | 44.7% |
| Lignite | 6,620 | 1.4% |
| Gas | 20,132 | 4.2% |

[*Source:* Central Electricity Authority (CEA)].

## 3.6.1 Non-Renewable Sources

### 3.6.1.1 Coal (Solid Nature of Fuel)

#### *3.6.1.1.1 Coal*

One of the oldest forms of energy known to mankind is coal-based energy. India majorly depends on coal to meet its energy requirements, i.e., around 55–60%. There are four types of coal found in India. The classification is on the basis of content of carbon in the coal. These are:

1. **Anthracite:** The best quality of coal with **85% carbon.** Anthracite is a very hard, compact, jet black coal with semi-metallic sheen.

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2. **Bituminous:** The most widely used coal is bituminous with **50–85% carbon,** which is dense, compact and brickle and is usually black in colour. Since it has a high proportion of carbon and low moisture content, its calorific value is remarkable. Jharkhand, Odisha, West Bengal, Chhattisgarh and Madhya Pradesh have high deposits of this type of coal.
3. **Lignite:** It is a lower grade coal, containing about **35–50% carbon,** and it is commonly known as brown coal. It hails to the intermediate stage during the alternation of wood matter into coal. Its colour varies from dark black to brown. The major lignite reserves are in Palna of Rajasthan, Neyveli of Tamil Nadu, Lakhimpur of Assam and Karewa of Jammu and Kashmir.
4. **Peat:** The first stage of transformation of wood into coal results in peat containing **less than 35% carbon.** It is sometimes sufficiently compact to be used as a good fuel without constricting into bricks. With no support, it burns like wood, giving less heat and fuming more smoke and leaving behind a lot of ash.

**COAL MINES (NATIONALISATION) ACT, 1973**

India's coal industry was predominantly driven by the private sector after Independence until the Indira Gandhi government decided to transfer all coal holdings to Coal India through the **Coal Mines (Nationalisation) Act, 1973.** India's coal blocks are allocated on an auction basis. India is still not self-sufficient in coal production to fulfil basic need for power generation.

India has a total coal reserve of 344.02 billion tonnes and is the second largest producer of coal in the world as per Press Release, PIB dated 21 December 2023. Indian coal has high ash content but the amount of sulphur content is very low. **Burning of coal (Thermal Power Plant) generates various pollutants in air like $CO_2$, $SO_2$, $NO_x$, etc. Mercury is also released into the air from the combustion of coal.** Some of the areas in India have used sea water at coal-based thermal power plants for its cooling. Approximately 40% of India's thermal power plants are in water

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scarce areas. In India, Adani Power Limited is the largest private thermal power producers with installed capacity of 17550 MW (2025).

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**COAL ASH: HAZARDOUS TO HUMAN HEALTH**

Coal ash is the waste that is left after combustion of coal. It includes fly ash (fine powdery particles that are carried up the smoke stack and captured by pollution control devices) as well as coarser materials that fall to the bottom of the furnace. Most coal ash comes from coal-fired electric power plants.

- Coal ash typically contains heavy metals including **arsenic, lead, mercury, cadmium, chromium** and **selenium**, as well as **aluminium, antimony, barium, beryllium, boron, chlorine, cobalt, manganese, molybdenum, nickel, thallium, vanadium** and **zinc**.
- **Environmental Protection Agency (EPA)** has found that living next to a coal ash disposal site can increase one's risk of cancer or other diseases.
- If one lives near an unlined wet ash pond (surface impoundment) and gets daily drinking water from a well, **he may have as much as 1 in 50 chance of getting cancer from drinking arsenic-contaminated water.**
- If eaten, drunk or inhaled, these toxicants can cause cancer and nervous system impacts such as cognitive deficits, developmental delays and behavioural problems. They can also cause heart damage, lung disease, respiratory distress, kidney disease, reproductive problems, gastrointestinal illness, birth defects and impaired bone growth in children.
- The EPA estimates that 140 million tons of coal ash are generated annually.
- **Arsenic is one of the most common, and most dangerous, pollutants from coal ash.**
- The EPA also found that living near ash ponds increases the risk of damage from cadmium, lead and other toxic metals.
- Coal ash recycling poses health risks, especially where the ash is exposed to water: for example, when sprinkled as cinders on snowy roads, spread as agricultural fertilizer, or used as a landfill or to fill abandoned mines. All these uses pose threat to ground water or surface water.

### *3.6.1.1.2 Fly ash from Coal-Based Power Plants*

Fly ash is obtained from the combustion process of coal, and is similar to the ash of a volcano. During the burning of coal in modern electricity generating plants, combustion temperatures reach 2800°F approximately. The non-combustible materials form two types of ashes, namely: bottom ash and fly ash. All light weight materials those fall to the bottom of the boiler and aggregates is called bottom ash. While fly ash is the material that

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is blown up with the flue gases, where it gets collected and can be stored in silos for testing and other benefits.

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## Concerns Associated with Fly Ash

- The impact is significant because of the high contribution of fly ash to air and water pollution and on crops being grown in villages around these plants.
- Fly ash is a major source of PM 2.5 (fine, respirable pollution particles) in summer. It becomes air borne, and gets transported to a radius of 10 to 20 km. It can settle on water and other surfaces. Fly ash contains heavy metals from coal, a large amount of PM 2.5 and black carbon (BC). Proper disposal of fly ash is still not happening in many places.

## What can be done?

There is a need for generating awareness and imparting education on how fly ash can be used in following ways:

- Fly ash, the end product of combustion during the process of power generation in the coal-based thermal power plants, is a proven resource material for many applications of construction industries and currently is being utilised in manufacturing of portland cement, bricks/blocks/tiles, road embankment construction and low lying area development, etc.

### COMPOSITION OF FLY ASH

It is a coal combustion product popularly known as **flue ash or pulverised fuel ash.**

- **Made up of particulates that are driven out of coal-fired boilers together with the flue gases.**
- The components of fly ash vary considerably depending upon the source and composition of the coal being burned, but **all fly ash includes substantial amounts of silicon dioxide ($SiO_2$), aluminium oxide ($Al_2O_3$) and calcium oxide (CaO), the main mineral compounds in coal-bearing rock strata.**
- **Minor constituents include:** arsenic, beryllium, boron, cadmium, chromium, hexavalent chromium, cobalt, lead, manganese, mercury, molybdenum, selenium, strontium, thallium and vanadium, along with very small concentrations of dioxins and PAH compounds. It also has unburnt carbon.

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**Health and Environmental Hazards of Fly Ash**

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- **Toxic heavy metals present:** All the heavy metals found in fly ash – nickel, cadmium, arsenic, chromium, lead, etc. – are toxic in nature. They are minute, poisonous particles which accumulate in the respiratory tract, and cause gradual poisoning.
- **Radiation**: For an equal amount of electricity generated, fly ash contains a hundred times more radiation than nuclear waste secured *via* dry cask or water storage.
- **Water pollution:** The breaching of ash dykes and consequent ash spills occur frequently in India, polluting a large number of water bodies.
- **Effects on environment:** The destruction of mangroves, drastic reduction in crop yields, and the pollution of groundwater in the Rann of Kutch from the ash sludge of adjoining coal power plants has been well documented.

**Application of Fly Ash**

- Concrete production, as a substitute material for portland cement and sand.
- Fly ash pellets which can replace normal aggregate in concrete mixture.
- Embankments and other structural fills.
- Cement clinker production – (as a substitute material for clay).
- Stabilisation of soft soils.
- Road subbase construction.
- As aggregate substitute material (e.g. for brick production).
- Agricultural uses: soil amendment, fertiliser, cattle feeders, soil stabilisation in stock feed yards, and agricultural stakes.
- Loose application on rivers to melt ice.
- Loose application on roads and parking lots for ice control.

- At present, 63% of the fly ash is being utilised and target is for 100% utilisation of the fly ash.
- Road contractors and construction engineers need to know the benefits of using fly ash in construction.
- Measures need to be taken to reduce the cost of construction of roads using fly ash and for the purpose subsidies need to be granted for encouraging the use of fly ash.
- Besides, there is a need to prevent the ash from coming to the power plant by washing the coal at its place of origin.
- The government should also come out with a policy to encourage fly ash use in cement plant.

**Fly ash and bottom ash, both are types of coal ash, but they are different in size and origin. Fly ash is a fine, powdery material carried up the smokestack, while bottom ash is coarser and heavier, remaining near the boiler.**

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**Table 3.2** Difference between Fly Ash and Bottom Ash

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| Characteristic features | Fly Ash | Bottom Ash |
|---|---|---|
| **Description** | Fine, powdery material, often described as "flying" up the smokestack due to its size. | Coarser and heavier ash than fly ash, remaining near the bottom of the boiler. |
| **Formation** | Created from the combustion of pulverized coal, the tiny particles remain suspended in the hot gases and are captured by filters like electrostatic precipitators. | Larger ash particles that do not rise with the flue gases and settle out near the bottom of the boiler. |
| **Composition** | Primarily made up of silica, alumina, and iron oxide, along with other trace elements. | Generally has a similar composition to fly ash but with a higher concentration of larger, unburnt coal particles. |
| **Uses** | Recyclable material, used | Often used as a landfill |

### *3.6.1.1.3 Bio-Bitumen*

Bio-bitumen is a sustainable **bio-based binder derived from renewable sources like crop stubble, vegetable oils, algae or lignin**.

**It is primarily extracted from lignocellulosic biomass** or **refined from residues of crude oil distillation.**

During refining, the heavier bitumen remains after lighter components like **gasoline and diesel are removed.** It can also naturally occur in deposits, such as in oil sands.

#### NH-44 Bio-Bitumen Stretch

- **Nagpur-Mansar Bypass** on **National Highway 44** in Maharashtra.

#### Production of Bio-Bitumen

- **Primary Source:** Lignin, a by-product of agricultural waste and plant-based materials.

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- **Process:** Biomass is processed to extract lignin, which is converted into bio-bitumen.

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## Characteristics of Bio-Bitumen

- **Eco-Friendly:** Reduces greenhouse gas emissions by 70% compared to petroleum-based bitumen.
- **Strength:**
  - ❑ Offers superior durability and load-bearing capacity.
  - ❑ Bio-bitumen is 40% stronger than conventional asphalt.
- **Sustainability:** Promotes the use of agricultural residues, reducing stubble burning.

## Applications of Bio-Bitumen

- **Road Construction:** Direct replacement for petroleum bitumen in asphalt pavements.
- **Modifier:** Enhances traditional bitumen properties.
- **Rejuvenator:** Restores aged asphalt's elasticity and functionality.
- **Industrial Use:** Applicable in waterproofing and adhesive materials.

## Objectives of Bio-Bitumen Production Initiative

- **Reducing Import Dependency:** The key objective is to transition from imported bitumen to domestically manufactured bio-bitumen within the next decade, thereby lowering foreign exchange expenditure.
- **Addressing Environmental Concerns:** Bio-bitumen production aims to **mitigate environmental issues** associated with stubble burning by utilising biomass and agricultural waste as feedstocks.
- **Promoting Sustainable Practices:** By leveraging bio-based materials, this initiative supports sustainable **road construction practices** and aligns with global environmental standards.
- **Technological Development and Pilot Study:** The **Central Road Research Institute (CRRI)** is collaborating with the **Indian Institute of Petroleum** to conduct a pilot study on a 1-km road stretch using bio-bitumen.

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## Challenges

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- **Cost-Effectiveness:** Currently, bio-bitumen production can be **more expensive** than traditional methods.
- **Long-Term Performance:** More extensive field trials are needed to assess the **long-term performance** and durability of bio-asphalt compared to traditional pavements.
- **Standardisation:** Establishing **clear standards and specifications** for bio-bitumen is necessary for its wider adoption in the construction industry.

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### Other Innovation Methods in Road Construction

**Steel Slag Technology** is a novel method of using steel slag, the waste generated during steel production, to build more robust and more durable roads.

- For example, **Steel Slag Road technology was first used in Surat.**

In Hamburg, Germany, companies developed **100% recycled asphalt pavement (RAP)** to meet reduce costs, save energy and lower carbon emissions.

India has built more than 2,500 km of **plastic roads** and, worldwide, **plastic roads** are proliferating in more than 15 countries.

- For example, it is mandatory to make use of at **least 10% of plastic waste** for **road construction in Ladakh.**

## *3.6.1.1.4 Coal Gasification and Liquefaction*

### (A) Gasification

It is **the process of converting any carbon-based raw material such as coal into fuel gas popularly known as syngas.**

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The coal is fed into a high-temperature pressurised container (gasifier) and combined with hot steam and controlled amounts of air or oxygen under high temperatures (up to 2600°F) and high pressures (up to 1200 psig/pound per square inch)) to generate synthetic gas or 'syngas'. Typically **syngas is a mixture of carbon dioxide, hydrogen, methane, nitrogen and carbon monoxide.**

### *Benefits of Gasification*

Transporting gas is cheaper than transporting coal.

1. Syngas burns much **cleaner** and **reduces air pollution**; it is renewable and does not depend on fossil fuels.
2. It has **greater efficiency** than conventional coal-burning because it can effectively use the gases twice: the coal gases are first cleansed of impurities and fired in a turbine to generate electricity. The exhaust heat from the gas turbine can be captured and used to generate steam for a steam turbine generator.

### *Concerns and Challenges*

- Coal gasification is one of the **more water intensive forms of energy production.**
- Also, there are concerns about water contamination, land subsidence and disposing of waste water safely.

### *Coal Gasification*

The coal gasification initiatives taken by the Government are as under:

(i) **On January 24 2024 the Government has approved an outlay of ₹8,500 crore as financial incentive, for promotion of coal/lignite gasification projects for both government PSUs as well as private sector.**

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(ii) **Government has also approved investment** by Coal India Limited (CIL) in joint ventures of CIL-BHEL and CIL-GAIL for undertaking coal gasification projects.

(iii) **In 2022, a new sub-sector**, '**Production of Syngas leading to coal gasification'**, was created under the non-regulated sector (NRS) linkage auctions policy to support this initiative.

(iv) 50% rebate in the revenue share for coal used in gasification has been introduced in commercial coal block auctions, provided that at least 10% of the total coal production is used for gasification purposes.

Coal is one of the most abundant natural resources in the country. Coal gasification technology enables conversion of coal into syngas (synthetic gas), which **can be used to produce downstream products like methanol, ethanol, ammonium nitrate, Synthetic Natural Gas (SNG) and Urea (Fertilizers) etc.**

Coal gasification technology provides alternative use of coal promoting environmental sustainability to align with vision of developed India 2047.

The Government has not conducted any specific impact assessment of the financial incentive scheme for coal gasification projects.

### *Underground Coal Gasification*

The Ministry of Coal has initiated a groundbreaking Underground Coal Gasification (UCG) pilot project in **Jharkhand's Jamtara district.**

### *Background*

In 2015, the Ministry of Coal introduced a dedicated policy framework for the development of UCG in coal and lignite-rich regions of India. In line with this policy, Coal India Limited (CIL) selected the **Kasta coal block** in **Jamtara** district to implement UCG technology, customized to suit India's unique geo-mining conditions.

### *What is Underground Coal Gasification (UCG)?*

UCG is an energy extraction process in which coal is gasified or chemically converted into synthesis gas (syngas) directly within the coal seam. The resulting syngas, similar to surface coal gasification gas, is primarily

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composed of methane ($CH_4$), carbon dioxide ($CO_2$), hydrogen ($H_2$), and carbon monoxide (CO).

**The process involves injecting steam and air or oxygen into the coal seam, followed by ignition at temperatures exceeding 1000°C**. The final composition of the gas depends on factors such as coal type, temperature, pressure, and whether air or oxygen is used for combustion.

### *Advantages of UCG*

- **Access to Unmineable Coal:** Enables the utilization of vast coal reserves that are otherwise unreachable through conventional mining.
- **Lower Capital Investment:** Eliminates costly components of traditional coal mining, such as extraction, transportation, and storage.
- **Higher Energy Density:** UCG requires less than 3% of the land needed for coal bed methane (CBM) extraction to produce an equivalent amount of energy.
- **Environmental Benefits**: Reduces reliance on imports, minimizes surface disruptions, and mitigates conventional coal mining hazards.
- **Optimized Use of High-Ash Coal**: UCG has the potential to harness energy from coal with high ash content, improving overall efficiency.

### *Government Initiatives for Coal Gasification*

- **National Coal Gasification Mission**: Targets gasification and liquefaction of 100 million tonnes of coal by 2030.
- **Coal/Lignite Gasification Promotion Scheme**: Offers financial incentives worth ₹8,500 crores to support coal and lignite gasification projects by public and private sector entities.
- **Joint Venture Agreements (JVA)**: Encourages partnerships like CIL and BHEL's coal gasification project for ammonium nitrate production.

### *Challenges Associated with UCG*

- **Induced Land Subsidence:** The formation of voids during gasification may cause surface deformation and instability.
- **Groundwater Contamination Risk:** Harmful byproducts like phenols and benzene could potentially seep into groundwater reserves.
- **Technology Gaps:** Limited expertise in converting Indian coal into syngas and the high cost of syngas production affect commercial feasibility.

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- **Process Variability:** UCG is inherently unsteady due to factors such as cavity expansion, variations in coal properties, water influx, and ash accumulation.

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### (B) Liquefaction

It is also called **Coal to Liquid (CTL) technology.** It is **an alternative route to produce diesel and gasoline** and makes economic sense only in a world of high crude oil prices.

- The **process involves** gasification of coal, which in turn will produce synthetic gas (a mix of $CO+H_2$). The synthetic gas can be liquefied to its fuel equivalent in presence of cobalt/iron-based catalysts at higher pressure and temperature.
- However, liquefied coal emits twice as much $CO_2$ as burning oil. It also emits a large volume of $SO_2$.

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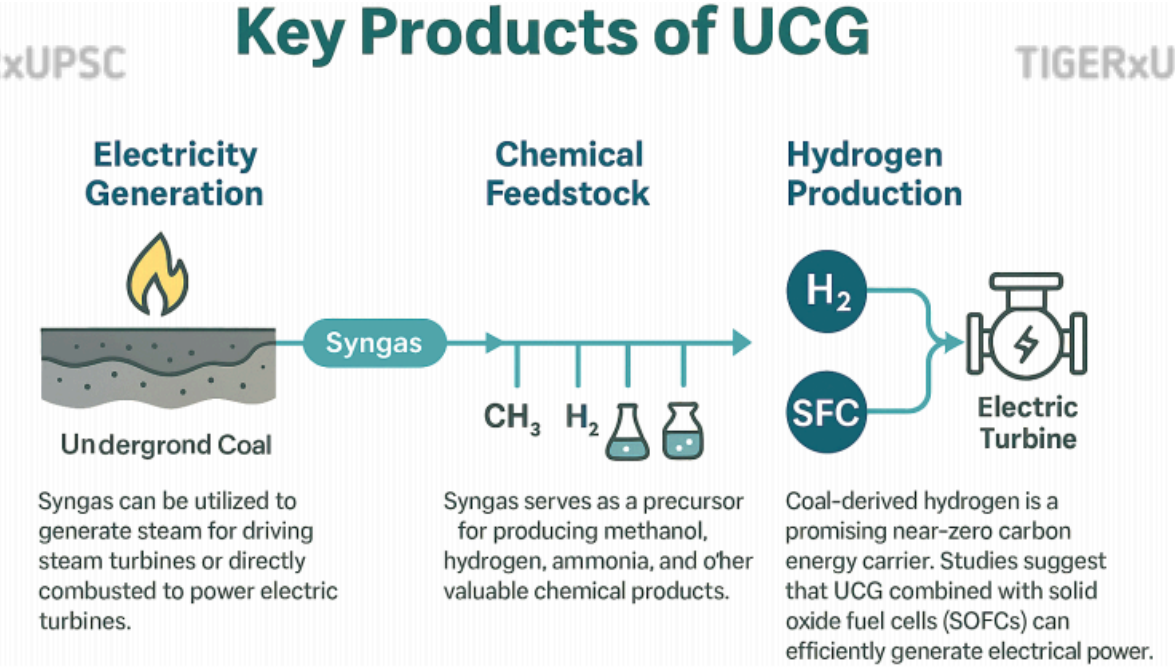


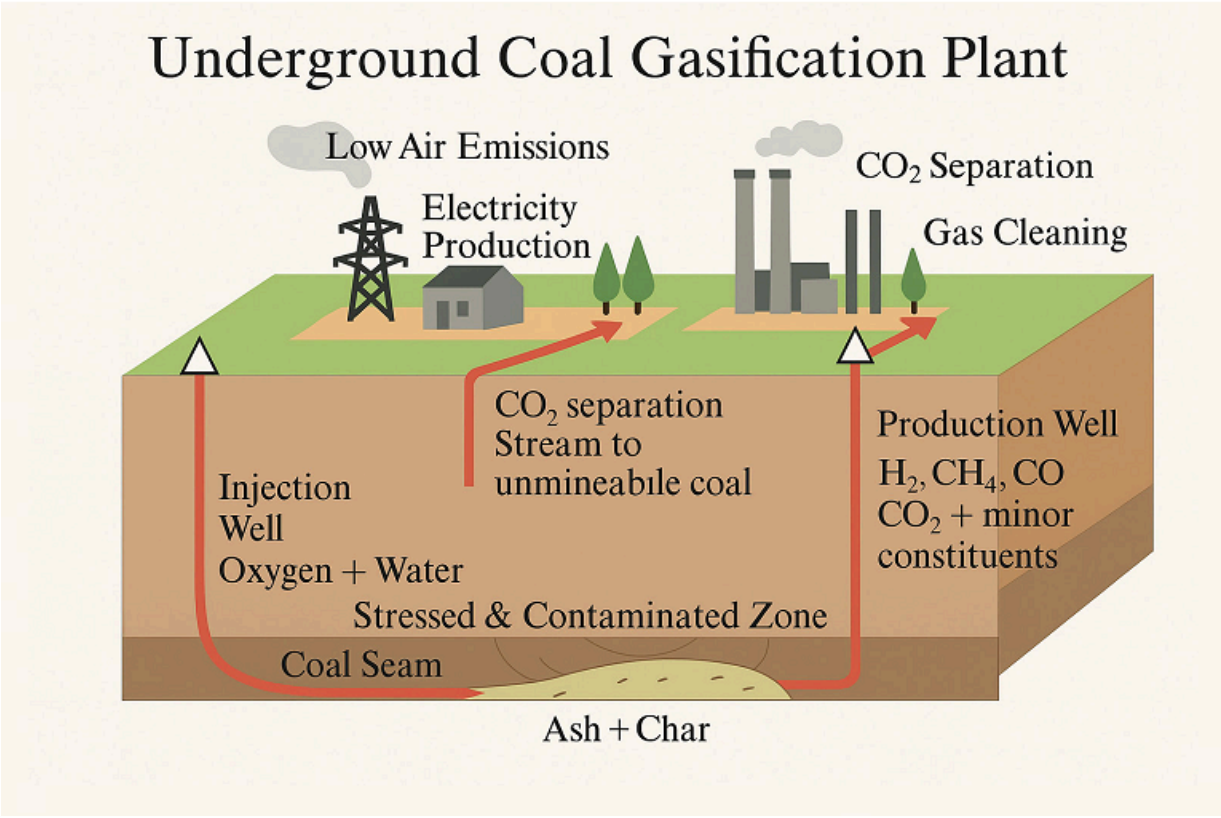


***Benefits of Liquefaction***

The CO2 emissions are more readily and cheaply captured from CTL plants than from conventional coal-fired power stations. The captured CO2 can be

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transported and injected into underground storage reservoirs (a procedure known as 'carbon capture and storage' (CCS) or 'geosequestration').

**Union Minister of Coal and Mines** made a target on Coal Gasification and Liquefaction in coming future**:**

- India aims for 100 million tonnes (MT) coal gasification by 2030 with investments worth over ₹4 lakh crores.
- For encouraging use of clean sources of fuel, government has provided for a concession of 20% on revenue share of coal used for gasification. This will boost production of **synthetic natural gas, energy fuel, urea for fertilisers** and **production of other chemicals**.

**HIGH VOLTAGE DIRECT CURRENT (HVDC)**

The transmission modes and wires we use for electricity transmission have more than 40% losses.

It is a highly effective mode for electricity transmission in long-range distribution to minimise transmission losses. Here, transmission mode of electricity is Direct Current (DC) in high voltage. This type of technology will be introduced in coming future.

## 3.6.1.2 Petroleum (Liquid Nature of Fuel)

### *3.6.1.2.1 Differences Among Different Natures of Fuel*

A 'crude oil', which is pumped out of the ground, is a black liquid. The constituents of this liquid are aliphatic hydrocarbons, or hydrocarbons composed of hydrogen and carbon, wherein the carbon atoms link together in chains of varying lengths.

Different lengths of hydrocarbons have different properties and behaviours. For instance, one carbon atom chain is the lightest chain, known as methane (CH4). Longer the chain, heavier is the hydrocarbon. The first four chains of hydrocarbons – CH4 (methane), C2H6 (ethane), C3H8 (propane) and C4H10 (butane) – are all gases. The chains up to C18H32 or so are all liquids at room temperature, while heavier chains above C19 are all solids at room temperature.

### *3.6.1.2.2 Premium Fuels: Fact versus Fiction*

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- **Research Octane Number (RON)** is a standard measure of the performance of a gasoline engine fuel.
- Fuels with a higher RON are mostly used in high-performance engines that have higher compression ratios.
- In India, most of the ordinary and premium petrol are 91 RON. The premium petrol is generally ordinary petrol with additives; hence they don't really change the value of RON.
- However, in some places, two variants – 93 Octane and 97 Octane – are also available with RON values of 93 and 97, respectively.
- In January 2017, 99 Octane was launched having RON 99 and it is the most advanced automotive fuel for sale in India till date.
- **Compression Ratio (CR)** is simply the ratio of the volume inside the cylinder of an engine, before and after compression. Therefore, a compression ratio of 10:1 means that an engine can compress the air and fuel mixture 10 times to that of the initial volume.
- Generally, with a higher compression ratio, you need a higher grade fuel to withstand the extreme pressures generated in the engine as a result. **A higher grade fuel in this aspect means a fuel with a higher value of RON.** A majority of the petrol engines have a compression ratio between 8:1 and 12:1. However, racing cars and bikes are exceptions, as they operate at very high CRs like 13:1 or even higher (**Table 3.3**).
- In a normal petrol engine, the air fuel mixture is heated due to being compressed and then is ignited rapidly by the spark plug. If this mixture is heated (or compressed) too much, it will self-ignite before the ignition system sparks. This will lead to much higher stresses than what the engine components are designed for, and can cause a **'knocking'** or **'pinging'** sound.
- Knocking can cause major engine damage or even seizures, if severe. The compression ratio is directly related to power and efficiency of an engine.
- Therefore, engines with higher compression ratios can extract more energy from the same quantity of fuel as compared to engines with lower compression ratios.
- Sports cars and bikes have engines that are designed to operate at a compression, and thus demand fuels of higher RON.

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**Table 3.3** High versus Low Compression Ratio

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| Higher Compression Ratio | Lower Compression Ratio |
|---|---|
| Increased power output | Lower power output |
| Improved mileage | Reduced mileage |
| Slightly shorter engine life | Slightly longer engine life |
| Requires high octane fuel | Requires normal fuel |
| Risk of engine knocking | Negligible risk of engine knock |

***Benefits of High Octane Fuel***

- Increased power output
- Improved fuel efficiency
- Lesser engine maintenance
- Better engine life

### *3.6.1.2.3 Asphalt and Bioasphalt*

Asphalt is brown/black, highly viscous hydrocarbon produced in the residue of petroleum distillation. This distillation process occurs naturally in asphalt lakes or under petroleum refinery by using crude oil.

Bioasphalt is an alternative of asphalt, made from non-petroleum-based renewable resources. These sources include sugar, molasses, rice, corn, potato, starch, gum, resins, latex, vegetable oil, cellulose, lignin, dried sewage effluents, etc. Bioasphalt is eco-friendly and is also used in road construction.

Asphalt is highly used in road construction, where it is used as the **glue or binder mixed** with aggregate particles to create asphalt concrete. Its other main uses are for bituminous waterproofing products, including production of roofing felt and for sealing flat roofs.

**TERMINOLOGY**

**Knocking and Anti-knocking Agent**

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Some fuels, when burnt, produce heat but this heat of the fuels does not transform into exact work and produces metallic sound, which is called knocking. Thus, those fuels, whose knocking is very low, are assumed the best fuels and vice-versa. To lower knocking, some compounds are added in the fuel to make the fuel good and such compounds are called anti-knocking compounds. The best anti-knocking compound is **Tetra Ethyl Lead (TEL).** Another ace anti-knocking compound is **Benzene Toluene Xylene (BTX)** and some other anti-knocking compounds are **Methanol** or **Ethanol, Alcohol** and **Tertiary Butyl Alcohol.** However, TEL is harmful for environment.

### Octane Number

The knocking of fuel is expressed by octane number. Octane number is high for low knocking, so fuel is good and *vice-versa*. Thus, for increasing the octane number of the fuels, anti-knocking compound is necessarily mixed up.

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Note: The increase in the carbon chain length leads to decrease in the Octane number.

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### Gasohol

It was firstly discovered in Brazil. The mixture of petrol (80–90%) and alcohol (10%) is called gasohol, and it is obtained by mixing petrol in extracted alcohol of sugarcane. By the use of gasohol as a fuel in the vehicles, pollutants like CO2, SO2, etc., can be reduced and minimised up to an extremely lower level. Thus, gasohol can be a better alternative option of the petrol because of great demand of petrol.

### Diesel

This is a type of oily liquid hydrocarbon obtained by the fractional distillation of petroleum. This is cheaper than petrol and a good fuel used in vehicles, machines, devices, etc. But during its consumption, various toxic gases like sulphur dioxide (SO2), etc., are evolved. **The gas SO2 creates some problems like irritation in eyes, headache, inhaling problems, etc., and sometimes skin diseases and respiratory system problems also come into existence.**

### Ultra Low Sulphur Diesel (ULSD) or City Diesel

This is an extremely pure form of diesel. Through the combustion of this diesel, less pollutants are spread in the atmosphere because of low sulphur presence in it. It is used frequently in almost all cities of Europe.

### Cetane Number of Diesel

Cetane number or CN is an indicator of the combustion speed of diesel fuel. The higher the Cetane number, the more easily the fuel will combust in a compression setting (such as a diesel engine).

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Notes: Additives alkyl nitrates (principally 2-ethylhexyl nitrate) and di-tert-butyl

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peroxide are used as additives to raise the cetane number.

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### 3.6.1.3 Natural Gas (Gaseous Nature of Fuel)

As suggested by the name, natural gas is a gaseous fuel which contains 87–92% of methane and a small percentage of other higher hydrocarbons. The different forms of gases under varied temperature and pressure include LPG, PNG, CNG and LNG.

**Problem with Gaseous Nature of Fuel**

Gaseous nature of fuel has maximum calorific value but still has many problems:

- Their **ignition temperature** (minimum temperature at which a fuel catches fire) is normal room temperature.
- They are **odourless**.
- They have **maximum chances of friction**, if not properly packed or processed.

#### *3.6.1.3.1 LPG, PNG, CNG and LNG*

The by-product of natural gas processing and petroleum refining is LPG. The components of LPG are gases at normal temperatures and pressures. Liquefied Petroleum Gas or LPG (also known as Auto gas) consists mostly of propane, propylene, butane and butylene in different mixtures. LPG is odourless and can burn at room temperature and hence, **Ethyl Mercapton** is added to it for smell, so as to identify its leakages *via* smell. Due to widely varied composition, LPG gives variable engine performance and cold starting performance, and this is seen as one limitation of LPG. At normal temperatures and pressures, LPG will evaporate, thus arises the need to store it in pressurised steel bottles.

**Natural gas** consists mainly of methane and is drawn from gas wells or in conjunction with crude oil production. Other constituents may include hydrocarbons such as ethane and propane as well as other gases such as nitrogen, helium, carbon dioxide, sulphur compounds and water vapour. Compressed Natural Gas or CNG is stored in high-pressure tanks – 20–25 mega Pascal (200–250 bar) on the vehicle. **For detection of leakage, sulphur-based odour is normally added to CNG.** As compared to

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gasoline, natural gas is effective in terms of safety as it is lighter than air and will normally disperse in case of leak.

A natural gas stored as a super-cooled (cryogenic) liquid is termed as **Liquefied Natural Gas or LNG.** To distil natural gas, the required temperature depends on its precise composition; however, it is typically between –120 and –170°C (–184 and –274°F). LNG is advantageous due to energy density offered, which is comparable to petrol and diesel fuels, extending range and reducing refuelling frequency.

However, the high cost of cryogenic storage on vehicles and the major infrastructure requirement of LNG dispensing stations, production plants and transportation facilities count towards its disadvantages. Currently, heavy-duty applications in places like the United States, Japan, the United Kingdom and some countries in Europe are using LNG. However, being a less cost-effective option, this is currently not a practical option for many developing nations.

### HYDROGEN-ENRICHED CNG (HCNG)

HCNG is a mixture of compressed natural gas (CNG) and some percentage of Hydrogen for heat or energy generation. HCNG which may be used as a fuel of Internal Combustion Engine (ICE) is considered a cleaner source of fuel, more powerful and offers more mileage than even CNG.

**Some Advantages of HCNG**

- HCNG reduces emissions of CO up to 70%.
- Enables up to 5% savings in fuel.
- Better performance due to higher Octane number.

In 2018, **Delhi Government** has tied up with **Indian Oil Corporation Limited (IOCL)** to study the technology and infrastructure needs to induct 50 HCNG buses on a trial basis. IOCL has plans to mix (18–20%) Hydrogen in these buses. Current cost of H2 is more than the cost of Natural Gas. So, HCNG is costlier than CNG.

***LNG Terminals in India***

- RGPPL LNG Terminal, Maharashtra
- Dahej Terminal, Petronet LNG Ltd., Gujarat
- Hazira Terminal, Shell Ltd., Gujarat

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**PROCESSING OF NATURAL GAS**

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LPG fractionator plants are used to process natural gas to dissipate it into different fractions as per different applications.

- **Methane** – Fuel and feedstock for urea plants and fuel for power plants
- **Ethane** – Production of petrochemicals
- **Propane** – LPG and industrial fuel
- **Butane** – Production of LPG
- **C5 and heavier** – Production of solvents and pentane

- Petronet LNG, Kochi
- Krishna Godavari LNG import terminal at Kakinada

**PNG (Pipelined Natural Gas)** is similar to CNG, except that, it is supplied through pipelines for household use.

### *3.6.1.3.3 Unconventional Gas Sources*

The extraction of these gas resources needs greater effort than other resources and these gas resources exist in the reservoirs. These resources require technology of specialised form, which is totally dependent on their location or reservoir type. Conventionally, the following gas sources are classified as unconventional sources: **Coal mine methane, Shale gas, Coal bed methane, Tight gas** (Tight gas is natural gas which is produced from reservoir rocks with so low permeability that massive hydraulic fracturing is necessary to produce the well at economic rates). However, the global demand of gas, which is expected to rise by 50% from 2010 to 2035, is likely to come from unconventional sources.

#### 3.6.1.3.3.1 Coal Bed Methane (CBM)

It is an unconventional form of natural gas found in coal deposits or coal seams. CBM is formed during the process of coalification, the transformation of plant material into coal.

***CBM Potential***

- India has the **fifth-largest coal reserves** in the world, and CBM has been looked at as a clean alternative fuel with significant prospects.

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Note: Ministry of Coal has asked the state-run coal miner Coal India Limited (CIL) to produce 2 MMSCB (million metric standard cubic metres) per day of coal bed methane (CBM) gas in the next 2–3 years.

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***Use of CBM***

- In power generation.
- As compressed natural gas (CNG) auto fuel.
- As feedstock for fertilisers.
- Industrial uses such as in cement production, rolling mills, steel plants and for methanol production.

***Advantages***

- High calorific value.
- Contains 95% pure methane and is hence a good fuel.
- Does not contain harmful gases like H2S.
- Environment friendly when compared with other conventional sources of energy.

***Disadvantages***

- Methane, being a greenhouse gas, becomes a matter of concern, keeping in view the emissions from coal mines (accounts approximately 8% of total global methane emission).
- Drilling of land for the extraction of CBM results in acute ecosystem damage.
- Wild life habitats are disturbed.
- Extraction of CBM involves disposal of highly salinized water, whose introduction to fresh water ecosystem poses a serious threat to the aquatic environment.
- Correct stimulation of gas content, thickness, reservoir pressure is a bit difficult.

## 3.6.1.3.3.2 GOBARdhan Scheme

The **Galvanising Organic Bio-Agro Resources Dhan (GOBARdhan) scheme** is implemented under **the Swachh Bharat Mission (Grameen)-Phase 2,** by the Department of Drinking Water and Sanitation under the Jal Shakti Ministry.

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***About the Scheme***

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- It was **launched in 2018.**
- The scheme aims to augment income of farmers by **converting biodegradable waste into compressed biogas (CBG).**
- The initiative aims at attracting entrepreneurs for establishing community-based CBG plants in rural areas.
- **The Ministry of Jal Shakti** has launched a unified portal on the government's **'Gobardhan' scheme.** Farmers can earn ₹1 lakh crore in 5 years from Jal Shakti Ministry's Gobardhan scheme, says Government.

***Benefits of the Scheme***

1. Helpful for the country as India is home to the highest cattle population in the world, close to 300 million in number, with a daily output of 3 million tonnes of dung.
2. Encourage farmers to consider dung and other waste not just as a waste but as a source of income.
3. It will be easier to keep the village clean and sanitised, livestock health will improve and farm yields will increase.
4. Increase self-reliance in energy utilised for cooking and lighting.
5. Provide a stable fuel supply in the market for oil companies and accessible credit in the market through government schemes and banks for entrepreneurs.

(Refer **Infographic – Biogas Cycle**)

### 3.6.1.3.3.3 Gas Hydrates/Clathrates/Methane Hydrates

Gas hydrates are naturally occurring ice-like combination of natural gas and water, firstly found in polar regions. Basically, these are solids, made up of hydrocarbon gas (mainly Methane) and liquid water. They resemble wet snow and can exist at temperatures above the freezing point of water. With pressurisation, they remain stable at temperatures up to 18°C. Hydrates belong to a form of complexes known as Clathrates. Clathrates are substances having a lattice-like structure, in which molecules of one substance are completely enclosed within crystal structure of another.

Hydrates are generally present in oceanic sediments (under the seafloor) along continental margins and in polar continental settings (Arctic tundra,

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etc.). Gas hydrates are restricted to shallow lithosphere (2000–4000 metre depth). Studies show that **gas hydrates found mainly in ocean bed (around 98%) and 2% on land**. Gas hydrate regions in India include **Western, Eastern and Andaman offshore areas**, **Krishna Godavari basin, Mahanadi basin** and **shallow areas of Bay of Bengal**. World's second largest gas hydrates reserve are located in **KG Basin** (Krishna Godavari basin) India.

### 3.6.1.3.3.4 Shale Gas

Shale formations contain a trapped natural gas, known as Shale gas. Fine-grained sedimentary rocks, which are rich resources of petroleum and natural gas are shale. **Horizontal drilling** on **combining with hydraulic fracturing**, over the past decade has enabled access to large quantity of shale gas that were previously not lucrative to produce. The **hydraulic fracturing (fracking)** technology's latest advancements cropped up shale gas around the well bores.

The US government's Energy Information Administration predicts that **by 2035, 46% of the United States' natural gas supply will come from shale gas**.

#### *Shale Gas in India*

Six basins are identified as potentially shale gas bearing, by the **Ministry of Petroleum and Natural Gas (MoPNG),** which include **Cambay, Assam–Arakan, Gondwana, Krishna–Godavari, Kaveri and the Indo-Gangetic plain.** Country-wise data of Shale gas reserves are given in **Table 3.4**.

**Table 3.4** Leading Countries with Technically Recoverable Shale Gas Resources

| Rank | Country | Technically Recoverable Shale Gas (Trillion Cubic Feet) |
|---|---|---|
| 1 | China | 1,115 |
| 2 | Argentina | 802 |

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| | | |
|---|---|---|
| 3 | Algeria | 707 |
| 4 | United States | 623 |
| 5 | Canada | 573 |
| 6 | Mexico | 545 |
| 7 | Australia | 429 |
| 8 | South Africa | 390 |
| 9 | Russia | 285 |
| 10 | Brazil | 245 |
| 11 | India | 96 |

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***Advantages***

- Greenhouse gas emission from shale gas is less.
- More environmental friendly than coal.
- Cost effective.
- Electricity generation from shale gas is cheaper than coal.
- Shale gas is a less-intensive carbon fuel than other fossil fuel.

***Disadvantages***

- Extraction process involves greenhouse gas emission (mainly methane), introduction of chemicals and wastes, environmental pollution and methane leakage.
- It produces photochemical smog.

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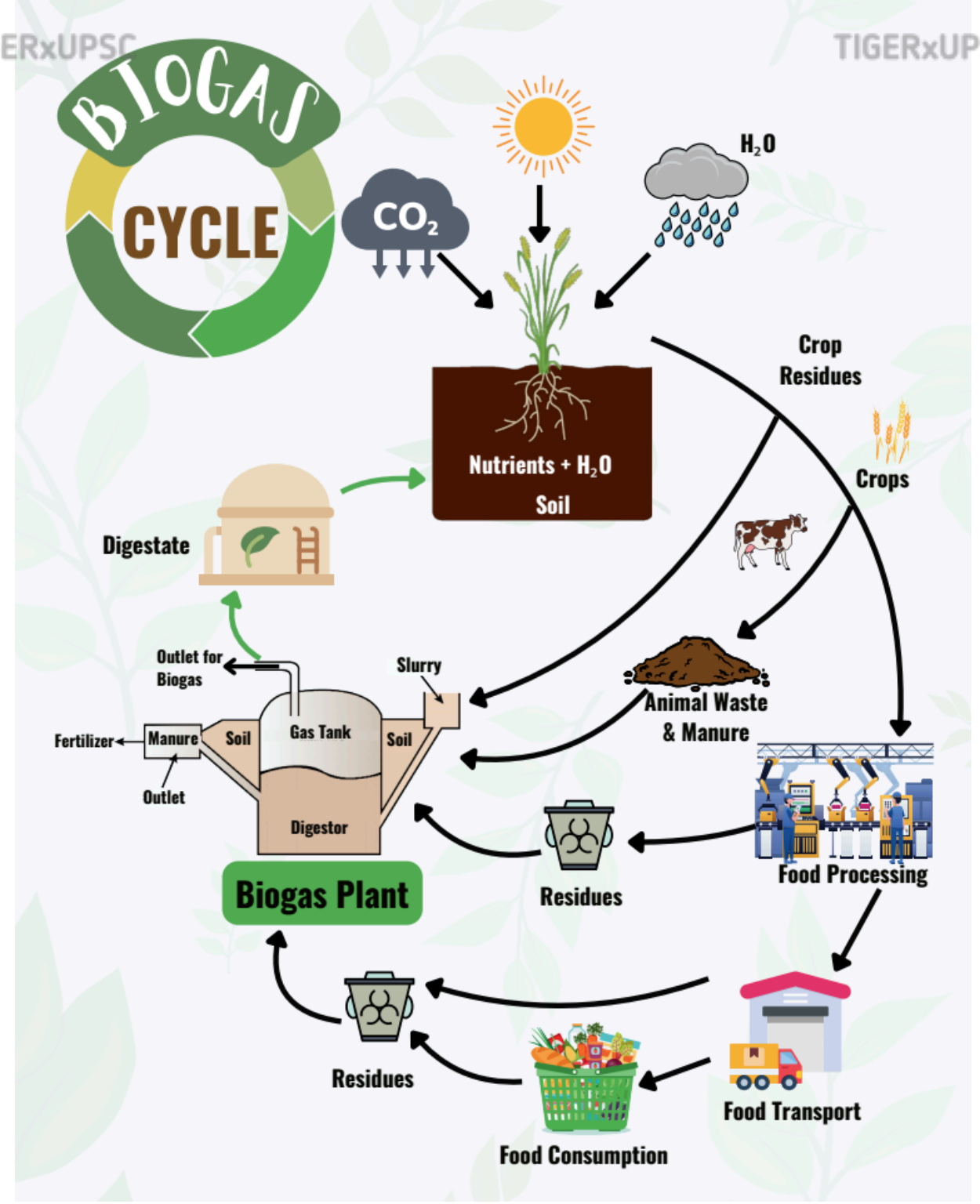


## Guar Beans and its Uses

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**Guar beans (cluster beans)** have been cultivated in north-enwestern India and Pakistan for at least several centuries. It is an **annual legume** that grows well in a variety of soil types and in **arid to semi-arid climates**. Farmers take advantage of the **nitrogen-fixing abilities** of the guar plant by using it as a green manure crop. Guar gum is water soluble and is used in **recipes as a thickener**. It is very similar to corn starch.

It is used to thicken dairy products such as **milk, yoghurt, cheese, ice cream** and **sherbet**. In processed foods, guar gum is used as a thickener in **salad dressing, sauces, ketchup, soup** and many other products.

During 2012–14, Guar gum was in **high demand to use in hydraulic fracturing technology** for shale gas exploration.

**Guar Gum Economics**

Primarily, most of the world's supply of guar beans was used as food for animals and people with a small amount going to the food-processing industry. However, in the last few years, the use of guar gum in hydraulic fracturing began to increase rapidly.

- Extraction processes like hydraulic fracturing may contaminate the ground water aquifers with methane, which consequently affect the aquatic eco-system.
- Hydraulic fracturing process may trigger seismic activities. Pollution can also occur through reinjection wells.

### 3.6.1.3.3.5 Hydrogen Fuel

***(A) Importance of Hydrogen Fuel***

Hydrogen is **the lightest and first element on the periodic table.** Since **the weight of hydrogen is less than air,** it rises in the atmosphere and is therefore **rarely found in its pure form,** those electric vehicles are powered by hydrogen fuel cell, generate water vapour and warm air those exhaust from its pipe emission. This process doesn't cause any environmental pollution.

At standard temperature and pressure, hydrogen is a non-toxic, non-metallic, odourless, tasteless, colourless and highly combustible diatomic gas.

Hydrogen fuel is **a zero-emission fuel** burned with oxygen. It is alternative fuel that can be produced from diverse domestic resources such as natural gas, nuclear power, biomass and renewable power like solar and

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wind. Although the market for hydrogen as a transportation fuel is in its infancy, it **can be used in fuel cells or internal combustion engines.** It is also **used as a fuel for spacecraft propulsion.**

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### *Storage*

**Hydrogen is abundant in our environment. It is stored in water ($H_2O$), hydrocarbons (such as methane, $CH_4$) and other organic matter. One of the challenges of using hydrogen as a fuel comes from being able to efficiently extract it from these compounds.**

- Hydrogen **can be stored physically as either a gas or a liquid.**
- Storage of hydrogen as a gas typically requires high-pressure tanks.
- Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen at one atmosphere pressure is –252.8°C.
- Hydrogen can also be stored on the surfaces of solids (by adsorption) or within solids (by absorption).

### *Potential of Clean Hydrogen Industry in Reducing Greenhouse Gas Emissions*

1. Hydrogen is a clean fuel that, when consumed in a fuel cell, produces only water.
2. Hydrogen is considered an alternative fuel. Efforts are underway by the government and industry towards clean, economical and safe hydrogen production and distribution for widespread use in **fuel cell electric vehicles (FCEVs)** as they are found to be more efficient than vehicles running on gasoline.
3. Light-duty FCEVs are now available in limited quantities to the consumer market in localised regions domestically and around the world.
4. The energy in 2.2 pounds (1 kilogram) of hydrogen gas contains about the same as the energy in 1 gallon (6.2 pounds, 2.8 kilograms) of gasoline.

### *GAIL Starts India's Maiden Project of Blending Hydrogen into Natural Gas System in Indore*

- In line with **National Hydrogen Mission,** in 2022 GAIL commenced **India's first-of-its-kind project of mixing hydrogen into the natural**

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**gas system** to establish the techno-commercial feasibility of **blending hydrogen in City Gas Distribution (CGD) network.**

- The project got initiated in Indore, Madhya Pradesh.
- GAIL started injecting grey hydrogen. This grey hydrogen would subsequently be replaced by green hydrogen.
- The hydrogen-blended natural gas is being supplied to Avantika Gas Ltd, one of GAIL's joint venture with HPCL, for retailing of CNG to automobiles and piped natural gas to households in Indore.

***The Goal***

The government is planning **to blend 15% green hydrogen with piped natural gas (PNG)** for domestic, commercial and industrial consumption.

***(B) National Hydrogen Mission (NHM)***

Energy transition is underway at an exceptional level and several countries are betting on **hydrogen to emerge as the top clean fuel with its high energy density and versatility.** Government of India's (GOI) **National Hydrogen Energy Mission (NHM) initiative** will capitalise on this.

***Efforts in this Regard***

- In the Union Budget 2020–21, the Finance Minister formally announced the NHM, with the aim to generate hydrogen from green power resources.
- The National Green Hydrogen Mission was **approved** by the **Union Cabinet on 4 January 2022**, with the intended objectives of '**Making India a leading producer and supplier of Green Hydrogen in the world'**.

***(C) Various Types of Hydrogen***

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**Based on the process of production**, hydrogen can be classified into **Green Hydrogen, Blue Hydrogen, Pink Hydrogen, Grey Hydrogen and Yellow Hydrogen**.

(Refer **Infographic – Green Hydrogen**)

### National Hydrogen Mission and Green Hydrogen

The Union Cabinet approved the National Green Hydrogen Mission on 4 January 2023, with an outlay of `19,744 crore. The overarching objective of the Mission is to make India a global hub for production, usage and export of Green Hydrogen and its derivatives, by targeting production of 5 MMT per annum of Green Hydrogen by 2030.

The **following components have been announced** as part of the Mission:

- Facilitating demand creation through exports and domestic utilization
- **Strategic Interventions for Green Hydrogen Transition (SIGHT) programme**, which includes incentives for manufacturing of electrolysers and production of green hydrogen
- **Pilot Projects** for steel, mobility, shipping, decentralised energy applications, hydrogen production from biomass, hydrogen storage, etc.
- Development of **Green Hydrogen Hubs**
- Support for **infrastructure development**
- Establishing a robust framework of regulations and standards
- **Research & Development programme** including through a **public–private partnership** framework for R&D
- **Skill development** programme
- **Public awareness** and outreach programme

The National Green Hydrogen Mission has an **outlay of ₹600 crore for the FY 2024–25** under various heads.

The Green Hydrogen production capacity envisaged **by 2030 is likely to leverage over `8 lakh crore in total investments in the Green Hydrogen industry**. This investment is estimated to **create 6,00,000 jobs by 2030.**

Green Hydrogen has the potential to replace the utilization of imported fossil fuels across various sectors, including fertiliser production, petroleum refining, the mobility sector, steel production and shipping propulsion applications.

The Mission is expected to **reduce a cumulative `1 lakh crore worth of fossil fuel imports by 2030**.

- The **Ministry of New & Renewable Energy (MNRE)** has increased the **annual allocation of Green Ammonia** for the **fertiliser sector** from **550,000 to 750,000 tonnes**.
- This move addresses rising demand and strengthens the focus on **Green Hydrogen adoption** in India.

***National Green Hydrogen Mission (NGHM)***

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- **Launch**: January 2023
- **Implementing Body**: Ministry of New & Renewable Energy (MNRE)
- **Goal**: To produce **5 million tonnes per annum** of **Green Hydrogen** by **2030**.

*Key Components of NGHM*

**SIGHT Programme** (Strategic Interventions for Green Hydrogen Transition):

1. Offers **incentives for manufacturing electrolysers**.
2. Promotes **Green Ammonia production**.
3. **Dedicated NGHM Portal**: Serves as a central platform for updates, data and guidance related to the Green Hydrogen ecosystem.
4. **Sector-Specific Initiatives**: Guidelines released for promoting **Green Hydrogen use** in **steel, transport and shipping** sectors.
5. **Innovation Support**: **Hydrogen Valley Innovation Clusters** launched by the **Department of Science and Technology** to boost research, innovation and ecosystem development.

*Government Initiatives*

- **Indian Oil Corporation has signed an agreement** with **Greenstat Norway** for **setting up a Centre of Excellence on Hydrogen (CoE-H)** which will promote R&D projects for the production of green and blue hydrogen between Norwegian and Indian R&D institutions/universities.
- **India and the US have set up a task force under the aegis of the Strategic Clean Energy Partnership (SCEP) to mobilise finance and speed up green energy development**.

*Steps taken by Private Sector*

**Reliance Industries Ltd. (RIL)** is investing `**75,000 crore** in its new business focused on clean energy, which includes solar and green hydrogen.

**The company will build four giga factories focusing** on solar, storage battery, green hydrogen and a fuel cell factory, which can convert hydrogen into mobile and stationary power.

(*Source: PIB Report, 24 JUL 2024*)

## Portal for National Green Hydrogen Mission Launched

The government held a **day-long workshop** on "**Quality Control in Green Hydrogen: Standards & Testing Infrastructure**" at Sushma Swaraj Bhawan, New Delhi on 8 **May 2024**.

The **workshop convened by the Union Ministry of New & Renewable Energy** deliberated on actions needed to create a homogeneous ecosystem for Green Hydrogen production processes through clear quality standards. It also discussed the steps to be taken to create a network of Green Hydrogen testing facilities, and for promoting Ease of Doing Business to accelerate the clean energy transition.

**On the occasion, a dedicated portal for the National Green Hydrogen Mission was launched, which would serve as a one-stop location for information on the Mission and steps taken for the development of the green hydrogen ecosystem in India.**

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The portal was launched jointly by Principal Scientific Advisor to Government of India, Prof. Ajay Kumar Sood, and Secretary, Ministry of New and Renewable Energy, Shri Bhupinder S. Bhalla.

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## STRATEGIC INTERVENTIONS FOR GREEN HYDROGEN TRANSITION (SIGHT) PROGRAMME

According to a report by the Union Power and New & Renewable Energy Minister, a strategic collaboration between India and Saudi Arabia holds the potential to capitalise on each other's strengths, thereby nurturing the growth of Green Hydrogen ecosystems in both nations. This collaboration could pave the way for mutual benefit, driving advancements in renewable energy technology and fostering sustainable growth. "Many countries and regions have announced new strategies to import Green/Clean Hydrogen and its derivatives, thus providing an opportunity for Indian producers."

**India and Saudi Arabia signed an MoU in the fields of Electrical Interconnection, Green/Clean Hydrogen and Supply Chains on 8.**

This strategic collaboration is expected to be advantageous for both countries, as they can leverage each other's capabilities to foster the growth of Green Hydrogen ecosystem within their respective nations.

The **Ministry of New & Renewable Energy is implementing the National Green Hydrogen Mission, approved** by the Union Cabinet on 4 January 2023, with an outlay of ₹19,744 crore from FY 2023–24 to FY 2029–30.

The National Green Hydrogen Mission has a **provision for supporting pilot projects for low-carbon steel, mobility, shipping, and ports**.

The Mission allocates funds for various sub-components, including SIGHT, Pilot projects, R&D, etc., to support specific projects that have been selected. However, it's important to note that the Mission does not allocate funds on a state-wise basis. Instead, funding is directed towards specific projects based on their merit and alignment with the objectives of the Mission.

The **Strategic Interventions for Green Hydrogen Transition (SIGHT) Programme**, is a **major financial measure with an outlay of ₹17,490 crore. The programme consists of two distinct financial incentive mechanisms to support domestic manufacturing of electrolysers and production of Green Hydrogen.**

Many countries/regions around the world like European Union, Singapore, South Korea, Japan among others have announced their new strategies to import Green/Clean Hydrogen and its derivatives, thereby providing an opportunity for Indian producers.

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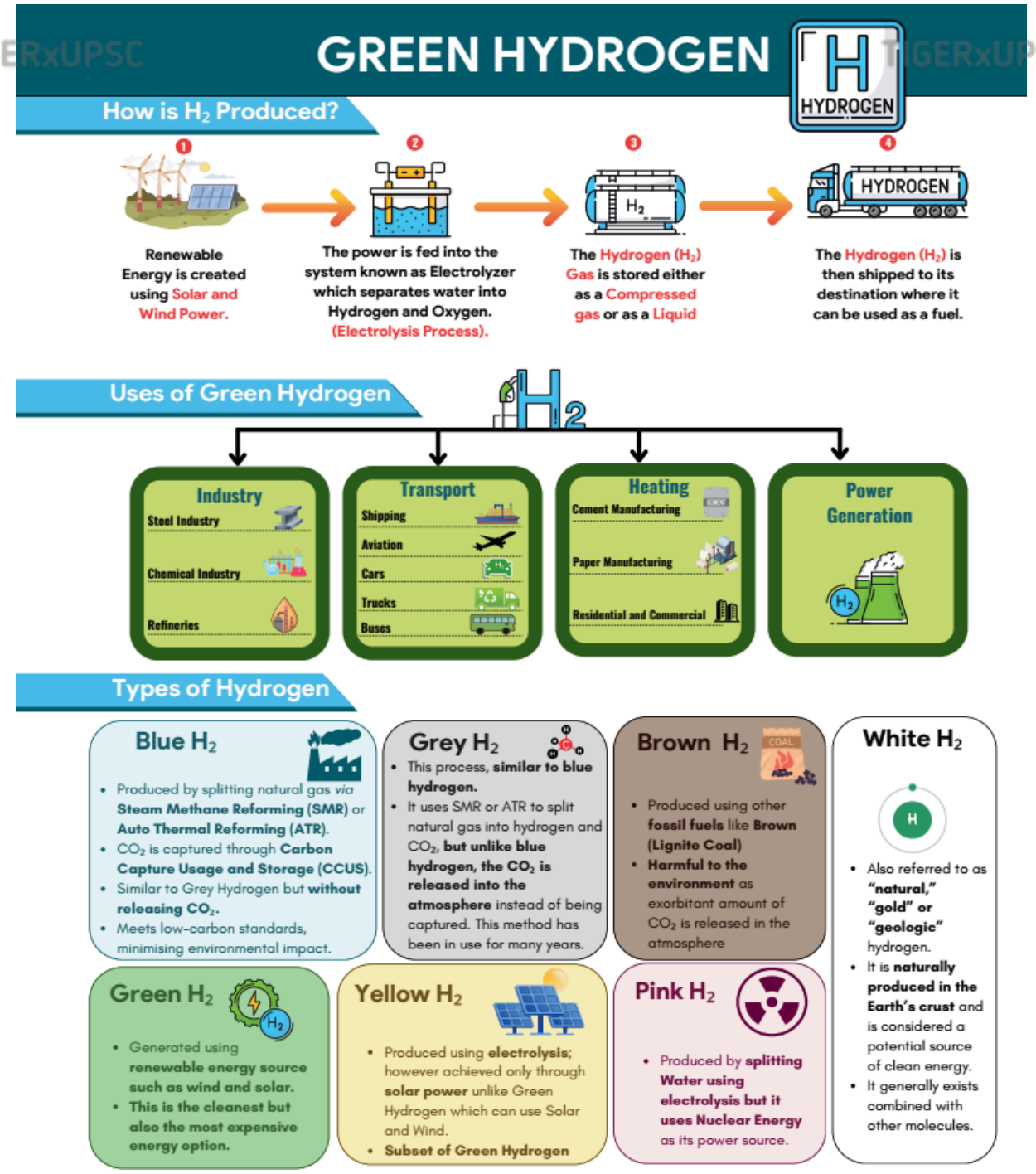


## 3.6.2 Renewable Sources of Energy

The energy that is collected from resources, which are naturally replenished on a human timescale, is termed as renewable sources of energy. They include sunlight, wind, rain, tides, waves and geothermal heat. The four

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important areas often supported by renewable energy are electricity generation, air and water heating/cooling, transportation and rural (off-grid) energy services.

Renewable energy in India comes under the purview of the Ministry of New and Renewable Energy (MNRE).

## Advantages and Disadvantages of Renewable Energy

### *Advantages*

- **Renewable Energy:** These sources are with infinite sustainability and we will never run out of it.
- **Environmental Benefits:** They are clean sources, causing little-to-no greenhouse gases and net carbon emissions, thus being eco-friendly.
- **Reliable energy source.**
- **Economic Benefits:** They are comparatively cheaper and more economically sound (being cost effective). As per estimations, renewable energy manufacturing will generate hundreds of thousands of stable jobs.
- **Stabilise Energy Prices:** They provide reliable pricing on energy.

### *Disadvantages*

- **Reliability of Supply:** One flaw is that sources heavily depend upon the weather for sources of supply: For instance, rain, wind and sunshine. Climatic factors may also cause issues.
- **Difficult to Generate in Large Quantity.**
- **Large Capital Cost:** Initial investments are quite high to establish renewable energy plants.
- **Large Tracts of Land Required:** The electricity requirement is high and to replace the non-renewable source to fulfil such high demand, more and more solar panels and wind farms need to be set up, which require large tracts of land.

## 3.6.2.1 Various Renewable Energy Sources in India

### *3.6.2.1.1 Solar Energy*

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Due to the location of India between the Tropic of Cancer and the Equator, the average annual temperatures range around 25–27°C, which shows India's huge solar potential. India is one of the few countries with plenty of sunshine, with approximately 300 clear sunny days per year. 0.1% of the solar energy coming to Earth, which can be utilised in different ways.

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### 3.6.2.1.1.1 Different Ways of Solar Energy Utilisation

There are different ways of solar energy utilisation:

1. Photosynthesis
2. Solar energy into thermal energy
3. Photovoltaic cells (solar energy into electrical energy)

**Photosynthesis** is a process through which plants use sunlight for food synthesis. Main source of life on Earth is the Sun due to autotrophic nature of plant.

**Solar Energy into Thermal Energy** This is a technology that will be used for the utilisation of solar energy for the generation of heat; e.g., **solar cooker, solar water heating system, solar air heating** and **solar pond.** Asia's first solar pond is located in Bhuj (Gujarat).

**Concentrated Solar Thermal Systems (CSP)** – The radiation of the sun is concentrated by CSP to heat a liquid substance, which is later used to drive a heat engine and even an electric generator. Alternating Current (AC) is generated by indirect method, which can be easily distributed on the power network.

**Photovoltaic Cells (PVCs)** Direct conversion of light into electricity, when light falls on certain elements like silicon the flow of electron starts. Photovoltaic (PV) solar panels, unlike solar thermal systems use sunlight through the 'photovoltaic effect' to generate direct current (DC) in a direct electricity production process rather than using the Sun's heat to generate power (as in solar thermal system). Here, the DC is then converted to AC,

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usually for which inverters are used, to be distributed on the power network.

**Silicon wafer is a material essential for manufacturing of semiconductors, used in photovoltaic units.** From last few years India has boost up manufacturing of silicon wafer in electronics. But India is not among top three manufacturing countries of silicon wafer.

(Refer **Infographic – Solar Photovoltaic & Solar Water Heater**)

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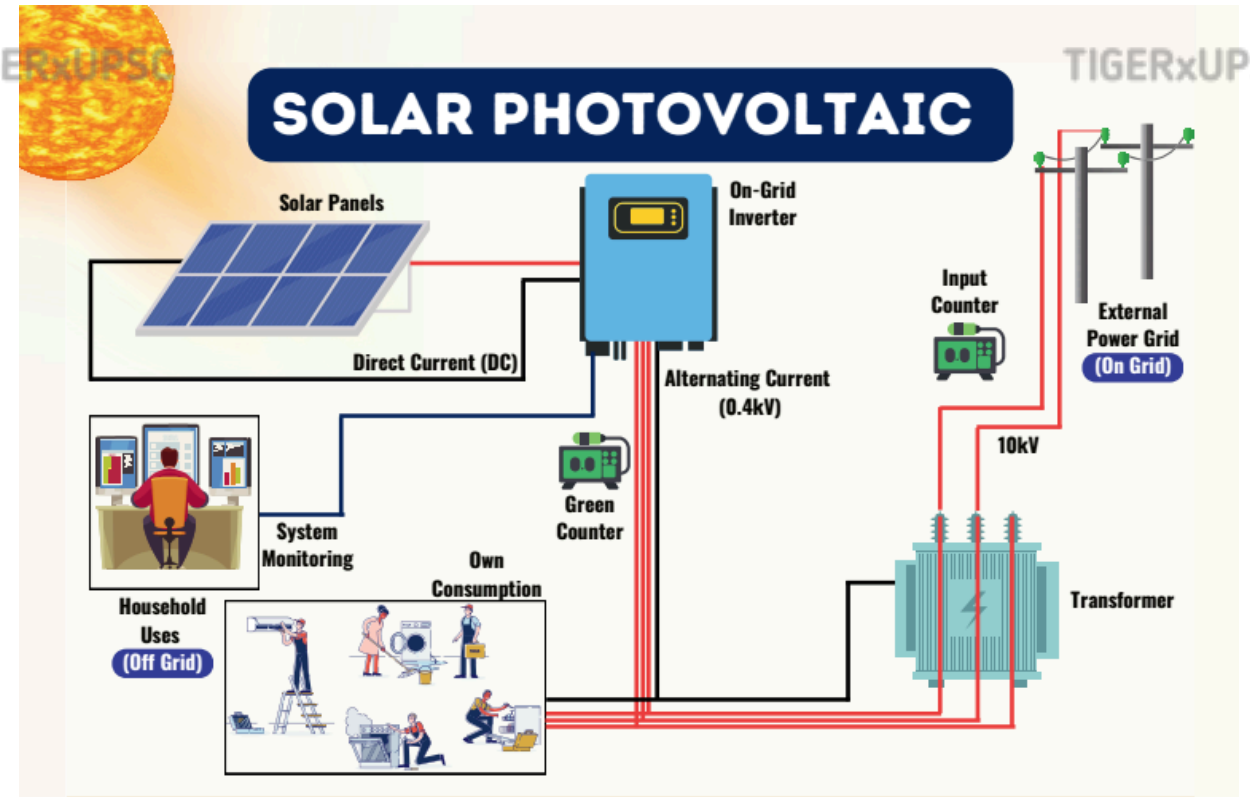


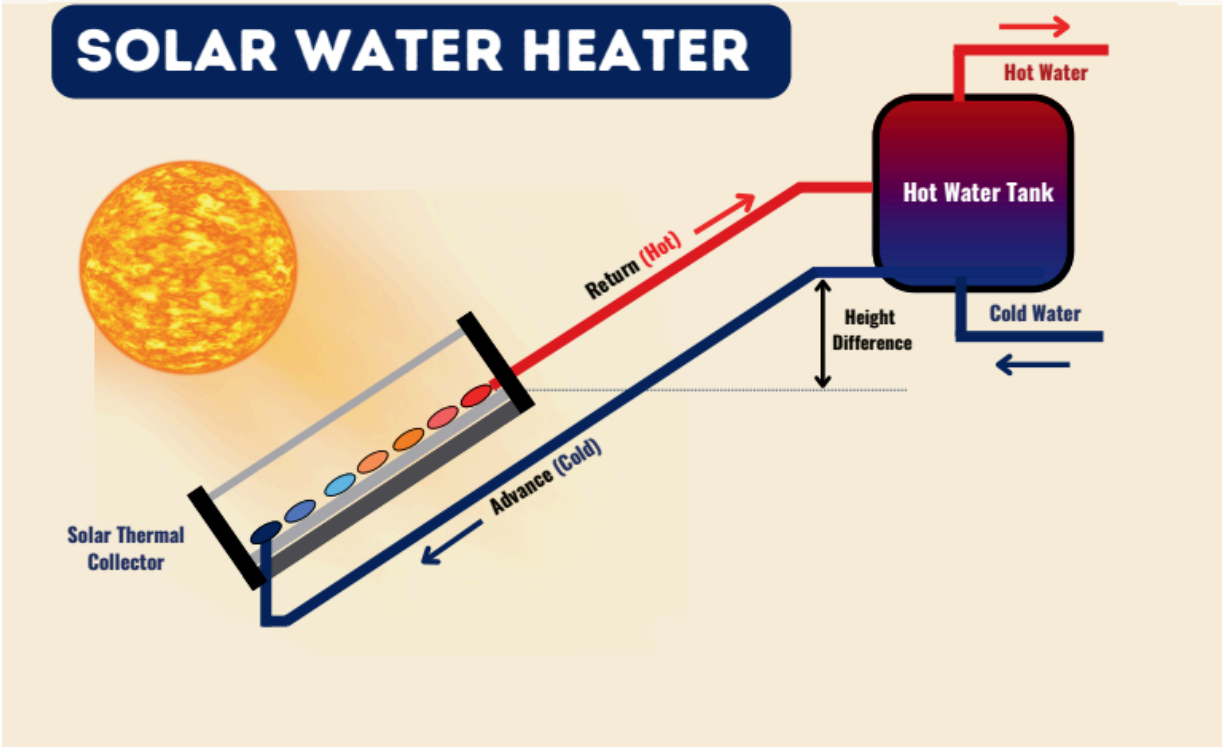


**GREEN ENERGY CORRIDOR**

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The Cabinet Committee on Economic Affairs in 2022 approved **the scheme on Green Energy Corridor (GEC)** Phase-II for **Intra-State Transmission System (InSTS)**.

### Green Energy Corridor Project for Intra-State Transmission System (InSTS)

- The **GEC Project** aims at synchronising electricity produced from renewable sources such as solar and wind with conventional power stations in the grid.
- The **GEC-InSTS** project was sanctioned in 2015–16, for evacuation and integration of the renewable energy capacity through setting up of transmission lines and increasing transformation capacity of substations.

### Phase I of the Project

- **It is being implemented by eight renewable-rich states** of Tamil Nadu, Rajasthan, Karnataka, Andhra Pradesh, Maharashtra, Gujarat, Himachal Pradesh and Madhya Pradesh.
- Under this phase, **the target is to** install 9700 circuit km of transmission lines and 22,600 MVA (MegaVolt-Amperes) transformation capacity of substations by 2022.
- The funding mechanism consists of a 40% Government of India Grant, 20% state equity and a 40% loan from KfW Bank, Germany.

### Phase II of the Project

- It is being implemented in seven states namely, Gujarat, Himachal Pradesh, Karnataka, Kerala, Rajasthan, Tamil Nadu and Uttar Pradesh.
- Under this phase, the target is to install 10,750 circuit km of transmission lines and 27,500 MVA transformation capacity of substations by 2025–26.
- The Centre will provide assistance at 33% of the cost of the project.

### Need for a Green Energy Corridor in India

- The Prime Minister has pledged **to increase the country's non-fossil fuel power generation capacity to 500 GW and meet 50% of its energy requirements from renewable sources** by the end of this decade. So there is a need for -integration.
- The scheme will help in **achieving the target of 500 GW installed RE capacity by 2030**.
- It will also contribute to **long-term energy security of the country** and promote ecologically sustainable growth by reducing the carbon footprint.
- It will **generate large direct and indirect employment** opportunities for both skilled and unskilled personnel in power and other related sectors.
- The GEC will **help in offsetting the intra-state transmission charges** and keep the power costs down. Thus, the government support will ultimately benefit the end users.
- It helps India to increase the share of non-fossil fuels-based electricity to 50% by 2030.
- The project is expected to help India meet **the climate commitments it made at the COP26 summit in Glasgow.**

## 3.6.2.1.1.2 Solar Power Tariff

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**Solar energy based power generation in per unit price is solar tariff.**

**While the Central Electricity Regulatory Commission regulates the tariff of generating companies owned or controlled by the central government, the Solar Energy Corporation of India Limited (SECI) has a power trading license.**

### 3.6.2.1.1.3 Ongrid and Offgrid System of Solar Power

***(A) Ongrid***

- Ongrid systems are solar PV systems that only generate power when the utility power grid is available. They must connect to the grid to function. They can send excess power generated back to the grid when you are overproducing, so you credit it for later use.
- This system is connected to the main power supply just like a regular home. This means that main power can still be used any time your solar panels generate less electricity than is needed.
- A solar electricity system without batteries is called grid-tied or grid-connected solar. It will enable every home to be hooked up to both the national grid and the Power Smart solar electricity system. Hence, the electricity will be available at the flick of a switch; even at night, or if it is cloudy. The household can install a solar electricity system of any size and the bill will be generated only if the consumption exceeds generation.
- As a 100% efficient system, this electricity will maintain free battery, on top of unlimited 'on-demand' energy, and hence grid-connected solar electricity appears attractive. On top of the benefits of electricity grid, it has an extra advantage of a portion of electricity bill never increasing in price.

***(B) Offgrid***

- These systems allow to store solar power in batteries for use when the power grid goes down or if are not on the grid. Hybrid systems provide power to offset the grid power whenever the sun is shining and will even send excess power to the grid for credit and for later use.

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- Saffron colour represents energy and Saffron Revolution focuses on renewable energy sources like **solar energy** to fulfil growing energy demand of India. India's solar potential is huge. Of the world's 20 top economies, it has the most sunlight, especially in the vast deserts.
- Also, solar energy is a way to leap progress over conventional fossil fuel power into twenty-first century's clear energy economy. It would also reduce India's hefty carbon footprint, which comes from relying on coal for 60% of its electricity.

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- These systems do not need to be connected to mains power, as there is no ready source of reserve power. They rely on storage batteries or generators to provide electricity on days when the solar panels generate less electricity than is required.
- System requirements – 1. Suitable solar mounting, 2. Offgrid solar inverter, 3. Solar power controller and batteries, 4. AC and DC safety isolation switches, 5. Grounding earth cables and clamps.

### 3.6.2.1.1.4 Solar Water Pump

The solar water pumping system uses solar energy to run pump water. There are three main components of any solar water system:

1. Solar PV panel
2. Any motor-pump sets, those will be equipped with the photovoltaic array:
   - ❑ Surface-mounted centrifugal pump set
   - ❑ Submersible pump set
   - ❑ Floating pump set
3. Pipes

***Working Mechanism of a Solar Water Pump***

This pump operates on solar PV (photovoltaic) system-based power generation. The photovoltaic array converts the solar energy into electricity, which is used for running the motor pump set. Any type of pumping system (either surface pump or submersible pump) draws water from the open well, bore well, stream, pond, canal, etc. The system requires a shadow-free area for installation of the Solar panel.

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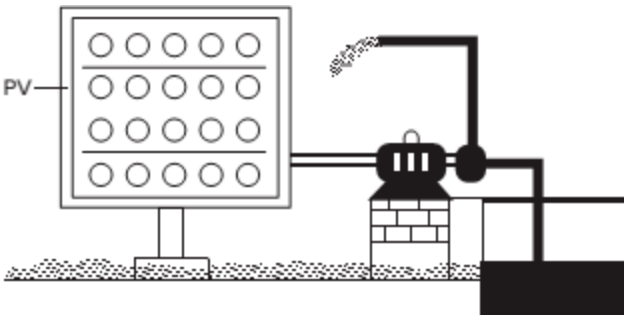


**Figure 3.1** Mechanism of Solar Water Pump

### *Utility*

A system with 1800 watt PV array capacity and 2 HP pump can give a water discharge of 1.4 lakh litres per day from a depth of 6 to 7 metres. This quantity of water is considered adequate for irrigating about 5–8 acres of land holding for several crops.

### *Advantages of a Solar Water Pumping System*

1. No fuel cost – as it uses available free sun light.
2. No electricity required.
3. Long operating life.
4. Highly reliable and durable.
5. Easy to operate and maintain.
6. Eco-friendly.

## APPROXIMATE COST

| Capacity of SPV system | Total Head (Suction & delivery) in m | Solar PV (watt) | Cost (₹) |
|---|---|---|---|
| 2 HP (surface pump) | 10 | 1800 | 2,90,000 |
| 4.6 HP (submersible pump) | 30 | 4800 | 7,15,000 |

This mission will create an enabling policy to make India a global leader in solar energy. To fulfil this vision, a National Solar Mission is

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being launched under the brand name 'Solar India'.

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***Mission Targets***

- To create an enabling policy framework.
- To create favourable conditions for solar manufacturing capability (especially solar thermal for indigenous production and market leadership).
- To increase solar thermal collector area.
- To deploy more solar lighting systems (Photovoltaics) for rural areas.

### 3.6.2.1.1.5 Solar Mission of India

The Jawaharlal Nehru National Solar Mission was launched on the 11 January 2010 by the UPA government. Earlier, the mission has setup the ambitious target of deploying 20,000 megawatts of electricity generations those extended 1,00,000 MW (100 GW) of grid connected solar power by 2022 (by NDA Government) with aim to reducing the cost of solar power generation in the country through

(i) Long-term policy in nation.
(ii) Large-scale deployment goals in India.
(iii) Aggressive research and development; and
(iv) Domestic production of raw materials, components and products to achieve grid tariff parity by 2022.

Note: **The 2023 budget identified renewable energy as one of the key seven priorities and the Indian government's commitment to increasing its renewable energy target to 500 GW by 2030 at the COP26 summit has provided great support to the industry and spurred unprecedented growth. However, there is still much work to be done, with a need to install 25–30 GW of solar energy each year for the next eight years to meet its 280 GW solar target.**

### 3.6.2.1.1.6 International Solar Alliance (ISA)

It is an international organisation with 116 Members and Signatory countries, which come together to enhance solar energy in the world. These countries are mostly between the Tropic of Cancer and the Tropic of Capricorn due to the high impact of solar radiation to reduce greenhouse gas emission in the atmosphere. This alliance is a treaty-based inter-

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governmental organisation. This initiative was brought forward by Narendra Modi at the India Africa Summit, and a meeting of member countries ahead of the 2015 United Nations Climate Change Conference in Paris in November 2015. The International Solar Alliance Framework Agreement opened for signatures in Marrakech, Morocco, in November 2016.

Under ISA, there has been an agreement between India and France in January 2016, which jointly laid the foundation stone at the National Institute of Solar Energy (NISE) in **Gwalpahari, Gurgaon.** The Indian government has dedicated five acres of land on the NISE campus for ISA as a future headquarters. France will contribute INR 1.75 billion (US$ 26 million) to build the ISA campus and for meeting expenditures for ISA's first five years. This alliance is also called **International Agency for Solar Policy and Application (IASPA).**

**India has already aimed to generate more than 500 gigawatts of renewable energy capacity by 2030 and reducing greenhouse gas emission intensity by 50% by 2030.**

***Objectives***

- It will promote solar technologies and investment in the solar sector to enhance income generation for the poor.
- It will formulate projects and programmes to promote solar applications.
- It will develop innovative financial mechanisms and reduce capital cost.
- It will unable common knowledge e-Portal build-up.
- It will facilitate capacity building for promotion and utilisation of solar technologies.
- It will promote research and development among member countries.

**ISA's Objectives**

The ISA's major objectives include global deployment of over 1,000 GW of solar generation capacity and mobilisation of investment of over **US$ 1000 billion** into solar energy by 2030.

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ISA brings together countries with rich solar potential to aggregate global demand, thereby reducing prices through bulk purchase, facilitating the deployment of existing solar technologies at scale, and promoting collaborative solar R & D and capacity building.

**ISA Framework Agreement entered into the force on December 6th, 2017, headquarters at Gurugram, India.**

**India announced a goal of obtaining 50% of its electricity from non-fossil fuels by 2030 at the Paris climate change summit, in pursuit of achieving its target of 500 GW by 2030.**

**Membership**

- All members of the United Nations can join the International Solar Alliance (ISA) under the recently amended ISA Framework Agreement.
- **Goal**: It has set a target of 1 TW of solar energy by 2030.

***Advantages for India***

- Launching of ISA has set up the stage for India's proactive and forward-looking leadership on climate change and the transition to a less carbon-intensive growth trajectory.
- It will also help in bringing down the price of solar technology, which will further help in accelerating the development of the country.
- It will help India to meet its Intended Nationally Determined Contribution target.

***Challenges***

- **Funding:** Although alliance talks about developing 'innovative financial mechanisms', the way to provide the capital is not fully and clearly addressed.
- **Technology Sharing:** There is need to create a comprehensive framework to share the modern solar technologies at low cost.

## 3.6.2.1.1.7 Solar Impulse

Solar Impulse is a Swiss long-range experimental solar-powered aircraft project. Solar impulse is also the name of the project's two operational aircrafts. The motives of the project were to make the first circumnavigation of the Earth by a piloted fixed-wing aircraft deploying solar power only, thereby focussing on clean technologies.

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The aircrafts are single-seat monoplanes, which are capable of taking off under their own power using photovoltaic cells. The prototype, often known as Solar Impulse 1, was designed to remain airborne for up to 36 hours. In December 2009, its first test flight was conducted and it flew an entire diurnal solar cycle, in July 2010.

A second aircraft, named Solar Impulse 2, was completed in 2014. It carries more solar cells and more powerful motors, besides other improvements. It departed from Abu Dhabi in United Arab Emirates, on 9 March 2015.

### 3.6.2.1.1.8 Solar Powerhouse

The Cabinet Committee on Economic Affairs approved phase 2 of **the grid-connected rooftop solar programme,** with a focus on the residential sector. **India has set an ambitious target of achieving 40 GW of rooftop solar capacity by 2022. According to a recent report, it is now supposed to be achieved by 2026.**

#### ONE SUN, ONE WORLD, ONE GRID (OSOWOG)

This initiative was announced in the **United Nations Framework Convention on Climate Change (UNFCCC) – COP26 [2021] convention** by **India and the United Kingdom** to tap solar energy and have it travel seamlessly across borders.

**About the Initiative**

- The **grid will be set up over the next few years by the International Solar Alliance (ISA),** another initiative authored by India initially, to transport solar power to different countries.
- The vision behind the OSOWOG is 'The Sun Never Sets' and is a constant at some geographical location, globally, at any given point of time.

**Implementation**

- A Ministerial Steering Group will work towards accelerating the making of large solar power stations and wind farms in the best locations, linked together by continental-scale grids crossing national borders.
- The Ministerial Steering Group includes France, India, the United Kingdom and the United States, and will also have representatives from Africa, the Gulf countries, Latin America and Southeast Asia.

**Significance of World Grid**

- With one worldwide grid, we can access clean energy at all places. The need to store energy would also lessen, and the viability of solar projects will increase.

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**Potential and Benefits of the Initiative**

- Following the mantra 'One Sun One Word One Grid' (OSOWOG), India would generate 50% of power from non-fossil fuels by 2030 and has called for connecting solar energy supply across borders.
- The proposed integration would lead to reduced project costs, higher efficiencies and increased asset utilisation for all the participating entities.
- This plan will require only incremental investment because it will not require a parallel grid infrastructure due to working with existing grids.
- It will help all the participating entities in attracting investments in renewable energy sources as well as utilising skills, technology and finances.
- Resulting economic benefits would positively impact poverty alleviation and support in mitigating water, sanitation, food and other socio-economic challenges.
- It will allow national renewable energy management centres in India to grow as regional and global management centres.
- At a global level, almost 2,600 GW of interconnection capacity may be possible up to 2050, delivering estimated power savings of 226 billion euros per year.

***Need of the Hour***

1. **Devise simple, well-designed and creative ways to disseminate information.**
2. **The local electricity linesmen, electricity inspectors, and other nodal officials in the electricity department also have key roles to play.**
3. **Objective information must be put out through various avenues,** so that it is accessible to all segments of the population and in local languages. Such awareness drives will reach larger audiences.
4. Information kiosks can be set up in public institutions like banks to offer information on the technology, as well as on practical issues such as guidance on selecting vendors.
5. **A robust feedback mechanism** can be put in place for consumers to share their experiences with others.
6. **Resident welfare associations (RWAs) can tie up with vendors to organise demonstration programmes,** so that consumers can observe, operate and understand how the system works.
7. Awareness building sessions need to **be socially inclusive and should take place during periods when consumers are likely to be at home.**

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## 3.6.2.1.1.9 Floating Solar Plant

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India's largest floating solar power plant is in **Ramagundam (Telangana)** and this place is known for its blistering summer heat. Locally, it is sometimes referred to as 'Agnigundam' (firewell) in Telugu language.

### *Floating Solar Plants – Significance*

- Floating solar plants are considered an **alternate option to tackle land availability issues.** The concept **involves setting up solar panels on floats placed on dams, lakes and similar water bodies.**

### *Challenges*

- In India, floating solar plant is likely to face challenges scaling up to the level of ground-mounted plants.
- Despite being land neutral, the cost of the floating systems including anchoring, installation, maintenance and transmission renders the overall cost of the floating solar systems much higher than the land-based systems at this initial stage of development.
- Besides the two major issues of corrosion and instability, other issues like the long-term impact of moist environment on modules, cables, safe transmission of power through the floats to the nearest feeder point, the environmental impact on the water body and the marine life, etc., needs to be addressed and make the system cost-effective.

### *Advantages of Floating Solar Power Projects*

Floating solar power projects deploy the photovoltaic panels on the surface of water bodies and hence have no issues of land acquisition, leaving the land for other essential uses like agriculture, etc. This improves the performance of solar photovoltaic panels by 5–10%, thus leading to significant cost savings over the time. It will also reduce water evaporation and do improvement of water quality like reduction in algal bloom, etc.

Thus, as a viable alternative to solar installations on land, India has also been making progress in floating solar power. Following are the top upcoming floating solar power projects in India – in planning stage, initial process, or partly under operation.

### *What Fails Floating Solar?*

- Non-availability of floats in India makes it an expensive option.
- Project costs higher by 30–50% than ground mounted solar.

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- Dependent on European or Chinese suppliers.

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### 3.6.2.1.1.10 Kisan Urja Suraksha Evam Utthaan Mahabhiyan (KUSUM)

Ministry of New and Renewable Energy (MNRE) has launched the Pradhan Mantri Kisan Urja Suraksha evem Utthan Mahabhiyan (PM KUSUM) Scheme for farmers for installation of solar pumps and grid connected solar and other renewable power plants in the country.

The Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyaan (PM-KUSUM) scheme has been extended. It was **initially launched in March 2019** and has been scaled up with the extension of the deadline to 31 March 2026.

The scheme aims to provide energy and water security to farmers, enhance their income, de-dieselise the farm sector and reduce environmental pollution.

### Target: Adding around 34,800 MW of solar capacity by March 2026

Key Points about the Extension:

**Extended Deadline:** The scheme's implementation period has been extended to 31 March 2026.

**Objective:** The PM-KUSUM scheme aims to promote solarisation of agricultural pumps, grid-connected solar power plants and feeder-level solarisation.

#### *Components*

The PM-KUSUM scheme has three components:

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- **Component A:** Setting up of Decentralised Ground/Stilt Mounted Grid Connected Solar Energy-based Power Plants by the farmers on their land.
- **Component B:** Installation of Stand-alone Solar Agriculture Pumps.
- **Component C:** Solarisation of Grid Connected Agriculture Pumps, including Feeder Level Solarisation.

**Financial Assistance:** The scheme provides financial assistance to farmers through Central Financial Assistance (CFA), procurement-based incentives (PBI) and State Government subsidies.

**Implementation:** The scheme is implemented by the Ministry of New and Renewable Energy (MNRE) and various State Implementing Agencies (SIAs).

***Objectives***

The objectives of the PM-KUSUM scheme are to:

- Incorporate renewable energy in farmers' irrigation practices.

**ELIGIBILITY**

The following entities are eligible under the PM-KUSUM scheme:

- Individual farmers
- Groups of farmers
- Farmer Producer Organisations
- Panchayats
- Cooperatives
- Water User Associations

- **Replace high-cost diesel** with less expensive solar energy.
- **Reduce the agriculture electricity subsidy** burden on States and improve the financial health of DISCOMs.
- Help farmers gain access to **solar water-pumps at subsidised rates**.
- Provide farmers an avenue to **utilise their barren land** by setting up solar power plants for energy generation.

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### *Benefits of PM-KUSUM Scheme*

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- **Reduces Reliance on Diesel:** Helps in **de-dieselisation** of the farm sector, reducing fuel costs and pollution.
- **Boosts Farmers' Income:** Farmers can **sell surplus power** generated from solar plants.
- **Enhances Energy Security:** Provides **reliable electricity** for irrigation, reducing dependence on erratic grid power.
- **Encourages renewable energy adoption:** Contributes to India's **solar capacity expansion**.
- **Supports water conservation:** Promotes **efficient irrigation** practices.
- **Reduces Burden on DISCOMs:** Decentralised solar power generation **lowers transmission losses**.

### *Scheme Benefits*

The scheme will open a stable and continuous source of income to the rural land owners for a period of 25 years by utilisation of their dry/uncultivable land. Further, if cultivated fields are chosen for setting up solar power project, the farmers will be able to continue to grow crops as the solar panels are to be set up above a minimum height.

The proposed scheme would ensure that sufficient local solar/other renewable energy-based power is available for feeding rural load centres and agriculture pump-set loads, which require power mostly during the day time. There will be reduced transmission losses for STUs and DISCOMs as these power plants will be located closer to the agriculture loads or to electrical substations in a decentralised manner

The solar pumps will not only save the expenditure incurred on diesel for running diesel pump and provide the farmers a reliable source of irrigation through solar pump but also will be preventing harmful pollution caused by running diesel pump.

In light of the long waiting list for electric grid connection, this scheme will benefit 17.5 lakh farmers over a period of four years, without adding to the grid load.

**Way Forward**

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**Speed up Implementation:** Address delays in fund disbursal and project approvals.

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**Strengthen Awareness:** Conduct **awareness campaigns** to educate farmers about **fraudulent websites** and **genuine application processes**.

**Improve Financial Support:** Offer **low-interest loans** and flexible financing to reduce farmers' initial investment burden.

**Enhance Technology Adoption:** Promote **advanced irrigation techniques** like **drip and sprinkler systems** alongside solar pumps.

**Expand Coverage:** Extend the scheme to **more states** and include **small & marginal farmers**.

**Encourage Public-private Partnerships:** Involve **private players** to boost efficiency and scale-up implementation.

**Strengthen Monitoring:** Develop **real-time tracking mechanisms** to ensure effective scheme execution and prevent fraud.

## 3.6.2.1.1.11 Solar Charkha Mission

***Specification***

- Launched in 2018.
- It is a Ministry of Micro Small & Medium Enterprises (MSME) initiative launched during June 2018.
- The Khadi and Village Industries Commission (KVIC) would implement the programme.
- It seeks to generate employment in rural areas and contribute to the green economy.
- The mission will entail a subsidy of ₹550 crore in the initial two years for 50 clusters and every cluster will employ 400 to 2000 artisans.
- The scheme also aims at linking five crore women across the country to the initiative.
- The mission is expected to create one lakh jobs during the first two years.

***Objectives***

- To ensure inclusive growth by generation of employment, especially for women and youth, and sustainable development through solar charkha clusters in rural areas.
- To boost rural economy and help in arresting migration from rural to urban areas.

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- To leverage low-cost, innovative technologies and processes for sustenance.

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***Significance***

- These solar charkhas are to be operated using solar power which is a renewable energy source. Being an environment friendly programme, not only will it help in development of Green Economy but will also generate sustainable employment for the artisans.

### 3.6.2.1.1.12 Pradhan Mantri Suryodaya Yojana and PM Surya Ghar Yojana

The government introduced the 'Pradhan Mantri Suryodaya Yojana' to **propel solar energy adoption by providing rooftop solar power systems to one crore households**.

This initiative ***aligns with India's ambitious target of achieving 40 GW of rooftop solar capacity***. By facilitating the installation of solar panels on rooftops, the **scheme aims to reduce electricity bills for households while simultaneously advancing India's goal of energy self-reliance**.

***Current Solar Capacity in India***

As of December 2023, India's solar power ***installed capacity stands at approximately 73.31 GW***, with rooftop solar contributing around 11.08 GW. **Solar energy holds a significant share in India's total renewable energy capacity, which is estimated to be around 180 GW**.

Among Indian states, **Rajasthan leads in total solar capacity, while Gujarat leads in rooftop solar capacity**.

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***PM Surya Ghar Yojana/Har Ghar Bijli Yojana***

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## Key Features

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| Feature | Description |
|---|---|
| Scheme Name | PM Surya Ghar Yojana/Har Ghar Bijli Yojana |
| Primary Goal | To promote rooftop solar in residential sector |
| Target Households | 1 Crore households (300 units/month) |
| Key Components | Subsidies, loans, capacity building, awareness |
| Training Focus | Installation, operation, maintenance, repairs etc |
| Skilled Manpower Target | More than 3 lakh individuals |
| Implementing Ministry | MNRE-Ministry of New and Renewable Energy |

### SOLAR ENERGY: FACTS CORNER

**(I) Current Top 5 Largest Solar Power Plants as of 2025**

1. **Golmud Solar Park**
   **Location:** Qinghai Province, China
   **Capacity:** Around 2800 MW
2. **Bhadla Solar Park**
   **Location:** Bhadla, Rajasthan, India
   **Capacity:** Approximately 2700 MW
3. **Pavagada Solar Park**
   **Location:** Pavagada, Karnataka, India
   **Capacity:** Nearly 2,050 MW
4. **Benban Solar Park**
   **Location:** Egypt
   **Capacity:** 1770 MW
5. **Mohammed bin Rashid Al Maktoum Solar Park**
   **Location:** Dubai, UAE
   **Capacity:** 1630 MW

**(II) List of Major Solar Parks of India**

- **Bhadla Solar Park:** Bhadla, Phalodi tehsil, Jodhpur district, Rajasthan, India.
- **Pavagada Solar Park:** Pavagada taluk, Tumkur district, Karnataka.
- **Kurnool Ultra Mega Solar Park:** Panyam mandal of Kurnool district, Andhra.
- **NP Kunta Ultra Mega Solar Park:** Also known as Ananthapuram Ultra Mega Solar Park, Ananthapur district of the Indian state of Andhra Pradesh.
- **Rewa Ultra Mega Solar:** Asia's Largest Single site solar plant.
- **Charanka Solar Park:** It is currently in the process of development, near Charanka village in Patan district, northern Gujarat.
- **Kamuthi Solar Power Project:** Madurai, in the state of Tamil Nadu, India.

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- **Kadapa Ultra Mega Solar Park:** Mylavaram mandal of Kadapa district, Andhra Pradesh.

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**(III) Floating Solar Power Projects in India**

1. **Ramagundam Reservoir:** Total capacity – 100 MW, State – Telangana
2. **Kayamkulam Project:** Total capacity – 92 MW, State – Kerala
3. **Rihand Dam:** Total capacity – 50 MW, State – Uttar Pradesh
4. **Simhadri Floating Solar PV Project:** Total capacity 25MW, State – Andhra Pradesh

Note: **Omkareshwar Reservoir (Madhya Pradesh) with a total capacity of 600 MW is the proposed biggest floating solar power plant in India.**

***Benefits of Solar Rooftop Scheme/PM Surya Ghar Yojana***

- Reduced electricity bills
- Long-term investment
- Job creation
- Environmental benefits
- Energy independence

***Need for Solar Energy Expansion***

India faces a substantial increase in energy demand, as projected by the International Energy Agency (IEA). To meet this growing demand sustainably, India must transition towards reliable and renewable energy sources, reducing dependence on coal. Despite efforts to boost coal production, **India aims to achieve 500 GW of renewable energy capacity by 2030**. This expansion in solar energy infrastructure is crucial to meet future energy needs efficiently while addressing environmental concerns.

## 3.6.2.1.1.13 World's Largest Solar Tree

***CSIR-CMERI develops world's largest 'Solar Tree'***

To promote the usage of solar energy, a leading CSIR institute **'Central Mechanical Engineering Research Institute' has developed 'world's largest solar tree' which was inaugurated in Ludhiana** in the recent past. It is officially certified by **Guinness World Records (GWR) as the 'Largest Solar Tree' in the world**. The entire solar PV panel surface area

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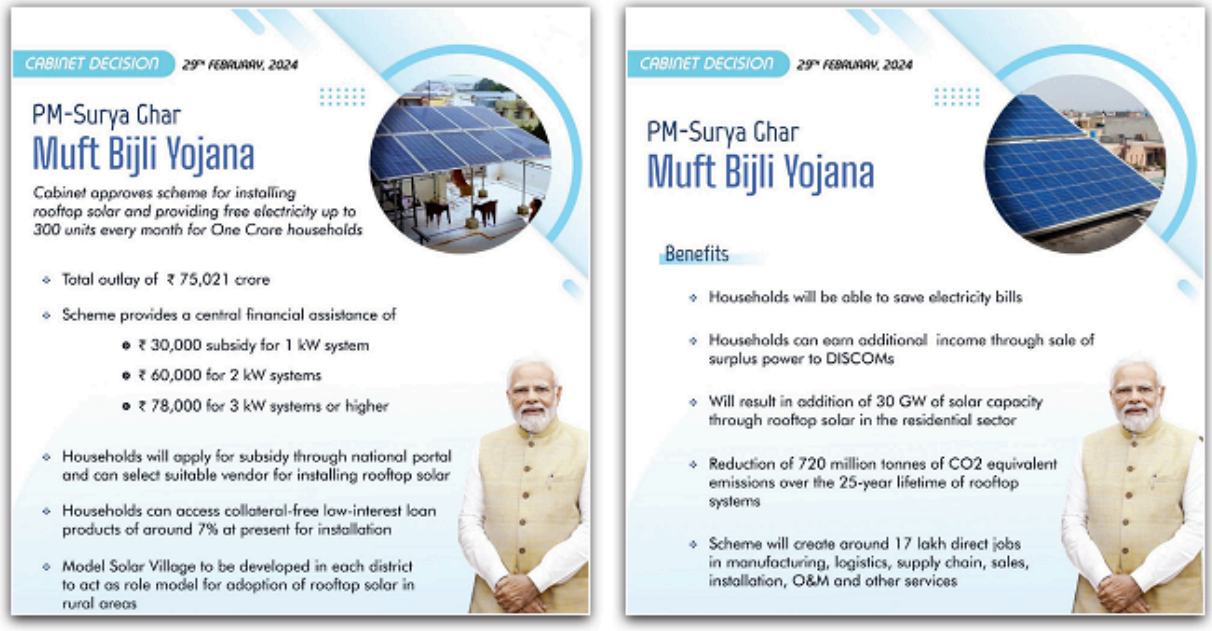
is 309.83 square metres and is used in mini grid application and rural electrification. It is designed to resemble a tree, where the 'leaves' of the tree are fitted with solar panels.




*Source:* PIB, 29 February 2024

### *Significance and Potential of Solar Trees*

- These Solar Trees can be **aligned with Agriculture** for **substituting price-volatile fossil fuels.**
- Each Solar Tree has the potential **to save 10–12 tons of $CO_2$ emissions** being released into the atmosphere as greenhouse gases when compared with fossil fuel-fired energy generation.
- Besides, the surplus generated power can be fed into an Energy Grid.
- This Agricultural Model can provide **a consistent economic return and help the farmers counter the effects of the uncertain variations** in agriculture related activities, thus, making farming an economic and energy sustainable practice.
- The solar tree also has **the capability to incorporate IoT-based features,** i.e., round-the-clock CCTV surveillance in agricultural fields, real-time humidity, wind speed, rainfall prediction and soil analytics sensors.
- The CSIR-CMERI-developed solar-powered e-Suvidha Kiosks may also be connected to the Solar Trees for real-time access to the vast

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majority of agricultural database as well as to the eNAM, i.e., National Agricultural Market which is the place for instant and real-time access to a unified online market.

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### 3.6.2.1.1.14 Mini Solar Grids

Solar energy mini-grids are **small-scale, decentralised electricity networks powered by solar energy.** Unlike traditional large-scale power grids, solar mini-grids are localised and can operate independently. They have the capacity to produce energy ranging from **a few kilowatts to 10 megawatts** and are primarily deployed to provide electricity to rural communities lacking access to centralised power grids.

#### *Developments in Mini Grid Usage*

- **Nigeria:** The World Bank aims to fund a thousand mini-grids in Nigeria to provide electricity to over 90 million people living without it. To date, approximately 20,000 mini-grids have been installed globally.
- **Cambodia:** Diesel-powered mini-grids have become integrated into the national grid, benefiting over 1 million consumers.
- **India:** Around ***700 solar mini-grids*** operated by private companies, mainly in Uttar Pradesh (UP), Bihar, and Jharkhand, contribute to sustainable development goals. Regulatory incentives in states like UP and Bihar encourage private-sector mini-grid setups, which are commercially viable and unsubsidised.

#### *Significance of Mini Grids*

- **Economic and Environmental Benefits:** Solar mini-grids offer a cost-effective and sustainable alternative to diesel generators, reducing reliance on fossil fuels and promoting renewable energy.
- **Decentralised Energy Systems:** Mini-grids are crucial in climate-vulnerable regions, providing resilience against climate shocks like droughts and floods in the absence of centralised grid infrastructure.
- **Development Accelerators:** Beyond electricity provision, mini-grids facilitate various rural development services such ***as mobile telephony, irrigation, agro-processing, and e-mobility, enhancing rural livelihoods and economic opportunities.***

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***Challenges of Installing Mini Grids***

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- **Funding Requirement:** Despite being cost-effective, insufficient investment in solar energy impedes the deployment of mini-grids. **Connecting 490 million people to mini grids by 2030 will require more than 210,000 mini grids and almost $220 billion in investment. [ESMAP. 2019. Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers.**
- **Executive Summary. Energy Sector Management Assistance Program (ESMAP) Technical Report 014/19. Washington, DC: World Bank.].**
- **Energy Bias:** There is a preference for large-scale solar projects over mini-grids, both nationally and internationally, which hampers decentralised energy solutions. International frameworks like the International Solar Alliance (ISA) may inadvertently favour large-scale projects.

**DISTRIBUTED ENERGY RESOURCES**

- Distributed energy resources are small, modular, energy generation and storage technologies that provide electric capacity or energy where you need it.
- Distributed energy resources (DER) refers to often smaller generation units that are located on the consumer's side of the meter.
- Examples of distributed energy resources that can be installed include: roof top solar photovoltaic units, wind generating units/wind turbines, battery storage, microgrids, energy storage, microturbines, diesel generators, diesel engines, duel fuel engine, natural gas engine, combustion engine, microturbine, fuel cell, Storage devices, hybrid system etc.

### *3.6.2.1.2 Wind Energy*

Wind turbine is capable to utilise wind for electricity generation. It is a specific property of blade which is developed with specific **aerodynamic shape** and other performance-enhancing equipment. To take advantage of faster and less turbulent winds, they need to be mounted at 100 feet (30 metres) or more above ground.

- When blowing wind turns the wing, it will utilise to generate electricity with the help of generator attached.

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- However, this rotation cannot be directly coupled to a generator, because the wind turbine blades connected with shaft, typically turn at very low rate of rpm due to the issues of its noise and mechanical strength.
- At this low speed of rotation, generator cannot produce any meaningful electricity. So before connecting to the generator the speed of shaft is increased in a gearbox. The gearbox uses a planetary gear set arrangement to increase the high speed ratio (up to 1:90).
- Minimum wind speed required for power generation is 8 km/hour (or 2 mt/sec) and in maximum speed which can be used for power generation is reported to be 36–54 km/hour. 10–12 km/hour is the average speed of wind in India.
- **Wind vane** is utilised to check wind direction. Wind direction can change at any time. A velocity sensor is fitted on the top of the nacelle which measures the wind speed and direction.
- The deviation in the winds direction is sent to an electronic controller which in turn sends an appropriate signal to the yawing mechanism to correct the error. Thus, the wind turbine will always be aligned with the wind direction.
- According to the wind-speed the relative velocity angle of the wind also changes.
- A blade tilting mechanism tilts the blade and guarantees a proper alignment of the blade with the relative velocity. Thus the blades are always at the optimum angle of the attack with the relative wind flow.
- Maximum limit that a wind turbine can achieve is called as **Betz's limit**. No wind turbine in this world can ever cross the efficiency limit of 59.3%.

***Characteristics***

- A clean and safe energy source, which is the most environment friendly.
- As compared to other plants, it has the lowest gestation period.
- Less time for set-up.
- Low operating costs.
- Low maintenance cost.

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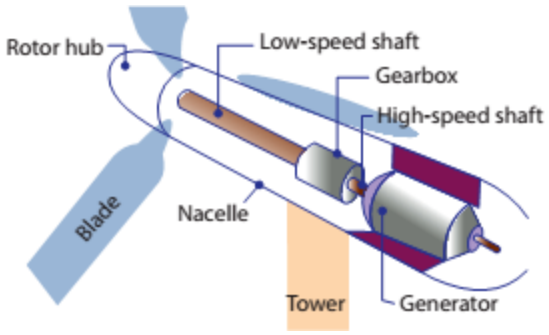


**Figure 3.2** Nacelle

(Refer **Infographic – Wind Energy Generation**)

**In 1986, the development of wind power in India began with the first wind farms being set up in coastal areas of Maharashtra (Ratnagiri), Gujarat (Okha) and Tamil Nadu (Tuticorin) with 55 kilowatts Vestas wind turbines.**

**MNRE** supported these demonstration projects.

China has the maximum wind power generation capacity in the world but America is the country, which has the maximum wind energy generation (production).

### OFFSHORE WIND POLICY

Establishment of offshore wind power projects and research and development activities, in waters, in or adjacent to the country, up to the seaward distance of 200 Nautical Miles (EEZ of the country) from the base line, etc., are all included under offshore wind energy development.

Initially, the 7600-kilometre long Indian coastline indicated prospects of development of offshore wind power and hence the development would smoothen the country's way to attain energy security and achievement of the NAPCC (National Action Plan on Climate Change) targets. As per the offshore wind potential availability, the scheme would be applicable throughout the country.

In India, **Tamil Nadu** is a leading state for wind energy generation. The maximum wind power capacity among Indian states is that of Tamil Nadu, with around 29% of India's total wind power capacity. Thereafter, realising the importance and need for renewable energy, a separate agency was set up

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by Tamil Nadu government, as early as in 1985, as a registered society. It was named the **Tamil Nadu Energy Development Agency (TEDA).**

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### *3.6.2.1.3 Geothermal Energy*

The heat from the Earth generates another renewable (that is clean and sustainable) form of energy called geothermal energy.

At a depth of more than 80 km, heat is available normally. In certain location at a depth of 300–3000 mt have also presented the presence of geothermal energy, i.e., called as **geothermal field.** World's first geothermal power plant was set up in **Larderello (Italy)** in 1911.

There are three types of Geothermal Power Plant setups worldwide:

1. Dry Steam Power Plant
2. Flash Steam Power Plant – Geothermally heated water (temperature over 350 Fahrenheit)
3. Binary Cycle Power Plant – Geothermally heated water (temperature up to 100 Fahrenheit)

Ranging from the shallow ground to hot water and hot rocks found a few miles beneath the Earth's surface, and down even deeper to the extremely high temperatures of molten rock called magma, various potential resources of geothermal energy are found.

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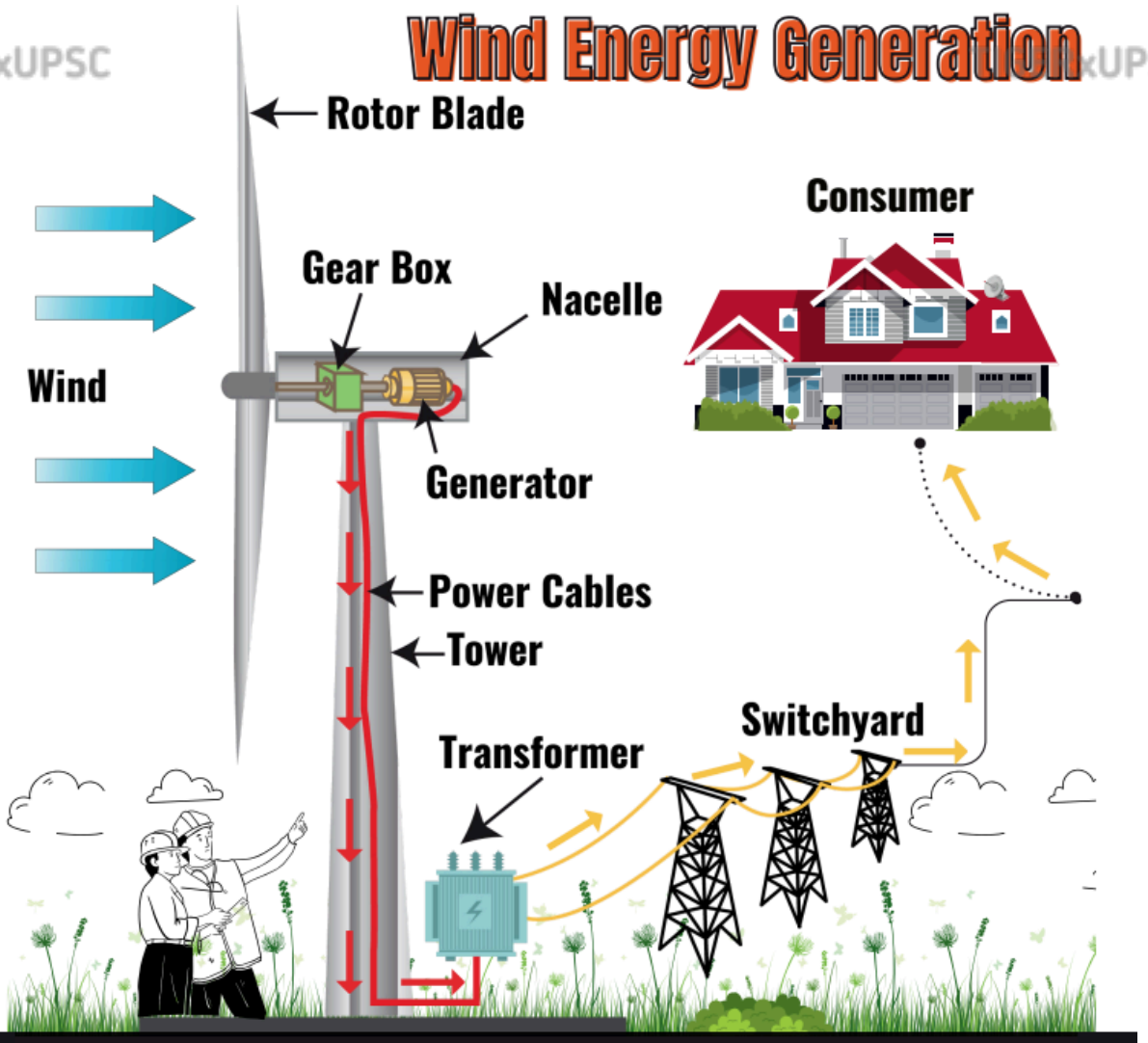


## STEPS OF WIND ENERGY FORMATION

1. Wind Capture: Turbine blades capture the kinetic energy from the wind.
2. Rotation: The wind's force rotates the blades connected to a rotor.
3. Gearbox Function: A gearbox increases the rotation speed for efficient electricity generation.
4. Electricity Generation: The fast-spinning rotor powers a generator, which converts mechanical energy into electrical energy through electromagnetic induction.
5. Power Conversion: A transformer adjusts the voltage of the electricity to suit the power grid.
6. Transmission: The electricity is sent to the grid and distributed to consumers.

## RECENT RENEWABLE ENERGY UPDATES

India is the 3rd largest energy consuming country in the world.

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India stands 4th globally in Renewable Energy Installed Capacity (including Large Hydro), 4th in Wind Power capacity & 5th in Solar Power capacity (as per REN21 Renewables 2024 Global Status Report).The country has set an enhanced target at the COP26 of 500 GW of non-fossil fuel-based energy by 2030. This has been a key pledge under the Panchamrit. This is the world's largest expansion plan in renewable energy.

India's installed non-fossil fuel capacity has increased 396% in the last 8.5 years and stands at more than 201.75 GW (including large Hydro and nuclear), about 45.3% of the country's total capacity (as of May 2024). India saw the highest year-on-year growth in renewable energy additions of 9.83% in 2022. The installed solar energy capacity has increased by 30 times in the last 9 years and stands at 84.27 GW as of May 2024. India's solar energy potential is estimated to be 748 GWp as estimated by National Institute of Solar Energy (NISE). The installed Renewable energy capacity (including large hydro) has increased by around 128% since 2014.

Up to 100% FDI is allowed under the automatic route for renewable energy generation and distribution projects subject to provisions of The Electricity Act 2003.

As of May 2024, Renewable energy sources, including large hydropower, have a combined installed capacity of 193.57 GW.

The following is the installed capacity for Renewables:

- Wind power: 46.42 GW
- Solar Power: 84.27 GW
- Biomass/Co-generation: 10.35 GW
- Small Hydro Power: 5 GW
- Waste To Energy: 0.59 GW
- Large Hydro: 46.92 GW

India has set a target to reduce the carbon intensity of the nation's economy by less than 45% by the end of the decade, achieve 50 percent cumulative electric power installed by 2030 from renewables, and achieve net-zero carbon emissions by 2070. India aims for 500 GW of renewable energy installed capacity by 2030.

India aims to produce 5 Mn Tonnes of green hydrogen by 2030. This will be supported by 125 GW of renewable energy capacity.

50 solar parks with an aggregate capacity of 37.49 GW have been approved in India. Wind Energy has an off-shore target of 30 GW by 2030, with potential sites identified.

### *1. Dry Steam Power Plant*

Dry steam ***power plants are the most common types of*** geothermal ***power plants, accounting for about half of the installed*** geothermal ***plants. They*** work ***by piping hot*** steam ***from underground reservoirs directly into***

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***turbines from*** geothermal ***reservoirs, which power the generators to provide electricity.***

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#### 2. Flash Steam Power Plant

*Flash steam power plants* ***take high-pressure hot water from deep inside the earth and convert it to*** *steam* ***to drive generator turbines. When the*** *steam* ***cools, it condenses to water and is injected back into the ground to be used again. Most geothermal*** *power plants* ***are*** *flash steam plants.*

#### 3. Binary Cycle Power Plant

*Binary Cycle* ***Power*** *Plant: Binary cycle* ***geothermal power generation*** *plants* ***differ from Dry Steam and Flash Steam systems in that the water or steam from the geothermal reservoir never comes in contact with the turbine/generator units.***

***A*** *binary vapour cycle* ***is defined in thermodynamics as a power*** *cycle* ***that is a combination of two*** *cycles,* ***one in a high temperature region and the other in a lower temperature region.***

(Refer **Infographic – Geothermal Power Energy Generation**)

**General characteristics of geothermal energy:** **1.** High accessibility and global availability, **2.** Environment friendly in nature, **3.** Low emission of sulphur, $CO_2$ and other greenhouse gases, **4.** Indigenous, **5.** Does not depend on weather and seasonal changes.

According to **India Energy Portal**, **India has a geothermal potential of 10,000 megawatts (10 GW).** The Tattapani (Himachal Pradesh) geothermal field is the most promising geothermal resource in central India. The promising geothermal sites for direct heat use applications are **Rajgir** in Bihar, **Manikaran** in Himachal Pradesh, **Surajkund** in Jharkhand, **Tapoban** in Uttarakhand and **Sohana** region in Haryana. More than 300 sites of geothermal energy sources are identified in India. There are seven geothermal provinces in India – **Himalayan (Puga, Chhumathang), Sahara Valley, Cambay Basin, Son–Narmada–Tapi (SONATA) lineament belt, West Coast, Godavari basin** and **Mahanadi basin.**

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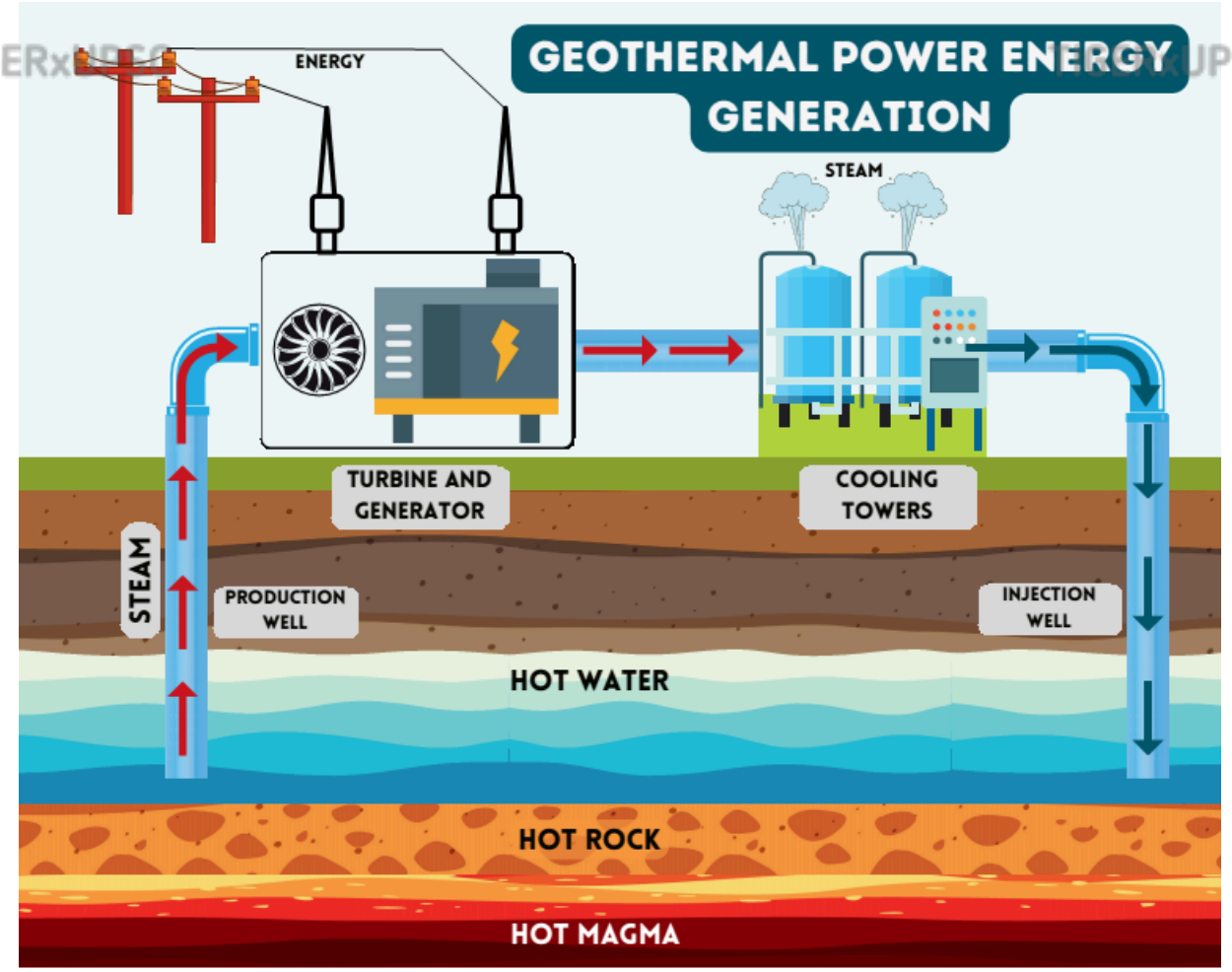


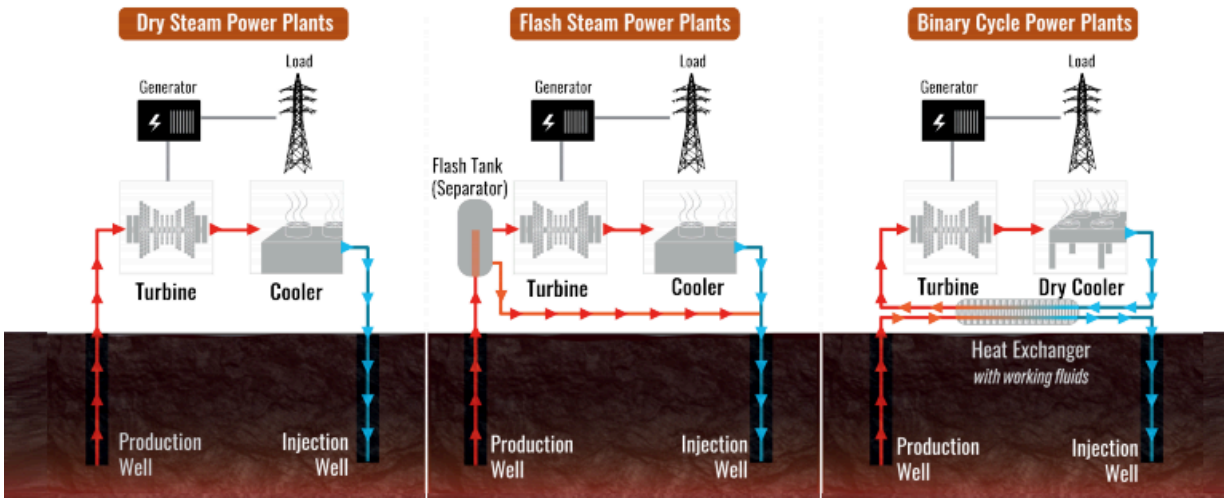


### *3.6.2.1.4 Ocean Energy*

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***Around 70% of Earth's surface is covered by ocean and hence they are the world's largest solar collectors.*** **Ocean has also become a source of energy generation in modern prospective.** ***Thermal energy from the Sun's heat and mechanical energy from the tides and waves are the reason of energy produced by ocean.***

**There are three forms of electricity generations from sea:**

1. **Tidal Energy**
2. **Wave Energy**
3. **Ocean Thermal Energy**

1. **Tidal Energy:** The gravitational pull of the moon is primarily responsible for tides in ocean. In this case a water reservoir is connected with ocean water. Due to up (high tide) and down (low tide) of water level, fluctuation start in water reservoir either water starts to enter inside the reservoir due to upliftment of water in ocean or it starts to move outside from the reservoir during downward movement (low tide) of ocean water. During both the processes, connected turbine or rotor will move and generate mechanical energy; those will further convert into electricity by the use of dynamo.
2. **Wave Energy:** In this type of ocean energy generation, **turbine is installed inside the ocean** and they will move during the effect of ocean current. This mechanical energy of turbine will be converted into electricity with the help of dynamo.
3. **Ocean Thermal Energy:** The surface water of ocean is heated more by Sun's heat, than the deep ocean water. The temperature difference thus created results in thermal energy. It is estimated that the whole world can be powered even with a small portion of the heat trapped in the ocean. In this system, hot water of upper layer of ocean will go in a chamber containing volatile liquid (Liquid Ammonia), which starts to vaporise (Gas) and finally leads to steam formation. This steam pressure will be used to move a turbine and mechanical energy generated can be utilised for electricity generation with the help of generator. Vaporised gas/steam will further cool down when it will come in contact with lower part of ocean which has low temperature. This type of system will work in those areas of ocean where

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temperature difference between hot water (upper surface of water) and cold water (down inside in ocean) should be at least 20°C.

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***Potential of Tidal Energy in India***

The **Gulf of Cambay** and the **Gulf of Kachchh** on the west coast, maximum tidal ranges are of 11 metres and 8 metres and an average tidal ranges are of 6.77 metres and 5.23 metres, respectively. These are the most attractive locations in India.

Even the **Ganges Delta in the Sunderbans** in West Bengal offers good locations for small-scale tidal power development. Sunderbans have the maximum capacity of approximately 5 metres, with an average tidal range of 2.97 metres.

(Refer **Infographic – Tidal Energy**)
(Refer **Infographic – Ocean Thermal Energy**)

### *3.6.2.1.5 Hydropower*

Hydroelectric power **or** hydropower represents electricity that is generated after capturing flowing water. Usually, the hydroelectric power plant uses a dam on a river for water storage in a reservoir. Water released from the reservoir flows through a turbine, resulting in spinning of turbine, producing mechanical energy to activate a generator, which finally converts it to electrical energy.

A pumped storage plant which can even store power is known as another type of hydroelectric power plant. Enough amount of electricity for a home, farm or a ranch can be produced by small or micro-hydroelectric power system.

**India is the 5th globally for installed hydroelectric power capacity.** One of the largest hydroelectric power potentials in the world is that of 1,48,700 megawatts at 60% load factor. **The hydroelectric power plants at Darjeeling (in 1898) and Shivanasamudra (in 1902) were among the first in Asia**. In terms of global hydro power development, India enjoys a dominant status, being blessed with immense amount of hydroelectric potential. It ranks fifth in terms of exploitable hydropotential on a global scale.

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# TIDAL ENERGY

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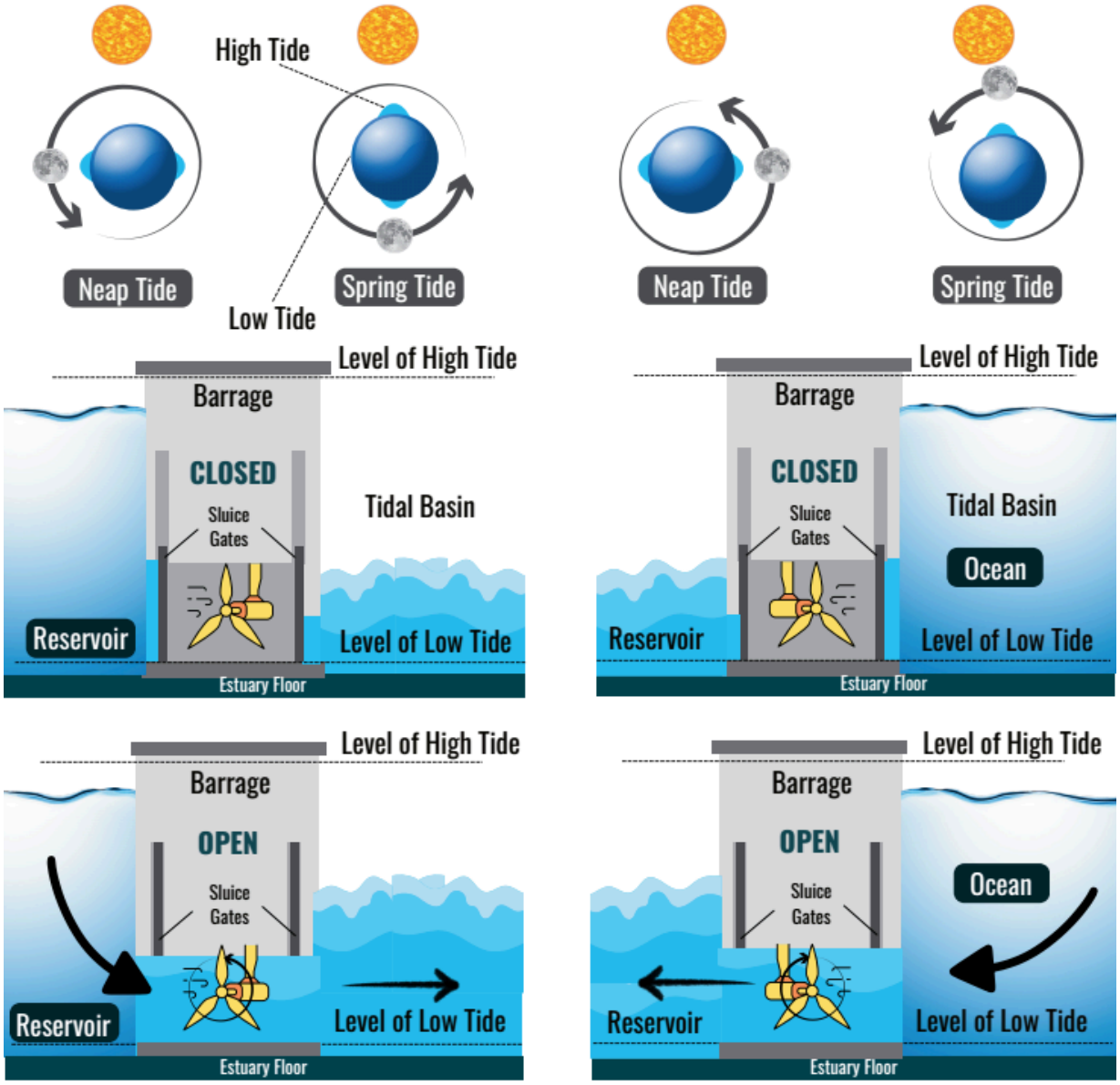


**A. During Low Tide: Outflow of water from water reservoir to ocean**

**B. During High Tide: Inflow of water from ocean to water reservoir**

**Note: Water flow drives the turbine, converting its kinetic energy into mechanical energy, which is then transformed into electrical energy by a dynamo or generator.**

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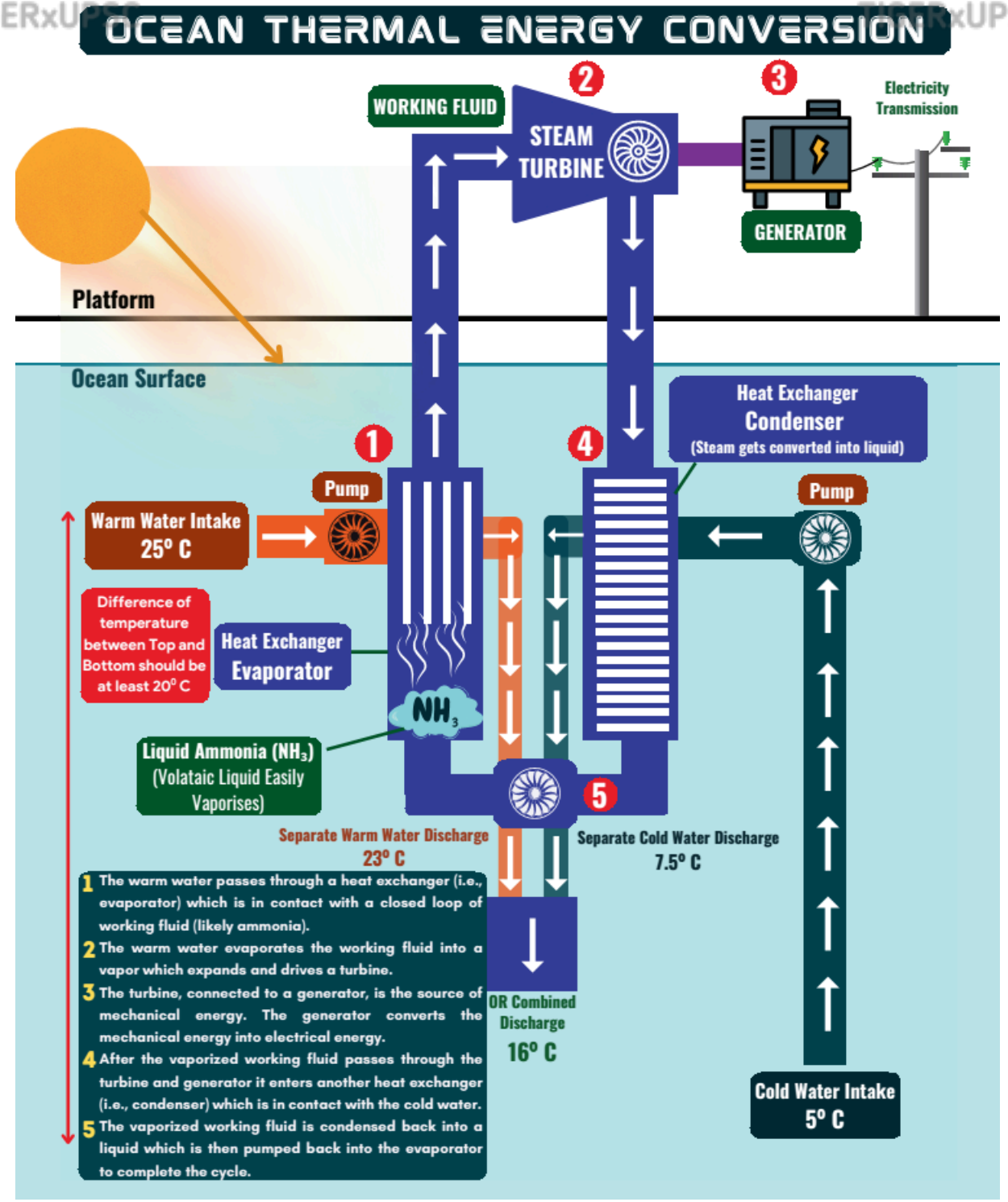


***Classification of Hydro Projects based on Installed Capacity***

- **Micro:** up to 100 kW.
- **Mini:** 101kW to 2 MW.

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- **Small:** 2 MW to 25 MW.
- **Mega:** Hydro projects with installed capacity ≥ 500 MW.

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#### *3.6.2.1.6 Biomass Energy*

It includes solid biomass (organic, non-fossil material of biological origins), biogas (principally methane and carbon dioxide that is produced by anaerobic digestion of biomass), liquid biofuels (bio-based liquid fuel from biomass transformation, mainly used in transportation applications) and municipal waste (wastes produced by the residential, commercial and public service sectors and incinerated in specific installations to produce heat and/or power).

## 3.7 Some Recent and Advanced Energy Generation Sources

### 3.7.1 Fuel Cell/Hydrogen Fuel Cell

A fuel cell converts the chemical energy from a fuel into electricity through a chemical reaction of positively charged hydrogen ions with an oxidising agent like oxygen.

***Difference & Similarity between Battery and Fuel Cell***

- The biggest difference between the battery and the fuel cell is that a battery stores energy, while a fuel cell generates energy by converting the available fuel.

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- A fuel cell can have a battery as a system component to store the electricity it is generating.
- An energy storage device is a battery, which stores its fuel internally. On the other hand, the fuel cell is an energy conversion device, wherein the fuel is supplied externally.
- A battery runs down as the electrodes dissolve, but a fuel cell operates for as long as oxygen and hydrogen flow and hence in terms of the durability, fuel cell is advantageous over battery.
- Both generate Direct Current (DC).

### *Working and Technology of Fuel Cell*

A fuel cell is an electrochemical device, in which the hydrogen fuel combines with oxygen to produce electricity, heat and water. The fuel cell is similar to a battery in that an electrochemical reaction occurs as long as fuel is available. Hydrogen is stored in a pressurised container and oxygen is taken from the air. Because of the absence of combustion, there are no harmful emissions, and the only by-product is pure water. It can be used at all levels of electrical appliances from small (e.g., phone battery) to large (e.g., Rocket).

### *Applications of Fuel Cells*

1. Fuel cells provide an all-weather solution that can keep providing power even though weather conditions prevent site visits.
2. Fuel cells can communicate data using our REMO (Remote Monitoring System).
3. Powering temporary buildings.
4. Transport (rail, road and air).
5. Materials handling.
6. Blue light and defence applications.
7. Cellular base stations.
8. Power generation.

**Electrolytes:** These are those substances which give ions when dissolved in water. They can be any acid, base or salts because all of them generate ions when dissolved in water. **Platinum is mostly used as electrolytes.** These solutions conduct electricity due to the mobility of the positive and negative

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ions, which are called cations and anions, respectively. Strong electrolytes completely ionise when dissolved, and no neutral molecules are formed in solution. For example, NaCl, $HNO_3$, $HClO_3$, $CaCl_2$, etc., are strong electrolytes. An ionisation can be represented by

$$NaCl(s) = Na^{+}(aq) + Cl^{-}(aq)$$

### *Advantages of a Fuel Cell*

- The cell will generate a voltage as long as it gets the fuel supply.
- The amount of oxygen and hydrogen (and not the size of the fuel cell) determines the total amount of energy generated.
- Recharging is not required.

### *Future of Fuel Cells*

1. For automotive engines, power stations and the power packs for portable electronics, fuel cells have the potential to become the dominant technology in future.
2. They will compete with different energy conversion devices.
3. The turbine and the gasoline engine (types of combustion engine) burn fuels and use the pressure created by the expansion of the gases to perform mechanical work. When required, the batteries convert chemical energy back into electrical energy. Both the tasks can be accomplished by the fuel cells more efficiently.
4. One of the most promising technologies is the **proton exchange membrane fuel cell (PEMFC)**. This type of fuel cell will enable powering cars, buses and maybe even houses.

### *How is Hydrogen Produced?*

Hydrogen does not exist on Earth as a gas, and hence is required to be separated from other compounds. Hydrogen can be usually produced by electrolysis (water splitting) and steam reforming. **The least expensive method is steam reforming**. In industries, hydrogen can be separated from carbon atoms in methane. The process of steam reforming causes greenhouse gas emissions, which contributes to global warming because methane is a fossil fuel.

The other process is **electrolysis**, which involves passing an electric current through water to separate water into its constituent elements –

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hydrogen and oxygen. At the negatively charged cathode, hydrogen is collected, while oxygen is collected at the positive anode. We get extremely pure hydrogen through electrolysis, leading to no emissions since electricity from renewable energy sources can be used. However, it is a very expensive process.

Other processes include **photo-electrolysis** and **biomass gasification**. Some **algae and bacteria have also been identified to produce hydrogen** under certain conditions, with the help of sunlight as their source of energy.

***The Future of Hydrogen***

In the future, it is expected that hydrogen as a fuel will play an important role as an energy carrier, because its production can be made by using renewable energy sources and it virtually does not cause pollution. Applicable for **'zero-emissions'** vehicles, to heat homes and offices, to produce electricity, and to fuel aircraft, hydrogen proves to be a potential fuel for future, hence reducing dependence on oil. However, to be used as an alternative to gasoline, it requires many new facilities and systems.

**Wind electrolysis system** is another method of hydrogen production. The way it works is: The hydrogen is compressed up to pipeline pressure to be fed into a transmission pipeline, which further transports the hydrogen to a compressed gas terminal where the hydrogen is loaded into compressed gas tube trailers. The tube trailers are delivered by a truck to a forecourt station. Here, the hydrogen is further compressed, stored and dispensed to fuel cell vehicles.

## 3.7.2 Microbial Fuel Cells (MFCs)

A microbial fuel cell is very similar to hydrogen fuel cell. In hydrogen fuel cell, platinum is used as an electrolyte/catalyst but in microbial fuel cells, organic waste decomposes/breaks in the presence of bacteria as a catalyst, that generates $H^+$/proton. **Bacteria have the ability to convert a variety of organic compounds into $CO_2$, water and energy.** The energy produced by the microorganism is utilised to grow and maintain their metabolism. By using MFC, we can harvest this energy in the form of electricity.

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**Components of MFC:** The MFC comprises of an anode, a cathode, a proton or a cation exchange membrane and an electrical circuit.

**Functional Activity of Microbial Fuel Cell:** The bacteria are put in an anode chamber where they convert organic compounds/substrates into $CO_2$, proton ($H^+$), electron, etc. Through the help of proton exchange membrane, protons will transfer from anaerobic chamber to aerobic chamber. In aerobic chamber, protons ($H^+$) bind with oxygen and form water as a by-product. In this process, the excess electrons left in anaerobic chamber start to move from anode to cathode through an electric circuit. This movement of electrons due to potential difference is responsible for the generation of electrical power (watt).

A platinum catalyst is required to obtain sufficient oxygen reduction reaction (ORR) rate in Hydrogen fuel cell but in **MFCs, this role is performed by bacteria as a catalyst. MFCs can also be applied in waste water treatment due to presence of organic substances.**

(Refer **Infographic – Hydrogen Fuel Cell and Microbial Fuel Cell**)

### 3.7.3 Lithium Ion (Li-ion) Battery

It is very popular and highly useful battery in modern prospective in many electronic equipment like mobile phones, laptops, e-readers and many other electronic gadgets as well as electric vehicles and power tools, etc.

Lithium ion batteries dominate the market and some experts favour green hydrogen, however, gravity batteries are also another way of storing the renewable energy.

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In comparison, gravity is free, clean and easily accessible, without the complications of producing hydrogen or the environmental and human rights concerns linked to some lithium mining.

(Refer **Infographic – Lithium Ion Battery**)

***Growth and Development of Li-Ion Batteries***

- Lithium battery technology has distinct advantages over other older rechargeable battery technologies such as Nickel cadmium and Nickel Metal Hydride.
- The idea for lithium ion battery technology was first proposed in the 1970s by M Whittingham who used titanium sulphide and lithium metal as the electrodes for his cell.
- Lithium ion battery has three main components: Cathode, Anode and Electrolytes.
- Due to its main charge carrier through lithium ions it is called as lithium ion battery. Lithium ions flow due to cathode and anode material into which and from which the lithium ions can flow. There are two processes involved during lithium ion movement.
- Separators are also used to safeguard this battery so no chances of short-circuit inside battery. **In lithium ion battery anode is made up of graphite; cathode is made up of lithium metal (cobalt/nickel) oxide.**
- In electrolytes, water molecule should be negligible. When we provide a load (An electrical load is an electrical component or portion of a circuit that consumes electric power).
- During discharge, the (positive) lithium ions move from the negative electrode (anode) (usually graphite – C6) to the positive electrode (cathode) (forming a lithium compound) through the electrolyte while the electrons flow through the external circuit in the same direction.
- When the cell is charging, the reverse occurs with the lithium ions and electrons move back into the negative electrode in a net higher energy state.
- This entire process is called as intercalation and de-intercalation. **Intercalation** is a process when battery is going to discharge but in **de-intercalation** process battery will charge again when lithium ions are extracted from the electrode.

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**Table 3.5** Types of Lithium ion batteries and its uses

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| Applications | Types of Lithium Ion Batteries |
|---|---|
| Cellphones, laptops, cameras | • Lithium Cobalt |
| Power tools, medical, hobbyist | • Lithium Manganese Oxide<br>• Lithium Iron Phosphate<br>• Lithium Nickel Manganese Cobalt Oxide |
| Electric vehicles and grid storage | • Lithium Nickel Cobalt Aluminium Oxide<br>• Lithium Titanate |

***Features of Li-Ion Battery Which Made it Better Among other Batteries***

- **Don't totally discharge:** Lithium ion batteries should be charged before the battery is completely discharged.
- **Non-use care:** If a Li-ion battery is not to be used for an extended period of time, it should ideally be brought to a charge level of between about 40% and 60% of full charge.
- **Keep cool:** Li-ion batteries should be kept cool. By keeping them cool, possibly in a refrigerator, the ageing process becomes slower.
- **Don't freeze:** Li-ion batteries should not be exposed to very low temperatures – most lithium ion battery electrolytes freeze at approximately –40°C.
- **Buy new batteries only when needed:** Li-ion batteries should be bought only when needed, because the ageing process begins as soon as the battery is manufactured.

Note: According to **International Energy Agency Report (2022)**: Contribution of different source of batteries, worldwide

1. 60% Lithium
2. 30% cobalt
3. 10% Nickel

More than 70 per cent of the world's cobalt is produced in the Democratic Republic of the Congo (DRC).

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# Hydrogen Fuel Cell vs Microbial Fuel Cell

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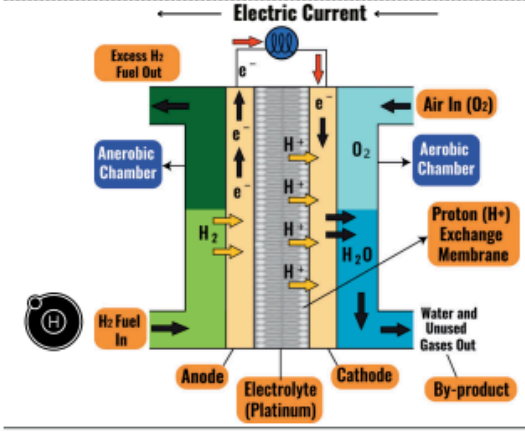


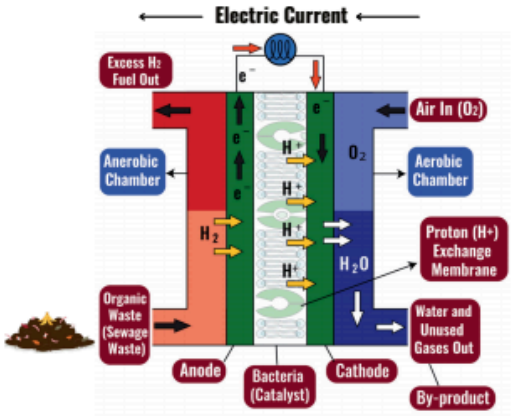


A fuel cell converts the chemical energy from a fuel into electricity through a chemical reaction of positively charged hydrogen ions with an oxidizing agent like oxygen. A fuel cell is an electrochemical device, in which the hydrogen fuel combines with oxygen to produce electricity, heat and water.

A microbial fuel cell is very similar to hydrogen fuel cell but in place of hydrogen fuel, organic waste (sewage water, etc) is used. In hydrogen fuel cell, platinum is used as an electrolyte/catalyst but in microbial fuel cells, organic waste decomposes/breaks in the presence of bacteria as a catalyst, that generates H+/proton.

The biggest difference between the battery and the fuel cell is that a battery stores energy, while a fuel cell generates energy by converting the available fuel.

Bacteria (are put in an anode chamber) have the ability to convert a variety of organic compounds into $CO_2$, water and energy. The energy produced by the microorganism is utilised to grow and maintain their metabolism. By using microbial fuel cell, we can harvest this energy in the form of electricity.

## PROCESS

1. Hydrogen Fuel enters the anode side of the fuel cell and oxygen (from air) enters the cathode side of the cell.
2. At the anode, a platinum catalyst splits the hydrogen into positive hydrogen ions (protons) and negatively charges electrons.
3. The Polymer Electrolyte Membrane (PEM) allows only the positively-charged ions to pass through it to the cathode. The negatively charged electrons travel along an external circuit to the cathode which creates an electrical circuit
4. At the cathode, the negative hydrogen electrons (having already produced electricity) reunite with the positively charged hydrogen ions, where they combine with oxygen to form $H_2O$, or water which simply flows of of the cell.

## PROCESS

1. The bacteria are put in an anode chamber where they convert organic compounds/substrates into $CO_2$, proton (H+), electron, etc.,
2. Through the help of Proton Exchange Membrane (PEM), protons will transfer from anaerobic chamber to aerobic chamber. In aerobic chamber, protons (H+) bind with oxygen and form water as a by-product.
3. The flow of electrons (Ampere) between the anode and the cathode due to potential difference between anode and cathode (Volt) results in the generation of electrical power (Watt).
4. The flow of protons is through the proton or cation exchange membrane to the cathode.

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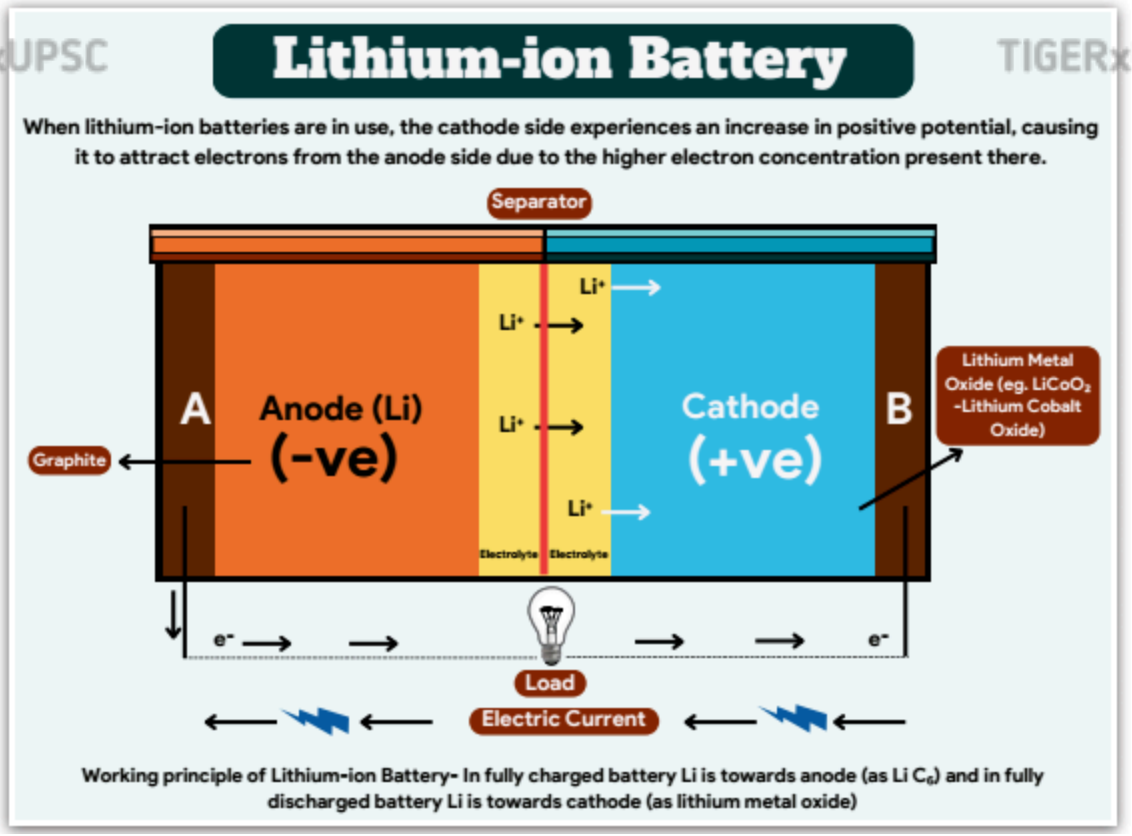


## 3.7.4 Sodium Ion Battery

**AR4 Tech, a start-up in Coimbatore, has partnered with Sodion Energy of Singapore to make sodium ion battery packs for domestic and export markets.** This collaboration aims to leverage the potential of sodium ion battery technology in the growing energy storage sector.

***AR4 Tech***

- **Motor Vehicle Manufacturing company**, Coimbatore, Tamil Nadu.
- **Start-up company** set up in 2021.
- **They have the Engineering expertise** and Manufacturing capabilities to convert any IC engine-driven vehicle into Electric Vehicle.
- They are **manufacturing Electric and Solar vehicle** related to Agricultural and Construction machinery.
- They do **customised Electric Vehicles Automotive Re-Engineering, Re-Purposing, Re-Designing, and Repair**.
- They do **servicing and spares for EVs.**

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- AR4 Tech will use the battery packs to **convert existing petroleum vehicles (mainly two wheelers) into electric vehicles.**
- The **life cycle of a battery is expected to be almost 10 years.** These battery packs can also go into uninterrupted power supply systems.

**Composition:** Sodium ion batteries operate through a chemical reaction involving *four main components*: an **anode, cathode, separator, and electrolyte**. In these batteries, **sodium ions replace lithium ions in the cathode, and sodium salts replace lithium salts in the electrolyte**.

***Operating Principle***

**During charging, sodium atoms in the cathode release electrons** to the external circuit, transforming into ions that migrate through the electrolyte to the anode. Here, they combine with electrons from the external circuit while interacting with the layered anode material.

This **process reverses** during discharge, generating electricity.

**LITHIUM RESERVES IN INDIA & ITS IMPORTANCE**

India has the **seventh largest resource of lithium (white gold) globally**, though it will take time to convert this into reserves. The Government of India's Ministry of Mines, in collaboration with the state-owned Khanij Bidesh India Ltd (KABIL), has initiated a preliminary exploration and development agreement with the Argentinian mining company CAMYEN. This agreement aims to explore the potential acquisition and development of several lithium blocks.

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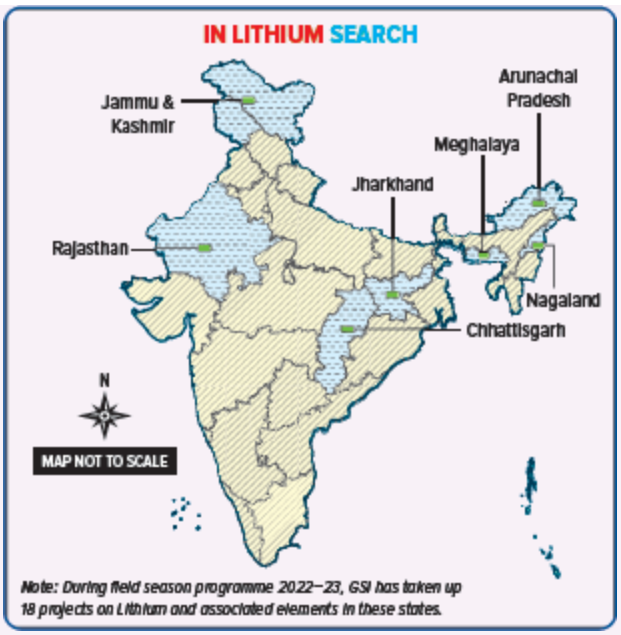


According to assessments by the Geological Survey of India, approximately 5.9 million tons of lithium resources have been identified in the rugged Salal-Haimana region of the Reasi district in the union territory of Jammu and Kashmir.

- Lithium is the source of powering the green future electric vehicles in lithium batteries.
- It is used in phones, laptops, digital cameras and electric vehicles & rechargeable batteries, etc.
- The pharma industry uses it as an **antidepressant, to treat bipolar disorder and depression**.
- Its usage extends to **space vehicles and military equipment** as well.
- Allied Market Research projects that the global electric vehicles (EVs) industry will surge from 163 billion in 2020 to a staggering 823 billion by 2030. This significant growth underscores the expanding opportunities within the sector. Moreover, the demand for lithium extends beyond EVs, encompassing areas like high-performance sport batteries.
- India's reliance on lithium imports, primarily sourced from countries like Australia, Argentina, and Chile, has been substantial. Government figures indicate an import bill of 173 crore for lithium. Establishing a foothold in lithium reserves could pave the way for self-reliance and reduce dependence on imports, aligning with the nation's goal of bolstering domestic resources.

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**Table 3.6** Sodium Ion Battery vs Lithium Ion Battery

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| Sodium Ion Battery | Lithium Ion Battery |
|---|---|
| **1.** Sodium is more abundantly available than Lithium hence, it is less expensive. | **1.** Lithium availability is limited to a few areas of the world, thus, it is comparatively expensive. |
| **2.** It is more environment friendly. | **2.** Lithium is less environment friendly than sodium. |
| **3.** A sodium ion battery uses aluminium which is cheaper. | **3.** Lithium ion battery uses copper which is 3 to 4 times more expensive than aluminium. |
| **4.** It has a higher operating temperature range. | **4.** It has a lower operating temperature range. |
| **5.** Sodium ion battery charges faster than Lithium ion battery and have a higher lifecycle. | **5.** It has a slow charge rate and comparatively small lifecycle. |

***Benefits of Sodium Ion Batteries***

1. **Readily Available Raw Materials:** Sodium ion batteries benefit from the **wide availability of their constituent raw materials**, making them a promising and sustainable energy storage option.
2. **Safety:** Sodium batteries ensure high levels of safety as cells based on this **chemical element are non-flammable** and **not prone to explosions or short circuits**, enhancing user safety and reducing risks associated with battery use.
3. **Low Cost:** The **raw materials required for sodium ion batteries are abundant in nature** and can be extracted at relatively low costs and with minimal energy consumption.
4. **Low-Temperature Resistance:** Sodium ion batteries exhibit **good performance even at low temperatures**, making them suitable for a wide range of applications and environments.

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# Electric Vehicles and Their Types/Different Types of Hybrid Vehicles (Alternative Options)

Electric vehicles (EVs) encompass various types, ranging from fully electric to hybrid models that use a battery to enhance efficiency or performance.

## 1. Battery Electric Vehicle (BEV)

A battery electric vehicle (BEV) is a fully electric vehicle powered by an electric motor and a rechargeable battery, without a petrol or diesel engine.

**Key Features**

- Zero tailpipe emissions.
- Rechargeable at home or public charging stations.
- Driving range depends on battery capacity (e.g., Tesla Model S: 412 miles, MINI Electric: 145 miles).

**Examples:**

Nissan Leaf, Renault Zoe, Tesla Model S.

**Pros:**

- ✓ Zero emissions
- ✓ Lower running costs
- ✓ Tax benefits
- ✓ Quiet operation
- ✓ Government incentives

**Cons:**

- × Higher initial cost
- × Longer charging times compared to refueling
- × Limited charging infrastructure (improving rapidly)

## 2. Plug-in Hybrid Electric Vehicle (PHEV)

A hybrid electric vehicle (PHEV) combines anelectric motor with a petrol or diesel engine and can be plugged in to charge.

**Key Features:**

- Provides around 40 miles of pure electric range.
- Switches to fuel engine once the battery depletes.
- Ideal for shorter commutes with the flexibility of a conventional engine for longer trips.

**Examples:**

Mitsubishi Outlander PHEV, Volvo XC60 Recharge, Volkswagen Passat GTE.

**Pros:**

- ✓ Shorter trips on electric power
- ✓ Backup of a fuel engine for long journeys
- ✓ Lower co-emissions than traditional fuel vehicles

**Cons:**

- × More expensive than conventional cars
- × Heavier battery reduces fuel efficiency on motorways
- × Limited pure electric range

## 3. Hybrid Electric Vehicle (HEV)

A hybrid electric vehicle (HEV) has both an electric motor and a fuel engine but cannot be plugged in. Instead, the battery is charged through regenerative braking and the engine.

**Key Features**

- Electric motor assists the fuel engine, improving efficiency.
- Can drive a few miles on electric power at low speeds.
- Primarily relies on petrol or diesel for longer trips.

**Examples**

Toyota Prius, Hyundai Ioniq Hybrid.

**Pros**

- ✓ Efficient in city driving
- ✓ No need to plug in
- ✓ More affordable than BEVs and PHEVs

**Cons**

- × Limited pure electric range
- × Less efficient on long journeys
- × Most new hybrid sales will end by 2030

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**4. Mild Hybrid Electric Vehicle (MHEV)**

An mild hybrid electric vehicle (MHEV) features a small battery and an integrated starter-generator to improve fuel efficiency and provide a slight boost to acceleration.

**Key Features**

- Cannot run solely on electric power.
- Improves fuel economy and reduces coemissions.
- Some models feature engine shut-off when coasting.

**Examples:**

Suzuki Swift, Ford Puma, Audi Q8.

**Pros:**

- ✓ Similar driving experience to conventional cars
- ✓ Slightly better fuel efficiency
- ✓ Comparable price to non-hybrid cars

**Cons:**

- × Not a true hybrid
- × Cannot operate in electric mode
- × Still dependent on petrol or diesel

**5. Range-Extended Electric Vehicle (RE-EV)**

An range-extended electric vehicle (RE-EV) features an electric motor for propulsion, but also includes a small petrol or diesel engine that generates electricity to recharge the battery.

**Key Features:**

- The fuel engine never drives the wheels, unlike a hybrid.
- Provides extended range when the battery runs low.
- Becoming obsolete due to advances in battery technology.

**Examples:**

BMW i3, Vauxhall Ampera, Chevrolet Volt.

**Pros:**

- ✓ No range anxiety
- ✓ Better efficiency than a standard hybrid
- ✓ Innovative technology

**Cons:**

- × Limited availability
- × Heavier engine reduces efficiency
- × Less range than a fully electric vehicle

**6. Hydrogen Fuel Cell Electric Vehicle (FCEV)**

A fuel cell vehicle (FCEV) uses hydrogen to generate electricity, emitting only water vapour.

**Key Features**

- Fast refuelling (similar to petrol/diesel cars).
- Longer driving range than many BEVs (Toyota Mirai: ~400 miles).
- Still limited by hydrogen infrastructure.

**Examples**

Toyota Mirai, Hyundai NEXO.

**Pros**

- ✓ Zero emissions
- ✓ Long driving range
- ✓ Quick refuelling time

**Cons**

- × Expensive technology
- × Limited hydrogen refuelling stations
- × Few vehicle options available

## 3.7.5 Gravity Battery

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***NTPC had signed an MoU for a long-term strategic partnership for the deployment of Energy Vault's EVx gravity-based energy storage technology and software solutions.***

A gravity battery is **a type of energy storage device** that uses the power of gravity to store and generate electricity.

(Refer **Infographic – Gravity Battery**)

***Common Types of Gravity Batteries***

- **Pumped-Storage Hydroelectricity (PSH):** This involves pumping water uphill to store energy and releasing it downhill through turbines to generate electricity.
- **Block Lowering Systems:** These systems lower a heavy block to produce electricity.

**PUMPED-STORAGE HYDROPOWER**

- It is the world's largest battery technology.
- It is accounting of approximately 94 per centof installed global energy storage capacity and ahead than lithium-ion battery.
- According to recent studies there has significant potential of global pumped hydro capacity, with more than 600,000 identified off-river sites.
- The technology absorbs surplus energy at times of low demand and releases it when demand is high.
- Pumped hydropower storage uses the force of gravity to generate electricity using water that has been previously pumped from a lower source to an upper reservoir.
- The water is pumped to the higher reservoir at times of low demand and low electricity prices. At times of high demand - and higher prices - the water is then released to drive a turbine in a powerhouse and supply electricity to the grid.

***Gravity Batteries vs Lithium ion Batteries***

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- **Durability:** Gravity batteries, made of sturdy components, tend to last longer than lithium ion batteries.
- **Cost:** Gravity batteries have lower long-term costs compared to lithium ion batteries, which may need frequent replacements.
- **Environmental Impact:** **Gravity batteries are cleaner and more sustainable** as they do not require materials like lithium, which can have environmental and human rights concerns.

## Energy Vault's 'EVx' Platform

**Energy Vault**, a **Swiss company**, has developed the 'EVx' platform, a gravity-based storage solution.

### *Working Mechanism*

The EVx platform uses a mechanical elevator system to stack heavy blocks, raising them with surplus electricity and lowering them to generate electricity when needed.

### *Advantages of the 'EVx' Platform*

- **Balancing Renewable Energy:** It helps balance out the variability in renewable energy generation by **storing excess energy for later use**.
- **Longevity:** The robust design of the EVx platform ensures a **longer lifespan compared to traditional batteries**.
- **Sustainability:** By using gravity instead of chemicals, it offers a ***cleaner and more sustainable energy storage solution.***

### *India's Push for Gravity Batteries*

**Renewable Energy Growth:** India is expanding its renewable energy capacity, making energy storage crucial to support this growth.

**India is the world's third largest producer of renewable energy** and **nearly 40% of installed electricity capacity comes from non-fossil fuel sources**.

**Collaborations:** **Energy Vault is collaborating with Indian entities like NTPC Ltd, Tata Power, and ReNew Power** to deploy its technology in the country.

### *Challenges and Future Outlook*

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- **Grid Integration:** India faces challenges in integrating renewable energy into its grid, making energy storage solutions like gravity batteries essential.
- **Future Prospects:** With the increasing adoption of renewable energy, gravity batteries could play a significant role in ensuring a stable and sustainable energy future for India.

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### 3.7.6 PM E-DRIVE SCHEME (Successor of FAME Scheme)

The **PM Electric Drive Revolution in Innovative Vehicle Enhancement (PM E-DRIVE)** Scheme was **launched by the Ministry of Heavy Industries (MHI)** on 29 September 2024 to **promote green mobility and strengthen India's electric vehicle (EV) manufacturing ecosystem.**

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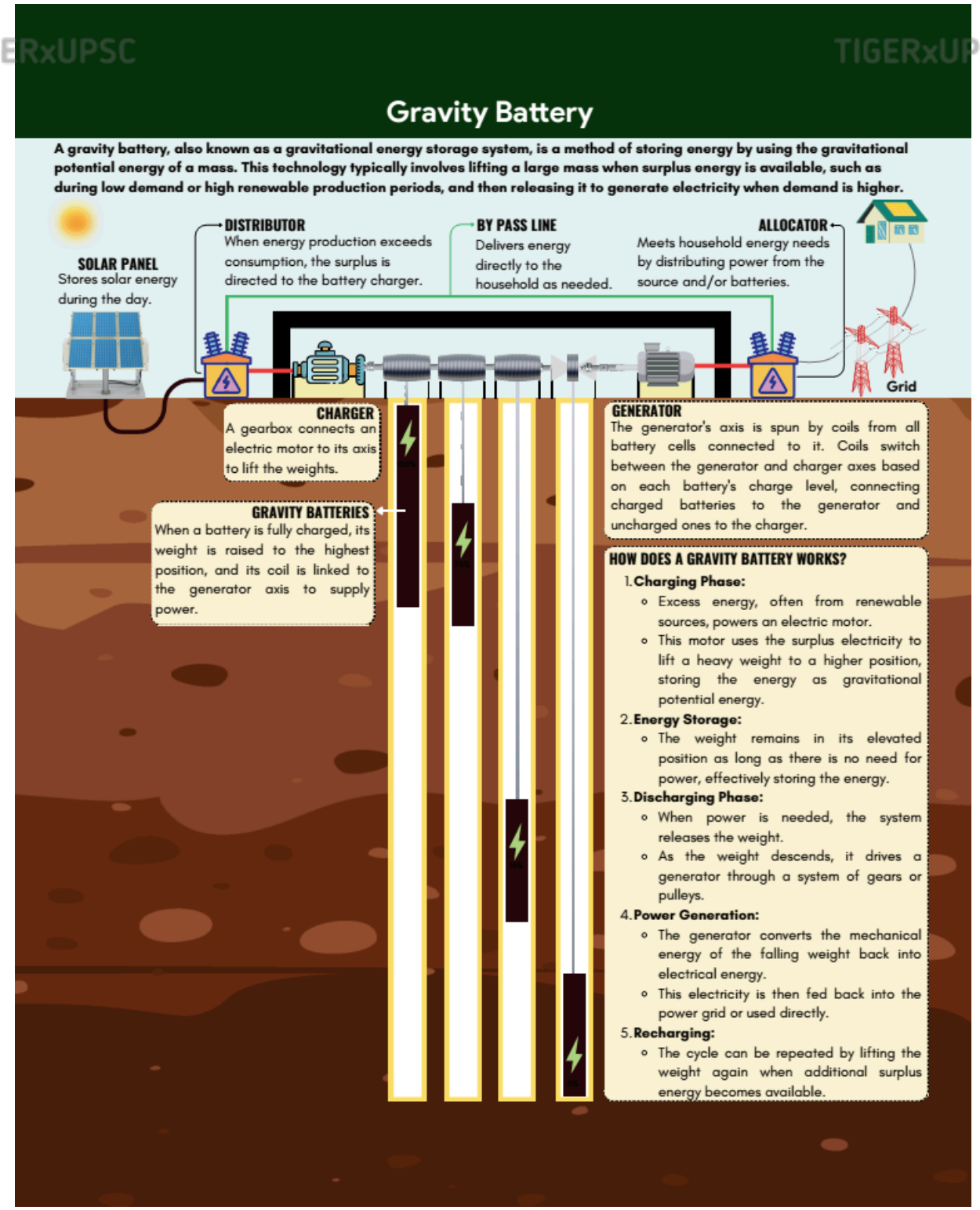


**Key Details:** Total Outlay: ₹10,900 crore

**Duration:** Two years (until March 31, 2026)

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**Predecessor:** The Electric Mobility Promotion Scheme (EMPS) 2024, which **ran from 1 April 2024 to 30 September 2024**, is now part of PM E-DRIVE.

### *Salient Features*

#### *E-Vouchers for Demand Incentives*

- EV buyers can avail subsidies through an E-Voucher system.
- **Individual Buyers:** Aadhaar e-KYC with face authentication via the PM E-DRIVE App.
- **Non-Individual Buyers:** E-Voucher issued through the PM E-DRIVE Portal.

### *Special Focus Areas*

- **E-Ambulances:** ₹500 crore allocated to promote e-ambulances for better patient transport.
- **E-Trucks:** ₹500 crore allocated to encourage electric trucks, reducing air pollution.
- **Scrapping Certificate Mandatory:** Buyers must provide a scrapping certificate from an MoRTH (Ministry of Road Transport and Highway) - approved vehicle scrapping center (RVSF) to avail the subsidy.
- **Vehicle Testing Infrastructure:** ₹780 crore set aside for upgrading vehicle testing agencies.

### *Financial Allocation & Components*

| Component | Fund Allocation | Purpose |
|---|---|---|
| **Subsidies** | ₹3,679 crore | Demand incentives for e-2W, e-3W, e-ambulances, e-trucks and emerging EV categories |
| **Capital Grants** | ₹7,171 crore | Support for e-buses, EV charging stations and vehicle testing infrastructure |

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| Component | Fund Allocation | Purpose |
|---|---|---|
| **Administration & IEC** | ₹50 crore | Scheme administration, awareness campaigns & |

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### *Impact & Adoption*

## EV Growth Since Scheme Launch

| Period | EVs Registered | Growth (%) |
|---|---|---|
| 1 April – 30 Nov 2023 | 10,39,408 | - |
| 1 April – 30 Nov 2024 | 13,05,926 | 25.64% |

### *EV Targets Under PM E-DRIVE*

The scheme aims to support the adoption of:

- 14,028 e-Buses
- 2,05,392 e-3 Wheelers (L5)
- 1,10,596 e-Rickshaws & e-Carts
- 24,79,120 e-2 Wheelers
- E-Trucks, E-Ambulances & Public Charging Stations

### *Make in India & Local Manufacturing*

- The scheme enforces a **Phased Manufacturing Programme (PMP)**. (Ministry of Heavy Industries, to promote domestic manufacturing of various goods, especially electric vehicles and their components.)
- **Mandates sourcing from Indian manufacturers**, including **small and medium enterprises (SMEs)**.
- Strengthens the **Make in India** initiative by boosting domestic production of EV components.

The Faster Adoption and Manufacturing of Electric Vehicles in India (FAME) Scheme was launched in 2015 to promote electric mobility. The scheme aims to reduce reliance on conventional vehicles and reduce pollution and fuel consumption.

The FAME Scheme has been replaced by the PM E-Drive Scheme, which focuses on producing electric vehicles for two- and three-wheelers,

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public transportation and more.

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| | FAME Scheme |
|---|---|
| **Objectives** | • Promote electric and hybrid vehicles, establish charging infrastructure and reduce pollution |
| **How it works** | • Offers incentives for purchasing electric vehicles and establishes charging infrastructure |
| **Phases** | • Phase I started in 2015 and Phase II started in 2019 |
| **Budget** | • Rs. 11,500 crore for Phase II, which ran from 2019 to 2024 |
| **What it supports** | • E-2Ws, e-3Ws, e-4Ws, e-buses and EV public charging stations |
| **Other impacts** | • Reduced GST on EVs, enabled state EV policies and supported policy initiatives |

### 3.7.7 Waste to Power/Energy

Any organic waste, which can be degraded, is a resource that can lead to energy generation. By adopting environment-friendly, waste-to-energy conversion technologies, the issues caused due to solid and liquid wastes can be significantly mitigated. This will enable treatment and processing of wastes before their disposal that can reduce the quantity of wastes and generate a substantial quantity of energy from them, substantially leading to environment pollution.

**INDIA'S LARGEST BATTERY ENERGY STORAGE SYSTEM (BESS)**

Battery storage, or battery energy storage systems (BESS), are **devices that enable energy from renewables, like solar and wind, to be stored and then released when the power is needed most.**

Under the auspices of the Ministry of New and Renewable Energy, the **Solar Energy Corporation of India Limited (SECI)** has achieved the **successful installation of India's largest Battery Energy Storage System (BESS)**, which **harnesses solar energy for storage**.

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This BESS, boasting a capacity of 40 megawatts (MW) / 120 MWh, is paired with a solar photovoltaic (PV) plant with an **installed capacity of 152.325 megawatt hour (MWh)** and a **dispatchable capacity of 100 MW AC (155.02 MW peak DC)**. It is **situated in Rajnandgaon, Chhattisgarh**.

The **energy generated will be procured by the state of Chhattisgarh**, thus aiding in meeting the state's peak energy demand through eco-friendly means and fulfilling its renewable purchase obligations.

**Types of Waste:** Wastes can be mainly classified into

- Urban waste
- Industrial waste
- Biomass waste
- Biomedical waste

**Energy from Waste Technologies:** The organic fraction of waste (biodegradable as well as non-biodegradable) can recover energy through thermal, thermo-chemical, biochemical and electrochemical methods.

1. **Thermal conversion:** Incineration is a major technological option; however, due to its emission features, it is losing attention these days.
2. **Thermo-chemical conversion:** Application is for wastes containing high percentage of organic non-biodegradable matter and low moisture content. **Pyrolysis** and **gasification** are main technological options under this category.
3. **Bio-chemical conversion:** Anaerobic digestion **(bio-methanation)** and **fermentation** are the major technological options under this category.
4. **Electrochemical conversion:** Electrochemical conversion in terms of energy waste refers typically to **microbial fuel cells (MFC)**.

### 3.7.8 Polycrack Technology

- The ***country's first Government-Owned Waste-to-Energy Plant*** was commissioned in 2020 at the Mancheswar Carriage Repair Workshop in ***Odisha.***
- The plant, a patented technology called *Polycrack, is first-of-its-kind in the Indian Railways and fourth in the country.*

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- It *converts multiple feed stocks into hydrocarbon liquid fuels, gas, carbon and water.*

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### What is Polycrack?

- It is ***the world's very first patented heterogeneous catalytic process which converts multiple feedstocks into hydrocarbon liquid fuels, gas, carbon as well as water.***
- The waste generated will become the feeder material for the waste to energy plant.
- The energy which will be produced at the plant will be in the form of light diesel oil and this oil will be used to light furnaces.

### The plant, having a capacity of 500 kg per batch, can be fed with the following

- All kinds of existing plastic
- Petroleum sludge
- Unsegregated MSW (Municipal Solid Waste) with moisture up to 50%
- E-waste
- Automobile fluff
- Organic waste including bamboo, garden waste
- Jatropha fruit and palm bunch

### Some of the Advantages of this Plant are as Follows

- ***The pre-segregation of waste is not required for processing in the plant.*** The waste as collected from the source can be directly fed into the polycrack plant.
- The plant has ***high tolerance to moisture hence drying of the waste after treatment is not required.***
- The waste is processed in the plant and reformed within a period of 24 hours.
- The biological decomposition is not allowed as the waste is treated in the plant as it is received.
- ***All the constituents are converted to valuable energy therefore, making it a zero discharge process through the plant.***
- The gas generated in the process of the plant is reused in order to provide energy to the system, hence making itself reliant and self-sufficient for its energy requirements. This also brings down the operating cost of the plant.

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- ***The plant does not cause atmospheric emission during the process*** unlike the other conventional methods except for the combustion of gases which have pollutants less than the prescribed norms across the world.

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### 3.7.9 Biomass Gasification

Coconut shell, coir pith, rice husk, firewood, coffee husk and other industrial wastes are those solid wastes which can be converted into producer gas by gasification route. The gas so produced can be used for heating and generation of electricity.

Biomass gasification follows **pyrolysis processes**, which represents the direct thermal decomposition of biomass under closed chamber (in the absence of oxygen).

(Refer **Infographic – Biomass Gasification Process**)

#### Biochar

Biochar is a high-carbon and fine-grained residue that is currently produced through modern pyrolysis processes (direct thermal decomposition of biomass in the absence of oxygen and preventing combustion). Biochar may increase soil fertility of acidic soils (low pH soils), increase agricultural productivity and provide protection against some foliar and soil-borne diseases.

##### *Benefits of Biochar*

- **Carbon Sink:** The burning and natural decomposition of biomass releases large amounts of carbon dioxide and methane to the Earth's atmosphere. The biochar production process also releases $CO_2$ (up to 50% of the biomass); however, the remaining carbon content is stable indefinitely.
- **Soil Amendment:** Biochar is recognised as offering a number of soil health benefits. The extremely porous nature of biochar is found to be effective at retaining both water and water-soluble nutrients. Its presence in the earth can improve water quality, increase soil fertility,

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raise agricultural productivity and reduce pressure on old-growth forests.

- **Water retention:** Biochar is hygroscopic. Thus, it is a desirable soil material in many locations due to its ability to attract and retain water.

**TERMINOLOGY RELATED TO BIOMASS GASIFICATION**

- **Pyogenic Carbonaceous Material (PCM)** are those which are produced by thermochemical conversion and contain some organic carbons, such as char, charcoal, biochar, ash, activated carbon, black carbon, soot, etc.
- **Char:** The incomplete combustion processes occurring in natural and man-made fires generate materials called as 'Char'.
- **Charcoal:** The thermochemical conversion from biomass (mainly but not exclusively wood) for energy generation produces charcoal.
- **Ash:** The operationally defined fraction of biomass or PCM, which typically includes inorganic oxides and carbonates.
- **Gasification:** This is a process of converting organic materials into carbon monoxide (CO), hydrogen (H), carbon dioxide (CO2) and methane (CH4) through really high temperatures, without combustion (burning).
- **Coal Gas:** It is a gaseous mixture of mainly hydrogen, methane and carbon monoxide together, made by the destructive distillation (i.e., heating in the absence of air) of bituminous coal and used as a fuel. Steam, when added to it, reacts with the hot coke, increases the yield of gas, which yields Coal tar and coke as by-products.

## 3.8 Biofuel

These fuel sources are directly or indirectly organic material, i.e., biomass (i.e., from plant materials and animal waste). Biofuel can exist in all three forms like solid, liquid or gas.

### *Primary Biofuel versus Secondary Biofuel*

Primary biofuels include processed form of organic matter directly such as fuel wood, wood chips and pellets, which are used primarily for heating, cooking and electricity production. Secondary biofuels include the

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processed materials of biomass; for instance, liquid fuels like ethanol and biodiesel.

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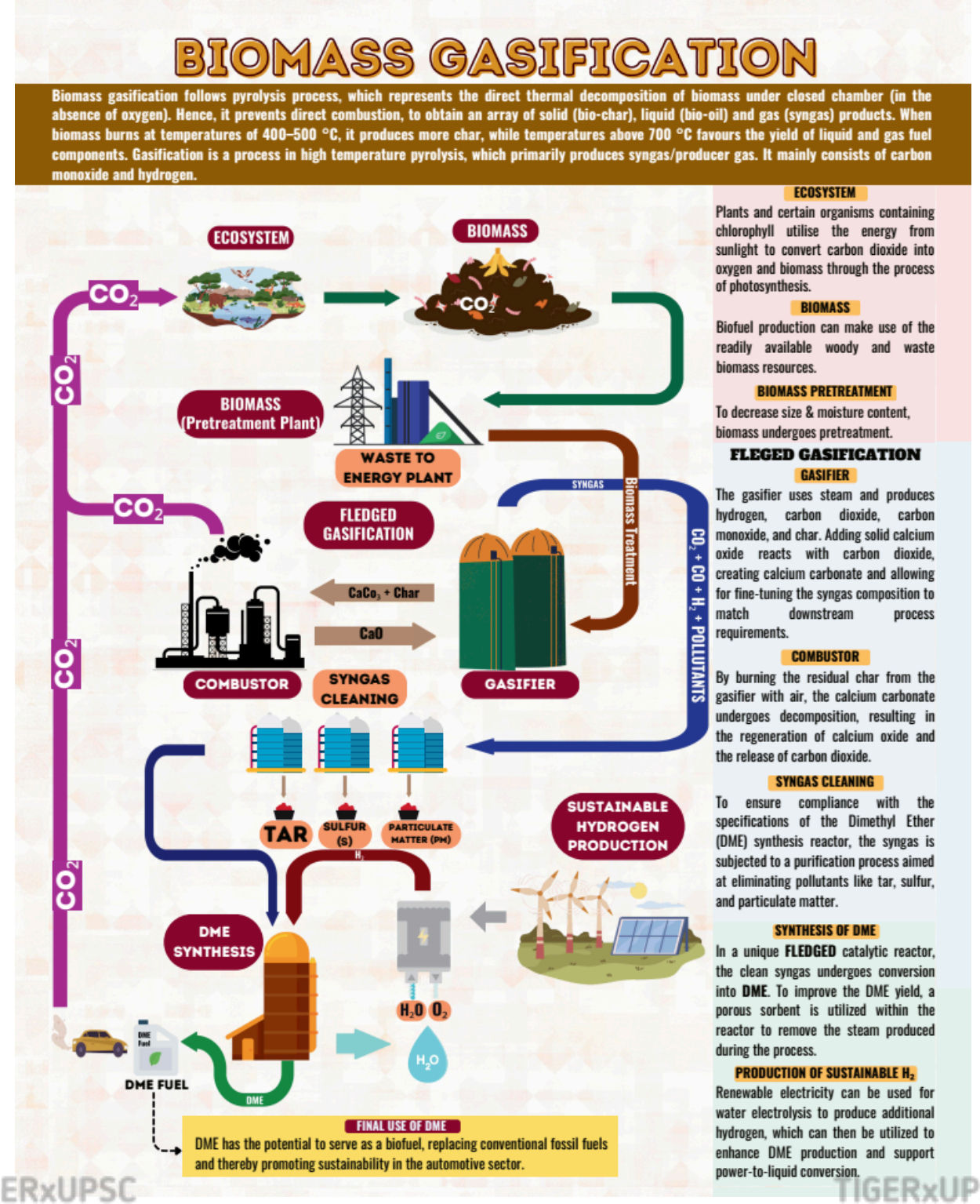


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### *Generations of Biofuel*

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1. **First-Generation Biofuel:** Food-crop derivatives directly used for the generation of biofuels like starch, sugar, animal fats and vegetable oil. The most commonly used first-generation biofuel feed stocks are corn, wheat, sugarcane, sugar beets, rapeseed, peanuts, soya bean, etc.

   **Ethanol Conversion Processes:** Ethanol is produced by a wide variety of carbohydrate material *via* fermentation process. These raw materials can be: **(a) Sugar-containing crops:** Sugar cane, wheat, beet root, fruits, etc. **(b) Starch- containing crops:** Grains such as wheat, barley, rice, corn, etc., and root plants like potato, cassava. **(c) Cellulosic biomass:** Wood and wood waste, cedar, pine, etc., agricultural residues and fibres. We can blend alcohol (ethanol) with petrol or burn it in nearly pure form in slightly modified spark-ignition engines. **Energy produced by 1 litre of ethanol is equivalent to two-thirds of the energy produced by 1 litre of petrol.** The combustion performance is improved by blending process and the emission of carbon monoxide and sulphur dioxide is consequently lowered.

   **Biogas** (60% methane + 40% $CO_2$ + some rare gases), cutting wood, grass, agriculture waste etc., are also considered under first-generation biofuel.

2. **Second-Generation Extraction Technology:** Feedstock of the first-generation biofuels is used in production of second-generation biofuels. These are generally **non-food crops (non-edible)**. All second-generation biofuels are not produced as a feedstock of the first-generation only, even a lot of plant cultivate is used as a direct biomass like – **Switch grass, Jatropha, Pangomia.**

   These are particularly **lignocellulose feedstock**, which tends to require several processing steps to obtain biofuel, like thermochemical conversion and biochemical conversion.

   ***(a) Thermochemical Conversion:*** **Gasification** and **pyrolysis** type of mechanisms are considered under this process (discuss in biomass gasification).

   ***(b) Biochemical Conversion:*** Different chemical reactions involved in the process of biofuel production, e.g., **transesterification.**

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Note: Second-generation feedstock should grow on marginal land (not very productive land like food crop) because second-generation feedstock does not require a huge amount of water or fertiliser to grow. There are different examples of second-stage feedstock: **Grasses** (Switch grass, Myscanthus, Indian grass and others), **biodiesel plants** (Jatropha/Ratanjot, Pongamia, etc.), other **seed crops** (Cammelina, Palm Oil, Rape seed oil, etc.), waste vegetable oil and municipal solid waste (landfill gas, human waste, grass and yard clippings).

Lignocellulosic material can also be converted into ethanol by some enzymatic activity. The recent discovery of fungus **Gliocladium roseum** points towards the production of diesel from cellulose. Recently, **neem seed** is being used for biodiesel production.

3. **Third-Generation Biofuel:** This generation is also called as Algal oil or algoil, because a specific group of plant (Algae) is used for biofuel production. Some microalgae species like **Botryococcus braunii** and **Schizochytrium** can contain up to 80% of their dry weight of lipids. Some algae like **Dunaliella** and **Chlorella** are also good sources of lipid constituents. Through trans esterification processes, these lipid substances can be converted into biodiesel.

***Why Algae for Biofuel (Biodiesel) Production?***

- ❏ The fastest growing plants in the world are algae and unlike seasonal crops they can be grown throughout the year.
- ❏ Agriculture land or clean water is not required in algae farming, and hence it does not compete with food crops.
- ❏ It is difficult to compare one energy crop to another energy crop, however per hectare of farming of algae is around 10–100 times more productive than soyabean, palm, corn, jatropha.
- ❏ The entire biomass produced can be utilised in end products, which is not the case with other energy crops.
- ❏ Algal meal has a higher protein content than other animal feed product.
- ❏ Unlike comparable crops, algae typically contain a sustainable amount of vested biomass, 100% of algal biomass can be utilised for creating new products.
- ❏ Jatropha takes 2–3 years for a commercial crop, whereas the algal form can start yielding within 2–3 days. While jatropha gives crop once a year, algae can give oil on a daily basis.

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Note: The list of fuels derived from algae includes Biodiesel, Butanol, Gasoline, Methane, Ethanol, Vegetable Oil and Jet Fuel.

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4. **Fourth-Generation Biofuel:** A combination of photovoltaic or inorganic water-splitting catalysts with metabolically engineered microbial fuel production pathways (Electro-biofuels/Microbial fuel cells – MFCs) has emerged to be powerful technology, enabling efficient production and storage of liquid fuels.

(Refer **Infographic – Generation of Biofuel**)

### 3.8.1 Ethanol

Ethanol, or ethyl alcohol, is a biofuel, that has several uses.

- At 95% purity, it is called rectified spirit and is used as the intoxicating ingredient in alcoholic beverages.
- Ethanol is basically alcohol of 99%-plus purity, which can be used for blending with petrol.
- Produced mainly from molasses, a by-product of sugar manufacture.

***Benefits of Ethanol Blending***

- Reduction in import dependency.
- Support to agricultural sector.
- Environmental friendly fuel.
- Additional income to farmers.

***About Ethanol Blended Petrol (EBP) Programme***

- Launched in 2003 on pilot basis.

**SPECIFICATION OF JATROPHA**

Scientific name of this plant is *Jatropha curcas* and belongs to Euphorbiaceae family. This plant is the **native of South America and Africa.** Jatropha curcas is well adapted to **arid** and **semi-arid conditions.** With low fertility and moisture demand, it can grow on stony, gravelly or shallow and even calcareous soils.

Jatropha is grown well in **low rainfall conditions (200 millimetres)**; however, it can also **tolerate high rainfall (up to 1,200 millimetres)** and hence it grows in a number of climatic zones in **tropical and sub-tropical regions** of the world and even in areas of low rainfall. Jatropha sheds most of its leaves, thereby reducing the transpiration loss and hence can stand long periods of drought. The **annual temperature range**

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**tolerable to Jatropa is of 18–28.5°C.** For seed germination, hot and humid climate is preferred. The flowers are induced in rainy season and fruits are borne in winter.

A multipurpose shrub or small size tree, Jatropha has significant importance due to its several industrial and medicinal uses. The various alkaloids contained in latex of Jatropha are **'Jatrophine', 'jatropham', 'jatrophone'** and **'curcain',** which bear **anti-cancerous properties**. The leaves of Jatropha contain galactogoue, rubefacient and anti-parasitic, which is used against **scabies**, **paralysis**, **rheumatism** and **hard tumours**. Seeds are aperient and used for **dropsy, gout, paralysis** and **skin diseases.** The oil is **applied to hair** as a growth stimulant, in Java. Other application include for sores on domestic livestock.

Jatropha have lipid substances that will convert into biodiesel through transesterification (TE) process in laboratory or industry.

## TYPES OF ALTERNATIVE FUELS

1. **Ethanol:** Primarily alcohol is produced by fermentation and further distillation of crops like corn, barley, sugarcane bagasse, wheat, etc. It is blended with gasoline to increase octane level and improve emission quality.
2. **Natural Gas:** It is already in use in several countries through utilities that provide natural gas to homes and it burns clean. Cars and trucks with specially designed engines run through natural gas with less harmful emissions.
3. **Propane:** The other name of propane is Liquefied Petroleum Gas (LPG); it is a by-product of crude oil refining and natural gas processing. It is widely used as cooking gas and a popular alternative fuel.
4. **Biodiesel:** It is one of the fuels whose source is always renewable. It can be produced from vegetable oils or animal fats and also the cooking wastes of restaurants. Vehicle engines can be converted to run by pure biodiesel and it is also blended with petroleum diesel to run in modified engines also.
5. **Methanol:** One of the present day important alternatives fuels – much attention has been paid on cost-effective production of methanol, which can be used as a source of alternative fuel. It is an important raw material for production of fuel cell grade hydrogen to run hydrogen powered vehicles.
6. **Electricity:** Electricity can be used as transportation alternative fuel for battery-powered electric and fuel-cell vehicles.
7. **Hydrogen:** It can be mixed with natural gas to create an alternative fuel which can be used in some types of internal combusting engines. Hydrogen is the fuel for fuel-cell vehicles and is produced by the petrochemical reaction.
8. **P-Series Fuels:** These are blends of ethanol, LNG and methyltetrahydrofuran (MeTHE) in different proportions. The P-Series alternative fuels are clear and have high octane number with flexibility of choice of vehicles. It has the option of addition to normal gasoline.

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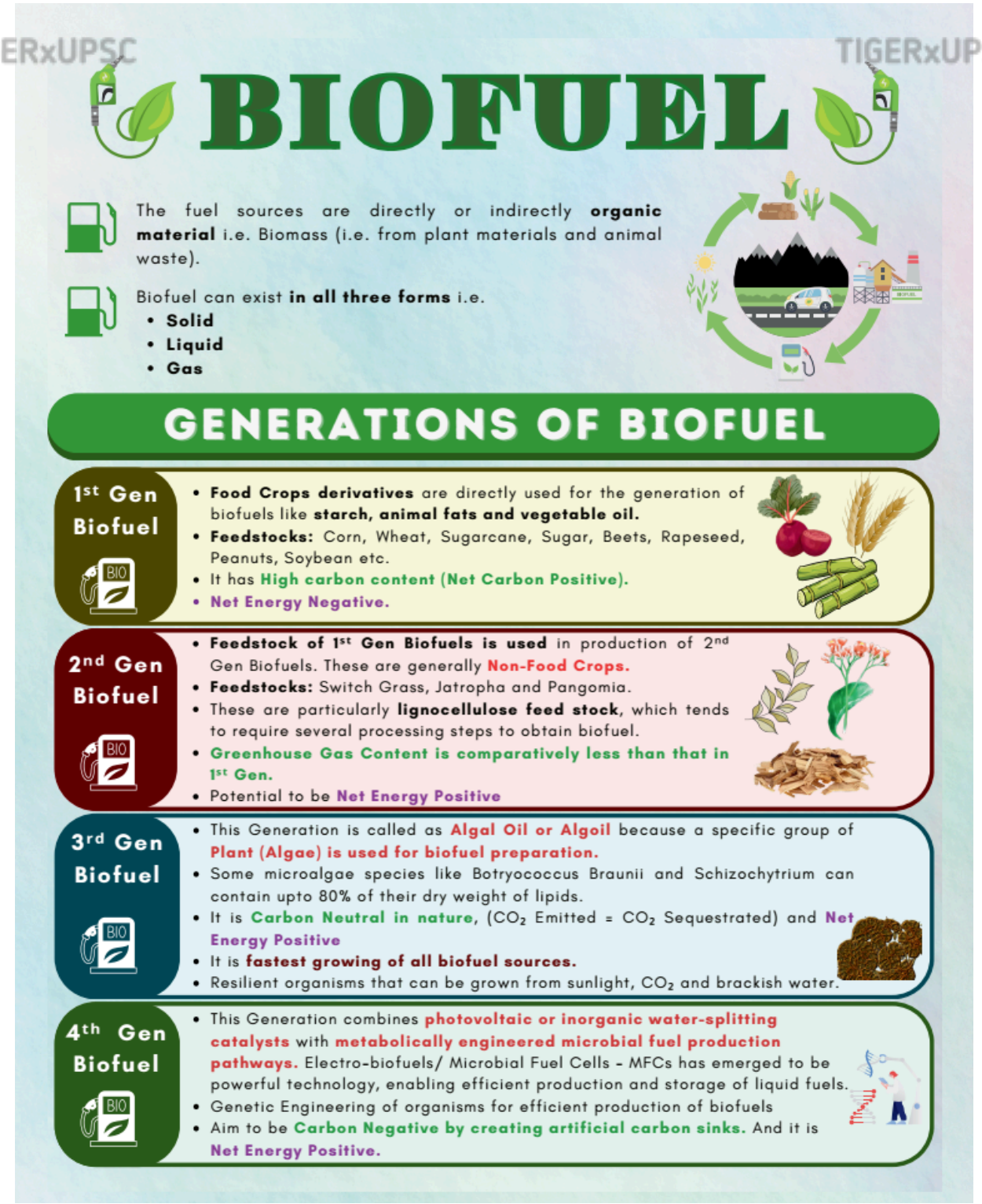


- The aim is to promote the use of alternative and environmental friendly fuels.
- Implemented by the Ministry of Oil Marketing Companies (OMCs).

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*Need*

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- India is the third largest consumer of energy in the world after China and the United States.
- India is dependent on imports of crude oil approximately 85%.
- India is expected to need 10 billion litres of ethanol annually to meet the 20% blending target in 2025 if petrol consumption continues to grow at the current pace.

### 3.8.2 Methanol-Blending in Petrol

The world over, methanol is being looked upon as a viable alternative fuel. The government of India's NITI Aayog has also drawn out a road map to replace 10% of crude imports by 2030 with Methanol. It has set up an apex committee and five task forces for carrying out R&D activities and developing a road map for implementing the methanol economy in India.

Methanol is a clear and colourless liquid produced from natural gas, coal biomass. It is a clean burning fuel that can replace petrol and diesel. It burns efficiently in all internal combustion engines causing almost zero pollution. **Its gaseous version – Dimethyl Ether or DME – can even be blended with LPG.**

The use of Methanol and DME as fuel has increased significantly during the last few years with the Methanol demand growing at 6–8% annually. Methanol accounts for almost 9% of transport fuel in China. Israel and Italy have adopted the **Methanol 15% (M15)** blending programme with petrol. Japan and Korea have extensive Methanol & DME usage and Australia has adopted GEM fuels (Gasoline, Ethanol and Methanol) and blends almost 56% Methanol.

**In India as per the plan prepared by the NITI Aayog, 20MT of Methanol can be produced every year by 2025 using coal, gas and biomass.** NITI Aayog's road map for Methanol Economy envisages utilisation of Methanol as well as DME in transportation – rail, road, marine and defence, industrial boilers, diesel gensets, power generation, mobile towers and domestic cooking fuel are other applications.

It is hoped that the advent of the Methanol Economy in the country would be a major step towards combating air pollution, reducing greenhouse gas emissions and promoting huge foreign exchange savings.

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- A study, in which **methanol (M-15) was blended with petrol and used in the existing BS-IV standard** cars, found that the carbon dioxide emission had reduced substantially.

***Methanol as an Alternative Fuel***

- Methanol is a promising fuel as it is **clean, cheaper than fossil fuels and a good substitute for heavy fuels. India imports methanol from Saudi Arabia and Iran at present.**
- Across the world, methanol is emerging as a clean, sustainable transportation fuel of the future.

***Why Methanol?***

- Methanol can be used as an **energy producing fuel, transportation fuel and cooking fuel,** cutting down India's oil import bill by an estimated 20% over the next few years. Unlike CNG, using methanol as a transportation fuel would require minimal alteration in the vehicles.
- Methanol is **a clean-burning fuel** that **produces fewer smog-causing emissions – such as sulphur oxides (SOx), nitrogen oxides (NOx) and particulate matter** – and can improve air quality and related human health issues.
- Methanol is **most commonly produced on a commercial scale from natural gas.** It can also be produced from renewable sources such as biomass and recycled carbon dioxide.
- As a high-octane vehicle fuel, **methanol offers excellent acceleration and power.** It also improves vehicle efficiency.

***Way Ahead***

- Methanol Economy, if adopted by India can be one of the best ways to mitigate the environmental hazards of a growing economy. NITI Aayog is preparing a road map for a full-scale implementation in the near future.

### 3.8.3 Sustainable Aviation Fuel (SAF) and Synthetic Petrol

The **National Biofuel Coordination Committee (NBCC)** has established the preliminary indicative goals for incorporating sustainable aviation fuel (SAF) into aviation turbine fuel (ATF).

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This determination aligns with the obligatory stage of the International Civil Aviation Organization's (ICAO) Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), slated to commence in 2027.

### *International Civil Aviation Organization (ICAO)*

The International Civil Aviation Organization (ICAO) is a **specialised agency of the United Nations established in 1947**, headquartered in Quebec, Canada. **It coordinates international air navigation principles and techniques, fostering safe and orderly growth in international air transport.**

### *Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA)*

The Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) is a **global market-based measure aimed at offsetting international aviation CO2 emissions to stabilise emission levels.**

CORSIA requires airlines worldwide to offset any growth in carbon dioxide emissions beyond 2020 levels, achieved through acquiring and canceling emissions units from the global carbon market.

### *National Biofuel Coordination Committee (NBCC)*

As per **National Policy on Biofuels-2018**, this **Ministry has constituted a National Biofuel Coordination Committee (NBCC)**, under the **Chairmanship of Minister, Petroleum & Natural Gas (P&NG)** to **provide overall coordination, effective end-to-end implementation and monitoring of biofuel programme**. In addition, **a Working Group has also been set up by this Ministry, consisting of eminent and technical experts in the field of biofuels** to monitor the implementation of biofuel programme.

### *Recent Decision on SAF (Synthetic Aviation Fuel)*

**NBCC approved initial indicative blending targets:**

(a) 1% SAF blending target in 2027 (initially for international flights).
(b) 2% SAF blending target in 2028 (initially for international flights).

**Using SAF-blended jet fuel is a method for carriers to keep emissions within permissible levels.**

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### *Sustainable Aviation Fuel (SAF)*

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It is a type of fuel **produced from biomass sources** such as used cooking oil, waste animal fat, manure and sewage, and various types of organic waste, including maize, foodgrains, rice, and waste wood. **These biomass materials are processed to create synthetic jet fuel substitutes.**

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#### *Benefits of SAF*

(a) ***Emission Reduction***
(b) ***Carbon Footprint Reduction***
(c) ***Negative Carbon Footprint***
(d) ***Compensating for Future Emissions***
(e) ***Waste Recycling***

#### *Roadblocks/Problems in Adoption*

- **High Cost**
- **Low Adoption**

#### *Synthetic Petrol*

Synthetic petrol, also known as ***synfuel***, is a liquid or gaseous fuel derived from syngas, which is ***a mixture of hydrogen and carbon monoxide.***

**Syngas can be obtained from biomass, coal gasification, or natural gas.**

## 3.9 'Flex-Fuel' Vehicles (FFV)

- An FFV is a modified version of vehicles that could **run both on gasoline and doped petrol with different levels of ethanol blends.**
- FFVs will allow vehicles to use all the blends and also run on unblended fuel.
- FFVs have compatible engines to run on more than 84% ethanol blended petrol.
- Union Government had issued new guidelines for use of **FFVs using flex engines**.
- The guidelines will specify engine configuration and other changes required in vehicles to conform to stipulated changes in fuel mix.

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- The **central government has released an expert committee report** on the **Roadmap for Ethanol Blending in India by 2025.**
- The roadmap proposes **a gradual rollout of ethanol-blended fuel** to achieve E10 fuel supply by April 2022 and phased rollout of E20 from April 2023 to April 2025.

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### *About*

- It is one of the principal biofuels, which is naturally produced by the fermentation of sugars by yeasts or via petrochemical processes such as ethylene hydration.
- These are currently being used successfully in Brazil, giving people the option to switch fuel (gasoline and ethanol) depending on the price and convenience.
- For India, FFVs will present a different advantage as they will allow vehicles to use different blends of ethanol-mixed petrol available in different parts of the country.

### *Blending Target*

- The Government of India has advanced the target for 20% ethanol blending in petrol (also called E20) to 2025 from 2030.
- Currently, 10% of ethanol is blended with petrol in India.

### *Benefits*

- Reduces the oil import bill.
- Can save the country USD 4 billion (₹30,000 crore) per annum.
- Benefits the sugarcane farmers.
- Encourages use of water-saving crops, such as maize, to produce ethanol, and production of ethanol from non-food feedstock.
- Decreases emissions such as carbon monoxide (CO), hydrocarbons (HC) and nitrogen oxides ($NO_x$).

### *Disadvantages/Challenges of using FFVs*

1. **Customer acceptance** will be a major challenge since the cost of ownership and running cost are going to be very high compared with 100% petrol vehicles.
2. **Running cost** (due to lower fuel efficiency) will be higher by more than 30% when run with 100% ethanol (E100).

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3. **Flex Fuel Engines cost more** as ethanol has very different chemical properties than petrol. Ethanol has very low (40%) Calorific value as compared to Gasoline, very high latent heat of vaporisation causing cooling of charge/combustion, etc.
4. **Ethanol also acts as a solvent and could wipe out the protective oil film inside the engine** thereby causing wear and tear.

#### *Possible Challenges that Might be Faced During the Implementation of the Program*

- High cost of the flex fuel engines.
- Running cost will be higher due to the lower fuel efficiency.
- Acceptance by the customers.
- Ethanol acts as a solvent and could wipe out the protective oil film inside the engine thereby causing wear and tear.

## Electrified Flex Fuel Vehicles

***Toyota Kirloskar Motor*** recently revealed the prototype of the world's first **Bharat Stage-6 (BS6) Stage-II Electrified Flex Fuel Vehicle.**

An Electrified Flex Fuel Vehicle combines a Flex Fuel engine with an electric powertrain, providing the flexibility to utilise flexible fuel, including petrol, diesel, electric, and ethanol blends of up to 100%.

**Flex Fuel Strong Hybrid Electric Vehicles (FFV-SHEV):** Integration of Flex Fuel technology with strong hybrid electric systems results in FFV-SHEVs, which offer improved fuel efficiency and can operate solely on electric or petrol modes.

#### *Understanding BS6 (Stage II) Norms*

**Overview:** The Bharat Stage (BS) norms serve as emission standards mandated by the Indian government to govern and mitigate air pollutant emissions originating from motor vehicles.

- **BS6 Stage II:** This phase of BS6 norms introduces even stricter emission limits, incorporating ***Real Driving Emissions (RDE), Corporate Average Fuel Economy (CAFE 2), and On-Board Diagnostics (OBD).***
- **RDE Test:** RDE testing **provides a more realistic estimation of vehicle emissions** under real traffic conditions, considering factors like speed variations and acceleration.

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- **OBD Systems:** Onboard Diagnostics **monitor and report the performance of vehicle subsystems, enhancing vehicle maintenance and emission control**.

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***Ethanol Blending Initiatives in India***

- **Ethanol Production:** Ethanol is a key biofuel produced through fermentation or petrochemical methods.
- **Ethanol Blending Programme (EBP):** India's EBP aims to decrease oil imports, reduce emissions, achieve energy self-sufficiency, and enhance farmers' income.
- **Government Targets:** The government aims to achieve 20% ethanol blending in petrol (E20) by 2025, advancing the previous target of 2030.
- **Increasing Ethanol Blending:** India has made significant progress in ethanol blending, increasing from 1.53% in 2013–14 to 11.8% in August 2023.
- **Other Initiatives:** Additional initiatives like the ***National Policy on Biofuels 2018***, ***E100 Pilot project, Pradhan Mantri G-VAN Yojana 2019***, and Repurpose Used Cooking Oil (RUCO) further promote ethanol blending and biofuel utilisation.

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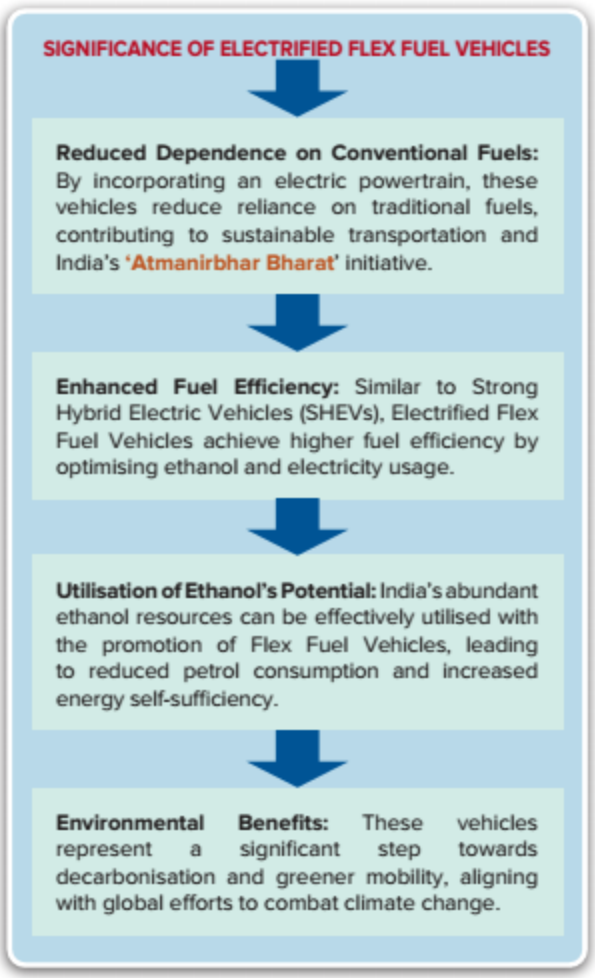


## 3.10 National Policy on Biofuels 2018

The Union Cabinet, chaired by the Prime Minister Shri Narendra Modi, has approved National Policy on Biofuels – 2018.

***Salient Features***

1. The **Policy categorises biofuels** as 'Basic Biofuels' viz. **First-Generation (1G)** bioethanol and biodiesel and 'Advanced Biofuels' – **Second-Generation (2G)** ethanol, Municipal Solid Waste (MSW) to drop-in fuels, **Third-Generation (3G)** biofuels, bio-CNG, etc., to enable extension of appropriate financial and fiscal incentives under each category.

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2. The policy expands the scope of raw material for ethanol production by allowing use of sugarcane juice, sugar containing materials like sugar beet, sweet sorghum, starch containing materials like corn, cassava, damaged food grains like wheat, broken rice, rotten potatoes, unfit for human consumption for ethanol production.
3. Farmers are at a risk of not getting appropriate price for their produce during the surplus production phase. Taking this into account, the policy allows use of surplus food grains for production of ethanol for blending with petrol with the approval of National Biofuel Coordination Committee.
4. With a thrust on Advanced Biofuels, the policy indicates a viability gap funding scheme for 2G ethanol biorefineries of ₹5000 crore in 6 years in addition to additional tax incentives, higher purchase price as compared to 1G biofuels.
5. The policy encourages setting up of supply chain mechanisms for biodiesel production from non-edible oilseeds, used cooking oil, short gestation crops.
6. Roles and responsibilities of all the concerned Ministries/Departments with respect to biofuels has been captured in the policy document to synergise efforts.

***Expected Benefits***

- **Reduce Import Dependency:** One crore litres of E10 saves ₹28 crore of forex at current rates.
- **Cleaner Environment:** One crore litre of E-10 saves around 20,000 tons of $CO_2$ emissions.
- **MSW Management:** It is estimated that, annually 62 MMT (Million Metric Tonne) of **Municipal Solid Waste** gets generated in India. There are technologies available which can convert waste/plastic, MSW to drop-in fuels. One ton of such waste has the potential to provide around 20% of drop-in fuels.
- **Infrastructural Investment in Rural Areas:** It is estimated that, one 100 klpd (kilo litres per day) biorefinery will require around ₹800 crore capital investment. At present Oil Marketing Companies are in the process of setting up twelve 2G biorefineries with an investment of

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around ₹10,000 crore. Further, addition of 2G biorefineries across the country will spur infrastructural investment in the rural areas.

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- **Employment Generation:** One 100 klpd 2G bio refinery can contribute 1200 jobs in plant operations, village level entrepreneurs and supply chain management.
- **Additional Income to Farmers:** By adopting 2G technologies, agricultural residues/waste which otherwise are burnt by the farmers can be converted to ethanol and can fetch a price for these waste if a market is developed for the same. Also, farmers are at a risk of not getting appropriate price for their produce during the surplus production phase. Thus conversion of surplus grains and agricultural biomass can help in price stabilisation.

(Refer **Infographic – National Policy on Biofuel 2018**)

### WORLD BIOFUEL DAY

- **World Biofuel Day is observed every** year on 10th August.
- **Aim:** to create awareness about the importance of non-fossil fuels as an alternative to conventional fossil fuels and to highlight the various efforts made by the Government in the biofuel sector.
- **Theme 2023:** 'Biofuels: Fuels for a sustainable future'.
- On this day in 1893, Sir Rudolph Diesel (inventor of the diesel engine) for the first time successfully ran mechanical engine with peanut oil.
- His research experiment had predicted that vegetable oil is going to replace the fossil fuels in the next century to fuel different mechanical engines.
- In India, Ministry of Petroleum and Natural Gas (MoPNG) celebrates the World Biofuel Day every year.

### Stardust 1.0, the first rocket to run on biofuel

Stardust 1.0 was launched on 31 January 2021 from Loring Commerce Centre in Maine, US.

- It has become **the first commercial space launch powered by biofuel,** which is non-toxic for the environment as opposed to traditionally used rocket fuels.
- Stardust 1.0 is a launch vehicle suited for student and budget payloads.

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## 3.11 Urja Sangam 2015/PAHAL

- The Prime Minister, Shri Narendra Modi, urged well-to-do sections of the Indian society to voluntarily give up LPG subsidy, for benefiting the poorer sections of society. During the inaugural address at global energy summit – **'Urja Sangam 2015'** – in New Delhi.
- '**Give it up**' campaign was launched by the Prime Minister for voluntarily giving up LPG subsidy. The Union Government targets to extend the reach of Piped Natural Gas (PNG) to 1 crore houses in the next five years.
- On the occasion of the platinum jubilee of India's independence, the Prime Minister called for enhancement of domestic production, which could reduce the energy import load of the country by at least 10% till 2022. He said that **PAHAL – the world's largest cash subsidy transfer programme** in the world, enabling transfer of LPG subsidy directly into bank accounts had consequently led to the elimination of leakages, and curbing corruption.
- He also mentioned deregulation of diesel prices, ethanol and ethanol-blended petrol in order to assist sugarcane farmers, promotion of bio-diesel and extension of the gas grid in eastern India, as some other major initiatives of the Government. According to the Prime Minister, over the next five years, the Union Government aims to extend piped natural gas (PNG) to one crore houses. According to recent report over 1.9 crore PNG connections reached across India (2023–2024).

## 3.12 Ujjwala Yojana

It was launched by the PM from Ballia on 1 May 2016. The scheme ideated to provide 8 crore cooking gas connections to Below Poverty Line families over the next three years (2016–2019). The scheme provides financial

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support for each LPG connection to the BPL household. The households will be identified in consultation with the state governments and UTs.

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***About Pradhan Mantri Ujjwala Yojana (PMUY)***

- Under the scheme, an adult woman member of a below poverty line family **identified through the Socio-Economic Caste Census** (SECC) is given a deposit-free LPG connection with **financial assistance of ₹1,600 per connection** by the Centre.
- Eligible households will be identified in consultation with state governments and union territories. **The scheme is being implemented by the Ministry of Petroleum and Natural Gas.**

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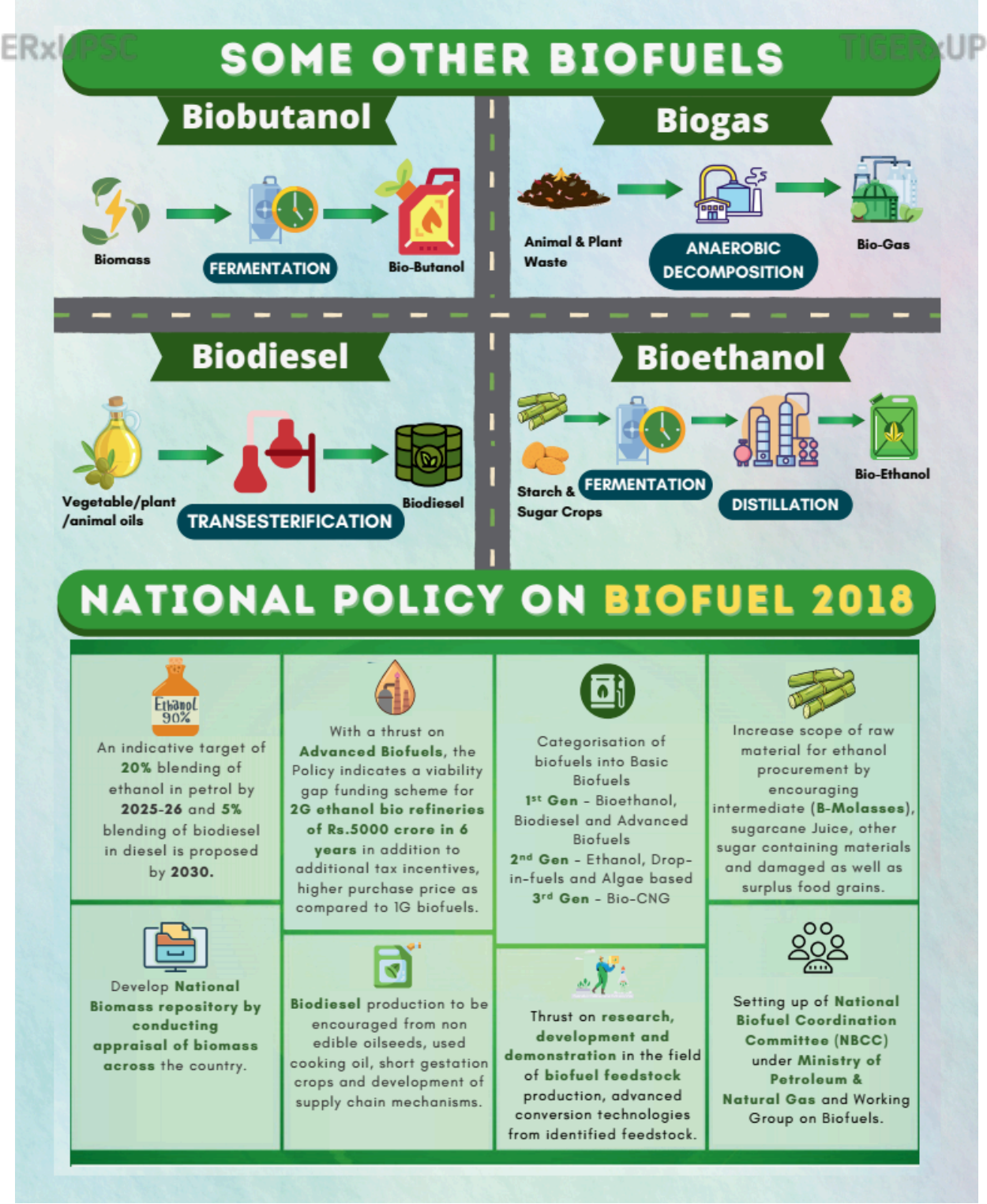


***Some objectives of the scheme are:***

- Empowering women and protecting their health.

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- Reducing the serious health hazards associated with cooking based on fossil fuel.
- Reducing the number of deaths in India due to unclean cooking fuel.
- Preventing young children from significant number of acute respiratory illnesses caused due to indoor air pollution by burning the fossil fuel.

***What makes LPG adoption necessary?***

- A large section of Indians, especially women and girls, are exposed to severe **household air pollution** (HAP) from the use of solid fuels such as biomass, dung cakes and coal for cooking. A report from **the Ministry of Health & Family Welfare places HAP as the second leading risk factor contributing to India's disease burden.**
- According to the World Health Organisation, **solid fuel use is responsible for about 13% of all mortality and morbidity in India** (measured as Disability-Adjusted Life Years), and causes about 40% of all pulmonary disorders, nearly 30% of cataract incidences, and over 20% each of ischaemic heart disease, lung cancer and lower respiratory infection.

Kerala is now set to become the **first smoke-free state in the country with public sector oil companies eyeing 100% LPG penetration here.**

- LPG is being supplied beyond commercial considerations with the **Pradhan Mantri Ujjwala Yojana initiated by the Ministry of Petroleum.**

## Ujjwala 2.0

Prime Minister Shri Narendra Modi had launched Ujjwala 2.0 by handing over LPG connections, at Mahoba Uttar Pradesh on 10th August, 2021 via video conferencing. During the event, the Prime Minister interacted with beneficiaries of the Ujjwala and also addressed the nation.

### *The Journey from Ujjwala 1.0 to Ujjwala 2.0*

During Ujjwala 1.0, a target was set to provide LPG connections to 5 crore women members of BPL households. Subsequently, the scheme was expanded in April 2018 to include women beneficiaries from seven more categories (SC/ST, PMAY, AAY, Most backward classes, tea garden, forest

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dwellers, Islands). Also, the target was revised to 8 crore LPG connections. This target was achieved in August 2019, seven months ahead of the target date.

In the Union budget for FY21–22, provision for an additional one crore LPG connection under the PMUY scheme was announced. These **one crore additional PMUY connections (under Ujjwala 2.0) aim to provide deposit-free LPG connections to those low-income families who could not be covered under the earlier phase of PMUY**.

## 3.13 Pradhan Mantri Urja Ganga

GAIL (Gas Authority of India Limited) is a pioneer in city gas distribution in India. 'Petroleum Ministry' launched a ₹1750 crore city gas distribution project between Bhubaneshwar and Cuttack on March 2017, taking a big step towards linking of eastern India to the natural gas grid connection through Jagdishpur-Haldia and Bokaro-Dhamra Natural Gas Pipeline (JHBDPL).

The gas pipeline project aims to **provide piped cooking gas to residents of Varanasi** and later to millions of people in states like **Bihar, Jharkhand, West Bengal and Odisha.** The government estimates that around 5 lakh gas cylinders will be sent to rural areas annually.

According to GAIL, with the Urja Ganga project, **20 lakh households will get PNG connections.** The project is said to be a major step towards collective growth and development of the Eastern region of India. GAIL has built a network of trunk pipelines covering the length of around 11,000 km. With Urja Ganga project, this number will further increase by 2540 km.

### Northeast Gas Pipeline Grid Project

Five public sector oil and natural gas companies have signed the joint venture agreement to implement the ₹6000 crore gas pipeline grid project in Northeast.

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The five companies are Indian Oil Corporation Limited (**IOCL**), **ONGC**, **GAIL**, **Oil India Limited** and **Numaligarh Refinery Limited** (NRL).

With the signing of the agreement the joint ventures company will develop, build, operate and maintain the natural gas pipeline grid.

- The proposed gas pipeline grid will connect Guwahati to the major Northeast cities and major load centres.
- The project is being implemented under **Prime Minister Narendra Modi's ambitious Urja Ganga Gas Pipeline Project.**
- Besides connecting all the state capitals in the region, the pipeline will also connect with the National Gas Grid through Barauni – Guwahati Gas Pipeline, which is being laid by GAIL.
- From Guwahati, the pipeline will extend to Numaligarh, Dimapur, Kohima and Imphal in one direction; Shillong, Silchur, Aizwal and Agartala in the second direction and to Itanagar in the third direction. Gangtok will be connected from Siliguri from the gas pipeline of GAIL coming from Barauni to Guwahati.

**KEY NOTES**

- The proposed gas pipeline grid will connect Guwahati to the major Northeast cities and major load centres.
- The project is being implemented under ambitious Urja Ganga Gas Pipeline Project.
- Besides connecting all the state capitals in the region, the pipeline will also connect with the National Gas Grid through Barauni–Guwahati Gas Pipeline, which is being laid by GAIL.
- The pipeline will enable the supply of piped cooking gas to households and CNG to automobiles, besides fuel to industry.
- The 1656-km-long natural gas pipeline NEGG (North East Gas Grid) Project has set a target of completing 70% of its works in the fiscal concluding by 31 March 2023, and the entire project by 31 March 2024.

## 3.14 Rural Electrification

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As per 2011 census, nearly 80% population of the country stays in the rural areas, including 167.8 million families. The backbone of the rural economy is often seen in rural electrification. The four facets of rural electrification are:

- Establishment of rural electrification infrastructure, thereby providing household connectivity.
- Desired power quality to be adequately supplied.
- Supplying electricity at reasonable tariff.
- Effective and efficient provision of environmentally friendly, clean and sustainable power.

### 3.14.1 Rural Electrification Policy 2005

The key for accelerating rural development is seen in rural electrification and hence, provision of electricity is essential to cater for many requirements of life. As per the National Electricity Policy, supply of electricity to all areas including rural, as mandated in Section 6 of Electricity Act, marks the key development objective of the power sector. This National Policy for Rural Electrification has been prepared and framed up through an extensive consultative process undertaken by the Ministry of Power.

#### *Aims of the Policy*

- Provision of access to electricity to all households by the year 2009.
- Quality and reliable power supply at reasonable rates.
- Minimum lifeline consumption of 1 unit per household per day by 2012.

However, these targets are yet to be achieved. Based on the certificate issued by Gram Panchayat with certain properties, the REP has changed the definition of electrified villages. For rural electrification to be achieved in a sustainable way, a massive focus on creation on income generation activities has to be given to boost the rural economy.

Note: Rural Electrification Corporation Limited (REC), a Government of India enterprise under Ministry of Power, is the nodal agency to implement Rural Electrification Programme.

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### 3.14.2 Rajiv Gandhi Grameen Vidyutikaran Yojna (RGGVY)

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RGGVY was launched by the central government on 4 April 2005, for attaining the National Common Minimum Programme (NCMP) objective of providing access to electricity to all households in the country in 5 years. The scheme aimed at electrification of over 1 lakh un-electrified villages and providing electricity connection to 2.34 crore rural households approximately. The estimated cost was ₹51,000 crores. A village or hamlet with a population of 100 or more is eligible for this scheme. This scheme is being implemented mostly to provide benefits to households below poverty line (BPL) through a free connection but chargeable consumption of power. Under this scheme, central government provides 90% of grant and 10% as a loan, which is provided by REC (Rural Electrification Corporation) to all state governments. The REC is the main nodal working agency for this scheme.

#### *Aim*

- Electrifying every village as well as habitation, according to new guidelines.
- Proper electricity to every rural household.
- To offer electricity connection to every BPL family without charge.

#### *Conditions and Implementation Methodology*

- Main work is preparation of various district boards for proper execution.
- Proper involvement of public sector.
- Daily supply of 6–8 hours of proper electricity in areas of RGGVY.
- Outline electronic fund transfer directly to contractor account.
- Three-tier system of quality monitoring management.

### 3.14.3 Deen Dayal Upadhyaya Gram Jyoti Yojna (DDUGJY)

DDUGJY is a Government of India scheme designed to give continuous power supply to rural India as one of a key initiative of the Modi Government. This scheme will replace existing RGGVY. This was launched

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on 25 July 2015 in Patna by PM Modi. This is one of the major programmes of the ministry of power and will facilitate 24 × 7 power supply.

It draws its inspiration from the similar pioneering scheme implemented by the Government of Gujarat. This will enable to initiate much awaited reform in rural areas. It focuses on feeder separations (rural household and agriculture) and strengthening of sub-transmission and distribution infrastructure. It also includes metering at all levels on rural areas.

Particularly, there are two major components in DDUGJY, i.e., **providing power till 2019 to all** and **feeder separation,** which were not included in the RGGVY. The government has started undertaking different measures and also working in the direction of supplying power to the country's last standing mile. The target of the Government is the completion of all electrification of villages by the year 2019.

***Objectives***

- Provide electrification to all villages.
- Feeder separation to ensure sufficient power to farmer and regular supply to other consumer.
- Sub-transmission and distribution framework to be improved to enhance quality and reliability of the supply.
- Metering to reduce losses.

***Benefits***

- All villages and households shall be electrified.
- Increased agriculture yield.
- Business of small and household enterprise shall grow consequently.
- Improvement in health, education, ATM services.
- Improvement in accessibility to radio, telephone, mobile, etc.
- Betterment in social security due to availability of electricity.
- Accessibility of electricity to schools, panchayats, etc.
- Rural areas shall get increased opportunities for comprehensive development.

### 3.14.4 Saubhagya: Sahaj Bijli Har Ghar Yojana

Union Ministry of Power had launched Pradhan Mantri Sahaj Bijli Har Ghar Yojana on 25 September 2017 (Birth ceremony of Pt. Deen Dayal

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Upadhaya). Its aim is to achieve universal household electrification (to all rural and urban poor households) in all parts of country by providing electricity connectivity. These households will be identified under Socio-Economic and Caste Census (SECC) data 2011.

Rural Electrification Corporation Limited (REC) is the nodal agency for operationalization of Saubhagya scheme throughout India. For this accelerated implementation, household survey will perform with the help of Mobile App. Public institutions/Gram Panchayat in rural areas will be authorised to collect application and documents. They will distribute and collect revenue in consultation with Panchayat Raj Institutions (PRIs) and Urban Local Bodies (ULBs).

For APL (Above poverty line) families, ₹500 will be payable in 10 equal monthly instalments but BPL cardholders will get free electricity connections.

In remote and inaccessible areas, solar power packs of 200 to 300 watts with battery will be provided. It will also comprise of 5 LED lights, 1 DC fan, 1 DC power plug with all Repair and Maintenance (R and M) for next 5 years.

This scheme had many outcomes like to improve health services, education services, various economic activities and jobs, improved quality of life especially for women, everyone can easily connect through radio, television, mobiles, etc., and also environmental upgradation (due to complete replacement of kerosene for lighting purposes).

## 3.15 Ujwal Discom Assurance Yojana (UDAY)

UDAY is the financial turnaround and revival package for power distribution companies of India (DISCOMs) and more importantly offers a sustainable permanent solution to the problem. It is a path breaking reform for realising the Prime Minister's vision of affordable and accessible 24 × 7 power for all. It is another decisive step, further being the landmark strides made in the power sector over the past years. Financially stressed DISCOMs could not function adequately, because the legacy issues trapped

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the DISCOMs in a vicious cycle with operational loses being funded by debt. A rise of vibrant and efficient DISCOMs through a permanent resolution of past as well as potential future issues of the sector is assured by UDAY. The scheme empowers DISCOMs to breakeven in the next 2–3 years by different initiatives.

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### *Four Initiatives of Scheme*

1. Operational efficiencies of DISCOMs improved.
2. Cost of power reduced.
3. Reduction in interest cost of DISCOMs reduced.
4. Financial disciplines on DISCOMs enforced, through alignment with state finances.

UDAY is a shining example of the utilisation of the best principles of cooperative and competitive federalism is being demonstrated by UDAY and has been evolved through discussion at highest levels with multiple states.

### *Features*

- State shall take over 75% of DISCOM debt on 30 September 2015. 50% DISCOM debt shall be taken over in 2015–16 and 25% in 2016–17.
- Government of India will not include debt taken over by states in calculation of fiscal deficit in financial years 2015–16 and 2016–17.
- States will issue non-SLR in market/FIs/Bank holding DISCOM debt.
- DISCOM debt not taken by state shall be converted to loan with interest rate not more than bank's base rate of .01%.
- States shall take over future losses of DISCOM.
- States accepting UDAY and performing as per it will be given additional funding through Deen Dayal Upadhyaya Gram Jyoti Yojana, Integrated Power Development Scheme (IPDS) and Power Sector Development Fund (PSDF).
- This scheme was announced by Minister of State of Power, Coal and New Renewable Energy in November, 2015.
- Scheme is under optional basis for states to join, and till now 31 states/UT have joined this scheme. Mizoram being the 27th state to join.

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### 3.15.1 Revamped Distribution Sector Scheme

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The scheme has been recently launched by the Government of India **for better operations and financial sustainability of all DISCOMs. RDSS sanctions 20.46 crores pre-paid smart consumers meters, 54 lakh smart DT meters and 1.98 lakh smart feeder meters on February 2023.**

#### *Highlights of the Scheme*

- It is **a reforms-based and results-linked scheme**.
- It seeks **to improve the operational efficiencies and financial sustainability** of all DISCOMs/Power Departments excluding Private Sector DISCOMs.
- The scheme envisages the provision of **conditional financial assistance** to DISCOMs for strengthening supply infrastructure.
- The **assistance will be based on meeting pre-qualifying criteria** as well as upon the achievement of basic minimum benchmarks by the DISCOM.
- The scheme involves **a compulsory smart metering ecosystem** across the distribution sector – starting from electricity feeders to the consumer level, including in about 250 million households.
- The Scheme also focuses on funding for feeder segregation for unsegregated feeders.
- The Scheme has **a major focus on improving electricity supply for the farmers and for providing daytime electricity** to them through solarisation of agricultural feeders.

#### *Implementation*

- Nodal agencies for the scheme's implementation are **Rural Electrification Corporation (REC)** Limited and Power Finance Corporation (PFC).

#### *Objectives of the Scheme*

1. Reduction of average aggregate technical and commercial loss to pan-India levels of 12–15% by 2024–25.
2. Narrow the deficit between the cost of electricity and the price at which it is supplied to zero by 2024–25.
3. Developing institutional capabilities for modern DISCOMs.

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4. Improvement in the quality, reliability and affordability of power supply to consumers through a financially sustainable and operationally efficient distribution sector.
5. These workers would only be required to submit a self-declaration of their residential address to get the gas connection.

## 3.16 Unnat Jyoti by Affordable LEDs for All (UJALA)

The Union Government has also initiated the nationwide programme for Home and Street LED-based Lighting, so as to conserve energy through the reduction of energy consumption. Subsequently, Energy Efficiency Services Ltd., a Government organisation, has launched the scheme for distribution of LED bulbs under **Domestic Efficient Lighting Programme (DELP)** in the year March 2015. Thereafter, in the year March 2016, this scheme got a fresh face and name termed as 'UJALA'.

The scheme's main objective is the promotion of efficient lighting, awareness enhancing on the usage of the equipment which efficiently reduces bills and aid in the environment preservation. Under this scheme, there is distribution of LED bulbs in a phased manner from the year March 2015. At present, the DELP scheme is continuing in the country's nine states – Uttarakhand, Himachal Pradesh, Andhra Pradesh, Jharkhand and Karnataka. Till 28 June 2016, there is distribution of around 123 million LED bulbs, which is being equivalent to energy saving of approximately 43 million kilowatts hour per day and further shunning away of peak demand of 3205 megawatts.

## 3.17 Challenges Behind Rural Electrification

Rural electrification faces a lot of challenges. The rural electrification promotion on the basis of **grid extension is facing main obstacles during**

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**DDUGJY and various different programmes,** such as high grid-extension cost and lesser recovery subjected to highly subsidised tariff, low tariff collection level amounting to return in a negative form, lack of rationing of power-based supply and high costs of maintenance and operation. The sustainable rural electrification achievement is necessary by the way of focusing on creation of activities of income generation, which enhances the economy in rural areas. These affordable services, in turn, will be helpful for the rural households and further the masses are ready to reimburse for those services, which are sustainable and wealthy. The National Solar Mission should visualise on the connectivity of micro mini grids with conventional grids and also should not lose its focus on rural electrification.

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### GRAM UJALA

Under this programme **Convergence Energy Services Limited (CESL),** a wholly owned subsidiary of Energy Efficiency Services Limited (EESL), will distribute high quality LED bulbs, at an affordable cost of 10 rupees per bulb in rural areas.

**Implementation**

7-watt and 12-watt LED bulbs with 3 years warranty will be given to rural consumers against submission of working incandescent bulbs.

- Each household will get up to 5 LEDs.
- Participating rural households will also have metres installed in their houses to account for usage.

**Financing Mechanism**

- The programme will be financed entirely through carbon credits and will be the first such programme in India.
- The revenue earned from carbon credits will contribute ₹60 per LED bulb piece, with the balance ₹10 to be paid by the rural consumer.

The programme of rural electrification should be focussed on sustainable economic development of the rural communities. Henceforth, the rural electrification scheme must be a part of overall agenda of rural development and should be linked with various other social schemes.

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# 3.18 'NELP' and 'HELP'

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## 3.18.1 'NELP' (New Exploration Licensing Policy)

National Exploration License Policy in 1997 became effective in February 1999.

### *Salient Features of New Exploration Licensing Policy (NELP)*

The main features of NELP are:

- Private participation for intensive exploration of Indian basins.
- Provide scope and opportunity for opening up of acreages in ultra-deep water and frontier areas.
- To introduce new generation and state-of-the-art technology in exploration and exploitation.
- Equal distribution of playing field to all participating companies.
- Transparency in bid evaluation system.

### *Current System and Problems with NELP*

It is at present the policy regime for exploration and production of oil and gas, as in existence for 18 years. The current system and various problems associated with it are as follows:

1. **Separate Policies for Different Hydrocarbons:** At present, there are separate policies and licences for different hydrocarbons, such as conventional oil and gas, coal-bed methane, shale oil and gas and gas hydrates. In practice, the present framework leads to inefficiencies in exploiting natural resources.
2. **Profit Sharing Revenue Model and Delays:** The present regime is based on the principle of 'profit sharing' and Production Sharing Contracts (PSCs) are awarded under NELP. According to this, if a contractor discovers oil or gas, he is expected to share the profit as per the percentage given in his bid with the government. On the other hand, only royalties and cesses are paid to the government if no profit is made.

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3. **Exploration Confined Only to Some Areas:** Exploration currently is confined to blocks, which the Government has put on tender.
4. **Unfair Pricing:** The producer price of gas is currently fixed by the government. This has led to loss of revenue, considerable litigation, disputes and arbitrations.
5. **Unfair Royalty Charges Under Some Cases:** While fixing royalties, the current system does not provide efficient distinction between shallow water fields with lower costs and risks and deep/ultra-deep water fields with higher risks and costs.

### 3.18.2 HELP (Hydrocarbon Exploration and Licensing Policy)

Hydrocarbon Exploration and Licensing Policy is a policy adopted by Government of India in March 2016 indicating new contractual and fiscal modal for award of hydrocarbon acreage towards exploration and production.

HELP replaces present policy regime for E and P of oil and gas, known as **New Exploration licensing policy (NELP),** which has been in existence for 18 years. HELP aims to provide for a uniform licencing system and to provide single licencing covering all hydrocarbons like oil, gas, coal-bed methane, etc.

#### *Features*

HELP has four main aspects:

- **Uniform licence:** Uniform licence for exploration and production of all forms of hydrocarbon.
- **Open Acreages:** Gives option to a hydrocarbon company to select exploration blocks throughout the year without waiting for government bid.
- **Revenue Sharing Model:** There will be no concern for the government regarding the cost incurred and will gain a share of gross revenue from sale of oil, gas, etc.
- **Marketing and pricing freedom** for crude oil and natural gas produce has been granted, subject to ceiling price limit, for new gas production from deep water. Provides pricing and marketing freedom to gas

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production from existing discoveries. Ceiling is calculated once in six months.

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***Significance of HELP***

It paves way for the modernisation of the oil and gas exploration policy. It is expected to boost new exploration activities for oil, gas and other hydrocarbons and reduce import dependence. It can enhance the employment opportunities, especially in the petroleum sector. It has potential to bring in substantial investment into this sector. Disputes, corruption, administrative delays, administrative discretion, arbitrations and litigation will reduce and result in growth.

## 3.19 Net Metering

Net Metering (or net energy metering – NEM) allows consumers who generate some or all of their own electricity to use that electricity any time, instead of when it is generated.

This is particularly important with wind and solar, which is non-dispatchable. Monthly net metering allows consumers to use solar power generated during the day and night or wind from a windy day later in the month.

Net metering is an enabling policy designed to faster private investment in renewable energy.

**History of Net Metering:** Net Metering originated in the United States, where small wind turbines and solar panels were connected to the electrical grid and consumers wanted to be able to use the electricity generated at a different time or date from when it was generated. Minnesota is commonly cited as passing the first Net Metering law in 1983.

**Impact of Net Metering in India:** Many Indian states have started implementation of Net Metering. States like Tamil Nadu, Karnataka and

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Andhra Pradesh have started implementation of Net Metering and the respective state electricity boards have announced the policy in 2014.

**Controversy of Net Metering:** Net Metering is controversial, as it affects different interest on the grid.

***Advantages of Net Metering***

- Giving customers control over their electricity bill.
- Creating job and encouraging private investment.
- Protecting the electric grid.

## 3.20 Bureau of Energy Efficiency Star Label

The Government of India has set up Bureau of Energy Efficiency (BEE) on 1st March 2002 under the provision of the Energy Conservation Act, 2001. The mission of BEE is to assist in developing policies and strategies with a thrust on self-regulation and market principles with the primary objective of reducing energy intensity of the Indian economy within the overall framework of the Energy Conservation Act, 2001. This will be achieved with active participation of all stakeholders, resulting into accelerated and sustained adoption of energy efficiency in all sectors.

Bureau of Energy Efficiency Star Label is the Standards and Labelling Programme to provide the consumer satisfaction at different levels about the energy and cost saving potential of various types of equipment (marketed, household and other equipments).

It is currently invoked for various equipments/appliances like Computer, Direct Cool Refrigerator, Stationary Type Water Heater, Submersible Pump Set, Frost Free Refrigerator, Room Air Conditioner, Ceiling Fan, Opens well Submersible Pump Set, Diesel Engine, Driven Mono set pumps. For Agricultural Purposes, Diesel Generator Set Distribution Transformer, Domestic Gas Stove, Monoset Pump, Solid State Inverter, Office Automation Products, Led Lamps, Inverter AC, Colour Television and General Purpose Industrial Motor.

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## 3.21 'One Nation, One Grid'

It will be high-voltage electric power transmission network in mainland India, connecting power stations and major substations and ensuring that electricity generated anywhere in mainland India can be used to satisfy demand elsewhere.

***Evolution of National Grid***

1. Grid management on regional basis started in 1961.
2. Initially, State grids were inter-connected to form regional grid and India was demarcated into five regions namely Northern, Eastern, Western, North Eastern and Southern region. In August 2006, North and East grids were interconnected thereby four regional grids Northern, Eastern, Western and North Eastern grids are synchronously connected forming central grid operating at one frequency.
3. On 31st December 2013, Southern Region was connected to Central Grid in Synchronous mode with the commissioning of 765 kV Raichur–Solapur Transmission line thereby achieving **'ONE NATION'-'ONE GRID' – 'ONE FREQUENCY'.**

***Benefits of a National Grid***

1. Better availability resulting in lesser power cuts.
2. More stability in power.
3. Better synchronisation.

## 3.22 Energy Challenges in India

Approximately a quarter of the India's population does not have any access to electricity. It has been recognised that energy poverty is the reason behind the slow economic development. Thus, access to energy for its entire population has been among top priorities of Indian policy makers. Energy security has a prime spot in government policy making. Earlier,

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weightage was given to electricity shortage and disappointed energy needs. Still, the high dependency rate on imported energy sources, such as oil, natural gas and coal, brought the subject into light. In India, energy security can be broadly described as supplying lifeline energy to all of our people, regardless of their capacity in paying as well as meeting its effectual demands for secure and suitable energy and satisfying its different requirement at reasonable prices all times with a set level of confidence with fair consideration to reasonably expected shocks and disruptions.

Even though the government took up the policy to diversify fuel, the main importance was given to the maximum utilisation of what is available in India (Thorium, Hydrocarbons, etc.). Dr. A.P.J. Abdul Kalam, the then president, announced a plan to realise zero dependence and move to renewable energy sources.

### *Challenges in India*

1. India's primary challenge is to maintain a steady economic growth. A well-developed energy sector is prime for its high functioning and financially sound economy. This will want an enhanced move towards an energy sector on the basis of market economy. The two prime challenges in the country's energy sector are:
   - ❑ **Shortage of electricity** as a result of energy shortages across different fuel sectors which, in turn, hampers economic and social development.
   - ❑ The **low domestic production** pushes it to a position of deficit and hence requiring import of energy resources.
   - ❑ **Enhancing Energy Security:** The main reason behind the dependency on the imports of energy sources is the high demand of the developing economy, which results in crises due to low domestic production. This dependency questions our policies of a future energy security, which is the most important challenge to the Indian Ministry of Power.
2. **Ensuring Energy Distributions and Supply:** In the country's power sector, the coal and natural gas shortages are important issues for power generation. The key causes for unsatisfactory supply of coal:

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- ❑ Flat domestic coal production due to delayed cases of mining. Slow acquisition of land as well as stern clearance in environment.
- ❑ Controlled coal import because of inadequate import infrastructures and also due to the high difference in price.

3. **Challenges of Energy Project Infrastructure:** Inadequate equipment and services in the power sector along with facilities that contain these equipments are a major disappointment.
4. **. Rural Electrification:** Electrification of rural India is the key to achieve high economic growth as well as employment, which would ultimately result in low poverty. There are huge number of rural areas that are still not privileged to have electricity.
5. **Development of Renewable Energy Sources:** Following challenges are major obstacles in our way to energy security:
   - ❑ Solar lanterns can be used for lighting in place of kerosene as a fuel, being a cleaner and brighter substitute.
   - ❑ Biomass technology, though is abundantly present in the country, India has unfortunately failed to manage in achieving the installations scale with regard to its potential.
   - ❑ India's rank is fifth in the world in terms of hydropower potential. However, the potential developed so far is even less than 25%.
   - ❑ Geothermal technology can power geothermal plants for application of projects of small-scale intensity. However, its application is faintly used so far.
6. **Making Use of Nuclear Power:** Despite consistent efforts, commitments and advances, India's nuclear power capacity remains small. The current generation capacity of nuclear power is 4.8 gigawatts and with this, India ranks thirteenth in the world, accounting to world's 12% nuclear capacity. All this is linked to the country's long separation from the nuclear global energy regime and its focus on the nuclear programmes, which are thorium-based. India's energy challenges can greatly be addressed by adequate and efficient application of nuclear energy.

The present challenges in the development and the utilisation of renewable sources of energy and nuclear energy have to be well tapped, so as to enhance India's chances of being energy sufficient and greatly solve the serious energy shortage. This, in turn, would also

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limit the country's growing import pressure and produce feasible business opportunities.

7. **Striving towards Sustainable Development:** One of India's top priorities is boosting the energy sector. There is close connection between energy security and economic growth and poverty alleviation. The biggest challenge is seen to acquire adequate economic growth with no harm to the environment, and supporting conversation of environment.

   Balanced energy mix, which means adequate balance between application of all non-renewable energy sources and an increasing renewable energy share, will enable the country to achieve multiple objectives like sustainable development challenges, meeting food, water and fuel basic needs for an increasing population, etc.

## 3.23 Energy Policy of India

India uses both conventional resources such as coal, lignite, petroleum and natural gas, as well as non-conventional energy sources such as hydro, wind, solar biomass and the cogeneration biogas.

**It aims that the energy needs of the country are achieved at the lowest possible cost in real resources, with attaining sufficient security and stability in reliability and supply with environment, social and other objectives of policy.**

## 3.24 Integrated Energy Policy

In order to address the ever-increasing demand and millions of people's aspirations, along with the accessibility of energy security and other vital resources, there is a need to formulate the country's integrated energy policy, with a vision for ensuring an unconstrained supply of energy and further replenishing the whole energy sources. This will ultimately ensure the country's sustainable development.

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**Vision of Integrated Energy Policy:** This policy, as depicted by India's Planning Commission/NITI Aayog, aims primarily **to make India self-reliant in terms of energy security, which can achieve the demands of all sectors and energy services, including the lifeline energy requirement of weak household.**

Through this policy, **it is aimed to supply all parts of the country with clean, safe and convenient energy at the minimum cost. The policies should be efficient, technically, viable, economically and environmentally sustainable. The ultimate energy policy goal is to supply energy security to all.**

### *Sustainable Development*

Sustainable development is directly related to environment protection and precisely, it is described as, the development meeting the needs of the present with no compromise on the ability of future generations to meet their own needs.

### *Challenges*

**Energy security and environmental protection, at the same time, pose the biggest challenge in India's path towards sustainable development.**

### *Role of Integrated Energy Policy in India*

The policy appears to be an effective promise towards fulfiling India's aspirations of becoming energy secured nation and status of responsiveness towards the environment.

## 3.25 International Energy Agency (IEA)

The **International Energy Agency (IEA)** has invited **India, the world's third-largest energy consumer,** to become its full-time member.

### *Implications*

The proposal if accepted will require New Delhi **to raise strategic oil reserves to 90 days requirement.** India's current strategic oil reserves equal 9.5 days of its requirement.

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### Background

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In March 2017, India became an associate member of IEA.

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### About IEA

- Established in 1974 as per the framework of the OECD, IEA is an autonomous intergovernmental organisation.
- Its mission is guided by four main areas of focus: energy security, economic development, environmental awareness and engagement worldwide.
- Headquarters (Secretariat): Paris, France.

### Roles and Functions

- Established in the wake of the 1973–1974 oil crisis, to help its members respond to major oil supply disruptions, a role it continues to fulfil today.
- IEA's mandate has expanded over time to include tracking and analysing global key energy trends, promoting sound energy policy, and fostering multinational energy technology cooperation.

### Composition and Eligibility

- It has 31 members at present. IEA family also includes eight association countries. A candidate country must be a member country of the OECD. But all OECD members are not IEA members.
- To become member a candidate country must demonstrate that it has:
    1. Crude oil and/or product reserves equivalent to 90 days of the previous year's net imports, to which the government has immediate access (even if it does not own them directly) and could be used to address disruptions to global oil supply.
    2. A demand restraint programme to reduce national oil consumption by up to 10%.
    3. Legislation and organisation to operate **the Co-ordinated Emergency Response Measures (CERM)** on a national basis.
    4. Legislation and measures to ensure that all oil companies under its jurisdiction report information upon request.
    5. Measures in place to ensure the capability of contributing its share of an IEA collective action.

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### Reports

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- Global Energy & CO2 Status Report.
- World Energy Outlook.
- World Energy Statistics.
- World Energy Balances.
- Energy Technology Perspectives.

## 3.26 PSUs and Other Organisations

| S.No. | PSU and other Organisation | Main Features |
|---|---|---|
| 1. | **National Thermal Power Corporation Limited (NTPC Ltd.) (1975)** | A Maharatna Company of the Government of India, it is the single largest power generator in India, with a vision of being world's largest and best power producer, capable of powering India's growth and a mission to develop and provide reliable power and related products and services at competitive prices. Besides this, integration of multiple energy sources with innovative and eco-friendly technologies, which can efficiently contribute to society is another crucial part of its vision. The Government of India holds 75% of its authorised share capital. NTPC has attained the position of global Number 1 Independent Power Producer in the world, among the top 250 |

## QUESTIONS AND ANSWERS

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## Objective Questions

**1. Consider the following activities:** **[UPSC (Pre) – 2025]**

**i.** Production of crude oil
**ii.** Refining, storage and distribution of Petroleum
**iii.** Marketing and sale of petroleum
**iv.** Production of natural gas

How many of the above activities are regulated by the Petroleum and Natural Gas Regulatory Board in our country?

(a) Only one
(b) Only two
(c) Only three
(d) All the four

**2. Consider the following substances:** **[UPSC (Pre) – 2025]**

**i.** Ethanol
**ii.** Nitroglycerine
**iii.** Urea

Coal gasification technology can be used in the production of how many of them?

(a) Only one
(b) Only two
(c) All the three
(d) None

**3. Consider the following statements about 'PM Surya Ghar Muft Bijli Yojana':** **[UPSC (Pre) – 2025]**

1. It targets installation of one crore solar rooftop panels in the residential sector.

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2. The Ministry of New and Renewable Energy aims to impart training on installation, operation, maintenance and repairs of solar rooftop systems at grassroot levels.
3. It aims to create more than three lakhs skilled manpower through fresh skilling and up-skilling, under scheme component of capacity building.

Which of the statement given above are correct?

(a) i and ii only
(b) i and iii only
(c) ii and iii only
(d) i, ii and iii

**4. Consider the following types of vehicles:** **[UPSC (Pre) – 2025]**

1. Full battery electric vehicles
2. Hydrogen fuel cell vehicles
3. Fuel cell electric hybrid vehicles

How many of the above are consider as alternative powertrain vehicles?

(a) Only one
(b) Only two
(c) All the three
(d) None

**5. In the context of electric vehicle batteries, consider the following elements:** **[UPSC (Pre) – 2025]**

1. Cobalt
2. Graphite
3. Lithium
4. Nickel

How many of the above usually make up battery cathodes?

(a) Only one
(b) Only two
(c) Only three
(d) All the four

**6. Which one of the following is the exhaust pipe emission from fuel cell Electric vehicles powered by hydrogen?** **[UPSC (Pre) – 2024]**

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(a) Hydrogen peroxide
(b) Hydronium
(c) Oxygen
(d) Water vapour

**7. Recently, the term "pumped-storage hydropower" is actually and appropriately discussed in the context of which one of the following?** **[UPSC (Pre) – 2024]**

(a) Irrigation of terraced crop fields
(b) Lift irrigation of cereal crops
(c) Long duration energy storage
(d) Rain water harvesting system

**8. Consider the following:** **[UPSC (Pre) – 2024]**

1. Battery storage
2. Biomass generators
3. Fuel cell
4. Rooftop solar photovoltaic units

How many of the above are considered "Distributed Energy Resources"?

(a) Only one
(b) Only two
(c) Only three
(d) All four

**9. Consider the Following materials:** **[UPSC (Pre) – 2024]**

1. Agricultural Residue
2. Corn Grain
3. Waste water treatment sludge
4. Wood mill waste

Which of the above can be used as feedstock for producing Sustainable aviation Fuel ?

(a) 1 and 2 only
(b) 3 and 4 only
(c) 1, 2, 3 and 4
(d) 1, 3 and 4 only

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**10. With reference to green hydrogen, consider the following statements:** **[UPSC (Pre) – 2023]**

1. It can be used directly as a fuel for internal combustion.
2. It can be blended with natural gas and used as fuel for heat or power generation.
3. It can be used in the hydrogen fuel cell to run vehicles.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

**11. About three-fourths of world's cobalt, a metal required for the manufacture of batteries for electric motor vehicles, is produced by** **[UPSC (Pre) – 2023]**

(a) Argentina
(b) Botswana
(c) the Democratic Republic of the Congo
(d) Kazakhstan

**12. Consider the following heavy industries:** **[UPSC (Pre) – 2023]**

1. Fertiliser plants
2. Oil refineries
3. Steel plants

**Green hydrogen is expected to play a significant role in decarbonising how many of the above industries?**

(a) Only one
(b) Only two
(c) All three
(d) None

**13. With reference to coal-based thermal power plants in India, consider the following statements:** **[UPSC (Pre) – 2023]**

1. None of them uses seawater.
2. None of them is set up in water-stressed district.
3. None of them is privately owned.

**How many of the above statements are correct?**

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(a) Only one
(b) Only two
(c) All three
(d) None

**14. In India, what is the role of the Coal Controller's Organisation (CCO)?** **[UPSC (Pre) – 2022]**

1. CCO is the major source of Coal Statistics in Government of India.
2. It monitors progress of development of Captive Coal/Lignite blocks.
3. It hears any objection to the Government's notification relating to acquisition of coal-bearing areas.
4. It ensures that coal mining companies deliver the coal to end users in the prescribed time.

Select the correct answer using the code given below:
(a) 1, 2 and 3
(b) 3 and 4
(c) 1 and 2
(d) 1, 2 and 4

**15. Consider the following statements:** **[UPSC (Pre) – 2022]**

1. Gujarat has the largest solar park in India.
2. Kerala has a fully solar powered International Airport.
3. Goa has the largest floating solar photovoltaic project in India.

Which of the statements given above is/are correct?
(a) 1 and 2
(b) 2 only
(c) 1 and 3
(d) 3 only

**16. According to India's National Policy on Bio-fuels, which of the following can be used as raw materials for the production of biofuels?** **[UPSC (Pre) – 2020]**

1. Cassava
2. Damaged wheat grains
3. Groundnut seeds

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4. Horse gram
5. Rotten potatoes
6. Sugar beet

Select the correct answer using the code given below:

(a) 1, 2, 5 and 6
(b) 1, 3, 4 and 6
(c) 2, 3, 4 and 5
(d) 1, 2, 3, 4, 5 and 6

**17. With reference to solar water pumps, consider the following statements:** **[UPSC (Pre) – 2020]**

1. Solar power can be used for running surface pumps and not for submersible pumps.
2. Solar power can be used for running centrifugal pumps and not the ones with piston.

Which of the statements given above is/are correct?

(a) 1 only
(b) Both 1 and 2
(b) 2 only
(d) Neither 1 nor 2

**18. Consider the following statements:**

1. Coal ash contains arsenic, lead and mercury. **[UPSC (Pre) – 2020]**
2. Coal-fired power plants release sulphur dioxide and oxides of nitrogen into the environment.
3. High ash content is observed in Indian coal.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 and 3
(c) 3 only
(d) 1, 2 and 3

**19. What is the use of biochar in farming?**

1. Biochar can be used as a part of the growing medium in vertical farming. **[UPSC (Pre) – 2020]**

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2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for longer time.

Which of the statements given above is/are correct?

(a) 1 and 2
(b) 2 only
(c) 1 and 3
(d) 1, 2 and 3

**20. Consider the following Statements:** **[UPSC (Pre) – 2019]**

1. Petroleum and Natural Gas Regulatory Board (PNGRB) is the first regulatory body set up by the Government of India.
2. One of the tasks of PNGRB is to ensure competitive markets for gas.
3. Appeals against the decisions of PNGRB go before the Appellate Tribunals for Electricity.

Which of the statements given above correct?

(a) 1 and 2
(b) 2 and 3
(c) 1 and 3
(d) 1, 2 and 3

**21. In the context of which one of the following are the terms 'pyrolysis and plasma gasification' mentioned?** **[UPSC (Pre) – 2019]**

(a) Extraction of rare earth elements
(b) Natural gas extraction technologies
(c) Hydrogen fuel-based automobiles
(d) Waste-to-energy technologies

**22. In the context of proposals to the use of hydrogen-enriched CNG (H-CNG) as fuel for buses in public transport, consider the following statements:** **[UPSC (Pre) – 2019]**

1. The main advantage of the use of H-CNG is the elimination of carbon monoxide emissions.

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2. H-CNG as fuel reduces carbon dioxide and hydrocarbon emissions.
3. Hydrogen up to one-fifth by volume can be blended with CNG as fuel for buses.
4. H-CNG makes the fuel less expensive than CNG.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 and 3
(c) 4 only
(d) 1, 2, 3 and 4

**23. Consider the following statements:** **[UPSC (Pre) – 2019]**

1. Coal sector was nationalised by the Government of India under Indira Gandhi.
2. Now, coal blocks are allocated on lottery basis.
3. Till recently, India imported coal to meet the shortages of domestic supply, but now India is self-sufficient in coal production.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 and 3
(c) 3 only
(d) 1, 2 and 3

**24. Which of the following statements are correct about the deposits of 'methane hydrate'?** **[UPSC (Pre) – 2019]**

1. Global warming might trigger the release of methane gas from these deposits.
2. Large deposits of 'methane hydrate' are found in Arctic Tundra and under the seafloor.
3. Methane in atmosphere oxidises to carbon dioxide after a decade or two.

Select the correct answer using the code given below:

(a) 1 and 2
(b) 2 and 3
(c) 1 and 3

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(d) 1, 2 and 3

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**25. With reference to solar power production in India, consider the following statements:** **[UPSC (Pre) – 2018]**

1. India is third largest in the work in the manufacture of silicon wafers used in photovoltaic units.
2. The solar power tariffs are determined by the Solar Energy Corporation of India.

Which of the statements given above is/are correct?

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**26. It is possible to produce algae-based biofuels, but what is/are the likely limitation(s) of developing countries in promoting this industry?** **[UPSC (Pre) – 2017]**

(i) Production of algae-based biofuels is possible in seas only and not on continents.
(ii) Setting up and engineering the algae-based biofuel production requires high level of expertise/technology until the construction is completed.
(iii) Economically viable production necessitates the setting up of large-scale facilities, which may raise ecological and social concerns.

Select the correct answer using the code given below:

(a) (i) and (ii)
(b) (ii) and (iii)
(c) Only (iii)
(d) All of these

**27. In which of the following regions of India are shale gas resources found?** **[UPSC (Pre) – 2016]**

(i) Cambay Basin
(ii) Cauvery Basin
(iii) Krishna–Godavari Basin

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Select the correct answer using the code given below:
(a) (i) and (ii)
(b) Only (iii)
(c) (ii) and (iii)
(d) All of these

**28. On which of the following can you find the Bureau of Energy Efficiency Star Label?**

(i) Ceiling fans **[UPSC (Pre) – 2016]**
(ii) Electric geysers
(iii) Tubular fluorescent lamps

Select the correct answer using the code given below:
(a) (i) and (ii)
(b) Only (iii)
(c) (ii) and (iii)
(d) All of these

**29. Consider the following statements:**

(i) The International Solar Alliance was launched at the United Nations Climate Change Conference in 2015. **[UPSC (Pre) – 2016]**
(ii) The Alliance includes all the member countries of the United Nations.

Which of the statements given above is/are correct?
(a) Only (i)
(b) Only (ii)
(c) All of these
(d) None of these

**30. Which one of the following is a purpose of 'UDAY', a scheme of the Government?**

(a) Providing technical and financial assistance to start-up entrepreneurs in the field of renewable sources of energy. **[UPSC (Pre) – 2016]**
(b) Providing electricity to every household in the country by 2018.
(c) Replacing the coal-based power plants with natural gas, nuclear, solar, wind and tidal power plants over a period of time.

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(d) Providing for financial turnaround and revival of power distribution companies.

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**31. 'Net Metering' is sometimes seen in the news in the context ofpromoting the** **[UPSC (Pre) – 2016]**

(a) Production and use of solar energy by the households/consumer
(b) Use of piped natural gas in the kitchens of households
(c) Installation of CNG kits in motor cars
(d) Installation of water metres in urban households

**32. With reference to 'fly ash' produced by the power plants using coal as a fuel, which of the following statements is/are correct?** **[UPSC (Pre) – 2015]**

(i) Fly ash can be used in the production of bricks for building construction.
(ii) Fly ash can be used as a replacement for some of the Portland cement contents of concrete.
(iii) Fly ash is made up of silicon dioxide and calcium oxide only, and does not contain any toxic elements.

Select the correct answer using the code given below:

(a) (i) and (ii)
(b) Only (ii)
(c) (i) and (iii)
(d) Only (iii)

**33. With reference to 'fuel cells' in which hydrogen-rich fuel and oxygen are used to generate electricity, consider the following statements:** **[UPSC (Pre) – 2015]**

(i) In fuel cell pure hydrogen is used as a fuel, fuel cell emits heat and water as by-products.
(ii) Fuel cells can be used for powering buildings and not for small devices like laptop computers.
(iii) Fuel cells produce electricity in the form of Alternating Current (AC).

Which of the statements given above is/are correct?

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)

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(d) All of these

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**34. With reference to the Indian Renewable Energy Development Agency Limited (IREDA), which of the following statements is/are correct?** **[UPSC (Pre) – 2015]**

(i) It is a Public Limited Government Company.
(ii) It is a Non-Banking Financial Company.

Select the correct answer using the code given below:

(a) Only (i)
(b) Only (ii)
(c) All of these
(d) None of these

**35. With reference to two non-conventional energy sources called 'coal-bed methane' and 'shale gas', consider the following statements:** **[UPSC (Pre) – 2014]**

(i) Coal-bed methane is the pure methane gas extracted from coal seams, while shale gas is a mixture of propane and butane only that can be extracted from fine-grained sedimentary rocks.
(ii) In India, abundant coal-bed methane sources exist, but so far no shale gas sources have been found.

Which of the statements given above is/are correct?

(a) Only (i)
(b) Only (ii)
(c) All of these
(d) None of these

**36. In India, cluster bean (Guar) is traditionally used as a vegetable or animal feed, but recently, the cultivation of this has assumed significance. Which one of the following statements is correct in this context?** **[UPSC (Pre) – 2014]**

(a) The oil extracted from seeds is used in the manufacture of biodegradable plastics.
(b) The gum made from its seeds is used in the extraction of shale gas.
(c) The leaf extract of this plant has the properties of anti-histamines.

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(d) It is a source of high-quality biodiesel.

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**37. Consider the following statements:** [UPSC (Pre) – 2014]

(i) Maize can be used for the production of starch.
(ii) Oil extracted from maize can be a feedstock for biodiesel.
(iii) Alcoholic beverages can be produced by using maize.

Which of the statements given above is/are correct?

(a) Only (i)
(b) (i) and (ii)
(c) (ii) and (iii)
(d) All of these

**38. With reference to technologies for solar power production, consider the following statements:** [UPSC (Pre) – 2014]

(i) 'Photovoltaics' is a technology that generates electricity by direct conversion of light into electricity, while 'Solar Thermal' is a technology that utilises the Sun's rays to generate heat, which is further used in electricity generation process.
(ii) Photovoltaics generates Alternating Current (AC), while Solar Thermal generates Direct Current (DC).
(iii) India has manufacturing base for solar thermal technology, but not for Photovoltaics.

Which of the statements given above is/are correct?

(a) Only (i)
(b) (ii) and (iii)
(c) All of these
(d) None of these

**39. With reference to the usefulness of the by-products of sugar industry, which of the following statements is/are correct?** [UPSC (Pre) – 2013]

(i) Bagasse can be used as biomass fuel for the generation of energy.
(ii) Molasses can be used as one of the feedstocks for the production of synthetic chemical fertilisers.
(iii) Molasses can be used for the production of ethanol.

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Select the correct answer using the codes given below:

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(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**40. Which of the following is/are the characteristic/characteristics of Indian coal?** **[UPSC (Pre) – 2013]**

(i) High ash content
(ii) Low sulphur content
(iii) Low ash fusion temperature

Select the correct answer using the codes given below:
(a) (i) and (ii)
(b) Only (ii)
(c) Only (ii)
(d) All of these

**41. Biomass gasification is considered to be one of the sustainable solutions to the power crisis in India. In this context, which of the following statements is/are correct?** **[UPSC (Pre) – 2012]**

(i) Coconut shells, groundnut shells and rice husk can be used in biomass gasification.
(ii) The combustible gases generated from biomass gasification consist of hydrogen and carbon dioxide only.
(iii) The combustible gases generated from biomass gasification can be used for direct heat generation but not in internal combustion engines.

Select the correct answer using the codes given below:
(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**42. Microbial fuel cells are considered a source sustainable energy. Why?** **[UPSC (Pre) – 2011]**

(i) They use living organisms as catalysts, generating electricity from certain substrates.
(ii) They use a variety of inorganic materials as substrates.

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(iii) They can be installed in waste water treatment plants to cleanse water and produce electricity.

Which of the statement(s) given above is/are correct?

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**43. In the Union Budget 2011–12, a full exemption from the basic customs duty was extended to the bio-based asphalt (bioasphalt). What is the importance of this material?** **[UPSC (Pre) – 2011]**

(i) Unlike traditional asphalt, bio-asphalt is not based on fossil fuels.
(ii) Bioasphalt can be made from non-renewable resources.
(iii) Bioasphalt can be made from organic waste materials.
(iv) It is eco-friendly to use bioasphalt for surfacing of the roads.

Which of the statements given above are correct?

(a) (i), (ii) and (iii)
(b) (i), (iii) and (iv)
(c) (ii) and (iv)
(d) All of these

**44. Hydrogen fuel cell vehicles produce one of the following as 'exhaust'** **[UPSC (Pre) – 2010]**

(a) $NH_3$
(b) $CH_4$
(c) $H_2O$
(d) $H_2O_2$

## Answer Key

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (b) | 2. (b) | 3. (d) | 4. (c) | 5. (c) | 6. (d) | 7. (c) | 8. (d) |
| 9. (d) | 10. (c) | 11. (c) | 12. (c) | 12. (d) | 14. (a) | 15. (b) | 16. (a) |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 17. (d) | 18. (d) | 19. (d) | 20. (b) | 21. (d) | 22. (b) | 23. (a) | 24. (d) |
| 25. (d) | 26. (b) | 27. (d) | 28. (d) | 29. (a) | 30. (d) | 31. (a) | 32. (a) |
| 33. (a) | 34. (c) | 35. (d) | 36. (b) | 37. (d) | 38. (a) | 39. (c) | 40. (a) |
| 41. (a) | 42. (c) | 43. (b) | 44. (c) | | | | |

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# CHAPTER 4
# NUCLEAR ENERGY, NUCLEAR FUEL, RADIOACTIVE ELEMENT AND RADIOACTIVITY

## IN THIS CHAPTER...


**4.1 Department of Atomic Energy**

The nuclear technology, including nuclear power and research, is taken care by this department

**4.2 Research and Development Sector in Energy Generation**

- 4.2.1 Bhabha Atomic Research Centre
- 4.2.2 Variable Energy Cyclotron Centre
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- 4.2.4 Indira Gandhi Centre for Atomic Research
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4.2.6 Atomic Minerals Directorate for Exploration and Research

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**4.3 Nuclear Energy**

4.3.1 Nuclear Power in India
4.3.2 Types of Nuclear Reactions
4.3.3 India’s Three-Stage Civil Nuclear Power Programme

Thorium is recognised by the Indian government, as a source capable of providing power to the country in the long term.

4.3.4 Nuclear Fuel
4.3.5 Nuclear Reactor

A nuclear reactor, or an atomic pile, is a device used to initiate and control a sustained nuclear chain reaction, used at nuclear power plants to generate electricity and in propulsion of ships.

4.3.6 Components of a Nuclear Reactor
4.3.7 List of Reactors

Kudankulam Nuclear Plant is India’s first Largest Indigenously Developed Nuclear Power Plant Unit in Gujarat

4.3.8 International Thermonuclear Experimental Reactor (ITER)/Artificial Sun Experiment
4.3.9 Advantages and Disadvantages of Nuclear Energy
4.3.10 Radiation and Radioactivity

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Radiation is the term referring to the propagation of particles and waves through space and includes both electromagnetic radiation and atomic and subatomic particle radiation

Radioactivity, as a term, refers to the property of spontaneous emission of electromagnetic radiation of particles, shown by certain materials

### 4.3.11 Nuclear Winter

Nuclear winter is the climatic effects of nuclear war.

### 4.3.12 Nuclear Medicine

The branch of medicine that deals with the use of radioactive substances in treatment, research and diagnosis is called as nuclear medicine.

## 4.1 Department of Atomic Energy

The **Department of Atomic Energy (DAE) was set up on 3 August 1954** with the vision of empowering India through making it energy independent, contributing to provision of sufficient, safe and nutritious food and better healthcare to our people through development and deployment of nuclear and radiation technologies and their applications. **The DAE works directly under the Prime Minister of India.** The nuclear technology, including nuclear power and research, is taken care of by this department.

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It is engaged to design, construct and operate nuclear power/research reactors and support nuclear fuel cycle technologies. On the whole, it will cover exploration, mining and processing of nuclear minerals, production of heavy water, nuclear fuel fabrication, fuel reprocessing and nuclear waste management. The department is also developing new elite crop varieties including oil seeds and pulses along with techniques for biological control of pests and pathogens, which ultimately protects the crops, radiation-based post-harvest technologies, radiation-based techniques for diagnosis and treatment of diseases, particularly cancer, technologies for safe drinking water, better environment and robust industry etc.

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**ATOMIC ENERGY COMMISSION**

The Indian Atomic Energy Commission was **first set up in August 1948 in the Department of Scientific Research. Important functions of the Atomic Energy Commission are:**

- To organise research for atomic scientists in the country.
- To train atomic scientists in the country.
- To promote nuclear research in the Commission's own laboratories as well as in India.
- To undertake prospecting of atomic minerals in India and to extract such minerals for use on an industrial scale.

**The following are significant contributions made by the DAE to national initiatives:**

- **Agriculture:** To enhance the production of oilseeds and pulses.
- **Education and Health:** Various institutions work for this promotion – Homi Bhabha National Institute (HBNI), Countrywide-Services in Cancer through Telemedicine, National Initiative on Undergraduate Science (NIUS).
- **Food and Nutrition Security:** Radiation-based processing of food and agro products enhances and saves these products.
- **Water Resources:** It also helps in the desalination of water in water-scarce areas along the sea coast.
- **Energy Security:** Electricity generation and its supply that will make India self-dependent.

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## 4.2 Research and Development Sector in Energy Generation

Various centres and research institutions are established that work for energy generation in India.

1. Bhabha Atomic Research Centre
2. Variable Energy Cyclotron Centre
3. Global Centre for Nuclear Energy Partnership
4. Indira Gandhi Centre for Atomic Research
5. Raja Ramanna Centre for Advanced Research
6. Atomic Minerals Directorate for Exploration and Research

### 4.2.1 Bhabha Atomic Research Centre

It is the premier multi-disciplinary research centre of India which has excellent infrastructure that **facilitates advanced research and development covering the entire spectrum of nuclear science** as well as areas related to engineering.

### 4.2.2 Variable Energy Cyclotron Centre

It is a premier centre for frontier research and development in the fields of accelerator science and technology, computer science and technology, nuclear science (theoretical and experimental), material science and other such relevant areas.

### 4.2.3 Global Centre for Nuclear Energy Partnership

The Global Centre for Nuclear Energy Partnership (GCNEP) is under construction at Kheri Jasaur, near Bahadurgarh, Haryana, India. The centre will consist of five schools for advanced research, study and training of nuclear systems and facilities.

### 4.2.4 Indira Gandhi Centre for Atomic Research

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In 1971, the Indira Gandhi Centre for Atomic Research (IGCAR) was established under the DAE. Since its inception, it has been engaged as multidisciplinary programmes of scientific research and advanced engineering, aiming towards the development of a fast breeder reactor (FBR) technology. Uniquely mixed plutonium–uranium carbide fuel (first of its kind in the world) and KAMINI Reactor (the only operating reactor in the world using U-233 fuel) are two important specialities of this reactor.

### 4.2.5 Raja Ramanna Centre for Advanced Research

The DAE established the Raja Ramanna Centre for Advanced Technology (RRCAT) in Indore, Madhya Pradesh, India. This enables India to expand the activities carried out at the Bhabha Atomic Research Centre (BARC), Mumbai, in two major areas of science and technology, namely Lasers and Accelerators. The first batch of scientists from the BARC, Mumbai, moved to the RRCAT in June 1986, and this initiated the scientific activities.

### 4.2.6 Atomic Minerals Directorate for Exploration and Research

Identification and evaluation of uranium resources required to successfully implement an atomic energy programme in the country is the mandatory factor for the Atomic Minerals Directorate for Exploration and Research.

## 4.3 Nuclear Energy

Nuclear energy utilises nuclear fuel to generate heat, which is used in steam turbines to produce electricity in a nuclear power plant. There are two modes of use of nuclear energy, i.e. nuclear fission and nuclear fusion (**Table 4.1**). Nuclear reactions such as nuclear fission (when a single heavy nucleus splits into two smaller nuclei) and nuclear fusion (when a single heavy nucleus is formed from the combination of two lighter atomic nuclei)

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release very high amounts of energy. In a nuclear fission reaction, radioactive decay happens in spontaneous breakdown of a radioactive atomic nucleus and results in the release of energy. Atom bomb and nuclear energy development programme of India are based on nuclear fission. Solar energy, hydrogen bomb and artificial sun research (ITER – International Thermonuclear Experimental Reactor) are examples of nuclear fusion. In the sun, the process of hydrogen fusion converts about 4 million tonnes of solar matter per second into electromagnetic energy, which is radiated into space. In 1896, nuclear energy or atomic energy, which is used as a power source, was first discovered by the French scientist Henri Becquerel.

**Table 4.1** Difference between Nuclear Fission and Nuclear Fusion

| Characters | Nuclear Fission | Nuclear Fusion |
|---|---|---|
| **Definition** | The splitting of a large atom into two or more smaller ones is known as nuclear fission. | The fusing of two or more lighter atoms to form a larger or heavier one is termed as nuclear fusion. |
| **Natural occurrence of the process** | It does not occur normally in nature and needs to be induced. | It occurs naturally and is the source of energy of stars, such as the sun. |
| **By-products of the reaction** | Many highly radioactive particles are produced. | Few radioactive particles are produced; however, through fission 'trigger', radioactive particles will result from that. |
| **Conditions** | Critical mass of the substance and high-speed neutrons are basic requirements. | High-density and high-temperature environment are basic requirements. |
| **Energy** | Splitting requires little energy in | Extremely high energy is |

Note: Very large and heavy atoms occurring in nature are unstable and hence show radioactivity. Examples include: atoms of the element uranium (atomic number or Z = 92), thorium (atomic number = 90), radon (atomic number = 86) and radium (atomic number = 88), among others. Many smaller atoms are artificially turned radioactive for specific uses.

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They include carbon (atomic number = 6), which often has a stable form but can be turned into non-radioactive form as well as unstable radioactive form.

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### 4.3.1 Nuclear Power in India

APSARA was the first nuclear research reactor in India. Its design was conceptualised by Dr Homi Bhabha (the Father of the Indian Nuclear Programme) in 1955. The reactor was built with assistance from the United Kingdom who provided the initial fuel. On 4 August 1956, the reactor reached criticality and was inaugurated on 20 January 1957. Further, another research nuclear reactor and its **first nuclear power plant were built with assistance from Canada.** In 1956, a 40-megawatt research reactor agreement was signed, and the Canadian-Indian Reactor, US (CIRUS) achieved first criticality in 1960. India got this reactor supplied on the assurance of applying for the military purposes, but without effective safeguards against such use. For India's first nuclear power plant at Rajasthan, RAPP-1 (Rajasthan Atomic Power Project), an agreement was signed in 1963, followed by RAPP-2 in 1966. To ensure no application of these reactors for a military programme, these reactors contain rigid safeguards unlike before. **The 200-megawatt RAPP-1 reactor was based on the CANDU (Canada Deuterium Uranium) reactor and its operation started in 1972.** However, owing to technical problems, the reactor was down rated from 200 megawatts to 100 megawatts. AECL (Atomic Energy of Canada Limited) provided free technical and design information to India. **After India's first nuclear explosion in 1974, the assistance from the United States and Canada was terminated.** The fourth largest source of electricity in India after thermal, hydroelectricity and renewable sources of electricity is nuclear power.

Note:

1. **Use of Nuclear Energy for Power Generation (PIB Report)*:** Presently, nuclear power is saving 41 million tonnes of $CO_2$ emissions annually, compared to the emissions that would be generated by equivalent electricity generation from coal-based thermal power plants. Currently, nuclear power provides 3% of electricity generation. Sufficient production and share of nuclear power is essential for ensuring the country's

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energy security. Current policy targets a three-fold rise in nuclear installed capacity by 2032.

It was pointed out that the present installed nuclear power capacity is set to increase from 7480 MW to 22480 MW by 2031 on progressive completion of projects under construction and accorded sanction. In the next 3 years, capacity addition of 5300 MW is planned on completion of KAPP 3&4 (2 × 700 MW), RAPP 7&8 (2 × 700 MW), KKNPP 3&4 (2 × 1000 MW) and PFBR (500 MW). (**KAPP** – Kakrapar Atomic Power Plant, **RAPP** – Rajasthan Atomic Power Plant, **KKNPP** – Kudankulam Nuclear Power Plant and **PFBR** – Prototype Fast Breeder Reactor)

* https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1988863 (last accessed in July 2024).

2. **Nuclear Vision of India:** India has a flourishing and largely indigenous nuclear power programme and expects to have 63,000 megawatts electric (MWe) nuclear capacities on line by 2032. It aims to supply 25% of electricity from nuclear power by 2050.

## 4.3.2 Types of Nuclear Reactions

Basically there are two types of reactions, which take place to release nuclear energy (see **Table 4.1**):

1. Nuclear fission
2. Nuclear fusion

### 4.3.2.1 Nuclear Fission

Nuclear fission is the process of splitting of a large nucleus into two smaller nuclei with the release of energy. In other words, fission is the process in which a nucleus is divided into two or more fragments and neutrons and energy are released. When any uranium nucleus ($_{92}U^{235}$) is bombarded with a neutron, the nucleus splits up into almost equal fragments with the release of some free neutrons and tremendous amount of energy per 92U235 nuclei. Such a nuclear reaction is called nuclear fission. For example: **Atom bomb, Nuclear Energy Development Programme of India**. The schematic equation for fission process may be written as:

$$_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{56}Ba^{144} + {}_{36}Kr^{89} + {}_{30}n^{1} + \text{Energy (200 MeV)}$$

### 4.3.2.2 Nuclear Fusion

When two or more light nuclei, moving with very high speeds, are fused together to form a single nucleus, then this process is called nuclear fusion.

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The mass of the product nucleus is less than the sum of masses of the nuclei, which are fused. The lost mass is converted into energy, which is released in the form of fusion energy. The number of deuterons (reactants of the fission reaction) in 1 gram of heavy hydrogen is much larger than the number of $U^{235}$ nuclei in 1 gram of uranium. Thus, the energy output per unit mass of the material consumed is much more in the case of fusion of the light nuclei than in the case of the fission of heavy nuclei. For example: **Solar energy (sunlight), Hydrogen bomb, International Thermonuclear Experimental Reactor (Artificial Sun Experiment)**. Fusion reactions are observed in the equation as below: ($Q$ = heat)

$$_{11}H^1 + {}_1H^1 \rightarrow {}_1H^2 + e + Q \qquad Q = 0.42 \text{ MeV}$$

$$_1H^2 + {}_1H^2 \rightarrow {}_2He^3 + {}_0n^1 \qquad Q = 3.3 \text{ MeV}$$

**NUCLEAR ENERGY DATA WORLDWIDE**

- On June 27, 1954, the world's first nuclear power station to generate electricity for a power grid, the Obninsk Nuclear Power Plant, commenced operations in Obninsk, in the Soviet Union.
- Nuclear energy now provides about 10% of the world's electricity from about 440 power reactors.
- Over 50 countries utilise nuclear energy in about 220 research reactors. In addition to research, these reactors are used for the production of medical and industrial isotopes, as well as for training.

### 4.3.3 India's Three-Stage Civil Nuclear Power Programme

In 1954, during the forum of the 'Development of Atomic Energy for Peaceful Purposes', Bhabha conferred the three-stage plan for the country's development and the same was also attended by the country's first Prime Minister, Jawaharlal Nehru. The three-stage plan was formally adopted by the Indian government after 4 years, that is, in 1958. Consequently, thorium was recognised by the Indian government, as a source capable of providing power to the country in the long term.

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#### 4.3.3.1 First Stage

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**Fuel –** Natural uranium

**Reactor –** PHWR (pressurised heavy-water reactor)

**Moderator –** $D_2O$ (heavy water)

**Coolant –** $D_2O$ (heavy water)

Pressurised heavy-water reactors (PHWRs), which are fuelled with natural uranium, generate electricity during the first stage of the programme, which **generates plutonium-239 as a by-product**.

The fissile isotope content is only 0.7% in natural uranium-235. However, most of the remaining 99.3% is uranium-238, which is although not fissile, in a nuclear reactor, can be converted to the fissile isotope plutonium-239. **Three per cent enrichment of uranium-235 is required for electricity generation but as an atom bomb, its enrichment should be more than 20% (it can go up to 90%)**.

#### 4.3.3.2 Second Stage

**Fuel –** Plutonium

**Reactor –** FBR (fast breeder reactor)

**Moderator –** Not in use/Rarely used

**Coolant –** Liquid sodium (liquid Na)

A mixed oxide (MOX) fuel is used in the FBRs prepared from plutonium-239, which is recovered by reprocessing spent fuel at the first stage, and natural uranium. This is the major process occurring during second stage. In FBRs, plutonium-239 experiences fission, producing energy. On the other hand, the mixed oxide fuel carrying uranium-238 transforms to extra plutonium-239. Hence, the stage-two FBRs are intended to 'breed' additional fuel than their consumption. **After the building of plutonium-239 inventory, thorium in a reactor is used as a blanket material and transformed to uranium-233. The U-233 so produced can be used in the third stage.**

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#### 4.3.3.3 Third Stage

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**Fuel –** Thorium

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**Reactor –** Advanced heavy-water reactor (AHWR) and thermal breeder reactor (TBR)

**Moderator –** Light water (demineralised water)

**Coolant –** $D_2O$ (heavy water)

An advanced or stage-three nuclear power system entails a self-sustaining series having fuelled reactor consisting of thorium-232 and uranium-233. According to this three-staged plan, the country's nuclear energy could grow to nearly 10 gigawatts with the help of PHWRs fuelled by domestic uranium. The nuclear energy growth over that could come only from FBRs till nearly 50 gigawatts. Only after achieving this capacity, the third stage is to be deployed.

India is currently focussing on the design of a reactor, which will enable more straightforward utilisation of thorium that is analogous with the three-stage sequential programme due to delay in direct utilisation of thorium. The accelerator-driven systems (ADS), advanced heavy-water reactor (AHWR) and compact high temperature reactor are the current three options that are being considered for the same. Even a molten salt reactor might be under deliberation, as per several reports. (Refer **Infographic – India's 3-Stage Nuclear Development Programme**).

## 4.3.4 Nuclear Fuel

### 4.3.4.1 First-Stage Fuel – URANIUM

Martin Klaproth, a German chemist, discovered uranium in 1789 in the mineral called **pitchblende**. It derives its name from Planet Uranus, which was discovered 8 years earlier. The melting point of uranium is 1132°C. It occurs in low concentration in soil, rock and water. Commercially, it is extracted from uranium-bearing minerals such as uraninite and pitchblende.

In nature, uranium is found as **uranium-238 (99.2742%), uranium-235 (0.7204%)** and a very small amount of **uranium-234 (0.0054%)**. Under certain conditions, the U-235 isotope can readily split, with a high yield of energy and thus adds to its importance. Hence, it is described as 'fissile' and

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the expression 'nuclear fission' is used. U-238 decays very slowly but **U-235 decays slightly faster**.

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#### *4.3.4.1.1 Enriched Uranium*

A type of uranium in which through the process of isotope separation, the per cent composition of uranium-235 has been increased is called enriched uranium. For both, civil nuclear power generation and military nuclear weapons, enriched uranium serves as a critical component. The monitoring and controlling of enriched uranium supplies and processes are attempted by the International Atomic Energy Agency (IAEA), in its efforts to ensure nuclear power generation safety and curb nuclear weapons proliferation.

From uranium ore to reactor fuel:

- **Low-enriched uranium** (Reactor Grade-Electricity): 3–5% $_{92}U^{235}$.
- **High-enriched uranium** (Weapon Grade): 90–95% $_{92}U^{235}$.

#### *4.3.4.1.2 Uranium in India*

The Uranium Corporation of India, a DAE subsidiary, is responsible for mining and processing of uranium at **Jaduguda** and **Bhatin** (since 1967), **Narwapahar** (since 1995) and **Turamdih** (since 2002) – all in Jharkhand.

### 4.3.4.2 Second-Stage Fuel – PLUTONIUM

The U-235 nucleus is 'fissile', while the nucleus of U-238 is said to be 'fertile'. It implies that it can capture one of the neutrons, which are flying about in the core of the reactor and become (indirectly) plutonium-239 that is the fissile form. Both Pu-239 and U-235 are similar, in terms of the fission process on hitting by a neutron, yielding enormous energy. Sometimes, on capturing a neutron, splitting of Pu-239 atom does not happen and instead it becomes Pu-240. **Due to the more unstable nature of Pu 239, it is used for atom bombs. Pu-240 has less stability so it is used in nuclear reactors for electricity generation.**

Note: After addition of a single neutron, Pu 239 will become Pu 240.

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**RADIOISOTOPE THERMOELECTRIC GENERATORS**

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Radioisotope Thermoelectric Generators (RTGs. RITEG), also known as radioisotope power systems (RPS), function as nuclear batteries utilising thermocouples to convert heat from the decay of radioactive materials into electricity. They have been employed to power satellites, space probes and unmanned remote installations like Soviet Union lighthouses in the Arctic Circle.

Unlike nuclear reactors, RTGs operate without moving parts and do not use fission or fusion processes for energy production. Instead, they harness the natural radioactive decay of materials such as plutonium-238 (non-weapons grade) to generate heat, which is then converted into electricity by thermocouples. These devices directly convert thermal energy into electrical energy.

RTGs are compact and lightweight, making them suitable for spacecraft power systems and localised power generation on Earth where sunlight is weak or absent. Their application is optimal in environments requiring small-scale, reliable power sources.

### 4.3.4.3 Third-Stage Fuel – THORIUM

Thorium was discovered in 1828 and named after Thor, the Norse god of thunder. It is a natural radioactive chemical element with the symbol $Th^{232}$ and atomic number 90. In nature, virtually all thorium is found as thorium-232, and through emission of an alpha particle, is decayed. Its half-life is of about 14.05 billion years (other, trace-level isotopes of thorium are short-lived intermediates of decay chains). **Thorium is estimated to be about four times more abundant than uranium** in the earth's crust and is a by-product, while extracting rare earth metals from monazite sands.

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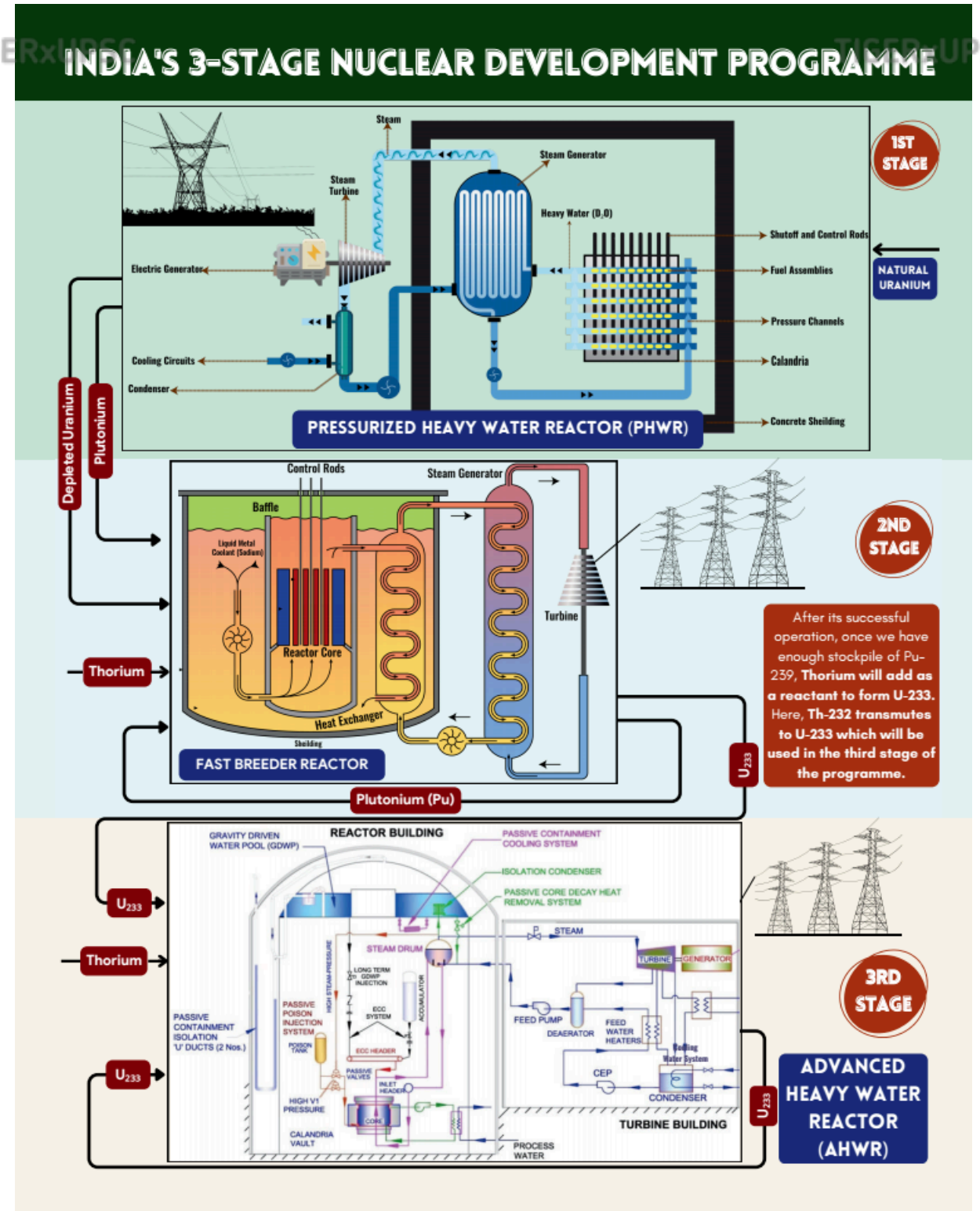


### *4.3.4.3.1 How Thorium is Better than Uranium?*

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Starting from the mining and purification, thorium is advantageous. However, a trace of naturally occurring thorium is $Th^{232}$, the isotope useful in nuclear reactors. It is much better than 3–5% of uranium that comes in the form we require. Unlike $U^{235}$, thorium is not fissile, that is, they will not start splitting on their own and exploding, no matter how many thorium atoms are packed together. So, to stop the reaction, the source of neutrons needs to be turned off, shutting the whole process down.

**Here's how thorium works:**

On absorbing a neutron, $Th^{232}$ becomes $Th^{233}$, i.e., the unstable form which decays into protactinium-233 and then into $U^{233}$.

### *4.3.4.3.2 Potential Benefits of Using Thorium*

If thorium reactors are as safe as claimed by the proponents, there may be no need to construct them as fortresses (which are expensive) because traditional reactors may cause disasters and involve radiations, which are harmful. Some thorium, obtained as a by-product of rare earth material mining of monazite, has a sustainability advantage, as it requires no additional mining to extract it, thus reducing further environmental destruction and financial cost.

### *4.3.4.3.3 Thorium in India*

India is capable of using thorium cycle-based processes for extraction of nuclear fuel, which is significant for the Indian nuclear power generation strategy because India has large reserves of thorium. Though the FBRs need higher cost for construction as compared to the PHWRs. India is focussing at the cheaper option – uranium fuel.

Note: India's first thorium-based commercial breeder reactor – 500 MW Prototype Fast Breeder Reactor (PFBR) – is working at the Indira Gandhi Centre for Atomic Research, Kalpakkam, Tamil Nadu. A total of 30% energy generation in India is targeted with the help of thorium reserves by 2050 in all nuclear energy generation targets.

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**PRESENCE OF THORIUM**

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**Worldwide Thorium Reserve (Estimated World Thorium Resources (in tonnes)] (IAEA Report)**

- India – 846,000
- USA – 595,000
- Brazil – 632,000
- Egypt – 380,000
- Australia – 595,000
- Turkey – 374,000

**In India (PIB 2014): Thorium Reserve**

It dominates the eastern coastal region in India. First rank – Andhra Pradesh (31%), Second rank – Tamil Nadu (21%), Third rank – Odisha (20%), Fourth rank – Kerala (16%), Fifth rank – West Bengal (10%), Sixth rank – Jharkhand (2%).

Kerala is largest producer of thorium in India.

**MONAZITE**

Monazite is a primarily reddish-brown phosphate mineral that **contains rare-earth elements**. Monazite is considered as a group of minerals due to its variability in composition. It is an atomic mineral that occurs naturally in the coastal sands of three districts: **Tirunelveli, Thoothukudi and Kanyakumari** in India.

Monazite is an important ore for thorium, lanthanum and cerium. Today, most of the world's monazite is produced in the offshore waters of India, Madagascar and South Africa.

### *4.3.4.3.4 Difference between Thorium and Uranium*

Thorium is more abundantly present in nature than uranium. Thorium is not fissile; and hence cannot go critical and generates a nuclear chain reaction. Thorium produces less radioactive waste products and generates more energy per tonne, so it is safe for the environment too. Thorium reactors do not produce plutonium, which is what you need to make a nuclear warhead. Thorium-derived $U^{233}$ is a better fuel on the basis of efficiency of energy generation in nuclear reactor, in comparison to $U^{235}$.

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#### *4.3.4.3.5 Other Uses of Thorium*

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- Thorium is found in the filaments of vacuum tubes, e.g. magnetron found in a microwave oven.
- Oxides of thorium known as 'thoria' are used in industry. Melting point of thoria is very high, so it is used in mantles of portable gas lights.
- When 'thoria' is added to glass, it increases the refractive index and decreases dispersion and is hence used in high-quality lens for cameras.
- As a chemical catalyst for converting ammonia to nitric acid in petroleum cracking and in production of sulphuric acid.

**DISTINCTION BETWEEN FISSILE AND FISSIONABLE**

Fissile materials can fission by absorbing a neutron with very low kinetic energy. The addition of neutron adds enough energy for the fissile material to undergo fission, owing to the odd number of neutrons in the nucleus, which will 'rearrange' themselves for releasing energy. U-235, Pu-239 and Pu-241 are fissile.

Fissionable materials can fission through absorption of a neutron if it has enough kinetic energy. The neutron needs to bring enough kinetic energy to the nucleus so that it can undergo fission. U-238 and Pu-240 are fissionable.

All fissile nuclides are fissionable, but only some fissionable nuclides are also fissile.

### 4.3.5 Nuclear Reactor

A nuclear reactor, or an atomic pile, is a device used to initiate and control a sustained nuclear chain reaction, used at nuclear power plants to generate electricity and in propulsion of ships. The heat generated from a nuclear fission is passed to a working fluid (water or gas), which runs through steam turbines. They either serve as a ship's propellers or help to turn electrical generators. Some reactors also serve to produce isotopes for medical and industrial uses, or for production of weapons-grade plutonium, while some are run only for research.

Note: A typical 1000 MWe reactor can provide enough electricity for a modern city with up to one million people.

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### 4.3.5.1 Types of Reactors

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There are different types of reactors. Some of the important reactors are given below:

#### *4.3.5.1.1 Pressurised Heavy-Water Reactor*

- A **pressurised heavy-water reactor (**PHWR) is a nuclear power reactor, commonly **using an un-enriched natural uranium as its fuel,** uses **heavy water (deuterium oxide $D_2O$)** as its coolant and moderator.
- The **heavy-water coolant is kept under pressure, allowing it to be heated to higher temperature without boiling, much as in a typical pressurised water reactor (PWR)**.
- While heavy water is significantly more expensive than ordinary light water, it yields greatly enhanced neutron economy, **allowing the reactor to operate without fuel enrichment facilities** (mitigating the additional capital cost of the heavy water) and generally enhancing the ability of the reactor to efficiently make use of alternate fuel cycles.

#### *4.3.5.1.2 Breeder Reactor*

It is a nuclear reactor generating more fissile material in fuel than the one consumed. Less than 1% of the natural uranium initiating the fuel cycle is consumed by a normal reactor. On the other hand, a breeder can burn almost all of it (minus re-processing losses); hence the same amount of energy is generated with lesser wastage. Breeders can be designed to use thorium. Two types of breeder reactors proposed include the following.

##### *4.3.5.1.2.1 Fast Breeder Reactor*

A fast breeder reactor (FBR) uses fast (unmoderated) neutrons to breed fissile plutonium and possibly higher transuranics from fertile uranium-238. The fast spectrum is flexible enough to breed fissile uranium-233 from thorium, if required.

FBRs have to be used in the second stage to support India's long-term goal of utilising its abundant thorium reserves by using the plutonium produced in PHWRs and breeding fissile uranium-233 from thorium. India

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has the world's largest thorium reserves, which can produce nuclear power for more than 10,000 years or as long as 60,000 years.

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### *Advantages of the FBR Programme in India*

1. **Breeding Concept:** FBR produces 30% more fuel than it consumes. It can obtain the high-fissile material production that can be used to fuel another reactor. Therefore, it is easy to alter the breeding ratio in the FBRs by enhancing the number of neutrons. FBRs use small cores, which is important to maintain chain reaction, and it does not require moderator for slowing down of neutrons as it uses fast neutrons.
2. **Economic Concept:** Due to high operating temperature of the FBRs, they are having high thermodynamic efficiency, which results in reduced thermal pollution to the environment. Moreover, FBRs generate more energy than the conventional thermal power plants such that 3 grams of uranium on fission can produce ten times the energy as generated by a tonne of coal.

**FAST BREEDER REACTOR**

## Fast Breeder Reactor (FBR)

- A **fast neutron nuclear reactor** that generates more nuclear fuel than it consumes while producing power.
- Utilizes **Uranium-Plutonium Mixed Oxide (MOX) fuel**.
- Surrounded by a **Uranium-238 'blanket**,' which undergoes **nuclear transmutation (the process of changing one element to another, either naturally or by nuclear reaction)** to produce additional fuel, hence the term '**breeder reactor'.**

## Significance of PFBR (Prototype Fast Breeder Reactor)

- Marks the commencement of **Stage II** of **India's three-stage nuclear power program**.
- Through transmutation, **Thorium will generate fissile Uranium-233**, which will be used as fuel in the third stage.

**NOTE:** India is also developing '**Bhavni', the world's first thorium-based nuclear plant**, using **Uranium-233**, at Kalpakkam.

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**Recently, Atomic Energy Regulatory Board (AERB) Grants Permission for First Approach to Criticality of 500 MWe PFBR.** In a nuclear reactor, 'critical' means the reactor is in a state of controlled, self-sustaining nuclear fission.

### About the *Prototype Fast Breeder Reactor (PFBR)*

- India's first indigenous **500 MWe sodium-cooled PFBR** is located in **Kalpakkam, Tamil Nadu**.

**Note:** MWe stands for **Megawatt electric**. It is a unit of electric power, specifically **one million watts**, used in the electric power industry.

- Commissioned by **Bharatiya Nabhikiya Vidyut Nigam Limited (BHAVINI)**, a government-owned entity under the **Department of Atomic Energy (DAE)**.

### Atomic Energy Regulatory Board (AERB)

- **Established:** 1983 by the President of India under the **Atomic Energy Act, 1962**.
- **Objective:** To ensure the safe use of ionising radiation and nuclear energy in India, minimising risks to public health and the environment.

In line with the true **spirit of Aatmanirbhar Bharat, PFBR** has been fully designed and constructed indigenously by BHAVINI with significant contribution from more than 200 Indian industries including MSMEs.

Once commissioned, **India will only be the second country after Russia (2016 onwards Russia started) to have commercial operating Fast Breeder Reactor.**

**Russia and India are the only two countries with commercial FBRs, with India's being developed and India plans to have six FBRs by 2039.**

The **Fast Breeder Reactor (FBR) will initially use the Uranium-Plutonium Mixed Oxide (MOX) fuel.**

3. **Waste Management:** In FBRs, the waste management burdens are nearly 200 times less than that of other conventional nuclear reactors. FBRs use the uranium waste from uranium processing plants and used fuel from other conventional nuclear reactors, which easily manages their nuclear waste. Due to the use of reused fuel in FBRs, the mining and processing expense is also minimised.

#### *4.3.5.1.2.2 Thermal Breeder Reactor*

Thorium-232 produces uranium-233 after neutron capture and beta decay. Thermal breeder reactors (TBRs) use thermal spectrum (moderated)

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neutrons to breed fissile uranium-233 from thorium (thorium fuel cycle). Since different nuclear fuels behave differently, a thermal breeder is thought commercially feasible only with thorium fuel, which avoids the build-up of the heavier transuranics.

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## 4.3.6 Components of a Nuclear Reactor

There are several components common to most types of reactors.

### 4.3.6.1 Fuel

Most common fuel used inside reactors are uranium, plutonium and thorium. Uranium is highly used worldwide for power generation.

Distribution of uranium worldwide: First rank – Australia (one-third uranium of the world), Second rank – Khazakhistan (12%) and Third rank – Canada (8%).

### 4.3.6.2 Moderator

Moderators are used to slow down the emitted neutrons, which have a high velocity range. Some common moderators are **graphite, heavy water, beryllium, beryllium oxide and some organic liquids.** Slowing down neutrons is also called thermalisation of the neutrons. Heavy water ($D_2O$) is one of the best moderators.

### 4.3.6.3 Control Rods

The control rods are used in initiating and stopping the nuclear fission reactions of the nuclear reactor. Due to large absorption cross-sectional area, **cadmium (Cd), boron (B) and Indium (In)** rods are used as control rods. When control rods are inserted in the reactor unit, they absorb the fast-moving neutrons and the chain reaction ceases.

### 4.3.6.4 Coolant

A coolant removes the tremendous amount of heat developed inside the reactor core. Through a heat exchanger, the coolant transfers heat to the

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secondary thermal system of the reactor. Water, steam, helium, $CO_2$, air, molten metals etc. are used as a coolant.

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### 4.3.7 List of Reactors

A total of six main types of reactors (see **Table 4.2**) have been developed worldwide. PWR is very common among them and highly used. Approximately 440 reactors are working worldwide 50% of which are PWRs.

**Table 4.2** Different Types of Reactors

| Reactor Type | Fuel | Moderator | Coolant |
|---|---|---|---|
| Pressurised water reactor (PWR) | Enriched $UO_2$ | Water | Water |
| Boiling water reactor (BWR) | Enriched $UO_2$ | Water | Water |
| Pressurised heavy-water reactor (PHWR) | Natural $UO_2$ | Heavy water | Heavy water |
| Gas-cooled reactor (GCR) | Natural U, enriched $UO_2$ | Graphite | Carbon dioxide |
| Light water graphite reactor (LWGR) | Enriched $UO_2$ | Graphite | Water |
| Fast breeder reactor (FBR) | $PuO_2$ and $UO_2$ | None | Liquid sodium |

#### 4.3.7.1 Nuclear Energy Parks (Future Reactor)

These are those places where nuclear power plants are set up for energy generation (**Table 4.3**). Various nuclear energy parks will be established in the future to fulfil the nuclear vision of India (**Table 4.4**).

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**Table 4.3** India's Operating Nuclear Power Reactors

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| Reactor | State |
|---|---|
| Tarapore | Maharashtra |
| Kaiga | Karnataka |
| Kakrapar | Gujarat |
| Kalpakkam | Tamil Nadu |
| Narora | Uttar Pradesh |
| Rawatbhata | Rajasthan |
| Kudankulam | Tamil Nadu |

**Table 4.4** List of Future Reactors

| Reactor | State |
|---|---|
| Kudankulam | Tamil Nadu |
| Jaitapur | Maharashtra |
| Mithivirdi (or Chhaya-Mithivirdi) | Gujarat |
| Kovvada | Andhra Pradesh |
| Haripur | West Bengal |
| Kumharia or Gorakhpur | Haryana |
| Bargi or Chuttka | Madhya Pradesh |
| Markandi (Pati Sonapur) | Odisha |
| Bhimpur | Madhya Pradesh |
| Madras | Tamil Nadu |
| Mahi Banswara | Rajasthan |

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**(Refer Infographic – Nuclear Power Plants in India)**

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### 4.3.7.2 Proposed Jaitapur Nuclear Plants

An agreement was signed on 6 December 2010 to construct a first set of two third-generation Europe-based pressurised reactors. During this agreement, the former French President Nicolas Sarkozy and the former Indian Prime Minister Manmohan Singh were present. The proposal is for construction of six European pressurised reactors, designed and developed by France, each of 1650 megawatts, thus totalling 9900 megawatts. These accord as the third-generation PWRs. If built, it would accord to be the **largest nuclear power generating station in the world** by net electrical power rating. However, it has largely been protested by locals and environmentalists who are concerned about the loss of land, the danger of radiation and destruction in the ecologically sensitive Western Ghats region.

### 4.3.7.3 Kudankulam Nuclear Power Plant

Kudankulam Nuclear Power Plant (KNPP) is a nuclear power station currently under construction in Kudankulam in the Tirunelveli district of the southern Indian state of Tamil Nadu. On 20 November 1988, an Inter-Governmental Agreement on the project was signed by the then Prime Minister Rajiv Gandhi and the then Soviet President Mikhail Gorbachev, for the construction of two reactors. The project remained in limbo for a decade, owing to the political and economic upheaval in Russia after the post-1991 Soviet breakup. On 14 January 2004, a small port became operational in Kudankulam. In 2008, the negotiations initiated in building four supplementary reactors on the location. Their capacity has not yet been declared; however, the expected capacity of each reactor will be around 1000 megawatts or 1 gigawatt, thereby bringing the total plant capacity to 9200 megawatts or 9.2 gigawatts.

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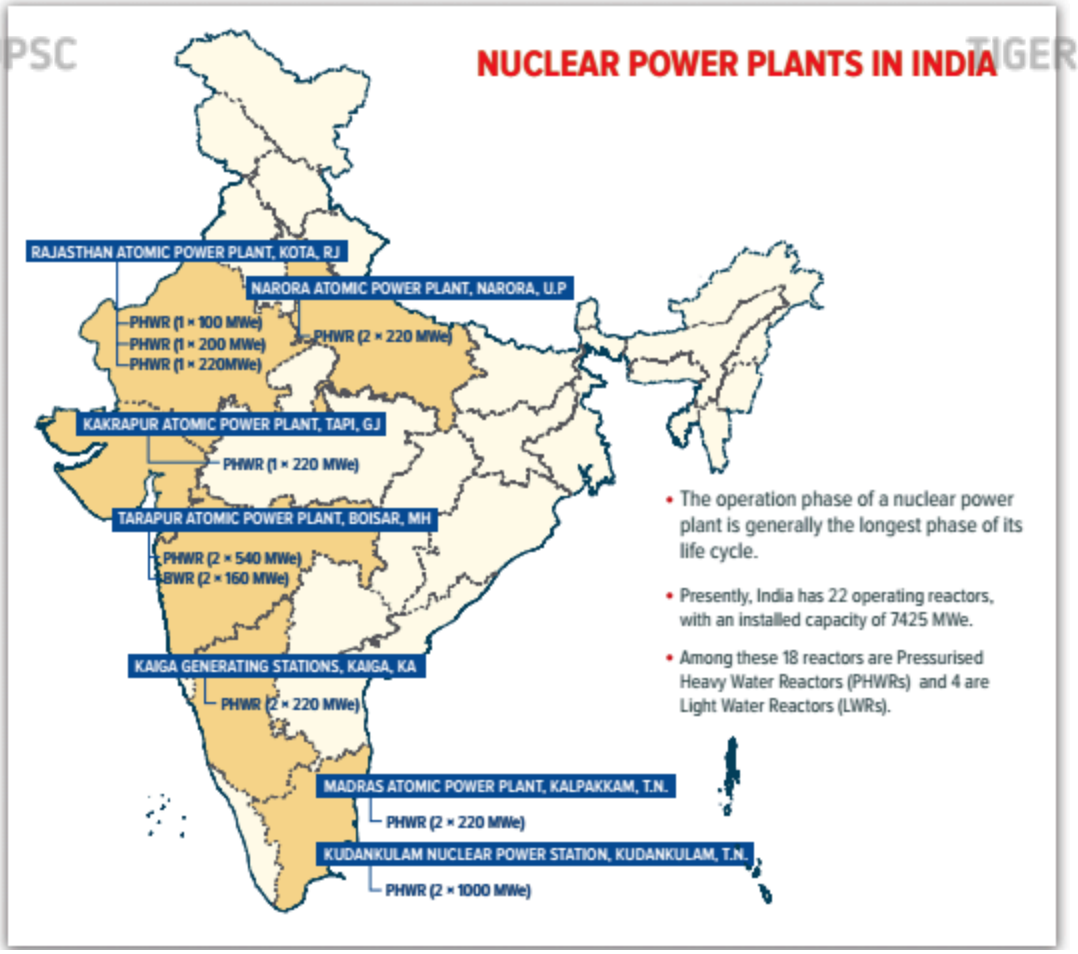


### *4.3.7.3.1 Technical Description of Kudankulam Nuclear Plant*

The Nuclear Power Corporation of India Limited (NPCIL) and Atomstroyexport are in the process of constructing two 1-gigawatt reactors of the VVER-1000 model (Voda Voda Energo Reactor – A Russian Reactor). One has been commissioned, and the second reactor is under construction. Both units are water-cooled, water-moderated power reactors.

### *4.3.7.3.2 Controversy with the Kudankulam Nuclear Plant*

Fearing the environmental impact of nuclear power, radioactive waste, nuclear accident similar to the radiation leak in March at Japan's Fukushima nuclear disaster were the reasons behind this protest. All these incidents contributed to the controversies in October 2011, which resulted in opposition to the Kudankulam Nuclear Power Plant project on several

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grounds. The Government of India constituted a central panel for surveying the safety features in the plant. It stated that the Kudankulam reactors are the safest, and fears of the people are not on scientific grounds. The plant was personally visited by the former President of India, A. P. J. Abdul Kalam, and he too reiterated that Kudankulam Nuclear Plant is safe and the plant was crucial and necessary for 'power hungry' India.

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## INDIA'S FIRST LARGEST INDIGENOUSLY DEVELOPED NUCLEAR POWER PLANT UNIT IN GUJARAT

The **third unit of the indigenously developed 700-megawatt electric (MWe)** nuclear power reactor at the **Kakrapar Atomic Power Project (KAPP3) in Gujarat** has commenced operations at full capacity, **marking a significant milestone in India's domestic civilian nuclear programme.**

### Significance of Kakrapar-3

- KAPP-3 is the **first 700 MWe unit and the largest indigenously developed variant of the Pressurised Heavy Water Reactor (PHWR)** in India.
- It surpasses the previous largest reactor size of 540 MWe PHWR deployed in Tarapur, Maharashtra.
- This achievement is a landmark event in India's nuclear energy sector, showcasing advancements in indigenous nuclear technology.

### Boost to Energy Security

- The operation of KAPP-3 contributes to India's efforts to enhance energy security by increasing nuclear power capacity.
- **India aims to ramp up its existing nuclear power capacity from 6,780 MWe to 22,480 MWe by 2031**, with the 700 MWe capacity playing a significant role in this expansion.
- **Currently, nuclear power constitutes around 2% of India's total installed capacity** of 417,668 MW as of May 31, 2023.

### Foundation Stone for 900 MWe Pressurised Water Reactor (PWR)

- India is preparing for the construction of a 900 MWe Pressurised Water Reactor (PWR) of indigenous design, representing the next frontier in the civilian nuclear sector.
- Experience gained from executing the larger 700 MWe reactor design will be valuable for this endeavour, particularly in manufacturing large-size pressure

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vessels and supplying enriched uranium fuel.

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### 4.3.8 International Thermonuclear Experimental Reactor (ITER)/Artificial Sun Experiment

**International Thermonuclear Experimental Reactor** (ITER) is a large-scale scientific experiment with the key objective to demonstrate the possibility of producing commercial energy from nuclear fusion.

#### 4.3.8.1 ITER Background and Indian Participation

In 1985, the US President Reagan and the General Secretary Gorbachev of the Soviet Union agreed to pursue an international effort that can help in the development of fusion energy to benefit all mankind, in Geneva, leading to the ITER project.

**WORLD'S FIRST FLOATING NUCLEAR PLANT IN RUSSIA**

- **Akademik Lomonosov** is the world's first 'floating' nuclear power plant (FNPP). The world's first FNPP, has successfully provided power to Russia's Chukotka region since 2019.
- An FNPP is basically a mobile, low-capacity reactor unit operable in remote areas isolated from the main power distribution system, or in places hard to access by land. They are designed to maintain both uninterruptible power and plentiful desalinated water supply in remote areas.
- **Russia offers FNPP Technology to India:** A Game Changer for Sustainable Energy – In a strategic move to bolster cooperation, Russia has extended an offer to India for the deployment of its advanced floating nuclear power plant (FNPP) technology.

(*Indian Defence Research Wing: Report dated 27 May 2024*).

#### 4.3.8.2 ITER Agreement is Signed

Ministers from the seven ITER members signed the ITER Agreement, in Paris on 21 November 2006. The initial signatories included: the former Soviet Union, the United States, the European Union (via EURATOM) and Japan. They were later joined by the People's Republic of China and the

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Republic of Korea in 2003, and by India in 2005. Over half of the world's population is represented by these seven nations, altogether.

In February 2007, a 'Broader Approach' agreement for complementary research and development was signed between the European Atomic Energy Community (hence, named by its initials EURATOM) and the Japanese government. Through this, a framework for Japan was established for conducting research and development, in support of ITER, over a period of 10 years.

ITER is being constructed in **southern France at Cadarache**, though it was scheduled to be operational by 2016 but the work is still in progress.

### 4.3.8.3 What is ITER?

- It is **an international nuclear fusion** research and engineering **megaproject,** which will be **the world's largest magnetic confinement plasma physics experiment.**
- It is an **experimental tokamak nuclear fusion reactor** that is being built in **southern France.**
- The goal of ITER is **to demonstrate the scientific and technological feasibility of fusion energy for peaceful use.**
- **The project is funded and run by seven member entities:**
  - ❑ The European Union, China, India, Japan, Russia, South Korea and the United States.

#### *What will ITER do?*

1. **Produce 500 MW of fusion power**
2. Demonstrate the integrated operation of technologies for a fusion power plant
3. Achieve a **deuterium–tritium plasma** in which the reaction is sustained through internal heating

##### *How is it achieved in the laboratory?*

- Most efficient fusion reaction in the laboratory setting is **the reaction between two hydrogen isotopes, deuterium (D) and tritium (T).**
- The **DT fusion reaction produces the highest energy gain at the 'lowest' temperatures.**

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### 4.3.8.4 India's Contributions to the ITER Project

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All the seven partners will contribute to build ITER, that is, each will build its share of ITER components through an appropriately formed Domestic Agency (DA) and industries and finally deliver them to ITER. The device will be finally assembled. India's contribution, like other partners except the host EU, was about 10% of the ITER construction cost (the EU pays about 40% since the ITER is being built in France and the remaining six members, including India, have to contribute 10% each). Most of this will be in the form of components made by the Indian industry and delivered to ITER, and a small part (1%) is to be paid in cash to a common fund for in-cash procurements by the ITER International Team.

**WHAT IS A TOKAMAK?**

- The tokamak is an experimental **machine designed to harness the energy of fusion.**
- Inside a tokamak, **the energy produced through the fusion of atoms is absorbed as heat in the walls of the vessel.**
- Just like a conventional power plant, **a fusion power plant will use this heat to produce steam and then electricity by way of turbines and generators.**
- **First developed by the Soviet research in the late 1960s,** the tokamak has been adopted around the world as the most promising configuration of magnetic fusion device. **ITER will be the world's largest tokamak** – twice the size of the largest machine currently in operation, with ten times the plasma chamber volume.

## Major Tokamak Projects and Milestones

### *Global Projects*

- **Joint European Torus (JET)**: First to achieve controlled fusion power.
- **International Thermonuclear Experimental Reactor (ITER):**
  - World's **largest Tokamak**, under construction in **France.**
  - A collaboration of **EU (27 countries), China, Japan, India, Korea, Russia and the USA.**
  - **ITER-India** handles components like **Cryostat**, **In-wall Shielding, etc.**

## Country-wise Achievements

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- **China**: Developed **HH70 (Eastern part of Shanghai, China),** world's first fully **high-temperature superconducting Tokamak**.

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- **EU & Japan**: Launched **JT-60SA**, the **largest and most advanced** Tokamak.
- **India**: Built **ADITYA** (indigenous) and **SST-1** (semi-indigenous steady-state Tokamak).

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## EAST Vs. HH70: FUSION Reactors (China)

| EAST | HH70 |
|---|---|
| • Chinese TOKAMAK Reactors | • Chinese TOKAMAK Reactors |
| • EAST is a **large, experimental reactor** aimed at achieving sustained high-temperature plasma experiment. | • HH70 is a **smaller, high-temperature superconducting tokamak** built by a commercial company. |
| • EAST is a key part of China's nuclear fusion development strategy and **serves as a testbed for the ITER project**. | • HH70, the world's first fully high-temperature superconducting tokamak, is **smaller and cheaper to build, potentially paving the way for more commercially viable fusion reactors.** |

### 4.3.8.5 India will Deliver Various Components in ITER Research

- Vessel ferromagnetic inserts
- Cryolines and cryo-distribution system
- Heat rejection and water cooling system components
- Some diagnostics systems
- Cryostat and vacuum vessel pressure suppression system
- Ion cyclotron heating and current drive sources, power supplies and control systems
- Start-up electron cyclotron heating source, power supply and control system
- Diagnostic neutral beam

### 4.3.8.6 Nuclear Fusion Research in India

Nuclear fusion research in India is primarily carried out at the **Institute for Plasma Research (IPR), Pune** (though other institutions like BARC are

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involved too). As an autonomous Physics research institute, it involves research in various aspects of plasma science, which includes basic plasma physics, research on magnetically confined hot plasmas and plasma technologies for industrial applications. Besides the basic research, the current process of the institute involves the building of a Steady State Superconducting Tokamak (SST-1). A tokamak represents a type of nuclear reactor (the ITER will also be a tokamak device). IPR has previously built a tokamak called ADITYA.

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## 4.3.9 Advantages and Disadvantages of Nuclear Energy

### 4.3.9.1 Advantages of Nuclear Energy

- Currently, the world derives 12–18% of the world's electricity through nuclear energy because of its advantages in terms of greenhouse gases. Greenhouse gases (carbon dioxide, methane, ozone, chlorofluorocarbon) are not released during nuclear reaction.

**CHINA'S 'ARTIFICIAL SUN'/EXPERIMENTAL ADVANCED SUPERCONDUCTING TOKAMAK (EAST)**

This project is a part of ITER that will be the world's largest nuclear fusion reactor by 2035. In the next goal EAST project will try to maintain the high temperature for a long period of time. The EAST device will replicate the nuclear fusion process carried out by the sun.

China's 'artificial sun' experiment set a new record after it ran at 120 million degrees Celsius for 101 seconds. For 20 seconds, EAST also achieved a peak temperature of 160 million degrees Celsius, which is over ten times hotter than the sun. This experiment was conducted at the **Institute of Plasma Physics of the Chinese Academy of Sciences (ASIPP),** in **Hefei**.

The ultimate goal of EAST is to create nuclear fusion experiments like the sun by using deuterium obtained from the sea. Deuterium from one litre of seawater is enough to produce energy equivalent to 300 litres of gasoline through a nuclear fusion reaction.

For nuclear fusion to occur, tremendous heat and pressure are applied to hydrogen atoms so that they fuse together. Unlike fission, nuclear fusion is considered a safer

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process with no nuclear waste. Nuclear fusion could potentially provide unlimited clean energy at very low costs.

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- Even a small amount of uranium generates huge energy through fission. As per current estimates, the reserves of uranium found on the earth may last for another 100 years. A single nuclear power plant can generate high amount of energy; also, nuclear fuel is comparatively inexpensive and can be transported easily.

#### 4.3.9.2 Disadvantages of Nuclear Energy

- Nuclear energy can lead to the production and proliferation of nuclear weapons; hence, fission, fusion or combination of both reactions can be misused for destructive purposes. This proves them to be a major threat to the world, causing large-scale devastation.
- A nuclear power plant requires large capital cost and is time consuming as well. Around 15–20 years are taken to develop a single plan thereby proving it not to be feasible. The functioning of nuclear reactor is possible only till the availability of the uranium.
- The fission reactions produce wastes containing unstable elements that are highly radioactive, proving them dangerous to the environment as well as human health, and their after effects also remain so, for thousands of years.
- Professional handling is mandatory for the nuclear reactor and it needs to be isolated from the living environments. Since, over a period of time, the radioactivity gets reduced, after decaying, there is a need for adequate and effective storage practices. The radioactivity of these elements reduces over a period of time.
- The Chernobyl disaster of 1986 in Ukraine was the worst nuclear power plant disaster. Hence, there stands the possibility of such disasters in future.

### 4.3.10 Radiation and Radioactivity

Radiation is the term referring to the propagation of particles and waves through space and includes both electromagnetic radiation and atomic and

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subatomic particle radiation; it was first discovered by Becquerel in 1986 during experiment on fluorescence of uranium.

Electromagnetic radiation has a broad, continuous spectrum of energy that includes visible light, radio waves, microwaves, gamma rays, X-rays, even the infrared and ultraviolet radiations. The neutrons, alpha and beta particles, protons and heavy ions are all included in the particle radiation. The energy and speed of the particle radiation depends upon the radiation source and any subsequent interaction between the particle and other matter.

For diagnosis and treatment of certain diseases or illness, radiation can also be generated. Besides, they can also be applied to eliminate or reduce harmful microorganisms as well as enhance the safety of the medical equipment, even the food supply, cooked food and in transmitting information (radio, cellular phones, television etc.).

Radioactivity, as a term, refers to the property of spontaneous emission of electromagnetic radiation of particles, shown by certain materials. The unstable atoms, while undergoing transition to a more stable state, emit these radiations. This transition is termed as radioactive decay. Naturally existing unstable atoms are said to be naturally radioactive. Examples of such naturally radioactive atoms include carbon-14, radon-222, potassium-42, uranium-235, uranium-238 and thorium-232. In addition to these, radioactive atoms can be produced when an interaction is conducted between the nucleus of an atom and a particle or electromagnetic radiations to produce an unstable nucleus. This process is typically conducted in nuclear reactors and particle accelerators.

### 4.3.10.1 Radioactive Decay Law

On emission of alpha and beta rays, which are usually but not invariably accompanied by gamma emission, the emitting parent nuclide transforms into a new daughter element; the daughter element may be again radioactive so that the process of successive disintegration continues till the original active parent nuclide gets transformed into a stable one.

According to the decay law, the number of active nuclides decreases exponentially with time. Therefore, an infinite time is theoretically needed for the complete disintegration of a radio element and in this respect, all

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radio elements are the same. The radioactive atoms undergo the transition to a stable state by emitting radiation in several forms as follows:

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- As charged particles (like alpha particles, beta particles and positrons)
- As uncharged particles (neutrons)
- As electromagnetic radiation (gamma rays and X-rays)

### 4.3.10.2 Sources of Radiation

Most of the radiations are received from the cosmic radiation and the sun. Thus, the individuals at higher elevations such as states of Colorado and adjacent Rocky Mountain receive comparatively greater radiation than those living at sea level. Besides this, many radiations are also derived from the rocks and soil around us.

### 4.3.10.3 Benefits of Radiation

Humankind can utilise radiation and radioactive materials in a number of ways:

1. **Agriculture:** There has been tremendous increase and improvement in the volume and quality, respectively, of grains as well as cereals by growing superior strains that are labelled by radioactive isotopes. Hence, they proved to be effective measures in alleviating famine in third-world countries.
2. **Environmental measurements:** Radioactive isotopes can greatly be utilised for the accurate measurement of movement of pollutants through the environment including its groundwater and rivers.
3. **Eradication of pests:** There is a drastic fall in the population of pests due to the sterilisation of male flies due to radiation.
4. **Food:** Food, such as chicken and beef, being sterilised through irradiation shows an increased shelf life and is also *E. coli*-free, which is a bacterium known to have taken the lives of many children due to the consumption of badly cooked fast-food hamburgers. The food irradiation extension could prove a saviour in such cases for many children and would be particularly of great help to the developing countries with less adequate refrigeration.

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5. **Generation of electricity:** Nearly 440 nuclear plants in and around the world have been reported to contribute around 16% of the world's electrical energy needs.
6. **Medical diagnostics:** The radiation uses have provided greater help from X-rays, through magnetic resonance imaging along with the usage of radioactive tracers, in order to diagnose various conditions like fault in thyroid glands or problem in bones. Moreover, the radioactive tracers have been known to replace invasive surgical diagnosis.
7. **Oil drilling:** There are wide uses of isotopes for measuring the steam quality before infusing into almost obsolete wells of oils so as to force out the residual supplies.
8. **Polymerisation of plastics:** It can be done through irradiation instead of the usage of destructive heat treatments.
9. **Quality control of metal parts:** Relying on metal parts such as the aircraft blades or turbines can be confirmed through radiophotography on a conveyor rather than destroying a blade's sample so as to ensure their intactness.
10. **Research in biology:** The usage of radioactive tracers enables the non-invasive elements and drugs tracking in the body, both for medicine and for studying metabolism.
11. **Space power:** When there is a requirement of small amounts of power in space, especially in regions of insufficient solar power (on the dark side or in regions where large solar panels are impossible), the ideal compact energy producers are plutonium batteries.
12. **Treatment of cancer:** Radioactivity can selectively kill cancerous cells, either in the direct beam form such as for breast cancer, or in the form of bullets of radioactive material, which are considered in the direct migration to the cancerous cells that need to be killed. The chemotherapy, being the sole alternative, involves the usage of invasive drugs, and is considered a difficult option for the patient.

Thus, the advantages of radiation continue. Hence, it can be concluded that for many aspects of our daily life, our society greatly depends on its assistance.

### 4.3.11 Nuclear Winter

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Nuclear winter, as the theory describes, is the climatic effects of nuclear war. The smoke from the fires started by nuclear weapons, mainly the sooty, black smoke from cities and industrial areas, would get heated by the sun, lofted into the upper stratosphere and thus spread globally, making it last for years. This results in a cool, dark, dry condition at the earth's atmosphere because it does not allow the sunlight to enter into the atmosphere, which would prevent crop growth at least for one growing season, resulting in mass starvation in the world.

Most frequently, it is suggested to manifest, as a result of the combined combustion and pollution from the burning of areas that are at least the size of 100 cities, at the intensity of a fire storm, wherein the smoke remained for years and thus caused temperature drops for as long as it remained, and this is what we called 'nuclear winter'.

#### 4.3.11.1 History

On 6 August 1945, a 15-Kt (kilotons) nuclear bomb was dropped on Hiroshima, Japan, killing 1,50,000 people at a time. Many of them died due to fires ignited by the bomb, which turned the city into a raging inferno and formed dense blackish clouds in the atmosphere. And then, three days later, another 20-Kt bomb was detonated on Nagasaki, Japan. This huge cloud formation cooled the global climate by 1°C and it produced a 'small nuclear winter'. In March 1991, due to Iraq's invasion of Kuwait, the smoke released due to the fires set to the oil reserves plumed to the stratosphere, serving as one of the main contributors to formation of soot, because the smoke from oil had a higher ratio of black soot, thus absorbing more sunlight.

### 4.3.12 Nuclear Medicine

The branch of medicine that deals with the use of radioactive substances in treatment, research and diagnosis is called as nuclear medicine.

The small amounts of radioactive materials used in medicine are known as radiotracers. These radiotracers are typically injected into the blood

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vessel, i.e., vein, inhaled or swallowed as capsules. The radiotracers travel through the area being examined and by dissolving directly in blood it gives off gamma rays. These gamma rays are detected by specific camera and computer, so as to create images of the inside of body and identify the disease.

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### 4.3.12.1 History

The father of nuclear medicine is John Lawrence. In 1936, he made the first application in patients of an artificial radionuclide by using **phosphorous-32 for treatment of leukaemia**. Due to successful treatment of the patient with **thyroid by use of Iodine-131**, nuclear medicine gained public recognition as a potential specialty on 7 December 1946. Among many radionuclides that were discovered for medical use, the most important discovery and development was Technetium-99m. C. Perrier and E. Segre discovered it in 1937 as an artificial element in order to fill space number 43 in the periodic table.

Today, it is the most utilised element in nuclear medicine.

### 4.3.12.2 Diagnosis

Depending on the types of nuclear medicine examination, a dose of radio tracer is injected into the vein or swallowed or inhaled as gas. It may take from 20 minutes to several hours or days to travel through the body. After reaching its destination, it gives off gamma rays and is detectable by specific cameras, which take a series of images. The end result of the nuclear medicine imaging is a data set having one more image. In a multi-image data set, the array of images repeat a sequence called ‘cardiac gate time sequence’ or ‘spatial sequence’. Areas of greater radiotracer intensity are called as hot spots, indicating high metabolic rate. Less intensity areas are called as cold spots, indicating low metabolic rate.

In radioactive iodine therapy for thyroid disease, radioactive iodine (I-131) is swallowed or injected into the gastrointestinal tract and is concentrated by the thyroid gland when it kills the cell.

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#### *4.3.12.2.1 Uses of Imaging Procedure in Varied Areas*

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Doctors use radionuclide imaging procedures to identify the structure of organs, their functions, bone and systems.

- **Brain:** It detects the early stage of Alzheimer's disease. It evaluates chemical disorder of Parkinson's disease. It provides an idea of the stages of tumour, surgical or radiational planning. It investigates disorders such as memory loss, abnormality in blood flow inside the brain, seizures etc.
- **Lungs:** It helps to detect lung transplant rejection. It scans lungs for respiratory and blood flow problems.
- **Heart:** It assesses the damage to the heart, following a heart attack. It evaluates treatment options such as bypass surgery and angioplasty. It detects heart transplant rejection. It evaluates spinal fluid flow and potential spinal fluid leaks.
- **Cancer:** Nuclear medicine helps in determining the presence and spread of cancer in various parts of the body. Doctors can plan for treatment by use of these medicines. It localises sentinel lymph nodes before the surgery in patients with breast cancer or skin as well as soft tissue tumours. In children, nuclear medicine is also used while investigating abnormalities in the oesophagus like oesophageal reflux or mortality disorders and helps in the evaluation of openness of ventricular shunts in the brain. Nuclear medicine therapies include radioactive antibodies used for treatment of certain cancer of the lymphatic system. I-131 MIBG is used to treat adrenal gland tumours in adults and nerve tissue tumours in children. It is used to treat painful tumour metastasis to the bones.

#### *4.3.12.2.2 Benefits*

For many diseases, nuclear medicine scans yield the most useful information needed to determine appropriate treatment. Nuclear medicine offers the potential to identify the disease in its earliest stage. PET (positron emission tomography) scans provide information that is helpful in radiation therapy, i.e. chemotherapy planning. PET scans identify the best biopsy location. It provides information about both function and anatomic structure of the body.

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#### *4.3.12.2.3 Risks*

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The nuclear medicine diagnostic procedures have been in use for more than five decades and so far no long-term adverse effects due to such low dose exposure are known. Allergic reactions to radiopharmaceuticals may be possible but are extremely rare and mild. Nuclear medicines are time consuming and can take several days. Also, the resolution of structures of the body with the nuclear medicine may not be as high as with the other imaging techniques such as CT and MRI. It is a very sensitive procedure compared to any other techniques. It is possible that a nuclear stress test could cause heart attack.

### 4.3.13 Atoms4Food (Atomic Energy for Food Security)

Bhabha Atomic Research Centre (BARC) Participated in International Atomic Energy Agency (IAEA) Scientific Forum 'Atoms4Food'.

#### *What is 'Atoms4Food'?*

- **Launched by:** IAEA and FAO at the **2023 World Food Forum** in **Rome, Italy**.
- **Goal:** To help countries improve **food security** using **nuclear and advanced technologies**.
- **Focus Areas**:
  - ❑ Enhancing **agriculture and livestock productivity**.
  - ❑ Reducing **post-harvest food losses**.
  - ❑ Combating **global hunger.**

#### *Why It Matters?*

- **Hunger projections** (FAO):
  - ❑ ~600 million people may be **chronically undernourished by 2030.**
- **Population trends** (UN):
  - ❑ By 2050, the global population will rise by **one-third**, primarily in **developing countries.**

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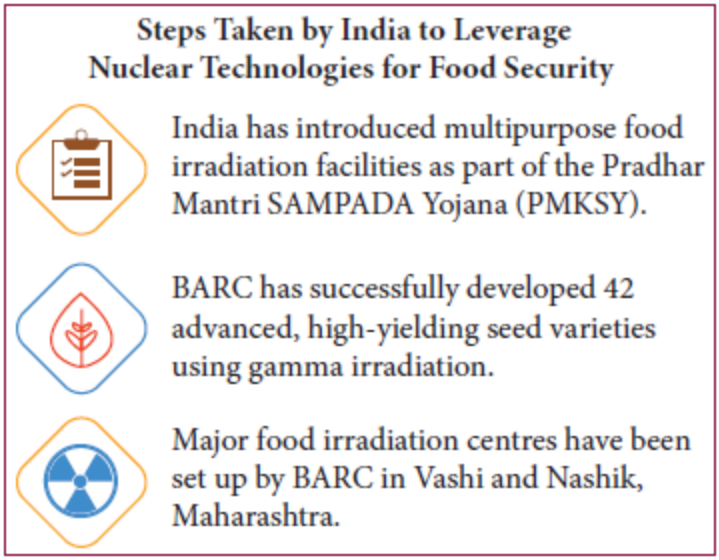


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### *Nuclear Technologies Supporting Agriculture*

1. **Irradiation**
   - Extends **shelf life** of food.
   - Reduces/eliminates **microorganisms and pests**.
2. **Fallout Radionuclide (FRN) Technique**
   - Measures **soil erosion patterns** by tracking radionuclide presence.
3. **Cosmic-Ray Neutron Sensor (CRNS)**
   - Measures **soil moisture** over large areas using cosmic ray reflections.
4. **Radioimmunoassay (RIA)**
   - Detects **animal hormone levels**.
   - Improves **timing for artificial insemination**.
5. **Sterile Insect Technique (SIT)**
   - Releases **sterile insects** to suppress pest populations.
6. **Other Nuclear Tools**
   - **Nitrogen-15 isotopes:** Track nitrogen fixation in crops.
   - **Isotope tracing:** Enhance **crop nutrition** and **water use efficiency**.

## 4.3.14 Nuclear Energy Summit

The **first-ever Nuclear Energy Summit**, a high-level meeting focused on nuclear energy, took place on 21 March 2024, in Brussels.

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- **Organised by:** International Atomic Energy Agency (IAEA) and **Belgium**.
- **Context:** Held after nuclear energy was included in the **Global Stocktake at COP28 (2023)**, calling for its accelerated use.
- **Goal:** Promote nuclear energy as a key to climate action.
- **Participation:** Attended by **30+ countries**, including **India** and the **European Union**.
- **Key Discussions:** The role of nuclear power in **achieving net-zero emissions** and **promoting sustainable development**.
- **Current Role:** Nuclear energy provides about **25% of global low-carbon electricity**

### *About IAEA (International Atomic Energy Agency)*

- **Established:** 1957, as part of the **UN system**.
- **Objective:** Promote **safe, secure and peaceful use** of nuclear technology.
- **Role:** Enforces safeguards under the **NPT** and other treaties to prevent nuclear weapons proliferation.
- **India:** A signatory to the **Additional Protocol** under IAEA.
- **Members:** 178
- **HQ:** Vienna, Austria
- **Key Initiative: Atoms4NetZero**

## Nuclear Energy: Relevance & Challenges

### *Relevance*

- **Low-carbon** source of electricity
- **Reliable** and always available
- **Advancements** like **Small Modular Reactors (SMRs)** improve feasibility (up to 300 MW/unit)

### *Challenges*

- **Safety risks** (e.g., Fukushima disaster)
- **Cybersecurity threats**
- **High initial costs** and **project delays**

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## Nuclear Power in Union Budget 2025–26

The **Union Budget 2025-26** outlines a significant push towards **nuclear energy** as part of **India's long-term energy transition strategy**.

The government has set an ambitious **target of 100 GW (1 lakh MW) nuclear power capacity by 2047**, positioning nuclear energy as a **major pillar** in India's energy mix.

This development aligns with the broader objectives of **Viksit Bharat, ensuring energy reliability and reducing dependency on fossil fuels**.

To achieve this goal, **strategic policy interventions** and **infrastructure investments** are being undertaken, with an emphasis on **indigenous nuclear technology** and **public-private collaborations**.

This initiative aims to **enhance domestic nuclear capabilities**, **promote private sector participation** and **accelerate the deployment of advanced nuclear technologies** such as **Small Modular Reactors (SMRs).**

## Small Modular Reactors (SMRs) and R&D Initiatives

A key highlight of the **Union Budget 2025–26** is the **launch** of a **Nuclear Energy Mission,** which is focused on research and development (R&D) of **Small Modular Reactors (SMRs).**

The government has allocated **₹20,000 crore** for this initiative, aiming to develop at least **five indigenously designed and operational SMRs by 2033.**

## Nuclear Energy Mission for Viksit Bharat

To facilitate the implementation of the **Nuclear Energy Mission**, amendments to the **Atomic Energy Act** and **Civil Liability for Nuclear Damage Act** will be taken up by the parliament. These amendments are expected to encourage **private sector investments** in nuclear power projects.

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These legislative changes are expected to create a more conducive environment for investment and innovation in the nuclear sector.

As of **30 January 2025, India's nuclear capacity is 8180 MW**.

The government will enter into **partnerships** with the **private sector** with the motive of:

1. Setting up **Bharat Small Reactors**
2. Research & development of **Bharat Small Modular Reactor**
3. Research & development of **newer technologies** for nuclear energy

**Note:** The private sector will be involved in the development of **Bharat Small Reactors (BSR)** and **Bharat Small Modular Reactors (BSMR).**

**BSR** refers to incrementally **modified forms of India's existing Pressurised Heavy Water Reactors (PHWR)** whereas **BSMR is a nascent technology being researched globally**.

## (A) SMALL MODULAR REACTOR

### Small Modular Reactors (SMRs)

SMRs are nuclear reactors with a maximum output of **300 MWe**, capable of generating **7.2 million kWh/day (Kilowatt – hour per day)** in contrast to large nuclear plants generate **over 1,000 MWe** and **24 million kWh/day**.

- **"Small"** – Compact size compared to conventional reactors.

- **"Modular"** – Factory-assembled and transported for on-site installation.
- **"Reactors"** – Use nuclear fission to produce energy.

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### Advantages of SMRs

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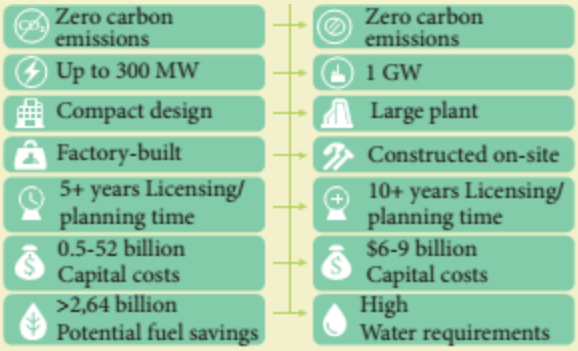
- Small physical footprint.
- Lower capital investment.
- Factory-built design (vs. on-site construction).
- Suitable for remote or space-limited locations.
- Scalable with **incremental power additions**.
- Enhanced **safety, security and non-proliferation** features.
- Ideal for **energy hubs** with mixed energy sources.
- Can provide **electricity, industrial heat, district heating** and **hydrogen production**.




## *(B) BHARAT SMALL REACTORS (BSRs)*

**Bharat Small Reactors (BSRs)** are a category of **Small Modular Reactors (SMRs)** tailored to India's energy demands. These are **compact, factory-manufactured nuclear reactors** with a generation capacity of up to **300 megawatts (MW)**.

**BSRs are engineered to be:**

- **Safe**
- **Cost-effective**
- **Low-carbon**
- **Easily deployable**, even in remote or space-constrained regions

**NOTE:** Their modular design allows for quicker construction and scalability compared to traditional large-scale nuclear power plants.

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The government is actively expanding its nuclear energy sector by developing **Bharat Small Reactors (BSRs)** and exploring **partnerships**

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with the **private sector**.

BSRs are **220 MW Pressurised Heavy Water Reactors (PHWRs)** with a proven safety and performance record.

These reactors are being upgraded to reduce land requirements, making them suitable for deployment near industries such as steel, aluminium and metals, serving as captive power plants to aid in decarbonisation efforts.

The plan involves private entities providing land, cooling water and capital, while the **Nuclear Power Corporation of India Limited (NPCIL)** handles design, quality assurance and operation and maintenance, all within the existing legal framework.

In addition to BSRs, the **Bhabha Atomic Research Centre (BARC)** is developing **Small Modular Reactors (SMRs)** for **repurposing retiring coal-based power plants** and meeting power needs in **remote locations**.

This strategic move signifies India's dedication to **reducing carbon emissions** and **enhancing its civil nuclear energy program,** with private sector participation playing a crucial role within the bounds of Indian laws and regulations.

This initiative aligns with India's commitment to achieving **500 GW** of non-fossil fuel-based energy generation by **2030** and meeting **50%** of its energy requirements from **renewable energy** by **2030**, as pledged at the **COP26 Summit** in **Glasgow** in **2021**.

India is actively exploring **Small Modular Reactors (SMRs)** as a crucial part of its energy transition strategy, aiming to achieve net-zero emissions while ensuring energy security.

SMRs, are advanced nuclear reactors with a power generation capacity ranging from less than **30 MWe to 300+ MWe**, provide a flexible, scalable and cost-effective alternative to conventional large nuclear reactors.

Given India's growing energy demands and the need for reliable, low-carbon power, SMRs can play a **transformative role** in complementing renewable energy sources and stabilising the grid.

Their modular design allows for **factory-based manufacturing**, reducing construction timelines and costs, making them suitable for both

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**on-grid** and **off-grid applications**, including deployment in **remote locations**.

**Note:** India's **expertise** in **Pressurised Heavy Water Reactors (PHWRs)** provides a **strong foundation** for the development and deployment of indigenous SMR designs.

By integrating SMRs into its energy mix, India can address land constraints, reduce dependence on fossil fuels and enhance its ability to meet international climate commitments under the **Paris Agreement** (**2015**) which India ratified in **October 2016**.

## (C) Government Initiatives for Enhancing India's Nuclear Capacity

India is actively enhancing its nuclear power capacity to meet growing energy demands and achieve environmental goals. The government has initiated steps to increase nuclear power capacity **from the current 8,180 MW** to **22,480 MW** by **2031–32**.

This expansion includes the construction and commissioning of **ten reactors,** totalling **8,000 MW,** across **Gujarat, Rajasthan, Tamil Nadu, Haryana, Karnataka** and **Madhya Pradesh**.

Additionally, pre-project activities for ten more reactors have commenced, with plans for progressive completion by **2031–32**.

Further, the government accorded in-principle approval to set up **6 × 1208 MW nuclear power plant** in cooperation with the **USA** at **Kovvada** in **Srikakulam** district in the state of **Andhra Pradesh.**

A significant milestone was achieved on **19 September 2024**, when the **Rajasthan Atomic Power Project's Unit-7 (RAPP-7), one of the country's largest and third indigenous nuclear reactors, reached criticality**, **marking the beginning of controlled fission chain reaction.**

**Note:** In a nuclear reactor, "critical" refers to a state of **self-sustaining fission** where the rate of neutron production equals the rate of neutron loss.

**Safety** remains a cornerstone of India's nuclear energy policy. India's nuclear power plants operate with **stringent safety protocols** and

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**international oversight**. The **radiation levels** at Indian nuclear facilities are consistently well **below global benchmarks**, underscoring the country's commitment to secure and sustainable nuclear energy. These efforts align with India's broader strategy to provide clean and reliable energy, contributing to long-term energy security and environmental sustainability.

## (D) Recent Developments in Nuclear Energy in India

A significant discovery of new deposit in **India's oldest Uranium Mine, the Jaduguda Mines**, has been made in and around the existing mine lease area. This will increase the life of an otherwise depleting mine by **more than fifty years**.

First two units of the indigenous **700 MWe PHWR** at **Kakrapar, Gujarat (KAPS - 3 & 4)** have started commercial operation in FY 2023–24.

**Closed fuel cycle** being the cornerstone of Indian nuclear power program, the **country's first Prototype Fast Breeder Reactor (PFBR 500 Mwe)** achieved many of the milestones in 2024, viz., Primary Sodium filling in Main Vessel, purification of the filled sodium and commissioning of all the four Sodium pumps (two primary sodium pumps and two secondary sodium pumps). Core loading was commenced with loading of first reactor control rod on 4 March 2024.

**NPCIL (Nuclear Power Corporation of India Ltd.)** and **National Thermal Power Corporation (NTPC)** have signed a supplementary **Joint Venture agreement** to develop nuclear power facilities in the country. The Joint Venture (JV) named **ASHVINI** will function within the existing legal framework of the Atomic Energy Act 1962 (amended in 2015) and will build, own and operate nuclear power plants, including the upcoming **4×700 MWe PHWR Mahi-Banswara Rajasthan Atomic Power Project.**

## Conclusion

The provisions for nuclear power in the Union Budget 2025-26 mark a **transformative shift** in India's energy landscape. By promoting nuclear

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energy as a sustainable, scalable and secure power source, the government aims to bolster **energy security** and meet the **nation's long-term economic** and **environmental goals**.

The Nuclear Energy Mission for Viksit Bharat is poised to accelerate nuclear power development, **positioning India as a global leader in advanced nuclear technology by 2047.**

## QUESTIONS AND ANSWERS

### Objective Questions

**1. With reference to radioisotope thermoelectric generators (RTGs), Consider the following statements:** **[UPSC (Pre) 2024]**

1. RTGs are miniature fission reactors
2. RTGs are used for powering the onboard systems of spacecrafts.
3. RTGs can use Plutonium-238, which is a by-product of weapons development.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

**2. Consider the following statements:** **[UPSC (Pre) 2023]**

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*Statement-I:* India, despite having uranium deposits, depends on coal for most of its electricity production.

*Statement-II:* Uranium, enriched to the extent of at least 60%, is required for the production of electricity.

Which one of the following is correct in respect of the above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
(c) Statement-I is correct but Statement-II is incorrect
(d) Statement-I is incorrect but Statement-II is correct

**3. With reference to India, consider the following statements:** **[UPSC (Pre) 2022]**

1. Monazite is a source of rare earths.
2. Monazite contains thorium.
3. Monazite occurs naturally in the entire Indian coastal sands in India.
4. In India, only Government bodies can process or export monazite.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1, 2 and 4 only
(c) 3 and 4 only
(d) 1, 2, 3 and 4

**4. India is an important member of the 'International Thermonuclear Experimental Reactor'. If this experiment succeeds, what is the immediate advantage for India?** **[UPSC (Pre) 2016]**

(a) It can use thorium in place of uranium for power generation.
(b) It can attain a global role in satellite navigation.

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(c) It can drastically improve the efficiency of its fission reactors in power generation.
(d) It can build fusion reactors for power generation.

**5. To meet its rapidly growing energy demand, some opine that India should pursue research and development on thorium as the future fuel of nuclear energy. In this context, what advantage does thorium hold over uranium?** **[UPSC (Pre) 2012]**

(i) Thorium is far more abundant in nature than uranium.
(ii) On the basis of per unit mass of mined mineral, thorium can generate more energy compared to natural uranium.
(iii) Thorium produces less harmful waste compared to uranium.

Which of the statement(s) given above is/are correct?
(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**6. The function of heavy water in a nuclear reactor is to:**
(a) Slow down the speed of neutrons **[UPSC (Pre) 2011]**
(b) Increase the speed of neutrons
(c) Cool down the reactor
(d) Stop the nuclear reaction

**7. In which one of the following locations is the International Thermonuclear Experimental Reactor (ITER) project to be built?**
(a) Northern Spain **[UPSC (Pre) 2008]**
(b) Southern France
(c) Eastern Germany
(d) Southern Italy

**Answer Key**

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| | | | | | | |
|---|---|---|---|---|---|---|
| 1. (b) | 2. (c) | 3. (b) | 4. (d) | 5. (d) | 6. (a) | 7. (b) |

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# CHAPTER 5
# NUCLEAR DIPLOMACY

## IN THIS CHAPTER...


**5.1 The International Atomic Energy Agency (IAEA)**

5.1.1 Scope of Safeguards

5.1.2 Aim of the IAEA

**5.2 Additional Protocol**


As of March 2023 Additional Protocols are into force with 141 states and Euratom (European Atomic Energy Community), and another 13 states are yet to bring it into force.


5.2.1 Status of the Additional Protocol

5.2.2 Goal of Additional Protocol

5.2.3 Strengthening Measures Under the Additional Protocol

5.2.4 Additional Measures

**5.3 International Treaties and Arrangements for the Nuclear Technology**


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5.3.1 Nuclear Non-Proliferation Treaty (NPT)

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The major countries which remain outside the NPT are Pakistan, Israel and India.

5.3.2 Nuclear Suppliers Group (NSG)
5.3.3 The Wassenaar Arrangement
5.3.4 The Australia Group
5.3.5 Missile Technology Control Regime (MTCR)
5.3.6 Comprehensive Nuclear-Test-Ban Treaty
**5.4 India's NFU Policy**
5.4.1 Why India Should Retain This Policy?
**5.5 Nuclear Deals of India and Nuclear Diplomacy**
5.5.1 Nuclear Energy Agreements Signed between India and Different Countries
**5.6 Nuclear Liability Act, 2010**

India likely to sell BrahMos missiles to Vietnam. BrahMos missile project has completed 25 years in 2023.

5.6.1 Necessity of This Act
5.6.2 Criticism
5.6.3 Suggestions and Way Forward
**5.7 Phasing Out of Nuclear Power in Europe**
**5.8 New START Treaty**
**5.9 Nuclear Security Summit**
**5.10 International Panel on Fissile Material (IPFM)**
**5.11 World's Major Nuclear Disasters (at Level 7, 6 and 5)**

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# 5.1 The International Atomic Energy Agency

The **International Atomic Energy Agency** (IAEA) serves as an intergovernmental forum for scientific and technical cooperation in the peaceful use of nuclear technology and nuclear power worldwide. **It was set up as the world's 'Atoms for Peace' organisation in 1957 within the United Nations family.** The agency works with its Member States and multiple partners worldwide to promote the safe, secure and peaceful use of nuclear technologies.

It seeks to promote the peaceful use of nuclear energy, and to inhibit its use for any military purpose, including nuclear weapons. The **IAEA reports to both the United Nations General Assembly and Security Council.** The IAEA has its headquarters in **Vienna, Austria.**

## 5.1.1 Scope of Safeguards

They prevent nuclear proliferation just like procedures of auditing for confidence building for having appropriate financial conduct, thereby preventing embezzlement. Specifically, they aim to validate if the nuclear material, which is declared, remains inside the established nuclear civil fuel cycle and its application is exclusively for peaceful purposes or not.

## 5.1.2 Aim of the IAEA

Conventional IAEA protection aims for the discouragement of nuclear material diversions for its usage in peaceful manner and hence, to maximise its early detection risk. The safeguards, thus, serve both the international community and individual states. Recognising it, by showing compliance with these commitments, will benefit them.

The inspections function as a vigilant system that warns the likely nuclear material diversion from peaceful activities. Such a system relies on the following:

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- **Material Accountability** – Collection and investigation of the nuclear material, on-site assessment, reassessment and substantiation of the functional records all come under material accountability.
- **Physical Security** – The nuclear materials access is confined at the use of its site, beyond which it is not permitted.
- **Containment and Surveillance** – Closures, automatic cameras and other such instruments are used for detection of unwanted shifting or tinkering with the materials, besides site spot checks.

Note: The five states, with weapons, and the non-NPT states such as India, Pakistan and Israel come under facility-specific safeguards application for the applicable plants. The IAEA inspectors regularly visit these facilities to keep an eye on any possible inconsistency in the records.

## 5.2 Additional Protocol

**The Additional Protocol is a safeguard agreement that serves as the legal document of the IAEA** safeguards agreement (as a tool for verification). As per this, the IAEA has the corresponding legal power for verification of a state's safeguard commitments. Hence, the Additional Protocol is intended for all the states with any kinds of safeguards agreement. **The Comprehensive Safeguards Agreements (CSAs) is involved in deciding to settle as well as take action for the additional protocols.**

### 5.2.1 Status of the Additional Protocol

As of March, 2023 Additional Protocols are into force with 141 states and Euratom (European Atomic Energy Community), and another 13 states are yet to bring it into force.

### 5.2.2 Goal of Additional Protocol

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The IAEA has granted the extended rights to access information and sites through Additional Protocol. The primary aim of the additional protocol is to seal the breaks in the information, given under the CSAs.

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### 5.2.3 Strengthening Measures Under the Additional Protocol

States with CSAs need to equip the IAEA with important supplementary substantiation measures. Such measures delivering broader access to information regarding the nuclear programme of the states, increases physical access by the IAEA as well as improved administrative measures.

### 5.2.4 Additional Measures

- Information to be provided by states, regarding all parts of its nuclear fuel cycle, ranging from uranium mines to the nuclear waste and other locations where the nuclear material intended for non-nuclear use is present;
- Information to be provided by states regarding all buildings on a site;
- Information to be provided by states regarding its nuclear fuel cycle, research and development activities that are not concerning nuclear material;
- Information to be provided by states regarding the production and trade of sensitive nuclear-related equipment and material, and access of IAEA to production and import locations of the states;
- IAEA examination of environmental samples outside the acknowledged locations, when deemed essential by the IAEA;
- A streamlined system for designation of IAEA inspectors, and the issuance of multiple entry/exit visas as well as IAEA use of - internationally recognised systems of communications.

According to the Additional Protocol, the IAEA can carry out complementary sweeping in order to be assured of the absence of undeclared nuclear material and the activities, to resolve any inconsistency regarding the information provided by a state, as well as to confirm the decommissioned status of any facility or Location Outside Facilities (LOFs) like hospitals where nuclear material was usually used.

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# 5.3 International Treaties and Arrangements for the Nuclear Technology

## 5.3.1 Nuclear Non-Proliferation Treaty

### 5.3.1.1 NPT Origins

The IAEA has been playing a vital role in the form of safeguards system, over the past four decades, under the Nuclear Non-proliferation Treaty (NPT). This safeguards system is a noticeable international success in the course of cutting the civil uranium diversion into the military usage. Its chief feature is the collaboration with the nuclear energy development and further ensuring the usage of plutonium, civil uranium and associated plants are only for peaceful purpose. The NPT was extended indefinitely in 1995, and it leads to widening its scope for inclusion of undeclared nuclear activities.

Nuclear weapons have been renounced by many nations, since they acknowledge the fact that the possession of such weapons would threaten, instead of enhancing the national security. They have, thus, embraced the NPT in the form of a public obligation for the peaceful usage of the nuclear materials and technology.

In the year 1968, a milestone achievement in the non-proliferation history was observed with the successful conclusion of negotiations of the NPT. After being enforced in 1970, its indefinite extension in May 1995 was the other landmark achieved. Essentially, it was an agreement among the five nuclear weapons states and included other countries that were interested in nuclear technology. The deal was on the grounds that the assistance and cooperation would be traded for pledges, which will be backed by international scrutiny, directing that no plant or material would, in any case, be diverted to weapons use. States or parties refusing to be a part of the deal would not be included from international cooperation or trade that involves nuclear technology.

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### 5.3.1.2 Members and Non-Members

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As per the current status, 191 states are NPT parties, inclusive of all declared **Nuclear Weapons States'** (five states). **These five states had tested a nuclear missile before the year 1967; they are France, China, the United Kingdom, the United States and the Russian Federation.** The major countries which remain outside the NPT are Pakistan, Israel and India. They all are equipped with nuclear programmes which have approached to maturity since the year 1970. Hence, they first need to renounce and dismantle their nuclear weapons in order to join NPT membership. For India, internationally particular arrangements were agreed, in the year 2008, taking it half-way, and its further ratification of the Additional Protocol in 2014 places it on a similar footing to the five nuclear weapon states (NWS).

**India declined to sign the NPT** because of the following reasons.

1. As per the NPT, the countries which tested nuclear devices before 1967 are termed as 'nuclear weapons states', which India cannot be a member of.
2. There was no mention of any fixed timelines for disarmament.
3. The NPT is an unjust treaty, as there was no obligation of nuclear weapon states for giving them up while non-nuclear states are not allowed to have them.

### 5.3.1.3 India's Emergence as *de facto* Weapons State Under NPT

**Being a nuclear-armed country, India could not join the NPT.** As per the current state of India and the associated lack of full-scope IAEA safeguards, it was pointed that India faced insult from the world trade by the Nuclear Suppliers Group; however, a clean chit to the trade restraint got approval in September 2008, as recognition to its impeccable non-proliferation credentials.

**KEY POINTS**

The NPT is a multilateral treaty, aimed at limiting the spread of nuclear weapons including three elements:

1. Non-proliferation

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2. Disarmament
3. Peaceful use of nuclear energy.

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These elements constitute a 'grand bargain' between the five nuclear weapon states and the non-nuclear weapon states. The treaty was signed in 1968 and entered into force in 1970. Presently, it has 191 member states.

### Implications

1. States without nuclear weapons will not acquire them.
2. States with nuclear weapons will pursue disarmament.
3. All states can access nuclear technology for peaceful purposes, under safeguards.

### Key Provisions

1. ***The Treaty defines Nuclear Weapon States (NWS)*** as those that had manufactured and detonated a nuclear explosive device, prior to 1 January 1967. All the other states are, therefore, considered ***Non-Nuclear Weapon States (NNWS).***
2. ***The five Nuclear Weapon States are*** China, France, Russia, the United Kingdom and the United States.
3. ***The Treaty does not affect the right of state parties to develop, produce and use nuclear energy for peaceful purposes.***

### Role of States

1. Nuclear Weapon States are not to transfer to any recipient whatsoever nuclear weapons and not to assist, encourage, or induce any NNWS to manufacture or otherwise acquire them.
2. Non-Nuclear Weapons States are not to receive nuclear weapons from any transferor, and are not to manufacture or acquire them.
3. NNWS must accept the IAEA safeguards on all nuclear materials on their territories or under their control.

**A regulation was passed in the US Assembly, in December 2006,** enabling nuclear trade with India and consequently, a nuclear collaboration accord was concluded with India in July 2007. This opened new means to enable the country's involvement in international business in nuclear equipment and fuel. As a result, India was required to lay mostly its nuclear power reactors under the safeguards of the IAEA and shut down the CIRUS research reactor by the year 2010, finally enabling the country to recycle the US-originated nuclear fuel that is sourced from other foreign countries at the new plant, under the safeguards of the IAEA. Hence, the fuel already used occurring from the 14 reactors is nominated as clearly civilian and under the full safeguards of the IAEA.

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The deal was accepted by the IAEA as a form of 'a creative break with the past' wherein the country was barred from the NPT. It faced much deferment in the country's Parliament, and ultimately, a fresh as well as safeguards agreement in a comprehensive manner by the IAEA, with a supplementary Protocol, was set up. An agreement was sanctioned by the board of the IAEA, in the year July 2008, after it had endangered to topple down the government. This accord is same as that among the non-nuclear weapons states and the IAEA, particularly **INFCIRC 66**.

Next came the unanimous resolution of the 48-member NSG for exempting India from trade prohibition with the NPT non-members. A trade joint agreement was, thus, made to the US Congress for its final sanction, which was followed by France and Russia. Mapping India, in level with China, along with its accountabilities and trade prospects is the primary aim behind it.

The introduction to safeguards agreement of India with the IAEA states access of India to assured deliveries of fresh fuel to be an 'essential basis' for the acceptance of the states of IAEA safeguards on certain areas of its reactors and that it is rightful to undertake curative measures to that, ensuring its nuclear civil reactors non-stop operation, in case there is foreign fuel supplies distraction. Besides this, it also states that India will 'give reassurance against the safeguarded nuclear material withdrawal from the civilian usage at any time'. Assurances about weapon testing were also provided during the course of the NSG.

**A nuclear trade agreement was contracted between India and France in October 2008 after the bill sanctioning civil nuclear trade with India was passed by US.**

**In early 2009, the safeguards agreement was signed between India and the IAEA, although the time-frame to get the extra reactors under safeguards was yet to be finalised. The IAEA Board agreed upon the Additional Protocol to the safeguards agreement in March 2009.** India signed the same in May 2009, that is, a couple of months later. The new government announced the approval with a list of 20 facilities that includes six at the Nuclear Fuel Complex, Hyderabad and the two stores at Tarapur and 12 reactors in June 2014. The enforcement of the Additional Protocol was carried out on 25 July 2014, providing the IAEA heightened access to civil power facilities of India.

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Thereafter, the UN Security Council was told by India that given its capability and inclination for the objectives of non-proliferation of the world, **it has obeyed the guidelines of the NSG as well as the Missile Technology Control Regime (MTCR).** India, as a country with the capability and willingness for promoting global non-proliferation objectives, believes that the next logical step is membership of India of the four export control regimes. **The other two 'regimes' include the informal Australia Group (re-chemical and bio-weapons) and the Wassenaar Arrangement on export control for the conventional arms and dual-use goods and technologies.** Not only this, India also backs the early commencement of the discussions in the Conference of Disarmament on a **Fissile Material Cut-off Treaty in Geneva.** Moreover, India will pursue association with the four export control regimes, in the course of following the sanctions of the Additional Protocol.

Besides the safeguards, a primary intent of the Fissile Material Cut-off Treaty is capping the weapons-grade fissile materials generation in the NPT Nuclear Weapons States as well as India, Pakistan and Israel. The country, such as India, has articulated its backing for a Cut-off Treaty, while the opposition is from China, Pakistan and Iran. The United States though is not keen on such a treaty being verifiable.

### 5.3.2 Nuclear Suppliers Group

It is a multilateral control regime group and nuclear supplier countries group, which identify to contain nuclear proliferation through the curbing of materials, equipment and technology exports used in the manufacturing of nuclear weapons. Currently, the NSG has 48 members, including five nuclear weapon states such as the United States, the United Kingdom, France, China and Russia. It is not a formal organisation, and its guidelines are non-binding in nature and membership decision for any nation can be made with censuses.

- Hence, India conducted its first nuclear test, Pokhran-I (Operation Smiling Buddha), in May 1974, under the leadership of Prime Minister Indira Gandhi.

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- The nuclear powers were convinced that the nuclear NPT alone would not be sufficient to contain the spread of nuclear weapons and accordingly, the **NSG was formed in 1974.**
- As per the current guidelines of the NSG, a country cannot be a member of the NSG if it is a non-NPT state, thereby, keeping India out of the group.
- In the year 1998, India again conducted the second nuclear test named under Operation Shakti termed as 'Pokhran-II'. However, new sanctions were imposed on India by Western countries, especially the United States.
- Pre-2005, the NSG was starved of fuel of the Tarapur Atomic Power station, while the United States used MTCR regulations to thwart the cryogenic engine technology transfer from Russia.
- In the year 2008, the country got a little help when it agreed to the **Indo-US civil nuclear deal,** in which India signed a civil-military separation plan and India-IAEA safeguard agreement and in return, the US diplomacy managed to get India an NSG waiver.
- At the time of visit to India in November 2010, US President Barack Obama announced support for India's participation in the Nuclear Suppliers Group, the Wassenaar Arrangement, the Australia Group and the MTCR, in a phased manner.
- Due to India's formal pledge for not sharing any sensitive nuclear technology or material and maintaining its voluntary suspension on testing nuclear weapons, the NSG-participating governments granted India a 'clean waiver' and made the country eligible to receive advanced nuclear technologies to enrich uranium and reprocess plutonium.
- However, the non-membership of the NSG Group has made the country devoid of the latest nuclear technologies. In 2016, India applied for NSG membership.
- The majority of the 48-member NSG group supported India's stand for NSG membership, but China along with New Zealand, Ireland, Turkey, South Africa and Austria were opposed to India's admittance into the NSG group. China insisted the signing of the NPT Treaty by India in order to get NSG membership and further, wants to have a non-discriminatory criterion for states' admission into the NSG Group.

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Basically, China's resistance clearly indicates its stand of supporting Pakistan's entry into the NSG group.

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### 5.3.3 The Wassenaar Arrangement

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The Wassenaar Arrangement consists of an elite group of countries, which is a Multilateral Export Control Regime (MECR), subscribing to arms export control, similar on the line of the NSG and MTCR. The agreement is based on Export Controls for conventional arms and dual-use goods and technologies consisting of 42 participating countries including all the permanent members of UN Security Council except China and consisting of many former Comecon (Warsaw Pact) countries. **The agreement was set up in order to contribute towards regional and international security and stability in transfers of conventional arms and dual-use goods and technologies through the promotion of transparency and greater accountability.**

**It was formed on 12 July 1996 in Wassenaar, near the Hague, Netherlands** and is the descendant to the Cold War Era Coordinating Committee for Multilateral Export Controls (CoCom). The Wassenaar Arrangement primarily focused on the lucidity of national export control regimes and is considered less strict than CoCom. Its administrative Secretariat is located in Vienna, Austria. Akin to CoCom, however, it is not a treaty, and therefore, is not legally binding. **As per the existing procedure, every six months, member countries exchange information on conventional arms deliveries to non-Wassenaar members, thereby, falling under eight categories such as battle tanks, Armoured Fighting Vehicles (AFVs), large-caliber artillery, military aircraft, military helicopters, warships, missiles or missile systems and small arms and light weapons.**

#### 5.3.3.1 India as a Member of the Wassenaar Agreement

**India was admitted as a 42nd member of the Wassenaar Agreement in December 2017.** This further polished the country's non-proliferation credentials, despite not being party to the NPT and further broadened its access to sensitive technologies, so as to cater to the demands of the space and defence sector. **Since the signing of the US-India Civil Nuclear Deal in 2005, India was constantly trying to acquire membership of the**

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**NSG, MTCR, Austrian Group and Wassenaar Agreement.** Resultantly, with the backing of the United States, **India got the membership of the MTCR in 2016** and thereafter, was admitted as a member of the Wassenaar Agreement in the year 2017. **With the WA membership, India is able to get easier access of dual use technologies and military equipment and its licensing which are basically banned for non-participating members. The country can also sell its nuclear reactors and other equipment and materials without attracting any adverse reaction and is in a better position to collaborate with other countries in the development of such capabilities.**

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### KEY POINTS ABOUT NSG

- **Brought in 1974–**in response to the Indian nuclear test (Smiling Buddha).
- It is a **multilateral export control regime.**
- It is a **group of nuclear supplier countries that seek to prevent nuclear proliferation** by controlling the export of materials, equipment and technology that can be used to manufacture nuclear weapons.
- The NSG first met in November 1975 in London, and is thus, popularly referred to as the '**London Club**'.
- It is **not a formal organisation, and its guidelines are not binding.** Decisions, including membership, are made by consensus.
- **Membership:** 48 supplier states.

#### Criteria for Membership

- Ability to supply items (including items in transit) covered by the annexes to parts 1 and 2 of the NSG Guidelines;
- Adherence to the Guidelines and action in accordance with them;
- Enforcement of a legally based domestic export control system which gives effect to the commitment to act in accordance with the Guidelines;
- Full compliance with the obligations of one or more of nuclear non-proliferation agreement.
- Support of international efforts towards non-proliferation of weapons of mass destruction and of their delivery vehicle.

#### Why the membership is important for India?

- Membership will **increase India's access to state-of-the-art technology from the other members of the Group.**
- Access to technology and being allowed to produce nuclear equipment will give a boost to the **Make in India programme.** That will, in turn, boost the economic

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growth of our country.

- As per India's INDC under the Paris Climate Agreement, commitment to reducing dependence on fossil fuels and ensuring that 40% of its energy is sourced from renewable and clean sources. In order to achieve this target, we need to scale up nuclear power production. This can only happen if India gains access to the NSG.
- India will get **an opportunity to voice its concern if in case of change in the provision of the NSG guidelines. Other benefits associated with NSG membership, once admitted, include the following:**
  - ♦ India will get timely information on nuclear matters.
  - ♦ It can contributes by way of information.
  - ♦ Its credentials will get confirmed.
  - ♦ It can act as an instrument of harmonisation and coordination.
  - ♦ It will be a part of a very transparent process.

### 5.3.4 The Australia Group

**It is a Multilateral Export Control Regime (MECR) and an informal countries group, set up in 1985, which seeks to identify the exports that do not contribute to the development and spread of chemical or biological weapons.** Initially, the group consisted of 15 members, which hosted its first meeting in Brussels, Belgium, in September 1989. With the inclusion of India on 19 January 2018, it has currently 43 members (42 members and the EU). Countries such as China, Pakistan, North Korea and Iran are not members. It is named after the Australian initiative to form the group and its secretariat is also managed by Australia. **There is no legally binding obligation on the member countries of the Australia Group. Primarily, the group was focused on chemical agents, but by 1990, the mandate was expanded to include biological agents.** In order to discuss the enhanced way of effectiveness of participating nations for preventing would be proliferators for getting CBW materials and its use, the group meets every year. The nation's parties to the Australia Group are also parties to the Chemical Weapons Convention and the Biological Weapons Convention.

#### 5.3.4.1 India and Australia Group

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**India was admitted as the 43rd member of the group on 19 January 2018.** The joining of India in the Australian Group will give a big boost to getting membership of the NSG Group. This will also help to raise the country's status in the field of NPT, despite being non-signatory to the Treaty and further will help in acquiring the latest and critical technologies. **The country is the first among the South Asian countries to acquire the membership of the Group.** To restrict China, which is curbing India's entry to the NSG Group, India's joining of the Australia Group gives an upper hand to India in becoming a member of the NSG Group.

**WARSAW PACT**

The Warsaw Pact, formally recognised as the 'Treaty of Friendship, Cooperation and Mutual Assistance', from 14 May 1955 to 1 July 1991 was a collective defence treaty, signed in Warsaw, Poland, among the Soviet Union and seven Soviet satellite states of Central and Eastern Europe such as Albania, Bulgaria, Czechoslovakia, East Germany, Hungary, Poland and Romania during the Cold War. The treaty provided for a united military command and for the maintenance of Soviet military units on the territories of the other participating states. The Warsaw Pact was the military complement to the Council for Mutual Economic Assistance (CoMEcon) and was established in order to counter NATO.

### 5.3.5 Missile Technology Control Regime (MTCR)

The Missile Technology Control Regime (MTCR) is a multilateral and voluntary partnership for the non-proliferation of missiles, capable of carrying chemical, biological and nuclear weapons of mass destruction and unmanned aerial vehicle technology, capable of carrying above 500 kilograms payload for more than 300 kilometres among 35 countries. It was established in April 1987 by the G7 countries, such as Canada, France, Germany, Italy, Japan, the United Kingdom and the United States, being a non-legally binding treaty. The MTCR controls the technologies and materials exports, which are involved in ballistic missile systems and unmanned aerial vehicles, particularly capable of carrying nuclear warheads of above 500 kilograms payload for more than 300 kilometres. The MTCR

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regime is a non-treaty member countries association, having certain guidelines, regarding the sharing of information, control laws and missile systems export policies and a regulation mechanism, so as to curb the transfer of technologies. Now, 35 full members, including India, are the members of the MTCR and four 'non-adherent members' – Israel, Macedonia, Romania and Slovakia, excluding China, which is not a member of this regime. These efforts are being strengthened by 'The International Code of Conduct against Ballistic Missile Proliferation', also known as the Hague Code of Conduct (HCOC), which was set up on 25 November 2002, as an arrangement to prevent the proliferation of ballistic missiles with 136 UN member countries, including India. Under this regime, the members must have national policies, which should govern the export of ballistic missiles, cruise missiles, unmanned aerial vehicles, space launch vehicles, drones, remotely piloted vehicles, sounding rockets and their components.

#### 5.3.5.1 India as a Member of the MTCR

India was admitted as a full member of the MTCR in June, 2016, with the consensus of other nations and also agreed to join the HCOC which will further strengthen its case for the membership of other MECRs such as the NSG. With this membership, there will be a boost to the defence sector and further enable India to purchase high-end missile technology and further facilitate India in the running of joint programmes for the development of unmanned aerial vehicles with other countries. Also, India will be able to sell its missiles easily to other non-member countries, such as selling of BrahMos to Vietnam, making India a significant arm exporter for the first time. With an entry in the MTCR, the probability of India with its joint venture with Russia and the United States for exporting of UAVs, such as Reaper and Global Hawk and Israel missile defence interceptor, such as Arrow II have enhanced. The ISRO will also get access to the forbidden cryogenic technology from Russia for use in space exploration operations with MTCR membership. MTCR membership can be used to put pressure

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on China for not being a member of the MTCR and clearing the way for India's membership for the NSG.

### 5.3.6 Comprehensive Nuclear-Test-Ban Treaty

The Comprehensive Nuclear-Test-Ban Treaty (CTBT) is the **treaty banning all nuclear explosions – everywhere, by everyone.** The Treaty was negotiated at the Conference on Disarmament in Geneva and adopted by the United Nations General Assembly. **It opened for signatures on 24 September 1996 but it is yet to enter into force because it must be signed and ratified by 44 specific nuclear technology holder countries, eight of which have yet to ratify it : China, Egypt, India, Iran, Israel, the Democratic People's Republic of Korea, Pakistan and the United States.**

- The CTBT is the last barrier on the way to develop nuclear weapons. It curbs the development of new nuclear weapons and the improvement of existing nuclear weapon designs.
- **When the Treaty enters into force, it provides a legally binding norm against nuclear testing.** The Treaty also helps to prevent human suffering and environmental damages caused by nuclear testing.

**The UN Chief Antonio Guterres reiterated his appeal to eight nations,** including India and the United States, to **ratify the CTBT,** saying the failure to bring it into force undermines global efforts to ensure a world free of atomic weapons.

## 5.4 India's NFU Policy

It refers to a pledge or a policy by a nuclear power, **not to use nuclear weapons as a means of warfare unless first attacked by an adversary using nuclear weapons.** Earlier, the concept had also been applied to chemical and biological warfare.

**India first adopted a 'No first use' policy after its second nuclear tests, Pokhran-II, in 1998.** In August 1999, the Indian government

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released a draft of the doctrine which asserts that nuclear weapons are solely for deterrence and that **India will pursue a policy of 'retaliation only'.**

The document also maintains that India 'will not be the first to initiate a nuclear first strike but will respond with punitive retaliation should deterrence fail' and that decisions to authorise the use of nuclear weapons would be made by the Prime Minister or his 'designated successor(s)'.

Note: The Defence Minister of India has given major hints about a possible review of the policy of 'no first use' of nuclear weapons and said it will depend on circumstances in future.

### 5.4.1 Why India Should Retain This Policy?

- Adopting a 'no first use' policy **enables New Delhi to keep the nuclear threshold high,** especially as Pakistan tries to lower the threshold by developing tactical nuclear weapons, the Hatf-9 with 60 km range.
- It must also be noted that New Delhi is not bordered by just one nuclear weapon state. China adopts a 'no first use' policy and in spite of calls for Beijing to revise its 'no first use' doctrine, it is unlikely to do so. Hence, **if New Delhi gave up its 'no first use' doctrine, it could give Beijing a chance to adopt a first-strike policy and shift blame on India.**
- In fact, India's adoption of a first-strike policy would be an **easy excuse for Beijing to give up its 'no first use' doctrine against the United States and Russia as well.**
- Moreover, India has always promoted herself as a responsible nuclear weapon state. Hence, **a first-strike policy would severely damage - India's reputation as a responsible nuclear weapon state.**
- Also, it is India's 'no first use' doctrine that has enabled both Pakistan and India to keep their nuclear arsenal in a de-mated posture rather than a ready deterrent posture. This means nuclear warheads are not mated with the delivery systems. **This reduces the chances of nuclear terrorism in Pakistan and also reduces the likelihood of an accidental launch of a nuclear weapon.** A first-strike policy by India may not have allowed Pakistan to keep their nuclear arsenal in a de-mated posture.

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- **A first-strike policy, coupled with a ballistic missile defence system, could provoke Pakistan to launch a nuclear pre-emptive strike** against India.
- By adopting a 'no first use' doctrine, New Delhi has also made it evident that nuclear weapons are indeed the weapons of last resort. **Abandoning this doctrine would make it evident that India considers the option of using nuclear weapons in the initial phases of the conflict.**

## 5.5 Nuclear Deals of India and Nuclear Diplomacy

Because of non-participation in the global forums and treaties like the Nuclear Suppliers Group, Comprehensive Test Ban Treaty (CTBT), the Nuclear NPT, etc., India was not a part of the nuclear commerce of the world, for a long time.

On signing the 123 agreement, also known as Indo-US civil nuclear deal, in 2008, a major breakthrough was achieved in this sector. Under the deal, it was agreed by India to isolate its civil nuclear facilities from the military nuclear facilities and thus keep all its civil nuclear facilities under the IAEA, that is, IAEA safeguards. In exchange for this, full civil nuclear cooperation with India was decided to be worked about from the end of the United States. However, the execution of this deal took more than three years. Numerous complex stages, like the revision of the US domestic law, as in the Atomic Energy Act of 1954, especially the civil-military nuclear Separation Plan in India, the India-IAEA safeguards (inspections) Agreement, as well as the allowance for an exemption for India by the NSG, the export-control cartel, that was formed chiefly in response to the first nuclear test conducted by India in 1974, were the major developments under the regime of this deal.

### 5.5.1 Nuclear Energy Agreements Signed between India and Different Countries

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- **France** became the first country to sign a civilian nuclear agreement with India on 30 September 2008, which happened only after the complete waiver was provided by the NSG. Nicolas Sarkozy, the French President, during his visit to India in December 2010, signed the General Framework Agreement and the early Works Agreement between NPCIL and M/s AREVA for the implementation of EPR (Extended Producer Responsibility) for the Jaitapur Nuclear Power Project (JNPP). Following the M/s AREVA's restructuring, French utility EDF was designated as the lead agency for negotiations and implementations of the JNPP so on 22 March 2016, EDF and NPCIL signed a revised MoU for the construction of six EPR units at Jaitapur of 1650 megawatts each.
- India has been assisted well by **Russia** to plan a nuclear plant for its nuclear submarine. Under a new treaty signed, it was accepted to provide India the full freedom to progress with the closed fuel cycle, including mining, fuel preparation for reactors and reprocessing of spent fuel. An agreement was signed between India and Russia, in 2010. As per this agreement, India will acquire 12 nuclear reactors from Russia that will include one to be commissioned in Kudankulam (KKNPP) (where a Russian reactor is already in operation) and for the rest, the site is yet to be decided. KNPP units 1 and 2 have already become operational and the construction of units 3 and 4 and 5 and 6 is going on. The plant is expected to start operating at full capacity in 2027.
- **India and Mongolia** signed the crucial civil nuclear agreement on 15 June 2009 regarding the supply of uranium to India.
- **India** signed a civil nuclear energy agreement with **Namibia** on 2 September 2009, permitting the supply of uranium from the African country.
- A civil nuclear agreement was signed between **India and Argentina**, enabling cooperation and strategic partnership. Both the countries have decided to encourage and upkeep scientific, technical and profitable collaboration for a mutual benefit in this field.
- **India and Canada** also signed a civil nuclear co-operation agreement on 28 June 2010 offering access to the growing nuclear market of India

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and also to fuel its reactors. They accord as the only users of heavy water nuclear technology.

- An agreement was contracted between **India and Kazakhstan** for support in peaceful uses of atomic energy. This contract visualises a legal structure to supply fuel, construct and operate atomic power plants. Besides these, exploration and joint mining of uranium, the give-and-take system of scientific and research materials, as well as reactor safety mechanisms and the use of radiation expertise for healthcare were included as other visions of the contract.
- **India and South Korea** also engaged into a nuclear agreement on 25 July 2011 enabling South Korea's participation in the nuclear expansion plan of India and to bid for erecting the nuclear power plants in India.
- **India-US** nuclear collaboration is also called as 123 Agreement.
- In 2014, a civil-nuclear agreement was signed between **India and Australia,** permitting the export of uranium to India. Australia stands as the world's third largest uranium manufacturer and hence, the agreement enables the supply of uranium for peaceful power generation for public usage in India.
- A nuclear deal was signed in 2015, between **India and Japan,** though there were apprehensions for Japan regarding the liability clause and the tracking clause, the two countries in November, 2016 signed a civil-nuclear deal over the peaceful uses of nuclear energy and with this India is the first non-NPT country to sign a nuclear deal with Japan. This landmark deal came into force in July 2017, which would enable Japan in exporting of nuclear power plant technology, as well as providing finance for the management of nuclear power plants in India. Through this deal, Japan will assist India in the management of nuclear waste and further undertake of joint manufacture of components of the nuclear plant under the Make-in-India programme.
- **India and Vietnam** signed a civil nuclear deal in December 2016 for the peaceful uses of nuclear energy and further strengthening the technical cooperation in nuclear energy. Vietnam is the fourteenth country with which India has signed a nuclear deal on the research areas of nuclear reactors. The Memorandum of Understanding asks for the cooperation between the DAE's Global Centre for Nuclear Energy

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Partnership (GCNEP) and the Vietnam Atomic Energy Institute (VinAtom).

- India signed a trilateral agreement in March 2018 with **Russia and Bangladesh** for the cooperation in the civil nuclear field. This will allow Indian companies in the construction and installation of nuclear power project on foreign soil. The parties will cooperate in the field of personnel training, consultation and material and equipment supply in non-critical category for undertaking the construction of Roppur nuclear power plant project in Bangladesh.

## 5.6 Nuclear Liability Act, 2010

Nuclear Liability Act 2010 is also known by another name as the **Civil Liability for Nuclear Damage Act 2010.**

**AIM:** To provide a civil liability for instances of nuclear damage and prompt compensation to the victims of the nuclear incident through a NO FAULT LIABILITY to the operator. The appointment of the claims commissioner and the establishment of nuclear damage claims commission.

NO FAULT LIABILITY implies 'absolute legal responsibility for an inquiry that can be imposed on the wrong doer without proof of carelessness or fault'.

This Act was one of the must steps, required to activate the 2008 Indo-US civilian nuclear agreement. Though, India is not a party to NPT; however, it is still allowed to carry nuclear commerce with the rest of the world. India becomes a member of the International Convention on Liability in the civil nuclear arena after passing this Act.

1. The maximum amount of liability is capped to ₹500 crore, to be paid by the operator of the Nuclear Plant, by the Act.
2. In case of the cost of damages exceeding this amount, the central government will pay up to ₹300 million.

### 5.6.1 Necessity of this Act

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The amount of electricity produced by the nuclear power plant is planned to be increased to 14,000 megawatts by 2020 and to 27,500 megawatts by 2032 and up to 25% of electricity will be generated by nuclear power plants by 2050, as per India's plans.

It also serves to bind operators and governments legally and financially that can provide relief, if any unwanted accident happens.

### 5.6.2 Criticism

- The Nuclear Power Corporation of India Limited, which functions as the operator, is a government-owned body. Hence, in case of any nuclear damage, the entire compensation has to be borne by the Government for the operator's side too.
- **Clause 17(a) and (b):** Only the operator can sue the manufacturers and suppliers and not the victims. Also, the recourse taken by the operator is only ₹1500 Crore.
- Too short time limit to make claim (within 10 years), which may not be sufficient, if there is any long-term damage.
- **Clause 35:** In the case of a nuclear accident, the operator or responsible persons will undergo the trial under the Nuclear Damage Claims Commission and besides that, no court is given the authority. This may divide the country into zones with each zone having a claims commissioner.
- On the contrary, the US counterpart, the Prime Anderson Act, enables lawsuit and criminal proceedings proceed under US court.
- **Clause 46:** The suppliers are exposed to domestic court law, under which citizens can drag them to court for settlement of Compensation. Clause 45 provides discretionary power to wave liability.

The United States has issues with the clauses 17(b) and 46 since they could enable to seek compensation from the supplier. This leads to the feeling that this is a big burden on them and that this could ultimately lead to unlimited liability on the suppliers.

### 5.6.3 Suggestions and Way Forward

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1. A Nuclear Insurance Pool needs to be established for the country through a Nuclear Liability Fund by operationalizing the normal surcharge of say 5 paise per unit of nuclear power that is produced. Even at this rate, at current production level, about ₹200 crore annually could be provided.
2. With a view to move ahead with the strategic development, the legacy of mistrust needs to be removed. The first thing is ratification of the convention on supplementary compensation, which was signed by India in 2010, post ratification; the Indian government will be strengthened to stand on the interpretation. This will enable the NPCIL to open preliminary techno-economic discussion with the US suppliers, while offering similar kind of assurances to other foreign suppliers.

## 5.7 Phasing Out of Nuclear Power in Europe

A nuclear power phase-out refers to the termination of usage of the nuclear power for energy production. It is often introduced due to concerns about nuclear power, it usually includes shutting down nuclear power plants and thereby, taking a step towards renewable energy and fossil fuels.

Three nuclear accidents have influenced the discontinuation of nuclear power, namely **the 1979 Three Mile Island partial nuclear meltdown, which happened in the United States; the 1986 Chernobyl disaster in the USSR; and the 2011 Fukushima nuclear disaster that happened in Japan.**

**Sweden (1980) accords to be the first country to begin a phase-out, which, in 2009, was repealed.** It was followed by Italy (in 1987), Belgium (in 1999) and Germany (in 2000). Austria and Spain have also enacted laws to cease the construction of new nuclear power stations. Many other European countries have debated phase-outs.

Following the March 2011 Fukushima nuclear disaster, eight of its 17 reactors were permanently shut down by Germany and it pledged to close the rest by the end of 2022. In late 2021, all but three of the remaining

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German nuclear power plants were shut down. **All German nuclear power plants were shut down in April 2023.**

Italy too voted overwhelmingly to keep their country non-nuclear. Moreover, the construction of new reactors has been banned by Switzerland and Spain.

As of June 2024, there were 210 nuclear reactors permanently shut down worldwide. The United States recorded the largest number of shutdowns, with 41 units followed by the UK with 36 reactors; the least being Pakistan with just one reactor. Shutdowns refer to a subcritical reactor state as defined by technical specifications and also require other factors like the reactor control key being secured.

## 5.8 New START Treaty

The **New START Treaty** was in between the United States and the Russian Federation for further reduction and limitation of strategic offensive arms.

This treaty has verification measures at on-site inspections and exhibitions, to monitor nuclear arms. Under this treaty, there is a provision to exchange all arms exchange data to each and other nation.

**New START Treaty** duration is for ten years, unless superseded by a subsequent agreement. After the termination of the 2002 Moscow Treaty, New START Treaty entered into force. This treaty will not restrict the testing, development or deployment of current or planned missile defence programmes of the United States.

**MT to MW (MegaTonne to MegaWatt):** In this provision, Russia will provide atom bomb based highly enriched uranium (in tonne weight) to the United States, which the United States will utilise for electricity generation (in unit of watt).

## 5.9 Nuclear Security Summit

**To warn the world against nuclear terrorism, Obama hosted the first Nuclear Security Summit (NSS) in Washington in 2010.** The main aim of

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this summit was to draw attention towards the use of secure nuclear material so we can prevent nuclear terrorism worldwide. **This summit is conducted on a periodical basis after every two years.** There were four summits in total – in Washington, D.C. in 2010, Seoul, South Korea in 2012, the Hague, Netherlands in 2014 and Washington, D.C. again in 2016.

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**NEW START DEAL**

- Signed on 8 April 2010 in Prague, and, after ratification entered into force on 5 February 2011.
- Replaced the Treaty of Moscow (SORT), which was to expire in December 2012.
- The New START treaty is a successor to the START I treaty, which expired in December 2009. The proposed START II treaty was signed but never entered into force, and the START III treaty negotiations were never concluded.

**Impact of the New START Treaty**

- Number of strategic nuclear missile launchers reduced by half.
- Replacement of SORT mechanism by a new inspection and verification regime.
- The number of deployed strategic nuclear warheads limited to 1,550 which is down nearly two-third from START I treaty.
- Number of deployed and non-deployed InterContinental Ballistic Missile (ICBM) launchers, Submarine Launched Ballistic Missile (SLBM) launchers and heavy bombers equipped for nuclear armaments limited to 800.

## 5.10 International Panel on Fissile Material

The International Panel on Fissile Material (IPFM) was established in 2006, as a group of 18 countries, which are independent nuclear experts with an objective to secure and sharply reduce stocks of highly enriched uranium and also separated plutonium. **Due to these initiatives, key materials of nuclear weapons will be protected and limit its further production.** (List of IPFM countries – Brazil, Canada, China, France, Germany, India, Iran, Japan, Netherland, Mexico, Norway, Pakistan, South Korea, Russia, Africa, Sweden, the United States and the United Kingdom).

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## 5.11 World's Major Nuclear Disasters (at Level 7, 6 and 5)

1. **Chernobyl, Ukraine, 1986 – Level 7:** It has been thought to be the worst nuclear accident in history, being one of the only two that are classified as the Level 7 event on the International Nuclear Event Scale (the other is the Fukushima Daiichi disaster in 2011). The official Soviet fatality count of 31 deaths has been dubious. Not only this, long-term effects like cancers and deformities are still being accounted for.
2. **Fukushima, Japan, 2011 – Level 7:** The Fukushima Daiichi nuclear disaster (the largest nuclear tragedy since the Chernobyl disaster of 1986) represents a series of catastrophes (Tohoku Tsunami) resulting in equipment failures, nuclear meltdowns and release of radioactive materials at the Fukushima Nuclear Power Plant, that took place on 11 March 2011. It is also accorded as the second disaster (along with Chernobyl) to size a Level 7 on the INES (International Nuclear and Radiological Event Scale).
3. **Kyshtym, Russia, 1957 – Level 6:** On 29 September 1957, a radiation pollution incident occurred at Mayak (a nuclear fuel reprocessing plant in the Soviet Union) and is known as the Kyshtymnuclear disaster.
4. **Windscale Fire (Sellafield), UK, 1957 – Level 5:** The worst nuclear disaster in the history of the Great Britain was Windscale Fire (Sellafield) and is measured as a level 5 on the INES scale.
5. **Three Mile Island Accident, Pennsylvania USA, 1979 – Level 5:** On 28 March, two nuclear reactor failures occurred, and hence, it was recorded as the worst disaster in commercial nuclear power plant history. Due to this disaster, radioactive gases and radioactive iodine were released into the environment. Fortunately, not a single type of cancer has been linked with the accident as per the studies that have been conducted.

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6. **Goiania Accident, Brazil, 1987 – Level 5:** A radioactive contamination accident harmed the Brazilian state of Goais on 13 September 1987. An old radiotherapy source was stolen from an abandoned hospital site in the city. Subsequently, many people carried and mishandled it, which ultimately cost four lives. Besides this adverse effect of the accident, 112,000 people were examined for radioactive contaminations with 249 having significant levels of radioactive material in or on their body.

## QUESTIONS AND ANSWERS

### Objective Questions

1. **In India why are some nuclear reactors kept under 'IAEA Safeguards' while others are not?** **[UPSC (Pre) – 2020]**
   (a) Some use uranium and others use thorium
   (b) Some use imported uranium and others use domestic supplies
   (c) Some are operated by foreign enterprises and others are operated by domestic enterprises
   (d) Some are State-owned and others are privately-owned

2. **What is/are the consequence/consequences of a country becoming the member of the 'Nuclear Suppliers Group'?** **[UPSC (Pre) – 2018]**
   1. It will have access to the latest and most efficient nuclear technologies.

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2. It automatically becomes a member of 'The Treaty on the Non-Proliferation of Nuclear Weapons (NPT)'.

Which of the statements given above is/are correct?

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**3. In the Indian context, what is the implication of ratifying the 'Additional Protocol' with the 'International Atomic Energy Agency (IAEA)'?** **[UPSC (Pre) – 2018]**

(a) The civilian nuclear reactors come under IAEA safeguards.
(b) The military nuclear installations come under the inspection of the IAEA.
(c) The country will have the privilege to buy uranium from the Nuclear Suppliers Group (NSG).
(d) The country automatically becomes a member of the NSG.

**4. Consider the following statements:** **[UPSC (Pre) – 2017]**

(i) The Nuclear Security Summits are periodically held under the aegis of the United Nations.
(ii) The International Panel on Fissile Materials is an organ of International Atomic Energy Agency.

Which of the statements given above is/are correct?

(a) Only (i)
(b) Only (ii)
(c) (i) and (ii)
(d) None of these

**5. Consider the following countries:** **[UPSC (Pre) – 2015]**

(i) China
(ii) France
(iii)India

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(iv) Israel
(v) Pakistan

Which among the above are Nuclear Weapons States, as recognised by the Treaty on the Non-Proliferation of Nuclear Weapons, commonly known as Nuclear Non-Proliferation Treaty (NPT)?

(a) (i) and (ii)
(b) (i), (iii), (iv) and (v)
(c) (ii), (iv) and (v)
(d) All of these

**6. The 'New START' treaty was in the news. What is this treaty?** **[UPSC (Pre) – 2011]**

(a) It is a bilateral strategic nuclear arms reduction treaty between the United States and the Russian Federation.
(b) It is a multilateral energy security cooperation treaty among the members of the East Asia Summit.
(c) It is a treaty between the Russian Federation and the European Union for the energy security cooperation.
(d) It is a multilateral cooperation treaty among the BRICS countries for the promotion of trade.

**7. Recently, the United States decided to support Indian's membership in MECRs called the 'Australian Group' and the 'Wassenaar Arrangement'. What is the difference between them?** **[UPSC (Pre) – 2011]**

(i) The Australia Group is an informal arrangement, which aims to allow exporting countries to minimise the risk of assisting chemical and biological weapons proliferation, whereas the Wassenaar Arrangement is a formal group under the OECD holding identical objectives.
(ii) The Australia Group comprises predominantly Asian, African and North American countries, whereas the member countries of Wassenaar Arrangement are predominantly from the European Union and American continents.

Which of the statements given above is/are correct?

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(a) Only (i)
(b) Only (ii)
(c) (i) and (ii)
(d) None of these

## Answer Key

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (b) | 2. (a) | 3. (a) | 4. (d) | 5. (a) | 6. (a) | 7. (b) | |

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**UNIT-III** Space Technology

# CHAPTER 6
# VARIOUS INSTITUTIONAL SETUPS FOR SPACE RESEARCH AND CONCEPT OF ORBITS AND LAUNCH STATIONS

## IN THIS CHAPTER...


6.1 Space Technology
6.2 Indian Space Research Programme (ISRP)
6.3 Institutional Setup
6.4 ISRO and Its Prominent Regional Centres
6.5 ANTRIX: Antrix Corporation Limited
6.6 New Space India Limited Council (NSIL)


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## 6.7 Different Types of Satellite Orbits

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6.7.1 Low Earth Orbit (LEO)

6.7.2 Medium Earth Orbit (MEO)

6.7.3 Geostationary Earth Orbit (GEO)

If the orbital period of satellite exactly matches the orbital period of Earth at an altitude of 35,863 kilometres above the Earth's surface and rotates in the equatorial plane of the Earth with angular velocity of 3075 metres per second, such an orbit is called geosynchronous orbit.

6.7.4 High Earth Orbit (HEO)

6.7.5 Polar versus Polar Sun-synchronous Orbit

This orbit also called as '**dawn-to-dusk orbit**' because satellite trails the Earth's shadow

6.7.6 Transfer Orbit

6.7.7 GPS Orbit/Semi-synchronous Orbit's

6.7.8 Geosynchronous Orbit versus Geostationary Orbit

6.7.9 Geosynchronous Transfer Orbit (GTO)

## 6.8 Satellite Launching Systems

After reaching into the space, the satellite does not fall down due to balance of centrifugal forces and Earth's gravity

## 6.9 Why Satellites do not Fall Down?

## 6.10 Basic Concepts

6.10.1 Satellite

6.10.2 Space Shuttle

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6.10.3 Spacecrafts

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6.10.4 Launch Vehicle
6.10.5 Rocket
6.10.6 Missile

## 6.1 Space Technology

It is a very dynamic technology of the space science or the aerospace industry, serving various applications like satellites, shuttles, spaceflights and various space exploration work. Space technology has a multidimensional role in human life and caters to everyday services like weather forecasting, remote sensing, DTH, business networking, GPS systems, etc. Astronomy and Earth Science (via remote sensing) also greatly benefit from space technology. It is also an income-generation industry to send various foreign satellites and also provides other benefits to many recipient nations.

## 6.2 Indian Space Research Programme (ISRP)

**Vikram Sarabhai** is the pioneer of the Indian Space Programme with various objectives in the promotion of science and technology application and development for the country's socio-economic benefit.

These objectives are:

1. Peaceful purposes of outer space.
2. Operational space services will establish in a self-sufficient manner.
3. In the beginning, Indian Space Programme was involved in the designing and fabrication of satellites and development of launch vehicles to launch various satellites.

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## 6.3 Institutional Setup

In the year 1962, **Indian National Committee for Space Research (INCOSPAR)** was formed under Dr. Vikram Sarabhai, to formulate Space Programme of India. On the recommendation of this committee, ISRO was formed (**Figure 6.1**). The formulation of ISRO, thus regularised the country's space activities. Now, it is under the control of the Department of Space (DoS) and reports to the Prime Minister.

### *Department of Space (DoS)*

Department of Space (DoS) is the nodal agency for space activities in India and have various features: space infrastructure, space transportation systems and at international level, it also involves in the outer space activities, governed by the United Nations (UN) treaties and principles evolved under **UN Committee on Peaceful Uses of Outer Space** (COPUOS) in particular.

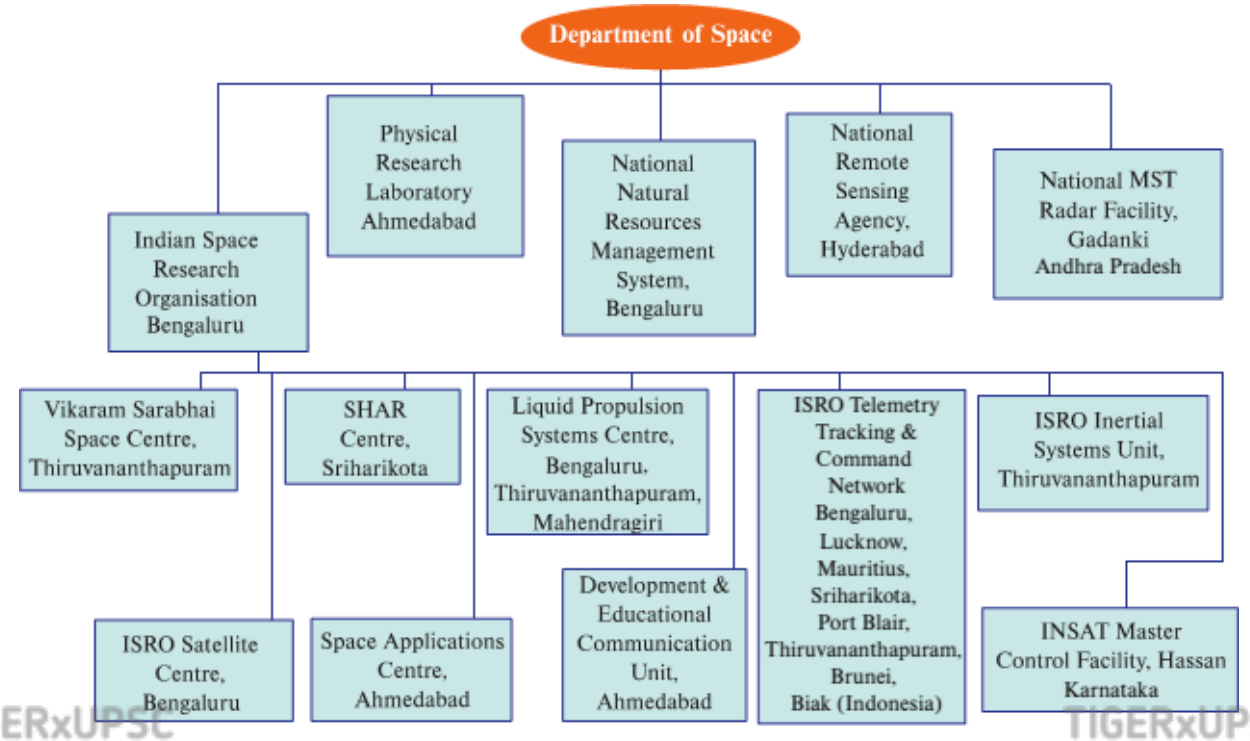

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Figure 6.1 Various Institutions and Organisations under the Department of Space (DoS)

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## 6.4 ISRO and Its Prominent Regional Centres

ISRO plays a leading role in India for space research with the help of various prominent regional centres (**Figure 6.2**). These are given below:

1. ISAC – ISRO Satellite Application Centre, Bengaluru.
2. VSSC – Vikram Sarabhai Space Centre, Thiruvananthapuram.
3. SDSC – Satish Dhawan Space Centre, Sriharikota.
4. SAC – Space Application Centre, Ahmedabad.
5. LPSC – Liquid Propulsion System Centre, Mahendragiri, Tamil Nadu.
6. ISTRAC – ISRO Telemetry, Tracking and Command Network, Bengaluru.
7. MCF – Master Control Facility, Bhopal (MP) and Hassan (Karnataka).
8. NRSA – National Remote Sensing Agency, Hyderabad.

1. **ISRO Satellite Application Centre (ISAC), Bengaluru:** It is responsible for designing and fabricating satellites, which include remote sensing satellite and geostationary satellite.
2. **Vikram Sarabhi Space Centre (VSSC), Thiruvananthapuram:** It serves as the major centre for developing satellite launch vehicle and further placing satellite in the Earth's orbit. It has successfully developed four generations of launch vehicles, namely SLV, ASLV, PSLV and GSLV.
3. **Satish Dhawan Space Centre (SDSC), Sriharikota (Andhra Pradesh):** It is the satellite launching station of the country. It was earlier called Sriharikota Range (SHAR) and renamed SDSC in 2002 (after Satish Dhawan, former chairman of ISRO, who also worked as the secretary in DoS in 2005).

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4. **Space Application Centre (SAC), Ahmedabad:** It is accountable for conceptualisation and carrying out of different space research projects. Besides, it also conducts research anchoring the practical space technology application for usage of civilians.
5. **Liquid Propulsion System Centre, Mahendragiri (LPSC), Tamil Nadu:** It is responsible for Satellite Propulsion Systems Integration for Geostationary Satellite and IRS programmes. India has developed its own cryogenic engine here.
6. **ISRO Telemetry, Tracking and Command Network (ISTRAC), Bengaluru:** It is responsible for providing various space operation services like spacecraft control, launch vehicle and low Earth orbiting spacecrafts with various deep space missions of ISRO and other space agencies around the world. The additional responsibilities of ISTRAC include development of RADAR systems for tracking and atmospheric applications and establishing Ground Segment Network for Indian Regional Navigation Satellite System.
7. **Master Control Facility (MCF), Hassan in Karnataka and Bhopal in Madhya Pradesh:** They observe and control all the geostationary satellites. They also carry out operations related to initial orbit rising of satellites, payload testing in orbits and various on-orbit operations of satellite throughout its life.
8. **National Remote Sensing Agency (NRSA), Hyderabad:** It strives to realise the Indian space vision and acts as a key player in Earth Observation Programme and Disaster Management. NRSC is also responsible for acquisition, processing and satellite-based remote sensing data. NRSA continuously explores the practical uses of remote sensing technology for multi-level applications. It helps with the required trained manpower via capacity building in remote sensing applications.

## 6.5 ANTRIX: Antrix Corporation Limited

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Antrix Corporation Limited is the commercial wing of ISRO. It helps in promotion and commercialisation of space products, technical consultancy services and also transfer of technologies developed by ISRO. It was incorporated as a private limited company in September 1992. It provides various services and space products to international customers worldwide. It was awarded ‘Miniratna’ status by the government in 2008.

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***Missions***

- Satellites and its related technology development and design for Earth observation, communication, navigation, meteorology and space science.
- Launch vehicles and its related technologies development and design for providing access of space.

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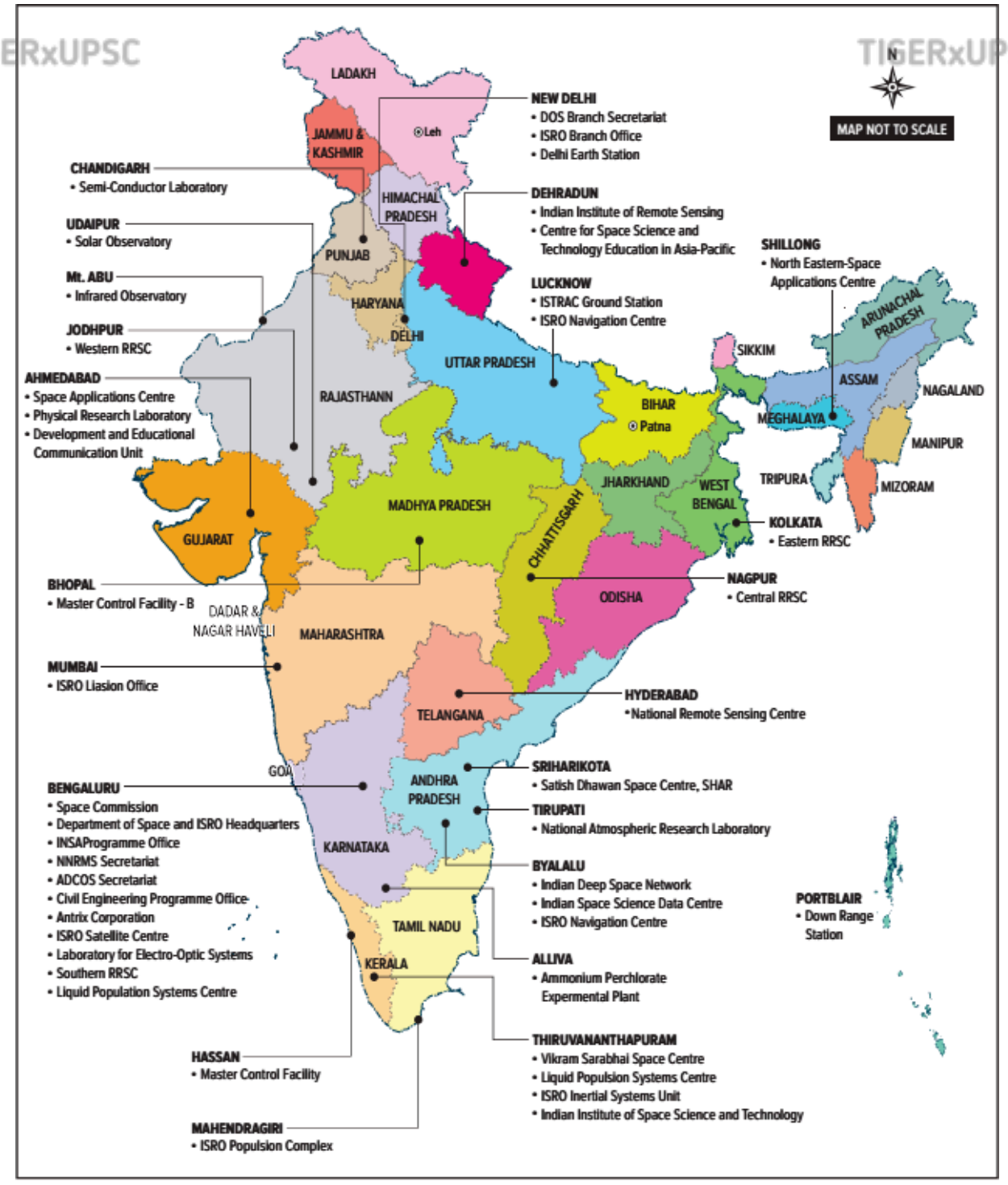


Figure 6.2 Various Sites of Indian Space Research Programme (Source: Department of Space)

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## SPACE-BASED PROGRAMME

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### 1. YUVIKA – Yuva Vigyani Karyakram (Young Scientist Programme)

- ISRO is organising a special programme for School Children called '**Young Scientist Programme**' YUVIKA, to impart basic knowledge on Space Technology, Space Science and Space Applications to the younger students in emerging trends in space science and technology amongst youngsters.
- The main thought behind the programme is to **'Catch Them Young'**.
- The programme is also expected to encourage more students to pursue in **Science, Technology, Engineering and Mathematics (STEM)**-based research/career.
- STEM education is an interdisciplinary programme that **provides a platform for innovation and creativity.** STEM education is now being made a part of the curriculum right from the elementary school. STEM education is different from regular science and mathematics. It talks about the **application of these techniques into the real world,** i.e., an integrating applied approach and cohesive learning techniques.

### 2. UNNATI – Unispace Nanosatellite Assembly & Training by ISRO

- ISRO, Department of Space, Government of India offers a **training programme on nanosatellite assembly and building** named UNNATI (Unispace Nanosatellite Assembly & Training).
- **Eligibility: Foreign Engineers/Scientists/Officials are eligible to get trained.**
- The **training provides an opportunity for the participating countries to strengthen their capabilities in assembling, integrating and testing nanosatellites.**
- India **announced this programme in 2018 in Vienna** during the Symposium to commemorate the 50$^{th}$ Anniversary of the First United Nations Conference on the Exploration and Peaceful Uses of Outer Space (UNISPACE+50).
- A total of 60 participants from 33 countries benefited from a 2-months residential programme.
- **The programme is conducted at ISRO's U.R. Rao Satellite Centre (URSC), Bengaluru.**

- Indian National Satellite (INSAT) programmes for telecommunication, television broadcasting and development applications.
- Space-based applications for societal development and disaster management support.
- Management of natural resources and monitoring of environment using space-based imagery by Indian Remote Sensing Satellite (IRS) programme.
- Research and development in space science and planetary exploration.

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## 6.6 New Space India Limited Council (NSIL)

New Space India Limited (NSIL) had been ***incorporated in 2019,*** as **a wholly owned Government of India Undertaking/Central Public Sector Enterprise (CPSE)** control of Department of Space (DoS) to commercially exploit the research and development work of Indian Space Research Organisation (ISRO) centres and constituent units of DoS. The company will spearhead ***various space products including production of launch vehicles, transfer of technologies and marketing of space products.***

***Role of NSIL***

- Small satellite technology transfer to industry, wherein the new company shall take licence from DoS/ISRO and sub-licence to industries.
- Manufacture of small satellite launch vehicle (SLV) in collaboration with the private sector.
- Product ionisation of Polar SLV through industry.
- Marketing of some spin-off technologies and products, both in India and abroad and any other subject which the Government of India deems fit.

**HOW NSIL IS DIFFERENT FROM ANTRIX?**

- ISRO already has a commercialisation arm Antrix Corporation Ltd. that looks at foreign markets. But as ISRO's capabilities expand and collaboration with industries increase, there is a need to expand the commercialisation activities as well as domestically.
- NSIL was kick started in May 2019 to monetise ISRO's research, products and services in space technology. Antrix focuses on foreign markets, but NSIL will focus on the domestic market. In the domestic space, there is a huge demand in earth observation, communication and navigation space.
- Antrix will handle ISRO's commercial deals for satellites and launch vehicles with foreign customers whereas NSIL will deal with capacity building of local industry for space manufacturing.

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- Product ionisation and marketing of space-based products and services, including launch and applications.

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## 6.7 Different Types of Satellite Orbits

Following are different types of satellite orbits (see **Table 6.1**):

- A satellite may orbit around the Earth in different planes. For instance, an orbit that is directly above the Earth's equator is known as equatorial orbit, while the one that passes over both poles is known as polar orbit.
- Other orbits are referred to as inclined orbits.
- The orbit may be circular (with the centre of the circle at centre of the Earth), or elliptical (with the Earth's centre at one of the two foci of the ellipse).
- The altitude of communication satellite can be Geostationary Earth Orbit (GEO), Medium Earth Orbit (MEO) or Low Earth Orbit (LEO). (Refer **Infographics – Types of Earth Orbits and Types of Orbits**).

Table 6.1 Different type of orbits and their distance from Earth's surface

| Orbit | Altitude |
|---|---|
| 1. Low Earth Orbit | 200–1500 kilometres (from the surface of Earth to 2000 kilometres distance is also considered as Low Earth Orbit) |
| 2. Medium Earth Orbit | 5000–12,000 kilometres (even 20,000 kilometres in the case of GPS) |
| 3. Geosynchronous Orbit | 35,786 kilometres (22236 miles) |
| 4. High Earth Orbit | Beyond 35,786 kilometres |

### 6.7.1 Low Earth Orbit (LEO)

- Low Earth Orbit satellites are often in polar orbit at altitude of 50–1500 kilometres and the orbital period in the range of 95–120 minutes.

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- LEO satellites try to ensure a high elevation for every spot on Earth, thus providing a high-quality communication link.
- The maximum time that LEO satellites are visible from a fixed point on Earth is up to 20 minutes. LEO is further classified into little LEO with Low Bandwidth Services (almost 100 bits per second), big LEO (almost 1 kilobits per second) and broadband LEO (almost 2 millibits per second).

***Advantages of LEO***

- It can provide true globe coverage and efficient use of spectrum.
- Due to lower path loss, use of smaller terminals becomes possible.
- The propagation delay is very low (10 milliseconds).

***Disadvantages of LEO***

- The mismatch of the orbital period of satellite and Earth results in serious consideration of Doppler effect.
- It requires large number of satellites for whole coverage.
- High probability of handover during a call.
- Failed satellites require to be replaced regularly because of their lower lifetimes.

### 6.7.2 Medium Earth Orbit (MEO)

- MEO satellite can be positioned somewhere between LEO and GEO, at an altitude of around 5000–12000 kilometres above the Earth's surface with inclination towards equator plane.
- So far, there have not been many satellites in this orbit, but some upcoming systems use this orbit.
- The orbital period of MEO satellite is about 6 hours. The visibility of satellite from a fixed point on Earth is about 60–100 minutes.

***Advantages of MEO***

- It can provide true globe coverage.
- It is possible to use smaller terminal because it provides lower path loss due to less distance.
- It provides medium propagation delay (80-milliseconds).
- MEO can cover large area, thus require less handover.

***Disadvantages of MEO***

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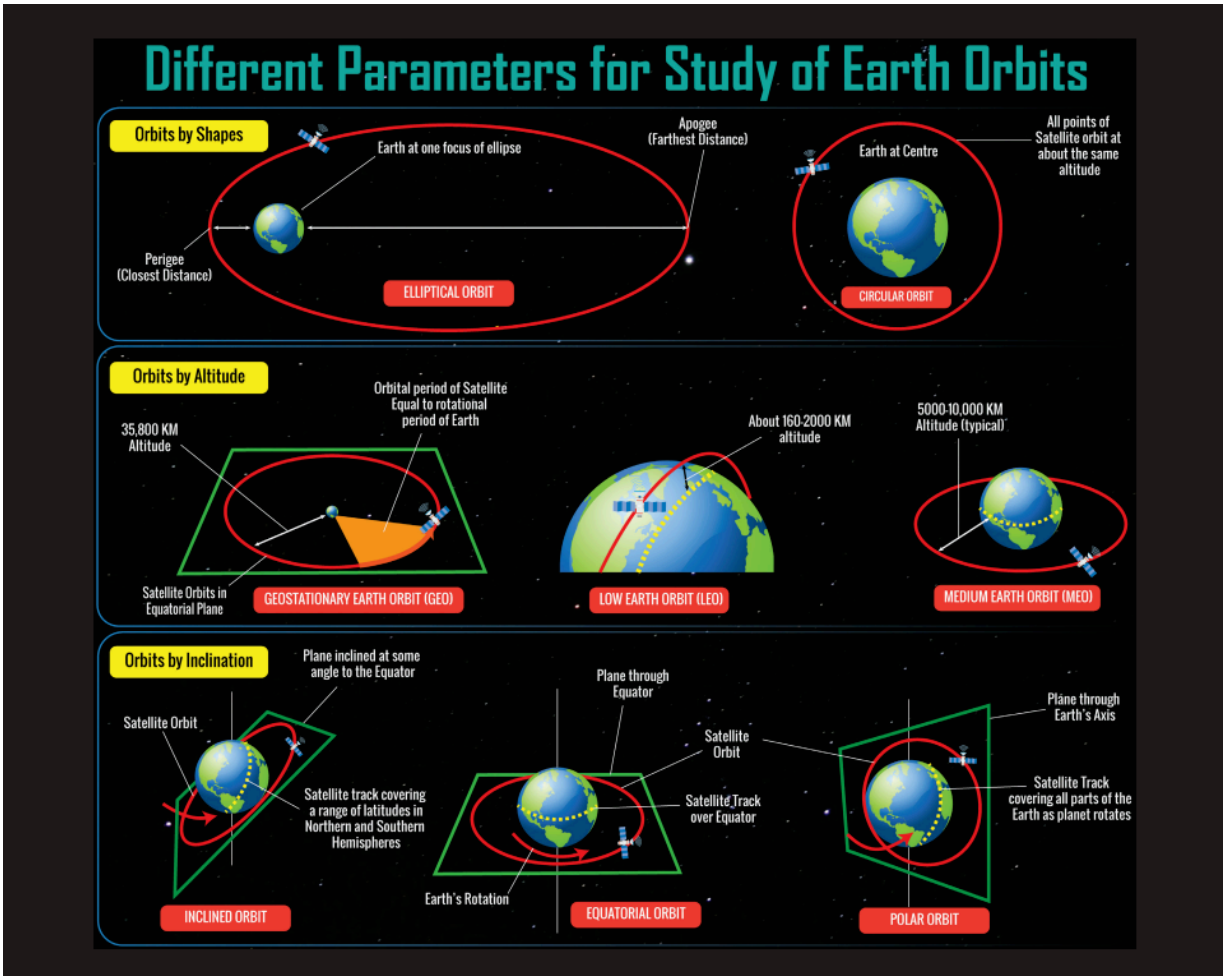
- It requires large number of satellites for whole coverage.
- Orbital period of satellite and Earth are mismatched, therefore Doppler Effect is considered.
- Required relatively long time to deploy a full constellation.
- Failed satellite needs to be replaced regularly due to their lower lifetime and larger number.

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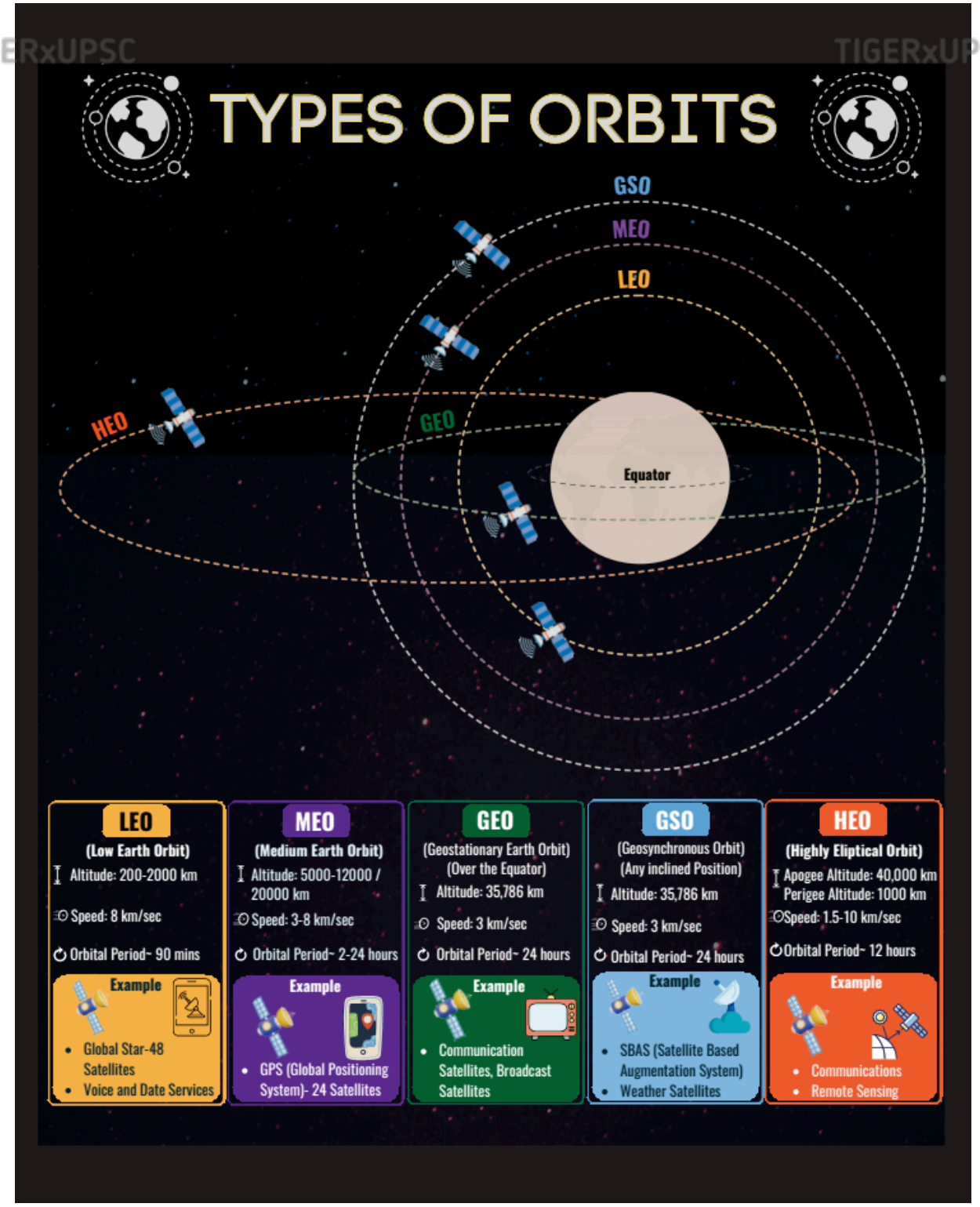


- The medium visibility of satellite leads to low probability of handover during call.

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### 6.7.3 Geostationary Earth Orbit (GEO)

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- The Geostationary Earth Orbit satellite is the most popular type of communication satellite.
- If the orbital period of satellite exactly matches the orbital period of Earth at an altitude of 35,863 kilometres above the Earth's surface and rotates in the equatorial plane of the Earth with angular velocity of 3075 metres per second, such an orbit is called geosynchronous orbit. When the inclination and the eccentricity of the orbit are zero, the satellite appears stationary to an observer on the ground. Thus, the orbit is termed as geostationary orbit.

***Advantage of GEO***

- Since the orbital period of satellite and Earth are almost same, the problem of Doppler Effect does not occur as relative motion is zero.
- The GEO satellite covers most of the inhabited portion of the Earth, excluding only the area near the north and south poles.

***Disadvantage of GEO***

- It provides relatively large propagation delay of 0.24 seconds. This affects voice and time data protocol.
- Relatively large path loss due to large distance travelled by signal.
- Relatively high launch cost.
- The polar region and far northern and southern hemispheres are poorly served by geostationary satellites.

### 6.7.4 High Earth Orbit (HEO)

Those satellite or space missions we send across the Earth's orbit (beyond 35,786 kilometres) are called as High Earth Orbit satellites, e.g. Chandrayaan-I, II, Mangalyaan I, Aditya Mission.

Note: LEO satellites are at polar position (Polar Satellite), MEO satellites are at inclined or transitional position and GEO satellites are at equatorial position for their own work.

### 6.7.5 Polar versus Polar Sun-synchronous Orbit

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Polar satellite cannot be used always to take images of the Earth's surface. Only for a certain time period, this satellite will be in front of Sun, for rest of the time, due to planetary movement of Sun and Earth position change satellite will see the shadow of the Sun on the Earth surface. The only solution to this problem is polar sun-synchronous orbit where the revolution of satellite is always in front of Sun as a result of changing their own inclination angle at every revolution. Since in this orbit, satellite appears to follow the path of Sun at every moment of time, it is called as **polar sun-synchronous orbit.**

This orbit also called as **'dawn-to-dusk orbit'** because satellite trails the Earth's shadow. The shadow is cast on the opposite side rather than on the side where the Sun shines. Since the satellite never moves into this shadow, the Sun appears to always shine on it. Due to closeness of satellite to the shadow, the part of the Earth on which the satellite is directly above, is always at sunset or sunrise. Hence, this orbit is called a dawn-to-dusk orbit. This enables the satellite to always have its solar panels towards the Sun. Generally, sun-synchronous orbits are medium or low orbits. **Radarsat (Canadian satellite) is an example of a satellite in a low sun-synchronous orbit. Radarsat is in orbit at 798 kilometres above Earth, at an angle of inclination of 98.6 degrees to the equator, as it circles the globe from pole to pole.** To keep its solar panels facing the Sun almost constantly, Radarsat relies on its dawn-to-dusk orbit. Radarsat can, henceforth, rely mostly on solar power rather than batteries.

(Refer **Infographic – Sun-Synchronous Orbit**)

### 6.7.6 Transfer Orbit

It is an intermediate orbit into which a satellite is released and from here, the satellite will subsequently be lifted up by making noise of its propulsion system.

### 6.7.7 GPS Orbit/Semi-synchronous Orbit's

**GPS Orbit/Semi-synchronous Orbits** (**Figure 6.3**) is an elliptical orbit at an altitude of about 5000–12,000 kilometres used for navigation purpose. A

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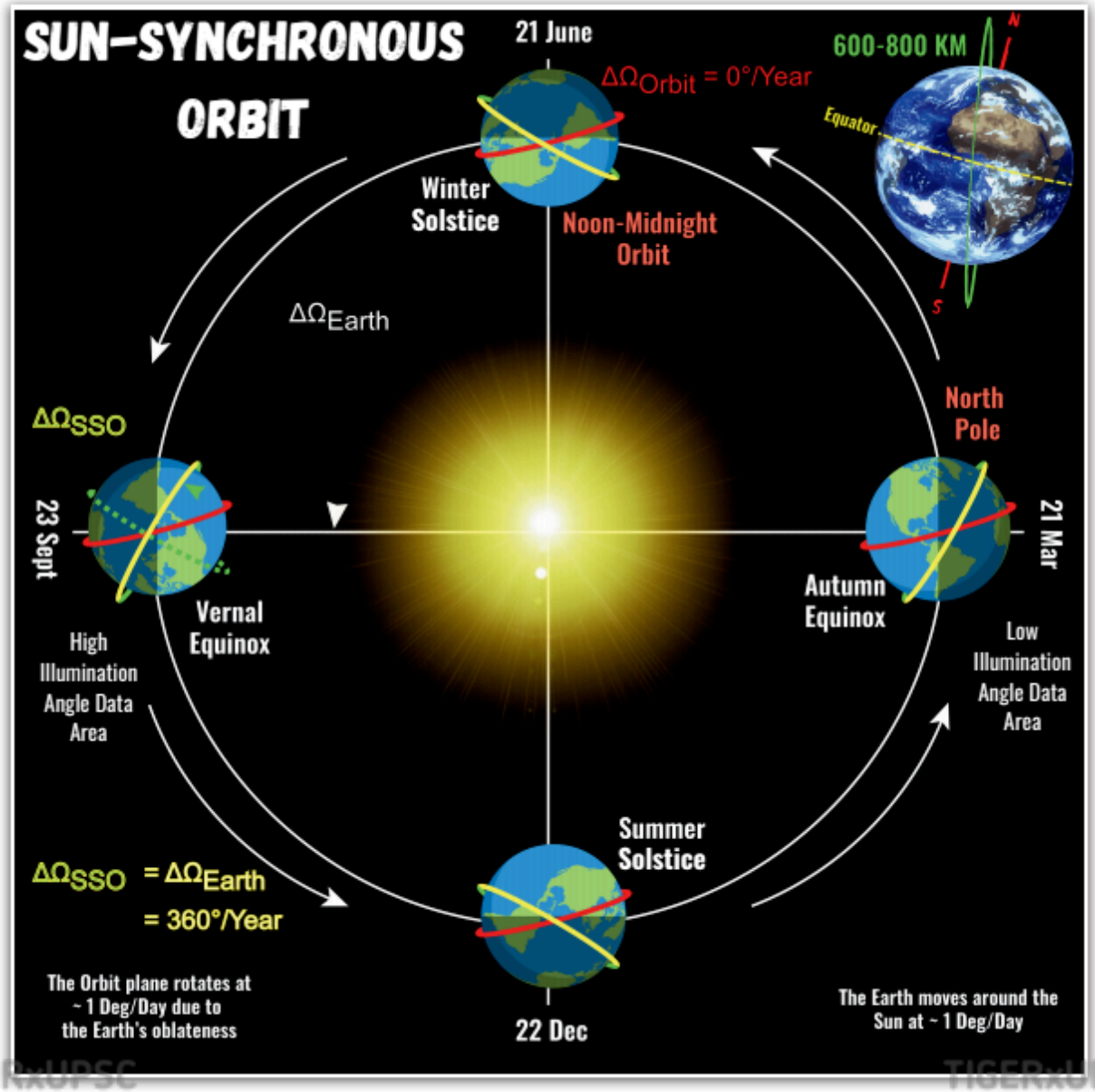
constellation of 24–36 satellites in MEO orbit (inclined or transitional in position) will be utilised to cover world navigation in Global Positioning Systems (e.g. **NAVSTAR** and **GLONASS)**. Three geostationary satellites can be used to cover the entire surface of Earth except polar area, so it cannot be used in global navigation system. In a country like India (those which are near to equator), for regional GPS, we can use GTO (Geosynchronous Transfer Orbit) and GEO (Geostationary Orbit) in combination, e.g. GAGAN and IRNSS.



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### 6.7.8 Geosynchronous Orbit Versus Geostationary Orbit

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A special case of geosynchronous satellite is the geostationary satellite, which has a geostationary orbit – a circular orbit directly above Earth's equator.

**Geosynchronous Orbit:** In this orbit, Earth's orbital period is equal to a sidereal day (23 hours, 56 minutes), in respect of its inclination. This implies that a person on a point on Earth will see a satellite in this orbit in the same place in the sky at the same time of the day, every day. Geosynchronous satellites have the advantage of remaining permanently in the same position in the sky, as viewed from a particular location on Earth.

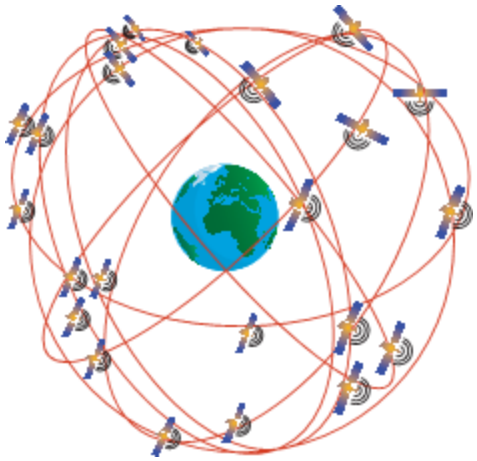

Figure 6.3 All GPS Satellites in Semi-synchronous Orbit

**Geostationary Orbit:** A *geosynchronous* orbit around Earth at 35,786 kilometres above the equator, thus appears stationary as seen from the Earth. Just like a star in the sky, a person at any point on Earth will see a satellite in this orbit stationary with respect to its position.

Note: Every geostationary orbit is a geosynchronous orbit but not vice-versa. A satellite will be geostationary if the revolution of a satellite is towards west to east, with respect to the rotation of Earth.

### 6.7.9 Geosynchronous Transfer Orbit (GTO)

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Geostationary Transfer Orbit (GTO) is an elliptical orbit, with an apogee (high point) of 35,784 kilometres and an inclination roughly equal to the latitude of the launch site, into which a spacecraft is initially placed before being transferred to a geosynchronous or geostationary orbit (**Figure 6.4**). [**Apogee:** This is a point in the orbit of a body (especially moon), or of a man-made satellite, at the farthest distance from the Earth].

## 6.8 Satellite Launching Systems

Since the eastern side of the east-coast has a water body, the satellite launching stations are generally located on the east-coast and as close as possible to the equator; for instance, Cape Canaveral on the south-east coast of Florida of the USA, northeast coast of French Guyana in South America (**Table 6.2**). The launch vehicle moves from west to east, same as the direction of Earth's rotation, so it is easy to up-thrust since Earth's rotation helps the satellite to launch it in the same direction.

Table 6.2 Some Important Rocket Launching Station of the World

| Rocket Launching Station | Country |
|---|---|
| Vandenberg Space Force Base | California |
| Rocket Lab Launch Complex 1 | Mahia, New Zealand |
| Al Anbar | Iraq |
| Baikonur | Kazakhstan (Formerly in the USSR) |
| Cape Canaveral | Florida, USA |
| Emamshahr | Iran |
| Edwards Air Force Base | California, USA |
| Jiquan Space Launch Centre | China |
| Kapustin Yar | Russia |

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| Rocket Launching Station | Country |
|---|---|

## 6.9 Why Satellites do Not Fall Down?

Satellite do not fall down to the surface of Earth even though a satellite is under the attraction of Earth (within Earth's orbit). It is defined on the basis of the orbital speed of satellites.

Because of Earth's gravity, everything thrown upwards or straight eventually comes down or falls down. However, in case of a satellite launched using a rocket, after reaching into the space, it does not fall down due to balance of centrifugal forces and Earth's gravity. This outward force (the centrifugal force) becomes greater, the faster the object swings. A certain 'orbital speed' is required to balance the centrifugal force of the satellite due to its motion around the Earth and pull of gravity so that the satellite is in proper orbit.

If a satellite is going farther from Earth, the weaker the gravity and hence, to balance the centrifugal forces and gravity, orbital speed of satellite will decrease (**Table 6.3**). The speed of a satellite in orbit is independent of the mass of the satellite.

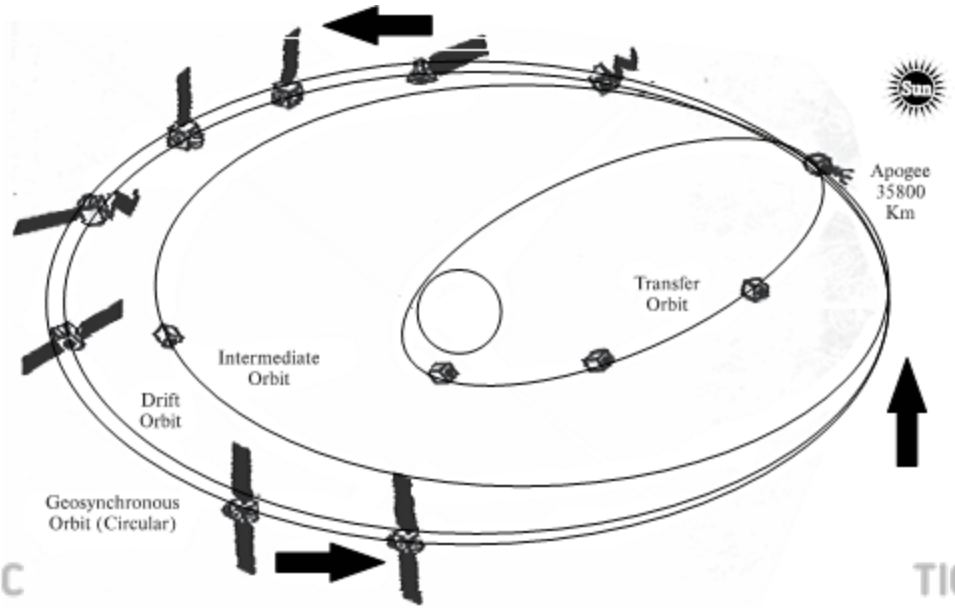


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Figure 6.4 Satellites follow different orbits to reach specific destinations – Transfer orbit, Intermediate orbit, Drift orbit and Geosynchronous orbit

Table 6.3 Selected values for the orbital speed, orbital period and altitude of satellites in circular orbits

| Altitude (km) | Orbital Speed (km/s) | Orbital Period (minutes) |
|---|---|---|
| 200 | 7.8 | 88.3 |
| 500 | 7.6 | 94.4 |
| 1,000 | 7.4 | 104.9 |
| 5,000 | 5.9 | 201.1 |
| 10,000 | 4.9 | 347.4 |
| 20,200 (Semi-synchronous) | 3.9 | 718.3 (12 hours) |
| 35,800 (Geosynchronous) | 3.1 | 1436.2 (24 hours) |

Satellites follow their own orbits as per Newton's first law of motion (Law of Inertia). According to Newton's first law of motion, in the absence of forces such as friction and air resistance, an object at rest will stay at rest and an object in motion will stay in motion. Consequently, when we put a rocket in motion, a satellite will stay in motion with the Earth's gravity bending its path from a straight line into an orbit. It means a satellite can stay in its own orbit for long periods of time, because it does not need to carry large amounts of fuel to keep moving.

Another key parameter to describe a satellite is the orbital period (total time covered in one revolution). As the altitude of the orbit increases, the satellite moves more slowly and must travel farther on each orbit, leading to an increase in the period with the altitude of the orbit. For low-altitude

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orbits (a few hundred kilometres altitude), the period is about 90 minutes; at higher altitudes, the period increases.

A satellite at an altitude of about 36,000 kilometres above the Earth's surface will roughly take 1,440 minutes to complete an orbit (at the same rate the Earth rotates). Such orbits are termed as geosynchronous. Those geosynchronous orbit-based satellites above the equator are also very unique, i.e. geostationary, because it stays over the same point on Earth. Such geostationary orbits have multiple uses.

## 6.10 Basic Concepts

### 6.10.1 Satellite

A satellite may be natural (moon) or human-made (machine) that orbits a planet or a star. For example, a moon is a satellite because it orbits Earth, similarly, Earth is a satellite as it orbits the Sun. An artificial 'satellite' refers to a machine that is launched into space and moves around Earth or other object in space. Satellites have numerous applications in the study of planets, like weather forecasting, study of galaxy, etc., to understand the solar system and universe better. All communication and information technology is also based on satellites. More than 20 satellites are required in a Global Positioning System (GPS). Soviet Union based Sputnik-1 was the first satellite in space launched in 1957.

### 6.10.2 Space Shuttle

Space Shuttle is a reusable rocket-launched vehicle that is designed to go into Earth's orbit. It is useful to send astronauts/cosmonauts and cargo to space. National Aeronautics and Space Administration (NASA) has developed the first space shuttle, formally called the STS (Space Transportation System), which lifted off into space first on 12 April 1981. A space shuttle consists of three major components:

- **Winged orbiter** – It carries both crew and cargo.

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- **An external tank** – It contains propellant (fuel such as liquid hydrogen and oxidiser as liquid oxygen for rocket engines).
- **A pair of large, solid-propellant, strap-on booster rockets** – During launch, the boosters and the orbiter's main engines are fired together to increase thrust. The boosters were returned to Earth by parachute mechanism for reuse after being jettisoned about two minutes just after lift-off.

### 6.10.3 Spacecrafts

Spacecraft is a machine or a vehicle developed to go into deep space. Space probes are robotic spacecrafts that support scientific research. These spacecrafts stay in trajectory around a planet and served as man-made satellites. Only few interstellar probes, such as *Voyager 1 and 2, New Horizons and Pioneer 10 and 11,* are on trajectories that leave the solar system.

***Type of Spacecrafts***

- **Orbiter Spacecraft:** This type of spacecraft is designed to go to distant planets or moon and enter their own orbit. They have high temperature resistance and power-storage capacity. For power generation, it is equipped with solar array (a group of solar panel). Examples: Galileo, Magellan and Mars Global Surveyor.
- **Manned Spacecraft:** It is used to carry astronauts into space. It requires life-support systems with crew compartment. They are designed to use either one time (like Soyuz) or reusable (like the space shuttle).
- **Flyby Spacecraft:** Flyby spacecrafts follow a continuous trajectory path to a target object. It is never sent into an orbit. Examples: Voyagers 1 and 2 and Pioneers 10 and 11.
- **Atmospheric Probe:** It is a small-sized instrumented craft that separates from the main spacecraft when it comes close to a planet. It involves the study of gaseous atmosphere. It is equipped with an aero shell to protect this probe during atmospheric entry and a parachute mechanism is adopted to slow its decent.

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- **Atmospheric Balloon Package:** It is used to study wind circulation patterns in a planet's atmosphere.
- **Lander Spacecraft:** Lander spacecraft is designed to reach the surface of a planet or moon as landing technology on the surface with longer survivability. It telemeters data back to the Earth. Examples: Surveyor, Mars Pathfinder, etc.
- **Surface Rover:** They mostly have wheels to rove on the surface of a planet or on other planetary bodies. It takes images and also does soil analysis for telemetering data back to Earth. India has planned to use wheeled rover in Chandrayaan-II mission.

### 6.10.4 Launch Vehicle

It is a rocket-powered vehicle used to transport a spacecraft into Earth's atmosphere, either into an inner orbit around Earth or an outer orbit into some other destination in outer space. Launch vehicles are being used to send unmanned space probes, manned spacecrafts and various satellites into space since the 1950s. India's used launch vehicles are as follows: RLV-TD, SLV, ASLV, PSLV and GSLV.

### 6.10.5 Rocket

Rocket engine is required to upthrust a missile, a spacecraft, an aircraft or other vehicles that are sent into space. Its efficiency is more in space than in atmosphere. They are lightweight, powerful and capable of generating large accelerations. Various uses of rockets are:

- **Sounding Rockets:** Sounding rockets are commonly used in science and research to carry instruments that take readings from 50 to 1,500 kilometres above the surface of the Earth.
- **Spaceflight:** For spaceflights, large-sized rockets are normally required to launch from a launch pad, providing stable support. These types of rockets are particularly useful if high speeds are required, such as orbital speeds at approximately 7,800 metres per second (28,000 kilometres per hour or 17,000 miles per hour) or more.

### 6.10.6 Missile

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A **missile** is a self-propelled precision-guided system. They **consist of four components:** targeting or missile guidance system, flight system, engine and warhead.

## QUESTIONS AND ANSWERS

### Objective Questions

1. **An artificial satellite orbiting around Earth does not fall down. This is so because the attraction of Earth:** **[UPSC (Pre) – 2011]**
   (a) does not exist at such distance.
   (b) is neutralised by the attraction of the moon.
   (c) provides the necessary speed for its steady motion.
   (d) provides the necessary acceleration for its motion.

2. **Satellites used for telecommunication relay are kept in a geostationary orbit. A satellite is said to be in such an orbit when:**
   1. The orbit is geosynchronous. **[UPSC (Pre) – 2011]**
   2. The orbit is circular.
   3. The orbit lies in the plane of the Earth's equator.
   4. The orbit is at an altitude of 22,236 km.

   Select the correct answer using the codes given below:
   (a) 1, 2 and 3
   (b) 1, 3 and 4
   (c) 2 and 4

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(d) 1, 2, 3 and 4

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**3. Which one of the following pairs is not correctly matched?** **[UPSC (Pre) – 2006]**

| | ***Well-known Place*** | | ***Country*** |
|---|---|---|---|
| (a) | Baikon our | : | Russia |
| (b) | Kourou | : | French Guiana |
| (c) | Borobudur | : | Indonesia |
| (d) | Cannes | : | France |

**4. Consider the following statements:** **[UPSC (Pre) – 2005]**

1. A geostationary satellite is at an approximate height of 10,000 km.
2. FM transmission of music is of very good quality because the atmospheric or man-made noises, which are generally frequency variations, can do little harm.

Which of the statements given above is/are correct?

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**5. *Assertion:* Artificial satellites are always launched from the Earth in the eastward direction.**

***Reason:*** **The Earth rotates from west to east and so the satellite attains the escape velocity.** **[UPSC (Pre) – 2002]**

(a) Both (A) and (R) are individually true and R is the correct explanation of A.
(b) Both (A) and (R) are individually true and R is not the correct explanation of A.
(c) A is true but R is false.

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(d) A is false but R is true.

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## Answer Key

| | | | | |
|---|---|---|---|---|
| 1. (d) | 2. (a) | 3. (a) | 4. (b) | 5. (b) |

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# CHAPTER 7
# SATELLITES AND LAUNCH VEHICLE TECHNOLOGIES OF INDIA WITH SPACE OBSERVATORY

## IN THIS CHAPTER...


**7.1 Experimental Satellites**

**7.2 Earth Observation Satellites**

7.2.1 NISAR: NASA-ISRO Synthetic Aperture Radar

**7.3 Indian Remote Sensing Satellite (IRS System)**


NISAR will be a future Earth observation satellite.


7.3.1 Applications of Indian Remote Sensing Satellites

7.3.2 Satellite Data Acquisition and Processing

**7.4 Communication Satellite–Indian Satellite (INSAT)/Geostationary Satellite (GSAT)**


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It is a combined initiative of the **India Meteorological Department, All India Radio (AIR), Department of Space (DoS), Department of Telecommunications (DoT) and Doordarshan.**

**7.5 Some Important Satellites**

7.5.1 India's First Private Satellite Constellation 'Firefly' Launched

7.5.2 Space-based Surveillance (SBS)

**7.6 Launch Vehicle Technology**

7.6.1 First Generation: Satellite Launch Vehicle

7.6.2 Second Generation: Augmented Satellite Launch Vehicle (ASLV)

7.6.3 Third Generation: Polar Satellite Launch Vehicle (PSLV)

PSLV-C37/Cartosat-2 Series Satellite – It has successfully injected 104 satellites into orbit – the highest number of satellites launched in a single mission so far.

7.6.4 Fourth Generation: Geosynchronous Satellite Launch Vehicle (GSLV)

7.6.5 Fifth Generation: GSLV MK- III

**7.7 Miscellaneous: Launch Vehicle and Missiles**

7.7.1 Small Satellite Launcher

7.7.2 Next-Gen Launch Vehicle (NGLV)

7.7.3 India's Anti-Satellite (A-SAT) Missile/Mission Shakti

7.7.4 Reusable Launch Vehicle – Technology Demonstrator (RLV-TD)

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The configuration of RLV-TD is similar to that of an aircraft and combines the complexity of both launch vehicles and aircraft.

#### 7.7.5 Pushpak: India's Reusable Launch Vehicle

Pushpak launch vehicle is India's efforts **to make space access more affordable.**


**7.8 Jet Engine Technology**
- 7.8.1 Normal Aircraft versus Jet Plane
- 7.8.2 Types of Jet Plane
- 7.8.3 Ramjet and Scramjet

**7.9 Indian Astronomical Observatory (IAO)**

**7.10 Project NETRA (Network for Space Object Tracking and Analysis)**


## 7.1 Experimental Satellites

Since its inception many years ago, Indian Space Research Organisation (ISRO) has launched a few experimental satellites for the purposes of remote sensing, orbital controls, atmospheric studies, etc. **Table 7.1** shows the list of such satellites launched by ISRO.

YouthSat is an Indo-Russian atmospheric satellite, built and developed by graduate, postgraduate and research scholars. This satellite is among the few satellites of ISRO that are built by students.

## 7.2 Earth Observation Satellites

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Besides the experimental satellites, ISRO has sent many missions with the objective of Earth observation, focussed mainly on climate, environment, disaster management, etc. At present, ISRO's IRS, i.e., Indian Remote Sensing system has one of the biggest satellite constellations that are presently operational. Starting from IRS 1A, HysIS, (Hyper spectral imaging satellite), ISRO launched its ScatSat 1 satellite, which is specifically launched for climate and environmental purposes. These satellites supply necessary data in a spectral, spatial and temporal resolution for global usage. In several applications covering agriculture, water resources, urban planning, rural development, forestry, ocean resources, mineral prospecting, environment and disaster management, such data is of immense use. Following is the most important Earth observation mission by ISRO.

### 7.2.1 NISAR: NASA-ISRO Synthetic Aperture Radar

In April 2012, India launched its first indigenous radar imaging satellite. With this satellite, India can easily observe visibility during both day and night and work in all types of weather and it will take images of Earth properly. Defining the roles and responsibilities of ISRO and NASA, a memorandum was signed between the two agencies in September 2014, by Barak Obama and Indian PM Narendra Modi (during Modi's visit to USA).

#### 7.2.1.1 NISAR Background

The first Joint Steering Group (JSG) meeting of NISAR was held on 21 July 2015.

**NASA Contribution:** In this joint mission, JPL/NASA will be responsible for the designing and development of L-band SAR, GPS system and data recorder.

**ISRO Contribution:** ISRO will be responsible for the designing and development of S-band SAR, spacecraft bus, data transmission system, spacecraft integration and testing, and launch using GSLV and on-orbit operations.

#### 7.2.1.2 Launch of NISAR

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The satellite will be launched from India in **2025–26 by GSLV Mark-II** launch vehicle of ISRO from Satish Dhawan Space Centre in Sriharikota, India. NISAR will be a future Earth observation satellite.

- **Orbit: Sun-synchronous/dawn-dusk orbit**
- **Altitude: 747 km**
- **Inclination angle = 98.5°**
- **Life span: 3 years**
- L-band (1.25 GHz, 24 cm wavelength) polarimetric SAR, to be produced by NASA
- S-band (3.20 GHz, 9.3 cm wavelength) polarimetric SAR, to be produced by ISRO
- **It will scan the globe every 12 days** (repeat cycle of 12 days) over the course of its **three-year mission** of imaging the **Earth's land, ice sheets and sea ice** to give an 'unprecedented' view of the planet.

**SYNTHETIC APERTURE RADAR (SAR)**

The name NISAR is shortened form of **NASA-ISRO-SAR.** SAR here refers to **the synthetic aperture radar** that NASA will use to measure changes in the surface of the Earth.

- Essentially, SAR refers to a technique for producing high-resolution images.
- Because of its precision, the radar can penetrate clouds and darkness, which means that it can collect data day and night in any weather.

A **synthetic aperture radar (SAR)** is an active sensor that first **transmits microwave signals** and then receives back the signals that are returned, or back scattered, from the Earth's surface. The instrument measures distances between the sensor and the point on the Earth's surface where the signal is backscattered.

SAR is a **coherent, mostly airborne or space borne, side looking radar system** which utilises the flight path of the platform to simulate an extremely large antenna, and that **generates high-resolution remote sensing imagery.**

Table 7.1 Different Experimental Satellite Launched by India

| Satellite Name | Launch Date | Launch Mass | Launch Vehicle | Orbit Type | Applic |
|---|---|---|---|---|---|
| INS-2TD | Feb 14, 2022 | | PSLV-C52/EOS-04 Mission | | Experime |

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| | | | | | |
|---|---|---|---|---|---|
| INS-1C | Jan 12, 2018 | | PSLV-C40/Cartosat-2 Series Satellite Mission | SSPO | Experime |
| INS-1A | Feb 15, 2017 | 8.4 kg | PSLV-C37/Cartosat-2 Series Satellite | SSPO | Experime |
| INS-1B | Feb 15, 2017 | 9.7 kg | PSLV-C37/Cartosat-2 Series Satellite | SSPO | Experime |
| YOUTHSAT | Apr 20, 2011 | 92 kg | PSLV-C16/RESOURCESAT-2 | SSPO | Student Satellite |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

- It will detect **movements of the planet's surface as small as 0.4 inches over areas** about half the size of a tennis court.
- NISAR will be equipped with the largest reflector antenna ever launched by NASA and its primary goals include tracking subtle changes in the Earth's surface, spotting warning signs, helping to monitor groundwater supplies and tracking the rate at which ice sheets are melting.

### 7.2.1.3 Key Features and Objectives

**Dual-frequency SAR:** NISAR is the first radar imaging satellite to use both L-band and S-band radar frequencies, allowing it to observe Earth's surface with high resolution and precision.

**Global Mapping:** The satellite will scan nearly the entire globe every 12 days, providing consistent data on various Earth processes.

**Monitoring Ecosystems:** NISAR will track changes in ecosystems, including vegetation patterns and biomass, and monitor the impacts of climate change.

**Ice Sheet Dynamics:** The mission will monitor the behavior of ice sheets, glaciers, and sea ice, providing insights into sea level rise and ice mass balance.

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**Natural Hazards:** NISAR will track and monitor natural hazards such as earthquakes, tsunamis, volcanoes, and landslides, helping scientists and disaster management agencies to better understand and respond to these events.

**Groundwater Monitoring:** The satellite will also provide data on groundwater resources, which is crucial for managing water resources and understanding hydrological cycles.

## 7.3 Indian Remote Sensing Satellite (IRS System)

India started its own Remote Sensing (IRS) satellite programme after successful tests of the Bhaskara 1 and Bhaskara 2 satellites. Further, National Natural Resources Management System (NNRMS) was established for the provision of active remote sensing data services. The Department of Space (DoS) was deemed the nodal agency. This data is received and distributed to various countries seeking privilege. The sectors of urban and infrastructural planning have seen a big push towards prosperity due to the applications of new high-resolution satellites and high performance applications.

Indian nomenclature is a bit straight forward. For instance, initially they used to name the satellites as 1A, 1B, so on. Later, the names were based on the area of application of the satellites (CartoSat, OceanSat).

### 7.3.1 Applications of Indian Remote Sensing Satellites

Indian Remote Sensing satellites are used for various applications of resources survey and management. Followings are the list of applications:

(a) In Space Based Inputs for Decentralised Planning
(b) In National Urban Information System
(c) In ISRO based Disaster Management Support Programme: Drought, flood, storm, Tsunami, etc.

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(d) In Hydro-geomorphological maps for locating underground water resources

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(e) In irrigation command area status monitoring

(f) Observation of snow-melt run-off estimates for planning water use in downstream projects

(g) In the land use and land cover mapping

(h) In urban planning

(i) In forest survey: Afforestation, Reforestation, Deforestation

(j) In the characterisations of biodiversity

(k) In the observation of crop area cover and production estimation of major crops

(l) In the drought monitoring and assessment

(m) In flood risk zone mapping and flood damage assessment

(n) In wetland mapping

(o) In environmental impact analysis

(p) In mineral prospecting

(q) In coastal studies

(r) Atmospheric Studies: Humidity, Pressure, Temperature, Pollution, etc.

- ❑ Land Productivity, Chlorophyll content in leaves
- ❑ Attack of locust
- ❑ Polar weather

All the data thus received from the satellites gets stored in Indian data centres and is available for public use *via* National Remote Sensing Centre (NRSC) data centres, and ISRO's Bhuvan Geoportal as open domain.

The Indian Institute of Remote Sensing and Centre of Space Science and Technology Education in Asia and the Pacific, both located in Dehradun, play an important role in the capacity-building programme of ISRO.

### 7.3.2 Satellite Data Acquisition and Processing

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**National Remote Sensing Centre (NRSC), Hyderabad,** is the nodal agency for achieving, receipting, handling and disseminating of remote

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sensing data in the whole country. The agency obtains and operates data from all the country's remote sensing satellites such as Resourcesat-1, IRS-1D, Oceansat-1, Cartosat-1, Cartosat-2 and foreign as well as Technology Experiment Satellites such as Terra, ERS and NOAA (**Table 7.2**).

Table 7.2 Different Earth Observation and Remote Sensing Satellites Launched by India

| | Launch Date | Launch Mass | Launch Vehicle | Orbit Type |
|---|---|---|---|---|
| EOS-07 | Feb 10, 2023 | | SSLV-D2/EOS- 07 Mission | |
| EOS-06 | Nov 26, 2022 | | PSLV- C54/EOS-06 Mission | |
| INDIA- BHUTAN SAT | Nov 26, 2022 | | | |
| EOS-04 | Feb 14, 2022 | | PSLV-C52/EOS-04 Mission | |
| EOS-03 | Aug 12, 2021 | | GSLV-F10/EOS-03 | GTO |
| EOS-01 | Nov 07, 2020 | | PSLV-C49 | LEO |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

Note: Earlier ISRO published both Remote Sensing Satellite (RSS) and Earth Observation Satellite (EOS) data separately, but now both Clubbed together in one table/list.

**COMPARATIVE STUDY OF EARTH OBSERVATION SATELLITES AND REMOTE SENSING SATELLITES**

Without being in direct contact with them, remote sensing technique is used for observing and analysing objects from a distance. During the Earth study, the term Earth observation is often used, which meant, the collected information from Earth by the observing satellites.

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The most widespread remote sensing systems measure reflected visible light, but the electromagnetic radiation can also be measured by them which is invisible to the human eye, e.g. infrared (IR), ultraviolet (UV) or microwave radiation. The reality is that the measurement done by satellites in different spectral wavelengths is used for acquiring information about the features and properties of the objects under study.

Some meteorological satellites are placed at around 36000 km above the equator, whereas other Earth resource and meteorological satellites circle Earth at about 800 km altitude. Some satellites view Earth infrequently with very high spatial resolution, even less than a metre, whereas other satellites image Earth frequently with low spatial detail.

Extensive range of phenomena and features can be monitored with the usage of Earth observation data, for example, phenology, vegetation biomass, land and sea surface temperature, ocean salinity, water quality and manifold, etc. Thus, Earth observation data present a powerful tool for the observance of processes which are human and nature induced, such as land cover changes, biogeochemical cycles, climate change and climatic variability.

## 7.4 Communication Satellite–Indian Satellite (INSAT)/Geostationary Satellite (GSAT)

ISRO initiated all-purpose geostationary series of satellites which accords to be the largest domestic communications system of the Asia Pacific, which is named as INSAT or Indian National Satellite System. It serves the objectives of broadcasting, meteorology and telecommunications, and search and rescue operations. It is a combined initiative of the **India Meteorological Department, All India Radio (AIR), Department of Space (DoS), Department of Telecommunications (DoT) and Doordarshan.** It includes various transponders for the various bands available for television and communications purposes serving the country. These satellites have the capability to pick up distress signals for undertaking operations of search and rescue operations in the regions of Indian Ocean and the South Asia.

**Established in 1983 with commissioning of INSAT-1B,** it initiated a major revolution in India's communication sector and sustained the same

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later. GSAT-17 joins the constellation of INSAT System consisting of 15 operational satellites, namely – INSAT-3A, 3C, 4A, 4B, 4CR and GSAT-6, 7, 8, 9, 10, 12, 14, 15, 16 and 18.

The INSAT system with more than 200 transponders in the C, Extended C and Ku-bands provides services to telecommunications, television broadcasting, satellite news gathering, societal applications, weather forecasting, disaster warning and search and rescue operations.

Earlier, the name of the satellite was INSAT but now GSAT is used. INSAT 1 and 2 series (except INSAT 2E) have expired but many of INSAT 3 and 4 series satellites are functional. List of communication satellites are mentioned in **Table 7.3**.

Table 7.3 List of Communication Satellite (INSAT/GSAT)

| | Launch Date | Launch Mass | Launch Vehicle | Application | R |
|---|---|---|---|---|---|
| GSAT-24 | June 24, 2022 | | | | |
| CMS-01 | Dec 17, 2020 | | PSLV-C50 | Communication | |
| GSAT-30 | Jan 17, 2020 | 3357 kg | Ariane-5 VA-251 | Communication | |
| GSAT-31 | Feb 06, 2019 | 2536 kg | Ariane-5 VA-247 | Communication | |
| GSAT-7A | Dec 19, 2018 | | GSLV-F11/GSAT-7A Mission | Communication | |
| GSAT-11 Mission | Dec 05, 2018 | 5854 kg | Ariane-5 VA-246 | Communication | |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

## 7.5 Some important satellites

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### 7.5.1 India's First Private Satellite Constellation 'Firefly' Launched

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#### Applications

The Firefly constellation has various applications, including:

- Monitoring crop health and early detection of crop diseases.
- Pinpointing water stress and managing water resources.
- Real-time deforestation monitoring and tracking ocean pollution.
- Supporting sustainable resource management and climate change efforts.

This allows for more detailed Earth observation, different types of vegetation, including monitoring crop health, detecting subtle changes in ecosystems, assessing environmental conditions and potentially tracking ocean pollution, Meterology.

#### Commercial Operations

Pixxel plans to integrate hyperspectral data with machine learning and AI tools to provide more actionable insights to industries and governments.

#### Future Expansion

Pixxel plans to expand the Firefly constellation to a total of 24 satellites.

In essence, the Firefly constellation represents a major step forward in India's space program and could revolutionize how we monitor and understand our planet

The "Firefly" is **India's first private satellite constellation**, launched by **Bengaluru-based space-tech company Pixxel.** The first three of the fireflies constellation launched on **15 January, 2025**.

It's designed to be the **world's highest-resolution hyperspectral satellite constellation**, offering significantly sharper imagery than existing hyperspectral satellites.

**Pixxel has successfully launched the first three satellites of its 'Firefly' constellation** using Elon Musk's company 'SpaceX's' Falcon-9 rocket, placing them **550 kilometers below Earth's orbit**.

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Here's a more detailed look at the Firefly constellation:

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### Hyperspectral Imaging

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- Firefly satellites **capture data across a wide spectrum of light,** unlike traditional satellites that rely on primary colors.

### High Resolution

- The satellites have a **5-meter resolution**, which is **six times sharper** than the 30-meter standard of most existing hyperspectral satellites.

## 7.5.2 Space-based Surveillance (SBS)

- The **Cabinet Committee on Security (CCS)** has approved **SBS-3** on October 2024, the **third phase of India's space-based surveillance programme**. The first batch of satellites is expected to be launched by 2027–2028.
- It will deploy **52 AI-enabled satellites** in **LEO and GEO orbits**, capable of **inter-satellite communication** for real-time **geo-intelligence**.
- Satellites in **GEO (36,000 km)** can prompt **LEO (400–600 km)** satellites to conduct closer inspections.
- Includes integration with **31 Predator armed drones** from the US to strengthen domain awareness.

**India's SBS Programme Evolution**

| Phase | Year | Focus | Satellites |
|---|---|---|---|
| **SBS-1** | 2001 | Basic Surveillance | Cartosat-2A, 2B, RISAT-2, Eros-B |

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| Phase | Year | Focus | Satellites |
|---|---|---|---|
| **SBS-2** | 2013 | Maritime Surveillance | Cartosat-2C, 2D, Microsat-1, RISAT-2A |
| **SBS-3** | 2024 | AI-based, multi-domain awareness | 52 satellites (LEO + GEO), integration with Predator drones |

## Components of a Space-Based Surveillance System

1. **Satellite Constellations:** Optical, Infrared (IR), and Synthetic Aperture Radar (SAR) satellites
2. **Ground Control Stations:** Satellite control, telemetry, and data reception
3. **Space Surveillance Networks:** Track objects, prevent collisions
4. **Tracking & Telemetry Systems:** Monitor satellite health, orbit, and performance

## India's Key SBS-Related Assets

- **EMISAT:** Electromagnetic Intelligence (signals, radar detection)
- **Cartosat Series:** High-resolution imaging for terrain mapping
- **GSAT-7 & 7A:** Communication satellites for Navy & Air Force
- **NETRA:** Tracks space debris for collision avoidance

## Global Comparisons

- **US:** Space-Based Infrared System (SBIRS), OPIR, and Space Force (est. 2019)
- **Russia:** Aerospace Forces (est. 2015)
- **China:** Advanced dual-use satellites; also tested ASAT weapons

## International Agreements

- Outer Space Treaty (1967)
- Registration Convention (1976)
- UN Debris Mitigation Guidelines

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- Artemis Accords

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## 7.6 Launch Vehicle Technology

### *Launch Vehicles*

The combination of thrusters and boosters that help in carrying the spacecraft to space is known as Launch Vehicles. Achieving a high accuracy in placing the satellites to their designated orbits requires adequate combination of accuracy, power, efficiency and immense to-the-point planning. ISRO's launch vehicle programme is the guiding arm of all the development of launch vehicles, which has various centres as per their speciality. **The designing and development of launch vehicles is taken up in Vikram Sarabhai Space Centre (VSSC), Thiruvananthapuram**. The liquid and cryogenic stages for these launch vehicles are developed at Liquid Propulsion Systems Centre, Valiamala, and ISRO Propulsion Complex, Mahendragiri. Satish Dhawan Space Centre (earlier known as SHAR), is the space port of India, responsible for integration of launchers. It houses two operational launch pads from where all flights are launched and a third launch pad is under development.

### *Sounding Rocket*

One- or two-stage solid propellant rockets used for the investigation of upper atmosphere and space research are called Sounding Rockets. These are a cheap platform for testing new prototypes that could later go in a launch vehicle or a satellite. **The launch of the first sounding rocket from Thumba near Thiruvananthapuram, Kerala on 21 November 1963, marked the beginning of the Indian Space Programme. It was imported from Russia.**

India used to import these rockets from Russia and France. The Russian **M-100** had a capability to carry 70 kilograms to an altitude of 85 kilometres. The **Centaure (France)** could carry a payload of about 30 kilograms to at least 150 kilometres. From 1965, ISRO started launching its own sounding rockets and in 1975, all these activities were consolidated

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under one roof and came to be known as Rohini Sounding Rocket (RSR) programme.

India has developed **five generations of launch-vehicle technologies**:

- **First Generation:** SLV (Satellite Launch Vehicle)
- **Second Generation:** ASLV (Augmented Satellite Launch Vehicle)
- **Third Generation:** PSLV (Polar Satellite Launch Vehicle)
- **Fourth Generation:** GSLV (Geostationary Satellite Launch Vehicle): GSLV MK I and GSLV MK II
- **Fifth Generation:** GSLV MK III / LVM III

### 7.6.1 First Generation: Satellite Launch Vehicle

Satellite Launch Vehicle-3 (see **Table 7.4**) was four-stage launch vehicle, consisting of all solid, nature of propellant, capable of placing **40 kilograms class payloads in Low Earth Orbit (LEO) at a distance of 400 kilometres.** SLV-3, a first in this generation, was **successfully launched on 18 July 1980 from Sriharikota High Altitude Range (SHAR). It successfully placed Rohini satellite (RS-1)** in its orbit. **This project was headed by Satish Dhawan who was responsible for hand-picking young, bright and dedicated professionals for ISRO, such as APJ Abdul Kalam who lead the project that developed SLV-3, India's first launch vehicle.** Now SLV is not in use.

Table 7.4 Different SLV used by India

| Launcher | Launch Date | Payload |
|---|---|---|
| SLV-3 | 17 April 1983 | Rohini Satellite RS-D2 |
| SLV-3D1 | 31 May 1981 | Rohini Satellite RS-D1 |
| SLV-3E2 | 18 July 1980 | Rohini Satellite RS-1 |
| SLV-3E1 | 10 August 1979 | Rohini Technology Payload (RTP) |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

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### 7.6.2 Second Generation: Augmented Satellite Launch Vehicle (ASLV)

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The ASLV Programme (see **Table 7.5**) was designed to augment the payload capacity to 150 kilograms in LEO. It was five-stage launch vehicle technology, in all had solid propellant. Under the ASLV programme, four developmental flights were conducted. It is now retired.

Table 7.5 Different ASLV used by India

| Launcher | Launch Date | Payload |
|---|---|---|
| ASLV-D4 | 05 May 1994 | SROSS-C2 |
| ASLV-D3 | 20 May 1992 | SROSS-C |
| ASLV-D2 | 13 July 1988 | SROSS-2 |
| ASLV-D1 | 24 March 1987 | SROSS-1 |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

### 7.6.3 Third Generation: Polar Satellite Launch Vehicle (PSLV)

India's one of the most reliable and the most popular launch vehicle (**Table 7.6**) has been in service for over thirty years, and has been the reason for some historic missions like Mars Orbiter Mission (MOM) and Indian Regional Navigation Satellite System (IRNSS).

It was a four-stage multi propellant launcher that has a height of 44 metres and a lift-off weight of 320 tonnes (PSLV XL) [First stage – **Solid (HTPB – Hydroxyl terminated polybutadiene)**, Second stage – **Liquid (UDMH – Unsymmetrical Dimethyl Hydrazine)**, Third stage – **Solid (HTPB)**, Fourth stage – **Liquid (MMH – Monomethyl hydrazine + MON – Mixed oxide of Nitrogen]**. Propellant is a mixture of fuel and oxidiser. Oxidiser is necessary for the burning of fuel. $N_2O_4$ (Nitrogen Tetra Oxide) is used as an oxidiser in propellant technology. PSLV has three variants – PSLV G, PSLV CA and PSLV XL. It is the "**workhorse of ISRO**".

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Table 7.6 Different PSLVs launched by India

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| Launch Vehicle/Satellite | Launch Date | Launcher Type | Orbit | Payload |
|---|---|---|---|---|
| PSLV-C61/EOS-09 Mission | 18 May 2025 | PSLV-XL | - | EOS-09 |
| PSLV C60/SPADEX Mission | 30 December 2024 | | - | SPADEX-A, SPADEX-B, POEM 4 |
| PSLV-C59/Proba-3 Mission | 5 December 2024 | PSLV-XL | - | Proba-3 |
| PSLV- C58/XPoSat Mission | Jan 01, 2024 | PSLV-DL | | XPoSat |
| PSLV- C57/Aditya-L1 Mission | Sep 02, 2023 | PSLV-XL | | Aditya-L1 payloads |

[**SSPO:** Sun Synchronous Polar Orbit/Polar Sun-sychronous Orbit, **GTO:** Geosynchronous Transfer Orbit, **LEO:** Low Earth Orbit, **HEO:** High Earth Orbit, **GTO:** Geostationary Transfer Orbit]

*Source:* https://www.isro.gov.in (last accessed in July 2025)

It is designed and developed at VSSC, Thiruvananthapuram. Most of the development is done in Thiruvananthapuram. ISRO's Inertial Systems Unit (IISU) develops the inertial systems. Liquid Propulsion Systems Centre (LPSC) looks after the propulsion systems for the second and fourth stages, and also takes care of the reaction control systems. The solid propellant motors are managed by Satish Dhawan Space Centre (SDSC), which also carries out the launch operations.

### 7.6.4 Fourth Generation: Geosynchronous Satellite Launch Vehicle (GSLV)

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GSLV was developed by ISRO for a capability to launch geosynchronous satellites into geostationary orbits. Indian space scientists also refer to it as

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"**Fat Boy**". This made India less dependent on other countries for their launch vehicles. The three variants include **GSLV Mark I (a)** with a capability of 1,500 kilograms, **GSLV Mark I (b)** with a capability of 1,900 kilograms and **GSLV Mark II** that uses Indian Cryogenic Engine, developed by Liquid Propulsion System Centre (LPSC), and is capable of launching 2,500 kilograms of payload to GTO (Geostationary Transfer Orbit) and up to 5,000 kilograms of payload to LEO. Different GSLVs launched by India are given in **Table 7.7**.

**Payload to SSPO (Sun Synchronous Polar Orbit): 1,750 kg**

PSLV earned its title 'the Workhorse of ISRO' through consistently delivering various satellites to Low Earth Orbits, particularly the IRS series of satellites. It can take up to 1,750 kg of payload to Sun Synchronous Polar Orbits of 600 km altitude.

**Payload to GTO: 1,425 kg**

Due to its unmatched reliability, PSLV has also been used to launch various satellites into Geosynchronous and Geostationary orbits, like satellites from the IRNSS constellation.

GSLV is a three-stage launch vehicle, where the first two stages are similar to PSLV with same solid and liquid natures of fuel but the third stage is completely different, i.e., **cryogenic stage**. Earlier, India used Russian-made cryogenic engine. However, now, **ISRO developed its own cryogenic engine in LSPC, Mahendragiri, Tamil Nadu. On 6 January 2014, India used the first cryogenic upper stage engine, those successfully used by GSLV-D5.**

**GSLV IS "NAUGHTY BOY" & "SMART BOY"**

The GSLV (Geosynchronous Satellite Launch Vehicle) rocket, once nicknamed the "naughty boy" due to its past performance issues, has been referred to as the "smart boy" after a successful launch of the INSAT-3DS satellite. This positive shift reflects improvements in the rocket's reliability and performance, particularly with its cryogenic engine.

**First Stage:** It is made of maraging steel and uses solid propellants.

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**Second Stage:** Vikas engine powers up this stage and UDMH is used as fuel along with Nitrogen Tetroxide ($N_2O_4$) as oxidiser.

**Third Stage:** Driven by an indigenous cryogenic engine, uses liquid hydrogen and liquid oxygen in separate chambers interconnected by an inter-stage.

**Cryogenic Engine:** Cryogenic is a low-temperature based technology. In this propellant system, $LH_2$ (Liquid Hydrogen) is used as a fuel and $LO_2$ (Liquid Oxygen) as an oxidiser. Oxygen liquefies at -183°C and Hydrogen at -253°C.

Table 7.7 Different GSLV launched by India

| Launch Vehicle/Satellite | Launch Date | Launcher Type | Orbit | Payloa |
|---|---|---|---|---|
| GSLV-F15/NVS-02 Mission | Jan 29, 2025 | GSLV | GTO (Geosynchronous Transfer Orbit) | NVS-02 |
| GSLV-F14/INSAT-3DS | Feb 17, 2024 | GSLV | GTO | INSAT-3D |
| GSLV-F12/NVS-01 Mission | May 29, 2023 | GSLV | GTO | NVS-01 |
| GSLV-F10 / EOS-03 | Aug 12, 2021 | GSLV | GTO | EOS-03 |
| GSLV-F11/GSAT-7A Mission | Dec 19, 2018 | GSLV | | GSAT-7A |
| GSLV-F08/GSAT-6A Mission | Mar 29, 2018 | GSLV | GSO | GSAT-6A |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

At such low temperatures, the propellants are to be pumped using turbo pumps running at around 40,000 rpm. It also entails complex ground support systems like propellant storage and filling systems, cryo-engine and stage test facilities, transportation and handling of cryo fluids and related

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safety aspects. As compared to solid and liquid propellant rocket stages, a cryogenic rocket stage is more efficient and provides more thrust for the burning of propellant. Specific impulse (a measure of the efficiency) achievable with cryogenic propellants (liquid Hydrogen and liquid Oxygen) is much higher compared to Earth-based storable liquid and solid propellants, thus it is advantageous in the form of substantial payload.

### 7.6.5 Fifth Generation: GSLV MK- III

GSLV Mk-III is a three-stage heavy lift launch vehicle developed by ISRO. GSLV Mk-III is intended for carrying **4 ton class of satellites into Geosynchronous Transfer Orbit** (GTO) or about **8 tons to Low Earth Orbit** (LEO), which being twice the GSLV Mk-II capability. The GSLV-MK-III, India's heaviest rocket has acquired the nick name **"Bahubali"**. **Table 7.8** lists various GSLV Mk-III launched.

(I) **First Experimental Flight of GSLV MK-III:** The first experimental LVM3 flight, the LVM3-X/CARE mission took off on 18 December 2014 from Sriharikota and successfully tested the flight atmospheric phase. In this flight, the **Crew Experiment module Atmospheric Re-entry** was also carried out. The module re-entered, as per the planning, deployed its parachutes and splashed down in the Bay of Bengal.

(II) **First developmental flight of GSLV Mk-III: GSLV MK-III D1** – successfully positioned GSAT-19 satellite on 5 June 2017 to a GTO from SDSC, Sriharikota.

(III) **Second developmental flight of GSLV Mk-III: GSLV Mk-III D2** – the second developmental flight of GSLV Mk-III successfully launched GSAT-29, a high throughput communication satellite in Geosynchronous transfer orbit, on 14 November 2018 from SDSC, Sriharikota.

(IV) **GSLV MkIII-M1,** successfully injected **Chandrayaan-2, India's second Lunar Mission**, into Earth Parking Orbit on July 22, 2019 from SDSC SHAR, Sriharikota.

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(V) **LVM3-M2/OneWeb India-1 Mission:** The LVM3-M2 mission is a dedicated commercial mission for a foreign customer OneWeb, through NSIL. It is the first multi-satellite mission with 36 OneWeb Satellites to the LEO as the heaviest payload mass of 5,796 kg of LVM3 till date.

***Salient Features of the Mission***

- ❑ First commercial mission of LVM3
- ❑ First multi-satellite mission with 36 OneWeb Satellites onboard
- ❑ First launch of LVM3 to LEO
- ❑ First Indian rocket with six ton payload
- ❑ First NSIL mission with LVM3
- ❑ First OneWeb mission with NSIL/DoS

(VI) **LVM3 M3/ OneWeb India-2 Mission:** LVM3 M3/ OneWeb as part of India-2 Mission, in its sixth consecutive flight successfully placed 36 satellites belonging to the OneWeb Group Company in their intended 450 km circular orbit with an inclination of 87.4 degrees. The vehicle took off with a total payload of 5,805 kg from the second launch pad at SDSC-SHAR, Sriharikota.

Table 7.8 List of GSLV Mk-III

| Title | Launch Date | Launcher Type | Payload |
|---|---|---|---|
| LVM3 M4 / Chandrayaan-3 Moon Mission | July 14, 2023 | LVM3 | Chandrayaan-3 |
| LVM3 M3 / OneWeb India-2 Mission | Mar 26, 2023 | LVM3 | OneWeb Gen-1 |
| LVM3 M2 / OneWeb India-1 Mission | Oct 23, 2022 | LVM3 | OneWeb Gen-1 |
| GSLV-Mk-III – M1/Chandrayaan-2 Mission | Jul 22, 2019 | Payload | Chandrayaan-2 |

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| | | | |
|---|---|---|---|
| GSLV Mk-III-D2/GSAT-29 Mission | Nov 14, 2018 | GSLV-MK-III | GSAT-29 |
| GSLV Mk-III-D1/GSAT- | Jun 05, 2017 | GSLV-MK-III | GSAT-19 |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

## CRYOGENIC TECHNOLOGY

Cryogenic technology is the development of useful applications based on cryogenic science, which is the study of temperatures below -150 °C, how to produce such low temperatures and the behaviour of materials at those temperatures. Cryogenic is a Greek word and means **'the production of freezing cold'.**

Various areas of cryogenic technology/science are:

- **Cryobiology:** It deals with the study of the effects of low temperatures on organisms, generally for the purpose of achieving cryopreservation.
- **Cryonics:** The emerging medical technology of cryopreserving humans and animals with the intention of future revival.
- **Cryoelectronics:** Research of super conductivity at very low temperatures.
- **Cryotrocins:** It is the real application of Cryoelectronics.

### Gases Used and Storage

Liquid nitrogen and liquid helium are used in cryogenic applications. Most often, liquid nitrogen is used in cryogenics. Liquid helium is also used and allows for the lowest attainable temperatures to be reached. A special type of container is used to keep these liquids known as Dewar flasks.

### Applications

1. **Superconductivity:** Cryogenic materials such as liquid nitrogen are useful for producing the temperatures necessary for superconductivity applications.
2. **Cryogenic Fuels:** Largely, liquid hydrogen has been used as rocket fuel. Liquid oxygen is used as an oxidiser of hydrogen, but oxygen is not strictly called as a fuel.
3. **Cryostorage:** Cryogenic gases are used in transportation of large masses of frozen food. When huge quantity of food must be transported to regions like war fields, earthquake-hit regions, etc., they must be stored for a long time. So cryogenic food freezing is used.
4. **Cryocooling:** Use of cryogenic gases for cooling (heat exchange) purposes in industrial applications, such as electrical cables.
5. **Magnetic Resonance Imaging:** It is a technique of imaging objects where a strong magnetic field is required that is generated by electromagnets. Elevated field strengths can be addressed by using superconducting magnets. Normally, liquid helium is used to cool the superconducting coils.

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6. **Cryogenic Processing:** This was developed during World War II when scientists found that metals frozen to low temperatures exhibited more resistance to wear.
7. **Cryomilling:** Some soft or elastic materials turn to hard and brittle at very low temperatures, which make cryogenic milling (cryomilling) an option for some materials thCC

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Note: **OneWeb Ltd** is a **global communication network** powered from space, **enabling internet connectivity for governments and businesses. India's Bharti Enterprises is a major investor and shareholder in OneWeb.**

**VIKAS ENGINES**

These are **liquid fuelled rocket engines** developed by the **Liquid Propulsion Systems Centre** in the 1970s. Earlier India has imported Vikas engines from French components. It is used in PSLV and GSLV series of expendable launch vehicles for space launch use.

Vikas engine is used to power the second stage of PSLV, boosters and second stage of GSLV Mark I and II and also the core stage of **GSLV Mark III**. The propellant loading for Vikas engine in PSLV, GSLV Mark I and II is 40 tons, while in GSLV Mark III is 55 tons.

(Refer **Infographic – PSLV, GSLV & GSLV-MK-III**)

## 7.7 Miscellaneous: Launch Vehicle and Missiles

### 7.7.1 Small Satellite Launcher

Small Satellite Launch Vehicle (SSLV) is ISRO's indigenous new launch rocket, is likely to have its much-delayed, maiden development flight this 3rd/4th quarter of 2022.

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One of the aims of the newly created ISRO's commercial arm, NewSpace India Limited (NSIL), is to use research and development carried out by ISRO over the years for commercial purposes through Indian industry partners.

"Manufacturing and production of Small Satellite Launch Vehicle (SSLV) and Polar Satellite Launch Vehicle (PSLV) through technology transfer," is one of the mandates of the new firm.

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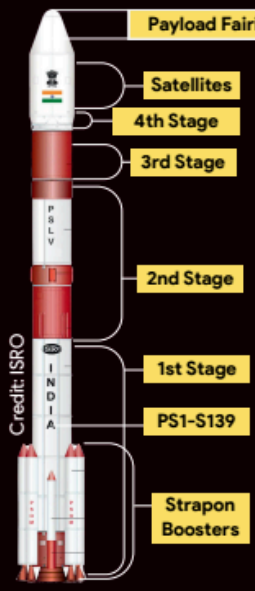
# 3rd Gen LV- PSLV (Polar Satellite Launch Vehicle)




1. Polar Satellite Launch Vehicle (PSLV) is the third generation launch vehicle of India. It is the first Indian launch vehicle to be equipped with liquid stages. After its first successful launch in October 1994, PSLV emerged as a reliable and versatile workhorse launch vehicle of India.
2. PSLV has also been used to launch various satellites into Geosynchronous and Geostationary orbits, like satellites from the IRNSS Constellation.
3. The vehicle successfully launched two spacecraft "Chandrayaan-1 in 2008 and Mars Orbiter Spacecraft in 2013" that later travelled to Moon and Mars respectively. Chandrayaan-1 and MOM (Mars Orbiter Mission) were feathers in the hat of PSLV.
4. PSLV earned its title 'the Workhorse of ISRO' through consistently delivering various satellites to Low Earth Orbits. It can take up to 1,750 kg of payload to Sun-Synchronous Polar Orbits of 600 km altitude.
5. First Flight: September 20, 1993 (Unsuccessful Mission)
6. Variants: 4 (PSLV (CA,DL,QL,XL))
7. **Stages of PSLV Launch**
   - 1st Stage- PSLV uses the S139 solid rocket motor that is augmented by 6 solid strap-on boosters.
   - 2nd Stage- PSLV uses an Earth storable liquid rocket engine for its second stage, know as the Vikas engine, developed by Liquid Propulsion Systems Centre.
   - 3rd Stage- The third stage of PSLV is a solid rocket motor that provides the upper stages high thrust after the atmospheric phase of the launch.
   - 4th Stage- The PS4 is the uppermost stage of PSLV, comprising of two Earth storable liquid engines.

# 4th Gen LV- GSLV (Geosynchronous Satellite Launch Vehicle)

Geosynchronous Satellite Launch Vehicle Mark II (GSLV MK II) is the launch vehicle developed by India, to launch communication satellites in geo transfer orbit using cryogenic third stage.

Initially Russian GK supplied cryogenic stages were used. Later cryogenic stage was indigenously developed and inducted in Jan 2014 from GSLV D5 onwards.

This operational fourth generation launch vehicle is a three stage vehicle with four liquid strap-ons. The flight proven indigenously developed Cryogenic Upper Stage (CUS), forms the third stage of GSLV MK II.

From January 2014, the vehicle has achieved six consecutive successes.

First Flight: April 18, 2001 | First Flight (With Indigenous Cryo Stage) : January 5, 2014

The name Chandrayaan means "mooncraft" in Sanskrit and Hindi. The mission "Chandrayaan 2" was launched on a Geosynchronous Satellite Launch Vehicle Mark III (GSLV MK-III) M1.

1st Stage (GS1)- The first stage of GSLV was also derived from the PSLV's PS1. The 138 tonne solid rocket motor is augmented by 4 liquid strap-ons.
2nd Stage (GS2)- One Vikas engine is used in the second stage of GSLV. The stage was derived from the PS2 of PSLV where the Vikas engine has proved its reliability.
3rd Stage (CUS)- Developed under the Cryogenic Upper Stage Project (CUSP), the CE-7.5 is India's first cryogenic engine, developed by the Liquid Propulsion Systems Centre. CE-7.5 has a staged combustion operating cycle.

**Stages of GSLV Launch**

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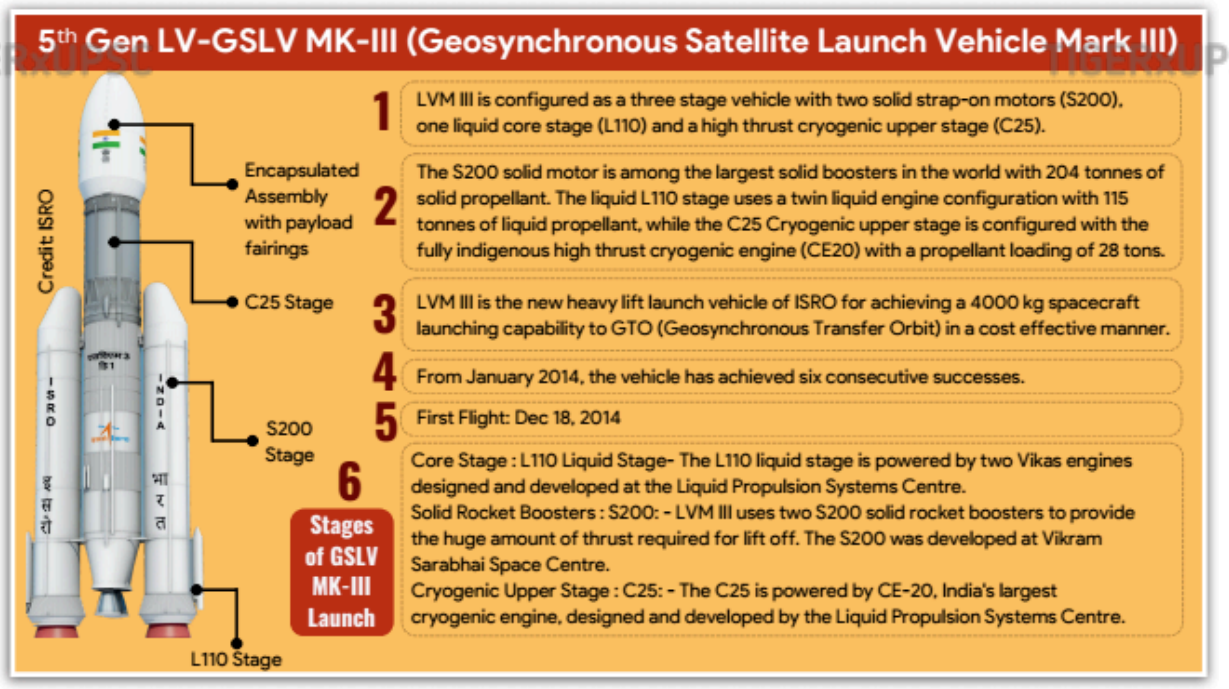


There are more than 500 industries contributing to ISRO programs at present, and more than half of the project budget outlay for space programs flows to these industries.

- The SSLV is intended to cater to a market for the launch of small satellites into low earth orbits, which has emerged in recent years on account of the need for developing countries, private corporations, and universities for small satellites.
- The launch of small satellites has until now been dependent on 'piggy-back' rides with big satellite launches on ISRO's workhorse – the PSLV, which has had over 50 successful, launches so far.

The launch of small satellites as a consequence has been dependent on the finalising of launch contracts for the larger satellites by ISRO. **Table 7.9** lists various SSLV launched till date.

***Features***

- SSLV is a 4-stage Launch Vehicle with the first 3 stages of the vehicle using HTPB-based solid propellant, with a fourth terminal stage being a Velocity-Trimming Module (VTM).
- The SSLV can carry satellites weighing up to 500 kg to a low earth orbit.
- The SSLV is the smallest vehicle at 110-ton mass at ISRO.

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- It will be an on-demand vehicle.
- SLV is perfectly suited for launching multiple microsatellites at a time and supports multiple orbital drop-offs.

Table 7.9 List of SSLV Launches

| Title | Launch Date | Launcher Type | Payload |
|---|---|---|---|
| SSLV-D3/EOS-08 Mission | Aug 16, 2024 | SSLV | EOS-08 |
| SSLV-D2/EOS-07 MISSION | Feb 10, 2023 | SSLV | EOS-07 |
| SSLV-D1/EOS-02 Mission | Aug 07, 2022 | SSLV | EOS-02 |

*Source:* https://www.isro.gov.in (last accessed in July 2025)

***Significance***

- **Less Time:** It will take only 72 hours to integrate, unlike the 70 days taken now for a launch vehicle.
- **Less Manpower:** Only six people will be required to do the job, instead of 60 people.
- **Saving Cost:** The entire job will be done in a very short time and the cost will be only around ₹30 crore.
- **Build Synergy:** The development and manufacture of the SSLV are expected to create greater synergy between the space sector and private Indian industries – a key aim of the space ministry.

### 7.7.2 Next-Gen Launch Vehicle (NGLV)

ISRO is developing a Next-Gen Launch Vehicle (NGLV) intended to replace PSLV.

***Features of NGLV***

- 3-Stage Launch Vehicle, reusable heavy lift vehicle
- Use semi-cryogenic propulsion for booster stages
- 10-tonne payload capability to Geostationary Transfer Orbit
- Cost efficient
- Simple, robust design allowing bulk manufacturing

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### *Potential Uses*

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Launching communication satellites, Deep space missions, Future human space flight and cargo missions, ISRO is intending to develop a business model for NGLV to launch commercial as well as national missions.

The Union Cabinet, chaired by Prime Minister Mr Narendra Modi, has approved the development of the **Next Generation Launch Vehicle (NGLV)**. This initiative marks a significant step towards the Government's vision of establishing and operating the **Bharatiya Antariksh Station**, as well as advancing **India's capability for a crewed lunar landing by 2040.**

NGLV will have **3 times the present payload capability** with **1.5 times the cost compared to LVM3**, and will also have **reusability** resulting in **low-cost access** to space and modular **green propulsion systems**.

NGLV is designed to have a maximum payload capability of 30 tonnes to Low Earth Orbit (LEO), which also has a reusable first stage. India has achieved self-reliance in space transportation systems, enabling the launch of satellites weighing up to 10 tonnes to LEO and 4 tonnes to Geo-Synchronous Transfer Orbit (GTO) using operational launch vehicles such as PSLV, GSLV, LVM3 and SSLV.

The NGLV development project will be carried out with extensive participation from the Indian industry, which is expected to invest in manufacturing capacity from the outset. This approach will ensure a smooth transition to the operational phase following the development stage.

NGLV will be demonstrated with **three development flights (D1, D2 & D3)** with a **target of 96 months (8 years)** for the completion of the development phase.

The total fund approved is **Rs. 8,240.00 Crore** and includes the development costs, three developmental flights, essential facility establishment, Programme Management and Launch Campaign.

### *Leap Towards Bharatiya Antariksh Station*

NGLV will facilitate **national and commercial missions**, including **human spaceflight** to the **Bharatiya Antariksh Station**, **lunar and interplanetary exploration** and the deployment of **communication and**

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**earth observation satellite** constellations in LEO. This initiative will strengthen India's space ecosystem by enhancing its capability and capacity.

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### 7.7.3 India's Anti-Satellite (A-SAT) Missile/Mission Shakti

A joint programme of DRDO and ISRO, **Mission Shakti**, as part of the mission, **an anti-satellite (A-SAT) weapon** was launched and targeted on an Indian satellite which had been decommissioned. It was carried out from DRDO's testing range in Balasore, Odisha.

After US, Russia and China, India became the fourth country to acquire such a specialised and modern capability, and the entire effort is indigenous. With this satellite we now have the capability to hit a live target in space.

### 7.7.4 Reusable Launch Vehicle – Technology Demonstrator (RLV-TD)

Developing the Reusable Launch Vehicle – Technology Demonstrator (RLV-TD) was a technologically challenging endeavour of ISRO, which India has achieved now. Access of reusable launch vehicle technology will minimise the cost for various space missions. In May 2016, it was successfully tested for its autonomous navigation, reusable thermal protection system, guidance and control, and re-entry mission management. The configuration of RLV-TD is similar to that of an aircraft and combines the complexity of both launch vehicles and aircraft.

(Refer **Infographic – RLV-TD**)

### 7.7.5 Pushpak: India's Reusable Launch Vehicle

The Indian Space Research Organisation (ISRO) **achieved a significant milestone with the successful launch of its third Reusable Launch Vehicle (RLV) mission** named **"Pushpak"** on **22 March 2024.**

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## Mission Objectives

- The primary goal of the Pushpak mission is to **develop essential technologies** for a fully reusable launch vehicle, **aiming to enable low-cost access to space.**

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## RLV-TD (Reusable Launch Vehicle – Technology Demonstrator)

Reusable Launch Vehicle – Technology Demonstrator (RLV-TD) is one of the most technologically challenging endeavors of ISRO towards developing essential technologies for a fully reusable launch vehicle to enable low cost access to space.

The configuration of RLV-TD is similar to that of an aircraft and combines the complexity of both launch vehicles and aircraft.

The winged RLV-TD has been configured to act as a flying test bed to evaluate various technologies, namely, hypersonic flight, autonomous landing and powered cruise flight.

In future, this vehicle will be scaled up to become the first stage of India's reusable two stage orbital launch vehicle.

RLV-TD was successfully flight tested on May 23, 2016 from SDSC SHAR Sriharikota validating the critical technologies such as autonomous navigation, guidance & control, reusable thermal protection system and re-entry mission management.

RLV-TD consists of a fuselage (body), a nose cap, double delta wings and twin vertical tails. It also features symmetrically placed active control surfaces called Elevons and Rudder.

### Objectives

- Hypersonic aerothermodynamic characterization of wing body
- Evaluation of autonomous Navigation, Guidance and Control (NGC) schemes
- Integrated flight management
- Thermal Protection System Evaluation

### Mission Profile

Credit: ISRO

### Data Communication

Credit: ISRO

- ISRO's objective is to **reduce the cost of space missions** and **minimise space debris** through the development of reusable launch vehicle

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technology.

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## Launch Details

- The Pushpak RLV was launched from the **Chalakere Runway in Karnataka's Aeronautical Test Range (ATR)** at approximately 7 a.m. on **22 March 2024.**
- **It was carried to an altitude of about 4.5 km by an Indian Air Force Chinook helicopter before being released to achieve predetermined parameters.**
- The launch vehicle autonomously approached the runway, made cross-range corrections, and landed precisely using its brake parachute, landing gear brakes, and nose wheel steering system.

## Significance of the Mission

- Pushpak launch vehicle is India's efforts **to make space access more affordable.**
- The reusable upper stage of the **RLV has the potential to enable activities such as refuelling in-orbit** satellites or retrieving satellites for refurbishment, **contributing to cost reduction** and sustainability in space operations.

## RLV-TD: A Technologically Challenging Project

- The **Reusable Launch Vehicle – Technology Demonstrator (RLV-TD)** project is one of ISRO's most technologically challenging endeavours.
- It aims to develop crucial technologies for a fully reusable launch vehicle, including hypersonic flight, autonomous landing, and powered cruise flight.

## Design and Configuration

- The Pushpak RLV is designed as an **all-rocket, fully reusable single-stage-to-orbit (SSTO) vehicle.**

## Previous Missions and Future Plans

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- ISRO had **conducted successful RLV missions in 2016** and **2023**, laying the groundwork for future developments.
- The organisation aims to scale up the RLV-TD to become the first stage of India's reusable two-stage orbital launch vehicle.

### Cost and Naming

- The **Pushpak project, named after the mythological spaceship in the Ramayana, is estimated to have cost over ₹100 crore** (approximately $13.5 million).
- The Indian Air Force's involvement, particularly providing the Chinook helicopter, was crucial for the mission's success.

## 7.8 Jet Engine Technology

### 7.8.1 Normal Aircraft versus Jet Plane

Normal aircrafts have light-weight piston engines or gas turbines but advanced jet planes have turbojet, ramjet and scramjet engines.

Normal aircrafts move through troposphere but jet planes move very easily and smoothly in stratospheric zone. The stratosphere lies above the troposphere (where the temperature increases with increase in altitude). The stratosphere extends to an altitude of about 50 kilometres. Compared to troposphere, the airflow in the stratosphere is much less turbulent. Since the air does not flow up and down, and flows parallel to the Earth in very fast moving air streams, this is the layer where most of the jet planes fly. There is significant difference in the temperature gradients, and hence the tropospheric air does not mix into the stratospheric air. The stratosphere is more layered or 'stratified' because of lack of mixing and less turbulence.

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## 7.8.2 Types of Jet Plane

Jet engine technology works on the basis of Newton's third law of action and reaction. Three principles are involved in any type of jet engine orderly compression, combustion and expansion. There are three types of jet engine technology evolved, namely turbojet, ramjet and scramjet.

A reaction engine which releases air with high velocity and results in the production of thrust through jet impulsion is known as jet engine. Jet engine is commonly referred as internal combustion air breathing engine. Such engines classically have a rotating air compressor, which is turbine powered, with the residual power providing thrust through a propelling nozzle and the process is termed as the Brayton thermodynamic cycle. These engines are used by jet aircrafts for long-distance travel.

The earlier jet aircrafts used turbojet engines, which were comparatively incompetent for subsonic flights. The current subsonic jet aircrafts generally make use of much complex high-bypass turbofan engines. These engines are offering superior speed and better efficiency of fuel over large distances in comparison to piston and propeller-featured aero engines.

The jet engines lacking inlet systems can only admit air at about half the sound speed. The only exception to this is scramjets. The inlet system slows the air movement and carries out the compression for transonic and supersonic aircrafts.

The flammability sets the upper altitude limit for engines, such as during very high altitudes, the air is too thin and it becomes too hot after compression. The **turbojet engines** can attain nearly 40 kilometre altitudes, whereas around 55 kilometres can be achieved for **ramjet engine**.

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Theoretically, **scramjets** can reach up to 75 kilometres. On the other hand, the rocket engines have no upper limit.

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Note: Rocket engines always carry own oxidiser because it goes in outer atmosphere where air is absent. Jet planes can always be up to that height where oxygen is present because they take atmospheric oxygen. Stratosphere is that part of atmosphere where oxygen is always present.

### 7.8.3 Ramjet and Scramjet

(Refer **Infographic – Turbojet, Ramjet & Scramjet**)

Note: ISRO's first experimental test of scramjets was successful on 28 August 2016. It tested its engine that later touched down at Bay of Bengal after a flight time of 300 seconds. During its 300-second course, the flight was successfully tracked and managed from ground stations. Hydrogen was used as fuel and the Ramjet breathed the atmospheric air of oxygen as oxidiser, thus attaining Mach 6. ISRO's Advanced Technology Vehicle (ATV) is an advanced sounding rocket, used for its lift off that together weighed 3277 kilograms at lift off. India is the fourth country to demonstrate the flight testing of a scramjet engine after US, Russia and European Union.

## 7.9 Indian Astronomical Observatory (IAO)

The Indian Astronomical Observatory is currently the world's third highest optical telescope at one of the world's highest sites (at an elevation of 4,500 metres) located near Leh in Ladakh, India. It works through infrared, gamma-ray and optical telescopes. The Indian Institute of Astrophysics, Bengaluru operates this observatory.

**AIR BREATHING PROPULSION SYSTEM**

**ISRO's Air Breathing Propulsion Technology (2nd Experimental Flight)**

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### *What Is Air Breathing Propulsion?*

- A rocket propulsion system that **carries fuel** but **uses atmospheric oxygen** instead of on-board oxidisers.
- Works **only within Earth's atmosphere** (up to ~70 km).
- Beyond that altitude, rockets require **conventional propulsion** with internal oxidisers.

### *Advantages*

- **Lighter rockets** (no oxidiser carried)
- **Higher fuel efficiency**
- Cost-effective for **lower atmosphere** flight stages

### *Limitations*

- **Ineffective in space** or thin atmospheric layers
- Requires **high speed for efficient operation**

### *Types of Air-Breathing Engines*

1. **Ramjet**
   - **No rotating parts**
   - Relies on **forward motion** to compress air
   - Effective at **supersonic speeds** (Mach 3–6)
   - Airflow remains **subsonic** inside combustion chamber
2. **Scramjet (*Supersonic Combustion Ramjet*)**
   - Advanced form of Ramjet
   - **Supersonic airflow and combustion**
   - Efficient at **hypersonic speeds** (>Mach 5)
   - India became the fourth country to test a scramjet engine in flight (2023)
3. **Dual-Mode Ramjet (DMRJ)**
   - Hybrid of **Ramjet + Scramjet**
   - Operates as **Ramjet at lower speeds** (Mach 4)
   - **Transitions to Scramjet** at higher speeds (Mach 8)
   - Works in both **subsonic and supersonic combustor modes**

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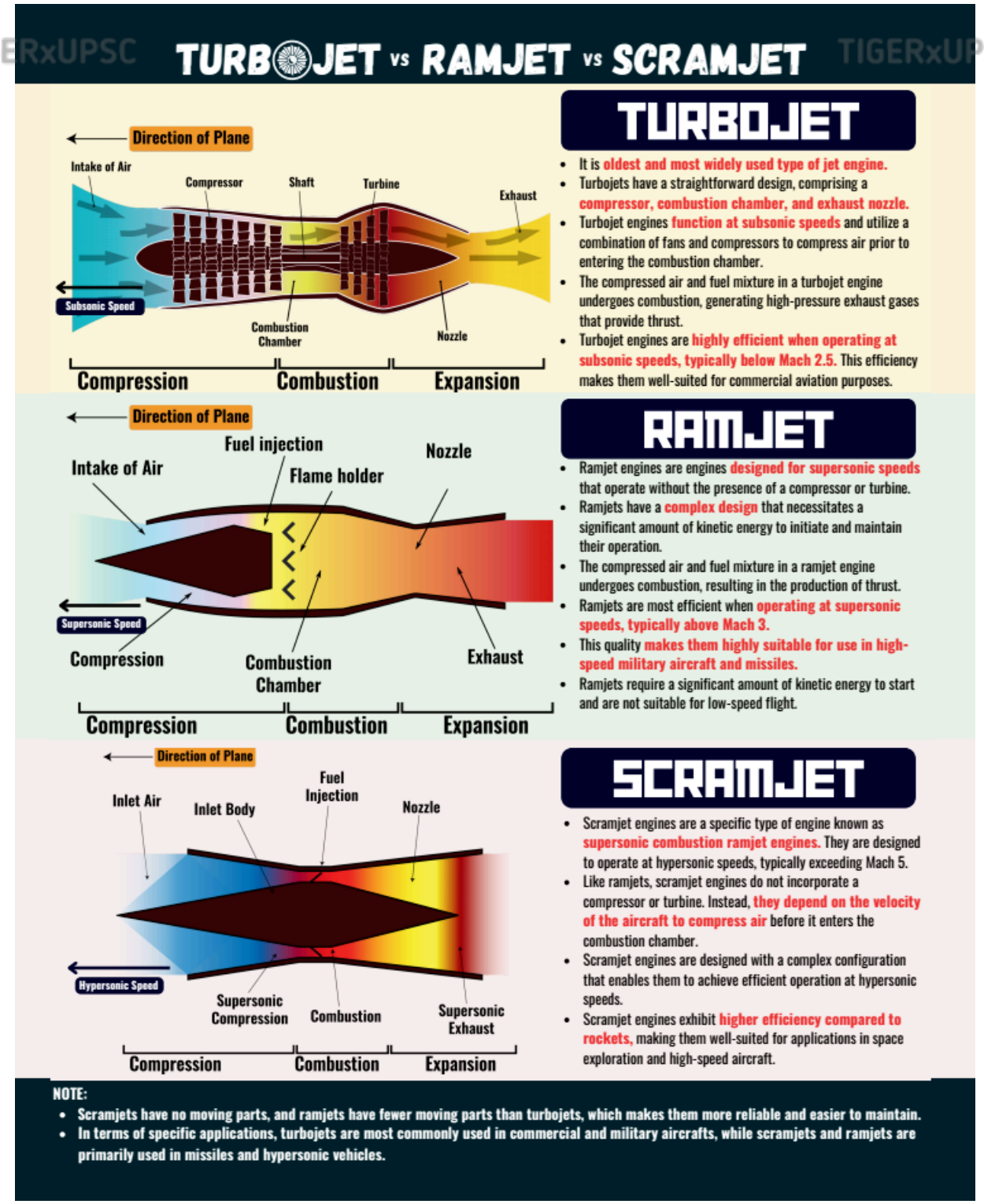


It has two dynamic telescopes – one is Himalayan Chandra Telescope (HCT), having a 2.01 metre diameter and the other is a high altitude Gamma Ray Telescope.

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### *Himalayan Chandra Telescope*

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The Himalayan Chandra Telescope is an optical-infrared telescope having 2.01 metres diameter, which is approximately 6.5 feet. This is developed by Electo-Optical Technologies System based at Arizona, USA. The telescope is operated remotely from the Centre for Research and Education in Science and Technology at Bengaluru with the help of a dedicated satellite link. The telescope is remotely operated *via* an INSAT-3B satellite link, which allows operation even in sub-zero temperatures in winter.

**CENTRE FOR RESEARCH AND EDUCATION IN SCIENCE AND TECHNOLOGY**

- The Centre for Research and Education in Science and Technology (CREST) is situated 35 km to the northeast of Bangalore near Hoskote town. The Centre houses the control room for the remote operations of the 2 m Himalayan Chandra Telescope (HCT) at the Indian Astronomical Observatory, Hanle, and the HCT data archive. The operations are controlled using a remote satellite link.

### *High Altitude Gamma Ray Telescope*

The High Altitude Gamma Ray Telescope (HAGAR) is an atmospheric Cerenkov experiment with seven telescopes setup in 2008. Each telescope has seven mirrors with a total area of 4.4 square metres. The telescopes are deployed on the periphery of a circle of radius 50 metres with one telescope at the centre.

### *List of major Astronomical Observatories of India:*

- **Kodaikanal Observatory:** Located in Tamil Nadu, this observatory is known for its solar observations.
- **Udaipur Solar Observatory:** Situated in Rajasthan, this observatory focuses on studying the sun.
- **Indian Astronomical Observatory (IAO), Hanle:** Located in Ladakh, this observatory is one of the highest in the world and is well-suited for astronomical observations due to its dry, clear skies.
- **Giant Metrewave Radio Telescope (GMRT):** Located in Pune, Maharashtra, this radio telescope is used for studying radio waves from celestial objects.

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**GROWTH-INDIA TELESCOPE**

The GROWTH-India telescope is a 0.7 metre wide-field optical telescope that had first light in 2018. It is the **country's first fully robotic research telescope**. It was set up as a part of the international GROWTH programme, and has been widely used for time domain astronomy. **The telescope is operated jointly by IIT Bombay and the Indian Institute of Astrophysics.**

- **Aryabhatta Research Institute of Observational Sciences (ARIES):** Located in Devasthal, Uttarakhand, this institute operates multiple telescopes for optical and infrared observations.

## 7.10 Project NETRA (Network for Space Object Tracking and Analysis)

ISRO has inked a pact with Indian Institute of Astrophysics (IIA) to pave the way for collaboration in establishing optical telescope facilities under ***Project 'NETRA'*** for tracking space objects.

ISRO has initiated ***'Project NETRA'*** as an early warning system in space to detect debris and other hazards to Indian satellites. Under the project, ISRO plans to put up many observational facilities: connected radars, telescopes; data processing units and a control centre. They can, among others, spot, track and catalogue objects as small as 10 cm, up to a range of 3,400 km and equal to a space orbit of around 2,000 km.

### *Significance of the Project*

- The project will give India its own capability in space situational awareness (SSA) like the other space powers, which is used to 'predict' threats from debris to Indian satellites.
- NETRA's eventual goal is to capture the GEO, or geostationary orbit, scene at 36,000 km where communication satellites operate.
- The effort would make India a part of international efforts towards tracking, warning about and mitigating space debris.

Currently there are many functional Indian communication satellites in the geostationary orbit of 36,000 km; remote sensing satellites in LEO of up

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to 2,000 km; and navigation satellites in medium earth orbits. Their protection is of utmost importance for India.

Space junk (Kessler Syndrome) is an ever-growing problem with more than 7,500 tonnes of redundant hardware now thought to be circling the Earth. The old rocket bodies and defunct spacecraft through to screws and even flecks of paint are such material that pose a collision hazard to operational missions. The rise in space debris increases the potential danger to all space vehicles, and especially to the International Space Station (ISS), space shuttles, satellites and other spacecraft.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. For the measurement/estimation of which of the following are satellite images/remote sensing data used?** **[UPSC (Pre) – 2019]**

1. Chlorophyll content in the vegetation of a specific location
2. Greenhouse gas emissions from rice paddies of a specific location
3. Land surface temperatures of a specific location

Select the correct answer using the code given below.

(a) 1 only
(b) 2 and 3 only
(c) 3 only
(d) 1, 2 and 3

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2. **With reference to India's satellite launch vehicles, consider the following statements:** **[UPSC (Pre) – 2018]**

   1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
   2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on earth.
   3. GSLV MKII is a four-stage launch vehicle with the first and third stages using solid rocket motors, and the second and fourth stages using liquid rocket engines.

   Which of the statements given above is/are correct?

   (a) Only 1
   (b) 2 and 3
   (c) 1 and 2
   (d) Only 3

3. **In which of the following activities are Indian Remote Sensing (IRS) satellites used?** **[UPSC (Pre) – 2015]**

   1. Assessment of crop productivity
   2. Locating groundwater resources
   3. Mineral exploration
   4. Telecommunications
   5. Traffic studies

   Select the correct answer using the code given below:

   (a) 1, 2 and 3
   (b) 4 and 5
   (c) 1 and 2
   (d) 1, 2, 3, 4

4. **The Jet aircraft fly very easily and smoothly in the stratosphere. What could be the appropriate explanation?** **[UPSC (Pre) – 2011]**

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1. There are no clouds or water vapour in the lower stratosphere.
2. There are no vertical winds in the lower stratosphere.

Which of the statement(s) given above is/are correct in this context?

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**5. Consider the following statements in respect of a jet engine and a rocket:** **[UPSC (Pre) – 2008]**

1. A jet engine uses the surrounding air for its oxygen supply and so is unsuitable for motion in space.
2. A rocket carries its own supply of oxygen in gas form and fuel.

Which of the statements given above is/are correct?

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**6. Low temperatures (Cryogenics) find application in**

(a) space travel, surgery and magnetic levitation **[UPSC (Pre) – 1999]**
(b) surgery, magnetic levitation and telemetry
(c) space travel, surgery and telemetry
(d) space travel, magnetic levitation and telemetry

## Answer Key

| | | | | | |
|---|---|---|---|---|---|
| 1. (d) | 2. (a) | 3. (a) | 4. (c) | 5. (a) | 6. (a) |

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**UNIT-III** Space Technology

# CHAPTER 8
# OUTER SPACE AFFAIRS AND VARIOUS NATIONAL AND INTERNATIONAL SPACE MISSIONS

## IN THIS CHAPTER...


**8.1 United Nations Office for Outer Space Affairs (UNOOSA): Roles and Responsibilities**

**8.2 Space Law**

**8.3 Scientific Space Craft of India**


India's Moon mission, Chandrayaan-3 created history by accomplishing a **'soft landing' on the Moon's south pole.** With this, India became the **first country to land near the South pole** of the Moon.


8.3.1 Successful Scientific Space Craft of India

8.3.2 Space Docking Experiment (SPADEX)

8.3.3 Future Scientific Space Craft of India

**8.4 World's GPS: NAVSTAR, GLONASS**


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**8.5 India's Regional Level Global Positioning System (GPS)**
- 8.5.1 GPS Aided Geo-Augmented Navigation Satellite System (GAGAN)
- 8.5.2 Indian Regional Navigation Satellite System (IRNSS) & NVS
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**8.6 Some Recent News**
- 8.6.1 AXIOM Mission
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**8.8 Kessler Syndrome/Space Junk/Space Debris Concept**

**8.9 Space Tourism**

**8.10 Problems Faced by Astronauts in Space**

**8.11 Extra Vehicular Activity/Spacewalk**

ISRO aims to establish the Indian space station by 2035

**projected to weigh only 20 tonnes**, in contrast to the 450 tonnes of the International Space Station and 100 tonnes of the Chinese Tiangong Space Station.

**8.12 International Space Station**

**8.13 Space History**

**8.14 Indian Astronauts and Their Achievements**
- 8.14.1 Rakesh Sharma
- 8.14.2 Kalpana Chawla
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**8.15 Some Important International Space Mission**
- 8.15.1 Important Jupiter Missions
- 8.15.2 Important Saturn Missions
- 8.15.3 Important Mars Missions
- 8.15.4 Important Mercury Missions
- 8.15.5 Important Solar Missions

The James Webb Space Telescope (JWST or Webb) is a large infrared telescope with a 6.5 meter primary mirror, launched on December 25, 2001 from French Guiana.

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The Hubble Space Technology, launched in 1990, is a space-based observatory that revolutionised astronomy by providing a clear view of the Universe above earth's atmosphere.

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8.15.6 Solar System and Beyond
8.15.7 Asteroid Missions in News
8.15.8 Space Craft Leaving Our Solar System/Interstellar Mission
8.15.9 important Moon Mission

Russia plans to exit the ISS by 2028 and focus on its own station to address scientific, technological, and national security needs.

8.15.10 Miscellaneous Missions in News
8.15.11 Commercial Crew Program

## 8.1 United Nations Office for Outer Space Affairs (UNOOSA): Roles and Responsibilities

- The UNOOSA is the United Nations office responsible for promotion of international cooperation in the peaceful uses of outer space.
- UNOOSA works as the secretariat for the General Assembly's only committee the United Nations "Committee on the Peaceful Uses of Outer Space" (COPUOS), dealing exclusively with international cooperation in the peaceful uses of outer space.
- UNOOSA conducts international workshops, training courses and pilot projects on topics including satellite navigation, satellite meteorology, remote sensing, tele-education and basic space sciences for the benefit of developing nations via the United Nations Programme on Space Applications.
- During disasters, it manages the United Nations Platform for **Space-based Information for Disaster Management and Emergency Response** (UN-SPIDER) and maintains a 24-hour hotline as the United Nations focal point for satellite imagery requests during disasters.
- The current secretariat of the International Committee on Global Navigation Satellite Systems (ICG) is UNOOSA **(United Nations**

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**Office for Outer Space Affairs)**.

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- It also prepares and distributes reports, studies and publications on various fields of space science and technology applications and those related to international space law. Documents and reports are available through its website in all official languages of the United Nations. UNOOSA is located at the United Nations Office at Vienna, Austria.

## 8.2 Space Law

- The body of law governing space-related activities is described as Space Law.
- It comprises of a variety of international agreements, treaties, conventions and United Nations General Assembly resolutions, along with rules and regulations of international organisations.
- "Space Law" is associated with the rules, principles and standards of international law appearing in the five international treaties and five sets of principles governing outer space, developed under the auspices of the United Nations.
- Space Law addresses a variety of matters, for instance, the preservation of the space and Earth environment, liability for damages caused by space objects, the rescue of astronauts. Other important applications include the settlement of disputes, the sharing of information about potential dangers in outer space, the use of space-related technologies, and international cooperation. The fundamental principles guiding the conduct of space activities include the notion of space as the province of all humankind, the freedom of exploration and use of outer space by all states without discrimination, and the principle of non-appropriation of outer space.

### *Space Law in India*

- An international dispute arose when debris of an Indian satellite from orbit fell back to Earth and landed on a fishing village in Japan. In another such instance, an Indian Space Research Organisation rocket launch failed and it blew up on the launching pad, which destroyed a dozen countries' satellites, which were on the launching stage.

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- The Government of India (GoI) unleashed reforms in space domain in 2020, opening the doors for enhanced participation of NGEs in carrying out end-to-end activities in the space domain and with an aim to provide them a level playing field. Subsequent to these reforms, the Government seeks to provide regulatory certainty to space activities by various stakeholders, in order to create a thriving space ecosystem.
- The new Indian Space Policy, 2023 made public on April 20, 2023 has been formulated as an overarching, composite and dynamic framework to implement the reform vision approved by the Cabinet.
- Vision: To augment space capabilities; enable, encourage and develop a flourishing commercial presence in space; use space as a driver of technology development and derived benefits in allied areas; pursue international relations, and create an ecosystem for effective implementation of space applications among all stakeholders; for the nation's socio-economic development and security, protection of environment and lives, pursuing peaceful exploration of outer space, stimulation of public awareness and scientific quest.
- Authority: IN-SPACe shall function as an autonomous Government organisation, mandated to promote, hand-hold, guide and authorise space activities in the country. For this purpose, IN-SPACe shall periodically issue guidelines and procedures, that would among other things promote ease of doing business.
- ISRO, as the National Space Agency, will focus primarily on research and development of new space technologies and applications, and for expanding the human understanding of outer space.
- NewSpace India Limited
- Department of Space
- Jurisdiction: This policy is applicable to any space activity to or from the Indian Territory or within the jurisdiction of India including the area to the limit of its exclusive economic zone. Notwithstanding anything contained in this policy, the GoI reserves its right to provide exemptions to the provisions contained herein on a case-to-case basis.

## 8.3 Scientific Space Craft of India

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## 8.3.1 Successful Scientific Space Craft of India

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### 8.3.1.1 Chandrayaan I

- ISRO's first mission to moon was an unmanned lunar spacecraft launched successfully on **22 October 2008** from SDSC, Sriharikota. On 12 November 2008, with 2 years of life span, the probe was successfully inserted into lunar orbit. Through this mission, several questions related to the moon's origin, evolution and chemical as well as physical structure got answered. More than `380 crores were spent on this mission, while a thousand scientists worked on it.
- Chandrayaan I is a cuboid shaped satellite **weighing 1,380 kilograms** at lift off. There were **11 payloads** on-board the satellite. It was a 3-axis stabilised spacecraft deploying two star sensors. The spacecraft was powered by solar arrays and by lithium ion batteries during eclipse. Out of the **11 payloads** on Chandrayaan I, most of them were foreign countries' payloads.
- The probe (as mentioned above), was said to have a **life span of 2 years**, but ISRO lost contact with it in August 2009 due to the failure of the star sensors, and unusually high levels of radiation fried the communications array, as per some scientists.
- According to ISRO officials, the mission had achieved 95% of its objectives before losing all contacts. The mission was officially decommissioned on 30 August 2009. The mission helped **map 50% of the lunar surface**, and **more than 90% of the mineralogy.** It was the first mission that had a satellite as low as 100 kilometres from the moon's surface. Chandrayaan I took around 70,000 photographs of the lunar surface with an unbeatable resolution of 5 metres, which helped to confirm the presence of **iron and iron-bearing minerals (pyroxene),** and **evidences of water** on the moon.
- **US, former Soviet Union, Japan** (albeit via a malfunction that sent its orbiter crashing onto the lunar surface) and the **European Space Agency** (17 nations) were other countries which have reached the Moon and India becomes the fifth member of this club.
- This mission bagged many other notable awards and recognitions, along with the 2009 Annual **AIAA Space Awards [American Institute**

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**of Aeronautics and Astronautics (AIAA)]** and one for accommodating most international payloads ever.

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### 8.3.1.2 Mars Orbiter Mission (MOM/Mangalyaan)

- Mars, among all the other planets, has been the centre of prime attraction since years of research. Its atmospheric values and somewhat Earth-like gravity may point towards the fact that it can be inhabited.
- India became the **first country to reach the Mars orbit successfully in the first attempt** via Mars Orbiter Mission of ISRO (which stands out to be first interplanetary mission making history).
- On 5 November 2013, it was launched from Satish Dhawan Space Centre by PSLV-C25 with an approximate launch mass of 1,350 kilograms.
- It was launched on Mars with an objective to explore the surface features, morphology, mineralogy and Martian atmosphere.
- It had **five scientific payloads**.
- Its team won the US-based National Space Society's "**Space Pioneer Award**" for science and engineering category for the year 2015. ISRO also bagged the 2014, **"Indira Gandhi Prize for Peace, Disarmament and Development"** for its significant contribution towards strengthening international cooperation in peaceful use of outer space. **Mangalyaan's success makes India only the fourth entity and the first Asian nation to put a spacecraft in Mars' orbit, after the United States, the Soviet Union, and the European Space Agency.**

### 8.3.1.3 Astrosat

- India's first dedicated **multi-wavelength space observatory** is ASTROSAT.
- It is ISRO's first satellite dedicated towards the study of celestial sources, or as commonly said – stars, in X-ray, optical, and UV spectral band with a **lift-off mass of 1,513 kilograms**.
- Its unique features include simultaneous multi-wavelength observations of various astronomical objects with a single satellite. It is said to have a minimum useful life of five years.

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- India joined the elite list of countries having its own space observatory after the **US, Japan, Russia and Europe**, with the successful launch of ASTROSAT.

**KEY POINTS**

- AstroSat is **India's first multi-wavelength space telescope, which has five telescopes** seeing through different wavelengths simultaneously – visible, near UV, far UV, soft X-ray and hard X-ray.
- Onboard the AstroSat is a 38-cm wide **Ultra Violet Imaging Telescope (UVIT)**, which is capable of imaging in far and near-ultraviolet bands over a wide field of view.
- It is **a multi-institute collaborative project**, involving IUCAA, ISRO, Tata Institute of Fundamental Research (Mumbai), Indian Institute of Astrophysics (Bengaluru), and Physical Research Laboratory (Ahmedabad), among others.

- In **September 2015**, it was launched by a **PSLV-C30**, into a 650-kilometre orbit.
- It aims at apprehending the high-energy processes in binary star systems containing neutron stars and black holes, and estimate the star's magnetic fields and study star birth regions and processes in star systems beyond our galaxy. Additionally, it helps to detect new bright sources of X-ray (expecting new stars) and, a deep field survey of the Universe in the Ultraviolet region.
- **Five payloads** of ASTROSAT are chosen to facilitate a deeper insight into the various astrophysical processes occurring in the various types of astronomical objects constituting our universe.

### 8.3.1.4 Chandrayaan-2

- In September 2008, the mission of Chandrayaan-2 was approved by the government for a cost of `425 crore. It is India's **second unmanned mission to the moon**. It aims **to explore the Moon's South Polar Region**.
- **The Chandrayaan-2 is an Indian Space Research Organization (ISRO) lunar mission *comprising an orbiter, lander (Vikram) and a rover (Pragyan), with a lift-off mass of 3877 kg. This was the first space mission to conduct a soft landing on the Moon's South Polar***

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***Region which has a lunar surface area much larger than that of the North Pole and remains in shadow.***

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### *Objectives of the Mission*

- The primary objective of Chandrayaan-2 is **to demonstrate the ability to soft-land on the lunar surface and operate a robotic rover on the surface. Scientific goals include studies of lunar topography, mineralogy, elemental abundance, the lunar exosphere, and signatures of hydroxyl and water ice**.
- **India's rocket** carrying **an orbiter, lander and a rover** in a stepped up moon mission launched into space successfully from the Sriharikota launch pad on ***22 July 2019***. The **Chandrayaan-2** was set on course to place the country in a niche league of nations populated by the ***US, Russia and China that have successfully conducted a moon-landing***.
- **The lander was to touch in the region of the south pole of the moon, an unexplored territory.** The target is to find more **evidence of water**, **clues about the moon's evolution** and **work the natural satellite** as a test bed for more space missions concerning the solar system. **The launch vehicle, the GSLV Mk-3, is the country's most powerful rocket to date.**
- **The *1400 kg Lander, Vikram*, will be alive for 14 earth days, or one day in the moon. It can communicate with the orbiter 100 km from the surface of the moon, and ISRO's earth station near Bengaluru.** But Vikram **lander** lost contact with ground control around 2.1 km from the lunar surface, during its soft landing.
- Chandrayaan-2 is loaded with **13 payloads (Orbiter – 8 Payloads, Lander – 3 Payloads & Rover – 2 Payloads)** that will carry out detailed study of topography, seismography, mineral identification and distribution, surface chemical composition, thermo-physical characteristics of top soil and composition of the tenuous lunar atmosphere for a new understanding of the origin and evolution of Moon.

(Refer **Infographic – Chandrayaan-1 Vs. Chandrayaan-2**)

**SOFT LANDING**

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- A soft-landing protects the object from impact while a hard landing doesn't.
- Soft-landing ensures that the object is able to carry out further experimentation on the target planet or satellite, mostly with the help of a rover vehicle.
- Soft-landing on any planetary surface is complicated. **Vikram was to use five thrusters –** four at the corners and one at the centre to make its final descent.
- Maintaining the required velocity with such thrusters is difficult as a fine balance among them needs to be maintained.
- Then there is the issue of moon dust which could wreck the engines of the thrusters.

## WHY EXPLORE THE LUNAR SOUTH POLE?

The South Pole is especially interesting because of the lunar surface area here that remains in shadow are much larger than that at the North Pole. There's a possibility of presence of water in permanently shadowed areas around it. Also, South Pole region has craters that are cold traps and contain a fossil record of the early Solar System.

**Chandrayaan Missions**

| | Chandrayaan-1 | Chandrayaan-2 | Chandrayaan-3 |
|---|---|---|---|
| Launch Date | October 22, 2008 | July 22, 2019 | July 14, 2023 |
| Objectives | Lunar surface mapping Detection of water ice | Soft landing attempt Studies of lunar topography | Soft landing attempt In-situ scientific experiments |
| Orbiter | Yes | Yes | Yes |
| Lander | No | Vikram Pragyan | Yes |
| Rover | No | Pragyan | Yes |
| Mission Result | Successful | Partially successful | Successful |

### *What was Supposed to Happen?*

- On the day of landing on 7 September 2019, **the Vikram lander** had to perform a series of complex manoeuvres, including imaging the landing site.
- **Vikram** was supposed to begin its descent from a height of 35 km above the lunar surface and at a velocity of around 6,000 kmph. In just over 10 minutes, the Vikram lander had to drop to a height of 7.4 km above the Moon altitude and lower its speed to around 526 kmph.
- Further, the lander had to reduce its speed to 0331.2 kmph and reach a height of 5 km above the lunar surface.
- At 100 metres above the lunar surface, the Vikram lander had to hover for about 25 seconds during which it was supposed to **choose between**

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**two pre-determined landing sites**.

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- Four hours after landing, the **Pragyan rover** had to unload from the Vikram lander.

**WHY EXPLORE THE LUNAR SOUTH POLE?**

It is a **NASA mission to the moon** within **the Lunar Precursor and Robotic Program (LPRP)** in preparation for future manned missions to the moon and beyond (Mars). LRO is **the first mission of NASA's 'New Vision for Space Exploration'**.

**The objectives of LRO are to:**

- Identify potential lunar resources.
- Gather detailed maps of the lunar surface.
- Collect data on the moon's radiation levels.
- Study the moons polar regions for resources that could be used in future manned missions or robotic sample return missions.
- Provide measurements to characterise future robotic explorers, human lunar landing sites and to derive measurements that can be used directly in support of future Lunar Human Exploration Systems.

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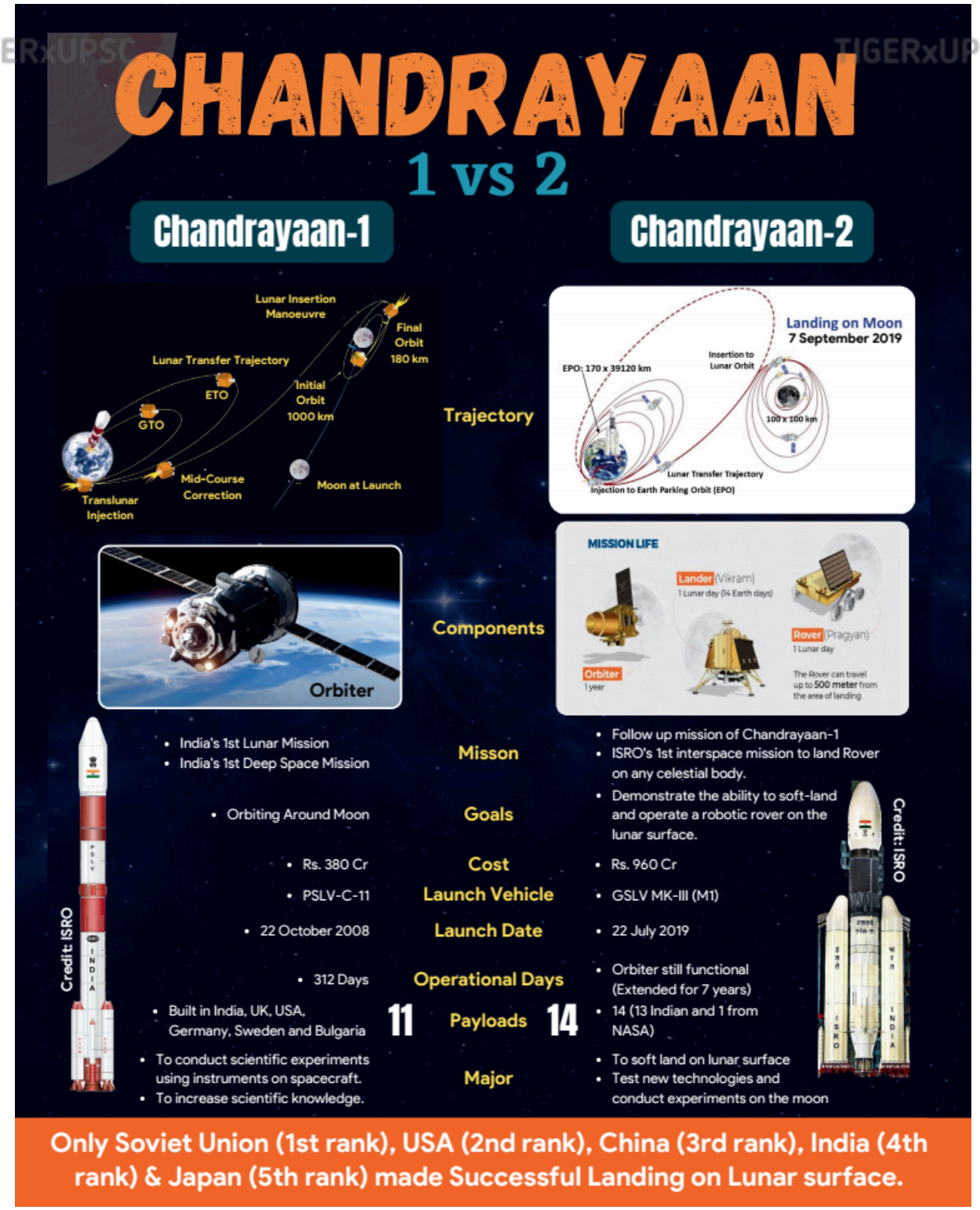


**ISRO** is attempting to figure out what happened to **Chandrayaan-2's Vikram** with the help of NASA's **Lunar Reconnaissance Orbiter (LRO)**.

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### 8.3.1.5 Chandrayaan 3

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India's historic Chandrayaan 3 Mission was successfully launched from SDSC on 14 July 2023 using a special rocket/launch vehicle LVM3-M4 (GSLV MKIII M4) and it successfully landed on moon surface on 23 August 2023 (6:04 PM).

India's Moon mission, Chandrayaan-3 created history by accomplishing a **'soft landing' on the Moon's south pole.** With this, India became the **first country to land near the South pole** of the Moon.

#### Facts: India's Ranking

- Study on Moon – Fifth Rank
- Soft Landing on Moon Surface – Fourth Rank
- Landing towards / near South Pole – First Rank

**Table 8.1**

| | |
|---|---|
| Mission Life (Lander & Rover) | One lunar day (~14 Earth days) |
| Landing Site (Prime) | 4 km × 2.4 km 69.367621 S, 32.348126 E |

**Chandrayaan-3 contains three modules with seven payloads:**

- **Lander Module (Vikram):** It was responsible for the soft landing on the Moon. It carried the rover and various scientific instruments to perform in-situ analysis. It contains **four scientific payloads**.
- **Rover Module (Pragyan):** It is equipped with payloads that will provide valuable data to the scientific community. It contains **two scientific payloads**.

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- **Propulsion Module:** It took the lander and the rover to the moon. This module did not land on the moon and instead settled in the parking orbit around the moon. It contains **one scientific payload**. **(Refer Infographic – Chandrayaan 3)**

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Note: **Slingshot Method** is used to make this project successful. The Slingshot method, scientifically known as **Hohmann Transfer Orbits**, is a technique to efficiently reach distant celestial bodies. It involves **utilising the gravitational forces** of celestial bodies to achieve the desired trajectory without the need for powerful and heavy-lifting rockets. The Slingshot method is an **energy-efficient way** to travel through space. It allows spacecraft to reach their destinations using limited resources and without the need for extremely powerful rockets.

Note: The point where the Vikram lander touched down on the lunar surface is named **Shiv Shakti** and the crash site of the Chandrayaan-2 mission (2019) is named **Tiranga Point**.

### 8.3.1.6 Aditya-L1 Mission

- ISRO's Aditya L-1 is successfully placed in a halo orbit around the L-1 point.
- Aditya-L1 is the first Indian space mission to observe the Sun and the solar corona.
- **L1 orbit allows** Aditya-L1 to look at the Sun continuously.
- It was launched aboard **a PSLV-XL launch vehicle** on **2 September, 2023** and **reached successfully on 6 January 2024 (approx 16:00 pm).**
- Aditya-L1, launched by ISRO, marks **India's second astronomy observatory-class mission**, following AstroSat (2015).

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## The Objective of the Mission

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- To study solar upper **atmospheric (chromosphere and corona) dynamics** and understand the physics of the solar corona and its heating mechanism.

## Significance of Exploring the Sun

1. **Understanding Our Solar System:** Exploring Our Solar System: Investigating the Sun offers valuable insights into the dynamics of our celestial neighbourhood.
2. **Forecasting Space Weather:** Solar phenomena like flares and coronal mass ejections have the potential to disrupt Earth's space environment, impacting communication systems, navigation, and power grids. By studying the Sun, we can better predict and mitigate these space weather hazards.
3. **Advancing Solar Science:** Examining the Sun's intricate behaviours, including its magnetic fields, heating mechanisms, and plasma dynamics, contributes to progress in fundamental physics and astrophysics.
4. **Enhancing Energy Studies:** As a natural fusion reactor, the Sun produces immense energy through nuclear fusion processes. Researching its core and nuclear reactions provides crucial knowledge for developing clean and sustainable energy sources on Earth.
5. **Optimising Satellite Operations:** Solar radiation and solar wind influence the performance and lifespan of satellites and spacecraft in orbit. Understanding these solar interactions enables the improvement of spacecraft design, shielding, and operational strategies, ensuring the reliability and durability of space missions.

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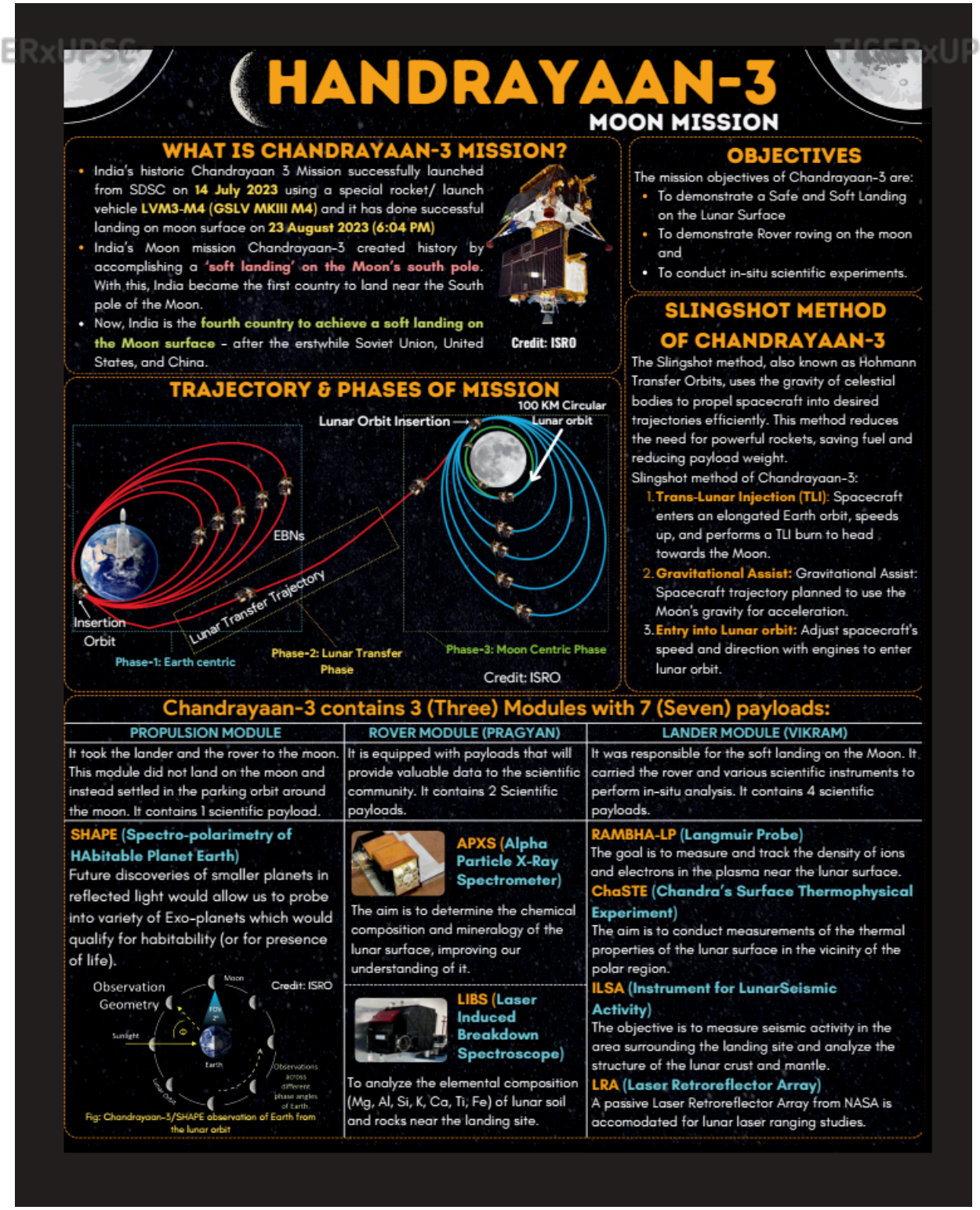


## Why do we study the sun and the solar wind?

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- The Sun is ***the only star we can study up close.***

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- The sun is ***a source of light and heat for life on Earth.***
- It is ***the source of the solar wind;*** a **flow of ionised gases from the sun that streams past Earth at a speed of more than 500 km per second** (a million miles per hour).
- ***Disturbances in the solar wind shake Earth's magnetic field and pump energy into the radiation belts,*** part of a set of changes in near-Earth space known as space weather.

The **major science objectives of Aditya-L1 mission** are:

- Study of Solar upper atmospheric (chromosphere and corona) dynamics.
- Study of chromospheric and coronal heating, physics of the partially ionised plasma, initiation of the coronal mass ejections, and flares.
- Observe the in-situ particle and plasma environment providing data for the study of particle dynamics from the Sun.
- Physics of solar corona and its heating mechanism.
- Diagnostics of the coronal and coronal loops plasma: Temperature, velocity and density.
- Development, dynamics and origin of CMEs (Coronal Mass Ejections).
- Identify the sequence of processes that occur at multiple layers (chromosphere, base and extended corona) which eventually leads to solar eruptive events.
- Magnetic field topology and magnetic field measurements in the solar corona.
- Drivers for space weather (origin, composition and dynamics of solar wind). [For details, visit www.isro.gov.in]

## Payloads

- The instruments of Aditya-L1 are tuned to observe the solar atmosphere, mainly the chromosphere and corona.
- There are **seven payloads** on board, with **four for remote sensing of the Sun (they directly view the sun)** and three for in-situ observation.
- **In-situ instruments will observe the local environment at the L1 point**.

**Table 8.2** Payloads with their major capability

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| Type | Sl. No. | Payload | Capability |
|---|---|---|---|
| **Remote Sensing Payloads** | **1** | Visible Emission Line Coronagraph (VELC) | Corona/Imaging & Spectroscopy |
| | **2** | **Solar Ultraviolet Imaging Telescope (SUIT)** | Photosphere and Chromosphere Imaging-Narrow & Broadband |
| | **3** | Solar Low Energy X-ray Spectrometer (SoLEXS) | Soft X-ray spectrometer: Sun-as-a-star observation |
| | **4** | High Energy L1 Orbiting X-ray Spectrometer (HEL1OS) | Hard X-ray spectrometer: Sun-as-a-star observation |
| **In-situ Payloads** | **5** | Aditya Solar wind Particle Experiment (ASPEX) | Solar wind/Particle Analyser Protons & Heavier Ions with |

*Source:* www.isro.gov.in

## The Solar Ultraviolet Imaging Telescope (SUIT)

The Solar Ultraviolet Imaging Telescope (SUIT), **developed by Pune's Inter-University Centre for Astronomy and Astrophysics (IUCAA)**, has been handed over to the Indian Space Research Organisation (ISRO).

**This cutting-edge space telescope was integrated into ISRO's ADITYA-L1 mission.**

## Overview of SUIT

- SUIT is designed to **study the Sun's ultraviolet (UV) emissions and capture high-resolution images of its atmosphere, particularly the corona, across various UV wavelengths**.
- **Operating in the far and near ultraviolet regions**, SUIT will cover wavelengths ranging from ***200 to 400 nanometers.***
- It focused on observing the Sun's hotter and more dynamic regions, such as the transition region and the corona, providing valuable insights

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into these critical aspects of solar activity.

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## Significance of SUIT

- The Sun poses unique challenges for scientific study due to its intense emissions and radiation.
- SUIT will play a pivotal role in **unravelling the mysteries of the Sun and its impacts on Earth and other celestial bodies.**

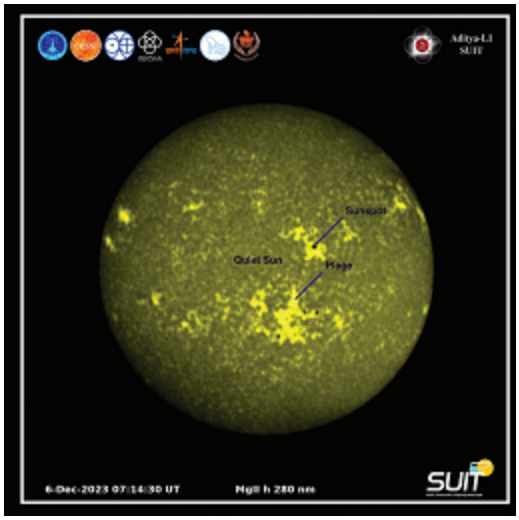


**Figure 8.1** Aditya-L1's SUIT

*Source:* www.isro.gov.in

- Additionally, **SUIT will measure UV radiation levels, contributing to the understanding of its potential hazards, including the risk of skin cancer.**
- By monitoring the Sun's activity***, SUIT will facilitate early warnings of solar flares and coronal mass ejections (CMEs),*** phenomena that can disrupt satellite operations, communication systems, power grids, and even pose risks to human health on Earth.

**WHAT IS HALO ORBIT?**

**How It Works:** The Lagrange point is just an equilibrium point with no gravitational pull and hence, the spacecraft does not technically orbit the actual Lagrange point, instead the spacecraft in a halo orbit moves in a circular path around the Lagrange

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point. Halo orbits are the result of a complicated interaction between the gravitational pull of the two planetary bodies and the coriolis and centrifugal accelerations on a spacecraft.

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**ADITYA-L1 SUPPORT CELL**

- Aditya-L1 Support Cell is a community service centre that has been set up to bring all data on board India's first dedicated solar space mission to a single webbased interface.
- It is **a joint effort of Indian Space Research Organisation and Aryabhatta Research Institute of Observational Sciences.**

(Refer **Infographic – Aditya-L1 Mission**)

### 8.3.2 Space Docking Experiment (SPADEX)

- **SpaDeX, or Space Docking Experiment,** is an initiative by the Indian Space Research Organisation (ISRO) aimed at demonstrating the technology for space docking.
- The **mission involved launching two satellites, SDX01 and SDX02, on December 30, 2024.** The **satellites docked on January 16, 2025,** making India the fourth country to achieve space docking.
- **ISRO successfully separated two satellites** – SDX01 (Chaser) and SDX02 (Target) – during **the SpaDeX mission**, achieving a **significant milestone in India's space exploration efforts**.
- With this achievement, **India becomes the fourth country**, following the USA, Russia and China, to successfully demonstrate space docking technology

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***About Mission***

| Feature | Details |
|---|---|
| **Launch** | 30 December 2024, via **PSLV–C60** from Sriharikota |
| **Satellites Involved** | **SDX01 (Chaser)** and **SDX02 (Target)** |
| **Mission Goal** | Develop and demonstrate **autonomous rendezvous and docking** capabilities |
| **Post-Docking Activities** | • Evaluate system stability and control<br>• Test power transfer between spacecraft<br>• Study combined maneuvering<br>• Explore life |

**In the context of space docking, "chaser" and "target" refer to the roles of two spacecraft during a rendezvous (meeting site) and docking maneuver (practices).**

The "chaser" spacecraft is the one that performs the maneuvers to approach and dock with the "target" spacecraft. **The target is the spacecraft that the chaser is docking with.**

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# ADITYA-L1 MISSION

**The first observatory-class space-based solar mission from India**

## About Aditya-L1

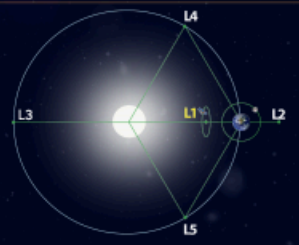
- First Indian Solar Mission: Aditya-L1 is **India's first mission to observe the Sun and the solar corona.**
- Orbit: **Successfully placed in a halo orbit around the Lagrangian point 1 (L1)** of the Sun-Earth system.
- Continuous Sun Observation: The L1 position allows uninterrupted observation of the Sun, free from occultation or eclipses.
- Launch Details:
- Launch Vehicle: **PSLV-XL.**
- Launch Date: **2nd September 2023.**
- Arrival at L1: **6th January 2024,** approximately 16:00 PM.
- Payloads:
  - Total: **Seven (7) payloads.**
  - Sun Observation: Four payloads to observe the photosphere, chromosphere, and corona.
  - In-situ Studies: Three payloads for studying particles and fields at L1.
- Particle and field propagation in the interplanetary medium.
- Significance: Marks India's second astronomy observatory-class mission, following AstroSat (2015).

## Major Science Objectives

- Study the **solar upper atmosphere** (chromosphere and corona).
- Understand **solar corona heating.**
- Analyze **Coronal Mass Ejections (CMEs).**
- Observe pre-flare and flare activities.
- Investigate space weather dynamics.

## Lagrange Points

- **Points of equilibrium** for small objects influenced by two massive orbiting bodies.
- Gravitational pull of the two large masses equals the centripetal force needed for a small object to stay in position.
- Useful for spacecraft to **minimize fuel consumption.**

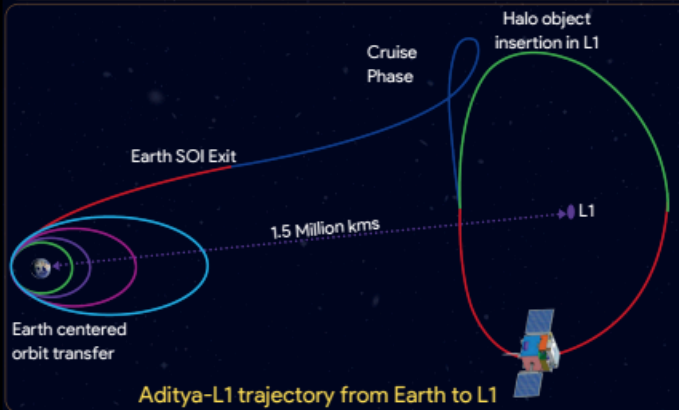



## Types of Lagrange Points

- **L1** (1.5 million km from Earth): Ideal for continuous solar observations; used by SOHO (ESA & NASA).
- **L2**: Excellent for observing the broader Universe without Earth's shadow; home to the James Webb Space Telescope.
- **L3**: Positioned behind the Sun, opposite Earth; useful for observing the Sun's far side.
- **L4 and L5**: Form stable equilateral triangles with Earth and the Sun; ideal for space observatories and studying asteroids.

## Significance of Exploring the Sun

1. Understanding Our Solar System 2. Space Weather Prediction
3. Advancing Solar Physics 4.Enhancing Energy Research
5.Improving Satellite Operations




Aditya-L1 trajectory from Earth to L1

## Why do we study the sun and the solar wind?

- The Sun is the only star we can study in detail up close.
- It provides light and heat essential for life on Earth.
- The Sun emits solar wind, a stream of ionized gases that travel past Earth at over 500 km per second (about a million miles per hour).
- Solar wind disturbances affect Earth's magnetic field, injecting energy into the radiation belts, contributing to changes in near-Earth space known as space weather.

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## SOLAR STORM (GEOMAGNETIC STORM)

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## What are Solar Storms?

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- **Solar storms** are intense bursts of energy originating from the Sun, often caused by large eruptions on its surface.
- These eruptions are commonly linked to **solar flares** and **coronal mass ejections (CMEs)**, which **propel charged particles at extremely high velocities**.
- These high-speed solar winds can lead to **geomagnetic storms** upon interacting with Earth's magnetic field.
- Solar storms are more frequent during periods of high solar activity, known as the **Solar Maximum**.

## Understanding Solar Flares and CMEs

**Solar flares and coronal mass ejections (CMEs)** are distinct but related solar events caused by the Sun's magnetic field.

**Solar flares** are sudden bursts of radiation, primarily electromagnetic energy, while CMEs are large expulsions of plasma and magnetic field into space.

- The particles ejected during a CME can travel at speeds exceeding **one million miles per hour**, potentially reaching Earth and affecting its magnetic environment.

## The Sun's Activity Cycle

- The Sun follows an 11-year cycle of rising and falling activity, marked by changes in the number of sunspots and the frequency of solar storms.

## Implications of Solar Storms

1. **Damage to Space Infrastructure**
   - Energetic solar particles can penetrate spacecraft shielding, potentially disrupting electronics and damaging sensitive components.
2. **Impact on Ground-Based Infrastructure**
   - Solar storms can induce currents in power lines, potentially leading to transformer failures and widespread **power grid disruptions.**
3. **Risks to Satellites**
   - Changes in atmospheric density and drag can alter satellite orbits, making it difficult to maintain proper satellite positioning and control.
4. **Disruption of the Ionosphere**
   - Solar activity affects the ionosphere, interfering with **high-frequency radio signals**, **GPS accuracy** and **aviation communications**, especially at high latitudes.
5. **Hazards for Astronauts**
   - Increased radiation levels during solar storms pose serious health risks to astronauts aboard the International Space Station and other missions beyond Earth's protective atmosphere.
6. **Auroras (Northern and Southern Lights)**
   - **When charged particles from the Sun interact with gases in Earth's upper atmosphere,** they create stunning natural light displays known as auroras,

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which are most visible near the poles.

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***Chaser:***

The chaser spacecraft is the one that initiates the rendezvous and docking process. It uses its propulsion system and navigation systems to approach the target.

***Target:***

The **target spacecraft is the stationary object** that the chaser is trying to dock with. It may also have a docking mechanism and may be a space station, another spacecraft, or a specific docking port.

***Rendezvous (meeting site):***

This is the **initial phase where the chaser spacecraft maneuvers (practices) to get close** to the target.

***Docking:***

This is the **final phase where the chaser spacecraft physically attaches to the target spacecraft.**

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Note: If you want to see docking process, kindly follow this Link: **https://www.isro.gov.in/mission_spadex_video_gallery.html**

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***Significance of SpaDeX***

1. **Space Capability Milestone:** Paves the way for advanced missions like:
    1. **Sample return missions**
    2. **Bharatiya Antariksh Station (BAS) –** India's own space station
    3. It provides platform for various **experiments in microgravity environment**
2. **Satellite Servicing:** Enables **in-orbit re-fuelling, repair or upgradation**
3. **Support for Complex Missions:** Useful in multi-stage missions and **modular spacecraft construction**
4. **Applications in Future:**
    1. **In-space robotics**

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2. Support for **Earth observation, vegetation studies, natural resource monitoring**

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Note: SpaDeX also facilitated research on plant growth and bacterial behavior, green propulsion systems and artificial intelligence experiments in microgravity.

## 8.3.3 Future Scientific Space Craft of India

### 8.3.3.1 Gaganyaan Programme

The Union Cabinet has approved the Gaganyaan Programme with demonstration of Indian Human Spaceflight capability to low earth orbit for a mission duration ranging from one orbital period to a maximum of seven days. ISRO aims to launch its maiden **Human Space Mission**, Gaganyaan in the first quarter of 2027.

*India could potentially become the fourth country to send a man in space*, after the erstwhile ***USSR, the US and China***.

- Its total budget cost is approximately `9023 crore.
- Its expected mass is approximately 8200 kg (includes Service Module).
- The Indian space agency displayed the Gaganyaan crew model and orange space suits at the Bengaluru Space Expo's 6th edition September 6–8, 2018.

### Gaganyaan Mission

- **Astronaut Training:** Four astronauts **underwent extensive training at Russia's Yuri Gagarin Cosmonaut Training Centre (Russian**

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**agency ROSCOSMOS Centre)** to prepare for the mission, **including simulations of weightless conditions**.

- The Gaganyaan mission involves **cooperation with various entities**, including the Indian Armed Forces, ISRO, academic institutes, industry partners and maritime agencies, among others.

### *Stages and Milestones of the Gaganyaan Mission*

- **Integrated Air Drop Test (IADT):** Conducted to test the **Parachute Deceleration System**, simulating the descent of the crew module from an altitude of 2.5 km.
- **Pad Abort Test (PAT):** A trial to **evaluate the spacecraft's launch abort system**, crucial for crew safety during launch.
- **Crew Abort Test Mission:** An **unmanned test mission to simulate abort conditions during ascent**, ensuring crew safety in emergency scenarios.
- **Vyomitra:** A humanoid robot expected to onboard the **second unmanned Gaganyaan mission, performing microgravity experiments and monitoring module parameters**.
- **Manned Flight:** The culmination of the Gaganyaan mission, where a human astronaut will be launched into space and safely returned to Earth, marking a historic achievement for India in human spaceflight.

### *Flight Schedule in Three Phases*

- **Phase 1 (2023):** Uncrewed Spacecraft Flight tested succesfully (Test Vehicle Abort Mission – 1/TV-D1)
- **Phase 2 (2025–2026):** Uncrewed Spacecraft Flight Test (TV–D2) + Vyommitra
- **Phase 3/Final Phase (2027):** Crewed Spacecraft

(Refer **Infographic – Mission Gaganyaan**)

### *Creation of Gaganyaan National Advisory Council*

Gaganyaan National Advisory Council was created in 2019 with members from different institutions and industries.

### *Human Space Flight Centre (HSFC)*

Human Space Flight Centre (HSFC) is created to carry out the activities related to human spaceflight. **HSFC shall be responsible for the implementation of Gaganyaan.** The project involves mission planning,

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development of engineering systems for **crew survival in space, crew selection and training and also pursue activities for sustained human space flight missions**. **HSFC will take the support of ISRO centres** to implement the first developmental [crewed] flight.

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### *Contribution of DRDO for Gaganyaan Mission*

ISRO, DRDO sign MoU to provide critical technologies for **Human Space Mission**. Under the agreement, the technological capabilities existing in DRDO labs for defence applications will be customised to meet the requirements of ISRO's human space mission. DRDO will be providing critical technologies to ISRO such as space crew health monitoring and emergency survival kit, space food, parachutes for the crew module's safe recovery and radiation measurement and protection.

### *Objectives of the Mission*

- Enhancement of science and technology levels in the country.
- A national project involving several institutes, academia and industry.
- Improvement of industrial growth.
- Inspiring youth
- Development of technology for social benefit
- Improving international collaboration

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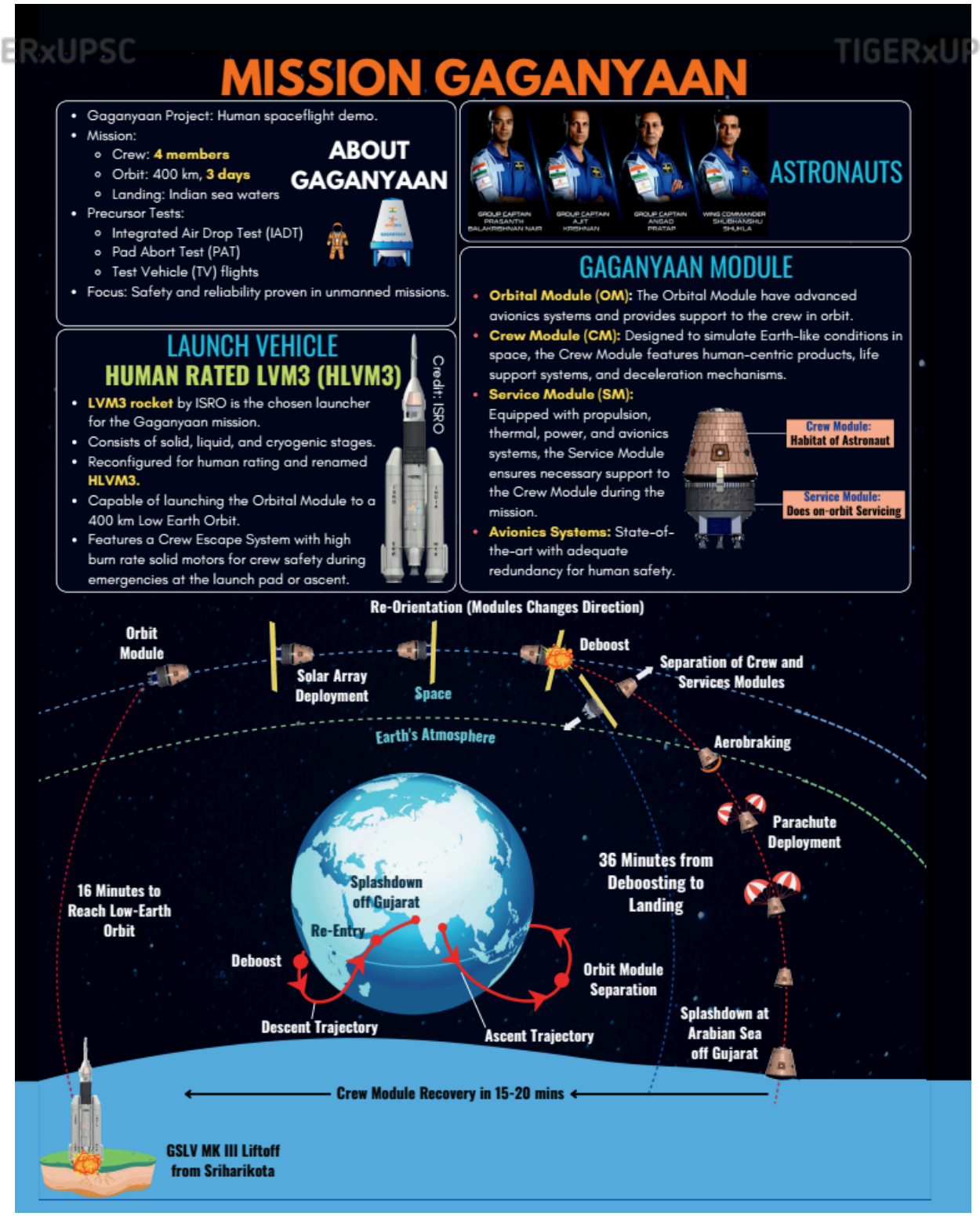


### *Challenges of Gaganyaan Programme*

- **Gravity field:** Transitioning from one gravity field to another is tricky. It affects hand-eye and head-eye coordination.

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- **Isolation:** In spite of providing well training to astronauts, behavioural issues may crop up. They may encounter depression, sleep and psychiatric disorders, etc.
- **Radiation:** In space stations, astronauts receive over ten times the radiation than what people are subjected to on Earth. It may increase the risk of cancer, etc.
- **Hostile Environment:** Space is hostile.
- In addition to lack of gravity and danger of radiation, there is no atmosphere. Human blood starts boiling if there is no pressure.

### *Relevance of a Manned Space Mission for India*

- **Boost to Industries:** The Indian industry will find large opportunities through participation in the highly demanding space missions. Gaganyaan Mission is expected will source nearly 60% of its equipment from the Indian private sector.
- **Employment:** According to the ISRO chief, the Gaganyaan mission would create 15,000 new employment opportunities, 13,000 of them in private industry and the space organisation would need an additional manpower of 900.
- **Technological Development:** Human Space flights are frontier field in the science and technology. The challenges the Human Space Flights provide to India, and the benefits accruing from taking up those missions will be very high and will lead to further thrust for technological developments in India.
- **Spurs Research and Development:** It will boost good research and technology development. With a large number of researchers with proper equipment involved, HSF will thrust significant research in areas such as materials processing, astrobiology, resources mining, planetary chemistry, planetary orbital calculus and many other areas.
- **Motivation:** Human space flight will provide that inspiration to the youth and also the national public mainstream. It would inspire young generation into notable achievements and enable them to play their legitimate role in challenging future activities.
- **Prestige:** **India will be the fourth country to launch human space mission**. The Gaganyaan will not only bring about prestige to the

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nation but also establish India's role as a key player in the space industry.

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**VYOMMITRA**

It is ISRO's ***first woman astronaut***. Vyommitra, equipped with a head, two arms and a torso, is built to ***mimic crew activity inside the crew module of Gaganyaan***.

It is capable of attaining launch and orbital postures, responding to the environment, generating warnings, replacing carbon dioxide canisters, operating switches, monitoring of the crew module, receiving voice commands, responding *via* speech (bilingual).

### 8.3.3.2 Shukrayaan Mission/Venus Orbiter Mission (VOM)

***Venus Orbiter Mission 'Shukrayaan'***

**Shukrayaan: Mission Overview**

ISRO receives government clearance for its Venus mission, **'Shukrayaan'**

**Objective:**

To explore Venus' **atmosphere, surface and interactions with solar radiation.**

***Key Goals:***

- Map **Venusian topography**
- Analyze **atmospheric composition**
- Study **solar X-ray interactions**
- Investigate **subsurface features**

**Launch Vehicle: LVM-3** (Launch Vehicle Mark-3)

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**Sanctioned Budget:** ₹1,236 crore

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**Science Phase Duration:** 5 years of exploration after reaching orbit

**Scientific Objectives:** The scientific objectives include studying surface processes, shallow subsurface stratigraphy, solar wind interactions with the Venusian ionosphere and analyzing the structure, composition and dynamics of Venus's atmosphere.

### *Scientific Features*

- **Total Payloads:** 19
  - ❑ 16 Indian instruments
  - ❑ 2 Indian-international collaborations
  - ❑ 1 fully international instrument
- **Orbit Maneuvering:** Use of **aerobraking** to reduce orbit to a low altitude (200 × 600 km) for detailed observations.
- **Aerobraking** is used when a spacecraft requires a low orbit after arriving at a body with an atmosphere, as it requires less fuel than using propulsion to slow down.

### *Other Missions to Venus*

#### *Historical Missions*

- **Venera Series (1961–1984)** – Soviet Union: Flybys, landers and balloons
- **Pioneer Venus (1978)** – NASA: Atmospheric and surface radar studies
- **Magellan (1989)** – NASA: Radar mapping of Venusian surface
- **Akatsuki (2010)** – JAXA: Atmospheric circulation studies

#### *Upcoming Missions*

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- **VERITAS (2026)** – NASA: Geologic radar mapping

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- **DAVINCI (2026)** – NASA: Probe of atmospheric structure and chemistry
- **EnVision (2030s)** – ESA: High-resolution radar imaging

### 8.3.3.3 LUPEX Mission (Lunar Polar Exploration Mission)

**India's 5th Lunar Mission** – Cleared by the National Space Panel

***Mission Objectives***

- LUPEX aims to **assess the quantity and quality of water** present on the Moon.
- It is designed to study the **far side** of the Moon, often referred to as the **dark side**, which **remains hidden** from Earth due to **tidal locking**.

## FUTURE MISSIONS OF INDIA

| Mission name | Expected launch Year | Spacecraft | Details |
|---|---|---|---|
| Gaganyaan | 2027 | Crewed spacecraft | ***Gaganyaan*** is an Indian crewed orbital spacecraft (jointly made by ISRO and HAL) intended to be the basis of the Indian Human Spaceflight Programme. The spacecraft is being designed to carry three people, and a planned upgraded version wi be equipped with rendezvous and docking capability. |
| Lunar Polar Exploration Mission | 2027–2028 | Lunar lander, rover | Lunar Polar Exploration Mission is a concept mission by JAXA and ISRO to explore the south pole region of the Moon ir |

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| Mission name | Expected launch Year | Spacecraft | Details |
|---|---|---|---|

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***International Collaboration***

- **ISRO:** Responsible for developing the **lander**
- **JAXA (Japan):** Will provide the **rover**
- **NASA & ESA:** Contributing **scientific instruments** to be mounted on the rover.

***Target Landing Site***

- **South Pole of the Moon**
  - ❑ This region is thought to contain significant deposits of water ice.
  - ❑ However, it presents challenges like **rugged terrain, low illumination** and **difficult communication**

***Significance***

- LUPEX is a **precursor to India's first lunar sample return mission (Chandrayaan 4)**
- It will also help lay the groundwork for **sending an Indian astronaut to the Moon by 2040**
- Follows the success of **Chandrayaan-3,** where the **Vikram lander** made India:
  - ❑ The first country to successfully land on the Moon's south pole.
  - ❑ The fourth nation to successfully land on the Moon, following the United States, Russia and China.

## 8.3.4.4 Chandrayaan 4

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After the success of **Chandrayaan-3**, ISRO has finalised the design of **Chandrayaan-4 on September 2024**, aimed at returning **lunar rock and soil samples** to Earth.

If successful, India will become the **fourth nation**, after the United States, Russia and China, to achieve a lunar sample return.

The total fund requirement for the technology demonstration mission "Chandrayaan-4" is estimated at ₹2,104.06 crore.

### *Key Objectives*

- Perform a **soft landing** on the Moon.
- Collect and **store lunar samples**.
- **Lift-off** from the Moon's surface.
- Execute **docking and undocking** in lunar orbit.
- **Transfer samples** between spacecraft.
- Return samples **safely to Earth**.

### *Mission Architecture: 5 Modular Components*

| Module | Function |
|---|---|
| **1. Propulsion Module** | • Carries the Lander and Ascender from Earth to the Moon |
| **2. Descender (Lander)** | • Executes lunar landing and soil sampling |
| **3. Ascender** | • Detaches after sampling, ascends to lunar orbit |
| **4. Transfer Module** | • The samples are received from Ascender and transferred to the Re-entry Module. |
| **5. Re-entry Module** | • Returns samples to Earth, endures atmospheric re-entry |

### *Launch Strategy: Dual-Rocket Deployment*

- **Launch Vehicle Mark-3 (LVM-3):**
  - ❑ **The Propulsion, Descender and Ascender modules will be carried.**
- **Polar Satellite Launch Vehicle (PSLV):**
  - ❑ Will launch the **Transfer and Re-entry modules** separately.
- Launches are expected **on different dates**, not before **2027 or 2028**.

### *Advanced Docking Technology – SPADEX*

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- **SPADEX (Space Docking Experiment)** is a precursor to Chandrayaan-4.
- Involves autonomous **docking between two spacecraft** in orbit.
- Demonstrates **rendezvous, formation flying and sample transfer**—all critical for Chandrayaan-4.

***Latest Developments***

- **Design is finalised;** awaiting final government clearance.
- **Two key technologies tested successfully** during Chandrayaan-3.
- **Chandrayaan-5** design is also complete, with details yet to be revealed.

### 8.3.3.5 Bharatiya Antariksh Station (BAS)

- **Union Cabinet approved the construction of India's first space station under the expanded scope of the Gaganyaan Programme.**
- **The revised programme includes:**
- **First BAS module launch by December 2028**
- Four crewed missions under Gaganyaan by **2026**
- **Total cost revised: From ₹12,000 Cr to over ₹20,000 Cr.**

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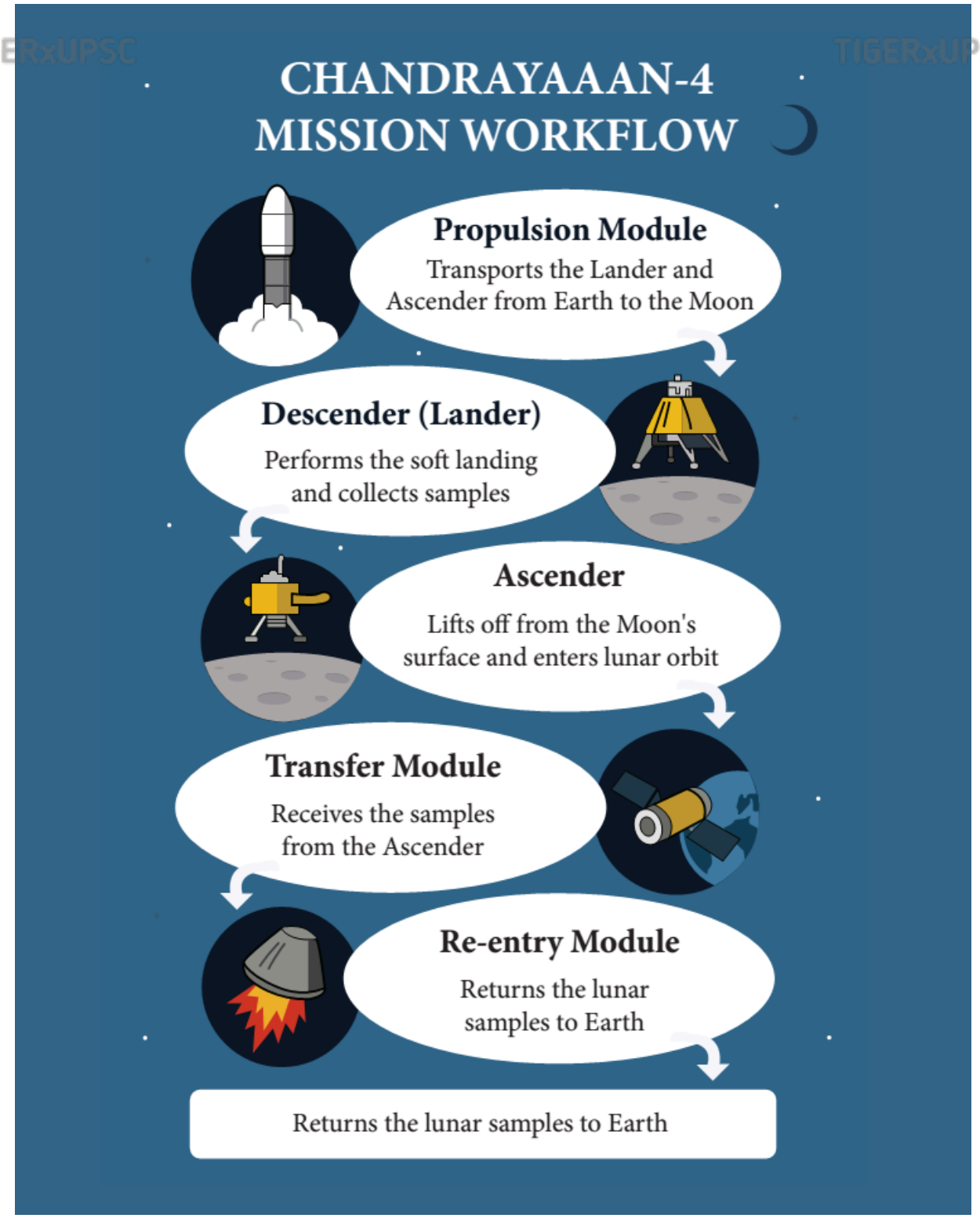


***About BAS***

- **Orbit Altitude:** ~400–450 km above Earth
- **Structure:** 5 modules to be launched in phases

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- **Operational Target:** Full functionality by **2035**
- **Current Stage:** Conceptualisation (studying design, docking ports, architecture, etc.)

***Significance of BAS***

| Area | Details |
|---|---|
| **Human spaceflight testing** | Life-support systems, health in microgravity, long-term habitation |
| **Earth Observation** | Disaster monitoring, better resolution, day-night coverage |
| **Microgravity Research** | Biomedical studies, e.g., muscle atrophy, bone density loss |
| **Innovation Boost** | Platform for startups to test space tech → job creation & entrepreneurship |
| **Technological Spin-offs** | Advanced materials, environmental sensors, aerospace applications |
| **International Prestige** | Enhances India's strategic autonomy and global space diplomacy |

***Challenges Ahead***

## Project-Level Challenges

- **Low R&D Budget:** Only **0.7% of GDP**, constraining tech development
- **Advanced Systems Required:** Life support, radiation shielding, orbital maintenance
- **Strategic Balancing:** Geopolitical implications with US, Russia, China, EU

**WHY STUDY VENUS?**

- Venus is often described as **the "twin sister" of the Earth** because of the similarities in size, mass, density, bulk composition and gravity.
- It is believed that both planets share a common origin, forming at the same time out of a condensing nebulosity around 4.5 billion years ago.
- Venus is around **30% closer to the Sun as compared to Earth resulting in much higher solar flux.**

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### Space Environment Challenges

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- **Astronaut health risks:**
  - ❑ Oxygen/pressure regulation
  - ❑ Brain fluid dynamics
  - ❑ Radiation exposure
  - ❑ Isolation and psychological effects
- **Space debris:**
  - ❑ Collision risks demand advanced tracking and avoidance systems

## 8.4 World's GPS: NAVSTAR, GLONASS

GPS is a satellite-based navigation system enabling the accurate position of the user on a map. It is extremely useful in marking the positions for an aircraft, submarine, or even an individual. To provide the accurate location, direction, speed, and time, it uses a group of 18 to 36 satellites. As of 2018, only the American **NAVSTAR (Navigation Satellite and Ranging)**, **Russian GLONASS (Global Navigation Satellite System)** and **China (Beidou/COMPASS)** are fully operational globally. Other big countries like **EU (GALILEO),** also expanded their regional navigational systems, and now it is also globalised. **NAVSTAR is the world's first GPS**. It has 31 satellites to itself with an accuracy of 3.5 to 7.8 metres. GLONASS, on the other hand, has 24 satellites operational at present (accuracy of 5 to 10 metres). GPS technology is not only used for navigation purpose but also for various other applications like operation of mobile phone in various banking operation, powergrid controls etc. Certain Countries have own regional GPS like **Japan (Quasi-Zenith Satellite System)** and **India (Gagan and IRNSS – Indian Regional Navigation Satellite System / NavIC – Navigation Indian Constellation)**.

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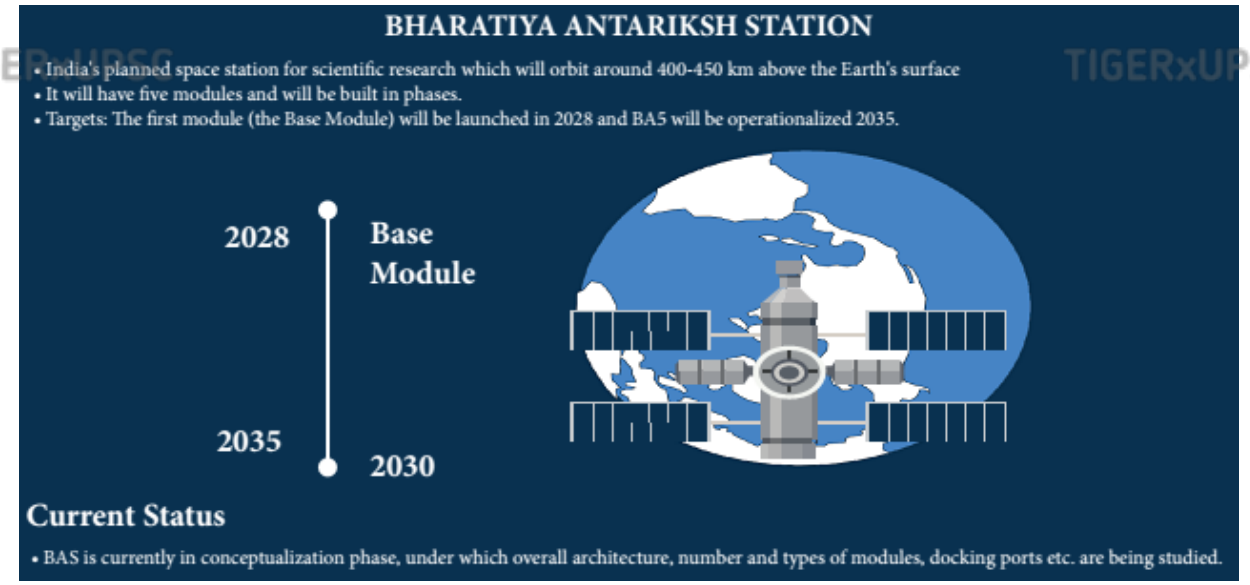


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**BEIDOU/COMPASS**

**China completed its BeiDou Navigation Satellite System constellation in June 2020.**

- It is Chinese Satellite Navigation System. The system uses a network of satellites and can **provide positional accuracies of under ten metres (GPS provides positioning accuracies of under 2.2 metres).**
- **China initiated BeiDou in 1994** with aims to integrate its application in different sectors, including **fishery, agriculture, special care, mass-market applications, forestry and public security.**
- BeiDou offers services including **accurate positioning, navigation and timing as well as short message communication.**

**How many satellites are there in the system?**

It consists of **27 satellites in medium Earth orbit, 5 in geostationary orbit and 3 in inclined geosynchronous orbits.**

# 8.5 India's Regional Level Global Positioning System (GPS)

## 8.5.1 GPS Aided Geo-Augmented Navigation Satellite System (GAGAN)

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GAGAN is a **satellite based on augmentation system (SBAS)**.

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It is the Airport Authority of India's first step towards introduction of modern communications and positioning system over Indian airspace, which included surveillance and air traffic management, enabling pilots to navigate in the Indian airspace with an accuracy of 3 metres. So far, three satellites are dedicated to this project.

**GSAT 8 (21 May 2011), GSAT 10 (29 September 2012)** with 12 Ku-band, 12 C-band, and 12 extended C-band transponders along with a GAGAN payload, and **GSAT 15 (11 November 2015**) with full India coverage by 24 Ku-band transponders and a GAGAN payload are the three satellites that made GAGAN a successful project. It is advantageous over the conventional radar-based system used by most of the world because of its precision.

Specially GAGAN has been developed to modernise air traffic with the association of AAI (Airport Authority of India) & ISRO (Indian Space Research Organisation). It is also useful for many other services like **Study of Natural Resources, Forest and Land Management, Civil Aviation, Road Asset Management System** (managing traffic in real-time helps prevent congestion and traffic jams on roads) **Improve Railways Signalling system** to enhance efficiency of train operations, **Tourism** (It can provide accurate location information for navigation, guidance and location-based services), **Atmospheric Studies** etc.

### 8.5.2 Indian Regional Navigation Satellite System (IRNSS) & NVS

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Having a global navigation system bolsters the ability of a nation to serve as ***a net security provider,*** especially through the guarantee of such assurance

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policies. It can also play a significant role in relief efforts post disasters such as the tsunami in the Indian Ocean region in 2004 and the Pakistan-India earthquake in 2005.

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**(Refer Infographic – IRNSS – Indian Regional Navigation Satellite System)**

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Note: ***Qualcomm Technologies*** has unveiled mobile chipsets supporting ***the Indian regional satellite navigation system – NavIC (Navigation in Indian Constellation)***. The release of chipsets will help accelerate the adoption of NavIC by smartphone Original Equipment Manufacturers (OEMs). The OEMs can now release any new models for the Indian market which are NavIC enabled, thus eventually making NavIC as a standard feature in the upcoming handsets, applications, processors, etc.

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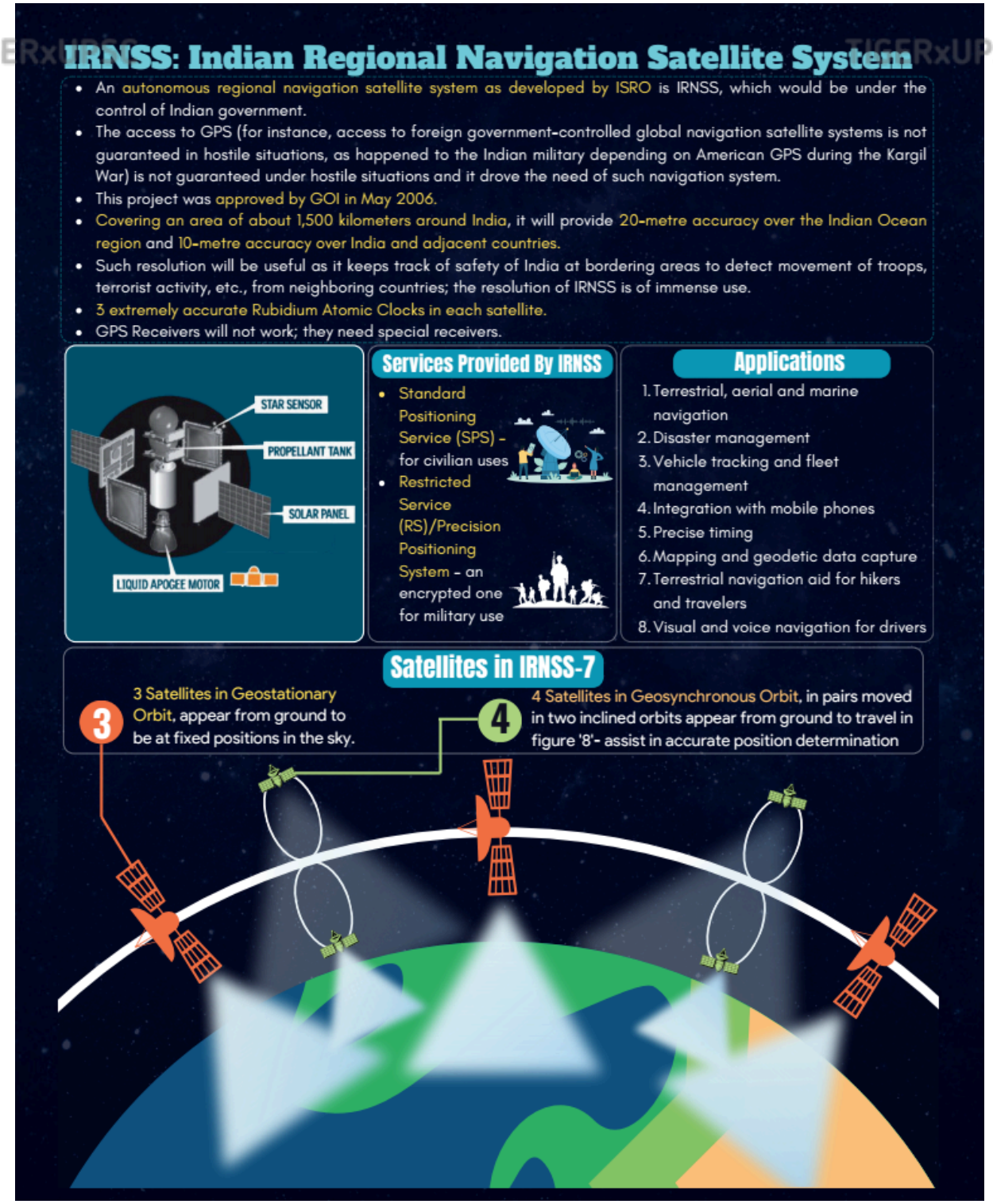


## 8.5.3 NavIC Updates

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- GSLV-F12/NVS-01 mission was accomplished successfully on **May 29, 2023**.
- This satellite represents the inaugural unit of the second generation within ISRO's NVS (Navigational Satellite) series of payloads.
- With a weight of 2,232 kg, it holds the distinction of being the heaviest among its cohort.
- NVS-01 is equipped with **navigation payloads** in the **L1, L5, and S bands**.
- Its primary objective is to ensure continuity for the **NavIC** services, which form an Indian Regional Navigation Satellite System akin to GPS, providing precise and real-time navigation within India and extending **up to a 1,500 km radius around the nation.**
- In the first generation, the Indian Regional Navigation Satellite System (IRNSS) constellation comprises seven satellites collectively known as NavIC, each weighing approximately 1,425 kg at liftoff.

This Geosynchronous Satellite Launch Vehicle (GSLV) mission deployed **NVS-01 (IRNSS 1J)** navigation satellite, **weighing about 2232 kg**, into a **Geosynchronous Transfer Orbit**.

NVS series of satellites will sustain and augment the NavIC with enhanced features. This series incorporates **L1 band signals additionally to widen the services. For the first time, an indigenous atomic clock will be flown in NVS-01**.

## Atomic Clock

The satellite will be equipped with a **Rubidium atomic clock**, a noteworthy technological achievement pioneered by India.

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- Several satellites within the navigation constellation have experienced a **loss of their ability to deliver precise location data** due to malfunctioning atomic clocks. Satellite-based positioning systems depend on accurate time measurements from atomic clocks to ascertain the locations of objects. Consequently, when these clocks fail, the satellites are unable to furnish accurate location information.

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## L1 Signals for better use in Wearable Devices

- It will send signals in a **third frequency, L1**, besides the L5 and S frequency signals that the existing satellites provide, **increasing interoperability with other satellite-based navigation systems**.
- The L1 frequency is **among the most commonly used in the Global Positioning System (GPS)**, and will increase the use of the regional navigation system in wearable devices and personal trackers that use low-power, single-frequency chips.

## Longer Mission Life

- It will have a **longer mission life of more than 12 years**. The existing satellites have a **mission life of 10 years**.

## IRNSS vs. NVS

**IRNSS:**
The initial name for India's regional satellite navigation system, signifying its developmental phase.
**NavIC:**
The operational name and acronym for the system, meaning 'Navigation with Indian Constellation'. IRNSS was the original concept and development phase, while NavIC is the operational name.
**NVS:**
Refers to the second-generation satellites being launched to modernise and enhance the NavIC system.

- **First Generation (IRNSS):** These satellites (e.g., IRNSS-1A, IRNSS-1I) formed the initial constellation and were primarily designed to provide regional coverage over India and its surrounding areas.

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- ❑ **IRNSS:** Launched using various versions of ISRO's Polar Satellite Launch Vehicle (PSLV).

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- **Second Generation (NVS):** The NVS series (e.g., NVS-01, NVS-02) are newer satellites with enhanced features, longer mission life and broader signal capabilities.
  - ❑ **NVS:** Launched using the Geosynchronous Satellite Launch Vehicle (GSLV).

***NVS – 01 & NVS – 02 Satellites***

NVS satellites are second-generation satellites in India's Navigation with Indian Constellation (NavIC) system, designed to augment existing NavIC services with enhanced features and improved coverage. The NVS series includes satellites like NVS-01 and NVS-02, which are equipped with L1 band SPS signals to widen the scope of NavIC services.

NVS-01, launched in May 2023, was the first in this series and featured an indigenous atomic clock. NVS-02, launched in January 2025, is designed to replace an older satellite (IRNSS IE) in the constellation and improve NavIC's capabilities.

**Key features and objectives of NVS satellites:**

- **Enhancements:** The NVS series offers enhanced features compared to the previous generation, including L1 band SPS signals to improve NavIC adoption.
- **Coverage:** NVS satellites contribute to expanding the coverage area of the NavIC system, extending the service beyond India.
- **Lifespan and Reliability:** The NVS-02 satellite, for example, has a longer lifespan of 12 years and incorporates indigenously developed, more accurate atomic clocks to enhance reliability.
- **Frequency Bands:** The navigation payload on NVS satellites operates in L1, L5 and S frequency bands, ensuring high accuracy.
- **Replacement:** NVS satellites are intended to replace existing IRNSS satellites as they reach the end of their lifespan, ensuring continuous navigation services.
- **Precise Timekeeping:** NVS satellites employ the Rubidium Atomic Frequency Standard (RAFS) for precise timekeeping, which is essential for accurate navigation.

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### 8.5.4 Sagar Sampark DGNSS

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The **Ministry of Ports, Shipping, and Waterways of India** has inaugurated **'SAGAR SAMPARK', a Differential Global Navigation Satellite System (DGNSS)**.

- SAGAR SAMPARK represents an indigenous Differential Global Navigation Satellite System (DGNSS), aimed at **revolutionising maritime navigation in India**.
- **Error Correction:** DGNSS corrects inaccuracies in the Global Navigation Satellite System (GNSS), **ensuring precise positioning information for maritime vessels**.
- **Impact:** The implementation of DGNSS is poised to **elevate safety standards and efficiency in port and harbour navigation**, marking a **significant milestone in India's maritime industry**.

#### Significance

- The implementation of DGNSS under SAGAR SAMPARK **ensures safer maritime navigation, reducing the likelihood of accidents and enhancing overall maritime security.**
- By providing accurate positioning information, **the system streamlines vessel movement in port and harbour areas, optimising operational efficiency**.
- SAGAR SAMPARK aligns with global maritime standards, **positioning India as a responsible stakeholder in maritime safety and navigation**.
- The introduction of indigenous DGNSS reflects India's commitment to digital transformation in the maritime sector, fostering innovation and technological advancement.

## 8.6 Some Recent News

### 8.6.1 AXIOM Mission

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**Shubhanshu Shukla,** an Indian Air Force (IAF) officer and ISRO astronaut, was recently selected as the pilot for Axiom Mission 4 (Ax-4).

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**ISRO is proud to announce the successfull launch of Axiom Mission 4 (Ax-04) on 25th June 2025.**

- **Ax-4** is the **fourth private astronaut mission** to the **International Space Station (ISS).**
- Organised by **Axiom Space,** in collaboration with **NASA.**
- Launched aboard **SpaceX's Dragon spacecraft** from **NASA's Kennedy Space Center,** Florida.

### *Crew Highlights*

- **Shubhanshu Shukla,** an **Indian Air Force officer** and **ISRO astronaut,** has been selected as the **pilot.**
- First Indian astronaut on ISS under a **NASA–ISRO partnership.**
- Includes **first astronauts from Poland and Hungary** to live aboard the ISS.

### *Mission Duration and Objectives*

- **Duration: Up to 14 days** aboard ISS.
- **Activities:**
  - ❑ **Scientific research**
  - ❑ **Technology demonstrations**
  - ❑ **Educational outreach**

## ATOMIC CLOCK

### What is an Atomic Clock?

- An atomic clock is a **highly precise device** that measures time based on the vibrations (oscillations) of atoms.
- **Key Atom Used:** Typically uses **cesium –133 atoms**, known for their stability and consistent oscillation frequency.
- **Accuracy:**
  - ❑ **Deviation:** Just **a few billionths of a second per day**.
  - ❑ Significantly **more accurate than mechanical or quartz clocks**.
- **Why More Accurate?**
- Atomic oscillations have **higher frequencies.**
- Much more **stable and consistent** over time.

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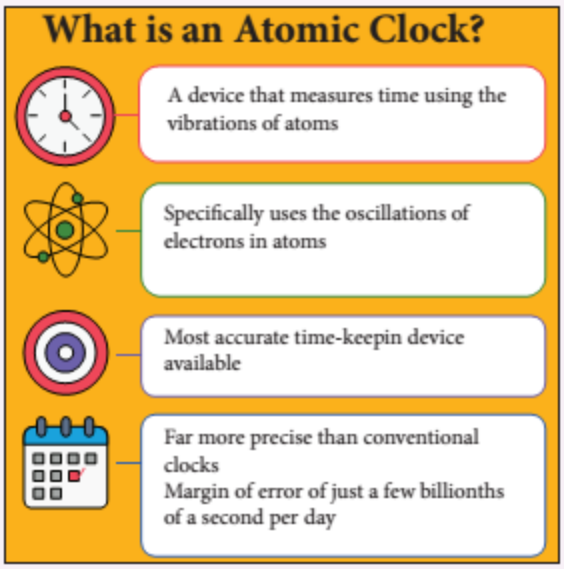


## *How Does an Atomic Clock Work?*

1. **Cesium Atoms as Timekeepers:**
   - Electrons in cesium atoms vibrate at a very specific frequency: **9,192,631,770 Hz.**
   - **This frequency defines 1 second in the International System of Units (SI).**
2. **Microwave Cavity System:**
   - Cesium atoms are exposed to microwave radiation inside a cavity.
   - At a precise frequency, cesium atoms absorb microwave energy and **transition between energy levels.**
3. **Detection & Synchronisation:**
   - The emitted frequency from atoms is compared with a reference.
   - Any difference is corrected using feedback loops to maintain accuracy.
   - This signal is used to **regulate the clock mechanism**.

## Cesium Atoms as Timekeepers

- Electrons in cesium atoms vibrate at a very specific frequency: **9,192,631,770 Hz**.
- This frequency defines **1 second** in the International System of Units (SI).

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***Research Domains:***

- Materials science
- Biology
- Earth observation
- Space-based innovation

***Significance***

- Boosts **India's human spaceflight credentials**
- Strengthens **international space collaboration**
- Advances **private space exploration**

## 8.6.2 CROPS Experiment

**(Compact Research Module for Orbital Plant Studies).**

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The **CROPS** (Compact Research Module for Orbital Plant Studies) Mission by **ISRO successfully germinated cowpea seeds within four days, achieving a significant milestone in space farming**.

**Note:** POEM-4, ISRO's fourth PSLV Orbital Experiment Module.

### *Why Grow Plants in Space?*

- **Sustainable Food Source:** Fresh crops can continuously replenish astronauts' diet.
- **Oxygen Generation:** Photosynthesis from plants helps maintain breathable air.
- **Waste Recycling:** Plants help recycle $CO_2$ and organic waste, creating a self-sustaining loop.
- **Mental Well-being:** Caring for plants can reduce stress and improve mood in isolated environments.

### *Challenges of Space Farming*

- **Root Growth Issues:** Microgravity affects directional root development.
- **Watering Problems:** Water behaves differently in space, making nutrient delivery tricky.
- **Radiation & Temperature:** Plants need protection from harmful radiation and extreme fluctuations.
- **Light Constraints:** Limited sunlight affects photosynthesis, reducing oxygen and food output.

### *Ideal Crops for Space*

- **Leafy Greens:** Fast-growing and packed with nutrients.
- **Legumes:** High in protein, improve soil/media quality.
- **Root Vegetables:** Compact and efficient.
- **Grains & Fruits:** Support long-term diet variety and nutrition.

| Importance | Techniques Used |
|---|---|
| • Space farming ensures **food security**, **life support** and **crew wellness** during long missions.<br>• Critical for future missions to **Moon**, **Mars** and beyond. | • **Hydroponics:** Nutrient-rich water replaces soil.<br>• **Aeroponics:** Roots hang in air and absorb nutrients from mist.<br>• **Soil Substitutes:** Porous materials with slow-release nutrients mimic Earth soil. |

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### 8.6.3 Electronic Toll Collection (ETC)

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The **NHAI (National Highway Authority of India)** plans to introduce a **GNSS-based electronic toll collection (ETC)** system alongside the existing **FASTag (RFID-based)** setup, using a **hybrid model** initially.

***About GNSS-Based ETC:***

- Uses **satellite navigation** to track vehicle movement and calculate tolls **based on distance travelled**, not fixed points.
- **Toll plazas** will serve as **enforcement gantries,** identifying GNSS-enabled vehicles.
- A **Centralised Toll Charger** will process data from **On-Board Units (OBUs)** in vehicles to compute and collect tolls.
- **OBUs** will be linked to payment systems similar to FASTag.

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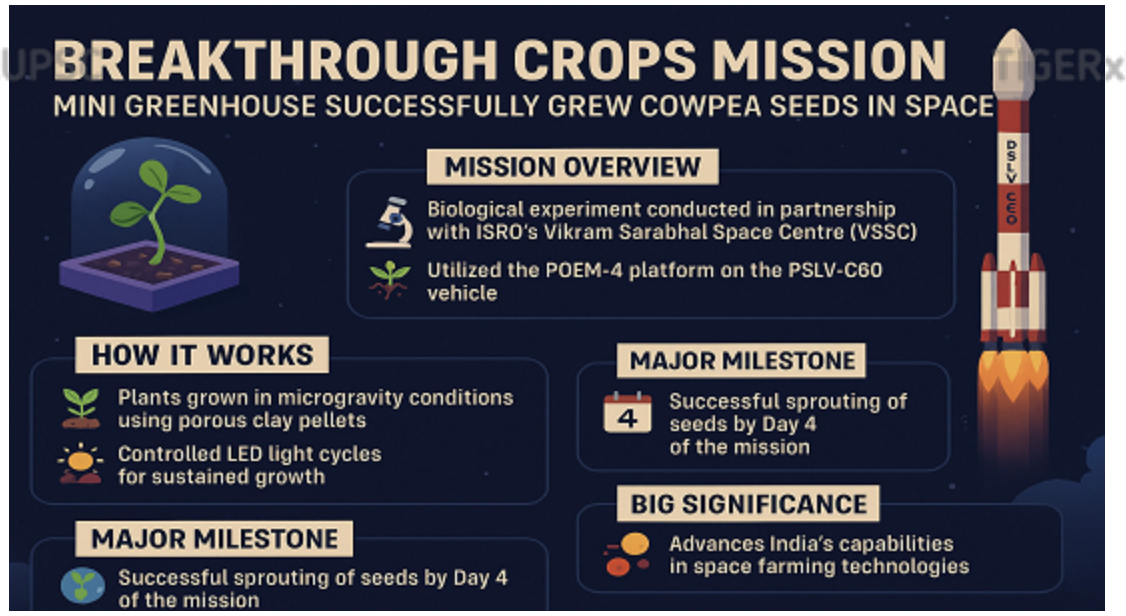
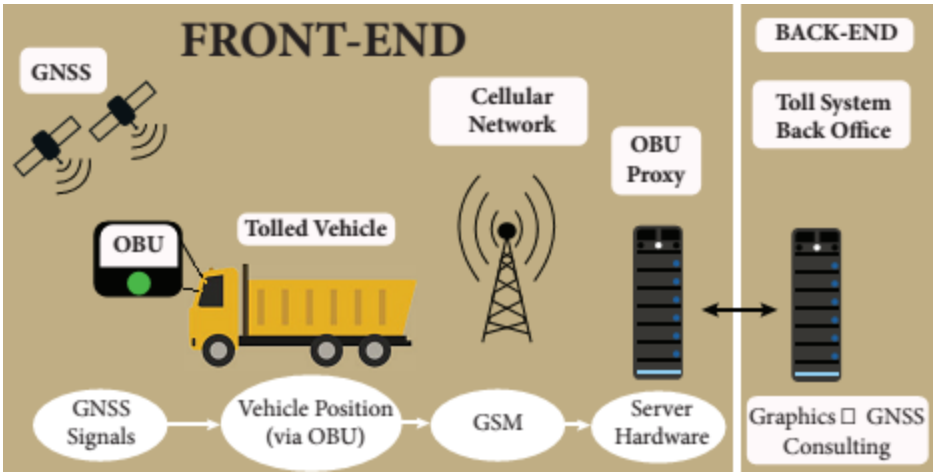



***Benefits:***

- **Barrier-free, distance-based tolling** ensures faster movement and reduced congestion.
- Improves transparency, reduces toll evasion and aligns with **international tolling standards**.




**On-Board Units (OBUs)** in vehicles

## 8.7 Bhuvan

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Bhuvan is an application developed by ISRO providing medium to high-resolution satellite imagery of virtually entire India over the Internet. On 12 August 2009, a prototype of this application was launched.

With spatial resolutions ranging from 10–100 metres, Bhuvan will offer superior imagery of Indian locations, as compared to other Virtual Globe software, such as Google and Wikimedia. The locations can be viewed from different perspectives and this will enable measurement of distances. The portal is designed to run on slow Internet connections and for security concerns, the imagery would steer clear of all sensitive military installations in India.

There are a number of interesting features of Bhuvan images, including weather information and even administrative boundaries of all states and districts, relevant only to the country. Though it is equally capable of offering images of the globe, the best resolution area includes India at the moment. The important role in the creation of this product was played by **National Remote Sensing Agency (NRSA)**. The data provided by Satellites like Resourcesat-I, Cartosat-I and Cartosat-II has been used by ISRO to get the best possible imagery for India.

### ISRO ANNOUNCES 'BHUVAN'

The Department of Space (DoS), under which ISRO comes, signed an MoU with **geospatial technology company CE Info Systems Pvt Ltd on 11 February 2021**.

- The collaboration will enable them to jointly identify and build a holistic **geospatial portal** utilising earth observation datasets, **'NavIC', Web Services and APIs (application programming interface) available in MapmyIndia**.
- The geospatial portals will be called **'Bhuvan', 'VEDAS' and 'MOSDAC'**.

**Key Facts**

**Bhuvan** is the national geo-portal developed and hosted by ISRO comprising geospatial data, services, and tools for analysis.

**VEDAS** (Visualisation of Earth observation Data and Archival System) is an online geo-processing platform that uses optical, microwave, thermal and hyperspectral data covering applications particularly meant for academia, research and problem solving.

**MOSDAC** (Meteorological and Oceanographic Satellite Data Archival Centre) is a data repository for all the meteorological missions of ISRO and deals with weather-related information, oceanography and tropical water cycles.

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## 8.8 Kessler Syndrome/Space Junk/Space Debris Concept

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The Kessler Syndrome/Kessler Effect/Collisional Cascading concept was proposed by NASA scientist, Donald J. Kessler in 1978. It is a scenario in which the density of objects in Low Earth Orbit (LEO) is increasing due to various collisions among objects and further a cascade of collision generates more and more of space debris. Every satellite, space probe and manned mission have the potential to produce a lot of space debris. There were about 2,000 commercial and government satellites orbiting the Earth, according to 2014 report. On an average, one satellite is destroyed each year. Now we have an estimation of six lakh pieces of space junk ranging from 1 cm to 10 cm in space. We have not yet adopted any procedure to solve this debris problem from space. In coming future they can lead to various problems in human life and will also be a congestion to launch satellites at proper orbit. In a movie, Gravity, space debris collision to a space shuttle is observed. This collision is dangerous and can destroy astronauts' life in space and also can harm to working satellites in space.

## 8.9 Space Tourism

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Space tourism allows everyday people to experience spaceflight and zero gravity as paying passengers.

- **Current Status:** Emerging industry led by companies like Blue Origin, Virgin Galactic and SpaceX
- **Market Size:** USD **851.4 million** in 2023
- **In India:** Limited involvement, but **Gaganyaan** may serve as a launchpad for future space tourism.

### *Types of Space Tourism*

| Type | Description |
|---|---|
| **Suborbital** | Reaches ~50–70 miles above Earth, crosses Kármán line briefly |
| **Orbital** | Goes beyond Kármán line and orbits Earth; may last days or weeks |
| **Others** | Includes Lunar tourism, interplanetary travel (e.g., Mars, Moon missions) |

### *Challenges*

- **High Costs:** Expensive spacecraft, medical and training requirements
- **Safety Concerns:** Risky launch and re-entry phases (e.g., STS-107 Columbia tragedy)
- **Legal Uncertainty:** Lack of regulation; unclear passenger rights vs astronaut status
- **Space Debris:** Risk of collisions increases, potential for **Kessler Syndrome**
- **Limited Demand:** Market limited to high-net-worth individuals

### *India's Space Tourism Potential*

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**Opportunities for Growth:**

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- **Gaganyaan** success can build trust and technological base
- **Encouragement of private participation**

**Recent Initiatives:**

1. **Space Policy 2023 –** Opens space sector to private players
2. **IN-SPACe –** Regulates & supports **non-government space startups**
3. **ISpA (Indian Space Association) –** Builds global links for **tech and investment**

***Way Forward***

- Formulate **global standards** (e.g., under **Outer Space Treaty**).
- Define **passenger rights and responsibilities**.
- Develop **sustainable space technology** to reduce **environmental impact and debris**.
- Create a **regulatory framework** for private sector and **ensure safety protocols**.

## NETRA

- To safeguard its space assets from space debris, **ISRO had set up a dedicated Space Situational Awareness (SSA) Control Centre named "Netra"** in Bengaluru in December 2020.
- Netra's key objective is to monitor, track and protect the national space assets and function as a hub of all SSA activities.
- Only the US, Russia and Europe have similar facilities in place to track space objects and share collision warnings.

## SPACE DEBRIS

- Human-created, inactive objects surrounding our planet in space or atmosphere.
- Includes:
  - ❑ Defunct (non-functional) satellites
  - ❑ Spent rocket stages
  - ❑ Fragments from explosions or collisions

**Types of Space Debris**

1. **Large Debris**
   - **Examples:** Old satellites, rocket boosters, fuel tanks
   - Often survive re-entry
2. **Small Debris**

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- Debris resulting from spacecraft impacts and deteriorated satellites

3. **Microscopic Debris**
   - Minuscule fragments of paint, dust and metal from degraded space equipment

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### SRO System for Safe and Sustainable Operation (IS4OM): Inaugurated on July 2022

- **'IS4OM to safeguard Indian space assets, mitigate collision threats from space objects'**
- The 'SRO system satellite' likely refers to the **Satellite Robotic Observatory (SRO)**, a type of observatory that uses robotic satellites to observe and track celestial objects.

### Project NETRA (Network for Space object TRacking and Analysis)

- **NETRA** is an ISRO initiative to establish a **Space Situational Awareness (SSA)** capability in India.
- It aims to **build an early warning system for detecting and tracking space debris** and other hazards to Indian satellites.
- The **project involves setting up observational facilities like radars, optical telescopes and a control center** to monitor space objects.

### *Legal Framework Governing Space Debris (Global)*

1. **Outer Space Treaty (1967)**
   - Countries are responsible for all space activities (public or private)
2. **Liability Convention (1972)**
   - States are **absolutely liable** for damage caused on Earth by their space objects
3. **UN Space Debris Mitigation Guidelines**
   - Non-binding safety recommendations for satellite disposal
4. **25-Year Rule (UN & IADC** - Inter-Agency Space Debris Coordination Committee)
   - Advises deorbiting satellites within 25 years
   - Compliance rate: Only ~30%
5. National Laws (US, EU, China)

- Mandate tracking and disposal plans
- Weak enforcement mechanisms

## Challenges in Space Debris Governance

- **No Binding Global Rules:** Guidelines are voluntary and lack enforcement.
- **Attribution Problems:** Difficult to identify the source of old or broken debris.

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- **Growing Space Traffic:** Mega-constellations like Starlink increase collision risk.
- **Uncontrolled Reentries:** No penalties for countries whose debris reenters Earth.
- **Lack of Compensation Enforcement:** E.g., delayed settlement after Cosmos 954 crash in Canada (1978).

## Way Forward

- **Introduce Binding Global Rules**
  - ❑ Make disposal and re-entry rules enforceable with penalties
- **Strengthen Tracking Systems**
  - ❑ Expand AI-based global monitoring networks
- **Enforce Deorbiting Plans**
  - ❑ Launch clear satellite disposal paths (e.g., graveyard orbit)
- **Set Up Global Liability Tribunal**
  - ❑ Quick and fair compensation settlements for damage caused
- **Promote Sustainable Space Practices**
  - ❑ Encourage reusable launch systems and debris-removal missions

## 8.10 Problems Faced by Astronauts in Space

The human body is uniquely designed to survive in Earth's gravity. In space, there is microgravity environment, which requires the human body to be adapted accordingly. There, astronauts need to float rather than walk in

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different parts of the spacecraft. This implies that the weight-bearing bones (those in the lower part of the body – the legs, hips and spine) experience a significant decrease in load bearing. This reduction leads to bone breakdown, releasing calcium that is reabsorbed by the body, leaving the bones more brittle and weak. There are fewer loads on the leg muscles and back muscles used for posture, during an extended spaceflight. Consequently, the muscles can begin to weaken or atrophy, subsequently leading to fall-related injuries and accidents during exploration missions. Currently, to overcome such issue, there is an exercise programme, but sooner nutritional interventions designed to reduce the muscle loss may be added as a complement to the spacecraft missions.

In space, there is no gravity to distribute the blood and other body fluids to the lower part of the body, especially the legs. This leads to redistribution of body fluids to the upper part of the body and away from the lower extremities. Due to this fluid shift, astronauts may experience puffy face and also a feeling of congestion.

In microgravity environment, the heart does not have to work hard and over time it gets deconditioned and decreases in size. Space radiation may affect endothelial cells, the lining of blood vessels, which might initiate or accelerate coronary heart disease.

The brain receives and interprets information from numerous sense organs, particularly the eyes, inner ear vestibular organs and deep senses from muscles and joints. An integrated set of neural circuits allows humans to maintain balance on Earth. The inner ear no longer functions as designed. During their stay in space, the body of astronauts adapt and hence when they return to Earth, they may experience problems standing up, stabilising their gaze, walking and turning. These disturbances are more profound with increase in microgravity exposure. The changes can affect operational activities including approach and landing, docking, remote manipulation, extra-vehicular activity and post landing normal and emergency egress. On top of this, the exploration crews will have to shift their ‘body clocks’ from Earth’s day/night cycle to that of their destination. By improving spacecraft lighting, sleep schedules and scheduling of work shift, such changes can be countered.

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## 8.11 Extra Vehicular Activity/ Spacewalk

Extra-Vehicular Activity (EVA) is the only activity performed ahead of the Earth's substantial atmosphere outside the spacecraft by an astronaut or a cosmonaut. The term generally refers to a spacewalk done external to a craft, which is orbiting the Earth. The cosmonauts have also undertaken deep-space Extra Vehicular Activities for retrieving film canisters from the exterior of the spacecraft on returning to Earth on each last of these three undertaken missions. In the year 1973, EVA was used in repairing of the launch damage to skylab, which is the US's first space station.

EVAs may be either tethered (the astronaut is connected to the spacecraft; oxygen and electrical power can be supplied through an unbiblical cable, no propulsion is needed to return to spacecraft) or untethered (here, astronaut are not connected to shuttle for life support system). Russia, US and China have demonstrated the capability to conduct an EVA.

The first EVA in deep space was made on August 1971 by American Al Worden to retrieve a film and data-recording canister from the Apollo 15 service module on return trip from the Moon.

American Bruke Mc Candless made his first journey with untethered EVA on 7 February 1984 during challenger mission STS-41B, using the Manned Maneuvering Unit.

First woman to perform an EVA was soviet Svellana Savitskaya, on 25 July 1984 while aboard the Salyut 7 space station. Her EVA lasted 3 hours 35 minutes.

## 8.12 International Space Station

It is an internationally developed research facility in space in Low Earth Orbit. This place has a microgravity environment in which astronauts conduct experiments in biology, human biology, astronomy, physics and

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meteorology. It is a unique environment for the testing of the spacecraft systems that will be required for missions to the Moon and Mars. International Space Station is a complete home to the humans beyond the Earth. It is the most expensive object ever constructed by man.

***Major Experiments***

- It observes effects of long-term space exposure on the human body.
- It is a place for ultrasound scans, various diagnosis and treatment of medical conditions in space.
- It will also be helpful in the study of near-weightless and microgravity environment and observe the evolution, growth, development, and internal processes of plants and animals.
- Investigation of superconductivity and the physics of fluids in microgravity will also open doors for various researches. This place will be also very helpful for the study of cosmic rays, cosmic dust, antimatter and dark matter in the universe.

The ISS is the ninth space station to be developed by crews, following the Soviet and later Russian Salyut, Almaz and Mir stations as well as Skylab from the US. After the US Space Shuttle programme ended in 2011, Soyuz rockets became the only provider of transport for astronauts at the International Space Station.

The ISS programme is a joint project among five participating space agencies: **NASA (USA), Roscosmos (Russia), JAXA (Japan), ESA (Europe)** and **CSA (Canada).**

**BHARTIYA SPACE STATION AND SPACE VISION**

The Indian Space Research Organisation (ISRO) outlined new targets for India's space exploration endeavours, including establishing the '**Bharatiya Antariksha Station**' (Indian space station) by 2035 and sending the first Indian to the Moon by 2040.

**Indian Space Station by 2035**

ISRO aims to establish the Indian space station by 2035, which is expected to be significantly smaller compared to existing space stations. It is **projected to weigh only 20 tonnes**, in contrast to the 450 tonnes of the International Space Station and 100 tonnes of the Chinese Tiangong Space Station.

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International Space Station is divided into two sections, the **Russian Orbital Segment (ROS)** and the **United States Orbital Segment (USOS)**, which is shared by many nations.

***How long do all the International Space Station partners plan to operate the complex?***

NASA and its international partners have maintained a continuous and productive human presence aboard the International Space Station for more than 21 years. Life extension analysis for the US Segment has been completed through 2028 with no issues that would prevent space station from being further extended. The United States has committed to extend International Space Station operations through 2030. NASA's space agency partners have all recommended International Space Station extension through 2030 with approvals pending through their own government processes.

[Refer **Infographic – International Space Station (ISS)**]

## 8.13 Space History

| Year | Achievements |
|---|---|
| 1942 | In 1942, the first rocket, designed by Wernher Von Braun, who later worked with NASA as the creator of the rockets that went to the moon. |
| 1947 | The first animals were launched into space in 1947. To study the effects of space travel on animals, fruit flies were used. Since they are more similar to humans than we may imagine, they were chosen for the purpose. The flies hac supply of corn to eat while travelling on the flight. |
| 1949 | The first monkey in space, was Albert II, a Rhesus monkey, a type of monkey, originally hailing from Asia. On 14 June 1949, Albert went into the space in a specially adapted American V2 rocket, that flew to a height of 83 miles from Earth. |
| 1957 | On 4 October 1957, Sputnik 1 was launched as the first satellite into space, marking the beginning of the space age. Sputnik was the first satellite to orbit around the Earth. |
| 1957 | The Russian space dog Laika was the first animal to orbit the Earth in a spacecraft known as Sputnik II, in November 1957. Laika means "Barker" in |

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| Year | Achievements |
| --- | --- |

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Note: The U.S. National Space Policy of 2010 made some goals for space exploration; to send humans to an asteroid by 2025 and to the planet Mars in the 2030s.

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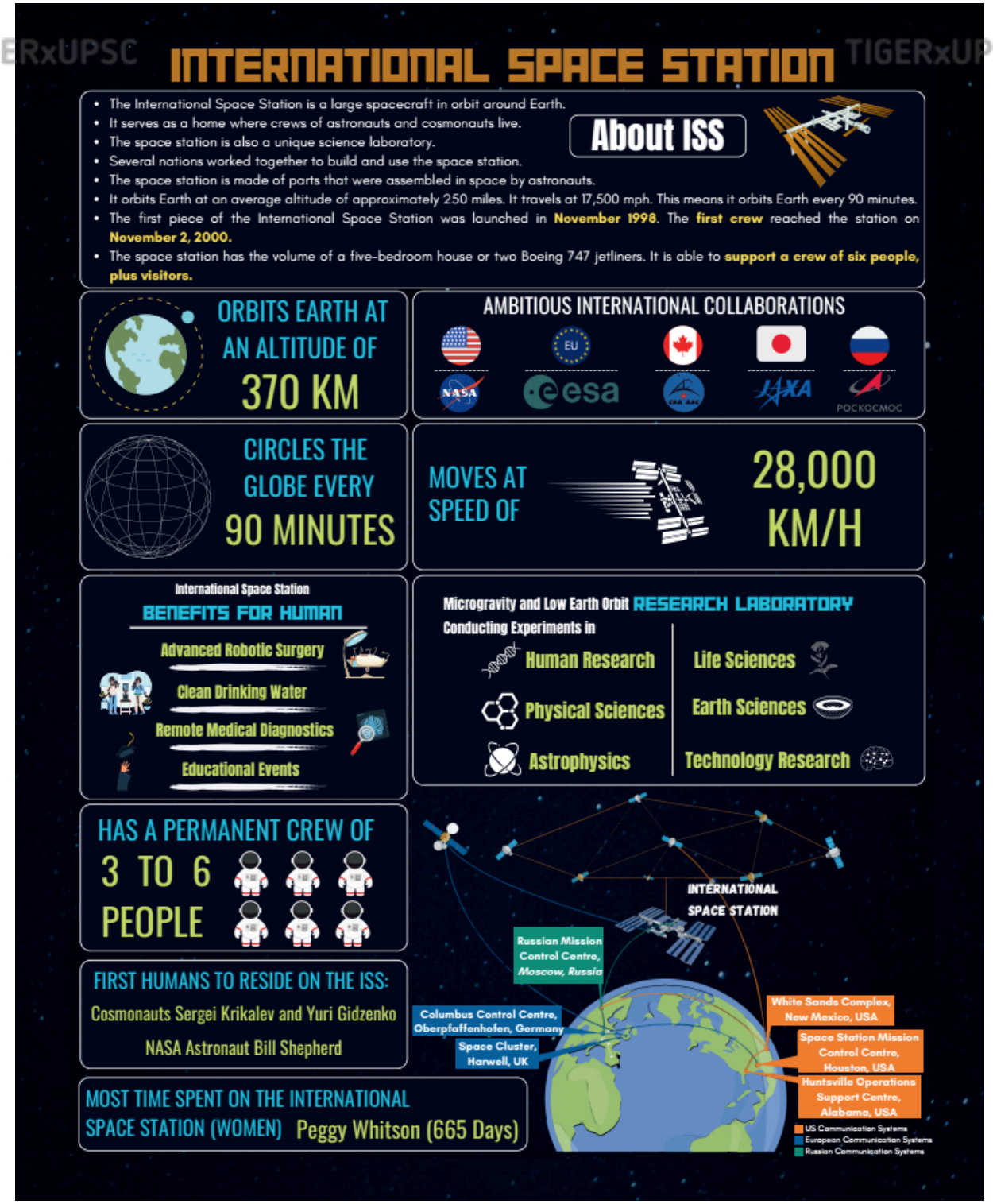


## 8.14 Indian Astronauts and Their Achievements

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### 8.14.1 Rakesh Sharma

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Rakesh Sharma was the first Indian to travel to space. He was born on 13 January 1949 in Patiala. Rakesh Sharma joined the Indian Air Force (IAF) in 1970; he was a test pilot in IAF and swiftly progressed to other levels. On 20 September 1982, he was selected by the Indian Space and Research Organisation (ISRO) to become a cosmonaut. He went to space as part of a joint mission between ISRO and the Soviet Intercosmos Space Programme, and thus became the first Indian to enter space.

After his return from space, he was given the **honour of Hero of Soviet Union**. The Indian government awarded him with an **Ashoka Chakra**.

### 8.14.2 Kalpana Chawla

Kalpana Chawla was the first Indian woman to go into space and also the first Indian–American astronaut. She was born in Karnal, India, in 1962. She had the determination to become an astronaut and that made her pursue many degrees in her field. She held a Ph.D in Aerospace Engineering. Her first flight to space was on Space Shuttle Columbia in 1997 as a mission specialist and primary robotic arm operator.

Unfortunately, in 2003, Kalpana Chawla was part of the Space Shuttle Columbia disaster and lost her life. Kalpana was honoured with the Congressional Space Medal of Honor, NASA Space Flight Medal and NASA Distinguished Service Medal.

### 8.14.3 Sunita Williams

The woman who holds the record for the longest single space flight (195 days) is Sunita Williams. She also holds the record of seven spacewalks, which total up to 50 hours and 40 minutes. She holds the fifth rank when it comes to experienced space walkers.

She was born to Gujarati parents in Euclid, Ohio. On 9 December 2006, Williams was launched into space in STS-116, where she served as the mission specialist.

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### 8.14.4 Ravish Malhotra

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Ravish Malhotra never went to space; he served as a backup for Rakesh Sharma on Soyuz T-11. Malhotra was born on 25 December 1943. He was awarded with the Kirti Chakra in 1984.

## 8.15 Some Important International Space Mission

### 8.15.1 Important Jupiter Missions

**Flyby Missions**

- *Pioneer* programme (1973 and 1974)
- *Voyager* programme (١٩٧٩)
- *Ulysses* (1992)
- *Cassini* (2000)
- *New Horizons* (2007)

**Orbiter Missions**

- *Galileo* (1995–2003)
- *Juno* (2016)
- *Jupiter Icy Moon Explore* (2023)

#### Juno

- **Launch Mass: 3,625 kilograms, Launch date: 5 August 2011, Launch site: Cape Canaveral**.
- It is a space probe **launched by NASA** and **orbiting around the planet Jupiter**.

***Objectives of Juno***

- Measurement of the composition, gravity, magnetic field and polar magnetosphere of Jupiter.
- The spacecraft will also search for clues about the formation of the planet, inclusive of having a rocky core, the amount of water it contains within the deep atmosphere, mass distribution and its deep winds reaching the speeds of 618 kilometres per hour. It is the second spacecraft after the nuclear powered Galileo Orbiter (1995–2003) to orbit Jupiter.

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***Achievements***

- Juno is the **first Jupiter's mission making the usage of solar panels in place of the radio isotope** based thermoelectric generators.

## Jupiter Icy Moon Explorer

ESA's Jupiter Icy Moon Explorer (JUICE) has been selected as part of ESA's Cosmic Vision science programme.

It was launched on 14 April 2023 and after a series of flybys in the inner Solar System is planned to arrive in 2031.

In 2012, the European Space Agency's selected the **Jupiter Icy moon Explorer (JUICE)** as its first large mission, replacing its contribution to EJSM (**Europa Jupiter System Mission**), the **Jupiter Ganymede Orbiter** (JGO).

### UNDERSTANDING THE BEGINNING

We can't understand the origin of the solar system – and how **Earth** came about, without understanding how Jupiter formed.

Scientist community believes that giant planets like Jupiter are the corner stones of planet formation. These planets were assembled early in the process, before their young stars had the chance to absorb or blow away the **light gases** in the huge cloud from which they were born.

Giant planets also play a big role in planet formation because their huge masses allow them to shape the orbits of other objects in their planetary systems, such as other planets, asteroids, and **comets**.

Although we have pieced together the basic story of Jupiter's origin, some important questions remain, and Juno's mission is to help us answer them. Exactly how early was Jupiter born? Jupiter might have formed at its current orbit, but some evidence also suggests that it could have formed farther from the sun before migrating inward. Because Jupiter formed at the same time as the sun, their chemical compositions should be similar. But Jupiter has more heavy elements – such as carbon and **nitrogen** – than the Sun.

Competing formation theories make different predictions about the content and mass of Jupiter's core, so measuring the core will allow us to eliminate ideas that are wrong. Determining the amount of water – and therefore oxygen – in the gas giant is important not only for understanding how the planet formed, but also how heavy

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elements were transferred across the solar system. These heavy elements were crucial for the existence of rocky planets like Earth – and life.

Since Jupiter is the best example of a gas giant that we have, learning its history will help us understand the hundreds of giant planets we've discovered orbiting other **stars**.

The partnership for the Europa Jupiter System Mission has since ended, but NASA will continue to contribute the European mission with hardware and an instrument.

## 8.15.2 Important Saturn Missions

**How many missions did Saturn have?**

**Four robotic spacecrafts have visited Saturn.** NASA's Pioneer 11 provided the first close look in September 1979. NASA's twin Voyager 1 and Voyager 2 spacecraft followed up with flybys nine months apart in 1980 and 1981.

### CASSINI – HUYGENS (Cassini Orbiter and Huygens: Lander)

**[Launch mass: 5,712 kilograms, Launch date: 15 October 1997, Launch site: Cape Canaveral]**

**Cassini – Huygens** is an **unmanned spacecraft sent to the planet Saturn.**

It is the **fourth space probe to visit Saturn** and the **first to enter orbit.**

- Team: **Sixteen European countries and the US** are responsible for designing, building, flying and collecting data from Cassini Orbiter.
- It got completely **destroyed on 15 September 2017.**

***Achievements***

- The first mission to successfully complete in various observations: depth, up-close study of Saturn and its realm from orbit.
- Titan was revealed to have Earth like conditions: rain, rivers, lakes and sea.
- Among the most surprising discoveries were geysers erupting on Enceladus.

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- First complete view of the North polar hexagon and discovery of giant hurricanes at both of the Saturn's poles.

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## Titan

**[Proposed launch date: between 2020 and 2029, Launch mass: 1613 kilograms]**

**Titan Saturn System Mission (TSSM)** was a joint **NASA–ESA** proposal for an exploration of Saturn and its moons **Titan** and **Enceladus**, where many complex phenomena were revealed by *Cassini*. TSSM was launched in 2020, get **gravity assists** from Earth and Venus, and arrive at the Saturn system in 2029.

This is a Saturn exploration mission proposed for the future. The mission has been proposed to study Saturn and its Moons – **Titan and Enceladus**.

The Titan Saturn system mission (TSSM) was officially decided in January 2009 by the **merging of ESA's Titan and Enceladus Mission with NASA's Titan Explorer study.**

The **TSSM mission consists of an orbiter and two titan exploration probes**: a '**hot air balloon**' **that will float in Titan's clouds** and a '**lander**' **that will splash down on one of its methane seas**.

## NASA to Launch Dragonfly

NASA plans to launch an unmanned nuclear-powered drone, Dragonfly as early as 2027 and arrive in 2034 to search for life on Saturn's largest moon, Titan.

**Titan is an analog to the very early Earth, and can provide clues to how life may have arisen on our planet.**

**Titan has a nitrogen-based atmosphere similar to Earth**. Unlike Earth, Titan has clouds and rain of methane. Other organics are formed in the atmosphere and fall like light snow. **The moon's weather and surface processes have combined complex organics, energy and water similar to those that may have sparked life on our planet**.

**Titan is larger than the planet Mercury and is the second largest moon in our solar system**.

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## 8.15.3 Important Mars Missions

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There **have been** around 50 **Mars missions** so far, of which about half **have been** successful.

**Mariner 9** was the **first** spacecraft that successfully entered into **Mars orbit on** 1971.

Some important Mars Missions are:

- **Mars Odyssey** (2001)
- **Mars Express** (2003)
- **Mars Spirit and Opportunity** (2003)
- **Mars Reconnaissance Orbiter** (2005)
- **Curiosity Rover** (2011)
- **Mangalyaan (Mars Orbiter Mission)** 2013
- **MAVEN** (2013)
- **ExoMars Trace Gas Orbiter** (2016)
- **InSight Lander** (2018)
- **Hope Mars Mission** (2020)
- **Perseverance Rover** (2020)
- **Tianwen-1** (2020)

### Mars Science Laboratory (MSL)

**[Launch mass: 3.839 kilograms, Launch date: 26 November 2011, Launch site: Cape Canaveral]**

MSL is a **robotic space probe mission to Mars** launched by **NASA**, which **successfully landed 'Curiosity,' a Mars rover, in Gale Crater** on 6 August 2012.

'Curiosity' is about **twice as long and five times as heavy as the previous Mars rovers 'Spirit' and 'Opportunity'** and carries over ten times the mass of scientific instruments.

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***Scientific Goals of MSL***

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1. **Determining the landing sites and its habitability**, including the **role of water**.
2. The **study of the climate and the geology of Mars**.
3. All of these are **important for preparing a future manned mission to mars**.

## Curiosity Rover

**[Launch mass: Rover only of 899 kilograms, Launch date: 26 November 2011]**

Curiosity is a **car-sized robotic rover exploring Gale Cater on Mars** as a **part of NASA's Mars Science Laboratory mission.**

Since landing on 6 August 2012, it is still working.

In December 2012, Curiosity's mission was extended to an indefinite period. **The goals and objectives remain same as those of Mars Science Laboratory.**

## Maven Spacecraft

The MAVEN mission was **launched on November 18, 2013**, and **went into orbit around Mars on September 21, 2014**.

**Mars Atmosphere and Volatile Evolution (MAVEN)** mission was **devised by NASA** for studying the Martian atmosphere during the Mars orbiting.

MAVEN was **launched aboard through an Atlas V launch vehicle**.

MAVEN has **demonstrated** that the **majority of the carbon dioxide (CO2) on the planet has been lost to space.**

## ExoMars Programme

It is a **joint endeavour** between **ESA and the Russian space agency, Roscosmos**. The primary goal of the ExoMars programme is to address the question of **whether life has ever existed on Mars**.

**The programme comprises two missions.**

- The **first** launched in March 2016 and consists of the **Trace Gas Orbiter (TGO) and Schiaparelli**, an entry, descent and landing demonstrator module. **TGO's main objectives are to search for**

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**evidence of methane and other trace atmospheric gases that could be signatures of active biological or geological processes**. The **Schiaparelli probe crashed during its attempt to land on Mars**.

- The **second, comprising a rover (ExoMars Rover) and surface platform, is planned for 2028.** Together they **will address the question of whether life has ever existed on Mars**.

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## InSight Lander

Launched in May 2018, with a successful landing in November 2018 of same year, the InSight Lander, short for '**Interior Exploration using Seismic Investigations, Geodesy and Heat Transport**' aims to **study the core of Mars and observe any possible seismic activity on the planet's surface**.

- InSight is part of **NASA's Discovery Program.**
- It will be **the first mission to peer deep beneath the Martian surface,** studying the planet's interior by measuring its heat output and listening for marsquakes, which are seismic events similar to earthquakes on Earth.
- It will **use the seismic waves generated by marsquakes to develop a map of the planet's deep interior.**

## HOPE Mission

**HOPE Mission –** the **first Arab interplanetary mission** – was announced in 2015 with the aim of creating mankind's first integrated model of the Red planet's (Mars) atmosphere. 'Hope' was developed by UAE scientists in the USA and was launched in **July 2020** from the Tanegashima Space Centre in Japan.

The Hope mission is **a Mars orbiter spacecraft,** which will study the thin atmosphere of Mars. The mission is officially named **the Emirates Mars Mission (EMM) and the orbiter has been named Hope or 'AlAmal'.** It is the first interplanetary mission for the Arab World.

***Significance of the Mission***

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1. It is a known fact that the Red Planet was once habitable, from signatures of flowing water and organic material that point to a past that could have supported living things.
2. An understanding of Mars past could help scientists understand the future of Earth.

Note: With the successful Mars orbit insertion, the **UAE becomes the fifth entity to reach the Red Planet, joining NASA,** the Soviet Union, the European Space Agency and India.

## MARS 2020: NASA's Perseverance's Mission on Mars

**It was** launched on **July of 2020** from Cape Canaveral Air Force Station in Florida. This was **the third constitutive launch to Mars on July 2020, following the UAE's Hope** and China's Tianwen**-1 spacecraft.**

### *Key Facts*

1. **Landing date – 18 February 2021**
2. The mission is planned to last for at least **one Mars year, which works out to about 687 days on Earth (**it takes longer for Mars to go around the sun).
3. **Landing site:** Jezero crater.
4. **Perseverance is loaded with seven instruments** chosen to help it achieve its mission objectives.

### *Some of the Key Mission Objectives*

1. Look for signs of ancient microbial life.
2. Collect Martian rock and dust samples for later return to Earth.
3. Deliver an experimental helicopter.
4. Study the climate and geology of Mars.
5. Demonstrate technology for future Mars missions.

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***Significance of This Mission***

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1. Perseverance will carry **a unique instrument, MOXIE or Mars Oxygen ISRU Experiment:** which for the first time will **manufacture molecular oxygen on Mars using carbon dioxide from the carbon-dioxide-rich atmosphere (**ISRU means In Situ Resource Utilisation: or the use of local resources to meet human needs or requirements of the spacecraft).
2. It will carry **Ingenuity, the first ever helicopter to fly on Mars.** This is the first time NASA will fly a helicopter on another planet or satellite.
3. It is **the planned first step to bring back rock samples from Mars for analysis in sophisticated laboratories on Earth** and do its **biosignatures** (try to detect past or present life).

***(A) NASA Mars Helicopter "Ingenuity" Makes First Flight on Another Planet***

- NASA's **Ingenuity mini-helicopter** has been dropped on the surface of Mars in preparation for its first flight.
- The helicopter had been fixed to the belly of **the Perseverance rover,** which touched down on the Red Planet on 18 February 2020.
- NASA successfully flew its tiny helicopter **Ingenuity** on Mars on 19th April 2020, **the first powered flight on another planet.**
- Ingenuity's goal is to demonstrate its technology works, and it won't **contribute to Perseverance's science goals.**

***(B) Perseverance Mission has Extracted Oxygen on Mars***

- A device aboard the rover was able to produce oxygen from the thin Martian atmosphere for the first time.
- It promises hope for future crewed missions that can rely on this technology for astronauts to breathe and return to Earth.

## Composition of Martian Atmosphere

- On Mars, carbon dioxide makes up ~96% of the gas in the planet's atmosphere. Oxygen is only 0.13%, compared to 21% in Earth's atmosphere.

***How did Perseverance Produce Oxygen on Mars?***

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- The **Mars Oxygen In-Situ Resource Utilisation Experiment (MOXIE)** produced 5 grams of **oxygen from carbon dioxide** in the Martian atmosphere, enough for an astronaut to breathe for 10 minutes.
- Like a tree on Earth, MOXIE inhales carbon dioxide and exhales oxygen.
- To produce oxygen, MOXIE separates oxygen atoms from carbon dioxide molecules. It does so by using heat at a temperature of around 800°C.
- In the process, it also produces carbon monoxide as a waste product, which it releases in the Martian atmosphere.

**Why is producing oxygen on the Red Planet so important?**

- Oxygen supply on Mars is essential for crewed missions to Mars – not just for astronauts to breathe but for rockets to use as fuel while coming back to Earth.
- As per NASA, for four astronauts to take off from Mars, a future mission would require around 7 metric tons of rocket fuel and 25 metric tons of oxygen–around the weight of an entire space shuttle.
- In contrast, **astronauts living and working on Mars would require far less oxygen to breathe,** maybe around one metric ton.
- Thus, it will be an enormous challenge to haul the 25 metric tons of oxygen from Earth to Mars for the return journey, and that their job would become significantly easier if the liquefied oxygen can be produced on the Red Planet.

## China's Tianwen-1 Probe

The mission was launched from the Wenchang Spacecraft Launch Site on 23 July 2020. China's Tianwen-1 probe has successfully entered orbit around Mars on 10 February 2021, after a 6-1/2- month journey from Earth.

In about three months, the Tianwen-1 will attempt to send a landing capsule carrying a 240-kilogram rover in a rapid seven-minute descent onto a massive plain in the northern hemisphere of Mars known as Utopia Planitia.

### *About Tianwen-1*

- China's first Mars probe is called Tianwen-1 (formerly Huoxing-1).
- The space craft consists of an orbiter, a lander and a rover.
- Launched last year on a Long March 5 rocket from Xichang, China.

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- **Landing site:** Somewhere in Utopia Planitia, a vast plain in Mar's northern latitudes and the same place NASA's Viking 2 mission landed in the 1970s.

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***Zhurong (China's first Mars rover)***

**China's first Mars rover** is named Zhurong after a traditional fire god. The rover is aboard the Tianwen-1 probe that arrived in Mars orbit on February 24, 2021.

The rover was designed to last 90 sols (93 Earth days), but was active for over 347 sols (358 days) after landing on May 14, 2021. Zhurong was expected to wake up in December 2022 after entering a planned sleep in May 2022, but did not due to a buildup of dust on its solar panels, which may have prevented it from generating enough power.

***Significance of the Mission***

- **China in 2019 became the first country to land a space probe on the little-explored far side of the moon** and in December returned lunar rocks to Earth for the first time since the 1970s.
- China would become **the third country after the former Soviet Union and the United States to put a robot rover on Mars.**

## 8.15.4 Important Mercury Missions

Only two spacecraft have ever visited Mercury as of 2023, and a third is on the way. It actually takes more energy for a spacecraft to reach Mercury than Pluto. Mercury is the fastest-orbiting planet in our solar system – it blazes around the Sun at 48 kilometres per second.

- **Mariner 10** (1973)
- **Messenger** (2004)
- **BepiColombo** (2018)

### Messenger Mission

MESSENGER was a **NASA robotic spacecraft that orbited the planet Mercury between 2011 and 2015**. The spacecraft was **launched aboard a Delta II rocket in August 2004** to study Mercury's chemical composition,

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geology, and magnetic field. MESSENGER **entered orbit around Mercury on 18 March 2011**, becoming the first spacecraft to do so.

The MESSENGER mission was planned to study **Mercury's features and environment** from orbit.

Particularly, the mission's scientific objectives were:

- Characterising the planet surface's chemical composition.
- Studying the geologic history of planet.
- Elucidating the global magnetic field nature (magnetosphere).
- Determining the core's size and state.
- Determining the poles' volatile inventory.
- Studying the planet's exosphere nature.

### BepiColombo

- BepiColombo is a joint **European-Japanese mission** to explore the planet Mercury. **It launched in 2018 and will enter Mercury orbit in 2025.**
- The mission will **study water ice at Mercury's poles and the planet's abnormally large core**. This will ultimately help us learn how Mercury formed, and what the early solar system was like.
- Mercury is a mysterious, contradictory world. **Despite being the closest planet to the Sun with surface temperatures of 430 degrees Celsius (800 degrees Fahrenheit), Mercury has water ice hidden in shadowed craters near its poles**.

**NOMENCLATURE**

BepiColombo is named in honour of Giuseppe "Bepi" Colombo (1920–1984), an Italian mathematicianengineer who explained Mercury rotating three times for every two orbits around the Sun.

## 8.15.5 Important Solar Missions

**First Phase: (1960–1969)**

- Pioneer 5 (1960)
- Pioneer 6 (A) (1960)

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- Pioneer 7 (B) (1960)
- Pioneer 8 (C ) (1960)
- Pioneer 9 (D) (1960)
- Pioneer E (1960)

**Second Phase: (1974–1997)**

- Helios A (1974)
- Helios B (1976)
- ISEE-3 (1978)
- ULYSSES (1994)
- WIND (1994)
- SOHO (1996)
- ACE (1997)

**Third Phase: (2000 – Present)**

- ULYSSES (2000)
- Genesis (2001)
- STEREO A (2006)
- STEREO B (2006)
- Ulysses (2007)
- DSCOVR (2015)
- Parker Solar Probe (2018–2025)
- Solar Orbiter (10 Feb, 2020 Launched)

## Parker Solar Probe

**NASA's historic Parker solar probe launched** on **12 August 2018 from the Cape Canaveral.**

It is a **small car sized spacecraft, which will travel at a distance of 4 millions miles from the Earth surface**. It will be **first ever mission to touch the Sun**.

**On 12 August 2019, NASA's Parker Solar Probe completed a year in service**.

It is **part of NASA's "Living With a Star" programme** that explores different aspects of the Sun-Earth system. The probe seeks to gather information about the Sun's atmosphere and NASA says that it "**will revolutionise our understanding of the Sun**". It is also the closest a human-made object has ever gone to the Sun.

## Solar Orbiter/SOLO Mission

Solar Orbiter is a space mission of **international collaboration between ESA (European Space Agency) and NASA.**

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- The spacecraft was launched from **Cape Canaveral on a United Launch Alliance Atlas V rocket** in **February 2020.**
- It was selected as **the first medium-class mission of ESA's Cosmic Vision 2015–2025 Programme**.
- This is **the first mission that will provide images of the sun's north and south poles**.
- It is **a seven-year mission** and will **come within 26 million miles of the sun**.
- It will be able to brave the heat of the sun because it has a custom **titanium heat shield coated in calcium phosphate** so that it **can endure temperatures up to 970 degrees Fahrenheit**.

**The European Space Agency** has released the closest pictures ever taken of the Sun captured by **the Solar Orbiter** that was **launched on 10 February 2020**. The spacecraft completed its **first close pass of the Sun** in mid-June, when **it flew within 48 million miles of the Sun** with 10 instruments turned on to snap the closest pictures of the giant star to date.

**Solar Flares**

- It is **a large explosion of magnetic energy in the Sun's atmosphere** which causes an intense burst of increased brightness.
- Flares occur in active regions around **sunspots**.
- **During solar flares**, the Sun releases bursts of **energetic particles that enhance the solar wind** that constantly emanates from the star into the surrounding space.
- When these energetic particles interact with Earth's magnetosphere, they can cause **magnetic storms that can disrupt telecommunication networks and power grids on the ground**.

**Corona**

- The solar corona is **the outermost layer of the Sun's atmosphere** that extends millions of kilometres into outer space.
- After many decades of studies, **the physical mechanisms that heat the corona are still not fully understood**, but identifying them is considered the 'holy grail' of solar physics.

## PUNCH Mission

**(Polarimeter to Unify the Corona and Heliosphere) NASA has selected Texas-based Southwest Research Institute to lead** its **PUNCH mission which will image the Sun**.

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This is a landmark mission that will image regions beyond the Sun's outer corona.

**Dipankar Banerjee**, solar physicist from **Indian Institute of Astrophysics is also a Co-Investigator** of the PUNCH mission.

**It is focussed on understanding the transition of particles from the Sun's outer corona to the solar wind that fills interplanetary space**. It will consist of a constellation of four microsatellites that, through continuous 3D deep-field imaging, will observe the corona and heliosphere as elements of a single, connected system. **The mission is expected to be launched in 2025**.

### 8.15.6 Solar System and Beyond

**In Progress**

- NASA's **Hubble Space Telescope** is a multipurpose astrophysics and planetary science observatory.
- ESA's **CHEOPS** precisely measures the diameters of known exoplanets.
- NASA's **TESS** hunts for exoplanets around a specific type of bright stars.
- NASA's **James Webb Space Telescope was launched on** December 25, 2021 from **ESA's launch site at Kourou in French Guiana**.

**Coming Soon**

- NASA's **Roman Space Telescope** will launch in 2025 and image Earth-sized exoplanets.

#### Hubble Space Telescope

Hubble Space Telescope is a large, space-based observatory, which has revolutionised astronomy since its launch and deployment by the space shuttle Discovery in 1990. Far above rain clouds, light pollution, and atmospheric distortions, Hubble has a crystal-clear view of the universe.

Scientists have used Hubble to observe some of the most distant stars and galaxies yet seen, as well as the planets in our solar system. Hubble's capabilities have grown immensely in its over 30 years of operation.

Telescopes have a particular range of light that they can detect. Hubble's domain extends from the ultraviolet through the visible (which our eyes see) and into the near-infrared. This range has allowed Hubble to deliver stunning images of stars, galaxies, and other astronomical objects that have

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inspired people around the world and changed our understanding of the universe.

**Hubble has made more than 1.4 million observations over the course of its lifetime**. Over 18,000 peer-reviewed science papers have been published on its discoveries, and every current astronomy textbook includes contributions from the observatory.

The telescope has tracked interstellar objects as they soared through our solar system, watched a comet collide with Jupiter, and discovered moons around Pluto. It has found dusty disks and stellar nurseries throughout the Milky Way that may one day become full-fledged planetary systems and studied the atmospheres of planets that orbit other stars.

Hubble has peered back into our universe's distant past, to locations more than 13.4 billion light years from Earth, capturing galaxies merging, probing the super massive black holes that lurk in their depths, and helping us better understand the history of the expanding universe.

In its over 30 years of operation, Hubble has made observations that have captured humanity's imaginations and deepened our knowledge of the cosmos. It will also continue in coming years.

## Transiting Exoplanet Survey Satellite (TESS)

The Transiting Exoplanet Survey Satellite (TESS) is a NASA mission that will look for planets orbiting the brightest stars in Earth's sky. It was led by the Massachusetts Institute of Technology with seed funding from Google.

The mission will monitor at least 200,000 stars for signs of exoplanets, ranging from Earth-sized rocky worlds to huge gas giant planets. **TESS will focus on stars that are 30 to 100 times brighter than those Kepler examined**. This will help astronomers better understand the structure of solar systems outside of our Earth, and provide insights into how our own solar system formed.

**KEPLER MISSION**

**Launched in the year 2009, for surveying the Milky Way galaxy region so as to discover hundreds of Earth-size and smaller planets in or around the habitable**

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**zone** and determining in our galaxy, the fraction of the hundreds of billions of stars that might have such planets.

As on March 2018, 2,342 planets had been found confirmed by the Kepler; added potential planets, and its finding of exo-worlds stands at 4,587.

Scientists have discovered a cache over 100 new exoplanets using data from NASA's Kepler Space telescope as well as ground-based observatories. The diverse planets are expected to play a large role in developing the research field of exoplanets and life in the Universe.

The Kepler Space Telescope has been officially retired by NASA. Its successor space telescope, called TESS (The Transiting Exoplanet Survey Satellite), has already started collecting data.

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It will use transit method to detect exoplanets. It watches distant stars for small dips in brightness, which can indicate that planet has passed in front of them. Repeated dips will indicate planet passing in front of its star. This data has to be validated by repeated observations and verified by scientists.

## James Webb Space Telescope

The James Webb Space Telescope (also called JWST or Webb) will be a **large infrared telescope** with a 6.5-meter metre primary mirror. The telescope was launched on December 25, 2021 from **ESA's launch site at Kourou in French Guiana**.

- JWST is **a joint venture between the US (NASA), European (ESA) and Canadian space agencies (CSA)**.
- It is an orbiting infrared observatory that will complement and extend the discoveries of **the Hubble Space Telescope** with longer wavelength

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coverage and greatly improved sensitivity.

- Webb was formerly known as **the "Next Generation Space Telescope" (NGST)** and it was renamed in 2002 after a former NASA administrator, James Webb.
- It will be a large infrared telescope with an approximately 6.5 meter primary mirror.

***Objectives and Functions of the Telescope***

1. It will look deeper into the cosmos and thus further back in time than is possible with Hubble.
2. It will do this with a much bigger mirror (6.5 meter in diameter versus 2.4 meter) and instruments that are tuned to the infrared.
3. Scientists hope this set-up can detect the light from the very first population of stars in the Universe to switch on more than 13.5 billion years ago.

***Orbit***

- **The Hubble Space Telescope** orbits around the Earth at an altitude of approximately 570 km.
- Webb will not actually orbit the Earth, instead it will sit at **the Earth-Sun L2 Lagrange point**, 1.5 million km away.
- At **the L2 point Webb's solar shield** will block the light from the Sun, Earth, and Moon which will help Webb stay cool, which is very important for an infrared telescope.

(Refer **Infographic – James Webb Telescope vs Hubble Space Telescope**)

## Roman Space Telescope/Wide-Field Infrared Survey Telescope (WFIRST)

On 17 February 2016, WFIRST was approved for development and launch but now it is renamed as Roman Space Telescope. The Roman Space Telescope is a NASA observatory designed to unravel the secrets of dark energy and dark matter, search for and image exoplanets, and explore many topics in infrared astrophysics. The Roman Space Telescope will have two instruments, the Wide Field Instrument, and the Coronagraph Instrument.

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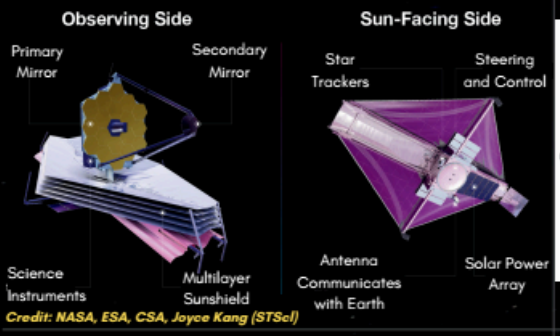
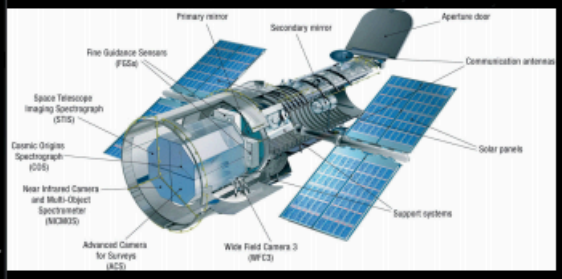
# James Webb Telescope VS Hubble Space Telescope




Credit: NASA, ESA, CSA, Joyce Kang (STScI)

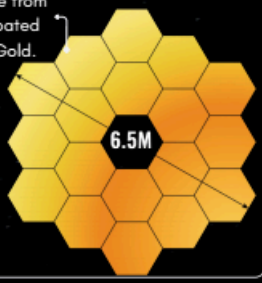



Credit: NASA, ESA, CSA, Joyce Kang (STScI)

**MIRROR**

Mirrors are made from Beryllium and coated with 48.25 gm Gold.

**21.3-Feet (6.5-meter) Primary Mirror**



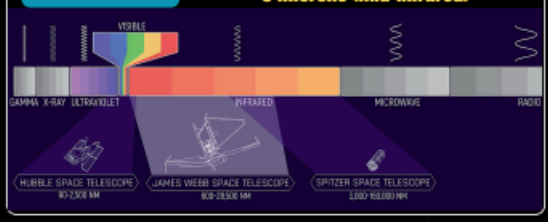

**MIRROR**

Hubble Space Telescope Primary Mirror

**7.9-Feet (2.4-meter) Primary Mirror**

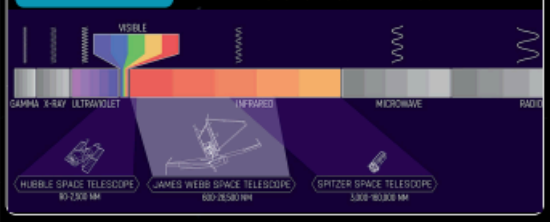
**WAVELENGTH RANGE**

**0.6 micron (Orange) to 5 microns (Mid-Infrared)**




**WAVELENGTH RANGE**

**0.2 micron (Ultraviolet) to 1.7 microns (Near-Infrared)**




**Infrared Field of View** *(Sky Seen by Telescopes)*

**2 times Hubble**

**Infrared Field of View** *(Sky Seen by Telescopes)*

**0.001 square degrees (Wide Field Camera 3)**

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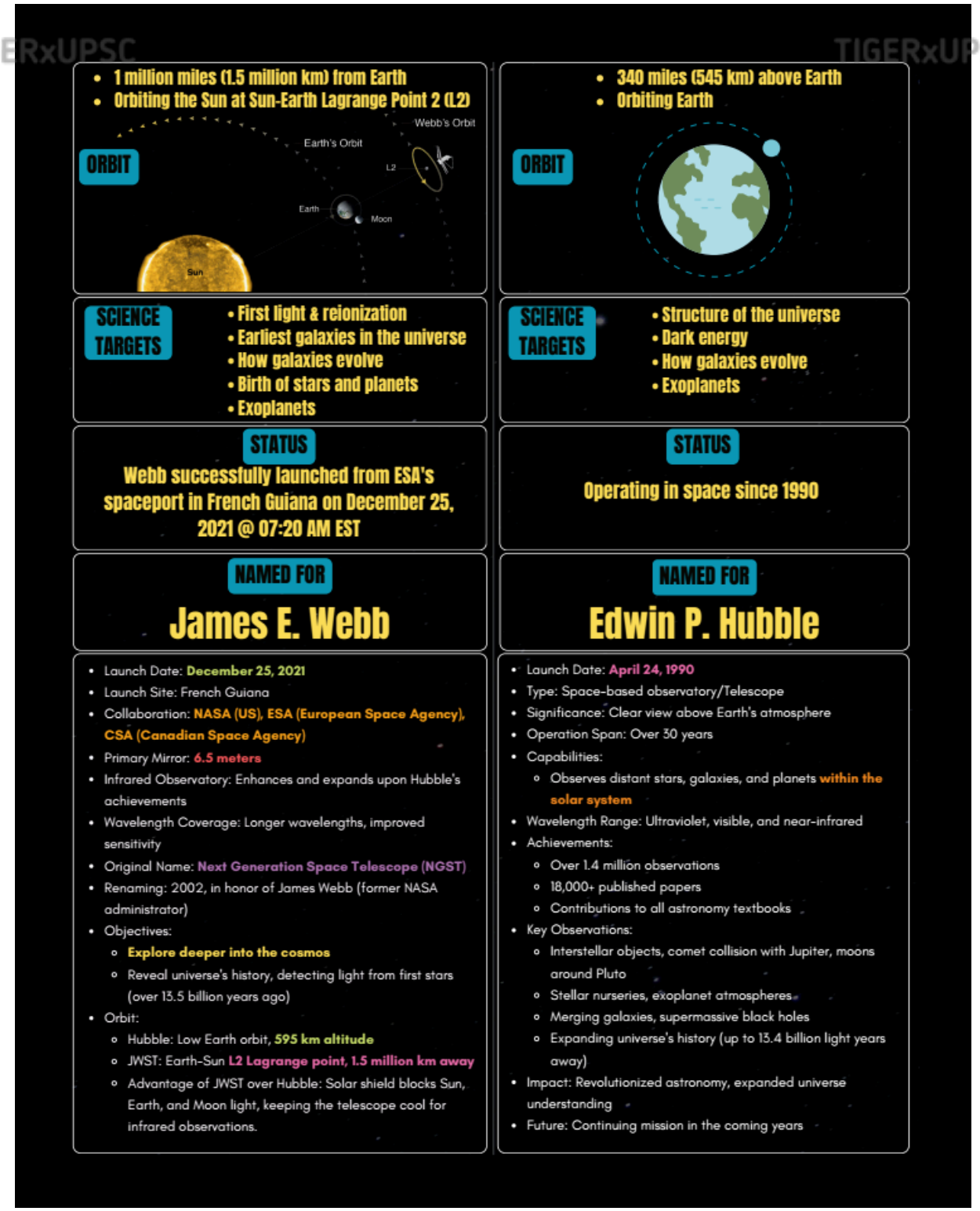


- The Roman Space Telescope has a 2.4 m telescope, the same size as Hubble's, but with a view 100 times greater than Hubble's.

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- The Roman Space Telescope was the top-ranked large space mission in the 2010 Decadal Survey of Astronomy and Astrophysics.
- The Roman Space Telescope is slated to launch in the mid-2020s.
- The Roman Space Telescope will have no proprietary data period and all observing time will be competed.

**Table 8.3** JSWT v Hubble

| Features | JSWT | Hubble |
|---|---|---|
| **Orbit** | **It will not orbit the earth but will sit at Lagrange Point 2 (over millions of km away):**<br>• This will allow it to shield from the light of the Sun, Moon, and Earth and thus avoid light interference. | **Orbits Low Earth Orbit (597 km)** |
| **Light Mirror** | Much bigger at 6.5 m diameter. This allows it to capture light **6 times greater** than Hubble.<br>It will be able to study **old and distant galaxies**, which is not currently possible with Hubble. | **At 2.4 m diameter** |
| **Strength** | It is **100 times** more powerful than Hubble.<br>• It will show images from **250 million years** after Big Bang-taking us **closer to the starting point** of the Universe. | Shows images formed **400 million years** after the Big Bang |
| **Spectrum** | **Infrared:** It will reveal clearer details about objects in distant places.<br>• **Analogically**, Hubble can see the equivalent of | Mostly **ultraviolet** (visible) light |

The Wide Field Instrument will have a field of view that is 100 times greater than the Hubble infrared instrument, capturing more of the sky with less observing time. As the primary instrument, the Wide Field Instrument will measure light from a billion galaxies over the course of the mission lifetime. It will perform a microlensing survey of the inner Milky Way to find approx 2,600 exoplanets. The Coronagraph Instrument will perform high contrast imaging and spectroscopy of dozens of individual nearby exoplanets.

The Roman Space Telescope will have a primary mission lifetime of 5 years, with a potential 5 year extended mission.

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## Chandra X-ray Observatory

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Chandra X-ray Observatory was launched on **23 July 1999** as NASA's flagship mission for X-ray astronomy, taking its place in the fleet of "Great Observatories".

This telescope is specially designed to detect X-ray emission from very hot regions of the Universe such as exploded stars, clusters of galaxies, and matter around black holes. Because X-rays are absorbed by Earth's atmosphere, Chandra must orbit above it, up to an altitude of 139,000 km (86,500 mi) in space.

The **Smithsonian's Astrophysical Observatory** in Cambridge, hosts the Chandra X-ray Centre which operates the satellite, processes the data, and distributes it to scientists around the world for analysis. Chandra carries four very sensitive mirrors nested inside each other.

Chandra has imaged the spectacular, glowing remains of exploded stars, and taken spectra showing the dispersal of elements. **Chandra has observed the region around the super massive black hole in the centre of our Milky Way, and found black holes across the Universe**. It has traced the separation of dark matter from normal matter in the collision of galaxies in a cluster and is contributing to both dark matter and dark energy studies. As its mission continues, Chandra will continue to discover startling new science about our high-energy Universe in future.

## Thirty Meter Telescope (TMT)

TMT is a planned ground-based mirror reflecting telescope that is backed by giant R&D spending countries, namely China, Japan, Canada, USA and India. Mauna Kea, Hawaii, was proposed as a feasible site for the large telescope.

It is planned to observe from near UV to mid-infrared wavelengths, and in addition, it corrects for the blurs caused by the atmosphere by its intuitive and adaptive optics system. It is supposed to be, the highest and will be the second largest telescope after the E-ELT (European Extremely Large Telescope), which uses an array of small 1.44-metre hexagonal mirrors. The design is far different from large mirror of Large Binocular Telescope (LBT) and Giant Magellan Telescope (GMT).

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### *Set to Begin Construction*

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The mega telescope completed its design and development phase in 2009, but legal challenges from Native Hawaiian activists, who treasure Mauna Kea for cultural and religious reasons have hounded the telescope.

### *About TMT*

- The Thirty Meter Telescope (TMT) is a proposed **astronomical observatory with an extremely large telescope (ELT)**.
- It is an **international project** being funded by scientific organisations of ***Canada, China, India, Japan and USA***.
- **Planned location**: ***Mauna Kea on the island of Hawaii*** in the US state of Hawaii.
- The TMT is designed for **near-ultraviolet to mid-infrared observations**.

### *Potential*

- TMT will enable scientists **to study fainter objects far away from us** in the Universe, which gives information about early stages of evolution of the Universe.
- It will give us **finer details of not-so-far-away objects like undiscovered planets and other objects in the Solar System** and planets around other stars.

Indian astronomers collaborated with Nobel laureate on Thirty Meter Telescope Project. **2020 Physics Nobel Laureate Prof. Andrea Ghez** had worked closely with Indian astronomers on the design of back-end instruments.

## Growth India Telescope (GIT)

The GROWTH-India telescope is part of a multi-country joint initiative, termed as the **Global Relay of Observatories Watching Transients Happen (GROWTH)**, for observance of universe's transient events.

- The **fully robotic telescope** is intended to confine cosmic events happening over relatively small periods of the cosmological timescale: years, days and even hours.
- **The US, UK, India, Japan, Germany, Taiwan and Israel Universities and research institutes are part of the initiative.**

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- **Their key research goal is time-domain astronomy, which brings about the study of explosive transients and changeable sources (of light and other radiation) in the universe.**
- **The 0.7 m GROWTH-India telescope at the Indian Astronomical Observatory located in Hanle, Ladakh, has made its first science observation which is a follow-up study of a nova explosion. Novae are explosive events involving violent eruptions on the surface of white dwarf stars, leading to temporary increase in brightness of the star**.

## Square Kilometre Array

On 4th Feb 2021, **the Square Kilometre Array Observatory (SKAO) Council** held its maiden meeting and approved the establishment of **the world's largest radio telescope**.

## What is Square Kilometre Array (SKA) Project?

1. The SKA project **aims to develop the world's largest radio telescope**.
2. It will **consist of a group of radio telescopes operating from South Africa and Australia** in two frequency ranges, with the Square Kilometre Array Observatory **headquartered in the UK**.

**Figure 8.2** Radio Telescope

## Objective

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- The project **seeks to explore galaxies in the cosmos, map visible galaxies, provide detailed data on our galaxy's evolution, search for signs of extraterrestrial life, and identify gravitational waves.**

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## Participating Countries

- Australia, Canada, China, India, Italy, New Zealand, South Africa, Sweden, the Netherlands, and the UK.

## India's Role

Recently, the Union Cabinet has granted approval for the **Square Kilometre Array (SKA) project**, with a financial **sanction of `1,250 crore**. [Press Release, PIB dated 3 January 2024]

1. India's primary contribution involves the development and operation of the Telescope Manager element, which monitors the telescope's functions. This element has been built by the Pune-based **National Centre for Radio Astrophysics (NCRA)** along with other international institutes.
2. India is also making significant **financial contributions** to the project.

## Construction Phases

- The SKA will be constructed in two phases, with the **first phase of SKA1 commenced construction in December 2022, and operations expected to start by 2029**.

## Significance

- The SKA project will offer profound **insights into the initial stages of our galaxy's evolution and actively explore the possibility of life beyond Earth**.

## Significance for India

1. India, as a **full-time member, will gain preferential access to SKA facilities**.
2. The focus on cutting-edge technologies in electronics, software, materials science, and computing within the SKA initiative will

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provide valuable learning opportunities for scientists, academics, and private industries in India.

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### Ariel (Atmospheric Remote-Sensing Infrared Exoplanet Large-Survey)

The **European Space Agency (ESA)** has formally adopted **Ariel. Ariel (Atmospheric Remote-sensing Infrared Exoplanet Large-survey)** will be launched in 2029. It will perform **a large-scale survey of over a thousand exoplanets over a period of four years**. The explorer that will **study the nature, formation and evolution of exoplanets**.

***Significance***

- Ariel is the **first mission of its kind dedicated to measuring the chemical composition and thermal structures of hundreds of exoplanets.**
- It will also help to answer one of the key questions of ESA's **Cosmic Vision Plan**, which is, "**What are the conditions for planet formation and the emergence of life?**".

## 8.15.7 Asteroid Missions in News

### Dawn Asteroid Mission

**Since its launch in the year 2007**, the unmanned spacecraft has travelled 4.3 billion miles (6.9 billion kilometres).

The **only spacecraft ever to orbit a cosmic body in 2011, in the main asteroid belt between Mars and Jupiter when it started orbiting around the asteroid Vesta**.

Dawn, after running out of fuel in 2018, being the only NASA spacecraft launched 11 years ago, that studied two of the largest objects in the asteroid belt, has ended its mission.

### Hayabusa and Hayabusa-2

Hayabusa is an **asteroid sample-return mission** operated by the **Japanese Space Exploration Agency (JAXA)**.

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**In mid-September 2005, Hayabusa landed on the asteroid Itokawa**, and managed to collect samples in the form of grains of a steroidal material. **It returned to Earth with the samples in June 2010, thereby becoming the first spacecraft to return asteroid samples to Earth for analysis**.

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***Hayabusa-2***

- It was l**aunched on 3 December 2014** and rendezvoused with **near-Earth asteroid Ryugu** on 27 June 2018 at 300 million km from Earth's surface.
- **It is in the progression of asteroid surveying for a year and a half, leaving in December 2019, and returned to Earth on 6 December 2020.** The soil samples and data from the asteroid could provide clues to the origins of the solar system.
- **Hayabusa-2 took numerous science payloads for remote sensing, sampling and four small rovers which will examine the asteroid surface so as to update the environmental and geological context of the samples collected.**

### NASA'S OSIRIS-REX

The NASA OSIRIS-REx mission **launch took place on September 8, 2016**. On **20th October 2020, NASA's OSIRIS-REx spacecraft briefly touched asteroid Bennu**, from where it is meant to collect samples of dust and pebbles and deliver them **back to Earth in 2023.**

**NASA's OSIRIS-REX spacecraft has discovered hydrogen and oxygen molecules traces, which being part of the water formula and thus the potential for life**.

**It entrenched in the asteroid Bennu's rocky surface, where it will grab a sample of rock and dust and bring it back to the Earth**.

## 8.15.8 Space Craft Leaving Our Solar System/Interstellar Mission

### Pioneer 10 & Pioneer 11

Pioneer 10 was launched toward Jupiter on **2 March 1972**. This spacecraft was the first one to fly to Jupiter, Saturn, the Milky Way Galaxy and stars.

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After Pioneer 10 emerged through the asteroid belt, Pioneer 11 was launched on a similar trajectory on **April 5, 1973**, like Pioneer 10, on top of an Atlas/Centaur/TE364-4 launch vehicle.

Pioneer 10 was the first spacecraft to pass through the asteroid belt, considered a spectacular achievement, and then headed toward Jupiter. Accelerating to a speed of 131,200 kilometres per hour, Pioneer 10 passed by Jupiter on Dec. 3, 1973. The spacecraft was the first to make direct observations and obtain close-up images of Jupiter. Pioneer 10 also took measurements of the gas giant's magnetosphere, radiation belts, magnetic field, atmosphere, and interior. In 1983, Pioneer 10 became the first human-made object to pass the orbit of Pluto, at that time considered the most distant planet from the sun.

Pioneer 10 spacecraft continued to make valuable scientific investigations in the outer regions of the solar system until its science mission ended on **March 31, 1997**.

Pioneer 10, Earth's first emissary into space, is **carrying a gold plaque** that describes what we look like, where we are and the date the mission began. Pioneer 10 will continue to coast silently as a ghost ship through deep space into interstellar space, heading generally for the red star Aldebaran, which forms the eye of the constellation Taurus (The Bull). Aldebaran is about 68 light years away.

Its sister ship, Pioneer 11, ended its mission **September 30, 1995**, when the last transmission from the spacecraft was received.

## Voyager Mission

The NASA's Voyager mission **was launched in the 1970's, to explore the outer planets**.

This probe is **still going in interstellar space**.

- The Voyager Interstellar Mission's (VIM) goal is to expand the NASA solar system exploration ahead of the outer planets neighbourhood to the outer limits of the Sun's sphere of influence, and possibly away from it.
- VIM was launched in **August (Voyager 2)** and **September (Voyager 1), 1977** from **Cape Canaveral, Florida**. Voyager 1 was at a distance

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of approximately 40 AU (Astronomical Unit – mean distance of Earth from the Sun, 150 million kilometres) from the Sun, and Voyager 2 was at a distance of approximately 31 AU, after **completion of its 12 years journey till 1989**.

- **On August 25, 2012, Voyager 1 flew beyond the heliopause and entered interstellar space, making it the first human-made object to explore this new territory**.
- **Voyager 2, which is travelling in a different direction from Voyager 1, crossed the heliopause into interstellar space on November 5, 2018**.
- The Voyagers have enough electrical power and thruster fuel to keep its current suite of science instruments on **until at least 2025**. By that time, Voyager 1 will be about 22.1 billion kilometres from the Sun and Voyager 2 will be 18.4 billion kilometres away.

## New Horizons

**[Launch Date: 19 January 2006, Launch Site: Cape Canaveral]**

It was launched by NASA, being an interplanetary space probe. The primary objective was to make the study of Pluto and a secondary mission in fly by study of one or more other Kupier belt objects.

- In January 2015: It started its approach towards Pluto. In July 2015, it flew 12,500 kilometres above the surface of Pluto to make it the first spacecraft to explore the dwarf planet.
- New Horizons was saluted for accomplishing the **first direct investigation of Kupier belt objects**.
- At Pluto, New Horizons found **signs of ice volcanoes, giant mountains**, and even a **liquid water ocean**. Then, the probe pushed on into the depths of the Kuiper Belt, where it **explored 486958 Arrokoth,** a primordial world of ice and rock that looks like two pancakes stuck together.
- But unlike its interstellar spacecraft kin, New Horizons doesn't carry a plaque or a golden record designed to teach aliens about the human race. And that was intentional.
- The New Horizons mission is currently extended through 2021 to explore additional Kuiper Belt objects.

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## 8.15.9 Important Moon Mission

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**In Progress**

- China's **Chang'e-5** returned lunar samples to Earth in 2020, and is on an extended mission to study the Sun and possibly asteroids.
- India's **Chandrayaan-2** orbiter maps the Moon's topography, studies its composition, and scans for water ice.
- China's **Chang'e-4** performed the first landing on the Moon's far side in 2018, where it studies an ancient region.
- NASA's **Lunar Reconnaissance Orbiter** studies the Moon with a high-powered camera.

**Coming Soon**

- NASA's VIPER rover launches to the Moon's south pole in 2024 to study water ice.
- Israeli non-profit SpaceIL launches Beresheet 2 to the Moon in 2024 to conduct scientific experiments and public outreach.

### Lunar Reconnaissance Orbiter (LRO)

**LRO** and the **Lunar** Crater Observation and Sensing Satellite (LCROSS) **were** launched on an Atlas V rocket on June 18, 2009, beginning a four-day trip to the **moon. LRO** spent its first three years in a low polar **orbit** collecting detailed information about the **moon** and its environment. After this initial orbit, LRO transitioned to a stable elliptical orbit, passing low over the lunar south pole. With a suite of **seven powerful instruments**, LRO has collected a treasure trove of data, making an invaluable contribution to our knowledge about the moon. With the Lunar Reconnaissance Orbiter (LRO), NASA has returned to the moon, enabling new discoveries and bringing the moon back into the public eye.

LRO is a robotic mission that set out to map the moon's surface and, after a year of exploration, was extended with a unique set of science objectives. LRO observations have enabled numerous ground breaking discoveries, creating a new picture of the moon as a dynamic and complex body. These developments have set up a scientific framework through which to challenge and improve our understanding of processes throughout the solar system.

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### Chang'e Mission

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The Chinese **Lunar Exploration Program is designed to be conducted in four phase**s.

**Phase-1:** Simply reaching lunar orbit, a task completed by Chang'e 1 in 2007 and Chang'e 2 in 2010.

**Phase-2:** Landing and roving on the Moon, as Chang'e 3 did in 2013 and Chang'e 4 did in 2019.

**Phase-3:** Collecting lunar samples from the **near-side** and sending them to Earth, with the help of Chang'e 5 **launched on 24 November 2020** and Chang'e 6 missions in future.

**Phase-4:** Consists of development of a robotic research station near the Moon's South Pole.

### *Chang'e-4 Mission*

Chang'e-4 mission achieved **humanity's first soft landing on the far side of the Moon, on 3 January 2019.** This mission will attempt to determine the age and composition of an unexplored region of the Moon, as well as develop technologies required for the later stages of the program.

### *Objectives*

An ancient collision event on the Moon left behind a very large crater, called the Aitken Basin, that is now about 13 km (8.1 mi) deep, and it is thought that the massive impactor likely exposed the deep lunar crust, and probably the mantle materials. If Chang'e 4 can find and study some of this material, it would get an unprecedented view into the Moon's internal structure and origins.

**China has launched Chang'e-4, a first probe ever to explore the dark side of the Moon, marking another milestone in its ambitious space programme.**

- The Moon is **tidally locked to Earth**, rotating at the same rate at which it orbits our planet, so the remote side is never visible from Earth. The probe, the Chang'e-4, is projected to make the first-ever soft landing on the far side of the Moon. Previous spacecraft have seen the Moon's far side, but none has landed on it.

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- The Moon's **far side termed as 'South Pole Aitken Basin'** still remains a mystery among space scientists and by sending a probe there, China will outshine the US and USSR historical achievements.
- Chang'e 4 is the country's fourth mission in the series of lunar mission which is being named after the **Chinese Moon goddess**.
- The main problem faced by the Chinese team was **difficulty in communication as they attempted to land on the Moon's other side** so China used options such as **radio telescopes for communicating in the absence of a transmitting medium**.

**WHY EXPLORE THE LUNAR SOUTH POLE?**

China has named the **lunar rover**, successfully deployed to carry out a string of experiments on the far side of the Moon, as 'Yutu-2'. The rover's touchdown is part of China Chang'e-4 lunar probe.

**Key Facts**

- The rover has been programmed to launch ground penetration radar that would help map the Moon's inner structures.
- It would **also analyse soil and rock samples for minerals, apart from activating a radio telescope to search for possible signals from deep space**.

### *Chang'e-5 Probe*

The Chang'e-5 was **launched on 24 November 2020** and a lander vehicle **touched down on the moon on 1 December 2020**. The mission was expected to take around 23 days in total. **The objective of the mission was to bring back lunar rocks, the first attempt by any nation to retrieve samples from the moon in four decades**.

- Chang'e 5, collected 1,731 gram (Approx. 2 kg) of lunar samples (including from a core 1 m deep), and returned to the Earth on 16 December 2020.
- Chang'e-5 mission is the first lunar sample-return mission conducted by humanity in **over four decades since the Soviet Union's Luna 24 in 1976**.
- **Chang'e 5** is the fifth lunar exploration mission of the Chinese Lunar Exploration Program, and China's first lunar sample-return mission.

### *About the Chang'e-5 Probe*

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- The probe is named after the mythical Chinese moon goddess.
- The rocket is comprised of four parts: an orbiter, a returner, an ascender and a lander.
- The objective of the mission is to bring back lunar rocks, the first attempt by any nation to retrieve samples from the moon in four decades.
- If successful, China will be only the third country to have retrieved samples from the moon, following the U.S. and the Soviet Union in the 1960s and 1970s.
- This will help scientists learn about the moon's origins, formation and volcanic activity on its surface.

## Artemis Program – Moon

**Artemis-Acceleration, Reconnection, Turbulence and Electrodynamics of Moon's Interaction with the Sun.** It is **NASA's next mission to the Moon**.

- With the Artemis program, **NASA will land the first woman and next man on the Moon by 2024**. **Artemis was the twin sister of Apollo** and **goddess of the Moon in Greek mythology**.

***Objective***

- To measure what happens when the Sun's radiation hits our rocky moon, where **there is no magnetic field to protect it**.

## Artemis Accords

**National Aeronautics and Space Administration (NASA)** and **Indian Space Research Organisation (ISRO)** will **collaborate** to **send Indian astronauts**, trained at the Johnson Space Centre in Houston, Texas, to the International Space Station (ISS) in 2024.

- Established by the U.S. State Department and NASA in 2020.
- It aims to set common principles for civil exploration and use of outer space, the moon, Mars, comets and asteroids, for peaceful purposes.
- This accord was **recently signed by India**, becoming the **27th country** to join.

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- Builds upon the **Outer Space Treaty of 1967**, emphasising peaceful use of space and prohibiting national appropriation.

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### *Commitments under the Accords*

- **Peaceful Purposes**: Conduct space activities in line with international law for peaceful purposes.
- **Common Infrastructure:** Recognise the importance of common exploration infrastructure.
- **Registration and Data Sharing:** Register space objects and share scientific data openly.
- **Preservation of Heritage:** Preserve historic landing sites and artifacts on celestial bodies.
- **Utilisation of Space Resources:** Utilise space resources sustainably and share information to prevent interference.
- **Mitigation of Debris:** Plan for safe disposal of spacecraft and limit harmful debris generation.

### *Main Missions under the Artemis Programme*

1. **Artemis-I:** Unmanned mission to the Moon launched in 2022.
2. **Artemis-II: Crewed Lunar Flyby Mission:** Scheduled for **2024**, Artemis-II will mark the first crewed mission under the Artemis programme. Four astronauts will be aboard the SLS as it performs multiple manoeuvres on an expanding orbit around Earth. The mission will also involve a lunar flyby and return to Earth.
3. **Artemis-III: Human Return to the Moon:** Set for 2026, Artemis-III will mark a significant milestone in human space exploration as astronauts return to the moon.
4. **Lunar Gateway Station:** Planned for 2029 to facilitate scientific research and astronaut docking.

### *Benefits for India*

- Access to **advanced training, technology, and scientific opportunities**.
- Potential to advance India's lunar exploration goals, including the Chandrayaan-3 mission.
- Enhancement of capabilities for the Gaganyaan human mission and future space endeavours.

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- Opportunity for mutual advancements in space exploration through collaboration with NASA.

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## 8.15.10 Miscellaneous Missions in News

### China's Artificial Moon Project

China is in the process of creating an "artificial moon" that would be bright **enough to replace the streetlights in the south-western city of Chengdu by 2022.**

- Chinese scientists intend for **sending three artificial Moons to space** in the **coming four years**, where **reflective material such as a mirror is estimated to orbit at 500 kilometres above the Earth** and **light up an area with a diameter of 10 to 80 kilometres**.
- There will be a **reflective coating on an artificial Moon** which can deflect sunlight back to Earth, akin to how the moon shines.
- The **enlightened satellite is said to be eight times brighter than the real Moon**.
- The three artificial moons would function alternately for significantly reducing infrastructural electricity consumption, especially at the time of winter. The illuminated satellite is intended to complement the moon at night.
- **Every year, light from the artificial moon in Chengdu covering 50 square kilometres could possibly save about 1.2 billion yuan ($240 million) in electricity costs**.
- It could also be used to lighten up the areas experiencing power breakdown due to **natural disasters such as earthquakes**.

Note: The thought for the artificial human made moon began from a French artist who imagined hanging a necklace in the sky made of mirrors for lightening up the Paris streets at night. Russia tried to initiate a 25-metre "space mirror" but the project was suspended in the year 1999.

### NASA'S Orion Spacecraft

NASA's Orion spacecraft is **designed to carry humans further than they've ever gone before**.

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Orion will function as the exploration vehicle which will take the crew to space, **provide capability for emergency abort**, sustain the crew at the time of travelling to space, and **provide safer re-entry from deep space return velocities.** Orion will be launched from NASA's new heavy-lift rocket, the System for Space Launch.

NASA has successfully finished the last test for qualifying Orion's space capsule's parachute system for onboard flights with astronauts, ahead of its mission in sending humans to the Moon and beyond.

The parachute system is the lone system that must assemble itself in mid-air and must be able to keep the crew safer in scenarios of numerous failures, for instance mortar failures which avoid deploying a single parachute type, or circumstances causing the failure of some of the components of parachute textile.

## ICESAT and ICESAT-2

**ICESat (Ice, Cloud and land Elevation Satellite) mission** was sent for **measuring ice sheet mass balance, cloud and aerosol heights, land topography and vegetation characteristics as an Earth Observing System**.

The Ice, Cloud and land Elevation Satellite-2 (ICESat-2) is the 2nd-generation of the laser altimeter ICESat-1 mission (January 13, 2003 to August 14, 2010).

ICESAT 2 was launched on 5 September 2018 from Vandenberg Air Force Base in California, into a near-circular, near-polar orbit with an altitude of approximately 496 km.

It was designed to operate for three years and carry enough propellant for seven years.

- NASA's ICESat-2 **will map ice sheets melting in Antarctica and the resulting rising sea level across the globe, thereby assisting in the improvement of climate forecasts**.
- **The satellite is measuring the sea ice height to inside an inch**, tracing the unmapped Antarctic valleys terrain, surveying of remote ice sheets, and peering through forest canopies and shallow coastal waters.

## ICON Mission

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**National Aeronautics and Space Administration (NASA)** has launched a satellite ICON to **detect dynamic zones of Earth's Ionosphere**. The satellite **Ionosphere Connection Explorer (ICON)** was **launched from** an aircraft over the Atlantic Ocean near the **Florida coast on 11 October 2019** with **two years lifespan**. The ICON satellite will study the Earth's Ionosphere. **It includes various layers of the uppermost atmosphere where free electrons flow freely**.

## Sentinel-6 Satellite

The **Copernicus Sentinel-6 Michael Freilich satellite** is designed to monitor oceans. It was launched on **22 November 2020** from California aboard a **SpaceX Falcon 9 rocket**. This is **a part of the next mission dedicated to measuring changes in the global sea level**.

### *What is the Mission?*

- The mission is called **the Jason Continuity of Service (Jason-CS) mission**.
- It is designed **to measure the height of the ocean**, which is a key component in understanding how the Earth's climate is changing.
- It has been **developed jointly by** the ESA, NASA, European Organisation for the Exploitation of Meteorological Satellites (Eumetsat), the USA's National Oceanic and Atmospheric Administration (NOAA) and the EU, with contributions from France's National Centre for Space Studies (CNES).

### *What will the Satellite do?*

- Provide measurements of global sea-level rise.
- Send pulses to the Earth's surface and measure how long they take to return to it, which will help scientists measure the sea surface height.
- Measure water vapour along this path and find its position using GPS and ground-based lasers.

### *Significance of the Mission*

- This data will allow improvements in both short-term forecasting for weather predictions in the two-to-four-week range (hurricane intensity

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predictions), and long-term forecasting, for instance for seasonal conditions like **El Niño and La Niña**.

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## Spacex Launches 143 Satellites, Breaks World Space Record

- Space-X broke a world space record by launching 143 satellites in quick succession, beating India's record of deploying 104 satellites in February 2017.
- The launch vehicle for the Space-X record-breaking flight was the Falcon 9 and the mission was designated as **Transporter-1**.
- The launch marks **the first dedicated mission for Space-X's Small Sat Ride share Program**, which enables small-satellite customers to book a ride to orbit with Space-X directly.

## Russia to Launch its own Orbital Station by 2030

Russia aims to establish the core of its new orbital space station by 2030, according to its space agency, Roscosmos. The agency's head, Yuri Borisov, approved the timeline, which includes launching an initial scientific and energy module in 2027 and adding three more modules by 2030, with two additional modules planned between 2031 and 2033. Previously collaborating on the International Space Station (ISS), Russia plans to exit the ISS by 2028 and focus on its own station to address scientific, technological, and national security needs.

### *About the International Space Station*

- Launched in **1998** and **involving Russia, the United States, Canada, Japan, and the European Space Agency**, the ISS is one of the most ambitious international collaborations in human history.
- ISS is a modular space station (habitable artificial satellite) in low Earth orbit.
- The ISS serves as a microgravity and space environment research laboratory in which scientific experiments are conducted in astrobiology, astronomy, meteorology, physics, and other fields.
- The ISS circles the Earth in roughly 93 minutes, completing 15.5 orbits per day.

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- **ISS is the ninth space station to be inhabited by crews**, following the Soviet and later Russian Salyut, Almaz, and Mir stations as well as Skylab from the US.

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### Inspiration4: SpaceX's First All-Civilian Space Mission

- **The mission involves circling the Earth for three days**, and then splashing down into the Atlantic Ocean. Inspiration4 **will orbit the Earth at 575 km**, higher than **the International Space Station** (408 km) and **the Hubble space telescope** (547 km). This **will be the farthest distance travelled by a crewed mission since 2009**, when astronauts last went to repair the Hubble.

***Facts***

- Entrepreneur Elon Musk's SpaceX announced '**Inspiration4**', **its first all-civilian, non-governmental spaceflight** on 16 September 2021.
- The **Crew Dragon spacecraft** was launched from **NASA's Kennedy Space Centre in Florida in the US**.
- It took a **group of four private citizens into space for three days**.
- All four seats on the spacecraft were **purchased by US billionaire Jared Isaacman**, founder of the fintech company Shift4 Payments.
- This was a part of an **effort to raise millions for the Tennessee-based St. Jude Children's Research Hospital**, a paediatric treatment and research facility that focuses on children's catastrophic diseases, particularly leukaemia and other cancers.

## 8.15.11 Commercial Crew Program

### Crew Mission

The main objective of this program is to make access to space easier in terms of its cost, so that cargo and crew can be easily transported to and from the ISS, enabling greater scientific research. Through this program, NASA plans to lower its costs by sharing them with commercial partners such as Boeing and SpaceX, and also give the companies incentive to design and build the Commercial Orbital Transportation Services (COTS).

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***Crew-1 Mission***

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**Crew-1 is the first of six crewed missions that NASA and SpaceX will operate as part of the Commercial Crew Program, whose objective is to make access to space easier in terms of cost**.

The mission was launched on 16 November 2020. Crew-1 is the first operational flight of the SpaceX Crew Dragon spacecraft on a Falcon 9 rocket to the ISS and is also the first of the three such flights scheduled over the course of 2020–21.

## What will members of Crew-1 do at the ISS?

- The Crew-1 team will conduct microgravity studies.
- Some of the research that the crew is carrying with themselves includes materials to investigate food physiology, which will study the effects of dietary improvements on immune function and the gut micro-biome and how those improvements can help crews adapt to spaceflight.
- Once in orbit, astronauts will collect samples to provide data to scientists back on Earth so that they can continue to study how dietary changes affect his body.
- Another experiment aboard is a student-designed experiment titled, "**Genes in Space-7**" that aims to understand how spaceflight affects brain function.

### *Crew-2 Mission*

**Four astronauts are sent** to the International Space Station (ISS) from Florida through a collaboration program of NASA and SpaceX under the Commercial Crew Program.

**SPACE X WILL LAUNCH NASA'S UPCOMING CREW-1**

- NASA has certified **SpaceX's Crew Dragon capsule and the Falcon 9 rocket**, making it **the first spacecraft certification provided by the space agency**.
- This means **SpaceX can now operate regular flights to the space station**.
- **SpaceX's Crew Dragon spacecraft** lifted off from NASA's Kennedy Space Centre in Florida on November 14, 2020.

### *What is the Crew-2 Mission?*

- The mission is second crew rotation of the SpaceX Crew Dragon and is first with international partners.

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- It is a six-month science mission.
- Astronauts include **two from NASA and one each from JAXA and ESA.**
- NASA's SpaceX Crew-2 mission consist a rocket booster that is recycled from a previous spaceflight.
- **The Falcon 9 rocket used in the Crew-2 mission was used to launch the Crew-1 into orbit five months ago.** This marks the **first time a previously flown rocket booster is re-used in a crewed launch**.
- The latest mission is the second "operational" space station team that was launched by NASA aboard a Dragon Crew capsule.

***What will Crew-2 do at ISS?***

- Crew-2 astronauts will join the members of Expedition 65.
- They will stay aboard the ISS for six months and will conduct science experiments in low-Earth orbit.
- Their central focus will be to continue a series of Tissue Chips in Space studies.
  - ❑ Tissue Chips are small models of human organs that contain multiple cell types that behave similarly to the human body.
  - ❑ These chips can speed up the process of identifying safe and effective drugs and vaccines.
  - ❑ Scientists will use these tissue chips in space to study diseases that affect specific human organs.
  - ❑ The main advantage is that experiments in space takes lesser times in comparison to earth.

## QUESTIONS AND ANSWERS

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## Objective Questions

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**1. GPS-Aided Geo Augmented Navigation (GAGAN) uses a system of ground stations to provide necessary augmentation. Which of the following statements is/are correct in respect of GAGAN?** **[UPSC (Pre) 2025]**

**i.** It is designed to provide additional accuracy and integrity.

**ii.** It will allow more uniform and high quality air traffic management.

**iii.** It will provide benefits only in aviation but not in other modes of transportation.

Select the correct answer using the code given below:

(a) i, ii and iii
(b) ii and iii only
(c) i only
(d) i and ii only

**2. Consider the following space missions:** **[UPSC (Pre) 2025]**

**i.** Axiom-4

**ii.** SpaDeX

**iii.** Gaganyaan

How many of the space missions given above encourage and support micro-gravity research?

(a) Only one
(b) Only two
(c) Only three
(d) None

**3. If a major solar storm (solar flare) reaches the Earth, which of the following are the possible effects on the Earth?**

1. GPS and navigation systems could fail. **[UPSC (Pre) – 2022]**
2. Tsunamis could occur at equatorial regions.
3. Power grids could be damaged.
4. Intense auroras could occur over much of the Earth.
5. Forest fires could take place over much of the planet.
6. Orbits of the satellites could be disturbed.

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7. Shortwave radio communication of the aircraft flying over polar regions could be interrupted.

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Select the correct answer using the code given below:

(a) 1, 2, 4 and 5 only
(b) 2, 3, 6 and 7 only
(c) 1, 3, 4, 6 and 7 only
(d) 1, 2, 3, 4, 5, 6 and 7

**4. In which of the following areas can GPS technology be used?** **[UPSC (Pre) – 2018]**

1. Mobile phone operations
2. Banking operation
3. Controlling the power grids

Select the correct answer using the code given below:

(a) Only 1
(b) 2 and 3
(c) 1 and 3
(d) 1, 2 and 3

**5. With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements:** **[UPSC (Pre) – 2018]**

1. IRNSS has three satellite in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq.km beyond its borders.
3. India will have its own satellite navigation system will full global coverage by the middle of 2019.

Which of the statements given above is/are correct?

(a) Only 1
(b) 1 and 2
(c) 2 and 3
(d) None of these

**6. With reference to 'Astrosat', the astronomical observatory launched by India, which of the following statements is/are correct?** **[UPSC (Pre) – 2016]**

1. Other than USA and Russia, India is the only country to have launched a similar observatory into space.

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2. Astrosat is a 2,000-kilogram satellite placed in an orbit at 1,650 kilometres above the surface of the Earth.

Select the correct answer using the code given below:
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**7. Consider the following statements regarding The Mangalyaan, launched by ISRO.** **[UPSC (Pre) – 2016]**

1. It is also called the Mars Orbiter Mission.
2. It made India the second country to have a spacecraft orbit Mars, after USA.
3. It made India the only country to be successful in making its spacecraft orbit Mars in its very first attempt.

Which of the statements given above is/are correct?
(a) Only 1
(b) 2 and 3
(c) 1 and 3
(d) 1, 2 and 3

**8. Which of the following pair is/are correctly matched?** **[UPSC (Pre) – 2014]**

| Spacecraft | | Purpose |
|---|---|---|
| 1. Cassini-Huygens | : | Orbiting Venus and transmitting data to Earth |
| 2. Messenger | : | Mapping and investigating Mercury |
| 3. Voyager I and II | : | Exploring the outer solar system |

Select the correct answer using the code given below.
(a) Only 1
(b) 2 and 3
(c) 1 and 3
(d) 1, 2 and 3

**9. In the context of space technology, what is "Bhuvan", that has recently been in the news?** **[UPSC (Pre) – 2010]**

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(a) A mini satellite launched by ISRO for promoting distance education in India.
(b) The name given to the next Moon Impact Probe, for Chandrayan-II.
(c) A geoportal of ISRO with 3D imaging capabilities of India.
(d) A space telescope developed by India.

## Answer Key

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (d) | 2. (c) | 3. (c) | 4. (d) | 5. (a) | 6. (d) | 7. (c) | 8. (d) |
| 9. (c) | | | | | | | |

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**UNIT-IV** Communication and Information Technology

# CHAPTER 9
# DIFFERENT REGULATORY BODIES AND GOVERNMENT PROGRAMMES ON ICT

## IN THIS CHAPTER...


9.1 Ministry of Communication and Information Technology
9.2 Department of Telecommunication (DoT)
9.3 Regulatory Framework-TRAI and TDSAT
9.4 IT Rules, 2021
9.5 Telecommunications Rules, 2024 (Procedures and Safeguards for Lawful Interception of Messages)
9.6 Draft Digital Personal Data Protection (DPDP) Rules 2025
9.7 Department of Electronics and Information Technology
9.8 Computer Emergency Response Team (CERT-IN)


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CERT-In has issued a high severity warning for iPhone and iPad users due to vulnerabilities that could allow a hacker to make one's device stop working, access sensitive information, and get around security measures.

**9.9 Bharat Broadband Network/BharatNet/National Optical Fibre Network (NOFN)**
**9.10 Centre for Development of Telematics (C-DoT)**
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**9.16 PRAGATI (Pro-Active Governance and Timely Implementation)**
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**9.19 JAM Trinity**

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JAM stands for three things – The Jan Dhan Yojana, the Aadhaar initiative of UIDAI and Mobile number

## 9.20 One Nation One FASTag

## 9.21 e-Learning (Education and Internet Technology)

The Union Minister of Education and Skill Development and Entrepreneurship launched the ‘SWAYAM Plus’ platform to offer employability and professional development-focused programmes developed with industry players and will be operated by IIT-Madras

### 9.21.1 Unnat Bharat Abhiyan

### 9.21.2 e-Pathshala

### 9.21.3 SARANSH

### 9.21.4 Indian MOOCs: “SWAYAM”

### 9.21.5 SANGAM: Digital Twins Initiative

## 9.22 Miscellaneous

### 9.22.1 CoWIN (Covid Vaccine Intelligence Network)

### 9.22.2 MANAS App

### 9.22.3 Aarogya Setu

### 9.22.4 DIKSHA (Digital Infrastructure for Knowledge Sharing)

### 9.22.5 Digi Yatra App

### 9.22.6 Sanchar Saathi Initiative

### 9.22.7 DIGIPIN web Application

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## 9.1 Ministry of Communication and Information Technology

The Ministry of Communication and Information Technology plays the leading role in Information and Communication Technology (ICT) regulation in the country with three associated departments: Department of Telecommunications (DoT), Department of Information Technology and Department of Posts.

## 9.2 Department of Telecommunication (DoT)

Its responsibilities include:

- Policy formulation; performance evaluation and review; monitoring; international cooperation; research and development and grant of licences to operators.
- Allocation of frequency and management of radio communications in close coordination with the international bodies.
- In recent years, **Indian telecom sector has grown at a steady rate and thus accords as the second largest telephone network in the world after China.**

## 9.3 Regulatory Framework-TRAI and TDSAT

At the first level of telecom dispute, matter is sent to TRAI (established in 1997) to resolve it. If TRAI could not resolve this issue, it will transfer to Telecom Dispute Settlement Appellate Tribunal. An ordinance amending TRAI Act led to the establishment of TDSAT for taking over the functions related to TRAI's arbitration and disputes, effective from 24 January 2000. Its primary aim was to adjudicate any dispute between licensor and a licensee, between two or more service providers and a consumers group and

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further appeals hearing and disposing against TRAI directions, decisions or orders. If TDSAT could not resolve this issue, Supreme Court's decision will be the final decision and mandatory for everyone.

***Composition of TDSAT***

- The Tribunal composed of a Chairperson and two members selected by the Central Government.
- The **Chairperson** should be or should have been a Supreme Court Judge or the High Court Chief Justice.

**TELECOM REGULATORY AUTHORITY OF INDIA (TRAI)**

Telecom Regulatory Authority of India (TRAI) was established on 20 February 1997 with the mission, to create and nurture conditions for the growth of telecommunications in India.

The main objectives of TRAI are to provide a fair and transparent environment in the market. TRAI regularly issues orders and directions on tariffs, interconnections, quality of services, Direct To Home (DTH) services and mobile number portability, etc.

From 1 January 2016, consumers have been compensated for call drops under TRAI Rule. In this case mobile users got compensation of ₹1 for every dropped call but it was limited to a maximum three dropped calls in a day.

TRAI is administered through a Secretariat headed by a secretary to organise the agenda for Authority meetings. (Consulting with the Chairman), prepares the minutes and issues regulations in accordance to the meetings.

**TRAI launched three new apps** (Mycall app, MySpeed app and 'Do not disturb' (DND 2.0) apps) and a **web portal** to ensure that the users are fully aware of the telecom services that are being offered to them, on 6 June 2017.

TRAI always tries to maintain transparency between what consumers are paying for and what telecom operators are promising to provide at a certain rate.

TRAI released their new app on 28 December 2018, called '"TRAI CHANNEL SELECTOR"' app. By using this app, we can add, remove and manage our channels.

In order to increase broadband penetration in India, TRAI has proposed WANI **(Wi-Fi Access Network Interface)** architecture which has been launched in August 2022 in Dehradun. According to the PM-WANI central registry, there are 198 Public Data Offices (PDO) aggregators, 111 app providers, and 199,930 public Wi-Fi hotspots, as of June 2024.

- A **member** should have held the Secretary post to the Government of India or any equivalent post in the Central Government or the State

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Government for a period of not less than two years or a person who is well versed in the field of technology, telecommunication, industry, commerce or administration.

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**PM-WANI**

- Public Wi-Fi plan 'PM Wani' approved in December 2020.
- The move is aimed at helping accelerate the uptake of broadband Internet services.
- It was first recommended by the Telecom Regulatory Authority of India (TRAI) involves in 2017.

**Key Features**

1. This has set up public Wi-Fi hotspots across the country *via* public data offices (PDOs).
2. It does not require the PDOs to get a license or pay a fee.
3. This will involve multiple players, including PDOs, Public Data Office Aggregators (PDOA), App Providers and a Central Registry.

**Significance of the Project**

Public Wi-Fi networks has 'democratised' content distribution and broadband access to millions at affordable rates. This is the UPI (unified payments interface) of connectivity services.

## 9.4 IT Rules, 2021

- It mandates **a grievance redressal system** for over the top (OTT) and digital portals in the country. This is necessary for the users of social media to raise their grievance against the misuse of social media.
- **Significant social media firms** have to appoint **a chief compliance officer** and have **a nodal contact person** who can be in touch with law enforcement agencies 24/7.
- **A Grievance Officer:** Social media platforms will also have to name **a grievance officer** who shall register the grievance within 24 hours and dispose it of in 15 days.

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- **Removal of Content:** If there are complaints against the dignity of users, particularly women – about exposed private parts of individuals or nudity or sexual act or impersonation etc. – social media platforms will be required to remove that within 24 hours after a complaint is made.
- **A monthly report:** They also will have to publish a monthly report about the number of complaints received and the status of redressal.
- There will be three **levels of regulation for news publishers** – self-regulation, a self-regulatory body, headed by a retired judge or an eminent person, and oversight from the Information and Broadcasting Ministry, including codes of practices and a grievance committee.

### *Significant Social Media Intermediary and its Benefits*

Social media companies with more than 50 lakh registered users will be considered 'significant social media intermediaries', as per the new norms.

### *What happens in case of noncompliance?*

- Social media giants such as Facebook, Twitter, Instagram and WhatsApp messenger **could face a ban** if they do not comply with the new Information Technology rules.
- They also run the risk of losing their status as '**intermediaries**' and may become liable for criminal action if they do not comply with the revised regulations.

## Recent Updates

### *2023 Draft Amendments*

- The draft Amendments propose to prohibit false/fake information and also regulate online gaming.
- Intermediaries must remove any information identified as false or fake by the fact check unit of the Press Information Bureau (PIB), or any other centrally authorised agency from their platforms, to avoid liability for such content.
- Online games are defined as games played on the internet with a deposit and the expectation of a winning. An online gaming intermediary has to ensure additional due diligence such as displaying a random number generator and a no-bot certificate.

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- Self-regulating bodies (SRBs) will also be empowered to regulate the content of registered games. SRBs must evolve a framework to test, verify and register games such that the sovereignty, integrity and security of the country is secured.

**UPDATION:** On 26 September 2024, the Bombay High Court's Division Bench delivered its final verdict striking down the **IT (Amendment) Rules, 2023, as unconstitutional**. The amendment allowed the Central Government to establish a Fact-Check Unit to label online content about government matters as fake or false

## 9.5 Telecommunications Rules, 2024 (Procedures and Safeguards for Lawful Interception of Messages)

The Union Government notified new rules under the **Telecommunications Act, 2023**, to govern lawful interception of messages (FOR UNFORSEEN CONDITION EVEN NOT DEFINED), replacing earlier rules under the **Indian Telegraph Act**.

**Competent Authority:** Joint Secretory level Officers – Union home secretory and State home secretaries

**Agency Authorisation:** Central Govt. under Section 20 (2) of the Telecommunication Act, 2023

***Legal Basis***

- **Enacted Under:** Section 56 of the **Telecommunications Act, 2023**
- **Supersedes:** Rule 419 and 419A of Indian Telegraph Rules, 1951
- Also governed by:
  - ❑ **IT Act, 2000:** Section 69 permits interception of digital transmissions
  - ❑ **IT Rules, 2009:** Provide safeguards and procedures for interception

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### *Key Provisions*

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| Provision | Details |
|---|---|
| **Authorised Agencies** | Can intercept messages upon approval from the **Competent Authority** |
| **Grounds** | Public emergency or public safety |
| **Devices Covered** | All telecom devices, including those using **electromagnetic systems** |
| **Retention** | Intercepted data can be retained **indefinitely** in some cases |
| **Consent Requirement** | Not applicable in case of interception orders |
| **Consent Manager Role** | Not included in interception context |

### *Judicial Precedent*

- **PUCL vs. Union of India (1996):** SC ruled that interception violates **Article 19(1)(a)** (freedom of speech)
  - ❑ Permissible only under reasonable restrictions defined in **Article 19(2)**

### *Concerns with the New Interception Rules*

| Issue | Explanation |
|---|---|
| **Broad Definition** | 'Telecommunication' includes almost all digital communication platforms |
| **Encrypted Platforms Targeted** | May extend to apps like WhatsApp, Signal, raising privacy alarms |
| **No Clear Definition** | Terms like 'public emergency' or 'public order' are **undefined** |
| **Executive Monopoly** | Both issuing and reviewing powers lie with executive, not judiciary |
| **Indefinite Retention** | Lacks time-limit on storage of intercepted messages |

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| | |
|---|---|
| **TSP Vulnerability** | No legal safeguards for **Telecom Service Providers** |

## 9.6 Draft Digital Personal Data Protection (DPDP) Rules 2025

The **Ministry of Electronics and IT** has issued the draft **DPDP Rules, 2025,** to implement the **DPDP Act, 2023**, aiming to balance **individual privacy with digital innovation** in India.

***Key Features of Draft DPDP Rules, 2025***

| Aspect | Details |
|---|---|
| **Purpose** | Operationalise the DPDP Act, 2023 to protect digital personal data and promote innovation |
| **Data Transfer** | Permits international transfer of personal data, subject to government approval |
| **Citizen Rights** | Users can demand **data erasure**, appoint **digital nominees** and access **easy consent systems** |
| **Data Fiduciaries** | Entities like social media, e-commerce, OTTs collecting personal data (e.g., Facebook, Amazon) |
| **Data Erasure Conditions** | Data must be deleted **after 3 years** of inactivity or per effective date; **48-hour prior notice** required |
| **Grievance Redressal** | Establishes **digital-by-design** Data Protection Board of India (DPBI) with civil court powers |
| **Graded Responsibilities** | Startups/MSMEs have lower compliance; **Significant Data Fiduciaries** (e.g. YouTube, Netflix) have higher compliance |

***Background***

- **2011:** Justice A.P. Shah Committee recommended a privacy framework
- **2017: Right to privacy** upheld as a **fundamental right** (Justice K.S. Puttaswamy case)

***Key Features of DPDP Act, 2023***

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| Feature | Explanation |
|---|---|
| **Right to Data Protection** | Control over personal data: access, correction and erasure rights |
| **Explicit Consent** | Mandatory clear consent for data processing |
| **Data Localisation** | Sensitive data must be stored/processed **within India** |
| **Regulatory Body** | Creation of **Data Protection Board of India (DPBI)** |
| **Data Breach Notification** | Mandatory alerts to both users and DPBI |
| **Penalties** | Heavy fines for non-compliance and breaches |

## 9.7 Department of Electronics and Information Technology

**Vision:** Developing India as the engine for transition into a developed nation and an empowered society.

**Mission:** e-Development of India through multi-pronged strategy of e-infrastructure creation to facilitate and promote e-governance, providing support for the Innovation/Research and Development (R&D), promotion of Electro Information Technology – Information Technology-Enabled Services (IT-ITeS) Industry, securing India's cyber space and building knowledge network.

***Objectives***

- **e-Government:** To provide e-infrastructure for delivery of e-services.
- **e-Industry:** To promote electronics manufacturing and IT-ITeS industry.
- **e-Innovation/Research and Development:** To provide support for creation of Innovation Infrastructure in the emerging areas of

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technology

- **e-Education:** To provide support for development of e-skills and knowledge network.
- **e-Security:** To secure India's cyber space.

## 9.8 Computer Emergency Response Team (CERT-IN)

It works under the Ministry of Electronics and Information Technology, deals with cyber security and serves as the national agency to perform certain functions.

**It intends to perform these duties:**

- Identify cyber incidents and issue information regarding it.
- Raise alerts for cyber security incidents.
- To take emergency measures during cyber security incidents.
- Coordinating cyber incident response activities.
- Issuing related guidelines, advisories, vulnerability notes and white papers.
- To report incidents on daily basis and alert computer system users about the vulnerability of the system. Recently, the Indian Computer Emergency Response Team (CERT-In) has issued a high severity warning for iPhone and iPad users due to vulnerabilities that could allow a hacker to make one's device stop working, access sensitive information, and get around security measures.
- CERT report new viruses and malwares time to time.

**APCERT:** APCERT (Asia Pacific Computer Emergency Response Team) is a coalition of CSIRTs (Computer Security Incident Response Teams), from 13 economies across the Asia Pacific region. APCERT organises an annual meeting called APSIRC conference, and the first conference was held in March 2002, Tokyo, Japan.

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It is an Indian trade organisation on Information Technology (IT) and Business Process Outsourcing (BPO) industry and was set-up in 1988, as a **non-profit organisation**. NASSCOM encourages software and services business and trade and the research enhancement in software technology. It is listed under the Indian Societies Act, 1860 and has regional offices in Bengaluru, Chennai, Hyderabad, Kochi, Kolkata, Mumbai, Pune and Thiruvananthapuram. The international IT trade body has more than 3200 members, of which more than 250 are companies from the China, EU, Japan, US and UK. NASSCOM's associate companies are in the dealing with software development, its services, products, IT-enabled/BPO services and e-commerce services.

## 9.9 Bharat Broadband Network/BharatNet/National Optical Fibre Network (NOFN)

The Government of India approved the National Optical Fibre Network (NOFN) initiative on **25 October 2011**. Under BharatNet Project there were provision to extend **optical fibre network to connect all 250,000 gram panchayats in the country covering nearly 625,000 villages.** On 25 February 2012 under Companies Act of 1956, a Special Purpose Vehicle (SPV) was incorporated to make this project successful as soon as possible. But still, between 2011 and 2014, project did not take off as planned, and only 350 km of optical fibre, out of 300,000 km optical fibre network needed for the Phase-I, was laid. **Between 2014 and 2017, the original Phase-I target of laying 300,000 km of optical fibre was completed.**

**The BharatNet project picked up pace under the Digital India initiative,** made several changes to expedite the project, significantly enhanced the BharatNet funding to several billion dollars under the Digital India, appointed government public sector units (BSNL, RailTel, and PowerGrid Corp) for the swift implementation and monitoring.

Note: **Idukki** district of Kerala became the first district in India to have high-speed Rural Broadband Network, i.e., National Optical Fibre Network (NOFN) Internet connectivity. In Idukki, 8 block offices and 52 village panchayats out of 53 have been connected on Optical Fibre and one Gram Panchayat, namely Edamalakudy, is connected through VSAT.

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## Recent Updates

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Union Cabinet has accorded approval for the revised implementation strategy of **BharatNet** through **Public-Private Partnership (PPP)** mode in 16 states of the country.

### *The strategy Includes*

1. The government will provide ₹19,041 crore as **viability gap funding** for the project.
2. BharatNet will now **extend up to all inhabited villages beyond the gram panchayats (GPs)** in the said states.
3. It includes creation, upgradation, operation, maintenance and utilisation of BharatNet by the concessionaire who will be selected by a competitive international bidding process.

### *Significance*

- The PPP model will leverage private sector efficiency for operation, maintenance, utilisation and revenue generation, and is expected to result in a faster rollout of BharatNet.
- Extension of reach of BharatNet to all inhabited villages with reliable, quality, high speed broadband will enable better access of e-services offered by various central and state government agencies.
- It will also enable online education, telemedicine, skill development, e-commerce and other applications of broadband.

### *The larger vision of the project is*

- To establish a highly scalable network infrastructure accessible on a non-discriminatory basis.
- To provide on demand, affordable broadband connectivity of 2 Mbps to 20 Mbps for all households and on demand capacity to all institutions.
- To realise the vision of Digital India, in partnership with States and the private sector.

### *Implementation*

- The project is **a Centre-State collaborative project**, with the States contributing free Rights of Way for establishing the Optical Fibre Network.

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- The entire project is being funded by **Universal service Obligation Fund** (**USOF**), which was set up for improving telecom services in rural and remote areas of the country.

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***Importance of BharatNet***

- The World Bank conducted a study, which stated that there is a 1.4% increase in GDP growth with every 10% increase in broadband penetration.
- The implementation of various e-governance initiatives such as e-health, e-banking and e-education, will be facilitated by NOFN, thereby enhancing inclusive growth.

***Education***

- Remote classrooms in rural government schools: Online vocational training courses delivered through rural ICT centres such as the Common Service Centres.

***Healthcare***

- Telemedicine centres for remote consultation and diagnostics, set up by private healthcare service providers or by rural entrepreneurs in collaboration with private healthcare service providers.

***Banking***

- Basic banking services to the unbanked population using, among others, the Post Office network.
- Doorstep banking services from various commercial banks through the Business Correspondent network maintained by a third-party private provider.

***Agriculture***

- Internet kiosks for providing agriculture-related information to farmers.
- Integrated application platform that pushes agricultural information over SMS and can also be accessed.

## 9.10 Centre for Development of Telematics (C-DoT)

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- The telecom technology development centre of GOI is named as C-DoT, which was established in 1984 as an autonomous body.
- Meant to provide a wide range of cost-effective indigenously developed and state-of-the-art total telecom solutions – satellite communications, IN, wireless broadband, GPON, NGN, ATM DWDM, NMS, and mobile cellular system.

***Telematics***

- Telematics, as an **interdisciplinary field, includes telecommunications, road safety, vehicular technologies, road transportation, electrical engineering (sensors, instrumentation, wireless communications, etc.) and computer science** (multimedia, Internet, etc.)
- This system is a suite of services powered by wireless communications, global positioning systems and on-board electronics. Telematics systems facilitate Internet and wireless cellular services that assist in services for location, navigation, traffic monitoring and control, toll collection, safety and traveller information.
- The technology enables sending, receiving and storing information *via* - telecommunication devices in conjunction with affecting control on remote objects.
- Global Navigation Satellite System (GNSS) technology integrated with computers and mobile communications technology in automotive navigation systems.
- The integrated use of telecommunications and informatics for application in vehicles.
- When used within road vehicles, it is also called vehicle telematics.

## 9.11 Telecom Public Sector Undertakings

Department of Telecommunication has four PSUs under its administrative control:

I. **Indian Telephone Industries Limited (ITI):** ITI Limited was incorporated as the first PSU in the country, by the DoT in 1950. Its headquarters is in Bengaluru, with manufacturing plants at Naini,

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Raebareli and Manka in UP, Palakkad in Kerala, and Srinagar in Jammu and Kashmir. ITI Limited was the sole equipment supplier to DoT and later to the corporatised entities, BSNL and MTNL, until the early nineties.

II. **Telecommunications Consultants India Limited (TCIL):** A wholly Government of India-owned enterprise (which was set up in March 1978).

Main objectives include:

- ❏ Globally enabling the world-class expertise and technology to the country, in all telecommunications and information technology fields.

III. **Mahanagar Telephone Nigam Limited:** It was first set up as a Navratna PSU in 1986. The primary objective is to expand quality telecom network and raise revenue for developing telecommunication facilities in India's key metros – Delhi and Mumbai. It acted as the forefront of technology induction by conversion of 100% of its telephone exchange network into the state-of-the-art digital mode. As of December 2023, the total number of wired MTNL subscribers in India was over 2.2 million users. Of these, 1.22 million users were in the Mumbai region while rest were in the Delhi region. There was a decrease in the number of users compared to the previous quarter of the corresponding year.

IV. **Bharat Sanchar Nigam Limited:** A comprehensive range of telecom services, viz., wire line, CDMA wireless, GSM wireless, Internet, Broadband, Carrier service, MPLS-VPN, VSAT, VoIP services, IN services, etc., are provided by BSNL, which was formed in 2000, throughout India except Delhi and Mumbai. The rural telephone system is one of the focus areas of BSNL and thus, it has remarkably provided Village Public Telephones (VPTs) in 5.82 lakh villages, as on 31 May 2013. It also introduced broadband services from January 2005. As of December 2023, Bharat Sanchar Nigam Limited had the highest number of wireline users in the state of Kerala with around 979

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thousand users. This was followed by the neighbouring state of Tamil Nadu with about 900 thousand users, which included Chennai and then by Maharashtra with around 550 thousand users. There were more than six million users across all the service areas.

## 9.12 Digital Convergence and Communication Commission of India: Issues

Digital convergence is convergence of the following four industries into one, i.e., ITTCE – Information Technologies, Telecommunication, Consumer Electronics and Entertainment. It is the ability to view the same multimedia content from different types of devices, thanks to the digitisation of contents (movies, pictures, music, voice, text) and the development of its connection methods.

***Use of Digital Convergence into Our Daily Life***

- **Reading emails on our TV via a connected smartphone.**
- **Watching a streaming movie on the home theatre connected to the Internet.**

***Objectives***

- To facilitate development of a national communication infrastructure, in order to provide a wide choice of services to consumers.
- To establish a regulatory framework that addresses the convergence of technologies, and defines the powers and roles of a signal regulatory and licensing licencing authority for broadcasting, telecommunications and multimedia.
- To establish a basis for codes and standards for broadcasting content.

## 9.13 e-Governance: Concept and Significance

Since 1990s, constant efforts are being made to improve the systems of public service delivery in order to reinvent governments worldwide. The

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recent innovations in ICT have facilitated the reinvention of governments and enabled them to serve the needs of a diverse society.

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- Moreover, the information age has redefined the fundamentals and transformed the institutions and mechanisms of service delivery forever. The vision is the articulation of a desire that can transform the functionality of the government and the way it relates to its constituents.
- Electronic governance, as a concept, is popularly called e-governance. The vision majority of democracies of the world share is e-governance, as a way of life.
- India has been at the forefront of the IT revolution, thereby making its contribution to the public administration systems, as we would see later in this unit. As a matter of fact, the potential of ICTs, if harnessed properly, has a lot of opportunities, especially for social and economic growth of the developing world.
- As per Ministry of Information and Technology, the role of e-governance goes far beyond mere computerisation of stand-alone back-office operation. It leads to fundamental changes in government operations; and new set of responsibilities for the legislature, executive, judiciary and citizens.
- It is providing public access to information *via* the Internet by government departments and their agencies.

### 9.13.1 Objectives of e-Governance

- Better service delivery to citizens
- Ushering in transparency and accountability
- Empowering people through information
- Improve efficiency within Government, i.e. between centre-state or inter-states
- Improve interface with business and industry

### 9.13.2 Pillars of e-Governance

- People
- Process
- Technology

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- Resources

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### 9.13.3 Types of Interaction in e-Governance

- G2G, i.e. Government to Government
- G2C, i.e. Government to Citizen
- G2B, i.e. Government to Business
- G2E, i.e. Government to Employees

### 9.13.4 Benefits/Outcomes of e-Governance

- Enhanced transparency and accountability
- Expanded reach of Governance
- Improved public administration
- Enables environment for promoting economic development
- Improved service delivery in the form of better access to information and quality services to citizens

### 9.13.5 Many e-Governance Programmes

- **Bhoomi:** Bhoomi is a project jointly funded by the **Government of India** and the **Government of Karnataka** to digitise the paper land records and create a software mechanism to control changes to the land registry in Karnataka.
- **Khajane:** **Its G2G-based e-Governance Platform launched in Karnataka.** It has been implemented mainly to eliminate systemic deficiencies in the manual treasury system and for the efficient management of state finances.
- **e-District:** The Mission Mode Project aims at delivery of high volume, citizen-centric services at the district level such as the issue of birth/death certificate, income and caste certificates, old age and widow pension, etc. It was launched by the **Department of Information Technology**.
- **MCA:** The project aims to provide electronic services to the companies registered under the Companies Act. Various online facilities offered include allocation and change of name, incorporation, online payment of registration charges, change in address of registered office, viewing

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of public records and other related services. The **Ministry of Corporate Affairs** launched it.

- **e-Seva:** All the services are delivered online to consumers/citizens by connecting them to the respective **government departments in Andhra Pradesh** and providing online information at the point of service delivery. The project has become very popular among the citizens especially for the payment of utility bills.
- **e-Courts:** It was launched by the **Department of Justice, Ministry of Law and Justice**. The Mission Mode Project (MMP) aims at utilising technology for improved provisioning of judicial services to citizens.
- **e-Office:** The Mission Mode Project aims at significantly improving the operational efficiency of the Government by transitioning to a "Less Paper work". **The Department of Administrative Reforms & Public Grievances** launched it.

Note: Now e-Governance is considered under the 9 Pillars of Digital India.

## 9.14 National e-Governance Plan (NeGP)

- The Department of Electronics and Information Technology (DeitY) and Department of Administrative Reforms and Public Grievances (DARPG) formulated the National e-Governance Plan (NeGP).
- The NeGP, comprising 27 Mission Mode Projects (MMPs) and 10 components, was approved by Union Government on 18 May 2006.
- **Mission Mode Projects:** A Mission Mode Project (MMP) is an individual project within the National e-Governance Plan (NeGP 2.0) that focuses on one aspect of electronic governance, such as banking, land records or commercial taxes, etc. Within NeGP 2.0, "mission mode" implies that projects have clearly defined objectives, scopes, and implementation timelines and milestones, as well as measurable outcomes and service levels. **NeGP 2.0 comprises of 44 MMPs**, which are further classified as state, central or integrated projects.
- It aims at improving delivery of government services to citizens and businesses with the vision: to ensure that all government services are

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easily available to everyone in their local area through common service centres, and to guarantee that these services are transparent, dependable, and efficient, all at affordable prices, to meet the essential needs of ordinary people.

In order to promote e-Governance in a holistic manner, various policy initiatives and projects have been undertaken to develop core and support infrastructure.

The major core infrastructure components are **State Data Centres (SDCs), State Wide Area Networks (SWAN), Common Services Centres (CSCs) and middleware gateways, i.e. National e-Governance Service Delivery Gateway (NSDG), State e-Governance Service Delivery Gateway (SSDG), and Mobile e-Governance Service Delivery Gateway (MSDG).**

The important support components include Core policies and guidelines on Security, HR, Citizen Engagement, Social Media as well as Standards related to Metadata, Interoperability, Enterprise Architecture, Information Security, etc. New initiatives include a framework for authentication, *viz*. e-Pramaan and G-I cloud, an initiative which will ensure benefits of cloud computing for e-Governance projects.

**STATE DATA CENTRES (SDCS)**

It is proposed to establish data centres in all the states/union territories to host state-level e-Governance applications/data *via* common and secure IT infrastructure. This will enable seamless delivery of Government-to-Government (G2G), Government-to-Citizen (G2C) and Government-to-Business (G2B) services, duly supported by State Wide Area Network (SWAN) and Common Service Centres (CSC), which are established at the village level.

**STATE-WIDE AREA NETWORKS (SWANS)**

SWANs act as a converged backbone network for data, voice and video communications throughout a state/union territory. It is expected to handle the information communication requirements of all the departments.

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**COMMON SERVICES CENTRES (CSCS)**

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The CSCs are ICT-enabled kiosks with broadband connectivity to enable various government, private and social services at the doorstep of citizens. The Government aims to create 1,00,000 CSCs across 600,000 rural and remote locations of India.

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### 9.14.1 Meghraj (GI Cloud)

DeitY has initiated a project called 'Meghraj', with a view to harness the benefits of cloud technology. Another name for Meghraj is the GI Cloud Initiative (GI – Government of India). This project aims at creating the Government of India Cloud (GI Cloud) computing environment at the national level.

This initiative primarily focuses on evolving a strategy and implementing various components including governance mechanism, thereby ensuring proliferation of cloud in government. Meghraj's role will be a common receptacle of cloud-based infrastructure resources and applications available on demand. It will enable our government for the effective delivery of e-services.

**CLOUD COMPUTING**

- Cloud computing is a tool for enabling ubiquitous, convenient, on-demand network access to a various shared configurable computing resources (e.g., networks, services, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.
- Cloud computing is defined as the storing and accessing of data and programmes over the Internet, instead of a computer's hard drive. On top of this, cloud in the term of Cloud Computing refers to a set of hardware, storage, services, networks, and interfaces combining to deliver aspects of computing as a service.
- So, over and above sharing information, even the infrastructure can be shared on real-time basis on the Internet.

#### 9.14.1.1 Components of Government's Cloud Computing

- Five essential characteristics of cloud computing – on demand self-service, ubiquitous network access, metered use, elasticity and resource pooling.

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- Three service models including infrastructure as a service, software as a service and platform as a service and four deployment models – public cloud, hybrid cloud, private cloud and community cloud.
- The initiative accelerates delivery of e-services provided by the government, thereby optimising ICT spending of the government.

#### 9.14.1.2 Advantages of GI Cloud

- Optimum utilisation of existing infrastructure.
- **Rapid Deployment and Reusability:** A software availed by any department in government of India can be made available to other departments with no additional costs.
- **Manageability and Maintainability:** A single point for maintenance of ICT infrastructure in India is provided by it.
- **Scalability:** Infrastructure of the government can be increased as a response to the demands from the citizens of India.
- Efficient service delivery.
- **Security:** A security framework for the entire GI Cloud will lead to less environmental complexity and less potential vulnerability.
- Increased user mobility.
- Reduced effort in managing technology.
- Ease of first-time IT solution deployment.
- Cost reduction.
- **Standardisation:** GI Cloud shall prescribe the standards around interoperability, integration, security, data security and portability, etc.

#### 9.14.1.3 Uses of Meghraj

- Not only through Internet but also *via* mobile connectivity, the GI Cloud will provide services to government departments, citizens and business.
- Besides accelerating the delivery of e-services to citizens and businesses, the government's cloud-based service delivery platform will also support other objectives such as increased standardisation, interoperability and integration, etc.

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#### 9.14.1.4 Main Issues of the GI Cloud Initiative

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- Lack of common policies among states and centres will challenge application reuse.
- Individual technology stacks and a lack of infrastructure standardisation will limit success.
- Lack of a clear mandate or incentives will affect G-Cloud uptake.

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### 9.14.2 e-Taal: Electronic Transactions Aggregation and Analysis Layer

e-Taal is a web portal developed by NIC (National Informatics Centre) for aggregation and analysis of e-transaction statistics of central and state level e-governance projects, which also includes MMPs. The transaction statistics is received from web-based applications periodically on near real-time basis and presents status on actual utilisation of various applications running at various locations.

It also assists with quick analysis of transaction counts in tabular form as well as graphical form to give quick view of the category and number of transactions done through e-governance projects. e-Taal provides visibility for the national/state level e-governance services.

### 9.14.3 UMANG (Unified Mobile Application)

UMANG is the acronym for Unified Mobile Application for New-age Governance, which is also popular with the name of mobile governance in India. After BHIM app, PM Modi launched UMANG app for government services.

**DND 2.0 AND DND 3.0**

**TRAI has now incorporated its Mobile App named as DND 2.0. DND (Do Not Disturb) Services App** facilitates smart phone users for registering their mobile number under DND and reporting spam messages or calls for avoiding Unsolicited Commercial Communication (UCC)/ Telemarketing Calls/SMS.

**TRAI DND 3.0 (Do Not Disturb)**

The new release (built ground up) has the following additional features:

- An intelligent spam detection engine (for SMS only) to assist the subscriber in reporting

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- Crowdsourcing of data about offending messages and calls to speed up detection of unregistered telemarketers
- Updates about action taken on complaints within the App
- Easier interface and set up.

TRAI DND App is now also available on UMANG App (Government of India Initiative).

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This app is developed by National e-Governance Division (NeGD) and Ministry of Electronics and Information Technology (MeitY) and was launched in November 2017. It encourages people to opt for digital payment transactions in the country. UMANG app is a single-window platform with multiple applications of pan India e-government services like income-tax filing, making Aadhaar, booking a gas cylinder, Passport Seva and provident fund queries, etc.

Here, people can access more than 100 services at the level of regional, state and central governments. Umang is a part of Digital India initiative to adopt a new-age governance policy. This application is also allowed through the web, SMS, or IVR (Interactive voice response).

## 9.15 Bharat Interface for Money (BHIM) App

***Bharat Interface for Money (BHIM)*** is a UPI-based payment interface, developed by ***National Payments Corporation of India (NPCI). It allows real time fund transfer. BHIM app was launched in December, 2016.***

The BHIM apps has three levels of authentication:

1. For one, the app binds with a device's ID and mobile number.
2. Second a user needs to sync whichever bank account (UPI or non-UPI enabled) in order to the conduct transaction.
3. Third, when a user sets up the app they are asked to create a pin which is needed to log into the app. The UPI pin, which a user creates with their bank account is needed to go through with the transaction.

### 9.15.1 What is UPI?

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***Unified Payments Interface (UPI)*** is a system that powers multiple bank accounts into a single mobile application (of any participating bank), merging several banking features, seamless fund routing & and merchant payments into one hood. It also caters to the ***'"Peer to Peer'"*** collect request which can be scheduled and paid as per requirement and convenience. Each Bank provides ***its own UPI App for Android, Windows and iOS mobile platform(s).***

***Achievements of UPI***

- Digital transactions through UPI **recorded phenomenal growth during the pandemic year 2020–21** and **several countries have evinced interest** to learn from Indian experience so that they could replicate the model.
- The value of transactions made using the UPI **crossed USD100 billion in a month for the first time in October, 2021** according to data from the NPCI, further cementing its position as India's most popular digital payments system.
- **India's digital payments industry** is likely to grow from ₹2,153 trillion **at 27% Compounded Annual Growth Rate (CAGR)** to ₹7,092 trillion by 2025.

**BHIM 2.0**

Government launched ***BHIM 2.0*** with new functionalities, additional language support.

**What is BHIM?**

- ***Bharat Interface for Money (BHIM) is a UPI-based payment interface***.
- Developed by ***National Payments Corporation of India (NPCI)***. Allows real-time fund transfer. ***Launched*** in December, 2016.

**What's new in BHIM 2.0?**

- Donation' gateway, increased transaction limits for high value transactions, linking multiple bank accounts, offers from merchants, option of applying in IPO, gifting money.
- It also supports three additional languages – Konkani, Bhojpuri and Haryanvi – over and above the existing 13.

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**BHIM UPI GOES INTERNATIONAL**

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- ***BHIM app*** has gone international with a pilot demo of ***BHIM UPI QR-based payments*** beginning at ***Singapore FinTech Festival 2019.***
- This QR code-based system would ***allow anyone with a BHIM app to scan the SGQR at NETS terminals for payments in Singapore.***
- The project is being ***jointly developed by NPCI and NETS of Singapore.*** It was targeted to go live by Feb 2020, but got delayed.

**After Singapore, Bhutan adopts India's BHIM-UPI**

- Bhutan has become **the first country to adopt India's Unified Payment Interface (UPI) standards for its quick response (QR) code. It is also the second country after Singapore to have BHIM-UPI acceptance at merchant locations.**
- Bhutan will also become **the only country to both issue and accept RuPay cards as well as accept BHIM-UPI.**

***Why UPI is outperforming e-wallets?***

- People are changing the way they transact, choosing bank-to-bank methods such as the Unified Payments Interface (UPI) over other instruments such as e-wallets. ***This is because UPI is completely interoperable and as such, it is unique in the world, where you have an interoperable system on the 'send' and 'receive' side.***
- The rapid growth of UPI is accompanied by a reasonably strong growth in the value of transactions done using e-wallets, but the latter's growth has not taken off much following the fillip it received in the aftermath of demonetisation in November 2016.

## 9.15.2 About NPCI

- NPCI is ***an umbrella organisation for operating retail payments and settlement systems in India***.
- It is ***an initiative of Reserve Bank of India (RBI) and Indian Banks' Association (IBA) under the provisions of the Payment and Settlement Systems Act, 2007***, for creating a robust Payment and Settlement Infrastructure in India.
- It has been ***incorporated as a not for profit company.***
- In 2016 ***the shareholding was broad-based to 56 member banks to include more banks representing all sectors.***

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## 9.16 PRAGATI (Pro-Active Governance and Timely Implementation)

It is a multi-purpose and multi-modal governance platform launched on 25 March 2015. An unique integrating and interactive platform, PRAGATI is aimed to address common man's grievances, and at the same time, monitor and review important programmes and projects of the GOI and projects flagged by state governments.

It is an innovative project in e-governance and good governance and accountability with real-time presence and information exchange among the key stakeholders.

Key features of the PRAGATI application are:

1. The PMO, Union Government Secretaries, and State Chief Secretaries constitute PRAGATI application and hence it is a three-tier system.
2. Designed in-house by the Prime Minister's Office (PMO) team with the help of National Informatics Centre (NIC).
3. Three latest technologies, including digital data management, geo-spatial technology and video-conferencing, are used to connect under PRAGATI.
4. To interact with the Government of India Secretaries and Chief Secretaries, the Prime Minister will hold a monthly programme once in every month on the fourth Wednesday at 3:30 PM. This programme (to be known as PRAGATI Day) is held through video-conferencing, enabled by data and geo-informatics visuals.
5. Prime Minister can directly discuss the issues with the concerned Central and State officials with full information and latest visuals of the ground-level situation.
6. The system is working on strengthening and re-engineering the databases of Centralised Public Grievance Redress And Monitoring System (CPGRAMS), along with Project Monitoring Group (PMG) and the Ministry of Statistics and Programme Implementation.

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## 9.17 Aadhaar

A revolutionary step in terms of digitalisation of proof of identity came in the form of Aadhaar. It is a 12-digit individual identification number issued by the Unique Identification Authority of India (UIDAI), on behalf of the Government of India. This 12-digit number serves as a proof of identity and address, anywhere in India. Aadhaar letter can be received *via* India Post, in form of e-Aadhaar downloaded from UIDAI website, both being equally valid.

Irrespective of age or gender, one can enrol for Aadhaar, being a resident of India and satisfying the verification process laid down by UIDAI. Single enrolment is required and that is free of cost. Each Aadhaar number is unique to an individual with lifelong validity. In due course of time, it is expected that *via* Aadhaar number one can access services like banking, mobile phone connections and other government and non-government services.

### *Basic Properties of Aadhaar*

1. Easily verifiable in an online, cost-effective way.
2. Unique and robust enough, thus eliminating the large number of duplicate and fake identities in government and private databases.
3. A random number generated, devoid of any classification, be it on basis of caste, creed, religion or geography.

Note: The UIDAI has brought in secure, digitally-signed QR Code on e-Aadhaar that will now contain photograph of the Aadhaar holder too, in addition to demographic details, to facilitate better offline verification of an individual. The UIDAI (Unique Identification Authority of India) has recently replaced existing QR code on e-Aadhaar having resident's demographic details now with a secured digitally-signed QR Code, which contains demographics along with photograph of the Aadhaar holder.

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## 9.18 Digital Locker

The Government of India launched DigiLocker, which is a digital locker service, in February 2015 to provide a secure dedicated personal electronic space for storing documents like PAN card, passport, marksheets, degrees and certificates. It is linked to Aadhaar number.

This locker maybe linked to a person's Aadhaar number. 10 megabytes of space is offered, which is further extended to 1-gigabyte space for all the documents, free of cost. It acts like a physical locker where the jewellery and documents can be stored, except the fact that it is digital, i.e., on website and will store digital information, in place of physical assets.

On 7 September 2016, Union Road Transport Minister, Nitin Gadkari initiated the integration of DigiLocker with the Driving License and Vehicle Registration system of the Ministry of Road Transport and Highways.

The Central Board of Secondary Education (CBSE) has issued mark sheets to class XII students in digital format along with hard copies for the students in their digital lockers from 2015.

The Ministry of Electronics & Information Technology (MeitY), Government of India, has appointed the **Digital Locker Authority (DLA)** to manage and administer the Digital Locker system to preserve and retain information within the system, ultimately facilitating efficient service delivery to users.

Digital locker minimises the usage of physical documents, thereby enabling sharing of e-documents across agencies. It will reduce the administrative overhead of government departments and agencies created due to paper work. It promotes the concept of paperless governance.

The Government of India launched this service to provide a secure dedicated personal electronic space for storing the documents of resident Indian citizens. The useful websites to access these benefits are: *https://digilocker.gov.in/ and https://digitallocker.gov.in/*

Other objectives of digital locker include:

1. Digital empowerment of residents by providing them with Digital locker on the cloud.

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2. Enable e-signing of documents and making them available electronically and online. Minimise the use of physical documents.
3. Ensure authenticity of the e-documents and thereby eliminating usage of fake documents.
4. Secure access to government-issued documents through a web portal and mobile application for residents.
5. Reduce administrative overhead of government departments and agencies and making it easy for the residents to receive services.
6. Anytime, anywhere access to the documents by the resident.
7. To support a well-structured standard document format via open and interoperable standards-based architecture that can support easy sharing of documents across departments and agencies.
8. Ensure privacy and authorised access to resident's data.

## 9.19 JAM Trinity

JAM stands for three things – **The Jan Dhan Yojana,** the **Aadhaar initiative of UIDAI** and **Mobile number.** These three things are often called the Trinity of reforms in India. The JAM Trinity holds the key to implement large-scale, technology-enabled, real-time direct benefit transfer (DBTS) to improve the economic condition of the India's poor. The first application of JAM was PAHAL scheme of transferring LPG subsidies directly to the account of BPL people. Though riddled with challenges of beneficiary identification, distributor-opposition and beneficiary financial inclusion, JAM trinity is expected to reduce leakages and provide more financial space to the government.

## 9.20 One Nation One FASTag

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***One Nation One FASTag scheme*** inaugurated by GoI mandated ***FASTags for all vehicles at all National Highways from 1 December 2019.***

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Finally, Union Road Transport and Highways Ministry made all lanes in the fee plazas on National Highways as 'FASTag lane of the fee plaza' from midnight of 15th/16th February 2021.

Any vehicle not fitted with FASTag will have to pay twice the fee for the category of the vehicle they are driving at electronic toll plazas across India. ANPR-Automatic Number Plate Reader is expected to replace Toll Plaza and FASTag on Highway.

### *Features of the Scheme – One Nation One FASTag*

- The scheme will be implemented from December 1, 2019, and can be availed upon activation by new cars having Radio Frequency Identification (RFID) tags on national and state highways throughout the country.

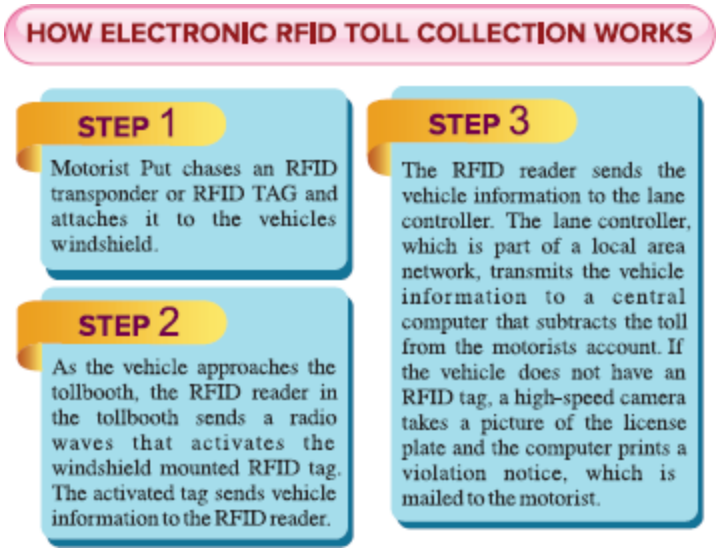


Figure 9.1 Working mechanism of FASTag

- The plan aims to integrate the collection of toll digitally and ensure seamless mobility of vehicles across India.
- The payment method is a part of ***the National Electronic Toll Collection (NETC) programme***. The ***National Payments Corporation of India (NPCI)*** collects the payments.

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***What is 'FASTag'? How they operate?***

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- Stickers affixed to the windscreen of vehicles.
- Use RFID technology to enable digital, contactless payment of tolls without having to stop at toll gates.
- The tags are linked to bank accounts and other payment methods.
- Sensors are placed on toll barriers, and the barriers open for vehicles having valid FASTags.
- A FASTag is valid for five years and needs to be recharged only as per requirement.

***Why do we need this scheme?***

- According to the National Highways Authority of India (NHAI), these devices will make passing through tolls considerably smoother since drivers will no longer have to carry cash or stop to make a transaction.
- Cameras at toll booths will take photos of passengers in a vehicle, which will be useful for the Ministry of Home Affairs as there will be a record of a vehicle's movement.

## 9.21 e-Learning (Education and Internet Technology)

The Internet and its World Wide Web have captured the imagination and interests of millions around the world, leading to use of terms like web-based learning, online learning, etc. The interests in the educational use of the web has been driven by higher demands for education and training, shift in the societal economy from labour-intensive to knowledge-intensive work, and need for earning while learning. As the terminology in e-learning needs further standardisation, many terms were used inter-changeably.

1. **Web-based Instruction:** It is a hypermedia-based instructional programme which utilised the attributes and resources of the World Wide Web to create a meaningful learning environment where learning is fostered and supported.
2. **Virtual Learning:** The educational process of learning over the Internet without having face-to-face contact is known as virtual

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learning. Virtual learning may also include tele-learning.

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3. **Online Learning:** It is synonymous with web-based learning, where learning is fostered *via* hypertext transfer protocol (http) on Internet or Intranet.
4. **e-Learning:** The term e-learning covers a wide set of applications and processes, including computer-based learning, web-based learning, virtual classrooms and digital collaboration.

The term e-learning is substituted for web-based learning or online learning, falling in line with e-commerce and e-business.

### 9.21.1 Unnat Bharat Abhiyan

The Ministry of Human Resource Development (MHRD) has launched Unnat Bharat Abhiyan with the objective of connecting institutions of higher education, including Indian Institutes of Technology (IITs), National Institutes of Technology (NITs) and Indian Institutes of Science Education and Research (IISERs), etc., with local communities to address the development challenges through appropriate technologies. Unnat Bharat Abhiyan was launched in 2014.

There are two main objectives of Unnat Bharat Abhiyan:

1. Building institutional capacity in institutes of higher education in research and training relevant to the needs of rural India.
2. Providing rural India with professional resource support from institutes of higher education, especially those that have acquired academic excellence in the field of science, engineering and technology and management.

Under this programme, 132 villages have been identified for intervention by the following 16 institutes of higher education: 14 IITs (Mumbai, Bhubaneshwar, Delhi, Guwahati, Hyderabad, Indore, Jodhpur, Kanpur, Kharagpur, Madras, Mandi, Patna, Roorkee, Ropar), IISER Bhopal and MNIT Jaipur.

### 9.21.2 e-Pathshala

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e-Pathsala is a web portal hosting educational resources for students, teachers, parents, researchers and educators. As a part of the Digital India campaign, the MHRD (GoI) and NCERT (National Council of Educational Research and Training) jointly launched 'e-pathshala' in November 2015, as a single point repository of e-resources containing NCERT text books and various other learning resources.

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It can be accessed through mobile app interface on Android, IOS and windows platforms. These resources are available in the form of text books and other e-books in various languages like Hindi, English and Urdu.

Students can get access to all educational material, including text books, audios, videos, periodicals and a variety of other print and non-print materials through e-Pathshala. The user can download these materials for offline use with no limits on downloads.

### 9.21.3 SARANSH

Saransh is a **web portal** launched with the objective of promoting information and communication technologies in schools. It serves as an interface to enhance communication between schools and parents as a self-review tool. This web portal was launched by the Human Resource and Development Minister for Central Board of Secondary Education (CBSE)-affiliated schools, in November 2015, to promote Communication and Information technology. **This portal will help to evaluate the strengths and weaknesses of children, which will help with the children's future.**

It is a progress of Digital India campaign and will bring transparency to the educational system. It intends to provide e-books of all standards and subjects, thereby encouraging sustainable development. It is a kind of one-on-one interaction platform between teachers and parents. It directly shares information about performance and attendance of students to parents. This portal will save three preceding years' data of all schools. Hence, a comparative year-on-year performance can be checked.

### 9.21.4 Indian MOOCs: "SWAYAM"

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SWAYAM or **Study Webs of Active-Learning for Young Aspiring Minds** programme is brought forward by the MHRD, Government of India in 2016, as a Massive Open Online Courses (MOOCs). Under this programme, various eminent Professors of IITs, IIMs and Central University, etc., will offer online educational courses.

It is an instrument for self-actualisation of learning offered free of cost. Here, learners can choose from hundreds of courses that are taught at the university/college/school level and these shall be offered by the best of the teachers in India and elsewhere. In this way, SWAYAM provides a unique educational opportunity to expand the horizons of knowledge at everyone's doorstep.

In the beginning, various IITs, IIMs, Delhi University, Jawaharlal Nehru University, IGNOU, individually as well as with the help of faculty from foreign universities will be offering courses in areas of engineering education, social science, energy, management, and basic sciences.

Recently in February 2024, Union Minister of Education and Skill Development and Entrepreneurship launched the 'SWAYAM Plus' platform to offer courses developed collaboratively with the industry. These employability and professional development-focused programmes have been developed with industry players and will be operated by IIT-Madras.

### 9.21.5 SANGAM: Digital Twins Initiative

#### What Is a Digital Twin?

A **virtual model** that mirrors a real-world object/system, tracking it **throughout its lifecycle** via **real-time sensor data**.

**Sangam: Digital Twin Initiative**

**Launched by:** Department of Telecommunications (DoT)

**First Announced:** 16 February 2024

**Last Updated:** 22 March 2024

***Key Features***

- **Two-way data flow** (unlike traditional simulations)
- Real-time **monitoring, analysis and prediction**
- Supports **decision-making** in dynamic conditions

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***Objective***

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To **revolutionise infrastructure planning and design** in India using:

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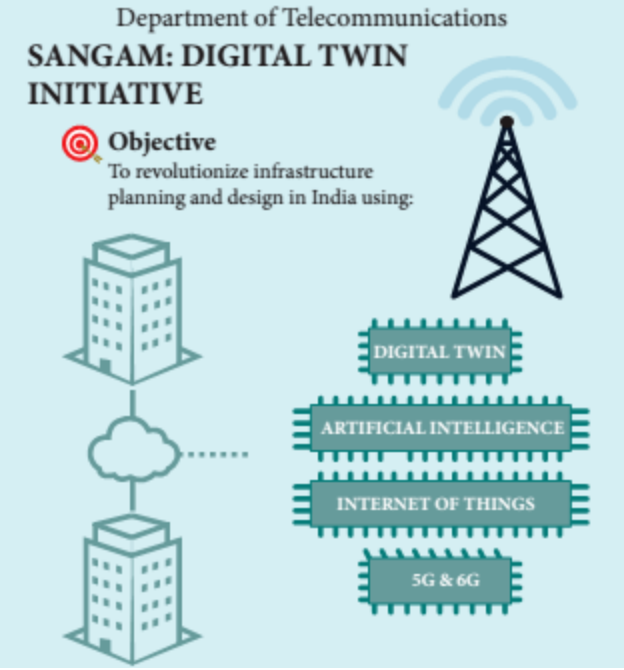
- **Digital Twin Technology**
- **Artificial Intelligence (AI)**
- **Internet of Things (IoT)**
- **5G & 6G**
- **Next-gen computational tools**

***Implementation Phases***

1. **Exploration Stage**
2. **Demonstration Stage**

   *To be piloted in one of India's major cities.*




***Applications and Benefits***

- **Urban Planning:** Simulate growth scenarios, traffic, housing, and so on
- **Hazardous Operations:** Remote monitoring of mines, reactors, etc.
- **Policy Making:** Enhanced forecasting tools for infrastructure and environment
- **Industry 4.0:** Use in **manufacturing, power generation, healthcare and automotive sectors**

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***Real-Life Example***

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***Digital Twin of Qutub Shahi Tombs, Hyderabad***

- Created using **drone scanners** by a private firm
- **Seven tombs** of Golconda kings, combining **Persian, Pathan and Hindu** architecture
- Built in **grey granite** with **stucco ornamentation**
- **Only site globally** where an entire dynasty is buried in one place

## 9.22 Miscellaneous

### 9.22.1 CoWIN (Covid Vaccine Intelligence Network)

- It is an Indian government web portal for COVID-19 vaccination registration.
- It is owned and operated by India's Ministry of Health and Family Welfare.
- It displays booking slots of COVID-19 vaccine available in the nearby areas and can be booked on the website.

### 9.22.2 MANAS App

- MANAS App is a well-being App that stands for **Mental Health and Normalcy Augmentation System.**
- It is endorsed as a national programme by the Prime Minister's Science, Technology, and Innovation Advisory Council (PM-STIAC).
- MANAS is a comprehensive, scalable, and national digital wellbeing platform and an app developed to augment the mental well-being of Indian citizens.
- MANAS App integrates the health and wellness efforts of various government ministries, scientifically validated indigenous tools with gamified interfaces developed/researched by various national bodies and research institutions.

### 9.22.3 Aarogya Setu

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- It is an Indian COVID–19 " contact tracing, syndromic mapping and self-assessment" digital service.
- It is primarily a mobile app, developed by the National Informatics Centre under the Ministry of Electronics and Information Technology (MeitY).

### 9.22.4 DIKSHA (Digital Infrastructure for Knowledge Sharing)

- It is a national platform for school education, an initiative of National Council for Education Research and Training (NCERT), Ministry of Education.
- DIKSHA was developed based on the core principles of open architecture, open access, open licensing diversity, choice and autonomy as outlined in the Strategy and Approach Paper for the National Teacher Platform released in May, 2017. DIKSHA itself was launched by the Hon' Vice President of India on Sept 5$^{th}$, 2017 and has since been adopted by 35 states/UTs across as well as CBSE and NCERT and by crores of learners and teachers.

### 9.22.5 Digi Yatra App

- The Airports Authority of India launched the Digi-Yatri service that will provide contactless entry to passengers at airports by using facial recognition technology.
- Digi Yatra – **Digital processing of passengers at the airports.** Passengers will be automatically processed based on facial recognition system at check points like, **Entry point check, Entry into Security Check, Aircraft Boarding.** Additionally, this will also facilitate self-Bag Drop and Check-in, using **facial recognition** to identify pax and data recall. Digi Yatra will facilitate paperless travel and avoid identity check at multiple points.
- Passenger can create Digi Yatra ID in a central system by providing the following: 1. Name, 2. Email ID, 3. Mobile Number, 4. Details of Identity (Voter Id, Driving Licence, Aadhaar etc.). On submission the Digi Yatra Id will be created. Pax can quote this number while booking the ticket. Pax data including Digi Yatra ID will be passed on to departure airport by the airlines.

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### 9.22.6 Sanchar Saathi Initiative

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The **Ministry of Communications** has introduced an innovative AI-based facial recognition tool called **Artificial Intelligence and Facial Recognition powered Solution for Telecom SIM Subscriber Verification (ASTR)** as part of the **Sanchar Saathi Initiative (A portal launched on May 2023)**.

#### About ASTR

- ASTR, launched under the Sanchar Saathi Initiative, is an AI-powered tool designed to detect SIM cards issued using fraudulent or forged documents.

#### About Sanchar Saathi Initiative

- The initiative **aims to empower citizens by enabling them to identify mobile connections** issued in their name and **report fraudulent activities**.
- **Developed by the Department of Telecommunications**, it comprises several modules, including **CEIR** (Central Equipment Identity Register) – for blocking or lost mobiles, **Know your mobile connections** – to know mobile phone registered in your name **and ASTR** – to identify fraudulent subscribers.

#### Technological Aspects

- ASTR utilises the **Param-Sidhhi Supercomputer** for processing large volumes of data.
- It employs **Convolutional Neural Network (CNN) models** for **effective facial recognition** within 10 seconds.

#### Facial Recognition Technology (FRT):

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- FRT is a **method of confirming an individual's identity** by analysing their facial features.
- It can be deployed in various scenarios, including photo and video identification, as well as real-time surveillance.
- Computer algorithms extract unique facial landmarks such as **cheekbone shape and lip contours** to create a numerical representation termed as faceprint.
- FRT relies heavily on AI techniques for processing and comparison, where the system matches the generated faceprint with the existing databases for verification or identification purposes.

### 9.22.7 DIGIPIN web Application

The DIGIPIN (Digital Postal Index Number) is an open-source, interoperable, geo-coded, grid-based digital address system developed by the Department of Posts in collaboration with IIT Hyderabad and NRSC, ISRO. It is a cornerstone of the Department of Posts' vision to offer Address-as-a-Service (AaaS) — an array of services associated with address data management to support secure and efficient interactions between users, government entities, and private sector organisations.

The 'Know Your DIGIPIN' portal enables users to:

- Retrieve their DIGIPIN based on precise geolocation.
- Input latitude and longitude coordinates to fetch corresponding DIGIPINs and vice versa.
- This initiative simplifies location mapping, enhances logistics and emergency response, and ensures last-mile delivery, especially in rural and underserved areas. The DIGIPIN system, officially reviewed and

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adopted by the Thematic Working Group on Address under the National Geospatial Policy 2022, is now available to all Ministries, State Governments, institutions for integration into their workflows, and users for their own use.

Incorporating DIGIPIN as an additional address attribute enables the leveraging of GIS capabilities, laying the foundation for future GIS-based digitalisation of service delivery across various organisations in a cost-effective manner. DIGIPIN will enhance location accuracy across multiple sectors by providing precise geographic coordinates, ensuring accurate service delivery and reducing emergency response times. A complete repository of the DIGIPIN technical documentation and source code has been made available on GitHub, promoting open collaboration and nationwide adoption.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. Consider the following:[UPSC (Pre) – 2022]**

1. Aarogya Setu
2. COWIN
3. Digi-Locker
4. DIKSHA

Which of the above are built on top of open-source digital platforms?

(a) 1 and 2 only

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(b) 2, 3 and 4 only
(c) 1, 3 and 4 only
(d) 1, 2, 3 and 4

**2. With reference to digital payments, consider the following statements:[UPSC (Pre) – 2018]**

1. BHIM app allows the user to transfer money to anyone with a UPI-enabled bank account.
2. While a chip-pin debit card has four factors authentication, BHIM app has only two factors of authentication.

Which of the statements given above is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**3. The identity platform 'Aadhaar' provides open "Application Programming Interfaces (APIs)". What does it imply?[UPSC (Pre) – 2018]**

1. It can be integrated into any electronic device.
2. Online authentication using iris is possible.

Which of the statements given above is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**4. Consider the following statements: [UPSC (Pre) – 2018]**

1. Aadhaar card can be used as a proof of citizenship or domicile.
2. Once issued, Aadhaar number cannot be deactivated or omitted by the Issuing Authority.

Which of the statements given above is/are correct?

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(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

**5. 'SWAYAM', an initiative of the Government of India, aims at [UPSC (Pre) – 2016]**

(a) Promoting self-help groups in rural areas
(b) Providing financial and technical assistance to young start-up entrepreneurs
(c) Promoting the education and health of adolescent girls
(d) Providing affordable and quality education to the citizens for free

**6. Regarding Digi Locker, sometimes seen in the news, which of the following statements is/are correct? [UPSC (Pre) – 2016]**

(i) It is a digital locker system offered by the Government under Digital India Programme.
(ii) It allows you to access your e-documents irrespective of your physical location.

Select the correct answer using the code given below:

(a) (i) only
(b) (ii) only
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

**Answer Key**

| | | | | | |
|---|---|---|---|---|---|
| 1. (d) | 2. (a) | 3. (c) | 4. (d) | 5. (d) | 6. (c) |

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# CHAPTER 10
# TECHNICAL ASPECT OF INFORMATION AND COMMUNICATION TECHNOLOGY

## IN THIS CHAPTER...


**10.1 Information and Communication Technologies (ICT)**

**10.2 Evolution of Telecommunications**

10.2.1 Wire Pairs

10.2.2 Wireless Transmission


International Telecommunication Union (ITU) divides this radio spectrum into 12 bands.


10.2.3 Analogue versus Digital Communications

10.2.4 Modem

10.2.5 Modulation


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Electromagnetic spectrum is the means of wireless transmission.

**10.3 Multiple Access/Multiplexing**

- 10.3.1 Frequency Division Multiple Access (FDMA)
- 10.3.2 Time Division Multiple Access (TDMA)
- 10.3.3 Code Division Multiple Access (CDMA)
- 10.3.4 A Comparative Study of GSM (Global System of Mobile Communication) and CDMA (Code Division Multiple Access)

**10.4 Ionosphere and Its Importance in Wireless Communication**

The 'Bluetooth' system, makes use of phase-shift keying for exchanging information amid various devices.

**10.5 Conventional Phone versus Satellite Phone**

**10.6 Radiospectrum**

Long-distance transmission or radio communication becomes possible due to ionosphere.

**10.7 Types of Transponders, Their Frequencies and Uses**

**10.8 Very Small Aperture Terminal (VSAT)**

**10.9 SIM Card (Subscriber Identity Module) and eSIM**

**10.10 IMEI (International Mobile Equipment Identity) Number**

**10.11 Generation of Mobile Phones**

- 10.11.1 0G/Radio Phone or Taxi Phone
- 10.11.2 1G FDMA: Frequency Division Multiple Access

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10.11.3 2G TDMA: Time Division Multiple Access and CDMA: Code Division Multiple Access
10.11.4 2.5G GPRS: General Packet Radio Services
10.11.5 2.75G EDGE: Enhanced Data Rates for GSM Evolution
10.11.6 3G UMTS: Universal Mobile Telecom Services, EDGE: Enhanced Data Rate for GSM Evolution and CDMA 2000/WCDMA
10.11.7 4G LTE: Long-Term Evolution, HSPA+: High Speed Packet Access Plus, VoLTE
10.11.8 5G
10.11.9 6G Technology

**10.12 Certain Services Available in Mobile Phone to Enhanced Internet Speed**

10.12.1 Long-Term Evolution (LTE)
10.12.2 High Speed Downlink Packet Access (HSDPA)
10.12.3 High Speed Packet Access Plus (HSPA+)

## 10.1 Information and Communication Technologies (ICT)

The technologies, which enable society to create, collect, consolidate, communicate, manage and process information in multimedia and various digital formats, render various purposes via the application of telecommunication techniques.

UNESCO defines ICT as **'the scientific, technological and engineering disciplines and the management techniques used to handle, transmit information with men and machines'.**

### *Impact of ICT on Society*

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Developments in ICT have greatly enabled exchange of information anytime, anywhere in the digital format lent by computers. It has enormous impact in everyday life, such as the use of email and cellular phones at home and the workplace and linked to all facets of society: business, education, communication, scientific exploration, knowledge management, etc.

## 10.2 Evolution of Telecommunications

Communication is a mode of voice system, which is two-way whereas information is a data trafficking system which is one way. **India stands at the second position in terms of both telecommunication and Internet users in the world.** Now, both voice and Internet facilities have been merged together with the help of smartphones. Worldwide, two types of transmission mediums are used– **guided media** such as copper wire and fibre optics and **unguided media** such as electromagnetic waves (radiowaves, microwaves, LASER, etc.) through the air.

### 10.2.1 Wire Pairs

#### 10.2.1.1 Coaxial Cable/Copper Wire

- A coaxial cable consists of a stiff copper wire as the core surrounded by an insulating material. Its application is for distance transmission. Additional advantage includes giving a good combination of high bandwidth, without distortion or loss of signal and excellent noise immunity.
- Earlier, they were used widely in telephone systems but are now replaced by fibre optics, on long-distance transmission.
- The coaxial cables that are used in analogue transmission are television cables, also termed as broadband coaxial cable.

#### 10.2.1.2 Optical Fibres

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- In fibre optics, the semiconductor laser transmits data at speed of light in the form of light along hair-like thin fibres of glass, without any considerable intensity loss over very long distances. Hence, it is a potential, cost-effective alternative for the future.
- There is a converter, which converts electrical signals into waves of light at the source end. These waves are transmitted over the fibre. With the help of photoelectric diodes, one more converter located near to sink changes the waves back to electrical signals. These electrical signals are then amplified and sent to the receiver.

Note: Photoelectric diode is a semiconductor device that converts light into current.

- They may be used to communicate either analogue or digital signals.
- The electromagnetic interference does not affect fibre optics transmission, thus reducing noise and distortion.
- They are most commonly used for end-to-end, one-way communication links because it is extremely difficult and expensive to tap a fibre optic cable at various points. This feature adds security against unauthorised tampering of information.

### *Fibre Optics and Its Applications*

- The use of Internet, e-commerce, computer networks, multimedia, voice, data and video, has been growing faster and thus, the need for a transmission medium with high bandwidth capabilities is necessary. Fibre optics, with its comparatively infinite bandwidth, has proven to be the solution.
- These installations are easy and simple due to large bandwidth, relatively smaller diameter and lightweight of optical fibre. Additionally, fibre optic solution are cheaper than copper.
- High-quality signal can be transmitted using optical fibre for longer distances.

For difference between Copper Wire and Optical Fibre, refer to **Table 10.1**.

**Table 10.1** Comparison between Copper Wire and Optical Fibre

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| Features | Copper Wire | Optical Fibre |
|---|---|---|
| **Constitution** | Copper | Silicon, Silicon oxide, etc. |
| **Rusting** | Yes | No |
| **Life** | Less | More |
| **Weight** | More | Less |
| **Connection through single wire** | Provides limited connection | Provides massive connection |
| **Heat resistance** | Low | High (up to 400°C, it does not melt) |
| **Preferred by Defence** | No | Yes |
| **Bandwidth** | Low | High |
| | Narrowband – Dial-up connection | Broadband – DSL, ADSL, Cable modem |

### 10.2.1.3 Mode of Internet Connection through Landline/Basic Phone/through Wire

(a) **Dial-up:** Dial-up Internet access is a form of Internet access that uses the facilities of the Public Switched Telephone Network (PSTN) to establish a connection to an Internet Service Provider (ISP) by dialling a telephone number on a conventional telephone line. Both phone call and dial-up connections do not work at the same time. During call, Internet connection is disabled.

(b) **Digital Subscriber Line (DSL):** Digital subscriber line (DSL) is a broadband digital transmission service offered on ordinary telephone lines. DSL works within the frequencies that the telephone does not. The telephone and DSL modem can work at the same time, enabling the use of Internet while making phone calls. There are two main types of DSL technologies:

1. ***Symmetrical DSL:*** The bandwidth for upload and download speeds is equal.
2. ***Asymmetrical DSL (ADSL):*** The most popular type of DSL connection. Most people upload less information but download

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more. Thus, an asymmetrical connection has more downstream bandwidth and less upstream bandwidth.

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### 10.2.2 Wireless Transmission

- Wireless is preferable for those areas where fibre running is difficult due to terrain constraints (mountains, jungles, swamps, etc.). It is noteworthy that modern wireless digital communication began in the Hawaiian Islands, where large chunks of Pacific Ocean separated the users and the telephone system was inadequate.
- Electromagnetic spectrum is the means of wireless transmission.

#### *Electromagnetic Spectrum*

- Electrons create electromagnetic waves capable of propagating through free space (even in a vacuum). These are invisible waves. James Clerk Maxwell first predicted these rays.
- The number of oscillations (a regular periodic variation about a mean position) per second of an electromagnetic wave is called its frequency (f) and the standard unit of measuring frequency is hertz (Hz). The wavelength denotes the distance between two consecutive maxima (the largest intensity) or minima (lowest intensity). It is denoted by Greek letter Λ (lambda).
- The electromagnetic waves can broadcast efficiently by attaching an antenna (an electrical device that sends or receives radio signals) of appropriate size to an electrical circuit and received by a receiver some distance away. All wireless communication work on this principle.
- In vacuum, all electromagnetic waves travel at the same speed, i.e. the speed of light, irrespective of their frequency.
- The electromagnetic spectrum consists of **radio, microwave, infrared** and **visible light.** They can all be used for transmitting information by modulating the amplitude, frequency or phase of the waves.
- Though ultraviolet light, x-rays and gamma rays would be better due to their higher frequencies, they are hard to modulate and do not propagate well through buildings and their harmful nature makes them of no good use for information transmission.

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### *Steps Taken by Government*

- **EMF emissions from mobile towers (they are non-ionising radio frequencies and not dangerous):** Electromagnetic Field (EMF) emissions norms from mobile towers in India are already ten times more stringent (even lower) than the safe limits recommended by the WHO.
- **Monitoring of EMF emissions:** Telecom Service Providers (TSPs) are to adhere to the prescribed norms including the submission of a **self-certificate** before the commercial start of the Base Transceiver Station (BTS) site.
- **EMF audit** by the field units of the Department of Telecommunications (DoT).
- Penalty and shut down of services if found non-compliant.

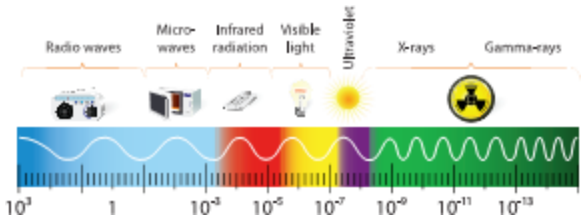


Figure 10.1 Types of Electromagnetic Radiation

#### 10.2.2.1 Microwave Transmission

- **Above 100-megahertz frequency**, microwave transmission is used. They travel in a **straight line/line of sight (directional waves)** and can therefore, be **narrowly focussed.** A fairly strong signal is achieved by concentrating all the energy into a small beam, using a parabolic antenna.
- Microwave is relatively inexpensive and can be used for long distance transmission.
- It is widely used for long-distance telephone communication, cellular telephones, television distribution, etc.

#### 10.2.2.2 Radio Transmission

- Radio waves are widely used for communication, both indoors and outdoors, as they be generated easily and can travel long distances and

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penetrate buildings easily.

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- Radio waves also are **omnidirectional,** that is they can travel in all directions from the source and are not in line of sight.
- At low frequencies, radio waves pass through obstacles well, but the power decreases. At high frequencies, radio waves tend to travel in straight lines and bounce off obstacles.
- They are also absorbed by rain. They are subject to interference from motors and other electrical equipment at all frequencies.

### 10.2.3 Analogue versus Digital Communications

- Based on communications signals, there are two types of communication systems – analogue (analogue signals) and digital communication systems (digital signals).
- In analogue signal, the signal varies continuously with respect to the data. In a digital signal, the data is encoded as discrete set of values (for example, ones and zeros).
- At the time of signal dissemination and reception, the data enclosed in analogue signals is unavoidably degraded by unwanted physical noise. Normally, the communication system noise can be articulated as addition or subtraction to the desirable signal in an entirely casual way, i.e. called as 'additive noise'. The noise, which is not being additive noise, is even more difficult in describing or analysing, hence, such noises will be lost here.
- However, until the disturbance in additive noise surpasses a definite threshold, the digit signals data will remain intact. Since the digital signals resist noise, it accounts as a key advantage of them over analogue signals.

### 10.2.4 Modem

- A special device termed as Modem undertakes the process of **modulation (converting of digital data into analogue)** and **demodulation (converting analogue data to digital).**
- During the usage of analogue facility for the communication of data between two digital devices (say a terminal and a computer), there is requirement of two modems, each near to the digital devices.

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- The digital signal produced at the terminal is transformed into analogue by the modulator and the analogue signal being transmitted through telephone line into digital form by the demodulator of the modem respectively located near to computer.
- The digital data being processed by the computer are modulated to analogue form and returned to terminal through the telephone line where the analogue signals are demodulated into digital for terminal display.
- Modems are used for:
  1. Allowing high transmission speed on an analogue line.
  2. Reducing the noise and distortion effects.
- A few modems consist of microprocessors and further allowing them for operating and functioning under a large number of different conditions and are termed as smart modems.
- Modems are generally provided by network operators (Department of Telecommunication of India) or by vendors who are not necessarily the manufacturers of computer systems. Hence, there is a need to standardise the interface between the modem and the computer equipment.
- Different types of modems are used, such as:
  (i) **Telephone Modem:** Means of connecting to the Internet from home, via telephone.
  (ii) **DSL (Digital Subscriber Line) Modem:** Used for high-speed data transmission.
  (iii) **Cable Modem:** Offer high-speed Internet access over TV cables.

**Cable Modems** offer high-speed Internet access over TV cables. Theoretically, they offer speeds from 3 to 30 megabits per second, but in the real world, actual speed ranges from 400 to 1440 kilobits per second. It is used in TV technology.

### 10.2.5 Modulation

- To convey information, the signal is shaped through a process known as modulation/keying. **'Keying'** is a phenomena of representing digital message in the digital form as an analogue waveform with the help of

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modulation. There are different types of keying, such as phase-shift, frequency-shift, and amplitude-shift keying, based on the changes brought in. **Modulation to the carrier wave is the process of changing amplitude or frequency or phase of a carrier wave.** There are three basic types of modulation: (i) Phase-shift modulation, (ii) Frequency modulation (iii) Amplitude modulation.

(i) **Phase-shift Modulation:**

- ❏ The modulation, where the phase of the carrier wave is varied by an amount proportional to the phase of modulating signal, with no change in amplitude of the carrier.
- ❏ For example, the 'Bluetooth' system, makes use of phase-shift keying for exchanging information amid various devices.

(ii) **Frequency-shift Modulation:**

- ❏ The change in frequency of carrier wave in accordance with the intensity of the signal is termed as frequency modulation with no change in the amplitude of the wave.
- ❏ These are used for FM Channels, those work in a small area (like a district or city) and Radio broadcasting, Telemetry, Radar, Medical application (EEG), Seismic prospecting, etc.

(iii) **Amplitude-shift Modulation:**

- ❏ Practically, all the natural and man-made noises consist of electrical amplitude disturbances. Hence, it becomes difficult for the receiver to distinguish between noise and desired signal. Therefore, reception is noisy.
- ❏ This is used in high-capacity digital radio communication systems at the big level, like All India Radio.

Note: In antenna-based television transmission, frequency modulation was used for sound signal and amplitude modulation was used for picture signal.

- *Modulation is used for transmitting low-frequency analogue signals data to higher frequencies. These being a valuable feature, as the low-frequency analogue signals are not being transmitted over free space effectively.* Therefore, prior to transmission, the low-frequency analogue data must be impressed into a higher-frequency signal, which is termed as the **'carrier wave'**. There are several different modulation programmes for this purpose and out of these, the most basic is

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amplitude modulation (AM) and the other one is frequency modulation (FM).

## 10.3 Multiple Access/Multiplexing

- **Multiplexing allows multiple mobile users to share a finite amount of radio spectrum simultaneously.** The sharing of spectrum should achieve high capacity by simultaneously allocating the available bandwidth to multiple users. So, multiple accesses enables multiple signals to occupy a single communication channel.
- For higher quality of communications, severe degradation in the performance of the system must be avoided.
- Any information, i.e. voice/data in this case can be sent to another party only by the use of communication channel.
- For efficient use, the communication channel is allotted to the users in multiple ways, which is termed as Multiplexing.
- Types of Multiplexing:
  (a) Code Division Multiple Access (CDMA)
  (b) Frequency Division Multiple Access (FDMA)
  (c) Time-Division Multiple Access (TDMA)

### 10.3.1 Frequency Division Multiple Access (FDMA)

FDMA is a technology where the total bandwidth available to the system is divided into frequencies. This division is done between two non-overlapping frequencies that are then assigned to each communicating pair (2 phones). **It is used mainly for analogue transmission.** In this technology, power consumption was very high. FDMA is considered as first-generation mobile phone technology.

### 10.3.2 Time Division Multiple Access (TDMA)

The division of calls happens on time basis, in TDMA. The calls are digitised initially, before combining conversations into a unified digital stream on a single radio channel. Each cellular channel is divided into three

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time slots under a single frequency, so three different users use one frequency at the same time. In TDMA technology, users transmit in rapid succession, each using their own time slot, thus allowing multiple stations to share the same transmission medium (e.g. radio frequency channel) while using only a part of its channel capacity.

### 10.3.3 Code Division Multiple Access (CDMA)

Unlike FDMA, CDMA separates calls by code. Every bit of a conversation is tagged with a specific and unique code. A unique code is allocated to that particular conversation and later, this data is split into small parts and is tagged with the unique code given to the conversation of which they are part of. This tagged data in small pieces is sent over a number of the discrete frequencies available for use at any time in the specified range. In the end, the system then reassembles the conversation from the coded bits and delivers. It is similar to recollecting our luggage after a flight journey by matching the codes on the bag.

Note: TDMA and CDMA are second-generation mobile phone technologies.

### 10.3.4 A Comparative Study of GSM (Global System of Mobile Communication) and CDMA (Code Division Multiple Access)

Both GSM and CDMA technologies have various differences (**Table 10.2**).

#### 10.3.4.1 GSM – Global System of Mobile Communication (The Euro-Asian Alternative)

CDMA was introduced in Hong Kong in 1995. Further, Time Division Multiple Access (TDMA) technology introduced in Europe to support the first available digital technology. **GSM is one of the variants of TDMA technology.**

GSM is a 'cellular' technology, where the entire coverage area is divided into various hexagonal-shaped cells (hence the popular name 'cell

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phones'). Every cell has a corresponding network tower, which serves the mobile phones in that cellular area. For instance: imagine a honeycomb on a tree in a hexagonal-shaped garden. The garden has many flowers. The honeybees collect the nectar from many flowers, and deposit it in the honeycomb. Similarly, our mobile phones are like the flowers, the network tower is like honeycomb and bees are like the signals.

**Table 10.2** Comparative study of CDMA and GSM

| S. No. | Parameter | CDMA | GSM |
|---|---|---|---|
| 1. | Technology | Code Division Multiple Access | Global System for Mobile Communication |
| 2. | Storage Type | Internal Memory (Pre-programmed handset) | SIM (Subscriber identity module) card |
| 3. | Global Market Share | Approx. 20% | Approx. 80% |
| 4. | Dominance | In the US | Worldwide except the US |
| 5. | Network | There is one physical channel | Every cell has a corresponding and a special code for every network tower, which serves the device in the coverage network mobile phones in that cellular. Using this code, the signal of |

### 10.3.4.2 CDMA – Code Division Multiple Access (US-Based Technology)

**CDMA's World War II Foundations:** CDMA, though had its roots in pre-World War II America, if put into an historical context, **it is a now patented technology that only became commercially available in mid-1990s.**

As the name suggests (Code Division Multiple Access), there are **multiple devices using the same spread spectrum** (hence, multiple access). Under CDMA, there is one physical channel and a special code for

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every device in the coverage network. The signal of the device is multiplexed and the same physical channel is used to send the signal (this codes may or may not change), by using this code. For instance, Assume a street with many buses, the bus will have many commuters and they will have to buy the tickets to travel on the bus. In CDMA, your mobile phone is like the commuter, your codes act as the tickets, the bus is a multiplexed carrier signal and the street is the spread spectrum.

**CDMA, incorporating spread-spectrum, works by digitising multiple conversations, attaching a code known only to the sender and receiver.**

This makes the communication more secure and hence the military loved CDMA, because coded signals with trillions of possible combinations resulted in extremely secure transmissions. **CDMA was launched commercially in Hong Kong in 1995.**

## 10.4 Ionosphere and Its Importance in Wireless Communication

- The ionosphere plays a crucial role in long-distance radio communication by enabling radio waves to travel beyond the line-of-sight of the transmitter through reflection and refraction. This enables us to communicate using our mobile phones.
- Four such layers are present in the daytime, i.e. D layer, E layer, F1 layer and F2 layer (**Figure 10.2**) respectively. At night-time, the absence of ultraviolet radiation and the high recombination rate in the dense lower atmosphere makes the D layer disappear.

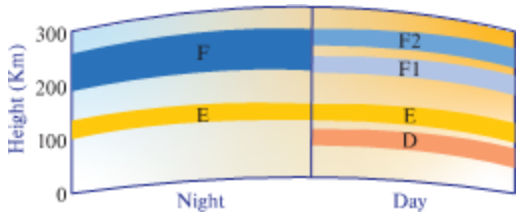


Figure 10.2 Ionosphere in day and night

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- The height of the ionosphere layers are not constant but vary both daily and seasonally, as the intensity of Sun's radiation fluctuates. At sunset, the F1 layer rises, merging with the next higher F2 layer and forming a common F layer. Thus, there are only two layers present in the night, i.e. E layer and F layer.
- The condition of the ionosphere being dependent upon solar radiation undergoes change during the course of the day, over different seasons of the year and also over different years. Its condition is also affected by the sun spot cycle and solar flares.
- The layers of the ionosphere maybe considered as mirrors or partial mirrors depending on the amount of ionisation present. Thus, the transmitted waves striking on ionosphere layer may be refracted back to the earth and received after one refraction. This is called **'single hop transmission'.**

Refraction: The change in direction of light when it passes from one medium to another obliquely is called refraction of light or the bending of light when it goes from one medium to another obliquely is called refraction of light. Two mediums involved are dense medium (more density) and rarer medium (less density).

**Reflection:** The process of sending back the light rays, which falls on the surface of an object is called reflection of light. Only one medium is involved here.

- In some cases, the waves may be refracted back from the Earth to the ionosphere and the process of refraction and reflection is repeated several times before reaching the distant receiver. This is called **'multi hop transmission'.**
- Long-distance transmission or radio communication becomes possible due to ionosphere.

## 10.5 Conventional Phone versus Satellite Phone

- Conventional phone works from tower to tower but satellite phones are directly connect through satellite to the Earth-bound network.

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- Satellite phone works everywhere even in the absence of tower range like: **Himalayas, Sahara desert, polar region, ship, aircraft,** etc.
- Conventional phones work due to the presence of ionosphere because it reflects electromagnetic waves and calls become possible from tower to tower. But in the case of satellite phone signal will go to satellite with the help of some carrier waves.
- Satellite Phone Network Provider: **Globalstar, Thurya, Iridium.**
- Limitations with satellite phone – Call delay and call minutes are costly.

## 10.6 Radiospectrum

It is a part of electromagnetic spectrum having frequency in the range of 3–3000 GHz (3 THz). In this frequency range, electromagnetic waves are called radiowaves, and those are highly used in telecommunication. **International Telecommunication Union (ITU) divides this radio spectrum into 12 bands with various uses.** Those are given in Table 10.3.

## 10.7 Types of Transponders, Their Frequencies and Uses

A satellite receives, modulates, amplifies and re-transmits an uplinked signal with the help of transponders. **Transponders are part of the payload of a telecommunications satellite.** Since the radio waves transverse long distances, the carrier signals are received by the satellite at a very low power. Hence, the power level of these signals need to be boosted by satellites significantly before re-transmission back to Earth, ensuring that they are detectable by an Earth-based receiver.

Satellites have multiple uses on the basis of their own payloads. A satellite is not only used for telecommunications but also for many other applications in **astronomy, weather forecasting, broadcasting, mapping, aircraft identifications, vehicle keys, Sonar, electronic**

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**toll collection system, magnetic labels on credit cards, motor sports, military applications, etc.** Many transponders are used in satellite with specific frequency bands for various applications. **If we use lower frequency band, there will be more congestion.** Different technological innovations are behind the use of these bands and now we have achieved the use of higher band without restrictions. Different types of transponders used in satellite for various purposes are given in Table 10.4.

**Table 10.3** Various Frequency Bands and Their Uses in ICT

| S. No. | Band Name | ITU Band | Frequency | Example Uses |
|---|---|---|---|---|
| 1. | **Extremely Low Frequency (ELF)** | 1 | 3–30 Hz | Communication with submarines |
| 2. | **Super Low Frequency (SLF)** | 2 | 30–300 Hz | Communication with submarines |
| 3. | **Ultra Low Frequency (ULF)** | 3 | 300–3,000 Hz | Submarine communication, communication within mines |
| 4. | **Very Low Frequency (VLF)** | 4 | 3–30 kHz | Navigation, time signals, submarine communication, wireless heart rate monitors, |

## 10.8 Very Small Aperture Terminal (VSAT)

A Very Small Aperture Terminal (VSAT) represents a **two-way satellite ground station** or may be called an **Earth-bound station.** Its application is in satellite communication of data, voice and video signals. **VSAT can handle up to 56 kilobytes per second. Its most common application is in**

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**point-of-sale transactions, such as applications of credit cards and RFID (Radio Frequency Identification).** The other extensive use of VSAT technology is by stockbrokers. **Across the globe, these services are used as method of providing access to broadband Internet sites, which do not have any ADSL or cable Internet access, generally in remote or rural locations.** Almost all VSAT systems are IP-based along with a very broad application range.

There are two typologies of VSAT: **Star VSAT** and **Mesh VSAT**. The deployment of these topologies depends on their commercial viability. Mesh VSAT systems are preferred where the number of users is high since the overall cost of the network is minimised. For broadband WAN or broadband Internet access, Star topology services can be deployed. They are also useful in the applications of front and back office and manage, store and forward solutions like digital signature and interactive distance learning.

**Table 10.4** Different Types of Transponders, their Frequencies and Uses

| Transponder Band | Range of Frequency | Application |
|---|---|---|
| **L** | **1 to 2 GHz** | Global Positioning System (GPS) carriers and also satellite mobile phones, such as Iridium and Inmarsat provide communications at sea, land and air; WorldSpace satellite radio. |
| **S** | **2 to 4 GHz** | Weather radar, surface ship radar and some communication satellites, especially those of NASA for communication with ISS and Space Shuttle. |
| **C** | **4 to 8 GHz** | Primarily used for satellite communications, for full-time satellite TV networks or raw satellite feeds, tropical rainfall, since it is less susceptible to rainfade than Ku band. |

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| | | |
|---|---|---|
| X | 8 to 12.5 GHz | Military Application and Metrological |

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## 10.9 SIM Card (Subscriber Identity Module) and eSIM

It is an integrated circuit chip, mainly comprising of unique serial and international mobile subscriber identity number, security authentication and ciphering information. The card consists of temporary information pertaining to the local network, user's accessible services list and two passwords, one is for ordinary use, which is a Personal Identification Number (PIN) and the other is a personal unblocking code for blocking of PIN. It is mainly used in GSM phones, and can be used in satellite phones. Network operator '**Radiolinja**' developed first SIM card in **Munich, Germany in 1991.** SIM have security number, i.e. IMSI (International Mobile Subscriber Identity Number).

### eSIM

An eSIM is an **embedded SIM**, essentially the same hardware as a regular SIM card chip, but now a permanently embedded part of the motherboard of a watch or smartphone. eSIMs were first established a decade ago in 2012, but despite their futuristic use-cases, have not completely made physical SIMs obsolete yet.

### *Advantages of eSIMs*

- Being attached to the motherboard also allows re-programming, letting users switch operators without having to replace any physical SIM cards.
- **Convenience:** The ability to store multiple SIM profiles in your eSIM also means you can switch between profiles easily.
- **Security:** There is no physical element to pull out and use in another device.

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### Disadvantages of eSIMs

- **Emergencies:** If your phone stops working, runs out of battery or simply, falls and gets a cracked screen, your communication is brought to a complete stand still with eSIMs.
- **Unusable in Countries with no eSIM Support:** You cannot use an eSIM phone in a country where the telecom operators simply do not support the technology yet.
- **Support Only Available in Premium Phones:** In India, eSIM support is currently available on more expensive devices like the Apple iPhones, and Google Pixel series.

## 10.10 IMEI (International Mobile Equipment Identity) Number

It is a 15-digit unique number used for identification of mobile phones, as well as some satellite phones. Usually, it is printed inside the battery compartment of the phone. It can also be displayed on the screen of the phone by dialling *#06# in the keypad on most of the phones to display the IMEI number of the respective phone.

The GSM network uses IMEI number to identify valid devices and hence, a stolen phone can be stopped from accessing the network in the respective country. For instance, in case a mobile phone is stolen, using the IMEI number, the owner can call his or her network provider to 'ban' the phone. This makes the phone useless in that country, no matter if the phone's SIM is changed or not.

### Security Concerns

In many countries, including India, it is compulsory for every phone to have an IMEI number. The 15-digit number is critical for tracking any handset whether it is lost or stolen or needs to be monitored by security agencies. It is possible to track an individual with the IMEI number of his phone, even if he changes his service provider. Without IMEI number, the mobile phones are considered a grave security threat and hence, several concerns

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have been raised over their usage for terrorist attacks. **The government had banned imports of mobile phones without IMEI numbers from China in June 2009.** In December 2009, the Government of India had asked mobile operators to discontinue services to the customers whose phones do not have an IMEI number.

## 10.11 Generation of Mobile Phones

Since early 1970s, the mobile wireless industry had started its technology creation, revolution and evolution as well. 0G to 4G have been experienced by mobile wireless technologies in the world. Cellular technology has evolved through generations, which are described below:

### 10.11.1 0G/Radio Phone or Taxi Phone

- In pre-cell days, people used to talk to mobile operator to set up the calls and there were only a handful of channels available.
- 0G refers to pre-cell phone-based mobile telephony technology, such as radio telephones or taxi phones that some had in cars before the advent of cell phones.
- In this technology, **transceiver was mounted in the vehicle and the 'head' (dial, display and handset) was mounted near the driver seat.** It was used only for basic voice communication. **This technology was firstly used in Finland, 1971.**

**CHRONOLOGY OF CELLULAR TECHNOLOGY**

- The evolution of cellular technology can be tracked back to **1979**, when the **first-generation mobile phones** using analogue networks evolved in **Japan**.
- The **second-generation systems** using digital technologies (such as GSM and CDMA) began in **Finland in 1991.**
- The **third-generation systems** (supporting high-speed data and voice services) again began in **Japan in 2001**.
- **Fourth-generation technology** firstly-released Long-Term Evolution (LTE) standard and has been commercially deployed in **Oslo, Norway and Stockholm, Sweden since 2009.**

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- **5G** deployments arguably started in **2018** in the **USA** but got properly underway in 2019 with many EU countries offering a limited service in cities. This developed throughout 2020 and by the end of the year, there had been commercial 5G launches in all EU countries with only four exceptions.

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### 10.11.2 1G FDMA: Frequency Division Multiple Access

- It refers to the first generation of wireless telephone technology (mobile telecommunication), which was first introduced in 1980s and completed in early 1990s.
- **This generation was setup with the help of tower and its range (i.e. a 'Cell').**
- Its **speed was up to 2.4 kilobytes per second.**
- 1G network uses analogue signal.

**Drawbacks:** Poor battery life, poor voice quality, large phone size, no security, limited capacity and poor hands off reliability.

### 10.11.3 2G TDMA: Time Division Multiple Access and CDMA: Code Division Multiple Access

- It refers to the second generation, which is based on TDMA and CDMA **(GSM is one of the TDMA Variant).**
- It was launched in 1991.
- This network uses **digital signals.**
- Its data speed was up to **64 kilobytes per second.**
- Features enable services such as **text messages** and provide better quality and capacity.

**Drawbacks:** It is totally based on network coverage work with strong digital signals at which mobile phones work. If there is loss of network coverage in any specific area, digital signals would weaken. These systems are unable to handle complex data such as videos.

### 10.11.4 2.5G GPRS: General Packet Radio Services

- This **technology works with GSM Network.**

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- It refers to the technology between second (2G) and third (3G) generation of mobile telephony.
- 2.5G can also be described as 2G cellular technology combined with 'General Packet Radio Services' (GPRS).

**Features:** Phone calls, send/receive email messages, **WAP (Wireless Application Protocol) services**: web browsing and surfing, MMS (Multimedia Messaging Services) and **speed: 64–144 kilobytes per second,** camera phones, take a time of 6–9 minutes to download a 3-minute MP3 song.

### 10.11.5 2.75G EDGE: Enhanced Data Rates for GSM Evolution

- This technology **also works in GSM networks.**
- It is an extended version of GSM, which allows faster transmission of data and information than GPRS. EDGE is a radio technology and it is also a part of third-generation technologies. EDGE technology evolved with certain handsets like **BlackBerry, N97 and N95 mobile phones.**
- **EDGE was three times faster than GPRS.** During the use of EDGE technology, there was no need to install any additional hardware and software. There are no additional charges for using this technology.
- A person having GPRS technology can utilise this technology without paying any other additional charges.

### 10.11.6 3G UMTS: Universal Mobile Telecom Services, EDGE: Enhanced Data Rate for GSM Evolution and CDMA 2000/WCDMA

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- 3G technology refers to third generation that was introduced in year 2000.
- Its data transmission speed increased from **144 kilobits per second to 2 megabits per second.** This is typically called as **smart phones** where features increased with bandwidth.

**Features:** Faster communication, high-speed web/ more security, video conferencing, TV streaming and send/receive large email messages.

**Drawbacks:** 3G networks have slower data speeds, limited coverage, the technology is more expensive making it less accessible and there are health concerns about radiation exposure from extensive use of 3G networks.

Note: Some more generations evolve between 3G and 4G, i.e. 3.5 G: HSDPA (High Speed Downlink Packet Access ) and **3.75 G: HSUPA** (High Speed Uplink Packet Access)

### 10.11.7 4G LTE: Long-Term Evolution, HSPA+: High Speed Packet Access Plus, VoLTE

- This generation started from late 2000s.
- It is capable of providing **100 megabits per second to 2 gigabits per second speed.**
- 4G is also described with the name of **MAGIC** where
  'M' stands for Mobile Multimedia,
  'A' stands for 'Anytime Anywhere',
  'G' stands for Global Mobility Support,
  'I' stands for Integrated Wireless Solution and
  'C' stands for Customised Personal Services.
- Higher data rate and also provides expanded multimedia services.
- Higher level of security.

**Drawbacks:** Battery usage is more, need very complicated hardware, and expensive equipment is required to implement these network.

### 10.11.8 5G

5G is the next generation of mobile broadband that will eventually replace, or at least augment 4G LTE connection.

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## Features and Benefits of the 5G Technology

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- Operate **in the millimetre wave spectrum (30-300 GHz)** which have the advantage of sending large amounts of data at very high speeds.
- Operate in **3 bands,** namely low, mid and high frequency spectrum.
- **Reduced latency** will support new applications that leverage the power of 5G, the Internet of Things (IoT), and artificial intelligence.
- **Increased capacity** on 5G networks can minimise the impact of load spikes, like those that take place during sporting events and news events.

## Challenges for 5G Roll-out in India

- **Procedural Delays:** India's telecom sector is greatly affected by the procedural delays and their multiple issues.
- **Affordability of Spectrum:** Many countries around the world may already roll out 5G connectivity to its users but in India, the 5G spectrum is yet to be allocated.
- **Last-mile Connectivity:** Catering to last-mile broadband connectivity in Tier-II, Tier-III cities and rural homes are challenging since India lacks optical f ibre infrastructure and greenf ield deployment which has immensely affected last-mile connectivity.
- **Affordable 5G Devices:** On the consumer front, affordable 5G devices are yet to take their place in the market.

## Significance of the Technology

- India's **National Digital Communications Policy 2018** highlights the importance of 5G when it states that the convergence of a cluster of revolutionary technologies including 5G, the cloud, Internet of Things (IoT) and data analytics, along with a growing start-up community, promise to accelerate and deepen its digital engagement, opening up a new horizon of opportunities.

## Potential Health Risks from 5G

- To date, and after much research performed, **no adverse health effect has been causally linked with exposure to wireless technologies.**

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- **Tissue heating** is the main mechanism of interaction between radiofrequency fields and the human body. Radiofrequency exposure levels from current technologies result in **negligible temperature rise in the human body.**
- **As the frequency increases, there is less penetration into the body tissues** and absorption of the energy becomes more confined to the surface of the body (skin and eye).

**Table 10.5** Comparison among Various Mobile Generations

| Characteristics | 2G | 3G | 4G | 5G |
|---|---|---|---|---|
| **Introduced year** | 1993 | 2001 | 2009 | 2018 |
| **Technology** | GSM | WCDMA | LTE, WiMax | mm waves (Millimetre waves) |
| **Frequency Band (Bandwidth)** | 25 MHz | 25 MHz | 100 MHz | 30 GHz – 300 GHz |
| **Speed** | 64 Kbps | Upto 3.1 Mbps | Upto 1 Gbps | Upto 10 Gbps |
| **Advantages** | Multimedia features (SMS, MMS), Internet Access, SIM | High Security, International Roaming | High Speed hadoffs, Global Mobility | Extremely High Speeds, Low Latency |

Note:

- **3G versus 4G:** 3G, a complete new innovation transformed the mobile telephony. It provided voice and data connectivity on a single network, for the first time, accelerating new ideas like e-learning, e-governance, etc. An improved bandwidth and speed of communication were achieved. 4G was highly concentrated on bandwidth enhancement and improved speed, further taking 3G to the next level. The two existing technologies in 4G are: 4G LTE and 4G WiMAX. The real difference between 3G and 4G is its difference in speed.
- **4G LTE versus 4G WiMAX:** Long-Term Evolution or LTE is a first-generation 4G technology termed as 'true 4G'. WiMAX stands for Worldwide Interoperability for

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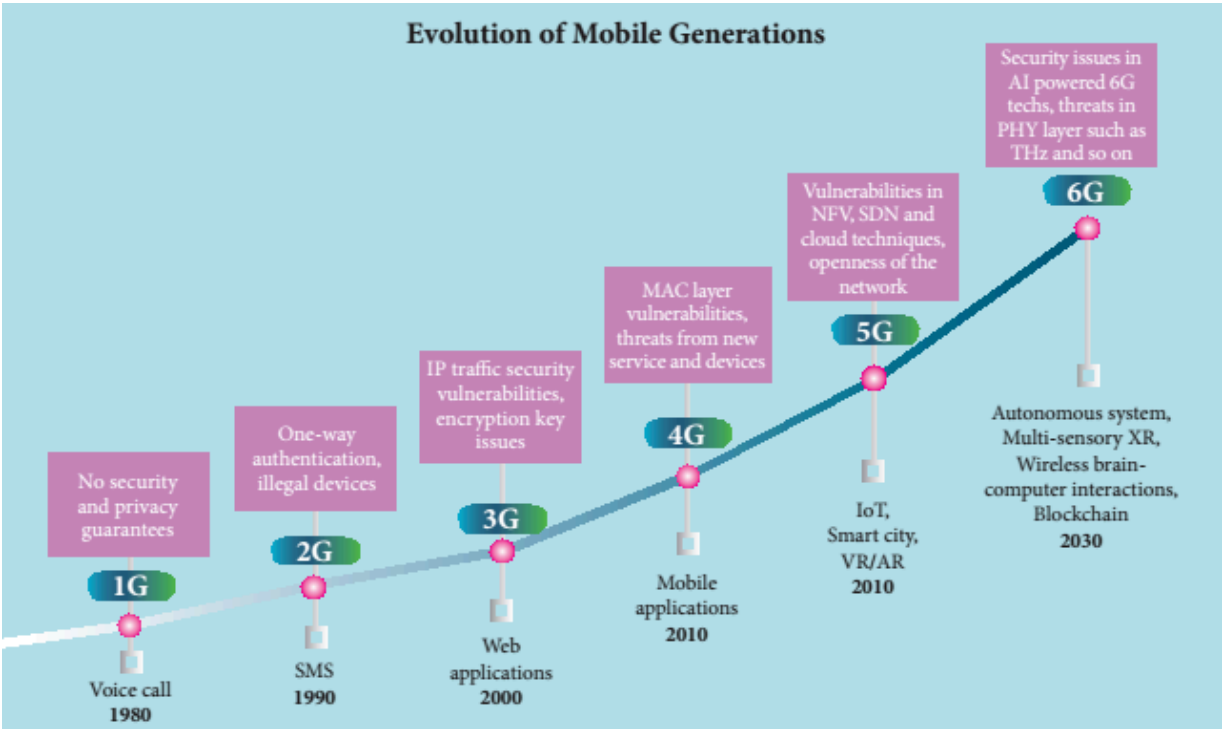
Microwave Access. LTE has higher bandwidth than WiMAX. LTE is compatible with existing networks but WiMAX works on new network. Overall, LTE is gaining lot of popularity as 4G network.

- **5G: Millimetre wave (MM wave)/ millimetre band** is the band of spectrum with wavelengths between 10 millimetres (30 GHz) and 1 millimetre (300 GHz). It is also known as the extremely high frequency (EHF) band by the International Telecommunication Union (ITU).
  **Advantages of millimetre wave:** Millimetre wave is a band of electromagnetic spectrum that can be used in a broad range of products and services, such as high-speed, point-to-point wireless local area networks (WLANs) and broadband access. In telecommunications, millimetre wave is used for a variety of services on mobile and wireless networks, as it enables higher data rates than at lower frequencies, such as those used for Wi-Fi and current cellular networks.

Scan this QR Code to watch a video lecture that simplifies this topic




- Provided that the overall exposure remains below international guidelines, **no consequences for public health are anticipated.**

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## International Exposure Guidelines

- Two international bodies produce exposure guidelines on electromagnetic fields. Many countries currently adhere to the guidelines recommended by:
  1. The International Commission on Non-Ionising Radiation Protection.
  2. The Institute of Electrical and Electronics Engineers, through the International Committee on Electromagnetic Safety.
- These guidelines are not technology-specific. They cover radio frequencies up to 300 GHz, including the frequencies under discussion for 5G.

## International Efforts – International Electromagnetic Fields (EMF) Project

- WHO established the International Electromagnetic Fields (EMF) Project in 1996. The project investigates the health impact of exposure to electric and magnetic fields in the frequency range 0–300 GHz and advises national authorities on EMF radiation protection.

## Where does India stand in terms of trials and launch?

5G services were first launched in India by Prime Minister Narendra Modi on October 1, 2022. At present, only Bharti Airtel and Reliance Jio are the two telecom operators that offer 5G networks in the country.

According to the recently released 'Ericsson Mobility Report':

- Fifth generation telecom services are likely to account for 39% of mobile subscriptions or about 500 million subscriptions in India at the end of 2027.
- The total number of smartphone subscriptions was expected to be 810 million at the end of 2021 and would be projected to grow at a compounded annual growth rate of 7%, exceeding 1.2 billion by 2027.
- 4G subscriptions are expected to reduce from 68% of mobile subscriptions in 2021 to 55% in 2027 as subscribers migrate to 5G.

Jio rolled out its 5G network in Bengaluru, Hyderabad, Gurugram, Noida, Ghaziabad and Faridabad in November 2022. Jio 5G was also made

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available in Pune in November 2022 and in all districts of Gujarat.

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## 10.11.8. 1 Voice Over New Radio

**Voice over New Radio (VoNR),** also known as Vo5G, is poised to **revolutionise voice calling in the era of 5G networks**.

### *Understanding VoNR*

- VoNR enables voice calls over 5G networks, leveraging the advancements of the fifth-generation mobile network technology.
- Unlike traditional voice calling standards that rely on 4G networks (VoLTE), **VoNR harnesses 5G's higher bandwidth, lower latency, and increased reliability to deliver superior voice communication experiences.**

### *Advantages of VoNR Over VoLTE*

- **Enhanced Call Quality:** VoNR utilises advanced audio codecs and the increased data capacity of 5G to deliver superior voice clarity and fidelity.
- **Faster Connection Times:** With reduced network latency inherent to 5G, VoNR promises quicker call connection times, ensuring seamless communication experiences.
- **Improved Reliability:** VoNR aims to address call drop issues prevalent in traditional voice calling standards by minimising packet loss and ensuring continuity of voice calls during network transitions.

### *Applications and Benefits of 5G Technology*

- **Ultra-High Internet Speeds:** 5G networks offer significantly higher data transfer speeds, facilitating seamless multimedia streaming and enhanced user experiences.
- **IoT Connectivity:** 5G enables efficient connectivity for millions of devices and sensors, driving advancements in sectors such as agriculture, healthcare, and smart cities.
- **Tech Revolution:** 5G technology fuels innovation in machine learning, robotics, and other programming modules, unlocking new possibilities

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for technological advancement.

- **Socio-Economic Impact:** The widespread adoption of 5G technology has the potential to create millions of job opportunities, bolster economic growth, and propel India's position as a global tech powerhouse.

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## BHARAT 6-G ALLIANCE

In June 2023, the **Department of Telecommunications (DoT) under the Ministry of Communications** **launched the Bharat 6G Alliance (B6GA)** to **promote innovation and leadership in 6G technology**.

### Bharat 6G Alliance (B6GA)

- Collaborative **platform comprising public and private entities, academia, research institutions, and standards development organisations**.
- **Objectives** include understanding the business and societal needs of 6G, fostering consensus, and **driving impactful research and development initiatives**.
- **Aims** to facilitate international collaboration and knowledge exchange by partnering with other global 6G alliances.
- India **aims to lead in 6G technology development and adoption, leveraging its strengths in software, hardware, manufacturing, talent pool, and market potential**.

### 6G Technology Overview

- Successor of 5G, **offering speeds up to 100 times faster**, **ultra-low latency**, **high reliability**, and **massive connectivity**.
- Enabling applications like **holographic communication, brain-computer interface, quantum internet,** and **artificial intelligence**.

### Evolution of India's Telecom Sector:

- Significant growth witnessed with **improvements in data costs, regulatory processes, BTS site (Cell tower site location) expansion, FDI, spectrum allocation, and technology exports.**
- India emerges as the **world's second-largest telecommunications market** with substantial subscriber base and investments.

### Telecom Technology Development Fund (TTDF) Scheme

- Launched by **DoT/Universal Service Obligation Fund (USOF) to fund R&D of telecom technologies**.

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- Allocates 5% of USOF collections to bridge the digital divide and **foster synergies among academia, start-ups, research institutes, and the industry.**

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### *Challenges and Considerations*

- **Infrastructure and Capital Requirements:** The widespread roll-out of 5G networks, especially in rural areas, necessitates strong technological infrastructure and sufficient capital investment from telecom companies.
- **Spectrum Pricing and Affordability:** The cost of 5G spectrum in India poses challenges in terms of affordability for customers, raising concerns about equitable access to advanced telecommunications services.
- **Regulatory and Interference Issues:** Regulatory frameworks and concerns about potential interference, particularly with aircraft navigation systems, need to be addressed for seamless deployment of 5G technology.

### *VoNR Roll-out in India*

- While VoNR is **not yet available in India**, leading carriers like **Reliance Jio have been testing the technology behind the scenes**.
- The roll-out of **VoNR is expected to enhance voice call quality and reduce latency for users switching** between 4G and 5G networks.
- The timeline for VoNR availability remains uncertain, but as 5G network continues to expand, VoNR is poised to become the standard for voice communication in India.

(Refer **Infographic – 4G vs. 5G**)
(Refer **Infographic – Standalone 5G vs Non-Standalone 5G**)

## 10.11.9 6G Technology

### 6G Technology Overview:

- The successor to **5G offers speeds up to 100 times faster, with ultra-low latency, high reliability and extensive connectivity**.
- Applications such as **holographic communication, brain-computer interfaces, quantum internet** and **artificial intelligence are enabled**.

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## Evolution of India's Telecom Sector:

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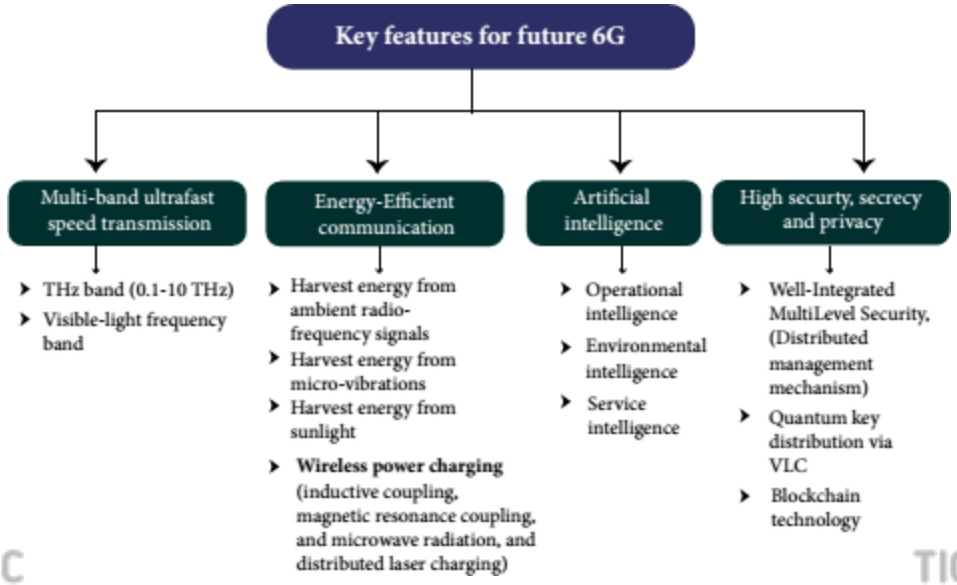
- Significant growth witnessed with **improvements in data costs, regulatory processes, FDI, spectrum allocation** and **technology exports**.
- **India has become the world's second-largest tele- communications market**, boasting a significant subscriber base and substantial investments.

## Telecom Technology Development Fund (TTDF) Scheme:

- **The Department of Telecommunications (DoT) and the Universal Service Obligation Fund (USOF) initiated a program to support research and development in telecom technologies.**
- The Universal Service Obligation Fund (USOF) allocates 5% of its collections to reduce the digital divide and promote collaboration among academia, start-ups, research institutions and industry.

**Holographic communication** is a technology that **allows for the transmission and display of three-dimensional images**, creating immersive and lifelike virtual interactions.

It goes beyond traditional video calls by providing a sense of spatial depth and presence, making remote interactions feel more natural and engaging.

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# 4G vs 5G

## What is 5G?

- 5G enables a new kind of network that is designed to connect virtually everyone and everything together including machine, objects and devices.
- Contributes to the development of Smart Communities which aim to provide digital innovative services for their citizens.

## Evolution of Wireless Technology

## 4G

- 4G refers to fourth generation of mobile communications standards.
- The latency varies from carrier to carrier.
- 4G is relatively slower than 5G in terms of peak speed.
- 4G offers a minimum peak download speed of 1Gbps

| | |
|---|---|
| Latency | 200 milliseconds |
| Potential Download Speed | 1 Gbps (Low) |
| Base Stations | Cell Towers |
| Channel | 20 Mhz Channels |
| Energy Efficiency | 1X |
| Density | 100K Connections/km² |

## 5G

- 5G is the fifth generation of wireless network technology.
- It is built on what's called millimeter waves, a section of very high frequency spectrum upwards of 20 GHz all the way up to 96 GHz.
- 5G offers highly reduced latency capabilities compared to previous generations of wireless technology.
- In terms of peak speed, 5G is nearly 20 times faster than 4G.
- 5G provides a minimum peak download speed of 20Gbps.

| | |
|---|---|
| Latency | Less than 5 milliseconds |
| Potential Download Speed | 20 Gbps (Max) |
| Base Stations | Small Cells |
| Channel | 100-800 Mhz Channels |
| Energy Efficiency | 8100X |
| Density | 1 Million Connections/km² |

## PROS of 5G

**GREATER TRANSMISSION SPEED**
100 times faster than 4G

**LOWER LATENCY**
10 times less latency than 4G

**INCREASED CONNECTIVITY**
100 times increase in traffic capacity

**ENERGY EFFICIENCY PLANS**
5G network aims to decrease energy consumption by 10% as compared to 4G networks.

**EFFICIENT BUSINESS PROCESS**
5G technology allows enterprises to coordinate all the devices under their control with infinite efficiency.

## CONS of 5G

**COST OF ESTABLISHMENT**
Replacement of existing mobile infrastructure and skilled workers to maintain a proper connection.

**LACK OF INFORMATION**
5G technology is still in developing phases and there is a lot left to be discovered.

**LIMITED COVERAGE**
5G waves- millimeter waves, generally have very low penetrating power and cannot pass through buildings, trees, walls or other barriers without signal being disrupted.

**OVERCROWDED RADIO FREQUENCY**
5G network is 6Ghz, which is already jam-packed with other signals

**SECURITY AND PRIVACY ISSUES**
Feature of allowing the transfer of high quality data and the presence of more entry points for attackers.

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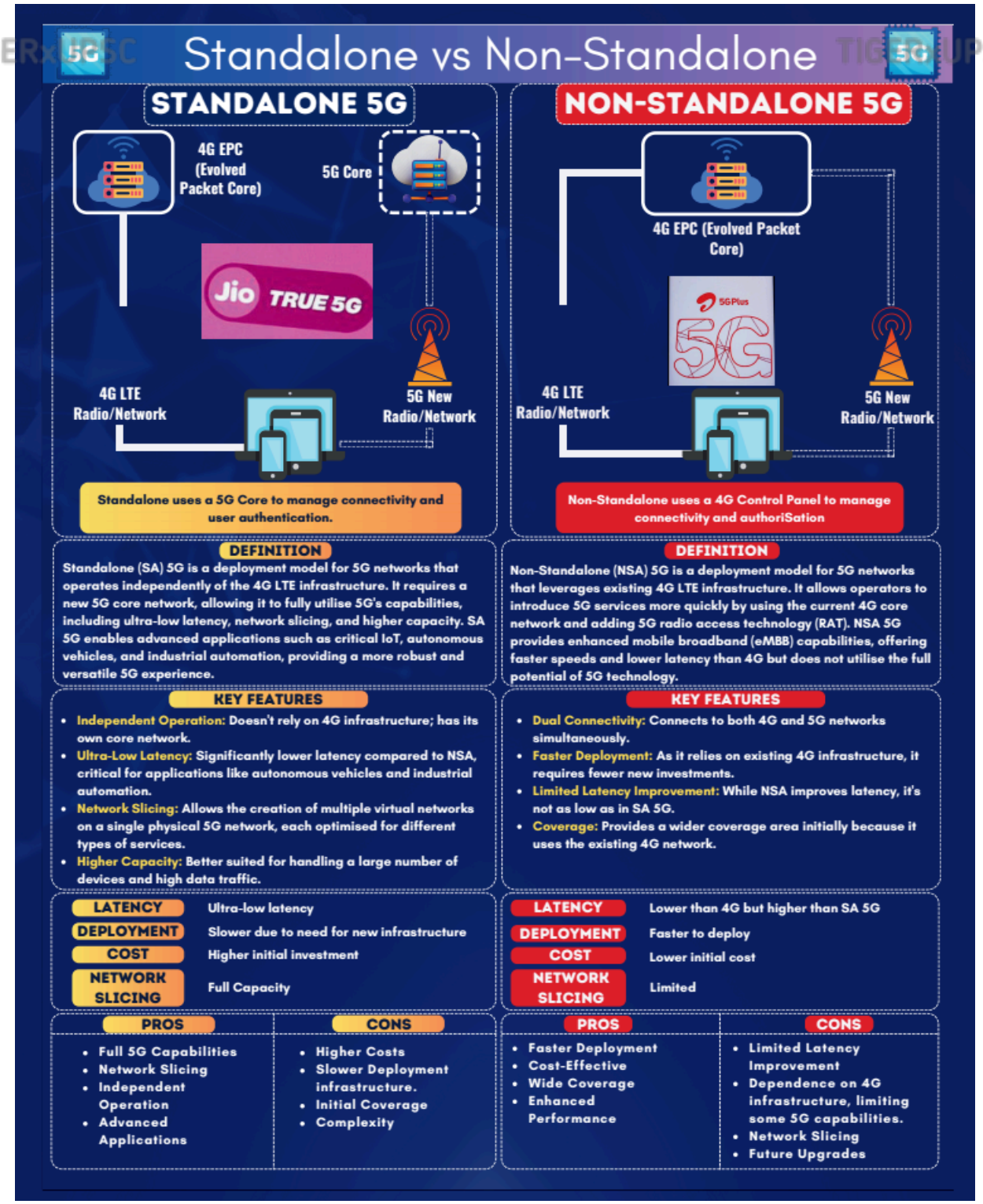


A **quantum internet** is a network that utilises quantum-mechanical principles, like entanglement and superposition, to transmit and process information.

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## 10.12 Certain Services Available in Mobile Phone to Enhanced Internet Speed

### 10.12.1 Long-Term Evolution (LTE)

LTE or Long-Term Evolution, commonly marketed as 4G LTE, is a standard for wireless communication of high-speed data for mobile phones and data terminals. For a true 4G technology, download speeds of 100 megabits per second and 1 gigabits per second should be available for moving (i.e., in a car) or pedestrian points, respectively.

The maximum download speed offered is of 300 megabits per second, although it has been controversial over the speeds provided by some operators running LTE networks. Sometimes the speed is lower than the supposedly 'inferior' HSPA (Plus) technology. Commercially available speed varies widely and using the (at the time of writing) recently launched UK LTE networks, tests have shown anywhere in between 8 and 50 megabits per second in available areas.

#### 10.12.1. 1 Difference between LTE and VoLTE

LTE is called '**Long Term Evolution'**. **Airtel launched the first LTE network service in India in 2012**. Generally, LTE is also called 4G. VoLTE

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stands for '**Voice over Long Term Evolution**' or 'Voice over LTE'. It means that unlike 4G LTE, even calls, SMS, web browsing will be possible using your Internet data plans. **Reliance Jio had started this VoLTE service in India in 2016**. Now every major carrier supports it and every new phone should support it (**Table 10.6**).

In April 2024, approximately 5.44 billion people around the world used the Internet.

**Table 10.6** LTE vs. VoLTE

| LTE | VoLTE |
|---|---|
| 1. It stands for 'Long Term Evolution'. | 1. It stands for 'Voice over Long Term Evolution'. |
| 2. External software like WhatsApp, Facebook Messenger and Skype are required by users to make video calls on LTE networks. | 2. Only the phone number is required to video call anybody. |
| 3. There is a possibility that LTE may not support voice calls and data services together. | 3. It supports voice calls and data services together. |
| 4. It targets increasing data rates on the 4G bandwidth only. | 4. It targets both internet data and voice calling without affecting each other. |
| 5. The time limit to connect a call on 3G networks is comparatively slow in case of LTE. | 5. The connectivity is comparatively high provided both the users are on the VoLTE network. |

### 10.12.2 High Speed Downlink Packet Access (HSDPA)

High Speed Downlink Packet Access (HSDPA) is 3G network-based technology, which can support speeds up to 7.2 megabits per second. Practically, it is likely to get a top speed of around 3 megabits, yet it is useful for mobile TV streaming and other high-end data transmissions. To use HSDPA, first the phone should be compatible for the technology plus

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the location of the device should be within range of a cell site that has been upgraded to offer the service.

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### 10.12.3 High Speed Packet Access Plus (HSPA+)

An evolved version of HSPA (HSDPA and HSUPA) standard is HSPA (Plus), which enables faster speeds. The maximum download speed allowed by the standard is 168 megabits per second, although in real sense, supporting network will offer 21 megabits per second download. This is due to the fact that the existing 3G network architecture operators would have deployed and made compatible was never designed to handle such massive bandwidth.

Hence, the operators need additional spectrum for improving the quality of services. The Government should formulate a spectrum policy promoting efficient use of spectrum via development of market incentives and differential pricing of spectrum in congested areas. The government can realise the best price for spectrum according to market forces, through open and transport auction format and simultaneously the telecom operators can minimise and efficiently use the spectrum.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and**

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**VoLTE (Voice over Long-Term Evolution)?** **[UPSC (Pre) – 2019]**

1. LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.
2. LTE is data-only technology and VoLTE is voice-only technology.

Select the correct answer using the code given below.

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**2. A layer in the Earth's atmosphere called Ionosphere facilitates radio communication. Why?** **[UPSC (Pre) – 2011]**

(i) The presence of ozone causes the reflection of radio waves to Earth.
(ii) Radio waves have a very long wavelength.

Which of the statement(s) given above is/ are correct?

(a) Only (i)
(b) Only (ii)
(c) Both (i) and (ii)
(d) None of these

**3. Which among the following do/does not belong to the GSM family of wireless technologies?** **[UPSC (Pre) – 2010]**

(a) EDGE
(b) LTE
(c) DSL
(d) Both EDGE and LTE

**Answer Key**

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| | | |
|---|---|---|
| 1. (d) | 2. (b) | 3. (c) |

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# CHAPTER 11

# INTERNET, GLOBAL INTERNET GOVERNANCE NETWORK AND THEIR VARIOUS APPLICATIONS

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**11.2 Internet**
**11.3 Modem (Modulator-Demodulator)**
- 11.3.1 Internal Modem
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**11.15 Evolution of Web (Web 1.0, 2.0, 3.0, 4.0 and 5.0)**

Web 3.0 is a decentralised internet to run on blockchain technology and will be based on Inter-Planetary File System (IPFS) which will replace the current DNS system (Web 1.0 and 2.0 work on DNS System).

**Web 5.0 (Intelligent web / symbiotic web)** also called the **sensory and emotive Web**, is designed to develop computers that interact with human beings.

**11.16 Computer Security**
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11.16.2 Rootkit
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11.17.1 Computer Virus
11.17.2 Worm
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**11.18 Global Internet Governance**
11.18.1 International Telecommunication Union (ITU)
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11.20.3 Issue
11.20.4 Telecom Commission Approved TRAI Recommendations on Net Neutrality
**11.21 Network**
11.21.1 Personal Area Network (PAN)
11.21.2 Local Area Network (LAN)

LAN connections use coaxial or CAT 5 cables.

11.21.3 Metropolitan Area Network (MAN)
11.21.4 Wide Area Network (WAN)

WAN uses satellites and microwave relays.

**11.22 Various Modes of Network Connection**
11.22.1 Bluetooth

Bluetooth 4.0 uses 128-bit encryption to protect your data while transmitting data between devices. On the other hand, the Wi-Fi Direct uses 256-bit encryption. Hence, Bluetooth 4.0 is remarkably secure than Wi-Fi direct. All Wi-Fi Direct operates in 2.4 GHz frequency band.

11.22.2 Wi-Fi Direct
11.22.3 Cordless Phone
11.22.4 Wi-Fi
11.22.5 Wi-max (Worldwide Interoperability for Microwave Access)
11.22.6 NFC (Near Field Communication)
11.22.7 Li-Fi (Light Fidelity)/Visible Light Communication (VLC)
**11.23 Virtual Private Network (VPN)**
**11.24 VoIP (Voice over Internet Protocol/Internet Telephony)**
**11.25 Hidden Part of Internet: Dark Net, Dark Web and Deep Web**

Through use of applications such as SecureDrop, darknets can facilitate anonymous communication between whistle-blowers, activists, journalists and news organisations.

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## 11.1 Evolution of Internet

- **1969 AD: Advanced Research Project Agency (ARPA)** of American defence agency started the Internet **ARPANET** by placing the four universities of USA into a network. It was developed for the research education and government institution. It had another aim too for the emergency that if all the another medium of connection had failed, the connection could be established through it. Up till 1971 the ARPANET had connected approximately two-dozen computers.
- **1972 AD:** e-mail service started.
- **1973 AD:** Transmission Control Protocol/Internet Protocol (TCP/IP) was designed. Till 1983, it became the mode of communication between two computers. In this Internet, user could download files by connecting to any computer.
- **1983 AD:** The military part of ARPANET was transferred to MILNET.
- **1986 AD:** US National Science Foundation (NSF) launched NSFNET. It was the first-ever wide-range network in which the Internet technique was used.
- **1988 AD:** In Finland, Jorcco Occirenene developed Internet chatting.
- **1989 AD:** Peter Deutsch firstly tried to prepare the index of Internet. Brewster Kahle of thinking machine corporation developed another index WAIS (World Area Information Service).
- **1989 AD Tim Berners Lee** of CERN (European Laboratory for Particle Physics) developed a new technique for sharing the information on Internet. It was called World Wide Web (www). It is based on the hypertext, which connects Internet users from different sites. It happens through hyperlink (word, buttons or graphics programmed in a distinguished manner).
- **1991 AD:** First user-friendly interface. GOFAR was developed in Minnesota University (USA) henceforth, GOFAR has been a famous interface. NFSNET opened for commercial traffic.

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- **1993 AD:** Marc Lowell Andreessen of National Centre for Super-computing applications developed a navigation system named 'Mosaic'.
- **1994 AD:** Netscape Communications and in 1995, Microsoft, rolled out their browsers in the market. Owing to these browsers, the use of Internet became very easy for users.
- **1995 AD:** Commercial sites were launched on Internet. Mass marketing campaign were started through e-mail.
- **1996 AD:** By 1996, the Internet became famous in the world. Number of Internet users reached up to 15 million.
- **1999 AD:** The concept of e-commerce gathered pace, due to which the shopping online through Internet became quite famous.
- **2003 AD:** In New Zealand, NIUE started a countrywide wireless access system in Internet.

## 11.2 Internet

Internet is the term used for the **wide area network/ global area network** that connects computers spread all over the world. It is the **network of networks**. Internet consists of interconnection of networks of various types. It is a public network. Through Internet, we can share text data, audio and video data. Nobody owns the Internet, it is governed by ISOC (Internet Society). Internet is also popular with the name of 'Network of Networks'.

Internet is the super network of computers worldwide. Through this, thousands of computers are connected worldwide. The journey of Internet was started by **Vint Cerf and Bob Kanh**. They devised a technique by which the formation could be divided into small packets and could be sent to another computer, so that the other computer could make a copy of this. It was a way of establishing communications between two computers. This technique was called **'Transmission Control Protocol' (TCP).**

The transmission of information can also be repeated when there are more than two computers in the network, because every computer has its own address in the network. This address is called **Internet Protocol (IP)**

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address, i.e., IP address. Internet Protocols are some numbers in fact, which are separated with a point from each other.

Information is sent to other computer in the broken form of small packets. When these packets reach a computer, the computer assesses whether this information is sent for it or not. It moves the data forward in the direction of that computer for which the data is sent. So, the transportation of data from one point to another is called Internet protocol (IP).

Often, the computers of an office are connected to each other and transmit information with each other. This is called LAN. Any computer connected in LAN is connected with a telephone line or a satellite. All the computers in the world are connected to each other. Internet is the network for worldwide computers, which can interconnect (See Figure 11.1).

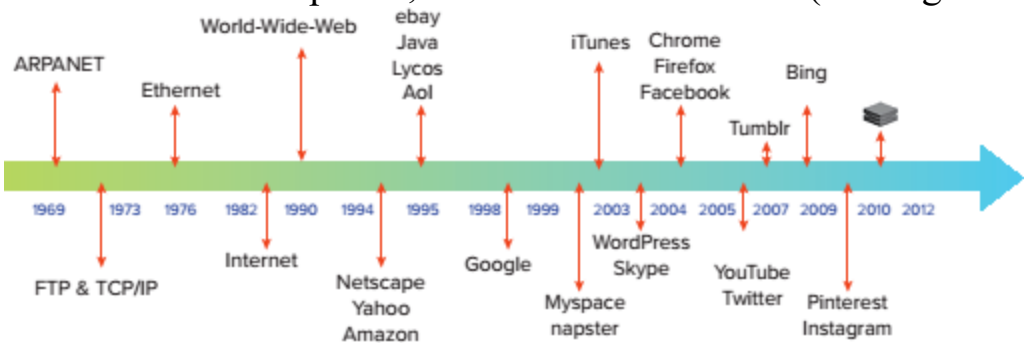


**Figure 11.1** Evolution of Internet Services

## 11.3 Modem (Modulator-Demodulator)

To connect the Internet through line or telecommunications line, a modem is required. It is a necessary link between the Internet service provider and the browser. Through telephone line, only analogue data can travel, whereas, a computer gives the digital data. To maintain compatibility in between computer system and a telephone line, a modem is needed, which converts digital signal to analogue and *vice versa*. The speed of modem is measured in bytes per second (BPs), kilobyte per second (Kbps), megabyte per second (Mbps), etc.

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### 11.3.1 Internal Modem

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It is the modem, which is already inside our desktop or laptop. The mobiles and other electronic devices in which we use Internet are the examples of gadgets having internal modem.

### 11.3.2 External Modem

The external modem has to be connected to our computer from outside. Such types of modems are used in data cards (Photon, Idea, etc.) or PCMCI. So modem is a device that lets your computer send signals back and forth along a telephone line.

### 11.3.3 Internet Software or Web Browser

Web is like a large book and web browser is a software, which connects computer to the Internet. Some web browsers are

- Netscape navigator
- Microsoft Internet explorer
- Mozilla Firefox
- Google chrome
- Opera

We can get the knowledge through connecting these browsers. URL (Universe Resource Locator) has both, the tools to be used and the address of Internet. In the URL *http://www.google.co.in* tool *http (hypertext transfer protocol)* is a protocol, means a predefined rule that sends the information to the given Internetaddress.

## 11.4 Internet Protocol

It represents the set of rules that govern sending/ receiving messages on Internet. The data sent from one computer to another, on the Internet,

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follows this protocol. The communicating end-point has no continuing connection, and is hence termed as connection-less protocol.

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Note: **Meaning of 'Protocol':** The set of rules or standards designed to enable computers to be connected with one another and to exchange information among them with very little error.

## Differences between IPv4 and IPv6

The Internet Protocol version 4 (IPv4) is for use on packet-switched Link Layer networks (e.g., Ethernet), which provides addressing capability of approximately 4.3 billion addresses.

The more advanced version is Internet Protocol version 6 (IPv6) with comparatively better features (**Table 11.1**). Capable of providing an infinite number of addresses, IPv6 is replacing IPv4, thereby accommodating the growing number of networks worldwide. This also helps to solve the IP address exhaustion problem.

**Table 11.1** Differences between IPv4 and IPv6

| Features | IPv4 | IPv6 |
|---|---|---|
| IP Address (No. of bits) | 32 | 128 |
| Format | Decimal | Hexa-decimal |
| Addresses Capability | 4.3 billion | No limitation (infinite number) |

A major difference between the two is the way the IP addresses appear. IPv4 uses four 1 byte decimal numbers, separated by a dot (i.e., 192.168.1.1). IPv6 uses hexadecimal numbers separated by colons (i.e., fe80:d4a8:6435:d2d8:d9f3b1:1).

## Advantages of IPv6 over IPv4

- The router's work got comparatively simpler *via* IPv6.
- IPv4 is less compatible to mobile networks than IPv6.
- The payload allowance of IPv6 is higher than that of IPv4.

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- In spite of all these remarkable differences, currently, IPv4 shares larger user portion (99 per cent) while the share of IPv6 is merely 1 per cent of the networks.

## 11.5 Ways of Accessing Internet

There are six common ways of accessing the Internet offered by service providers.

### 11.5.1 Dial-up Internet Access (Dial-up Modem)

This one has slowest speed of connection (about 60 kilobits per second). The user connects to the Internet *via* a modem that connects to the telephone line by dialing one of the numbers supplied by the ISP. A big disadvantage is that the telephone line is tied up when a dial-up modem is in operation.

### 11.5.2 Cable Internet Access (Cable Modem)

By using a cable network modem, local cable television operators give a user access to the Internet through their own cable networks.

### 11.5.3 Internet Access *via* DSL(Broadband Modem)

DSL broadband connections maintain the fastest download speeds (at least 11,000 kilobits per second).

- It is often offered with wireless interface, which requires an Access Point and a router.
- ISPs usually have a download/upload limit (e.g., 10 gigabyte of data) as part of the contract.
- Through the help of this system, Voice over IP (VoIP) and online chat rooms can be used effectively.

### 11.5.4 Wi-Fi Broad Band

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This facility requires a Wi-Fi card attached to the computer and a wireless modem.

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### 11.5.5 Satellite Services

Broadband Internet access in rural or outlying areas is taken care by satellites, as these places are still not fully accessible by wired phone lines or cable connections.

- The broadband option requires an unobstructed aiming path between a small dish and the orbiting satellite. Hence, it is similar to a satellite TV service, except it runs on different antennas and receiving equipment.
- The antenna is connected to a modem device *via* wires and then connected to your computer.
- The bandwidth and uptime can be interrupted by the same forms of service interruption degrading satellite TV signals, including rain, snow and high wind.

### 11.5.6 Mobile Methods

The mobile methods like full-featured smart phones and tablet devices enable us to conduct online activities through wireless access.

- These services can reach speeds that rival or equal traditional wired broadband, inclusive of DSL and cable connections.
- These networks require a location close enough to a cellular service tower to provide enough signal strength to make a connection.
- A USB modem device is plugged to the laptop or computer to access wireless service, or the phone is connected to the computer to share the bandwidth. Such sharing process is known as tethering, which involves additional fee payment to the mobile phone service provider.

## 11.6 Server

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A programme that uses the client/server model and the World Wide Web's hypertext transfer protocol (http) is termed as a web server. This programme services the files that form web pages to web users. Every computer containing a website on the Internet must have a web server programme. Apache is the most widely installed web server. Apache and Microsoft's Internet Information Server (IIS) are two leading web servers.

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## 11.7 Website

It is a collection of web pages, typically common to a particular domain name or sub-domain on the World Wide Web.

## 11.8 Web Page

A Web Page is a resource on the World Wide Web, usually in HTML/XHTML format with hypertext links enabling navigation from one page or section to another. Web Page contains dynamic content (e.g., images, text, form fields, etc.) that can change/move without reloading the web page.

## 11.9 Services on Internet

### 11.9.1 Chatting

Instant messaging or chatting is a type of communication, wherein the participants type messages to other chat group participants and receiver responses instantly, which can be read on the screen. Recently, some sites have also come up with additional features like audio/video chat that may be performed with the help of webcam.

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### 11.9.2 Upload and Download

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You can send (upload) and receive (download) large files over the net from one computer to another through FTP (File Transfer Protocol). This includes Word, EXCEL or of any other format file.

### 11.9.3 Gopher

All the available resources on the Internet are shown by Gopher in the form of menus. The concept of Gopher is based on the clients and servers.

### 11.9.4 Search Engine

A programme that searches documents for specified keywords and returns a list of documents where the keywords are found is known as search engine.

Note: Search engines introduced earlier: LYCOS (1993), WEB CROWLER in (1994), Yahoo and ALTA VISTA in 1995, GOOGLE (1998). Microsoft introduced BING as its search engine.

## 11.10 Email Protection against Virus

In digital communication, the security over Internet becomes most important and given below are some characteristics of e-mail messages and attachments indicating that they are unsafe to open:

- e-mail from an unknown sender;
- e-mail from a known sender, but subject line does not make sense;
- Message attachment with filename extension associated with viruses;
- Messages containing links to destinations unrecognisable to the user.

Users will successfully avoid virus infection by applying knowledge of these e-mail characteristics and taking appropriate action when reading e-mail (i.e., open message or delete it).

Various types of problem appear through Internet.

### 11.10.1 E-mail Spoofing

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When the sender's address and other parts of the e-mail header are altered to appear as though the e-mail originated from a different source, it may indicate E-mail spoofing. Since the core SMTP does not provide any authentication, impersonation and forging the emails becomes easy.

In terms of network security, a spoofing attack represents a situation in which one person or programme successfully masquerades as another by falsifying data and thereby gaining an illegitimate advantage.

### 11.10.2 Spywares

Spyware is installed with or without permission on the personal computers to collect information about users, their computer or browsing habits track each and everything that is done without the user's knowledge and sends it to remote user. Other malicious programmes can also be downloaded via spyware from Internet and installed on the computer. Similar to adware in function, but it is usually a separate programme installed unknowingly while installing another freeware-type programme or application.

### 11.10.3 Adware

Generally, it is a software application, which displays advertising banners during the running of any programme. While browsing any website, adware can automatically get downloaded to the system and can be viewed through pop-up windows or through a bar appearing on a computer screen automatically. Companies use adwares for marketing purposes.

### 11.10.4 Spam

When Internet is flooded with copies of the same message, it is termed as spamming. Mostly, spams include commercial advertisements sent as an unwanted email to users. It also includes electronic junk mails or junk newsgroup postings.

### 11.10.5 Tracking Cookies

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It is a plain text file stored in given computer in cookies folder, thereby storing data about the browsing session. Many websites deploy cookies to track visitors' information. Since cookies keep track of the browsing history, it can be dangerous to access personal important details like bank account details, credit card information, etc., because *via* cookies, the hackers and companies may access all such personal details.

## 11.11 Uniform Resource Locator (URL)

A formatted text string called URL is used by Web browsers, to identify a network resource on the Internet network resources such as a file that can include plain Web pages, other text documents, graphics or programmes.

## 11.12 Types of Domain

Various types of domains are used on the Internet to distinguish different types of websites.

- .com commercial
- .edu or ac educational
- .mil military
- .net belonging to networking or Internet service provider company
- .org organisational
- .gov government
- .info informational sites

## 11.13 Hyper Text Transfer Protocol (HTTP)

A file can be transferred over the World Wide Web *via* FTP (file transfer protocol) or HTTP (Hypertext Transfer Protocol). HTTP represents the set

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of rules for transferring files (text, graphic images, sound, video; and other multimedia files) on the World Wide Web. The moment the Web user opens the Web browser; HTTP comes into play. HTTP is an application protocol running on top of the TCP IP suite of protocols, while FTP is used to transfer bulk files.

Note: HTTP is the primary protocol used for communication between web browser and servers.

## 11.14 Web Browser

Web browser is used to access the World Wide Web. Since they get information from a server, they are also referred to as Web clients. When WWW browser is started or a hyperlink is followed, the browser (acting like a client) sends a request to a site on the Internet. That site (acting like a server), then returns a file, which the browser then has to display. The different types of browsers available include **Mosaic, Netscape navigator, Internet Explorer, Mozilla Firefox, Safari, Opera, Google chrome, etc. First web browser Mosaic was developed in 1993.** Netscape Navigator was the second browser developed in 1994 and Microsoft introduced Internet Explorer as a browser in 1995.

## 11.15 Evolution of Web (Web 1.0, 2.0, 3.0, 4.0 and 5.0)

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## Web 1.0

- Early internet that persisted until about 2000.
- It was the version of the internet that existed before social media and video streaming.
- Websites were just places you could read the information posted on servers and interact with such servers in simple ways.
- There were search engines, and there were e-commerce sites like AltaVista, Netscape, Amazon and eBay.

## Web 2.0

- It started roughly around 2003 when the term was coined by Dale Dougherty, Vice President at O'Reilly Media.
- It is far more interactive, far more collaborative, and far more capable as users were not mere idle visitors, they could create and upload their own content because of the improved bandwidth available to users, and servers.
- It is this generation of the web that gave us smart phones and mobile computing.

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- Web 2.0 could support near real-time interactions and thus collaborative activity was feasible.
- Social networks like Facebook, Twitter and Instagram became part of this and also graphical multiplayer games came into picture.
- It also included the birth of Big Data and the machine learning algorithms that sifted through it.

## Web 3.0

- It is defined by intelligence.
- This intelligence is not just in interactions between people and websites, but between software and software.
- The model is a decentralised internet to run on blockchain technology and it would be different from the versions in use i.e. Web 1.0 and Web 2.0.
- In Web 3.0 users will have ownership stakes in platforms and applications unlike now where tech giants control the platforms.
- Web 3.0 will be based on Inter-Planetary File System (IPFS).
- IPFS will replace the current DNS system (Web 1.0 and 2.0 work on DNS System).
- IPFS is also a postal system, but it is not centralised around a group of root servers like the HTTP protocol.

## Properties

### *1. Digital Identities*

With Digital IDs, individuals can only have one verifiable identity, since each ID has to be linked to a unique credential, like a birth certificate.

Web 2.0 is infested with cybercrime – dark deeds of every description from identity theft to click fraud can all be avoided or atleast reduced by digital identities.

### *2. Semantic Web*

Tim Berners-Lee coined the term for a web of data that can be processed by machines.

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The semantic web will help teach the computer what the data means and that will evolve artificial intelligence which can utilise that information.

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The core idea is to create a spiderweb of knowledge across the internet which will help it to understand the meaning of words to generate, share, and connect content through search and analysis.

### 3. Artificial Intelligence (AI)

AI will allow websites to filter and present users the best data possible.

Example: A group of people may get together and give positive ratings to an undeserving movie, boosting their ratings up.

Artificial intelligence can learn how to separate the good from the bad and gives us reliable data.

### 4. 3D Graphics

Web 3.0 is going to change the future of the internet developing from the simple 2D web into a more realistic three-dimensional cyberworld.

The three-dimensional design is being used extensively in websites and services in Web 3.0 such as online games, e-commerce, real-estate industry, etc.

### 5. Ubiquitous

Ubiquitous means the idea of existing or being everywhere, especially at the same time i.e., omnipresent.

Development of mobile devices and internet access will make the Web 3.0 experience available anywhere at any time.

The internet will no longer be only on your desktop like with Web 1.0, or your smartphone, like Web 2.0, it will be omnipresent.

Web 3.0 may as well be called the web of everything and everywhere, as most things around are connected online with Internet of Things.

### Advantages

- Increased information linking: Semantic web will help in the connectivity of online data.
- Efficient searching.

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- Better marketing.
- More efficient web browsing.
- Effective communication.

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### *Disadvantages*

- Less advanced devices will not be able to handle Web 3.0.
- Web 1.0 websites will seem much more obsolete.
- It can be very complicated for newcomers to understand.

## Web 4.0

Web 4.0, known as the Symbiotic Web, focuses on the interaction between humans and machines. It is also known as 'Intelligent Web'.

## Web 5.0

Recently, **former Twitter CEO Jack Dorsey announced his vision** for a new decentralised web platform that is being called Web 5.0.

Web 5.0 is aimed at "**building an extra decentralised web** that puts **you in control of your data and identity**" that is return of "**ownership of data and identity to individuals**".

- It is being developed by Dorsey's Bitcoin business unit "**The Block Head" (TBH)**.
- According to TBH, "**The web democratised the exchange of information, but it's missing a key layer: identity. Today, identity and personal data have become the property of third parties**."
- This is what is the **purpose of Web 5.0 giving the identity ownership to an individual who owns it rather than third parties**.
- Web 5.0 is **Web 2.0 plus Web 3.0** that will **allow users** to '**own their identity**' on the Internet and 'control their data'.

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Note: Web 3.0 is not truly decentralised or owned by its users, but is instead controlled by various "venture capitalists and limited partners".

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- **Web 5.0 (Intelligent web / symbiotic web)** also called the **sensory and emotive Web**, is designed to develop computers that interact with

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human beings.

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- This **relationship will become a daily habit for many people** in the coming future with the augment of Web 5.0.
- Web 5.0 will see **applications able to interpret** information on **more complex levels, emotionally as well as logically.**
- **It will focus on the individual**, perhaps allowing a website to convey a different experience for each different person.
- **It could perceive the emotions of an individual** and respond appropriately, and it could detect subtleties that enable more powerful interactions.

**Cases for Web 5.0:** There are two use cases that Web 5.0 will change things in the future –

1. **Changes the "Control of Identity":** An individual can switch apps whenever she wants, taking her social persona with her. E.g. A person's detail on various social media.
2. **Gives Users Control Over Their Own Data:** An individual can keep his data in his decentralised web node. E.g. Music on various apps will identify your choices without feeding information.
   (Refer **Infographic – Web 1.0 vs Web 2.0 vs Web 3.0 vs Web 4.0 vs Web 5.0**)

## 11.16 Computer Security

'The protection of data against accidental or intentional destruction, disclosure or modification is termed as Data or Information Security'.

All the technological safeguards and managerial procedures applicable to computer hardware, software and data is included under computer data - security. It ensures that organisational and individual privacy are protected. Various types of computer security tools used.

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## Web 1.0 vs 2.0 vs 3.0 vs 4.0 vs 5.0

- Web 1.0 represents the **initial version of the internet**, characterised by:
  - **Static Pages:** Minimal interactivity
  - HTML 3.2 Elements: Frames and tables for layout.
  - HTML Forms: Sent *via* email.
  - Server File System: Content from the server's file system, not databases.
  - Graphics: **Use of GIF buttons and simple graphics.**
  - **"Read-Only" Internet:** Users could view and read content with limited interaction.
  - Digital Encyclopedia-Like: Interconnected static pages using hyperlinks.
- **Web 1.0 functioned like a digital encyclopedia, providing static content with minimal user interaction.**

- Web 2.0 represents the second generation of the internet, characterised by:
  - **Dynamic Content: Pages that change in response to user input.**
  - User-Generated Content: Platforms for users to create and share content (e.g., blogs, social media).
  - Interactivity: Enhanced interaction through technologies like JavaScript, AJAX, and HTML5.
  - **Social Media: Enables users to create, share, and discuss content (e.g., Facebook, Twitter).**
  - Collaborative Projects: Multiple users can contribute and edit content (e.g., Wikipedia).
  - Rich User Experiences: **Improved graphics, multimedia, and interactive elements.**
  - **APIs and Mashups:** Integrates different services and data sources to create new functionalities.
  - Tagging and Folksonomies: Users categorise and organise content using tags.
- Web 2.0 marks the **shift from the "read-only" web to a more interactive and participatory web,** fostering community-building and collaboration online.

- Web 3.0, also known as the **Semantic Web**, represents the next evolution of the internet with the following key features:
  - Semantic Web: Technologies like RDF and OWL enable machines to understand and interpret data.
  - **Artificial Intelligence:** AI and machine learning provide personalised and relevant user experiences.
  - Interoperability: Seamless data exchange between different systems and platforms.
  - **Decentralisation: Blockchain technology** creates a more secure, transparent, and decentralised web.
  - Ubiquitous Connectivity: Accessible across various devices and platforms.
  - Enhanced User Experiences: More intuitive, personalised, and interactive experiences.
  - Data Ownership: Users own and control their data, enhancing privacy and security.
  - **Smart Contracts:** Blockchain-based contracts execute automatically when conditions are met.
- Web 3.0 aims to create an **intelligent, connected, and decentralised web**, making data more meaningful and accessible, and empowering users with greater control over their digital interactions.

- Web 4.0, known as the **Symbiotic Web**, focuses on the **interaction between humans and machines**, characterised by:
  - Symbiotic Interaction: Autonomous, proactive systems driven by self-learning agents and advanced semantics.
  - **Automation and AI:** Utilises deep learning models and natural language processing for insights and decision-making.
  - Integration: **Combines hardware, software, and data for personalised experiences.**
  - **VR and AR:** Expands possibilities with virtual and augmented reality applications.
  - Business Opportunities: Offers personalised experiences, AI-driven automation, new products, and innovative customer engagement.
  - Future Potentials: Natural interactions with autonomous agents, advanced AI systems, powerful hardware/software, and immersive VR/AR experiences.
- Web 4.0 represents an evolution that will revolutionise human-machine interaction and unlock new potentials.

- Web 5.0, developed by **Dorsey's Bitcoin business unit**, aims to create a decentralised web with the following key features:
  - User Control: Empowers users to own their online identity and control their data.
  - **100% Decentralisation:** Aims to overcome limitations of previous versions and eliminate censorship and major outages. (That's why it is 100% decentralised)
  - **Combines Web 2.0 and Web 3.0:** Integrates user-generated content and decentralised systems.
  - **Emotional Intelligence:** Fosters a deeper connection between humans and computers.
  - **Blockchain Technology:** Implements decentralised identity control, data ownership, and improved privacy.
  - User-Centric Experience: Strives for a more efficient, transparent, and user-focused internet.
- Web 5.0 seeks to empower users, providing greater control, privacy, and a more personalised internet experience.

## 11.16.1 Firewalls

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A device that guards the entrance to a private network and keeps out unauthorised or unwanted traffic is termed as Firewall. It can be implemented in both hardware and software, or a combination of both. Firewalls are frequently deployed to prevent unauthorised Internet users from accessing private networks connected to the Internet, especially intranets. Firewalls, as a security policy, distinguished 'good' traffic from 'bad' traffic.

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### IMPORTANT MILESTONES

**TCP is developed:** Vinton Cerf and Robert Kahn designed TCP during 1973.

**First commercial network:** In 1974, the commercial version of ARPANET, known to be Telnet, was introduced and many considered it to be the first Internet Service Provider (ISP).

**Ethernet is conceived:** Bob Metcalfe developed the idea of Ethernet in 1973.

**HTML:** Tim Berners-Lee developed HTML in 1990. It made a huge contribution to how we navigate and view the Internet today.

**USB Dialers:** Various Internet Service Providers like BSNL, Reliance and Tata, etc., are providing their services through USB Dialers. A pen drive-type small device, with a SIM card fitted in it, employs CDMA technology and offers a good speed. It may be connected to any Laptop or Desktop for Internet access.

## 11.16.2 Rootkit

A rootkit is designed to conceal the compromise of a computer's security. As a set of programmes, it works to subvert control of an operating system from its legitimate operators. It may include the replacements for system; binaries are included in rootkit for immobilising the genuine user during the detection of intruder presence on the system by looking at process tables.

Hacker: A 'clever programmer' or 'someone who tries to break into computer systems' is known to be a Hacker. A hacker can be an expert programmer. There are two types of Hackers: (1) Ethical Hacker(2) Malicious Hacker.

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Note: Malicious Hacker, known as cracker, refers to those who break into computers without authorisation.

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## 11.17 Common Security Breach Methods

### 11.17.1 Computer Virus

Computer virus (a software programme) is developed by a programmer to infect other programmes. The viruses adversely affect the computer system in the following manner: Destructs file allocation table (FAT); Erases specific programme and data on discs; Alters contents of files; Suppresses execution of RAM; Destroys part of the programme/data held on disc *via* bad sectors creation; Reduces free space on disc; Formats discs or tracks on discs in a different way.

### 11.17.2 Worm

A 'worm' is also a self-replicating programme, just like the virus with major difference of not creating copies of itself on one system. It propagates through computer network.

### 11.17.3 Trojan Horses

A Trojan horse is a malware programme, which seems to mislead the users by doing one thing, but in reality, another thing is being done and is used in the setting up of a rear door into the system, thereby enabling the intruder in getting access to users' personal information afterwards.

### 11.17.4 Spoofing Attack (Phishing)

A programme, system, or website successfully masquerading as another by falsifying data and thereby being treated as a trusted system by a user or another programme, is included in spoofing attack. It is aimed in deceiving

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the systems, programmes or users through disclosing confidential data, such as user names and passwords, to the hackers.

### 11.17.5 Spyware

It is a software programme that gathers personal information secretly and sends it from the computer without the user's knowledge through the Internet connection. Adware software programme is an example of a spyware.

## MALWARE OR MALICIOUS SOFTWARE

Malware is any programme or file that is harmful to a computer user. It includes computer viruses, worms, Trojan horses and spyware. They create various types of problems like stealing of data, encrypting or deleting sensitive data, altering or hijacking core-computing functions and they start to monitor computer without permission.

## IMPORTANT LAUNCHES

- Wikipedia was introduced by Jimmy Wales on 15 January 200l.
- **YouTube,** launched by Javed Karim, Chad Hurley, Steve Chen in 15 February 2005.
- The **Pandora Internet radio service** was launched on 25 August 2005. Bing was launched by Microsoft as a search engine on 3 June 2009.
- **Orkut** was launched officially on 24 January 2004 by Google engineer Orkut Buyukkokten.
- **Google:** It was first incorporated as a privately held company on 4 September 1998. **Founders**: Sergey Brin and Larry Page.
- **Yahoo:** (Yet Another Hierarchical Officious Oracle) started in Jan 1994 by Jerry Yang and David Fillo.
- Vinton Cerf and Robert Kahn invented **TCP/IP protocol** in the 1970s and Ray Tomlinson introduced network email in 1972.

## 11.18 Global Internet Governance

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The way in which the Internet is managed is known as Internet governance.

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### 11.18.1 International Telecommunication Union (ITU)

The International Telecommunication Union (ITU) is the United Nations specialised agency, which was originally founded as the International Telegraph Union in 1865. Its headquarters is at Geneva, Switzerland. It holds responsibilities, which are concerned with information and communication technologies. The ITU not only manages the usage of shared radio spectrum globally but also undertakes the promotion of worldwide collaboration in the assignment of satellite orbits. It significantly performs on infrastructure improvisation in respect of telecommunication towards development and coordination of worldwide technical standards.

#### *The ITU consists of three sectors:*

- **Radio Communication (ITU-R)** – ensuring effective, fair and balanced usage of the radio frequency (RF) spectrum.
- **Telecommunication Standardisation (ITU-T)** – prepares the standardising telecommunication operations recommendations worldwide.
- **Telecommunication Development (ITU-D)** – aids countries in development and maintenance of internal communication operations.

There are **193 ITU Member states, consisting of all UN member states excluding the Republic of Palau, plus the Vatican City.**

**ITU membership is open to only UN members,** which may join as Member States, **and also to private organisations such as equipment manufacturers, carriers, funding bodies, organisations of research and development and international and regional telecommunication organisations, which may join as nonvoting Sector Members.**

#### *Functions*

- The ITU defines and circulates rules and principles significant to electronic communication and broadcasting technologies of all types such as radio, television, satellite, telephone and the Internet.

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- The organisation carries out working parties, study groups and meetings for addressing present and future issues and for resolving disputes. Every four years, the ITU arranges and holds an exhibition and forum termed as the **'Global TELECOM'**.
- Another significant feature of the ITU's mandate is to assist developing countries for establishing and developing telecommunication systems of their own.
- Even though the ITU recommendations are non-binding, the majority countries adhere to them for keeping up a useful environment of international electronic communication.

Note: The International Telecommunication Union (ITU) has declared that it will set up its South Asia Area Office and Technology Innovation Centre at New Delhi.

### 11.18.2 ICANN

It stands for **Internet Corporation for Assigned Names and Numbers** and plays an important role in the global Internet's system coordination of sole identifiers, particularly, ensuring the operation in a secure and stable way. It was formed in 1998 as a non-profit partnership of people from all over the world to secure Internet. ICANN develops policy on the Internet's unique identifiers but it does not control content on the Internet.

ICANN is an organisation incorporated under the law of the State of California (USA). ICANN is accountable by the judicial system, i.e., ICANN can be taken to court. ICANN has a board of directors, who are legally responsible for their duties under this law. It is a non-profit public benefit corporation.

Global internet body, ICANN and Indian IT industry body NASSCOM are to develop identifier technology that can be used for managing devices and infrastructure using the internet.

It will first focus on updating the Internet of Things (IoT) devices using domain name system (DNS) even in the situation when the manufacturer or supplier has closed down the business.

This collaboration provides a structure to jointly identify research projects, particularly in new technologies related to the internet's system of

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unique identifiers.

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**FOUR MAJOR FUNCTIONS OF ICANN**

- **Companies' approval** that can become recognised registrars for domain names.
- **Decision making** in regard to the new Top-level Domains (TLDs) additions to the Root system.
- **Manage technical factors** for maintaining widespread connectivity.
- **Forming a Uniform Domain Name Dispute Resolution Policy (UDRP)** for contending domain names.

### 11.18.2.1 Domain Name System (DNS)

The DNS is an arranged disseminated system for naming computers, services or any resource, which is linked to the Internet or any private network. The DNS is used for the translation of simply memorised domain names into the IP addresses in the numerical form. The DNS is an essential component of the functionality of the Internet, as it provides a worldwide, distributed keyword-based redirection service.

#### *Domain Name*

A domain name, such as 'abc.com', works as a string of classification, for defining the administrative autonomy, authority or control area on the Internet. These are assigned according to its existing rules and procedures and any name, which is DNS registered is a domain name. Normally, it represents an Internet Protocol resource and is used for the naming and addressing purpose and in the context of networking.

### 11.18.2.2 What is the Issue?

Though there is no sole owner of Internet, it is mainly controlled by USA because of ICANN; (Recently, the control over ICANN was renounced by USA) but other countries are advocating oversight of United Nations for Internet Governance.

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As a result of recent surveillance programmes of USA, the right to control Internet available in the respective country is being demanded by many countries, which includes India (for security purpose). On the other hand, global powers like Russia and China want Internet controls as a demonstration of their super power.

### 11.18.2.3 Electronic Surveillance Disclosures

- Electronic surveillance by USA created much buzz across the world.
- Brazilian President Dilma Rousseff, during her address in the United Nation General Assembly, complained about supposed surveillance and asked the UN for taking a leading role in an attempt to regulate the conduct of states in respect of new 'multilateral mechanisms for the worldwide network'.

### 11.18.2.4 USA's View

- Freedom of Internet is one of the columns of foreign policy of USA.
- Internet freedom will be defended at any cost (i.e., by controlling and keeping the Internet rights only with USA).
- The present proposal of transferring Internet control to United Nations would provide UN with greater power.
- If the Internet control will be handed over to UN, then it will direct to greater suppression and a spectacular anonymity reduction.

### 11.18.2.5 The Real Debate

- The real debate is whether to possess an open Internet globally or a national Intranets patchwork.
- It is also being discussed that UN be given Internet control and the individual nations be allowed the Internet addresses management in their own areas. At present, there are two groups, the demand of one group is multilateral platform consisting of India, Russia and China and the other group's demand is multi-stakeholder platform consisting of USA.
- Countries are demanding more sovereign control over web addresses.

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- The threat expected is that once every country is allowed to manage their own Internet address books, sites seen as troublesome by the governments could be easily and silently eliminated by removing them from the index and making them permanently inaccessible to the outside world.
- Already, the governments at their discretions are having powerful tools with them, such as the regulatory authorities over telecommunications companies such as China's control over the Internet that gives it an edge for paralysing the Internet services when it wants. Then why is there a need of an extra control on Internet?

#### 11.18.2.6 Why India is Worried?

- India was authorised to contain root servers (computers through which all Internet data originating or coming to a particular country passes through) but their management was not given to the country. No servers' control and management is allowed to India.
- This type of policy is of no use to the country. For the country's security reasons, it requires their control.
- India is concerned about the proximity of China, Russia and USA while deciding on issue of Internet governance.
- India believes that these countries would formulate a strategy, which will ultimately be forced upon other countries. Hence, the country is pressing its demands and voice in order to meet all the country's problems before the formulation of any strategy.
- India is inclined towards a pre-dominate joint approach (each country be provided a power to speak over this issue) on Internet governance issue rather than multi-stakeholder approach (only few countries are authorised to speak), which is mostly followed by the West.
- India has the world's second largest number of Internet users, with nearly 820 million active internet users, just after China, which is 1.05 billion.

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## 11.19 Internet of Things (IoT)

- IoT is the network of physical objects, which is accessed through the Internet, as defined by technology analysts and visionaries, comprising of embedded technology to interact with internal states or the external environment.
- IoT is the way of connecting new places like transportation systems, manufacturing floors, energy grids and healthcare facilities through the Internet. Through the help of this technology, an object can represent itself digitally and can be controlled from anywhere easily. This connectivity will bring more ways to increase efficiency and improve safety and security.
- Through the help of the IoT, all objects, animals or people are provided with unique identifiers and easily transfer data over a network without requiring any interaction, be it human-to-human or human-to-computer. IoT has evolved from the convergence of Internet, wireless technologies and micro-electromechanical system (MEMS).
- IoT has a broad scope; a heart patient can monitor his/her heart easily; farm animals can be easily tracked with the help of biochip transponder; an automobile that has built-in sensors can alert driver when tire pressure is low. Any natural or man-made objects that can be assigned by an IP address have the ability to transfer data over a network. Nowadays, IoT has been most closely associated with machine-to-machine (M2M) communication and various other uses like power, oil and gas utilities. IoT is to provide ubiquitous connectivity with many features like- **Online monitoring, positioning and tracking, alarm linkage, command and display functionalities, plan management, security and privacy controls, remote maintenance, and Online upgrades, etc.**

***India's first 'Internet of Things' policy: One step closer to Smart and Digital India***

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Department of Electronics and Information Technology (DeitY) now called as the Ministry of Electronics and Information Technology had drafted India's first 'Internet of Things Policy'.

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## *India's First IoT policy*

Two major efforts taken by the Government of India, which will lead to a rapid growth of IoT industry are Smart Cities Project and Digital India Programme. These two major initiatives by the GoI will definitely boost the IoT industry and pave the way forward for it.

## *What the policy had in-store for the IoT industry?*

- As part of this policy, GoI will fund research and development for specific applications in IoT. Also, DeitY plans to initiate an innovative project named 'International IoT Research Collaboration scheme (IIRC)' to get investments from private sector for research and development.
- In order to support the growth and boost investment in IoT, government will promote venture funds, which provide support to organisations in IoT-related domains like memory, processors, sensors, low-power devices and solar electronics.
- This policy will be supporting high-quality manufacturing of IoT devices in India without affecting the environment. 100 per cent duty benefit will be available on raw materials imported for manufacturing IoT devices, thereby helping organisations to maintain their expenses.
- Also, domestically purchased raw materials will get excise and CST reimbursement. These benefits will lower the costs of IoT devices on the whole, benefitting the organisations that are manufacturing as well as using them.
- Development of human resource is important for any industry to be successful and with this objective, DeitY will be responsible for creating an IoT Education and Awareness programme with the aim to develop skillsets required for IoT at various levels to allow smooth functioning of the industry.

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## *India's first IoT platform*

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- Home-grown technology and IoT company **Smartron** unveiled **'tronX'** – an Artificial Intelligence (AI)-powered IoT platform that would help make users' daily life easier and smarter. Terming it as one of the first global technologies being developed in India, the company said **'tronX' is an intelligent ecosystem that helps connect a range of devices.**
- Built on the world of 'Internet of Trons', the ecosystem allows instant access to profile, data, content, services, community and other IoT devices whether you are at home, in the car or at the office. 'Smartron has been working for more than two years on creating a new connected ecosystem fuelled by AI-powered "IoT" and "tronX" is at the core of this brave new world'.
- To begin with, Smartron announced four verticals that are in the works at its research and development facility and already have built-in capabilities that integrate in our lives: **'tronX Personal'**, **'tronX Health'**, **'tronX Home'** and **'tronX Infra'**. **'tronX Personal'** offers personalised entertainment, travel, shopping, financial, health, event, communication and collaboration experiences across devices and contexts.
- With 'tronX Health', the system monitors and manages medical history, insurance data and offers intuitive and preventive solutions from a range of partners, who are already on board. 'tronX Home' helps control locking mechanisms, security updates, home appliances, services like water, electricity and payment modules for these services, all available through a Voice Activated Assistant.
- 'tronX Infra' is Smartron's B2B vertical, offering an AI-assisted system that covers a range of monitoring and management services, thus increasing productivity and security for enterprises. Smartron was founded with a vision to build India's first true global OEM brand to revolutionise the country's ecosystem and create experiences that are at par with global innovations.

[Refer **Infographic – IoT (Internet of Things)**]

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# (Internet of Things)

## What is Internet of Things?

The Internet of Things is a **system made up of physical devices connected by the internet and capable of accumulating and sharing data across a network.** While the platforms used to develop IoT devices can differ, they all have the same core components in common when it comes to their contribution.

## Components of IoT

1) IoT Sensors
2) Connectivity/Network
3) Cloud
4) Hyper Decision Framework
5) User Interface

## IoT Examples

## IoT Growth

- 2015: IoT technology was not a priority for small and mid-sized businesses.
- 2018: 7 billion devices connected to the internet.
- 2020: Projected to reach 10 to 14 billion connected devices.
- 2021: Average person expected to have at least 4 connected devices.
- 2025: IoT market value expected to exceed $1.5 trillion.

## How IoT Works?

1

"The Data needs to be sent to cloud to be analysed, but it needs a way to get there."

### Data Ingestion

IoT devices and sensors gather data from the surrounding environment, ranging from basic measurements such as temperature and humidity to more complex information like full video feeds. The amount of data produced by IoT devices every day is staggering, reaching nearly quintillions of data bytes.

2

"To ensure the data security protocols such as Bluetooth, LORA, Zigbee, etc. are used."

### Data Transmission

The data collected by IoT devices is sent to the cloud using telemetry devices called gateways. These gateways employ a combination of cellular and satellite communication to transmit the data.

3

### Data Processing

Once the data arrives in the cloud, it is processed by an IoT platform. The processing may involve simple operations like checking if the temperature falls within the acceptable range, or it can be more intricate, such as applying computer vision algorithms to identify objects in a video feed.

4

### Data Visualisation

Processed data is converted into useful information for users through alert notifications *via* email, text messages, or notifications. Users can also utilise an application or interface to proactively monitor and engage with the system.

5

"One can make intelligent business decisions based on the insights and predictions generated from the data."

### Data Analysis & Prediction

Data analytics utilises historical data to derive actionable insights, enabling the prediction of future events. For instance, by analysing data, it becomes possible to anticipate potential machinery malfunctions.

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# 11.20 Net Neutrality and Internet Fast Lanes: The Debate

## 11.20.1 Net Neutrality

Net Neutrality is also called as network neutrality or Internet neutrality is the guiding principle of equality that Internet service providers and governments should follow. It implies that all Internet data should be given equal treatment with no discrimination or differential charges for users, content, site, platform, application, type of attached files or mode of communication.

In simple words, it implies no restriction to access any type of content on the web, no restrictions on downloads or uploads, and no restrictions on communication methods (email, chat, etc.). It also concludes that access will not be blocked, slowed down or sped up on the basis of owner of the access point(s).

## 11.20.2 Why is Net Neutrality Important?

- **Growth:** Since its creation by Sir Tim Berners-Lee (in 1991), it became the chief reason of the phenomenal rate of web growth.
- **Creativity:** Creativity, innovation and unbridled inventiveness have given us Wikipedia, YouTube, Google, Torrents, the Internet Movie Database, and many more.
- **Communication:** It is because of this net neutrality we enjoy the access and ability to communicate with people on a personal basis, government leaders, business owners, celebrities, work colleagues, medical personnel, family, etc., with no restrictions.

## 11.20.3 Issue

- The US Federal Communications Commission recently jeopardised the prospects for 'net neutrality', a principle requiring non-discrimination by Internet service providers between their client websites, when it

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voted in favour of a proposal to allow ISPs to charge higher fees to those who seek to deliver higher-quality content in the US.

- This raised a debate of Internet Governance and Free access of Internet by the users.

**KEY POINTS**

- Net neutrality means that users get unrestricted access to Internet traffic without any discrimination.
- In its real sense, net neutrality is 'Freedom of Speech and Expression without partiality'.
- In recent past, there have been violations of net neutrality by certain governments and companies.
- The Federal Communications Commission (FCC) enacted the Open Internet Order in 2010 in order to prevent large telecommunication firms from stifling competition and innovation online.
- The FCC, in its order, stated that net neutrality rules were intended to 'preserve the Internet as an open platform, enabling consumer choice, freedom of expression, end-user control, competition, and the freedom to innovate without permission'.
- Few countries like Chile have enacted laws to protect net neutrality.
- In India, there is no net neutrality law and it is not regulated by government. India is yet to come up with a transparent and impartial law.

### 11.20.4 Telecom Commission Approved TRAI Recommendations on Net Neutrality

- **Telecom Commission (a group within India's Department of Telecom)** has agreed to the recommendations given by **TRAI** on net neutrality in 2018.
- The move plans in ensuring that all web traffic is treated in a fair manner, and that Internet service providers won't obstruct, choke or favour any material or services (with a few reasonable exceptions).

#### *The rules*

- According to the country's net neutrality rules, mobile operators, Internet providers and social-media and Internet companies cannot indulge in, or ask for, special treatment as from now there will be prohibition on interference of any kind in the content treatment,

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consisting of practices such as **jamming, mortifying, slowing down or giving way differential speeds or treatment to any content.**

- Any efforts to build **'zero rated platforms' have now been stopped.** Zero-rated platforms, which had earlier been attempted (by companies such as Airtel and Facebook) but banned, offer free only a certain category of services and websites, thus forming paid layers and stifling competition and modernism.
- Under net neutrality, **access of online contents is not restricted and non-discriminatory.** The only exceptions are fresh and rising services such as **autonomous driving, tele-medicine or remote-diagnostic services, which may need focussed Internet lanes and faster-than-normal speeds.**
- A **committee will look after** into the probable exceptions for 'critical services' which will also be defined considering the basic doctrine of net neutrality.

## RECENT NET NEUTRALITY ISSUE

In recent past, many **start-ups have written to Telecom Regulatory Authority of India (TRAI) opposing Telecom Service Providers' (TSPs)** push for regulating Over the Top (OTT) services.

### Over The Top (OTT) Service and Issues

- An OTT service refers to any content, service, or application that is delivered to end users over the public Internet.
- OTT services are typically classified into two main groups:
  - **OTT communications services**, which include messaging platforms like **WhatsApp and Telegram**.
  - **OTT application services**, encompassing a wide range of **services such as media streaming** (e.g., Netflix), e-commerce (e.g., Amazon), cloud services, and social media platforms like Facebook.

### Issues

- Recently, the **Telecom Disputes Settlement and Appellate Tribunal (TDSAT) ruled that OTT streaming platforms fall outside the jurisdiction of the Telecom Regulatory Authority of India (TRAI)**.
- **OTT services do not require any specific permission or license from the Union government to operate**.
- These services are governed by the **Information Technology Rules, 2021**, which were notified by the **Ministry of Information Technology**.

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## 11.21 Network

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- A computer network or data network is as a telecommunication network allowing computers to transfer data in and out. In computer networks, networking devices exchange data with each other using a data link. The connections between nodes are established using either a cable media or wireless media. The best-known computer network is the Internet.
- Various kinds of computer networks can be characterised by their size and purpose. The geographical area occupied and the number of computers forming the network express the size of network. Anything ranging from a handful of devices to millions of devices spread across the whole globe can be covered under network. Here are few networks based on size:
  1. Personal area network, or PAN
  2. Local area network, or LAN
  3. Metropolitan area network, or MAN
  4. Wide area network, or WAN

### 11.21.1 Personal Area Network (PAN)

- Personal area network or PAN implies a network covering a very small area; for example, a small room. Bluetooth is the best-known wireless PAN network technology, while the most popular wired PAN is USB. The wireless headset, printer or the smartphones may not appear as components in a network, but they definitely communicate with each other. Many peripheral devices act as computers in their own right.
- A PAN interconnects information technology devices within the range of an individual person.
- It usually works within a range of 10 metres.
- The concept of a PAN was first developed by Thomas Zimmerman.
- Examples: Bluetooth, NFC, Share-it, Xender.

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### 11.21.2 Local Area Network (LAN)

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- The network that operates within a limited physical area is termed as LAN. It enables linking of several computers within a building, companies, offices and factories in order to share resources and information in close proximity, e.g., Wi-Fi, Hotspot.
  1. Two or more PCs can form a LAN to share files, folders, printers, applications and other devices.
  2. Connections use coaxial or CAT 5 cables.
  3. Since there are short distances, the errors and noises are minimum.
  4. Data is transferred at the rate of 10–100 mbps.
  5. Distinction on their transmission media and topology.
  6. A computer lab in a school, Wi-Fi are common examples of LAN.
  7. Range is nearly 1 km.

### 11.21.3 Metropolitan Area Network (MAN)

- MAN is larger than LAN but smaller than WAN, e.g., Wi-Max, FM, STD code of a city (011-for Delhi).
  1. Designed to extend over a large area.
  2. Resources are shared by connecting number of LANs forming larger network.
  3. The usual range is up to 5–50 km.
  4. The owner can be an organisation or an individual.
  5. Compared to LAN, the rate of data transfer is low.
  6. Organisation with different branches located in the city.

### 11.21.4 Wide Area Network (WAN)

- It covers a wider area, larger than LAN and MAN. It can have multiple ranges like a network connecting multiple buildings on a corporate or college campus to satellite links connecting offices in different countries. The Internet is one of the most popular WAN. In true sense, it represents a collection of other networks, including other LANs and WANs, hence, the name is derived.
- WANs can be wired (using fibre-optic cable) or wireless. A wireless WAN might use microwave or infrared (IR) transmission technology. Even a satellite can be used for transmission. To save money, an

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organisation may opt for wireless technology or lease lines from a third party. E.g., Internet, AM, +91 (Nation Code).

1. Includes country and worldwide network with multiple LANs and MANs.
2. The geographical range is the most distinguishing feature.
3. Uses satellites and microwave relays.
4. The ISP provider guides the data transfer rate, which varies over the location.

## 11.22 Various Modes of Network Connection

### 11.22.1 Bluetooth

- Bluetooth standard derives its name from a tenth century Scandinavian king, **Harald Bluetooth**. Modern Bluetooth technology was developed by Jaap Haartsen and Sven Mattis-son in 1994, working at Ericcson in Sweden. In 1998, the original group of Promoter companies – Ericsson, Intel, Nokia, Toshiba and IBM – came together to form the Bluetooth Special Interest Group (SIG).
- A wireless technology standard (i.e., Bluetooth) enables data exchange over small distances (using UHF short-wavelength **radio waves** ranging from **2.4 to 2.485** GHz) through mobile and fixed devices, thereby forming a kind of Personal Area Networks (PANs).
- A headset or watch, as a Bluetooth product, has a tiny computer chip with a Bluetooth radio and software, which makes the connection easy to connect. The two Bluetooth devices should be paired while communicating to each other.
- Bluetooth is a packet-based protocol with a **master-slave structure (Piconet)**, that is, a master Bluetooth device communicates with seven devices maximum in a piconet (an ad-hoc computer network using Bluetooth technology); however, all the devices do not reach this maximum.
- There are two versions of Bluetooth (**Table 11.2**), which are differentiated on the basis of data sharing distance and power

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consumption.

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**Table 11.2** Bluetooth Versions

| Bluetooth (Version) | Maximum Speed | Data Sharing Distance |
|---|---|---|
| 4.0 | 25 Mbps | 200 feet (60.96 m) |
| 5.0 | 50 Mbps | 800 feet (243.84 m) |

### 11.22.2 Wi-Fi Direct

Wi-Fi direct is the chief competitor of Bluetooth 4.0. Wi-Fi Direct and Bluetooth, both operate at **2.4-gigahertz frequency band.**

***Comparison of Bluetooth 4.0 and Wi-Fi Direct:***

- The speeds offered by **Wi-Fi Direct are up to 250 Mbps** while Bluetooth 4.0 lags behind at a paltry-seeming 25 Mbps.
- The stated maximum distance of **Wi-Fi Direct is more than 600 feet** while Bluetooth 4.0 has 200 feet.
- Only the battery life is less in the case of Wi-Fi Direct in comparison to Bluetooth 4.0.
- Bluetooth 4.0 uses 128-bit encryption to protect your data while transmitting data between devices. On the other hand, the Wi-Fi Direct uses 256-bit encryption. Hence, Bluetooth 4.0 is remarkably secure than Wi-Fi direct. All Wi-Fi Direct operates in 2.4 GHz frequency band.

Note: Share-it and Xender types of application (App) connect through Wi-Fi direct.

### 11.22.3 Cordless Phone

Cordless phone (or portable phone) serves as a replacement of the handset cord with a radio link, which is connected with base station and also

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connected with a fixed telephone line. This technology is based on **radio waves** and **amplitude modulation.** In 1994, bandwidth of the cordless phone was 900 megahertz but today it is **2.45 gigahertz.** One phone is connected with the other when both are located in the radio radar. Today cordless telephones use PHS (Personal Handy-phone System) and DECT ((Digital Enhanced Cordless Telecommunications).

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### 11.22.4 Wi-Fi

Wi-Fi stand as Wireless Fidelity. This type of technology allows electronic devices to connect wireless LAN (Local Area Network). **Vic Hayes** is called the father of Wi-Fi. He created Wi-Fi 802.11 standard in 1997. It is **radio wave** based technology of bandwidth **2.45-5 gigahertz**. It is carried to maximum distance of **100 metres** and a speed of 11 megabytes per second (802.11b standard) – 1.3 gigabytes per second (802.11ac standard). There are various standards of Wi-Fi developed – 802.11b, 802.11a, 802.11g, 802.11n and 802.11ac.

#### WiFi 7 Technology

**Qualcomm, a US-based corporation**, recently **recommended India to implement the most recent Wi-Fi 7 technology.**

#### *What is Wi-Fi 7?*

- Wi-Fi, short for **wireless fidelity, is a technology enabling wireless internet connectivity and inter-device communication.**
- It utilises **radio waves** for high-speed wireless internet and network connectivity.
- Wi-Fi signals are broadcast through three mediums: **base stations, routers, and accessing devices like phones and laptops.**
- The latest Wi-Fi standard, **Wi-Fi 7, is based on IEEE 802.11be, designed for** exceptionally high throughput (EHT).

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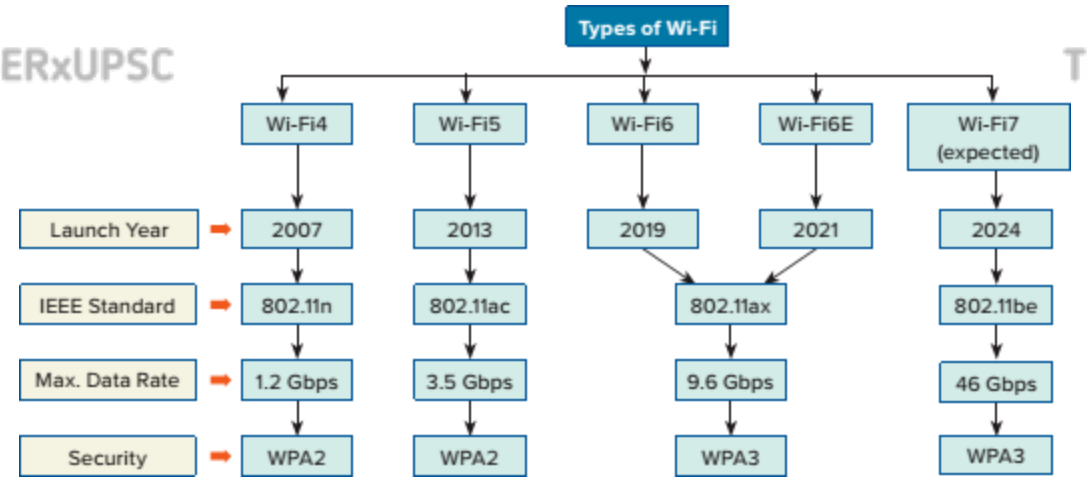


- The Institute of Electrical and Electronics Engineers (IEEE), **the largest technical professional association globally, is dedicated to developing technology** for the betterment of humanity.

## *Key Features of Wi-Fi 7*

1. Allows devices in the 2.4 GHz, 5 GHz, and 6 GHz bands to connect without requiring additional hardware.
2. Combines multiple frequencies from the available spectrum to offer a comprehensive solution, rather than being limited to a single frequency.
3. **Reduced Latency:** Speeds up cloud-based activities like gaming and file transfers.
4. **Multi-Link Operation (MLO):** Enhances network performance by leveraging multiple channels across various frequencies.

## *Benefits of Wi-Fi 7*

- **Alignment with India's Tech Adoption:** Facilitates **rapid adoption of wireless and cloud-based applications**, along with bolstering government digital security and privacy efforts.
- **Positive Impact on India:** Holds promise for enterprise digital transformation, smartphone technology, fixed wireless connections, and artificial intelligence (AI), particularly Edge AI.
- **Meeting Emerging Technology Needs:** Capable of accommodating the growing number of wearables, smart home appliances, and industrial IoT applications.

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- **Advancements in Medical Field:** Enables telesurgery and telediagnostics, enhancing medical practices remotely.
- **Enhanced User Experiences:** Enables smoother streaming of high-resolution media, such as 8K movies, and boosts efficiency in AR/VR, cloud gaming, and other data-intensive tasks.
- **Logistics Management:** Assists in the efficient management of shipping and logistics operations.

### 11.22.5 Wi-max (Worldwide Interoperability for Microwave Access)

Wi-max is the hottest broadband wireless technology. It is expected to provide services to residential and enterprise customers in an economical way. **Its characteristics are similar to Wi-Fi but it can accommodate larger distances and greater number of users.** It carries MAN network bandwidth of 1 gigabits per second. It is a long-term system covering many kilometres in the cases of Internet connection. It needs a base station where service is provided in greater number and carries the **microwave access.** Initially, it provided 30–40 megabits per second but from 2011 it is providing up to **1 gigabits per second**.

### 11.22.6 NFC (Near Field Communication)

Near Field Communication (NFC) represents a set of **short-range wireless technologies** that typically requires a distance of **4 cm or less to initiate a connection.** This network carries **point-to-point range of up to 20 cm** and operates on **13.56 megahertz frequency** at rates ranging from **106–424 kilobits per second** (whereas Bluetooth 4.0 is 25 megabits per second). NFC always works on initiator and target principle wherein the initiator actively generates an **RFID** field that can power a passive target. Firstly, Sony was approved for NFC but now so many mobile companies are interested about NFC.

Note: In 1983, the first patent with the abbreviation 'RFID' (Radio frequency identification)

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was granted to Charles Walton.

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### *Comparison between NFC and Bluetooth*

Both, NFC and Bluetooth represent short-range communication technologies, which are available on mobile phones. NFC does not require pairing and operates at slower speeds (as compared to Bluetooth), but consumes far less power. NFC needs less time to set up than standard Bluetooth; however, the transfer rate is slower than Bluetooth. The connection between two NFC devices is automatically established in less than 0.1 second, rather than Bluetooth *via* manual configurations to identify devices.

## 11.22.7 Li-Fi (Light Fidelity)/Visible Light Communication (VLC)

University of Edinburgh Professor, **Harald Haas** coined the term Li-Fi in 2011. His invention was demonstrated using an ordinary table lamp, which successfully transmitted data at speeds exceeding **10 megabits per second** using **light waves** from LED light bulbs to a computer located below the lamp. He periodically blocked the beam of light, which caused the connection to drop. This proved that the light bulb was the source of the data stream. The LED bulb will hold a microchip that processes the data. We can manipulate the light intensity to send data by tiny changes in amplitude. The visible spectrum of light, a part of the electromagnetic spectrum (that is still not greatly utilised) is used by this technology. The inventor of Li-Fi, Harald Haas called the light used to transmit the data as **D-light**. In future, data for laptops, smartphones and tablets can be

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transmitted through the light in a room by using Li-Fi. Li-Fi is a **Visible Light Communications (VLC)** system running wireless communications travelling at very high speeds, boasting speeds of **up to 224 gigabits per second.** Li-Fi is many times better than Wi-Fi (**Table 11.3**).

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## Li-Fi Technology

**LiFi** is a **wireless communication technology** that transmits data using **visible light, particularly emitted by LED bulbs.**

- Functions similarly to WiFi but offers:
  - ❑ **Faster speeds**
  - ❑ **Lower latency**
  - ❑ **Wider bandwidth** (430–770 THz)

## How LiFi Works

1. **LED Modulation:** LEDs flicker at high speeds to encode binary data.
2. **Transmission:** Light carries the modulated data.
3. **Reception and Decoding: Photodiodes** receive the light signals and convert them into digital data.
4. **Bidirectional Communication:** Allows data transmission both ways using separate light paths.
5. **Suitable Spectrum:** Primarily uses the **visible light spectrum**, but infrared/UV can also be used.

## Key Components of LiFi

- **LED Transmitter:** (LiFi-enabled LED fixtures acting as transmitters)
  - Sends encoded data via modulated light.

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- **Photodiode Receiver:** Captures modulated light and decodes the data.
- **Controller Circuit:** Manages signal modulation and timing.
- **Software Stack:** Handles encoding/decoding, encryption and error control.

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## *LiFi Ecosystem*

- **Optical Access Points:** LiFi-enabled LED fixtures acting as transmitters.
- **LiFi Dongles:** Attach to devices to receive LiFi signals.
- **Power Lines:** Use existing power infrastructure to enhance data coverage.
- **Optical Wireless Heads (LiFi Femtocells):** Offer dense, localised connectivity.
- **Management System:** Controls network performance and handovers.
- **Backhaul Integration:** Connects access points to LAN/internet.

## *LiFi Vs WiFi*

| Feature | LiFi | WiFi |
|---|---|---|
| Technology | Visible Light Communication | Radio Waves |
| Frequency Band | 430–770 THz | 2.4 GHz, 5 GHz |
| Max Speed | Up to 224 Gbps | Up to 9.6 Gbps (WiFi 6) |
| Latency | Very low (<1 ms) | Higher (>10 ms) |
| Security | High – confined area | Moderate – can penetrate |
| Energy Efficiency | Very high – uses LEDs | Moderate |
| Interference | Minimal | Prone to EM interference |
| Outdoor Use | Challenging (sunlight) | Very good |
| Deployment Cost | Low (uses lights) | Higher (routers) |

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## Applications of LiFi

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- **Indoor Broadband:** Malls, offices, homes
- **Hospitals and Aircraft:** No EM interference
- **Underwater communication**
- **Smart Cities:** Lighting + connectivity
- **Vehicles:** Inter-vehicular communication
- **AR/VR and Industry 4.0** support
- **Rural/Remote Connectivity**
- **Military and Disaster Response**
- **Wireless Charging + Data Transfer**

## Challenges for Adoption

- Line-of-sight requirement
- Interference from ambient light
- Lack of device compatibility
- Initial higher costs
- Dependency on lighting systems
- Standardisation lacking
- Latency still improving
- Security threats (LED flicker manipulation)

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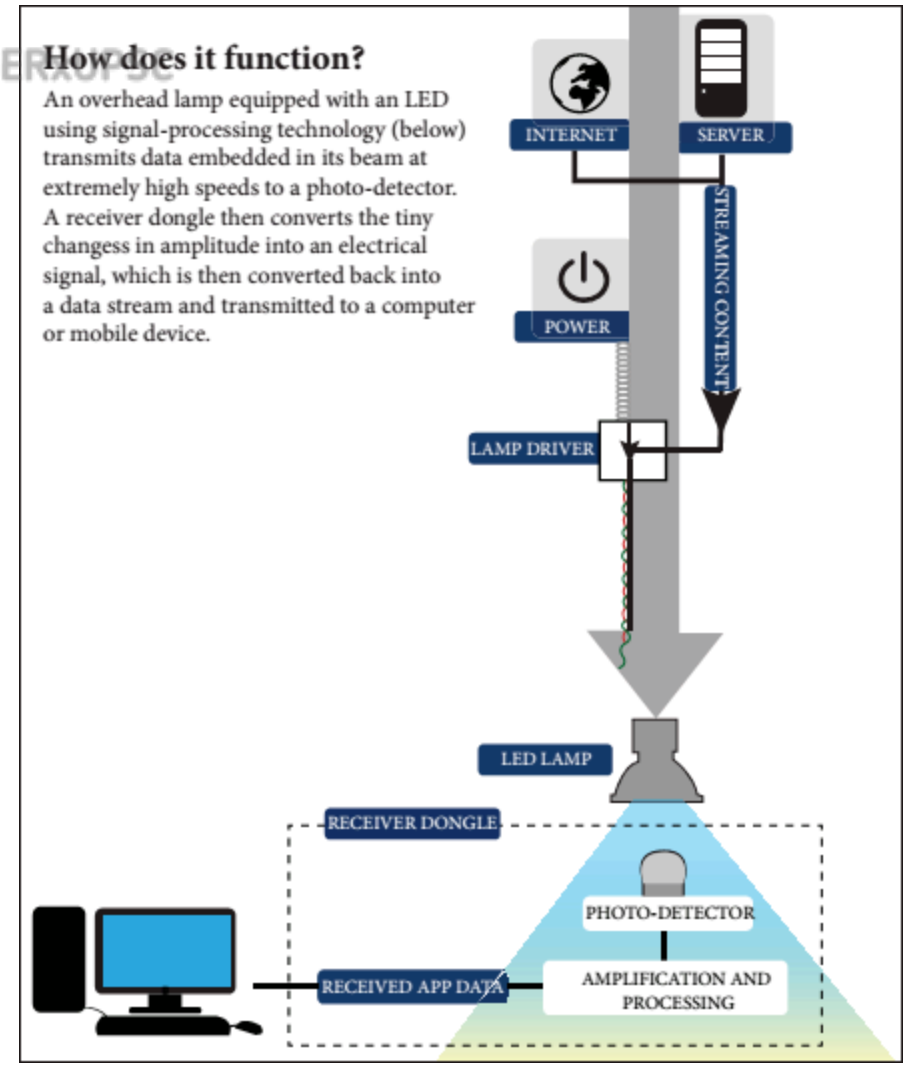


## SHORT RANGE DEVICES (SRD)

Short Range Devices (SRD) are radio devices those have low risk of interference with other radio services because of their transmitted power, and hence their range, is low. 'Short Range Device' may be applied to many different types of wireless equipment, including various forms of:

- Access control (including door and gate openers)
- Alarms and movement detectors
- Closed-circuit television (CCTV)
- Cordless audio devices, including wireless microphones
- Industrial control
- Local Area Networks
- Medical implants
- Metering devices
- Remote control

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- Radio frequency identification (RFID)
- Road Transport Telematics
- Telemetry (the process of recording and transmitting the readings of an instrument)

Short range devices often benefit from a relaxed regulatory regime compared with other radio communication equipments. As a general principle, a user is licence free to operate such equipment, some specific cases may require an individual licence.

However, like all radio equipment, SRDs have to meet the Radio Equipment Directive (RED) to be placed on the market within the European Community. The operation of the equipment is subject to the frequency management regulations of the relevant member state.

## Advantages

1. Li-Fi can be used in traffic signals, which will communicate with the LED lights of the cars, thus reducing the accident numbers. In order to transfer the data, thousand and millions of street lamps can be transferred to Li-Fi lamps.
2. Even the petroleum or chemical plants can use Li-Fi, where other transmission or frequencies could be hazardous.
3. Certain electromagnetic sensitive areas like aircraft cabins, hospitals and nuclear power plants can use Li-Fi with advantage of no electromagnetic interference.
4. In terms of capacity, Li-Fi has almost no limitations. The visible light spectrum is 10,000 times larger than the entire radio frequency spectrum.
5. Researchers have reached data rates of over 224 gigabits per second, which is much faster than a typical fast broadband.
6. In terms of cost-effectiveness, Li-Fi is expected to be ten times cheaper than Wi-Fi.

## Disadvantages

1. Effective for short range
2. Low reliability
3. The installation cost is high

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## 11.23 Virtual Private Network (VPN)

- VPN stands to be a popular way to **support remote workers**, especially in fields where privacy is paramount, like healthcare.
- An additional feature to the process is Remote Desktop Virtualisation. The entire desktop and applications run on a remote server, and are accessed from a client, which can run on a conventional laptop, mobile devices like tablets or even the smartphones. Such a feature makes virtual desktops great for supporting BYOD (bring your own device) schemes. Since the date lives on a central server, even if the device gets lost or stolen, you can be relieved about the safety of the data. The biggest known vendors of virtual desktops are Citrix and VMware.
- When VPN gets connected to a computer (or different device, such as a tablet or a smart-phone), the computer behaves to be on the similar local network as that of VPN. The whole traffic of network is sent to the VPN over a secured connection. The behaviour of computer on the similar network, further assists us in securing access to resources of local network even though we are on the other side of the world. Being present at the location of VPN, we can access Internet along with certain benefits if we are making use of pubic Wi-Fi or accessing the geo-blocked websites.

### *Uses for VPNs*

In spite of being a simple tool, it is used for doing a wide range of things:

- **Accessing a Business Network during Travelling.**
- **Accessing Home Network During Travelling.**
- **Hiding Your Browsing Activity from Local Network and ISP.**
- **Accessing of Geo-Blocked Websites:** Being an American, if you try to access your Netflix account during travelling from USA or want to use American media sites such as Pandora, and Hulu, a VPN connection located in the USA enables you to access these region-restricted services.

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- **Files Downloading:** Many people use VPN connections for file downloads *via* Bit Torrent. It is useful when you are downloading totally legal torrents, even if your ISP is choking Bit Torrent and further extremely slowing it.

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## 11.24 VoIP (Voice over Internet Protocol/Internet Telephony)

VoIP is an IP-enabled voice calling technology over the Internet, e.g., Skype, Yahoo messenger, MSN messenger. It requires broadband connectivity to make a call along with IP-enabled devices like Computers, Smartphone, etc. The voice is converted into digital packets and transmitted to destination over packet switched network.

### *Advantages of VoIP are:*

- Cost of calling is cheaper than normal phone.
- No need to carry a dedicated device for calling if you just have a computer with you.
- Its uses existing LANs so there is no need of dedicated wiring features and hence reduces the complexity of calling.
- Call anywhere anytime; do not worry about roaming features and cost.
- One payment, two services: voice calling and broadband data usage.

### *Disadvantages of VoIP are*

- It is dependent on broadband network connectivity: no Internet, no calling.
- The quality of voice depends on broadband bandwidth and speed.
- Power shortage can hamper VoIP calling as it is totally dependent on power-enabled devices.
- No emergency calling features like normal and Smartphone.
- Highest disadvantage of VoIP is security. It is really tough to trace the source and identity if any imposter is at work.

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- Threats like phishing, spoofing and sniffing, call tampering, etc., are very common.

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## 11.25 Hidden Part of Internet: Dark Net, Dark Web and Deep Web

Internet is **very vast** and what we use on daily basis is only a small part of it.

The **internet** is the global system of **interconnected computer networks** that uses the internet protocol suite (TCP/IP) to **communicate between networks and devices.**

The Internet carries a vast range of information resources and services, such as the **inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, telephony, and file sharing.**

### Web

- The **Web (WWW)**, is an **information system where documents and other web resources are identified** by Uniform Resource Locators (URLs, such as https://example.com/), which may be interlinked by hyperlinks, and are accessible over the Internet.
- The **resources of the Web are transferred via the Hypertext Transfer Protocol (HTTP), may be accessed using *web browser***, and are published on ***web server.***
- The World Wide Web is built on top of the Internet, which pre-dated the Web by over two decades.

(Refer **Infographic – Dark Web vs Deep Web vs Surface Web**)

### Surface Web

- Also called the **Visible Web**, **Indexed Web**, **Indexable Web** or **Lightnet.**
- It is the portion of web which is **readily available to the general public and searchable** with standard web search engines
- The **Surface Web is made with a collection of public Web pages on a server** accessible by any search engine.

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## Deep Web

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- It is also called **invisible web** or **hidden web**
- It is the **part of the web which cannot be accessed by the normal search engines**, such as Google, Bing and Yahoo. Example – government private data, bank data, cloud data etc.
- Contents of Deep Web are **not indexed by standard web search-engines.**
- **These data are sensitive and private** and thus is kept **out of reach of normal public.**
- It is used to **provide access to a specific group of people.**
- The contents of the deep web range from pages that were not indexed by search engines,**Example** – paywalled sites, private databases and the dark web.

## *What is meant by no indexing?*

- No index means that a **web page shouldn't be indexed by search engines** and therefore shouldn't be shown on the search engine's result pages.
- The **robots meta tag is a piece of code** in the head section of a web page.
- **It tells search engines how to crawl** and whether to index a page.

## Dark Web

- The **dark web** is the World Wide Web content that exists on darknets.
- The dark web is a small part of the deep web which is the part of the Web not indexed by web search engines.
- Private computer networks can communicate and conduct business anonymously through dark web, without revealing information, such as a user's location.

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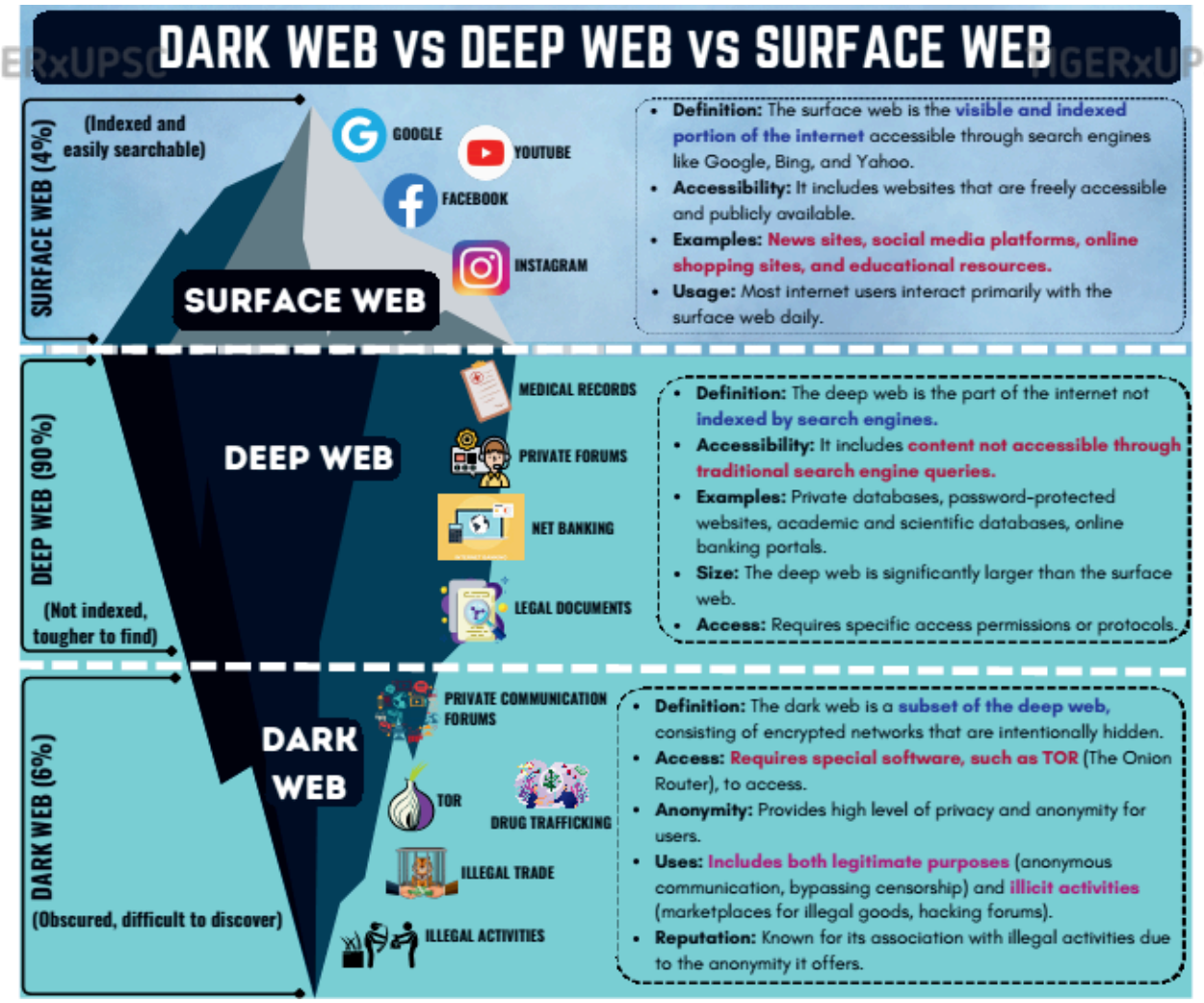


- Dark Web is the most hidden part of the Internet, not much people know about it.
- The Dark Web is also called the black world of the internet because many illegal acts like hacking, online robbery, human trafficking, arms sales and drugs supply etc. takes place on the Dark Web.
- One example is a website called the silk road which was on darknet called TOR, used to sell drugs and was taken down by FBI.
- Anyone can use the Dark Web, but the number of people doing illegal work are more on Dark Web. The main reason behind this is that no one can trace who uses the Dark Website as it lies on darknet.

## Darknet

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It is a **network build over the internet which is encrypted** that is it can **only be accessed with specific software, configurations, or authorisation and often uses a unique customised communication protocol.**

**Two typical darknet types** are:

1. **Social networks (usually used for file hosting with a peer-to-peer connection):** Friend-to-Friend is another kind of darknet where two-person transfer data between them anonymously.

   Only concerned people have access to it and it is encrypted and password protected.

2. **Anonymity proxy networks [via an anonymised series of connections]**
   - ❑ One such darknet is **TOR** (The Onion Router).
   - ❑ A special software is required to enter this network like "TOR browser, is required to enter into the TOR's network".
   - ❑ TOR can access regular website also, a website on this network has .onion address.

**Freenet is also a darknet** which is used for file transfer anonymously; there are many other darknets available.

Darknet **provides a user with anonymity** and thus, **many hidden services are provided on darknet.**

When a service [Now, popular as dark web] was introduced which allowed someone to host a website on the darknet and remained anonymous, **this attracted people who undertook illegal activities to sells things without getting caught on darknets using darkweb.**

Technology such as TOR, I2P, and Freenet which was intended to defend digital rights by providing security, anonymity, or censorship resistance are used today for both illegal and legitimate reasons.

Though it might seem a bit scary darknet is very useful for which it was created and that is to provide anonymity to people like government official, journalist etc.

Through use of applications such as SecureDrop, darknets can facilitate anonymous communication between whistle-blowers, activists, journalists and news organisations.

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## QUESTIONS AND ANSWERS

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## Objective Questions

**1. With reference to Web 3.0, consider the following statements:** **[UPSC (Pre) – 2022]**

1. Web 3.0 technology enables people to control their own data.
2. In Web 3-0 world, there can be blockchain based social networks
3. Web 3.0 is operated by users collectively rather than a corporation.

Which of the statements given above are correct?

(a) 1 and 2 only
(a) 2 and 3 only
(c) 1 and 3 only
(c) 1, 2 and 3

**2. Consider the following communication technologies:** **[UPSC (Pre) – 2022]**

1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered Short-Range devices/technologies?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only

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(d) 1, 2 and 3

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**3. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?** **[UPSC (Pre) – 2020]**

(i) VLC uses electromagnetic spectrum wavelengths 375 to 780 nm.
(ii) VLC is known as long-range optical wireless communication.
(iii) VLC can transmit large amounts of data faster than Bluetooth.
(iv) VLC has no electromagnetic interference.

Select the correct answer using the code given below:

(a) (i), (ii) and (iii) only
(b) (i), (ii) and (iv) only
(c) (i), (iii) and (iv) only
(d) (ii), (iii) and (iv) only

**4. When the alarm of your smartphone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a message to your office accordingly. In the context of emerging communication technologies, which one of the following terms best applies to the above scenario?** **[UPSC (Pre) – 2018]**

(a) Border Gateway Protocol
(b) Internet of Things
(c) Internet Protocol
(d) Virtual Private Network

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5. **With reference to 'Li-Fi, recently in the news, which of the following statements is/are correct?** **[UPSC (Pre) – 2016]**

(i) It uses light as the medium for high-speed data transmission.

(ii) It is a wireless technology and is several times faster than 'Wi-Fi'.

Select the correct answer using the code given below:

(a) Only (i)

(b) Only (ii)

(c) Both (i) and (ii)

(d) Neither (i) nor (ii)

6. **With reference to 'Near Field Communication (NFC) Technology', which of the following statements is/are correct?** **[UPSC (Pre) – 2015]**

(i) It is a contactless communication technology that uses electromagnetic radio fields.

(ii) NFC is designed for use by devices, which can be at a distance of even a metre from each other.

(iii) NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below:

(a) Only (i) and (ii)

(b) Only (iii)

(c) Only (i) and (iii)

(d) (i), (ii) and (iii)

7. **What is the difference between Bluetooth andWi-Fi devices?** **[UPSC (Pre) – 2011]**

(a) Bluetooth uses 2.4 GHz radio frequency band, whereas Wi-Fi can use 2.4 GHz or 5 GHz frequency band.

(b) Bluetooth is used for Wireless Local Area Networks, (WLAN) only, whereas Wi-Fi is used for Wireless Wide Area Networks (WWAN) only.

(c) When information is transmitted between two devices using Bluetooth technology, the devices have to be in the line of sight

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of each other, but when Wi-Fi technology is used, the devices need not be in the line of sight of each other.

(d) The statements (a) and (b) given above are correct in this context.

**8. What is 'Virtual Private Network'?** **[UPSC (Pre) – 2011]**

(a) It is a private computer network of an organisation where the remote users can transmit encrypted information through the server of the organisation.

(b) It is a computer network across a public Internet that provides users access to their organisation's network while maintaining the security of the information transmitted.

(c) It is a computer network in which users can access a shared pool of computing resources through a service provider.

(d) None of the statements (a), (b) and (c) given above is a correct description of Virtual Private Network.

**9. Consider the following:** **[UPSC (Pre) – 2010]**

(i) Bluetooth device
(ii) Cordless phone
(iii) Microwave oven
(iv) Wi-Fi device

Which of the above can operate between 2.4 and2.5 GHz range of radio frequency band?

(a) (i) and (ii)
(b) (iii) and (iv)
(c) (i), (ii) and (iv)
(d) All of these

**10. WiMAX is related to which one of the following?** **[UPSC (Pre) – 2009]**

(a) Biotechnology
(b) Space technology
(c) Missile technology
(d) Communication technology

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**11. MCA-21 is a major initiative taken up by the Government of India in which one of the following areas?** **[UPSC (Pre) – 2007]**

(a) Foreign direct investment in India
(b) Attracting international tourists
(c) E-governance
(d) Modernisation of airports

## Answer Key

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (d) | 2. (d) | 3. (c) | 4. (b) | 5. (c) | 6. (c) | 7. (a) | 8. (b) |
| 9. (d) | 10. (d) | 11. (c) | | | | | |

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# CHAPTER 12
# ICT-BASED ISSUES

## IN THIS CHAPTER...


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- 12.1.1 Spectrum
- 12.1.2 Agencies Allocating Spectrum
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**12.2 Broadband Policy of India**
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**12.4 Digital Personal Data Protection Act, 2023**
**12.5 Data Localisation**
**12.6 Digital Arrest**
**12.7 National Supercomputing Mission (NSM)**
**12.8 Cloud Computing**
- 12.8.1 Basics of Cloud Computing

**12.9 Grid Computing**


It is an intelligence network of information sharing. The 11 central agencies that can take advantage of this network include: Research and Analysis Wing, Intelligence Bureau, Central Bureau of Investigation, Financial Intelligence Unit, Central Board of Direct Taxes, National Investigation Agency (NIA), Enforcement Directorate, Directorate of Revenue Intelligence, Narcotics Control Bureau, Central

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Board of Excise and Customs and Directorate General of Central Excise Intelligence.

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**12.10 National Intelligence Grid (NATGRID)**
**12.11 Encryption**
**12.12 Wearable Technology**
**12.13 Biometrics**
**12.14 Digital Lending**
**12.15 Digital Signature**
**12.16 Edge Computing**
**12.17 BSNL Launched Direct – to – Device (D2D) Satellite Connectivity**
**12.18 Space Internet**
12.18.1 Starlink Project
12.18.2 OneWeb: Internet
12.18.3 Jio Platforms to Provide Satellite Broadband Services
**12.19 Virtual Reality (VR) and Augmented Reality (AR)**
**12.20 Metaverse**

Metaverse is a combination of three key aspects – Presence, Interoperability and Standardisation.

**12.21 DNA Computer**
**12.22 Non-Fungible Tokens (NFT)**

They are digital assets that represent ownership or proof of authenticity of a unique item or piece of content.

**12.23 Traditional Knowledge Digital Library (TKDL)**
**12.24 Cryptography**
**12.25 Digital India Initiative**
**12.26 Telemedicine**
**12.27 Project Loon**
**12.28 TRP Rating Controversy**

# 12.1 Spectrum Policy in India

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## 12.1.1 Spectrum

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Spectrum represents a collection of different types of electromagnetic radiations of different wavelengths. All communication signals work on these spectrums. Different types of services like space communication, mobile communication, broadcasting, radio navigation, mobile satellite service, aeronautical satellite services, defence communication, etc., all use and rely on radio frequencies in India. Radio frequency is a natural resource but unlike other natural resource, its major limitation is that it will deplete on use. Additionally, it will be wasted if not used efficiently.

Indian telecom operator uses the spectrum, which is the most crowded and inadequate thing to accommodate the usage by 1.15 billion wireless mobile subscribers as on date. Hence, the quality of customer service deteriorates and leads to poor voice quality, call drop and undelivered messages of mobile services in India.

### 12.1.2 Agencies Allocating Spectrum

For international purposes, the international body called International Telecommunication Union (ITU) allocates spectrum. For domestic purposes, it is done by Wireless Planning and Coordination (WPC) wing of the Ministry of Communications, created in 1952. It is the National Radio Regulatory Authority responsible for Frequency Spectrum Management, which includes licensing for the needs of all wireless users in the country. The licences to operate wireless stations are allotted by it.

#### *Reserve Price*

It is the minimum amount set by the government from where auction begins, i.e., it is the starting amount or base price from where the auction starts.

#### *Auction of Spectrum*

- Spectrum is a scarce resource. It needs to be managed efficiently.
- Spectrum cannot be used by many people. It has to be allocated to some pErsons who can manage the services under it. Hence, it is auctioned.

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- Government auctions it because spectrum is a resource of the Government of India. It is not a private property, so government auctions it.
- A lot of revenue is generated by selling the spectrum and that money can be used for developmental programmes in India.

### 12.1.3 The Challenges of India's Spectrum Policy

To get advantage of new possibilities for social and economic development, India requires more spectrums. The mobile technology demand is likely to rise quickly in the coming years. Hence, the country needs the spectrum deployment for utilising its commercial in order to accommodate the mobile broadband growth. The country faces various challenges and difficulties in respect of its spectrum policy, which is summarised as follows:

1. **Lack of Available Spectrum:** There are insufficient bands for the growing mobile demands in India, leading to the issue of availability.
2. **High Costs:** India has one of the highest spectrum costs in the world. India also suffers from high telecommunications costs.
3. **Fragmented and Inflexible Usage:** The spectrum quality is a topic of discussion, as some bands are booked for particular applications, resulting in issues related to the quality of spectrum. For instance, Defence Ministry occupies too much spectrum, at times even more than required and hence, is not fully utilised, thereby limiting the access of the said resources to private.
4. **Auction Outcomes:** Besides the above major issues, another severe problem is cost and availability problems contributed by past spectrum auctions. The dual objectives of the spectrum are resourceful spectrum allotment to the full utilisation and maximising the revenue of government and hence, the appropriate design of the auctions becomes critical. Due to auction, there are huge level of company debts, unreasonable prices and subsequently, high charges for consumers.

### 12.1.4 Spectrum Auctions

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A spectrum auction is a process whereby a government uses an open auction system to sell the rights to transmit signals over specific bands of the electromagnetic spectrum and to assign scarce spectrum resources.

- Devices such as cellphones and wireline telephones require signals to connect from one end to another. These signals are carried on airwaves, which must be sent at designated frequencies to avoid any kind of interference.
- The Union government owns all the publicly available assets within the geographical boundaries of the country, which also include airwaves.
- To sell these assets to companies willing to set up the required infrastructure to transport these waves from one end to another, the central government through the DoT auctions these airwaves from time to time.
- **These airwaves are called spectrum**, which is subdivided into bands which have varying frequencies. All these airwaves are sold for a certain period of time, after which their validity lapses, which is generally set at 20 years.

#### 12.1.4.1 Main Objective of Auction

- To obtain a market-determined price for the spectrum on offer.
- Ensure efficient use of spectrum and avoid hoarding.
- Stimulate competition in the sector and maximise revenue procedure.

#### 12.1.4.2 Spectrum Auctions in India

Devices such as cell phones and wireline telephones require signals to connect from one end to another. These **signals are carried on airwaves,** which must be sent at designated frequencies to avoid any kind of interference.

- The Union government owns **all the publicly available assets within the geographical boundaries of the country, which also include airwaves.**
- With the expansion in the number of cellphone, wireline telephone and Internet users, the need **to provide more space for the signals arises from time to time.**

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- **To sell these assets to companies** willing to set up the required infrastructure to transport these waves from one end to another, **the central government through the DoT auctions** these airwaves from time to time.
- **These airwaves are called spectrum,** which is subdivided into bands which have varying frequencies.
- All these airwaves are sold for a certain period of time, after which their validity lapses, which is generally set at 20 years.

## 12.2 Broadband Policy of India

Broadband is a high-capacity transmission technique, which uses a wide range of frequencies, enabling communication of a large number of messages simultaneously through coaxial cable, optical fibre, radio or twisted pair.

As the demand for high speed and quality Internet and telecommunication service increases and having the potential of widespread broadband services, **India has taken its first Broadband Policy in 2004 aimed at enhancing the quality of life through societal applications which includes tele-medicine, tele-education, e-governance, entertainment, as well as employment generation through high-speed access to information and web-based communication.**

In a competitive environment like today, appropriate steps ensuring required bandwidth available in a time-bound manner within their licence framework is required by service providers. A major cost component for Broadband services is contributed by the bandwidth costs. The DIT, Government of India has established National Internet Exchange of India (NIXI) to ensure that Internet traffic, originating and destined for India, should be routed within India.

## 12.3 Section 66A of the Information Technology Act

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## Section 66A

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- Section 66A defines the punishment for sending "offensive" messages through a computer or any other communication device like a mobile phone or a tablet.
- If convicted, it can lead to a maximum of 3 years in jail and fine.
- It empowers the police to make arrests over what policemen, in terms of their subjective discretion, could construe as "offensive" or "menacing" or for the purposes of causing annoyance, inconvenience, etc. to the victim.

## Shreya Singhal Case

The Supreme Court had in its judgment in the *Shreya Singhal* case struck down Section 66A.

## SC strike down Section 66A

The Supreme Court had noted that **Section 66A** arbitrarily, excessively and disproportionately invades the right of free speech, under **Article 19(1)(a) of the Constitution**, and upsets the balance between such right and the reasonable restrictions that may be imposed on such right and the definition of offences under the provision was open-ended and undefined.

The Delhi High Court has asked the Centre to consider a petition seeking to remove provisions from the statute such as Section 66A of the Information Technology Act, that has already been declared unconstitutional.

## Issues

There are several criminal law sections that were struck down by the Supreme Court but continue to be used by police officers. In some cases, trial courts went ahead with framing charges under the defunct IT Act provisions, even after taking cognizance of the Supreme Court's 2015 judgment.

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## 12.4 Digital Personal Data Protection Act, 2023

On August 11, 2023, India enacted the Digital Personal Data Protection Act, 2023 ("DPDP Act"), following extensive public consultations on the Digital Personal Data Protection Bill, 2022 released by the Ministry of Electronics and Information Technology on November 18, 2022. This Act focuses exclusively on digital personal data and supersedes Section 43A of the Information Technology Act, 2000 and the Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data of Information) Rules, 2011, once its provisions are phased into force.

### *Key Highlights of the DPDP Act*

1. **Applicability:** The DPDP Act applies solely to digital personal data, encompassing data collected digitally or subsequently digitised non-digital data. It extends overseas to data processed outside India concerning the offering of goods or services to Indian data subjects but excludes personal data used for personal or domestic purposes or made publicly available under legal obligations.
2. **Data Protection Principles:** It establishes principles such as purpose limitation, ensuring data processing aligns with lawful purposes consented to by the data principal and collection limitation, permitting only necessary data collection.
3. **No Sub-classification of Personal Data:** Unlike current regulations, the DPDP Act treats all personal data equally, without sub-categorisation distinctions like sensitive or critical personal data.
4. **Consent and Notice:** The Act mandates affirmative consent for data processing, withdrawal rights equivalent to consent granting ease and provision of detailed notices preceding consent requests, including multi-language options.
5. **Legitimate Uses:** It introduces "legitimate uses" for processing personal data without consent, encompassing specific scenarios like emergency responses or legal compliance.

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6. **Obligations of Data Fiduciaries:** Data fiduciaries must comply with the Act, including overseeing data processing by processors, ensuring accuracy, and deleting data upon consent withdrawal or no longer serving its purpose.
7. **Notification of Breaches:** Breach notifications to the Data Protection Board (DPB) and affected data principals are mandatory.
8. **Cross-border Data Transfers:** Personal data may be transferred abroad unless restricted by the Central Government, with provisions allowing for exemptions or stricter sectoral regulations.
9. **Significant Data Fiduciaries:** Certain entities may be designated as significant data fiduciaries, necessitating additional compliance measures such as appointing data protection officers and conducting audits.
10. **Rights of Data Principals:** Data principals hold rights to access, correct and erase personal data, nominate representatives for data rights and seek grievance redressal through designated mechanisms.
11. **Data Protection Board of India:** The DPB will enforce the Act, overseeing breaches, imposing penalties and ensuring compliance, with appeals handled by designated tribunals.
12. **Penalties:** Monetary penalties up to INR 250 crores may be levied for violations, with exemptions barring compensation claims by data principals for breaches.
13. **Voluntary Undertaking and Exemptions:** The Act permits voluntary undertakings to mitigate enforcement actions and grants exemptions for state instrumentalities and specific processing activities deemed essential.

The DPDP Act represents a comprehensive framework aimed at safeguarding digital personal data in India, ensuring compliance through stringent regulations and rights protection mechanisms.

## 12.5 Data Localisation

Data localisation is the **act of storing data on any device that is physically present within the borders of a specific country where the**

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**data was generated.**

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## *Why is Data Localisation Necessary for India?*

- For securing citizen's data, data privacy, data sovereignty, national security and economic development of the country.
- Recommendations by the RBI, the **committee of experts led by Justice BN Srikrishna,** the draft e-commerce policy and the draft report of the cloud policy panel show signs of data localisation.
- The extensive data collection by technology companies has allowed them to process and monetise Indian users' data outside the country. Therefore, **to curtail the perils of unregulated and arbitrary use of personal data, data localisation is necessary.**
- Digital technologies like Machine Learning (ML), Artificial Intelligence (AI) and Internet of Things (IoT) can generate tremendous value out of various data. It can **turn disastrous if not contained within certain boundaries.**
- With the advent of cloud computing, Indian users' data is outside the country's boundaries, leading to a conflict of jurisdiction in case of any dispute.
- Data localisation is an **opportunity for Indian technology companies to evolve an outlook from services to products.** ***International companies will also be looking at the Indian market, and this will benefit the growth of the local ecosystem.***
- More data centres in India mean new, **power-hungry customers for India's renewable energy market.** That means data localisation could boost India's renewable energy.

## *Policies that Imply Data Localisation*

- The **Srikrishna Committee** wants to localise data for law enforcement to have easy access to data, to ***prevent foreign surveillance*** and to ***build an artificial intelligence ecosystem*** in India and also because undersea cables through which data transfers take place are vulnerable to attacks.
- **Reserve Bank of India has also imposed *a hard data localisation mandate on payment systems providers to store payment systems data***

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***only in India.***

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- Barring limited exceptions, telecom service providers are not allowed to transfer user information and accounting information outside India.
- Goals set in the **Draft National Digital Communications Policy 2018 and the Guidelines for Government Departments for Contractual Terms related to Cloud Storage 2017,** draft e-commerce policy and the draft report of the cloud policy panel show signs of data localisation.

## Concerns/Challenges

- Several recommendations including the draft e-commerce policy, falter on a key ground like they gloss over the negative economic impact of data localisation. This approach exhibits lack of evidence-based policy making.
- Having data in India does not mean that domestic companies will be able to access this data. Localisation might aid the growth of the data centres and the cloud computing industry in India, but as a matter of wider public policy, such an approach is extremely myopic.
- Mandating localisation is less of a solution for data protection and might be less relevant to promote e-commerce.
- Given the comparative trade advantages enjoyed by one section of Indian industry in this context, mandating a strict data localisation regime could be perceived as a restrictive trade barrier and spur retaliatory measures.
- There is a possible rise in prices of foreign cloud computing services in case of a data localisation and its impact on MSMEs as well as start-ups relying on these services.
- The possibility of triggering a vicious cycle of data localisation requirements by other countries as a response to India's possible data localisation will be detrimental for the global data economy.
- Growth will be restricted if data cannot be aggregated internationally. Infrastructure in India for efficient data collection and management is lacking.

## Need of the Hour

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- There is an urgent need to have an integrated, long-term strategy for policy creation for data localisation.
- Data localisation needs **to integrate a wide range of social, political and economic perspectives.**
- Creating an opportunity ***for local data centres all over the country.***
- Devising an optimal ***regulatory and legislative framework for data processors and data centres operating in the country.***
- ***Adequate infrastructure*** in terms of energy, real estate and Internet connectivity also needs to be made available for India to become a global hub for data centres.
- ***Adequate attention needs*** to be given to the interests of India's **Information Technology Enabled Services (ITeS) and Business Process Outsourcing (BPO) industries**, ***which are thriving on cross border data flow.***

### *India Digital Personal Data Protection Act, 2023*

This legislation introduces a dedicated legal framework in India, representing a significant milestone as the country's inaugural privacy law designed to safeguard citizens' personal data. It emphasises the role of the Data Protection Board of India, outlines key provisions and defines the rights and responsibilities of both organisations and individuals.

The Act aims primarily to regulate the processing of digital personal data, respecting individuals' rights to safeguard their data while acknowledging the lawful necessity of processing and using such information. Its language is clear and accessible, ensuring broad comprehension. Furthermore, the Act seeks to establish a comprehensive legal structure governing the protection of digital personal data in India.

## 12.6 Digital Arrest

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Over **92,000 Indians** have been targeted in **digital arrest scams** in 2024. Cybercriminals **impersonate law enforcement** and extort money under the pretense of legal or tax issues.

## *What Are Digital Arrest Scams?*

- **Modus Operandi:** Scammers pretend to be Officials from agencies like the **CBI, ED, Narcotics Bureau** or **State Police.**
- **Tactics Used:**
  - ❑ Initiate **audio and video calls** from fake courtrooms, airports or police stations.
  - ❑ Display photos of **fake officers, judges or lawyers** on social media.
  - ❑ Send **forged legal notices, arrest warrants** or documents via email or messaging apps.
- **Accusations:** Victims are falsely accused of **money laundering, drug smuggling** or other serious crimes.
- **Manipulation Techniques:**
  - ❑ Use of **AI-generated voices**, **professional logos** and **simulation of authority**.
  - ❑ Isolation tactics: Victims are discouraged from seeking help or verifying information.

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- Emotional pressure using **fear**, **shame** and **urgency.**

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## Status of Cybercrime in India

- **Source:** Indian Cyber Crime Coordination Centre (I4C)
- **Complaints Rising:**
  - **2021:** 1.35 lakh
  - **2022:** 5.14 lakh
  - **2023:** 11.31 lakh
- **Total Loss (2021–2024):** ₹27,914 crore

## Major Scams & Losses

| Scam Type | Loss (₹ Crore) | Complaints |
|---|---|---|
| Stock Trading Scams | 4,636 | 2,28,094 |
| Ponzi Scheme Scams | 3,216 | 1,00,360 |
| Digital Arrest Frauds | 1,616 | 63,481 |

## Money Laundering Methods

- Withdrawals via **ATMs, fintech apps, CBDCs, crypto wallets**
- Use of **mule accounts** (4.5 lakh frozen by I4C)

## About I4C (Indian Cyber Crime Coordination Centre)

- **Established:** 2020 by Ministry of Home Affairs
- **Purpose:** A centralised platform to tackle all types of cybercrimes

## Key Objectives

- Support cybercrime detection and prevention
- Aid law enforcement with early warning systems
- Educate the public on cyber hygiene
- Provide **National Cyber Crime Reporting Portal** for citizens

## Challenges in Combating Cyber Scams

- **Anonymity:** Use of **VPNs and encrypted apps** hides criminal identity.

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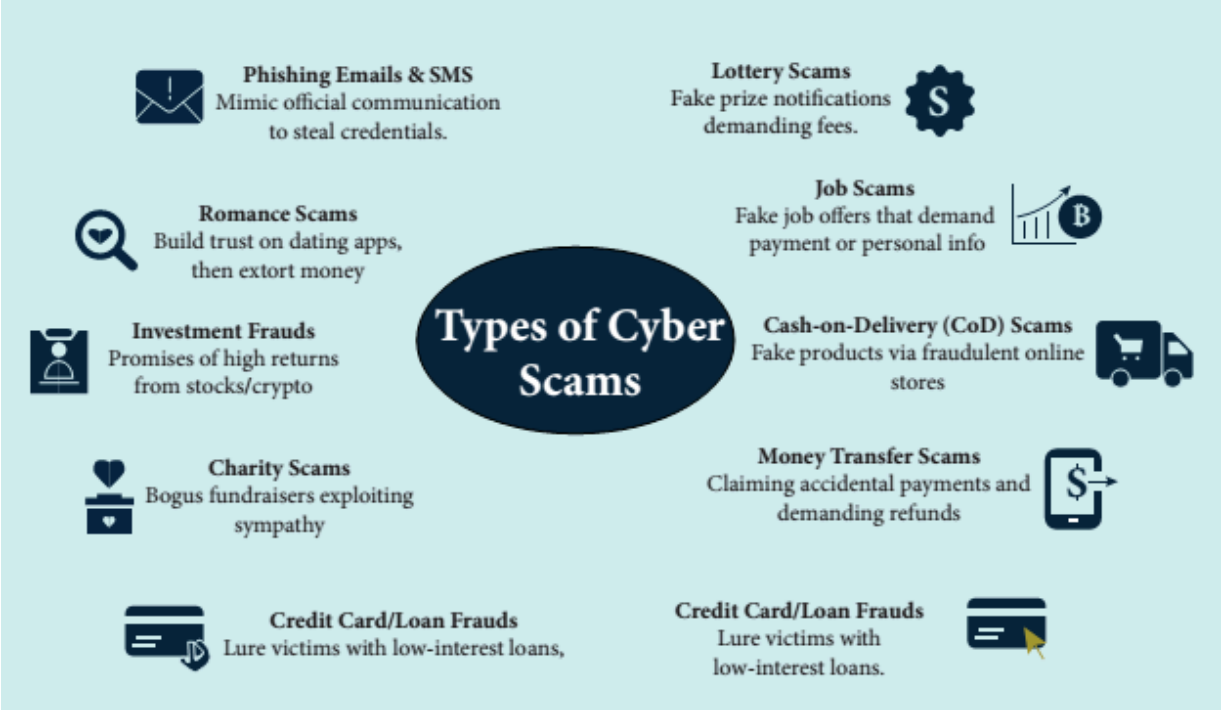
- **Cross-border Nature:** Scams often originate from **China and Southeast Asia**.
- **Advanced Techniques:** Phishing, AI voice manipulation, malware, social engineering.
- **Weak Global Laws:** No unified regulation or intelligence sharing mechanisms.
- **Growing Digital Market:** Rise of **e-commerce, digital payments and fake websites**.




## *Key Government Initiatives*

- National Cyber Security Policy
- CERT-In (Computer Emergency Response Team – India)
- Cyber Surakshit Bharat
- Cyber Swachhta Kendra
- NCIIPC (Critical Infrastructure Protection Centre)
- Digital Personal Data Protection Act, 2023
- Citizen Financial Cyber Fraud Reporting System
- Cyber Crime Reporting Portal (https://cybercrime.gov.in/)

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## 12.7 National Supercomputing Mission (NSM)

### *Introduction*

- **Launched:** 2015
- **Objective:** To create a robust **supercomputing ecosystem** for academic institutions, research bodies, startups and MSMEs.
- **Led by:** Department of Science & Technology (DST) and Ministry of Electronics and IT (MeitY).
- **Executed by:** Centre for Development of Advanced Computing (**C-DAC**, Pune) and **Indian Institute of Science (IISc), Bengaluru**.
- **Key Applications:** Include fields like **computational biology**, **nuclear simulations**, **national security** and **seismic activity analysis**.
- **Goal:** Install **73 indigenous supercomputers** across India to meet growing demands of research and innovation.
- **Network:** To form a **National Knowledge Network (NKN)** linking all HPC (High Performance Computing) centers.
- **Focus:** Enhancing both **capacity** and **capability** in high-performance computing for:
  - ❑ Academia
  - ❑ Scientific research
  - ❑ National development

The Prime Minister unveiled **three new PARAM Rudra supercomputers** along with **two high-performance computing (HPC) systems**, with a combined value of ₹130 crore.

Developed under the **National Supercomputing Mission (NSM)**, these systems are **indigenously built** and strategically deployed across:

- **Pune:** At the **Giant Meter Radio Telescope (GMRT)** for advanced exploration of phenomena like **Fast Radio Bursts (FRBs)** and other astronomical events.
- **Delhi:** At the **Inter-University Accelerator Centre (IUAC)** to boost research in **material sciences and atomic physics**.
- **Kolkata:** At the **S.N. Bose Centre**, supporting high-end research in **physics, cosmology and earth sciences**.

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Additionally, the PM inaugurated **two HPC systems**:

- *'Arka'* **and** *'Arunika'*, which aim to significantly enhance **weather prediction capabilities**—especially for **tropical cyclones**, **heavy rainfall** and **thunderstorms**.

## About Supercomputers

- Supercomputers are **high-capacity mainframe systems** capable of solving extremely complex problems by **dividing tasks and processing them simultaneously**.
- Their performance is measured in **FLOPS (Floating-Point Operations Per Second)**.

## About C-DAC

- **Established:** 1988
- A top-tier R&D organisation under MeitY.
- Known for its **PARAM series** of supercomputers, including:
  - **PARAM 8000:** India's first Giga-scale supercomputer (1990)
  - **PARAM 10000:** Launched in 1998 and used in various scientific fields

## Indian Supercomputers in Global Top 500 List

| Rank | Site | Name | Rmax (TFlop/s) | Rpeak (TFlop/s) |
|---|---|---|---|---|
| 75 | Centre for Development of Advanced Computing (C-DAC) | **AIRAWAT – PSAI** | 8.5 | 13.17 |
| 131 | Centre for Development of Advanced Computing (C-DAC) | **PARAM – Siddhi AI** | 4.62 | 5.27 |
| 169 | Indian Institute of Tropical Meteorology | **Pratyush (Cray XC40)** | 3.76 | 4.01 |
| 316 | National Centre for Medium-Range Weather Forecasting | **Mihir (Cray XC40)** | 2.57 | 2.81 |

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**Rmax:** maximum performance achieved by a system, Rmax is a practical measure, indicating how well the system performs in real-world tasks.

**Rpek:** theoretical peak performance of the system, Rpeak is a theoretical ceiling based on hardware capabilities.

## 12.8 Cloud Computing

### 12.8.1 Basics of Cloud Computing

Cloud Computing uses the Internet and central remote servers to maintain data and applications. The consumers can access their personal files at any computer with Internet access. This technology is more efficient because it enables computing through centralised storage, memory, processing and bandwidth.

Yahoo mail or Gmail are simple examples of cloud computing. It does not require software or a server to use them. The consumers just require an Internet connection to send emails. All the management of server and email is done on the cloud (Internet) and it is overall managed by the cloud service provider like Yahoo, Google, etc. It is more advantageous since the consumer gets to use the software alone and avail the benefits.

**Two main types of clouds** are **private** and **public.** Services sold by public cloud are available for anyone on the Internet. Now, the largest public cloud provider is Amazon Web Services. On the other hand, a private cloud represents a proprietary network or a data centre, supplying hosted services to limited number of people. A virtual private cloud can be created by using public cloud resources to establish a private cloud. Be it private or public, cloud computing aims at easy, scalable access to IT services and computing resources.

#### *Categories of Cloud Computing*

Anything that includes delivering hosted services over the Internet comes under the broad umbrella of Cloud Computing. The services so provided are broadly of three categories, as follows:

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1. Infrastructure-as-a-Service (IaaS)
2. Platform-as-a-Service (PaaS)
3. Software-as-a-Service (SaaS)

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The Internet in flowcharts and diagrams is often represented by the symbol of cloud and this inspired the technology to be named as Cloud Computing.

### *Various types of Clouds*

1. Public Cloud
2. Private Cloud
3. Community Cloud
4. Hybrid Cloud

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Note: **iCloud**

The cloud storage and cloud computing service from Apple Inc. is named as iCloud. The users can store data, such as music files on remote computer servers for downloading in multiple devices, such as iOS-based devices and personal computers running OS X or Microsoft Windows, using this service. It also serves as a placement for Apple's Mobile Me service, which acts as a data-syncing centre for email, contacts, calendars, bookmarks, notes, to-do lists and other data. Up to 5 GB, the users enjoy free storage facility, above which it is a paid service.

Note: Apple Inc. announced iCloud on 06 June 2011 at the Apple World-Wide Developers Conference (WWDC).

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(Refer **Infographic – Cloud Computing & Cloud Hosting**)

## Network-as-a-Service (NaaS)

### *What Is NaaS?*

**Network-as-a-Service (NaaS)** is a **cloud-based model** that allows businesses to **lease networking services** from cloud providers instead of building and maintaining their own physical network infrastructure.

### *Key Features of NaaS*

- **Virtual Networks:** Businesses can **create and manage networks virtually** using software-based tools – no need for physical routers, firewalls or load balancers.

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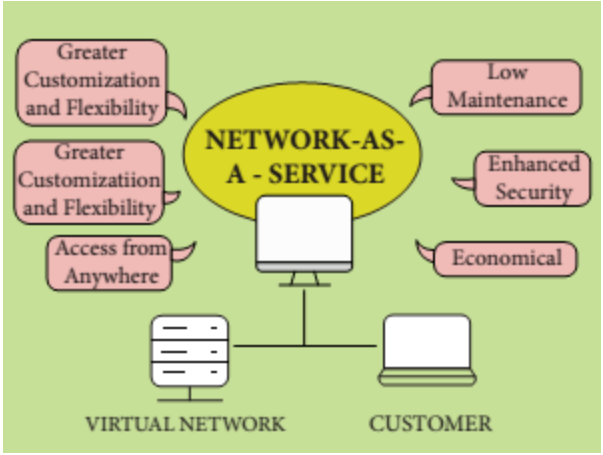
- **Internet-Driven:** NaaS operates entirely via **internet connectivity**, making it a modern alternative to traditional setups like **VPNs**, **MPLS** and on-premise networking devices.

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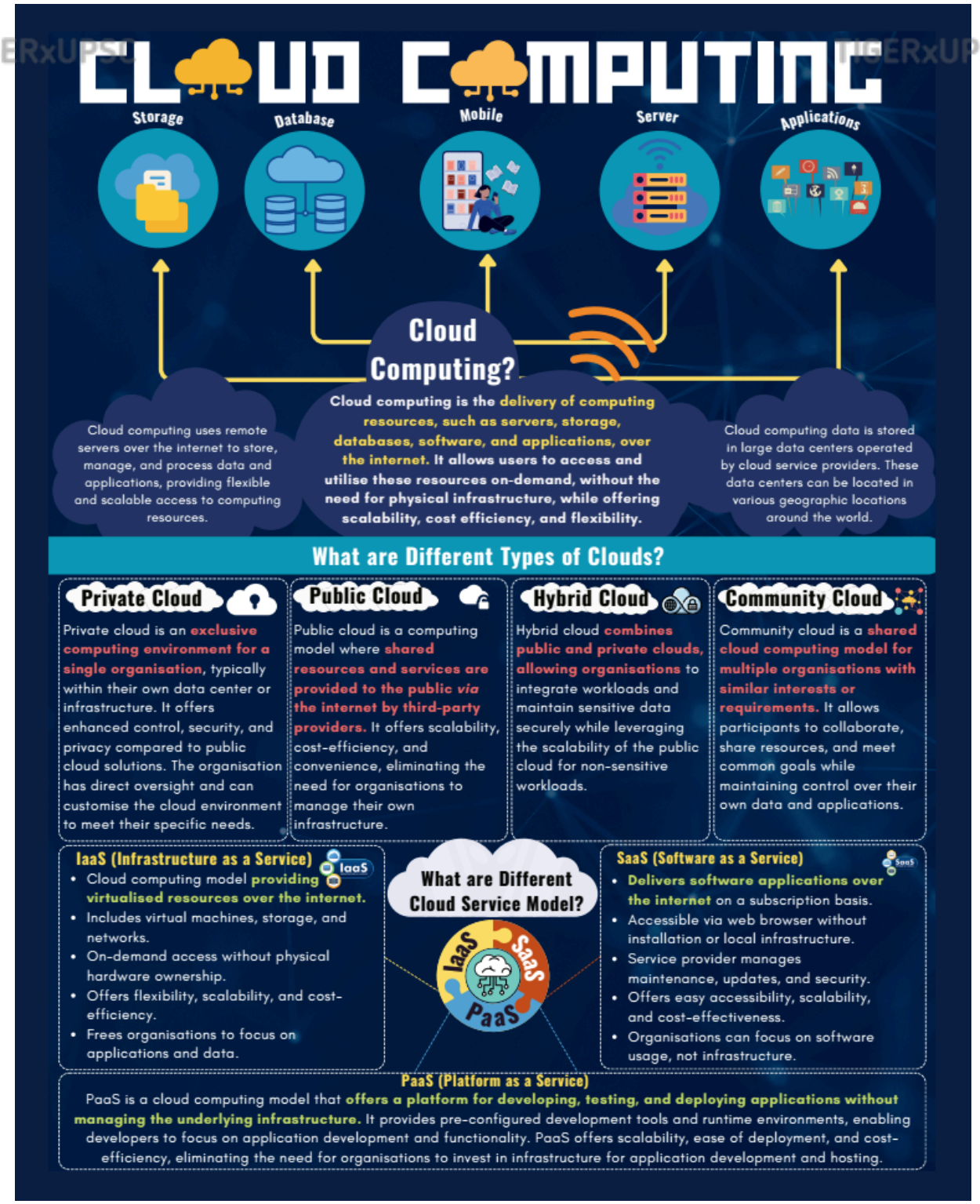


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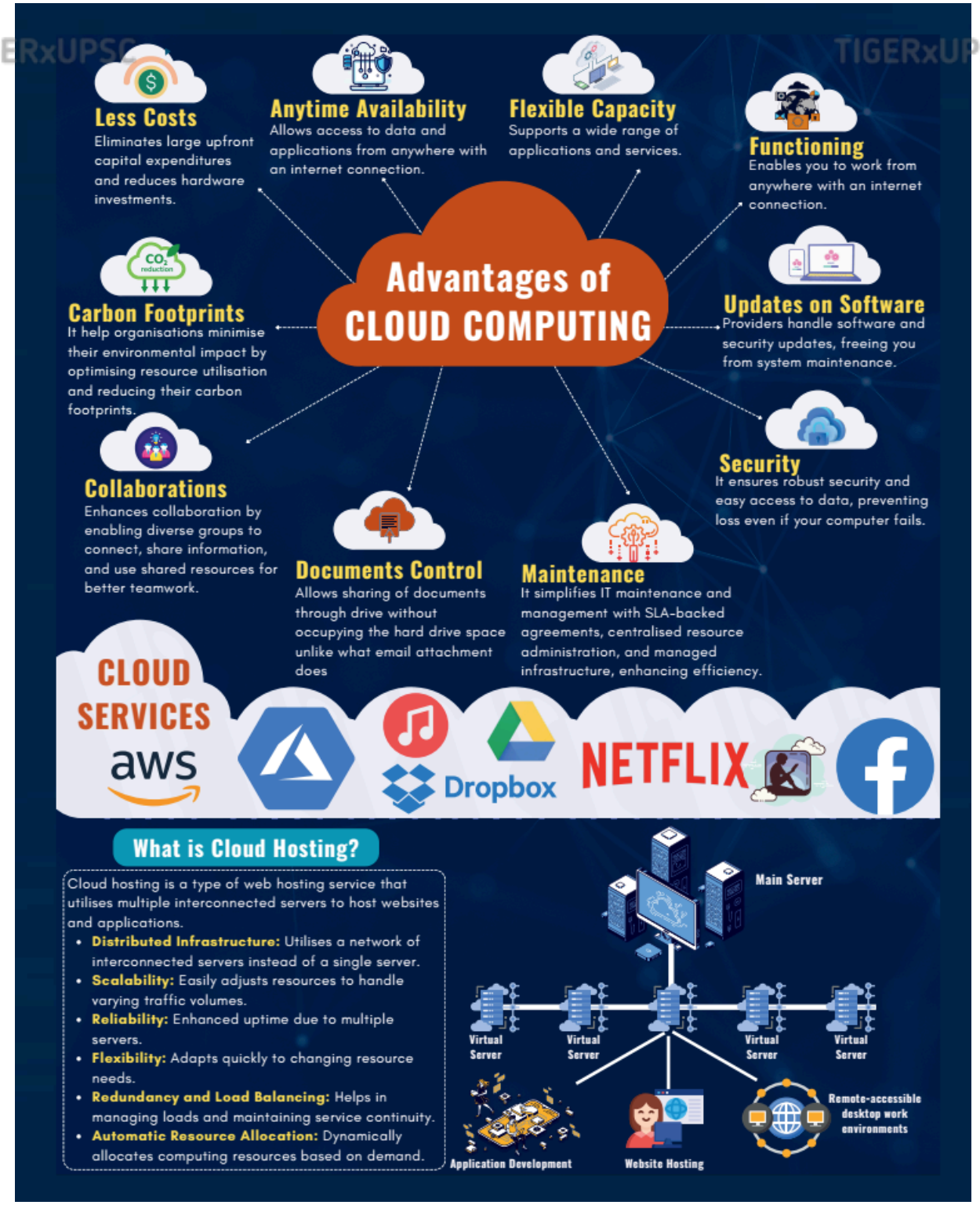


- **Self-Service Model:** Users can set up and oversee networks using **web-based dashboards or APIs,** which are sets of rules and protocols

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enabling communication and data exchange between software applications.

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## Benefits of NaaS

| Advantage | Description |
|---|---|
| **Flexibility & Customisation** | Networks can be reconfigured easily using software tools. |
| **Scalability** | Services can be expanded on-demand through subscriptions – no hardware limits. |
| **Remote Access** | Access the network from anywhere – no need for traditional VPNs. |
| **Low Maintenance** | Cloud vendors manage updates, upgrades and troubleshooting. |
| **Stronger Security** | Integrated security and network management result in improved protection. |
| **Cost Efficiency** | No need for expensive infrastructure – reduces upfront and ongoing costs. |

## Challenges of NaaS

| Challenge | Description |
|---|---|
| **Legacy Compatibility** | Older systems may not align with modern cloud infrastructure. |
| **Migration Complexity** | Moving from traditional to virtual networks can be technically demanding. |
| **Vendor Lock-in Risk** | Heavy reliance on one provider could be risky if pricing or service changes. |

## Relevance in India's Digital Growth

As India's internet economy rapidly expands, **NaaS offers scalable, secure and cost-efficient network solutions**, making it ideal for enterprises and startups that are part of the digital transformation journey.

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## 12.9 Grid Computing

Grid systems serve for collaborative sharing of resources. It may also be used as distributed and large-scale cluster computing. A grid basically uses the processing capabilities of different computing units for processing a single task. The way it works is as follows: The task is divided into multiple sub-tasks; and each machine on a grid is assigned a task. On completion of the sub-tasks, they are sent back to the primary machine (taking care of all the tasks). As a final step, they are combined or clubbed together as an output.

### *Cloud versus Grid Computing*

1. The services offered by cloud are more than that of grid computing. As a matter of fact, cloud provides almost all the services on the Internet, ranging from web hosting, multiple operating systems, DB support and much more.
2. In grid system, server computers are still needed to distribute the pieces of data and collect the results from participating clients on grid.
3. Grids tend to be more loosely coupled, heterogeneous and geographically dispersed when compared with conventional cluster computing systems.

## 12.10 National Intelligence Grid (NATGRID)

NATGRID is a unified intelligence grid through which the government's core security agencies' databases are interlinked. This enables collection of comprehensive patterns of intelligence that the intelligence agencies can readily access. Soon after the attack of 26/11 in 2008, the requirement of such agency was realised, and hence, it was first proposed by the

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government of India, in July 2016. Ashok Patnaik was appointed as the Chief Executive Officer of the National Intelligence Grid.

American Lashkar operative David Coleman Headley had visited the country many times, before the attack of 26/11 and performed a comprehensive study of the places that later became the places of attack. In spite of travelling numerous times to India and further returning to the US through Pakistan or West Asia, Indian agencies did not have any doubt, due to the lack of a system which did not identify a pattern in his strange travel itineraries. It is observed that if a system similar to the NATGRID was there before the attack, Headley would have been easily detained.

## KEYNOTE

- It is an ambitious **counter terrorism programme.**
- It will **utilise technologies** like Big Data and analytics to study and analyse the huge amounts of data from various intelligence and enforcement agencies to **help in tracking suspected terrorists and prevent terrorist attacks.**
- It will **connect, in different phases, data providing organisations and users** besides developing a legal structure through which information can be accessed by the law enforcement agencies.
- NATGRID is a post Mumbai 26/11 attack measure.

NATGRID is an intelligence network of information sharing. It collates the data from individual databases of the different agencies and ministries. It acts as an efficient counter-terrorism measure by further combining the data from different government databases. Such databases include bank and tax account details, transactions of credit card, records of visa and immigration and itineraries of rail and air travel. The **11 central agencies** that can take advantage of this network include: Research and Analysis Wing, Intelligence Bureau, Central Bureau of Investigation, Financial Intelligence Unit, Central Board of Direct Taxes, National Investigation Agency (NIA), Enforcement Directorate, Directorate of Revenue Intelligence, Narcotics Control Bureau, Central Board of Excise and Customs and Directorate General of Central Excise Intelligence.

### *Criticisms*

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- NATGRID is facing opposition on charges of possible violations of privacy and leakage of confidential personal information.
- Its efficacy in preventing terror has also been questioned, given that no state agency or police force has access to its database, thus reducing chances of immediate and effective action.
- According to few experts, digital databases such as NATGRID can be misused. Over the last two decades, the very digital tools that terrorists use have also become great weapons to fight the ideologies of violence.
- Intelligence agencies have also opposed amid fears that it would impinge on their territory and possibly result in leaks on the leads they were working on to other agencies.

### *Why do we need NATGRID?*

- The danger from not having a sophisticated tool like the NATGRID is that it forces the police to rely on harsh and coercive means to extract information in a crude and degrading fashion.
- After every terrorist incident, it goes about rounding up suspects – many of who are innocent. If, instead, a pattern search and recognition system were in place, these violations of human rights would be much fewer.
- NATGRID would also help the Intelligence Bureau keep a tab on persons with suspicious backgrounds.
- The police would have access to all his data and any movement by this person would also be tracked with the help of this data base.

NATGRID was opposed on grounds of possible violations of privacy and leakage of confidential personal information. Even its efficacy in preventing terror has been questioned, given that no state agency or police force has access to its database; thus reducing immediate actions sensing threat due to terrorism, its efficiency in preventing terror has been questioned.

## KEYNOTE

**The National Intelligence Grid (NATGRID) has signed a memorandum of understanding with the National Crime Records Bureau (NCRB) to access the centralised online database on FIRs and stolen vehicles.**

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### Who can access the data?

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- **It will be a medium for at least 11 Central agencies** such as the Intelligence Bureau (IB) and the Research and Analysis Wing (R&AW) to access data on a secured platform.
- **The data will be procured by NATGRID from 21 providing organisations such as the telecom, tax records, bank, immigration, etc**.

## 12.11 Encryption

The most effective way of providing data security is Encryption. Encrypted data represents a type of code text. You must have access to a secret key or password, which can enable you to decrypt it, thus helping to read an encrypted file. On the contrary, an unencrypted data represents a plain text.

It is a very effective tool for the security of important data by excluding the unauthorised access. It can be done through providing the decryption code only to the authorised persons.

Military and governments have been using encryption since long, in order to facilitate secret communication. In recent times, it is more commonly used in protecting information within several kinds of civilian systems including WhatsApp, Viber, which use end-to-end encryption (E2EE). Here, except the sender and the receiver no one can decrypt the encrypted data.

There are two types of **encryption: Symmetric encryption,** a simple type of encryption where only private key is used; and **Asymmetric encryption,** which is a relatively complex type of encryption. Here, a public key, which is known to all recipients is used to decrypt the message and also the private key. But, **WhatsApp uses more complex types of asymmetric encryption.** It uses three types of public key and here, the private key varies in each message sent and gets encrypted without any intervention of the user.

### *Encryption Level and Issue of Security*

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Encryption is a challenge, not only for India but also for the world to enforce law. In India, the telecom sector is limited to the encryption of 40 bits; however, the **2015 draft encryption policy (now this policy is withdrawn) recommended the use of 256-bit key for encryption and promoting the use of Digital Signature, thereby envisioning a secure cyberspace.**

After the Pathankot Attack in 2016, it raised questions on the government's regulation policy regarding encryption. Keeping in mind that a strong regulation can hamper the innovation of encryption technology, the Indian government has to make a policy that can control the terrorist and criminal activities.

## 12.12 Wearable Technology

Wearable electronics are the next big innovation in the world of technology after the invention of smartphones. **Wearables are electronic technology or devices incorporated into items that can be comfortably worn on a body.** These wearable devices are used for tracking information on real time basis. They have motion sensors that take the snapshot of your day-to-day activity and sync them with mobile devices or laptop computers.

Wearable technology appears prominently in consumer electronics with the popularisation of the smartwatch and activity tracker. Apart from commercial uses, wearable technology is being incorporated into navigation systems, advanced textiles and healthcare. Hearing aid wearable devices are more common and older than the popular smartwatches today.

Even before the wearable technology entered the consumer market, these wearable devices were used in the field of military technology. In fact, these devices were an integral part of the medical and healthcare sector in the military forces. Devices like **'Wearable Motherboards'** or **'Smart Shirts'** used to monitor the health and well-being of the patients and send back information to the hub station on real-time basis.

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## 12.13 Biometrics

The measurement and statistical analysis of people's physical and behavioural characteristics comes under Biometrics. The term 'biometrics' is derived from the Greek words 'bio' meaning life and 'metric' meaning to measure. The technology enables identification and access control, or for identification of individuals that are under surveillance. The basic premise and principle that guides biometric authentication is that everyone is unique and hence, an individual can be identified by his or her intrinsic physical or behavioural traits.

The individuals are labelled and described using biometric identifiers, which are the distinctive, measurable characteristics. They are often classified as physiological and behavioural features.

**Physiological characteristics** are those, which are connected to the body shape. Common examples are fingerprint, palm veins, face recognition, palm print, hand geometry, iris recognition, retina and odour/scent. Even DNA is a very important characteristic.

**Behavioural characteristics** relate to the pattern of behaviour of a person, including but not limited to typing rhythm, gait and voice. Some researchers have coined the term behaviour metrics to describe these features.

### *Methods of Biometrics*

A biometric system can be of two types: An 'identification' system and a 'verification' system.

**Identification:** One to Many: A person's identity can be determined using biometrics, even without his knowledge or consent. For example, scanning a crowd with a camera and using face recognition technology, matching against a known database, can be determined, which has become very useful in recent times.

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**Verification:** A person's identity can be verified using biometrics. For instance, one can grant physical access to a secure area in a building by using finger scans or can grant access to a bank account at an ATM by using retinal scan. Biometric authentication needs to compare a registered or enrolled biometric sample (biometric template or identifier) against a newly captured biometric sample (for example, the one captured during a login). It is a three-step process (Capture, Process and Enrol), after which Verification or Identification process follows.

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## *Characteristics of Biometrics*

During the first phase of processing, number of biometric characteristics may be captured. However, that the biometric characteristics should satisfy the following characteristics for automated capturing and automated comparison with previously stored data:

1. **Universal:** The characteristic/attribute should be possessed by everyone and should be such that it is universal and seldom lost to accident or disease.
2. **Invariance of Properties:** They should be constant over a long period of time and not subjected to significant differences on the basis of age, either episodic or chronic diseases.
3. **Measurability:** The properties can be captured with ease and in less time, waiting time and the attribute data can be gathered passively.
4. **Singularity:** Each expression of the attribute must be unique to the individual with sufficient unique properties to distinguish one person from other. Height, weight, hair and eye colour are all attributes that are unique, assuming a particularly precise measure; however, not enough points of differentiation except in categorising usage.
5. **Acceptance:** A large percentage of the population should accept the way to capture. Invasive technologies, particularly, are excluded because they require a part of the human body to be taken or which (apparently) impair the human body.
6. **Reducibility:** The captured data should be capable of being reduced to an easy-to-handle file.

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7. **Reliability and Tamper-Resistance:** The attribute should be practically impossible to mask or manipulate to ensure high reliability and reproducibility.
8. **Privacy:** The process should not violate the privacy of the person.
9. **Comparable:** The attribute can be reduced to a state, which is digitally comparable to others. The less probabilistic the matching involved, the more authoritative the identification.
10. **Inimitable:** The attribute cannot be reproduced by other means. The lesser the reproducibility of the attribute, the more likely it will be authoritative.

### *Biometrics in India*

India, being perpetually under the threat of terrorism and security breach, is gearing up to adopt various biometric methods.

It is used in higher level of security and authentication:

- In Defence and Security Industry
- In Police Administration
- In Finance Ministry
- In Home Affairs

## 12.14 Digital Lending

It consists of lending through **web platforms or mobile apps,** by taking advantage of technology for authentication and credit assessment. **India's digital lending market has seen a significant rise** over the years. January 1, 2024 Reports suggest that India's digital lending market is projected to reach $515 billion by 2030, up from $38.2 billion in 2021, with a compound annual growth rate (CAGR) of 33.5 percent. **Banks** have launched their **own independent digital lending platforms** to tap in the digital lending market by leveraging existing capabilities in traditional lending.

### *Significance of Digital Lending*

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**Financial Inclusion:** It helps in meeting the huge unmet credit need, particularly in the microenterprise and low-income consumer segment in India.

**Reduce Borrowing from Informal Channels:** It helps in reducing informal borrowings as it **simplifies the process of borrowing**. Indians continue to borrow from family and friends and moneylenders, sometimes at unreasonably high interest rates, primarily because these loans are more flexible and convenient.

**Time Saving:** It decreases time spent on working loan applications in-branch. Digital lending platforms have also been known to **cut overhead costs by 30–50%.**

### *Issues with Digital Lending Platforms*

Growing nn of **unauthorised digital lending platforms** and mobile applications as:

- They charge **excessive rates of interest and additional hidden charges.**
- They adopt **unacceptable and high-handed recovery** methods.
- They **misuse agreements to access data on mobile phones** of borrowers.
- **Steps Taken by RBI:**
  - **Non-Banking Financial Companies (NBFCs)** and **banks** need to **state the names of online platforms** they are working with.
  - RBI has also **mandated that digital lending platforms which are used on behalf of banks and NBFCs should disclose the name of the Bank(s) or NBFC(s)** upfront to the customers.
  - The Central Bank had also **asked lending apps to issue a sanction letter to the borrower on the letter head of the bank/NBFC concerned** before the execution of the loan agreement.
  - Legitimate public **lending activities can be undertaken by banks, NBFCs registered with the RBI** and other entities who are regulated by state governments under statutory provisions.
- **India's Digital Ecosystem:**
  - Banks have partnered with Fintechs to serve their customers better.

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- Government of India has taken many initiatives such as **Unified Payments Interface, Jan Dhan Yojana, Aadhaar enabled Payment System**, etc., to promote digital environment in the country especially after demonetisation.

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India is on the verge of a **digital lending revolution and making sure that this lending is done** responsibly and can ensure the fruits of this revolution are realised. As several players have access to sensitive consumer data, **there must be clear guidelines around, for example, the type of data that can be held, the length of time data can be held for and restrictions on the use of data.**

Digital lenders should proactively develop and commit to a **code of conduct that outlines the principles of integrity, transparency and consumer protection,** with clear standards of disclosure and grievance redressal. An **agency can be created that tracks all digital loans and consumer/lender credit history.** Apart from establishing technological safeguards, **educating and training customers to spread awareness about digital lending is also important.**

## 12.15 Digital Signature

A Digital Signature means an electronic signature, useful to authenticate the identity of the sender of a message or the signer of an electronic document. This ensures that the original content of the document is not changed. Digital signatures are advantageous, as they are easily transportable, cannot be imitated by someone else and can be automatically time-stamped. The sender cannot easily repudiate it later, which ensures that the original signed message is received.

Certain types of encryptions make digital signature effective to ensure authentication. Encryption takes all the data that one computer is sending to another and encodes it into a form that only the other computer will be able to decode. Authentication verifies that information is coming from a trusted source. These two processes work hand-in-hand for digital signatures.

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### How does It work?

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Let us say, you are going to send the draft of a contract to your lawyer in another town. To assure your lawyer that it is unchanged from what you have sent and that it is really from you, this is how you proceed with the process:

1. Copy-and-paste the contract into an email note.
2. By using special software, you obtain a message hash (mathematical summary) of the contract.
3. You then use a private key that you have previously obtained from a public–private key authority for encryption of hash.
4. The encrypted hash acts as your digital signature of the message. (Note: It will be different each time you send a message.)

### At the other end, your lawyer receives the message

1. To make sure it is intact and from you, a hash is made by your lawyer, of the received message.
2. Your lawyer then uses your public key to decrypt the message hash or summary.
3. The received message is valid only if the hashes match.

### Significance

Just like a handwritten signature authenticates printed documents, a digital signature authenticates electronic documents; hence, this signature cannot be forged and it asserts that a named person wrote or otherwise agreed to the document, which attaches that signature. On the whole, digital signatures enable the 'authentication' and 'non-repudiation' of digital messages, which assures the recipient of a digital message in terms of both, identity of the sender and the integrity of the message.

A Certification Authority (CA) issues a digital signature, which is signed with the CA's private key. A digital signature typically contains the: Owner's public key, the owner's name, expiration date of the public key, the name of the issuer (the Certification Authority that issued the Digital ID), serial number of the digital signature and the digital signature of the issuer.

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### Legal in India

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India is among those few nations that have the Digital Signature Legislation. This Act grants digital signatures (issued by the Licensed Certifying Authority in India) the same status as a physical signature. Digital signatures deploy the Public Key Infrastructure (PKI) technology.

### Digital Signatures vs. Digital Certificates

A Digital Signature is an electronic method of signing an electronic document whereas a Digital Certificate is a computer-based record which:

- Identifies the Certifying Authority issuing it.
- Has the name or the identity of its subscriber.
- Contains the subscriber's public key.
- Is digitally signed by the Certifying Authority issuing it.

So, digital signature is not a record, and the identification of certifying authority is ascertained from the digital certificate, not digital signature.

A digital certificate is a form of an electronic credential for the Internet. Similar to a driver's licence, employee ID card, a digital certificate is issued by a trusted third party to establish the identity of the certificate holder. The third party who issues the digital certificate is known as the Certifying Authority (CA).

- **Digital Signatures** provide Authentication, Privacy, Non repudiation and Integrity in the virtual world. IT Act 2000 in India gives legal validity to electronic transactions that are digitally signed. Therefore, we need digital signatures for secure messaging, online banking applications, online workflow applications, e-tendering, supply chain management, etc.
- **Digital Certificates** are digital documents attesting to the binding of a public key to an individual or specific entity. They allow verification of the claim that a specific public key does, in fact, belong to a specific individual. Digital certificates help to prevent someone from using a phony key to impersonate someone else.

## 12.16 Edge Computing

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According to a research, by 2025, companies will generate and process more than 75% of their data outside of traditional centralised data centres – that is, ***at the 'edge' of the cloud.*** As the use of ***Internet of Things (IoT)*** devices expands and the need to process IoT data quickly increases, many IT leaders are considering or beginning to employ ***edge computing*** options.

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## What is Edge Computing?

- Edge Computing ***enables data to be analysed, processed and transferred at the edge of a network. Meaning, the data is analysed locally,*** closer to where it is stored, in real-time without latency, rather than sending it far away to a centralised data centre.
- It allows for quicker data processing and content delivery.

## How is Edge Computing Different from Cloud Computing?

- The basic difference between edge computing and cloud computing lies ***in where the data processing takes place.***
- ***Internet of Things (IoT)*** systems perform all of their computations ***in the cloud using data centres.***
- In ***Edge Computing,*** massive amounts of data generated by IoT devices are stored and processed locally. ***That data doesn't need to be sent over a network as soon as it processed; only important data is sent*** – therefore, an edge computing network reduces the amount of data that travels over the network.

## Challenges

According to experts, ***edge computing introduces several new security risks.***

1. One of the most prominent concerns is the physical security of the devices, which are more vulnerable to malicious attacks and mishaps of all kinds than typical office equipment and technology safely held within corporate walls.
2. ***Micro data centres,*** such as those being deployed by telecommunication companies – in some cases, at the base of cell

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towers-introduce a level of physical vulnerability that didn't exist with corporate data centres and large cloud providers.

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3. Meanwhile, ***many organisations will be challenged to understand, track and monitor what data they have and where, what protections are required at the various points based on the data and vulnerabilities specific to each endpoint and how to govern what could soon be a sprawling infrastructure at many companies.***

### *Benefits*

- Edge computing may ***offer some protection against a catastrophic attack where a single incident can compromise large amounts of a company's data.***
- In some ways, ***it's more resilient, because instead of one or two or even three data centres,*** you have distributed data and compute on the edge, which makes it much more resilient to malicious and nonmalicious events.

### *Way Ahead*

- Experts believe the true potential of edge computing will become apparent when 5G networks go mainstream in a year from now. Users will be able to enjoy consistent connectivity without even realising it.

## 12.17 BSNL Launched Direct – to – Device (D2D) Satellite Connectivity

**BSNL launched** India's First Direct-to-Device (D2D) Satellite Connectivity.

BSNL has become the **first Indian telecom provider** to launch **Direct-to-Device (D2D) satellite connectivity**, enabling satellite-based communication **directly on smartphones**, eliminating the need for terrestrial towers.

This development brings India in line with global initiatives like:

- **AST SpaceMobile – United States**
- **Lynk Global – American Company**

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- **Constellation Global – London, United Kingdom**
- **SpaceX Starlink**

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### *How D2D Satellite Technology Works*

- **Core Principle:** Satellites function like space-based cell towers.
- **Transmission:** Signals are sent directly from **geostationary satellites** to devices without relying on traditional ground infrastructure.
- **NTN (Non-Terrestrial Network):** Enables two-way communication between satellites and mobile devices.
- **Technology Partner:** BSNL uses **Viasat's Geostationary L-band satellites**, stationed 36,000 km above Earth.

### *Key Benefits*

- **Seamless connectivity** in remote and underserved areas
- **High-speed internet** with wider coverage
- **Digital empowerment** through support for **UPI payments**
- **Emergency Use:** Enables **SOS messaging** and calls during disasters or in isolated areas
- Ideal for **travelers, hikers and remote field operations.**

### *Challenges*

- **Latency:** High orbit distances can cause delays in voice/video communication.
- **Regulatory Hurdles:** Issues with **cross-border data** transmission and permissions.
- **Spectrum Allocation:** Requires appropriate and interference-free frequencies.
- **Device Compatibility:** Needs updates in smartphone hardware/software.
- **Signal Propagation:** Vulnerable to **environmental interference** and terrain obstacles.

## 12.18 Space Internet

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### 12.18.1 Starlink Project

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***SpaceX,*** the world's leading private company in space technology, has launched many satellites into ***the low earth orbit,*** under the ***Starlink Network Project.***

## What is It?

- The Starlink network is one of several ongoing efforts to ***start beaming data signals from space.***
- Under the project, the company intends ***to evolve into a constellation of nearly 12,000 satellites.***
- The aim is ***to provide low-cost and reliable space-based Internet services to the world***.
- The project was announced in 2015.

## Significance of the Project

- The project ensures that reliable and uninterrupted Internet services are universally available in every part of the globe.
- About 4 billion people, more than half of the world's population, do not have access to reliable Internet networks.
- This is because ***the traditional ways to deliver the Internet – fibre-optic cables or wireless networks – cannot take it everywhere on Earth.***
- In many remote areas or places with difficult terrain, it is not feasible or viable to set up cables or mobile towers.

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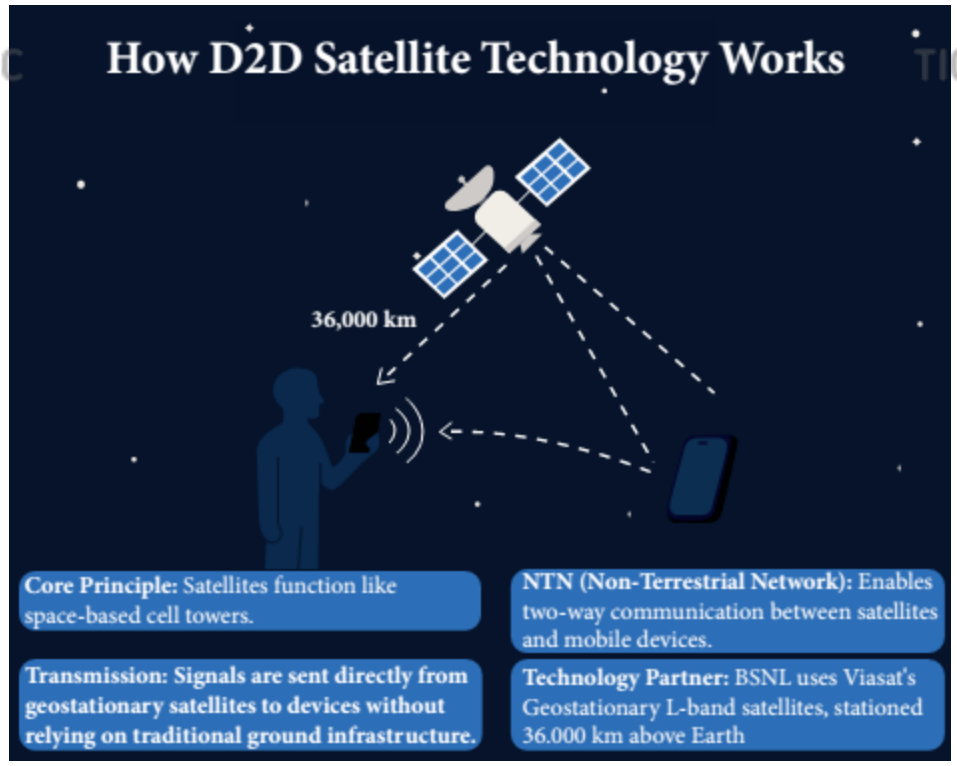


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- ***Signals from satellites*** in space can overcome this obstacle easily.

## *Starlink*

- Starlink satellite internet is currently available in over 100 countries and territories.
- It has been launched in areas across the US, Europe, South America, Africa, and Southeast Asia.
- Some countries are still on a waitlist or are slated to have Starlink services coming soon.

Countries with Starlink services:

**United States:** Starlink is widely available in many areas, with some cities experiencing high demand and waitlists.

**Europe:** Several European countries have Starlink services.

**South America:** Starlink is available in various countries in South America.

**Africa:** Starlink is available in a growing number of African countries.

**Southeast Asia:** Starlink is available in several countries in Southeast Asia.

Countries pending regulatory approval or with services coming soon:

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**India:** SpaceX is waiting for regulatory approvals before launching Starlink services in India.

**Pakistan:** Starlink is also awaiting regulatory approval in Pakistan.

**Other Countries:** Several other countries, including Saudi Arabia, Thailand, Turkey, Vietnam, and the United Arab Emirates, are listed as having services coming soon or pending regulatory approval.

### *Viasat Company*

ViaSat, an American communications company headquartered in Carlsbad, California, operates additional facilities across the United States and globally.

Viasat is a global communications company focused on satellite technology, providing high-speed internet and connectivity solutions across various sectors, including residential, commercial, government, and military markets.

They offer services on land, in the air, and at sea, connecting individuals, businesses, and communities globally

## 12.18.2 OneWeb: Internet

OneWeb has a partnership formed with debris-removal start-up Astroscale under **ESA's Sunrise program**.

### *About OneWeb's LEO Internet Programme:*

- **OneWeb is a Low Earth Orbit (LEO)** satellite communications operator.
- Using LEO satellites OneWeb seeks to offer connectivity across **the UK, Alaska, Northern Europe, Greenland, the Arctic Seas and Canada.** The company expects the service to be switched on before the end of this year.
- It calls this programme the **'Five to 50' service** of offering internet connectivity to **all regions north of 50 degrees latitude**.

### *Potential*

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- LEO satellite broadband is preferable in areas that cannot be reached by fibre and spectrum services. The target market will therefore be rural populations and military units operating away from urban areas.
- OneWeb's chief competitor is **Starlink**, a venture led by Elon Musk's SpaceX. Starlink currently has 1,385 satellites in orbit and has already started beta testing in North America and initiating pre-orders in countries like India.

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### 12.18.3 Jio Platforms to Provide Satellite Broadband Services

- Jio has formed a joint-venture with Luxembourg-based SES to deliver satellite broadband services across India.
- The joint venture will use **multi-orbit space networks** that is a combination of GEO **(geostationary equatorial orbit) and MEO (medium earth orbit) satellite constellations** capable of delivering multi-gigabit links and capacity to enterprises, mobile backhaul and retail customers across the length and breadth of India and neighbouring regions.

#### *How is this different from what Starlink or OneWeb offers?*

- SES primarily has satellites in the GEO and the MEO, while those of Elon Musk-led Starlink and Bharti Group's **OneWebare** in **low earth orbit (LEO)**.
- While **GEO satellites are positioned at an altitude of 36,000 km, MEO and LEO are lower at altitudes of 5,000–20,000 km and 500–1,200 km,** respectively.
- **The altitude of the satellite is directly proportional to the area of earth that it covers**. Therefore, the higher a satellite is positioned, the larger an area it covers.

## 12.19 Virtual Reality (VR) and Augmented Reality (AR)

#### *Virtual Reality*

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Virtual Reality or Virtual Realities (VR) is/are also called as immersive multimedia or computer-simulated reality. It is a computer technology replicating an environment, real or imagined, wherein, a user's physical presence and environment are simultaneously with user interaction.

- Virtual realities artificially create the experience of senses, including sight, touch, hearing and smell.
- Either a computer screen or a special virtual reality headset (head mounted display) display the most up-to-date virtual realities. Some simulators consist of additional sensory information and focus on real sounds with the help of speakers or headphones.
- Currently, some advanced haptic systems consist of physical information, which is usually termed as force feedback in applications of medical, gaming and military.
- Through the usage of standard input devices like keyboard and mouse or with the help of multimodal devices, such as a wired glove or omnidirectional treadmills, there is coverage of the remote communication environments, thereby enabling the user's virtual presence by deploying the telepresence and telexistence concepts or a virtual artefact. The mesmerising environment can be similar to the real world in order to create a real type experience which can be beneficial in cases like pilot or combat training simulations.

## *Augmented Reality*

- **Augmented Reality (AR)** is a live direct or indirect view of a physical, real-world environment. The elements of AR are augmented (or supplemented) by computer-generated sensory input like sound, video, graphics or GPS data.
- It is related to a more general concept called mediated reality, in which modification is done by computer as a view of reality. Consequently, the technology enhances one's current perception of reality. Contrary to this, virtual reality replaces the real world with a simulated one.
- Augmentation by convention is in real time and in semantic context with environmental elements. The examples include sports scores on TV during a match.

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- The real world is overlaid with information about the environment and its objects. This can be virtual or real, such as seeing other real sensed or measured information, such as the over lading of electromagnetic radio waves in correct alignment with where they actually are in space.

### *Applications*

- **Health Care:** Virtual reality exposure therapy (3-dimensional computer simulation), when combined with physiological monitoring and feedback, enables treatment of panic and anxiety disorders.
- **Education Sector:** Education sector has adopted virtual reality for teaching and learning situations. It is advantageous because it enables large groups of students to interact with each other; in fact, within a three-dimensional environment.
- **Tourism:** Tourism is another industry in the use of virtual reality, i.e., in museums and historical places. These settings employ interaction for communicating information to the general public in new and exciting ways.
- **Other Sets of Applications are:** Business, engineering, sports, media, military and scientific visualisation.

(Refer **Infographic – VR/AR/MR & AR**)

## 12.20 Metaverse

Metaverse is a combination of "meta" (meaning beyond) and the stem "verse" from "universe", denoting the next-generation Internet in which the users, as avatars, can interact with each other and software applications in a three dimensional (3D) virtual space. In other words it can also be explained as an **interoperable network of 3d virtual worlds** that can be accessed simultaneously by millions of users, who can **exert property rights over virtual items.**

Think of a combination of immersive virtual reality, a massively multiplayer online role-playing game and the web.

Metaverse is a successor of today's internet. It's only a vision at this point, but technology companies like Facebook are aiming to make it the

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setting for many online activities, including work, play, studying and shopping.

If we have to answer the question – what is metaverse, it could be answered with the help of three key aspects – presence, interoperability and standardisation.

**Presence** is the feeling of being in a virtual space, with virtual others. It will improve the quality of online interactions. This sense of presence is achieved through virtual reality technologies such as head-mounted displays.

**Interoperability** means being able to seamlessly travel between virtual spaces with the same virtual assets, such as avatars and digital items. Zoom meetings through apps like Animaze, blockchain technologies such as cryptocurrencies and non-fungible tokens facilitate the transfer of digital goods across virtual borders.

**Standardisation** is what enables interoperability of platforms and services across the metaverse. International organisations such as the Open Metaverse Interoperability Group define these standards.

## 12.21 DNA Computer

DNA computing is a branch of computing, which uses DNA, biochemistry and molecular biology hardware, in place of the traditional silicon-based computer technologies. Research and development in this area is concerned with theory, experiments and applications of DNA computing.

The term ‘molectronics’ has sometimes been used; however, it was already used for an earlier technology, a then-unsuccessful rival of the first integrated circuits. This term has also been used for molecular-scale electronic technology, more generally.

The potential of DNA-computer to make a high amount of multiple parallel computations at a high speed compensates its slow processing speed (the response time is measured in minutes, hours or days, and not in milliseconds). Thus, it enables the system to consume similar amount of time for a complex calculation as for a simple one.

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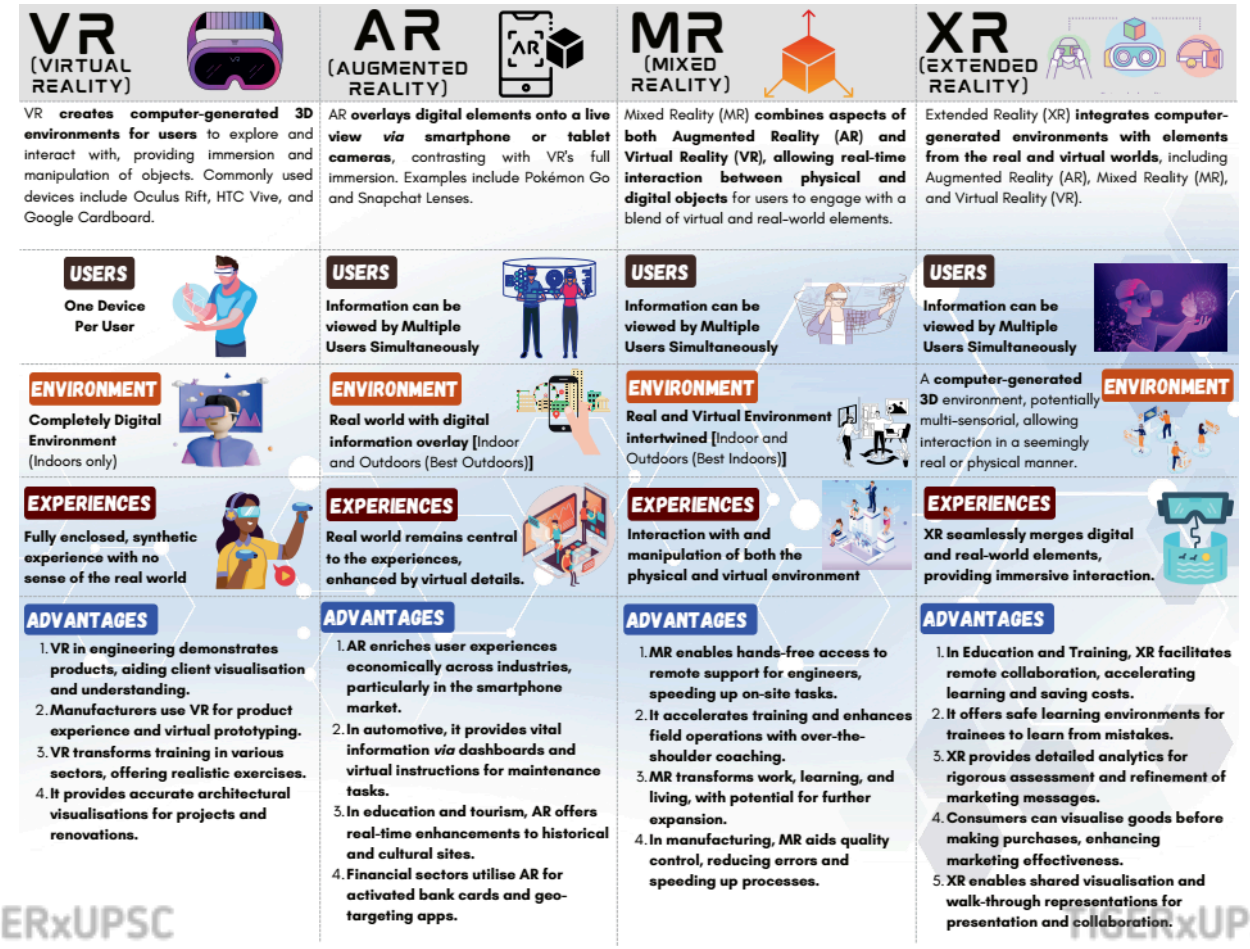
This is achieved considering that millions or billions of molecules interact with each other simultaneously. However, the analysis of the answers, given by aDNA-computer is much harder as compared to a digital one.

## 12.22 Non-Fungible Tokens (NFT)

The global NFT market is projected to reach $608.6 million in 2025. Revenue is expected to show an annual growth rate (CAGR 2023-2027) of 18.55% resulting in a projected total amount of US$3,162.00m by 2027. As the crypto asset exploded in popularity, fuelled by the rising interest of celebrities and tech evangelists. Some experts believe NFTs are a bubble which might pop.

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### Features

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- NFT can be anything that can be converted into a digital form.
- Everything from drawings, photos, videos, GIF, music, in-game items, selfies, and even a tweet can be turned into an NFT, which can then be traded online using cryptocurrency.
- NFTs are unique from other digital forms as it is backed by Blockchain technology.
- NFTs are gaining massive popularity now because they are becoming an increasingly popular way to showcase and sell your digital artwork.

### Why are people willing to spend millions on something they could easily screenshot or download?

When NFT is listed on a marketplace, you need to pay something called a gas fee (transaction fee) for using the Blockchain, following which digital art is then recorded on Blockchain, mentioning that you (your address) own the particular NFT.

This gives you full ownership, which cannot be edited or modified by anyone, including the marketplace owner. An NFT is thus created to get exclusive ownership rights. NFTs can have only one owner at a time.

Apart from exclusive ownership, NFT owners can also digitally sign their artwork and store specific information in their NFTs metadata which will only be viewable to the individual who bought the NFT.

### Working Mechanism of NFTs

NFT works on blockchain as it gives users complete ownership of a digital asset. For example: consider a sketch artist who converts digital asset to an NFT what he gets in return is proof of ownership, powered by Blockchain.

### Blockchain Technology

A blockchain is a digital ledger of transactions that is duplicated and distributed across the entire network of computer systems on the blockchain.

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Each block in the chain contains a number of transactions, and every time a new transaction occurs on the blockchain, a record of that transaction

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is added to every participant's ledger. In short, here the transactions are transparent and can be seen by anyone and cannot be changed or modified once recorded.

### Cryptocurrency

Cryptocurrency also called "crypto" is any form of currency that exists digitally or virtually and uses cryptography to secure transactions. Cryptocurrencies does not have a central issuing or regulating authority. It instead uses a decentralised system to record transactions and issue new units.

It's a peer-to-peer system that can enable anyone anywhere to send and receive payments.

### NFT vs. Cryptocurrency

NFTs and cryptocurrencies are very different from each other. The only similarity is that both are built on Blockchain.

Cryptocurrency is a currency which is fungible meaning that it is interchangeable. For instance, if you hold one crypto token, say one Ethereum, the next Ethereum that you hold will also be of the same value.

While NFTs are non-fungible that means the value of one NFT is not equal to another. Every art is different from other, making it non fungible, and unique.

### Who can Buy NFTs?

- Anyone who holds a cryptocurrency wallet can buy an NFT.
- That is the only prerequisite to purchase an NFT.
- No need of any KYC documents to purchase an art.
- All that is needed is a cryptocurrency wallet powered by Metamask, and an NFT marketplace where you can buy and sell NFTs.

[Refer **Infographic – Non Fungible Token (NFT)**]

### Risks Associated with NFTs

- Several incidents of NFT scams like emergence of fake marketplaces, unverified sellers often impersonating real artists and selling copies of

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their artworks for half prices.

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- Also sharing of potential phishing link leading to draining of cryptocurrency wallet.
- Hacking of NFT's is another potential threat which can leave you bankrupt.
- It also negatively impact the environment in order to validate transactions crypto mining is done, which requires high powered computers that run at a very high capacity, affecting the environment ultimately.

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## 12.23 Traditional Knowledge Digital Library (TKDL)

TKDL is an Indian digital knowledge repository of traditional knowledge, especially dealing with medicinal plants and formulations used in Indian systems of medicine. TKDL was established with the collaboration of the

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Council of Scientific and Industrial Research (CSIR) and the Department of Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy (Department of AYUSH), the Ministry of Health and Family Welfare, Government of India, in 2001. It aims at shielding the country's ancient and traditional knowledge from the exploitation of bio-piracy and unethical patents, by electronically documenting and characterising it, as per the patent systems of international classification. In addition to this, the non-patent database based on the traditional knowledge serves in the promotion of modern research, which eventually helps in the simplification of the naturopathy's vast knowledge access, including those of traditional remedies or practices.

**Background:** In the late 1990s, subsequent to the granting of patents of turmeric and basmati rice by United States Patent and Trademark Office (USPTO) and the neem patent granted by European Patent Office (EPO), respectively, the Indian Government successfully achieved their repudiation. Thereafter, the biopiracy and unethical bioprospecting issue made headlines. This led to coming of more number of patent claims. Due to the existence of India's vast traditional medicine knowledge in languages such as Sanskrit, Hindi, Arabic, Persian, Urdu and Tamil, it became impossible for patent examiners for the verification of such claims at the International patent offices. Thus, the Department of AYUSH, Government of India was encouraged for the creation of TKDL in 2001.

**Latest Achievements:** TKDL protects Indian traditional medicinal knowledge, converting texts into international languages and setting standards. It enhances prior art search with TKRC, accessed by thirteen patent offices. Over 230 patent (as on July 2024) applications affected. Recognized for deterring bio-piracy, TKDL sets global standards in proactive traditional knowledge protection.

TKDL also signed an agreement with the world's leading international patent offices, such as the European Patent Office (EPO), the United Kingdom Trademark and Patent Office (UKPTO) and the United States Patent and Trademark Office (USPTO). Such agreements are signed with a

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primary objective in the protection of conventional knowledge from biopiracy and further giving access to the database of TKDL by the patent examiners at the International Patent Offices for enabling the purpose of patent search and examinations.

## 12.24 Cryptography

Cryptography is defined as a process of converting ordinary plain text into unintelligible text and vice-versa. Through this method, the data can be stored and transmitted in a particular form that can be read and processed by only those for whom it is intended. Cryptography protects data from theft or alteration and in addition, can also be used for user authentication.

Earlier, cryptography was effectively synonymous with encryption but nowadays, it is mainly based on computer science practice and mathematical theory.

- Internet communication could be secure or insecure, depending upon the technique used for transmitting digital information over the network.
- With the advancement of Information Technology, most of the information, be it private information or public information, is being transmitted with high speed and frequency.
- But, is it secure? There have been frequent cases of data theft and information leaks.
- Cryptography is an encoding technique of communication where the actual message is encrypted into unreadable format using various algorithms. This text is called cipher text.
- It is then sent over the network. Even if someone sniffs and intercepts the message, they will not be able to interpret the exact meaning of the text.
- The receiver receives the message and decrypts the information into plain text.
- There are two broad techniques of implementing cryptography: Symmetric key technique and Public key technique.

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- Cryptography implements the different aspects of information security including data confidentiality, data integrity, authentication and non-repudiation.

### *Application*

- It is now used in our day-to-day actions like Internet banking, ATM transaction, and online shopping.
- Our email communication is encrypted.
- Personal chats and messaging systems like WhatsApp, Facebook messaging are also encrypted.

Cryptography is used in many applications like banking transactions, cards, computer passwords and e-commerce transactions.

## 12.25 Digital India Initiative

Government of India launched Digital India campaign for the availability of the Government services electronically to the citizens through the improvisation of online infrastructure and enhancing the Internet connectivity. It also aims to make the country digitally empowered in the technology field. DeITY or Department of Electronics and Information Technology is an umbrella organisation, which cooperates and coordinates implementation of this programme. All the digital initiatives are like National Broadband Plan and Domestic Manufacturing Policy are integrated and brought under this initiative.

Prime Minister Narendra Modi launched Digital India on 01 July 2015. The initiative involves the strategy of linking rural areas with the network of high-speed Internet. Its three core components are:

1. Creating digital infrastructure
2. Service delivery in digital manner
3. Digital literacy

Through this, Government of India expects growth on multiple fronts. Specifically, the nine 'Pillars of the Digital India' as identified and targeted by GOI are:

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- Broadband Highways
- Universal Access to Internet
- Public Internet Access Programme
- e-Governance Reforming Government through Technology
- eKranti – Electronic Delivery of Services
- Information for All
- Early Harvest Programme
- IT for Jobs
- Electronics Manufacturing

Undoubtedly, it is one of the grandest projects till date, embracing all the government machinery and departments in its fold. It primary aim is to create a truly participatory democracy with the application of digital technology. Other targets include connecting more than 120 crore Indians with their government so that they could avail services and benefits of government services in realtime and also get quick grievance redressal.

### *The Vision of Digital India*

- Infrastructure development that would enable each and every citizen to utilise it to the fullest. Some of these infrastructures include high-speed Internet facilities, digital identity, mobile and phone banking and safe and secure cyber space.
- Governance and online services that citizens can use to avail services, which remain away from their reach. Some of the facilities include digital support system, services available in real time, services that can enable ease-of-doing business, electronic and cashless financial transactions.
- Digital Empowerment of Citizens: Only digital literacy can harness the benefits of digitisation. Some of the steps would be universal accessibility of digital resources and regional compatibility of digital resources.

### *Challenges in the Implementation of Digital India*

However, implementation of this scheme stands as a great challenge before the government;

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**Firstly** connecting 2,50,000 Gram Panchayats through National Optical Fibre Network is not easy in any manner and hence, will require a lot of efforts and dedication from BSNL, the nodal agency that lays down the cables in India.

**Secondly**, IT literacy needs huge improvement, which is again a great task ahead because the project can succeed only when masses know how to utilise the digital services.

**Thirdly**, security of data is mandatory as digital medium is still very unsafe and data vulnerability is a big issue, especially during financial transactions.

**Lastly**, creation of awareness and knowledge about Digital India scheme and its benefits among common masses is also a great challenge, considering high population of India.

**Overall**, great efforts are required from the entire government machinery and people concerned for the success of this project.

## *What is the Digital India Act (DIA) 2023?*

In 2022, the Indian government introduced the Digital India Act (DIA) to establish a contemporary legal framework for India's rapidly evolving digital landscape.

The DIA incorporates the Digital India Goals of 2026, aiming to achieve a $1 trillion digital economy. With the largest number of internet users globally, India intends to influence global technological advancements through these goals. The legislation emphasises ensuring the openness, safety, trust and accountability of the internet while safeguarding citizens' rights.

While promoting India's digital economy growth, the DIA addresses challenges posed by emerging technologies such as data privacy and cybersecurity. The removal of the safe harbor principle may draw criticism from major technology firms. Implementation will necessitate specialised expertise and robust infrastructure for law enforcement, addressing uncertainties related to technologies like AI, deep fakes and dispute resolution outlined in the legislation. Defining territorial jurisdiction is crucial due to the borderless nature of online interactions and information flow.

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The Act is built on principles of transparency and accountability, striving to balance the interests of stakeholders including users, large technology companies, government entities, businesses and civil society.

## 12.26 Telemedicine

The communications and information technology usage in delivering clinical care is termed as Telemedicine. It is a rapidly developing application of clinical medicine, which transfers medical information through interactive audio-visual, media over a distance. The primary aim of telemedicine is consultation, and sometimes, remote medical procedures or examinations.

Telemedicine is the interaction between patient and doctor without physically facing each other through a medium of communication.

- The medium of communication is usually IT and telecommunication networks.
- Health care facilities like consultation, diagnosis, status report, report analysis, etc., can be provided through it.

At times, the terms 'e-health' and 'telehealth' are wrongly interchanged with telemedicine. Similar to the terms 'medicine' and 'health care', telemedicine often refers only to the provision of clinical service. On the contrary, telehealth can refer to clinical as well as non-clinical services, including medical education, administration, and research; while e-health, particularly in the UK and Europe, is used as an umbrella term including telehealth, electronic medical records and other components of health IT.

### *Types of Telemedicine*

The three main categories of telemedicine are as follows:

- **Store-and-forward telemedicine** implies acquiring medical data (like medical images, bio signals, etc.), which is later transmitted to a doctor or a medical specialist for offline assessment at a convenient time. Both parties are not required simultaneously. The common specialities that are conducive to asynchronous telemedicine include dermatology,

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radiology and pathology. For this transfer, a properly structured Medical Record (preferably in electronic form) is required.

- **Remote monitoring** or self-monitoring/testing is a technology enabling medical professionals to monitor a patient remotely using various technological devices. Through this method, chronic diseases or specific conditions, such as heart disease, diabetes mellitus or asthma can be managed. These services are advantageous in terms of comparable health outcomes to traditional in-person patient encounters, greater satisfaction to patients and cost-effectiveness.
- For real-time interactions between patient and provider, Interactive telemedicine services are developed. They include phone conversations, online communication and home visits. Many activities such as history review, physical examination, psychiatric evaluations and ophthalmology assessments are some activities that can be conducted. Additionally, the cost of 'clinician-interactive' telemedicine services is much lesser than in-person clinical visits.

### *Benefits and Uses of Telemedicine*

For populations living in isolated communities and remote regions, telemedicine proves to be a boon of technical advancement. It is currently being applied in virtually all medical domains. Telemedicine serves as an effective tool for communication between a general practitioner and a specialist available at a remote location.

#### IMPORTANCE OF TELEMEDICINE

- Remote area connectivity for medical facilities.
- Urgency of consultation like eye treatment, etc.
- Treatment and consultation of contagious disease like Ebola, etc.
- In India, Panchayat's Primary Health Centres (PHCs) can be connected through telemedicine facilities.
- Cost benefit for patients.
- Speedy diagnosis and treatment facility.
- Can provide health care to the neglected sections of society like elderly people, children and women.
- Issue with telemedicine.
- There is need for security and privacy. Private information need not to be made public.

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- Electronic versions of sensitive documents such as 'X-ray' or 'Doctor's notes' must be as secure as paper ones.

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On the other hand, the emerging new way of practicing telemedicine in developing countries is called as Primary Remote Diagnostic Visits, whereby devices are used to remotely examine and treat a patient. This proves to be promising for countries like India because Primary Remote Diagnostic Consultations monitors an already diagnosed chronic disease with additional promise to diagnose and manage the diseases for which a patient will typically visit a general practitioner.

**Telehealth:** delivers health-related services and information *via* telecommunication technologies. In simpler words, it is like two health professionals discussing a case over the telephone, or as sophisticated as using videoconferencing between providers at facilities in two countries, or even as complex as robotic technology.

The expansion of telemedicine is Telehealth. It encompasses preventive, promotive and curative aspects (focus of telemedicine). Today, telehealth stresses a myriad of technology solutions in addition to administrative or educational functions related to telemedicine. For instance, the physicians can communicate with patients, prescribe drugs and provide other health services *via* email.

**eHealth:** eHealth is a relatively recent term that involves healthcare practices done with the help of electronic processes and communication.

**mHealth:** mHealth or mobile health defines the practice of medical and public health, with the help of mobile devices, like using mobile phones and PDAs, for health services and information. It works as a sub-segment of eHealth. It is another important application of Information and Communication Technology (ICT), deploying computers, mobile phones, communications satellite, patient monitors, etc., for health services and information.

Stemmed from the rapid rise of mobile phone penetration in low-income nations, eHealth has emerged largely in developing nations.

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## 12.27 Project Loon

It is a research and development project developed by Google X with a network of balloons travelling on the edge of space. It is designed for connecting people in rural and remote areas, thus enabling to fill coverage gaps, and bring people back online post disasters. The balloons float in the stratosphere, approximately 20 kilometres above the Earth's surface that is twice as high as airplanes and the weather. Stratosphere has many layers of wind, and each layer of wind varies in terms of direction and speed. Loon balloons move rising or descending into a layer of wind blowing in the desired direction of travel, when needed. The balloons can be arranged to form one large communications network with moving wind, then the signal is passed across the balloon network and back down to the global Internet on Earth and ultimately connected to the balloon network directly from the phones and other LTE-enabled devices.

In June 2013, Project Loon began with an experimental pilot basis in New Zealand, where this technology was tested. The stratosphere is situated (between 10 km and 60 km in altitude) with unique engineering challenges like low air pressure of 1 per cent at sea level, being so thin that it offers less protection from UV radiation and dramatic temperature swings, which can reach as low as –80 degrees centigrade. The balloon envelope has been carefully designed to withstand these conditions, and hence, Project Loon is able to take advantage of the stratosphere's steady winds and remain well above weather events, wildlife and airplanes. A balloon envelope is the inflatable part of the balloon made from sheets of polyethylene plastic. To allow a balloon to last around 100 days in the stratosphere, it needs to be carefully designed.

### *The Benefits on Google Loon*

It would be easy to use; no need to setup wired network and other complexities would be eliminated.

- High-speed Internet all over globe without glitches.

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- It will increase Internet penetration and subscriber base.
- The biggest advantage would be its extension to those areas where Internet expansion is either complex or not possible; like forests, deserts, mountains, hilly areas, etc.
- Also the remote areas will get affordable Internet communication, which they have been deprived of till now.
- Hence, it will bridge the gap of Internet distribution and provide equal accessibility to all.

## 12.28 TRP Rating Controversy

Television Rating Point system is an indicator of the popularity of a channel. It rates channels on the basis of viewership. TRP rating is calculated on the basis of duration of time and the channel viewed by the viewer on a daily basis and finally, at the end of the month, it is averaged. This data is important for advertisers, as they have to pay on this basis. The sole agency which measures viewership and TRP ratings is known as Television Audience Measurement (TAM) Media Research.

### *What is the Controversy?*

- Based on the TRP, advertising rates are determined.
- Major broadcasting firms like NDTV, MSM, etc., have blamed TAM for manipulating the TRP and called it as flawed data that pays them nothing.
- With the increase in household televisions, TRP should go up but it has decreased, indicating that the viewership has decreased.
- This decreasing viewership data is a major concern, which fetches less advertisement fees from advertising companies, thus reducing their revenue collection.

### *Solution*

- TAM has decided to shift rating system to a new system based on absolute number of viewers.

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- Television viewership will now be measured on the basis of Television Viewership in Thousands (TVTs), which is based on absolute numbers of viewers rather than TRP, which was based on averaged points.

All stakeholders' broadcasters, advertisers, ad-agencies and TAM, later agreed to it.

## QUESTIONS AND ANSWERS

### Objective Questions

1. **Which one of the following words/phrases is most appropriately used to denote "an interoperable network of 3D virtual worlds that can be accessed simultaneously by millions of users, who can exert property rights over virtual items?** **[UPSC (Pre) – 2024]**
   (a) Big data analytics
   (b) Cryptography
   (c) Metaverse
   (d) Virtual Matrix

2. **With reference to "Software as a Service (SaaS)", consider the following statements:** **[UPSC (Pre) – 2022]**
   1. SaaS buyers can customise the user interface and can change data fields.
   2. SaaS users can access their data through their mobile devices.
   3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

   Which of the statements given above are correct?

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(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

**3. Which one of the following is the context in which the term "qubit" is mentioned?** **[UPSC (Pre) – 2022]**

(a) Cloud Services
(b) Quantum Computing
(c) Visible Light Communication Technologies
(d) Wireless Communication Technologies

**4. With reference to Non-Fungible Tokens (NFTs), consider the following statements:** **[UPSC (Pre) – 2022]**

1. They enable the digital representation of physical assets.
2. They are unique cryptographic tokens that exist on a blockchain.
3. They can be traded or exchanged at equivalency and therefore can be used as a medium of commercial transactions.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

**5. In the context of digital technologies for entertainment, consider the following statements:** **[UPSC (Pre) – 2019]**

1. In Augmented Reality (AR), a simulated environment is created and the physical world is completely shut out.
2. In Virtual Reality (VR), images generated from a computer are projected onto real-life objects or surroundings.
3. AR allows individuals to be present in the world and improves the experience using the camera of smartphone or PC.

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4. VR closes the world, and transposes an individual, providing complete immersion experience.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 3 and 4
(c) 1, 2 and 3
(d) 4 only

**6. Consider the following statements: A digital signature is:** **[UPSC (Pre) – 2019]**

1. An electronic record that identifies the certifying authority issuing it.
2. Used to serve as a proof of identity of an individual to access information or server on Internet.
3. An electronic method of signing an electronic document and ensuring that the original content is unchanged.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 and 3 only
(c) 3 only
(d) 1, 2 and 3

**7. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?** **[UPSC (Pre) – 2019]**

1. Location identification of a person
2. Sleep monitoring of a person
3. Assisting the hearing impaired person

Select the correct answer using the code given below.

(a) 1 only
(b) 2 and 3 only
(c) 3 only
(d) 1, 2 and 3

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**8. Which of the following is/are the aim/aims of 'Digital India' Plan of the Government of India?** **[UPSC (Pre) – 2018]**

1. Formation of India's own Internet companies like China did.
2. Establish a policy framework to encourage overseas multinational corporations that collect Big Data to build their large data centres within our national geographical boundaries.
3. Connect many of our villages to the Internet and bring Wi-Fi to many of our schools, public places and major tourist centres.

Select the correct answer using the code given below:

(a) 1 and 2
(b) Only 3
(c) 2 and 3
(d) 1, 2 and 3

**9. 'Project Loon', sometimes seen in the news, is related to:** **[UPSC (Pre) – 2016]**

(a) Waste management technology
(b) Wireless communication technology
(c) Solar power production technology
(d) Water conservation technology

**10. In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?** **[UPSC (Pre) – 2014]**

(i) Iris scanning
(ii) Retinal scanning
(iii) Voice recognition

Select the correct answer using the code given below.

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**Answer Key**

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (c) | 2. (d) | 3. (b) | 4. (a) | 5. (b) | 6. (c) | 7. (d) | 8. (b) |
| 9. (b) | 10. (d) | | | | | | |

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# CHAPTER 13
# CYBER SECURITY

## IN THIS CHAPTER...


**13.1 Cyber Security and Its Related Issues**

13.1.1 Cybercrime
13.1.2 Measures to Control Cybercrime in India
13.1.3 Cyber Security
13.1.4 Cyber Check
13.1.5 Cyberterrorism


Distributed Denial of Service (DDoS) involves sending a large number of web requests to servers, overloading them and ultimately leading to their collapse. This results in serious disruption. It results in immense pecuniary and strategic loss to government and its agencies.


13.1.6 Combating Cyberterrorism
13.1.7 Cyberwarfare
13.1.8 National Cyber Security Policy, 2013
13.1.9 Cyber Crime Coordination Centre

**13.2 lockchain Technology**


Blockchain generates a proficient and cost-effective database that is practically tamper-proof, by preserving an absolute and chronological planned record of all

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actions and files ('blocks') associated together ('chain') in a circulated and decentralised database.

13.3 Cryptocurrency and Regulation of Official Digital Currency Bill, 2021
13.4 Toolkit
13.5 Cyber Insurance
13.6 WhatsApp's New Policy being Examined by GoI
13.7 Cyberdome
13.8 Spyware-Pegasus
13.9 Cryptojacking
13.10 Bluebugging

## 13.1 Cyber Security and its Related Issues

### 13.1.1 Cybercrime

Any crime that involves a computer and a network is termed as Cybercrime. Here, the computers may or may not have played an instrumental part in the commission of a crime.

Cybercrime spans activities like hacking, spam, online fraud, obscene or offensive content, harassment, cyberterrorism and cyber warfare. It usually includes high-profile crimes, particularly those surrounding hacking, copyright infringement, child pornography and child grooming. The major issue of privacy arises when confidential information is lost or intercepted, in a lawful manner or otherwise.

### 13.1.2 Measures to Control Cybercrime in India

**The Information Technology (Amendment) Act, 2008, serves as the legal framework for tackling cybercrime in India.** The original Act of 2000 contained provisions on cybercrimes related to online transactions. The legal framework was required to be strengthened and amended due to several new forms of computer crime, misuse and fraud taking place. To tackle new forms of cybercrimes like sexually explicit material in electronic form, video terrorism and breach of confidentiality and leakage of data by

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intermediary and e-commerce frauds, the amended Act of 2008 came into force. As part of cyber security efforts of the government, a holistic effort approach is being followed, involving research and development in cyber security. Several research and development projects have been initiated for basic research and indigenous development of technology in this area.

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## 13.1.3 Cyber Security

### 13.1.3.1 Basics of Cyber Security

The processes and mechanisms ensuring protection of computer-based information against unauthorised access, transfer, modification or destruction (whether accidental or intentional) is termed as Cyber Security. It deals with security threats such as hacking, malware, cybercrime, cyberterrorism, data theft, etc. It studies these threats and the features of computer systems and networks that make them vulnerable to these threats and then try to come up with measures to prevent these threats and find solutions after the computer is already affected. Cyber security companies **produce anti-virus programmes** to prevent infection from viruses and heal infected computers. Cyber security is also concerned with **developing threat-resistant firewalls, encryption** and **passwords** for protection against hacking and data theft.

### 13.1.3.2 Significance of Cyber Security

With the expansion of Internet and computer systems, the responsibility of these is getting more and more complex and interdependent. Thus, cyber security has become a serious concern as new security threats emerge every day. Poor cyber security may make government, business and personal computers vulnerable to these threats. New viruses are known to cripple businesses worldwide, causing billions of dollars of damages every year. Cyber security is not just an issue that concerns individuals and businesses but also government, as it has national security dimensions. For example, if

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military computers are not adequately protected against hacking, enemy countries may steal sensitive data.

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### 13.1.3.3 Cyber Security in India

In India, because of the thriving IT industry, large-scale use of computers by governments, especially e-governance and the growing usage of personal computers, cyber security has emerged as an important issue that requires a national-level response. The legal framework for cyber security in India is the Information Technology (Amendment) Act, 2008. **The Union Government had set up an Inter Departmental Information Security Task Force (ISTF) with National Security Council as the nodal agency to highlight the growing threat to information security in India and focus related actions.**

The issues were studied and deliberated by Task Forces relating to cyber security and recommendations were made. Considering the recommendations of the **ISTF, the following initiatives** have been taken by the Government:

1. **Indian Computer Emergency Response Team (CERT-In)** has been established to respond to the cyber security incidents and take steps for prevention of recurrence of the same.
2. **Public Key Infrastructure or PKI** (i.e., user and verifier): A system for the creation, storage and distribution of digital certificates which are used to verify that a particular public key belongs to a certain entity.
3. To support implementation of Information Technology Act and **promote use of Digital Signatures,** adequate infrastructure has been set up.
4. Research and Development activities have been supported by GOI through premier academic and Public Sector Institutions in the country.
5. **Government cyberspace and critical infrastructure have been developed.**

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6. Information Security Education and Awareness Programme has been launched all over India.
7. The Government is also following a holistic strategic approach for the long term. The approach includes research and development, legal framework, security incidents, early warning and response, best security policy compliance and assurance, international cooperation and security training.
8. The **Crisis Management Plan was formulated** in April 2010 to deal and counter cyber-attacks and cyberterrorism. It is to be implemented by all ministries and departments of central and state governments.

### 13.1.4 Cyber Check

It is an **advanced cyber forensics tool kit** for analysing and viewing evidence files, indigenously **developed by the Centre for Development of Advanced Computing (CDAC),** for use by law enforcement agencies. A branch of forensic science pertaining to obtain legal evidence found in computers and digital storage media is termed as computer forensics.

### 13.1.5 Cyberterrorism

#### 13.1.5.1 Basics of Cyberterrorism

Cyberterrorism describes the acts of deliberate, large-scale disruption of computer networks, especially personal computers connected to the Internet. It is performed by the means of tools such as computer viruses.

Cyberterrorism is a controversial term. Some authors choose a very narrow definition, relating to deployments, by known terrorist organisations. By this narrow definition, it is difficult to identify any instances of cyberterrorism. Hence, a more general definition includes, for example, as 'The premeditated use of disruptive activities, or threats against computers and/or networks, with the intention to cause harm or further social, ideological, religious, political or similar objectives or to intimidate any person in furtherance of such objectives'.

#### 13.1.5.2 Different Forms of Cyberterrorism

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Cyberterrorism can take various forms:

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1. **Terrorist Propaganda on the Internet:** The ultimate aim of a physical terrorist attack is to terrorise a target population. It involves spreading a feeling of fear and therefore, adequate coverage in the media is very important for terrorists to succeed. The Internet is one of the media that is used by terrorists to spread terror, promote terrorist ideologies, organise terrorist activities by connecting terrorists in different parts of the world, etc.
2. **Privacy Violation:** A lot of private information is present on the Internet and computers. This includes passwords, encryption keys, images and videos, sensitive files, etc. Cyber terrorists may steal these resources or threaten it. Right to privacy is part of the Right to Life of Article 21 of the Constitution. Various provisions of the Information Technology Act, 2000, protect the online privacy rights of the citizens.
3. **Attacks on Government Properties:** Governments maintain large databases and a large number of government computers are connected to the Internet. This makes them susceptible to hostile attacks mainly in the form of theft of sensitive government information, valuable government secrets including those relating to defence and national security, passwords and other private information of government officials, etc. E-governance systems may be hacked and disrupted, causing great inconvenience to public and even threatening day-to-day government activities. A similar attack may be made by cyber terrorists on computers and networks of private companies and individuals.

### 13.1.5.3 Distributed Denial of Service Attack

The cyber terrorists may also use the method of Distributed Denial of Service (DDoS) to overburden the computer systems and networks of governments, armed forces and businesses. This involves sending a large number of web requests to servers, overloading them and ultimately leading to their collapse. This results in serious disruption. It results in immense pecuniary and strategic loss to government and its agencies.

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### 13.1.6 Combating Cyberterrorism

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In India, **the Information Technology Act, 2000, has laws regarding privacy violations, information theft, DDoS attacks, network damage and disruptions.** The Council of Europe Convention on Cybercrime, established in 2004, is the first International treaty for fighting against computer crime.

### 13.1.7 Cyberwarfare

#### 13.1.7.1 Basics of Cyberterrorism

Politically motivated hackers aiming to conduct **sabotage and espionage** lead to Cyber warfare. It being an information warfare is sometimes seen analogous to conventional warfare, although this analogy is controversial in terms of its accuracy and its political motivation.

Thus, cyber warfare is described as 'actions by a nation-state to penetrate another nation's computers or networks for the purposes of causing damage or disruption'. Cyber warfare can be described as **'the fifth domain of warfare',** which has become just as critical to military operations as land, sea, air and space. In spite of its growing threat, it has been the least common warfare and so far has not been used effectively.

**President Barack Obama, in 2009, declared America's digital infrastructure to be a 'strategic national asset',** and thereafter, the Pentagon set up its new US Cyber Command (USCYBERCOM) in May 2010, to defend American military networks and attack other countries' systems. A cyber-security and 'operations centre' was also established then by United Kingdom.

#### 13.1.7.2 Cyberwarfare Consists of Many Different Threats

1. **Cyber Espionage:** The includes any practice meant to obtain secret information from individuals, competitors, rivals, groups, governments and enemies to take military, political or economic advantage.
2. **Cyber Vandalism:** This usually involves **hacking enemy websites** and defacing the web pages or plant propaganda messages.

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3. **DDoS Attack:** Attempt to make a computer resource unusable by persons or groups who disrupt an Internet site or service. In cyber warfare, DDoS (distributed denial-of-service) attacks mainly involve targeting of government and military websites.
4. **Sabotage:** Military activities using computers and satellites for coordination are often at risk of equipment disruption. Power, water, fuel, communications and transportation infrastructure, all of them may be vulnerable to disruption, as they are increasingly controlled by computer systems. Particularly, power transmission systems and telephonic systems including cell phones are susceptible to cyber warfare. Even interception or replacement can affect orders and communications. Massive power or communication outages caused by a cyber-attack could disrupt the economy, distract from a simultaneous military attack or create a national crisis.

#### 13.1.7.3 Cyberwarfare and India

##### *13.1.7.3.1 India's Vulnerability*

In recent times, cyber-attacks resemble political conflict, leading to warning from experts of the unfolding of a 'Cyber Cold War'. It is said that massive cyber-attacks may precede future wars. India appears to be extremely vulnerable due to huge growth in government state-wide area networks and 25 mission e-governance mode projects, in addition to electrical grid, cell phones, oil and gas infrastructure. This is on top of the private industry, especially the banking and finance sector deploying huge number of online transactions.

##### *13.1.7.3.2 Response*

The main agency dedicated to defend India's IT infrastructure is **Computer Emergency Response Team (CERT)** within the Department of Information Technology. It performs its functions in coordination with the National Informatics Centre, the government's service provider. Dedicated cyber warfare groups operate in the **Defence Intelligence Agency (DIA)**

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and India's technical intelligence organisation **National Technical Research Organisation (NTRO).**

The Indian Army announced that an impenetrable and secure wide area network exclusively for army's function and use has been established for its functioning. The army has its own cyber security policy, a Computer Emergency Response Team (CERT) and its own cyber audit process, which is conducted by cyber security personnel.

***SAMBHAV (Secure Army Mobile Bharat Version).***

The Indian Army has created SAMBHAV (Secure Army Mobile Bharat Version), a comprehensive and secure mobile ecosystem. SAMBHAV aims to provide secure communication with instant connectivity, even in mobile situations, representing a notable advancement in India's defence capabilities. This initiative signifies a shift where traditional military strengths are not the sole indicator of prowess.

### *13.1.7.3.3 Shortcomings Exist in India's Cyber Warfare Capabilities*

1. Dedicated cyber warfare groups operate in the DIA and NTRO but these are tasked only with **offensive cyber warfare and not with defending** against cyber-attacks.
2. **India spends less than $1 million on offensive cyber warfare.** There are a handful of experts available. On the other hand, China has an annual budget of $55 million and employs thousands of hackers. The US and some European nations are even ahead.
3. **India cannot counterattack,** but various nations can. For example, **a secure operating system 'Kylin' has been developed by the Chinese** as an effective protection from such attacks. Not only this, **a secure microprocessor has also been developed by them.** This microprocessor, unlike the US-made chips, which most computers in India use, is known to be hardened against external access.
4. India, in spite of being an IT-superpower, is almost **entirely dependent on external sources,** be it hardware or software. **Even for all anti-virus programmes, network protocols and network**

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**hardware components,** India depends on external sources. Such vulnerable operating systems could lead to massive DDoS attacks.

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#### *13.1.7.3.4 A Long-Term Response Involves in Cyber Security*

1. **A dedicated cyber security organisation,** which employs police and armed forces personnel that can initiate developing a strategy to protect national information infrastructure.
2. **Increasing the cyber warfare budgets** and inducting more personnel.
3. **Setting up a national centre for coordinated response to cyber-crime** between various law-enforcement agencies, Internet Service - Providers (ISPs) and high-quality researchers within the country.
4. **Carrying out extensive information security audits of government networks and encouraging the private sector to do the same.** The idea would be to identify security gaps to determine exactly what the problem is and try to address its security requirement.
5. A software and hardware policy intended for development of **secure indigenous microprocessors and operating systems,** so that more and more computers use these and make computers in India, in general, more resistant to cyber-attacks.
6. There is a view among officials and experts that the **Indian government needs to evolve a national security policy,** which should be implemented in all the government departments.

In May 2010, the Indian government announced several measures designed to address the cyber security challenge:

- A task force was formed by the government to devise a plan for building open source, indigenous operating systems, intended at reducing dependence on foreign hardware and software vendors.
- The government established a Crisis Management Plan against cyber-attacks, which all central ministries, state governments and critical sectors are implementing.

In August 2010, at a high-level security meet chaired by the **National Security Advisor (NSA), the government made several more proposals:**

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1. **Set up a cyber-warfare infrastructure,** manned by a small army of software professionals, for spying purposes on the classified data of hostile nations by hacking into their computer systems.
2. According to the proposal, the responsibility for creating cyber-offensive capabilities is on the National Technical Research Organisation (NTRO) and Defence Intelligence Agency (DIA).
3. Further plans of the government include amplification of the efforts to strengthen its cyber armour. It is proposed that to check all types of hardware and software being sourced by departments for spyware, **a National Testing Centre will be established.** It will prevent India's computers from coming under attack.
4. The NSA has also asked the DRDO and DIA to **magnify efforts against electromagnetic-pulse bombs that can interrupt wireless signals inside the country.** As per NSA's directives, the DIA has to harden its **Transient Electro Magnetic Pulse Emanations Standards,** known as TEMPEST in military parlance. Hardening TEMPEST includes lower chances of interception of data transferred by defence agencies through Internet.
5. The introduction of cyber security in the curriculum of NTs and educational institutes is as per the directives of the National Security Council Secretariat to the HRD and IT ministries.

### 13.1.8 National Cyber Security Policy, 2013

Indian Government announced National Cyber Security Policy, 2013 with these important features:

- To form a secure and flexible cyber space.
- To have a safe and secure cyber ecosystem and generate belief and trust in IT transactions.
- To setup a 24 × 7 National Critical Information Infrastructure Protection Centre (NCIIPC).
- To test all Information and Communication Technology (ICT)-based products and certifying them.
- To create workforce of 500,000 professionals in the field of ICT.

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### 13.1.9 Cyber Crime Coordination Centre

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The ***Indian Cyber Crime Coordination Centre (I4C)*** was inaugurated by the government in January 2020. The scheme to set up I4C was ***approved in October 2018,*** to deal with all types of cybercrimes in a comprehensive and coordinated manner.

#### *About the Indian Cyber Crime Coordination Centre*

It was set up under the newly created ***Cyber and Information Security (CIS) division*** of the MHA (Ministry of Home Affair).

It has seven components:

1. National Cyber Crime Threat Analytics Unit,
2. National Cyber Crime Reporting Portal,
3. National Cyber Crime Training Centre,
4. Cyber Crime Ecosystem Management Unit,
5. National Cyber Crime Research and Innovation Centre,
6. National Cyber Crime Forensic Laboratory Ecosystem, and
7. Platform for Joint Cyber Crime Investigation Team.

#### *Functions*

- The I4C will assist in centralising cyber security investigations, prioritise the development of response tools and bring together private companies to contain the menace.

#### *Objectives*

- To act as a nodal point in the fight against cybercrime.
- Identify the research problems/needs of LEAs and take up R&D activities in developing new technologies and forensic tools in collaboration with academia/research institutes within India and abroad.
- To prevent misuse of cyber space for furthering the cause of extremist and terrorist groups.
- Suggest amendments, if required, in cyber laws to keep pace with fast changing technologies and International cooperation.
- To coordinate all activities related to implementation of Mutual Legal Assistance Treaties (MLAT) with other countries related to

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cybercrimes in consultation with the concerned nodal authority in MHA.

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***Need for Surveillance***

- Asia is the region most targeted by cyber-attackers, resulting in significant economic losses. As the region continues to play a key role in the global economic market, these cyber threats are expected to increase. Over 460 million people in India currently use the Internet, leaving them vulnerable to online criminals – both individuals and organised syndicates.

***Way Ahead***

- The government has decided to hire IT experts from premier public and private institutes, including IITs, to help fight new age crimes like online fraud, hacking, identity theft, dark net, trafficking, child pornography, online radicalisation and cyber-terrorism and prepare a roadmap for Indian Cyber Crime Coordination Centre.

## 13.2 Blockchain Technology

Blockchain is a decentralised digital ledger technology (follow principle of open ledger) introduced with **Bitcoin (2009)** by **Satoshi Nakamoto**. It combines cryptography and distributed computing to securely record transactions in chained data blocks.

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### *How It Works?*

- A client initiates a transaction
- The transaction is shared with the **peer-to-peer (P2P) network**
- Nodes **validate** it using consensus mechanisms
- Validated data is **added to the blockchain**
- All participants can verify the transaction

They are ***a new data structure that is secure, cryptography-based and distributed across a network.*** The ***technology supports cryptocurrencies such as Bitcoin, and the transfer of any data or digital asset.*** Spearheaded by Bitcoin, blockchains ***achieve consensus among distributed nodes, allowing the transfer of digital goods without the need for centralised authorisation of transactions.***

## Blockchain

### *Core Properties*

- **Smart Contracts:** Automatically executed agreements
- **Distributed:** All trusted members share the same ledger
- **Immutable:** Records cannot be altered once verified
- **Timestamped:** All entries are recorded with time data
- **Consensus-driven:** Validations require network-wide agreement
- **Encrypted and Secure**

### *Types of Blockchain*

1. **Public Blockchain:** Open to everyone (e.g., Bitcoin, Ethereum)
2. **Consortium Blockchain:** Managed by a group of authorised entities
3. **Private Blockchain:** Controlled by a single entity

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### *Applications of Blockchain*

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- **Cryptocurrencies:** Securing digital currency transactions
- **Supply Chain:** Ensuring traceability and transparency
- **Voting Systems:** Enhancing trust and verifiability
- **Intellectual Property:** Protecting patents and trademarks
- **Health Records:** Unified, tamper-proof patient data
- **Others:** Law enforcement, banking, IoT, crowdfunding

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### *How does it Operate?*

1. The technology allows transactions to be simultaneously anonymous and secure, peer-to-peer, instant and frictionless.
2. It does this by distributing trust from powerful intermediaries to a large global network, which through mass collaboration, clever code and cryptography, enables a tamper-proof public ledger of every transaction that is ever happened on the network.
3. A block is the 'current' part of a blockchain which records some or all of the recent transactions, and once completed, goes into the blockchain as permanent database.
4. Each time a block gets completed, a new block is generated. Blocks are linked to each other (like a chain) in proper linear, chronological order with every block containing a hash of the previous block.
5. As a public ledger system, blockchain records and validates each and every transaction made, which makes it secure and reliable.
6. All the transactions made are authorised by miners, which makes the transactions immutable and prevent it from the threat of hacking.
7. Blockchain technology discards the need of any third-party or central authority for peer-to-peer transactions.
8. It allows decentralisation of the technology.

### *Benefits of Blockchain Technology*

- In terms of a public ledger system, Blockchain account and authenticate each and every made transaction, thereby making it safe and dependable.
- All the done transactions are endorsed by miners, which make them absolute and avoiding it from the hacking risk.

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- Blockchain Technology rejects any third-party or central authority requirement for peer-to-peer transactions.
- It allows technology decentralisation.

***Usage of Blockchain in Public Administration***

- Blockchain has the potentiality of optimising the public services delivery; further country's fight against corruption and produce ample citizen's value.
- Blockchain generates a proficient and cost-effective database that is practically tamper-proof, by preserving an absolute and chronological planned record of all actions and files ('blocks') associated together ('chain') in a circulated and decentralised database. By executing this, it undertakes building more clear, liable and able governments.
- In addition to this, Blockchain can also assist in building a more honest government. A public Blockchain, similar to Bitcoin usage, reports all information and transactions on the decentralised database - permanently, publicly and most important of them securely. Blockchain can hold state and local actors answerable for any embezzlement, by permitting governments for tracking the government funds movement.
- Blockchain not only dissuades corruption with the help of accountability but it can also undertake so by entirely evading the middleman.

The West Bengal government has been preparing for introduction of blockchain technology for safeguarding its documents from cyber-attacks. The state government's planned Cyber Security Centre of Excellence would be assigned in executing the new 'blockchain' system at various departments. In July 2023, Airchains has partnered with New Town Kolkata Development Authority (NKDA) to enable blockchain-based tracking and recording of land mutations to enhance efficiency and security.

The cyber security centre will fetch the best in academic, law enforcement and other sections under single roof for undertaking the finest practices for countering Cybercrimes. The centre will also undertake research.

In 2017, computers at a few offices of the West Bengal State Electricity Distribution Company Limited were disabled by 'WannaCry' virus, a global

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ransomware. Ransomware is a kind of malicious software intended to block computer system access till the asked sum of money is paid.

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Note: The **Tech Mahindra** and the Government of **Telangana** have signed an accord for establishing a country's first Blockchain district in Hyderabad, a first of its kind Centre of Excellence for Blockchain.

***Possibilities of Blockchain are***

- Confidential communication of cryptocurrency.
- Safe, cost-effective and fast bank transactions.
- Secure legal documents, health data, notaries and personal documents.
- Distribution of land records and government financial assistance.
- Cloud storage, digital identification, smart communication and digital voting.

***National Informatics Centre (NIC)*** has set up a Centre of Excellence (CoE) in ***Blockchain Technology*** in Bengaluru, which will provide Blockchain as a service and allow stakeholders to benefit from shared learning, experiences and resources.

***Functions***

1. The Centre of Excellence will facilitate various government departments in building proof of concepts for use of Blockchain technology in different dimensions of governance, leading to large-scale deployment of some such applications.
2. With National Informatics Centre (NIC) providing a robust and an agile infrastructure, the CoE shall also provide Blockchain as a -Service (BaaS) for efficient hosting of Blockchain network, says an official release.

## Blockchain

***Challenges in Blockchain Adoption***

- **Performance:** Replication across nodes slows speed
- **Scalability:** Resource-intensive and complex architecture
- **Storage:** Data replication demands high capacity
- **Energy Use:** High computational needs lead to high consumption
- **Interoperability:** Incompatibility across different blockchain systems

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- **Legal Gaps:** Existing IT laws lack blockchain-specific provisions
- **Localisation Conflicts:** Public blockchain's global data replication may violate data localisation norms

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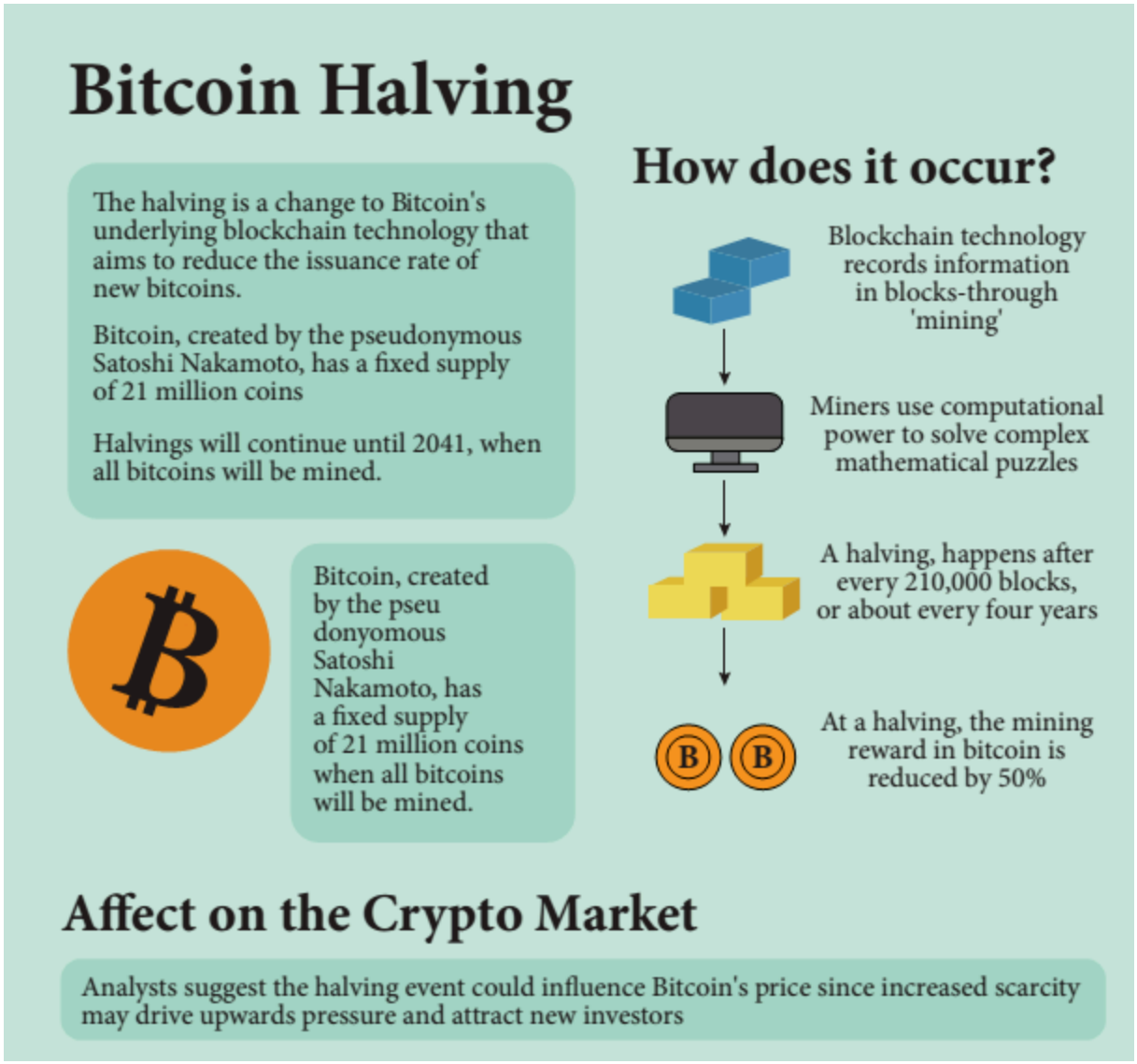
## Bitcoin halving




### *Regulation in India*

- The current debate in India has, unfortunately, focussed too heavily on trading and speculation, looking at cryptocurrencies as an investment tool, rather than understanding ***the potential of core blockchain***

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***technology and the basic role of cryptocurrencies as an incentive mechanism to secure decentralised transactions.***

- Prevailing cyber laws in India touch almost all aspects of transactions and activities involving ***the Internet, www and cyber space (IT Act, 2000 and amended in 2008, section 463 of IPC and section 420).*** But in today's techno-savvy environment the world is becoming more and more digitally sophisticated and so are the crimes. India's cyber laws are lacking in this respect.
- There are sufficient global examples of countries that have taken cautious steps in regulating the technology and are focussing on stopping illegal activity without hurting innovation.

## Vishvasya [Blockchain-as-a-Service (BaaS)]: National Blockchain Technology Stack

### *About Vishvasya*

**MeitY** introduced the **Vishvasya-Blockchain Technology Stack,** offering Blockchain-as-a-Service through a geographically distributed infrastructure designed to support various permissioned blockchain applications.

- It is part of the broader **National Blockchain Framework (NBF)** under India's **National Strategy on Blockchain.**
- NBF is intended to promote blockchain technology across various domains such as **healthcare, agriculture, education, finance** and more.

### Key Features of Vishvasya BaaS

- Quick deployment of blockchain apps
- Secure, pre-audited containers
- Specialised security reviews for blockchain components
- Distributed infrastructure for resilience
- Improved documentation for easy adoption

### *Why BaaS Matters*

- Enhances **trust** through distributed, consensus-driven digital architectures.

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- Simplifies blockchain adoption for different user groups—**infra providers, app developers and smart contract coders**.
- Delivers **security assurance** across the blockchain development stack.

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The **Ministry of Electronics and Information Technology (MeitY)** has launched **Vishvasya, India's National Blockchain Technology Stack**, to strengthen and promote a robust blockchain ecosystem.

In addition to this, **MeitY also launched:**

- **NBFLite:** A blockchain sandbox specifically **designed for startups and academic institutions** to support quick prototyping, research and skill development.
- **Praamaanik:** A blockchain-powered solution aimed at **verifying the origin of mobile applications**, developed under the National Blockchain Framework.
- **National Blockchain Portal:** A **content management platform to host and manage all resources** related to the National Blockchain Framework (NBF).

## RANSOMWARE CASE WORLDWIDE

**WannaCry** is a ransomware worm that spread rapidly through, across several computer networks in May of 2017. After infecting a Windows computer, it encrypts files on the PC's hard drive, making them impossible for users to access, then demands a ransom payment in bitcoin in order to decrypt them.

**Petya** is a ransomware strain discovered in 2016 that infects Microsoft Windows-based computers. Like other forms of ransomware, Petya encrypts data on infected systems. The data is unlocked only after the victim provides the encryption key, usually after paying the attacker a ransom for it.

**Eternalblue** allowed the ransomware to gain access to other machines on the network. On May 12, 2017, the worldwide WannaCry ransomware used this exploit to **attack** unpatched computers.

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## 13.3 Cryptocurrency and Regulation of Official Digital Currency Bill, 2021

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The government of India was scheduled to introduce new cryptocurrency regulations during the Winter Session of Parliament. This was the second time that the Cryptocurrency bill was listed but got delayed. The first time it happened was during the Budget Session of Parliament in 2021. It prohibits all private cryptocurrencies and provides for **an official digital currency to be issued by the Reserve Bank of India.**

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# BLOCKCHAIN

## Encrypted Information

All information transferred *via* blockchain is encrypted and every occurrence is recorded, this implies that once the block is created and added to the chain, it cannot be altered.

## Peer-to-Peer

Blockchain allows direct exchange of currency with a peer without an intermediary. It removes the middle man.

## Data Sharing

Blockchain can also be used to share contracts, records and any other type of data.

## A Digital Ledger

Blockchain is a digital ledger which keep records of all transactions taking place at peer-to-peer network

## Decentralisation

Blockchain is decentralised, so there isn't a need for a Central Certifying Authority.

## How does Blockchain works?

Someone requests a transaction

The requested transaction is broadcast to a **P2P network consisting of computers known as Nodes.**

**VALIDATION**
The Network of nodes validates the transaction and the user's status using known algorithms.

**A verified transaction** can involve cryptocurrency, contracts, records or other information.

Once verified, the transaction is combined with other transactions **to create a new block of data for the ledger**

**The Transaction is complete.**

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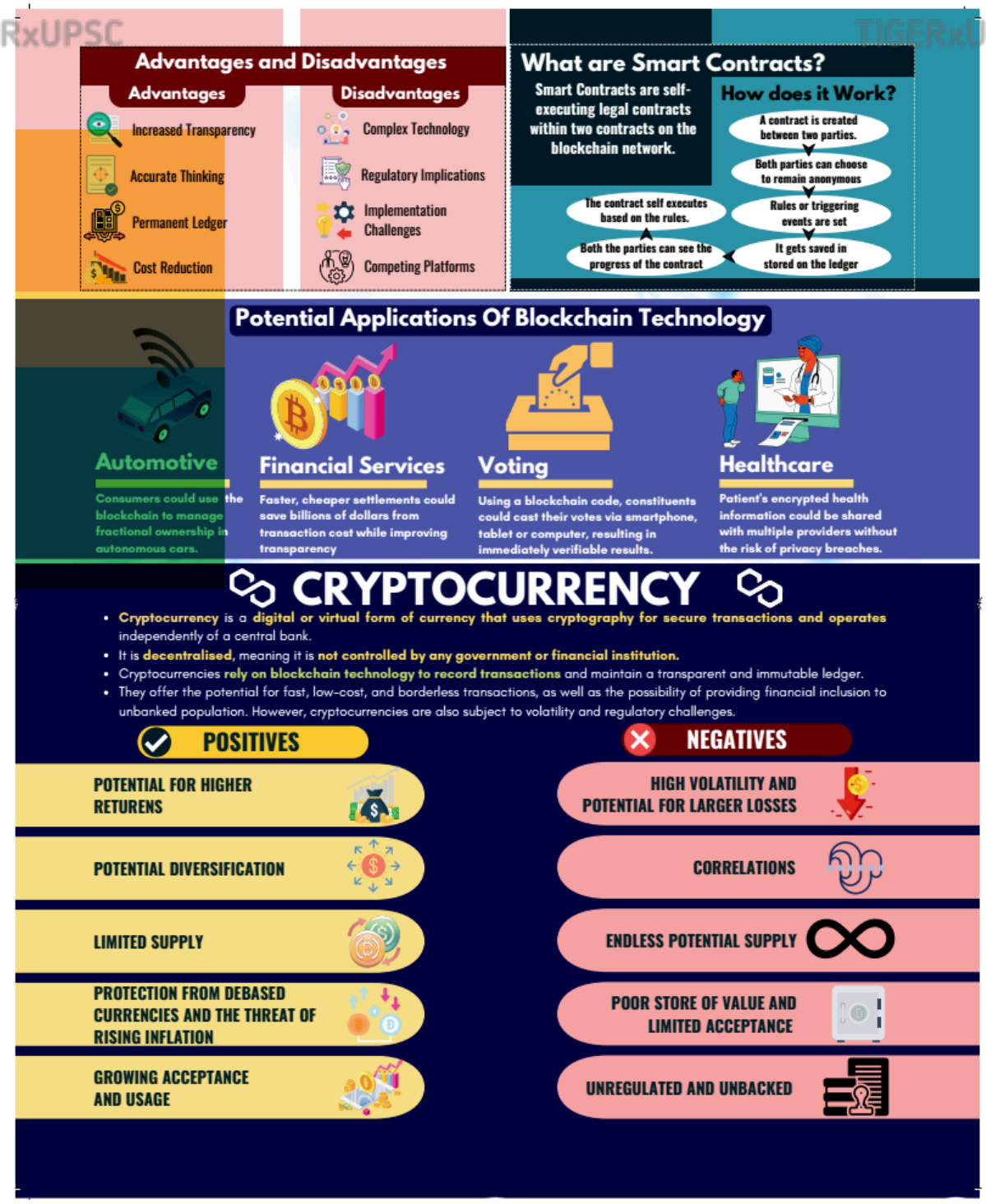


**The purpose of the law has been described as:**

1. To create a facilitative framework for an official digital currency issued by the RBI.
2. To 'prohibit all private cryptocurrencies in India'.

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The Bill also seeks to prohibit all private cryptocurrencies in India, however, it allows for certain exceptions to promote the underlying technology of cryptocurrency and its uses.

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## INDIA AND CRYPTOCURRENCY

2009: First cryptocurrency, Bitcoin launched in 2009 by Satoshi Nakamoto.
2018: RBI banned banks and other regulated entities from supporting crypto transactions.
2019: Inter-ministerial committee recommended banning all private cryptocurrencies.
2020: SC struck down the ban on the trading of cryptocurrency as unconstitutional.
2021: Cryptocurrency and Regulation of Official Digital Currency Bill, 2021 introduced.
Under this, a plan to ban private digital currencies favours RBI backed currency.
A 3-6 month exit period prior to banning the trading, mining and issuing of cryptos.
Finally, Cryptocurrencies, though unregulated, are not illegal in India.

### *Background*

The Supreme Court, set aside on April 6, 2018, circular of the Reserve Bank of India (RBI) that prohibited banks and entities regulated by it from providing services in relation to Virtual Currencies (VCs).

### *What did the Court Say?*

1. RBI has not come out with a stand that any of the entities regulated by it namely, nationalised banks/scheduled commercial banks/cooperative banks/NBFCs, have suffered any loss or adverse effect directly or indirectly, on account of Virtual Currencies (VCs).
2. Hence, the RBI circular is 'disproportionate' with an otherwise consistent stand taken by the central bank that VCs were not prohibited in the country.
3. Besides, the court found that the RBI did not consider the availability of alternatives before issuing the circular.
4. Besides, the court referred to the Centre's failure to introduce an official digital rupee despite two draft Bills and several committees.

### *What are Cryptocurrencies?*

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Cryptocurrencies are digital currencies in which encryption techniques are used to regulate the generation of units of currency and verify the transfer of funds, operating independently of a central bank.

**Examples:** Bitcoin, Ethereum, etc.

## 13.4 Toolkit

A toolkit has become a handy tool in sustaining a movement or campaign in times of social media influences.

- It is a document created as an explainer on an issue as a guide to everybody who is associated with the campaign or can be roped in to give a fillip to the campaign.
- It also provides a roadmap of how to take forward the campaign or agitation explaining what needs to be done, when and how.

## 13.5 Cyber Insurance

The policy provides protection in case of theft of funds due to **cyber** event/hacking of **insured** bank account/credit card/debit card/mobile wallet **by** a third party. It also provides protection in terms of defence costs for claims made against the **insured by** the third or affected party to identity theft fraud.

Cyber insurance for individuals policies are designed to cover expenses related to post factor actions of a cyber-attack, such as the cost incurred during the prosecution process and defence costs are paid by the insurance company relating to the types of cyber risks mentioned in the documentation of the policy.

Other costs such as financial losses, online loss of money suffered by the policyholder due to cyber-attacks are also covered in the policy up to the sum assured as mentioned in the policy. Expenses incurred for counselling treatments post a cyber-attack is also paid for including damages against third-party liabilities and restoration costs. It also takes

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care of consultant fees, court expenses and legal fees concerning the cyber-attack.

If the attack was intentionally and deliberately done, the policy will not pay any claim. Insurers do not cover any fraudulent, dishonest or malicious act. These policies do not provide any cover for any prior actions of the attack that the policyholder has faced before buying the policy; nor do they cover for personal data or lost pictures in the policy. The insurers also take measures to avoid any cyber-attack to their customers by prompting for backing up their data, updating anti-virus and having a proper set of passwords. If such measures are not taken properly, in case of a cyber-attack, your claim can be jeopardised.

## 13.6 WhatsApp's New Policy being Examined by GoI

The Central government has informed the Delhi High Court that it was examining WhatsApp's controversial new privacy policy, which was scheduled to come into effect from 15 May, 2021, at the highest level.

### *Issue*

*The court was hearing a petition claiming that WhatsApp's new privacy policy violates **the right to privacy** guaranteed under the Constitution.*

- The plea said the new policy 'virtually gives a 360-degree profile into a person's online activity' without there being any supervision of the government.
- The plea sought direction to issue guidelines to ensure such change in privacy policy by WhatsApp are carried out.

### *Key Features of the Policy*

1. **Information Sharing with Third-Party Services:** When users rely on third-party services or other Facebook Company Products that are integrated with the services, those third-party services may receive information about what someone or others share with them.
2. **Hardware Information:** WhatsApp collects information from devices such as battery level, signal strength, app version, browser

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information, mobile network and connection information (including phone number, mobile operator or ISP), among others.

3. **Deleting the Account:** If someone only deletes the WhatsApp app from their device without using the in-app 'delete my account' feature, then that user's information will remain stored with the platform.
4. **Data Storage:** WhatsApp mentions that it uses Facebook's global infrastructure and data centres including those in the United States to store user data. It also states that the data in some cases will be transferred to the United States or other parts where Facebook affiliated companies are based.
5. **Location:** Even if a user does not use their location-relation features, Whatsapp collects IP addresses and other information like phone number and area codes to estimate one's general location (city, country).
6. **Payment Services:** WhatsApp says if anyone uses their payment services, they will process additional information about a person including payment account and transaction information.

### *Centre Moved Court over WhatsApp's New Privacy Policy*

The Ministry of Electronics and Information Technology has asked the Delhi High Court to step in and restrain WhatsApp from rolling out its new privacy policy.

### *Why does the IT Ministry want the High Court to Restrain WhatsApp?*

**Supreme Court judgements had placed a responsibility upon the Centre to come out with a 'regime on data protection and privacy',** which would 'limit the ability of entities' such as WhatsApp to issue 'privacy policies which do not align with appropriate standards of security and data protection'. Therefore, WhatsApp must be stopped from rolling out the services.

The IT Ministry has listed major violations of the current IT rules that the new privacy policy of WhatsApp, if rolled out, could entail. They are:

1. WhatsApp failed to specify the type of sensitive data being collected by it.

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2. WhatsApp has failed to provide the user an option to review or amend the users' information being collected by it.
3. The policy fails to provide users an option to withdraw consent on data sharing retrospectively.
4. It also fails to guarantee non-disclosure by the third parties.

***Other Issues and Concerns***

1. The new Whatsapp policy contradicts the recommendations of **the Srikrishna Committee Report,** which forms the basis of the Data Protection Bill, 2019.
2. The principle of **Data Localisation,** which aims to put curbs on the transfer of personal data outside the country, may come in conflict with WhatsApp's new privacy policy.
3. With the updated privacy policy, **WhatsApp can now share one's metadata, essentially everything** beyond the conversation's actual text.
4. **If users disagree with the messaging platform's updated privacy policy,** they will have to quit WhatsApp when the new terms of service are set to come into effect.

### SANDES

- The National Informatics Centre has launched an instant messaging platform called Sandes on the lines of WhatsApp.
- Like WhatsApp, the new NIC platform can be used for all kinds of communications by anyone with a mobile number or email id.
- The limitation, however, is that the app does not allow the user to change their email id or registered phone number. The user will have to re-register as a new user in case they wish to change their registered email id or phone number on the app.

## 13.7 Cyberdome

**Cyberdome** is a pioneering **project** as it brings together Government Departments, Law Enforcement Agencies, Industry, Academia,

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International Organizations and experts from the public domain for collaborating on cyber security to enhance the capabilities of the state in dealing with cyber threats.

Cyberdome is technological research and development Centre of Kerala Police Department. It aims to meet the long-term security challenges of bridging the gap between the latest innovations in the **cyber** space and the skill set of Kerala Police, in combating the emerging **cyber** threats.

***Work of Cyberdome***

- Child pornography pages have been removed.
- Movie piracy has been stopped by blocking the websites.
- Tracking system developed by the team to identify the location from where data is sent.
- During Ransomware attack in 2017, it also protected lot of computers in Kerala against cyber threat.

In 2018, Assam government launched Assam Police Cyberdome and Digital Intelligence and Training and Analysis Centre (DITAC) in Kahilipara, Guwahati.

Thus, Cyberdome project can be termed a unique experiment in our current administrative system. Success of this project depends on its implementation. There are chances to expand this project to include other parts of our governance system too.

## 13.8 Spyware-Pegasus

Spyware refers to malicious software or programs that are designed to covertly gather information about a user or their activities without their knowledge or consent.

(Refer **Infographic - Spyware, Pegasus, Cryptojacking and Bluebugging**)

## 13.9 Cryptojacking

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Cryptojacking is the act of hijacking a computer to mine cryptocurrencies against the user's will, through websites, or while the user is unaware.

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(Refer **Infographic – Spyware, Pegasus, Cryptojacking and Bluebugging**)

## 13.10 Bluebugging

Bluebugging is a type of malicious attack that allows hackers to gain access to your Bluetooth-enabled devices. Bluebugging is conducted by exploiting a security flaw in the Bluetooth protocol.

(Refer **Infographic – Spyware, Pegasus, Cryptojacking and Bluebugging**)

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## SPYWARE

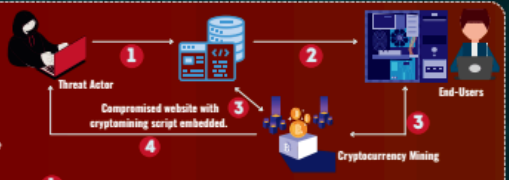
- **Definition:** Malicious software designed to gather information without user consent.
- **Purpose:** Steals sensitive data like personal info, credentials, and financial details.
- **Infection Methods:** *Via* phishing, malicious downloads, or bundled with software.
- **Types:**
  - **Keyloggers:** Capture keystrokes.
  - **Adware:** Displays ads and collects browsing data.
  - **Trojans:** Disguised as legitimate software.
  - **System Monitors:** Track user activity.
- **Symptoms:** Slows performance of machine, frequent pop-ups, changes browser settings, increases data usage.
- **Risks:** Identity theft, financial loss & privacy breach.
- **Detection:** Use antivirus/anti-spyware software, update systems, monitor network activity.
- **Prevention:** Avoid untrusted downloads, suspicious links, use strong passwords, enable 2FA.
- **Legal/Ethical Issues:** Unauthorised spying is illegal and a privacy invasion.

## WHAT IS PEGASUS SPYWARE?

- Pegasus is a sophisticated spyware developed by NSO Group.
- It targets mobile devices, including iOS and Android smartphones.
- Pegasus is advanced spyware that can infiltrate devices silently, using "sero-click" or "nearly sero-click" techniques, requiring minimal or no interaction from the user.
- Controversy surrounds its use, including alleged targeting of journalists, activists, and dissidents.
- Pegasus has been associated with government and state-sponsored surveillance.
- Once installed, Pegasus gains complete access to device contents.
- The collected data is transmitted to a remote server for analysis.
- Pegasus can activate the device's microphone and camera for real-time surveillance.
- It can monitor calls, messages, emails, and collect data like photos and location information.

## CRYPTO JACKING (MALWARE)

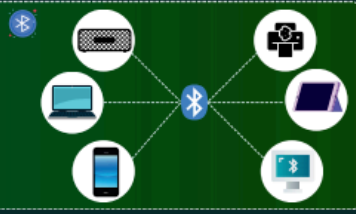
- **Definition:** Unauthorised cryptocurrency mining by hijacking someone else's device.
- **Infection Methods:** *Via* vulnerabilities, malicious downloads, infected websites, or phishing emails.
- **Operation:** Malware runs in the background, using device resources for mining.
- **Impact:** Profits go to the attacker; owner suffers performance issues and increased costs.
- **Protection:**
  - Keep software updated.
  - Use reputable security software.
  - Be cautious online.
- **Detection:** Monitor system performance for signs of excessive resource usage.




1. The threat actor compromises a website.
2. Users connect to the compromised website and the crypto mining script executes.
3. Users unknowingly start mining cryptocurrency on behalf of the threat actor.
4. Upon successfully adding a new block to the blockchain, the threat actor receives a reward in cryptocurrency coins.

## WHAT IS A BLUEBUGGING (HACKING) ?

- Bluebugging targets Bluetooth-enabled devices.
- It exploits Bluetooth vulnerabilities for unauthorised access and control.
- Attackers can make calls, send messages, or access data without the owner's knowledge.
- It may allow eavesdropping or monitoring of activities.
- Keep Bluetooth off when not in use and update devices for protection.
- Be cautious of untrusted Bluetooth connection requests.



## QUESTIONS AND ANSWERS

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## Objective Questions

**1. With reference to 'Blockchain Technology', consider the following statements:** **[UPSC (Pre) – 2020]**

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that depend on basic features of blockchain can be developed without anybody's permission.

Which of the statements given above is/are correct?

(a) 1 only
(b) 1 and 2 only
(c) 2 only
(d) 1 and 3 only

**2. In India, under cyber insurance for individuals, which of the following benefits are generally covered, in addition to payment for the loss of funds and other benefits?** **[UPSC (Pre) – 2020]**

1. Cost of restoration of the computer system in case of malware disrupting access to one's computer.
2. Cost of a new computer if some miscreant wilfully damages it, if proved so.
3. Cost of hiring a specialised consultant to minimise the loss in case of cyber extortion.

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4. Cost of defence in the Court of Law, if any third party files a suit.

Select the correct answer using the code given below:

(a) 1, 2 and 4 only
(b) 1, 3 and 4 only
(c) 2 and 3 only
(d) 1, 2, 3 and 4

**3. In India, the term 'Public Key Infrastructure' is used in the context of:** **[UPSC (Pre) – 2020]**

(a) Digital security infrastructure
(b) Food security infrastructure
(c) Health care and education infrastructure
(d) Telecommunication and transportation infrastructure

**4. The terms WannaCry, Petya and EternalBlue sometimes mentioned in the news recently are related to:** **[UPSC (Pre) – 2018]**

(a) Exoplanets
(b) Cryptocurrency
(c) Cyber-attacks
(d) Mini satellites

**5. In India, it is legally mandatory for which of the following to report on cyber security incidents?** **[UPSC (Pre) – 2017]**

(i) Service Provider
(ii) Data Centres
(iii) Body Corporates

Select the correct answer using the code given below:

(a) Only (i)
(b) (i) and (ii)
(c) Only (iii)
(d) (i), (ii) and (iii)

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**Answer Key**

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| | | | | |
|---|---|---|---|---|
| 1. (d) | 2. (b) | 3. (a) | 4. (c) | 5. (c) |

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# CHAPTER 14
# FUNDAMENTALS OF COMPUTER

## IN THIS CHAPTER...


**14.1 Computer**

**14.2 History of Computer**

**14.3 Types of Computers**


A bit is single digit number base-2 (0 or 1). It is the smallest unit of computer data. About 16,000 bits are contained in a full page of text.


14.3.1 On the Basis of Function

14.3.2 On the Basis of Processing Speed and Storing Capacity

**14.4 Units Used in Computers to Store and Process Data**

**14.5 Uses of Computer in Different Disciplines**

**14.6 Characteristics of the Computer System**

**14.7 Basic Structure of a Computer System**

**14.8 Data Representation Computer**

**14.9 Components of Computer System**

14.9.1 Hardware

14.9.2 Software

**14.10 Computer Language**


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A computer language is the means by which instructions and data are transmitted to computers.

14.10.1 Low Level Language
14.10.2 High Level Language
**14.11 Analogue and Digital Transmission**
**14.12 Data Flow**
**14.13 LAN Hardware and Software**
14.13.1 LAN Hardware (The Basic Hardware Components of LAN)
14.13.2 LAN Software/Operating System
14.13.3 Metropolitan Area Network (MAN)
14.13.4 Wide Area Network (WAN)
14.13.5 WAN Devices: Bridges, Router
**14.14 Terminology**
14.14.1 Node

Different terminals which are attached to the network and share the resources of the network are called nodes.

14.14.2 Server
14.14.3 Workstation
14.14.4 Network
14.14.5 Network Interface Unit

The interface which helps in the communication between the server and different nodes is called Network Interface Unit.

14.14.6 Architecture

## 14.1 Computer

Computer is a combination of two words 'compute' and 'er'. **'Compute' means calculation and 'er' means device. Computer is an electronic device. It can perform all types of mathematical and logical operations.**

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**It can accept data, store data, process data, retrieve data and print data.**

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## 14.2 History of Computer

The ***Abacus*** **Idea** was first introduced between 1000 BC and 500 BC. A series of moveable beads or rocks were used in the apparatus. The positions of these beads were changed while entering a number and again to perform mathematical operations.

The Abacus or counting frame serves as a calculating tool used in primarily parts of Asia for performing arithmetic calculations. It is made of beads or stones placed in grooves or strung on rods. *Abacus* is a Latin word which originated from the Greek word abax or abakon (meaning 'table' or 'tablet'). The Chinese invented it around 2600 B.C. Leonardo da Vinci may have conceived the world's first mechanical calculator around 1502. **In 1834, Charles Babbage introduced analytical engine and showed it at the Paris Exhibition in 1855.**

## 14.3 Types of Computers

### 14.3.1 On the Basis of Function

*On the basis of function, computers are divided into three parts:*

1. **Analogue Computer:** In analogue computers, numerical data are represented by measurable physical variables, like electrical voltage, pressure, temperature, etc. For instance, a thermometer is a simple analogue computer, where mercury moves with variation in temperature.
2. **Digital Computer:** A computer that accepts and processes data in the form of numbers (0–9) and characters (A–Z) that has been converted into binary code. Most of the computers are digital.
3. **Hybrid Computer:** The feature of analogue and digital machines is combined to create hybrid computers.

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### 14.3.2 On the Basis of Processing Speed and Storing Capacity

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On the basis of processing speed and storing capacity, computers are divided into four parts:

1. **Microcomputers:** Microcomputers known as personal computers (PC). It has lowest storing and processing speed. It is designed to be used by individuals.
2. **Mini Computers:** Mini computers are more powerful computers as compared to microcomputers. It has high memory and processing speed and can be used as server. A minicomputer is a multiprocessing system, which can support 4 to 200 users at the same time.
3. **Mainframe Computers:** A mainframe computer is different from micro and mini. It has very high memory and processing speed and is used as server (can support thousands of users). A mainframe is high-performance computer that is used for large-scale computing purposes.
4. **Supercomputers:** Supercomputer is a different type of computer and can be used for complex applications. Supercomputer is a powerful and high-performance machine used for mostly scientific computations. Cray-1 was the first supercomputer developed in 1976. Supercomputer is used for scientific research, weather forecasting, weapon designing, etc. India's first supercomputer PA RAM-10000 was developed by C-DAC, Pune. Other supercomputers are EKA, SAGA-220, Anurag, etc.

Note: Disk space, data storage space and system memory all are described on the basis of various parameter. Few years ago we were describing hard drive space using the term Megabytes, on the basis of data storage capacity. Today, Gigabytes and Terabytes are the most common terms being used to describe the size of a hard drive (**Table 14.1**).

**Bit** The term 'bit' is derived from phrase Binary Digit. A bit is single digit number base-2 (0 or 1). It is the smallest unit of computer data. About 16,000 bits are contained in a full page of text.

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## 14.4 Units Used in Computers to Store and Process Data

**Table 14.1** Units used in computers to store and process data

| 1 nibble | 4 bit |
|---|---|
| 1 byte | 8 bit |
| 1 kilobyte | 1024 byte |
| 1 megabyte | 1024 kilobyte |
| 1 gigabyte | 1024 megabyte |
| 1 terabyte | 1024 gigabyte |
| 1 petabyte | 1024 terabyte |
| 1 exabyte | 1024 petabyte |
| 1 zettabyte | 1024 exabyte |
| 1 yottabyte | 1024 zettabyte |

## 14.5 Uses of Computer in Different Disciplines

Computers have *become very popular in all fields. Here are some of areas where they are widely used.*

- **Education:** Computers give students more flexibility with their studies. Computers are also used by teachers to prepare lessons, report card and as reference tools.
- **Medicine:** A large number of computerised equipment are used for medical tests in hospitals and clinics. They can be used for storing medical records of patients visiting big hospitals. Doctors can access these records of patients to diagnose.
- **Business:** Every company or organisation requires computers for budgeting, accounting, billing, reporting, presentations, etc. This

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information must be constantly maintained and updated. Computers are also used for sales forecasting, production, planning, etc.

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- **Science and Technology:** Scientists and Engineers use computers as a tool to experiment, design and develop their ideas/projects. Architects use computers to design structures. Nuclear tests can be simulated without damaging the environment. Hence, the more popular forms of today's time are Computer-Aided Designing (CAD) and Computer--Aided Engineering (CAE).
- **Communication:** Many offices and homes have computers and hence, the sharing of data and programmes among various computers needs to be in accordance with the advancement of data-communication facilities. Using Internet facility, you can send e-mail to your friends and relatives.
- **Banking:** Computers are being used in banks for carrying out everyday transactions like online enquiry of customers' balance, cheque verification, updating of balance, calculating interests, etc. All progressive banks have installed Automated Teller Machines (ATM) to enable the customers to withdraw money from their accounts, money transfer, etc.
- **Weather Forecasting:** Data are collected from weather stations and satellites all over the world. Changes in weather and direction of winds can be analysed with the help of computers. Timely prediction may avoid damage due to natural disasters.
- **Entertainment:** Through computers, you can play various games, create your own music, watch cartoons or films, listen to your favourite music, etc. Cartoon films are created very easily through computer animation. Special effects like fire, battle, earthquake, etc., can be created for films.
- **Defence:** In defences, computers can be very useful. Modern weapons and missiles are totally computer-controlled.

## 14.6 Characteristics of the Computer System

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The characteristics of the computer system are as follows:

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- **Speed:** A computer works with much higher speed and accuracy compared to humans while performing mathematical calculations. Computers can process millions (1,000,000) of instructions per second. The time taken by computers for their operations is microseconds and nanoseconds.
- **Accuracy:** Computers perform calculations with 100% accuracy. Errors may occur due to data inconsistency or inaccuracy.
- **Diligence:** A computer can perform millions of tasks or calculations with the same consistency and accuracy. It doesn't feel any fatigue or lack of concentration. Its memory also makes it superior to that of human beings.
- **Versatility:** Versatility refers to the capability of a computer to perform different kinds of work with same accuracy and efficiency.
- **Reliability:** A computer is reliable as it gives consistent result for similar set of data, i.e., if we give same set of input for any number of times, we will get the same result.
- **Automation:** Computer performs all the tasks automatically, i.e., it performs tasks without manual intervention.
- **Memory:** A computer has built-in memory called primary memory where it stores data. Secondary storage are removable devices such as CDs, pen drives, etc., which are also used to store data.

## 14.7 Basic Structure of a Computer System

**Computer follows** input + process + output cycle (IPO CYCLE) in order to perform a task (**Figure 14.1**).

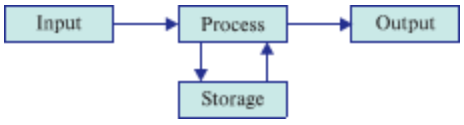


**Figure 14.1** Basic Structure of Computer System

**IPOS Cycle:** It is how computer intakes data, processes the data, outputs information and then saves the information. I stand for input, P stands for processing, O stands for output and S stands for storage.

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**Input:** Input is the raw data, entered into a computer from the input devices. It is the collection of numbers, letters, images, etc. Keyboard, mouse, scanner, webcam, etc., are some examples of the input devices.

**Process:** Process is a total internal process, which represents the operation of data or information as per given instruction. The main processing device of the computer is the CPU (Central Processing Unit).

**Output:** The result of processed data is given in the form of 'output', post data processing. We can save these results in the storage devices for the future use. Monitor, printer and speaker are the main output devices.

## 14.8 Data Representation Computer

***Most Computers are Digital***

- Computer recognises only two discrete states: On or Off (**Figure 14.2**).
- Computers are electronic devices powered by electricity, which has only two states, On or Off.

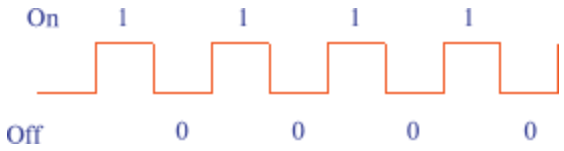


**Figure 14.2** Digital Signals

Data in computers are represented in binary form. To understand data representation, one has to know four number system.

1. **Decimal Number System:** In this system we use ten different symbols to represent any number. Therefore, its base is ten. Following are the symbol used in this system. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
2. **Binary Number System:** In this system, we use only two symbols, i.e., 0 and 1. Therefore, its base is two.
3. **Octal:** In this system eight symbols are used, i.e., 0, 1, 2, 3, 4, 5, 6, 7. Therefore, its base is eight.
4. **Hexadecimal:** In this system sixteen symbols are used 0123456789 ABC DE F. Therefore, its base is 16.

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## 14.9 Components of Computer System

A computer consists of two basic components: (1) Hardware and (2) Software.

### 14.9.1 Hardware

The entire physical components that are grouped together and form a complete system are known as hardware, e.g., PC-Case, motherboard, RAM, ROM, hard disk, monitor, keyboard, mouse, etc.

Hardware is divided into three parts:

1. CPU
2. Memory Unit
3. Peripheral Device

#### 14.9.1.1 Central Processing Unit

It is known to be the brain of a computer system. It is also named as microprocessor, when constructed on a single chip. A complete CPU consists of Arithmetic Logic Unit (ALU) and control unit along with registers. The input/output unit, CPU and memory unit together constitute a computer system.

#### 14.9.1.2 Microprocessor

They are made up of millions of transistors to process the instruction. These are tiny electronic devices carrying an electric charge. **The first microprocessor 4004** (4 bit microprocessor) **was invented by INTEL in 1971.** It was designed for specific applications. The first general purpose microprocessor 8008 was introduced in 1974 which was 8 bit microprocessor. Currently, 32/64-bit processors are available in the market.

Note: All processors are not CPUs but all CPUs are known as microprocessors. The CPU controls all the components of the computer system, while the function of a processor is to

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process, the instruction.

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### 14.9.1.3 Control Unit

The flow of information within the computer system is controlled by Control Unit. It functions as in-charge of the computer as it decodes machine instructions and generates control signal. It may also control some external devices.

### 14.9.1.4 ALU (Arithmetic Logic Unit)

The arithmetic and logical operations of the computer are performed by arithmetic/logic unit (ALU).

### 14.9.1.5 Memory Unit

It serves as the main memory, also known as primary memory or semiconductor memory. It is an internal memory, which includes the following two parts:

#### *1. RAM (Random Access Memory)*

It is a 'read and write' memory, which is a volatile and temporary memory. Two types of RAM are as follows:

1. **Static RAM:** It is called so since it continues to hold information without refreshment and does not need to refresh.
2. **Dynamic RAM:** It is called dynamic as refreshment is required and it loses information with time.

#### *2. ROM (Read Only Memory)*

Read only memory is non-volatile and hence, permanent in nature. With ROM, the data and instruction can only be read and the user is not allowed to write anything on it. The manufacturer writes the contents of ROM at the time of manufacturing.

#### *3. Flash Memory*

Flash memory, like ROM, is non-volatile memory. However, it is an integrated circuit with no continuous power requirement to retain the data.

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Flash memory is comparatively more expensive than magnetic storage. However, flash memory is widely used with car radios, cell phones, digital cameras, PDAs, MP3 players and printers.

### 14.9.1.6 Peripherals Device

The input device, output device, storage device, communication device or other functions under the direct control of a computer, altogether constitute auxiliary equipment or peripheral devices (**Figure 14.3**).

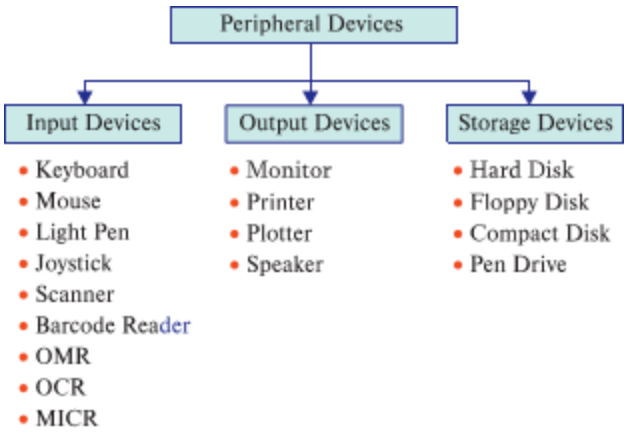


**Figure 14.3** Outline of Peripheral Device

#### *1. Input Device*

It is a computer hardware or device that can accept data or instructions. Examples include **keyboard, scanner, mouse, joy stick, bar code reader, microphone,** etc.

#### *2. Output Device*

It is a computer hardware or device which displays the result or output of the processed data. Examples include **monitor** (VOU), **printer, plotter, speaker,** etc.

#### *3. Storage Device*

A computer hardware device that stores information permanently, hence, also called secondary memory, e.g., **hard disk, floppy disk, compact disk, pen drive,** etc.

#### *4. Communication Device*

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- **Microphone, modem** and **router**

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#### *5. Keyboard*

It serves as standard input device. Keyboard was introduced with 84 numbers of keys. Now-a-days, minimum number of keys present on keyboard is 101 keys.

#### *6. Mouse*

It serves as a standard pointing device. Mouse can only be used in GVI. Douglas Engelbart invented it in 1963.

#### *7. Light Pen*

It is another pointing device, used for digital signature. A light pen only works with CRT-based monitor and not with LCD screens, projectors or other display devices.

#### *8. Joy Stick*

It also serves as a pointing device, which provides better control during computer games.

#### *9. Scanner*

A scanner enables to capture documents or pictures, which are in printed form on paper. After capturing, they are turned into digital format to be viewed on a computer.

#### *10. Microphone*

A microphone serves to input sound information. We can record sound and voice to place it onto the computer via microphone.

#### *11. Barcode Reader*

The different groups of vertical bars are known as Barcodes and to read them on a computer, we need a barcode reader. Today we can find barcodes printed on nearly every product available in the market.

### 14.9.1.7 Output Device

#### *1. Monitor*

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The standard output device (**Figure 14.4**) is known as monitor. Another name for monitor is visual display unit. It displays the computer's user

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interface and opens programmes, thus, enabling the user to visualise the inputs.

#### 2. Printer

Printer prints text or image on paper, hence, it is an output device. The output given out by the printer on a paper is called a hard copy. Printers are of two types:

1. **Impact Printer:** The oldest printer still in action is known as impact printer. With this type of printer, a character is formed when something strikes the paper and ribbon together. Examples include dot matrix printer, chain printer, etc.
2. **Non-Impact Printer:** They do not involve striking the paper. Rather, they use ink spray or toner powder to form the character.

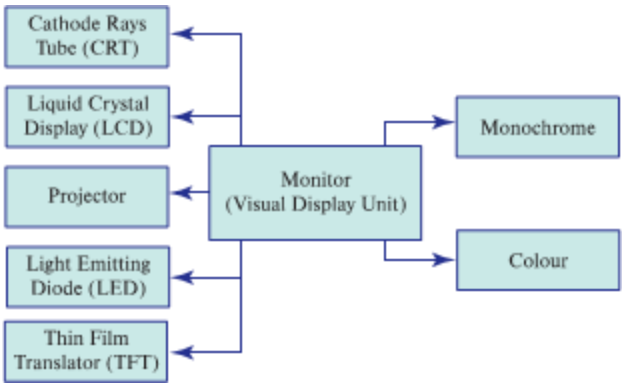


**Figure 14.4** Various Output Devices

### 14.9.1.8 Storage Device (Secondary Memory)

#### 1. Hard Disk

It is a magnetic disk on which you can store computer data. Hard disk is a magnetic disk, which serves as the computer's main storage device. In terms of storage capacity and speed, it is better than other storage devices. On 13 September 1956, the hard drive was first introduced.

#### 2. Floppy Disk

A small portable magnetic disk used to store and transport computer data is known as floppy disk or a diskette. Floppy disk is of two types. In 1978, a 5.25-inch disk was introduced, with storage capacity of 360 KB of data.

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Later revisions of the 5.25-inch floppy disk was developed which could store 1.2 MB. Drive B was used for 5.25-inch floppy. In 1987, the 3.5-inch HD (High Density) floppy disk was released, with storage capacity of 1.44 MB. Drive A was used for 3.5 inch (small floppy). IBM made the first floppy.

#### *3. Compact Disc (CD)*

It is also known as optical disc. A compact disc is an optical storage medium with digital data (in the form of audio, video or computer information). Initially CD was read only, however, technology enabled the user to record data on it. A newer technology, the digital versatile/video disk (DVD), offers better storage capacity in same space. **James T. Russell invented CD in 1960.**

#### *4. Pen Drive*

Pen drive represents small portable USB flash drive. It consists of flash memory data storage device integrated with USB. A plug and play device or pen drive is known with different names such as USB flash drive, jump drive, thumb drive, etc. Dov Moran introduced and invented USB Flash Drive in 1996.

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Note: Direct Access Devices: Hard disk, Floppy disk, Compact disk, etc.

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### 14.9.1.9 Primary Memory (RAM)

#### *1. Disk Cache*

It is used between hard disk and main memory. Its main objective is to reduce the speed gap between main memory and hard disk. It is logically created inside the disk.

#### *2. Cache Memory*

The fastest and most expansive memory is Cache, which increases the efficiency of central processing unit and lies between CPU and main memory. The cache memory is used by CPU to store repeatedly required instructions, while running a programme. Additionally, it improves system

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speed. The cache memory is advantageous since the CPU does not have to use the motherboard's system bus for *data* transfer.

The *data* frequently used by CPU is instantly available via cache. A fetch is made from the main memory, only if the required information is not located in the cache.

The performance of cache memory is frequently major in terms of quantity called hit ratio. When CPU refers to memory to find the words in cache, it is 'HIT'. If the word is not found in cache, it is in main memory, then it is counted as a 'Miss'.

#### *3. Virtual Memory*

When the space on a Hard Disk Drive (HDD) is used to simulate additional main memory, it is termed as virtual memory. It allows software (with greater size than computer's RAM) to run in a memory space (i.e., a logical memory). In other words, if your computer lacks the RAM required for running a programme or operation, virtual memory helps windows to compensate.

### 14.9.1.10 Generation of Computer (Hardware)

The development of computer through different generations is detailed in **Table 14.2** as follows:

**Table 14.2** Different Generation of Computer System

| Generation of Computer | Hardware |
|---|---|
| First-Generation (1940–1956) | Vacuum Tube |
| Second-Generation (1956–1963) | Transistors |
| Third-Generation (1964–1970) | IC (Integrated Circuit) |
| Fourth-Generation (1971–Present) | LSI (Large-Scale Integration),<br>VLSI (Very Large-Scale Integration),<br>ULSI (Ultra Large-Scale Integration) |
| Fifth-Generation (Today) | Still under development phase |

Note: Today's computers are the fifth-generation computers. This generation is based on parallel processing hardware and AI (Artificial Intelligence) software. AI is an emerging

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branch in computer science, which interprets the means and method of making computers think like human beings. All the high-level languages like C and C++, Java, .Net, etc., are used in this generation.

Artificial Intelligence is also known as knowledge information processing system.

***Generation of Computers***

C – Common

O – Operating

M – Machine

P – Purposely

U – Used for

T – Technological

E – Educational

R – Research

## 14.9.2 Software

**Software:** It is a set of programmes which can be developed by a software programmer with the help of computer language and a set of instructions.

Software is classified into three categories as discussed below:

1. Application software
2. Utility software
3. System software

### 14.9.2.1 Application Software

Application software is a computer programme designed to help the user performing a certain type of task. These softwares are written or developed by a programmer for enabling the computer to perform specific tasks such as processing words, handling calculations, etc.

### 14.9.2.2 Utility Software

A utility programme functions for the maintenance of your computer's health hardware or data. It is a helpful programme that ensures smooth

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functioning of the computer. It is specifically designed to help manage and time the computer hardware, operating system or application software, e.g., disk defragmentation, antivirus programme, file/folder management, compression, etc.

### 14.9.2.3 System Software

System software controls internal computer operations, with variety of programmes supporting the operation of the computer. This software enables the user to focus on his tasks without caring how the machine works internally. For example, user operating system, linker loader, language processor, etc.

**Operating System** – A set of programmes that helps in controlling and managing the hardware and the software resources of a computer system is termed as operating system. It establishes the relationship between hardware and software. It gives the platform to run various softwares. It has three main responsibilities:

- Performs basic tasks including recognising input from the keyboard, sending output to the display screen and keeping track of files and directories on the disk. It also controls peripheral devices.
- Ensures that different programmes and users are running simultaneously and they do not interfere with each other.
- Provides a software platform to run other programme.

**Table 14.3** 19th Century Computers

| | First-Generation Computers | Second-Generation Computers | Third-Generation Computers | Fourth-Generation Computers |
|---|---|---|---|---|
| Year | 1942–1954 | 1952–1964 | 1964–1972 | 1972–1989 |
| Technology | Vaccum Tubes | Transistors | Integrated Circuits with SSI and MSI technology | Integrated Circuits with VLSI technology |

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| | First-Generation Computers | Second-Generation Computers | Third-Generation Computers | Fourth-Generation Computers |
|---|---|---|---|---|
| Memory | Electromagnetic relay memory | • Magnetic cores memory<br>• Magnetic | Larger magnetic cores | Microprocessor semiconductor memory |

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**MAYA OPERATING SYSTEM**

The **Defence Ministry's recent decision to transition from Microsoft's operating system to Maya OS marks a significant shift towards bolstering cybersecurity and promoting indigenous technological development.** Maya OS, a locally developed operating system, offers a robust alternative to traditional platforms, emphasizing enhanced security features and seamless usability.

**Key Features of Maya OS**

1. **Open-Source Platform:** Maya OS is built on an open-source platform, allowing for transparency, community collaboration, and customisation according to specific requirements.
2. Despite being a new entrant, Maya OS boasts an interface and functionality akin to Windows, ensuring a smooth transition for users accustomed to conventional operating systems.
3. **Developed by Government Agencies:** Maya OS is the result of concerted efforts by government development agencies, developed within a remarkably short timeframe of six months. This highlights the commitment to promoting indigenous technological solutions.
4. **Cybersecurity Focus:** The primary objective behind adopting Maya OS is to fortify cybersecurity defenses, particularly against malware attacks and other cyber threats that have witnessed a surge in recent years.

## 14.10 Computer Language

A computer language is the means by which instructions and data are transmitted to computers. Languages are divided into two categories:

### 14.10.1 Low Level Language

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**Machine Language:** In machine language, programmes can be developed in binary language (0 or 1). It is also known as binary language.

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**Assembly Language:** It is known as symbolic language. It is a low-level programming language using human readable instruction of the CPU. It is written as:

Move ab, cd

Move ax, 50

### 14.10.2 High Level Language

Programmer can write code in simple English language. It is very user friendly as compared to low-level language, e.g., C, C++, Pascal Fortran, COBOL, basic, prolog, java, etc.

**Loader:** A set of programme that loads the machine language translated by the translator into the main memory and makes it ready for execution.

**Linker:** The main purpose of linker is to link the external module or functions which are not defined in the given programme to the current running programme.

## 14.11 Analogue and Digital Transmission

**Analogue Signal:** An analogue signal (**Figure 14.5**) is continuous with respect to time and may take on any value within a given range of values. Analogue signals are produced when human voice, videos and music are converted to electrical signal using a suitable device. Analogue signal is used in landline phones.

**Figure 14.5** View of Analog Digital

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**Digital Signal:** Only a discrete set of values is taken by digital signal (**Figure 14.6**) within a given range. Most of the computers and computer-related equipments are digital.

**Figure 14.6** View of Digital Signal

## 14.12 Data Flow

Different forms of communication between two devices can be simplex (**Figure 14.7**), half-duplex and full-duplex (**Figure 14.8**).

- **Simplex:** It is one way data transmission, e.g., pager, radio and T.V.

**Figure 14.7** Simplex

**Figure 14.8** Duplex

**Half-Duplex:** In half-duplex, each station can both transmit and receive but not simultaneously. The half-duplex mode resembles one-lane road with traffic allowed in both direction, e.g., Wireless, Walky-talky.

**Full-Duplex:** In full-duplex mode, data can be transmitted and received at the same time. It is a simultaneous two-way data transmission, e.g., telephone, video-conferencing, etc.

## 14.13 LAN Hardware and Software

### 14.13.1 LAN Hardware: The Basic Hardware Components of LAN are

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#### 14.13.1.1 Transmission Channel

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Four types of channels are used for data communication LAN. They are as follows: (1) Twisted pair cable, (2) Coaxial cable, (3) Fibre optics and (4) Radio waves.

#### 14.13.1.2 Network Interface Unit (NIU)

It connects each device in the LAN network to shared transmission device. The LAN contained in NIU contains the rules and logic to access the LAN. Additional applications involve implementing LAN protocols and for device attachment. Its function depends on the type of topology used in LAN.

#### 14.13.1.3 Servers

LAN is beneficial while sharing expensive resources such as storage devices, printer, etc. This is achieved through providing servers on LAN. It is dedicated to control one or more resources. Both hardware and software interface for LAN is contained in it. Three major categories of servers.

##### *14.13.1.3.1 File Server*

It shares storage space for files used for taking periodical backup and also provides gateway to other servers within and between LANs.

##### *14.13.1.3.2 Printer Server*

Printer server handles printing works of all workstations that are connected in the network.

##### *14.13.1.3.3 Modem Server*

In LAN environment also, we require modem for connection (or simply use a telephone) to other network. To share resource by all connected workstations in a network, we need a modem server.

#### 14.13.1.4 Workstations

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In a network, computers connected with servers to the main computer is known as workstation/terminal/client/slave.

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### 14.13.2 LAN Software/Operating System

Main software of LAN is an operating system. It manages workload with number of various types of servers attached to it. It has two aspects (1) Server software and (2) Work station software. LAN operating system facilitates the sharing of expensive resources such as printer storage space, etc. There are various types of LAN operating systems. Such examples include Novel, Netware and Win N1: Win 2000 server, Unix, etc.

### 14.13.3 Metropolitan Area Network (MAN)

It is designed for a city or town. MAN, as a computer network, is smaller than WAN but larger than LAN.

### 14.13.4 Wide Area Network (WAN)

WAN covers large geographical area, e.g., different cities, states and countries. One LAN can be connected to other LAN over any distance via telephone and radio waves. A system of LANs so connected is called WAN.

### 14.13.5 WAN Devices: Bridges, Router

While searching a path between two computing devices that are to communicate and for forwarding the data packets on this path, a routing device is required. Bridges, router and gateway are some of the devices that provide this routing function.

**Bridges:** Bridges connect two LANs using identical protocols over a wide area. The bridge acts as an address filter which picks up packets from one LAN to be transmitted to a destination on another LAN, and thereby, passing these packets on the network.

**Router:** Routers may connect dissimilar networks. They provide connectivity between two LANs or two WANs over large geographical

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distances.

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## 14.14 Terminology

### 14.14.1 Node

Different terminals which are attached to the network and share the resources of the network are called nodes. In the network, the nodes are computers.

### 14.14.2 Server

A computer programme or software package provides a specific kind of service to client software that runs on other computers. A dedicated computer provides function to the network. A client is a computer which requires any of the resources provided by the server. The advantage of client/server model of computing is the control which can be exercised on the content of the server.

### 14.14.3 Workstation

It is that device which is deployed to communicate with data communication network. The workstation may be a terminal printer. In telephone or in other communication devices, workstation is called terminal client or slave.

### 14.14.4 Network

A way or means of transmission or receiving of information from one or more sources is termed as network.

### 14.14.5 Network Interface Unit

The interface which helps in the communication between the server and different nodes is called Network Interface Unit.

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### 14.14.6 Architecture

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The broader classification of network is either *via* a peer to peer or client/server Architecture.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. Consider the following statements:**

**[UPSC (Pre) – 2004]**

1. Adam Osborne produced the first portable computer.
2. Ian Wilmut created the first cloned sheep.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**2. Param Padma, which was in news recently, is:**

**[UPSC (Pre) – 2004]**

(a) A new Civilian Award instituted by the Government of India.
(b) The name of a supercomputer developed by India.
(c) The name given to a proposed network of canals linking northern and southern rivers of India.
(d) A software program to facilitate e-governance in Madhya Pradesh.

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3. **Which of the following pairs is not correctly matched?** **[UPSC (Pre) – 2004]**
   (a) ReffiKoId Messner: Computer Technology
   (b) Harlow Shapley: Astronomy
   (c) Gregor Mendel: Hereditary Theory
   (d) Godfrey Hounsficld: CT Scan

4. **The memory of a computer is commonly expressed in terms of kilobytes or megabytes. A byte is made up of:** **[UPSC (Pre) – 2000]**
   (a) eight binary digits
   (b) eight decimal digits
   (c) two binary digits
   (d) two decimal digits

5. **For reproducing sound, a CD (Compact Disc) audio player uses a:** **[UPSC (Pre) – 2000]**
   (a) quartz crystal
   (b) titanium needle
   (c) laser beam
   (d) barium titanate ceramic

**Answer Key**

| | | | | |
|---|---|---|---|---|
| 1. (c) | 2. (b) | 3. (a) | 4. (a) | 5. (c) |

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**UNIT-V** Defence Technology

# CHAPTER 15
# DEFENCE TECHNOLOGY

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## 15.1 Defence

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‘Defence’ for a country is protecting its integrity, maintaining sovereignty, protecting international borders, maintaining unity and protecting the geo-economic interests of the nation and not allowing any kind of threat to enter into the country through land, sea or air.

## 15.2 Indian Defence

**The supreme commander of the Indian defence system is the President of India.** The whole administrative control of Armed Forces lies in **Ministry of Defence.** Indian Defence System has been divided into three services – Army, Air Force and Navy.

***Officer Ranks of the Indian Army, Navy & Air Force***

| Army | Navy | Air Force |
|---|---|---|
| Field Marshal | Admiral of the Fleet | Marshal of the Indian Air Force |
| General | Admiral | Air Chief Marshal |
| Lieutenant General | Vice Admiral | Air Marshal |
| Major General | Rear Admiral | Air Vice Marshal |
| Brigadier | Commodore | Air Commodore |
| Colonel | Captain | Group Captain |
| Lieutenant Colonel | Commander | Wing Commander |
| Major | Lieutenant Commander | Squadron Leader |
| Captain | Lieutenant | Flight Lieutenant |
| Lieutenant | Sub-Lieutenant | Flying Officer |

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## 15.3 Departments Under Ministry of Defence

Ministry of Defence provides the policy framework and the execution of approved programmes within the allocated resources to the Department of Defence, Department of Defence Production, Department of Defence Research and Department of Ex-Servicemen.

### 15.3.1 Defence Research Development Organisation (DRDO)

The scientific adviser of Defence Ministry is the secretary and the authority, which heads the DRDO. It was established in 1958.

**55 per cent of our defence system needs are fulfilled by DRDO.** The vision of DRDO is to make India prosperous through world-class establishments of science and technology.

In 1980, a separate Department of Defence Research and Development was formed. Later on, it was involved in administering DRDO and its 52 laboratories/establishments. The first major project started by DRDO was in SAM (Surface-to-Air Missile) known as **Project Indigo** in 1960s. In later years, Indigo got discontinued without achieving the complete success. It led to **Project Devil,** along with **Project Valiant,** for development of short-range surface-to-air missile and ICBM in the year 1970. Under IGMDP in the year 1983, Project Devil led to the later development of the Prithvi missile. An Indian Ministry of Defence programme between the early 1980s and 2007 was named **IGMDP (Integrated Guided Missile Development Programme).** It was launched to develop a comprehensive range of missiles, which includes the Agni missile, Prithvi ballistic missile, Akash missile, Trishul missile and Nag Missile.

DRDO is responsible for the development of technology for use by the Indian armed forces.

### 15.3.2 Defence Acquisition Council (DAC)

**In order to tackle corruption and accelerate decision-building in military procurement, the Union government, in the year 2001, decided**

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**to form a comprehensive DAC,** being headed by the Defence Minister. Its objective is ensuring **speedy procurement of the approved requirements of the Armed Forces,** in respect of sought capabilities and prescribed time frame, through effective utilisation of the allocated budgetary resources. The **DAC is responsible for providing policy guidelines** to acquisitions, on the basis of long-term procurement plans. It also clears all acquisitions, inclusive of both imported and indigenously manufactured or under a foreign licence.

## 15.4 Integrated Guided Missile Development Programme (IGMDP)

In 1983, the Government of India launched the Integrated Guided Missile Development Programme to achieve self-sufficiency in the development and production of wide range of Ballistic Missiles. The IGMDP, as an Indian Ministry of Defence programme between the early 1980s and 2007, worked to develop a comprehensive range of missiles, which includes the **intermediate-to-long-range Agni Missile (Surface-to-Surface)** and **short-range missiles like the Prithvi Ballistic Missile (Surface-to-Surface), Akash Missile (Surface-to-Air), Trishul Missile (Surface-to-Air) and Nag Missile (Anti-Tank).**

On **May 07, 2008,** the closing of the Strategic Integrated Guided Missile Programme was announced by India. It was also mentioned that the development and production would, henceforth, be taken up with foreign partnerships, for most of the futuristic weapons systems.

***Ballistic Missile vs Cruise Missile***

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| Feature | Ballistic Missile | Cruise Missile |
|---|---|---|
| **Flight Path** | Parabolic; exits atmosphere | Atmospheric; straight or guided |
| **Propulsion** | Only initial phase | Continuous throughout |
| **Speed** | Very fast; often hypersonic | Slower; subsonic/supersonic |
| **Guidance** | limited mid-course control | Advanced GPS/radar-based |
| **Range** | Very long; ICBMs possible | Generally medium/short range |
| **Accuracy** | Lower; large-scale targets | High; pinpoint strikes |
| **Payload** | Nuclear; conventional; strategic | Conventional/nuclear; tactical |
| **Detection** | Easier (heat/smoke during launch) | Harder (low-altitude stealth) |

### 15.4.1 Prithvi

Under the IGMDP programme, the first missile to be developed was Prithvi. It was inducted into the Indian Army in 1994. Prithvi Missiles have three variants (**Table 15.1**).

**Table 15.1** Different Variants of Prithvi Missile

| Missile | Version | Distance | Warhead |
|---|---|---|---|
| Prithvi-I | Army | 150 km | 1000 kg |

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| | | | |
|---|---|---|---|
| Prithvi-II | Air force | 250 km | 500–1000 kg |
| Prithvi-III | Navy | 350 km | 1000 kg |

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Note:

**PRAHAR**

- Prithvi I Missile will be replaced by Prahar Missile soon.
  PRAHAR – Prahar is a solid-fuelled surface-to-surface guided short-range Ballistic Missile indigenously developed by DRDO. It could be used for hitting both strategic and tactical targets, being equipped with omni-directional warheads, and has a range of about 150 kilometres. It was test-fired successfully on 21 July, 2011 from the Integrated Test Range (ITR) at Chandipur. Prahar is to replace Prithvi I Missile soon.
- 'Prahar', manufactured by the DRDO, has the capability of bridging the gap between the multi-barrel rocket system 'Pinaka' and medium-range ballistic missile 'Prithvi'. It can also undertake numerous targets in different directions.
- It is 150 km in range and 35 km in flight altitude.
- It is a solid-fuelled short-range missile integrated with inertial navigation system.
- It is a responsive, all-weather, all-terrain and extremely precise battleground support strategic weapon system.

**Pradyumna Ballistic Missile**

- The Pradyumna Ballistic Missile is the name of Prithvi Air Defence Missile, developed to use as an anti-Ballistic Missile system.

**Dhanush**

- It is a variant of surface-to-surface or ship-to-ship Prithvi III Missile, Dhanush which has been developed for the Indian Navy. It is capable of carrying both types of warheads like conventional as well as nuclear with payload capacity of 500–1000 kg and can strike targets in the range of 350 km. This missile got various test-fired successfully. Depending on the range, the Dhanush Missile can be used for destroying land targets and as an anti-ship weapon as well.

***Indian Army Plans to Phase Out Prithvi Missiles for Modern SRBM Solutions***

The Indian Army and Strategic Forces Command (SFC) are planning a complete phase-out of the Prithvi-1 and Prithvi-2 short range ballistic missiles (SRBMs) by 2030. This move signifies India's commitment to modernising its missile arsenal and deploying next-generation solutions.

## BALLISTIC VERSUS CRUISE MISSILE

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- Ballistic Missiles always follow parabolic path but Cruise Missiles are horizontal to the surface. Ballistic Missiles change their own atmosphere but Cruise Missiles are always in same atmosphere.
- Ballistic Missiles can vary widely in terms of range and use. Hence, their division can often be based on range. Different countries use different schemes for categorising the ranges of Ballistic Missiles:
  1. **Tactical Ballistic Missile:** Its range is between 150 and 300 kilometres.
  2. **Short-Range Ballistic Missile (SRBM):** Its range is between 300 and 1,000 kilometres.
  3. **Medium-Range Ballistic Missile (MRBM):** Its range is between 1,000 and 3,500 kilometres.
  4. **Intermediate-Range Ballistic Missile (IRBM)** or **Long-range ballistic missile (LRBM):** Its range is between 3,500 and 5,500 kilometres.
  5. **Intercontinental Ballistic Missile (ICBM):** Its range is greater than 5,500 kilometres

### Intercontinental Ballistic Missiles or ICBMs:

- Intercontinental Ballistic Missiles or ICBMs are guided missiles which can deliver nuclear and other payloads. ICBMs have a minimum range of 5,500 km, with maximum ranges, varying from 7,000 to 16,000 km.
- Only a handful of countries, including **Russia, United States, China, France, India and North Korea,** have ICBM capabilities.

## DR. A.P.J. KALAM IS THE MISSILE MAN OF INDIA

- In 1965, he started to work on an expandable rocket project independently at DRDO. Later on, Kalam single-handedly got the government to expand his independent programme and include more engineers. There came Kalam's first stint with aerospace engineering in 1969.
- He became the Project Director for India's first Satellite Launch Vehicle (SLV-III) at Indian Space Research Organisation (ISRO), which launched the Rohini Satellite in 1980.
- Kalam worked on developing the Polar Satellite Launching Vehicle (PSLV) and SLV-III projects between the years 1970 and 1990. Both projects were met with success.
- The two secret missile programmes, which primarily earned Kalam the alias 'the Missile Man of India', were Project Devil and Project Valiant. Both projects were top secret liquid-fuelled missile projects, indigenously developed under the directorship of Kalam and with special permission by Prime Minister Indira Gandhi. Later on, these projects served as the foundation in the development of Ballistic Missile Prithvi. Then came the `388 crore project, known as Integrated Guided Missile Development Programme (IGMDP), headed by Kalam as the chief executive. This was the project where Kalam played a monumental role in developing supersonic missiles Agni (an intermediate range Ballistic Missile) and Prithvi.

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- Kalam played an intensive political and technological role, being the Chief Scientific Advisor to the PM and Secretary of Defence, during the Pokhran-2 nuclear tests that were conducted between 1992 and 1999.

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### 15.4.2 Trishul Missile System

It is a short-range surface-to-air missile with a range of 9–12 km and fitted with a 5.5/15/130 kg warhead. The Trishul Missile project was officially shut down on 27 February, 2008.

Note: **Maitri Missile:** Due to the failure of Trishul missile, Maitri Missile will be launched. It is believed to be a blend of French Mica and DRDO Trishul. This missile has low-level quick reaction time.

**Last Update:** As of 2020, the project was expected to be cancelled as DRDO has instead taken up the development of an alternative missile known as **VL-SRSAM** (Vertical Launch Surface to Air Missile) derived from the **Astra MK1 air-to-air missile** for use by the **Indian Navy and QRSAM** (Quick Reaction Surface-to-Air Missile) for the **Indian Army**.

### 15.4.3 Nag Missile System

Nag is an indigenously built third-generation 'Fire-and-Forget' anti-tank missile. The three different types of guidance that have been available with Nag include a wire-guided version, an infrared version and a millimetric wave version.

Note: **Fire and Forget:** It means the Missile guidance does not require further guidance after the launch and still hits the target without the launcher being in line-of-sight of the target.

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### MAN PORTABLE ANTI-TANK GUIDED MISSILE (MPATGM)

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In a major boosting of Army, DRDO has effectively test-fired indigenously produced, low weight, fire and forget Man Portable Anti-Tank Guided Missile (MPATGM). About MPATGM:

- It is third-generation Anti-Tank Guided Missile (ATGM), indigenously manufactured by DRDO.
- It has striking range of 2.5 km and weighs approximately 15.5 kg for maintaining man portability. It is also proficient of being fired from shoulder and can be fired during day and night. It has least lateral centre and gravity offset.
- It works on the principle of fire and forget. It is known for its capabilities of top attack. It is useful against together stationary and moving targets and will be positioned in the Indian Army's infantry and parachute battalions.

The missile has a weight of 42 kg and ranges between 4 and 5 km. With the claim of being the first anti-tank missile with complete fiberglass structure, this missile marked a significant evolution in development of the missile system of India. A tracked vehicle known as **NAMICA (Nag Missile Carrier)** was deployed to test the land version of this missile. Its configuration is to enable its use on the Advanced Light Helicopter (ALH) and the Hindustan Aeronautics Limited (HAL) Light Combat Helicopter (LCH). This version will be called as **HELINA (HELIcopter based NAg).** Nag was successfully test-fired from the test range at Pokhran, Rajasthan, for the second day in a row on 8 August, 2008. This marks the completion of its developmental tests.

The Agni Missile represents a family of medium to intercontinental range of indigenously developed Ballistic Missiles. Being surface-to-surface Ballistic Missile, Agni Missiles are nuclear weapons capable. Under the Integrated Guided Missile Development Programme, the first missile of the series Agni-I was developed, and tested in 1989. Comparison of different variants of Agni missiles are mentioned in **Table 15.2**.

## Anti-Tank Guided Missile (ATGM) – Nag Mk 2

***Latest Update:*** Successfully completed **field evaluation trials** at **Pokhran Range**, Rajasthan, demonstrating **high precision and reliability**.

***About Nag Mk-2***

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- ***Type:*** All-weather, *fire-and-forget*, *lock-on after launch* third-generation ATGM.
- ***Developed by:*** Defence Research and Development Organisation (**DRDO**).
- ***Launch Platform:*** **NAMICA** (Nag Missile Carrier).
- NAMICA is based on **SARATH BMP-II** – an **amphibious infantry combat vehicle** used by the Indian Army.

### Specifications

| Feature | Details |
|---|---|
| **Range** | 7–10 km |
| **Earlier Version (Nag Mk-1)** | 4 km |
| **Warhead** | **Tandem High-Explosive Anti-Tank (HEAT)** warhead |
| **Target Capability** | **Defeats Explosive Reactive Armour (ERA)** on modern tanks |

**Explosive Reactive Armour (ERA)** is a type of armour used on tanks and other vehicles to protect against anti-tank weapons.

#### *Key Features*

- **Third-Generation Fire-and-Forget** Technology: No need for operator guidance post-launch.
- **Lock-on After Launch (LOAL):** Can track target post-firing.
- **Versatile Use:** Neutralises heavily armoured targets.
- **High Mobility:** Launch-ready via NAMICA across diverse terrains.

#### *Significance*

- Enhances India's indigenous **anti-tank warfare** capabilities.
- Bridges a key gap in long-range precision targeting of armoured vehicles.
- Supports **Aatmanirbhar Bharat** by reducing reliance on foreign ATGMs.

## 15.4.4 Agni Missile

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**Agni-I** It is capable to cover the range of 700–1200 km and, at the same time, capable of carrying a conventional warhead of 1000 kg or nuclear warhead at a **speed of 2.5 km per second.** It consists of one stage in the short range and two stages in the intermediate range.

**Agni-II** It is capable to cover the range of 2000–2500 km and can carry the conventional or nuclear warhead up to 1000 kg at a **speed of 3.9 km per second or Mach 12.** It is built upon solid propellant in both stages.

**Agni-III** It has capability to cover the range of 2500–3500 km and can have a warhead of 1.5 tonnes, at the **speed of Mach 13.** It is built upon the powerful solid propellant in its both stages.

**Agni–IV** Earlier it was known as Agni-II Prime. It has capability to cover the range of 3000–4000 km and capable to carry a warhead weighing more than 800 kg. It has two stages, powered by solid propellant.

**Agni-V** It is a solid-propelled Intercontinental Ballistic Missile (ICBM) and it will cover a range of up to 5000–8000 km. It has been designed with third composite stages and a semi-cryogenic engine has been used in the third stage (not officially declared). It can carry a warhead approximately weighing 1500 kg at a **speed of Mach 24.**

**Table 15.2** Different Variants of Agni Missile

| Missile | Warhead (kg) | Range (km) | Fuel Stages (Propellant) |
|---|---|---|---|
| Agni – I | 1000 | 700–1200 (Short Range to Medium Range) | Single – solid |

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| | | | |
|---|---|---|---|
| Agni – II | 750–1000 | 2000–2500 (Medium Range to Intermediate Range) | Two – solid, solid |
| Agni – III | 1500 | 2500–3500 (Intermediate Range) | Two – solid, solid |
| Agni – IV | 800–1000 | 3000–4000 (Intermediate Range) | Two – solid, solid |
| Agni – V | 1500 | 5000–8000 (Intercontinental Range) | Three – solid, solid, solid |
| Agni – VI | 1000 | 6000–12000 (Intercontinental Range) | Three – solid, solid, solid |

**Mission Divyastra: On 11 March, 2024, India** achieved a significant milestone in missile technology by successfully testing Agni-V ballistic missile which is a part of Mission Divyastra. India marked its entry into the elite group of nations possessing **Multiple Independently Targetable Re-entry Vehicle (MIRV)** capabilities.

***Key Facts About MIRV Technology***

- **Inception:** MIRV technology **originated in the United States in 1970** with the deployment of MIRVed Intercontinental Ballistic Missiles (ICBMs).
- It enables **a single missile to carry multiple warheads (usually 3–4)**, capable of targeting different locations independently.

**AGNI PRIME (AGNI – P)**

- The Agni Prime or the Agni-P is a nuclear-capable next generation advanced variant of the Agni class of missiles.
- This **medium-range missile** was developed by DRDO to **replace Agni-I and Agni-II missiles**.
- This 10-meter-long **two-stage** canisterised missile has a maximum range of 2,000 kilometres.
- **It weighs less than 50 per cent of the Agni 3 missiles.**
- It is equipped with **new guidance and propulsion systems**.
- The **missile can be fired from mobile launchers and has a high survival rate**.

| Feature | Detail |
|---|---|

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| | |
|---|---|
| **Range** | 1,000 to 2,000 km |
| **Stages** | Two |
| **Launch Platform** | Road-mobile canister |
| **Propulsion** | Solid-fuel |

**Key Technology**

- **MIRV (Multiple Independently Targetable Re-entry Vehicle):**
  - ❑ Enables single missile to **deliver multiple warheads** to **different targets**.
  - ❑ Enhances **survivability** and **first-strike capability**.

**Strategic Significance**

- Enhances India's **credible minimum deterrence** posture.
- **Supports rapid deployment** and **quick-response capability**.
- Demonstrates progress in **indigenous missile technology** and **nuclear delivery systems**.

- This technology **enhances a missile's effectiveness** by increasing the number of potential targets it can engage.

### 15.4.5 Akash Missile System

Akash is a medium range, surface-to-air missile. It marks to be the most expensive missile project ever undertaken by the Government of India in the 20th century. Its interception range is 30 km. **Its launch weight is 720 kg.** Akash moves with supersonic speed and reaches around **Mach 2.5.** The altitude which it can reach is 18 kilometres.

Akash Missiles have **integrated ramjet propulsion system.** The missile is supported by a multi-function and multi-target phased array fire control radar, which is known as the **'Rajendra'.**

## 15.5 Quick Reaction Surface-to-Air Missiles (QRSAM)

India has effectively test-fired two indigenously developed Quick Reaction Surface-to-Air Missiles (QRSAM) from a test range off the Odisha coast.

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- It has been developed to replace the 'Akash' missile defence system and has 360-degree coverage.
- It uses solid fuel propellant and has a range of 25–30 km with the ability to strike multiple targets.
- It has the ability of striking the low flying objects.

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## 15.6 VL-SRSAM – Vertically Launched Short-Range Surface-to-Air Missile

- Developed indigenously by **DRDO.**
- Designed for **quick reaction** against aerial threats, including **sea-skimming targets.**
- Initially for **Indian Navy** (40 km range); now extended to **80 km.**
- Being adapted for the **Indian Air Force** to protect **air bases.**

***Key Features***

- **Length:** 3.93 m
- **Diameter:** 178 mm
- **Wingspan:** 508 mm
- **Weight:** ~170 kg
- **Speed:** Up to **Mach 4.5**
- **Max altitude:** 16 km
- **Propellant:** Solid fuel

**Vertical Launch:** It's launched vertically from a land-based vertical launcher.

### Anti-Ballistic Missile System / Ballistic Missile Defence Program

The Indian Ballistic Missile Defence Programme conceived and developed **interceptor systems** like the Prithvi Air Defence system and the Advance Air Defence system.

## 15.7 Shaurya Missile System

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Shaurya is a **short-range, surface-to-surface, hypersonic, Ballistic Missile** to be used by the Indian Army. It has a range of 600 km and can carry a warhead, weighing up to one-tonne (conventional or nuclear). It represents a land version of the under-water launched K-15 Missile, Sagarika (Missile). It can reach to an altitude of 40 km and distance of this missile is 800 km with a speed of 7.5 times than speed of sound. On 3 October 2020, DRDO successfully test-fired an advanced version of the Shaurya with an 800 km range from Balasore as part of user trials.

## 15.8 Anti-Ballistic Missile System/ Ballistic Missile Defence Program

The Indian Ballistic Missile Defence Programme is a significant initiative to develop and deploy a multi-layered Ballistic Missile defence system to enable protection from Ballistic Missile attacks.

***Indian Ballistic Missile Defence (BMD) Programme***

### *Overview*

- **Launched:** 2000, post-Kargil War.
- **Objective:** Defend India against ballistic missile threats from **Pakistan and China**.
- **Approach: Multi-layered defence** using two interceptor systems:
  - ❑ **PAD** (Prithvi Air Defence) – High altitude interceptor (exo-atmospheric)

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❑ **AAD** (Advanced Air Defence) – Low altitude interceptor (endo-atmospheric)

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### *Key Components*

- **Interceptor Missiles:** Destroy enemy missiles in flight.
- **Phased Array Radars:** e.g. **Swordfish** for long-range tracking.
- **Command and Control:** Integrated Air Command and Control System (IACCS).
- **Sea-based Defence:** e.g. tests from **INS Anvesh**.
- **Communications Network:** Real-time link between radars, launchers and control centres.

| System | Range | Altitude | Features |
|---|---|---|---|
| **PAD** | 2,000 km | 50–80 km (exo) | Two-stage, Mach 5+, kinetic kill vehicle |
| **AAD (Ashwin)** | ~100 km | 15–30 km (endo) | Single-stage, Mach 4.5, solid-fuel |

#### *Future Enhancements*

- **Phase 2:** Anti-ballistic systems like **THAAD**.
- **AD-II Project:** Intercepts missiles up to **5,000 km**.
- **Hypersonic Glide Vehicles (HGVs)** and enhanced **radar tech** under development.

#### *Key Technologies*

- **Interceptor Missiles:** e.g. Pradyumna, Ashwin.
- **Radars:** Swordfish (phased array), multi-object tracking.
- **Guidance:** Inertial and radar-based homing.
- **Kill Method:** Explosive warheads or kinetic hit-to-kill.

#### *Global Collaboration*

- **S-400 Triumf System** acquired from **Russia**.
- Tech exchanges and cooperation with **USA** and other allies.

#### *Significance*

- **National Security:** Shields major cities and assets from missile attacks.
- **Deterrence:** Supports India's **second-strike nuclear capability**. (**second-strike capability** is the ability of a nuclear-armed state to

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respond to an enemy nuclear strike with a nuclear counterattack)

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- **Indigenous Capability:** Boosts self-reliance and defence innovation.
- **Geopolitical Credibility:** Enhances India's strategic standing.

***Challenges***

- **Technological Complexity:** Integration of radars, guidance, interceptors.
- **Limited Coverage:** ~80 km interception radius; not full country coverage.
- **Multi-Vector Threats:** Difficulty in handling simultaneous multi-front attacks.
- **High Costs:** Expensive to maintain and expand BMD (Ballistic Missile Defence) shield.
- **Restricted Tech Access:** Limited by international non-proliferation agreements.

## ENDO AND EXO ATMOSPHERIC SYSTEMS

**1. Advanced Air Defence (AAD)**

- This is an **endo-atmospheric Anti-Ballistic missile** and can intercept and destroy a target in lower atmospheric layer.
- It is a **single stage solid fuelled missile** with a maximum range from **150 to 200 km** and **maximum altitude of interception range is 30 km**.
- Maximum speed is nearly about 4.5 Mach.

**2. Prithvi Defence Vehicle (PAD)**

- This missile serves as an **exo-atmospheric anti-ballistic missile**, capable of intercepting targets outside the Earth's atmosphere.
- It shares a similar range as the **PAD system (300–2000 km)** but boasts an enhanced **flight altitude of 150 km**.
- Equipped with a more precise guidance system utilising a Ring Laser Gyro-based inertial navigation system.

## INDIA JOINS US, RUSSIA, CHINA HYPERSONIC MISSILE CLUB

Recently, India successfully flight tested the indigenously developed **Hypersonic Test Demonstrator Vehicle (HSTDV)** and became the fourth country after the United States, Russia and China to develop and successfully test hypersonic technology.

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**Significance and implications of this test flight:**

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- This indigenously developed technology paves to develop hypersonic weapons that can travel six times faster than the speed of sound (Mach 6).
- The HSTDV is an **unmanned scramjet demonstration aircraft for hypersonic speed flight.**
- The HSTDV cruise vehicle is mounted on a solid rocket motor, which will take it to a required altitude, and once it attains certain much numbers for speed, the cruise vehicle will be ejected out of the launch vehicle. Subsequently, the scramjet engine will be ignited automatically.

**Way ahead**

It has utility for long-range cruise missiles of the future and it can be also used for launching satellites at low cost too.

## 15.9 Cruise Missiles

### 15.9.1 Nirbhay

Nirbhay is an all-weather, long-range Cruise Missile. It can carry conventional and nuclear warheads with subsonic speed (0.8 mach). With a range of more than 1000 km, its weight is about 1500 kg. Missile will be for use by all the three defence services, the Indian Army, Indian Navy and the Indian Air Force. It can be launched from multiple platforms on land, sea and air.

### 15.9.2 BrahMos

An agreement was signed by India with Russia to design, develop, manufacture and market a Supersonic Cruise Missile System in 1998. By 2006, it has been successfully accomplished. It was the world's fastest Cruise Missile with speeds of Mach 2.5 to 2.8 (supersonic speed), first developed in India. In that time, in terms of speed, it was about three and a half times faster compared to the American subsonic Harpoon Cruise Missile.

It can be launched from submarines, ships, aircrafts or land. It was a joint venture between India's DRDO and Russia's NPO Mashinostroeyenia.

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They have together established BrahMos Aerospace Private Limited. The acronym BrahMos is perceived from the two rivers – the Brahmaputra of India and the Moskva of Russia – which represents the confluence of the two nations. This land-based and ship-launched missile can carry a 200-kg warhead. On the other hand, the aircraft-launched variant (BrahMos A) is with a capacity to carry a 300-kg warhead.

Brahmos has a two-stage propulsion system: it has a solid-propellant rocket and a liquid-fuelled ramjet enabling sustained supersonic cruise.

India has a plan to have BrahMos II by 2028, as a hypersonic Cruise Missile. The laboratory testing of the missile has started. As India became a member of the MTCR in 2016, India and Russia are now looking forward to jointly develop a new generation of BrahMos Missiles with 600 km+ range and the capability of hitting protected targets with pin-point accuracy.

**TYPES OF HYPERSONIC WEAPONS**

- The **two types of hypersonic weapons** systems are **Hypersonic Glide Vehicles (HGV)** and **Hypersonic Cruise Missiles**.
- The HGV are **launched from a rocket before gliding** to the intended target while the **hypersonic cruise missile is powered by air breathing high speed engines or 'scramjets'** after acquiring their target.

## 15.10 Astra BVR Air to Air Missile

It is the indigenous Beyond Visual Range (BVR) air-to-air developed missile. It is an **all-weather; high-tech missile manufactured by DRDO and can engage and demolish enemy aircraft at supersonic speed (1.2 Mach to 1.4 Mach).** The **3.8 metres tall Astra is a radar homing missile** and the **DRDO-developed smallest missile and can be launched from different altitudes. It has a target range of 110 km when launched from a 15 km altitude, 44 km when launched from an 8 km altitude and 21 km when fired from sea level.**

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## 15.11 Medium-Range Surface-to-Air Missile

The Indian Army and DRDO have signed an MoU with the development of the advanced Medium-Range Surface-to-Air Missiles (MRSAM). It is an all-weather, mobile, land-based air defence system and capable of multiple aerial targets, with range of more than 50 km.

In this project, Israel will help India, with the involvement of private sectors. This system will have majority of indigenous content, to boost up the Make-in-India initiative. MRSAM is a land-based variant of the Long-Range Surface-to-Air Missile (LRSAM) or Barak-8 Naval Air Defence System, which is designed to operate from naval vessels and has a range of up to 100 km.

Note: Barak-8 was also jointly developed with Israel.

## 15.12 Mission Shakti: First Anti-Satellite Missile (ASAT) System of India

India successfully carried out its first **anti-satellite (ASAT)** missile test in space while destroying a low earth orbit satellite through the usage of a missile which covered a **300 km** distance for engaging the target.

The test places India in a selected group alongside the **US, Russia and China** – countries with demonstrated anti-satellite capability.

This test does not violate any international law or treaty obligation and is much less harmful than the Chinese ASAT test in 2007 that led to a large-scale scattering of debris in space that threatened other satellites. 'The test was done in the lower atmosphere to ensure that there is no space debris'.

This missile system was under development since 2016, after the government gave a go-ahead and is a demonstration that the Ballistic Missile Defence Programme is progressing at a good pace. DRDO said that the test, called 'Mission Shakti', was carried out from the APJ Abdul Kalam

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Island in Odisha and the interceptor was a three-stage missile with two solid rocket boosters.

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## 15.13 Sudarshan (Laser-Guided Bomb)

In order to occupy the niche of a precision delivery mechanism, India's first laser-guided bomb, Sudarshan was developed. It is the latest weapon system and can fit to a 1000-pound gravity bomb. It uses lasers to guide it to the target with a CEP (Circular Error Probability) of 10 metres.

## 15.14 NASAMS-II (National Advanced Surface-to-Air Missile System-II)

India is in negotiation with **United States** for procuring National Advanced Surface-to-Air Missile System-II (NASAMS-II), a **superior air defence system** at the outlay of **$1 billion** for **shielding National Capital Region (NCR) from airborne attacks.**

NASAMS-II is an upgraded version of the NA-SAMS, developed by Raytheon (US defence contractor and company), in association with Norway KONGSBERG Defence and Aerospace and is operational since 2007.

It is equipped with radars of latest 3D mobile surveillance and 12 missile launchers for immediate reaction. **NASAMS-II is extremely adjustable mid-range answer for any requirement of working air defence.** It offers modern defence system which **maximises the capability for rapidly identifying, engaging and destroying recent and evolving enemy aircraft, UAV or emerging threats of Cruise Missile.**

NASAMS-II is **equipped with 3D Sentinel radars, launchers, short and medium-range missiles, centres of distribution and command-and-control units for rapidly detecting, tracking and shooting down multiple airborne threats.** It is a division of the air defence network,

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safeguarding Washington DC, US capital city and also positioning in numerous NATO countries.

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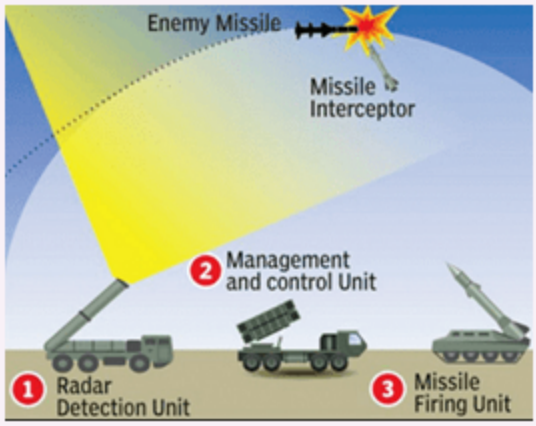
**NATIONAL ADVANCED SURFACE-TO-AIR MISSILE SYSTEM-II**




- Armed with 3-D Sentinel radar, short- and medium-range missiles, launchers, fire-distribution centres.
- Quickly detects, tracks and shoots down multiple airborne threats such as Cruise Missiles, aircraft and drones.
- Deployed to guard US and several NATO countries, Israel, Russia and European militaries also deploy them.
- India's own shield is in the final stages of development by DRDO. It will be deployed to protect cities like Delhi and Mumbai.

India's acquisition of NASAMS-II **will assist in prevention of 9/11-type on NCT Delhi.** It will also enhance the country's other systems under procurement such as the medium and long-range Surface-to-Air Missile (SAM) method. **By this, India will join nations league consisting of the US, Russia and Israel, etc., which possess their own missile defence systems for protecting their respective national capital regions.**

## 15.15 Nuclear Test of India

India's first nuclear test was held on 18 May 1974. Since then, another series of tests have been conducted by India in 1998. India has extensive civil as well as military nuclear programmes. Besides that, India has heavy

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water production facilities, a uranium-enrichment plant, fuel-fabrication facilities, as well as extensive nuclear-research capabilities. In 1998, temporary economic sanctions were imposed by the United States and Japan on India in response to the continuing tests conducted by India.

### 15.15.1 Command and Control of Atom Bomb

In terms of civil leadership, the CCS (Cabinet Committee on Security) is the only authorised body, which can order a nuclear strike against another offending strike. In effect, the Prime Minister is the real authorising body who has his finger 'on the button'.

**Table 15.3** Nuclear Programme of India

| | |
|---|---|
| Nuclear programme start date | 1967 |
| First nuclear weapon test | 18 May 1974 (Smiling Buddha) |
| First fusion weapon test | 11 May 1998 (disputed) |
| Last nuclear test | 13 May 1998 |
| Total tests | 6 |
| Maximum missile range | 3,500 km (Agni-III) |

## 15.16 Popular Bombs

There are two types of bombs used worldwide: conventional warheads and nuclear warheads. Some bombs that are very popular worldwide are given below:

### 15.16.1 Atomic Bombs

Atomic bomb is a fission bomb. It develops its energy from the fission of heavy, unstable nuclei (radioactive element) like uranium and plutonium, which are especially susceptible for fission. This is described in Nuclear Energy chapter. Little Boy and Fat Man were the names of the two atom

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bombs dropped on Hiroshima and Nagasaki by America-based superfast bomber plane B-29 (of these two bombs, one is uranium and the other is plutonium-based).

### 15.16.2 Atomic Bombs

Hydrogen bomb works on the principle of nuclear fusion of hydrogen isotopes. It is highly explosive in comparison to an atom bomb. Fission is required before fusion reaction because higher temperature, in the order of millions of degrees centigrade, is required for the fusion of two hydrogen isotopes. Hydrogen bomb is easy to use in missiles due to its small size.

### 15.16.3 Thermobaric Bomb

US in 2017 dropped a **G-43 Massive Ordnance Air Blast (MOAB) bomb,** which is popularly known as **'Mother of all bombs',** on Afghanistan. It utilises oxygen from the surrounding air to generate an intense high-temperature blast wave that generates incredible amount of energy into a small, localised area. GBU-43 has **8,000 kg of explosives.** Russia also possesses a thermobaric bomb that is popularly known as 'Father of all bombs', which is four times more powerful than the US weapon.

## 15.17 Nuclear Submarine of India – INS Arihant, INS Arighaat and INS Aridhaman

These are submarines powered by a nuclear reactor. They are considerably advantageous, in terms of their performance over 'conventional' (typically diesel-electric) submarines. The nuclear reactor generates enormous amount of power, which enables nuclear submarines to operate at high speed for a longer span of time. Current generations of nuclear submarines need not to be refuelled throughout their life spans of 25 years.

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### *Recent Development*

- **India launched its 4th SSBN (nuclear-powered ballistic submarine),** *INS Aridhaman (S4)*, at the **Ship Building Centre (SBC), Visakhapatnam**.
- Equipped with **K-4 SLBMs (Submarines Launched Ballistic Missiles) (3,500 km range)** via **vertical launch systems**.
- **~75% indigenous content**.

### *INS Aridhaman Highlights*

- **Stretched variant** of INS Arihant.
- Displacement: ~1,000 tons more than Arihant class.
- Enhanced capacity for **more SLBMs**.
- Critical for **India's second-strike capability**.

### *Kalam Missile Series (SLBMs)*

| Missile | Range | Launch Platform |
|---|---|---|
| **K-15** | 750 km | INS Arihant |
| **K-4** | 3,500 km | Arighaat, Aridhaman |
| **K-5** (under development) | > 5,000 km | Future SSBNs |

**WHAT IS AN SSBN?**

- **SSBN = Ship Submersible Ballistic Nuclear.**
- A nuclear-powered submarine carrying **submarine- launched ballistic missiles (SLBMs)** with **nuclear warheads.**
- Functions as a **stealthy nuclear deterrent**; survivable even after a first strike.

### *Strategic Significance*

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- Part of India's **nuclear triad**: Land-based, Air-based, Sea-based nuclear delivery.
- SSBNs provide a **credible second-strike capability.**
- India joins **elite nations (USA, Russia, China, UK, France) operating SSBNs.**

### *Future Outlook*

- India to lease **another Akula-class submarine** from Russia (by 2028).
- Three new **advanced diesel attack submarines** planned at **Mazagon Dockyard** in partnership with **French Naval Group.**
- India's growing underwater nuclear deterrent ensures **strategic stability** in the Indo-Pacific.

India's SSBNs are under the direct control of the PMO and the Strategic Nuclear Command, which oversee their construction, testing and status.

### *India Currently has Three SSBNs Operational*

**INS Arihant** was quietly **commissioned into service in August 2016.** It has a displacement of 6,000 tonnes and is powered by an 83 MW pressurised light-water reactor with enriched uranium.

| Submarine | Name | Missile Type | Missile Range | Status |
|---|---|---|---|---|
| **S1** | INS Chakra (Akula Class – leased from Russia) | Not disclosed | Not disclosed | Returned to Russia in 2021 |
| **S2** | INS Arihant | K-15 SLBM | 750 km | Operational; on patrol |
| **S3** | INS Arighaat | K-4 SLBM | 3,500 km | Commissioned Aug 2024; on patrol |
| **S4** | INS Aridhaman | K-4 SLBM (K-5 under development) | 3,500–5,000+ km | Commissioned in Feb 2025 |

The **second SSBN, INS Arighaat,** which retains the same reactor and dimensions with several technological upgrades, was **commissioned**

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**August 2024.**

The **3rd SSBN Aridhman** is an upgraded Arihant-class submarine, launched in October 2024, and is **commissioned in February 2025.**

***INS Arihant (S2)***

- The first nuclear-powered ballistic missile submarine (SSBN) in India's Arihant class.
- Commissioned in August 2016 and entered active service in 2018.
- Intended as a technology demonstrator.
- Powered by an 83 MW pressurized light-water reactor.
- Can carry K-15 SLBMs with a range of 750 km.
- **SLBM – Submarine Launched Ballistic Missiles**

***INS Arighaat (S3)***

- The second SSBN in the Arihant class, commissioned in August 2024.
- Originally known as INS Aridhaman, but renamed upon launch.
- More operationally focused than INS Arihant.
- Can carry K-15 SLBMs with a range of 750 km and K-4 SLBMs with a range of 3,500 km.

***INS Aridhaman (S4)***

- The fourth SSBN in the Arihant class, launched in October 2024.
- Significantly more advanced than INS Arihant, with increased indigenous content.
- Equipped with K-4 SLBMs with a range of 3,500 km.
- Also carries K-15 SLBMs with a range of 750 km.

***Other Arihant-class Submarines in Future***

- India plans to build four nuclear-powered ballistic missile submarines (SSBNs) under its Advanced Technology Vessel (ATV) program.
- A fifth SSBN, designated S-5, is also planned, which will carry K-5 SLBMs with a range of 5,000 km.

## 15.18 Frigates, Destroyers and Aircraft Carriers

Besides the submarines, ships like destroyers, modified patrol crafts and frigates are also maintained by India. These are capable to launch nuclear-

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capable Ballistic and Cruise Missiles.

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### 15.18.1 Destroyer

The largest type of surface combatant that is currently available is destroyer. They have steadily grown in size (now 5,000 to 10,000 tonnes) and capability. A destroyer, generally, is considered to be a ship with all of the sensors (which includes sophisticated phased array radars), combat systems and weapons required to operate in a high-threat environment.

### 15.18.2 Frigates

A frigate, a medium-sized surface combat (between 2000 and 5000 tonnes) is either suited for a specific role (anti-submarine warfare or anti-air warfare) or has lesser all-round capabilities compared to a destroyer. A frigate is generally the smallest surface combatant that can conduct the extended blue-water missions in environment that involves high-threat.

### 15.18.3 Aircraft Carrier

An aircraft carrier is a warship, serving as a sea-going airbase, equipped with a full-length flight deck. It also facilitates carrying, arming, deploying and recovering the aircraft. They are expensive to build and, hence, are critical assets.

'INS Vikramaditya', 45,400 tonnes, is a modified Kiev-class aircraft carrier. On 20 January 2004, after an extensive upgrade at Sevmash Enterprise, Russia agreed to sell Admiral Gorshkov to India. On 16 November 2013, the carrier, now christened INS Vikramaditya got commissioned into the Indian Navy. In 2017, INS Viraat, which was previously called as the Royal Navy's HMS Hermes, retired from the active service. Hence, India, at present, has only one aircraft carrier.

## 15.19 INS in News

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### 15.19.1 INS Astradharni

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It is the first indigenously built warship by Indian Navy. It is a Torpedo Launch and Recovery Vessel (TLRV) and is a replacement of INS Astravahini, which was decommissioned in July 2015. It was designed by Naval Science and Technological Laboratory (NSTL), IIT Kharagpur and Shoft shipyard. It can easily carry two officers, 27 sailors and 13 scientists.

## 15.20 First Indigenously Built Floating Dock

The Indian Navy launched its first indigenously built Floating Dock (FDN-2) at the shipyard in **Kattupalli**, near Ennor Port, Chennai. This dock is 185-metres long and 40-metres wide. This will enable docking of all kinds of vessels, including naval ships and submarines of up to 8,000 tonnes displacement.

## 15.21 Different Generation of Aircraft

**First-Generation Aircrafts (mid-1940s to mid-1950s)** E.g.: Messerschmitt Me 262, MiG-15 and MiG-17.

These have subsonic speeds, straight wings, lack of radar and machine guns as primary weapon and unguided bombs.

**Second-Generation Aircrafts (mid-1950s to early 1960s)** E.g.: MiG-21, Su-9 and F-106, F-104 star fighter, F-5, MiG-19 and Sukhoi-7, etc.

They had turbojet engines, are equipped with radars and utilise air-to-air missiles as the primary weapon. They are enabled to reach and sustain supersonic speeds in level flight by the-aerodynamic engine. Nuclear warfare environment distinguishes jet fighters into two categories: Interceptors (MiG-21, SU-9, F-106) that are specialised in preventive missions against enemy bombers and aerial reconnaissance flights, which were fighter-bombers (F-105, SU-7) providing air superiority and ground attack.

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**Third-Generation Aircrafts** E.g.: Mirage F1, MiG-23, MiG-25, F-4 and F-5 are the third-generation aircrafts that are equipped with advanced avionics and aerodynamic performance. Moreover, they also include air guided missiles, air-to-air missiles and radar systems. However, due to poor accuracy of weapons and electronic countermeasure (device, which are designed to trick detection system), it was not easy to win an air fight.

**Fourth-Generation Aircrafts** E.g.: MiG-29, MiG-27, Mirage-2000, Su-27, F-16 and F-18.

Since the movements of flight controls are converted to electronic signals, transmitted by wires, they are termed as fly-by-wire. The 'fly-by-wire system' also enables automatic signals, sent by the aircraft's computers for operating functions without the pilot's input, just like systems that automatically help in stabilising the aircraft, or preventing unsafe operation of the aircraft outside its performance envelope. 'Thrust vectoring technology' was introduced to enhance a fighter's turning ability further, as introduced in Soviet fighters.

**5$^{th}$ Generation Fighter Planes (late 1990s and into the 2005)** It is equipped with Active Electronically Scanned Array (AESA) radar system. Radar absorbent materials, thrust vector controlled engines, greater weapons carriage capacity (Supercruise) are some significant features of this class. Addition of stealth technology is also possible. E.g.: Eurofighter Typhoon, Saab JAS 39 Gripen and Dassault Rafale (Medium Multi-Role Combat Aircraft, MMRCA). This category of fighter aircrafts has a greater scope to conduct multi-role missions.

**Fifth-Generation Jet Fighter** This jet fighter is the most advanced jet fighter generation, as of 2015. They have all-aspect stealth technology, Low Probability of Intercept Radar (LPIR), high-performance airframes, advanced avionics features and highly integrated computer systems that can network with other elements within the battle space for situation awareness.

F-22 Raptor is the only current combat-ready fifth-generation fighter. It was introduced to service with the United States Air Force in 2005. The Lockheed Martin American F-35 Lightning II, Sukhoi PAK FA, Chengdu J-

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20 and Shenyang J-31 are currently undergoing various stages of testing and development.

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**WHY WAS RAFALE IN NEWS?**

The issue with Rafale is the per-unit cost. As per Brazilian media, Brazil is also buying 36 Rafales at the cost of 4.2 billion USD and an additional 4 billion USD for short-period maintenance with transfer of technology. So, one Rafale would roughly cost $209 million without maintenance. Brazil had doubts about the quality of the AESA radar, which enables the aircraft to switch quickly from air-to-air to ground mode in flight and about the helmet-mounted heads-up display. In Indian context, the deal was signed in 2009 for 126 Rafale at 10 billion USD and now the cost is increased to 22 billion USD, which roughly means 238 million USD per aircraft, compared to the 209 million USD per aircraft quoted for Brazil. Terminating the deal will now be disruptive and somehow, international politics are also a reason for creating such a scenario; hence, India also decreased its order from 126 Rafale to 36 Rafale.

## 15.22 Active Indian Military Aircraft

**Table 15.4**

| Aircraft | Origin | Type | Variant |
|---|---|---|---|
| Dassault Rafale | France | Multirole | DH |
| | | | EH |
| Dassault Mirage | France | Multirole | 2000H |
| | | | 2000TH |
| HAL Tejas | India | Multirole | Mk.1 |
| | | Conversion trainer | Mk.1 Trainer |
| | | Multirole | Mk.1A |
| Mikoyan-Gurevich MiG-21 | Soviet Union | Interceptor | MiG-21 Bison |
| Mikoyan MiG-29 | Soviet Union | Fighter | MiG-29 Fulcrum |
| | | | MiG-29 Fulcrum B |

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| | | | |
|---|---|---|---|
| SEPECAT Jaguar | United Kingdom | Attack | IB |

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## 15.23 Transport Aircraft of IAF (Indian Air Force)

***Transport Aircraft of IAF (Indian Air Force)***

The Indian Air Force (IAF) utilises a diverse fleet of transport aircraft, encompassing strategic airlifters and tactical airlifters. These aircraft play a crucial role in military operations, personnel and cargo movement and humanitarian assistance.

***Key Transport Aircraft in the IAF's Fleet***

- **Strategic Airlifters:**
  - **C-17 Globemaster III:** A large, long-range aircraft used for strategic airlift, capable of transporting heavy equipment and large personnel.
  - **Ilyushin Il-76:** A heavy transport aircraft, also capable of strategic airlift operations.
  - **Airbus A400M Atlas:** A medium-to-heavy tactical airlifter with high capacity for troop and equipment transport.
- **Tactical Airlifters:**
  - **C-130J Super Hercules:** A versatile tactical airlifter capable of operating in challenging environments, often used for special operations and humanitarian assistance.
  - **Antonov An-32:** A medium-range transport aircraft, used for various missions, including cargo and personnel transportation.
  - **Dornier 228:** A versatile turboprop aircraft, primarily used by the Indian Coast Guard for maritime patrol and surveillance.
  - **Embraer C-295:** A medium-to-heavy tactical airlifter, being inducted to replace older fleets.
- **Other Transport Aircraft:**
  - **Mil Mi-17:** A workhorse helicopter used for transport, VIP movement and other missions.

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- ❑ **HAL Dhruv:** A multi-role helicopter developed by HAL, used for various tasks including transport and search and rescue.
- ❑ **Ilyushin Il-78:** A mid-air refuelling tanker, supporting the IAF's fighter aircraft.

- **VIP Transport:**
  - ❑ **Boeing 737 BBJ (Business Jet):** Modified Boeing 737 aircraft used for transporting VIPs, including the President and Prime Minister.

## 15.24 Helicopters of India

**Table 15.5** Airforce Helicopters

| Aircraft | Origin | Type | Variant |
|---|---|---|---|
| Mil Mi-17 | Russia | Utility | Mi-17V-5 |
| Mil Mi-24 | Russia | Attack | Mi-24/25/35 |
| HAL Dhruv | India | Utility | |
| HAL Rudra | India | Attack | |
| Alouette III | France | Liaison | Chetak |
| SA 315B Lama | India | Utility | Cheetah |
| Boeing AH-64 | United States | Attack | AH-64E |
| CH-47 Chinook | United States | Transport | CH-47F |

**Table 15.6** Army Helicopter

| Helicopter | Origin | Type | Variant |
|---|---|---|---|
| Alouette III | France/India | liaison | Chetak |
| Boeing AH-64 | United States | attack | AH-64E |
| CH-47 Chinook | United States | transport | CH-47F |
| HAL Prachand | India | attack | |
| HAL Dhruv | India | utility | |
| HAL Rudra | India | attack | |

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| Helicopter | Origin | Type | Variant |
|---|---|---|---|
| HAL Light Utility Helicopter | India | utility | |
| Mil Mi-17 | Russia | utility | Mi-17V-5 |
| Mil Mi-24 | Russia | attack | Mi-24/25/35 |

**Table 15.7** Navy Helicopters

| Helicopter | Origin | Type | Variant |
|---|---|---|---|
| HAL Dhruv | India | utility | |
| Kamov Ka-31 | Russia | AEW | |
| Kamov Ka-27 | Russia | ASW | Ka-28 |
| SH-60 Seahawk | United States | ASW / SAR | MH-60R |
| Westland Sea King | United Kingdom | SAR / utility | Mk.42B/C |
| HAL Chetak | France/India | liaison / utility | |

Note: AEW: Air-borne Early Warning helicopter, ASW: Anti-Submarine Warfare, SAR: Search and Rescue

## 15.25 Biological Weapons Convention (BWC)

As a supplement of the 1925 Geneva Protocol, a multilateral disarmament treaty (of indefinite duration that is open to any country) that bans the production of an entire category of weapons for establishing a new instrument against biological weapons, is in the form of BWC. This Convention completely prohibits the development, production and stockpiling of biological weapons. **As per Geneva Protocol, possession or development of chemical and biological weapons is not prohibited but only the use is prohibited**.

The British submitted this draft of the BWC that was opened for signature on 10 April 1972 and entered into force on 26 March 1975. As of

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2018, it commits total 180 party states, as legally binding.

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However, without any formal verification regime to monitor compliance, it meets limited effectiveness.

The BWC defines permitted purposes of biological activities like prophylactic, protective and other similar peaceful purposes.

The Biological Weapons Convention (BWC) has been ratified by India. In October 2002, reiterating the latter point, the then President Dr. A. P. J. Abdul Kalam asserted that 'India will not make biological weapons. It is cruel to human beings'.

## 15.26 Chemical Weapons Convention (CWC)

The Chemical Weapons Convention (CWC) represents a multilateral treaty to ban chemical weapons in no fixed duration. It is more comprehensive as compared to the 1925 Geneva Protocol that outlaws the use of chemical weapons; however, their possession is not outlawed. CWC negotiations started in 1980 in the UN Conference on Disarmament and were opened for signature on 13 January 1993 and it entered into force on 29 April 1997.

Chemical Weapons Convention is implemented by the Organisation for the Prohibition of Chemical Weapons (OPCW). The OPCW receives detailed declarations of chemical weapons-related activities or materials and its relevant industrial activities from state parties, which will be inspected and monitored by the convention. As of 2018, over 96 per cent of the world has declared to destroy their chemical weapons stockpiles. All nations can be party of CWC; it now has 193 states-parties as its members.

### 15.26.1 Prohibitions Under CWC

These are the following prohibitions under CWC:

- Development, production of chemical weapons and acquiring, stockpiling or retaining them.
- Direct or indirect transfer of chemical weapons.
- Their use for military purposes.

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- Assisting, encouraging or inducing other states to follow CWC-prohibited activities.

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In 1992, CWC was signed by India, denying possession of chemical weapons and the capacity or capability to manufacture chemical weapons. This made India as one of the original signatories of the CWC in 1993, and it ratified the same on 2 September 1996.

### 15.26.2 Organisation for the Prohibition of Chemical Weapons (OPCW)

The OPCW is a self-governing, autonomous international organisation that has a working association with the United Nations. The organisation is the **Chemical Weapons Convention (CWC)** executing body, which came into force in the year 1997.

The organisation was honoured with the **2013 Nobel Peace Prize** 'for its wide-ranging efforts in eliminating chemical weapons'.

The OPCW member states share the joint goal of avoiding usage of chemical weapons ever yet again being used in warfare, thus, strengthening the international security. To conclude this, the Convention contains **four key requirements:**

- Demolishing all present chemical weapons under globally verified organisations by the OPCW.
- Supervising chemical industry for preventing fresh weapons from re-emergence.
- Giving aid and security to state parties against chemical threats.
- Promoting worldwide collaboration for strengthening the implementation of the Convention and promoting the peaceful usage of chemistry.

## 15.27 Indian Tanks

A tank is a large, heavily armoured fighting vehicle with tracks and a large tank gun, that is designed for front-line combat. Modern tanks are a mobile land-weapon platform, mounting a large-caliber cannon in a rotating gun

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target. Tanks are important for supporting infantry. Tanks used/being used by the Indian Army (**Table 15.8**) are as follows:

**Table 15.8** List of Different Tanks

| | |
|---|---|
| Arjun MBT | India |
| T-90S 'Bhishma' and T-90M | Russia and India |
| T-72, Ajeya Combat and Improved Ajeya | Soviet Union, Poland and India |

*Source:* DRDO, (GOI Sites)

1. **AMX-13:** It is a French Light tank, produced from 1953 to 1985, was used by 25 nations, including India. Indian Army has used this tank during the war in 1965 with Pakistan. At that time, Pakistan's troops were using M-48 Pattons, which were far superior than AMX-13.
2. **Vijayanta:** It was the main battle tank, built in India based on a licensed design of the Vickers NK-I. It was the first indigenous tank of the Indian Army and was in service during 1965–2008. It has now been supplanted by the T-72 MI in Indian service.
3. **T-72:** It is a Soviet-designed main battle tank. It has two variants: T-72 N and T-72 MI. Its indigenous name is Ajeya Tank and has been used by the Indian Army from late 1990s to early 2000s.
4. **T-90:** Its indigenous name is Bhishma. In 2001, around 310 T-90S tanks were imported from Russia. It had features almost similar to T-72, with some improvements. Indian scientists are currently producing a customised and improved version of T-90 – the T-90 M Bhishma. As of now, a grand total of 500 T-90 and T-90 M tanks are being operated by the Indian Army.
5. **Tank Ex or NBT Ex:** It is India's ambitious defence project to build a battle tank after the MBT Arjun. DRDO is working on this since 2002. It will be an improved version of T-90 MI and its name would be **'Karna'.** It can withstand a nuclear, biological and chemical (NBC) attack and would be more superior and much lighter than the Arjun NBT.

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**MUNTRA (INDIA'S FIRST UNMANNED TANK)**

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It is the first unmanned, remotely piloted tank of India that has been developed by the DRDO for various surveillance missions, mine detection and reconnaissance in the areas with nuclear and biological threats. It can be used in Naxal-hit area as well. This is equipped with an integrated camera along with laser range-finder, which can spy on a ground target that is 15 km away. There are different types of Muntra tank for various levels of surveillance:

Muntra S: For surveillance mission,

Muntra M: For detecting mines and

Muntra N: For operation in areas where there is risk of nuclear or bio-weapons.

6. **Arjun Tank MK-2:** The Arjun is a third-generation main battle tank, developed by DRDO for the Indian Army. It is powered by a single MTU multi-fuel diesel engine and can achieve a maximum speed of 67 km per hour and a cross-country speed of 40 km per hour. Arjun MK-I entered service with the Indian Army in 2004, but failed to impress the army and improvements were required in order to make it combat-ready. The new Arjun MK-2 is the improved version, made available in 2012 and would now be replacing T-55 and T-72 M.
7. **Arjun Tank:** The **Defence Research and Development Organisation (DRDO)** has unveiled the prototype of **Zorawar**, India's **first indigenous light tank**, now set for **extensive trials**.
   ***Jointly developed by:*** DRDO and **Larsen & Toubro (L&T)** in collaboration with several **Micro, Small and Medium Enterprises (MSMEs)** under the **Make in India** initiative.
8. **Zorawar Tank:**
   - **Need Identified:** Post the **2020 standoff with China in Eastern Ladakh**, the Army recognised the **need for agile, lightweight tanks** that can operate in **high-altitude and rugged terrain**.
   - **Mobility Focus:** Light tanks offer **rapid deployment**, ideal for regions with **difficult terrain** like the Himalayas.

| Feature | Details |
|---|---|
| **Weight** | Less than **25 tonnes** |
| **Transportability** | **Air-transportable** for quick deployment (capable of being transported by air : |

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| | |
|---|---|
| | designed to be transported by air) |
| **Firing Capability** | Can **fire at high angles**, suitable for **limited artillery roles** |
| **Mobility** | Designed for **tactical & operational flexibility** in constrained terrains |

## 15.28 Unmanned Aerial Vehicles (UAV)

UAV is a pilotless targeted aircraft, i.e., Remotely Piloted Vehicle (RPV), which typically falls into various functional categories (although multi-role airframe platforms have been becoming more prevalent):

- **Target and Decoy:** Providing ground and aerial gunnery a target, stimulating an enemy aircraft or missile.
- **Reconnaissance:** To provide battlefield intelligence.
- **Combat:** To provide attack capability for high-risk missions.
- **Logistics:** In delivering cargo.
- **Research and Development:** Advancement of UAV technologies.
- **Civil and Commercial UAVs:** In the field of agriculture, aerial photography and data collection.
- There are many unmanned aerial vehicles, developed by DRDO, namely, Nishant, Lakshya, Rustom, etc.
  - ❑ ***Lakshya:*** Efforts are being made to develop PTA (Pilotless Target Aircraft) further, with an improved, all-digital flight control system and a better turbojet engine.

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- ***Nishant:*** Serves as a hydraulically launched short-ranged UAV, serving the tactical battle area. Currently, the Indian Navy and the Indian Paramilitary forces are evaluating it.
- ***Panchi:*** It is wheeled version of UAV Nishant and has been designed and developed by ADE.
- ***Dakshya:*** It is a land-based drone used for bomb disposal developed by DRDO.

The DRDO is also moving ahead with its plans for the development of a new class of UAVs. They are commonly referred with HALE (High Altitude Long Endurance) and MALE (Medium Altitude Long Endurance) designations. The MALE UAV has been tentatively named as the Rustom. India has developed many types of drones: Abhyas, AURA, Fluffy, Imperial Eagle, Kapothaka, Lakshya, Netra, Nishant, UAV, Rustom and Ulka.

**UAVS FEATURES**

**Different launching modes of UAVs:**

1. **Vertical Takeoff and Landing (VTOL) UAVs:** These UAVs are designed to take off and land vertically, similar to helicopters, which eliminates the need for runways or large takeoff areas.
2. **Horizontal Flight Capability:** Many VTOL UAVs also have fixed wings and can transition to horizontal flight once airborne, allowing them to travel long distances efficiently, similar to fixed-wing aircraft.
3. **Hybrid Designs:** VTOL UAVs often combine the best features of both vertical flight and horizontal flight, providing flexibility and adaptability for various missions.

**Automated Hovering vs. Horizontal Landing**

1. **Automated Hovering:** Rotary-wing UAVs are designed for vertical lift and can hover in place using their rotors. This allows them to pause in flight, which is useful for various tasks like surveillance, inspection or search and rescue.
2. **Horizontal Landing:** Many UAVs are designed for horizontal takeoffs and landings, meaning they land on a flat surface, like a runway or a ship deck.

Note: UAVs are not limited to only **battery power**. Other power sources like gasoline engines, hybrid systems (combining batteries with other sources), fuel cells and solar power are also used.

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| Type | Class | Role | Notes |
|---|---|---|---|
| Harpy | Israel | Attack | A small UCAV capable to home into enemy radar emissions – thereby, destroying both itself and the enemy radar. In service with the IAF. |
| Harop | Israel | UAV | A small UCAV capable to home into enemy radar emissions – thereby, destroying both itself and the enemy radar. In service with the IAF. |
| Heron | Israel | | In service with the IAF and Indian Navy. 15 more were ordered in December 2013. |
| Searcher | Israel | | It is in service with Indian Army. |
| Nishant | India | | It is in service with Indian Army. |
| Lakshya | India | Aerial target | A small target drone, which is in |

## 15.29 Drone

The formal name of drones is Unmanned Aerial Vehicles (UAV). Essentially, a drone is a flying robot. The aircraft may be controlled remotely or can fly autonomously with the help of software-controlled flight plans in their embedded systems that work in conjunction with GPS.

The first city to have a margarita pizza delivered via a drone was Mumbai, in May 2015. As per current regulations, the use of drones commercially is still illegal in India. Drones are becoming a serious issue in India, both in terms of commercial and military purposes.

### 15.29.1 India's Military Drones

During the 1999 Kargil War with Pakistan, India first used military drones. Heron and Searcher drones were discreetly supplied to Indian Air Force by Israel. They proved to be quite useful in acquiring target information along

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the Line of Control. India has procured a number of Israeli military unmanned aircrafts since Kargil War. Currently, India's arsenal includes the Israel Aerospace Industry's Harpy and Harop, unmanned combat aerial vehicles, along with Searcher and Heron, unmanned aerial vehicles.

DRDO has also developed its own domestic UAV programme. The project is aimed at developing a domestic arsenal that can replace and augment the existing fleet of IAI vehicles (Israel Aerospace Industry). There are few completed and pending DRDO project as drone of India like Lakshya, Nishant, Aura and Rustom.

### 15.29.2 Civilian Drones in India

Drones have been used to assist in services ranging from disaster relief, aerial photography and security and surveillance. Many civilian drones are developed – Airpix, Garuda Robotics, Edall systems, etc.

**Social Drones:** A start-up involved in 'designing and manufacturing user-friendly, high-performance drones that can be used easily for social and unconventional applications'. During disaster-relief operation in Uttarakhand, after severe flooding in 2013, the Social Drones were used and hence, made news. Drones were used for the areas that were deemed unsafe for conventional relief methods and also for the areas where relief efforts have been stalled, the company's drones were used to airdrop first-aid kits.

### 15.29.3 Multiple Uses of Civilian Drones

Advancements in fields such as automation, robotics, miniaturisation, materials science, spectral and thermal imaging and light detection and ranging have resulted in drone-enabled solutions in areas as diverse as the **agriculture, power, infrastructure and telecom sectors, as well as crowd and disaster management.** UAV activity will impact proprietary interests because common law has not clearly demarcated the commons from owned airspaces. It will also raise huge privacy concerns, considering the potential

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deployment of drones for massive data capture and analytics. No clear guidance exists on the liability standards for midair collisions and injury to property or persons in the event of untoward incidents.

This may seem like an imaginary scenario now, but with the rapid pace in the development of drone and its software, drone spraying and spreading is not inconceivable in the near future. Manufacturers are pushing ahead with the developments to make drone spraying and spreading a practically reality. Drones can capture images of erupting craters. The main factor is the **safe accessibility of dangerous terrain.** Researchers from the German Research Centre for Geosciences, known by the German abbreviation GFZ, **revealed amazingly detailed data from an active volcano using drones with visual and thermal-imaging cameras.** Scientists are using drones to sample whale breath and snot.

### 15.29.4 Updates on Drones

The **Ministry of Civil Aviation (MoCA)** has sent a note on use of drones across various sectors to different ministries in the Centre.

***Where can drones be effectively utilised?***

- **Ministry of Home Affairs:** For surveillance, situational analysis, crime control, VVIP security, disaster management, etc.
- **Ministry of Defence:** Drones for combat, communication in remote areas, counter-drone solutions, etc.
- **Ministry of Health and Family Welfare:** Delivery of medicines, collection of samples from remote or epidemic/pandemic-affected areas.
- **The Petroleum and Natural Gas, and Power Ministries:** For real-time surveillance of assets and transmission lines, theft prevention, -visual inspection/maintenance, construction planning and management, etc.
- **Environment, Forests and Climate Change Ministry:** Anti-poaching actions, monitoring of forests and wildlife, pollution assessment, and evidence gathering.
- **Ministry of Information and Broadcasting:** For high-quality videography of events and difficult to reach places at a fraction of the

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cost and approvals required. This move would also facilitate low-altitude shooting without noise, and prevent dust pollution and risk of accidents.

- **Other Areas:** To undertake disaster management, incidence response, inspection/maintenance works and project monitoring.

### *Drone Rules, 2021*

- As on 31 December 2021, nine **remote pilot training organisations** have been set up by entities under government or private ownership.
- As per **Drone Rules, 2021,** any person who intends to obtain the authorisation to establish **a Remote Pilot Training Organisation (RPTO)** shall submit **an application to the Director General of Civil Aviation** in Form D5 on **the Digital Sky Platform,** along with the specified fees.

### *New Drone Rules*

- **Digital sky platform** shall be developed as a business-friendly single-window online system.
- **No flight permission required upto** 400 feet in green zones and up to 200 feet in the area between 8 and 12 km from the airport perimeter.
- **No pilot licence required for** micro drones (for non-commercial use), nano drones and for R&D organisations.
- **No restriction on drone operations by foreign-owned companies registered in India.**
- Import of drones and drone components to be regulated by DGFT.
- **No security clearance required** before any registration or licence issuance.
- **No requirement of certificate of airworthiness,** unique identification number, prior permission and remote pilot licence for R&D entities.
- **Coverage of drones under Drone Rules, 2021 increased from 300 kg to 500 kg.** This will cover drone taxis too.
- **Issuance of Certificate of Airworthiness** delegated to Quality Council of India and certification entities authorised by it.
- **Manufacturer may generate their drone's unique identification number** on the digital sky platform through the self-certification route.
- **Maximum penalty** under Drone Rules, 2021 reduced to `1 lakh. This shall, however, not apply to penalties in respect of violation of other

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laws.

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- **Drone corridors** will be developed for cargo deliveries.
- **Drone promotion council** to be set up to facilitate a business-friendly regulatory regime.

***Drone Categories in India***

- **Nano:** Less than or equal to 250 grams
- **Micro:** From 250 grams to 2 kg
- **Small:** From 2 kg to 25 kg
- **Medium:** From 25 kg to 150 kg
- **Large:** Greater than 150 kg

## 15.30 BHARGAVASTRA

India has successfully tested **Bhargavastra**, the **first indigenous micro-missile system** specifically designed to neutralise **swarm drone threats**.

***About Bhargavastra***

- ***Developed by:*** *Economic Explosives Ltd.*
- ***Purpose:*** Designed for **Army Air Defence** to counter **drone swarms** using micro missiles.
- ***Deployment:*** Can be rapidly deployed from **mobile platforms**.
- ***Range:*** Targets can be engaged from **over 2.5 km** away.

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- ***Detection Capability:*** Detects **small UAVs/drones** from **6 km**.
- ***Firepower:*** Can launch **64+ micro missiles simultaneously**.
- ***Terrain Flexibility:*** Functions in **all terrains**, including **high-altitude regions**.

### *What are Swarm Drones: Definition*

- **SWARM:** *Smart War-Fighting Array of Reconfigured Modules*
- Swarm drones involve the **coordination of multiple drones** (from 3 to thousands) to execute missions **cooperatively** with **minimal human intervention**.

### *Key Highlights*

| Feature | Description |
|---|---|
| **Autonomous Collaboration** | Drones work together via **local sensing**, **AI algorithms** and **communication protocols** |
| **Redundancy & Resilience** | If one drone fails, others **adapt and continue** the mission |
| **Versatile Missions** | Used for **military ops**, **disaster response**, **wildfire control**, etc. |
| **Control Methods** | Can operate via: |

- Pre-programmed paths
- Centralised control
- Distributed/self-coordinated swarm control

### *Significance of Bhargavastra*

- Represents a **major leap** in India's **anti-drone defence** capabilities.
- Enhances India's preparedness against **unconventional aerial threats**, especially **swarm drone attacks** used in modern warfare.

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- Reinforces the drive towards **indigenous defence technology** under *Aatmanirbhar Bharat.*

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## 15.31 AWACS Systems (Airborne Warning and Control System) or AEW and C (Airborne Early Warning and Control)

An AEW&C system is an airborne RADAR system, which is designed to detect aircrafts, ships and vehicles that are at long ranges and perform command and control of the battlespace during an air engagement by directing fighter as well as attacking aircraft strikes.

### *Phalcon*

Phalcon, an Airborne Early Warning, Command and Control (AEWC&C) system, was manufactured by Israel. It is one of the most powerful systems of its kind in the world which provides real-time surveillance for a few hundred kilometres of territory and also renders services in command and control. It can pick up or track a low-flying aircraft, a missile and hence, provide advance warning after correlation. It will assist the Indian Air Force to maintain air superiority during a battle. It can pick up an incoming object in any weather condition, reportedly up to some 300 miles. On top of this, it can pick up all sorts of communications, be it from air, sea or land. All the information, so collected, can be easily conveyed live to the control centre of the headquarters.

This system has four sensors for coordinating with each other. If any sensor picks up an incoming object or enemy communication, the other sensors are automatically co-related and reconfirm it.

### *How is Phalcon Different from Other Airborne Early Warning Systems (AEWS)?*

Most of the other AEWS are mechanically rotating antennas, or rotodomes that are mounted on an aircraft. On the other hand, Phalcon is with a stationary dome that is mounted on the aircraft and is based on an Active Phased Array Electronic Scanning Technology. It would be capable of

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continuously tracking any fast, high-manoeuvring objects in a lesser time period as compared to the rotodome-based AWACS. Being a full-fledged command and control centre, it can be deployed mid-air for directing the aircraft and providing them a picture of the whole battlefield. This feature enables them superiority over enemy fighters. Pakistan and China do not have Phalcon.

## 15.32 NETRA: A Surveillance Technology of India

In 2017, India has tested an air-to-air refuelling of the Embraer transport aircraft, which is equipped with NETRA. NETRA is the first indigenously developed, Airborne Early Warning and Control System (AEW&C), developed by DRDO. AEW&C is also popular with the name of eye-in-the-sky, due to its capability of long-range surveillance. The range of NETRA is 200 km, to detect aerial threats from incoming aircrafts and missiles with 240-degrees coverage. It can scan the area on both sides of the aircraft. For its multiple features, it is equipped with state-of-the-art active electronically scanned radar and secondary surveillance radar. It will use in electronic and communication counter measures and connect with line of sight and beyond line of sight data link. It is also equipped with a voice communication system and self-protection suit. Now, India stands at the fourth position in the world after the United States, Russia and Israel that have such technologies of their own.

Note: 1. NETRA Mark-2 is to be equipped with DRDO's Uttam, advanced AESA radar, having enhanced detection capabilities of more than 500 km.

2. Currently, Indian Air Force is using three Israeli Phalcon AWACS (Airborne Warning and Control System) that have a range of 400 km and 360-degree coverage.

## 15.33 Stealth Technology

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Stealth technology represents a sub-discipline of military tactics and passive electronic counter-measures. It, thus, covers a range of techniques used by personnel, aircrafts, ships, submarines and missiles, so as to make them less visible (ideally invisible) to radar, infrared, sonar and other methods of detection. It is also known as LO technology (Low Observable technology). Two methods adopted for low observation are:

### 15.33.1 Non-Metallic Airframe

Dielectric composites show more transparency to radar. On the other hand, electrically conductive materials (like metals and carbon fibres) show reflection of electromagnetic energy incident on the material's surface. Composites may also contain ferrites in order to optimise the dielectric and magnetic properties of the material that may enhance its application.

### 15.33.2 Radar-Absorbing Material

Radar Absorbent Material (RAM), such as paints, is especially used on the edges of metal surfaces. While the material as well as thickness of RAM coatings is classified, the material employs absorption of radiated energy from some ground or air-based radar station into the coating, and then converting it to heat instead of reflecting it back.

## 15.34 Anti-Satellite Weapon

India had identified development of ASAT weapons, under the DRDO management, for destroying satellites in both LEO (2,000-km altitude above earth's surface) and the higher geosynchronous orbit, electronically as well as physically, as a thrust area in the DRDO's longterm integrated perspective plan (2012–2027).

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## 15.35 Interceptor

An interceptor aircraft or simply an interceptor represents a type of fighter aircraft, which is specifically designed for intercepting and destroying enemy aircraft, particularly the bombers, which usually rely on great speed. Many such aircrafts were built in the period that started just prior to the World War II and ended in the late 1960s, since they became less important due to the shift in the strategic bombing role to Inter Continental Ballistic Missiles (ICBMs).

## 15.36 SONARS (Sound Navigation and Ranging)

SONAR systems are used to detect an object under water with the help of ultrasonic sound waves. A range of Sonar systems, to serve the Indian Navy, have been developed by the DRDO, BEL (Bharat Electronics Limited), as well as the Indian Navy.

- **APSOH** (Advanced Panoramic Sonar Hull mounted)
- **HUMVAD** (Hull-Mounted Variable Depth)
- **HUMSA** (Follow on of the APSOH series; the acronym HUMSA represents Hull-Mounted Sonar Array)
- **NAGAN** (Towed Array Sonar)
- **PANCHENDRIYA** (Submarine sonar and fire control system)

Some other sonars like 'Mihir', which is an airborne sonar, are in trial phase, while work is being proceeded apace on a new generation of sonars. Since the Indian Navy's most powerful ships currently rely on DRDO-made sonars, they may be considered as one of the most successful achievements of the DRDO. The standard fit for the frontline naval ship would include the HUM-SA-NG hull-mounted sonar, as well as the Na-gan towed array sonar. Mihir, a dunking sonar, is meant to be used by the Naval ALH (Advanced Light Helicopter). The Panchendriya, which is currently in production, is for the Kilo-class submarine upgrades.

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## 15.37 Torpedoes

TORPEDO is an **underwater missile system,** which is self-propelled. It is designed with an objective to be fired from a ship or a submarine. It can also be dropped into the water from an aircraft and can explode on reaching a target. The DRDO is currently engaged in the development of multiple torpedo designs, which include a lightweight torpedo. This lightweight torpedo has been accepted by the Navy and got clearance for production. The heavy weight wire-guided torpedoes are named **Varunastra** and **Thakshak.** The electrically powered Varunastra, reportedly, is now also in production. **Shyena** is an experimental advanced torpedo. It is developed by the Naval Scientific and Technological Laboratory, that is, India's DRDO wing.

## 15.38 Electronic Warfare System

Electronic Warfare (EW) systems use electromagnetic spectrum or directed energy for attacking the enemy or impeding enemy assaults via the spectrum. EW can target humans, communications, radar or other assets. It can be applied from air, sea, land and space via manned and unmanned systems. **'Samyukta'** is a mobile-integrated electronic system of India. DRDO, Bharat Electronics Limited, Electronics Corporation of India Limited and Corps of Signals of Indian Army have jointly developed it.

It is not limited to radio or a radar frequency, as it also includes IR, visible, ultraviolet and other portions of the EM spectrum, which are less used. Some other major features are self-protection, standoff, and escort jamming, and anti-radiation attacks. EW is a specialised tool, enhancing many air and space functions at several levels of conflict. Electronic warfare systems have following major subdivisions:

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**Electronic Attack (EA),** for attacking personnel. It is equipped for degrading, neutralising, or even destroying enemy combat capability, inclusive of human life.

**Electronic Protection (EP)** includes actions that are taken for protection of personnel, facilities and equipment from any effects of friendly or enemy application of the electromagnetic spectrum, which degrade, neutralise, or even destroy friendly combat capability. Notably, Jamming is not a part of EP; instead, it is an EA measure.

**Electronic Warfare Support (ES)** includes actions tasked by or under direct control of an operational commander for searching, intercepting, identifying and locating sources of intentional and unintentionally radiated electromagnetic (EM) waves. Its major objectives are immediate threat recognition, targeting, planning, as well as conduct of future operations.

## 15.39 NTRO (National Technical Research Organisation) Under Intelligence Act

The NTRO was created after the 1999 Kargil conflict, as a dedicated **'technical intelligence agency'.** It was finally constituted in 2004. It reports to the Prime Minister's Office (PMO) and the National Security Advisor (NSA). It functions under the National Security Advisor. It also includes National Institute of Cryptology Research and Development.

The Home Ministry issued a notification listing National Technical Research Organisation (NTRO) under the Intelligence Organisations (Restriction of Rights) Act, 1985.

Note: Intelligence Organisations (Restriction of Right) Act, 1985, aims to prevent leakage of information by intelligence agencies.

Various intelligence agencies, setup for the safety and security of India from various external threats, are Research and Analysis Wing (RAW), Intelligence Bureau, Directorate of Revenue Intelligence, Narcotics Control Bureau, etc.

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# 15.40 RADAR (Radio Detection and Ranging)

RADAR utilises **radio waves** similar to those in wireless networks and mobile phones. It transmits electromagnetic waves as short pulses that can be partially reflected back by objects in their path. When these pulses hit precipitation, some of the energy scatters back to the RADAR. Low frequency radio-waves can be used to detect narcotics and explosives.

It typically operates within the microwave region of the electromagnetic spectrum, between 400 MHz and 40 GHz, though it can also function at lower frequencies and in optical and infrared ranges for long-range applications.

RADAR is employed to track and identify satellites and spacecraft, ensuring safe docking and landing. Satellites equipped with RADAR are also used for remote sensing and identification.

## Application and use of Radar in Various Disciplines

### *Remote Sensing*

RADAR detects various wavelengths to **monitor environmental conditions** and track changes in planets, asteroids, and other celestial bodies.

### *Aircraft Navigation*

Planes use ground-mapping RADAR and collision-avoidance RADAR to navigate safely, ensuring the area is clear of obstacles that could endanger the flight.

### *Ship Navigation*

Ships use RADAR systems set up along coasts to navigate safely, detect hazards, and measure the distance and speed of their journey through the sea, even in adverse weather conditions.

### *Air Traffic Control*

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RADAR helps manage air traffic, especially in poor weather, guiding planes for safe landings and takeoffs. It also detects planes' approach and ascent.

#### *Law Enforcement*

Law enforcement agencies, particularly highway police, use RADAR to **measure vehicle speed during pursuits**. RADAR is also used when satellites cannot provide clear traffic views due to environmental conditions.

#### *Military*

RADAR is extensively used in military operations for surveillance, tracking, and target acquisition. It is crucial in weapon control and missile guidance systems.

RADAR is utilised for detecting, locating, tracking, and viewing objects over long distances by directing electromagnetic radiation at targets and analysing the reflected signals. It can target planes, ships, vehicles, celestial objects, migratory birds, insects, and water. RADAR is widely used across various industries, including aviation, meteorology, military, navigation, law enforcement, agriculture, and space exploration. It can also determine the size, shape, position, area, and speed of these objects.

## List of Indian Military Radar

#### *Airforce*

**LCA MMR installed on:** HAL Tejas Mk1

**Netra installed on:** ERJ 145 aircraft

#### *Navy*

**XV-2000 3D:** for Dornier 228 maritime patrol aircraft

**Revathi 3D:** Shivalik-class frigates

#### *Army*

**Swordfish:** Long-range tracking radar for Ballistic missile

**Arudhra:** Medium power radar for airspace surveillance upto 300 km

**ADTCR:** Medium power radar for airspace surveillance

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**Ashwini:** For Low level air targets up to ranges of 200 km
**INDRA:** Medium range airspace surveillance radars
**Rajendra:** Medium range radar for Akash SAM
**Bharani:** Low level aircraft tracking radar
**Swathi:** Weapon Locating Radar

## 15.41 Recent News

### 15.41.1 Cluster Munitions

Cluster munitions are weapons that can be delivered by rockets, missiles, and aircraft. They open in mid-air and disperse dozens and even hundreds of smaller submunitions, also called bomblets, over an area the size of a city block. Many submunitions fail to explode on initial impact, leaving duds that act like landmines, posing a threat to civilians for years and even decades.

### 15.41.2 Rudram I & II: Anti-Radiation Missile

The **Defence Research & Development Organisation (DRDO)** successfully conducted a **flight test of RudraM-II** from a **Su-30MKI aircraft** of the Indian Air Force (IAF) off the coast of **Odisha.**

***About RudraM-II***

- ***Type:*** Air-to-surface missile system.
- ***Propulsion:*** Solid-fuel propelled.

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- ***Range:*** Capable of striking targets up to **350 km.**
- ***Platform:*** Designed for launch from **Sukhoi Su-30MKI** fighter jets.
- ***Purpose:*** To neutralise various enemy ground-based assets such as **radars**, **communication systems** and **command centres.**
- ***Development:*** Incorporates several advanced **indigenous technologies** developed by DRDO laboratories.

### *RudraM Missile Series Overview*

- **Family:** New-Generation Anti-Radiation Missiles (**NGARMs**) developed by DRDO.
- **RudraM-I:** First tested in **October 2020**, has a range of **150 km.**
- Equipped with **INS-GPS navigation** and a **passive homing head** for precise targeting.
- **Objective:** Designed for **Suppression of Enemy Air Defences (SEAD)** to allow IAF strike aircraft to operate with minimal threat from ground-based defences.

### *What Is an Anti-Radiation Missile?*

- ***Function:*** Tracks and destroys **enemy radar systems**, **communication hubs** and other **radio frequency (RF) sources**.
- ***Used For:*** Targeting components of **air defence networks** to degrade the enemy's situational awareness and coordination.

### *Navigation and Guidance System*

1. **Inertial Navigation System (INS):**
   a. Uses internal motion sensors to track movement and position.
2. **GPS (Global Positioning System):**
   b. Satellite-based navigation for precision guidance.
3. **Passive Homing Head:**
   a. Detects and locks onto RF emissions from enemy systems.
   b. Capable of engaging targets across **wide frequency bands.**

### *Rudram Anti-Radiation Missile*

- Successfully test-fired recently from a Suk-hoi-30 fighter aircraft.
- The missile has been developed by the DRDO.
- It has a strike range of around 100 to 150 km.
- It is the first indigenous air-to-ground missile developed by the DRDO, after the supersonic BrahMos, which has been developed jointly with

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Russia.

- The missile has been designed for Suppression of Enemy Air Defence (SEAD).
- The missile can be launched from a varying range of altitudes for destroying enemy surveillance radars, tracking and communication systems.

### 15.41.3 S-400

The S-400 Triumf is an air defence missile system developed as an upgrade of the S-300 series of surface-to-air missile system (SAM) designed by Russia. Rosoboronexport signed a contract with India to deliver S-400 missile system. It is the most dangerous operationally deployed modern long-range SAM (MLR SAM) in the world, considered much ahead of the **US-developed Terminal High Altitude Area Defense system (THAAD).**

S-400 can engage all types of aerial targets including aircraft, unmanned aerial vehicles (UAV) and ballistic and cruise missiles within the range of 400 km, at an altitude of up to 30 km. The system can simultaneously engage 36 targets.

#### *Significance for India*

From India's point of view, China is also buying the system. In 2015, China signed an agreement with Russia to purchase six battalions of the system. Its delivery began in January 2018.

China's acquisition of the S-400 system has been viewed as a 'game changer' in the region. However, its effectiveness against India is limited.

Note: Russia to deliver last two squadrons of S-400 air defence missiles by 2026. Russian Ambassador to India, Nikolai Kudashev recently confirmed that both India and Russia are 'committed' to completing their contract for the S-400 missile system, due to be delivered to India at the end of the year. He said both countries opposed **US sanctions on the issue.**

#### *Issues*

The S-400 deal could attract sanctions under US' CAATSA law. The US has already sanctioned China and Turkey over similar purchases.

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#### *India gets S-400 training equipment*

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- **S-400** training equipment and simulators have arrived in India from Russia.
- However, there is a delay in the delivery of the second regiment of S-400 from Russia due to the ongoing war in Ukraine.

***Concerns for India***

There is the threat of U.S. sanctions under **CAATSA (Countering America's Adversaries Through Sanctions Act)** on India for imports from Russia. However, nothing is clear as of now.

***What is CAATSA, and how did the S-400 deal fall foul of this Act?***

- Countering America's Adversaries through Sanctions Act (CAATSA)'s core objective is to counter **Iran, Russia and North Korea through punitive measures.**
- Enacted in 2017.
- Includes sanctions against countries that engage in significant transactions with Russia's defence and intelligence sectors.

***What sanctions will be imposed?***

1. Prohibition on loans to the sanctioned person.
2. Prohibition of Export-Import bank assistance for exports to sanctioned persons.
3. Prohibition on procurement by United States government to procure goods or services from the sanctioned person.
4. Denial of visas to persons closely associated with the sanctioned person.

***Significance of the deal***

The S-400 decision is a very strong example of how advanced our defence and strategic partnership is, and how strong Indian sovereignty is, to choose its international partners, especially when it comes to issues of national interest and national security.

## 15.41.4 Countering America's Adversaries through Sanctions Act (CAATSA)

Countering America's Adversaries through Sanctions Act (CAATSA) is a United States federal law that imposed sanctions on Iran, North Korea and

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Russia. It was enacted in 2017 and includes sanctions against countries that engage in significant transactions with Russia's defence and intelligence sectors.

***What sanctions will be imposed?***

1. Prohibition on loans to the sanctioned person.
2. Prohibition of Export–Import bank assistance for exports to sanctioned persons.
3. Prohibition on procurement by United States Government to procure goods or services from the sanctioned person.
4. Denial of visas to persons closely associated with the sanctioned person.

## 15.41.5 Directed Energy Weapons (DEWs)

***What Are DEWs?***

Part of **Electronic Warfare**, used to **disable, damage or destroy** enemy assets.

- **Ranged weapons** that use **concentrated energy (laser, microwave, or particle beams)** instead of kinetic projectiles.

***How DEWs Work***

- Emit **electromagnetic energy at light speed**.
- **Target materials** are penetrated or disrupted based on **wavelength and power output**.

| Type of DEWs | Mechanism | Best For |
|---|---|---|
| **HEL (High Energy Laser)** | Uses laser beams | **Long-range, precise targeting of drones, missiles** |

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| | | |
|---|---|---|
| **HPM (High Power Microwave)** | Uses microwaves | **Disables electronics in a wide area** |
| **Millimetre Wave** | Non-lethal crowd control | **Border security, riot control** |
| **Particle Beam** | Accelerated particles | **Experimental, high-energy damage** |

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### *Applications*

- **Military:** Intercept drones/missiles, disable electronics.
- **Law Enforcement:** Crowd control using non-lethal force.
- **Space:** Defence of satellites from debris or attacks.

### *Advantages of DEWs*

- **Low cost per shot:** e.g., UK's *DragonFire* costs < £10/shot.
- **Instantaneous targeting** (speed of light).
- **No physical ammo (ammunition)** = easier logistics.
- **High precision**, unaffected by gravity or weather.
- **Silent and stealthy** (especially lasers).
- **Great for drone swarms and cheap rockets**.

### *Challenges*

- ***Weather-dependent:*** Fog, rain disrupts lasers.
- **Cooling systems and energy demand** are major tech limits.
- **Friendly fire risk** in wide-area DEWs (e.g., HPMs).

(**Wide-area Directed Energy Weapons (DEWs)** utilise focused beams of energy, like lasers or microwaves, to affect targets over a large area.)

- **Ethical concerns and health effects** of exposure.
- **High R&D cost**.
- **Countermeasures:** Reflective surfaces can deflect energy.

### *India's Initiatives in DEWs (Directed Energy Weapons)*

| Project | Description |
|---|---|
| **DURGA-II** | DRDO's flagship 100-kW laser weapon project |
| **2kW BEL System** | For anti-drone operations |
| **Tri-Netra (LASTEC)** | Direct energy weapon lab by DRDO |

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| Project | Description |
|---|---|
| **KALI** | Electron accelerator weapon by DRDO **and** BARC |

***Global Examples***

- **USA:** HELIOS, THOR, HELWS
- **UK:** *DragonFire*
- **Israel:** *Iron Beam*
- **Others:** Russia, China, Germany, France – all developing DEWs

## 15.41.6 Pralay Missile and Pinaka Rocket

Tactical ballistic missile Pralay and long-range Pinaka rocket system were showcased at the Republic Day parade (26th Jan 2025) in the Capital, focusing on indigenously developed systems.

***Pralay Tactical Ballistic Missile***

- ***Type:*** Short-range, quasi-ballistic surface-to-surface missile.
- ***Developed by:*** DRDO (Defence Research and Development Organisation).

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- ***Payload:*** 500–1,000 kg.
- ***Propulsion:*** Solid-fuel powered.
- ***Flight Features:***
  - ❑ Low trajectory hyperbolic path.
  - ❑ Capable of evasive manoeuvres (changing lane quickly) with **thrust vector control**.
  - ❑ Warhead can **manoeuvre during terminal phase (before end/ death)**, similar to cruise missiles.
  - ❑ **Stealth-enhanced**: Equipped with small fins to reduce radar signature.
- ***Strategic Role:*** Designed to evade missile defence systems and strike precision targets.

### *Pinaka Multi-Barrel Rocket Launcher (MBRL)*

- **Type:** Long-range artillery rocket system.
- **Developed by:** ARDE, DRDO.
- **Features:**
  - ❑ Fires **12 rockets in 44 seconds**.
  - ❑ All 12 can be pre-programmed in **20 seconds**.
  - ❑ Covers **700 ´ 500 sq m area in 48 seconds**.
  - ❑ Equipped with **Inertial Navigation System (INS)** for accurate targeting.
  - ❑ "**Shoot and scoot**" feature: Quick relocation post-firing to avoid enemy retaliation.
- **Variants:**
  - ❑ **Pinaka Mk-I:** Up to 40 km range.
  - ❑ **Enhanced Version:** Over 75 km range (tested successfully).
- **Global Reach: Armenia** is the first international buyer.

### *Significance of Pralay and Pinaka*

- Reflect India's advancement in **indigenous defence technologies**.
- Their inclusion in the **Republic Day parade** symbolises growing strategic capability and **Aatmanirbhar Bharat** in defence production.

### *Ballistic Missiles of India*

- **Prithvi Series:** Short-range tactical missiles (150–350 km).
- **Agni Series:** Medium to ICBMs (Agni-V > 5,000 km).
- **Dhanush:** Ship-launched ballistic missile (150–500 km).

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- **Pralay:** Fills the operational gap between Prithvi and Agni series.

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### 15.41.7 Hellfire Missile

### 15.41.8 Fractional Orbital Bombardment System (FOBS)

A Fractional Orbital Bombardment System (FOBS) is a warhead delivery system that uses an LEO towards its target destination. Just before reaching the target, it de-orbits through a retrograde engine burn. The idea behind FOBS is, a warhead is put into a stable orbit and it deorbits over the target.

If the target and the launch position is lined up and the warhead keeps going round, it will complete a circle. Obviously, with the earth moving during this, it is more of a spiral than a ring, but the horizontal movement is countered in such a way that the warhead still goes over the target.

#### *Flight Tests of FOBS*

In 1968, after its twentieth test the Soviet succeeded in FOBS. This FOBS project was called R-360.

#### *Features*

It granted unlimited striking range. It was capable of striking from any direction. It was smart to hide the target location till the payload was

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dropped. According to the Soviet Union, the FOBS were capable of trumping the anti-ballistic missiles of the US.

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### *Outer Space Treaty*

The Outer Space Treaty signed in 1967 stated that the signatories should not place nuclear weapons in the orbit around the earth. Even after the treaty, FOBS stayed in the space. The Soviet Union claimed that FOBS takes a partial orbit and can thus stay in space.

### *SALT II*

The SALT II agreement was signed between the Soviet Union and the US in 1979. SALT stands for Strategic Arms Limitation Talks. The agreement prohibited the deployment of FOBS. The US Government did not ratify SALT II. However, the Soviet Union complied and decommissioned FOBS in 1983.

### *Why was FOBS ended?*

One of the reasons is SALT II. Apart from this, the system faced technical disadvantages. The system demanded high level of energy to load a weapon into orbit. Also, it was less accurate than ICBM. ICBM stands for Intercontinental Ballistic Missile. ICBM was developed during Second World War. The countries that hold operational ICBMs are India, North Korea, USA, China, Russia, France and the UK.

### *Recent Developments*

In 2021, Frank Kendal III (the US Secretary of Air Force) stated that the Chinese Government is developing and testing FOBS.

## VARIOUS JOINT EXERCISES (NAVY, AIR FORCE & ARMY)

### Indian Navy Joint Exercises

1. **Varuna:** France and India.
2. **SLINEX:** Sri Lanka and India.
3. **INDRA:** Bi-annual military exercise conducted by India and Russia.
4. **Exercise Malabar:** A bilateral of the United States and India.
5. **Simbex:** Indian and the Republic of Singapore.
6. **IBSAMAR:** India, Brazil and South Africa.

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7. **KONKAN:** A bilateral naval exercise between Indian navy and the Royal Navy of Britain.
8. **AUSINDEX:** Indian and Australian navies' bilateral maritime exercise.
9. **Sahyog-kaijin:** Coast guards of Indian and Japan.
10. **Naseem Al Bahr:** India and Oman.

**Indian Air Force Exercises**

1. The joint India–UK air exercises are known as the **'Indradhanush'** or 'Rainbow'.
2. India-France air exercise **'Garuda'.**
3. **Avia Indra 2014:** India–Russia maiden aerial exercise.

**Indian Army Joint Exercises**

1. **Mithra Shakti exercise:** India and Sri Lanka.
2. **Hand-in-hand:** India and China.
3. **Exercise Shakti:** India and France.
4. **Exercise Nomadic Elephant:** India and Mongolia.
5. **Exercise Yuddh Abhyas:** Series of joint exercises between India and the USA since 2005.
6. **Surya Kiran:** India and Nepal.
7. **LAMITYE:** India and Seychelles.
8. **Prabal Dostyk:** India and Kazakhstan.

**OPERATION MITRA SHAKTI 2023**

- It is an army exercise between India and Sri Lanka.
- "Mitra Shakti-2023," commenced on November 16, 2023, in Aundh (Pune).
- The scope of the exercise includes synergising joint responses during counter-terrorist operations.
- The Indian contingent, of 120 personnel, is being represented mainly by troops from the MARATHA LIGHT INFANTRY Regiment.

The Sri Lankan side is being represented by personnel from 53 Infantry Division. 15 personnel from Indian Air Force and five personnel from Sri Lankan Air Force are also participating in the exercise.

## 15.41.9 Operation 'SADBHAVANA'

As part of 'Operation SADBHAVANA', the Indian Army is undertaking multiple welfare activities such as running of Army Goodwill Schools, Infrastructure Development Projects and Education Tours, etc. in Ladakh Region. More than 2,200 students are currently studying in these schools.

In addition, a total of `8.82 crore of 'Operation SADBHAVANA' funds has been allotted for the Union Territory of Ladakh during the financial year

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2022–23.

TIGERxUPSC The funds are being utilised for various activities like Human Resource/Skill Development, Sports, Healthcare, National Integration, Infrastructure Development, Ecology, Environment and Education. TIGERxUPSC

## 15.41.10 Dirty Bomb

A dirty bomb combines radioactive material with explosives, intended to cause harm by **spreading radioactive material** that can contaminate an area and poison the human body.

### Material used in Dirty Bomb

Cesium-137, RDX (Explosives) like Cobalt-60 (High-energy gamma emitters).

### India's Position to Detect Dirty Bombs

- **Bhabha Atomic Research Centre (BARC):** It has developed systems such as the **aerial gamma spectrometry system** which can be used to detect even shielded and hidden devices.
- **Other Technology:** **BARC has developed technology that can detect elements from solid and liquid industrial wastes**, as most of these radioactive elements can dissolve in water.

## 15.41.11 Explosive Weapons

**Explosive weapons**, a type of conventional weapon, rely on the rapid expansion of gases resulting from the detonation of high explosives to create destructive effects. These effects include blast waves, fragmentation and heat, causing damage to targets and personnel.

**Types of Explosive Weapons:**

- **Bombs:** Aerial bombs, naval bombs and other types of bombs.
- **Artillery:** Shells, rockets and mortars.
- **Grenades:** Hand grenades and grenade launchers.
- **Missiles:** Various types of missiles, including anti-tank guided missiles.
- **Landmines:** Anti-personnel and anti-tank mines.

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- **Improvised Explosive Devices (IEDs):** Devices made from readily available materials, often used by non-state actors.

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## Types of Explosives Used

### *High Explosives*

Explosives like RDX (Cyclonite), TNT, PETN and C4 are commonly used in military weapons. CL-20 **(China Lake 20)** is a polycyclic nitroamine explosive with a high energy density. It's considered one of the most powerful explosives, surpassing even HMX and RDX in performance. CL-20 is used in rockets and missiles.

**HMX (High Melting Explosive) i**s also called octogen, HMX is a nitroamine explosive chemically related to RDX. It is widely used in military applications, including as a detonator in nuclear weapons, in polymer-bonded explosives and as a solid-rocket propellant.

**LLM-105** is another high-energy explosive developed at **Lawrence Livermore National Laboratory, Livermore (California).** It›s characterised by its high density and energetic properties, making it suitable for various military applications.

### *Low Explosives*

Gunpowder and other less sensitive explosives can also be used.

### *Mixtures*

Commonly used in IEDs, mixtures like ANFO (Ammonium Nitrate Fuel Oil) are a fuel and oxidiser combination.

### *Initiating/Primary High Explosives*

These are very sensitive to shock or heat and are used to detonate secondary high explosives.

### *Secondary High Explosives*

These are less sensitive and require an initiating high explosive to detonate.

## Effects of Explosive Weapons:

- **Blast Wave:** A wave of pressure that radiates from the detonation.

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- **Fragmentation:** Material from the weapon is propelled outwards, creating high-velocity fragments.
- **Heat:** The detonation generates high temperatures.

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## 15.41.12 GAURAV Project

- ***Defence Research and Development Organisation (DRDO) has conducted successful release trials of the indigenously developed Long-Range Glide Bomb (LRGB) 'Gaurav' from the Su-30 MKI aircraft.***
- Designed for **precision air-to-surface strikes** from stand-off distances, staying out of enemy air defence range.

### *Key Highlights*

| Feature | Description |
|---|---|
| **Type** | Air-launched, precision-guided, long-range glide bomb |
| **Range** | Demonstrated ~100 km;<br>**operational range:** 30–150 km |
| **Weight** | Gaurav (Winged): 1,000 kg<br>Gautham (Non-winged): 550 kg |
| **Navigation** | Combines **Inertial Navigation System (INS)**, **satellite guidance** and **digital control systems** |
| **Purpose** | Neutralise land targets with **high accuracy** from long distances |

### *What is a Glide Bomb?*

- A **precision-guided munition** that travels significant distances without engine thrust, using **aerodynamic lift**.
- Guided by systems like **INS**, **GPS** or **laser**.
- Allows launch aircraft to **remain at a safe distance**, improving survivability and mission success.

### *Su-30MKI Platform*

- A **multirole combat aircraft**, co-developed by **Russia's Sukhoi Design Bureau** and **India's HAL**.
- Key platform for deploying indigenous weapons like 'Gaurav'.

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### *Strategic Significance*

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- Boosts India's **indigenous precision strike capabilities**.

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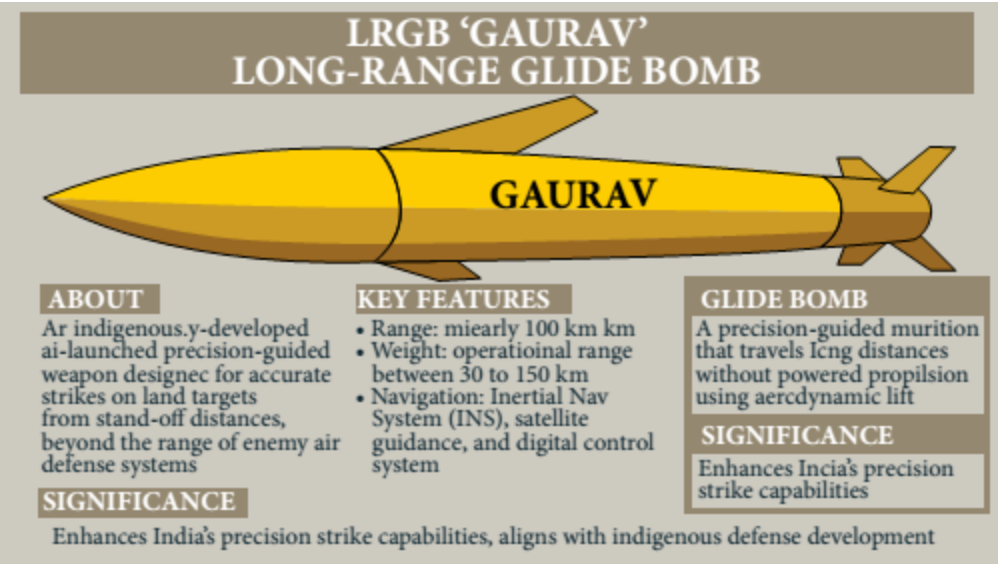


- Aligns with the **Aatmanirbhar Bharat** initiative in defence production.
- Increases effectiveness in **deep-strike missions** without exposing pilots to frontline risks.

## 15.41.13 Sanjay System (Battlefield Surveillance System)

The Defence Minister **flagged off the 'Sanjay'**, an advanced Battlefield Surveillance System (BSS), from south block, New Delhi on **January 24, 2025**.

***What is SANJAY?***

- **Sanjay** is an advanced automated **Battlefield Surveillance System (BSS)**.

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- Designed to **integrate real-time data** from ground and aerial sensors.

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- Creates a **Common Surveillance Picture (CSP)** for better situational awareness.
- Enables **quick, informed decision-making** in both conventional and sub-conventional warfare.

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### *Deployment Timeline*

- Induction planned in **three phases (March – October 2025)**.
- Will be deployed across **all operational brigades, divisions and corps** of the Indian Army.

### *Development Details*

- **Developed by:** Indian Army in collaboration with **Bharat Electronics Limited (BEL)**.
- **Procurement Category:** Under the ***Buy (Indian)*** initiative to promote indigenous defence capabilities.
- **Cost of Development:** ₹2,402 crore.
- **Strategic Alignment:** Supports the Government's *Aatmanirbhar Bharat* (self-reliance) mission.

### *Objectives*

- **Enhance battlefield transparency** through real-time surveillance.
- **Strengthen intelligence, surveillance and reconnaissance (ISR)** capabilities across terrains and borders.
- **Enable swift and precise decision-making** at all levels of command.
- **Showcase India's defence technology innovation** during the 'Year of Technology Absorption'.

### *How It Works*

1. **Data Integration:** Collects inputs from diverse battlefield sensors (ground + aerial).
2. **Processing:** Filters and verifies data to avoid duplication and errors.

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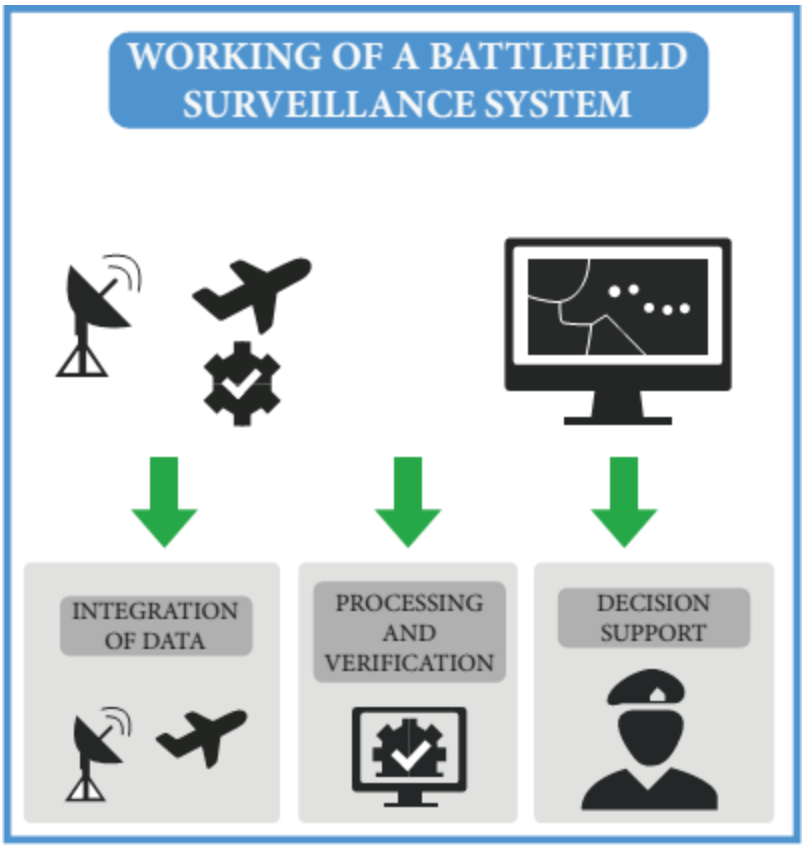


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3. **Common Surveillance Picture (CSP):** Generates a unified, real-time view accessible through secure Army networks.
4. **Decision Support:** Provides actionable intelligence to field commanders and Army HQ through a secure web application.

Key Features

- ***Real-Time Integration:*** Consolidates sensor data into a single, live operational picture.
- ***Advanced Analytics:*** Uses AI-driven analytics for threat detection and situational evaluation.
- ***Border Surveillance:*** Detects intrusions and enhances monitoring across frontiers.
- ***Network-Centric Operation:*** Operates within secured data and satellite communication networks.
- ***Fully Indigenous:*** Entirely developed in India to boost local defence capabilities.
- ***Operational Versatility:*** Suitable for both **conventional military engagements** and **counter-insurgency operations**.

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## 15.42 Iron Dome

***What is Iron Dome Defense System?***

1. Enemy Rockets are Fired
2. Radar System detects Rockets and track
3. Control Systems estimates Rocket Impact Time (Battle Management and Control (BMC))
4. Launcher fires Missile to intercept
5. Missiles explodes near Rocket, thus destroying it

**Purpose:** Protects civilian population and critical infrastructure from short-range rockets and artillery shells.

**Development:** Created by Israeli defense companies, primarily Rafael Advanced Defense Systems.

**Capabilities:** Highly effective in intercepting and destroying incoming projectiles.

**Strategic Importance:** Crucial component of Israel's defense strategy.

**Impact:** Enhances security and safety of Israeli citizens amidst regional threats.

**Recognition:** Gained international attention for its technological achievements and effectiveness.

## QUESTIONS AND ANSWERS

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## Objective Questions

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**1. With reference to Unmanned Aerial Vehicles (UAVs), consider the following statements:**

**i.** All types of UAVs can do vertical landing. **[UPSC (Pre) 2025]**

**ii.** All types of UAVs can do automated hovering.

**iii.** All types of UAVs can use battery only as a source of power.

How many of the statements given above are correct?

(a) Only one
(b) Only two
(c) All the three
(d) None

**2. What is the common characteristic of the chemical substances generally known as CL-20, HMX and LLM-105, Which are the sometimes talked about in media?** **[UPSC (Pre) 2025]**

(a) These are alternatives to hydro-fluorocarbon refrigerants.
(b) These are explosives in military weapons.
(c) These are high-energy fuels for cruise missiles.
(d) These are fuels for rocket propulsion.

**3. With reference to India's defence, consider the following pairs:** **[UPSC (Pre) 2025]**

| ***Aircraft type*** | ***Description*** |
|---|---|
| Dornier-228 | Maritime Patrol Aircraft |
| IL – 76 | Supersonic combat Aircraft |
| C-17 Globemaster III | Military Transport Aircraft |

How many of the pairs given above are correct matched?

(a) Only one
(b) Only two
(c) Only three
(d) None

**4. Consider the following aircraft:** **[UPSC (Pre) – 2024]**

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1. Rafael
2. MiG-29
3. Tejas MK-1

How many of the above are considered fifth generation fighter aircraft ?

(a) Only one
(b) Only two
(c) All three
(d) None

**5. Which of the following statements about "Exercise Mitra Shakti – 2023" are correct?** **[UPSC (Pre) – 2024]**

1. This was a joint military exercise between India and Bangladesh.
2. It commenced in Aundh (Pune).
3. Joint response during counter-terrorism operations was a goal of this operation.
4. Indian Air Force was a part of this exercise.

Select the answer using the code given below:

(a) 1, 2 and 3
(b) 1, 2 and 4
(c) 1, 3 and 4
(d) 2, 3 and 4

**6. Operation undertaken by the Army towards upliftment of the local population in the remote areas to include addressing of their basic needs is called:** **[UPSC (Pre) – 2024]**

(a) Operation Sankalp
(b) Operation Maitri
(c) Operation Sadbhavana
(d) Operation Madad

**7. Consider the following activities:** **[UPSC (Pre) – 2024]**

1. Identification of narcotics on passengers at airports or in aircraft.
2. Monitoring of precipitation.
3. Tracking the migration of animals.

In how many of the above activities can the radars be used?

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(a) Only one
(b) Only two
(c) All three
(d) None

**8. Which of the following is/are correctly matched in terms of equivalent rank in the three services of Indian Defence forces?** **[UPSC (Pre) – 2024]**

| | *Army* | *Airforce* | *Navy* |
|---|---|---|---|
| 1. | Brigadier | Air Commodore | Commander |
| 2. | Major General | Air Vice Marshal | Vice Admiral |
| 3. | Major | Squadrone Leader | Lieutenant Commander |
| 4. | Lieutenant Colonel | Group Captain | Captain |

Select the answer using the code given below:
(a) 1 and 4
(b) 1 and 3
(c) 2, 3 and 4
(d) 3 only

**9. Consider the following statements:** **[UPSC (Pre) – 2023]**

1. Ballistic missiles are jet-propelled at subsonic speeds throughout their flights, while cruise missiles are rocket-powered only in the initial phase of flight.
2. Agni-V is a medium-range supersonic cruise missile, while BrahMos is a solid-fuelled intercontinental ballistic missile.

Which of the statements given above is/are correct?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**10. Which one of the following statements best reflects the idea behind the "Fractional Orbital Bombardment System" often**

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**talked about in media?** **[UPSC (Pre) – 2022]**

(a) A hypersonic missile is launched into space to counter the asteroid approaching the Earth and explode it in space.
(b) A spacecraft lands on another planet after making several orbital motions.
(c) A missile put into a stable orbit around the Earth and deorbits over at target on the Earth.
(d) A spacecraft moves along a comet with the same speed and places a probe on its surface.

**11. Consider the following activities:** **[UPSC (Pre) – 2020]**

1. Spraying pesticides on a crop field.
2. Inspecting the craters of active volcanoes.
3. Collecting breath samples from spouting whales for DNA analysis.

At the present level of technology, which of the above activities can be successfully carried out by using drones?

(a) 1 and 2 only
(a) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

**12. Which one of the following is the best description of 'INS Astradharini', that was in the news recently?** **[UPSC (Pre) – 2016]**

(a) Amphibious warfare ship
(b) Nuclear-powered submarine
(c) Torpedo launch and recovery vessel
(d) Nuclear-powered aircraft carrier

**13. With reference to 'Organisation for the Prohibition of Chemical Weapons (OPCW)', consider the following statements:** **[UPSC (Pre) – 2016]**

(i) It is an organisation of European Union in working relation with NATO and WHO.
(ii) It monitors chemical industry to prevent new weapons from emerging.
(iii) It provides assistance and protection to States (Parties) against chemical weapons threats.

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Which of the statements given above is/are correct?

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii), and (iii)

**14. With reference to Agni-IV Missile, which of the following statements is/are correct?** **[UPSC (Pre) – 2014]**

(i) It is a Surface-to-Surface missile.
(ii) It is fuelled by liquid propellant only.
(iii) It can deliver one-tonne nuclear warheads from about 7500 km away.

Select the correct answer using the code given below.

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii) and (iii)

**15. In the context of Indian defence, consider the following statements:** **[UPSC (Pre) – 2009]**

(i) The Shourya missile flies with a speed of more than Mach 8.
(ii) The range of Shourya missile is more than 1600 km.

Which of the statements given above is/are correct?

(a) Only (i)
(b) Only (ii)
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

**16. Which one of the following pairs is not correctly matched?** **[UPSC (Pre) – 2005]**

(a) Arjun: Indigenously produced Main Battle Tank (MBT).
(b) Phalcon: Cruise Missile supplied by Russia to India.
(c) Saras: Indigenously developed civilian passenger aircraft.
(d) Operation Seabird: New Indian Naval Base at Karwar.

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**Answer Key**

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (d) | 2. (b) | 3. (b) | 4. (d) | 5. (d) | 6. (c) | 7. (b) | 8. (d) |
| 9. (d) | 10. (c) | 11. (d) | 12. (c) | 13. (b) | 14. (a) | 15. (d) | 16. (b) |

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**UNIT-VI** Artificial Intelligence, Robotics and Nanotechnology

# CHAPTER 16
# ARTIFICIAL INTELLIGENCE

## IN THIS CHAPTER...

**16.1 Concept of Artifiial Intelligence**
**16.2 Strong AI and Weak AI**

To test the artificial general intelligence of machines, some tests such as Turing Test (human and machine talk), Coffee Test (a machine can make coffee at your home), College Admission Test (a machine can enrol admission, join class and clear exam to get degree) and Employment test (machine is given an economically important job) have been suggested.

**16.3 From Artifiial General Intelligence to Applied AI**

NLP is a technology that **machines use to understand, analyse and interpret human languages**. It plays an important role in various applications, such as chat boards, virtual assistant speech to text, machine translation, auto correct and much more.

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**16.4 Generative AI and Large Language Models (LLMs)**

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16.4.1 Nobel Prize in Physics 2024
16.4.2 Neuromorphic Computing (Brain-inspired Analog Computing)
**16.5 Some Recent News of AI**
16.5.1 ChatGPT

Eugene Goostman is a chatbot that some regard as having passed the first Turing test.

Turing Test is named after **Alan Turing**, the founder of the Turing Test and an English computer scientist, cryptanalyst, mathematician and theoretical biologist. Turing proposed that a computer can be said to possess artificial intelligence if it can mimic human responses under specific conditions.

16.5.2 Gemini – AI Model by Google

Google has launched Project Gemini, designed **to enhance the capabilities of Google's AI-powered chatbot, Bard**, developed by **Google DeepMind, the AI division** which emphasises Gemini's problem-solving skills in areas such as mathematics and physics, potentially leading to scientific breakthroughs.

16.5.3 Deepfake
16.5.4 World's AI Safety Summits
16.5.5 S.A.R.A.H.
**16.6 Artifiial Intelligence in India**

YUVAi, launched by the **National e-Governance Division (NeGD) in partnership with Intel India, aims to educate school students** about AI and empower them to become human-centric designers and users of AI.

**16.7 Global Partnership on AI (GPAI 2024) / New Delhi Declaration**
**16.8 BharatGen Programme**
**16.9 AI in Education Sector**
16.9.1 Difference between AI Tutor and Robot Teacher
16.9.2 AI and National Education Policy (NEP) – 2020
**16.10 AI in Medical Science**
**16.11 AI in Defence**
**16.12 Advantages of Artifiial Intelligence**
**16.13 Disadvantages of Artifiial Intelligence**
**16.14 Is AI moving towards Super Intelligent Machines?**

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## 16.1 Concept of Artificial Intelligence

Artificial intelligence generally implies machines exhibiting traits of human intelligence. The important traits of human intelligence include observing, taking inputs from the observations, reasoning and rationalising, decision making, manipulations, learning from behaviour and patterns, communicating, analysing complex situations and performing tasks to achieve goals and targets.

At present, the 'subfields of AI' like machine learning, autonomous systems, natural language processing, robotics and artificial creativity are popular areas of research (**Table 16.1**).

**Table 3.2:** Thrust Areas of Contemporary AI Research

| | |
|---|---|
| **Machine Learning** | This is the field in which computers are given the ability to learr |
| **Autonomous Systems** | A system that learns on its own to perform various tasks. |
| **Language Processing** | The field of interaction between machines and human languag |
| **Robotics** | Robotics in entirety is not just AI but there is a field of AI-powe |
| **Artificial Creativity** | Creativity using computer, e.g., Aiva is an AI-powered system, project Magenta is all about AI applications in arts. |

## 16.2 Strong AI and Weak AI

There are two types of Artificial Intelligence: Strong Artificial Intelligence (SAI) and Weak Artificial Intelligence (WAI). What has been actually called strong AI is actually 'full AI'.

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A full AI means a machine becoming capable of performing all intelligence tasks that a human being can perform. In this, a machine would not exhibit selective fragments of human intelligence but would rather behave like a human being in terms of consciousness.

Full AI is also known as 'artificial general intelligence'. Human beings reason, use strategy, solve problems, make judgements, use common sense, plan, learn, communicate and integrate all these to achieve their goals.

To test the artificial general intelligence of machines, some tests have been suggested. These are Turing Test (human and machine talk), Coffee Test (a machine can make coffee at your home), College Admission Test (a machine can enrol admission, join class and clear exam to get degree) and Employment test (machine is given an economically important job). There are different thrust areas of AI research those are mentioned in **Infographic – Artificial Intelligence**.

## 16.3 From Artificial General Intelligence to Applied AI

AI research began in the United States with particular interest in developing machines that could translate from Russian to English and vice-versa. This was in context of Cold War with the then Soviet Union.

The applications that are attracting maximum attention are machine learning, robotics, autonomous systems and natural language processing. Autonomous systems are systems with the ability to acquire or create content, interpret and develop knowledge from that content to make a decision on the basis of that interpretation.

They are now applied in the development of driverless cars, drones and fraud detection. Prominent examples of driverless cars include Waymo of Google S, model X and model 3.

### Artificial Intelligence, Machine Learning and Deep Learning

In the context of artificial intelligence, two other terms often used interchangeably with artificial intelligence are machine learning and deep

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learning. These three terms are not the same, but are interrelated.

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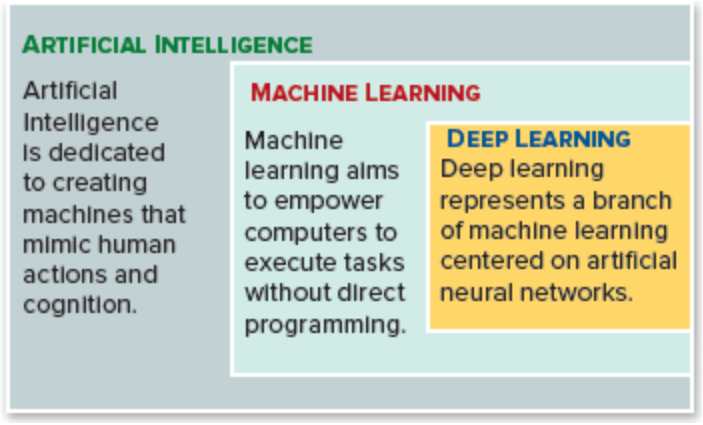


**Figure 16.1** Machine Learning and Deep Learning are sub-sets of Artificial Intelligence & deep learning in a sub-sets of machine learning.

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Neurons are a tiny messenger in our brain that sends and receives messages throughout our body, allowing us to think, feel, move and interact with the world around us.

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## NATURAL LANGUAGE PROCESSING (NLP)

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## NATURAL LANGUAGE PROCESSING (NLP)

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In deep learning projects, **Natural Language Processing (NLP)** has emerged as an important area.

It is an area of artificial intelligence **concerned with giving machines the ability to interpret, written and spoken language similarly, to humans.**

Search engines' predictive typing feature assists users by suggesting the next word as they type keywords, while spell-checking features exemplify Natural Language Processing (NLP).

It is a technology that **machines use to understand, analyse and interpret human languages**. NLP plays an important role in various applications, such as chat boards, virtual assistant speech to text, machine translation, auto correct and much more.

NLP integrates principles from computer science, linguistics, machine learning and deep learning. Its purpose is to enable computers to analyse unstructured text or voice data and extract meaningful information from it.

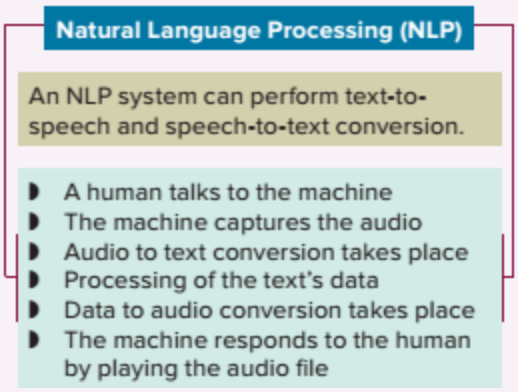


**Figure 16.2** Interaction between Computers and Humans using NLP

## 16.4 Generative AI and Large Language Models (LLMs)

### *Generative AI*

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- **Definition:** AI that can generate original content—text, images, videos, audio or code, based on a prompt.

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- **Foundation Model:** Relies on **Large Language Models (LLMs)** which simulate human-like learning and content generation using **deep learning algorithms**.

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### *Large Language Models (LLMs)*

- A large language model is an advanced artificial intelligence system that processes, understands and generates human-like text based on massive amounts of data.
- These models are usually constructed using deep learning methods like neural networks. They undergo training using large datasets that encompass text from diverse sources like books and websites, facilitating natural language processing.
- One of the most well-known examples of LLMs is OpenAI's ChatGPT.

### *Traditional AI vs Generative AI*

| Feature | Traditional AI | Generative AI |
|---|---|---|
| **Key Focus** | Data analysis, automation, decision-making | Creating new data like text, images, music |
| **Learning Type** | Rule-based, algorithmic | Data-driven, using neural networks |
| **Output** | Predictions, classifications, structured responses | Novel content, creative outputs |
| **Adaptability** | Requires manual updates | Self-improving with data and use |
| **Analogy** | Chef following a recipe | Chef inventing new dishes |
| **Applications** | Efficiency, accuracy, reasoning | Creativity, content generation, storytelling |

## 16.4.1 Nobel Prize in Physics 2024

### Artificial Neural Networks and Machine Learning

### *Awarded To:*

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**# John J. Hopfield # Geoffrey E. Hinton**

**Field: Artificial Neural Networks (ANNs) and Machine Learning (ML)**

## Awarded Discoveries

**John J. Hopfield**

- **Invented:** *Hopfield Network*, a type of **Recurrent Neural Network (RNN)**.
- **Purpose:** Acts like a memory system; stores and reconstructs patterns (images, signals).
- **Based on: Hebbian learning** – 'neurons that fire together, wire together'.
- **Applications:** Pattern recognition, image reconstruction, error correction.

**Geoffrey E. Hinton**

- **Developed:** *Boltzmann Machine* – one of the earliest **generative models**.
- **Breakthrough:** Discovered hidden patterns in data without explicit programming.
- **Impact:** Laid foundation for **deep learning**, used in chatbots, finance, healthcare, NLP.

## Artificial Neural Networks (ANNs)

- **Definition:**
- ANNs are **machine learning models** inspired by the structure and functioning of the **human brain**. They consist of artificial neurons that process data and learn patterns.

## Biological Analogy

| Biological Brain | Artificial Neural Network |
|---|---|
| Neurons | Artificial Neurons (Nodes) |
| Synapses | Weighted Connections |

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| | |
|---|---|
| Learning via signals | Learning via training data |

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### *Structure of ANN*

1. **Input Layer:** Receives raw data (e.g., pixels, words).
2. **Hidden Layers:** Perform complex transformations.
3. **Output Layer:** Produces final decision (classification, prediction).

### *Major Types*

| Type | Key Features | Use-Cases |
|---|---|---|
| **Deep Neural Networks (DNNs)** | Multiple hidden layers | General AI tasks |
| **Convolutional Neural Networks (CNNs)** | Designed for visual data | Image classification, face recognition |
| **Recurrent Neural Networks (RNNs)** | Use sequential memory | Language translation, speech recognition |
| **Generative Adversarial Networks (GANs)** | Generator + Discriminator | Fake image generation, deepfakes |

## Machine Learning (ML)

A **subset of AI**, ML allows computers to learn from data and improve performance without being explicitly programmed.

### *Working Mechanism*

1. **Training on Data:** Learns from large datasets.
2. **Pattern Identification:** Recognises features or trends.
3. **Prediction/Decision:** Applies learning to new data.

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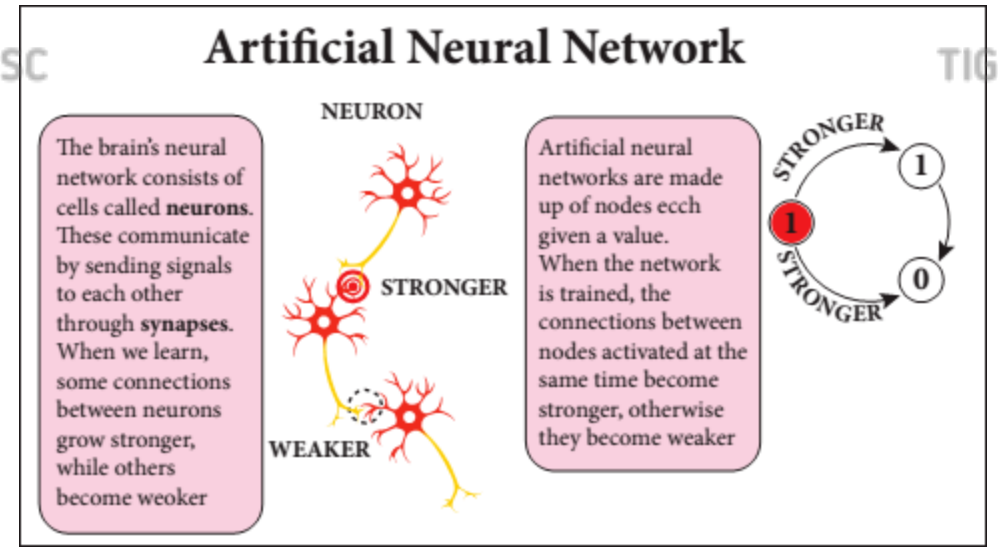


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## Applications of ML

| Sector | Examples |
|---|---|
| Scientific Research | Discovery of **Higgs boson**, **exoplanets** |
| NLP | ChatGPT, speech-to-text, language translation |
| Computer Vision | Object detection, face recognition |
| Healthcare | Cancer diagnostics, drug discovery |
| Financial Systems | Fraud detection, algorithmic trading |

## Challenges and Ethical Concerns

| Issue | Explanation |
|---|---|
| **Explainability** | Neural networks are 'black boxes'; difficult to interpret their logic. |
| **Bias & Discrimination** | Human biases can be embedded in data and models. |
| **Accountability** | Who is liable if AI (e.g., self-driving car) makes a wrong decision? |
| **Privacy & Security** | Sensitive data can be misused or leaked. |

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| | |
|---|---|
| **Superintelligence Risk** | AI may surpass human control in future (theoretical concern). |

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### 16.4.2 Neuromorphic Computing (Brain-inspired Analog Computing)

Neuromorphic computing is a process designed to **mimic the structure and function of the human brain** using **artificial neurons and synapses**.

It represents a significant shift from **traditional binary computing** to **neuromorphic computing,** allowing systems to **learn from the environment**.

#### *Working Mechanism*

- It involves the use of **Artificial Neural Networks (ANN)** made up of millions of **artificial neurons**, similar to those in the human brain.
- These neurons **pass signals to each other in layers**, converting input into output through **electric spikes or signals,** based on the architecture of **Spiking Neural Networks (SNN)**.
- This allows the **machine to mimic the neuro-biological networks** in the human brain and perform tasks efficiently and effectively, such as visual recognition and data interpretation.

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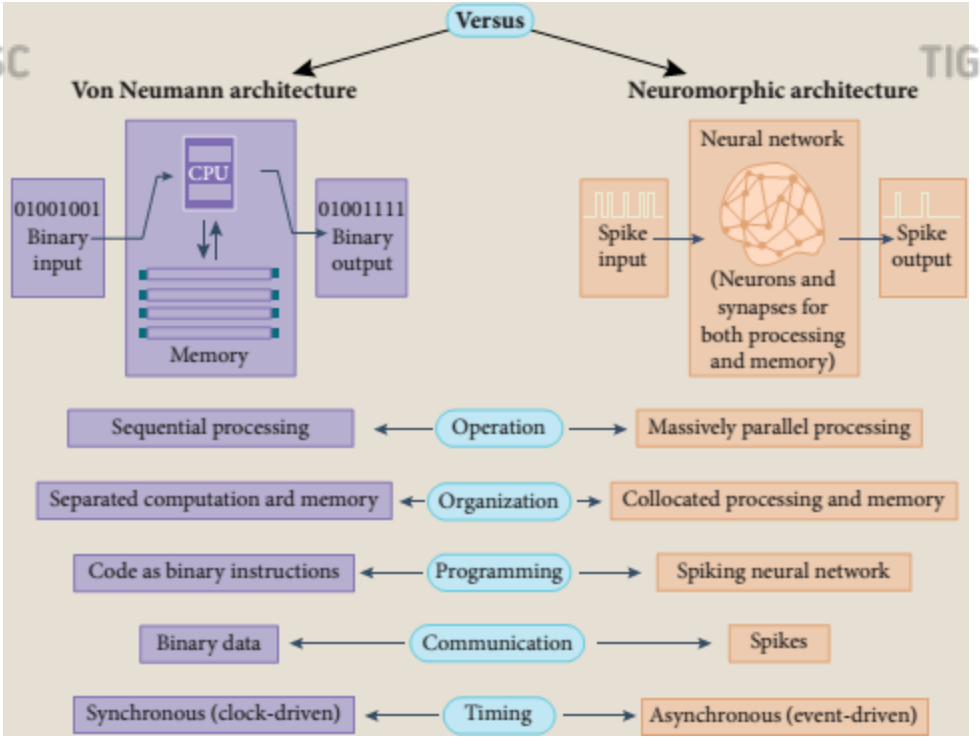


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## *Differences from Traditional Computing*

- **Parallel Processing:** Neuromorphic computers can **process multiple streams of information simultaneously**, unlike traditional computers that operate sequentially.
- **Energy Efficiency:** They **consume less power** by computing only when relevant events occur, making them ideal for applications requiring real-time data processing.
- **Traditional binary computing** operates with **bits in two states: 0 or 1**, similar to a light switch being on or off. In contrast, **analog computing uses continuous values,** similar to a dimmer switch with a range of brightness levels.

**Neuromorphic computing (Computer is design to work similar to human brain)**, also called **neuromorphic engineering**, is a cutting-edge branch of computer science that designs both **hardware and software systems to emulate the structure and functioning of the human brain**.

## *Historical Background*

- Originated in the **1980s**, pioneered by **Carver Mead** and **Misha Mahowald**

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- They introduced **silicon-based neurons, synapses and artificial retinas**, laying the foundation for brain-like machines.

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## Core Components

| Components | Functions |
|---|---|
| **Neurons** | Store and process data like biological brain cells. |
| **Synapses** | Create connections between neurons and adjust signal strength over time. |
| **Spiking Neural Networks (SNNs)** | Use event-driven spikes (action potential) to transmit data (electrical impulses) — mimicking real brain activity. |

## Key Algorithms

- **Deep Learning Conversion:** Translates deep neural networks into **spiking models**.
- **Evolutionary Algorithms:** Uses **biologically inspired** methods to optimise networks.
- **Neural Plasticity:** Allows synaptic weights to change dynamically, just like in the brain.

**Neural Plasticity:** The brain's ability to reorganise and adapt its structure and functions in response to stimuli or injury.

## Advantages of Neuromorphic Computing

- **Real-Time Learning:** Systems can adapt and solve problems on the go.
- **Energy Efficient:** Only active neurons consume power, making it ideal for low-power applications.

## Challenges

| Issue | Description |
|---|---|
| **Accuracy Loss** | Precision may drop when converting standard models to spiking networks. |

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| | |
|---|---|
| **No Common Standards** | Lack of standardised testing frameworks and performance benchmarks. |
| **High Complexity** | Requires knowledge across multiple fields including biology, electronics and AI. |

## *Applications*

| Sector | Use Case |
|---|---|
| **Autonomous Vehicles** | Faster reaction times for navigation and obstacle avoidance. |
| **Cybersecurity** | Detects anomalies and threats in network activity. |
| **Edge AI Devices** | Powers smart sensors, wearables and IoT devices with low energy needs. |
| **Robotics** | Enhances decision-making capabilities and adaptability in robots. |

## *Related Concepts*

### Edge AI

- Deployment of AI models directly on local devices (like IoT sensors), enabling real-time processing without needing the cloud.

### Latency

- Time lag between input and response—critical in real-time systems.

## *Understanding Neurons and Synapses*

| Term | Description |
|---|---|
| **Neuron** | A nerve cell that transmits electrical and chemical signals. |
| **Cell Body (Soma)** | Contains the cell's nucleus. |
| **Dendrites** | Receive incoming signals from other neurons. |
| **Axon** | Sends signals to other neurons or muscles. |

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| | |
|---|---|
| **Synapse** | The junction where neurons connect and transmit signals using neurotransmitters. |

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## 16.5 Some Recent News of AI

### 16.5.1 ChatGPT

- GPT stands for **Generative Pre-trained Transformer**, a kind of computer language model that **relies on deep learning techniques to produce human-like text-based inputs.**
- **It is a second AI already passed the Turing test (Human–machine talk). It is developed by** ChatGPT founder and OpenAI CEO **Sam Altman.** Eugene Goostman is a chatbot that some regard as having passed the first Turing test.
- The test is named after **Alan Turing**, the founder of the Turing Test and an English computer scientist, cryptanalyst, mathematician and theoretical biologist. Turing proposed that a computer can be said to possess artificial intelligence if it can mimic human responses under specific conditions.
- ChatGPT is an open AI chatbot that uses natural language processing to create human-like conversation. It is a **conversational chatbot powered by artificial intelligence.** The language model can respond

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to questions and compose various written content, including articles, social media posts, essays, code and emails.

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- It is also a **neural network**, as a **large network of computers** can fine-tune its output of words based on feedback.
- ChatGPT is a powerful tool that can assist in the coding process. By using machine learning algorithms to analyse and understand code snippets, ChatGPT can generate accurate and efficient code solutions. While it does not replace human programmers, it can significantly improve productivity and coding accuracy.
- After the success of OpenAI's ChatGPT, several companies have launched their own versions of chatbots to grab a share of the market.
- **META, which loves to monopolise the 'Internet Chat' world, has launched its new chatbot called LLaMA.** Meta Llama 3, Meta AI is one of the world's leading AI assistants, that we already have in our phone, for free. And it's starting to go global with more features. The users can use Meta AI on Facebook, Instagram, WhatsApp and Messenger to get things done, learn, create and connect with the things that matter to them.
- **Generative AI**, like other forms of AI, learns to take actions from past data. However, it goes beyond simply categorising or identifying data and **creates brand new content, such as text, images, and computer code, based on that training**. One famous example of generative AI is ChatGPT.
- ChatGPT is insanely good at generating human-like responses to queries and solving problems through textual conversations. On the other hand, Google is really good at finding information, videos, images, products and almost anything on the internet.

GPT has been used in a variety of applications in chat systems, including:

1. **Generating Responses for Chatbots:** GPT can be used to generate coherent and believable responses for chatbots, allowing them to engage in natural and engaging conversations with users.
2. **Language Translation:** GPT can be trained to translate text from one language to another, making it useful for chat systems that need

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to support multiple languages.

3. **Text Summarisation:** GPT can be used to generate concise summaries of long texts, which can be useful in chat systems for providing quick and concise responses to users.
4. **Dialogue Systems:** GPT can be used to power dialogues between chatbots and humans, allowing for more natural and engaging conversations.
5. **Sentiment Analysis:** GPT can be trained to classify the sentiment of text, which can be useful in chat systems for detecting and responding to negative or positive sentiment.
6. **Knowledge Base Creation:** GPT can be used to create a knowledge base for a chatbot by generating responses to a large number of prompts and questions.

### 16.5.2 Gemini – AI Model by Google

- Google has launched Project Gemini, an AI model aimed at **replicating human-like behaviour,** sparking interest and discussions about its implications.
- Gemini is designed **to enhance the capabilities of Google's AI-powered chatbot, Bard**, particularly in tasks involving planning.
- Developed by **Google DeepMind, the AI division**, Gemini emphasises its problem-solving skills in areas such as mathematics and physics, potentially leading to scientific breakthroughs.
- Concerns have been raised about the **potential consequences of AI**, including job displacement, the spread of misinformation, and the possibility of triggering harmful behaviours.

**Google DeepMind**, a subsidiary of Google specialising in artificial intelligence research, **introduced Gemini AI in December 2023.**

- This groundbreaking technology **represents a new generation of Large Language Models (LLMs).**
- It is accessible globally through Bard (Google's AI chatbot platform), some developer tools, and the new Pixel 8 Pro phones.
- It understands, explains, and generates high-quality code in popular programming languages (**Python, Java, C++, Go**).

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# About Google's Gemini AI

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## *Model Types*

| Variant | Purpose |
|---|---|
| **Gemini Ultra** | High-end, for data centres and research |
| **Gemini Pro** | Integrated into Bard and Google apps |
| **Gemini Nano** | Optimised for smartphones/devices |

## *Key Features*

- **Multimodal AI:** Understands and processes **text, code, images, audio and video.**
- **Real-Time Response:** Can **speak**, **see** and **react live**.
- **5× faster** than GPT-4 in many tasks.
- **First model to beat humans on** Massive Multitask Language Understanding **(MMLU)** with **90% score.**

**Massive Multitask Language Understanding (MMLU)** is a ground-breaking approach in the field of natural language processing (NLP) and artificial intelligence (AI) that aims to enhance how machines interpret and process human language significantly.

## *Benchmark Performance*

| Domain | Gemini Ultra vs GPT-4 |
|---|---|
| **Math (GSM8K, MATH)** | Ultra outperforms GPT-4 |
| **Code Generation (Python)** | Ultra leads |
| **Reasoning (BBH, DROP, HellaSwag)** | Ultra scores higher |
| **Academic Benchmarks** | Ultra leads in 30/32 tests |

## *Applications and Advantages*

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- **Scientific Discovery:** Recognises patterns, proposes hypotheses.

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- **Faster AI Model Training:** Real-time learning with massive datasets.
- **AI Integration:** Gemini Pro is integrated with **Bard**, **Google Workspace**, etc.
- **Free Advantage:** Gemini Pro (free) > ChatGPT-3.5 (free version)

## CHATGPT-4

### Strengths

- Excellent natural language understanding and generation.
- Strong at summarizing text, writing different creative text formats and answering questions.
- Conversational fluency and structured responses.
- More reliable in text-based tasks and research.

### Conclusions

**Choose ChatGPT-4 for:**

Text-based tasks, writing, creative content, answering questions and situations where reliable and clear language is crucial.

## GOOGLE GEMINI

### Strengths

- Multi-modal capabilities, including text, images, audio, video and code.
- Strong in complex reasoning, understanding and research.
- Can access real-time information from the web.
- Better for creative tasks and image generation.

### Conclusions

**Choose Google Gemini for:**

Tasks involving multiple data types (text, images, video), complex reasoning, **real-time information access** and creative tasks.

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### 16.5.3 Deepfake

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- **Deepfakes** are **AI-generated fake media content**—images, audio or videos—that convincingly mimic real people or events.
- The term combines:
  - ❑ **"Deep"** from deep learning (a branch of AI)
  - ❑ **"Fake"** meaning not real or truthful

- **Deepfakes are digital media** – video, audio and images **edited and manipulated using Artificial Intelligence (AI)** to **inflict harm on individuals and institutions.**
- **AI-generated synthetic media or deepfakes** have clear **benefits** in certain areas, such as accessibility, education, film production, criminal forensics and artistic expression.
- It can be **exploited** (hyper-realistic digital falsification) **to damage the reputation**, fabricate evidence, defraud the public and undermine trust in democratic institutions with **fewer resources** (cloud computing, AI algorithms and abundant data).
- **Deepfakes, powered by Artificial Intelligence (AI), have emerged as a grave threat to democracy and social institutions worldwide.** These manipulated media pieces, often disseminated through social media platforms, pose significant challenges to societal integrity and individual privacy.

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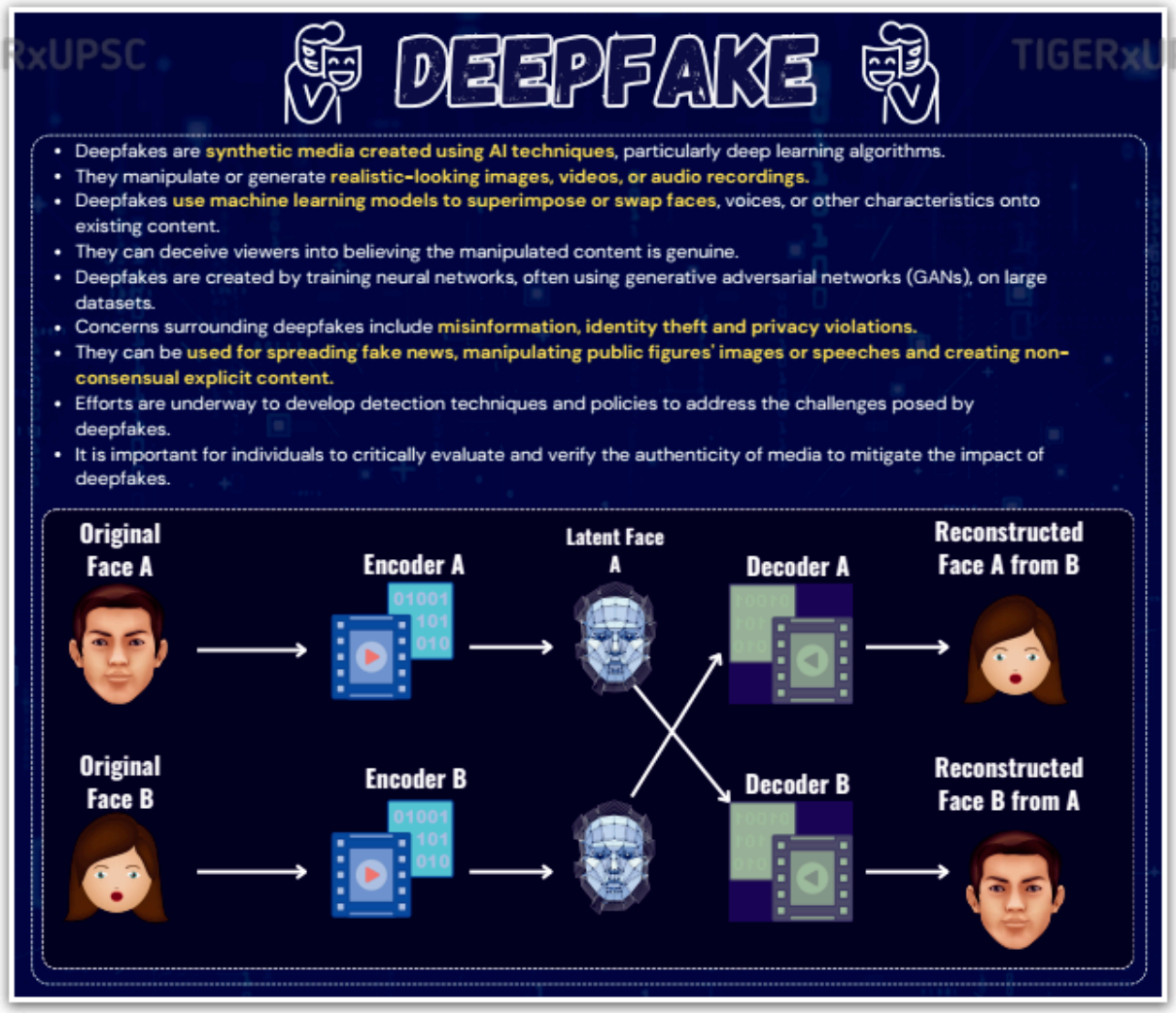


- Deepfakes refer to **synthetic media crafted** through AI techniques, **manipulating visual and audio content to deceive or mislead individuals**. Created using **generative adversarial networks (GANs),** deepfakes blend real and fake elements seamlessly, often collected without consent from various online sources.

## *How Do Deepfakes Work?*

### 1. Training Phase:

(i) A **neural network** is trained on hours of real footage/images of a person.

(ii) The AI learns **facial expressions, angles, lighting variations,** etc.

### 2. Synthesis Phase:

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(i) AI merges the learned facial features with **computer graphics** and superimposes them on another person's body.
(ii) The result: A **realistic but entirely fake video, image, or audio.**

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## *Uses and Misuses*

While deepfake technology holds potential for positive applications such as **film dubbing, education, art, message amplification, and public safety**, its misuse is rampant.

From **spreading propaganda and fake news to blackmailing individuals and tarnishing reputation**, deepfakes undermine trust in media and **disrupt societal harmony.**

| Legitimate Uses | Misuses |
|---|---|
| Film dubbing, animation | Celebrity porn |
| Gaming, virtual reality | Scams & fraud |
| Education, accessibility (e.g. lip-syncing) | Election manipulation |
| Digital resurrection (revival) in films | Social engineering & identity theft |
| Satire (blandness) & entertainment | Disinformation campaigns |

## *Why Deepfakes Are Dangerous*

- **Hard to Detect:** Unlike photoshopped images or basic edits, deepfakes are generated with high precision.
- **Erosion of Trust:** Threatens the **credibility of audio-visual evidence**.
- **Legal and Ethical Concerns:** Compromises **privacy, consent and digital rights.**

## *Concerns and Challenges*

1. **Procedural Concerns:** Deepfakes pose significant procedural challenges, including the spread of misinformation, election interference, and reputational damage to individuals and organisations.

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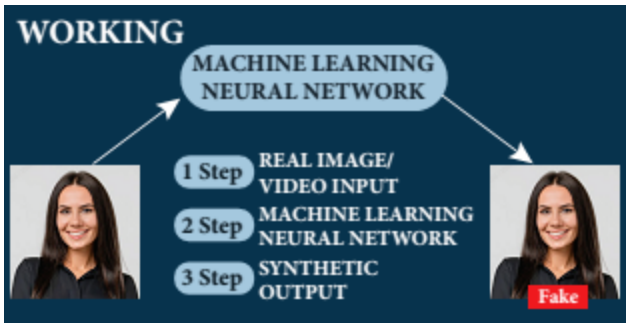
2. **Legal Concerns:** Legal frameworks addressing deepfakes vary globally, with complexities surrounding defamation claims and freedom of expression. While existing laws offer some protection, comprehensive legislation specific to deepfakes is necessary to address emerging threats effectively.
3. **Ethical Concerns:** Ethically, deepfakes **raise questions about consent, privacy and the proliferation of false narratives**. Their **misuse, including revenge porn and hacking facial recognition systems**, underscores the urgent need for ethical guidelines and responsible AI practices.




## *Global Approaches to Combat Deepfakes*

1. **India: While India lacks specific legislation on deepfakes, existing laws address certain aspects of their misuse.** Calls for an ethical AI framework and the regulation of personal data highlight the need for comprehensive legal measures.
2. **United States:** The **Deepfake Task Force Act and bipartisan efforts** aim to **counter deepfake technology and safeguard national interests.**
3. **China:** China's **comprehensive regulation on deep synthesis, effective from 2023,** aims to curb disinformation and ensure transparency in content creation and usage.
4. **Tech Companies:** Major tech companies have introduced measures to combat deepfakes, including **content identification tools and metadata embedding**, yet vulnerabilities persist.

(Refer **Infographic – Deepfake**)

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### 16.5.4 World's AI Safety Summits

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**AI Safety Summits** are international meetings hosted by States to discuss the safety and regulation of artificial intelligence, particularly advanced AI systems.

#### *I. First AI Safety Summit*

The **first AI Safety Summit** was convened by the United Kingdom at Bletchley Park in November 2023.

**Highlights: Bletchley Park Declaration**

- The first-ever global pact addressing frontier AI risks.
- Reflects consensus among major AI players like the United States, China, India and the EU.
- Acknowledges AI's potential for enhancing human well-being while recognising associated risks.
- Calls for international cooperation and collaboration among various stakeholders.
- Establishes a regular AI Safety Summit for ongoing dialogue and collaboration.

#### *II. Second AI Safety Summit*

On 21–22 May 2024, the Republic of Korea and the United Kingdom cohosted the second AI summit at Seoul (South Korea). The Seoul summit saw 16 top AI companies making voluntary commitments to develop AI transparently.

#### *III. Third AI safety Summit*

The **third AI safety Summit** was held on 10–11 February 2025, at the Grand Palais in Paris. It was a global event focused on artificial intelligence, with the goal of establishing scientific foundations, solutions and standards for more sustainable AI. The summit was co-chaired with India and brought together heads of state, industry leaders, researchers and civil society representatives from around the world.

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### *Key objectives of the summit included*

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1. **Providing access to independent, safe and reliable AI**
2. **Developing more environmentally friendly AI**
3. **Ensuring global governance of AI**

Fifty eight countries, including India, China, Brazil, France and Australia, signed a **joint statement on Inclusive and Sustainable Artificial Intelligence for People and the Planet** at the AI Action Summit in Paris. The statement was **not signed by the United States and the United Kingdom.**

## 16.5.5 S.A.R.A.H.

**(Smart AI Resource Assistant for Health)**

- **WHO launched S.A.R.A.H.: Smart AI Resource Assistant for Health**
- **Theme:** "My Health, My Right" – *World Health Day Initiative*

### *What is S.A.R.A.H.?*

A **Generative AI-powered digital health assistant** developed by WHO to:

- Provide **24/7 multilingual health information**
- Enable **empathetic, personalized conversations**
- Ensure **equitable access** to evidence-based healthcare guidance
- Supports **8 languages**, available on multiple platforms

### *Key Features and Capabilities*

**Health Information Access**

- Preventive care tips (lifestyle, diet, vaccines)
- Mental health support
- Non-communicable diseases: cancer, diabetes, heart, lung
- Tobacco cessation, physical activity, stress management

**Real-Time, Human-Like Interaction**

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- Context-aware and empathetic
- Tailored advice without judgment

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- Available anytime, anywhere

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**AI-Driven Scalability**

- Based on **Soul Machines' Biological AI**
- Accessible across devices and digital channels

**Public Health Decision Support**

- Informs users of major health risk factors
- Aligns with **WHO's latest guidelines**
- Promotes actionable, preventive behaviour

**Strategic Impact on Global Health**

- Reduces healthcare inequality
- Combats health misinformation
- Boosts global health literacy
- Empowers self-care and decision-making
- Strengthens public health infrastructure

**Ethical Considerations**

WHO emphasises:

- **Equity** in access to digital health
- **Data privacy and safety**
- **Accuracy and reliability**
- **Bias mitigation**

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Ongoing research and ethical evaluation are key to its future.

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## 16.6 Artificial Intelligence in India

India ranks third or fourth (depending on China's position in AI) in the size of AI clusters. India has already taken a lead in this regard, thanks to several AI start-ups. In this industry, the private sector has been vibrant.

According to a Canada based company's report, Global AI Report 2019, India stood at the ninth position in terms of the number of the AI specialists working in the field. The US, China and the UK topped the list.

However, the 2018 Union Budget, presented on 1 February 2018 signalled the government's intention to push for the development of machine learning and artificial intelligence.

The government allocated $480 Mn to fifth generation technology startups like artificial intelligence, machine learning (ML), Internet of Things (IoT), 3D printing and blockchain.

The NITI Aayog has even been asked to prepare a roadmap for a nationwide programme in artificial intelligence, including research and development of its applications.

The government mandated NITI Aayog to establish the National Program on AI with a view to guiding research and development in new and emerging technologies.

The government has also promised to offer incentives to start-ups and venture funds that undertake application-oriented research on artificial intelligence across key sectors, including banking, insurance, education, health, retail and transportation. The Department of Service and Technology is also looking into setting up centres of excellence that will push investments in 'research, training and skilling in robotics, artificial intelligence, digital manufacturing, big data analysis, quantum communication'. Different types of AI start-ups in India in the last few years are mentioned in **Table 16.2**.

**Table 3.2:** AI Startups in India

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| | |
|---|---|
| **Active AI** | Providing conversational banking service, aims to bring automation. Natural language processing and advanced machine intelligence are the focal areas. |
| **AIndra Systems** | Working on devices that can detect and identify people and objects. |
| **Artificial** | Deep learning. |
| **Artivatic Data labs** | Ingestible computing, which is used in making very tiny robots inside a capsule that can be ingested to collect data from the body. |
| **Arya.AI** | Neural network that will resemble human brain in functioning. |
| **Fluid.AI** | Facial recognition; present in problem-solving for banking, finance, web and government sectors. |
| **Morph.AI** | Chatbots for marketing and customer services. |
| **Sig Tuple** | In medical diagnosis, AI-powered visual analysis of medical data |

India's Defence Research and Development Organisation (DRDO) has developed a robot called Daksh, which can destroy life-threatening objects. Apart from applications like these, AI-powered robotics has been developed for entirely different uses.

It is estimated that AI will add 957 billion dollars to India's GDP by the year 2035 boosting India's annual growth by 1.3% points.

In 2019, NITI Aayog circulated the cabinet note to establish a cloud computing platform called AIRAWAT (Artificial Intelligence Research, Analytics and Knowledge Assimilation Platform).

In Budget 2019–20, the government had announced setting up of a National Sports Education Board under "Khelo India" project to prepare youth for new age skills, Artificial Intelligence, IoT, Big Data, 3D Printing, Virtual Reality etc.

Apart from these vibrant private sector clusters, India's Centre for Artificial Intelligence and Robotics (CAIR), a laboratory of the Defence Research and Development Organisation (DRDO), has developed a number of products that are powered by AI. Many of them are under development. Most of the products are in the areas of robotics, autonomous systems and

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intelligent systems that would do analyses without any human effort to give quick results.

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Note: **AI Supercomputer 'AIRAWAT' puts India among top supercomputing league. 'AIRAWAT' ranks 75th in top 500 Supercomputing List**.

**Interdisciplinary Cyber Physical Systems (IM-ICPS) has suggested the following four-tier framework for promoting AI research:**

(a) ICON (International Centres of New Knowledge)
(b) CROSS (Centre for Research On Sub-Systems)
(c) CASTLE (Centre for Advanced Studies, Translational research and Leadership): focusing on development and deployment of application based research
(d) CETIT (Centre of Excellence in Technology Innovation and Transfer)

## *AI Research in India*

Two-tier integrated approach to boost both core and applied research in AI is proposed:

1. **COREs (Centres of Research Excellence in Artificial Intelligence):** COREs will focus on core research of AI, and will take on the mantle of executing the responsibilities of both ICON and CROSS as per the IM-ICPS framework.
2. **ICTAI (International Centre for Transformational Artificial Intelligence):** ICTAIs willprovide the ecosystem for application based technology development and deployment, and will take on the mantle of executing the responsibilities of both CASTLE and CETIT as per the IM-ICPS framework

## *Academic Institutes and Centres for AI*

- Centre for Artificial Intelligence, IIT Kharagpur
- Centre for Artificial Intelligence & Robotics (CAIR), DRDO
- Robert Bosch Centre for Data Science and Artificial Intelligence, IITM
- The Artificial Intelligence Group, IISc, Bangalore
- Department of AI, IITH
- Academia-industry Collaboration on Artificial Intelligence

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- Laboratory of Statistical Artificial Intelligence and Machine Learning (LSAIML), IITR

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### CHATBOT ASKDISHA2.0

**Indian Railways AI Chatbot AskDisha 2.0 powered by CoRover. AI, an improved version of the original AsDisha Chatbot supports voice command and** converse with customers in **the Hindi, English and Hinglish languages. It also offers users a range of services like booking tickets and getting refunds and is available on both IRCTCs website and mobile app.**

**Keynote**

- It is **an Artificial Intelligence-based chatbot**.
- It is **a special computer programme** designed to simulate conversation with users, especially over the Internet
- Initially launched in the English language in October 2018.
- Developed by **Indian Railway Catering and Tourism Corporation (IRCTC)**.
- The aim is **to resolve queries of railway passengers over the Internet pertaining to various services offered**.

**Services provided**

- Since its initial launch, passengers seeking help on the reservation of tickets, cancellation, enquiry of refund status, fare, PNR search, train running status, enquiry about retiring rooms and tourism products have been benefited.

## 16.7 Global Partnership on AI (GPAI 2024) / New Delhi Declaration

- India, as the lead chair of **GPAI in 2024**, hosted the summit, where the '**New Delhi Declaration**' was adopted by the European Union.
- **Key discussions revolved around national AI initiatives**, such as **India's AI portal AIRAWAT**, and **concerns regarding the misuse of deepfake technology.**
- **YUVAi, a youth-focused AI initiative**, was prominently featured, showcasing AI models and solutions developed by winners and startups.

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- The **Prime Minister emphasised using AI to make digital services accessible in local languages to enhance digital inclusion.**
- The summit featured sessions on **responsible AI, data governance, future of work and innovation and commercialisation**, along with various side events like industry panel discussions and workshops.

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## Delhi Declaration of the GPAI

- Acknowledges **the need to harness opportunities and mitigate risks associated with AI development and deployment.**
- **Emphasises upholding human dignity, rights, and democratic values**, fostering trust, transparency, accountability, and inclusiveness in AI.
- **Recognises AI's potential to contribute to UN Sustainable Development Goals** and global challenges.
- **Supports international cooperation on AI research, innovation, and policy**, and advocates for a comprehensive framework of shared principles for safe and trusted AI.
- Endorses India's proposal for a **Global Digital Public Infrastructure Repository (GDPIR)** to share digital public goods.
- Calls for **further dialogue on AI governance and ethics among** stakeholders.

## AIRAWAT

- AIRAWAT is a **cloud computing platform established by NITI Aayog** to **advance AI research and applications in sectors like education, health, and agriculture**.

## Deepfake

- Deepfakes are **synthetic media created by using AI** to **manipulate or generate visual and audio content, often with deceptive intentions**.
- Deepfakes utilise **generative adversarial networks (GANs)** to produce realistic-looking fake images or videos.

## YUVAi Initiative

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- YUVAi, launched by the **National e-Governance Division (NeGD)** in **partnership with Intel India**, **aims to educate school students** about AI and empower them to become human-centric designers and users of AI.
- The initiative provides **applied learning experiences for students to understand AI's potential in addressing critical problems and promoting inclusive development**.

## 16.8 BharatGen Programme

**Ministry of Science and Technology** launched BharatGen, a **pioneering initiative in generative Artificial Intelligence (AI).**

### *What Is BharatGen?*

BharatGen is a **Multimodal Large Language Model (LLM)** initiative designed to develop generative AI systems capable of producing **high-quality text, audio and image content** in **multiple Indian languages**.

- **Project Objective:** To strengthen public service delivery, enhance citizen engagement and position India as a global leader in generative AI.
- **Implementing Agency: Technology Innovation Hub (TIH)** Foundation for IoT and IoE – Internet of Every things (Section-8 non-profit company by **IIT Bombay**) under the **National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)**.
- **Project Timeline:** Expected to be completed by **July 2026** (two-year timeline).

### *Key Features of BharatGen*

- **Multi-lingual and Multi-modal Capabilities:** Supports content generation in diverse Indian languages across text, speech and images.
- **Trained on Indian Datasets:** Utilises datasets tailored to India's cultural, linguistic and societal contexts.

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- **Open-Source Platform:** A transparent and publicly accessible AI development ecosystem.
- **AI Research Ecosystem:** Fosters innovation through a collaborative network of researchers, developers and institutions.

### *Strategic Research Pillars of BharatGen*

1. **Data and Model Innovation:** Enhancing data efficiency and model performance.
2. **Advanced Capabilities and Applications:** Expanding AI utility for societal impact.
3. **Interactivity and User Experience:** Personalising AI interactions with transparency and explainability.
4. **Ethical AI and Privacy:** Ensuring fairness, privacy and decentralised model training.
5. **Integration and Future Technologies:** Combining AI with technologies like **IoT and AR/VR**.
6. **Evaluation and Benchmarking:** Establishing robust frameworks for performance and progress assessment.

### *Bharat Data Sagar*

- A central repository to store vast **India-centric datasets** that fuel BharatGen's AI models.
- Ensures the AI systems are **deeply rooted in Indian realities**—from languages to societal needs.

## BHASHINI – AN AI TOOL

- Bhashini, an AI Tool is a real-time Artificial Intelligence-based **translation tool.**
- **Nodal Ministry:** It falls under the jurisdiction of the Ministry of Electronics and Information Technology (MeitY).
- **Objective:** The tool facilitates communication between individuals speaking different Indian languages by enabling real-time translation.

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### 'Bhasadaan' Section

BhashaDaan is an initiative to crowdsource language inputs for multiple Indian languages as part of Project BHASHINI. It calls upon citizens to help build an open repository of data to digitally enrich his/her own language.

- **Suno India:** Users can contribute by transcribing audio content or validating transcriptions made by others.
- **Bolo India:** Participants can donate their voice through sentence recordings and validate audio recordings contributed by others.
- **Likho India:** Users can contribute by translating provided text and validating translations submitted by others.
- **Dekho India:** Participants can contribute by typing text seen or labelling images and validate images contributed by others.

### Importance of Bhashini AI Tool

- **Digitisation of Services in Local Languages:** With India's diverse linguistic landscape, Bhashini encourages the development of digital services in local languages, reducing the dominance of English content online.
- **Potential Roadmap for Local Languages:** The project presents an opportunity to create a roadmap for internet access in local languages, as suggested by a report from Analytics India Magazine.
- **Promoting Financial Inclusion:** Bhashini has the potential to promote financial inclusivity by providing digital financial services in native languages, extending accessibility to a broader population.

## 16.9 AI in Education Sector

### 16.9.1 Difference between AI Tutor and Robot Teacher

An **AI tutor typically refers to software programs or algorithm** designed to provide a personalised learning experiences and assistance to students.

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These tutors often utilise artificial intelligence to **adapt to individual learning styles**, offer feedback, and track progress, usually through digital platforms or applications.

On the other hand, robot teachers, which have not yet been developed as physical robots programmed to perform teaching tasks in educational settings. They have the **capacity to engage with students, impart lessons, and evaluate comprehension**. They are already being used in some schools, such as Kerala's Makerlabs Edutech, which recently unveiled India's first AI-generated school teacher robot, named Iris. However, robot teachers are not yet widely used in most parts of the world.

### 16.9.2 AI and National Education Policy (NEP) – 2020

In view of the immense potential of artificial intelligence in education, the NEP-2020 has recommended its integration into the education system at all levels in India. It is believed that such an integration would transform education by making it more efficient, effective and accessible to students worldwide.

In this regard, the **CBSE has already introduced AI and the Internet of Things (IoT) in the school curriculum from classes VI to X**. The purpose of introducing AI in the school curriculum is to usher in more efficiency, personalised learning, streamline administrative tasks, etc.

**Figure 16.3** Transition from Traditional Education to AI-based Education

Even **India's first AI university has already started functioning in Karjat, near Mumbai**. This university shall impart degree and postgraduate specialised artificial intelligence courses. In the future, the focus of both school and higher education will be on developing AI skills in

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students. These skills refer to the competencies and knowledge required to develop, implement and manage AI systems and applications.

Some of the top AI skills that are going to be in demand are programming languages (Python, R), machine learning and deep learning, data analytics, maths and statistics, etc.

Note: **India's first AI university has started in Mumbai from 1 August 2023.**

## 16.10 AI in Medical Science

Artificial intelligence (AI) refers to computer programs that can learn from and make decisions based on data. These programs are designed to perform tasks that typically require human intelligence, such as recognising patterns, analysing images and making predictions.

### Potential of AI in Healthcare

#### *1. Medical Imaging*

- AI in MRI and CT imaging allows deep learning of every organ, aiding in early disease detection.
- **AIIMS in Delhi uses AI to detect oral cancer in its early stages**.

#### *2. Analysis of Electronic Health Records (EHRs)*

- AI can identify patients at high risk of chronic diseases, improving preventive care.

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- **Apollo Hospitals collaborated with Microsoft to create an AI-powered cardiovascular disease risk score**.

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### *3. Addressing the Healthcare Crisis*

- AI can mitigate the shortage of healthcare professionals and improve accessibility to healthcare across the country.
- India has only 64 doctors per 100,000 people, well below the global average of 150.

### *4. Reducing Out of Pocket Expenditure*

- Using AI for diagnoses can significantly reduce treatment costs and improve health outcomes.
- AI diagnoses **may reduce treatment costs by up to 50%** and improve health outcomes by 40%.

## AI Healthcare Startups in India

- **Tricog:** Offers **cloud-connected devices for ECG and echocardiogram analysis.**
- **Niramai:** Patented device **Thermalytix detects early-stage breast cancer.**
- **PharmEasy: Utilises AI for real-time data and user communication.**
- **HealthifyMe: Monitors calorie intake and provides dietary recommendations** using AI.

## Significance of Integrating AI in Healthcare

### *1. Accuracy in Patient Treatment*

- AI-powered **RPA (Robotic Process Automation) solutions minimise errors and streamline workflows for accurate patient treatment.**

### *2. Early Diagnosis of Diseases*

- AI in wearable devices continuously monitors vital signs, **identifying potential health risks or early signs of diseases.**

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### 3. Promoting Medical Value Travel (MVT)

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- AI contributes to medical value travel by improving treatments and procedures, **attracting patients from abroad.**

### 4. Precision Medicines

- AI algorithms **personalise treatment plans** based on **individual patient data**, improving efficiency and efficacy.

### 5. Quick Disease Prediction

- AI predicts disease outbreaks, **optimises treatment plans**, and revolutionises drug discovery, making healthcare more personalised and effective.

### 6. Healthcare Management

- AI **optimises hospital operations, predicts patient admissions, manages bed allocations**, and streamlines supply chain logistics.

### 7. Chatbots and Virtual Assistants

- Enhance patient engagement and streamline administrative tasks in healthcare facilities.

## 16.11 AI in Defence

### Evaluating Trustworthy AI (ETAI) Framework – Indian Armed Forces

- **Launched by** the **Chief of Defence Staff** to **assess and regulate AI integration** in India's defence sector.
- Developed as a **risk-based framework** to ensure **reliable, responsible and secure AI** use in critical military systems.

### Five Core Principles of Evaluating Trustworthy AI (ETAI)

1. **Reliability and Robustness (Durability)**
2. **Safety and Security**

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3. **Transparency**
4. **Fairness**
5. **Privacy**

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## Use of AI in Defence: Applications

### *Supporting Functions*

- **ISR—Intelligence, Surveillance** – Observation**, Reconnaissance** – Early Analysis**)**
- Navigation
- Decision support (Command and Control – C2)

### *Offensive Functions*

- **Target selection** and **engagement** (e.g., drone swarms)
- **Cyber operations** (AI-powered malware, hacking)
- **Precision (accuracy) targeting** in active combat

## Key Use Cases and Examples

| Area | Application | Example |
|---|---|---|
| **Autonomous Weapons** | Seek & destroy targets | **Israeli Harop, Harpy drones** |
| **Target Recognition** | Identifies missile systems, avoids civilians | **Shahed-136 drones** in Russia-Ukraine war |
| **Data Analysis** | Real-time surveillance processing | **Project Maven**, USA |
| **Simulation & Training** | Virtual training modules | Sukhoi 30MKI flight simulators |
| **Crime Prediction** | AI in C4ISR for attack prediction | (Adversary Network Analysis Tool) **ANANT Tool by BEL** (Bharat Electronics Ltd.) |
| **Cyber Defence** | Preemptive detection of attacks | **Project Seeker**, Indian Army |

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## Challenges of Using AI in Defence

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### *Security and Legal Concerns*

- Use by **non-state actors** (e.g., Islamic State AI manuals)
- **Generative malware** (e.g., *BlackMamba* evading EDR)
- **No binding international law** on AI weapon use

**Non-state actors** are individuals or organisations that exert influence in international relations but are not officially part of any government or state.

### *Ethical and Human Rights Issues*

- **Privacy Violations** (e.g., China's surveillance of Uyghurs)
- **Automation Bias:** Risk of hitting civilians
- **Proportionality and Predictability:** Difficulty assessing military-legal balance
- **Dehumanisation:** Quick, AI-led attacks increase collateral damage tolerance

### *Indian Initiatives*

- **Defence AI Council (DAIC)** and **Defence AI Project Agency (DAIPA)** – established 2019
- **Strategic Implementation Task Force** – chaired by C. Chandrasekaran
- **AI in Defence (AIDef)** symposium 2024 – 75 AI tools launched
- **AI-powered systems** under development in surveillance, simulation, targeting

### *International Regulation Efforts*

| Initiative | Purpose |
|---|---|
| **UN GGE (2016)** | On Lethal Autonomous Weapons (CCW) |
| **UN First Committee Resolution (2023)** | Limits algorithmic control over life decisions |
| **UNIDIR Guidelines (2024)** | National AI strategies for defence |

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| | |
|---|---|
| **ICRC Advocacy** | Calls for global legal norms on AI weapons |

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## Way Forward

### *National Strategy*

- Strengthen **public–private AI collaboration**
- Invest in both **defensive and offensive cyber warfare**

### *Global Approach*

- Establish **international law** for AI use in warfare
- Develop an **arms control regime** for AI weapon systems
- Codify **responsible use principles** via multilateral platforms

## 16.12 Advantages of Artificial Intelligence

1. There is utilisation of machines for taking on complex and stressed work that would else be undertaken by humans.
2. Machines finish the task sooner than a human if assigned with the same task.
3. They can be used in discovering unknown landscapes and outer space which is useful in our daily life.
4. As the task is being done by the machines, it causes less stress and injuries to humans.
5. Providing assistance to people who are visually and hearing impaired.
6. Through artificial intelligence, understanding complex software can be made easier.
7. Less number of errors and defects.
8. Time and resources are not wasted but smartly used to achieve the goal.

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## 16.13 Disadvantages of Artificial Intelligence

1. AI lacks human sensation; human qualities are ignored.
2. AI has the ability to replace a human job. This gives rise to humans feeling insecure of losing their jobs.
3. A machine can replace human capabilities and therefore, it can promote inferiority feelings among workers and staff.
4. There can be malfunctioning in artificial intelligence and there is possibility of the reverse happening as they are programmed to do so.
5. The younger generations may be corrupted.
6. No information filtration.
7. There can be misuse of such technology and thereby, causing a mass-scale destruction.

## 16.14 Is AI moving towards Super Intelligent Machines?

The question is pertinent since the revival of intelligent machines has not just created euphoria over its potential benefits but has also given rise to fears that it would replace humans.

Even as robots are being made for performing simple human tasks that do not require complex problem-solving, there is a worry that planet Earth would turn into a place where humans would be slaves to machines that would be more intelligent than them. Artificial Intelligence is definitely trying to create machines and systems that will be more intelligent than their predecessors. So, this march of AI may lead to a future where machines are extremely intelligent but the fears may turn out to be exaggerated.

### PROJECT BRAINWAVE

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Microsoft has launched 'Project Brainwave', a deep-learning acceleration platform for real-time AI. It uses the massive field-programmable gate array (FPGA) infrastructure.

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## 2024 & ONWARDS: THE YEAR OF AI-ON-PHONE

On 17 January 2024, a significant milestone in mobile phone evolution occurred **at a global event in San Jose**, in the heart of the US Silicon Valley with **Samsung's unveiling of the Galaxy S24**, dubbed the **world's first 'AI Phone'**. This handset integrates artificial intelligence directly into the device itself, rather than relying on cloud resources. Notably, it is **manufactured in India** at the Noida factory, the world's largest smartphone production facility with a capacity of 120 million handsets annually.

The **'AI Phone' introduces innovative features such as Generative AI-powered photo editing**, an enhanced Chat Assist for natural conversational tones, and Transcript Assist using AI and Speech-to-Text for transcription, summarisation, and translation of recordings. Additionally, it features a gesture-driven Circle to search with Google, enabling users to interact directly with the screen for precise search results.

This advancement is made possible by new editions of microchips, such as Qualcomm's Snapdragon Gen 3, which integrate AI capabilities directly into the hardware. These chips empower the Galaxy S24 to perform tasks previously reliant on web tools like ChatGPT, setting a new standard in mobile AI technology.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. Consider the following statements regarding AI Action Summit held in Grand Palais, Paris in February 2025: [UPSC (Pre) 2025]**

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i. Co-chaired with India, the event builds on advances made at the Bletchley Park Summit held in 2023 and the Seoul Summit held in 2024.

ii. Along with other countries, US and UK also signed the declaration on inclusive and sustainable AI.

Which of the statements given above is/are correct?

(a) i only
(b) ii only
(c) both i and ii
(d) neither i nor ii

**2. With the present state of development, Artificial Intelligence can effectively do which of the following?** **[UPSC (Pre) – 2020]**

1. Bring down electricity consumption in industrial units.
2. Create meaningful short stories and songs.
3. Disease diagnosis.
4. Text-to-speech conversion.
5. Wireless transmission of electrical energy.

Select the correct answer using the code given below:

(a) 1, 2, 3 and 5 only
(b) 1, 3 and 4 only
(c) 2, 4 and 5 only
(d) 1, 2, 3, 4 and 5

**Answer Key**

| | |
|---|---|
| 1. (a) | 2. (b) |

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# CHAPTER 17
# ROBOTICS

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'**Daksh**' is an electrically powered and remote-controlled robot. It's a military robot used to locate, handle and destroy risky objects safely and can even climb stairs. The main role of this military robot is to recover improvised explosive devices.


17.7.1 Defence


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In October 2017, Sophia was also given **citizenship of Saudi Arabia** – the first robot to receive citizenship of any country.

**17.13 Biorobotics**

## 17.1 Historical Background of Robotics

In 1950s Karl Capek in his play *Rossum's Universal Robots* indicated the origin of a superior race that has intelligence similar to that of humans. Afterwards, Issac Asimov launched his robot laws and lastly Eric Elenberger, who is the father of robotics, pioneered real-time robots in the world.

A robot is an automatic control machine that substitutes human efforts, despite the fact that the robots may or may not look like in appearance to human being or performing function in the similar manner as that of human being. In the field of Science and Technology, robots deal with the design, manufacture and application and make use of computer for their manipulation and processing.

**HISTORY OF ROBOTICS**

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In 1928, Japanese robot **GAKU TENSORU** was manufactured. In 1952, IBM's first commercial computer IBM701 was introduced, having a numerically controlled machine. The development of NC machine tools has been a turning point in the development of robotics. In 1954, George C. Devol invented the first truly modern digital Robot and this is called **UNIMATE**.

**WABOT-1** was created by George Devol and constructed by Unimation, the first robot manufacturing company. Waseda University initiated the WABOT project in 1976, and in 1972 completed the WABOT-1, the world's first full- scale humanoid intelligent robot.

Then in 1956, John Macarty and Marwin Minsky coined the phrase AI. In 1973, **FUMULUS** Robot was manufactured by Germany. It was the **first 6 degree of freedom of computer; those can move 360°.** In 2000, HONDA built a humanoid robot **ASIMO**, that walked like a human being.

## 17.2 Definition

According to the Robot Institute of America, 1979, 'A robot can be defined as a reprogrammable, multifunctional manipulator designed to move parts, tools, material or specialised devices through various programmed motions for the performance of a variety of tasks'.

## 17.3 Laws of Robotics

Sir Isaac Asimov gave the laws of robotics;

**First Law:** A robot must not harm a human being or, through inaction, allow one to come to any harm.

**Second Law:** A robot must always obey human beings unless it is in conflict with the first law.

**Third law:** A robot must protect itself from harm unless that is in conflict with the first and/or second laws.

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## 17.4 Robotics in India

In ancient times, a robotic water organ was invented by Ctesibius, an Alexandrian inventor and mathematician. Already, the number of robots in use worldwide multiplied three-fold over the past two decades, to 2.25 million. Trends suggest the global stock of robots will multiply even faster in the next 20 years, reaching as many as 20 million by 2030. Due to automation, there is a major impact of robotics on the economy of the world. **'Automation'** pertains to robots in which a computer programme dictates its actions. A robot comprises a repertoire of sensors and actuator, together with a **'Brain'** (a microprocessor) and communication capability.

(Refer **Infographic – Robotics**)

## 17.5 Components of Robots

### 17.5.1 Structure

It is mostly mechanical and usually referred to as a kinematic (Kinematics as a field of study is often referred to as the 'geometry of motion') chain formed of links, actuators and joints, which can allow one or more degrees of freedom (in particular, any of the directions in which independent motion can occur). Robots used as manipulators have end effectors mounted on the last link that could be a welding device or a mechanical hand used to manipulate the environment.

### 17.5.2 Essential Components of Humanoid

- Hands of robots are called **End effectors.**
- Arm of robots are called **Manipulators.**
- End effectors are termed as **grippers**, which consist of just fingers that open for picking and close for releasing a variety of small objects. It

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may be a vacuum gripper where large electronic components such as car and windscreen are manipulated.

- **Actuators** are the robot's muscles and joints.

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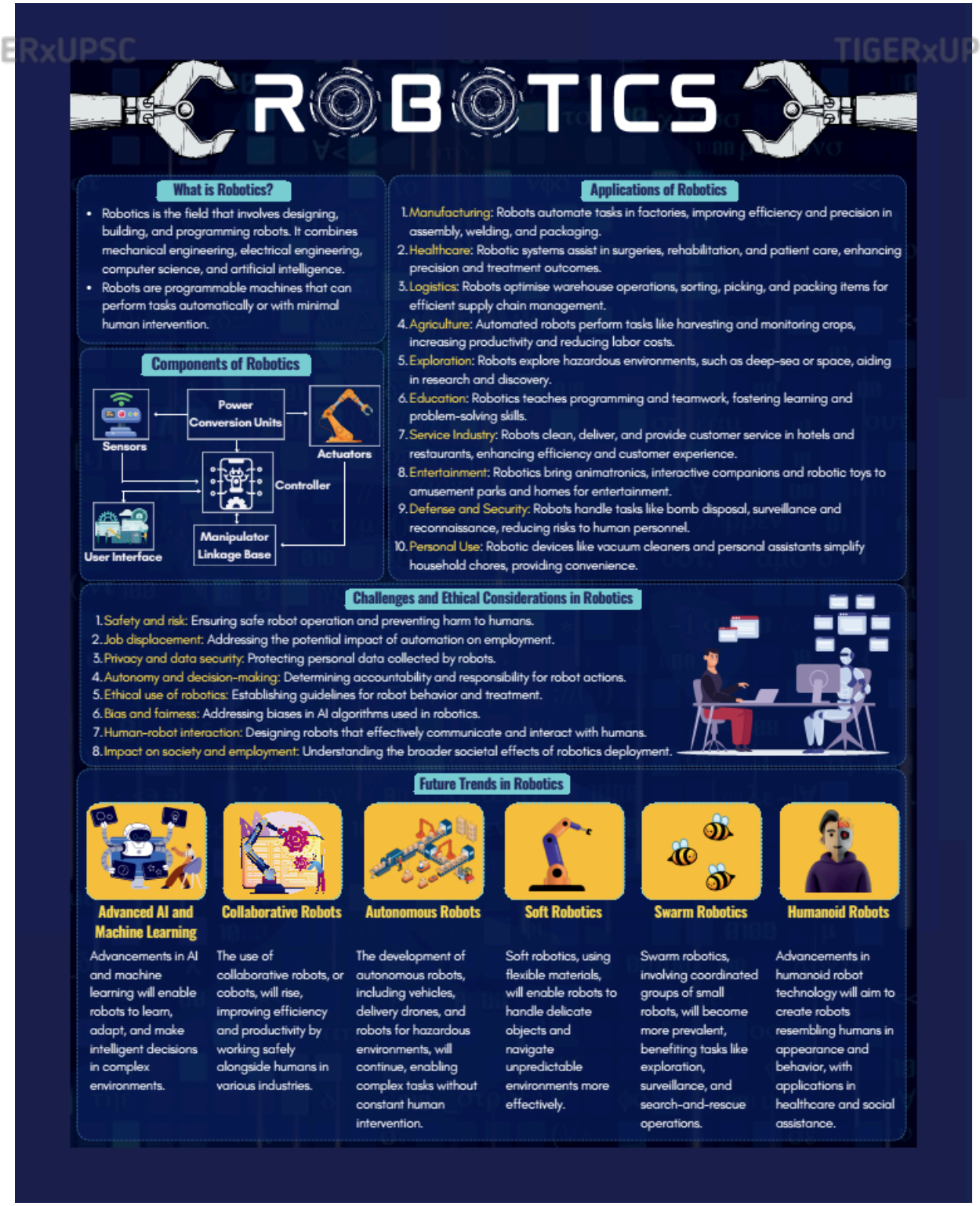


## 17.5.3 Planning, Controls and Sensing in Humanoid Robots

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The robots' planning and control are focussed on path planning, self-collision, detection and avoidance of obstacles by collecting information from the three sensors.

1. **Proximity Sensors:** Used in the measurement of relative distance in the surroundings between the sensors and objects.
2. **Exteroceptive Sensors:** These sensors assist in interacting with the outer world by providing information to robots about the surrounding environment.
3. **Proprioceptive Sensors:** Their usage is done in sensing humanoid robots' position, orientation and speed.

## 17.6 Robotics Systems and Robot Anatomy

A robot is a system, as it combines many subsystems that interact among themselves as well as with the environment in which the robot works.

1. Base: fixed or mobile
2. Manipulator arm
3. End effector
4. Drives or actuators
5. Controller
6. Sensors
7. Interfaces

1. **Base – Fixed or Mobile:** Without wheel robot is fixed.
2. **Manipulator Arm:** The manipulator is located relative to the ground on either a fixed or on a movable base. In 6 degrees of freedom robot, the first three links of the manipulator constitute the body and they help to place the end-effector at a desired location inside its work environment. At the end of manipulator, the wrist is located. Three rotational freedoms of the wrist are usually designated as pitch, yaw and roll.
3. **End-effectors:** Robot's end effector is the grip-per of end-arm tooling mounted on the wrist of the robot manipulator. It is designed to grip materials, parts and tools to perform specific tasks. Example:

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mechanical gripper, lifting gripper, vacuum cups, magnetic grippers, tool handling grippers, special grippers.

4. **Actuator:** Actuator is a component that converts energy extracted out of fluid, i.e., electric current to mechanical work. It is responsible for moving or controlling the system.
5. **Controller:** It acts as the brain of a robot. Based on the signal from the controller, manipulators do its work.
6. **Sensors:** Sensor is a transducer used to make measurements of physical variables. By this sensor, robot receives necessary manipulation, sends signals and interacts with peripheral equipment. It is of two types: Contact sensors and Non-contact sensors. Contact sensors respond to physical contact, such as touch, slip, force and torque. Non-contact sensors are relying on the response of a detector to variations in acoustic or electromagnetic radiation. Proximity and vision sensors are examples of non-contact sensors.
7. **Interfaces:** It acquires information from the surrounding environment by means of external sensors and takes decisions with the knowledge acquired and sends the necessary signals to the machine for user-friendly work.

## 17.7 Applications of Robotics

### 17.7.1 Defence

- Robots that could make decisions, such as to fire upon a target without any human commands and to replace the human soldiers in dangerous missions.
- Many of the most advanced Indian Defence Technologies in time to come would comprise robots. As is evident, for supporting military operations through land, sea, air, the US deployed thousands of robots in Iraq and Afghanistan.
- Unmanned aerial, ground or underwater vehicles often achieve intelligence gathering. Deployment of unmanned autonomous vehicles for the purpose of reconnaissance such as detection of anti-ship mines

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in littoral waters, coastal waters monitoring for adversaries such as pirate ships and accurate air strikes on evasive targets.

- Boston Dynamic's robotic **'Big Dog':** It was created in 2005 by Boston Dynamics with Foster-Miller, the NASA Jet Propulsion Laboratory, and the Harvard University Concord Field Station. It looks similar to a big dog and is a remote-controlled robot; thereby, enabling soldiers for transportation of heavy equipment over narrow steep foothills, ice, and dense forest areas.
- **'DARPA'** has developed six-legged robots, which are an all-terrain and all-weather robots having the capability of passing intelligence from far-flung areas to the control stations with the help of wireless communications.
- Robot-powered exoskeletons will make human soldiers with super-human strength.
- Explosive-disposal robots named as 'Talon' are keeping soldiers away from the physical danger by destroying mines and ultimately saving human life.
- **'Andros II'** and **'Mini-Andros'** are deployed for their usage in bomb squads worldwide; they can also monitor in the dark and further helps in the defusing or detonating bombs without the risk of human life.
- **'Daksh'** is an electrically powered and remote-controlled robot. It is one of the most current military robots of India. It is used to locate, handle and destroy risky objects safely and can even climb stairs. The main role of this military robot is to recover improvised explosive devices.
- A very high level of intelligent robot enable us to differentiate an enemy or a friend. These types of robots can be deployed in difficult warfare zones, like the Line of Control (LOC) – a step that would help avert the loss of human lives.
- The *'Robotic Soldier'* would be at the front end in the battlefield and the human soldiers would be assisting him from far away.

### 17.7.2 Manufacturing

- From cars to computers, automation and robotics play a vital role in the production of different types of consumer goods. Automotive and

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appliance manufacturers and bottling plants have increased the usage of industrial robots in their assembly lines. Robots make manufacturing processes more efficient and consume less time by assembling the product faster and making the process more reliable.

- They can work in different conditions and also do heavy lifting. Robots reduce the physical size of technological machinery, which is shrinking now at an amazing rate.
- Today, various manufacturing industries in the world have the most innovative technologies and nanotechnologies demand for the requirement of high human precision. For the creation of synthetic diamond, there is requirement of human hands, which can consistently place one carbon molecule on another but with the usage of robots, it can be more accurately done.

### 17.7.3 Medicine

- Since the 1980s, robots are helping the doctors in the removal of gallbladders, hysterectomies and repairing knee ligaments.
- Through little and better robotic tools and software systems at the time of surgery, there is a requirement for smaller incision wounds and shorter duration under anaesthesia for the patients than with conventional methods of surgery. This will result in the reduction of healing times, lesser pain, reduced infection risk and lesser loss of blood so there is no need of blood transfusion. “Davincy Surgical system” is a robotic surgery device.
- Through remote surgery (telesurgery), rural patients can be operated by the doctors in the city, thereby, decreasing the costs of transportation and increasing the alternatives for both patients and doctors. In the current world of surgery, good doctors are limited to a geographic location and population and hence, are unable to perform at a broad range of services.
- In case of bio-terrorist attack or an epidemic, medical robotic immune personnel can be the best alternative solution for the curing of infected individuals.
- Remote surgery will also be helpful to provide medical care more quickly than under current conditions.

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- Robotic limb can also be helpful to soldiers injured on the battlefield; whereby, the soldier, after some practice or recouping from the injury, can again be sent to war, depending on the gravity of the injury or there can be a robot who can replace the soldier on ground.
- **Da Vinci Surgical System** in Robotic Surgery Device used in various hospitals in India.

### 17.7.4 Disaster Preparedness, Response and Relief

- At the time of any national disasters, roads may be ruined or flooded and telephone or Internet lines may not work. Thus, to help in these types of situation, Denel Dynamics has established a 'homing pigeon' UAV, which can fly and supply the required relief material quickly to the disaster-prone area and return to base without any human intervention, being faster than conventional carriers.
- At the time of bio-terrorist or biochemical assaults, the said UAVs can possibly deliver face masks or antidotes without risking a pilot's health. The usage of UAVs could create self-contained networks of communications.
- Currently, we are using satellite communication networks. After huge explosions, in weather and other conditions, the existing technology could be useless. Swiss Institute of Technology in Lausanne [École polytechnique federale de Lausanne (EFPL)] has planned a group of 'Micro Air Vehicles', which have the capability of autonomously establishing an emergency wireless network, thus, connecting ground users in a disaster area.
- After an earthquake, or tsunami or any missile attack, rescue through the assistance of robots will be the country's most effective course of action. Cloud burst in Uttarakhand had claimed many lives and wrecked havoc. In this area, rescue operations took lot of time, which was crucial for the stranded people waiting for help. In this type of situation, robots can quickly find trapped victims, send their images to rescuers in real time without needlessly putting responders at risk and thus save many innocent lives. Taking a clue from this natural calamity, we should now be prepared with a Disaster Preparedness, Response and Relief plan, incorporating robots to carry out relief operations.

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### 17.7.5 Humanoid Robots

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The development of Honda's ASIMO and Sony's SDR-3X and now the more advanced SDR-4X refers to robots having the human resemblance. These humanoid robots, with remarkable grace, walk on their two feet, similar to the human-walking movements. Sony's engineers have provided SDR-4X with advanced features such as the ability to remember and express emotions, recognition from facial features, besides better voice and speech recognition technology.

**Table 3.1** World's Important Humanoid Robots

| S. No. | Humanoid Robots (Company) | Features |
|---|---|---|
| 1 | **SOPHIA (Hanson Robotics)** | • most advanced humanoid robot who made her debut in 2016 |
| 2 | **ATLAS (Boston Dynamics)** | • developed in 2013 with a prime focus on locomotion, first walking in the lab, then walking on every type of unstable terrain, and performing some sick Parkour tricks |
| 3 | **AMECA (Engineered Arts)** | • world's most advanced, most realistic humanoid robot created in 2021 |
| 4 | **Nadine (Kokoro)** | • an empathetic robot developed in 2013 responds to greetings, makes eye contact and remembers all comments and conversations with her |
| 5 | **ASIMO (Honda)** | • created for search and rescue missions<br>• it responds to various questions by nodding and speaking in different languages and can recognise almost ten different faces and address them by name |

### 17.7.6 Agriculture Robot

*The advantage of Agri-robot, which is* developed in India, is that it can become an admirable agriculture aid for farmers. The high labour cost and the increased need of assisted living have led to the development of the service robotics market. This automated system will be the most essential appliance for our society as India is the second largest producer of

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vegetable crops in the world. The developed model avoids the wastage of precious seeds, as it is designed to pick and drop only one seed at an instant; consequently, it increases the percentage of germination, which leads to profit for the farmers, who are the backbone of our nation's economy. As a result, the farmers/society will be the beneficiaries. The concept of automation incorporated in the intended system avoids the labour cost and reduces the process time. Consequently, it is cost-effective.

### 17.7.7 Robotics for Environmental Monitoring

Scientists, allowing new perspectives and a better understanding of the planet and its environmental processes, are increasingly utilising robotic systems as fundamental data-gathering tools. In the past, their use in environmental sciences has been relatively subtle and not highly publicised, but some disruptive events (such as the 2010 eruption of volcanoes in Iceland, the massive oil spill in the Gulf of Mexico from the Deepwater Horizon incident and the tragic 2011 earthquake and tsunami in Japan) have reinforced and publicised the importance and abilities of robotic systems to efficiently and precisely measure and potentially reduce environmental events at scales that were never possibly thought earlier.

With rising public awareness of robots, collecting pertinent scientific information and environment interaction, there are rising new opportunities for major environmental monitoring.

The benefit of robotics usage in environmental sciences is allowing the monitoring of events and sampling that is impossible or dangerous to undertake by humans.

In one article, which refers to 'Volcanic Environments', there is discussion of their long experience history in the terrestrial robotic systems application and development for active volcano monitoring. They emphasise the requirements of system and the further challenges for the collection of scientifically relevant data in these hostile environments. With this in situ volcanic monitoring process capability, volcanologists are now empowering for the better understanding and investigating the precursor of volcanic eruptions, further allowing a more precise activity prediction.

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The arrangement of navigation and chemical sensors for estimating the environmental parameters such as wind speed and direction can be measured with the usage of robotics. A single robotic vehicle's capability in capturing the temporal and spatial dynamics like harmful cyanobacterial blooms of developing aquatic processes can be a serious public health issue.

In spite of the unbelievable developments of research and commercially available robotic systems, regulators and authorities are still inflicting considerable restrictions on scientists in the usage of these systems for monitoring of environment, mainly near populated areas.

## 17.8 Challenges of Robotics in India

Among the many challenges that has plagued the field of Robotics in India, the primary among them have to do with the high capital cost, talent availability and procurement of hardware components. Robotics is a multidisciplinary field with four to five engineering subjects; so it's difficult to gain expertise in this field. There is a scarcity of skilled faculties to teach robotics. Except a few regions and institutions in the country, Robotics as a subject is not introduced in engineering courses.

## 17.9 India's Advancement in Robotics

- R&D organisation initiated in India the development of mini robots and about 101 master slave manipulators were made till April 1987.
- BARC (Bhabha Atomic Research Centre), Bombay, supported research projects on tele-robots.
- The Department of Electronics of Government of India also successfully initiated a project of fins welding of the turbine blades at BHEL (Bharat Heavy Electronics limited), Hyderabad, during 1994–97.
- By 2013–14, agricultural robots were introduced in Indian agricultural sectors.

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- From the information in **National Institute of Science and Technology Policy (NISTEP)** 2030 report, it is said that, by 2013–20, our researchers will introduce nano robots in medical sectors for surgery.

Developments related to robotics from the **NISTEP (National Institute of Science and Technology Policy) 2030 report:**

- By 2013–2014 – Agricultural robots
- By 2013–2017 – Robots that care for the elderly people
- By 2013–2020 – Nano robot
- By 2017 – Medical robots to perform low-invasive surgery
- By 2017–2019 – Household robots (cleaning of house, keep eye on intruders, etc.)
- By 2035 – Completely autonomous robot soldiers on the battlefield
- SASTRA's (It is a robotic private limited company in Kochi, Kerala) SCARA robot can perform repetitive tasks, for uses such as moving products, milling, 3D-paintings and is being used by Bosch to test touch screens. **'AcYut' is India's first indigenously developed humanoid robot.**
- **'Daksh'** is an electrically powered and remotely controlled robot used for locating, handling and destroying hazardous objects safely in military war. It was developed by DRDO in December 2011 and by 2035, DRDO is going to have completely autonomous robot soldiers on the battlefield.
- The premier institution in India for Robotic research is **Centre for Robotics and Mechatronics, Kanpur.**
- Various IITs are offering courses in computer sciences with specialisation in AI, which is an integral part of robotics.
- Centre for AI and Robotics in Bengaluru has developed a **'Chatur Robot'** – an intelligent robot with vision sensor, which can pick up objects in visual field.
- Scientists from University of Ulster and IIT-Kanpur have teamed up to investigate how to develop intelligent robotics devices to help people with severe disabilities, making them achieve greater independence.
- In AIIMS New Delhi, there is successful performing of **robotic surgery** in which the thymus gland was removed from a patient suffering from 'Myasthenia Gravis' (a progressive muscular weakness

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disease that can sometimes be life threatening). Thymectomy is the removal of thymus gland early in the course of the disease.

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- India is emerging as a hub for industrial robots and many Americans, Koreans and Japanese are using them.
- Ahmedabad-based 'Grid Bots' launched **'Robo Grad',** a robot that can clean homes and keep an eye on intruders.

***India's First Robotic System Performs Telesurgeries – SSI Mantra***

India achieved a groundbreaking milestone in healthcare technology with the successful completion of robotic cardiac surgeries via telesurgery using the ***SSI Mantra 3 Surgical Robotic System.***

**Central Drugs Standard Control Organisation (CDSCO)** is India's national regulatory body for drugs, cosmetics and medical devices.

**Mitra Robot**

Mitra Robot is a Humanoid Robot designed and developed by the Indian start-up company Invento Robotics. Mitra Robot made its first appearance in November 2017 at the Global Entrepreneurship Summit in 2017.

The five feet tall robot, Mitra-2 is India's first humanoid robot built and designed to engage in hospitality management and workplace productivity.

- Its body is made of fibreglass equipped with a touchscreen with battery backup of up to 8–10 hours. This robot has facial and speech recognition those are designed to recognise speech in multiple languages.
- Mitra-3 is an improved version of Mitra which has three degrees of freedom (shoulder, elbow, and finger movements) so it can welcome guests with a 'namaste' and also has other hand gestures.
- Mitra Robot is utilised in various disciplines like banks, wedding, hotels, malls, airports, cinema halls and hospitals, etc.

**Shalu Robot**

Robot Shalu is a multilingual, social and educational humanoid robot made-up of waste material (aluminium, plastic, cardboard, wood, newspapers, and other things readily available in the local market), and can speak 47 languages developed by Dinesh Kunwar Patel. During the Corona pandemic, it spread a message "Wear your mask, keep physical distance and follow government guidelines".

She can recognise people and remember them as well as identify many common objects. She can not only understand human emotions like anger, happiness, laughter, jealousy, etc. but can also behave accordingly. The total weight of the robot is only 4 kg as it has only an upper body.

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Robot Shalu is dedicated to Indian daughters towards *Beti Bachao Beti Padhao* mission, and this creation is a tiny step towards *Atmanirbhar Bharat* and *Digital India Mission*.

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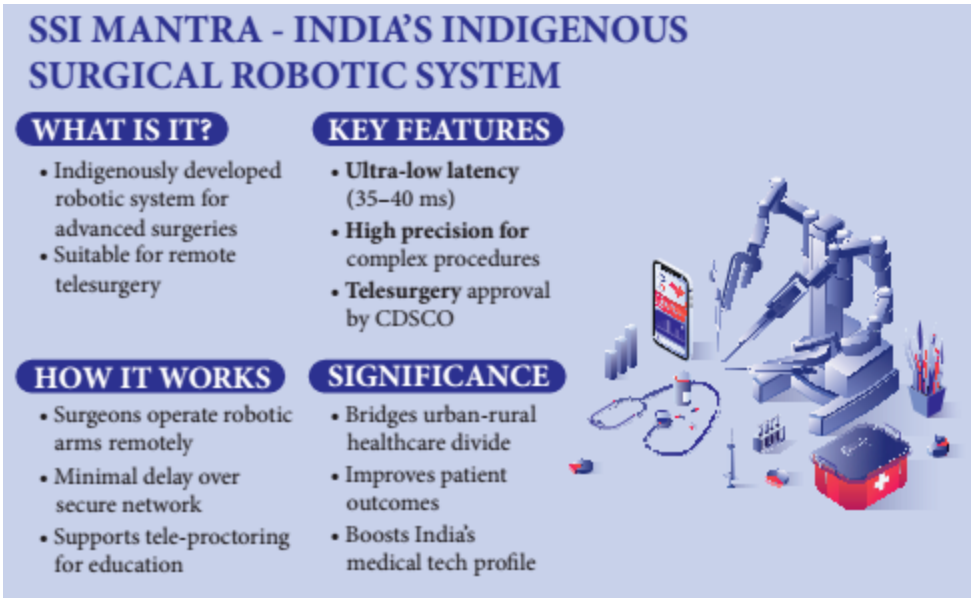
## 17.10 Disadvantages

- Employing a robot in factories has fewer limitations, as the humans performing the tasks require creativity, decision-making skills, time for adaptation and learning the job. Thus, the employees having no experience in dealing with robots require training programmes to interact with the new robotic equipment, which will take time and cost huge amounts of money in the financial input.
- The robot helps the workers' protection from various hazards. But, they themselves can create safety problems, thereby causing new danger, which need to be taken into consideration.
- The preference of industries in the utilisation of robots instead of human workers will increase unemployment rate, hence, creating social disturbances.




- Robots are costly in terms of both production and purchasing. Robots in the home could also cause conflicts because not everyone is going to

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financially be able to own it. Due to regular use of robots, the humans will become physically lazy, which might lead to many diseases.

- The major drawback of a robot is that it has no emotions. So, it cannot understand human values, ethics and cannot take decision based on philosophical or social values. It has no thinking power analysing of what is wrong or right and a day will come, when robots will be more powerful than humans and rule the whole world.

## 17.11 Bionic Man

Rex Bionics Limited is the global technology leader in robotic exoskeletons (Rex). Uniquely, Rex provides independent mobility to wheelchair users and other mobility-impaired persons, using advanced robotic technology, custom-designed electromechanical actuators, precision engineering and specialised networking systems.

For more detail refer **Infographic – Bionic Man and Sophia**.

## 17.12 Sophia

'Sophia' is a **social humanoid robot** developed by Hong Kong-based company, Hanson Robotics. With the ability to display more than 62 facial expressions, Sophia has participated in many high-profile interviews. In October 2017, Sophia was also given **citizenship of Saudi Arabia** – the first robot to receive citizenship of any country.

(Refer **Infographic – Bionic Man and Sophia**)

## 17.13 Biorobotics

With an understanding of biomechanics, engineers can develop biologically inspired robots, with improved and enhanced capabilities over traditional robots, which are robotic! Biologically inspired robots have greater

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mobility and flexibility than traditional robots and often possess sensory abilities.

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Biorobotic technologies are often utilised to provide assistance to accommodate a deficiency – either as fully functioning robots or highly advanced prosthetics; the latter represents one area in which neural engineering and biorobotics intersect, as both disciplines are required in order to first signal and then generate movement. Such devices may also be used to measure the state of disease, track progress or offer interactive training experiences that can speed recovery from an injury or stroke.

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# BIONIC MAN

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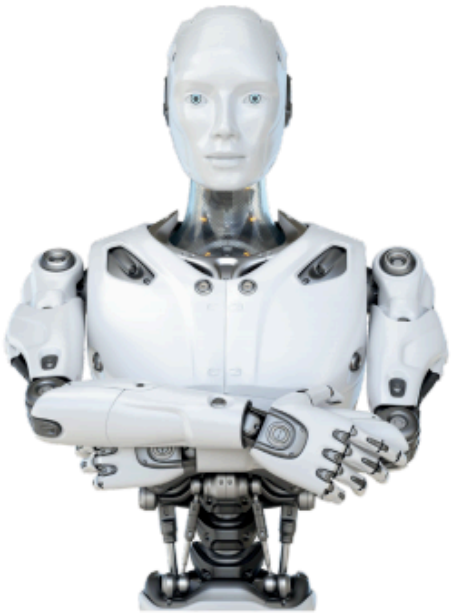
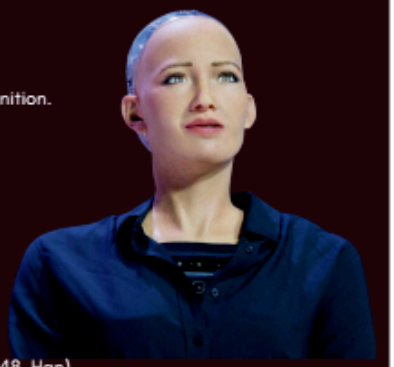
- In robotics, **the "Bionic Man" concept involves creating systems like bionic limbs that mimic or enhance human functions**, enabling mobility restoration through advanced, thought-controlled neural interfaces.
- Researchers are developing full-body exoskeletons, robotic suits that enhance strength and endurance, with **applications in healthcare, manufacturing, and military sectors.**
- Advancements in humanoid robotics aim to develop robots with human-like capabilities for interaction and task performance, focusing on practical assistive technologies and specialised applications rather than superhuman abilities.
- **Rex Bionics Limited specialises in the Rex robotic exoskeleton**, designed to enhance mobility for wheelchair users. First conceptualised in 2003, its prototype was completed in 2007, launched in 2010, and became commercially available in 2011.
- **The term "Bionic Man" became popular in the 1970s through the TV show "The Six Million Dollar Man,"** featuring Steve Austin, a character rebuilt with artificial parts. Today, engineers have developed a robot with similar features, including artificial organs and limbs.
- **This advanced bionic creation features a functioning artificial heart, various artificial organs, and prosthetic limbs with integrated systems for circulation and blood regulation.** It also includes retinal implants for vision and uses artificial intelligence to interact through speech, facilitated by cochlear implants and a speech generator.



# SOPHIA

- **Sophia Development:** Created by **Hanson Robotics**, a **social humanoid robot.**
- **Facial Expressions:** Displays over 62 facial expressions.
- **Saudi Citizenship: Gained in October 2017, the first robot to receive citizenship.**
- **Model:** Based on **Audrey Hepburn, using AI**, visual data processing, and facial recognition.
- **Capabilities:**
  - Cameras in eyes with computer algorithms for vision.
  - Can follow faces, **maintain eye contact**, and recognise individuals.
  - Processes speech and converses using Google Chrome voice recognition.
  - **Imitates human gestures and facial expressions.**
  - Answers questions and makes simple conversations on predefined topics.
- **Design Purpose:**
  - Companion for the elderly in nursing homes.
  - Assistance at large events or parks.
- **Siblings: Nine other humanoid robots** created by Hanson Robotics (e.g., Alice, Bina-48, Han).
- **Applications:** Suitable for healthcare, customer service, therapy and education.
- **India Visit:**
  - Appearances at IIT-B and WCIT in Hyderabad.
  - Discussed global issues and women's status.
- **Significance:** Demonstrates the advancement and potential future impact of AI and robotics.



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# CHAPTER 18
# NANOTECHNOLOGY

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The nanorobots are capable of undertaking human tasks as well as tasks that are not being performed by the humans. The reconstruction of the depleted ozone layer could potentially be performed.

**18.12 Pitfalls of Nanotechnology**

## 18.1 Nano and Technology

The term 'Nanotechnology' came from nanometre, which is a length unit in the metric system, equivalent to one billionth of a metre ($10^{-9}$). It is the manipulating study matter on an atomic scale. Nanotechnology's periodical development is summarised in **Figure 18.1**.

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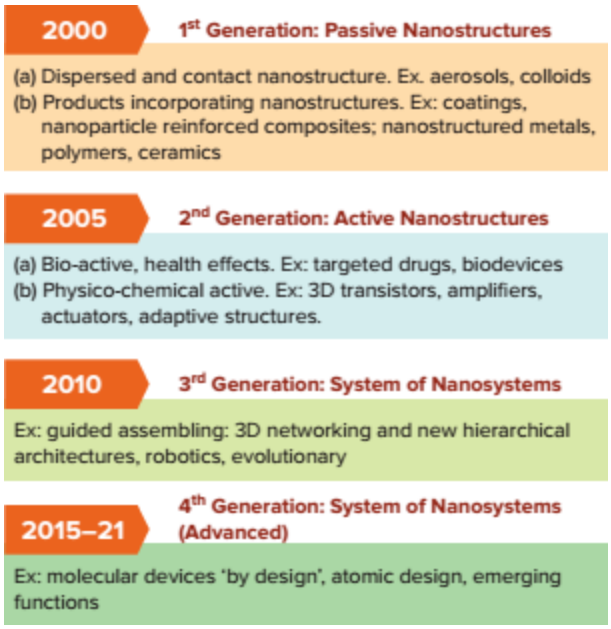


**Figure 18.1** Timeline of Nanotechnology

## 18.2 History

- In the year 1959, the first theory on Nanotechnology was developed by the famous physics professor, **Dr. Richard P. Feynman.**
- In 1981, Eric Drexler published the first technical paper on 'Molecular Nanotechnology'. After that, in 1985, Richard Smalley and Robert Curl, with astronomer Sir Harry Kroto, discovered a new form of carbon (3-1) nanoscale carbon cages.
- The **scanning tunnelling microscope invention** in 1981 and the **fullerene discovery** in 1985 led to the emergence of nanotechnology. Fullerene is a C60 Carbon made compound.
- Dr. Nori Taniguchi, a Japanese scientist coined the term "nanotechnology" in 1974.
- **In 1996, the Nobel Prize in Chemistry was awarded jointly to Sir Harold W. Kroto, Richard E. Smalley and Robert F. Curl, 'for their pioneer discovery of fullerenes'.**
- **Fullerence:** A fullerene is a molecule of sixty carbons in the form of a hollow sphere, ellipsoid, tube and in many other shapes. Spherical

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fullerenes are also called buckyballs and cylindrical structures are called Bucky tubes or carbon nanotubes (**Figure 18.2**).

Different number of carbon and shape form different types of substances in nanotechnology – 0-dimensional Structure is Buckyball (Synthetic), 1-dimensional Structure is Carbon Nanotubes (Synthetic), 2-dimensional structure is Graphene (Synthetic) and in 3-dimensional structure has Graphite (natural and synthetic both), which is an allotrope of diamonds (natural).

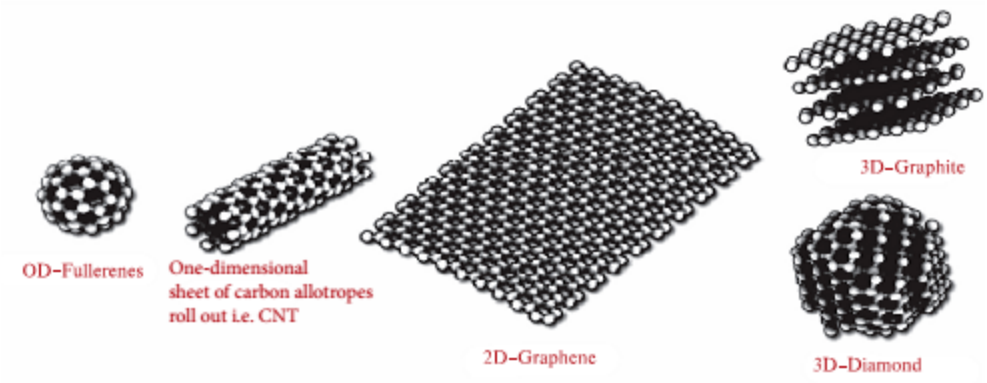


Figure 18.2 Evolution of different dimensional structures of Nanotechnology

Note: If a chemical element or compound exists in two or more than two different forms, all are called allotropes, and this type of behaviour is called allotropy.

## 18.3 Fullerenes

The Buckminsterfullerene was the first fullerene to be discovered. It was named after the American architect, Buckminster Fuller, as it has resemblance to the geodesic dome designed by him. The Buckminsterfullerene is an allotrope of carbon. Due to its relative ease of synthesis, fullerene of C60 remains popular and a lot of research for its potential applications has been carried out. The C60 buckyball is made of 60 carbons at 60 vertices that makes a spherical structure.

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These molecules have very high tensile strength and bounce back to their original shape. Due to the unique properties of this allotrope of carbon, it has a number of applications – a few of them have been discussed below:

## 18.4 Carbon Nanotube

Carbon is the only element in nature found with varying physical and chemical characteristics. In 1991, Sumio Iijima discovered a new form of pure carbon with few nanoscopic threads. He found closed graphite structures including nanoparticles and became known as 'NanoTubes'. The world's tiniest carbon nanotube that has been produced is 0.4 nm wide in NEC Japan. Carbon nanotubes are exceptionally strong against any crack or damage. As the nanotubes are hollow, tubular and caged molecules, they are ideal for gas and lithium storage for drag delivery. It has vast amount of capacity to conduct electricity and heat than diamond. In India, scientists started research on carbon nanotubes in the early 1990s.

***Properties***

- Generally, the ratio of length-to-diameter of carbon nanotubes is up to 132,000,000:1.
- These are the fullerene structural family members. One atom-thick sheets of carbon with their long, hollow structure derives carbon nanotube. Carbon nanotube is a one-dimensional structure of carbons.
- They posses the highest strength-to-weight ratio, and assists in creation of *lightweight spacecraft.*
- It can easily penetrate through membrane so it will be helpful in *cancer treatment.*

## 18.5 Graphene

It is defined as a one-atom thick layer. It is a two-dimensional network of carbon atoms and is the fundamental element of different allotropes such as

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charcoal, graphite, fullerenes and carbon nanotubes. It is the one of the strongest and thinnest materials known. Nobel Prize in Physics 2010 has been awarded to Andre Geim and Konstantin Novoselov "for ground-breaking experiments regarding the **two-dimensional material graphene**".

(Refer **Infographic – Nanotechnology**)

## 18.6 Comparison between Diamond and Graphite

Table 18.1 Diamond vs Graphite

| Diamond | Graphite |
|---|---|
| It occurs naturally in free state. | It occurs naturally and is manufactured artificially also. |
| It is the hardest natural substance known. | It is soft and greasy to touch. |
| It has high relative density (about 3.5). | Its relative density is 2.3. |
| It is transparent and has high refractive index (2.45). | It is black in colour and opaque. |
| It is non-conductor of heat and electricity. | It is a good conductor of heat and electricity. |
| It burns in air at 900° C to give $CO_2$. | It burns in air at 700–800° C to give $CO_2$. |
| It occurs as octahedral crystals. | It occurs as hexagonal crystals. |
| It is insoluble in all solvents. | It is insoluble in all ordinary solvents. |

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## 18.7 Nano Material

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### 18.7.1 Nanobots (Nano + Robots)

- Its size is close to $10^{-9}$ scale.
- It is mainly in research and development phase.

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# NANOTECHNOLOGY

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## What is Nano Technology?

- **Meaning:** Nanotechnology is the manipulation of matter at the atomic and molecular scale, typically below 100 nanometers, to create materials with unique properties and functions.
- **Scale:** Deals with the size of individual atoms and small molecules.
- **Unique Properties:** Materials **exhibit different properties at the nanoscale** compared to their bulk forms.
- **Goal:** Exploit unique properties to design and engineer materials and devices at the atomic and molecular levels.

## What is Nano Scale?

- **Meaning:** Length scale of **nanometers (one billionth of a meter).**
- **Unique Properties:** Quantum effects and increased surface area-to-volume ratio at this scale.
- **Nanotechnology Focus:** Manipulate properties by working with materials at the nanoscale.
- **Engineering Nano-materials:** Control size, shape, composition, and surface properties.
- **Applications:** Enables novel technologies in various fields.

## Applications of Nano Technology

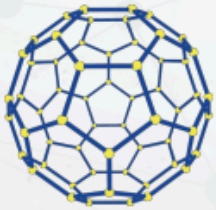
- Electronics and Computing
- Medicine and Healthcare
- Energy
- Environmental Remediation
- Agriculture and Food Industry
- Materials Science and Engineering
- Water Treatment and Desalination

## Environmental Impacts of Nano Technology

- Exposure and Toxicity
- Environmental Fate and Behavior
- Ecotoxicity and Ecosystem Impact
- Life Cycle Assessment
- Regulatory Framework
- Sustainable Nanotechnology

## What is Fullerene?

- **Meaning: Class of carbon molecules with atoms in closed cage structures.**
- **Key Example: $C_{60}$, known as buckminsterfullerene** or buckyball, resembling a soccer ball.
- **Discovery: Identified in 1985, named after architect Buckminster Fuller.**
- **Structure:** Carbon atoms **arranged in hexagons and pentagons**, like a soccer ball.
- **Properties: High tensile strength, electrical conductivity, and chemical stability.**
- **Applications:** Used in materials science, electronics, medicine, and energy as catalysts, drug delivery vehicles, superconductors and in solar cells.
- **Research Significance:** Subject of ongoing research in nanotechnology for their unique structure and properties.



**Fullerene ($C_{60}$Molecule)**

## Applications of Fullerene

1. **Antioxidants:** They act as efficient radical scavengers.
2. **Cytoprotectors:** They protect cells from ultraviolet damage.
3. **Cosmetics:** Water-soluble derivatives prevent skin damage and aging.
4. **Antiviral Agents:** They show activity against viruses like HIV and Hepatitis C.
5. **Drug and Gene Delivery:** They can transport drugs and DNA sequences to target cells.
6. **Photodynamic Therapy:** They are used to target and treat malignant cells.
7. **Solar Cells:** They enhance energy conversion in organic solar cells.
8. **Protective Eyewear:** They can be used in optical limiters for eye protection.
9. **Hydrogen Storage:** Fullerenes have potential for storing and releasing hydrogen.
10. **Hardening Agents:** They strengthen lightweight metals without compromising ductility.
11. **Diamond Films:** Fullerenes can be converted into diamond films for electronic applications.
12. Fullerenes offer **diverse possibilities in medicine**, but further research is needed for practical implementation.

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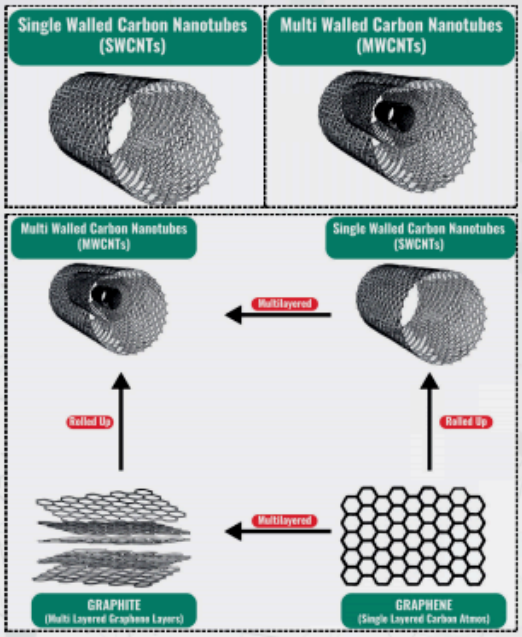
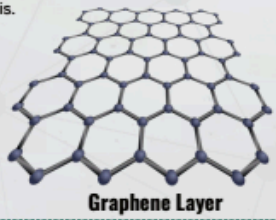
## Carbon Nano Tubes

- **Carbon Nanotubes (CNTs):** **Cylindrical structures made of carbon atoms.**
- **Types:**
  - **Single-walled (SWCNTs):** One cylindrical layer, known for exceptional electrical and mechanical properties.
  - **Multi-walled (MWCNTs):** Multiple concentric layers, offering enhanced mechanical strength and durability.
- **Properties:**
  - **Strength:** Extremely high strength-to-weight ratio.
  - **Electrical Conductivity:** Efficient electricity conduction, suitable for electronics.
  - **Thermal Conductivity:** High levels, useful for heat management.
  - **Structural Flexibility:** Can be bent and twisted without breaking.
  - **Nanoscale Applications:** Ideal for devices, sensors, and other nano-applications.
- **Challenges:** Issues with large-scale production, cost, and potential health/environmental impacts.
- **Applications:**
  - Used by Easton-Bell Sports in bicycle components.
  - Zyvex Technologies for manufacturing lightweight boats.
  - Potential replacement for silicon in transistors, beneficial for reducing heat emission.
  - Improvements in electric cables and wires to reduce electricity losses.
  - Advances in communication technology *via* tiny CNT transistors.
  - Applications in photovoltaics like solar cells and in innovative fabrics.
- **Medical and Biomedical Applications:**
  - Degradable by oxidative enzymes, opening new medical applications.
  - Used in biomedical applications such as artificial implants and drug delivery systems.
  - NASA uses them in miniaturized electronics for highly sensitive biomarker detection in diagnostics.




## Graphene

- **Graphene**: **A two-dimensional sheet of carbon atoms in a hexagonal lattice.**
- **It is also called "Miracle Material".**
- Properties:
  - **Strength and Flexibility:** Extremely strong and flexible, can stretch and bend without breaking.
  - **Electrical Conductivity:** Exceptionally high, beneficial for electronic applications.
  - **Thermal Conductivity:** Excellent, efficiently transfers and dissipates heat.
  - **Transparency:** Highly transparent, allowing over 97% of light to pass through.
  - **High Surface Area:** Large surface area relative to weight, useful in energy storage and catalysis.



Graphene Layer

## Applications of Graphene

- **Electronics:** Graphene is pivotal in developing faster, more efficient electronic components, including transistors, semiconductors, and flexible displays, due to its exceptional electrical conductivity.
- **Energy Storage and Conversion:** It enhances the performance of batteries and supercapacitors owing to its high surface area and conductivity, and is also utilised in solar cells for improved photoelectrochemical energy conversion.
- **Composite Materials:** Leveraged for its strength and lightweight nature, graphene is ideal for creating stronger, lighter composite materials used in automotive, aerospace, and sports equipment industries.
- **Biomedical Applications:** Utilised for drug delivery systems, DNA sequencing, and bionic devices due to its biocompatibility, graphene plays a significant role in healthcare.
- **Filtration:** Graphene oxide membranes are employed for water desalination and purification, and in air filters to remove contaminants and allergens.
- **Optical Electronics:** Its optical transparency and electrical conductivity make graphene suitable for touchscreens, light displays, and photovoltaic cells.
- **Sensors:** Graphene is used in highly sensitive sensors for detecting gases, chemicals, and biological agents.
- **Anti-bacterial Materials:** Graphene oxide exhibits antimicrobial properties, utilised in coatings and packaging to improve hygiene.
- **Ethanol Distillation:** Graphene oxide membranes facilitate selective water vapor passage, aiding in biofuel and alcohol production.

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## What is Nano Material?

- **Definition:** A material is considered a nanomaterial **if at least one of its dimensions is within the nanorange (1 nm to 100 nm).**
- **Composition:** Nanomaterials can be made from various substances, including silicon, gold, silver, titanium dioxide, zinc, and carbon-based materials like fullerenes, as well as metal-based materials and semiconductors.
- **Nanomaterial Structure:**

### Quantum Well

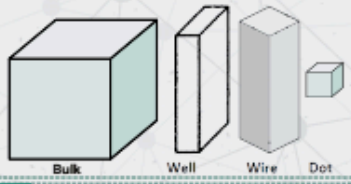
- **Dimensionality:** Two-dimensional (2D) confinement.
- **Structure:** Any material that has only one dimension in nano range.
- **Properties:** Electrons and holes are confined in the direction perpendicular to the layer, allowing motion only within the plane. This changes the energy spectrum of the carriers, leading to quantised energy levels.
- **Applications**: Used in lasers, photodetectors, and high-electron-mobility transistors (HEMTs).

### Quantum Wire

- **Dimensionality:** One-dimensional (1D) confinement.
- **Structure:** Any material those have two dimensions in nano range.
- **Properties:** Exhibits even more pronounced quantum effects due to greater confinement, leading to discrete energy levels.
- **Applications:** Used in quantum computing, advanced electronic devices, and sensors.

### Quantum Dots

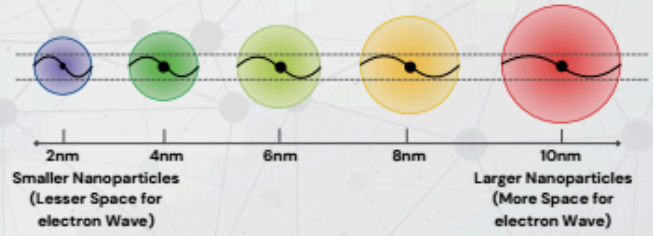
- **Dimensionality:** Zero-dimensional (0D) confinement.
- **Structure:** Any material has all dimensions in nano range.
- **Properties:** Similar to atoms but with tunable properties, quantum dots have discrete, atom-like energy levels. They can emit light of specific frequencies when energised, with the color dependent on the size of the dot.
- **Applications:** Potential applications of quantum dots include single-electron transistors, solar cells, LEDs, lasers, single-photon sources, second-harmonic generation, quantum computing, cell biology research, microscopy, and medical imaging. Due to the reflectivity of rods, it can also be used in display technologies. Quantum dots will be also useful in cancer therapeutics.

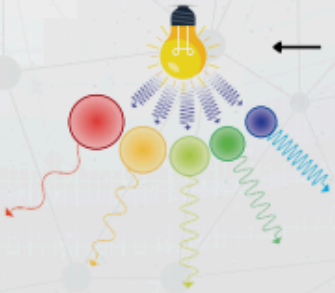



## What are Quantum Dots?

- **Quantum dots (Q-dots) or nanocrystals are nanoscale particles less than 10 nm.**
- **Fabricated from metals or semiconducting materials.**
- **Considered artificial atoms due to a fixed number of electrons confined in a small area.**
- **Roughly spherical, containing electrons ranging from one to several thousand.**
- **Band gap changes with size; smaller Q-dots have larger band gaps.**
- **Electromagnetic radiation wavelength changes with size, producing colors from blue to infrared.**
- **Q-dots emit white light in the visible spectrum, similar to sunlight, used in LED bulbs.**
- **Common materials for LEDs: cadmium sulphide (CdS) and zinc selenide (ZnSe).**
- **The Nobel Prize in Chemistry 2023 was awarded to Moungi G Bawendi, Louis E Brus, and Alexei I Ekimov for their groundbreaking discovery and synthesis of quantum dots.**

**When particles are only a few nanometers in diameter, the space available for electrons becomes limited. This confinement affects the particle's optical properties.**



**Quantum dots capture light and re-emit it at a different wavelength. The emitted color varies based on the particle size.**

- 1.5 nanometre-sized nanobots have the capability of counting specific molecules in a chemical sample.

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- Due to microscopic size of nanobots, it would probably be necessary for very large numbers of them to work together to perform microscopic and macroscopic tasks.

***Applications***

- It will be useful for detection of toxic components in environment.
- In drug delivery system.
- In biomedical instrumentation and utilisation.

## 18.8 Approaches in Nanotechnology

### 18.8.1 Bottom Up

In this approach, the manufacturing of different materials and products is done from their own molecular components. They accumulate themselves in a chemical manner through their own breed recognition. The bottom-up approach calls for assembly of nanostructures from atoms and molecules. Self-assembly of nanotubes has been found to be a useful feature.

### 18.8.2 Top Down

In this approach, the nanomaterials and objects are produced by bigger bodies without its atomic reactions bouncing. Generally, there is more usage of the bottom-up than the top-down approach. The top-down approach refers to the building of nanostructures of progressively smaller and smaller dimensions using larger parts. DNA technology utilises the specificity Watson–Crick base preparing to construct well-defined structures out of DNA.

## 18.9 Application of Nanotechnology

### 18.9.1 Nanotechnology in Drugs (Cancer)

- It provides innovative methods for drug delivery and therapies.

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- It assists in the accurate delivery of drugs in the body to the right location and further for controlling the infection; the drug doses are released on a fixed schedule for treatment.
- The drug will be delivered through a nano-sized carrier for becoming concentrated at the site of disease, i.e., cancer tumour. At the final destination, at a specific location, they release the medicine and kill the tumour cells in human body. This type of treatment is popular for drug targeting. Recent treatment is given through radiotherapy or chemotherapy, which not only kills cancerous cells but also kills normal cells that are necessary to the human life.
- Nanobots can clear the blockage in arteries and solve heart attack risk in human body.
- Nano-optics could allow for an increase in precision of pupil repair and other type of laser eye surgery.

### 18.9.2 Nanotechnology in Fabrics

- Some cloth producers are developing clothing which can strain and be water-repellent, through the usage of nano-sized whiskers in the fabric, thereby, causing water to drop from the cloths' surface.
- These fabrics can also be used in manufacturing bulletproof jackets.
- It makes spill and dirt-resistant clothes and also develops antimicrobial and antibacterial fabrics.

### 18.9.3 Nanotechnology in Mobile

- Nokia Research Centre (NRC) and the University of Cambridge (UK) combinedly developed **'Morph',** a nanotechnology concept-based device. **Morph** is hydrophobic and dirt repellent in nature. It also has the capability to charge itself with accessible light sources due to the covering of photovoltaic nanowire on its surface.
- Nanoscale electronics technology also allows to stretch the handset. These nanoscale fibre mesh allows our mobile handsets to change into different shapes such as stretched, bent and folded into any number of imaginable shapes.

### 18.9.4 Nanotechnology in Electronics

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- The electrodes made up of nanowires assist the flat panel displays to be flexible as well as skinny.
- Chip fabrication can be done by Nanolithography (this technology involves fabricating nanoscale structures).
- The transistors are made up of nanowires, which brought revolution in communication and information technology.
- Nanotechnology helps in the manufacturing of e-paper, displays on sunglasses, etc.
- Recently available solar cells have much lower efficiencies (15–20%). Nanotechnology could help increasing this efficiency up to 40%.
- Nanobatteries are rechargeable batteries and the nanomaterials used here could be helpful for the battery disposal.

### 18.9.5 Nanotechnology in Computers

- Transistors based on carbon nanotubes can replace silicon transistors in computers.
- Nanorods for display technique are the future technology, which consume less electricity and emit less heat.
- Due to nanotechnology, microprocessors' size has been reduced to a great extent.
- According to the researchers at North Carolina State University, nanodots are growing magnetic nanoparticle arrays.
- Nanowires coated with titanium dioxide can be used as memory devices.
- Memristor-made memory chips can achieve higher memory density due to their small size.
- Before 'i' series processors, intel used chips of size ranging between 65 nm and 45 nm. Later on, due to nanotechnolgy, the size of chip reduced to 22 nm, which itself is a milestone.

### 18.9.6 Other Uses

- As cutting tools, nanocrystalline materials, such as tungsten carbide, tantalum carbide and titanium carbide, are more wear and erosion-resistant and have good life than other conventional counterparts.
- Silver nanocrystal materials kill bacteria and prevent infection.

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- Synthetic bone has been developed by the molecular level manipulation of calcium and phosphate through the nanoparticulate.
- The usage of lightest identified solid material, termed as 'Aerogels' in space suits and further proposed its usage in spacecraft due to good insulating properties are also importance of nanotechnology.

### 18.9.7 Nano Bubble Technology

***What Is Nano BUBBLE Technology?***

- A technology that uses **ultra-small gas bubbles (<200 nanometers)** suspended in water to **purify and treat** it effectively.

**WORLD'S FIRST MOLECULAR ROBOT**

Scientists from the University of Manchester have created the world's first 'molecular robot' – millionth of a millimetre in size. It can be used to build molecules and may help discover novel drugs.

1. The molecular robot is millionth of a millimetre in size.
2. It is made up of just 150 carbon, hydrogen, oxygen and nitrogen atoms, which are basic building blocks required to form molecules.
3. It can be programmed to move and build molecular cargo using a tiny robotic arm.
4. The molecular robot operates by carrying out chemical reactions in special solutions, which can then be controlled and programmed by scientists to perform the basic tasks.
5. It responds to a series of simple commands that are programmed with chemical inputs.

**Significance**

The molecular robot is so small, that it massively reduces the demand for materials, dramatically reduces the power requirements, can accelerate and improve drug discovery and rapidly increases the miniaturisation of other products. Such molecular robots can be used for medical purposes, advanced manufacturing processes and even building molecular factories and assembly lines.

- These **nanobubbles** remain **suspended for months**, offering **extended action** and **superior gas transfer**.

***Key Properties of Nanobubbles***

| Property | Description |
|---|---|

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| | |
|---|---|
| **Size** | 70–120 nanometers (2,500× smaller than a salt grain) |
| **Surface Charge** | Strong negative charge prevents merging, supports microbial stimulation |
| **Neutral Buoyancy** | Stay suspended longer for extended water contact |
| **Hydrophobic** | Repels water to lift contaminants and organics |

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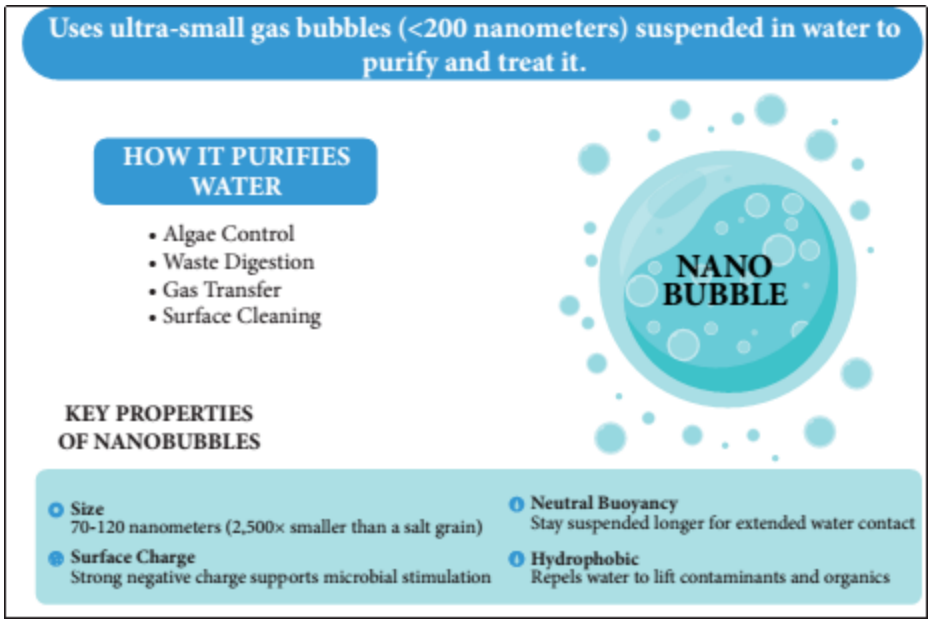
### *How It Purifies Water*

- **Algae Control:** Breaks down and inhibits algae in stagnant waters.
- **Waste Digestion:** Removes biological waste, oils and grease.
- **Gas Transfer:** Hyper-efficient **oxygenation** improves water health.
- **Surface Cleaning:** Cleans without toxic chemicals.

### *Significance and Applications*

- **Improves Aquatic Health:** Prevents disease and supports ecosystems.




- **Eco-Friendly: Chemical-free**, sustainable purification method.
- **Multi-Industry Use:** Applied in **wastewater treatment**, **fermentation** and **biological processing**.

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**Nanobubbles facilitate efficient gas transfer, including oxygen, into liquids, which is beneficial for various applications.**

- **Water Treatment:** They can effectively remove algae, organic waste and even oil and grease from water, and improve water quality.
- **Agriculture:** Nanobubbles enhance root growth and plant health by improving irrigation water quality.
- **Aquaculture:** They increase dissolved oxygen levels, promoting fish growth and can reduce the need for chemical treatments.
- **Other Industries:** They find applications in oil and gas extraction, wastewater treatment and more.

### 18.9.8 Nano Fertilisers

- Indian scientists developed **nano-coated muriate of potash (potassium chloride)** using **nanoclay-reinforced binary carbohydrates**.
- **Nanoclay-reinforced binary carbohydrates** are used to create a coating for fertilisers, enhancing nutrient efficiency and crop production. **This coating, made from nanoclay and a combination of two carbohydrates (like chitosan and lignin),** helps reduce the amount of fertiliser needed while improving crop yields.
- Improves **Nutrient Use Efficiency (NUE)** with **slow-release**, **biodegradable** and **hydrophobic properties**.

#### *Nano Fertilisers*

- **Definition:** Fertilisers coated or infused with **nanomaterials (1–100 nm)** for **controlled and efficient nutrient release**.
- **Purpose:** Improve nutrient delivery, reduce environmental loss and support sustainable farming.

#### *Types of Nanomaterials Used*

**Inorganic:**

- **Metal Oxides:** ZnO, TiO, MgO, AgO
- **Silica Nanoparticles:** Biocompatible, high surface area
- **Hydroxyapatite:** Supplies calcium and phosphorus

**Organic:**

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- **Chitosan:** Biodegradable nutrient carrier
- **Carbon Nanomaterials:** CNTs, fullerenes enhance germination, protein, chlorophyll content

### *Types of Nano Fertilisers*

| Type | Description |
|---|---|
| **Nanoscale Coating** | Nutrients coated in nanoparticles for slow release |
| **Nanoscale Additive** | Nutrients added to nano-adsorbents |
| **Nanoporous Carriers** | Nutrients loaded into nanoporous material |

### *Applications in Agriculture*

- **Precision Farming:** Efficient use of water and fertilisers
- **Soil Health:** Boosts germination, metabolism and stress tolerance
- **Sustainability:** Maintains long-term soil fertility

### *Advantages*

- **Higher Nutrient Efficiency:** Less leaching, better absorption
- **Improved Yields:** Controlled release ensures better growth
- **Eco-Friendly:** Reduces runoff and pollution
- **Biofortification:** Micronutrient delivery (Zn, Fe, I)
- **Cost-Effective:** 85–90% efficiency (vs. 25% in traditional urea)
- **Supports Biofertilisers:** Enhances Rhizobium and Azotobacter function

### *Challenges*

- **Environmental Risks:** Potential **ecotoxicity**
- **Human Health:** Nanoparticles may penetrate biological systems
- **Soil Microbe Disruption:** Harm to beneficial microorganisms
- **Lack of Regulation:** No standard laws or safety protocols
- **Bioaccumulation:** Long-term persistence in the food chain
- **Yield Drop:** Studies show reduced wheat (–21.6%) and rice (–13%) yields using nano urea

### *Way Forward*

- **Farmer Outreach:** Use KVKs, awareness programs for small farmers
- **Research:** Invest in nano-toxicity and plant interaction studies

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- **Standardisation:** Frame clear regulatory and safety frameworks
- **Biodegradable Materials:** Promote plant/microbe-based nanocarriers
- **Affordable Innovation:** Utilise phosphate rocks to produce cost-effective nano fertilisers

## 18.10 Nanotechnology in India

Today, nanotechnology has great relevance, as it has the potential to address the needs of all innovative technological solutions.

### 18.10.1 Significance

The Indian government is looking towards nanotechnology as a means of accelerating agricultural productivity in the country. According to the Planning Commission, nanotechnologies such as nanosensors and nano-based smart delivery systems could help ensure natural resources like water, nutrients and chemicals; those are used efficiently in the field of agriculture. Nowadays, nanobarcodes and nano-processing could also be used to help in monitoring the quality of agricultural products.

Nanotechnology can help India in ensuring food security with increasing crop yields and reducing consumption. Nanotechnology can be used to supply water and nutrients to crops, thereby increasing the yield and reducing wastage. It can be used in preventing damage to stored food grains.

Nanotechnology can be used in public health, in India, with better medical practices and affordability. With the advancement of nanomedicine, researchers had developed nanobioceramics that repair bones and tissues and an aerosol spray that uses nanoparticles to deliver lung cancer drugs. A nano-based water filter and the ‘Sens’ (a sensor) are also developed in India as an affordable and user-friendly nanodiagnostic tool that anticipates heart attacks. It is expected to become commercially available soon. With its vast scientific and industrial base, India can become a leader in nanoscience and nanotechnology research.

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### 18.10.2 Government Efforts

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Nanotechnology is a knowledge-intensive and 'enabling technology' that can influence a wide range of products and processes with far-reaching implications for national economy and development. The government has been taking various initiatives to promote nanotechnology. Some of the prominent initiatives are:

1. **Nano Science and Technology Initiative (NSTI):** The Department of Science and Technology, in October 2001, had launched a modest programme in Nano Science and Technology, called the Nano Science and Technology Initiative (NSTI). Under the NSTI, the DST supported a number of activities in Nano Science and Technology. Based on the success of the NSTI, the second phase of DST's efforts was launched as the Nano Mission.
2. **Nano Mission:** The Government of India in May 2007 has approved the launch of a Mission on Nano Science and Technology (Nano Mission).

   The Department of Science and Technology (DST) is the nodal agency for the implementation of Nano Mission in India. An allocation of ₹1000 crore for 5 years has been made by DST for this research work. Two advisory groups guide the technical programmes of the Nano Mission:

   ❑ (NSAG): Nano Science Advisory Group

**MISSION ON NANO SCIENCE AND TECHNOLOGY (NANO MISSION)**

- It was launched in 2007.
- It is as an 'umbrella capacity-building programme'.
- The Mission's programmes will target all scientists, institutions and industry in the country.
- It will also strengthen activities in nanoscience and technology by promoting basic research, human resource development, research infrastructure development and international collaborations among others.
- It will be anchored in **the Department of Science and Technology** and steered by a **Nano Mission Council, chaired by an eminent scientist.**

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- ❑ (NATAG): Nano Applications and Technology Advisory Group

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***Outcomes and Significance of the Mission***

- ❑ As a result of the efforts led by the Nano Mission, today, ***India is amongst the top five nations in the world, in terms of scientific publications in nanoscience and technology (moving from 4$^{th}$ to the 3$^{rd}$ position).***
- ❑ The Nano Mission itself has resulted in about 5000 research papers and about 900 PhDs and also ***some useful products like nano hydrogel based eye drops, pesticide removal technology for drinking water, water filters for arsenic and fluoride removal, nano-silver-based antimicrobial textile coating, etc.***
- ❑ The Nano Mission has thus helped establish a good ecosystem in the country to pursue front-ranking basic research and also to seed and nurture application-oriented R&D, focussed on useful technologies and products.
- ❑ **The International Conference on Nano Science and Nano Technology (ICONSAT)** under the aegis of **Nano Mission,** Department of Science and Technology (DST) was held at ***Kolkata*** (March 5–7, 2020) focussing on **the recent advances in this frontier research field.**

3. **Nanotechnology Initiatives Programme:** The Department of Information Technology (DIT) and Ministry of Communications and Information Technology (MCIT) has inaugurated the Nanotechnology Initiatives Programme in 2004. It is mainly concentrating on nanoelectronics aspects of the nanotechnology. The focus of the programme is on capacity building and human resource development, infrastructure creation and research and development in nanoelectronics, nanotechnology and nanometrology. This programme emphasises on the establishment of major centres on nanoelectronics and nanometrology in India. This programme brings a number of R&D projects at the academic level of R&D organisations across the country and also involves in Indian nanoelectronics user programmes.

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4. **National Nanotechnology Institutes:** The government has identified three cities – Bengaluru, Mumbai and Kolkata – for the promotion and development of nanotechnology, by setting up the Indian Institute of Nano Sciences (IINS) at these three locations. The institutes will primarily focus on advanced scientific research in nanotechnology.
5. **Vision Group for developing a National Nanotechnology Policy:** India formed a Vision Group in January 2008, consisting of about a dozen researchers from academics, industries and research spheres to develop a National Nanotechnology Policy in India.
6. **Bengaluru Nano (Bengaluru India Nano Conference):** Bengaluru Nano is organised by the Government of Karnataka and the Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bengaluru. It is a biannual event. The event comprises exhibition, of conference and provides a meeting place for global scientists, industry, academia and government.
7. **International Collaboration:** Besides launching the Nano Science and Technology Initiative and Nano Mission, India has also put its step into bilateral nanotechnology programmes with the European Union, Germany, Italy, Taiwan and the United States. India is a participant in the IBSA (India–Brazil–South Africa) nanotechnology initiative. It was started as a collaborative programme between the Department of Science and Technology of three participating countries, with a goal of developing trilateral collaborative projects of common relevance and interest.
8. **Nanotechnology Regulatory Board:** In February 2010, the Nano Mission Council stated that India will have a Nanotechnology Regulatory Board to regulate the industrial nanotech products that are used in day-to-day life. It is felt that India should have an efficient regulatory body and well-defined protocols for dealing with the deployment of nanotechnology like handling, movement and approval of nanoproducts and also to maintain an adequate database on its toxicity.

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9. **Institute of Nano Science and Technology (INST)** at Mohali, Chandigarh is the **first Nano-Science Institute** in the country set up to undertake research and generate products/devices and technology in the area of Nanoscience and Technology for the benefit of the country.
10. Refer **Infographic – India Graphene Engineering and Innovation Centre (iGEIC)**

### 18.10.3 Objectives of Nanotechnology

- Promotion of basic research in nanotechnology.
- To provide infrastructure for Nanoscience and technology research.
- To enhance Nanoapplications and technology through promotion of application-oriented research and development projects with special efforts, either directly through industrial sector into nanotechnology research and development or through Public Private Partnership (PPP) ventures.
- Human Resource Development – Nano Mission should focus on providing effective education and training to researchers and professionals in diversified fields those enhance Interdisciplinary culture for nanoscale science, engineering and technology.
- International collaborations through exploratory visits of scientists and involvement in various workshops and conferences under various joint research projects.

## 18.11 Possibilities for the Future

- With the help of nanotechnology, there comes the possibility of manufacturing lighter, stronger and programmable materials, which require less energy than conventional material to generate and guarantee greater fuel efficiency in land, ships, aircrafts, transportation and space vehicles.

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# India Graphene Engineering and Innovation Centre (iGEIC)

Recently, Union **Ministry of Electronics and Information Technology (MeitY)** launched **India Graphene Engineering and Innovation Centre** (IGEIC) under the vision of **Viksit Bharat@2047.**

## About iGEIC

- **Launched by:** Ministry of Electronics and IT (MeitY) under **Viksit Bharat@2047** vision.
- **Registered as:** Not-for-profit (Section 8 Company). (Not-for-fit: a non-profit organisation registered under the Companies Act, 2013)
- **Purpose:** Commercialising **Graphene technology** through innovation and R&D.
- **Locations:**
  - ☐ R&D Hub – **Trivandrum, Kerala**
  - ☐ Business Hub – **Bengaluru, Karnataka**

## Potential Applications

1. **Energy: Supercapacitors and batteries** with high surface area (2630 $m^2/g$)
2. **Electronics: Faster and efficient** than silicon for semiconductors and transistors
3. **Water Treatment:** Nanoporous graphene membranes for **desalination and filtration**
4. **Biomedical: Graphene oxide (GO)** used for: (a) drug delivery, (b) biosensors, (c) cancer diagnostics and (d) tissue engineering
5. **Defence:** Strong yet lightweight material for **armour and ballistic protection**
6. **Environment:** Absorbs oils and solvents Helps in **wastewater purification**

## Challenges in Graphene Development

- **Toxicity:** Can damage human cell membranes
- **High Cost of Production**
- **Band Gap Problem:** Limits use in semiconductors
- **Mass Production Issues:** Hard to maintain quality in large-scale production

## Key Properties of Graphene

| Property | Detail |
|---|---|
| Strength | 200× stronger than steel, 6× lighter |
| Transparency | Absorbs only 2.3% of visible light |
| Thermal Conductivity | Up to 5000 W/m/K |
| Impermeability | No gas, not even hydrogen, passes through |
| Quantum Effects | Shows **quantum Hall effect**, useful in metrology and quantum computing |

## Global Graphene Scenario

- **Top Innovators:** China, United States, United Kingdom, Japan, South Korea, Russia
- **Top Producers:** China, Brazil
- **India's Output:** ~1/20th of China

## India's Graphene Initiatives

- **Graphene-Aurora Program:** Industry-startup support
- **India Innovation Centre for Graphene (IICG)** – JV of Digital University Kerala, C-MET and Tata Steel

## Research Institutions

- **IIT Roorkee:** Ultracapacitor tech
- **CeNS Bengaluru:** Nanomaterials and composites

**NANOPHARMACEUTICALS**

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They are a relatively new class of *therapeutic-containing nanomaterials* that often have unique 'nanoproperties' (physiochemical properties) due to their small size (compared with their bulk-phase counter parts), a high surface-to-volume ratio and the possibility of modulating their properties.

- *Nanopharmaceuticals present novel reformulation opportunities for active agents* (e.g., single molecule drugs, proteins, nucleic acids, etc.) that were previously insoluble or could not be targeted to a specific site of the body where they were needed.
- Nanopharmaceuticals can *also increase drug half-life* by reducing immunogenicity and diminishing drug metabolism.
- With these advantages, *nanopharmaceuticals have the ability to extend the economic life of proprietary drugs*, thereby creating additional revenue streams.

A Guideline was released for Evaluation of *Nanopharmaceuticals* in India. *Those are developed by* DBT, ICMR and Central Drugs Standard Control Organization (CDSCO). Developed *in line with the provisions of Schedule of Drugs and Cosmetics Rules, 1945 as well as Second Schedule of the New Drugs and Clinical Trials Rules, 2019* with specific requirements for nanopharmaceuticals.

### Need for these guidelines

*Nanocarrier-based targeted drug delivery* is an emerging field with introduction of *nanopharmaceuticals* in the market. These nanoformulations have higher efficacy, lower toxicity and are safer than the conventional drugs. Therefore, to provide *transparent, consistent and predictable regulatory pathways for nanopharmaceuticals in India* these guidelines were necessary.

### Application of these guidelines

The guidelines apply to the nanopharmaceuticals in the form of *finished formulation as well as Active Pharmaceutical Ingredient (API)* of a new molecule or an already approved molecule with altered nanoscale dimensions, properties or phenomenon associated with the application of nanotechnology intended to be used for treatment, in vivo diagnosis, mitigation, cure or prevention of diseases and disorders in humans.

### Significance

- These guidelines mark one of the most important steps for delineating quality, safety and efficacy assessment of the novel nanoformulations.
- The guidelines will pave the way for significant benefits through such cutting edge technology and contribute to the mission on '*Affordable Health Care for All*'.
- Private investments would also be attracted since *these guidelines would strengthen the regulatory system*.

- Nanorobotics has a good future in medicine as a drug targeting and gene therapy.

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- The nanorobots are capable of undertaking human tasks as well as tasks that are not being performed by the humans. The reconstruction of the depleted ozone layer could potentially be performed.
- Nanosurgical field helps in the treatment of everything from natural aging to diabetes, etc. Everything will be repaired easily with the introduction of nanosurgery.

## 18.12 Pitfalls of Nanotechnology

Nanoparticles can easily enter into the human body through food, skin or lungs and produce free radicals, thereby causing cell damage. If nanoparticles enter the bloodstream and cross the blood–brain barrier, they will become more dangerous to human life. Nano-bomb is the most dangerous outcome of the use of nanotechnology that contains engineered self-multiplying deadly viruses that are dangerous at different levels, like community, country or even an entire civilisation.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. With reference to carbon nanotubes, consider the following statements:** **[UPSC (Pre) – 2020]**

1. They can be used as carriers of drugs and antigens in the human body.

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2. They can be made into artificial blood capillaries for an injured part of human body.
3. They can be used in biochemical sensors.
4. Carbon nanotubes are biodegradable.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2, 3 and 4 only
(c) 1, 3 and 4 only
(d) 1, 2, 3 and 4

**2. With reference to the use of nanotechnology in health sector, which of the following statements is/are correct? [UPSC (Pre) – 2015]**

(i) Targeted drug delivery is made possible by nanotechnology.
(ii) Nanotechnology can largely contribute to gene therapy.

Select the correct answer using the code given below:

(a) Only (i)
(b) Only (ii)
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

**3. Graphene has been frequently in news recently. What is its importance? [UPSC (Pre) – 2012]**

(i) It is a two-dimensional material and has good electrical conductivity.
(ii) It is one of the thinnest, but strongest materials tested so far.
(iii) It is entirely made of silicon and has high optical transparency.
(iv) It can be used as 'conducting electrodes' required for touch screens, LCDs and organic LEDs.

Which of the statements given above are correct?

(a) (i) and (ii)
(b) (iii) and (iv)
(c) (i), (ii) and (iv)

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(d) (i), (ii), (iii) and (iv)

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## Answer Key

| | | |
|---|---|---|
| 1. (c) | 2. (c) | 3. (c) |

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**UNIT-VII** Intellectual Property Rights (IPR) and Related Rights

# CHAPTER 19
# INTELLECTUAL PROPERTY RIGHTS (IPR) AND RELATED RIGHTS

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CIPAM (Cell for IPR Promotion and Management), a professional body created under DIPP, has been entrusted with the implementation of the National IPR Policy 2016.

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The Department of Industrial Policy and Promotion has been made the nodal agency for all IPR-related issues and issues linked to copyrights have also been shifted to DIPP from that of the Human Resource Development Ministry.

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## 19.1 Introduction

Intellectual Property is a **collection of ideas and concepts,** which can be protected through trademarks, copyrights and patents. IPR, in literal sense, means something that emanates from human brain. **IPR are the rights which allow the patent holder, trademark carrier or copyrighted work to benefit them for their brainwork or investment.** These rights have been outlined under Article 27 of Universal Declaration of Human Rights (UDHR). The importance of IPR was first recognised in the Paris Convention for the Protection of Industrial Property (1883) and Berne Convention for the Protection of Liberty and Artistic Works (1886) (both administered by WIPO).

**Intellectual property can be better understood with various types.**

### 19.1.1 Copyrights

It is the exclusive rights bestowed or given upon the creator, writer, author, artist or any one of their original works to print, publish or record for a

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**fixed period of time.**

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### 19.1.2 Trademarks

It is a distinguishable **symbol, signature** or **design** of goods or products or services **representing a particular company.** It deals with the reputation of the Company's brand, essentially acting as a guarantor or indication of the origin of the service or products provided. Example: Coca Cola.

### 19.1.3 Patents

Patents can be termed as **legal monopoly** granted to an individual or a firm to make use of their brainwork by others, which can be done by selling their invention and making it a restricted property for use. An invention can also be sometimes rendered as a solution to any specific technological problem that can be a product or a process.

There are three following conditions to recognise any invention:

(a) It has to be a new invention,
(b) It should not be obvious, and
(c) Their needs should hold industrial application.

### 19.1.4 Industrial Design

Industrial design trains one to blend art with applied science in order to improve a product in terms of **visual aspects,** its application and usage.

### 19.1.5 Trade Secrets

A trade secret is the protecting or concealing of certain **recipes, processes, formulae or designs, patterns, information of certain product** by a particular company or business firms, thereby giving themselves an edge and advantage over competitors.

### 19.1.6 Trade Dress

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Trade dress, like Trademark, is a source identifier or indicator to the customers but the difference is that it is not only the symbol or short phrase of the company but everything that a company does to their product, **presenting itself to the customer distinctively, from giving visual appeal to a product or from packaging that signifies the source of the product to consumers.**

### 19.1.7 Plant Varieties

These are the rights given for the usage of a new plant species commercially, which is original and different and in order to get them registered, the variety's propagating material assessment is to be scrutinised.

### 19.1.8 Geographical Indications (GI TAG)

A geographical indication (GI) is a name or sign used on certain products which **correspond to a specific geographical location or origin (e.g., a town, region or country).**

India, as a member of *the* ***World Trade Organization (WTO), enacted the Geographical Indications of Goods (Registration and Protection) Act, 1999 which came into force with effect from 15 September 2003.***

GIs have been defined under Article 22(1) of the WTO Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement as: 'Indications which identify a good as originating in the territory of a member, or a region or a locality in that territory, where a given quality, reputation or characteristic of the good is essentially attributable to its geographic origin'.

The GI tag ensures that none other than those registered as authorised users (or at least those residing inside the geographic territory) are allowed to use the popular product name. **Darjeeling tea became the first GI tagged product in India, in 2004–05.**

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## 19.2 Need for Intellectual Property Rights

Intellectual Property Rights (IPR) are needed for the following reasons:

- They protect and safeguard the capacity of humans in their creation or creativity and their invention of new works in different fields or areas.
- They legally safeguard and protect new inventions and conceptions, thereby motivating the resources for more novelty.
- Protection of intellectual property **enhanced economic growth,** thereby creating new jobs and opportunities in industries. Efficient management of intellectual property rights can act as a driving force for economic, social and cultural well-being of the people.
- Intellectual Property (IP), in a way, **helps in maintaining a balance between the interests of creators and the public, providing** an environment for creativity and innovation to flourish.

[Refer **Infographic – Intellectual Property Rights (IPR)**]

**KAPILA**

- The Government has launched a campaign namely **Kalam Program for Intellectual Property Literacy and Awareness Campaign (KAPILA)** for Intellectual Property Literacy and creating patent awareness.
- The objectives of the scheme include creating awareness regarding IPR in Higher Education Institutions (HEIs), enabling IP protection of inventions originating from faculty and students of HEIs, development of Credit Courses on IPR, training programmes, etc.

## 19.3 History of Intellectual Property Rights in the World

Intellectual Property can be traced to its roots in England about 300 to 400 years ago, with British Statute of Anne 1710 and statute of monopolies originally created not as intellectual property statute but ultimately as a way

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to encourage learning and to keep the monarchy, which was so relatively powerful that time from being able to censor and control what was published.

In **1883, Paris Convention** for the **Protection of Intellectual Property** was set forth as an international agreement **to help creators in ensuring that their intellectual works are being protected in other countries.**

In **1887, Berne Convention** for the **Protection of Literary Work and Artistic Works** was approved and agreed to give rights to the creator to control and receive the payments of their work on the international level and arena.

In 1893, BIRPI (the United International Bureaux for the Protection of Intellectual Property) was formed with the amalgamation and submergence of the two secretariats set up to administer Berne and Paris Conventions.

In 1970, **BIRPI transformed into WIPO** (World Intellectual Property Organisation), which is an intergovernmental organisation with 193 member countries based in Geneva, Switzerland.

Later in **1974, WIPO, by joining the UN,** becomes a specialised agency thereby increasing the relevance of intellectual property.

### WORLD INTELLECTUAL PROPERTY ORGANISATION (WIPO)

- **UN specialised agency (est. 1967)** headquartered in **Geneva, Switzerland**
- Administers **26 international treaties** related to IP
- **India joined WIPO in 1975**
- Publishes:
  - ❏ Global Innovation Index (GII)
  - ❏ World Intellectual Property Indicators (WIPI)
  - ❏ WIPO Technology Trends
- **World Intellectual Property Day: 26th April**
  This day raises awareness about the role of intellectual property rights (IPRs) in encouraging innovation and creativity.

#### India's Membership in Major WIPO Treaties

- Paris Convention (1998) – Patents and Designs
- Berne Convention (1928) – Copyrights
- Budapest Treaty (2001) – Microorganism patents
- Marrakesh Treaty (2016) – Access for visually impaired
- Madrid Protocol (2013) – Trademarks

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- Patent Cooperation Treaty (1998) – Global filing
- Others: Nairobi, Washington, Phonogram Duplication, etc.

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WIPO was basically established to deal with issues of intellectual property particularly work involving creativity and innovations and the protection of traditional knowledge and culture.

## 19.4 IPR in India

### 19.4.1 Historical Perspective in India

Intellectual property in India has gone through a series of developments, which can be better understood as follows:

- In 1856, this Act of 1856 gave protection to inventions based on British Patent Law (1852), which gave certain exclusive privileges to inventors for a period of 14 years.
- In 1911, Patents and Designs Act was introduced.
- After India became independent in 1947, the government saw the need to update its patent laws. In 1949, a committee led by Justice **Dr. Bakshi Tek Chand** was formed to review these laws. They wanted to make sure the laws helped India's development. Based on their advice, **changes were made in 1950 and 1952 to allow licenses for certain inventions like medicines and agricultural products**. In 1957, another committee led by **Justice N. Rajagopala Ayyangar** suggested keeping the patent system, despite its flaws. This led to a proposed Patents Bill in 1965, which wasn't passed. Finally, in 1970, the Patents Act replaced the old law from 1911. It came into effect in 1972 with the publication of Patent Rules 1972.
- **In 1995, India joined World Trade Organisation (WTO).**
- **In 1998, India joined Paris Convention.**
- In 1999, Patents Amendment Act came into force allowing patents for pharmaceuticals and agro-chemicals.
- In 2002, The Patents Amendment Act 2002 came into force introducing new rules of 2003.
- In 2003, Second Amendment in Patents Act was proposed with terms of patent as 20 years after 18 months of publication and setting up of patent tribunal in Chennai.

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- **An Ordinance in 2004 was enforceable from 1 January 2005, which later got superseded by Patents Amendment Act, 2005.**

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Note: On March 15, 2024, the Ministry of Commerce and Industry (Department For Promotion of Industry and Internal Trade) notified the Patents (Amendment) Rules, 2024 that immediately came into effect.

### *India's Legal and Policy Framework*

| Law/Policy | Scope |
|---|---|
| **Copyright Act, 1957** | Literary, musical, artistic works |
| **Patents Act, 1970** | Inventions, processes |
| **Trade Marks Act, 1999** | Brand identity, logos |
| **Designs Act, 2000** | Industrial designs |
| **GI Act, 1999** | Geographical Indications |
| **PPVFR Act, 2001** | Plant varieties and farmers' rights |
| **National IPR Policy (2016)** | Unified, TRIPS-compliant, promotes awareness (NIPAM, KAPILA) |

### *Types of IPRs and Legal Provisions in India*

| IPR Type | Protection Scope | Law | Duration |
|---|---|---|---|
| **Copyright** | Expression of ideas | Copyright Act, 1957 | Variable |
| **Patent** | Invention (product/process) | Patents Act, 1970 | 20 years (from the date of filing the application) |
| **Trademark** | Business sign/logo | Trade Marks Act, 1999 | 10 years (can be renew Indefinite) |
| **Trade Secret** | Confidential info | No registration needed | Unlimited |
| **GI** | Origin-based product qualities | GI Act, 1999 | 10 years (renewable) |

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| | | | |
|---|---|---|---|
| **Industrial Design** | Aesthetic design | Designs Act, 2000 | 10 years |

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***India's Innovation and IPR Initiatives***

- **Make in India:** Promotes domestic manufacturing innovation
- **National IPR Awareness Mission (NIPAM):** IP literacy
- **KAPILA Scheme:** IP literacy among students
- **Digital Filing System:** Unified IP filing
- **Rank in GII 2024: 39th** (up from 81st in 2015)

***Socio - Economic Impact of Strong IPRs***

- Boosts local business competitiveness
- Generates high-skill employment
- Enhances India's global innovation reputation
- Encourages foreign investment

***Challenges and Solutions***

| Challenges | Solutions |
|---|---|
| **Bureaucratic delays** | Digitise and streamline IP processes |
| **Low awareness in MSMEs** (Micro, Small and Medium Enterprises) | Conduct IP literacy campaigns |
| **Weak R&D investment** | Public–private funding, incubation hubs (entrepreneurial ventures) |
| **Costly global filings** | Subsidies and international partnerships |

## 19.4.2 National Intellectual Property Right Policy

**It is the first policy of India on IPR,** which was revealed by the Finance Minister (Union Government) in accordance with TRIPS in May 2016. So far, only specific Act regulates the IPR in India. **CIPAM (Cell for IPR Promotion and Management), a professional body created under DIPP, has been entrusted with the implementation of the National IPR Policy 2016.** It is a vision document, which aims to create and exploit synergies between all forms of intellectual property, concerned statutes and agencies.

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The policy, which is in compliance with WTO's (World Trade Organisation) agreement on TRIPS (Trade Related aspects of IPRs), aims to sustain entrepreneurship and boost **'Make in India'** scheme.

**MAIN OBJECTIVES OF IPR POLICY IN INDIA**

- IPR Awareness and outreach
- Stimulating the IPR generation
- Strong legal and legislative framework
- Modernise and strengthen service-oriented IPR administration
- Commercialisation of IPR
- Enforcement and adjustment for combating IPR adjustment
- Human Capital Development for teaching, training, research and skill-building in IPRs

### 19.4.2.1 Highlights of New Intellectual Property Policy

- The said Policy aims towards the protection of public interest through the promotion of IPR as a profitable financial asset; and further encouraging the innovation and entrepreneurship.
- There will be **reviewing of plan after every five years** in discussion with stakeholders.
- Various steps, such as reviewing of the present IP laws, their revision and improvisation for the removing of abnormalities and discrepancies, is required to be undertaken in order to have strong and effective IPR laws.
- There is complete conformity of policy in accordance with WTO's agreement on TRIPS.
- There should be special focus on creation of IPRs awareness and its effective enforcement, in addition to persuading of IP merchandising with the help of different incentives.
- India will participate gainfully in international treaties and agreements negotiations in discussion with stakeholders. According to the policy, there will be evaluation of the country's consent to few multilateral treaties, which are in its interest, by the government and become a party to those treaties, which the country has actually implemented for facilitating it in its decision-making process.

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- As per the policy proposal, for all the IPR related issues, the nodal agency will be the **Department of Industrial Policy and Promotion (DIPP).** Moreover, **DIPP will also be responsible for tackling of all the copyrights related issues in place of the Human Resource Development (HRD) Ministry.**
- There is a rationalisation of **trademark offices for reducing the examination and registration time to just 1 month by the year 2017.** There is hiring of approximately 100 new examiners for trademarks by the government and there is **reduction of trademarks evaluation time from 13 months to**
  **8 months, and ultimately bringing it to one month by March 2017.**
- There will be the coverage of films, music and industrial drawings under the copyright.
- The policy helps in the filing of domestic IPR from IPR generation to commercialisation and further helps in promoting the research and development through tax benefits.
- The **policy also facilitates** in the creation of **an effective loan guarantee scheme** for the promotion of start-ups.
- **As per the policy, the country will persist** in the utilisation of the legislative space and **flexibilities offered in international treaties and the TRIPS Agreement.** These flexibilities consist of the **usage of provisions like Section 3(d) and compulsory licences (CLs) under the country's sovereign rights to ensure the availability of essential and life-saving drugs at affordable prices.**
- Due to the policy, the patent laws of the country remain intact and there is no opening up of
  Section 3(d) of the Patents Act, setting the standards in the country for reinterpretation.
- **In the field of compulsory licencing, the CL has been issued in the country only for a cancer drug.** As per Mr. Jaitley, the country rarely exercises this power. This being considered as significant statement as the developed countries, such as the US, have raised their apprehensions over the country's issuance of the CL. According to the standards of the WTO, a government in public interest can raise CL allowing a company for making a patented product without the patent owner's permission. **As per the Indian Patents Act, a CL can be**

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**raised in the category of a drug if any medicine seems to be not reasonable, and the permission for drug manufacturing is given by the government to the qualified generic drug manufacturer.**

- In view of the financial support on the export and sale of the items on the basis of IPRs produced from the research being publically funded is the outcome of the IPR policy, ultimately benefitting the government.

### 19.4.2.2 Compulsory Licencing

Compulsory licenses, **authorised by the Controller General, allow a third party to use patented products or processes** without the patent owner's permission.

This is recognised nationally and internationally, covered in the **Indian Patents Act, 1970 and TRIPS agreement.**

**Application for compulsory licence can be made after 3 years** if certain conditions are met, as per
Section 83 of the Act.

The **Controller can also issue licenses suo moto under Section 92** upon a government notification for reasons like **national emergency, extreme urgency or public non-commercial use**.

#### *First Licence Issued*

- **India's first ever compulsory licence was granted** by the Patent Office **on March 9, 2012,** to **Hyderabad-based Natco Pharma for the production of generic version of Bayer's Nexavar,** an anti-cancer agent used in the treatment of liver and kidney cancer.

#### *Global Perspective on Compulsory Licencing*

- This phenomenon of compulsory licencing is a hugely debated issue. **Many developing countries are giving importance to the compulsory licencing because of the unavailability and unaffordability of the medicines,** and they are continuously granting more and more compulsory licences. **The developed countries of Europe, USA are opposing this view as it would make innovation difficult for the pharmaceutical companies.**

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Note: During the COVID-19 pandemic, compulsory licensing became a hot topic. It highlighted the importance of health and the need to regulate patents and intellectual

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property rights. Even major economies faced shortages of essential medicines. High prices made these medicines unaffordable, especially for underdeveloped countries. Limited resources prevented them from manufacturing or buying these medicines. Pharmaceutical companies' patent protections led to monopolies, driving up costs. If a third person or manufacturer tries to manufacture the patented products, they can seek compensation and file a suit for injunction. This is where the concept of compulsory licensing came into picture whereby a third party is granted a licence by the authority concerned, to manufacture, sell and distribute the patented product without the permission of the patentee. Since **'Gilead Sciences' (a pharmaceutical company)** anti-viral drug **Remdesivir had shown efficacy in treating COVID-19 patients, compulsory license was issued for the manufacturing of an affordable generic version of the medicine**.

Note: While the cost of manufacturing Remdesivir for a full course – as worked out by experts – is less than $10 or ₹750 in the US and about ₹100 in India.
Gilead, by virtue of its patent monopoly, held the world to ransom by asking a price that was hundreds of times its cost.

### 19.4.3 Steps Taken by Government to Improve the Intellectual Property Rights Ecosystem in India

- A comprehensive national IPR policy has been put in place.
- Integrated approach and synergy had been adopted through transferring various IP offices and Acts under DIPP and also merging Copyrights Board with Intellectual Property Appellate Board.
- Cell for IPR Promotion and Management has also been established for assisting in simplifying and streamlining of IP processes as well as creating IPR awareness, commercialisation and enforcement.
- A scheme for facilitating Start-up Intellectual Property Protection (SIPP) has been launched for encouraging innovation and creativity of Start-Ups.
- **India has also become the 90$^{th}$ member of Madrid Protocol.**

**MADRID PROTOCOL**

- It is an international treaty that **allows a trademark owner to seek registration in any of the countries that have joined Madrid Protocol by filing a single application.**
- International Bureau of the World Intellectual Property Organisation administers the international registration system.

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- The Department of Industrial Policy and Promotion (DIPP) has inked agreement with Punjab State Council of Science and Technology to establish India's first TISC (Technology and Innovation Support Centre).
- TISC is WIPO's (World Intellectual Property Organisations) programme that provides innovations in developing countries with access to locally based high-quality technology, information and related services. The programme will help innovators to fully exploit their creative potential and also protect their IPRs.
- CIPAM (Cell for IPR Promotion and Management) has been designated as the national focal point for the TISC networks.
- Scheme for IPR Awareness Creative India; Innovative India has been launched by CIPAM to raise IPR awareness across India.
- CIPAM has also launched 'IPrism', an Intellectual Property Competition for college and university students to foster a culture of innovation and creativity.

### CELL FOR IPR PROMOTION AND MANAGEMENT (CIPAM)

With the slogan of 'Creative India; Innovative India', the Government approved CIPAM, a professional body under the aegis of DIPP to take forward the implementation of the National IPR Policy in May 2016. **CIPAM has been created to work towards public awareness of IPRs, promotion in filing of IPRs, to provide various inventors to commercialise their IP assets.** CIPAM also works for implementing the National IPR Policy, in collaboration with various Government Ministries/Departments and other stakeholders.

### 'IPRISM' AND ITS ISSUE IN INDIA

It is an Intellectual Property (IP) Competition to promote a culture of innovation and creativity in the younger generation at college and university-level students. **Iprism promotes young creators and provides unique opportunity in their creations.** Iprism launched various bodies: The Cell for IPR Promotion and Management (CIPAM), Department of Industrial Policy and Promotion (DIPP), in collaboration with ASSOCHAM and ERICSSON India. This competition encourages students in submitting piracy and counterfeiting-based films under dual 30 and 60 seconds categories and further on gaming mobile applications. Due to the piracy and counterfeiting, the industry is encountering a considerable loss and thereby posing serious safety threats to the consumers. This type of

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awareness will help tackle such types of crimes as the little knowledge on piracy and circulation of fake items leads to negative socio-economic impacts on the society.

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## 19.5 Limitations of IPR

According to the critics of intellectual property, it is harming health of new creations, preventing progress and providing benefits to concentrated masses only.

As per the critics, intellectual property harms the public interest by its wide control of copyright extensions and other patents.

Some critics also believed that if property rights in information and ideas are allowed, then artificial scarcity would be created. Another opinion is that intellectual property needs to be expanded, in duration as well as in scope.

## 19.6 TRIPS Agreement

TRIPS agreement is an agreement at the international level, governed and administered by the World Trade Organization (WTO), which sets some basic and minimum standards for Intellectual property. The very first of its kind, an agreement was prepared to provide International Law for Intellectual Property and the agreement was on TRIPS. It comes under Article (4) of WTO provisions, and provides protection to all the member countries of WTO in terms of copyrights, producers of broadcasting organisations, trade dress, geographical indication, designs and all other patents, etc. It was concluded in Uruguay round of GATT 1994.

## 19.7 Copyright and Related Rights

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### 19.7.1 Provision of Copyright and its Related Rights

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Copyright law protects the works of authors, artists, and creators. **Related Rights**, a related area, offer similar protections but for shorter durations. It **covers actors, musicians, phonogram producers and broadcasting organisations**. Administered by the HRD Ministry, the Copyright Act, 1957 now falls under the Department of Industrial Policy and Promotion (DIPP), established in 1995 under the Ministry of Commerce.

**DEPARTMENT OF INDUSTRIAL POLICY AND PROMOTION (DIPP)**

It was set up in the year 1995 and the Department of Industrial Development was merged with DIPP in the year 2000 working under the Ministry of Commerce and Industry, Government of India. Considering the national priorities and socio-economic goals, **DIPP facilitates the industrial sector growth through the formulation and implementation of promotional and developmental measures and also liable for assisting and raising the country's FDI flows.** The department is also responsible for the items intellectual property rights in relation to patents, designs, trademarks and geographical indication and further manage the actions in linkage to their encouragement and protection. DIPP is now called as Department for Promotion of Industry and Internal Trade (DPIIT).

Following are the main areas of DIPP:

- India's Industrial Policy Formulation and Implementation.
- The FDI policy Formulation and Amendment.
- Industrial Promotion.
- Intellectual Property Rights Policy Formulation and Implementation.

### 19.7.2 Rights Provided and Assured by 'Copyright and Related Rights'

Copyright law grants certain rights to copyright holders as follows:

- They have the exclusive right to permit or deny others from using their work.
- They can authorise or restrict reproduction, including printing, recording, broadcasting, and translation.

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These rights are vital for creators to succeed financially. Often, creators transfer these rights to companies in exchange for compensation, typically a

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percentage of revenue as royalty. **Economic rights of creator last around 50 years after the creator's death (as per the Copyright law-as relevant in the WIPO Treaty)**, though national laws may differ. Related rights protect works for shorter periods, usually 50 years after performance, recording, or broadcasting. Copyright holders can enforce their rights through civil suits, administrative actions, or criminal prosecution, including orders for item destruction or inspections. These methods help ensure creators and their successors benefit from their creations and protect their intellectual property.

### 19.7.3 Benefits of Protecting Copyright and Related Rights

Protection of these rights is important for nurturing and growth of creativeness in humans with improved innovation. It gives recognition along with fair economic reward to creators and also increases their willingness to do such works in future. So, the companies and individual with the assurance of rights granted invest more faithfully in the creation. This, in turn, helps in encouraging and pushing towards socio-economic development of every nation.

### 19.7.4 Delhi High Court Copyright Judgement (DU Photocopy Case)

On December 9, 2016, the ruling was no restriction books can be photocopied, as long as 'justified by the demands of the course'. Under Section 52(1)(i) of the Copyright Act, photocopying material, which is copyrighted for using/usage purpose 'in the course of instruction', is permitted. The court concluded that 'qualitative and quantitative' tests for fair use are not included in the law.

The large publishing companies such as Oxford University Press and Cambridge confronted the University of Delhi and its photocopier for violation of copyright. As per the Delhi High Court ruling, the course pack photocopying for the education point is within the restrictions of law.

Photocopying, in itself, does not lead to violation of copyright. According to the court, copyright is to create equilibrium between

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challenging ideas of private and public interests.

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The software programming can be considered as a 'literary work'. According to the Copyright Act, 1957, there is inclusion of software programmes, tables and compilations, including databases as literary work. In order to register the software products' copyright, along with the application, source code has also to be submitted.

### 19.7.5 ANI Vs OpenAI (Copyright issue)

As artificial intelligence (AI) continues to evolve, it raises new legal and ethical challenges—particularly in **how AI systems are trained using vast datasets.**

**A landmark legal dispute has emerged in India, where news agency ANI has taken OpenAI to court, alleging 'unauthorised use of its copyrighted content'**.

This case is not only significant for India's legal landscape but also echoes similar global debates on the rights of content creators in the age of AI.

***Case Background***

- **ANI's Allegation:** Established in 1971, ANI claims that OpenAI used its copyrighted news content without permission to train AI models like ChatGPT. It argues this violates India's Copyright Act of 1957 and spreads misinformation through false attribution.
- **OpenAI's Response:** Represented by advocate Amit Sibal, **OpenAI denies wrongdoing**, citing industry standards, fair use principles and lack of jurisdiction due to no physical presence or servers in India. OpenAI also noted it had blocked ANI's content from future use following a legal notice in September 2024.
- The **case was filed by Asian News International ('ANI'),** an Indian news agency, alleging that OpenAI used its copyrighted news content without authorisation to train its large language model (LLM), Artificial Intelligence chatbot ChatGPT.
- **ANI contends that OpenAI had no right to use its content for profit, even if the material was freely accessible online.**

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- The agency is seeking ₹2 crore in damages and an injunction to prevent OpenAI from further using its content.
- This case spotlights the ongoing conflict between the intellectual property rights of content creators and AI companies' use of publicly available data for model learning.

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### Legal Arguments

***ANI's Position:***

- **Copyright Violation:** Unauthorised use of ANI's content infringes on its exclusive rights under Indian law.
- **False Attribution/Allegation:** ChatGPT occasionally attributes fabricated or incorrect content to ANI.
- **Refused Licensing:** OpenAI declined ANI's formal licensing offer in October 2024.

***OpenAI's Defense:***

- **Jurisdictional Challenge:** Operates internationally with no legal footprint in India.
- **Fair Use Claim:** Asserts that the data was used in a transformative way under fair use.
- **Public Data Argument:** Maintains that only publicly accessible information was used.

### Current Court Status

- The **Delhi High Court** acknowledged the case's complexity and **appointed an *amicus curiae*** to assist.
- The court **denied interim relief** to ANI, citing the need for deeper examination of AI and copyright intersections.

### Global Context

- **USA:** OpenAI faces similar lawsuits from *The New York Times*, Universal Music Group and authors like Sarah Silverman and George R.R. Martin.
- **EU:** The GDPR and the proposed AI Act emphasise ethical AI use and copyright-safe data training.
- **OECD and Global Conferences:** Stress the importance of transparency, ethical AI development and international cooperation.

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### Key Legal and Ethical Challenges

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| Issue | Concern |
|---|---|
| **AI Ownership** | Current laws (e.g., India's Copyright Act) don't recognise AI as a creator or copyright holder. |
| **Data Mining** | Many datasets include copyrighted material without explicit consent. |
| **Fair Use Ambiguity** | Fair use application in AI is unclear and often contested. |
| **Lack of Transparency** | AI firms often don't disclose what data was used to train models. |

***Implications for India's AI Ecosystem***

- **Greater Legal Scrutiny:** Startups may face more stringent review of their data practices.
- **Higher Compliance Costs:** Licensing and legal safeguards may increase costs for smaller firms.
- **Barriers to Innovation:** New players may struggle to match the data advantage of big tech.
- **Opportunities:** The case could encourage a structured licensing market and push for regulatory clarity.

## 19.8 Trademark

As per WIPO, a trademark is a sign capable of distinguishing the goods or services of one enterprise from those of other enterprises. This implies that a trademark can be a **word, signature, name, device, label, numerals or combination of colours** used by one undertaking on goods or services or other articles of commerce to distinguish it from other similar goods or services belonging to a different undertaking.

The legal requirements to register a trademark under the Trade Marks Act, 1999 are:

1. The selected mark should be capable of being represented graphically (that is in the paper form).

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2. It should be capable of distinguishing the goods or services of one undertaking from those of others.
3. It should be used or proposed to be used as a mark in relation to goods or services for the purpose of indicating or so as to indicate a connection in the course of trade between the goods or services and some person have the right to use the mark with or without identity of that person.

### 19.8.1 Different Types of Trademarks that may be Registered in India

- Any name (including personal or surname of the applicant or predecessor in business or the signature of the person), which is not unusual for trade to adopt as a mark.
- An invented word or any arbitrary dictionary word or words, not being directly descriptive of the character or quality of the goods/service.
- Letters or numerals or any combination thereof.
- The right to proprietorship of a trademark may be acquired by either registration under the Act or by use in relation to particular goods or service.
- Devices, including fancy devices or symbols.
- Monograms.
- Combination of colours or even a single colour in combination with a word or device.
- Shape of goods or their packaging.
- Marks constituting a 3-dimensional sign.
- Sound marks when represented in conventional notation or described in words by being graphically represented.

### 19.8.2 Registrar

- The Controller General of Patents, Designs and Trade Marks heads the TRADE MARKS Registry offices and functions as the Registrar of TRADE MARKS.

### 19.8.3 Intellectual Property Appellate Board (IPAB)

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Established on September 15, 2003, by the Indian Government, the Intellectual Property Appellate Board (IPAB) resolves appeals against Registrar's decisions under the Indian Trade Marks Act, 1999, and the Geographical Indications of Goods (Registration and Protection) Act, 1999.

Since April 2, 2007, IPAB also hears appeals on Patent Controller's decisions under the Patents Act, transferring pending High Court appeals to itself.

**Organisation of an IPAB Bench:** Each IPAB Bench comprises a Judicial and a Technical Member, with technical member's qualifications outlined in the relevant Acts.

**Jurisdiction:** Appeals from Registrar's decisions must be filed within three months from the date on which the order/decision was communicated to the appellant, extendable with valid cause.

The IPAB **dismissed Red Bull's plea to remove a trademark held by Dr. Reddy's Laboratories Limited.**

### 19.8.4 Red Bull's Plea on Trademark Rejected

- In its petition, Red Bull claimed to be a pioneer in using the concept of an animal/human acquiring wings after consumption of the Red Bull Energy Drink and hence for the tagline **'Gives You Wings'** they are the registered proprietor in India.
- **They pointed out that another trademark which was registered in November 2001 with the tagline 'Your Wings to Life', in the name of Dr. Reddy's Laboratories Limited** have not undertaken any activity to commence use of the impugned registration and hence sought for its cancellation.
- IPAB stated 'The rival trademarks are neither deceptively similar nor identical with. The applicant mark 'Your Wings to Life' is not the main mark. It is the sub-brand. Therefore, the goodwill and reputation alleged by the applicant is not pertaining to the main mark Red Bull'. Hence, both trademarks are not similar and the petition got dismissed.

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## 19.9 Patent

### 19.9.1 Concept of Patent

It is a special right given for an invention. The invention can be a process or a product that brings an unusual way of doing something, or that provides a scientific solution to a problem. Duration for protection of the invention is limited and generally restricted to 20 years.

### 19.9.2 Importance of Patent

Patent rewards individuals by recognising their creativity and by offering incentives to their inventions and improving the quality of human life by motivating and encouraging further innovation.

### 19.9.3 Protection of Different Types of Patents

It safeguards inventions by giving the rights to the owner and making sure that without their consent, he invention cannot be commercialised. Court holds the enforcement rights of the patents and the authority to stop infringement of the patent. The court also holds the right to declare a patent as invalid after successful challenge by a third party.

### 19.9.4 Rights of Patent Owners

The owner has the right to take decisions on who can use its invention on mutually agreed terms for any particular time period. It also has the right to sell their invention, if they wish to do so. After some fixed duration of time, patents get expired; which means that there is ending of patent protection and further, the entering of invention into the public realm is termed as becoming off patent. After that, there are no limited rights of owner on the invention, and so the invention is easily available for commercial exploitation by the public.

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### 19.9.5 Importance of Patents in Everyday Life

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Patented inventions are deeply intact with human life, from sewing machines (patented held by Howe and Singer), to magnetic resonance imaging (MRI) (patented by Damadian) to electric lightning (patented by Edison and Swan) and the iPhone (patented by Apple). All patent owners, for enriching the total technical knowledge in the world, publically disclose information of the invention. This will enhance more creation and innovation. Hence, the patents provide valuable information and inspiration for future generations of researchers and inventors with the protection of the patent owner.

### 19.9.6 Patent Granting

To secure a patent, an application must be filed with clear details of the invention's title, technical field, description, and background. Visual aids like diagrams are helpful. Claims defining the patent's scope are crucial. Regional or national patent offices, like African Intellectual Property Organisation (AIPO) and European Patent Office (EPO), grant patents for individual or multiple countries. WIPO's Patent Cooperation Treaty (PCT) simplifies international patent filing. In India, patents are administered by the Department of Industrial Policy and Promotion, Ministry of Commerce under The Patents Act, 1970.

### 19.9.7 Protection of Invention

Conditions for an invention to be protected by a patent are:

- Invention should be of practical usage.
- Patent should be novel.
- An 'inventive step' needs to be shown by the invention.

### 19.9.8 Highlights of India's Patent Policy 2016

The policy aligns with WTO's TRIPS agreement, focusing on IPR awareness, enforcement, and commercialisation incentives.

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- DIPP manages all IPR and copyright matters.
- Trademark office streamlining aimed to reduce examination time to one month by 2017. It accelerated trademark evaluation to 8 months, then one month by March 2017.
- The policy aids in domestic IPR filing, commercialisation, R&D promotion *via* tax benefits and startup loans.
- It ensures the use of international treaty flexibilities like Section 3(d) and CLs to maintain affordable drug availability.
- In the year 2013, the Supreme Court had upheld the decision of Intellectual Property Appellate Board regarding the rejection of giving patent protection to the application of Novartis for a new drug form termed as imatinib but branded as Glivec. Novartis had challenged Section 3(d) of the country's patent act, which guarantees that the companies do not get the patent protection by merely altering the already existing drug without there being any subsequent changes in its healing effects. At the time of giving decision, Section 3(d) of the Indian Patents Act, 1970 was validated by the Supreme Court. Moreover, the court noted the product named as beta crystalline had already been identified earlier to the year 1995, with the past patent being held by the Novartis. The outcome of the decision is that the country's patent law was completely upheld and the granting of patents will be undertaken only for authentic innovation and there will be no acceptance of any litigative patenting. This means that the patent cannot be granted if any old molecule is kept in a new substance, just creating a minor modification, and presenting it as an entirely new invention.
- The creation of trivial modifications into the original product for the claiming of another patent to extend the period for the validation of patent is termed as the ever-greening of patents.
- The compulsory licencing (CL) has only been issued once for a cancer drug, Sorafenib to-sylate (Nexavar). As per the Indian Patents Act, a CL can be raised in the category of a drug if any medicine seems to be not reasonable, and the permission for drug manufacturing is given by the government to the qualified generic drug manufacturer.

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The patent system aims to encourage innovation and technological development. India's current Patents Act, 1970, effective from 1972, updated the previous Patents and Designs Act of 1911. According to Section 3(j), plants and animals (excluding microorganisms), seeds, varieties, species, and biological processes for their production or propagation are not considered inventions and are thus not patentable.

The Patent (Amendment) Act 2005, effective from January 1, 2005, expanded product patents to include all technological fields such as food, drugs, chemicals, and microorganisms. It also introduced pharmaceutical product patents for the first time in India. The amendment clarified that patents must meet the criteria of novelty, non-obviousness, and industrial applicability. Discoveries, which uncover something that already exists in nature but was previously unknown, are not patentable under Section 3 of the Indian Patent Act 1970.

### 19.9.9 Patent Prosecution Highway Programme

The Union Cabinet approved the Indian Patent Office's adoption of the Patent Prosecution Highway (PPH) programme, initially piloted with the Japan Patent Office (JPO) for three years, it continues in the pilot stage till date.

#### What is PPH?

PPH expedites patent prosecution by sharing information between participating patent offices. Indian applications in specified fields like Electrical, Electronics, and Computer Science can benefit, while JPO accepts applications from all fields.

***Benefits for the Indian IP office*** include faster application processing, reduced backlog, enhanced search and examination quality, and accelerated examination opportunities for Indian inventors, especially MSMEs and startups, in Japan.

### 19.9.10 Patenting in India

- The Union Government has the powers to formulate rules for the implementation of the Act and amending the administration of patent, according to Section 159 of the Indian Patents and Designs Act, 1970, and therefore the Patents Rules, 1972, were declared and amended under the Patents Amendment Rules, 2006.

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- The government has made amendments to the rules and introduced methods for facilitating the start-ups as well as entities' patent applications evaluation, preferring India for the first patent filing. Moreover, there are around lakhs of patent applications pending in the country.

Note: India issued one lakh patents in 2024, ten years back it was just 6000.

### 19.9.11 Secondary Patents and Its Recent Issue for India

Every year there are many rejections of the applications for pharmaceutical patents at the Indian Patent Office that sought protection in the form of secondary patents for blockbuster medicines.

***Secondary Patents***

- The owners market is being offered by the patents exclusivity for a restricted time period. In case of medicines, the exclusiveness should last as long as the primary patent, which relates to the active pharmaceutical ingredient (API) of the medicine and is effective, particularly for 20 years.
- The patent cliff is being described as the ending of patent exclusivity as the prices of the drug subsequently drop sharply by 80 per cent due to generic competition.
- The pharmaceutical companies discover new ways due to the risk of their huge profit reduction, which made them delay their exclusivity through the filing of secondary patents for derivatives and variants of the API, such as a physical variant of the API, a new formulation, a dosage regimen, or a new method of administering the medicine.
- The secondary patents prop up before the expiry of a primary patent, thereby stretching the exclusivity beyond, a practice that is called 'evergreening'. This strategy is most lucrative when employed in the context of so-called blockbuster medicines, which reap annual revenues exceeding $1 billion.
- These provisions also extend to biologics, the new big players in the therapeutics marketplace. Biologics, due to their complex structure, offer more opportunities in secondary patenting for extending patent terms.

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### 19.9.12 PepsiCo's Patent Issue

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The **Protection of Plant Varieties and Farmers' Rights** (PPV&FR) Authority has revoked PepsiCo India Holding's (PIH) Plant Variety Protection (PVP) certificate for the potato variety **FL-2027**, citing several grounds. These include **inaccuracies in the information provided** during the application process, **eligibility issues** regarding the recipient of protection, and considerations **against public interest**.

#### *What's the issue?*

In 2019, PepsiCo had initiated legal actions against farmers in Gujarat for cultivating the FC5 potato variety, known for its suitability in snack production due to its low moisture content. However, the lawsuits were withdrawn later that year with PepsiCo expressing a desire to resolve the matter amicably. Subsequently, activist Kavitha Kuruganti petitioned the PPV&FR Authority to revoke the intellectual property protection granted to PepsiCo's FC5 variety, arguing that Indian regulations prohibit patents on seed varieties under Section 39 of the PPV&FR Act, 2001.

#### *PPV&FR Act*

The PPV&FR Act, enacted by India in 2001, establishes a *sui generis* system in line with the International Union for the Protection of New Varieties of Plants (UPOV), 1978. This legislation aims to protect plant breeders' rights, recognise farmers' contributions to conservation and improvement of plant genetic resources, and promote agricultural development by facilitating investment in research and development.

#### *Rights Under the Act*

Under this Act, breeders have exclusive rights over their varieties, researchers can use registered varieties for experimentation and research with certain conditions, and farmers have rights to save, use, exchange, and sell farm produce, including seeds, provided they do not sell branded seeds of protected varieties.

Overall, the PPV&FR Act seeks to balance the interests of breeders, researchers, and farmers while fostering agricultural innovation and

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ensuring availability of high-quality seeds for farmers.

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## 19.10 Computer-Related Inventions (CRI)

- In February 2016, the Controller General of Patents, Designs and Trademarks Office declared a modified set of guidelines for the evaluation of Computer – Related Inventions.
- The latest guidelines are in order with the provisions of the Patents Act, 1970, excluding mathematical and business methods, algorithms and computer programmes from patentability.
- In order to promote stability in the patents field on computer-related inventions, the Controller General of Patents, Designs and Trademarks Office introduces Manuals and Guidelines for its evaluation at times.

The earlier guidelines in the field of patent made possible that some software advancement could get a patent in comparison to the previous art, which could lead to the overflow of patents to be given in the software field. This could have created an adverse impact on the Indian IT industry, particularly the Indian start-ups, product companies, research organisations, a range of academics, organisations such as Society for Knowledge Commons, Free Software Movement of India, which are at vanguard of innovation and have accepted the guidelines, which will further encourage innovation in the country.

## 19.11 Intellectual Property Index

The report on **International Intellectual Property Index 2024** (IIPI-2024) is released by the US Chamber of Commerce's Global Intellectual Property Centre (GIPC). It is an annual index, which examines a country's Intellectual Property (IP) framework across eight categories of indicators: patents, copyrights, trademarks, trade secret and market access, enforcement commercialisation of IP assets, systemic efficiencies and ratification of international treaties. It consists of 90 per cent of global gross

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domestic product and grades countries on patents, trademarks, copyrights, trade secrets, enforcement and international treaties.

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***Major Findings of the IIPI 2024***

- The US Chamber of Commerce in February 2024 released the 12th edition of its International IP (intellectual property) Index.
- **India ranked at 42 out of the 55 countries in the index** with an overall score of 38.64 per cent.
- The top 5 economies in the world for IP rights are (i) United States (95.48%), (ii) United Kingdom (94.12%), (iii) France (93.12%), (iv) Germany (92.46%) and (v) Sweden (92.12%).

## 19.12 Industrial Design

### 19.12.1 Basic Concept of Industrial Design

According to WIPO, 'A design may consist of three-dimensional features, such as the shape or surface of an article, or two-dimensional features, such as patterns, lines or colour'. A wide variety of products are included in industrial designs such as technical and medical instruments, appliances, vehicles, house ware, watches, etc. The condition for the protection under national law is that industrial design must be original/new and non-functional, which means that an industrial design is primarily of an aesthetic nature, and any technical features of the article to which it is applied are not protected by the design registration. However, a patent could protect those features.

### 19.12.2 Protection of Industrial Designs

- This gives an exclusive right against unofficial copying of the design by other third parties.
- It also promotes fair competition and honest trade practices, which benefit consumers too. Protection encourages the people to create more useful products, which benefit the people in general.

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- It may also ensure a good return on investment, thereby encourage creativity in manufacturing as well as in traditional arts and crafts and industries. It also expands the export market of that nation.
- The small- and medium-sized enterprises as well as the craftsmen and individual artists, in both developed and developing countries, should have access to industrial designs.

Industrial design registration is under industrial design law for its protection. But, those designs which are 'new' or 'original' can only be registered. 'New' means no design is known to have existed previously with the similar or identical design. Protection is given generally for 5 years, with renewal possibility normally for 15 years.

Industrial design protection is generally bound to the concerned nation. The Hague Agreement, which concerns the International Registration of Industrial Designs and a WIPO-administered treaty, laid down the international designs registration procedure. The applicant needs an application to be filed in WIPO or in the regional office or national office of a state, which is a party to the treaty.

## 19.13 Geographical Indication

A GI tag is a **designation used on certain products indicating their specific geographical origin**.

- It ensures that products possess qualities, reputation or characteristics unique to their place of origin.
- Only authorised users or producers from the geographical area are permitted to use the product's name with the GI tag.
- It protects the product from unauthorised use, imitation, or misuse by others.
- The **validity of a registered GI tag is for 10 years.**
- **GI registration is overseen by the Department for Promotion of Industry and Internal Trade (DPIIT) under the Ministry of Commerce and Industry**.

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- The **Geographical Indications of Goods (Registration and Protection) Act, 1999** governs the registration and protection of geographical indications in India.
- It aligns with the **WTO Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS).**
- The **Paris Convention acknowledges and emphasises** the importance of protecting geographical indications as integral components of intellectual property rights.

### 19.13.1 Geographical Indications at the International Level

- Geographical Indications are covered as a component of intellectual property rights (IPRs) under the Paris Convention for the Protection of Industrial Property.
- GI is also governed by the World Trade Organisation's (WTO's) Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS). In India, Geographical Indications registration is administered by the Geographical Indications of Goods (Registration and Protection) Act, 1999 which came into force with effect from September 2003. The first product in India to be accorded with GI tag was Darjeeling tea in the year 2004–05.
- Geographical indication rights are territorial. This means that these rights are limited to the country (or region) where protection is granted. At present, no 'world' or 'international' geographical indication (GI) right exists.

### 19.13.2 Basmati Rice GI Tag: Advantage India as EU republishes Pakistan Plea under a New Clause

In May 2024, the European Union (EU) republished Pakistan's application for a geographical indication (GI) tag for basmati rice under a new clause, which could give India an advantage in its own application for the tag. India filed its application in July 2018 and it's listed under Article 50 (2) of EU regulation. According to trade analysts, the republication of Pakistan's application is a sign that India is the rightful owner of basmati rice.

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***Background***

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- The statute of Geographical Indications (Registration and Protection) Act in Pakistan is the basis of Pakistan opposing Indian application for registration of Basmati rice exclusive rights. India has said that it is an Indian-origin product in its application, published in the EU's official journal on September 11, 2020.
- As per the EU's official journal, any country can oppose the application for registration of a name pursuant to Article 50(2)(a) of Regulations of the European Parliament and of the Council on quality schemes for agricultural products and foodstuffs within three months from the date of publication.
- Pakistan has claimed to be a major grower and producer of Basmati rice and hence pleading to declare India's application for exclusivity as unjustified.

**GI TAG SOUGHT FOR INDIA'S COSTLIEST MUSHROOM**

A geographical indication (GI) tag has been sought for one of the costliest mushrooms in the world that grows in Jammu and Kashmir's Doda district by an NGO, Borderless World Foundation.

**Key Points**

- Locally called **Gucchi, or Morel,** the mushroom, priced at INR 30,000 a kg, grows at an altitude of 2000 metres in cold environment of coniferous forest and collected by local people.
- It is said to have medicinal and nutritional properties and is a rich source of protein.

- In May 2010, GI status was given to basmati grown in Punjab, Haryana, Delhi, Himachal Pradesh, Uttarakhand and parts of western Uttar Pradesh and Jammu & Kashmir.

Note: Frequently, the GI and trademark are erroneously mistaken for each other and each year GI tag is being accorded on different varieties of new articles.

### 19.13.3 Summary of the Basmati GI Tag Issue

- The Agricultural and Processed Food Products Export Development Authority, being a statutory body with the duty of promotion of export

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and development of a products variety, had submitted an application for the Basmati rice registration in respect of 13 districts in Indo-gangentic plain situated on the Himalayan foothills, under the Geographical Indications of Goods (Registration and Protection) Act, 1999.

- The Intellectual Property Appellate Board has instructed for the issuance of GI tag to Basmati Rice by the Chennai-based GI registry.
- The APEDA has included Madhya Pradesh for GI Basmati tag and subsequently, an application was filed with the registrar by Madhya Pradesh in resistance to APEDA.
- Contrary to that, the Intellectual Property Appellate Board was approached by APEDA for appealing against the order. In February 2016, there was a reversal of registrar's order by the IPAB regarding retreating the GI tag from APEDA. This order of IPAB was challenged in Madras High Court by the government of Madhya Pradesh.
- As per the Madhya Pradesh government's profession, there is cultivation of Basmati rice in 13 out of its 51 districts and they are qualified to get the GI tag. Further, as per the recommendation of Agricultural and Processed Food Products Export Development Authority, the seven states were granted the Basmati GI tag.

### 19.13.4 Role of WIPO in the Protection of Geographical Indications

It administers international agreements like the Paris Convention and the Lisbon Agreement, dealing partially or entirely with protection of geographical indications (GI) at the international level.

### 19.13.5 Geographical Indications in India

The total products registered under GI in India are approximately 643 as per March 2024 Press Release by PIB. 'Darjeeling Tea' was the first product to be recognised in 2004. GI is administered under The Geographical Indications of Goods (Registration & Protection) Act, 1999. Till now, more than approximately 554 products were applied for the GI tag. (For all details and updates of GIs you may go to the official website of GI Registry:

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*http://www.ipindia.nic.in (last accessed in July 2024)*

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Some examples of GI from India are as follows:

1. Lucknow Chikan Craft
2. Mango Malihabadi Dusherai
3. Banaras Brocades and Sarees
4. Lucknow Zardozi Craft
5. Kannauj Perfume
6. Moradabad Metal Craft
7. Hyderabadi Haleem
8. Narayanpet Handloom Saree
9. Madurai Malli
10. Eartomozhy Tall Coconut
11. Thanjavur Veenai
12. Saleem Silk
13. Temple Jewellery of Nagercoil
14. Madurai Sungudi
15. Makarana Marble
16. Bikaneri Bhujia
17. Ganjam Kewada Flower
18. Naga Mircha
19. Nasik Grapes
20. Mahabaleshwar Strawberry
21. Tirupati Laddu

Note: Till March 2024, 643 products from different states in India were given GI Tag with Uttar Pradesh leading the list with highest number of tags.

## 19.14 Plant Breeder's Rights

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These rights provide legal protection for the investment in plant breeding to develop new varieties. It is on the will of the breeders whether to apply for the plant breeder's rights or not, which provide royalties if any one uses the protected variety. It is a form of Intellectual Property Right, which protects the new variety of plants. Protection of new developed varieties of plants is an incentive to the breeder to get the benefits; moreover, it encourages the people to develop superior variety. Improved varieties are cost effective and high yielding, which are economically good for the farmers.

## 19.15 Indian Supreme Court Says Seeds, Plants and Animals are not Patentable

### 19.15.1 A Victory for the Planet

With the Supreme Court refusal to overrule a High Court decision of upholding Article 3(j) of Indian's Patent Law, has resulted into a key success in respect of biodiversity, planet, farmers seeds.

Article 3(j) excludes from patentability 'plants and Trade linked animals in whole or in any part thereof other than microorganisms but including seeds, varieties, and species, and essentially biological processes for production or propagation of plants and animals'.

The above article is an execution of TRIPs 27.3 b article. In a Treaty of Trade, Monsanto has brought life patents, whereas the government of India, Brazil and Argentina has omitted the patentability, as well as a sui generis option for varieties of plant protection. Monsanto tried to amend the TRIPs agreement for its own advantage, allied towards corporate interest and against the farmers, nations and their citizens interest.

### 19.15.2 Farmers Rights

India's representatives at the GATT succeeded in securing exclusions for plants and animals, as well as a sui generis option in the TRIPS agreement and in India's Patent Act and its implementation. Therefore, Monsanto is producing a false claim in its SC appeal that the decision of High Court is in

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breach to the TRIPS agreement. Monsanto is contravening science in reference to science ad nauseam.

### 19.15.3 Genetic Modification is a Method not a Product

In Monsanto's argument, a gene is just a chemical and a plant is a machine where the chemical can be brought and removed mechanically as per our will. Further, as per Monsanto's misleading argument, Bt cotton is not a plant and modified Bt genes modified are not a plant or plant variety's division and their patent is for a 'product', not a 'process'.

However as per the Patent title clarification, patent is for 'methods for transforming plants to express *Bacillus thurengensis* delta endotoxins' and therefore Monsanto is arguing falsely that patent is a product patent and not a process patent.

As per the Monsanto, a chemical is being invented which transforms plants into superplants and the seed is just the box of the chemical.

Monsanto is deforming science at the basic level and further corrupting our laws by wrongly arguing patents on Bt Cotton Seeds initially as plants patents, thereafter as genes patents and presently as chemicals patents. Monsanto is concealing from the courts that a gene in its chemical structure becomes plant's part and its future production by referring Bt genes as a technology and currently as a chemical product. Monsanto claim for a Bt Cotton seed patent is false because its patent was for a lab method for joining Bt gene to an existing cotton plant.

Now after legal assertion, Monsanto did not have a Bt Cotton patent and it has accumulated royalties from farmers in an illegal way, confining them in debt, and forcing thousands of cotton farmers to suicide. Now Monsanto has to pay.

## 19.16 IPR and Indian Agriculture

There are lots of debates regarding the consequences of IPR in agriculture. But the fact is that a proper IPR regime cannot protect foreign inventors and domestic players. Some prominent instances of IPR are given below:

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### 19.16.1 Turmeric

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In the year 1995, the US Patent Office granted the claim made by the two scientists in the invention was of using turmeric powder at the site of injury by oral intake or tropical applications. But the Council for Scientific and Industrial Research (CSIR) challenged the claim, thereby giving the 32 references from ancient texts showing the previous knowledge on turmeric usage in the form of powder. The US Patent office re-examined the claim, stated that it lacks novelty and hence revoked the patent.

### 19.16.2 Neem

The US and European laboratories held more than 40 patents on Neem pesticides. W. R. Grace, a US-based company, was accorded a patent for the usage of Neem in pesticide. Indian scientists were experimenting with it but were unable to challenge the claim as the patent was not earlier applied for the said process. Out of around 80 patents on Neem, accorded to the country, 48 are of Indian origin. Neither any royalty was paid nor these patents have restricted Indians from the usage of Neem.

### 19.16.3 Cotton

In October 1992, the W. R. Grace-based subsidiary, named as Agracetus Inc., was accorded the patent rights on all the genetically engineered cotton varieties along with the usage of all the techniques in its production. The same was challenged in court and subsequently in the year 1994, the US Patent Office cancelled the claim.

### 19.16.4 Basmati

In September 1997, RICETEC patented the name “Basmati” for aromatic rice grown outside India, promoting it under brands like texmati and kasmati. They claimed this rice as a Basmati type, using specific scientific parameters like grain dimensions, starch content, and cooking characteristics. However, genuine Basmati, characterised by unique soil,

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climate, and genetic factors, is traditionally only found in India and Pakistan. RICETEC argued that these traditional characteristics were irrelevant to product quality. This issue mirrors geographical indications (GI), such as Champagne and feta cheese, which are tied to specific regions. India lacks GI laws but could use this argument for products like Darjeeling tea, similar to Champagne's protected status.

## 19.17 World Intellectual Property Organisation (WIPO)

### 19.17.1 WIPO Structure

The WIPO is a United Nations self-funding agency serving intellectual property services, information, policy and cooperation. The WIPO, at present has 193 member states, with its main aim/objective for developing a balanced and effective international intellectual property (IP) system that supports creativity/creativeness and innovation for everyone. WIPO's mandate, governing bodies and procedures are set out in the WIPO Convention, which was established in 1967.

- The WIPO helps in realising the advantages of Intellectual Property to different governmental and non-governmental bodies.
- The WIPO sets out a programme on the member states and budget for all the proposed activities, objectives, measures and performances and reports, on the basis of each year's results in the Program Performance Report.
- WIPO submits biennial reports on financial management and financial statements annually to the Member States Assemblies, formulated as per the International Public Sector Accounting Standards.
- They work closely worldwide, in collaboration with the offices of Intellectual Property, NGOs, IGOs and other public and private stakeholders for delivering the programmes. By providing a stable environment for marketing products protected by intellectual property, it also oils the wheels of international trade. The WIPO works closely

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with its member states and other constituents to ensure the intellectual property system remains a supple and adaptable tool for prosperity and well-being, crafted to help in the realisation of the full potential of created works for present and future generations.

### 19.17.2 WIPO's Promotion in the Protection of Intellectual Property

The WIPO functions as a medium for its member states for the establishment of rules and practices in order to safeguard the intellectual property rights, act as a global registration systems for trademarks, industrial designs, a global filing system for patents and appellations of designs. WIPO's members regularly review this system.

Member states and other stakeholders are also determined for improved and better services. The intellectual property protection systems of many countries are very old, as they are not updated. Therefore, a lot of developing countries are in the course of formulating their legal structure related to patent and copyright trademarks and intellectual systems of property. WIPO performs an important role in accommodating these systems for their development with the help of treaty negotiation, technical and legal assistance, various forms of trainings, with the inclusion of the enforcement field in the era of the ever-increasing trade globalisation and rapid transformation in technological advancement.

WIPO performs globally along with its member countries for ensuring better recognition, easy understanding, accessibility and availability of intellectual property information from the grassroots level to the business sector and policymakers.

### 19.17.3 Funding of WIPO Audience

The WIPO is an autonomous organisation, in which the maximum budget is generated with the help of its global systems of registration and filing utilised by the various countries and member states, besides its services of publications and arbitration and mediation. Afterwards, the member countries donate the remaining funds.

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Note: N.B. Indian Patent Act, 2005, Geographical Indications of Goods (Registration and Protection) Act, 1999, and Trademark Act, 1999, are administered by the Department of Indian Policy and Promotion (DIPP) under the Ministry of Commerce and Industry. The Copyright Act, 1957, was administered by the HRD Ministry, but after approval of National Policy, this Act was administered under DIPP in 2016.

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## QUESTIONS AND ANSWERS

### Objective Questions

**1. Consider the following statements:** **[UPSC (Pre) – 2019]**

1. According to the Indian Patents Act, a biological process to create a seed can be patented in India.
2. In India, there is no Intellectual Property Appellate Board.
3. Plant varieties are not eligible to be patented in India.

Which of the statements given above is/are correct?

(a) 1 and 3 only
(b) 2 and 3 only
(c) 3 only
(d) 1, 2 and 3

**2. India enacted the Geographical Indications of Goods (Registration and Protection) Act, 1999 in order to comply with the obligation to** **[UPSC (Pre) – 2018]**

(a) ILO
(b) IMF

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(c) UNCTAD
(d) WTO

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3. **With reference to the 'National Intellectual Property Right Policy', consider the following statements:** **[UPSC (Pre) – 2017]**

(i) It reiterates India's commitment to the Doha Development Agenda and the TRIPS Agreement.

(ii) Department of Industrial Policy and Promotion is the nodal agency for regulating intellectual Property rights in India.

**Which of the above statement is/are correct?**

(a) Only (i)
(b) Only (ii)
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

4. **Which of the following has/have been accorded 'Geographical Indication' status?** **[UPSC (Pre) – 2015]**

(i) Banaras Brocades and Sarees

(ii) Rajasthani Daal-Bati-Churma

(iii) Tirupathi Laddu

**Select the correct answer using the code given below:**

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii) and (iii)

## Answer Key

| | | | |
|---|---|---|---|
| 1. (c) | 2. (d) | 3. (c) | 4. (c) |

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UNIT-VIII Universe and Particle Physics

# CHAPTER 20
# UNIVERSE

## IN THIS CHAPTER...


**20.1 Terminology**

**20.2 Universe, Space and Cosmos**

20.2.1 Origin and Expansion of the Universe

**20.3 Cosmic Inflation Theory**

**20.4 Red Shift and Blue Shift**

**20.5 Cosmic Microwave Background Radiations**

**20.6 Missing Links in Universe**

**20.7 Matter and Antimatter**


When matter and antimatter come into contact, they annihilate each other and produce vast amounts of energy.


**20.8 Dark Matter and Dark Energy**

**20.9 Galaxy**


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**20.10 Nebulae**
**20.11 Life Cycle of Stars**
**20.12 Planets**
**20.13 Dwarf Planets**

There are currently five dwarf planets, namely: **Pluto, Eris, Makemake, Haumea, Ceres.** These are all present in the Kuiper belt.

**20.14 Exoplanets**
**20.15 Goldilocks Zone (or Habitable Zone)**
**20.16 Solar System**
20.16.1 Sun
**20.17 Magnetic North Pole Drifting Fast Towards Russia**
**20.18 Earth's Magnetic Field Protects Us from Space Radiations**
**20.19 Gamma-Ray Bursts**

## 20.1 Terminology

**Proxima Centauri:** It is a red dwarf star. It is very faint, hence can be seen through a telescope. It was discovered in 1915.

**Big Bang Theory:** This is the cosmological theory that states that all the matter and energy in the universe originated from a state of enormous density and temperature that exploded at a finite moment in the past. This theory successfully explains the expansion of the universe.

**Cosmic Background Radiation:** Scientists have detected a faint radio signal; present in space, which they believe, is a faint glow from the Big

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Bang's superhot fireball.

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**Black Hole:** A black hole forms when a large dying star collapses or explodes. When a star has run out of fuel, it cannot stop gravity pulling its gas together, squeezing it tighter, until it forms a black hole. Black hole emits no light. Radio astronomy equipment detects its presence.

**Worm Hole:** A worm hole is a short-lived portal. It joins two black holes in different locations.

**Antiparticle:** This is a subatomic particle that has the same mass as another particle and equal but opposite values of some other properties.

## 20.2 Universe, Space and Cosmos

The **universe** is a collection of things in space that are similar to a space-time fabric and have its contents. It includes many components like stars, galaxies and the contents of intergalactic space, planets, moons and minor planets. Our universe is made up of matter and energy that was generated/released by the Big Bang. The size of the entire universe is still unknown.

Space is the part of the universe that is outside of Earth's atmosphere. Earth is described as a part of the universe but by definition, Earth is not a part of space.

Physical **space** is a linear three-dimensional structure, although it has a fourth-dimensional structure, i.e., time, those all together known as **space-time.** The **cosmos** is a complex and orderly system in universe.

### 20.2.1 Origin and Expansion of the Universe

According to most of the astronomers; 'the origin of universe' involved a gigantic explosion of matter and energy, about 13.6 billion years ago, called the 'Big Bang' theory. **'Big'** means large matters and **'Bang'** means strike violently. Now striking of big matters is 'Big Bang'. In 1920, Edwin

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Hubble provided evidence that the universe is expanding, i.e., also called 'Expanding Universe Hypothesis'.

Two discoveries have helped in promoting the 'Big Bang' theory and expansion of the universe concept: First, is a measured redshift in the light radiated from distant stars. This property of starlight is similar to the lowering in pitch of a departing train whistle, known as Doppler's effect.

Second is the presence of weak microwave radiations throughout the space. This background radiation detected by Arno Penzias and Robert Wilson is found to have a characteristic temperature just three degrees above absolute zero.

Fragments of matter and the space itself have been expanding outward like a display of fireworks. Stars, galaxies and planets gradually formed from these fragments in a purely mechanistic fashion. As time passed, galaxies shifted more and more apart. Distance between them was found to be increasing, so it is believed that the universe is expanding.

## 20.3 Cosmic inflation Theory

In the year 1981, the Theory of Cosmic Inflation was given by Alan Guth. As per the theory, ripples were sent **(primordial gravitational waves), which are responsible for the polarisation of the universe, thereby resulting in the widening of cosmic space and compressing** due to a swift universe expansion phase, immediately after the Big Bang, occurring at a pace faster than the light.

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## 20.4 Red Shift and Blue Shift

The light, having unique spectral features coming from distant stars and galaxies, when examined under VIBGYOR, tilted towards the spectrum's red end **(red shift)** has the largest wavelength, thereby pointing towards the movement of galaxies away from us. The similar phenomena, which points towards the distant objects moving nearer to the observer is termed as the **blue shift.**

## 20.5 Cosmic Microwave Background Radiations

These are the thermal radiations, which remained after the Big Bang. The European Space Agency's Planck spacecraft effectively prepared a map of the cosmic microwave radiation in the sky, which pictured the oldest light in our universe, termed as the cosmic microwave background (CMB). In each direction, it is filling the universe and its existence in itself is an evidence of Big Bang theory and accordingly, an expanding universe.

There is an in-depth understanding of composition of the universe with the help of the cosmic microwave background radiation. As per the CMB observation, the universe is mostly made up of dark energy, which, being a mystifying force, drives the expansion of the universe and further, the dark matter interacts with the remaining universe merely with the help of its gravity.

## 20.6 Missing Links in Universe

Mass, time and initial expansions of universe are some of the desired missing links in the 'Big Bang' theory.

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**Concept of Missing Mass:** If an expanding universe were to consist of sufficient material and unlimited time, gravity would eventually stop the outward motion and pull everything back together again into a cataclysmic fireball.

**Time is a Missing Link:** Many observations indicate a recent creation of universe; only thousands of years ago instead of the assumed billions of years of history. There is also a missing explanation for the initial formation of stars. The calculations done by James Clerk Maxwell show that a gas cloud in space will simply not collapse itself into a star. Instead, the gas dissipates outward due to thermal pressure, in accordance with the 'Second Law of Thermodynamics', the universal tendency towards disorder.

**Confirmations:** In late 1940s, George Gamow suggested that the initial explosion might have left some traces that are still observable. He reckoned that the universe, so hot and dense, would have emitted a lot of light. In 1965, Arno Penzias and Robert Wilson discovered that the static continued, no matter where they pointed the antenna. When they measured the radiation, they found a value close to that expected for cosmic background radiation – 2.7 Kelvin (close to absolute zero). This confirms the 'Big Bang' theory and they won the Noble prize in 1978.

### *Does the theory explain everything?*

It has passed through various stages, all of which can be considered as cosmological theories: the flat Earth, the geocentric mode, heliocentricity, the big bang. Each model explains what was known at the time and what the measures could confirm. It cannot be said that those theories were wrong; perhaps, they were incomplete.

## 20.7 Matter and Antimatter

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**Concept of Zero:** In the beginning of universe, matter and antimatter, both were parallelly formed during the incidence of Big Bang (from a highly dense body). At the time of Big Bang, the temperature was very high. [As zero is the sum of (+1) + (-1) = 0; Here, +1 is matter and -1 is antimatter]. Matter and antimatter have equal mass but opposite charge; for example, electron (-e) and positron (+e), both have same mass but opposite charges. If matter and antimatter are created and destroyed together, the universe should contain nothing but left-over energy.

Spontaneous transformations have been observed by researchers, between particles and their antiparticles, occurring millions of times per second before they decay. Some unknown entity intervening in this process in the early universe could have caused these 'oscillating' particles to decay matter more often than they decayed antimatter.

Corresponding to most kinds of particles, there is an associated **antiparticle with the same mass and opposite electric charge.** For example, the antiparticle of the electron is the positively charged anti-electron or positron, which is produced naturally in certain types of radioactive decay. A hydrogen atom is composed of a proton (a positively charged particle) and a much less massive electron (a negatively charged particle). A positron and an antiproton can form an anti-hydrogen atom (antimatter) in the same way that an electron and a proton form a normal matter hydrogen atom (matter). **When matter and antimatter come into contact, they annihilate each other and produce vast amounts of energy.**

For reasons unknown to physicists, **very little antimatter exists in our universe.** Theoretically, when the universe was formed, there should have been equal amounts of matter and antimatter; however, our universe consists primarily of matter. There is a considerable speculation in theoretical physics as to why the observable universe is apparently almost entirely made up of matter. But at this time, this apparent asymmetry of matter and antimatter in the visible universe is one of the greatest unsolved mysteries in physics. There is also research into whether there exists other places that are almost entirely made of antimatter instead, and what might be possible if antimatter could be harnessed.

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If antimatter can be produced artificially and controlled, the matter–antimatter reaction can be used to provide vast amounts of energy and can propel the spaceships of the future. But as of now, little is understood about antimatter. It is expensive to produce antimatter by current technology. **Tiny amounts of antimatter can be produced in particle accelerators such as the Large Hadron Collider.** It takes far more energy to produce antimatter today than one can get from the annihilation reactions of this antimatter. Also, the antimatter so produced cannot be controlled and made use of.

(Refer **Infographic – Matter & Antimatter**)

**Brookhaven National Laboratory –** Brookhaven National Laboratory is a **United States Department of Energy National Laboratory** located in **Upton, New York.** Mission of Brookhaven National Laboratory is to perform **advanced fundamental research in nuclear and particle physics** to **gain a deeper understanding of matter, energy, space and time; apply photon science and nanomaterial research to energy challenges of critical importance to the nation; and perform cross-disciplinary research on climate change, sustainable energy and Earth's ecosystems.** Many Indian scientists are also working here to find some results and unravel the mysteries of the universe.

## 20.8 Dark Matter and Dark Energy

### *Dark Energy and Dark Matter*

Astronomical observations suggest that **a significant part of the universe is made up of dark matter which interacts with the rest of the universe only through the gravitational pull.** Many large lab experiments have

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tried to detect elementary particles that could be candidates for dark matter. However, such dark matter particles have not been detected until now.

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### Assessing Dark Matter

The researchers use the non-observation of the lensing signatures to assess what fraction of the dark matter could be made of black holes. **Gravitational lensing is useful to cosmologists because it is directly sensitive to the amount and distribution of dark matter.**

### Dark Energy

- More is unknown than is known. We know how much dark energy there is because we know how it affects the universe's expansion. Other than that, it is a complete mystery. But it is an important mystery. It makes up about 68% of the universe.
- **Dark Energy is a hypothetical form of energy that exerts a negative, repulsive pressure, behaving like the opposite of gravity.**
- **It is causing the rate of expansion of our universe to accelerate over time, rather than to slow down**. That's contrary to what one might expect from a universe that began in a Big Bang.

### Difference between Dark Energy and Dark Matter

- Everything we see – the planets, moons, massive galaxies – makes up less than 5% of the universe. **About 27% is dark matter and 68% is dark energy.**
- While **dark matter attracts and holds galaxies together, dark energy repels and causes the expansion of our universe.**
- The existence of dark matter was suggested as early as the 1920s, while dark energy was not discovered until 1998.

### XENON1T Experiment

- It is **the world's most sensitive dark matter experiment** and was operated deep underground at the **INFN Laboratori Nazionali del Gran Sasso** in **Italy.**
- It uses the **dual-phase (liquid/gas) xenon technique** and is located underground at the Laboratori Nazionali del Gran Sasso of INFN, Italy.

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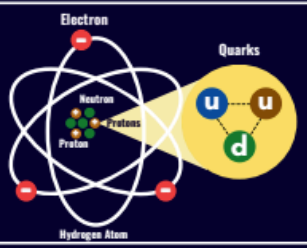
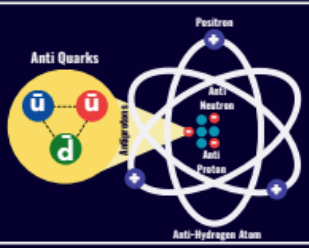
# MATTER vs ANTIMATTER

| ASPECT | MATTER | ANTI-MATTER |
|---|---|---|
| DEFINITION | Composed of particles known as fermions, such as protons, neutrons, and electrons. | Composed of antiparticles, such as antiprotons, antineutrons, and positrons. |
| CHARGE | Particles have specific charges: protons (positive), electrons (negative), and neutrons (neutral). | Antiparticles have opposite charges: antiprotons (negative), positrons (positive), and antineutrons (neutral). |
| COMPOSITION | Made up of atoms (protons, neutrons, electrons). | Made up of antiatoms (antiprotons, antineutrons, positrons). |
| INTERACTION WITH NORMAL MATTER | Normal interactions observed in daily life. | Annihilates with matter upon contact, releasing energy as gamma rays. |
| CONSERVATION LAWS | Interactions follow the conservation of electric charge. | Annihilation satisfies the conservation of electric charge by combining opposite charges to zero. |
| EXISTENCE | Much more abundant in the observable universe. | Rare compared to matter; observed in high-energy processes. |
| ROLE IN PHYSICS | Forms the basis of all physical objects and phenomena. | Used in research and medical applications (e.g., PET scans). |
|  | |  |





## COMET

A comet has a solid nucleus and a glowing coma. Near the Sun, it forms dust and gas tails. Originating from the Kuiper Belt and Oort Cloud, comets offer early solar system insights.

## ASTEROID

An asteroid is a rocky or metallic object orbiting the Sun, mostly in the belt between Mars and Jupiter. They vary in size and composition, providing insights into the early solar system.

## METEORITE

A meteorite is a space object that lands on Earth, offering insights into asteroids and the early solar system.

## METEOR

A meteor is a space object creating a bright streak in Earth's atmosphere. Known as shooting stars, they are fragments of comets or asteroids. Meteor showers occur from comet debris.

## METEOROID

A meteoroid is a small space object. In Earth's atmosphere, it becomes a meteor (shooting star). If it lands, it's a meteorite. Meteoroids contribute to meteor showers and reveal celestial composition.

## FIREBALL

A fireball is a bright, luminous meteor caused by large meteoroids entering Earth's atmosphere. It appears as an intense ball of light, can produce sonic booms and may leave meteorite fragments.

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## 20.9 Galaxy

There are millions of stars in the universe, together forming a group called galaxy. Stars are not uniformly distributed in the universe. Galaxy is a gravitationally bound system of stars, dust, interstellar gas and dark matter.

Each galaxy may contain as many as hundred million stars. There are millions of galaxies in the universe. Our Sun belongs to a galaxy called the 'Milky Way Galaxy'. It is believed that the Earth's own galaxy (Milky Way) alone contains as many as 1,00,000 million stars.

These stars are so far away from us that they cannot be distinguished individually and hence, appear as a white band.

The galaxies are calculated in thousands of light years spreading over large distances and their diameters range from 80,000 to 1,50,000 light years. A galaxy is formed by the hydrogen gas amassing in the form of a very large cloud termed as 'Nebula'. In due course, a growing nebula expands into a localised gas mass, which continues growing into even denser gaseous bodies. There are three main types of galaxies, i.e., elliptical, spiral and irregular.

### SARASWATI (SUPERCLUSTER OF GALAXIES)

A team of Indian astronomers identified a previously unknown, extremely large supercluster of galaxies located in the direction of constellation, Pisces. The supercluster of forty-three galaxies, which they named **'Saraswati',** is one of the largest known structures in the nearby universe, and is four billion light years away from us and may contain the mass equivalent of over twenty million billion Suns. The Saraswati supercluster, for instance, extends over a scale of 600 million light years.

A team of Indian astronomers from **Inter-University Centre for Astronomy and Astrophysics (IUCAA),** Pune, Indian Institute of Science Education and Research (IISER), Pune National Institute of Technology (NIT), Jamshedpur and Newman College (Thodapuzha) identified a massive supermega river of galaxies. The new discovery has been named Saraswati, which in Sanskrit literally means 'ever-flowing stream with many pools'. The discovery is forcing the astronomers to rethink about

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the early evolution of the universe and provides vital clues about the mysterious dark matter and dark energy.

Collision of gases and dust creates new stars within galaxies. Many galaxies are found in galaxy clusters, with thousands of members. Our galaxy, the Milky Way (Akashganga) belongs to a cluster of about thirty galaxies called the 'Local Group'. It takes light 1,00,000 years to pass from one edge of the Milky Way to the other. Light from the centre of the Milky Way takes 25,000 years to reach us on the Earth.

In a few billion years, the Milky Way and Andromeda galaxies will likely collide and form a combined galaxy. Edwin Powell Hubble studied galaxies and devised a classification system for galaxies based on their shapes.

## 20.10 Nebulae

Stars are born in an enormous cloud of interstellar dust and hydrogen gas called a nebula. Nebulae represent the building blocks for stars, galaxies and planets in the universe. New stars are being born in one of the nebulae. Other nebulae are the remains of dead or dying stars. Emission nebulae are hot, discrete clouds of primarily ionised hydrogen that glow with their own light. Reflection nebulae emit a bluish glow by reflecting the light of nearby stars.

Nebulae may be bright or dark. The bright nebulae shine. Some shine by reflecting the light of nearby stars. Others shine because the gases in them start glowing on being heated by nearby stars. The dark nebulae do not shine. They block light and look like dark patches when seen against a brighter region of space. Dark nebulae are also known as absorption nebulae.

Nebulae form when stars die. When the stars are near their end, their outer layers will heat, swell and eventually blow off. The hot, dead core will create a glowing nebula, which will in turn become a nursery for new stars. The shapes and spectral characteristics of planetary nebulae are unique and

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visible from a great distance. Some of the well-known nebulae are horsehead, lagoon, eagle and cat's eye, which have acquired these names inspired by their shapes.

Nebulae are major contributors to the chemical enrichment of galaxies, especially nitrogen and carbon. They also act as multi-wavelength laboratories for understanding atomic, molecular, dust and plasma processes in different astrophysical environments. Horsehead nebula, one of the dark nebulas, is situated in the Orion constellation in space.

## 20.11 Life Cycle of Stars

Refer Infographic – Life Cycle of a Star.

## 20.12 Planets

The word 'PLANET' is derived from a Greek word **Planetai,** meaning **wanderer.** Planets are so named because they revolve around the Sun. Planets do not have their own light, as they are made-up of solid material and gases. They shine with the light reflected from the Sun. Planets are developed by the gravitational force present in the stars. The stars are localised gas lumps inside a nebula. The force of gravity inside the lumps leads to the gas cloud core formation, which results in the creation of a huge rotating gas and dust disc around its core.

The gas cloud gets condensed and develops into small-rounded objects known as **Planetesimals.** They are a large number of smaller bodies held together by gravitational attraction. At last, these huge tiny planetesimals accumulate and result in the formation of a fewer large bodies in the form of planets. The planets have two types of movements, viz., rotation and revolution. Each planet has its own axis about which it rotates or spins. Two planets, Venus and Uranus, rotate from East to West. All other planets rotate in the anticlockwise direction, that is from West to East. The planets also

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revolve around the Sun in fixed paths. The path of revolution of each planet is egg-shaped or elliptical and is known as its orbit.

Each planet takes a different time period to complete one rotation about its axis and one revolution around the Sun. The formation of all the planets occurred in the same period, around 4.6 billion years ago. All the planets are spherical in shape. According to IAU (International Astronomical Union), a planet is a celestial body that (i) orbits a star, (ii) has acquired a roughly round shape by its own gravity, (iii) contains enough mass that its gravitational field has cleared its orbit of debris.

## Classification of Planets

Planets can be classified into two types, according to the position of their orbits with respect to the Earth's orbit.

1. **Inferior Planets:** Their orbits lie within the orbit of the Earth, e.g., Mercury, Venus.
2. **Superior Planets:** Their orbits lie outside the orbit of the Earth, e.g., Jupiter, Saturn, Uranus and Neptune.

Planets are also classified on the basis of their distance from the Sun:

1. **Inner Planets:** They are closer to the Sun.
   - ❏ They are small in size and have relatively high densities.
   - ❏ Gravity is low, hence, they could cannot hold the escaping gases.
   - ❏ They are positioned in between the Sun and the belt of asteroids.
   - ❏ They are made up of rocks and metals and are also called terrestrial planets, means Earth-like.
   - ❏ The formation of terrestrial planets occurred in the close surroundings of the parent star.
   - ❏ Near to the Sun, the solar winds are most intense, thereby, blowing off huge amounts of gases and dust.
   - ❏ They take a short time to complete one revolution, e.g., Mercury, Venus, Earth and Mars.
2. **Outer Planets:** They are far away from the Sun.
   - ❏ They are very large in size and have relatively low densities.
   - ❏ Gravity is comparatively high, hence, can hold the escape gases.
   - ❏ They lie outside the belt of asteroids.
   - ❏ They are made up of gases and are also called gas giants.

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- ❏ They are also known as Jovian planets, means Jupiter-like. Jovian planets are formed at quite a distant location.
- ❏ Solar winds are not intense to cause removal of gases from the Jovian planets.
- ❏ They take a long time to complete one revolution, e.g., Jupiter, Saturn, Uranus, Neptune.

In descending order of their size, the planets are Jupiter, Saturn, Uranus, Neptune, Earth, Venus, Mars and Mercury.

## 20.13 Dwarf Planets

The term 'Dwarf Planet' was adopted in 2006 due to the discovery of objects beyond the orbit of Neptune; that were comparable in size to Pluto. According to the definition adopted by IAU (International Astronomical Union), a **dwarf planet should follow various criteria**:

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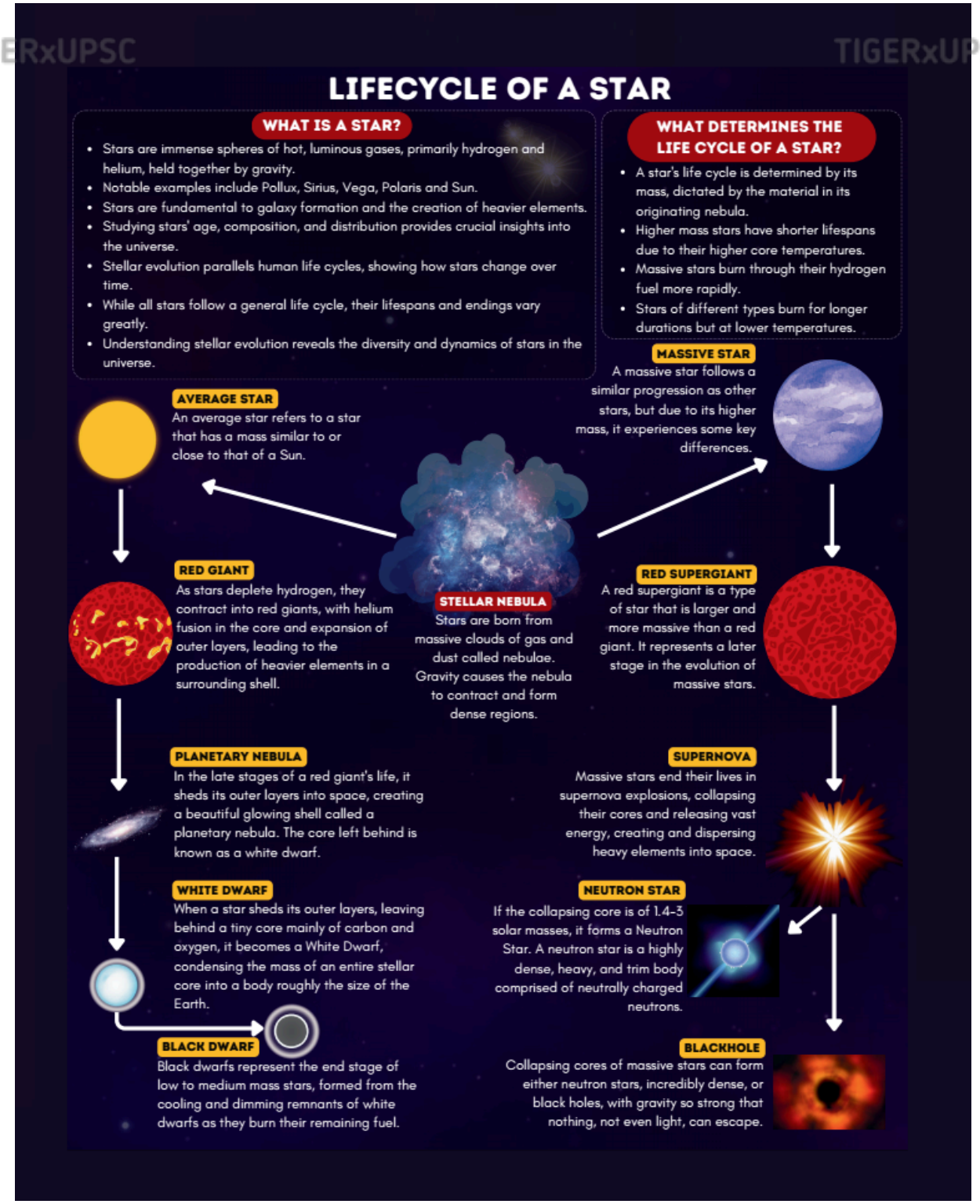


- It should be in orbit around the Sun.
- It should have sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round

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shape).

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- It does not have a clear neighbourhood around its orbit.
- It has no satellite.
- It has a weak gravitational pull.
- It has debris within its orbit.

There are currently five dwarf planets, namely: **Pluto, Eris, Makemake, Haumea, Ceres**. These are all present in the Kuiper belt (**Figure 20.1**).

It is possible that there are another forty known objects in the solar system that could be rightly classified as dwarf planets.

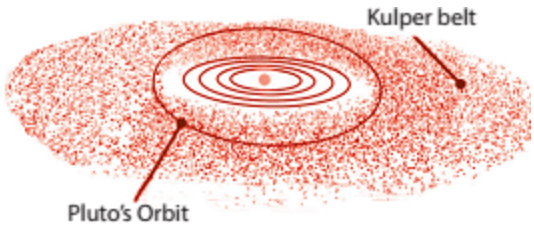


Figure 20.1 Position of Pluto in Kuiper belt

## 20.14 Exoplanets

Exoplanets, by definition, are planets beyond our solar system. Scientists also call them extra-solar planets.

- With the launch of Kepler mission in 2009, more than 2,300 of the exoplanets were confirmed.

## 20.15 Goldilocks Zone (or Habitable Zone)

On the basis of Kepler space mission data, astronomers reported that there are about forty billion Earth-sized planets orbiting in the 'habitable zones' or 'life zones' of Sun-like stars and red dwarf stars within the Milky Way Galaxy. Terrestrial planets are also more likely to lie in the Goldilocks

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Zone. Goldilocks region is an area of space in which a planet is at just the right distance from its home star so that its surface is neither too hot nor too cold. If we observe Venus, it has a lot of greenhouse effect but Mars exists as a frozen, arid world. In between both these planets, the conditions are just right so that liquid water remains on the surface of the planet without freezing or evaporating into the space. Hence, the possibility of life is maximum in this zone. Now, the search is on to find another planet in the Goldilocks zone of another solar system.

## 20.16 Solar System

Sun and its family are together called as the solar system.

### 20.16.1 Sun

It is the closest star to our planet. Like all stars, it is a ball of hydrogen gas that radiates heat and light, fed by constant explosions. It generates power by nuclear fusion (smaller nuclei combined to form larger nuclei and produce energy).

- Sun converts nearly four million tonnes of matter into energy every second.
- Sun was born five billion years (approximately) ago. It is so huge that it will continue to burn for another five billion years.
- It is composed of about seventy-four per cent hydrogen and twenty-five per cent helium, with traces of iron, carbon, calcium and sodium.
- The Sun is the centre of the solar system. Sun is only around eight light minutes away from the Earth. Due to the reflection of sunlight, we are able to see other things. Sun is the source of energy for our planet.
- The Sun accounts for ninety-nine per cent of the total matter in our solar system.
- Like other planets in the solar system, the Sun spins on its axis.
- The Sun takes up to 250 million years to complete one revolution around the Milky Way.

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- With the help of its different layers such as photosphere, chromosphere and corona, Sun can be easily described. Photosphere is the visible surface of the Sun, from which the emitted sunlight reaches the Earth. Chromosphere is the layer above photosphere and below the corona. Corona is the outermost region of the Sun's atmosphere (consisting of plasma or hot ionised gases), which is visible as a white halo during solar eclipse. The heat of corona is caused by magnetic field and undergoes constant changes.

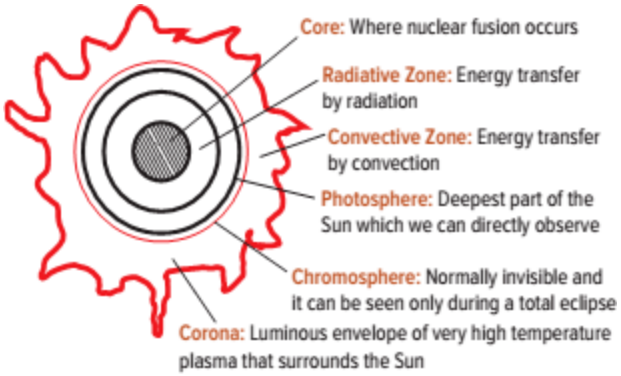


Figure 20.2 Sun Structure

- Solar activity occurs in the outer layers when a strong magnetic field breaks through the photosphere and emits intense radiation.

### 20.16.1.1 Solar Wind

The Sun emits a stream of invisible particles called the solar wind. It creates shimmering lights when passed through the polar regions of the Earth.

(i) It consists mostly protons and electrons with tiny amounts of silicon, sulphur, calcium, chromium, nickel, argon and neon.
(ii) Solar wind produces auroras (stunning light at polar regions) and also makes the tails of comets that point away from the Sun.
(iii) Heavy solar winds can disrupt telecommunication on the Earth.

### 20.16.1.2 Solar Tsunami

A solar tsunami is a powerful shockwave on the Sun's surface, triggered by explosions in its atmosphere that result in large amounts of superhot

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plasma, containing electrically charged particles, being expelled into space. The solar tsunami is technically referred to as a Magnetohydrodynamical Wave (MHD wave), as the explosions that causes it result from release of magnetic energy in the Sun's atmosphere. It is also called as a More-ton wave. The release of the superhot plasma is technically referred to as a 'Coronal Mass Ejection' (CME).

Solar tsunamis were discovered in 1997 by the Solar and Heliospheric Observatory (SOHO) spacecraft of the NASA, USA. The SOHO confirmed their reality in February 2009, when sunspot 11012 unexpectedly erupted, hurling a huge coronal mass ejection into space and sending a massive shockwave across the Sun's surface. This shockwave came to be known as a solar tsunami in general and MHD wave in scientific parlance.

### 20.16.1.3 Solar Flares

Solar flares are sudden eruptions on the surface of the Sun. They spurt fiery gases beyond the Sun's corona and into the space. Solar flares release energy of the order of billions of tonnes, at more than 600 miles per second into the space. Solar flares last only for a few minutes. The charged particles of solar flares sometimes extend to the Earth's magnetic field, which can cause auroras, geomagnetic storms and disruption to satellite communication.

### 20.16.1.4 Sunspot

Sunspots are the cooler and darker areas (cooler as compared to other parts of the Sun's surface) where the field breaks through the photosphere. The regions which appear to be darker on the surface of the Sun are termed as Sun spots, being 1500 K cooler than surrounding photosphere. As the sunspots are darker relatively, thus, a sunspot removed from the bright background of the Sun would glow quite brightly. The frequency with which sunspots appear and their location on the Sun's surface changes over time in a pattern is called the sunspot cycle.

***Cause Behind the Sunspots***

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Sunspots are temporary phenomena on the Sun's photosphere, thereby, appearing clearly as dark spots in comparison to the surrounding regions. The extreme magnetic action inhibiting convection by an effect analogous to the eddy current brake, leading to the formation of reduced surface temperature is the major reason for its occurrence. They also have two poles similar to that of magnets. Even though the temperature of Sun spots is nearly 3000–4500 K (2727–4227°C), their contrast with the temperature of surrounding material at nearly 5,780 K makes them clearly visible as dark spots, as the heated black body's luminous intensity is a function of fourth power of the temperature.

## Basic Features of Sunspots

- The diameters of largest experienced sunspots is nearly 50,000 kilometres, making them big enough to be spotted with the naked eye.
- Usually, the sunspots are spotted in groups with nearly hundred in a group, despite the fact that sunspot groups greater than ten are relatively rare. In order to measure the sunspots, there are well-established methods.
- Sunspots remain for hours to months, and move around the Sun's surface through its rotation.
- A typical sunspot consists of a dark, central region called the umbra and somewhat lighter surrounding region called the penumbra.

### MOON

Moon is the natural satellite of Earth. Different types of features of moon have been identified in the recent past.

**Blue Moon:** When two full moons appear in the same calendar month, the second is termed as 'blue moon'. First full moon occurred on 01 January 2018. On 31 January 2018, a rare blue moon event was experienced on large parts of the globe. It was a rare moment, as a blue moon, a supermoon and a total lunar eclipse fell on the same day after more than 150 years.

**Super Moon:** It occurs when the full moon is at the closest point of its orbit to the Earth, which is also called the perigee. The moon appears thirty per cent brighter and fourteen per cent bigger than the apogee full moon.

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**Blood Moon:** The Moon turns into red colour during the height of the eclipse, as some light does reach it even though the Moon is in the shadow of the Earth. Fine particles in the atmosphere scatter (Rayleigh scattering) the blue component of solar spectrum, and what reaches us is the longer wavelength red light.

A lunar eclipse happens whenever the moon passes through the Earth's shadow, also known as its umbra.

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## 20.17 Magnetic North Pole Drifting Fast Towards Russia

***'The magnetic North pole is drifting fast from the Canadian Arctic and towards Russia'*** has been published in many news reports.

### *Earth's Magnetism*

- The Earth's magnetism originates in its outer core, a more than 2,000-km layer of liquid iron and some other metals like nickel that surround the central core, or the innermost part.
- Due to Earth's rotation, liquid iron is in constant motion and this motion produces a magnetic field.

### *What's the Issue Now?*

- Currently, the magnetic North Pole is located somewhere over northern Canada, a fact discovered in 1831 by Sir James Clark Ross. The magnetic North Pole has been moving across the Canadian Arctic towards Russia, and has moved hundreds of miles over the last several decades. This phenomenon is known as the Polar Shift Theory.
- Scientists have now understood that the speed of this movement has speedily increased, quite significantly, from about 14–15 km per year till the 1990s to about 55 km per year in the last few years. This has led to updating of World Magnetic Model (WMM) that tracks this movement. It was a year ahead of schedule.

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### *Why it is Moving Faster?*

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- Scientists do not have full clarity on the matter. The movement of liquid iron and other metals in the outer core of the Earth is known to influence the magnetic field, but this movement is chaotic and turbulent. Scientists do not fully understand how or why the movement happens.
- The study of the phenomena happening inside the Earth can only be done indirectly or through computer modelling, because of the extremely hot temperatures prevailing there. Scientists hope that this acceleration in the shifting of magnetic North pole would throw some new insights into the phenomena happening deep inside the Earth's surface.

### *What is Magnetic North Pole? How is it Different From Geographic North Pole?*

The Earth has ***two north poles: geographic and magnetic***

- **Geographic North Pole:** It is the Northern axis around which the Earth rotates and is fixed. The North pole is located in the middle of the Arctic Ocean.
- **The Magnetic North Pole** (also known as the North Dip Pole) is a point on Ellesmere Island in Northern Canada where the Northern lines of attraction enter the Earth. A compass needle will point to the Magnetic North Pole – which is different from the **geographic North.**

## 20.18 Earth's Magnetic Field Protects Us from Space Radiations

Dangerous particles, which are in the Earth's atmosphere, do not hit the Earth's surface because they are forced by the magnetic field around the Earth. Particles do enter at the funnels over the poles or they gain entry far downstream from the Earth. The particles that enter downstream or at the magnetotail (a narrow and elongated region of the magnetosphere of the Earth or of another planet that extends in the direction away from the Sun) cause the auroral lights.

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The Aurora is an incredible light showcaused by collisions between electrically charged particles released from the Sun that enter the Earth's atmosphere and collide with gases such as oxygen and nitrogen. The lights are seen around the magnetic poles of the Northern and Southern hemisphere. Auroras that occur in Northern hemisphere are called **Aurora Borealis** and those that occur towards the Southern hemisphere are called **Aurora Australis** or Southern lights.

***Angle:*** The angle at which the magnetic field lines intersect the surface of the Earth ranges from 0 degrees at the equator to 90 degrees at the poles.

***Intensity:*** The magnetic field also varies in strength over the Earth's surface. It is strongest at the poles and weakest at the **equator.**

## 20.19 Gamma-Ray Bursts

**Gamma-ray bursts** (GRBs) are **extremely energetic explosions that have been observed in distant galaxies.** They are the brightest electromagnetic events known to occur in the universe. These bursts can last from ten milliseconds to several hours. These waves are generated by radioactive atoms in nuclear explosions. Gamma-rays can kill living cells; a fact which medicine uses to its advantage, using gamma-rays to kill cancerous cells. Gamma-ray bursts were discovered in the late 1960s by the US Vela nuclear test detection satellites. Gamma-rays come from solar flares (eruptions on the Sun's surface), pulsars (rotating objects which regularly emit radiation), nova and supernova explosions (explosions that cause stars to become extremely bright). They are strongly penetrating ionising radiations which means that they create charged radicals in any material they travel through. In the human body, they cause mutations in DNA and damage cellular mechanisms. In large doses, they are enough to kill cells and cause radiation poisoning.

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## QUESTIONS AND ANSWERS

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## Objective Questions

1. **Which one of the following is a reason why astronomical distances are measured in light-years?** **[UPSC (Pre) – 2021]**
   (a) Distance among stellar bodies do not change.
   (b) Gravity of stellar bodies does not change.
   (c) Light always travels in straight line.
   (d) Speed of light is always same.

2. **The term 'Goldilocks Zone' is often seen in the news in the context of** **[UPSC (Pre) – 2015]**
   (a) the limits of habitable zone above the surface of the Earth
   (b) region inside the Earth where shale gas is available
   (c) search for the Earth-like planets in outer space
   (d) search for meteorites containing precious metals

3. **Which of the following is/are cited by the scientists as evidence/evidences for the continued expansion of universe?** **[UPSC (Pre) – 2012]**
   (i) Detection of microwave
   (ii) Observation of redshift phenomenon in space
   (iii)Movement of asteroids in space
   (iv)Occurrence of supernova explosions in space

   Select the correct answer using the codes given below:

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(a) (i) and (ii)
(b) Only (ii)
(c) (i), (iii) and (iv)
(d) None of these

4. **Electrically-charged particles from space travelling at speeds of several hundred kilometres per second can severely harm living beings if they reach the surface of the Earth. What prevents them from reaching the surface of the Earth?**
**[UPSC (Pre) – 2012]**
(a) Earth's magnetic field diverts them towards its poles.
(b) Ozone layer around the Earth reflects them back to the outer space.
(c) Moisture in the upper layers of atmosphere prevents them from reaching the surface of the Earth.
(d) None of the statements given above are correct.

5. **What is the difference between asteroids and comets?[UPSC (Pre) – 2011]**
(i) Asteroids are small rocky planetoids, while comets are formed of frozen gases held together by rocky and metallic material.
(ii) Asteroids are found mostly between the orbits of Jupiter and Mars, while comets are found mostly between Venus and Mercury.
(iii)Comets show a perceptible glowing tail, while asteroids do not.

Which of the statement(s) given above is/are correct?
(a) (i) and (ii)
(b) (i) and (iii)
(c) Only (iii)
(d) (i), (ii) and (iii)

**Answer Key**

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| | | | | |
|---|---|---|---|---|
| 1. (d) | 2. (c) | 3. (a) | 4. (a) | 5. (b) |

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# CHAPTER 21
# PARTICLE PHYSICS

## IN THIS CHAPTER...


**21.1 Particle Physics**

**21.2 Standard Model of Physics**

**21.3 Fundamental Particles**

- 21.3.1 Fundamentals of Bosons
- 21.3.2 Quarks
- 21.3.3 Leptons
- 21.3.4 Higgs Boson/God Particle


**Satyendra Nath Bose is also called the Father of God Particle.** God particle is also popular with the name of **'Higgs Boson particle'.**


**21.4 CERN (European Council for Nuclear Research)**

- 21.4.1 Large Hadron Collider (LHC)
- 21.4.2 Purpose of LHC
- 21.4.3 CERN Experiment


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21.4.4 India's Contribution at CERN
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**21.5 Neutrino**
21.5.1 Concept of Neutrinos and Its Importance
21.5.2 Neutrinos Fog
21.5.3 Neutrinos Oscillation
21.5.4 IceCube
21.5.5 Baikal-GVD (Gigaton Volume Detector)

LISA consists of three spacecraft forming in space an equilateral triangle where the triangle sides are also termed as the LISA's 'arms', thereby extending nearly a million miles.

21.5.6 Indian Neutrino Observatory (INO)
**21.6 Theory of Relativity**
**21.7 Gravitational Waves**
21.7.1 LISA Pathfinder, LISA and eLISA

The **most powerful gravitational waves originate from cataclysmic occurrences like the merging of black holes, the explosive deaths of massive stars known as supernovae, and the collisions of neutron stars.**

It is **among the world's three largest neutrino detectors together with the IceCube at the South Pole and ANTARES** in the Mediterranean Sea.

**21.8 Daksha Project**

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## 21.1 Particle Physics

Particle physics is a branch of physics in the study of the elementary constituents of matter, radiation and the interactions between them. It is also popular with the name of 'high energy physics' because these elementary particles do not occur under normal circumstances in nature. It can be created and detected during energetic collisions of other particles due to particle accelerators.

Modern particle physics research is focussed on the basis of subatomic particles such as electrons, protons and neutrons (protons and neutrons are actually composite particles, made up of quarks), particles formed by the process of radiation and scattering, such as photons, neutrinos and muons, and a large number of exotic particles. The dynamics of particle physics are governed by quantum mechanics.

Various theories and models are given for the understanding of universe like **Event Horizon, Singularity, Standard Model, String Theory, etc.** Among these, standard model gained a lot of popularity.

## 21.2 Standard Model of Physics

All particles and their interactions are observed and described by a quantum field theory (under Standard Model).

The Standard Model classifies elementary particles into Quarks, Leptons, Bosons, etc., those that can combine and be found as various composite particles. The physics standard model concerns with the various particles' interactions, properties and characteristics at the subatomic level.

Scientists have been wondering from a long time regarding the evolution of the universe. To solve these mysteries, theoretical models are prepared to observe these realities. The standard model of physics explains the evolution of our universe, as it is observed today.

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There are four fundamental forces exerted in nature: electromagnetic, strong (which binds atomic nuclei), weak (which governs radioactive decay and some fusion reactions) and gravitational forces. 'Theory of everything' unifies these four forces in one equation. Standard model unified the description of first three forces but not the existence of gravitational waves.

Often, the standard model is in news through recent advancements and latest discoveries. In this context, the basic knowledge is the key for understanding the recent discoveries such as gravitational waves or the neutrino particle and its masses.

The standard model of physics has been accepted globally, since the discovery of the Higgs boson.

After the latest discovery of gravitational waves and successful simulations of Big Bang at CERN, in all the analyses, the particle research has played the most important role. India has also gained an associate membership of CERN.

## 21.3 Fundamental Particles

### 21.3.1 Fundamentals of Bosons

The famous physicist Satyendra Nath Bose worked with Albert Einstein to develop the Bose–Einstein statistics, according to which bosons can overlap and coexist with other bosons.

All fundamental particles in nature can be divided into two categories, fermions or bosons.

- Particles that make up matter are called **Fermions.**
  **Examples:** electrons, protons, leptons, quarks, neutrinos.
- Particles that carry force are called **bosons.** They obey Bose–Einstein statistics.
  - ❑ Bosons are sometimes called force particles because they control the interaction of physical forces.
    **Examples:** photons, helium atoms, gluons, W Boson, Z Boson.

### 21.3.2 Quarks

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In standard model of physics, quarks are the proposed elementary particles having six major quarks and six anti-quarks, carrying the fractional charge. The atomic particles are formed by the combination of different quarks. As the combination of two or more quarks leads to the formation of an integer (–1, 0, 1...) valued particle, termed as Hadron and the Hadron consists of three quarks which are called Baryons. Its examples are protons and neutrons. The protons are made from a combination of two up-quarks and one down-quark and neutrons are made from the combination of one up-quark and two down-quarks, respectively.

**TERMINOLOGY**

**Event Horizon:** Under general relativity is the concept of an event horizon. It is a boundary in spacetime beyond which there is no possibility of any events, e.g., the points at which the gravitational pull becomes so great as to make escape impossible, even for light, i.e., Black hole.

**Singularity:** It is a location in space-time where the gravitational field of a celestial body becomes infinite. When a star collapses beyond a certain point it would form a black hole, inside which a singularity (covered by an event horizon) would be formed. According to modern general relatively, the initial state of the universe, at the beginning of the Big Bang, was a singularity.

**Standard Model:** Our current knowledge about the subatomic composition of the universe is summarised in what is known as the **Standard Model** of particle physics. It describes both the fundamental building blocks out of which the world is made, and the forces through which these blocks interact. There are twelve basic building blocks. Six of these are **quarks** – they go by the interesting names of **up, down, charm, strange, bottom and top.** (A proton, for instance, is made of two up quarks and one down quark.) The other **six are leptons** – these include the electron and its two heavier siblings, the **muon** and the **tau**, as well as **three neutrinos**. Similar to standard model, String theory also explains about the study of universe.

### 21.3.3 Leptons

Leptons are also made up of six different types as similar to quarks, but they do not carry any fractional charge. **They consist of electron, muon, tau and three neutrino types.**

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### 21.3.4 Higgs Boson/God Particle

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Bosons are the combination of the particles carrying forces. Firstly, an Indian scientist, **Satyendra Nath Bose** discovered this type of particle but he could not get the attention of the scientist community. Later on, **Peter Higgs** further analysed these type of particle and got attention of the scientific community across the world. Due to first observation of God particle, **Satyendra Nath Bose is also called the Father of God Particle.** God particle is also popular with the name of **'Higgs Boson particle'.** Due to the presence of God particle, every particle has mass. Gravity as a natural force has still not been accepted by the standard model, owing to its failure in discovering its boson. The strongest existing force is the nuclear force, whereas the force of gravity is the weakest. Within all fundamental particles (like proton, neutron, etc.) and in universe there is an existence of Higgs boson particle/God particle, for the existence of mass.

### Higgs Boson

***Higgs Boson (God Particle)***

| Feature | Description |
|---|---|
| **Type** | **Boson (force-carrying particle)** |
| **Mass** | Approximately **125 GeV** (**giga electron volts**/billion electron volts)<br>It is **130 times more massive than a proton**. |
| **Spin** | **Scalar particle**, **spin = 0** (no angular momentum) |
| **Lifetime** | **Extremely short, decays rapidly** |
| **Detection** | Indirect—via **decay products like photons**, W/Z bosons |
| **Importance** | Confirms the **Standard Model**, offers clues to **dark matter**, **early universe** conditions |

Higgs boson particle is present everywhere, as in proton and neutron and in the surroundings of quarks. According to scientists, **Higgs field is also present in each and every planet. This is analogous to electric and**

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**magnetic field**. Without this field, atoms, molecules, planets, stars and living organisms would not exist. Higgs field completely fills the universe.

Higgs boson is one of the fundamental particles, which on interaction with Higgs field gives mass to all particles. **It was experimentally proved in the Large Hadron Collider at CERN in 2013**.

(Refer **Infographic – CERN Experiment**)

## 21.4 CERN (European Council for Nuclear Research)

In the year 1952, as an interim agency, the CERN was set up for establishing the world's biggest nuclear and particle physics laboratory, with the aim of creating a world-class fundamental physics research organisation, near Geneva, along the **Swiss-France border.**

At CERN, the fundamental particles are forced to collide collectively in the collider at light speed, thereby providing clues to the physicists regarding the interaction of particles at such high speed and further providing the view of the creation of the universe by recreating the Big Bang.

### BOSE EINSTEIN CONDENSATE (BEC) – FIFTH STATE OF MATTER & BOSE EINSTEIN STATISTICS (BES)

At extremely cold temperatures close to absolute zero, all particles in a system come together into a single state known as a Bose–Einstein condensate. This is considered the fifth state of matter, following solid, liquid, gas, and plasma. This idea was first proposed in 1924, but the technology at that time couldn't reach such low temperatures needed to test it.

Nearly 70 years later, in 1995, Eric Cornell and Carl Wieman successfully created a Bose–Einstein condensate using a small group of rubidium atoms cooled to just a few hundred billionths of a degree above absolute zero. Their achievement confirmed the existence of the condensate as predicted decades earlier. In recognition of their breakthrough, they were awarded the Nobel Prize in Physics in 2001, sharing it with Wolfgang Ketterle for his independent work on the same phenomenon.

- Over the years, experiments related to Bose–Einstein Statistics were carried out. Resultantly, all the fundamental particles were classified into two categories, the

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first one is 'Fermions', and the second one is 'Bosons' on the name of Indian scientist Prof. Bose.

- Thus, 'Bose–Einstein Statistics' helped in improving the benchmark in terms of Standard Model of Particle Physics.

All Bose-Einstein condensates are made up of bosons, but not all bosons are part of a Bose-Einstein condensate. Bosons refer to a class of particles with specific quantum properties, while a Bose-Einstein condensate is a unique state of matter that certain bosons can form under specific conditions of low temperature and high density.

It revolutionised the field of Physics in three ways:

- Improved the benchmark in terms of Particle Physics model.
- Laid the foundation of 'God Particle' and helped Peter Higgs in winning a Nobel Prize.
- Enhanced the benchmark of Standard Model of Particle Physics.

**Discovery of God Particle**

- Peter Higgs carried out further research in the year 1960 using Bose-Einstein Statistics and wrote a paper about a very unstable particle that survives for a fraction of a second after its birth, and then it breaks apart producing other fundamental particles. He termed this particle 'Higgs Boson Particle' or 'God Particle'. If scientists could produce such 'God Particle' in laboratory, it could further improve the Standard Model and our understanding of the universe.
- In early 2000s, world's largest particle accelerator called 'Large Hadron Collider (LHC)' was set up at CERN's laboratory in Europe. They carried out experiments and successfully discovered the God particle and confirmed the theory of Peter Higgs.
- Subsequently, Peter Higgs was given a Nobel Prize in Physics (2013). In April 2024, he passed away at the age of 94.
- The Bose–Einstein Statistics helped CERN scientists discover God particle and thus improved the Standard Model of Particle Physics and our understanding of the Universe.

Since early 1960s, Indian scientists were involved in the research through the assistance of Department of Atomic Energy and Department of Science and Technology. The country was given the status of Observer at CERN in 2003 but India fully joined it as an Associate Member in 2016. Being the member of CERN, it will assist in increasing young scientists and engineers' participation in different projects and further help in bringing back the knowledge for its usage in the country's local programmes. This will give further opportunities to Indian companies for getting involved directly in CERN projects.

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(Refer **Infographic – CERN Experiment**)

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### 21.4.1 Large Hadron Collider (LHC)

The world's largest and highest-energy particle accelerator is the Large Hadron Collider (LHC). It is likely to solve the most basic question of physics, further facilitating the understanding of nature's laws through recreating the conditions of the 'Big Bang'.

The LHC is positioned in a tunnel with a circumference of 27 kilometres, and depth of 175 metres, under the border of France and Switzerland, near Geneva in Switzerland, established in 2008. This accelerator is designed for colliding the opposite particle beams of either protons or lead nuclei at very high energy. The hadrons are the particles consisting of quarks.

The LHC was created by CERN for testing different high-energy physics assumptions, consisting of the hypothesis of Higgs boson testing and other large new particles as predicted by different theories. It was formed in association with over 10,000 scientists and engineers from 100 countries, as well as hundreds of universities and laboratories.

**FUNDAMENTAL PARTICLES**

Everything in the universe is made up of atoms. But what is an atom made of? Subatomic particles – Protons, Electrons and Neutrons, right? But there is a world beyond them. Experiments have confirmed that the subatomic particles are not indivisible but they can be further divided into 17 elementary or fundamental particles. These 17 elementary particles are grouped in matter particles (12) and force particles (5).

- Quarks are the fundamental constituents of matter. They are of six types: up quark, down quark, top quark, bottom quark, strange quark and charm quark. They combine to form composite particles called Hadrons. The hadrons are of two types: Baryons and Mesons. The baryons are made up of three quarks and the mesons are made up of one quark and one antiquark.
- The most common examples of baryons are protons and neutrons. The up quarks and down quarks build up the protons and neutrons. An Up Quark (UQ) carries a charge of +2/3 and a Down Quark (DQ) carries a charge of -1/3. So, we can see, even protons and neutrons could be broken down into smaller units.
- A proton is made up of two up quarks and one down quark. Hence, its resulting charge is +1(+2/3+2/3-1/3).

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| Elementary particles | | | | |
|---|---|---|---|---|
| **Matter Particles** | | | **Force Particles** | |
| Quarks | Leptons | | Bosons | |
| Unitary | Charged | Neutral | Vector bosons | Scalar bosons |
| 1. Up | 7. Electron | 10. Electro Neutrino | 13. Photon | 17. Higgs Boson |
| 2. Down | 8. Muon | 11. Muon Neutrino | 14. W boson | |
| 3. Top | 9. Tau | 12. Tau Neutrino | 15. Z boson | |
| 4. Bottom | | | 16. Gluon | |

**(Charge of 1UQ = +2/3; Charge of 1DQ = -1/3; I Proton = 2UQ + 1DQ = +2/3+2/3-1/3=+1)**
**A neutron is made up of one up quark and two down quarks. Hence, its resulting charge is 0 (+2/3-1/3-1/3).**
**(Charge of 1 UQ = +2/3; Charge of 1DQ = -1/3; 1 Neutron = 1UQ + 2 DQ = +2/3-1/3-1/3=0)**

- Leptons are of six types and are grouped in two categories: charged leptons (electrons, muons and taus) and neutral leptons or anti-leptons (electron neutrinos, muon neutrinos and taus neutrinos). Electrons are the most stable type of leptons. The other types of leptons – muons and taus – also exist but they are very high energy particles and are not stable. Hence, they transform to their lower energy states, the electrons.
- Neutral particles called neutrinos also exist.
- Some people also theorise the existence of anti-neutrinos but nothing has been confirmed yet.
- Quarks and leptons both complete the picture of the atom.
- Bosons are also called force particles as there is a Boson 'causing' every fundamental force. Bosons are of two types: vector boson and scalar boson. There are four vector bosons: photons, W boson, Z boson and gluon. Photon (not proton) is the boson responsible for electromagnetic interaction. We commonly know it as the 'particle of light' which is nothing but an electromagnetic wave.

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- W and Z bosons are responsible for the weak nuclear interaction or the force which binds the nucleus and the electrons.
- Gluons (derived from the word ‘glue’) are responsible for the strong nuclear interaction or the force that binds the protons and neutrons.
- A scalar boson called the Higgs Boson was discovered. ATLAS was involved in this jigsaw puzzle without which the world of elementary particles would be incomplete. The behaviour of Higgs Boson with other particles is still a mystery to be solved.
- These particles are responsible for imparting mass to other particles and what about the most fundamental force, gravity! Scientists are still not quite clear how to accommodate this force in the standard model. They have conceptualised a particle called ‘gravitation’ which they say is responsible for gravity but they are not quite sure.
- These 17 elementary particles and gravitation together complete the picture of the atom and its interactions as per the standard model of particle physics.

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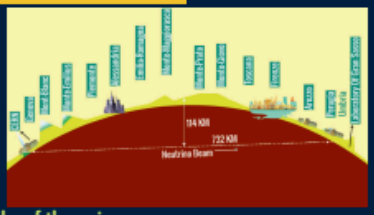
# CERN EXPERIMENT

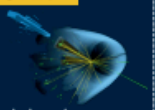
## WHAT IS CERN EXPERIMENT?

- **CERN and the LHC:**
  - CERN (European Organisation for Nuclear Research) is located in **Switzerland.**
  - Home to the Large Hadron Collider (LHC), the **world's largest particle accelerator.**




- **Objective:**
  - **Probe the fundamental building blocks of the universe.**
  - Recreate conditions similar to those just after the Big Bang by colliding particles at high energies.
- **Discovery of the Higgs Boson:**
  - Discovered in 2012 at CERN.
  - Theorised to give mass to other particles through interactions with the Higgs field.
- **Impact on Particle Physics:**
  - Revolutionised our understanding of particle physics.
  - Helped unravel mysteries of the universe's origins.

## WHAT IS THE HIGGS BOSON OR GOD PARTICLE?



- **Higgs Boson:**
  - **Quantum particle associated with the Higgs field.**
  - Discovery confirms the existence of the Higgs field and the theory.
- **Discovery Announcement:**
  - On **4 July 2012**, ATLAS and CMS collaborations at CERN announced the observation of a particle consistent with the Higgs boson.
  - Subsequent analyses confirmed it matches the theoretical Higgs boson.
- **Search History:**
  - Began in the late 1980s at CERN with the Large Electron-Positron (LEP) collider.
  - Continued in the 1990s with experiments at Fermilab's Tevatron collider.
  - Culminated with the discovery at the Large Hadron Collider (LHC) in 2012.

## WHAT IS THE BROUT-ENGLERT-HIGGS (BEH) MECHANISM?

- **BEH Mechanism:**
  - **Theoretical framework explaining how particles acquire mass.**
  - Proposed in 1964 by Robert Brout, François Englert, and Peter Higgs.
- **Higgs Field:**
  - Particles interact with the Higgs field, which permeates the universe.
  - The strength of interaction determines particle mass.
  - Stronger interaction with the Higgs field results in greater mass.
  - Particles not interacting with the field remain massless (e.g., photon).
- **Confirmation:**
  - **Higgs field and Higgs boson confirmed by experiments at CERN's LHC in 2012.**
- **Significance:**
  - **Explains the origin of mass in the universe.**
  - Resolves the issue in electroweak theory by accounting for the masses of W and Z bosons.
  - Fills a crucial gap in understanding fundamental particles and their interactions.

## MEMBER STATES OF CERN

**MEMBER STATES (TOTAL-23)**

Austria, Belgium, Bulgaria, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Israel, Italy, Netherlands, Norway, Poland, Portugal, Romania, Serbia, Slovak Republic, Spain, Sweden, Switzerland and United Kingdom.

**ASSOCIATE MEMBER STATES**

- Cyprus, Estonia, and Slovenia are Associate Member States in the pre-membership stage at CERN.
- Croatia, **India**, Latvia, Lithuania, Pakistan, Türkiye, and Ukraine are **Associate Member States.**

**OBSERVER STATUS**

- Japan and the United States of America have Observer status for the LHC, and the United States also has Observer status for the HL-LHC.
- The European Union and UNESCO are international organisations with Observer status at CERN.
- The Observer status of the Russian Federation is currently suspended as per the CERN Council Resolution of 8 March 2022.

## LARGE HADRON COLLIDER (LHC)

- **LHC Overview:**
  - **World's largest particle accelerator.**
  - Started on **10 September 2008.**
  - 27-kilometre ring with superconducting magnets.
- **Operation:**
  - Two particle beams travel near light speed.
  - Collide in ultrahigh vacuum beam pipes.
  - Superconducting magnets, cooled to **-271.3°C**, guide beams.

- **Magnets:**
  - 1232 dipole magnets (15 meters) to bend beams.
  - 392 quadrupole magnets (5-7 meters) to focus beams.
  - Additional magnets "squeeze" particles before collision.
- **Collision and Detection:**
  - Collisions occur at four points around the ring.
  - Detectors: ATLAS, CMS, ALICE, and LHCb.
- **Control:**
  - Managed by CERN Control Centre.

## 21.4.2 Purpose of LHC

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The formation of the LHC has given a hope to physicists for answering the basic questions relating to the fundamental laws, which govern the elementary particles' interactions and forces, the profound space and time structure and specifically, the quantum mechanics and general relativity intersection, where the present knowledge and theories are ambiguous. Moreover, there is a requirement of data from the high-energy particle testing for ascertaining the correctness of different scientific models' versions, such as those among the Standard Model of Physics and Higgsless model and validating their assumptions for further theoretical development. As per the expectation of many theorists, there will be emergence of new physics at the TeV-scale beyond that of the Standard Model on the basis of its unsatisfactory properties.

### 21.4.3 CERN Experiment

CERN performs experiments through various particle accelerators and detector (Cyclotron, Synchrotron and Large Hadron Collider), where we increase the speed of fundamental particles up to the speed of light and fuse them in opposite direction at last within LHC. Within a second, millions billions times have fusion of these hadron particles become possible and we observed the presence of fundamental particles through various detectors. Seven types of detectors are used inside Large Hadron Collider: ALICE, ATLAS, CMS, LHCb. In this experiment, the presence of God particle/Higgs boson particle was proved. In this CERN experiment, the presence of antimatter was also proved during this collision. But, development of this antimatter through artificial means is very costly. In the study of LHCb detector, it proved that standard model is not accurately correct in many aspects like dark matter, etc.

A particle accelerator is a device that uses electromagnetic fields to propel charged particles to high speeds and to contain them in well-defined beams. An ordinary CRT television set is a simple form of accelerator. Particle accelerators come in two basic types: Linear – Particles travel down a long, straight track and collide with the target; e.g: Linac; and Circular – Particles travel around in a circular orbit until they collide with the

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target; e.g: Cyclotron, Synchrotron and Large Hadron Collider (LHC). **Superconducting magnet** is used in all accelerators to increase the speed of proton (hadron) so collision will proceed with very high potential to give a better result inside the detector.

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## WHITE RABBIT (WR) TECHNOLOGY

### White Rabbit (WR) Technology

White Rabbit (WR) technology, developed at CERN, is **used for high-precision time synchronisation across networks**, with **applications ranging from particle physics to medical imaging, financial markets, Smart Grid, Astronomy, Meterology and various industries. It achieves sub-nanosecond accuracy, surpassing traditional methods and allowing for precise timing and data transfer in distributed systems.**

**It solves the challenge of establishing a common notion of time across a network.**

Indeed, at a scale of billionths of a second, the time light takes to travel through a fibre-optic cable and the time the electronics take to process the signal are no longer negligible. **To avoid potential delays, the co-inventors of White Rabbit designed a new ethernet switch.**

First used in 2012, the application of this fully open-source technology has quickly expanded outside the field of particle physics.

**In 2020, it was included in the worldwide industry standard known as Precision Time Protocol (PTP), governed by the Institute of Electrical and Electronics Engineers (IEEE).**

The **objective is to maintain a high-performance open-source technology** that meets the needs of users and to facilitate its uptake by industry.

### Charge-Parity (CP) Violation

Physicists at CERN's **Large Hadron Collider beauty (LHCb) experiment** have recently confirmed the presence of **Charge-Parity (CP) violation** in a category of particles known as **baryons.**

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| Feature | Matter | Anti-matter |
|---|---|---|
| **Composition** | Atoms (protons, electrons, neutrons) | Anti-atoms (anti-protons, positrons, antineutron- $\bar{n}$) |
| **Charge** | Normal | Opposite |
| **Interaction** | Stable in our Universe | Annihilates with matter (→ γ-rays) |
| **Creation** | Big Bang (equal amounts) | Created **naturally** (cosmic rays) and in labs **artificially** (LHC) |

### Annihilation

**Annihilation** is the **process where matter and anti-matter particles collide** and **transform into energy.**

This occurs when a particle and its anti-particle meet, causing them to disappear and release energy in the form of other particles, like photons.

A **common example is the annihilation** of an electron and a positron, producing two or more photons.

### What Is CP Violation?

**CP Symmetry:** In theory, **matter and anti-matter** should behave identically under CP.

**CP Violation = Matter behaves differently than anti-matter**

**CP violation refers to the asymmetry between matter and anti-matter, meaning the laws of physics are not identical for a particle and its anti-particle.**

***Significance:*** Explains why the **universe is dominated by matter** despite equal Big Bang production.

### Hadrons

**Hadrons** are subatomic particles **made of quarks** and they are classified into **two main categories:**

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1. **Baryons (made up of three quarks) e.g., proton and neutron, other examples (lambda, sigma, omega, delta, etc.)**
2. **Mesons (made up of quark and antiquark)**

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### Baryons

**In particle physics, a baryon is a type of composite subatomic particle that contains an odd number of valence quarks, conventionally three.**

- Made of **three quarks** (e.g., **proton**, **neutron**)
- **Lambda-b (Λb):** A baryon made of up (u), down (d) and bottom (b) quarks
- **Anti-Baryon:** Made of **three anti-quarks**

### Large Hadron Collider Beauty (LHCbs) Discovery

| Feature | Details |
|---|---|
| **Experiment** | CERN's LHCb (Large Hadron Collider beauty) |
| **Particle Studied** | **Lambda-b (Λb)** baryon |
| **Decay Observed** | Into **proton, kaon and 2 pions** |
| **Result** | **First confirmed CP violation in baryons** |

***Why It Matters?***

- **Supports asymmetry** between matter and anti-matter
- Offers clues to **why the universe exists in its current form**
- Helps advance **particle physics beyond the Standard Model**

## 21.4.4 India's Contribution at CERN

Indian scientists have been collaborating with CERN for the LHC experiment since 1991. India contributes about 100 scientists to the LHC. Indian scientists have put together crucial components to activate the LHC much below the sanctioned costs. All precision-made jacks on which the entire machine rests were made in India. Nearly 2,000 corrector magnets and circuit breakers were also made in India. The development of the

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machines gave the Indian scientists an opportunity to work on technologies of the highest level, validating the country's capabilities. India's involvement in the project has put the country in the big league. CERN granted India 'observer' status. Work on LHC's sequel, the 33-kilometre-long International Linear Collider, amply indicates the high value that Indian scientists now command.

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## 21.4.5 Brookhaven and the Belle-II Experiment

### 21.4.5.1 Brookhaven National Laboratory (BNL)

- Brookhaven National Laboratory (BNL) is a multi-disciplinary research facility located in Upton, New York, United States.
- It is operated by the U.S. Department of Energy (DOE) and is known for its contributions to various scientific fields, including nuclear and particle physics, materials science, and biology.
- BNL is home to several major research facilities, such as the Relativistic Heavy Ion Collider (RHIC) and the National Synchrotron Light Source II (NSLS-II). It collaborates with scientists from around the world and conducts cutting-edge research in fundamental and applied sciences.
- Atoms of antimatter have been trapped and stored for the first time by the ALPHA collaboration, an international team of scientists working at CERN, the European Organisation for Nuclear Research near Geneva, Switzerland. Scientists from the U.S. Department of Energy's Lawrence Berkeley National Laboratory and the University of California at Berkeley have made key contributions to the ongoing international effort.

### 21.4.5.2 Belle II Experiment

- Belle II is the successor to the Belle experiment and commissioned at the SuperKEKB accelerator complex at KEK in Tsukuba, Ibaraki prefecture, Japan.

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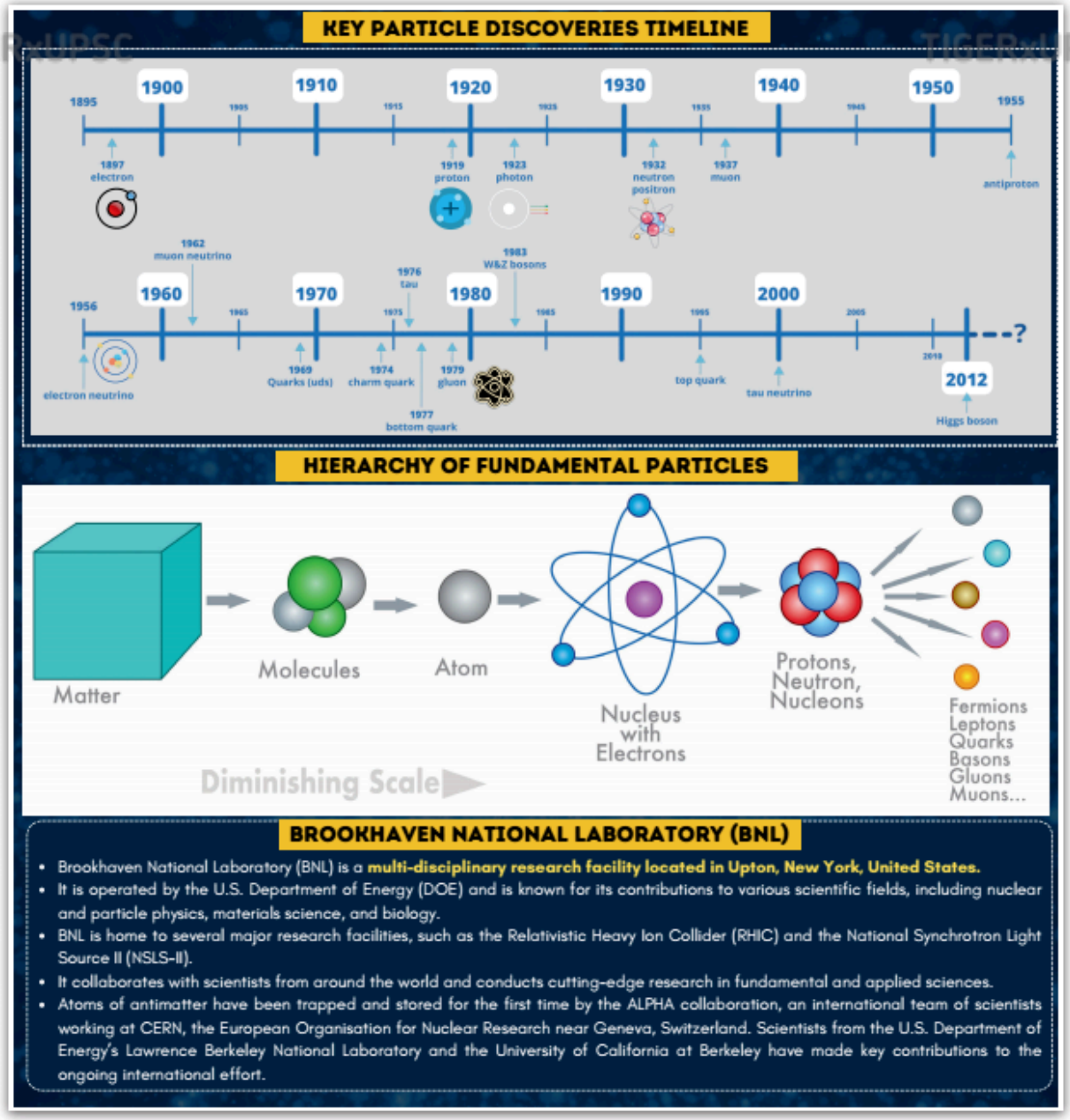


- In the big bang, matter and antimatter should have been created in equal amounts. But why is the universe today filled almost only with matter? There are so many other mysteries possible to search with the help of this experiment.
- The Belle II experiment is a **particle physics experiment** designed to **study the properties of B mesons** (heavy particles containing a beauty quark) and other particles.

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- The particles being created in the collisions and the decay products formed are measured and analysed in the Belle II experiment.

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## 21.5 Neutrino

### 21.5.1 Concept of Neutrinos and Its Importance

Neutrinos are the fundamental particles that were first formed at the beginning of the early universe, even before atoms could form. They are also continually being produced in the nuclear reactions of stars, like our Sun, and nuclear reactions here on Earth.

They are formed from the radioactive elements' decay and are the most abundant particles in the cosmos after the photons. They proliferated into solid matter over large distances and are very small in size equivalent to the tiny mass fraction of the proton having mass. Neutrinos are of three types, termed as flavours consisting of electrons, muons and tau particles. The major difference between them is that neutrinos are electrically neutral, whereas the electron, muon and tau are electrically charged.

It is very hard to detect the neutrinos, as they do not interact with other matter. But with the usage of special equipment situated in profound underground laboratories, it has become easier for scientists in the detection and discovery of neutrino and its various other properties, where it is not impossible for the penetration of other cosmic particles. The neutrino was postulated first by **'Wolfgang Pauli'** in 1930 to explain how beta decay could conserve energy, momentum and angular momentum (spin).

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Neutrinos are so abundantly available around us that each second, nearly greater than 100 trillion of them passing right through us without our observation. Due to this very reason, INO is required to be constructed deep into the Earth, nearly 1,300 metres in depth. The existence of neutrino in the universe is from almost 13.6 billion years ago, same as the age of universe.

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There is occurrence of three different types of neutrinos, such as νε, νμ and ντ and are also divided in terms of different masses. From the recent experiments, neutrinos are known to have a tiny mass, but the order of their mass states is unknown and is one of the major unanswered questions nowadays. This big challenge may be solved by the INO, thereby concluding our neutrino image.

There are three different reasons through which the neutrinos are very much vital for our scientific growth and technological advancement. **Firstly,** they are abundantly available. **Secondly,** they have very tiny mass and no charge and hence can easily move irrespective of any interaction through planets, stars, rocks and human bodies. In reality, a trillion neutrinos can move thousands of kilometres throughout a rock before the occurrence of any interaction between the neutrino and a single rock atom. **Thirdly,** there is a huge information pool hidden under them, which would create a new panorama in astronomy and astrophysics, communication and in medical imaging fields.

Moreover, the neutrinos have the following additional significance.

- **Firstly,** in the field of nuclear non-proliferation, neutrinos play a key role through the **monitoring of nuclear reactors.**
- **Secondly,** the usage of neutrinos assists in the **detection of mineral and oil deposits deep inside the Earth.** The Neutrinos have the property of altering their ‘flavour’, depending on their length of movement and the quantity of matter in the way they have passed through, which can help us in identifying the untimely **geological defects deep inside the Earth,** thereby answering our question of an **early warning system against earthquakes.** In this field only, the **Geo-neutrinos** are applicable. Their swift analysis through neutrino monitoring stations, which is termed as a process called Neutrino Tomography, can give us significant seismological data, thereby detecting early turbulence and vibrations created by earthquakes.

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- **Thirdly,** as the neutrinos can travel through the earth, which can unlock the faster way of sending data than the current 'around the Earth' model, with the **usage of towers, cables or satellites.** This system of communication may be transmission losses free as the usage of - neutrinos hardly react with any atoms on their way, thus establishing a **new panorama for telecom and Internet services.** As per few scientists, if there is any sign of any extraterrestrial life form, then the quickest way for making communication with them is neutrinos.
- **Fourthly,** the neutrinos are the **carriers of information regarding the universe and India's effort at INO in examining the neutrinos can facilitate in opening the universe's deepest mystery.**

As 95 per cent of universe is made up of dark matter and dark energy, and as per the belief of few scientists, the **neutrino research will assist in understanding the dark matter.**

The only way to solve the biggest mystery of dark matter is neutrinos, which can further alter our concepts in the understanding of the universe and physics. The probing of this dark matter can only be undertaken in INO. Due to this, the scientists consider neutrino is the only mode in accessing some of the most incredible technologies, and consequently, with INO, India is confident of undertaking its correct place at the helm of neutrino research.

**SOURCES OF NEUTRINO**

1. **Artificial:** The major source of human-generated neutrinos is the nuclear reactors. Anti-neutrinos are formed during the fission process in the beta-decay of neutron-rich daughter fragments.
2. **Geological:** Neutrinos are part of the natural background radiation.
3. **Atmospheric:** Atmospheric neutrinos result from the interaction of cosmic rays with atomic nuclei in the Earth's atmosphere, creating showers of particles, many of which are unstable and produce neutrinos when they decay.
4. **Solar:** Solar neutrinos originate from the nuclear fusion powering the Sun and other stars. When four protons fuse to become one helium nucleus, two of them have to convert into neutrons, and each such conversion releases one electron neutrino.
5. **Supernovae:** Neutrinos are an important product of Types Ib, Ic and II (core-collapse) supernovae. In such events, the density at the core becomes so high

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that the degeneracy of electrons is not enough to prevent protons and electrons from combining to form a neutron and an electron neutrino.

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## 21.5.2Neutrinos Fog

***LUX-ZEPLIN (LZ) and Dark Matter Detection***

- **LUX-ZEPLIN (LZ):** LUX-ZEPLIN (LZ) experiment is a **large-scale, underground experiment** designed to directly detect dark matter particles, specifically weakly interacting massive particles (WIMPs).
- LZ is a 7-tonne liquid xenon detector.
- By detecting these rare interactions, **LZ aims to provide insights into the nature of dark matter,** including its mass and how it interacts with ordinary matter.

This experiment is located 1.5 kilometres underground at the **Sanford Underground Research Facility (SURF) in South Dakota** to minimise interference from cosmic rays and other background radiation.

***What is Neutrino Fog?***

- **Definition:** Background noise in dark matter detectors caused by **neutrinos**—extremely light and elusive particles emitted by the **Sun,** supernovae and cosmic rays.
- **Problem:** Neutrinos interact **weakly**, just like dark matter particles, thus **mimicking their signals.**
- 'Neutrino fog' refers to the interference caused by neutrinos, elusive subatomic particles, in dark matter detection experiments. **Since neutrinos rarely interact with matter, they can pass through detectors and create background noise, making it difficult to distinguish dark matter signals from neutrino interactions.**
- This **interference is referred to as 'neutrino fog' or 'neutrino floor'**," and it presents a significant challenge in dark matter detection experiments.
- They can interact with the detectors, creating background noise that can **interfere with the search for dark matter signals.**

***Why Neutrino Fog Matters***

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| Aspect | Impact |
|---|---|

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| | |
|---|---|
| **Detection Confusion** | Neutrinos produce similar weak signals, making it hard to isolate actual dark matter interactions. |
| **Scientific Barrier** | **Differentiating between neutrinos and WIMPs** is one of the **major hurdles in future dark matter research.** |

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***Significance of LZ's Findings***

- **Calls for Innovation:** Pushes scientists to develop **discrimination techniques** to filter out neutrino noise.
- **Global Impact:** Helps refine the direction of **dark matter exploration** worldwide.

### 21.5.3 Neutrinos Oscillation

The bulk neutrinos are received by the Earth from the sun itself. Over the years, scientists have been finding out an anomaly between the observed and the theoretical data of the neutrinos. As per the studies undertaken by Japan's **Super-Kamiokande detector,** up to two-third of the neutrinos were lost in measurements performed on Earth. This phenomenon is explained by the process termed as neutrino oscillation in which three neutrinos 'metamorphosise' into one another, implying that neutrinos have mass, though, very little.

### 21.5.4 IceCube

It is a particle detector positioned at the South Pole, which records the interactions of subatomic particles having no mass, termed as the neutrinos.

(Refer **Infographic – Neutrino & Antineutrino**)

### 21.5.5 Baikal-GVD (Gigaton Volume Detector)

**The world's biggest underwater neutrino telescopes named as Baikal-GVD (Gigaton Volume Detector) is currently being launched by the Russian Scientists** in Serbia's **Lake Baikail, the world's deepest lake.**

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- It is **among the world's three largest neutrino detectors together with the IceCube at the South Pole and ANTARES** in the Mediterranean Sea.
- It aims **to study the mysterious fundamental particles termed as neutrinos in detail** and to probably find out their sources.

**REASON BEHIND OPPOSITION**

Tamil Nadu has made clear to the Supreme Court that it does not want **the Indian Neutrino Observatory (INO)** to be set up in a sensitive ecological zone in the Western Ghats at great cost to wildlife, biodiversity, and by ignoring the local opposition and public agitations to the project.

**Why is it being opposed?**

- The project in question falls exactly on the hill slopes of the part of the Western Ghats, which aligns within it a significant tiger corridor, namely **the Mathikettan-Periyar Tiger corridor.**
- This corridor links the **Periyar Tiger Reserve** along the Kerala and Tamil Nadu borders and **the Mathikettan Shola National Park.**
- The proposed project area also ecologically links to the eastern habitats, where **Srivilliputhur Meghamalai Tiger Reserve** is located. It hosts tigers from this region and helps in genetic dispersal.
- The area is a significant watershed and catchment zone for **the rivers Sambhal and Kottakudi.**

### 21.5.6 Indian Neutrino Observatory (INO)

The **India-based Neutrino Observatory (INO)** Project is **a multi-institutional effort** aimed at building a world class **underground laboratory** with a rock cover of approx.1200 m **for non-accelerator-based high energy and nuclear physics research in India.** The initial goal of INO is to **study neutrinos.**

(Refer **Infographic – Neutrino & Antineutrino**)

## 21.6 Theory of Relativity

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German scientist Albert Einstein proposed the 'Theory of Relativity' under two heads, i.e., Special Relativity (1905) and General Relativity (1915). A lot of assumptions underlying earlier physical theories were ousted by Einstein, enumerating the basic concepts of space, time, matter, energy and gravity. Specifically, relativity gives the fundamentals that are essential for the cosmic processes and understanding the universe's structure. The Nobel Prize in Physics 1921 was awarded to Albert Einstein 'For his services to Theoretical Physics and especially for his discovery of the law of the photoelectric effect'.

Observers who are experiencing constant velocity are accounted for by Special Relativity, whereas General Relativity covers those who are experiencing acceleration.

**Special Theory of Relativity:** 'Special Relativity' is restricted to those objects, which move at constant speed in a straight line, termed as the inertial motion. A frame of reference, either at rest or moving with a uniform velocity, is known as the inertial frame. As per the theory of special relativity – the speed of light is a maximum limit, which can be approached but not by any material object – forms the basis of the most famous physics equation that is $E = mc^2$.

**General Theory of Relativity:** 'General relativity' is linked with gravity being the universe's most fundamental force. Gravity describes the macroscopic behaviour, and consequently general relativity explains the physical phenomena such as planetary dynamics on larger scale. The applications of relativistic effects are there in basic science related to civilian and military technology. The theory $E = mc^2$ is vital in the study of subatomic particles, thereby determining the energy needed to form particles or for converting them from one type to another and the releasing of energy when the particles get destroyed. A 2016 research confirms Einstein's theory of relativity. Einstein compared the universe's shape to a single fabric. According to theory of relativity, the force of gravity is the result of curvature in this space-time. The theory of relativity describes that

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the laws of physics are similar everywhere. In short, it can predict everything from the existence of black holes, to light bending due to gravity, to the behaviour of Planet Mercury in its orbit.

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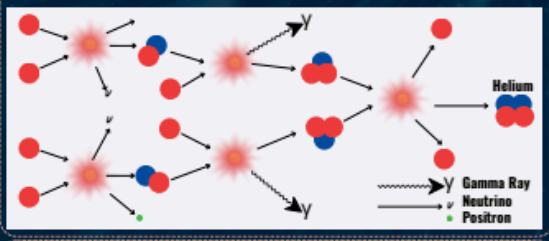
# NEUTRINO & ANTI-NEUTRINO

## NEUTRINO

- **Subatomic Particles:**
  - Belong to the **lepton family (includes electrons, muons, and tau particles).**
  - Fundamental particles, not composed of smaller particles.
- **Properties:**
  - **Electrically neutral (no electric charge).**
  - Interact very weakly with matter.
  - Possess an incredibly small mass (non-zero, confirmed by recent experiments).
- **Standard Model:**
  - Essential components of the Standard Model of particle physics.
  - **Three types (flavors): electron neutrinos, muon neutrinos, and tau neutrinos.**
  - Each flavor is associated with a specific charged lepton (electron, muon, or tau).
- **Production:**
  - **Produced in natural processes** (nuclear reactions in the Sun's core, radioactive decays, supernovae).
  - Created in particle accelerators and nuclear reactors.
- **Detection Challenges:**
  - Can pass through matter almost unaffected due to weak interactions.
  - Detected indirectly using specialised detectors (massive underground or underwater experiments).
- **Significant Discoveries:**
  - Neutrino oscillation: **Neutrinos can change their flavor** as they travel through space.
- **Research Importance:**
  - Crucial for understanding the fundamental properties of matter.
  - Provides insights into the origins of the universe and particle behaviour in extreme environments.
  - Implications for astrophysics, cosmology, and the fundamental laws of the universe.

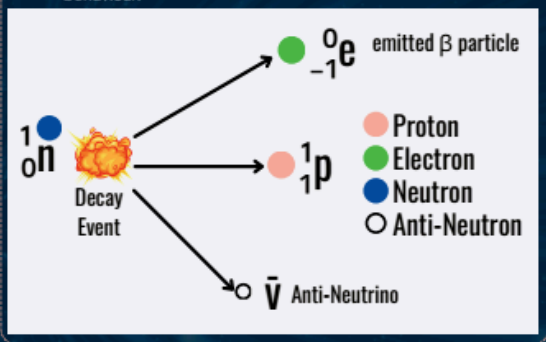



## WHAT IS ICECUBE NEUTRINO OBSERVATORY TRYING TO FIND?

- **IceCube detects blue light given off by fast-moving particles created when a neutrino hits a proton or neutron in an atom. This collision creates other particles that move faster than light in ice, emitting blue Cherenkov radiation. The detected neutrinos act like clues, helping us understand where they came from and what caused them.**

## ANTI-NEUTRINO

- **Antiparticles:**
  - Antiparticles of neutrinos.
  - **Electrically neutral, very low mass.**
- **Lepton Flavors:**
  - Opposite lepton flavors to neutrinos.
  - **Three types: electron, muon, tau anti-neutrinos.**
- **Production:**
  - Emitted in processes like beta decay (neutron to proton, emitting anti-neutrino and beta particle).
- **Interactions:**
  - **Weak interactions with matter,** passing through large amounts of material.
  - Detected via weak nuclear force in specialised experiments.
- **Research Importance:**
  - Insights into antimatter and matter-antimatter symmetries.
  - Crucial for particle physics, astrophysics, and nuclear physics.
  - Enhances understanding of fundamental particles and universe behaviour.




## WHAT IS ICECUBE NEUTRINO OBSERVATORY?

- **Overview:**
  - First detector of its kind, **located at the South Pole.**
  - Operated by the **IceCube Collaboration (300 physicists, 58 institutions, 14 countries).**
- **Design and Purpose:**
  - Encompasses a cubic km of ice.
  - Observes **high-energy neutrinos to study violent astrophysical events** (exploding stars, gamma-ray bursts, black holes, neutron stars).
- **Components:**
  - Includes surface array IceTop and dense infill array DeepCore.
  - Multipurpose experiment for various physics questions (dark matter, neutrino properties).
- **Scientific Contributions:**
  - **Observes cosmic rays interacting with Earth's atmosphere.**
  - **Reveals new structures and phenomena.**
- **Funding and Management:**
  - Primary funding by the National Science Foundation (NSF).
  - Partner funding agencies worldwide.
  - University of Wisconsin-Madison leads maintenance and operations.

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## WHY IS ICECUBE IMPORTANT?

- **Fundamental Research:** Aims for groundbreaking discoveries.
- **Exploration Areas:** Dark matter, dark energy and neutrinos.
- **Goals:** Uncover unexpected phenomena and enhance understanding of the universe's composition.

## WHY BUILD ICECUBE AT THE SOUTH POLE?

- **Material Need:** Large volume of transparent material (water or ice) to detect light from neutrino interactions.
- **Location Choice:** South Pole offers vast quantities of clear, pure, stable ice with necessary research infrastructure.
- **Ice Quality:** South Pole ice is ultra-transparent due to compression, which removes air bubbles.
- **Radiation Shielding:** Sensors (DOMs) buried 1,500 meters deep to avoid surface radiation.
- **Optimal Conditions:** South Pole station provides ideal conditions for building and operating IceCube.

## HOW BIG IS ICECUBE?

- **Volume:** 1 cubic kilometer.
- **Area:** Approximately 1 square kilometer.
- **Depth:** Detector array extends from 1,500 meters to 2,500 meters below the surface.

## WHO IS IN CHARGE OF THE DETECTOR?

- The IceCube Collaboration runs the project and the University of Wisconsin–Madison (UW) is the lead institution. UW led construction of the IceCube proof-of-concept project called AMANDA, and in evaluating proposals from several institutions, the National Science Foundation determined that UW had the experience and success to oversee IceCube.

# INDIA-BASED NEUTRINO OBSERVATORY (INO)

## WHAT IS INO?

- **Stage:** Feasibility study.
- **Collaboration:** Over 50 scientists from 15 Indian institutions.
- **Group:** National Neutrino Collaboration Group (NNCG).
- **Task:** Develop proposal for an underground neutrino laboratory.

## WHY INO?

- **Non-zero Mass:** **Neutrinos are not massless** as previously thought.
- **Oscillation:** Different neutrino flavors mix and oscillate.
- **Significance:** Evidence for physics beyond the Standard Model.
- **Impact:** Affects nuclear and particle physics, astrophysics, and cosmology.
- **Research Need:** Detailed study of neutrino interactions.
- **Underground Labs:** Maximises sensitivity to weak interactions.

## LOCATION OF INO

- **Location:** **Bodi West Hills, near Pottipuram village, Theni district, Tamil Nadu.**
- **Geographical Advantage:** Near Earth's equator, beneficial for solar neutrino studies.
- **Site Selection Criteria:**
  - Over 1,000 meters underground to reduce cosmic radiation.
  - Favorable geological conditions: dense gneiss rock.
  - Access to water, power, and easy site accessibility.
- **Proximity:** 110 km from Madurai.

## DETECTOR AT INO

- **Objective:** Study various aspects of neutrino physics.
- **Phase One Focus:** Atmospheric neutrinos from cosmic ray interactions.
- **Detectors:** Multiple types, including an iron calorimeter (ICAL).
- **ICAL Function:**
  - Identifies neutrino and antineutrino interactions.
  - Horizontal layers trigger upon charged particle passage.
  - Solenoidal coils create a magnetic field to bend particles.
  - Measures charge, energy, and momentum of particles.

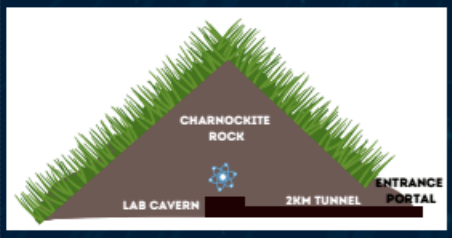
## ICAL & NEUTRINOS

- **Neutrino Oscillation:** **Aims to provide clear evidence by observing both depletion and enhancement in the neutrino spectrum.**
- **Muon Spectra:** Studies muon spectra from atmospheric neutrinos to establish oscillation patterns.
- **Oscillation Parameters:** Determines oscillation parameters accurately.
- **Neutrino/Antineutrino Distinction:** Studies Earth-matter effects and explores poorly-known oscillation parameters.

## WHY UNDERGROUND LABORATORY?

- **Detection Challenge:** Weak interactions with matter, high background noise from cosmic rays and natural radioactivity.
- **Underground Locations:** Reduce background noise, enhance detection.
- **Historic Lab:** Kolar Gold Field (KGF) mines in India (closed).
- **Current Labs:** Sudbury (Canada), Kamioka (Japan), Gran Sasso (Italy), Soudan (USA).
- **INO Project:** Aims to continue KGF's pioneering neutrino research.
- **Advancement:** Improved neutrino detection and research capabilities.

## WHY DETECT NEUTRINOS?




- **Non-zero Masses:** Exact values for the three types are uncertain.
- **Cosmic Impact: Small masses influence the Universe's evolution via gravitational effects.**
- **Research Importance:**
  - Essential for understanding the Universe's origin and stellar energy production.
  - Enables neutrino tomography to study Earth's deep interiors.

**EINSTEIN'S THEORY OF RELATIVITY**

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As per the Theory of General Relativity, **under the condition of no air resistance, all objects fall at the same rate in the same gravitational field.** For instance, if you drop a bowling ball (heavy item) and a tiny marble (lighter item), off a building, both will hit the ground at the exact same time. **The difference in mass and composition do not affect the timing of the fall at all.** The researchers found in a year long research that **no detectable difference indicating no alternative theories of gravity were in motion.** These results were consistent with Einstein's theory of relativity.

### The theory of general relativity

- In 1915, Albert Einstein propounded his theory of general relativity, which proposed that gravity itself was the result of warping of space-time by massive objects like stars and planets.
- Einstein's theory of relativity indicates that all objects fall the same way regardless of mass or composition.

### Things predicted by General relativity include

- ***Light bends towards the Sun when it gets closer to the Sun.***
- ***The perihelion of the planet Mercury rotates along its orbit more than is expected:*** General relativity accounts for the difference between what is seen and what is expected without it.
- ***Redshift from gravity:*** The light is stretched into longer wavelengths when lights move away from an object with gravity (moving away from the centre of the valley).
- ***The Shapiro delay:*** Light seems to slow down when it passes close to a massive object.
- ***Gravitational waves:*** They were first observed on 14$^{th}$ September 2015.

### 100 Years of Theory of Relativity

Theory of relativity completed its centennial anniversary in 2015. It was given by Albert Einstein in the year 1915 and is considered the most revolutionary theory to human kind.

## 21.7 Gravitational Waves

Refer **Infographic – Gravitational Waves - Ripples in Space Time**.

Gravitational wave observatories are installed in many places (**Figure 21.1**) and also work in future for better understanding of gravitational waves: **Hanford** in Washington (Operational), **Livingston** in Louisiana (Operational), GEO600 in Germany

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(Operational), **Virgo** in Italy (Operational), **KAGRA** in Japan (Operational), India (Planned).

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**GRAVITATIONAL LENSING**

- Gravitational lensing is an **effect of Einstein's theory of general relativity** – simply put, **mass bends light**.
- The gravitational field of a massive object will extend far into space, and cause light rays passing close to that object (and thus through its gravitational field) to be bent and refocused somewhere else.
- The more massive the object the stronger its gravitational field and hence the greater the bending of light rays – just like using denser materials to make optical lenses results in a greater amount of refraction.

**Benefits**

- Gravitational lensing is useful to cosmologists because it is directly sensitive to the amount and distribution of dark matter.
- Lensing can help astronomers work out exactly how much dark matter there is in the universe as a whole and also how it is distributed.

Lensing has also been used to help verify the existence of dark matter itself.

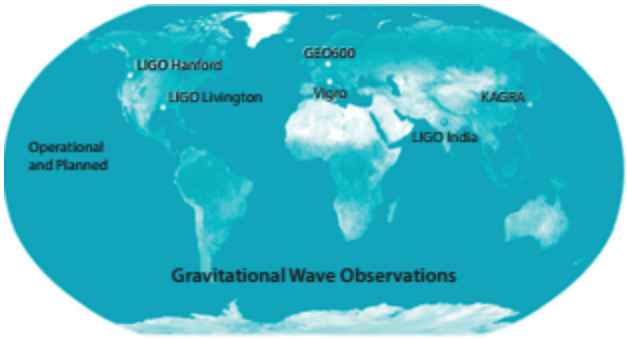


Figure 21.1 Gravitational Wave Observatories

### 21.7.1 LISA Pathfinder, LISA and eLISA

LISA mission detects the gravitational waves which **might also approach from other remote systems** such as Extreme Mass Ratio Inspirals (EMRIs) **which are smaller stellar mass black holes** orbiting supermassive black holes. Gravitational waves study provides **vast potential for the discovery of those universe parts** such as black holes, the Big Bang, unknown

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objects, etc., **which are invisible** by other means. **LISA will supplement our information about the universe's beginning, evolution and structure.**

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### *LISA Pathfinder Mission*

**LISA Pathfinder, a LISA proof-of-concept mission was initiated on December 3, 2015** which tests within the spacecraft, the free-floating test masses that are small enough in comparison to an anticipated gravitational wave signal. After the completion of its **mission in 2017, LISA Pathfinder has demonstrated that the outstanding noise levels exceeds the original requirements by a factor of ~100**. The mission also surpassed the full LISA mission requirement, indicating that major technology is in progress for LISA.

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# GRAVITATIONAL WAVES – RIPPLES IN SPACETIME

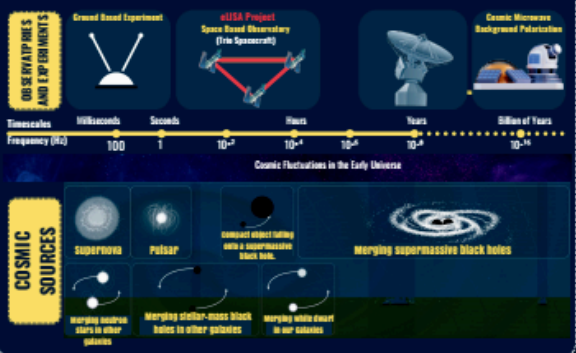
- **Origin and Theory:**
  - Gravitational waves are **disturbances in the fabric of space-time.**
  - **Theorised by Albert Einstein in 1916** through his general theory of relativity.
  - **Arise from the acceleration of massive objects like black holes and neutron stars.**
- **Essential facts:**
  - **Travel at the speed of light.**
  - **Carry valuable information about their sources.**
  - **Provide insights into the nature of gravity.**
- **Sources of Gravitational Waves:**
  - **Merging black holes.**
  - **Explosive deaths of massive stars (supernovae).**
  - **Collisions of neutron stars.**
  - **Irregular rotation of neutron stars.**
  - **Possible remnants of gravitational radiation from the Big Bang.**

## DISCOVERY OF GRAVITATIONAL WAVES

- Einstein's Prediction and Initial Confirmation
  - Predicted by Einstein in 1916.
  - **Confirmed in 1974 by Russell Hulse and Joseph Taylor using the Arecibo Radio Observatory.**
  - **Observed a binary pulsar emitting gravitational waves** as predicted by general relativity.
  - Hulse and Taylor awarded the Nobel Prize in Physics in 1993.
- Key Observations and Studies
  - Taylor, Weisberg, and Fowler tracked binary pulsar emissions.
  - Confirmed orbital period changes matched gravitational wave emission predictions.
  - Pulsar radio-emission studies provided further indirect confirmations.
- LIGO's Landmark Detection
  - **On September 14, 2015, LIGO directly detected gravitational waves.**
  - **Detected waves from a black hole collision** 1.3 billion light-years away.
  - Gravitational waves detected were a thousand times smaller than an atom's nucleus.
  - LIGO designed for extremely precise measurements.

## SOURCES OF GRAVITATIONAL WAVES

- Sources of Gravitational Waves:
  - Produced by any **accelerating physical object.**
  - Detectable waves require massive objects outside our solar system.
- Key Massive Objects:
  - **Orbiting black holes or neutron stars.**
  - **Interactions between neutron stars and black holes.**
  - **Exploding massive stars (supernovae).**
- Types of Gravitational Waves:
  - **Continuous:** Steady signals from single, spinning neutron stars.
  - **Compact Binary Inspiral:** From orbiting pairs of black holes or neutron stars.
  - **Stochastic:** Background noise from many distant sources.
  - **Burst:** Short, strong signals from cataclysmic events like supernovae or mergers.




## TYPES OF GRAVITATIONAL WAVES

### CONTINUOUS GRAVITATIONAL WAVES

- From spinning neutron stars with shape imperfections.
- Constant frequency and amplitude, like a sustained musical note.
- Detected waves have been simulated by researchers.

### COMPACT BINARY INSPIRAL GRAVITATIONAL WAVES

- Detected by LIGO, originating from orbiting pairs of dense, massive objects.
- Includes Binary Neutron Star (BNS), Binary Black Hole (BBH), and Neutron Star-Black Hole (NSBH) systems.
- Objects emit gravitational waves, spiral closer, and merge due to energy loss.

### STOCHASTIC GRAVITATIONAL WAVES

- LIGO isn't worried about simultaneous significant sources.
- Small waves from all directions create a random pattern.
- May include remnants from the Big Bang.
- Detecting these waves reveals new insights into the Universe's history.

### BURST GRAVITATIONAL WAVES

- Exploration of the unexpected, with many unknowns.
- LIGO has not yet detected these waves.
- Nature and sources are uncertain; no predetermined properties.
- Requires recognising unfamiliar patterns and signals.
- Detection could reveal groundbreaking insights about the Universe.

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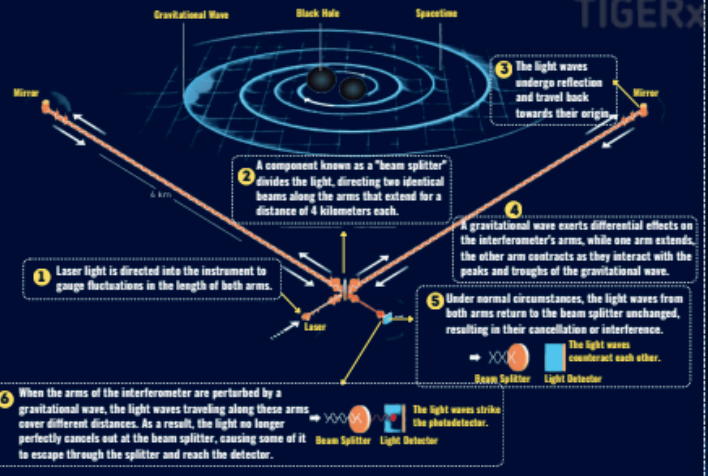
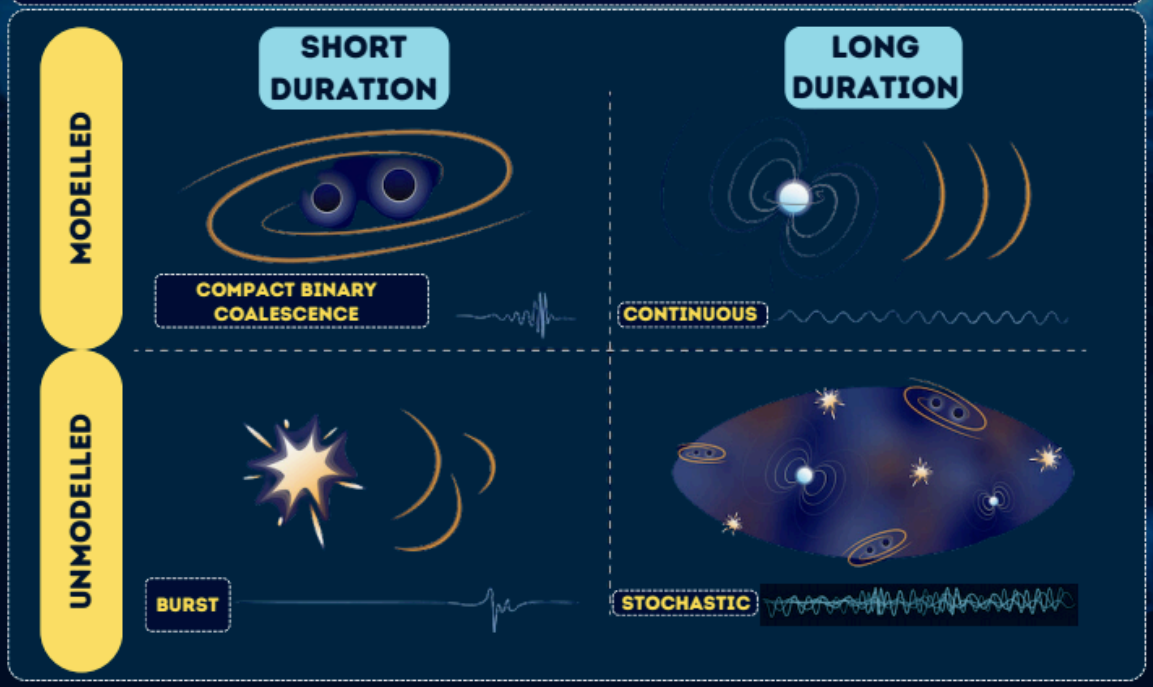
## WHY DETECT GRAVITATIONAL WAVES?

- **Traditional vs. New Messengers:**
  - Traditionally relied on EM radiation.
  - Now exploring neutrinos and gravitational waves for new insights.
- **Gravitational Waves: A New Perspective:**
  - Provide a different view from EM radiation.
  - Comparable to gaining the sense of hearing along with sight.
- **LIGO's Role:**
  - Functions as a gravitational wave antenna.
  - Detects space-time vibrations from events like colliding black holes.
  - Gravitational waves travel undistorted, carrying pure information.
- **Sources and Insights:**
  - Originate from cataclysmic events like merging neutron stars and the birth of the Universe.
  - Offer unprecedented insights into unseen cosmic phenomena.
- **Impact on Science**
  - LIGO's detection has revolutionised physics, astronomy, and astrophysics.
  - Unveils the mysteries of the Universe through groundbreaking research.




## LIGO - A GRAVITATIONAL-WAVE INTERFEROMETER

**LIGO, the world's largest gravitational wave observatory, uses two laser interferometers 3000 kilometers apart to study gravitational waves through light and space properties.**

**Advanced LIGO (aLIGO), built from 2008 to 2015, replaced Initial LIGO to be 10 times more sensitive, following its predecessor's operation from 2002 to 2010.**

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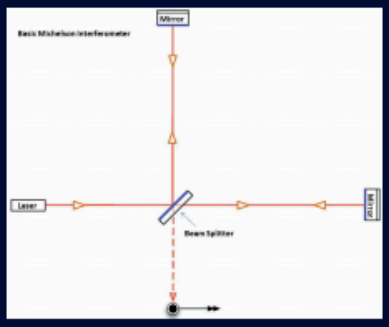
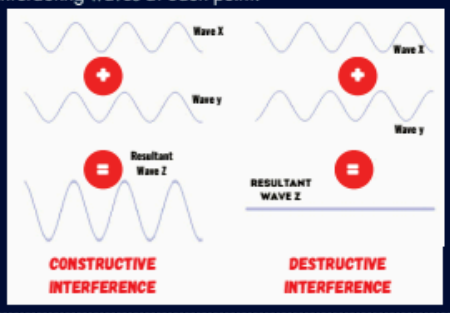
## WHAT IS AN INTERFEROMETER?

- **Versatile Tools:**
  - Interferometers used in various scientific and engineering fields.
  - **Combine light sources to create interference patterns for valuable data.**
  - Effective for highly precise measurements.
- **LIGO's Interferometers:**
  - **Detect gravitational waves by measuring distances as small as 1/10,000th of a proton's width.**
- **Historical Development:**
  - Developed in the 19th century by scientists like Fizeau, Hoek, Mascart, Airy, and Ketteler.
  - Initially studied light wave properties and "luminiferous ether."
- **Michelson-Morley Interferometer:**
  - Invented in 1887, disproved the existence of the ether.
  - Contributed to Einstein's theories of relativity.
  - Configuration ideal for detecting gravitational waves since the late 1960s.
- **Modern Interferometric Detectors:**
  - **Michelson laser interferometer includes a laser, beam-splitter, mirrors, and photodetector.**
  - Fundamental in LIGO, Virgo, Geo, and KAGRA detectors.




## WHAT IS AN INTERFERENCE PATTERN?

- **Interference Basics:**
  - Interference occurs when **multiple waves intersect and interact.**
  - Creates an **interference pattern from the sum of interacting wave heights and depths.**
- **Principles of Interference:**
  - **Total constructive interference:** identical waves moving in the same direction align perfectly, forming a larger wave.
  - **Total destructive interference:** identical waves moving in opposite directions align to cancel each other out.
- **Real-World Applications:**
  - **Real waves are rarely identical and perfect alignment is uncommon.**
  - Resulting wave at **intersections represents the sum of their heights or depths.**
- **Complex Interference Patterns:**
  - Varying amplitudes and frequencies lead to complex patterns.
  - Fundamental principle: adding the heights or depths of interacting waves at each point.

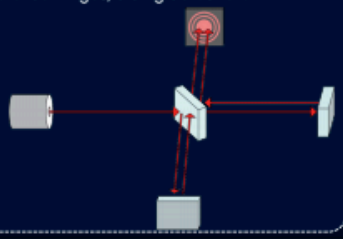



## WHAT IS LIGO?

- **LIGO Overview:**
  - Stands for **Laser Interferometer Gravitational-wave Observatory.**
  - **Specialises in detecting gravitational waves, not electromagnetic radiation.**
- **Operational Differences:**
  - Detection Method: Detects gravitational waves, not part of the electromagnetic spectrum.
  - Structure: Uses long vacuum tubes in an "L" shape, not optical telescopes or radio dishes.
  - Signal Confirmation: Requires multiple detectors to confirm a signal; initial detection needed both Hanford and Livingston detectors.
- **Collaborations and Discoveries:**
  - Successful detection of numerous gravitational wave signals.
  - Collaborations with other observatories like Virgo.
  - Significant discoveries include the first binary neutron star merger detection.

## LIGO'S INTERFEROMETER

- **LIGO's Michelson Interferometer Core:**
  - **Based on the Michelson Interferometer** design from the 1880s.
  - **L-shaped structure with significant enhancements.**
- **Scale and Complexity:**
  - LIGO's arms are 4 km long, compared to the original's 1.3 meters.
  - Larger size enables detection of extremely small distance changes (10,000 times smaller than a proton).
- **Fabry-Perot Cavities:**
  - Additional mirrors in each arm near the beam splitter.
  - Laser light bounces, effectively increasing arm length from 4 km to 1200 km.
  - Enhances sensitivity to gravitational waves.
- **Laser Power:**
  - Power Recycling Mirrors reflect laser light back, boosting beam power inside the cavities.
  - Increases visibility of interference fringes, aiding detection.
- **Signal Recycling and Vibration Damping:**
  - Signal Recycling Mirrors enhance the received signal.
  - Advanced vibration damping mechanisms minimise noise.



Since 1970, the thought of **space-borne large-scale interferometer** was there and the Laser Interferometer Space Antenna (LISA) idea was originated. Its main aim is the

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**identification of low frequency gravitational waves which are unreachable from the Earth surface. LISA created by satellites constellation which are 5 × $10^6$ Km apart was an interferometer having shape of an equilateral triangle.**

Better frequency range can be discovered by the Space-borne interferometers than their complement on Earth. Precisely, **eLISA can take gravitational radiation of less than 1 Hz dimensions.**

LIGO detected the **gravity wave which grows from 35 Hz up to 250 Hz.** There are inevitable gravitational noise sources on Earth **in the shape of, weather phenomena, moving objects and seismic motion.** As a result, only high frequency gravity waves will be noticeable while **interferometers in space can escape these noise sources. The bigger the interferometer, the smaller the detectable strain.**

The **mother satellite** flames two Lasik signals which get reflected by the opposite vertex satellites and thereby creating the two interferometer arms. There is no beam splitter in mother **satellite but although it generates two independent phase-locked laser signals. When these signals return to the mother satellite, any present phase shift is registered.**

The **complete arrangement is positioned 20° at the rear of our planet and rotates on an Earth like orbit around the sun** which is nearly 50 million Km far away from the Earth.

There is a requirement of each satellite maintaining a constant distance with respect to the other satellite, if they have to function as an interferometer because every satellite pair is an interferometer arm and it would be ineffective if they changed length (i.e., without the perturbation of a gravitational wave).

## LISA: Taking the search for gravitational waves into space

Due for launch in the mid-2030s, LISA (Laser Interferometer Space Antenna) will be the first dedicated space-based gravitational wave detector. LISA will consist of a constellation of three identical spacecraft arranged to form a triangle with ‘arms’ one million kilometres long.

LISA will work rather like LIGO, which made the first historic detection of gravitational waves, but, with arms 600,000 times longer than LIGO, it will extend our capabilities to ‘listen’ to new kinds of dark phenomena in the Universe.

The distance between the spacecraft is precisely monitored so that even the tiniest changes in the lengths of the ‘arms’ caused by a passing gravitational wave can be measured.

## LISA Pathfinder

Launched in 2015, LISA Pathfinder paves the way for the LISA mission by testing the concept of a space-based gravitational wave detector in miniature.

It serves as a test bed for the technology by mimicking one arm of the future LISA constellation.

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LISA Technology Package (LTP) onboard LISA Pathfinder is a miniature version of one arm of LISA. Instead of three full-size spacecraft separated by one million kilometres, it has two test masses – identical 46 mm cubes – separated by around 40 centimetres. They are housed in individual vacuum enclosures.

### LISA Technology Package (LTP)

A laser beam (red) probes the test mass positions and is recombined with a reference beam (blue) to detect any movement.

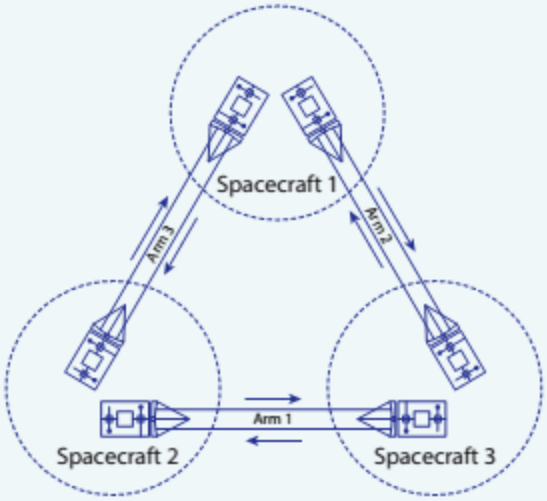


Figure 21.2 Trio Space Craft

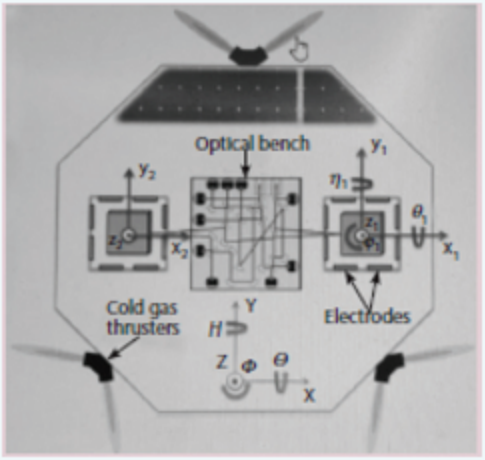


Figure 21.3 Interferometer

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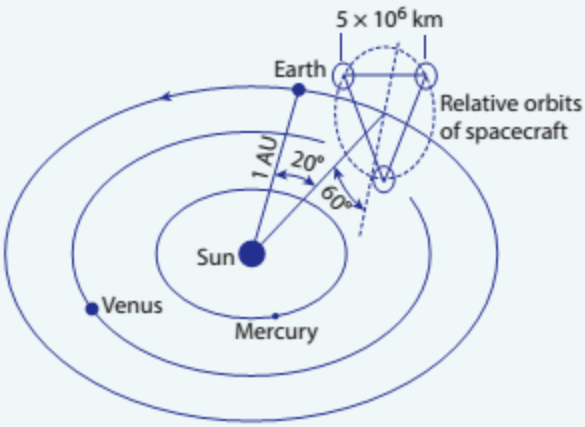


Figure 21.4 Position of All Three Spacecrafts

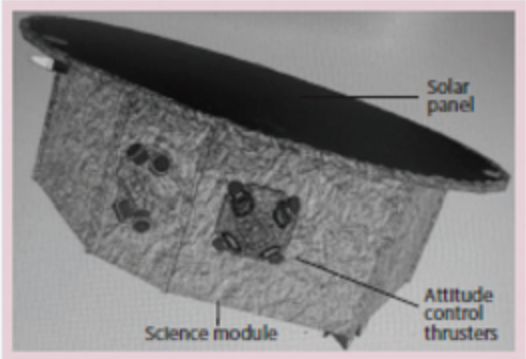


Figure 21.5 Space Craft Outer View

The test masses' movements must only be influenced by gravity so they have to be protected from other forces – such as the pressure of sunlight, magnetic fields, temperature fluctuations and the LISA spacecraft's own gravity.

**LIGO-INDIA**

- Scheduled for **completion by 2030,** it is located in **Hingoli, Maharashtra,** chosen for its low seismic activity.
- **Geographical Impact:** Spans 430 acres, impacting six villages and boosting the local economy.
- **Equipment:** Includes **4-km arms with vacuum chambers and highly reflective mirrors for precise detection.**
- **Precision Engineering:** Mirrors polished to nanometer scales, vacuum chambers maintained at near-absolute vacuum.

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- Strict access control ensures a controlled environment for sensitive experiments.

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**Future Prospects and Benefits**

- LIGO-India will foster international collaboration, attracting researchers for groundbreaking experiments.
- With an **operational life of 30 years**, it will contribute significantly to scientific advancements.
- Expected to **generate vast amounts of data valuable for astronomy and astrophysics research.**

**Outreach**

- The proposed science outreach centre, 'Light & Gravity,' promotes scientific curiosity and knowledge in the local community.

***A Different Frequency Range from Different Objects***

A wide frequencies range are being covered by the gravitational wave spectrum. In comparison to LIGO's frequency of 10 Hz to 1000 Hz, LISA functions in the frequency range between 0.1 mHz and 1 Hz which is very low. The differentiation pertains to the concept that the LISA is searching for waves having a much longer wavelength, consequent to objects which are much heavier and in wider orbits than those for which LIGO is seeking for, thereby opening up the detection sphere to a wider range of sources of gravitational wave.

**LISA consists of three spacecrafts forming in space an equilateral triangle where the triangle sides are also termed as the LISA's 'arms', thereby extending nearly a million miles.** Hence, in space, **LISA can shun the noise and access spectrum regions which are inaccessible from Earth owing to these particularly long arms.** LISA discovery of the gravitational wave sources consists of Galaxy's ultra-compact binaries, supermassive black hole mergers, and extreme mass ratio inspirals, between other unusual possibilities.

**TERMINOLOGIES**

**LISA Pathfinder, eLISA & LISA (Laser Interferometer Space Antenna)**

**LISA Pathfinder, Elisa** and **Laser Interferometer Space Antenna (LISA)** are all related to gravitational wave detection, but they represent different phases of the technology and mission. LISA Pathfinder was a technology demonstration mission,

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Elisa is a proposal for a future gravitational wave observatory and LISA is the actual mission that is designed to detect and study gravitational waves.

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**I. LISA Pathfinder**

**Purpose:** To test the technologies needed for a future space-based gravitational wave observatory.

**Key Feature:** Placed two test masses in a nearly perfect gravitational free-fall and controlled their relative motion with high precision.

**Launch:** Launched in ٢٠١٥.

**Outcome:** Successfully demonstrated the technology needed for LISA.

**II. eLISA**

**Purpose:** A proposed mission to explore the Gravitational Universe from space by detecting gravitational waves.

**Key Feature:** A constellation of three spacecraft forming an equilateral triangle with sides measuring about a million kilometres.

**Status:** A proposal for a future mission.

**III. Laser Interferometer Space Antenna (LISA)**

**Purpose:** To detect and measure gravitational waves from space.

**Key Feature:** A trio of spacecraft positioned in a heliocentric orbit, forming a triangular configuration with arms stretching about a million kilometres.

**Launch:** Planned for ٢٠٣٥.

**Goal:** To study gravitational waves and their sources.

## 21.8 Daksha Project

### *Daksha Project:* India's High-Energy Space Telescope Initiative

***About the Project***

- **Led by:** Indian scientists in collaboration with **Physical Research Laboratory (PRL)**, **TIFR**, **Raman Research Institute (RRI)**
- **Mission:** Build **two advanced space telescopes** to monitor high-energy astrophysical phenomena
- **Instruments:** Each satellite will carry **three types of sensors** to detect from **low to high-energy gamma rays**

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***Key Objectives***

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- **Detect and study** high-energy signals from **gravitational wave sources**
- **Track Gamma Ray Bursts (GRBs)** – short-lived, ultra-energetic light bursts
- **Localise cosmic events** like **neutron star mergers**
- **Probe Primordial Black Holes (PBHs)** – ancient black holes from early universe

### *Design Significance*

- **Two satellites** in **opposite Earth orbits ((Prograde—same direction of earth) and (Retrograde—Opposite direction of earth))** for **full-sky coverage**
- **Better precision and localisation** than current gamma-ray missions
- Enables rapid response to gravitational wave alerts

### *Scientific Impact*

- Improve multi-messenger astronomy (gravitational waves + gamma rays)
- Aid understanding of **cosmic evolution**, **black holes** and **extreme physics**
- First potential probe of **Primordial Black Holes (PBH) mass window**

### *Related Missions*

| Mission | Agency | Focus/Observation/Study |
|---|---|---|
| **Astrosat** | ISRO (India) | Multi-wavelength (X-ray, UV, optical) |
| **Fermi Telescope** | NASA | Broad-spectrum gamma-ray observation |
| **Swift Observatory** | NASA | Gamma-ray burst detection and afterglow study |

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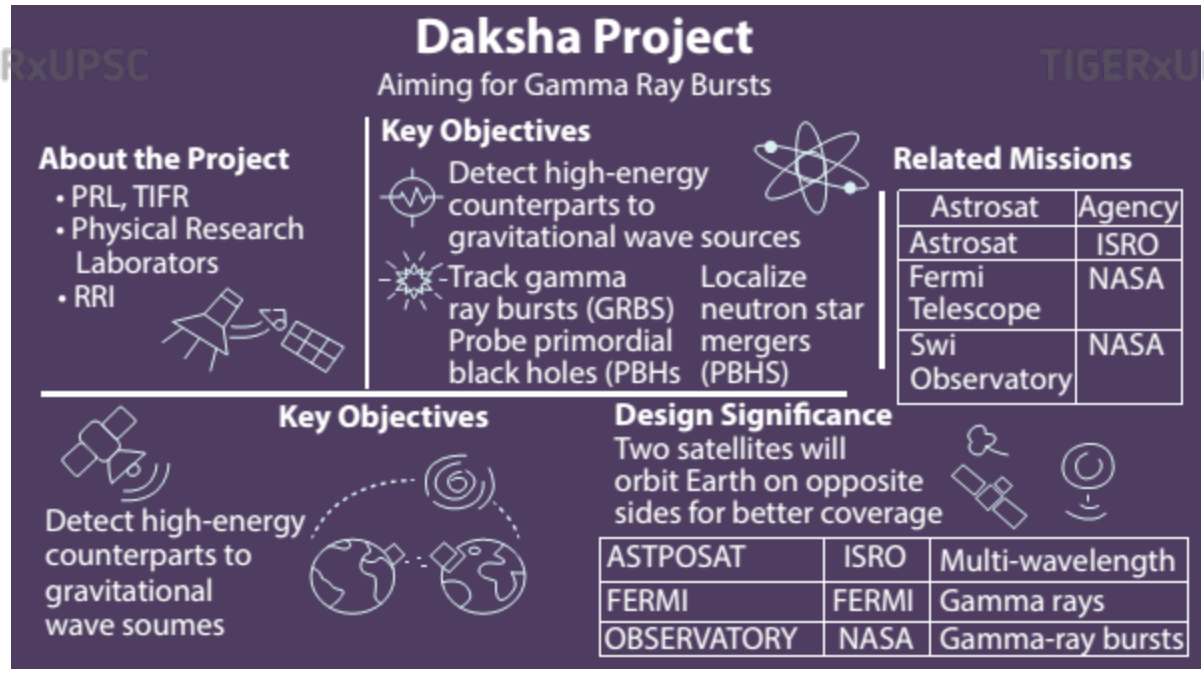


## QUESTIONS AND ANSWERS

### Objective Questions

**1. The experiment will employ a trio spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometres long, with lasers shining between the craft. The experiment in question refers to** **[UPSC (Pre) – 2020]**

(a) Voyager-2

(b) New Horizons

(c) LISA Pathfinder

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(d) Evolved LISA

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**2. Recently, scientists observed the merger of giant 'blackholes' billions of light-years away from the Earth. What is the significance of this observation? [UPSC (Pre) – 2019]**

(a) 'Higgs boson particles' were detected.

(b) 'Gravitational waves' were detected.

(c) Possibility of inter-galactic space travel through 'wormhole' was confirmed.

(d) It enabled the scientists to understand 'singularity'.

**3. Consider the following phenomenon: [UPSC (Pre) – 2018]**

1. Light is affected by gravity.
2. The Universe is constantly expanding.
3. Matter wraps its surrounding space-time.

Which of the above is/are the prediction/predictions of Albert Einstein's General Theory of Relativity often discussed in media?

(a) 1 and 2

(b) Only 3

(c) 1 and 3

(d) All of these

**4. The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of [UPSC (Pre) – 2017]**

(a) Observation and understanding of the universe

(b) Study of the solar and lunar eclipses

(c) Placing satellite in the orbit of the earth

(d) Origin and evolution of living organism

**5. What is the purpose of 'evolved Laser Interferometer Space Antenna' (eLISA) project? [UPSC (Pre) – 2017]**

(a) To detect neutrinos

(b) To detect gravitational waves

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(c) To detect the effectiveness of missile defence systems
(d) To study the effect of solar flares on communication systems

**6. In the context of modern scientific research, consider the following statements about 'Ice Cube', a particle detector located at South Pole, which was recently in the news: [UPSC (Pre) – 2015]**

(i) It is the world's largest neutrino detector, encompassing a cubic kilometre of ice.
(ii) It is a powerful telescope to search for dark matter.
(iii) It is buried deep in the ice.

Which of the statements given above is/are correct?

(a) Only (i)
(c) (i) and (iii)
(b) (ii) and (iii)
(d) (i), (ii) and (iii)

**7. The efforts to detect the existence of Higgs boson particle have become frequent news in the recent past. What is the importance of discovering this particle? [UPSC (Pre) – 2013]**

(i) It will enable us to understand as to why elementary particles have mass.
(ii) It will enable us in the near future to develop the technology of transferring matter from one point to another without traversing the physical space between them.
(iii) It will enable us to create better fuels for nuclear fission.

Select the correct answer using the codes given below.

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii) and (iii)

**8. A team of scientists at Brook-Haven National Laboratory including those from India created the heaviest antimatter (anti-**

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**helium). What is/are the implications of the creation of antimatter?** **[UPSC (Pre) – 2012]**

(i) It will make mineral prospecting and oil exploration easier and cheaper.

(ii) It will help probe the possibility of the existence of stars and galaxies made of antimatter.

(iii) It will help understand evolution of the universe.

Select the correct answer using the codes given below:

(a) Only (i)

(b) (ii) and (iii)

(c) Only (iii)

(d) (i), (ii) and (iii)

**Answer Key**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (d) | 2. (b) | 3. (d) | 4. (a) | 5. (b) | 6. (d) | 7. (a) | 8. (b) |

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UNIT-IX Biotechnology

# CHAPTER 22
# BASIC CONCEPTS OF BIOTECHNOLOGY

## IN THIS CHAPTER...


**22.1 Cell**

**22.2 Types of Cell**


The smallest cell is Mycoplasma Laidlawii (0–1 to 0–3 μ), also called PPLO (Pleuro – pneumonia like organisms). The PPLO are also called as **'Joker of Plant Kingdom'**. The largest cell is an egg of Ostrich (170 × 155 mm).


22.2.1 Prokaryotic Cells

22.2.2 Mesokaryotic Cells

22.2.3 Eukaryotic Cells

**22.3 Cell Structure**


The most striking difference between plant and animal cells is the absence of cell wall in the animal cells. Cell wall is dead

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and permeable.

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22.3.1 Cell Wall
22.3.2 Plasma Membrane/Cell Membrane
22.3.3 Endoplasmic Reticulum (E.R.)
22.3.4 Golgi Complex
22.3.5 Plastids
22.3.6 Mitochondria
22.3.7 Ribosomes
22.3.8 Lysosomes
22.3.9 Vacuoles
22.3.10 Nucleus, Chromosome, Nucleic Acids and Cell Division

In a DNA, the bases of two chains are linked together by $H_2$ bonds which are necessary to stabilise the double helical structure.

RNA was the first nucleic acid of this planet.

## 22.1 Cell

The study of cell is called **cytology.** The study of cell along with metabolic aspects is called **cell biology.**

**Cell Theory:** **H. J. Dutrochet** (1824) for the first time gave the idea of cell theory. The cell theory was, formulated by two German scientists **M. J. Schleiden** (Botanist) and **T. Schwann** (Zoologist) in 1839. According to cell theory, "cell is the basic structural and functional unit of an organism."

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## 22.2 Types of Cell

On the basis of presence of nucleus, cells are of the following types (Table 22.1).

### 22.2.1 Prokaryotic Cells

They lack nuclear membrane and nucleolus or a definite nucleus. The primitive nucleus is called a **nucleoid** or incipient nucleus. The cell organelles like mitochondria, chloroplasts, golgi bodies and endoplasmic reticulum are absent. Genetic material is histone-free and made up of double stranded DNA only and is also called a **genophore,** e.g., mycoplasma, bacteria and blue-green algae.

### 22.2.2 Mesokaryotic Cells

They have definite nucleus and show peculiar type of division called dinomitosis. They lack deoxyribo-nucleohiston, e.g., Dinoflagellates (a group of algae).

Table 22.1 Comparison between Prokaryotic and Eukaryotic Cells

| Prokaryotic *vs* Eukaryotic |
|---|

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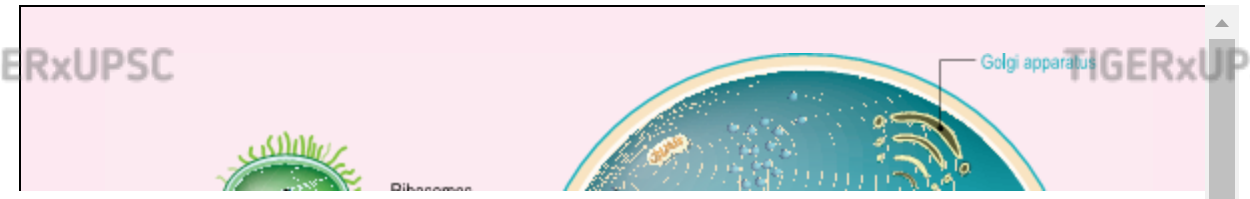


### 22.2.3 Eukaryotic Cells

They have true or definite nucleus (with nuclear membrane and nucleolus). DNA is associated with histone protein in the chromosomes. Membrane bounded cell organelles such as Mitochondria, Plastids, E.R. and Golgi bodies are present in cells of higher organisms like plants and animals.

Table 22.2 Comparison between Plant Cells and Animal Cells

| Plant Cells | Animal Cells |
|---|---|
| Outer layer cell wall and inner layer Cell membrane (Plasma Membrane). | Cell wall is absent so outer layer is cell membrane. |
| Chloroplasts are present in plant cells. | Animal cell do not possess chloroplasts. |
| Plant cells are almost rectangular in shape. | Animal cells are round in shape. |
| Plant cell possess large vacuoles but mostly one in number. | Animal cells have many small vacuoles. |
| Cilia rarely occur in plant cells. | Cilia are present in animal cells. |
| Higher plants do not possess centrioles. | Animal cells contain centrioles. |
| Plant cells have no or less number of lysosomes. | Animal cell possess many lysosomes. |

The smallest cell is Mycoplasma Laidlawii (0–1 to 0–3 μ), also called PPLO (Pleuro – pneumonia like organisms). The PPLO are also called as **'Joker of Plant Kingdom'**. The largest cell is an egg of Ostrich (170 × 155 mm).

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## 22.3 Cell Structure

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A cell consists of various components. They are briefly described below.

### 22.3.1 Cell Wall

The plant cells have a definite and rigid cell wall. The most striking difference between plant and animal cells is the absence of cell wall in the animal cells. Cell wall is dead and permeable (**Table 22.2** and **Figure 22.1**).

**Middle Lamella** is the cementing layer between two cells which consists of mainly pectic substances (Ca pectate and some Mg pectate). During maturation of fruits the pectic substances in the middle lamella become soluble due to the activity of pecteolytic enzymes; hence ripe fruits become soft.

***Functions***

(i) The cell wall represents the exoskeleton of cell and provides mechanical strength to cell.

(ii) Cell wall provides a definite shape to plant cell.

(iii) It is involved in the movement of material in and out of the cell.

### 22.3.2 Plasma Membrane/Cell Membrane

Every cell is enclosed by cell membrane. It is living and selectively permeable. In cell membrane lipids are 40%, protein 52% and carbohydrates (oligosaccharides) are in least proportion (5 to 8%).

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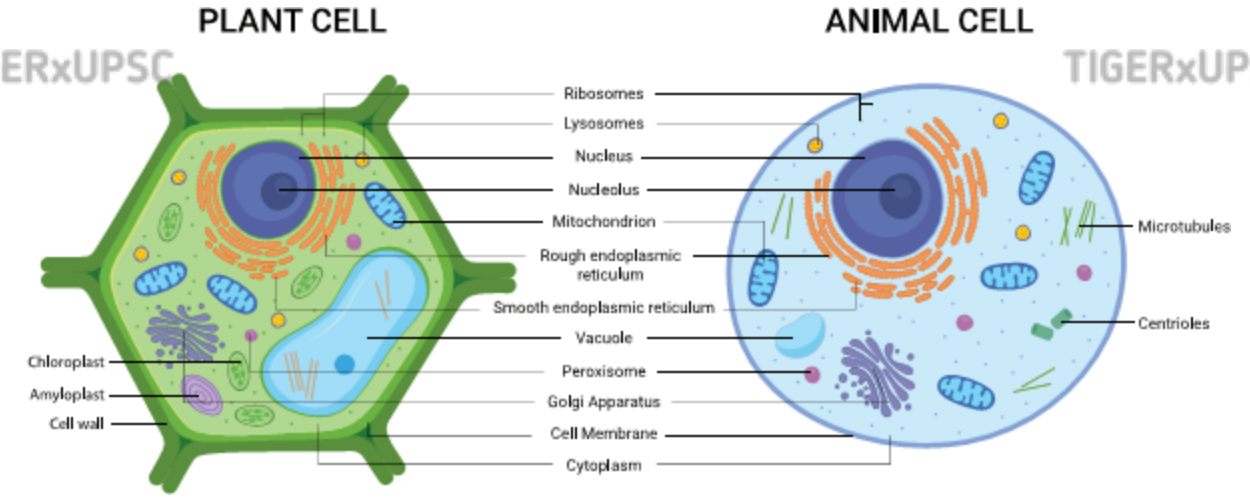


Figure 22.1 Pictorial view of Plant and Animal Cell

***Functions***

(i) **Endocytosis and Exocytosis:** Macro-molecules are ingested into the cell from outside through **endocytosis,** which includes both **phagocytosis** (when solid particles are ingested) and **pinocytosis** (when fluid droplets are ingested), and are secreted out from the cell through **exocytosis.**

(ii) **Passive Transport:** (diffusion and facilitate transport). In passive transport movement of molecules or ions is from high to low concentration in a chemical gradient or in an electrochemical gradient.

(iii) **Active Transport and ATP-Linked Systems:** The basic difference between active transport and facilitated transport is that the active transport allows transport against a concentration gradient; such that the cells can obtain considerable quantities of certain metabolites even when they are present at a very low concentration in the surrounding medium.

### 22.3.3 Endoplasmic Reticulum (E.R.)

It is network of delicate strands and vesicles in the cytoplasm. E.R. extends from nucleus throughout the cytoplasm in the form of flattened sacs or cisternae, which are surrounded by double unit membrane.

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***E.R. may be***

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(A) **Rough or Granular E.R.:** It bears ribosomes on its surface. It is mainly found in cells actively engaged in protein synthesis.

(B) **Smooth or Agranular E.R.:** It does not bear ribosomes on its surface. This type of E.R. is associated with **lipid** and **steroid** synthesis. It also helps in cell detoxification.

***Functions***

(i) It provides internal support to cytoplasm or form endoskeleton or cytoskeleton of the cell.

(ii) It helps in the intracellular transport of substances.

(iii) It takes part in the synthesis of lipo-protein and glycogen.

(iv) Rough E.R. helps in protein synthesis.

(v) Smooth E.R. helps in steroid or lipid synthesis.

## 22.3.4 Golgi Complex

Golgi apparatus is also known as dictyosome, golgiosome, idiosome or lipochondria. In plant cells the golgi complex consists of many unconnected units known as **dictyosomes.**

Chemically golgi complex is made up of protein (65–75%), phospholipids (25–30%) and some enzymes such as galactosyle transferase, thiamin pyrophosphatase, and adenosine diphosphatase.

***Functions***

(i) Synthesis of carbohydrates such as cellulose and hemicellulose.

(ii) Formation of acrosome.

(iii) Formation of lysosomes (secretion of lysosomal enzymes). That is why they are abundant in secretory cells.

(iv) Synthesis of cell plate and cell wall.

(v) It plays a major role in glycosidation of lipids and proteins (synthesis of glycolipids and glycoproteins).

## 22.3.5 Plastids

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Plastids are the characteristics of plant cells. Plastids can be divided into three groups:

(A) **Chromoplasts:** They are pigmented plastids. They mostly contain carotenoids. They may be yellow, red or orange in colour. They are mostly found in the petals, fruits and roots.

The red colour of tomato is due to carotenoid lycopene. Carrots are very rich in β carotene (derivatives of vitamin A). Chromoplasts are derived from chloroplast.

(B) **Leucoplasts:** They are colourless plastids and are mostly found in the plant parts which are not exposed to the light. On the basis of nature and function leucoplasts are of following types:

(i) **Amyloplasts:** Store starch

(ii) **Elaioplasts:** Store fats or oils

(iii) **Aleuroplasts or Proteinoplasts:** Store protein

(C) **Chloroplast:** Chloroplast is the largest cell organelle, and is commonly found in eukaryotes and absent in prokaryotes. The chloroplasts are bounded by double unit membrane. DNA which is present in chloroplasts or plastids is called **Plastidome.**

**Function:** Photosynthesis.

### 22.3.6 Mitochondria

Mitochondria are present in aerobic organisms only and are absent in bacteria and blue green algae (prokaryotic). Mitochondria are generally rod shaped. Mitochondria are surrounded by two unit membranes. **Nass and Nass** discovered the presence of DNA in mitochondria. About 60–70% enzymes have been found in mitochondria (the largest amount). Mitochondria are semi-autonomous bodies, hence are referred as 'cell within the cell' (genetically autonomous). Mitochondria are commonly called as **powerhouse of the cell,** as the energy rich compounds (ATP) are formed by them.

**Function:** Aerobic respiration.

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### 22.3.7 Ribosomes

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Ribosomes are present in all plants and animal cells. Ribosomes are the smallest cell organelles. These are ultra microscopic structures. Their presence was revealed by electron microscope. These are amembranous (without membrane) cell organelles. In eukaryotic cells they are either free in cytoplasm or attached to outer membrane of E.R. through **ribophorins** (glycoproteins) by their larger subunit 60S. Ribosomes are negatively charged particles. They are made up of ribonucleoprotein. In the cytoplasm they are present in groups called as **polysome or polyribosome**. On the basis of sedimentation value, ribosomes are of two types:

(i) **70S Ribosomes:** e.g., bacteria, blue green algae (prokaryotic cells) and cell organelles such as chloroplasts and mitochondria. They consist of smaller subunit 30S and larger subunit 50S.

(ii) **80S Ribosomes:** e.g., eukaryotes. They consist of smaller subunit 40S and larger subunit 60S.

The ribosomes contain 40–60% of rRNA (ribosomal RNA) and 30–60% proteins with Arginine and lysine.

**Functions:** They are the seats of protein synthesis (protein factories).

### 22.3.8 Lysosomes

These are single membrane bound cell organelles. They are mostly found in animal cells and rarely in plant cells. Lysosomes contain hydrolytic or digestive enzymes (about 36 enzymes), e.g., Glycosidases, Phosphatases, Lipases, Nucleases and Proteases. These enzymes are capable of lysis of cells, so lysosomes are known as **suicidal bags** or lytic bodies of the cells. The hydrolytic enzymes are active at acidic pH (pH-5).

**Tay-Sachs disease** and **Niemann-Pick disease** of human brain are caused by lysosomes. These diseases are due to the lack of certain enzymes which could digest the sphingolipids.

***Functions***

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(i) Digestion of extracellular substances (Exocytosis).

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(ii) Digestion of intracellular substances (Endocytosis).

(iii) Autophagy or cellular digestion. So they are known as cellular scavengers.

Serious genetic diseases like **Hurler's disease** (bones are deformed) is the result of accumulation of glycosaminoglycan (mucopolysaccharide) in lysosomes because the patients suffering from this disease do not have the hydrolytic enzyme α-L-iduronidase in their lysosomes.

### 22.3.9 Vacuoles

Mature plant cells have one and large-sized vacuole but animal cells have small and many in numbers.

The watery fluid enclosed in the vacuole is called cell sap. A vacuole is fluid-filled sac bounded by a single membrane called the tonoplast (unit membrane).

Cell sap contains water (mainly), mineral salts, sugars, organic acid, oxygen, carbon dioxide and pigments (anthocyanin) as well as some wastes and secondary products of metabolism.

Vacuoles are osmoregulatory in function. Most of the water in mature plant cells occurs in the vacuoles.

### 22.3.10 Nucleus, Chromosome, Nucleic Acids and Cell Division

#### 22.3.10.1 Nucleus

Nucleus is the controlling centre of the cell. **Robert Brown (1835)** recognised a spherical body, an important biological entity and termed it as **nucleus**. Nucleus consists of following parts:

(A) **Nuclear Membrane:** It is the outermost layer of nucleus.

(B) **Nucleolus:** Nucleolus mainly consists of rRNA (ribosomal RNA) and proteins.

(C) **Nucleoplasm:** It is proteinaceous substance. It contains enzymes such as DNA and RNA polymerase and ribosomes.

(D) **Chromatin Network:** The thread like chromatin fibres are embedded in nucleoplasm. During cell division, these chromatin fibres break into

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chromosomes.

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**Significance of nucleus: Hammerling** demonstrated the role of nucleus in heredity.

### 22.3.10.2 Chromosomes

The haploid set or basic set of chromosomes in diploid organisms is described as **genome.** Usually the gametes contain only one copy of each chromosome. The somatic cells of most organisms contain two sets of chromosomes (a pair of each homologous chromosome).

- Minimum number of chromosomes in plants is 2 n = 4 in *Haplopappus gracilis* (Asteraceae), Maximum number of chromosomes in plants is 2n = 1262 in *Ophioglossum* (fern).
- Minimum number of chromosomes in animals is 2n = 2 in Ascaris, Maximum number of chromosomes in animals is 2 n = 1600 in *Aulacantha*.

When chromosomes or chromatin materials are stained with **Orcein** or **Acetocarmine** at prophase two regions become differentiated.

(i) **Euchromatin:** It is light stained region. It is active part of chromatin which helps in RNA synthesis.

(ii) **Heterochromatin:** Darkly stained region as heterochromatin, it is inactive and late replicating.

#### *Structure of Chromosome*

In chromosomes, DNA and histone are bounded to form **deoxyribonucleoprotein.** The most accepted model for the arrangement of DNA and histone protein is **Nucleosome** model. The term nucleosome was given by **P. Outdet (1975)**.

**Functions:** The specified function of chromosome is to carry the genetic information from one generation to another generation.

### 22.3.10.3 Nucleic Acids

Nucleic acid (DNA) was first isolated in 1868 by a Swiss scientist **Friedrich Miescher**, from the nuclei of pus cells. He called it **nuclein.**

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Nucleic acids are made up of the following three components:

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(A) **Nitrogenous Bases:** They are of two types:

(i) **Purines:** They are dicyclic in nature, e.g. Adenine **(A)**, Guanine **(G)**

(ii) **Pyrimidines:** They are monocyclic, e.g. Thymine **(T)**, Cytosine **(C)**, Uracil **(U)**

(B) **Sugars:** They are also of two types:

(i) Deoxyribose sugar

(ii) Ribose sugar

(C) **Phosphoric Acid:** $H_3PO_4$

**Nucleosides:** Nucleoside is formed by purine or pyrimidine nitrogenous bases and sugar.

**Nucleotides:** In a nucleotide purine or pyrimidine nitrogenous base is joined by deoxyribose sugar or ribose sugar, which is further linked with phosphate group to form nucleotide.

**Polynucleotides:** Nucleic acids are **polynucleotides. A number of nucleotides join through phosphodiester bond** and form polynucleotide.

### 22.3.10.4 Structure of DNA

DNA is mainly localised in nucleus as the part of chromosomes (DNA protein complex is called chromatin).

Of the total amount of DNA in plant cell, 90% occurs in nucleus (in chromatin), 8% in mitochondria and 0.5 to 2.0% in chloroplasts, hence cytoplasmic DNA is 1–10% (**Figure 22.2**).

**Wilkins** and **Franklin** in 1952 for the first time (on the basis of X-ray diffraction studies) indicated double helical structure of DNA. In 1953, **J. Watson** and **F.C. Crick** gave the double helical model of DNA. **Watson, Crick** and **Wilkins** in 1962 got Nobel Prize for the structure of DNA.

(i) The amount of adenine is always equal to that of thymine and the amount of guanine is always equal to that of cytosine (i.e., A = T and G = C).

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(ii) The amount of purine and pyrimidines are equal (i.e., A + G = T + C).

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(iii) The ratio of A + T / G + C is not constant or may vary from one species to another. In higher plants and animals A + T and G + C ratio is 1.40:1 and in viruses, bacteria and lower plants A + T and G + C ratio is 0.6:1.

(iv) Purines need much space while pyrimidines occupy little space.

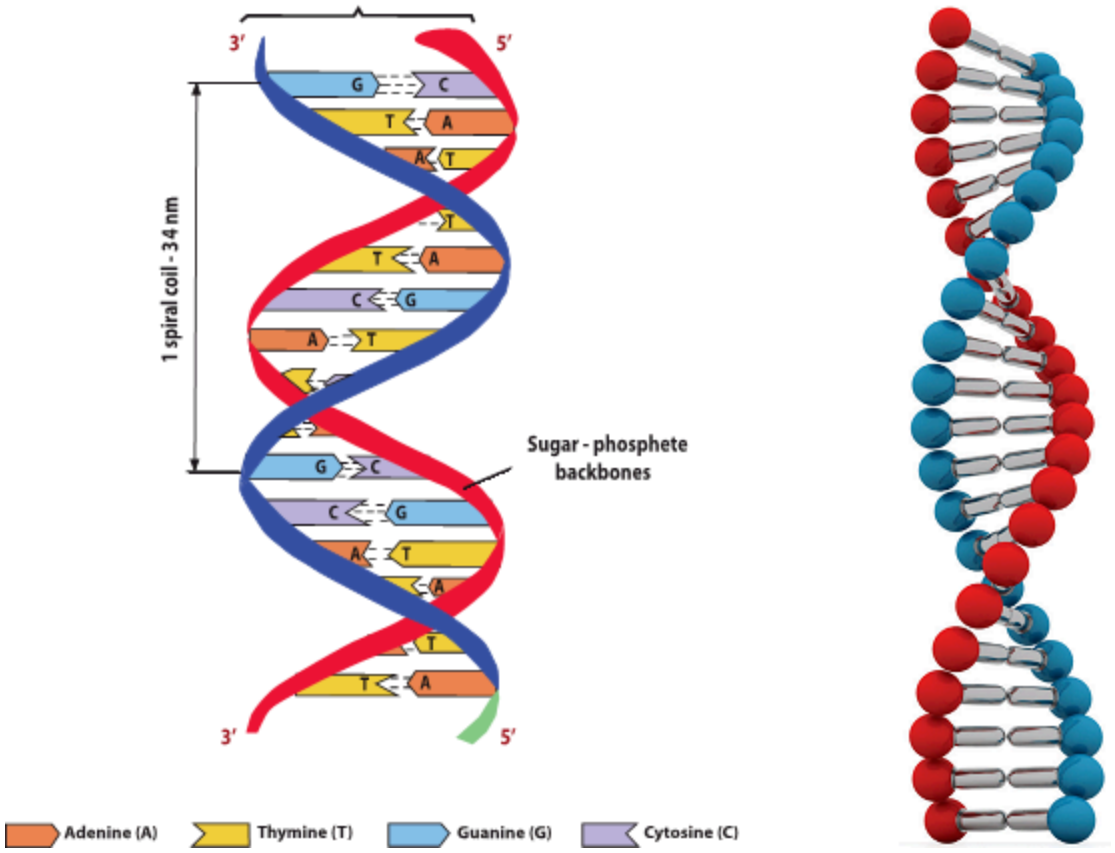


Figure 22.2 DNA Structure

The bases of two chains are linked together by $H_2$ bonds which are necessary to stabilise the double helical structure.

Adenine–thymine pair has two hydrogen bonds and guanine–cytosine pair has three hydrogen bonds. If one strand has adenine another would have thymine. So, the two strands are known as complementary strands. These complementary strands show that sugar molecule in one strand is in upward direction while the sugar molecule in another strand is in downward

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direction. Such a condition is known as antiparallel. Thus two strands of DNA are **complementary** and **antiparallel.**

DNA molecule consists of two right-handed long chains in the form of a double helix with a diameter of 20 Å.

The double helix makes a turn after every 34 Å. Generally, a turn has 10 nucleotides or 10 base pairs. So, the distance between two base pairs is 3.4 Å.

The DNA is mostly right-handed, i.e., B-DNA, and the DNA with left-handed coiling is called **Z-DNA** (Table 22.3).

Table 22.3 The Difference between B-DNA and Z-DNA

| B-DNA | Z-DNA |
|---|---|
| 1. The helical coiling is right-handed. | 1. The helical coiling is left-handed. |
| 2. The length of one turn is 34 Å which contains mostly 10 nucleotides. | 2. The length of one turn is 45 Å which contains mostly 12 nucleotides. |
| 3. Distance between two nucleotides is 3.4 Å. | 3. Distance between two nucleotides is 3.7 Å. |
| 4. Repeating unit is a mononucleotide. | 4. Sugar residues have alternating orientation, so the repeating unit is a dinucleotide. |
| 5. Diameter is 20 Å. | 5. Diameter is 18 Å. |
| 6. The angle of twist per repeating unit is at 36°. | 6. The angle of twist per repeating unit is at 60°. |

### 22.3.10.5 RNA (Ribonucleic Acid)

RNA is the polymer of ribonucleotides. These nucleotides join together to form single-stranded RNA. RNA was the first nucleic acid of this planet. There are different types of RNA:

(A) **Messenger RNA (mRNA):** The term mRNA was given by **Jacob** and **Monod** (1961). It carries genetic information from DNA to ribosomes where protein synthesis occurs under its direction. It is 5 to 10% of the total RNA.

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(B) **Transfer RNA (tRNA) or Soluble RNA (sRNA):** It is about 10–20% of the total RNA. The molecules are smallest (having 75 to 80 of ribonucleotides). It is responsible for carrying amino acid molecules to the site of protein synthesis. **Holley** described the molecular structure of yeast alanyl tRNA and shared Nobel Prize with **Nirenberg** and **Khorana** in 1968. The molecular model of this tRNA is popularly known as **clover-leaf model**. The main function of IRNA is selection of correct codon and amino acid, The 5' end of tRNA always ends in the base guanine while the 3' end always ends in the base sequence CCA. The nucleotide sequence of the rest of the molecule is variable. The amino acids are attached to CCA end or terminus.

(C) **Ribosomal RNA (rRNA):** It consists of about 80% of the total RNA. It is the most stable type of RNA. Molecular weight is 12,00,000. Ribosomes consist of 65–80% of rRNA.

***Functions***

(i) The mRNA carries information from nucleus to cytoplasm in the form of codons.

(ii) The rRNA gives site for the attachment of mRNA.

(iii) The tRNA carries amino acids up to the site of ribosomes.

Table 22.4 The Difference between DNA and RNA

| DNA | RNA |
|---|---|
| 1. Double-stranded structure. | 1. Single-stranded structure. |
| 2. The sugar is Deoxyribose type. | 2. The sugar is Ribose type. |
| 3. DNA is always genetic. | 3. RNA is mostly non-genetic. |
| 4. The pyrimidine bases are cytosine and thymine. | 4. The pyrimidine bases are cytosine and uracil. |

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**Awarded to Victor Ambros and Gary Ruvkun for Discovery of *microRNA*** and its role in **post-transcriptional gene regulation**

In **1993**, Victor Ambros and Gary Ruvkun discovered **microRNA (miRNA)** while studying mutant ***Caenorhabditis elegans*** **nematodes** with developmental defects.

- Before this, it was believed gene regulation only occurred via **transcription factors** acting directly on DNA.
- Their work showed **gene expression is also regulated after transcription (post transcription mechanism)**, through small non-coding RNA molecules—**microRNAs**.
- **MicroRNA (miRNA) is a short, single-stranded RNA molecule that regulates gene expression by binding to messenger RNA (mRNA).**
- This **redefined the understanding of gene regulation**, especially in multi-cellular organisms.

**What Is microRNA (miRNA)?**

| Feature | Description |
|---|---|
| **Type** | Small, single-stranded, non-coding RNA |
| **Function** | **Regulates gene expression post-transcription by:**<br>• Binding to mRNA<br>• Preventing translation into protein<br>• Degrading mRNA |
| **Location** | Functions in the cytoplasm |
| **Importance** | Over 1,000 miRNA genes in humans; miRNA-mediated regulation is **universal among multi-cellular organisms** |

### 22.3.10.6 Replication, Transcription, Reverse Transcription and Translation

**DNA to DNA formation is called replication or duplication**. The formation of mRNA from DNA in the presence of RNA polymerase enzyme is described as **transcription**. However, the formation of DNA upon RNA template was discovered by **Temin** and **Baltimore.** That is why this process

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is called **teminism** or **reverse transcription.** In this way all these three RNAs help in protein synthesis. The formation of protein (polypeptide chain) at the site of ribosomes is called **translation**.

### 22.3.10.7 Gene

**Johanson** (1909) coined the term 'gene'. Johanson described gene as "unit of heredity". The genes are located on chromosomes. One chromosome has one DNA. A specific DNA base pair sequence those are responsible for the formation of specific protein is called gene. All characteristics of human body are an expression of certain protein. Genes have full control over protein synthesis and are responsible for hereditary characters those are going from generation to generation.

**S. Benzer** has proposed the following different terms to explain different aspects of gene on the basis of functional activities:

(A) **Cistron:** It is the functional unit of DNA chain. It is also called as structural gene.

(B) **Recon:** It is the unit of recombination.

(C) **Muton:** It is the unit of mutation.

The gene is the part of DNA. The artificial synthesis of gene was done by **H.G. Khorana.** This gene was of yeast alanyl tRNA and made up of 77 nucleotide pairs.

**Jumping or Motile or Floating Genes:** These genes were discovered by **Barbara McClintock** in corn. For this brilliant work **McClintock** was awarded Nobel Prize in 1983. The term **transposon** was used for "jumping genes" by **Hedges** and **Jacob**. The jumping genes change their position on the same chromosome and also move from one chromosome to the other.

**Pleiotropic gene**: Single gene which controls the expression of more than one character is called **pleiotropic gene.**

### 22.3.10.8 Genetic Code

The term genetic code was given by **George Gamow.**

The sequence of nitrogen bases on the mRNA which stores information for linking the amino acids in a definite sequence during synthesis of

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proteins is called genetic code.

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There are 64 triplets (4 × 4 × 4 = 64) of nitrogen bases (codons) and 20 amino acids for protein synthesis (**Figure 22.3**).

The first clue to codon assignments came when **M. W. Nirenberg** and **J. H. Mathai** synthesised mRNA using only uracil, so that all along the length of mRNA, there was no other base and the only possible triplet was UUU. When such a poly-U (RNA) was used in cell free synthesis of polypeptide, only polyphenyl-alanine was synthesised, meaning that the only amino acid coded was phenylalanine. So, it was concluded that triplet UUU coded for the amino acid **phenylalanine.** This news that the genetic code had been cracked appeared in newspapers in 1961. Subsequently, poly AAA gave **polylysine** and poly CCC to **polyproline.** Thus, the following three codons could be assigned, using homopolymers of RNA UUU-phenylalanine, AAA-lysine, CCC-proline.

After getting success with homopolymers, **Nirenberg** and co-workers used co-polymers (RNA synthesised by two or more bases) and worked out about 45 codes using AA-tRNA binding technique.

**H.G. Khorana**, using synthetic DNA prepared polynucleotides with known repeating sequences, e.g., CU CU CU CU CU CU CU = CUC / UCU / CUC / UCU (co-polymers of repetitive sequences). It produced only two amino acids Leucine/Serine in alternating sequence.

It helped in deciphering the code.

### *Features of Genetic Code/Codon*

1. The code is **triplet** (a group of three bases).
2. **Starting codon (Initiation Codon)** is AUG (for methionine amino acid) or rarely GUG: Protein synthesis starts from this codon on RNA during translation.
3. **Termination or non-sense codons** are UAA (Ochre), UAG (Amber), UGA (Opal): Translation process stops with this codon.

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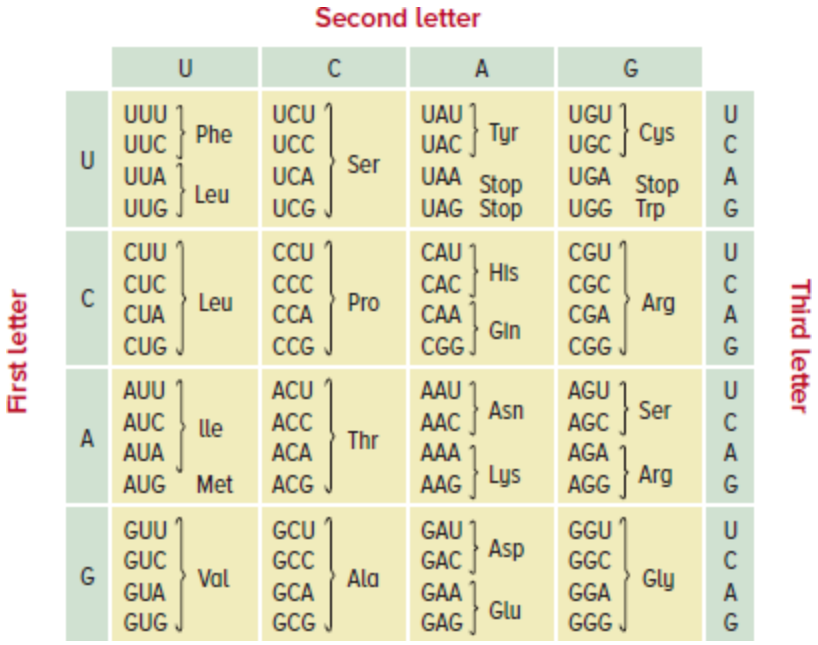


Figure 22.3 Genetic Dictionary

4. The **Degeneracy of Codons:** An amino acid can be coded by more than one codon.
5. **Universality of Codons:** Genetic codon is universal meaning it is same in every organism except in some cases.
6. **Comma Less Codons**: No comma used between two codons.
7. **Non-Overlapping Codons**: One base cannot participate in the formation of more than one codon.
8. The code is **non-ambiguous**: Thus one codon cannot specify more than one amino acid.

### 22.3.10.9 Cell Cycle

It consists of interphase and M-Phase (cell division).

(A) **Interphase:** It is resting phase but nucleus is metabolically most active. It is the **regulatory phase** and **control phase** of cell cycle.

(B) **M-Phase:** It is a **cell division phase** or **response phase** in cell cycle. It is of two types:

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(1) **Mitotic Division or Mitosis:** The term mitosis was given by **W. Flemming.** Mitotic division occurs in somatic or vegetative cells. The two daughter cells, which are formed are identical to their parents. The chromosome number remains same in the daughter cells.

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***Significance of Mitotic Division***

(1) It helps in growth and development.

(2) **Meiotic Division or Meiosis:** The term meiosis was used by **Farmer and Moore**. It occurs in reproductive or germ cells. Four haploid cells are produced after this division which are not identical to their parent cell. The chromosome number is reduced to half. Meiosis includes two nuclear and one chromosome division.

***Significance of Meiotic Division***

(1) It is responsible for recombination of characters in the progeny due to crossing over.

(2) It maintains the original number of chromosomes or constant number of chromosomes.

Colchicine, an alkaloid isolated from Colchicum autumnale (liliaceae) is used in plant breeding for inducing polyploidy. It inhibits or interferes in the formation of spindle fibres and cell plate formation.

Sometimes there is replication of DNA as well as the chromosome but no division of nucleus, so the somatic number of chromosomes in nucleus is increased. This process is called as **endopolyploidy** or **endomitosis** or **c-mitosis**. Endopolyploidy can be induced by colchicine chemical.

X-rays and UV rays inhibit or retard cell division. Antibiotics, e.g., chloromycetin, tetracyclin, puromycin, block cell division and interfere with protein synthesis.

## QUESTIONS AND ANSWERS

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### Objective Questions

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**1. Which of the following statements are correct regarding the general difference between plant and animal cells?**

**[UPSC (Pre) – 2020]**

1. Plant cells have cellulose cell walls whilst animal cells do not.
2. Plant cells do not have plasma membrane unlike animal cells which do.
3. Mature plant cell has one large vacuole whilst animal cell has many small vacuoles.

Select the correct answer using the code given below:

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

**Answer Key**

1. (c)

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# CHAPTER 23
# ADVANCEMENTS IN BIOTECHNOLOGY

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**23.2 Biotechnology in India**

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- 23.2.2 Advisory Committees

**23.3 Application of Biotechnology**

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**Part (B): Gene, Genome and Biotechnological Tools**

**23.7 Gene**

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**23.8 DNA Structure and its Composition**

- 23.8.1 DNA Fingerprinting or Genomic Fingerprinting
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23.8.4 Environmental DNA (e-DNA)
23.8.5 Genetic Marker
23.8.6 Jumping Genes/Transposons

**23.9 Genome**

23.9.1 Genome or DNA or Nucleotide Sequencing
23.9.2 Specific Applications of Genome Research

The main goal of Human Genome Sequencing was to determine the sequence of the genes present in human DNA.

23.9.3 Human Genome Project (HGP)/Human Genome Sequencing
23.9.4 DNA Technology Bill
23.9.5 Genome India Initiative
23.9.6 National Genomic Grid (NGG)

**23.10 Recombinant DNA Technology (r-DNA Technology)**

r-DNA Technology is the definite protocol or method deployed to introduce foreign DNA segment (DNA of other species) into host species.

23.10.1 Applications of Recombinant DNA Technology (r-DNA Technology)

**23.11 Transgenesis**

23.11.1 Transgenic Animals
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**23.12 Genetically Modified Organism (GMO)**

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**23.17 IVF (In-vitro Fertilisation) and Test Tube Baby**
**23.18 Surrogate Mother**

The Act came into force on 25 January 2022. Surrogacy is permissible only for **altruistic purposes** or for couples with proven infertility or disease but **Commercial surrogacy** is not allowed.

23.18.1 Surrogacy Regulation Act, 2021
**23.19 Pronuclear Transfer/Three Parents Baby/Mitochondrial Gene Therapy**
23.19.1 Procedure of Mitochondrial Gene Therapy
23.19.2 Legalisation
23.19.3 Many Pros and Cons in Mitochondrial Gene Therapy
**23.20 Tissue Culture**
**23.21 Gene Therapy**
**23.22 Gene Editing and CRISPR Technology**
**23.23 Gene Doping**
**23.24 Cloning**
23.24.1 Types of Cloning
**23.25 Stem Cells**
23.25.1 Stem Cell Treatment

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23.25.2 Stem Cell Controversy
23.25.3 India's Top Centres for Stem Cell Research
**23.26 Epigenetics**
**23.27 Synthetic Biology and Synthetic Life**
**23.28 Hayflick Limit/Telomeres: The Biological Clock**
**23.29 BioE3 Policy (Biotechnology for Economy, Environment and Employment)**
**Part (C): Biological Terminology**
**23.30 Biological Terminology**
23.30.1 Biosensor
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23.30.3 Bio Signature
23.30.4 Bio Generics
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23.30.8 Bioleaching
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23.30.10 Bioinformatics
23.30.11 Biosparging

# Part (A): Concepts of Biotechnology

## 23.1 Origin and Development

Biotechnology is the use of biological system (microbes, plants and animals) and the technique to produce substances, which are beneficial to people. It includes the manufacturing of antibiotics, vitamins, vaccines, plastics, etc., disposal of toxic waste using bacteria, pollution control and production of new fuels. Various techniques deployed in biotechnology are

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genetic engineering or recombinant DNA technology, hybridism technology, cell and tissue culture and protoplasm fusion. Others can be germplasm development, embryo transfer technology, enzyme and protein engineering, fermentation, bioconversion and immobilisation of cells and cellular products. On the basis of level at which the transfer is made, biotechnology can be classified into two types: gene biotechnology involving gene transfer and non-gene biotechnology involving cells, tissue or the whole organism.

From 5000 BC, hybridisation, has been the best process in biotechnology experiment to improve varieties of plants and animals. During 1970s, Norman Borlaug (The Nobel Laureate 1970), an American agriculturist, significantly contributed towards increasing supply of food through various techniques such as hybridisation, selective breeding and gene transfer. The term 'hybridisation' refers to the technique or method involving crossing two or more individuals having opposite sex of various species with desirable characters. This is undertaken for producing off-springs of mixed character (improved variety). Indian Scientist, Dr. M.S. Swami Nathan (1967) started and developed technologies for raising the country's food production. Hence, he earned the name 'Father of Economic Ecology', through his valuable efforts in this field.

In 1960, another important discovery was made by Werner Arber in 1960 in the form of enzymes known as Restriction Enzyme in bacteria. These enzymes were advantageous as they could slice the organism's DNA strand at a particular point. Moreover, in the year 1973, Stanley Cohen and Herbert Boyer successfully removed a particular gene from a bacterium and placed it into another bacterium using this enzyme. This resulted in formation of Recombinant DNA Technology or generally termed as 'Genetic Engineering'. By using genetically engineered bacteria, later, the scientists successfully produced Human Insulin in 1978, being the market's first biopharmaceutical product for controlling *Diabetes Mellitus*.

## 23.2 Biotechnology in India

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From ancient times, people used microorganisms in various applications and uses such as improving a variety of plants, fermentation products, curd formation, etc. With time, gradually, the scope of biotechnology expanded. It also enabled to develop crops that can withstand the harsh weather conditions. This, in turn, greatly helped the poor farmers to retain their yield and the government to retain the revenue from it, simultaneously ensuring food security. India, with a large population, offers a huge market for the products and services, taking advantage of these opportunities over many other countries of the world. India's remarkable demographic nature has formed roughly an ideal environment for making the biotech companies to shift their bases to the country.

### 23.2.1 Department of Biotechnology (DBT)

In 1986, a separate Department of Biotechnology (DBT) was established under the Ministry of Science and Technology to promote the development in the field of modern biology and biotechnology in the country. Several research and development projects, demonstrations and creation of infrastructural facilities are directly or indirectly based on this field. The department has contributed significantly to the growth and application of biotechnology in the areas of agriculture, health care, animal sciences, environment and industry.

### 23.2.2 Advisory Committees

The two apex level committees functioning with the department are advisory in nature. They are the Scientific Advisory Committee – DBT (SAC-DBT) and the Standing Advisory Committee Overseas (SACO). The committees suggest new areas, assist in priority setting and review monitoring of large inter-institutional, inter-disciplinary projects.

## 23.3 Application of Biotechnology

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Biotechnology has vast application but the major areas are medical, agricultural, industrial and environmental. The specific applications including general applications of biotechnology can be categorised as follows:

(a) **Genetic Engineering:** Stanley Cohen and Herbert Boyer (1973) conducted an experiment, where they removed a particular gene from a bacterium and placed the same gene into another bacterium. This led to the evolution of Recombinant DNA technology, which is also called as Genetic Engineering. The organisms so formed with altered genetic makeup, with the help of DNA recombinant technology or genetic engineering, are known as Genetically Modified Organisms or GMOs. Transgenic Organism is an organism in which a foreign gene has been inserted. Such an organism could be a plant, an animal, bacteria or fungi. Foreign genes have also been injected into the embryo or the fertilised egg of an organism.

***Importance of Genetic Engineering***

- ❑ Successful alterations in the genetic makeup of an organism
- ❑ Valuable tool for increasing food production from plants and animals
- ❑ Diagnoses of diseases and improvements in medical treatment

(b) **Plant Tissue Culture:** It represents the method of growing tissues of plants in sterile conditions, in known artificial nutrient medium in laboratory conditions, for producing new plants.

(c) **Monoclonal Antibody (MAb):** The monoclonal antibodies are particular protein-type molecules formed in the laboratory. These are a kind of defence workers, which protect us from different types of diseases. Our body produces them naturally when invaded by any bacteria or virus. Many diagnostic tests for bacteria and viruses also use them, besides their application for fighting against cancer. Interferon is a specific protein synthesised from genetically engineered bacterial species *E. coli.* The human body produces them

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naturally to fight proteins for opposing viral infections. The biologically created interferon has been tested and found successful.

(d) **Synthesis of Enzymes:** The various enzymes synthesised *via* biotechnology are used in various processes such as stain removing, fabric softening, digestible food preparation, meat processing and cancer treatment.

(e) **Synthesis of Biodegradable Plastics/Bioplastics:** Biodegradable plastic, unlike other plastics, can be broken down into simpler substances by microorganisms.

(f) **Blood Substitute:** Biotechnologists are trying to synthesise artificial blood. This can bridge the gap between the need of blood for transfusion and available blood during accidents and disease outbreak. Biotechnologists like Mary L. Nucci and Abraham Abuchowski (1998) are expected to be successful in this regard in their experiments.

(g) **Biotechnology and Mining:** Specific bacteria work on the particular metallic compounds found inside the earth. Hence, some bacteria are being currently employed in the extraction of different metals such as copper, zinc, lead and other metals. They also serve during the isolation of respective metals.

(h) **Cloning:** Cloning is application of biotechnology to produce cells or organisms being originally derived from a single organism or cell through asexual method in laboratory conditions. The organisms formed at the time of cloning have similar genes and are termed as clones. A clone is an individual cell or organism developed by a single cell of body and is genetically similar to its parental cell. In 1950s, scientists initiated the cloning of frogs and until the 1980s, mice were cloned. In the year 1996, Ian Wilmot was successful for the first time in the cloning of an adult sheep. He succeeded with his team of researchers and the clone was named as Dolly.

(i) **Food and Beverage:** A huge number of food products are manufactured industrially through application of fermentation

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technology. Some popular examples of those products include wine, yoghurt, cheese, bread, probiotic foods etc. Not only this, biotechnology also enabled production of different vitamins, amino acids and vinegar.

**Probiotics**

Probiotics are living microorganisms that are intended to have health benefits when consumed by the human body. They can be found in yogurt and other fermented foods and dietary supplements. Yogurt is a well-known food source of probiotics. Certain strains of bacteria in yogurt have ß-D-galactosidase, which is an enzyme that helps **break down lactose in dairy products into sugars (glucose and galactose)**. Thus probiotics help in the digestion of milk sugars. The most common place linked to beneficial microbes is your gut (caecum).

Good bacteria keep humans healthy by supporting their immune function and controlling inflammation. There are various uses of probiotics:

- ❏ Helps body to digest food.
- ❏ Keeps bad bacteria from getting out of control and making a person sick.
- ❏ Synthesises the vitamins.
- ❏ Helps support the cells present in the line of your gut to prevent bad bacteria that you may have consumed (through food or drinks) from entering your blood.
- ❏ Breakdown and absorb medications.

There are two specific types of bacteria that are common probiotics found in stores. Probiotics are also made up of good yeast. These include:

- ❏ *Lactobacillus* (Bacteria)
- ❏ *Bifidobacterium* (Bacteria)
- ❏ *Saccharomyces boulardii* (Yeast)

(j) **Pharmaceuticals:** Important examples are as follows:

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- ❑ Human insulin produced from non-human sources, production of interferon, cytokinin, steroids and human growth hormones.
- ❑ Gene-therapy to prevent and control diseases
- ❑ Vaccines and antibodies development.

(k) **Pollution Control:** The aerobic and anaerobic bacteria are being used for making manure from crop residues and animal wastes, in order to combat the problem of organic solid waste. The species of bacteria such as *Pseudomonas aeruginosa* is capable of eating away the oil spilled on the sea surface. Solid wastes such as crop residues can form biogas, being the most appropriate rural energy source. In environment, the atmospheric carbon dioxide is controlled by green plants during photosynthesis. Therefore, fast-growing species of trees developed through biotechnology will be an important tool to control the rising $CO_2$ level.

(l) **Waste Water Treatment:** The complex organic and nitrogenous waste can be degraded through biotechnology by employing bacteria, fungi, algae, etc. The process involves passing of the effluent through a microbial slime layer. The layer contains microbes, which help in the degradation of the organic and nitrogenous waste, releasing carbon dioxide and nitrogen dioxide.

(m) **Degradation of Pesticides:** Bacteria species such as Pseudomonas sp., Flavobacterium, Azotobacter, *E. coli,* etc., can degrade different pesticides. These microorganisms are genetically created for degrading a large number of pesticides.

(n) **Bioremediation:** Bioremediation is defined as the reclamation of environment through application of biotic agents like microorganism.

(o) **Industrial Applications:** Industrial processes like production of stain remover, detergents, bread, biotech-polyester, vitamins, stone-washed denims, bleached paper, etc., all deploy biotechnology. The enzyme cellulose is used for fading denims.

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## 23.4 Achievements of Biotechnology in Different Fields

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### 1. Biomedical

Diagnosis and therapy account for 68% of the biotechnology industry in the USA, 43.7% in Canada and about 43% in Europe. For example:

- Recombinant drugs, like human insulin, growth hormone, interferon, erythropoietin.
- Recombinant diagnostic kits (e.g., testing HIV, or pregnancy)
- Recombinant vaccines (e.g., against Hepatitis B and AIDS)
- Production of medicines by transgenic plants or animals (in milk or urine)

### 2. Agriculture

- Genetically modified crops that are resistant to pests, viruses, drought, etc.
- Genetically modified crops that ripen slowly during transport
- Use of organic material to produce biodegradable plastics, fuel, fertiliser
- In vitro fertilisation of farm animals using selected sperm and eggs
- Use of recombinant growth hormone to increase milk and meat production
- Genetically modified crops with better nutritional qualities

### 3. Marine Biotechnology

- Fish farming (aquaculture)
- Seaweed farming to produce fatty acids, etc.
- Production of adhesives from mussels and barnacles
- Use of enzymes from thermophile bacteria (usually living in dark, sulphurous, hot environments in the deep sea) for waste removal, or in sequencing DNA.

### 4. Environmental Biotechnology

- Fighting pollution using enzymes or microbes (e.g., oil spills)
- Use of plants to remove contamination by heavy metals

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- Water treatment
- Treatment of air pollution

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### *5. Forestry*

- Production of tree clones from tissue culture
- Production of wood pulp for the paper industry

### *6. Space Biotechnology*

**ISRO and DBT collaborate to advance Space Biotechnology in recent past. Space missions face critical biological and environmental challenges**:

- Nutrient scarcity and food preservation
- Waste management inefficiencies
- Exposure to microgravity and radiation
- Health issues: muscle atrophy, bone loss, cataracts and cancer risks

To tackle these, **Space Biotechnology** explores how living organisms adapt, survive and evolve in extreme space conditions.

## Key Applications of Space Biotechnology

| Application Area | Description | Significance |
|---|---|---|
| **Microgravity Research** | Utilizes microgravity to grow purer protein crystals | Aids in **precision drug development** |
| **Radiation Biology** | Studies cellular response to ionizing radiation | E.g., **BioSentinel** (NASA, Artemis I, 2022): Understand DNA repair mechanisms |
| **Environmental Monitoring** | Uses microbes for **bioremediation** of planetary regolith | Helps convert Martian/Moon soil for plant growth |
| **Disease Modelling** | Simulates disease in space conditions | Improves astronaut health and models rare disease behaviour |
| **Bioregenerative Life Support Systems** | Closed-loop systems for waste recycling and food | Supports long-duration missions and **Earth** |

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### Notable Initiatives

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**India**

- ***Gaganyaan Mission:*** India's first human spaceflight programme, incorporating space life science experiments.
- ***Axiom-4 Mission:*** Collaborative commercial mission expected to include Indian experiments in low-earth orbit.

**Global**

- ***NASA's Space Biology Program:*** Pioneering experiments on ISS on plant growth, cell biology and immune response.
- ***ESA's Biolab:*** Dedicated lab on ISS for studying effects of space on microorganisms, tissues and cells.
- ***China's Space Breeding Programs:*** Use space environment to induce beneficial genetic mutations in crops.

#### *7. Other Applications*

- Production of energy from biomass
- Biological sensors and switches for electronic processes
- Conservation of endangered species using cloning

## 23.5 Terminology Related to Modern Biotechnology

- Blue biotechnology describes the applications of biotechnology in the field of marine and aquatic fields.
- Green biotechnology represents application of biotechnology in agricultural processes like selection and domestication of plants *via* micro propagation, designing and then growing transgenic plants under specific environments (cold, hot, dry, humid), in the presence (or absence) of chemicals. It might produce more eco-friendly solutions than the conventional industrial agriculture.
- Red biotechnology applies to medical processes. For example, the organisms designed for producing antibiotics and genetic cures with the help of genetic manipulation.

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- White biotechnology or industrial biotechnology is used in industrial processes, which includes designing of an organism for creating a useful chemical. The enzymes are used an industrial catalysts for manufacturing either valuable chemicals or destroying hazardous/ polluting chemicals. Bioeconomy represents economical output and venturing all the applied biotechnologies types.

Biotechnology is defined in different colours on the basis of various applications:

| Colour Type | Area of Biotech Activities |
|---|---|
| Red | Health, Medical, Diagnostics |
| Yellow | Food Biotechnology, Nutrition Science |
| Blue | Aquaculture, Coastal and Marine Biotech |
| Green | Agricultural, Environmental Biotechnology – Biofuels, Biofertilisers, Bioremediation, Geomicrobiology |
| Brown | Arid Zone and Desert Biotechnology |
| Dark | Bioterrorism, Biowarfare, Biocrimes, Anticrop warfare |
| Purple | Patents, Publications, Inventions, IPRs |
| White | Gene-based Bioindustries |
| Gold | Bioinformatics, Nanobiotechnology |

## 23.6 Genetic Engineering

Genetic engineering is a technique of manipulating the organism's genome with the help of biotechnology to add one or more traits that are not found in the organism naturally. The technique is also called gene manipulation/genetic modification. This process takes a number of steps: the first of which is the constitution of DNA segment from nucleic acid

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molecules in a test tube. The DNA so made is inserted into a vector (vector could be virus or some other suitable organism) that can serve as a molecular ferry to carry it into the host organism. The inserted DNA molecule expresses itself and also multiplies. Genetic engineering enhances the plant and animal food production, diagnosis of disease, improvement of medical treatment, and produces vaccines and other useful drugs. Additionally, it helps in the disposal of industrial wastes and the plants and animals' selective breeding, hybridisation and recombinant DNA. Various techniques are involved in genetic engineering, like isolation of genes, synthesis of genes, recombinant DNA and gene cloning, etc.

***Hazards of Genetic Engineering***

1. If a wrong DNA segment is inserted and if it gets expressed, it can cause new diseases in human beings.
2. It can be used in biological warfare.
3. Genetic modification of existing species/recreation of extinct species can lead to disasters.
4. Scientists are prone to making mistakes – new strains of bacteria, fauna, etc., can come out of labs, which could be hostile to human beings.
5. Even in a single species, genetic engineering could lead to elimination of varieties; if some new disease comes up, the entire species may be wiped out.

# Part (B): Gene, Genome and Biotechnological Tools

## 23.7 Gene

Gene is hereditary unit of living organism. It is a small segment of DNA (deoxyribonucleic acid) or RNA (ribonucleic acid) located on chromosome,

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which codes for a protein with specific function in the organism. This information passes through genetic traits to the off-spring. An allele is an alternate form of gene, present on homologous chromosome. The gene (specific base pair sequence of DNA) undergoes transcription, a process to produce messenger RNA. This mRNA, through translation process, synthesises proteins. Transcription and translation together constitute central dogma. So the gene is processed ultimately to produce protein.

### 23.7.1 National Gene Bank (NGB)

It was established in 1996 to preserve the seeds of Plant Genetic Resources (PGR) for future generations. It has the capacity to preserve about one million germplasm in the form of seeds. It stores different crop groups such as cereals, millets, medicinal and aromatic plants, and narcotics, etc. In 2021, the National Gene Bank (NGB) was protecting 4.52 lakh accessions, of which 2.7 lakh were Indian germplasm while the rest were imported from other countries.

NGB has four kinds of facilities to cater to long-term as well as medium-term conservation namely:

1. Seed Gene bank (-18°C).
2. Cryo gene bank (-170°C to -196°C).
3. In-vitro Gene bank (25°C).
4. Field Gene bank.

#### *Need for Gene Bank*

It will make the farmers of the country self-reliant and the government has been making every effort in this direction.

Note: The world's second-largest refurbished state-of-the-art National Gene Bank was inaugurated in 2021 at the National Bureau of Plant Genetic Resources (NBPGR), Pusa, New Delhi.

#### *Significance*

1. Researchers or farmers can withdraw samples from these "gene" banks to help rebuild populations of rare plant varieties and animal breeds or

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to help increase genetic diversity within species.

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2. Gene banks also preserve cells or organisms that host unusual gene variants – genes with special traits. Those genes might later prove useful if some epidemic strikes, in case of climate changes or when other factors threaten the survival of plants or animals.
3. Farmers could use the banked deposits, stored cells or tissues to restore genetic diversity or to introduce traits from other breeds or varieties.

## 23.8 DNA Structure and its Composition

The genes are encoded in long strands of DNA for majority of living beings. DNA consists of a double helical chain made of four types of nucleotide subunits [adenine (A), guanine (G), cytosine (C), and thymine (T)], sugar (deoxyribose), and a phosphate group.

Nucleotide = Nucleoside + Phosphoric acid

Nucleoside = Nitrogenous base + Sugar

Nitrogenous Base: Purine (Adenine, Guanine) and Pyrimidine (Cytosine, Thymine and Uracil)

### 23.8.1 DNA Fingerprinting or Genomic Fingerprinting

DNA fingerprint is a characteristic feature of every individual. It is a technology which is adopted by the forensic scientists to identify the individuals by their respective DNA profiles. It is also called as DNA profiling, DNA testing, DNA typing, or genetic fingerprinting or gene mapping or genography. It serves an important role as legal evidence in court to prosecute or defend alleged criminals. The chemical structure of the DNA of every individual is same but the difference is in the base pairs' ordering. The information enclosed in DNA is calculated mainly by the letters sequence along the zipper.

The encrypted sets of numbers showing a person's DNA makeup are called DNA profiles. It can also be used in the identification of individuals and in parental testing and investigation of criminals. In paternity cases,

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DNA fingerprinting technology can be used to determine persons as biological parents of a child. Although 99.9 per cent of human DNA sequences are similar in every person, only 0.1 per cent of human DNA is unique. Sir Alec Jeffreys at the University of Leicester in England first reported DNA profiling technique in 1984. The DNA fingerprinting advancement is undertaken for quicker and precise identification of criminal suspects during investigations. The DNA fingerprints of all tried and convicted criminals are stored in a database for identification.

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#### 23.8.1.1 DNA Fingerprinting is Performed in the following Research Institutes in India

1. Centre for Cellular and Molecular Biology (CCMB), Hyderabad
2. Centre for DNA Fingerprinting (CDFD), Hyderabad
3. National Bureau of Plant Genetic Resource (NBPGR), New Delhi
4. National Institute of Plant and Genetic Research (NIPGR), New Delhi

#### 23.8.1.2 Process of DNA Fingerprinting

It includes the following steps:

1. DNA isolation.
2. **Cutting, Sizing and Sorting:** Specific enzymes known as restriction enzymes are used for cutting of the DNA at specific places.
3. **Transfer of DNA to Nylon:** The DNA pieces distribution is moved to a nylon sheet by keeping the sheet on the gel and soaking them overnight.
4. **Probing:** When radioactive or coloured probes are attached to the nylon sheet, a pattern is produced termed as the DNA fingerprint.
5. **DNA Fingerprint:** The final DNA fingerprint is produced by the usage of several probes (510 or more) at the same time.
6. Blotting techniques used in different segment measurements include:
   - ❑ Southern Blotting – DNA
   - ❑ Northern Blotting – RNA
   - ❑ Western Blotting – Protein

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#### 23.8.1.3 Applications of DNA Fingerprinting

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1. **Paternity and Maternity:** Analysis of VNTR (Virtual Number Tandem Repeat) pattern of parent–child has been used to solve the father-identification cases.
2. **In Forensic Sciences:** DNA isolated from blood, hair, skin cells or other genetic evidence left at the scene of a crime can be used to identify criminals. Police from all over the world are using this technique to link the suspects with their biological evidence.
3. **Personal Identification:** DNA fingerprints can be used as a genetic bar code to identify individuals.
4. **Diagnosis of Inherited Disorders:** It can help to diagnose inherited disorders in both prenatal and newborn babies. Disorders included under these are cystic fibrosis, haemophilia, Huntington's disease, familial Alzheimer's, sickle cell anaemia, thalassaemia and many others.

### 23.8.2 Junk DNA

Susumu Ohno introduced the term 'Junk DNA' in 1972 as a provisional label for the portions of a genome sequence, which have no discernible function.

There are several evidences indicating that 'junk DNA' sequences are likely to have unidentified functional activity. Though current researches indicates that this DNA is not 'junk' at all, in fact it has been formulated that it has various regulatory roles. It implies that this so-called 'non-coding DNA' influences the behaviour of the genes of the 'coding DNA', in important ways. However, we know very little about this DNA and especially regarding non-coding DNA and the DNA of genes.

### 23.8.3 DNA Barcoding

A technique for specifying the organisms' species with the usage of a short sequence of DNA situated in the genome is termed as DNA barcoding. Its primary objective is to identify an unknown sample in terms of a known

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classification rather than determine classification. And hence, it differs significantly from molecular phylogeny. The barcode DNA sequences are too short in respect of the complete genome and hence reasonably available and are cheaper too. The cytochrome c oxidase subunit 1 mitochondrial region (COI) is emerging as the standard barcode region for higher animals. It is 648-nucleotide base-pairs long in most groups, a very short sequence as compared to 3 billion base-pairs in the human genome. Applications of DNA barcoding are given below:

1. Identification of plant leaves even in absence of flowers or fruit,
2. Identification of insect larvae (with typically fewer diagnostic characters than adults),
3. Identification of the diet of an animal based on stomach contents or faeces, and
4. Identification of products in commerce (e.g., herbal supplements or wood).

DNA barcodes allow non-experts to objectively identify species even from small, damaged, or industrially processed material. Just as the unique pattern of bars in a universal product code (UPC) identifies each consumer product, a “DNA barcode” is a unique pattern of DNA sequence that can potentially identify each living thing.

Chloroplast DNA and Mitochondrial DNA barcoding protocols are being used more and more in the food industry and food supply chains for food labelling, not only to support food safety but also to uncover food piracy in freshly commercialised and technologically processed products. DNA barcoding helps to identify undesirable animal or plant materials in processed foods.

Many scientists criticise DNA barcoding due to various reasons:

1. Lack of reliable information above the species level from DNA barcoding.
2. Some resent what they see as a gross oversimplification of the science of taxonomy.
3. As per others, recently diverged species might not be distinguishable based on their COI sequences.

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### 23.8.4 Environmental DNA (e-DNA)

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Aerial Metagenomics is the part of environmental DNA (e-DNA). Recent studies found that DNA floating in the air can boost biodiversity conservation efforts across the world. It is also shown that e-DNA can potentially identify and monitor terrestrial animals.

Animals shed DNA through their breath, saliva, fur or faeces into the environment. These samples are called e-DNA.

***Significance***

- e-DNA could help understand the composition of animal communities and detect the spread of non-native species.
- In its current form, this method cannot provide accurate data for population census but will soon overcome this limitation in the future.
- The results of study indicate a big potential for airborne e-DNA to be used for conservation of terrestrial vertebrates.
- This method will work with the current techniques to monitor endangered species after some fine-tuning.
- This method allows for biomonitoring without requiring the collection of the living organism.
- It allows creating the ability to study organisms that are invasive, elusive or endangered without introducing anthropogenic stress on the organism.
- Access to this genetic information makes a critical contribution to the understanding of population size, species distribution and population dynamics for species not well documented.
- Importantly, e-DNA is often more cost-effective compared to traditional sampling methods.

### 23.8.5 Genetic Marker

A short sequence of DNA or microsatellites with a recognised position on a chromosome, useful in identifying cells, individuals or species is known as a genetic marker. A mutation or alteration in the genomic loci may cause this variation. It may be a **small DNA sequence**, such as a sequence around a single base-pair change termed as single nucleotide polymorphism, or a

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long sequence such as minisatellites. From many years, in most organisms, the gene mapping was restricted. The usage of genetic makers can be undertaken for studying the connection between an inherited disease and its genetic cause.

Microsatellites are one of the best suitable genetic markers for analysing pedigree, population structure, genome variation, evolutionary process or demographic history, and fingerprinting purposes. Genetic markers are basically of two types – Protein and DNA (molecular).

**GREEN FLUORESCENT PROTEIN**

A gene from a jellyfish (*Aequorea victoria*), which encodes a fluorescent protein called Green fluorescent protein (GFP), may be linked physically and thus together expressed with the genes of mammals. This enables the identification of the location of the protein encoded by the GFP-tagged gene in the mammalian cell. It is a bioluminescent protein. GFP can be included into a range of biological systems for the purpose of functioning as a marker protein. GFP has significantly contributed to multiple research fields like monitoring gene expression, localisation, mobility, traffic, interactions between various membrane and cytoplasmic proteins, etc.

## 23.8.6Jumping Genes/Transposons

**Jumping genes**, scientifically known as **transposons** or **transposable elements**, are DNA sequences that **can move from one location to another within a genome.**

This movement can cause mutations, alter cell identity and change genome size.

**Barbara McClintock**, who discovered these elements, won the **Nobel Prize in 1983 for her work**.

Here is a more detailed explanation:

### *What Are transposons?*

Transposons are mobile genetic elements that can move within the genome of a cell.

They can be found in various life forms, including bacteria, plants and animals.

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### *How Do They Move?*

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Transposons move via two main mechanisms:

- **Cut-and-Paste:** They remove themselves from their original location and insert themselves elsewhere.
- **Replicative:** They make a copy of themselves and insert the copy in a new location, leaving the original in place.

### *Types of Transposons*

- **DNA Transposons:** These DNA sequences can move from one location to another within a genome.
- **Retrotransposons:** These first transcribe their DNA into RNA, and move in genome as RNA intermediate, which is then reverse transcribed into DNA and integrated into a new location.

### *Impact of Transposons*

- **Mutations:** They can disrupt genes and cause mutations, leading to genetic changes.
- **Evolution:** They can contribute to genome diversity and evolution by introducing new genetic variations.
- **Disease:** In some cases, jumping genes can be associated with diseases, like haemophilia and cancer.
- **Gene Therapy:** Scientists are exploring the use of transposons as tools for gene therapy and other genetic engineering techniques.

## 23.9 Genome

A genome is an entire set of DNA of an organism, which contains hereditary information. It consists of both the DNA or RNA genes and the non-coding sequences. Prokaryotes usually store their genomes on a single chromosome and eukaryotes store their genome on one or more chromosomes. As we know that DNA is present in nucleus as well as some organelle (mitochondria, chloroplast) and based on this, the genome term can be applied specifically for specifically storing a complete nuclear DNA set but can also be applied for storing within organelles. The 'genome' of any given individual (except for identical twins and cloned organisms) is

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unique. Genome that can chemically be synthesised in lab is known as artificial or synthetic genome.

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### 23.9.1 Genome or DNA or Nucleotide Sequencing

Genome sequencing is a laboratory process through which scientists get the complete DNA sequence (in terms of A, T, G, C) of an organism's genome at a time. DNA-containing cells such as saliva, epithelial cells, bone marrow, hair (as long as the hair contains a hair follicle), seeds, plant leaves, are used as samples for sequencing. Genome sequencing is done by an instrument called as automated DNA sequencers.

### 23.9.2 Specific Applications of Genome Research

Some current and potential applications of genome research include:

1. Medicine
2. Evolutionary biology and anthropology
3. Energy and environmental applications
4. DNA forensics
5. Agriculture

#### *1. Medicine*

- Improved diagnosis of disease and earlier detection of genetic predispositions to disease
- Rational drug design
- Gene therapy
- Assess health damage and risks due to exposure of radiation, exposure to mutagenic chemicals and cancer-causing toxins
- Reduce the possibility of heritable mutations

#### *2. Evolutionary Biology and Anthropology*

- Study of evolution
- Study of migration of different human population groups

#### *3. Energy and Environmental Applications*

- Creating new sources of energy (bio-fuels)

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- Use microbial genomics for creating environmental monitoring techniques for the detection of pollutants, safe and efficient environmental remediation, and carbon sequestration
- Detect bacteria and other organisms that may pollute air, water, soil and food

#### *4. DNA Forensics*

- Identification of possible suspects whose DNA may match proof remained at crime scenes
- Acquitting those persons who are falsely accused in crimes
- Identification of victims of crime and catastrophe during transplantation, matching organ donors with recipients

#### *5. Agriculture*

- Development of productive, and disease, insect and drought-resistant crops (GM crops)
- Developing healthier, more productive, disease-resistant farm animals (Transgenic)
- Foods with higher nutritional content (GM foods)

### 23.9.3 Human Genome Project (HGP)/Human Genome Sequencing

The Human Genome Project (HGP) was a 13-year effort, which formally began in 1990 and completed in 2003 as a global project of scientific research. The main goal was to determine the sequence of the genes present in human DNA.

#### 23.9.3.1 Project Goals

1. Identification of approximately 24,000 human DNA genes
2. Determining the 3 billion chemical base pairs (300 Mega base pairs/3 $\times 10^9$ base pair) sequences consisting of human DNA
3. Information storage in databases
4. Improving tools for analysis of data
5. Transfer-related technologies to the private sector

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6. Addressing of the ethical, legal and social issues (ELSI) arising from the project

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The participating countries were the US, the UK, France, Germany, Japan and China.

The human DNA sequence is piled up in databases, which are freely available on the Internet. **'GenBank'** is the databank of The US National Center for Biotechnology Information and its sister organisations based at Europe and Japan, which contains sequence of genes along with recognised and imaginary genes and proteins sequences. For the data analysis, various computer programmes are created since it is difficult for the data interpretation itself without such programmes.

### 23.9.3.2 Human Genome Project – Read and Write (HGP-Read and HGP-Write)

#### *1. HGP-Read*

- This was an international and multi-institutional effort that took 13 years [1990–2003] and $2.7 billion to produce a blueprint of the human genome.
- The HGP has revealed that there are probably about 20,500 human genes composed of over 3 billion base pairs.
- India did not participate in HGP-read.

#### *2. HGP-Write*

- This project was launched in 2016 to write or build an artificial human genome from scratch with sophisticated bioengineering tools.
- It aims to address a large number of human health problems of which some of the potential challenges are rising human organs transplantation, devising viruses immunity in cell lines through genome-wide recoding, devising cancer resistance into new therapeutic cell lines, and increasing high-productivity, cost-efficient and pharmaceutical development through the usage of human cells and organoids.
- The encouragement of project property through the accessing of broad intellectual property by the use of patent pooling.
- But, the HGP-write will require public involvement and consideration of ethical, legal and social implications.

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### 23.9.3.3 HGP and its Benefits

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The goal of sequencing and identifying all three billion chemical units in the human genetic instruction set and determining the genetic roots of disease for further development of treatments lead to Human Genome Project. In April 2003, the sequencing of the human genome was successfully completed by Human Genome Project, which sequenced about 99 per cent of the human genome's gene-containing regions with an accuracy of 99.99 per cent of the 3 billion DNA letters in the human genome.

1. Post sequencing, the next step is identification of the genetic variants which are involved in increasing the risk for common diseases like cancer and diabetes. The interpretation of genome data, as a task, is still in its initial stages. It is expected that detailed knowledge of the human genome will serve with new avenues for advances in medicine and biotechnology.
2. New therapeutic procedures may be determined by deeper understanding of disease processing at molecular biology level. There is establishment of DNA significance in molecular biology and understanding its central role in calculating the basic cellular processes operation, which will facilitate medical advances in various fields of clinical interest.
3. The data from this project is expected to answer several questions involving similarities and differences between humans and our closest relatives (the primates and the other mammals).
4. They offer numerous benefits for biological scientists, as in, a researcher investigating a certain form of cancer can narrow down his/her search to a particular gene. By visiting the human genome database on the World Wide Web, more data about this particular gene and its properties can be used.
5. Identification of predictive and predisposition markers would be accelerated by harnessing the huge amount of genetic variability

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available for common and complex diseases genes as well as individual responsiveness to medications and different environments.

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### 23.9.3.4 Missing in Genome Sequencing

- The genetic sequence made available in 2003 from the **Human Genome Project**, which is an international collaboration between that happened between 1990 and 2003, contained information from a region of the human genome known as the **euchromatin**. Here, the chromosome is rich in genes, and the DNA encodes for protein.
- **The 8% chromosome that** was left out was in the area called **heterochromatin**. This is a smaller portion of the genome, and does not produce protein.
- There were at least two key reasons why heterochromatin was given lower priority. This part of the genome was thought to be "**junk DNA**", because it had no clear function. Besides, the euchromatin contained more genes that were simpler to sequence with the tools available at the time.
- Now, the fully sequenced genome is the result of the efforts of a global collaboration called the **Telomere-2-Telomere (T2T) project.** The invention of new methods of DNA sequencing and computational analysis helped complete the reading of the remaining 8% of the genome.
- **Genome sequencing** is the process that involves deciphering the exact order of base pairs in an individual. This 'deciphering' or reading of the genome is what sequencing is all about. Cost of sequencing differ based on the methods employed to do the reading or the accuracy stressed upon in decoding the genome.

### 23.9.3.5 Need for Genome Sequencing

- Ever since the human genome was first sequenced in 2003, it opened a fresh perspective on the *link between disease and the unique genetic make-up of each individual*.
- Nearly 10,000 diseases – including cystic fibrosis, thalassemia – are known to be the result of a single gene malfunctioning.

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- While genes may render some insensitive to certain drugs, genome sequencing has shown that cancer too can be understood from the viewpoint of genetics, rather than being seen as a disease of certain organs.

### 23.9.3.6 Importance of Genome Sequencing

- A genetic test, which is commercially available at several outlets in the country, usually involves analysing only a portion of the genome that is known to contain aberrant genes linked to disease.
- It is important to ensure that India, with its unparalleled human diversity, is adequately represented in terms of genomic data and develops indigenous capacity to generate, maintain, analyse, utilise and communicate large-scale genome data, in a scalable manner.
- The human genome has about 3.2 billion base pairs and just 10 years ago cost about $10,000. Now prices have fallen to a tenth.

(Refer **Infographic – Human Genome Project**)

### 23.9.3.7 Indian Human Genome Project/India's Efforts towards Genome Sequencing

CSIR scientists at the Institute of Genomics and Integrative Biology (IGIB), Delhi, have sequenced the anonymous healthy Indian citizen Human Genome.

Due to lack of resources, India could not be a part of the Human Genome Project in the early nineties. After the first human genome sequence completion in India, the country is presently party to the league of few countries, such as United States, China, Canada, United Kingdom and South Korea who have shown their capability of sequencing and assembling complete human genomes.

The first Human Genome Sequence effort took more than a decade and huge capital i,e. around billion US dollars, CSIR scientists at IGIB accomplished complete sequencing and assembly in a comparatively much shorter time. Within 10 weeks, from September 25 to December 4, 2009, the feat was achieved by the IGIB scientists. CSIR was able to achieve this feat through the adoption of latest technologies and through successfully combining its high estimating abilities.

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It opened newer vistas for low cost affordable healthcare and predictive medicine in future for the masses. Consequently, understanding several diseases important to India and other developing countries would get improved, adding to the global knowledge on genetic variations and predisposition to various diseases. The genetic diversity of the Indian population had been successfully completed by CSIR along with the genome sequence of a fish variety, called Zebra fish, popularly used by the scientific community as an organism for modelling human diseases.

### *Indian Genome Variation Initiative*

The Indian Genome Variation initiative is a network programme initiated in 2003 for 5 years, with the help of six constituent laboratories of the Council of Scientific and Industrial Research (CSIR), with funding from Indian Government. It aims to provide data on the sub-population's variations, which will be useful for investigators in order to carry out the study of specific candidate gene in any part of the country. This is the first large-scale comprehensive study of the structure of the Indian population with wide-reaching implications.

### *Indian Genome Variation Database (IGVdb)*

Being a consortium activity, the Indian Genome Variation database (IGVdb) is a part of the Indian Genome Variation initiative, established in 2003. It helps in understanding the intrinsic genetic variability of human sub-population being the first step in identifying vulnerable biomarkers for any disease.

**INDIGEN GENOME PROJECT**

- The IndiGen initiative was undertaken by CSIR in April 2019, which was implemented by the CSIR-Institute of Genomics and Integrative Biology (IGIB), Delhi and CSIR-Centre for Cellular and Molecular Biology (CCMB), Hyderabad.
- The objective is to enable genetic epidemiology and develop public health technologies applications using population genome data.
- This has enabled benchmarking the scalability of genome sequencing and computational analysis at population scale in a defined timeline.
- The ability to decode the genetic blueprint of humans through whole genome sequencing will be a major driver for biomedical science.

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IndiGen programme aims to undertake whole genome sequencing of thousands of individuals representing diverse ethnic groups from India.

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### 23.9.4 DNA Technology Bill

The ***DNA Technology (Use and Application) Regulation Bill, 2019,*** which seeks to control the use of DNA technology for establishing the identity of a person, was referred to a parliamentary standing committee for examination.

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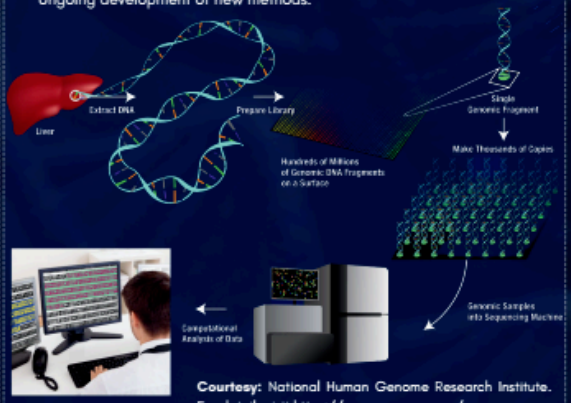
# THE HUMAN GENOME PROJECT

## WHAT IS HUMAN GENOME PROJECT?

- The Human Genome Project (HGP) was an international scientific effort to map and sequence the entire human genome.
- **Launched in 1990**, it involved countries like the USA, UK, France, Germany, Japan, and China.
- **Primary goals:**
  - **Determine the sequence of DNA base pairs.**
  - **Identify approximately 20,000-25,000 human genes.**
- **Objectives:**
  - Enhance understanding of genetics and human biology.
  - Provide insights into the genetic basis of diseases.
  - Pave the way for personalised medicine.
- **Initial projection: 15 years; completed ahead of schedule in 2003.**
- **Achievements:**
  - Comprehensive reference for the human genome.
  - Crucial for studying genetic variation, human evolution, and disease genetics.
- **Post-HGP advancements:**
  - Next-generation sequencing technologies.
  - Faster, more accessible, and less expensive genome sequencing.
- **Impact:**
  - Breakthroughs in personalised medicine.
  - Advances in genetic diagnostics.
  - Significant contributions to medicine, genetics, and biotechnology.

## WHAT IS DNA SEQUENCING?

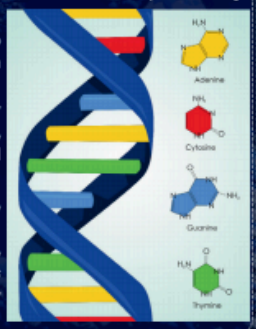
- DNA sequencing determines the **exact sequence of nucleotides (A, T, C, G) in a DNA molecule**, essential for understanding gene function; sequencing by synthesis uses labeled nucleotides and polymerase to generate and assemble 125-nucleotide reads with fluorescent signals, identifying genetic variations and aiding in disease research, with ongoing development of new methods.




**Courtesy:** National Human Genome Research Institute.
For details visit https://www.genome.gov/

## WHAT IS DNA?

- Deoxyribonucleic acid (DNA) contains the instructions for developing and directing the activities of nearly all living organisms.
- DNA molecules consist of two twisting, paired strands, forming a double helix.
- Each strand is made of four nucleotide bases: adenine (A), thymine (T), guanine (G), and cytosine (C).
- Bases pair specifically: A with T, and C with G.
- The sequence of these bases (A, T, C, G) encodes genetic information, similar to how letter order determines word meaning.



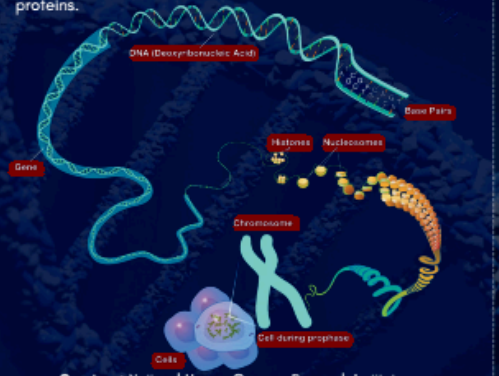

## COLLABORATIVE EFFORT

## KEY FACTS

1. Total DNA base pairs: **~3 billion.**
2. Human genome: **23 pairs of chromosomes.**
3. Number of genes: **~20,000-25,000.**
4. Project Duration: **1990-2003 (13 years).**
5. DNA sequencing machines: Essential for sequence data.
6. Sequencing multiple individuals: Captures genetic variations.
7. Comparative genomics: Human genome vs. Other species.
8. Disease-related gene discoveries: Identifies disease-associated genes.

## WHAT IS GENOME?

- The genome is an organism's complete set of DNA, containing about 3 billion base pairs in humans.
- DNA carries instructions to build and maintain an organism.
- Genes are specific DNA units that instruct protein production.
- The human genome has an estimated 20,000 to 25,000 genes, each coding for an average of three proteins.
- Proteins form structures, control chemical reactions, and transmit signals in the body.
- Mutated DNA can produce abnormal proteins, causing diseases like cancer.
- Genes are located on 23 pairs of chromosomes in the cell nucleus.
- Protein production process:
  - Enzymes copy gene information into messenger RNA (mRNA).
  - mRNA travels to the cell's cytoplasm.
  - Ribosomes read mRNA and link amino acids to form specific proteins.




**Courtesy:** National Human Genome Research Institute.
For details visit https://www.genome.gov/

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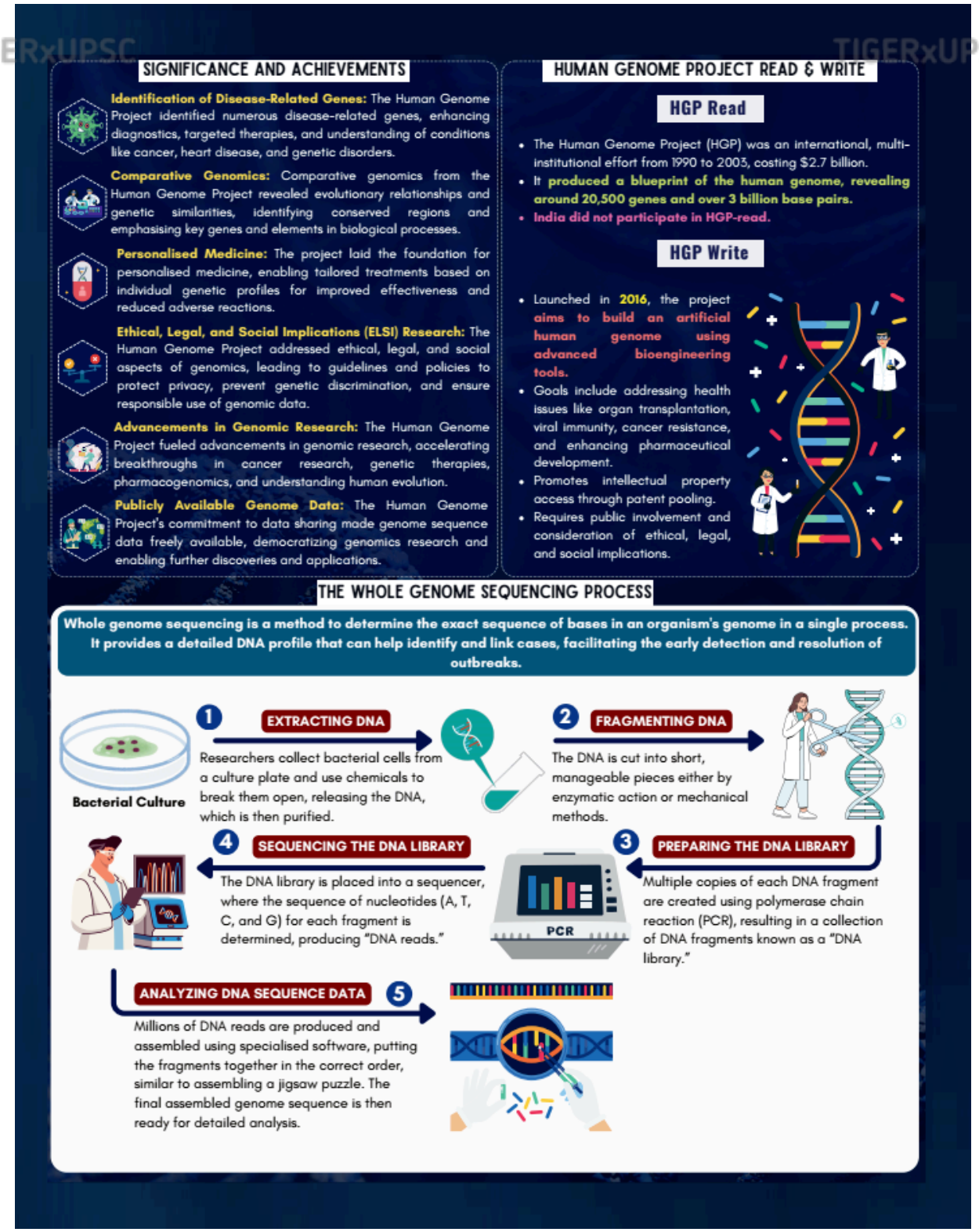


The committee submitted its report in February 2021, recommending several changes in the draft. But instead of introducing a fresh Bill with

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changes, the government decided to withdraw it altogether, its decision made easy by the fact that the main provisions of the Bill have already been enacted as part of another law, the Criminal Procedure (Identification) Act, that was passed by both houses of Parliament last year.

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### *Need for the Legislation and its Significance*

The utility of DNA-based technologies for solving crimes, and to identify missing persons, is well recognised across the world. Therefore, the new bill ***aimed to expand the application of DNA-based forensic technologies to support and strengthen the justice delivery system of the country.***

### *Highlights of the Bill*

- As per the Bill, ***national and regional DNA data banks*** will be set up for maintaining a national database for identification of victims, suspects in cases, undertrials, missing persons and unidentified human remains.
- **Punishment:** According to it, those leaking the DNA profile information to people or entities who are not entitled to have it will be punished with a jail term of up to three years and a fine of up to ₹1 lakh. Similar, punishment has also been provided for those who seek the information on DNA profiles illegally.
- **Usage:** As per the bill, all DNA data, including DNA profiles, DNA samples and records, will only be used for identification of the person and not for 'any other purpose'.
- ***The bill's provisions will enable the cross-matching between persons*** who have been reported missing on the one hand and unidentified dead bodies found in various parts of the country on the other, and also for establishing the identity of victims in mass disasters.
- The Bill establishes a ***DNA Regulatory Board*** to accredit the DNA laboratories that analyse DNA samples to establish the identity of an individual.

### *Benefits of the Bill*

- By providing for the mandatory accreditation and regulation of DNA laboratories, the Bill seeks to ensure the proposed expanded use of this technology in the country.

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- Assurance that the DNA test results are reliable and the data will remain protected from misuse or abuse in terms of the privacy rights of our citizens.

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### 23.9.5 Genome India Initiative

The Department of Biotechnology (DBT) plans to scan nearly 20,000 Indian genomes over the next five years, in a two-phase exercise, and develop diagnostic tests that can be used to test for cancer.

**100K GENOMEASIA PROJECT**

**100K GenomeAsia Project** is a project with the presence of Indian Scientists and Companies which is being done at **Nanyang Technological University (NTU), Singapore, to sequence the whole genomes of 100K Asians, including 50,000 Indians.**

**About 100K GenomeAsia Project**

- A non-profit consortium project which has an ambitious plan to sequence 100,000 Asian individuals in hopes of fast-tracking precision medicine applications for Asian populations is termed as GenomeAsia 100K.
- Having influence on big data analytics and advances in data science and artificial intelligence. The participants will be selected from 12 South Asian countries and at least 7 North and East Asian countries.
- The first step of the project will be on creating phased reference genomes for all major Asian ethnic groups – representing a major step forward in understanding the population history and substructure of the region. There will be pairing of 100,000 sequenced individual genomes with microbiome, clinical and phenotype information which will allow deeper analysis of diseased and healthy individuals in the context of inferred local ancestries.

**Significance**

- The recent insights into the genome diversity of Asian ethnicities show the possibility to understand the biology of disease in the currently under-studied Asian populations which represent 40% of humankind.
- Also, provides a valuable source of clinical insights that should enhance our understanding of several rare and inherited diseases, as well as chronic diseases such as cancer, diabetes and cardiovascular disease by the unique genetic diversity prevalent in South, North and East Asia.



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- The first phase involves sequencing the complete genomes of nearly 10,000 Indians from all corners of the country and capture the biological diversity of India.
- In the next phase, about 10,000 'diseased individuals' would have their genomes sequenced. These vast troves of data sets would be compared using machine learning techniques to identify genes that can predict cancer risk, as well as other diseases that could be significantly influenced by genetic anomalies.

***Significance***

- The data generated would be accessible to researchers anywhere for analysis. This would be through a proposed National Biological Data Centre envisaged in a policy called the **'Biological Data Storage, Access and Sharing Policy',** which is still in early stages of discussion.
- As the genetic landscape differs across the world, it is necessary that genetic data is shared in order to derive greater knowledge from information and serve the purpose of enabling better treatment outcomes.
- The initiative will pave the way for identifying genes and genetic variations for common diseases, treating Mendelian disorders, enabling the transformation of the Precision Medicine landscape in India, and thus improving the healthcare of the general population in our country.

***Need for Genome Sequencing***

- Ever since **the human genome was first sequenced in 2003,** it opened a fresh perspective on the link between disease and the unique genetic make-up of each individual.
- **Nearly 10,000 diseases** – including cystic fibrosis, thalassemia – **are known to be the result of a single gene malfunctioning.** While genes may render some insensitive to certain drugs, **genome sequencing has shown that cancer too can be understood from the viewpoint of genetics, rather than being seen as a disease of certain organs.**

## 23.9.6 National Genomic Grid (NGG)

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The government has announced to set up **a National Genomic Grid (NGG).**

***About NGG***

- It will study genomic data of cancer patients from India.
- It will collect samples from cancer patients, through a network of pan-India collection centres by bringing all cancer treatment institutions on board.
- The grid to be formed will be in line with the National Cancer Tissue Biobank (NCTB) set up at the Indian Institute of Technology, Madras.

***NCTB***

- **National Cancer Tissue Biobank (NCTB)** is **a joint initiative of the Department of Science and Technology (DST), Government of India and Indian Institute of Technology, Madras.**
- It collects cancer tissue samples with consent from patients diagnosed with cancer.
- The aim is to provide researchers with high quality of cancer tissues and the patient data in order to facilitate cancer research that will lead to improvements in cancer diagnosis and treatment.
- This research is carried out through the technique of Genome Sequencing.

***Significance***

- Through NGG, the government seeks to boost cancer research and make treatment viable for people of different economic classes.
- NGG will help to study genomic factors influencing cancer and identifying the right treatment modalities for the Indian population.

## 23.10 Recombinant DNA Technology (r-DNA Technology)

Recombinant DNA technology is the definite protocol or method deployed to introduce foreign DNA segment (DNA of other species) into host species. The process followed in gene cloning is also called recombinant

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DNA technology. Recombinant DNA represents DNA from a foreign source.

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### 23.10.1 Applications of Recombinant DNA Technology (r-DNA Technology)

Every biological technology, western pharmacy, doctor's or veterinarian's office, medical testing laboratory, and biological research laboratory is using this technology. The products of r-DNA technology include Recombinant proteins. r-DNA serves as a tool for identification, mapping and sequencing of genes, subsequently determining their functions. Various applications of r-DNA technology are given below:

- **Recombinant Human Insulin:** Insulin-dependent diabetes has seen major breakthrough in its treatment with the production of recombinant insulin. It is cheaper and easily available as compared to insulin obtained from animal sources.
- **Recombinant Human Growth Hormone:** The patients with pituitary glands generating insufficient quantity of hormone for normal growth and development, it proves to be a boon. Recombinant HGH solved this issue, and is now used therapeutically. Some athletes have been found misusing it as a performance-enhancing drug.
- **Recombinant Vaccines:** Recombinant vector vaccines are bacteria or viruses that are genetically engineered to carry extra genes derived from a pathogen. A recombinant vaccine is a vaccine produced through recombinant DNA technology. This involves inserting the DNA encoding, an antigen (such as a bacterial surface protein), that stimulates an immune response into bacterial or mammalian cells, expressing the antigen in these cells and then purifying it from them.
- **Recombinant Blood Clotting Factor VIII:** A blood-clotting protein that helps in curing patients with bleeding disorder, haemophilia, due to inability to produce factor VIII in sufficient amount that can support normal blood coagulation. Earlier, when it was not developed, the protein in large quantities of human blood was obtained from multiple donors. And hence, there was a high risk of transmission of blood-borne infectious diseases like HIV and hepatitis B.

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- **Recombinant Hepatitis B Vaccine:** Recombinant hepatitis B vaccine can control infection of hepatitis B, which carries a form of the hepatitis B virus surface antigen, produced in yeast cells. Hepatitis B virus, unlike other common viruses like polio virus, cannot be grown in vitro, it marks to be an important and necessary development.
- **Golden Rice:** It is a recombinant variety of rice expressing the enzymes responsible for β-carotene biosynthesis. It is advantageous, being capable to reduce the deficiency of vitamin A in the world's population. Golden rice is not currently in use, due to pending resolution of intellectual property, environmental and nutritional issues.
- **Herbicide and Insect-Resistant Crops:** Commercial varieties of important agricultural crops (like soyabean, maize/corn, sorghum, canola and cotton) have been developed. Such varieties integrate a recombinant gene that cause the resistance to the herbicide glyphosate, and also simplifies weed control by glyphosate application. Several countries use these crops commercially.

## 23.11 Transgenesis

It a subset of GMOs – Organisms that have inserted DNA that originated in a different species. Transgenesis is the process in which an exogenous gene (external gene) is introduced, called a transgene, into a living organism. This enables the organism to exhibit a new property and transfer the same to the next generation. To facilitate transgenisis, liposomes, plasmid vectors, viral vectors, pronuclear injection, protoplast fusion and ballistic DNA injections are usually deployed. The genetic code is similar for all organisms and hence, there is expression of foreign genes by a transgenic organisms. This implies that similar proteins will be coded by a specific DNA sequence in all organisms.

### 23.11.1 Transgenic Animals

Through alternation of DNA or insertion of DNA to an animal (sheep, pigs and rats), certain proteins useful for medical treatment can be produced. For

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instance, Human-alpha-1-antitrypsin, developed in sheep, treats humans with this deficiency and transgenic pigs with human histocompatibility have been studied with the primary objective to check if organs will be suitable for transplant with low chances of rejection.

A **knockout mouse** is a genetically engineered mouse, in which an existing gene has been knocked out or removed by replacing it or disrupting it with an artificial piece of DNA. The manipulation in the gene results in changes in a mouse's phenotype, which include appearance, behaviour and other observable physical and biochemical characteristics.

### 23.11.2 Transgenic Plants

To implant desirable traits, like resistance to pests, herbicides, or harsh environmental conditions, improved product shelf life, and increased nutritional value, etc., transgenic plants are being developed. In 1996, the first commercial cultivation of genetically modified plants was performed, post which many modifications are done such as tolerance to the herbicides, glufosinate and glyphosate, resistance to virus damage as in Ring spot, virus-resistant GM papaya, which is grown in Hawaii, and the production of Bt toxin, an insecticide that is non-toxic to mammals. Most GM crops growing today provide benefits mainly to farmers. GM plants are developed by both private companies and public research institutions like International Maize and Wheat Improvement Centre are involved in development of GM plants. Monsanto, an American Company is an example of such a private company.

## 23.12 Genetically Modified Organism (GMO)

A Genetically Modified Organism (GMO) or Genetically Engineered Organism (GEO) is an organism with altered genetic material due to application of RDT/genetic engineering techniques. The way it works is: The desirable DNA segment from different sources are combined into one

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molecule and this creates a new set of genes. This DNA is then transferred into an organism, consequently modifying the genome of the organism. An organism so developed is considered to be a genetically modified organism. GMO bear their application in preparation of medicine, food, etc. The first genetically modified crop was produced in 1982, an antibiotic-resistant tobacco plant. The first field trial was done in France and USA in 1986.

***Methods of GM Crop Development***

**1. Agrobacterium Tumefaciens–Mediated Transfer**

- *A. tumefaciens* causes crown gall disease.
- Tumour-inducing plasmid (Ti plasmid) is modified to insert desired genes without causing disease.

***A. tumefaciens*** is a **soil bacterium** that naturally transfers a segment of its DNA, called T-DNA (Ti Plasmid), into the genomes of susceptible plant cells.

This process, known as *Agrobacterium*-mediated gene transfer, is a common method for creating transgenic plants.

**2. Direct Methods**

- **Gene Gun (Biolistics):** DNA-coated metal particles are shot into plant cells.
- **Electroporation, Microinjection:** Other techniques to directly transfer genes.

## 23.12.1 Genetically Modified Plants

### 23.12.1.1 Flavr Savr Tomato

It was the **first commercially grown genetically modified (GM) food to be granted human consumption licence**. It was developed by the Californian company 'Calgene', with the aim to **slower the process of ripening** in tomato and thus preventing it from softening. It also enables the tomato to retain its natural colour, making it as safe as tomatoes bred by conventional means. Polygalacturonase (PG) is an enzyme produced in plants, which controls the ripening process and restricts the attack of certain bacteria and fungi, which play a role during rotting process.

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### 23.12.1.2 Golden Rice

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Beta carotene (Pro-vitamin-A) is required as a source of Vitamin A. So, golden rice is a precursor of Vitamin A. Golden rice is a genetically modified plant, which can be used in those areas where Vitamin A deficiency is common, so it prevents night blindness. Some scientists are against golden rice, as they think it will crossbreed with the wild rice and destroy natural variety. The dark yellow colour shows more β-carotene.

### 23.12.1.3 Bt Crops

The Belgian company 'Plant Genetic Systems' emerged as the first company (in 1985), aimed to develop genetically engineered tobacco plants with insect tolerance by expression of Cry genes from *Bacillus thuringiensis* bacteria.

Bt stands for *Bacillus thuringiensis,* a Gram-positive bacteria (Gram-positive bacteria do not have an outer cell membrane found in Gram-negative bacteria) that lives in soil and commonly used as a biological alternative to a pesticide. A trans-gene (those gene transfer, i.e., Cry-I AC gene) producing Cry toxin protein extracted by *B. thuringiensis* and introduced in desirable plant to develop Bt-crops. Bt-crops protect them from insect named Lepidoptera (Boll worm/white fly). In 1995, potato plants producing Bt-toxin was the first pesticide producing crop to be approved safe by the Environmental Protection Agency USA.

Note: The first and only transgenic crop approved by Genetic Engineering Approval Committee (GEAC) for commercial cultivation in 6 states, namely Andhra Pradesh, Gujarat, Karnataka, Madhya Pradesh, Maharashtra and Tamil Nadu is named the Bt Cotton. India is in the top ranks, accounting for 20 per cent of the total area planted under cotton.

***Bollgard I and Bollgard II Technology***

Bollgard I Bt cotton (single-gene technology) is India's first biotech crop technology approved for commercialisation in India in 2002, followed by Bollgard II–double-gene technology in mid-2006, by the Genetic Engineering Approval Committee (GEAC), the Indian regulatory body for biotech crops.

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Bt cotton expressing a *Cry1Ac* gene is grown over a large area in many places of the world, however, there is a major concern that bollworms may develop resistance. Here, a durable resistance strategy against bollworms is used by developing a double gene construct containing *Cry1Ac* and *Cry2Ab* for cotton transformation.

**Bollgard I cotton** provides in-built protection for cotton against destructive American Bollworm Heliothis infestations and contains an insecticidal protein from a naturally occurring soil microorganism, *Bacillus thuringiensis*.

**Bollgard II technology** contains a superior double-gene technology– Cry1AC and Cry2Ab which provides protection against bollworms and Spodoptera caterpillar, leading to better boll retention, maximum yield, lower pesticides costs and protection against insect resistance.

Both Bollgard I and Bollgard II insect-protected cotton is widely planted around the world in an environment-friendly way of controlling bollworms.

#### 23.12.1.4 GM Mustard/DMH-11

**Dhara Mustard Hybrid-11 (DMH-11)** is a *genetically modified* hybrid variety of the mustard species *Brassica juncea*. It was developed by the *University of Delhi* , with the aim of reducing India's demand for edible oil imports.

DMH-11 was created through *transgenic technology*, primarily involving the bar, *barnase* and *barstar* gene system.

The barnase gene confers male sterility, while the Barstar gene restores DMH-11's ability to produce fertile seeds.

The insertion of the third gene bar, enables DMH-11 to produce *phosphinothricin-N-acetyl-transferase*, the enzyme responsible for *Glufosinate* resistance.

This hybrid mustard variety has come under intense public scrutiny, mainly due to concerns regarding DMH-11's potential to adversely affect the environment as well as consumer health. DMH-11 was found not to pose any food allergy risks, and has demonstrated increased yields over existing mustard varieties. Conflicting details and results regarding the field

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trials and safety evaluations conducted on DMH-11 have delayed its approval for commercial cropping.

The Government had approved the environmental release of Genetically Modified (GM) mustard hybrid DMH-11 and its parental lines during 147th meeting of Genetic Engineering Appraisal Committee (GEAC) on 18 October 2022 for its seed production and testing as per existing ICAR guidelines, conditions imposed by the GEAC while recommending the environmental release of GM mustard hybrid DMH-11 and its parental lines; and other extant rules/regulations prior to commercial release.

The hybrid DMH-11 had been tested against the check varieties Varuna and RL-1359 during Biosafety Research Level Trials (BRL) I and II, where it gave 28.15% high over Varuna during 2010–11 to 2014–15. Field trials for three years were conducted to assess the impact on human health and environment as per the stipulated guidelines and applicable rules. Extensive studies carried out on toxicity, allergenicity, compositional analysis, field trials and environmental safety studies of GM mustard lines against their non-transgenic comparators have provided evidence that the GE mustard hybrid DMH-11 and its parental lines are safe for cultivation and for food and feed use.

## 23.12.2 Advantages and Disadvantages of GM Crops

***Advantages of GM Crops***

- Faster and precise breeding in comparison to conventional methods.
- Pest resistance reduces pesticide use and pollution.
- Virus and drought resistance, flood resistance, disease resistance, increase productivity, etc.
- Herbicide tolerance reduces tilling and erosion.
- Enables **biofortification** and **sustainable farming.**
- Enhances **nutritional value (eg. Golden rice etc.)** and **food security.**

***Disadvantages and Concerns***

- **Horizontal gene transfer—**Antibiotic/pesticide resistance in other organisms.
- **Health risks** and ecological disturbances.
- **Pest resistance** (e.g. Bt resistance).
- **Herbicide overuse—**Resistant weeds (e.g. Glyphosate-resistant).

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- **Loss of biodiversity** in local crop varieties.
- **Pollinator risks** (e.g. honeybees).

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**India and GM Crops**

**Current Status:**

- **Bt Cotton (2002):** Only commercially approved GM crop.
- **Bt Brinjal:** Approved in 2009 but under moratorium.
- **GM Mustard (DMH 11):** Awaiting commercial approval.

**Other Crops on Trial:**

- Tomato, maize, chickpea, etc.

## 23.12.3 Applications of GM Technology

- **Biofortification:** e.g. *Golden Rice* (β-carotene).
- **Edible vaccines:** Lower cost, fewer side effects.
- **Biofuels:** Fourth-gen from GM algae/cyanobacteria.
- **Phytoremediation:** Cleanup of pollutants by engineered plants.

## 23.12.4 Many Reasons Behind the Protest of Genetically Modified Plants in India

- Consumption of Bt-crop **decreases immunity of human beings**, so it is not safe for human health and it also affects surrounding diversity due to loss of some other important insect pollinators those also die due to consumption of Bt-crop.
- According to a research, it is observed that there are **33 side effects of Bt-Brinjal on human body** and that is why it is now banned in India (earlier it was banned in USA, UK, Canada, Australia).
- People also protest Bt-Mustard in India due to its crossbreed with the native mustard crop and **destroy natural variety.**
- It has been observed that due to the Bt-crop seeds, there is **stagnant yield**, which leads to **new pest attacks and disease** these days.
- In the recent past, **white fly has attacked Bt-cotton variety** in Punjab and Haryana and has caused a rural crisis, so many farmers have committed suicide and protested to demand state aid.
- Approximately 50–60 per cent of the honey is produced in India through mustard crop. Its GM crop will **harm honeybees.**

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**SWAMINATHAN CALLS GM CROPS A FAILURE**

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Leading agriculture scientist M.S. Swaminathan, in a research paper, has described Bt cotton as a 'failure'. The findings were published in paper 'Modern Technologies for Sustainable Food and Nutrition Security'. It is a review of crop development in India and transgenic crops – particularly Bt cotton, the stalled Bt brinjal as well as DMH-11, a transgenic mustard hybrid.

**Key Observations**

The paper notes that GE (genetically engineered) Bt cotton has failed in India. It has failed as a sustainable agriculture technology and has, therefore, also failed to provide livelihood security for cotton farmers who are mainly resource-poor, small and marginal farmers. Besides, the precautionary principle (PP) has been done away with and no science-based and rigorous biosafety protocols and evaluation of GM crops are in place. The paper also raises questions on the genetic engineering technology itself on the grounds that it raises the cost of sowing. Also, the insertion of foreign genes (in the plant) could lead to 'molecular and cellular events not precisely understood'.

**Way Ahead – Prof. Swaminathan's Suggestions**

The government should only use genetic engineering as a last resort. Genetic engineering technology is supplementary and must be need-based. Only in very rare circumstances (less than 1%) may there arise a need for the use of this technology.

### 23.12.4.1 Bt-Brinjal Ban in India

Bt-Brinjal plant has a resistance against lepidopteron insects like the Brinjal Fruit and Shoot Borer and Fruit Borer. The ingestion of the Bt toxin by the insect disrupted the digestive processes, which ultimately led to the death of the insect. In India, Bt-Brinjal is being developed by M/s Mahyco [Maharashtra Hybrid Seeds Company] but now neither Bt. Brinjal nor Bt. Mustard is commercially cultivated in India.

### 23.12.4.2 State Govt Bans Sale of Mahyco Bt-Cotton Seeds

The state government's Department of Agriculture was receiving complaints regarding the substandard bacillus thuringiensis (Bt) cotton seeds. So they withdrew the licence of Mahyco-Monsanto Biotech (MMB). This being the first Bt-cotton seed manufacturing company to be banned by

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the state as the seeds have resulted in a stagnant yield and led to new pest and disease attacks.

### 23.12.5 Monsanto

Monsanto is an American multinational agricultural biotechnology corporation. It is one of the pioneer companies in the field of applying biotechnology to agriculture. It is a leading producer of the herbicide glyphosate and genetically engineered (GE) seed. Its headquarters is situated in Creve Coeur, Missouri.

The standard biotechnology industry business model, in which patent rights play a key role was brought up by this company. Conflict created when this business model was applied to agriculture, along with a growing movement to create a global, uniform system of plant breeder's rights in the 1980s. The role of Monsanto in these changes in agriculture (which include its litigation and its seed commercialisation practices) has generated controversy for Monsanto. Monsanto has patent of Cry-I Ac gene.

## 23.13 Terminator Gene and Terminator Seeds

It is a precise genetic sequence inserted into a seed's DNA. A synthetic chemical (tetracycline) activated the Terminator gene to produce toxins, which will kill the embryo and the seed will become sterile. The technology in the patent could be applied in a number of ways. Farmers plant the seeds, grow plants and harvest the crops, but due to Terminator gene, the seeds that are harvested will be sterile and cannot be used again for agricultural purpose. This will have adverse effect on the farmers of developing countries, since it forces the farmers to buy new seeds every season.

Note: In March 2006, at the Cop-8 meeting of the UNCBD (United Nation Convention on Biodiversity) terminator seeds were totally prohibited. India and Brazil have passed national laws to prohibit this technology due to farmers' concerns. So, in India, terminator seeds never grow for any crop. Terminator seeds were initially developed as a concept by United States Department of Agriculture and Multinational seed companies. They had not been

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commercialised anywhere in the world due to opposition from farmers, indigenous people, NGOs and some governments.

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## 23.14 Genetic Engineering Appraisal Committee (GEAC) and its Importance in India

Genetic Engineering Appraisal Committee, constituted under Ministry of Environment, Forest and Climate Change, with a role to manufacture, use, import, export and store hazardous micro-organisms/genetically engineered organisms. GEAC was setup under the Environment Protection Act, 1986.

This committee is responsible for judging the importance of hazardous living micro-organisms and recombinants in research and industrial production from the environmental perspective. And also looks for the product in market involving therapeutic applications before and after release. GEAC also plays a vital role in releasing the genetically modified product for trial, i.e., in an experiment field as well as a final product. This committee or any member also has a right to punish someone found guilty for environmental damage under the Environment Protection Act, 1986.

***Regulatory Framework in India***

| Body | Role |
|---|---|
| **RDAC — Recombinant DNA Advisory Committee** | Monitors biotech development |
| **RCGM —** Review Committee on Genetic Manipulation | Reviews high-risk GM research |
| **GEAC — Genetic Engineering Appraisal Committee** | Approves environmental release of GMOs (under MoEFCC) |
| **SBCC —** State Biotechnology Co-ordination Committee | State-level monitoring of GM organisms |

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## 23.15 Molecular Farming (Biopharming) and Edible Vaccine

Under molecular farming, genetically engineered crops are used to produce various chemical compounds in plants with the help of recombinant DNA technology for therapeutic value (drug production).

Different types of therapeutic proteins form, like antibodies, blood products, growth factors, hormones, recombinant enzymes and various human and veterinary vaccines.

Edible vaccine is derived through genetically engineered expression of an antigenic protein by an edible plant. In its consumption, these recombinant proteins involve in the immune of living beings.

Plant-derived vaccine production system has many advantages over animal-derived vaccines that are used in the generation of immunoglobulin and interferon (antiviral proteins).

First edible vaccines were produced in tobacco in 1990. In the leaves and seeds of transgenic tobacco, recombinant proteins were synthesised which are used against Hepatitis B antigen. In potato, banana, carrot, tobacco, maize, etc., various types of edible vaccines are developed. Vaccine against cholera is also successfully developed in potato.

***Various Advantages of Edible Vaccines***

- These are easy to produce at bulk level.
- Their production is cheap and much easier in consumption.
- Compared to injections, edible vaccines are stable at room temperature but traditional vaccines have a need of refrigeration.

## 23.16 Silenced Genes/Gene Silencing/Post Transcriptional Gene Silencing (PTGS)

Gene silencing is a unique phenomenon of **switching off the expression of a gene.** It is a method which reduces or eliminates the protein production

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from its corresponding gene.

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- Proteins are vital molecules, which carry out various functions such as signalling between cells, speeding up biochemical reactions, and providing structural support for the cell. Each gene is responsible for producing a corresponding protein in a two-step process.
- Initially, the information copy programmed in a gene is prepared in the messenger RNA form (mRNA), by a process known as transcription. This occurs in the nucleus of the cell, the cellular structure where all of the cell's genetic material is contained. The mRNA subsequently travels out of the nucleus, and the genetic information it carries is used to produce a specific protein, a process known as translation.
- Gene silencing is often considered the same as gene knockout. **When genes are silenced, their expression is reduced, whereas when genes are knocked out, they are completely erased from the organism's genome and, thus, have no expression.**
- Methods using gene silencing are often considered better than gene knockouts, since they allow researchers to study essential genes that are required for the animal models to survive and cannot be removed.
- There are **many different transcriptional and post-transcriptional processes that can lead to gene silencing.** Generally, gene silencing can be of two types: Transcriptional and Post-Transcriptional.
- **Post Transcriptional Gene Silencing (PTGS) has a major role in plants as a defence mechanism or resistant against, bacteria, viruses and viroids etc.,** for genome stability, and regulation of endogenous gene expression.
- Some of the major applications of the PTGS mechanisms include the antisense RNA approach for the development of Flavr tomato, the development of blue roses by using iris gene, making cotton seeds edible by silencing the genes involved in the production of toxin gossypol, shutting down of the lachrymatory factor synthase gene, in order to produce tearless onion, repression of theobromine synthase coding gene to produce decaffeinated coffee.
- Post-Transcriptional Gene Silencing (PTGS) in plants is an RNA-degradation mechanism.

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### 23.16.1 Post-Transcriptional Gene Silencing (PTGS) Mechanisms in Crops

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**Blue rose** Over many years, breeders have tried to produce true blue roses. The CSIR-developed RNAi technology has been used for **removing the gene encoding enzyme dihydroflavonol reductase (DFR)** in roses during its initially commercial plants applications. The **'blue' was the delphinidin gene that was cloned from a pansy,** for directing the synthesis of pigment in the rose into the 'blue' pathway. The silencer gene exploited a cellular phenomenon called **RNA interference (RNAi), which switched off the DFR gene in a red rose,** to block the cyaniding pathway, and then install the delphidin gene – plus a new DFR gene to complete delphinidin synthesis.

**Edible Cotton Seeds:** To make cotton seed edible, researchers **reduced its toxic gossypol level** by RNA interference (RNAi). The Texas A&M University scientists have worked upon reducing the toxic chemical levels in cotton, thereby moulding the seeds edible and transforming the cotton into an important new source of food.

**Tearless Onion:** Gene-silencing technology, RNAi allows retargeting the plant's own natural regulation system without expression of foreign proteins in the plant. **Through RNAi genes, lachrymatory factor synthase gene stop the synthesis of sulphur compounds** those get converted to the tearing agent.

**Decaffeinated Coffee:** The demand for decaffeinated coffee is increasing as the caffeine stimulatory effects are adversely affecting susceptible individuals through **triggering palpitations, increasing blood pressure and insomnia. The coffee plants were modified, in which expression of theobromine synthase (CaMXMTi) is repressed by RNA interference (RNAi). The plants' caffeine content is decreased up to 70 per cent.**

**Opium Poppy:** Opium poppy (*Papaver somniferum*) is also an example in gene silencing usage. The silencing of **codeinone reductase (COR) gene**

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**in the opium poppy using RNA interference (RNAi) is designed to silence** all members of the multigene COR family for causing the high-yield accumulation of the non-narcotic alkaloid reticuline.

**Flavr Savr Tomato:** It is a genetically changed tomato produced by Calgene and carries an **antisense RNA, thereby inhibiting the gene expression causing the softening of fruit;** thus, it stays firm longer. This gives the producers a larger transportation time and the mechanical harvesting opportunity with little bruising.

## 23.17 IVF (In-vitro Fertilisation) and Test Tube Baby

### *In Vitro Fertilisation*

In Vitro Fertilisation (IVF) is the fertilisation of an egg and sperm outside the mother's body. The process involves stimulating a women's adulatory process, removal of egg or eggs from her ovaries and letting the sperm fertilise them in a liquid in a laboratory and hence requires efficient monitoring. The resultant zygote is cultured for 2–6 days in a growth medium (blastocyst stage) and is then transferred to the same or another women's uterus, with the intent to establish a successful pregnancy.

It can help overcome female infertility caused due to problems with the Fallopian tubes (it is a natural place inside mother body where fertilisation occurs), which makes fertilisation difficult. It can also serve to be useful in cases of male infertility caused due to defect in sperm quality. It helps infertile couples so that they can have a baby of their own (with their own eggs and sperm). Without using donor egg or donor sperm, DNA of the zygote will be of intended parents only.

This treatment is a good option for those who are facing either of the following mentioned problems:

- Blocked Fallopian tubes
- Low sperm numbers or motility
- Infrequent or absent ovulation

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- Endometriosis
- Unexplained infertility
- Age-related infertility
- Failure of other fertility treatments

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### *Test Tube Baby*

Test Tube Baby (also called in vitro fertilisation) is a revolutionary treatment which enables couples with different types of infertilities, to conceive a baby. Developed by **Robert G. Edwards** and **Patrick C. Steptoe of Great Britain**, the technique was devised to help in case of damaged female's Fallopian tubes or when a male's sperm count is low. In recent times, its application has increased, which now includes prospective parents with other reproductive problems (e.g., inability to produce eggs, poor sperm quality, etc.).

Test tube baby refers to a child, conceived outside the women's body. Eggs removed from the mother's ovary are incubated with the sperm from the father. Now, they are allowed to divide until the Blastula stage (64–128 cell structure), which usually takes 3–4 days, which is then transferred to the mother's or a surrogate's uterus to develop normally. Drugs are induced to the patient's ovaries to grow several mature eggs rather than a single one that develops each month. If more than one egg is fertilised and transferred to a uterus, chances of success increases.

In 1978, Steptoe and Edwards successfully carried out the birth of the world's first test tube baby, 'Louis Joy Brown' in United Kingdom, whose mother had Fallopian tube blockages. The second successful test tube baby occurred in India, after 67 days of Louis's birth, named Durga.

Nobel Prize in Physiology or Medicine was awarded to **Robert G. Edwards** in 2010, for co-developing the treatment together. Egg donation and IVF enable women who are past their reproductive years to get pregnant. The record of oldest woman to give birth using IVF and donated egg was held by Adriana Iliescu in 2004, at the age of 66.

(Refer **Infographic: In-vitro Fertilisation & Test Tube Baby**)

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## 23.18 Surrogate Mother

In this, a woman carries a baby for a couple who are not able to conceive or carry a child themselves because of medical or physical reasons. The supposed parents become the legal parent(s) of a child born with the help of surrogacy. It is of two types.

**Traditional surrogacy** is a pregnancy condition where the surrogate is linked genetically to the baby and becomes pregnant with the help of artificial insemination. It used to be common, but today, host surrogacy is most prevalent. **Host surrogacy** is done through IVF, either through the usage of the supposed mother's eggs or with donor eggs. Thus, the surrogate mother does not make the use of her own eggs and is genetically not linked to the baby. There are three stages in host surrogacy, which are as follows:

- **Egg Extraction:** The intended parent (female) undergoes a special procedure to extract a number of eggs.
- **Fertilisation:** The eggs are fertilised with sperm in the laboratory, resulting in embryos.
- **Embryo Transfer:** Transfer of embryo into the womb of the surrogate mother.

Surrogacy was initially disclosed to help infertile couples. But, with time, the humanity associated with its discovery is lost. The greediness in humans makes both the 'child' and the surrogate mother a 'commodity in demand'. With time, many ethical issues were discovered related to surrogacy and loop holes in laws allowed the unethical behaviour flourish.

Even before the IVF technology surrogacy was known and traditional surrogacy was popular, this kind of surrogacy not only objectified the child and the borrowed womb but also sidelined the legal mother. As the surrogate mother was not gestational mother but also the genetic mother of the child, this was more of an emotional and mental suffering for the legal mother as her existence has no role in the development (Zygote Embryo Infant) of the child.

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Once IVF gained popularity, the exploitation of womb shot up. People rented womb by force, development of agencies for surrogacy, etc., increased. With this, India became a hub as willingness for surrogacy was maximum and price charged for this was very low (compared to other nations) and the success rate was also high.

And then, the society witnessed issues such as citizenship of the child (when the nation of the surrogate and the nation of the legal parents both deny to accept the child as a citizen), acceptance of the child (in cases of those couples who went for surrogacy got divorced before delivery and denied to accept the child and even the surrogate refuse to accept), no payment after (especially in the cases where the child is born still or dead or born with disorder either mental or physical) families with financially low background forcing their women to take surrogacy and so on.

Looking at this, India has recently altered its surrogacy law to stop the commercialisation and exploitation of surrogacy both by normal citizens and people from medical stream.

### 23.18.1 Surrogacy Regulation Act, 2021

**The Act came into force on 25 January 2022.**

**Permissibility:** Surrogacy is permissible only for **altruistic purposes** or for couples with proven infertility or disease but **Commercial surrogacy** is not allowed.

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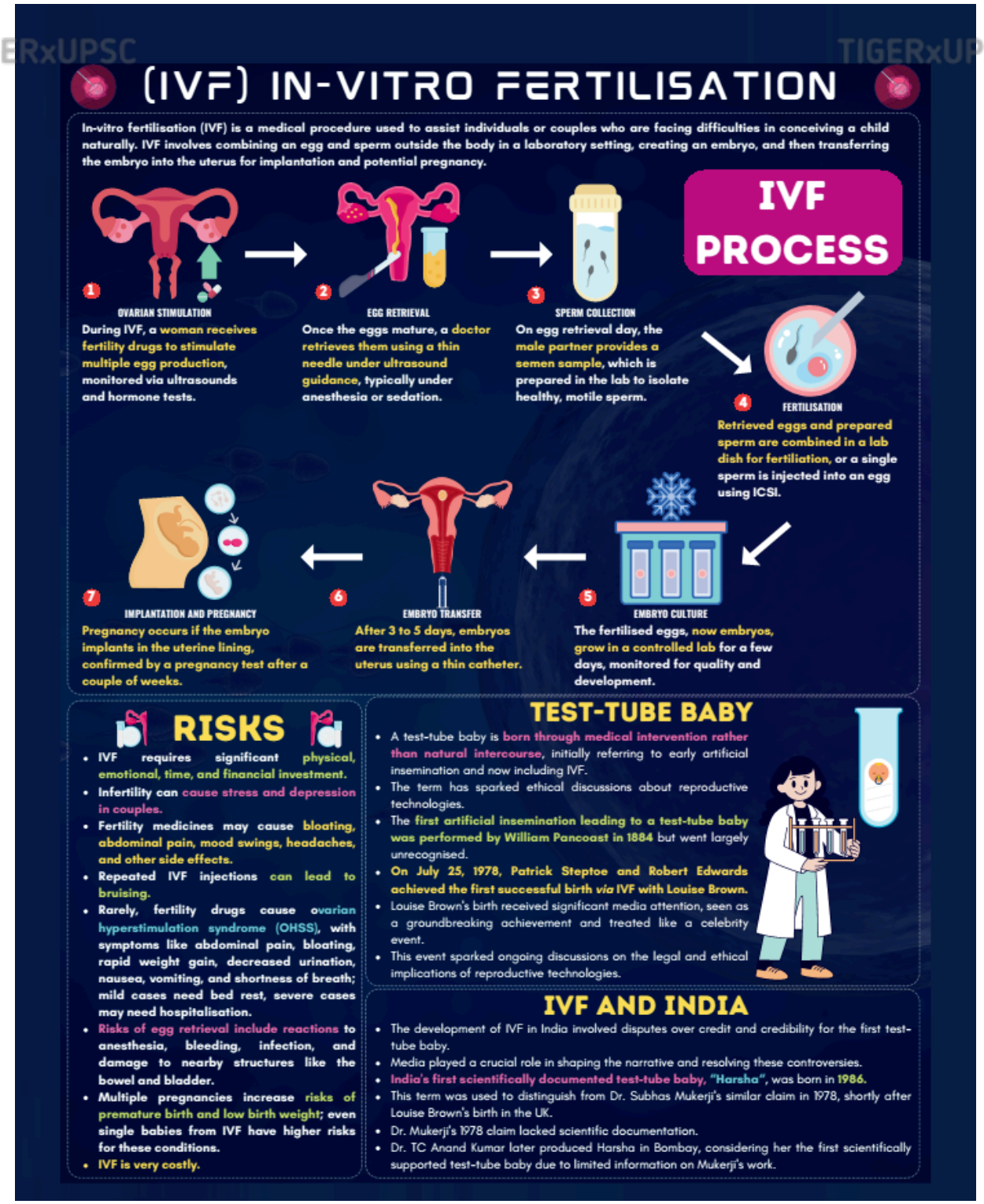


***Eligibility Criteria for Intending Couples [Section 4(c)]***

- Couples must be married for at least 5 years and must be nationals of India.

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- The wife must be between **25 and 50 years**, and the husband between **26 and 55** years.
- The couple must not have any living child, whether **biological, adopted, or through surrogacy,** except in cases of children with disabilities or life-threatening disorders.

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***Eligibility Criteria for Surrogate [Section 4(b)]***

- The surrogate mother must be a close relative of the couple.
- She must be a married woman with at least one child of her own.
- Her age must be between **25 and 35 years on the date of implantation**.
- She must have only been a surrogate once in her life.
- She must obtain a certificate of physical and psychological fitness for surrogacy and surrogacy treatments before proceeding with the procedure.

***Parental Status upon Birth***

Upon birth, the child is legally recognised as the biological child of the intended couple.

Abortion of the foetus requires consent from both the surrogate mother and the relevant authorities, following the provisions of the **Medical Termination of Pregnancy Act. [Section 3(vi)]**

The Surrogacy Regulation Act of 2021 was amended on March 14, 2023, through a gazette notification. The amendment includes the following provisions:

- **Donor Gametes:** Couples undergoing surrogacy must use their own gametes, meaning donor eggs or sperm are prohibited. However, if either spouse has a medical condition that prevents them from having biological children, they may be able to use donor gametes if certified by the District Medical Board.
- **Single Women:** Single women (widows or divorcees) undergoing surrogacy must use their own eggs or self-eggs, along with donor sperm.

| Before | After |
|---|---|

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|---|---|
| • Initially restricted surrogacy only to intending couples' own gametes (reproductive cells).<br>• This requirement stopped couples with specific health issues from pursuing biological parenthood through surrogacy. | • Allows surrogacy with donor gametes if one spouse is medically certified to require them.<br>• Divorced or widowed women must use their own eggs with donor sperm. |

***Recent Changes to the Provisions***

The Act nowhere mentions that a surrogate mother should be a close relative or blood relative. But the way Section 2(1) is worded, it was interpreted as if only a close relative or blood relative of the couple or the woman who wants to have a surrogate child can be a surrogate mother.

**KEY FEATURES OF SURROGACY REGULATION ACT**

1. It allows any 'willing' woman to be a surrogate mother and proposes that widows and divorced women can also benefit from its provisions, besides infertile Indian couples.
2. Surrogacy will be regulated by establishing National Surrogacy Board at the central level and State Surrogacy Board and appropriate authorities in states and Union Territories, respectively.
3. The insurance cover for the surrogate mother has now been increased to 36 months from 16 months provided in the earlier version.
4. Commercial surrogacy will be prohibited, including sale and purchase of human embryo and gametes.
5. Ethical surrogacy to Indian married couples, Indian origin married couples and Indian single woman (only widow or divorcee between the age of 35 and 45 years) will be allowed on the fulfilment of certain conditions.

## 23.19 Pronuclear Transfer/Three Parents Baby/Mitochondrial Gene Therapy

In the concept of a three-parents baby, a baby is conceived using genetic material from three people in which mitochondrial DNA (a small part in a

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cell that helps your baby to turn food into energy) from a female is used to replace faulty material mitochondria and thus prevent inherited diseases.

In this technique, doctors will replace an egg's defective mitochondrial DNA (those are present in egg cytoplasm) with a female donor's healthy DNA. This will prevent children from suffering from debilitating conditions such as muscular dystrophy, etc., which are inherited through cytoplasm.

Mitochondria is one of the cell organelles that is passed from mother to child due to cytoplasm inheritance (all male and female receive mother's cytoplasm and egg nucleus + father sperm nucleus, inside zygote). Mutations in the 37 genes in the mitochondria can produce fatal inherited diseases. There is no cure or effective treatment for many of the mitochondrial diseases.

### 23.19.1 Procedure of Mitochondrial Gene Therapy

***Where can we Produce Three-Parent Baby?***

- Clinics in Britain can now apply to create three-parent babies after the fertility final approval for the procedure.
- The British scientists created a fertility technique, which assists the doctors in replacing an egg's defective mitochondrial DNA with a female donor's healthy DNA. This will prevent children from suffering debilitating conditions such as muscular dystrophy.
- They said that procedure could go ahead when options like screening and others have proved futile.

### 23.19.2 Legalisation

The UK is the first country to legalise the procedure. Although, in 2016, the first baby was born using mitochondrial replacement in Mexico.

The world's first three-parent baby has been born in **Mexico** with the help of a controversial new fertility technique that incorporates DNA from three people. The 'three-parent' technique allows parents with rare genetic mutations to have healthy babies. **The boy's mother carries genes for Leigh syndrome, a fatal disorder that affects the brain, muscles and nerves of developing infants.** Genes for the disease reside in DNA in the mitochondria, which provide energy for our cells. The resulting egg – with nuclear DNA from the mother and mitochondrial DNA from a donor – was then fertilised with the father's sperm.

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### 23.19.3 Many Pros and Cons in Mitochondrial Gene Therapy

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- However, critics said the technique was 'dangerous', 'needless' and 'ethically reckless'.
- Dr. David King, director of the secular pro-choice watchdog group, Human Genetics Alert said 'This decision to approve experiments on babies, using this dangerous and medically unnecessary technology risks all our children's futures'.
- It opens the door to designer babies.
- Throughout the process, the campaigners for approval of this technology have relied on shameless emotional blackmail.

(Refer **Infographic: Pronuclear Transfer/Three Parents Baby**)

## 23.20 Tissue Culture

Tissue culture is a method used to **produce plants under sterile conditions** also known as **micro propagation.** In this technique, a plant explants (plant parts) or seeds are used, which have been sterilised before. They are put in containers with a growing medium (usually a gel) with some nutrients added.

Plant tissue culture enables to grow exact copies of the donor plant. Those plants that have genetically desirable traits can lead to several clones of a particular plant, at a faster rate than traditional propagation methods like cuttings, pulling or divisions.

'Toshio Murashige' (a scientific pioneer in this field), Professor Emeritus of the University of California at Riverside defined four stages of plant tissue culture.

| | |
|---|---|
| **Stage I** | Prepare an aseptic (sterile) culture. |
| **Stage II** | The multiplication of propagules (any part of a plant used to make or become new plants). |
| **Stage III** | Preparation for successful transfer of propagules to soil, for successful transfer to soil (rooting and 'hardening' |

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| | (acclimating) outside of sterile conditions in regular growing media). |
| **Stage IV** | Establishment in soil (or other appropriate growing medium). |

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***Animal Tissue Culture***

It is the biological technique in which scientists take a piece of tissue from an animal specimen, grow and maintain the cells from that tissue in an in vitro or artificial environment. Stem cell technology will also be considered under tissue culture. Contamination-free laboratory conditions should be maintained to mimic the original environment. It grows in a flask containing media, which is enriched with nutrients and growth factors to provide nutrition, which are kept in incubators that regulate temperature and humidity. Mimicking the environment in the lab allows the scientists to study them in controlled conditions.

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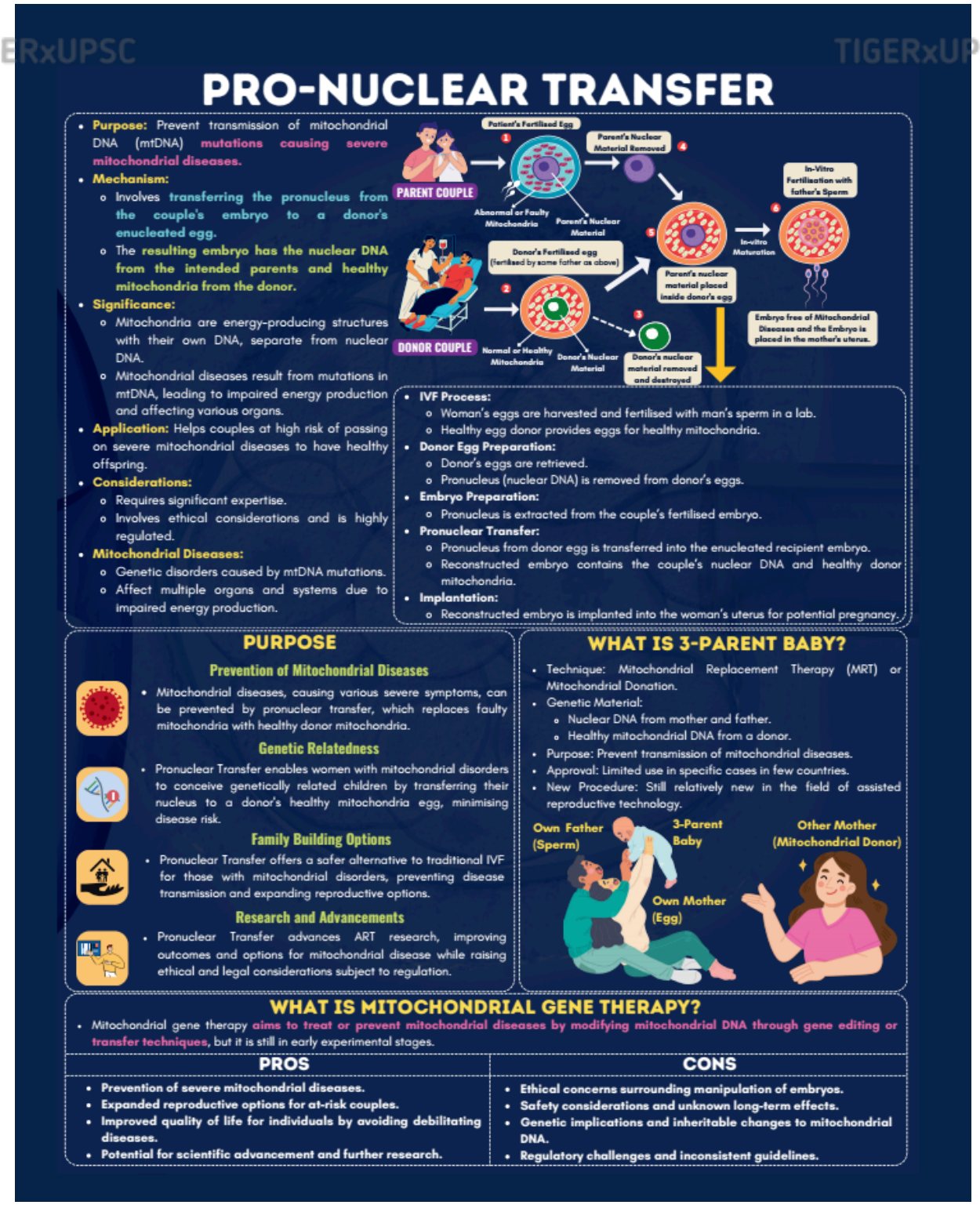


It is used for culturing cells in laboratory experiments. It is helpful in learning more about cells, organ structures and functions, diagnose viruses,

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the effects and toxicity of drugs, developing the treatments to organ failure and much more. Products developed from cell culture include vaccines, hormones, enzymes and anticancer drugs, etc.

## 23.21 Gene Therapy

In the process, genetic material is transferred into a person's cells to provide immunity against diseases. Gene therapy is an extremely useful tool for numerous diseases, like severe combined immune-deficiencies, haemophilia, Parkinson's disease, cancer, etc., and a **carrier known as a 'vector' is used to deliver a gene into a cell. Viruses are the most common type of vectors used in gene therapy.** Before using in therapy, vectors should be altered to make them safe, although some risks still exist with gene therapy.

Gene therapy is the method of introduction, manipulation, or deletion of genes within an individual's cells or biological tissues for the treatment of diseases. Most common form of gene therapy is the insertion of functional genes to replace a mutated gene, and in some cases, directly correcting the mutation or modifying normal gene to cure the diseases, such as cystic fibrosis, haemophilia, muscular dystrophy and sickle cell anaemia. Gene therapy can be classified as follows:

### *1. Germ Line Gene Therapy*

In this type of therapy, **Germ cells, (i.e., sperm or eggs), are modified by the insertion of functional genes, which are added into their genomes.** The change occurring in the genome due to therapy would be heritable and would be passed on to later generations. This new technique is highly effective to fix the genetic disorders and hereditary diseases. On the other side, many jurisdictions forbid the use of this application, in human beings, at least for the present, due to numerous technical and ethical reasons.

### *2. Somatic Gene Therapy*

**The therapeutic genes are introduced into the somatic cells of a patient, in this gene therapy.** Any modifications will be restricted to individual

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patient only, without any chances of being inherited by the next generations of the patient.

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***Problems and Ethics of Gene Therapy***

- **Short-Lived Nature of Therapy:** The therapeutic DNA transferred into target cells must remain functional and the cell containing therapeutic DNA need to be stable and long lived. Problems with inserting therapeutic DNA into the genome and the fast dividing nature of many cells check gene therapy from achieving any long-term benefits; as a result, patients undergo multiple rounds of gene therapy.
- **Immune Response:** Whenever a foreign entity is introduced into human body (cell or tissues), the immune system attacks the invader. When a foreign segment of DNA is introduced in the body for gene therapy, there is always a possibility of risk that immune system reduces the gene therapy's effectiveness. Besides this, the immune system has also improved the response against invaders that it has seen before and thus, it becomes difficult for gene therapy to be repeated in patients.
- **Problems with Viral Vectors:** Viruses are the carrier of choice in most gene therapy studies. When we choose virus as a vector for gene therapy, it shows many problems to the patient, such as toxicity, immunity and inflammatory responses, and gene control, etc. There is always the probability that the viral vector, inside the patient, may recover its ability to cause disease.
- **Chance of Inducing a Tumour:** If the DNA segment is introduced in the incorrect place in the genome, for instance in a tumour suppressor gene, a tumour can be induced.

Note: Misuse of gene therapy in the form of gene doping, which athletes resort to, for increasing their physical stamina.

## 23.22 Gene Editing and CRISPR Technology

***Gene Editing***

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The bio-information that defines any individual is contained in genes. Physical characteristics like skin or hair colour, more subtle features and even behavioural traits can be ascribed to information encoded in the genetic material.

With ability to alter this information, scientists try to control some of these features. It is not a new phenomenon but sometimes expressed with terms like genetic modification, genetic manipulation or genetic engineering.

### *CRISPR-Cas9*

The **clustered, regularly interspaced, short palindromic repeats,** or CRISPR/CRISPR-associated protein 9 (Cas9) (CRISPR-Cas9) system has made genetic manipulations and gene editing simpler, faster and easily available to most laboratories.

CRISPR technology, a gene-editing technology can be utilised for different purposes like altering genetic expression or changing the genome of an organism.

- Targeting specific stretches of an entire genetic code or editing the DNA at particular locations can be done through this technology.
- CRISPR technology is an important tool for genome editing. Now researchers will be able to easily alter DNA sequences and modify gene function.
- It can be used for correcting genetic defects, treating and preventing the spread of diseases and improving crops. However, its promise also raises ethical concerns.

### *How it Works?*

This technology works like a cut-and-paste mechanism on DNA strands that contain genetic information.

- The DNA strand identifies the specific location of the genetic codes that need to be changed, or 'edited' and then, using the Cas9 protein, which acts like a pair of scissors, that location is cut off from the strand. Once the DNA strand is broken, it has a natural tendency to repair itself.

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- During the auto-repair process, scientists intervene by supplying the desired sequence of genetic codes that binds itself with the broken DNA strand.

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### *Issues*

Study by Stanford University, US, says that unexpected off-target (outside of the intended editing sites) effects in mice have been introduced by the CRISPR-Cas9 system. There is a fear that the CRISPR system is being impulsively ran for clinical use lingers.

- Studies highlighted that edited cells might trigger cancer.
- It may also increase the risk of mutations elsewhere in the genome in those cells.
- Although CRISPR-Cas9 technology has been successfully used to cure several diseases however, there remains many a things which are not clear like how we can determine which disease or traits are suitable for gene editing.
- There are ethical concerns related to manipulation of human embryos for own interest.

### *Ethical Challenges*

- ***Bioethicists expressed concern over the clinical application of such research.*** These are still early days in a new frontier of genome engineering.
- ***Bioethicists fear abuse of gene editing***, not just by misguided governments but also by the private sector preying on a parent's desire to create a perfect child.
- It can create ***unforeseen changes in the genome which are undesirable.***
- ***It has also been ethically questioned*** whether editing gene to create babies that parents desire would make them more like commodities.
- **Safety:** Safety is of primary concern because of the possibility of off-target effects (edits in the wrong place) and mosaicism (when some cells carry the edit but others do not),
- **Informed Consent:** Some people have a concern that it is impossible to obtain up-to-date consent for germline therapy because the patients affected by the edits are the embryo and future generations.

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- Researchers and bioethicists are also concerned about the possibility of obtaining truly informed consent from prospective parents as long as the risks of germline therapy are unknown.
- **Justice and Equity:** As with many new technologies, there is concern that only wealthy people will have the reach to genome editing which will increase existing disparities in access to health care and other interventions. Some worry that a class of individuals defined by the quality of their engineered genome can be created through germline editing.

However, the CRISPR technology is a path-breaking technology, to alter genes in order to tackle a number of conventional and unconventional problems, especially in the health sector. However, experiments and tests to validate its use shall be subjected to appropriate scrutiny by the regulators, and commercial misuse must be prevented.

(Refer **Infographic – CRISPR Technology**)

## 23.23 Gene Doping

Altering genes to enhance athletic performance is the reason behind gene doping. In the era of gene therapy, a simplest way in delivering proteins and hormones to an athlete's tissues and organs turned into reality. A substance can change the basic genetic DNA expression such as large growth of muscles, quick recovery and extra forcefull contraction.

The muscles, which are not doped cannot be noticed by anti-doping laboratories, which would be perfect for taking advantage of competition. Gene doping is of two types:

**Ex Vivo Gene Doping** The method of external gene doping, consists of initial transferring of gene to cells in culture, then transferring the tissue into the recipient. Once the genes are embedded into the athlete's cell; it releases biochemicals or hormones, thereby increasing the person's performance during competitions.

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**In Vivo Gene Doping** The new gene's deliverance into an athlete can be with the help of physical, biological or chemical methods. There is biological customisation of viruses, thereby inserting the artificial gene into cells in a particular organ or tissue or into cells right through the competitor's body.

Gene doping is a hazard to the sports integrity and athletes' health, and as the **WADA (World Anti Doping Agency)**, being a worldwide organisation accountable for encouragement, coordination and monitoring the fight against doping in every form of sport, is providing vital resources and awareness to the methods for enabling the detection of gene doping.

## 23.24 Cloning

Cloning represents a copy of production of cells or completely desired organisms. Vegetative and asexual reproductions are colonel propagation because they produce a same organism with similar genotype and phenotype. There are different levels of cloning:

1. **Gene/Molecular Cloning:** This cloning is the amplification of DNA fragments that contain whole genes.
2. **Cellular Cloning:** This type of cloning is used in stem cell research, where a population of identical cells form a single cell.
3. **Organism Cloning:** In this process, a new multicellular organism is created, which is genetically identical to other organism. It is an asexual method of reproduction. Scientists have made some major achievements in this area, which includes the asexual reproduction of sheep and cows. There is a lot of ethical debate over using cloning. Dolly sheep is an example of organism cloning.

**In February 1997, Ian Wilmut and his colleagues at the Roslin Institute**, successfully cloned a sheep, from the mammary glands of an adult female. Dolly was the first mammal (although the first clones were frogs) to be effectively cloned from an adult cell. She was cloned at **Scotland's Roslin Institute** and stayed there since her birth in 1996 until

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her death in 2003. **Scientists at India's National Dairy Research Institute (NDRI), Karnal (Haryana), produced the first cloned buffalo in February 2009, using DNA from the ear of an adult female**. Unfortunately, it died of pneumonia soon after its birth.

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In genetics and developmental biology, **Somatic Cell Nuclear Transfer (SCNT)** is a method for the creation of a clonal embryo, with the usage of an ovum with a donor nucleus (Refer **Infographic – Cloning**). This application has lot of application in embryonic stem cell research and in regenerative medicine, where it is sometimes referred to as 'therapeutic cloning'. This was the first step in the reproductive cloning process.

### 23.24.1 Types of Cloning

***There are Two Types of Cloning***

(a) **Reproductive Cloning:** In this type of cloning, the egg cell is placed into a uterus after few divisions in it. The cell is then allowed to develop into a foetus that is genetically identical to the donor of the original nucleus.

(b) **Therapeutic Cloning:** The egg is kept in a petridish for developing into embryonic stem cells, showing potentials for treating several ailments.

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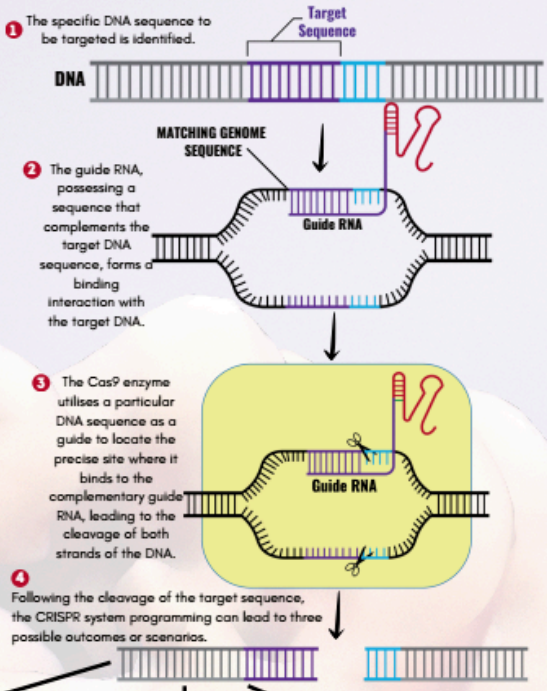
# CRISPR-CAS9

- **Definition:** CRISPR-Cas9 is a **gene-editing technology** enabling precise DNA changes in organisms.
- **Components:**
  - **CRISPR:** **Clustered Regularly Interspaced Short Palindromic Repeats, a bacterial immune defense mechanism.**
  - **Cas9:** An **enzyme** acting as **"molecular scissors" to cut DNA** at specific locations.
- **Function:**
  - Guide RNA (gRNA): **Designed to match target DNA sequence,** directing Cas9 to the specific site.
  - DNA Cutting: **Cas9 binds to target DNA** and creates a break in both strands.
  - DNA Repair: Natural repair mechanisms can be exploited to introduce desired genetic changes.
- **Applications:**
  - Medicine: Potential for treating genetic disorders, engineering cells for immunotherapy, and developing accurate disease models.
  - Agriculture: Enhancing crop yields, improving nutritional content, and increasing resistance to pests and diseases.
- **Ethical Concerns:**
  - Human Germline Editing: Raises debates about the implications of altering human traits and characteristics.
  - Regulations: Ongoing efforts by the scientific community and regulatory bodies to establish responsible usage guidelines.




## Applications of CRISPR Cas9

**Gene Therapy**

CRISPR-Cas9 enables gene editing for treating genetic disorders like sickle cell anemia, cystic fibrosis and certain cancers.

**Disease Modeling and Drug Discovery**

CRISPR-Cas9 allows precise genetic modifications for better disease models, aiding in studying disease mechanisms, drug target identification, therapy development and faster drug discovery.

**Agriculture and Food Security**

CRISPR-Cas9 enhances crop yield, nutrition and resilience, promoting sustainable agriculture and addressing food security challenges.

**Livestock Improvement**

Cas9 improves livestock by enhancing disease resistance, productivity and desirable traits through targeted genetic modifications.

**Conservation Biology**

It aids conservation by addressing genetic issues, increasing diversity and mitigating harmful mutations in endangered species.

## Ethical Considerations of CRISPR-Cas9 Gene Editing

## Future Perspectives of CRISPR-Cas9 Gene Editing

***Therapeutic Cloning/Biomedical Cloning***

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The initial stages of this procedure are similar to adult DNA cloning. It aims in producing a healthy replica of a sick person's tissue or organ for transplantation. This technique would be better to rely on for transplanting organs from other people. The tissue or organ contains the sick person's original DNA and the patient does not require to take immunosuppressant drugs for the remaining life, which is currently required during transplants. In fact, there is less danger of organ rejection.

It is also called as 'Somatic Cell Nuclear Transfer or Research Cloning'. In this technique, the resultant embryo is allowed to grow for 14 days. Its stem cells would then develop into human tissue or a complete human organ for transplant. The first successful therapeutic cloning was performed in November 2001 by Advanced Cell Technology, a biotech company in Worcester.

### *Uses of Therapeutic Cloning*

1. Overcomes the problem of immune rejection, which is a major concern in tissue transplantation or organ transplantation as patients' own cells are used (immunosuppressant will not be used anymore then).
2. Cells, which are removed (Pluripotent), can give rise to all cells in the body except embryo, i.e., it can treat diseases by replacing damaged cells.
3. It can help in studying of stem cells and future medical importance to treat common diseases like diabetes, heart disease and Parkinson's diseases.
4. It can help in understanding the process of ageing and ways to postpone it.
5. It can help in understanding the process of cancer formation.
6. It can help in plastic, reconstructive and cosmetic surgeries. Breast 'implant surgery' is possible through Therapeutic Cloning.

### *Opposing Views on Human Cloning*

The Centre for Genetics and Society also presents viewpoints for both reproductive and therapeutic human cloning.

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- President George W. Bush's council on Bio-ethics released its publication in July 2002 – *Human Cloning and Human Dignity*.
- The United States National Bioethics Advisory Committee on 9 June 1997 submitted its full report on cloning of human being. Several commissioned papers by various experts are included in the report and address scientific, religious, legal and ethical considerations. In October 2000, this committee was dissolved.

The main site to find articles in favour of human cloning with likes to news, commentary and discussion forums is reproductive cloning; however, the site now appears to be taken down.

**Religious Perspective** Different religious gurus are also against complete cloning of human beings.

(Refer **Infographic – Cloning**)

## 23.25 Stem Cells

Stem cells are those having the ability to continuously divide and differentiate (develop) into various other kind(s) of cells/tissues.

### *Properties of Stem Cells*

Stem cells differ from other kinds of cells in the body. All stem cells, no matters what their source is, show three general properties:

- These are **unspecialised cells** (blank cells).
- They have the **ability of proliferation and renewal** (These are capable of dividing and renewing themselves for long periods of time).
- **Have the potential to give rise to specialised cell types** (differentiation).

### *Types of Stem Cells*

1. **Embryonic stem cells (ESCs)**
2. **Induced pluripotent stem cells (iPSCs)**
3. **Adult stem cells**
4. **Amniotic stem cells/ Perinatal stem cells**
5. Fetal stem cells

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(Refer **Infographic – Stem Cells**)

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As per their potential to differentiate into other types of cells, stem cells are of various types:

- Totipotent
- Multipotent
- Pluripotent
- Oligopotent
- Uni potent

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# CLONING

## WHAT IS CLONING?

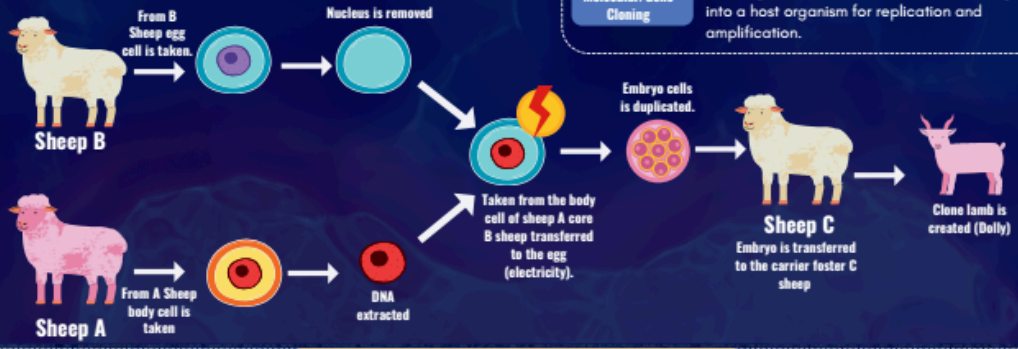
- Cloning is the **process of creating a genetically identical copy of an organism, cell or DNA** fragment.
- It involves **replicating the genetic information** of a specific individual or entity.
- Cloning can produce genetically identical copies of various biological entities.
- The copied material, having the same genetic makeup as the original, is called a clone.
- Researchers have cloned genes, cells, tissues, and entire organisms, such as a sheep.

## TYPES OF CLONING

**Reproductive Cloning** — It creates a genetically identical organism by transferring donor cell DNA into an empty egg, stimulating embryo growth, and implanting it into a surrogate mother.

**Therapeutic Cloning** — Therapeutic cloning (SCNT) produces embryonic stem cells for medical use by transferring a non-reproductive cell nucleus into an enucleated egg, providing stem cells for research and therapies.

**Molecular/Gene Cloning** — Molecular cloning replicates DNA fragments by inserting them into a vector and introducing it into a host organism for replication and amplification.




## APPLICATIONS OF CLONING

### Reproductive Cloning

- Dolly the Sheep: First mammal cloned from an adult somatic cell.
- Agricultural Cloning: Reproduces high-quality livestock with desirable traits.
- Conservation Cloning: Preserves endangered species by creating genetic replicas.

### Therapeutic Cloning

- Stem Cell Research: Generating embryonic stem cells for regenerative medicine.
- Tissue and Organ Transplantation: Providing genetically matched cells and tissues for transplantation.
- Disease Modeling: Creating animal models to study and develop treatments for diseases.

### Cellular Cloning

- Biotechnology and Drug Development: Producing recombinant proteins and therapeutic substances.
- Cloning for Research: Replicating and studying specific cell types in laboratories.

### Artificial Reproduction

- Assisted Reproductive Technology (ART): Assisting couples with infertility using donor sperm or eggs for IVF.

## ADVANTAGES

1. **Genetic Preservation:** Maintains valuable traits in plants and animals.
2. **Medical Advancements:** Aids regenerative medicine, organ transplants and disease research.
3. **Agricultural Benefits:** Enhances food production with high-quality livestock.
4. **Species Conservation:** Helps preserve endangered species and revive extinct ones.

## CHALLENGES

1. **Low Success Rates:** Inefficient reproduction limits cloning application.
2. **Ethical Concerns:** Raises dilemmas about genetic manipulation and animal welfare.
3. **Health Issues:** Clones may face health problems and disease susceptibility.
4. **Public Perception:** Varies due to concerns about safety, morality, and misuse.

## ETHICAL CONSIDERATIONS

- Human Dignity
- Animal Welfare
- Playing God
- Public Perception
- Reproductive Cloning
- Unintended Consequences
- Consent & Autonomy
- Equality and Justice

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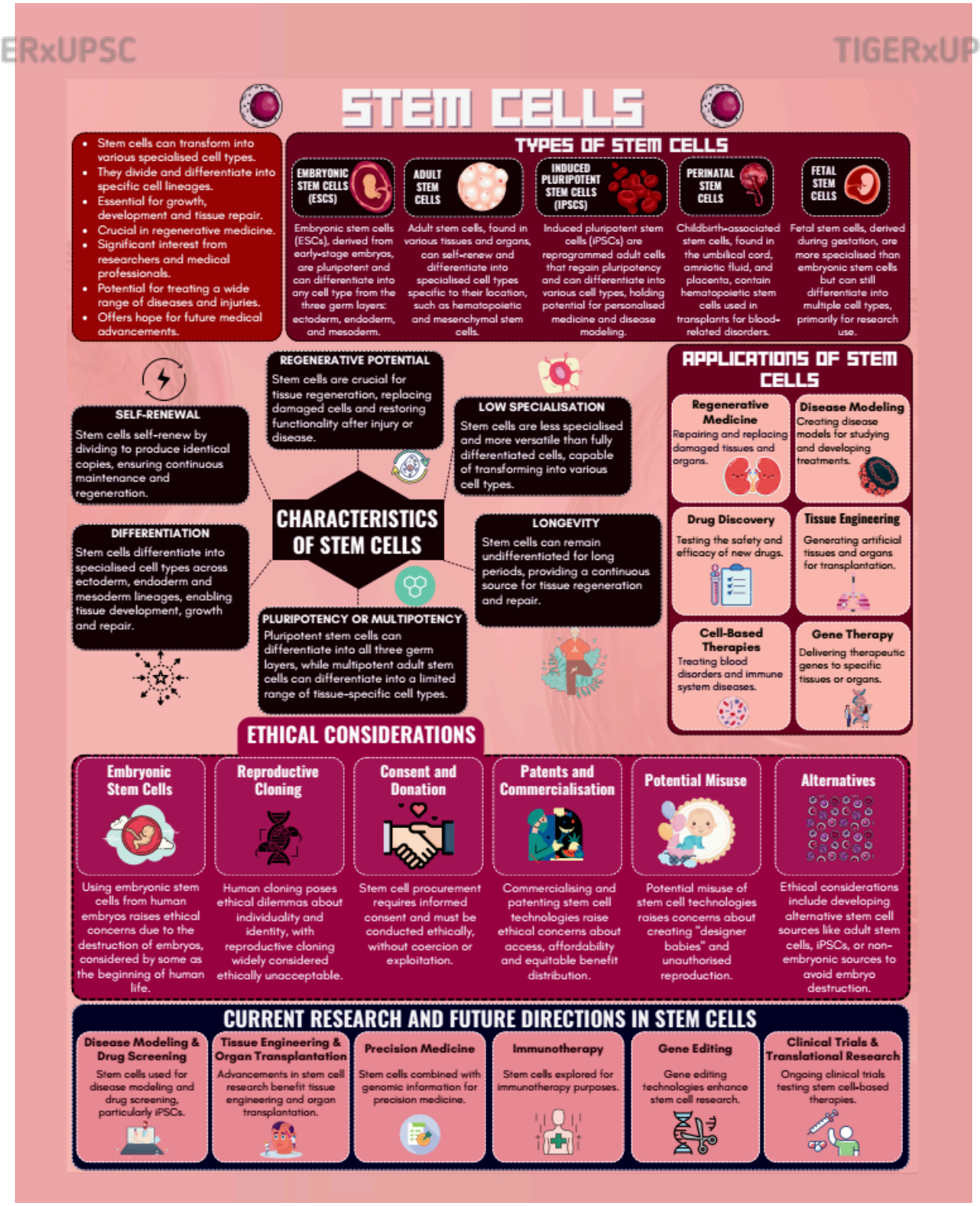


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| Stem Cell Type | Description | Example |
|---|---|---|

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| | | |
|---|---|---|
| Totipotent | Each cell can develop into a new individual | Cells from early (1–3 days) embryos |
| Pluripotent | Cells can form any (over 200) cell types | Some cells of blastocyst / inner cell mass-ICM (5 to 14 days) |
| Multipotent | Cells differentiated, but can form a number of other tissues | Foetal tissue, cord blood and adult stem cells |

**Totipotent:** These types of stem cells have the ability to develop a complete organism. The fertilised egg (zygote) and the cells produced by the first few divisions of the fertilised egg cells, all are totipotent in nature.

**Pluripotent:** These types of stem cells **can differentiate into nearly all cells,** i.e., Blastocyst-based stem cell/Inner cell mass (ICM).

**Multipotent:** These stem cells **can differentiate into many types of cells** that are closely related cells, i.e., bone marrow, haematopoietic stem cells, adipose tissue, etc.

**Oligopotent:** Stem cells can **differentiate into only a few cells like** lymphoid or myeloid stem cells, squamous epithelium.

**Unipotent:** These cells can **differentiate into only one cell type,** but have the property of self-renewal, distinguishing them from non-stem cells, though it is unclear if they actually exist or not, i.e., muscle stem cells.

### STAGES OF EMBRYOGENESIS

Embryonic stem cell lines are derived during embryogenesis process. After fusion of sperm and egg, the zygote cell starts to divide mitotically in different embryonic stages – Marula stage (4–16 cell structure – 2 days), Blastula stage (128–256 cell structure – 5 days). This blastula/blastocyst stage at which pluripotent embryonic stem cell lines are generated. After blastocyst stage, the tissues of the embryo further grow and the cells become multipotent in nature

**Derivation and Use of Embryonic Stem Cell Lines**

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The inner cell mass (the part that would form the foetus) of the embryo is separated and disrupted to form embryonic cell lines. In this process, embryo die. These cells are put under specific culture medium, to differentiate in different cell types. These differentiated cells have a lot of advantages: for repairing or replacing defective cells or tissues in human beings.

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### 23.25.1 Stem Cell Treatment

It is an intervention strategy type for inducing new cells into damaged tissue for treating disease or injury. The self-renewing ability for giving rise to consequent generations with variable capacities offers vital potential for generating tissues, which can possibly replace the body's diseased and damaged cells. The risk of rejection and side effects will also be minimal in this case.

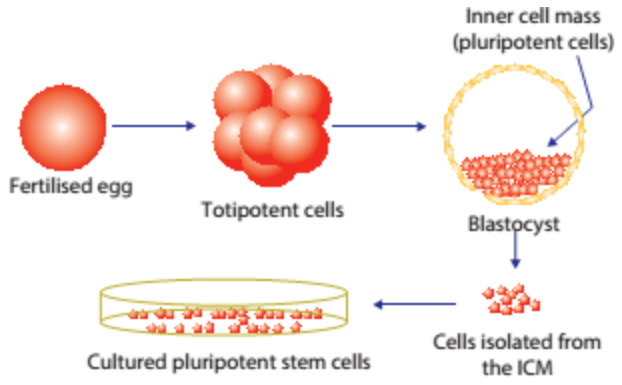


Figure 23.1 Culture of Stem Cells and Blastocyst

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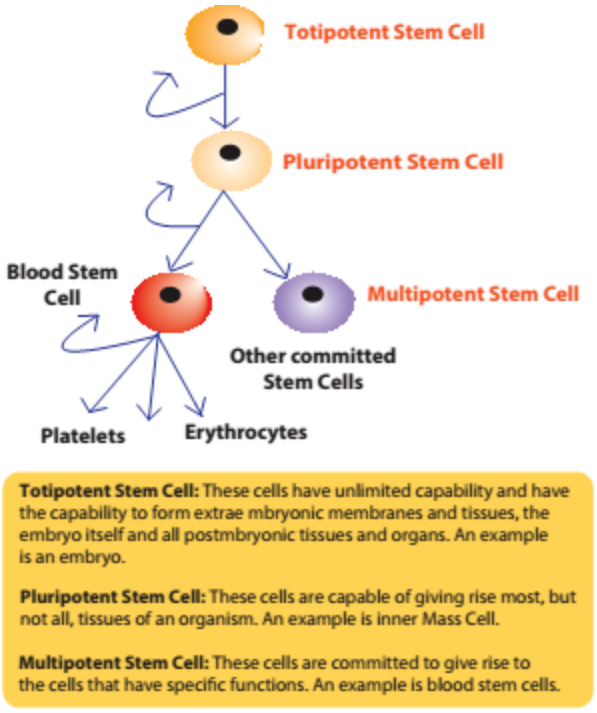


Figure 23.2 Pictorial Presentation of Totipotent, Pluripotent and Multipotent Nature of Stem Cells

Large number of stem-cell therapies are existing; however, most of them are at experimental stage or costlier, with the noteworthy exception of bone marrow transplantation. **It is anticipated that adult and embryonic stem cells will soon be able to treat cancer, Type 1 diabetes mellitus, spinal cord injury, Parkinson's disease, Huntington's disease, Celiac Disease, arthritis, leukaemia and other cancers, cardiac failure, muscle damage and neurological disorders, and many others.** Stem cells are also used in the testing of new drugs and faster development of drug.

### 23.25.2 Stem Cell Controversy

This controversy is over ethical debate due to killing of human embryos in embryonic stem cells. This is not in all other different nature of stem cell those involves in the creation, usage and destruction of human embryos. For example, adult stem cells, amniotic stem cells and induced pluripotent stem cells have no issue because using or destroying of human embryos is very rare.

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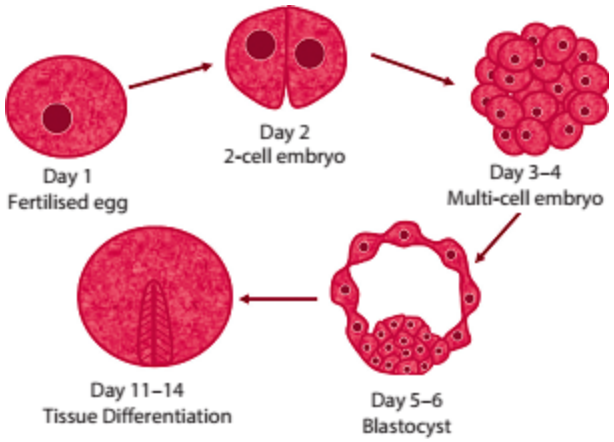


Figure 23.3 Stages of Embryogenesis

### 23.25.3 India's Top Centres for Stem Cell Research

- Institute for Stem Cell Biology and Regenerative Medicine, Bengaluru
- Centre for Stem Cell Research (CSCR), Christian Medical College (CMC) Vellore, Tamil Nadu

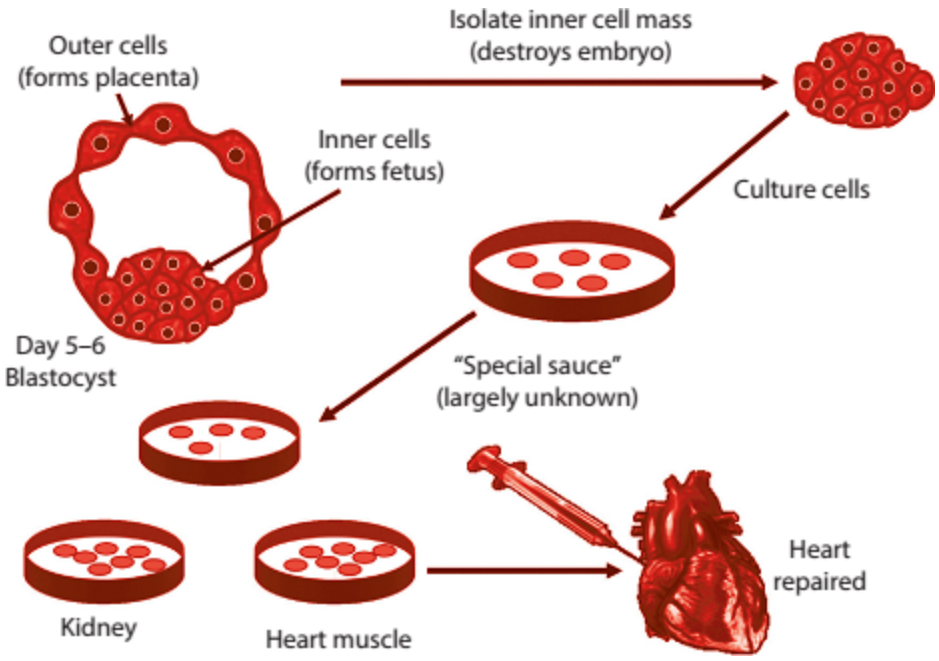


Figure 23.4 Derivation and Use of Embryonic Stem Cell Lines

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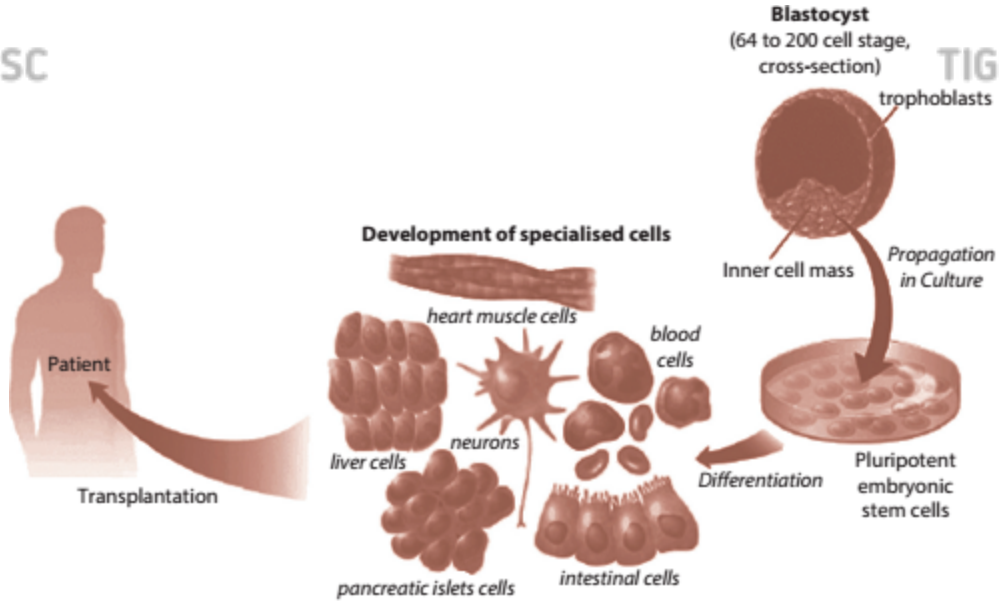


Figure 23.5 Stem Cell Treatment

- School for Regenerative Medicine, Manipal, Karnataka
- Centre for Stem Cell Science, Hyderabad
- International Centre for Stem Cell, Cancer and Bio-technology, Pune
- National Centre for Cell Science, Pune
- L.V. Prasad Eye Institute, Hyderabad

**DRAFT GUIDELINES FOR STEM CELL RESEARCH**

Indian Council of Medical Research (ICMR) and Department of Biotechnology in 2019 released revised guidelines on stem cell research 2017. **All stem cell research will evolve in an ethical and scientifically responsible manner.** These guidelines have also provision of various stakeholders in research related to human stem cells and have also options for sharing IPRs information. All guidelines will ensure through National Apex Committee for stem cell research and therapy. This committee currently monitors and oversees stem cell research activities in India.

(Refer **Infographic – Stem Cell**)

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## 23.26 Epigenetics

Concept of epigenetics was given by **Conrad Waddington** in 1942 as 'the branch of biology'. He has done his work to study the interactions between genes and their expression, those are **responsible for phenotypic expression.** It has emerged to bridge the gap between nature and nurture. In the 21st century, the 'the study of heritable changes in genome function that occur without a change in DNA sequence' is the most acceptable definition of this branch.

## 23.27 Synthetic Biology and Synthetic Life

Synthetic biology is not only considered to design and fabricate biological components and systems, which are not already existing in the natural world but also re-design and fabricate existing biological systems.

The term 'synthetic biology' was first used on genetically engineered bacteria, created with recombinant DNA technology. Later, the term 'synthetic biology' was used as a means to **redesign life.**

Diagnostic tools for diseases such as HIV and hepatitis viruses as well as devices from bio molecular parts with interesting functions have been produced by synthetic biology, which count as its major achievements so far. More trials are being made to assemble unnatural components that can support Darwinian evolution.

***Significance***

- It provides a complementary perspective to consider, analyse and ultimately understand the living world.
- It also attracts physicist, chemists and others, being an approach with which probing the behaviour of molecules and their activity inside living cells becomes easy.

***Challenges***

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1. **Opposition to Synthetic Biology:** Civil society groups oppose Synthetic Biology and have called for a global moratorium on developments in the field and for no synthetic organisms to be released from the lab. Taking into view the societal impacts of synthetic biology, they dub it as 'Extreme Genetic Engineering'.
2. **Safety and Security:** Besides numerous scientific and technical challenges, questions raised on synthetic biology are for bio-security, bio-safety, involvement of stakeholders and intellectual property rights. For instance, if on one hand, the study of synthetic biology may lead to more efficient ways to produce medical treatments (e.g., against malaria), on the other hand, it may also lead to synthesis or redesign of harmful pathogens (e.g., smallpox) by malicious actors. The International Association Synthetic Biology has proposed an initiative for self-regulation, which suggests some specific measures to be implemented by the synthetic biology industry, especially DNA synthesis companies.
3. **Social and Ethical:** At the SYNBIOSAFE forum, online discussion of 'societal issues' took place on issues related to ethics, safety, security, IPR, governance and public perception. In order to engage social issues, 'upstream' focus on the integral and mutually formative relations among scientific and other human practices, some efforts have been made. Such efforts render an attempt to invent ongoing and regular forms of collaboration among synthetic biologists, ethicists, political analysts, funders, human scientists and civil society activists.

### *Synthetic Life*

Synthetic life is **artificial life created from nonliving (abiotic) substances. It is artificial life created in vitro from bio-chemicals and their component materials.** It belongs to the discipline of synthetic biology. It is usually distinguished from mechanical life that usually belongs to the discipline of robotics. For synthetic life to be life, it must have the properties of self-replication (able to make copies of itself through

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cell division), and it must be capable of self-perpetuation, it must function autonomously, just as a normal life form would.

However, creating synthetic life is a great challenge and a complex task. **It entails the creation of an artificial genetic code and an artificial DNA.** It would also involve creating cellular organelles such as mitochondria, ribosome and nucleus, completely anew. Creating a fully synthetic cell would also be expensive. These constraints can be bypassed and a new organism can be created by just replacing the genome in an existing, natural cell with a different genome created by gene synthesis. This is one way around. The creation of Synthia, in 2010, marked the first instance of achieving it.

## 23.28 Hayflick Limit/Telomeres: The Biological Clock

Renowned **biomedical researcher Leonard Hayflick**, who discovered the **limit to human cell division**, has passed away. His work reshaped our understanding of **cellular aging**.

### *What Is the Hayflick Limit?*

- The **Hayflick Limit** refers to the **maximum number of times a normal human somatic cell can divide**, typically **40–60 times**.
- After reaching this limit, the cell enters a **senescent phase (gradual decline in cellular and organismal functions),** followed by **programmed cell death (apoptosis)**.

### *Three Phases of Cell Division (Hayflick's Model)*

1. **Phase I—Rapid Growth:** Healthy and active cell division.
2. **Phase II—Slowed Division:** Mitosis becomes less frequent.
3. **Phase III—Senescence:** Cells **stop dividing**, but remain metabolically active. Eventually, they undergo **apoptosis**.

### *Telomeres: The Biological Clock*

- Telomeres are **protective DNA caps** at the ends of chromosomes.
- With every cell division, **telomeres shorten**.

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- Once critically shortened, they **signal the cell to stop dividing**—this enforces the **Hayflick**

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**Limit:**

- Though **shortened telomeres are associated with aging**, their exact role in **lifespan determination** is still under study.

**Significance:**

- Key to understanding **aging**, **cancer biology** and **age-related diseases**.
- Forms the foundation of research in **regenerative medicine** and **telomere-targeted therapies.**

**Regenerative medicine is a field focused on repairing or replacing damaged tissues and organs in the human body to restore or improve their function.**

## 23.29 BioE3 Policy (Biotechnology for Economy, Environment and Employment)

### *BioE3 Policy (2024)*

- ***Approved By:*** Union Cabinet
- ***Implemented By:*** Department of Biotechnology (DBT)
- ***Goal:*** Achieve a US $300 billion bioeconomy by 2030 through sustainable, innovation-driven biomanufacturing.

### *What Is Biomanufacturing?*

The **use of biological system like engineered microbial, plant or animal cells** to produce **commercially valuable products** (e.g., enzymes, bioplastics, proteins) like pharmaceuticals, food additives, biofuel, etc., **with precision and scale.**

### *What Is Bioeconomy?*

**Defined by FAO** as the **production, use and conservation of biological resources** (including knowledge and technologies) to deliver products and services in sectors like:

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- Agriculture and aquaculture

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- Bioplastics and biodegradable clothing
- Functional foods, medicines and biofuels

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### *Key Features of BioE3 Policy*

| Feature | Description |
|---|---|
| **Focus Areas** | Supports R&D and entrepreneurship across six thematic sectors |
| **Target** | Sustainability, food security, climate resilience and bioeconomic growth |
| **Vision** | High-performance biomanufacturing with global standards and local innovation |

### *Six Thematic Sectors of Biomanufacturing*

1. Bio-based Chemicals and Enzymes
2. Functional Foods and Smart Proteins
3. Precision Biotherapeutics
4. Climate Resilient Agriculture
5. Carbon Capture and Utilisation
6. Futuristic Marine and Space Research

### *Catalytic Enablers*

- **Bio-AI Hubs:** Use AI with genomics, proteomics and imaging for disease research, agriculture and biomedicine.
- **Biomanufacturing Hubs:** Shared pilot and manufacturing facilities for startups and researchers.
- **Regulatory Frameworks:** Harmonize biosafety, biosecurity and global compliance.
- **Data Governance:** Promote **open science** while protecting **IPR**.

### *Why BioE3 Is Needed?*

| Challenge | How BioE3 Helps |
|---|---|
| Sustainability | Enables green, bio-based production alternatives |

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| | |
|---|---|
| Nutrition & Food Security | Supports smart proteins, soil microbiome studies & crop resilience |
| Healthcare Revolution | Promotes gene therapy, mRNA therapeutics & monoclonal antibodies |
| Climate Mitigation | Focus on CO- capture and microbial conversion |
| Space Missions | Develops space-friendly nutrition & bio-based packaging |
| Skill Gaps | Bio-hubs as training centres for bioengineering & AI-biology skills |

***Other National Initiatives Boosting Bioeconomy***

- National Mission on Bioeconomy (2016)
- National Biopharma Mission (2017)
- National Biotechnology Development Strategy (2015–2020)
- National Policy on Biofuels (2018)

***Way Forward***

- Adopt **Circular Bioeconomy** principles (reuse, repair, recycle)
- Learn from **US startup support** for scaling biotech
- Enable **Single Window Clearance** for biomanufacturers
- Retain top **STEM talent** in India
- Expand **international collaborations** with countries like USA, Japan and Finland

# Part (C): Biological Terminology

## 23.30 Biological Terminology

### 23.30.1 Biosensor

It is an analytical device, comprising a biological component and a physicochemical detector component, which is used for the detection of an analyte.

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In this process, a **sensitive biological element** is used (tissue, microorganisms, organelles, cell receptors, enzymes, antibodies, nucleic acids, etc.) under the **transducer or the detector (it is an electronic or signal processor)** and appears in the form of certain signals for detection.

***Features of a Biosensor***

- Small size that should be appropriate for use
- Cheap and producible in large quantity
- Rapid result
- Economical
- Blood glucose biosensor is an example of commercial biosensor. It breaks the blood glucose down by using the enzyme glucose oxidase
- Industry – process monitoring and control, particularly food and drink
- Medicine – diagnostics, metabolites, hormones
- Military – battlefield monitoring of poison gases, nerve agents and people
- Domestic – home monitoring of non-acute conditions

***Some Examples are***

- Sugar monitoring in diabetic people
- Used to solve environmental problems like detection of pesticides
- Remote sensing of airborne bacteria in counter-bioterrorist activities
- Pathogens detection
- Determining of intensity of toxic substances before and after bioremediation
- Presence of organophosphates
- Detection of toxic metabolites such as mycotoxins
- Pregnancy test (Prega news, etc.)

## 23.30.2 Biochip

Biochip was developed in 1983 for the monitoring of fisheries. It is an outcome of the field of computer science, electronics and biology.

- It is a bio-security device for exact tracking of information.
- A single electronic card may replace everything in the wallet including, cash, credit cards, ATM card, ID cards, insurance, medical records.
- It is a microchip of a computer, used for identification purposes.

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- It is a radio frequency identification (RFID) system and works on low-frequency radio signals for the communication between the bio-chip and the reader. It is inserted under the skin for tracking purpose.

***Applications***

1. A person/animal can be tracked anywhere in the world.
2. Stores and updates data of financial, medical, demographics of a person.
3. Used as secured e-commerce systems.
4. Medicinal implementations (used as glucose detector, Biochip as oxygen sensor, used as blood pressure sensor).

The development of this technique is a major push of the rapidly growing biotechnology industry, which encompasses a wide range of research, which includes genomics, proteomics, and pharmaceuticals, etc. Genomics is a branch of genetics dealing with the study of genomes of organisms.

### 23.30.3 Bio Signature

Any substance such as an element, isotope, molecule or phenomenon providing scientific proof of past or present life is known as a bio-signature. Some related branches such as geo-chemistry, geo-biology, and geo-microbiology frequently use bio-signatures to determine the presence of living organisms in a sample.

Bio-signatures, which are produced from life process are nucleic acids, lipids, proteins, amino acids, kerogen-like material and different morphological features. Microbes that participate in the geochemical processes leave some features in the rock record, indicative as bio-signatures. A potential bio-signature is a phenomenon which can be produced from living sources, but it can also be originated from abiotic sources.

For example, the types of bacteria and archaea living in particular environment can be detected by measuring a sample of fatty acids. The long-chain fatty alcohols, which consist of more than 23 atoms are

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produced by planktonic bacteria. Geochemists used the term biomarker for such types of indications.

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***Atmosphere***

Over the past many years, a fog of chemicals were created in the process of life, similar to the one in an ordinary chemical equilibrium. For instance, life on Earth generates a large amount of oxygen and small amount of methane. According to some scientists, life on Mars could be detected by measuring the ratio of hydrogen and methane levels on the planet, as the presence of methane in the atmosphere, gives a hint that there must be an active source on the planet, because it is an unstable gas.

## 23.30.4 Bio Generics

Bio generics are the biological drugs made from living organisms and are usually used to treat serious diseases such as cancer, HIV and AIDS and Alzheimer's disease, etc.

These drugs are normally very expensive to produce and purchase because they are derived from the living organisms. In India, bio generics is a booming sector with a very high growth rate. Domestic companies are manufacturing recombinant insulin, monoclonal antibody (MAb) products, interferon alpha, erythropoietin and various types of recombinant vaccines. All the Western-made bio-pharma products are very highly priced, which make them unaffordable to most of the world. Some of the Western-made bio-pharma products will nearly lose their patent protections and India wants to grasp this opportunity. Consequently, the government has streamlined its few drug application and also reviewed some procedures, further providing strong tax incentive to the Indian Biotech Industry. Now, industries are heavily investing in the development of facilities. Very soon, Indian bio generics makers are expected to obtain their first approval for manufacturing products for the European market.

Various Indian companies like Dr Reddy's, MJ Biopharma, Biocon, Intas Pharmaceuticals, etc., have the capability to build and produce recombinant biotech products such as Hepatitis B vaccine, streptokinase, human insulin, interferon and MAbs (Monoclonal Antibodies) for cancer and other diseases.

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### 23.30.5 Bio Computer

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A Bio computer is a biologically derived molecular (e.g., DNA and proteins) system for computational calculations. These calculations are storing, retrieving and processing of various data. A pathway or series of metabolic pathways, which involve biological materials that are engineered to perform in a certain manner based upon the conditions (input) of the system, all come under bio computer. An output is constituted by the resulting pathway of reactions taking place, based on the engineering design of the bio computer. It can also be interpreted as a type of computational analysis.

Nano biotechnology provides the means to synthesise the multiple chemical components essential for creation of such a system. The sequence of amino acids – the chemical building blocks of proteins – dictates the chemical nature of the proteins and a specific sequence of DNA nucleotides – the building blocks of DNA molecules – in turn, dictate these sequences. Also, the synthetically designed DNA molecules themselves may function in a particular bio computer system. Thus, implementing Nano biotechnology to design and produce synthetically designed proteins and the design and synthesis of artificial DNA molecules can enable the construction of functional bio computers, e.g., Computational Genes.

Researchers in Nano-medicine hope to build molecular-scale computing devices that could be embedded in our bodies to monitor health and treat various diseases before its progress. This computer is made up of biological materials, so it has the ability to speak about biochemical feature of life.

#### *Economics*

Bio computers can be highly efficient and relatively inexpensive because all of the necessary proteins can be synthesised inside a biological cell from a single DNA molecule, which could itself be replicated many times over. We can produce bio computers in large quantities from cultures, without any machinery assemblage.

#### *Advance Level of Bio Computer Technology*

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Currently, there is existence of bio computers with various functional capabilities, including logic operation and mathematical calculations. The first suggestion of a biochemical computing scheme was given by **Tom Knight** of the MIT Artificial Intelligence Laboratory. As per him, protein concentrations will work as binary signals and will perform logical operations. With the help of this method, logical operations can be performed by Bio computers. Hence, in other words, the appropriate binary output serves as a logically derived conclusion from a set of initial conditions, serving as premises from which the logical conclusion can be derived. Apart from these logical operations, bio computers can also be used for the demonstration of mathematical computations.

### 23.30.6 Bioplastic or Biopolymer (Biodegradable Plastic)

Bioplastics are composed of chemical compounds, which are synthesised from microbes such as bacteria or by genetically modified plants. Conventional plastics are produced from petroleum products but bioplastics are formed from renewable resources and are biodegradable.

In 1926, with a bacteria *Bacillus megaterium,* the first known bioplastic, **Polyhydroxybutyrate** (PHB) was developed. The petroleum crisis of the mid-1970s drew attention towards finding alternate options to petroleum-based products. It gave rise to molecular genetics and recombinant DNA technology.

Various renewable and non-renewable sources are used as feedstock sources for modern plastic materials. Non-renewable sources produced petroleum-based plastics and are unbreakable by glass or carbon. Renewable resources include microbial-grown and starch-extracted polymers. These materials can be strengthening with the help of natural fibres like flax, jute, hemp and other cellulose.

**Polyhydroxyalkanoate** (PHA) and **Poly Lactic Acid** (PLA) are also bioplastics. Starch-based fermentation produces lactic acid by bacterial activity. Polymer of lactic acid is a type of bioplastic, i.e., PLA.

Although biodegradable plastics are scientifically sound, they required a better infrastructure to expand for commercial use. For the further growth

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of biopolymer, there is a need to change the current view of the society by making them realising their environmental responsibility.

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## 23.30.7 Biological Weapons and Bioterrorism

Biological weapons are artificially manufactured toxic substances, which are produced from pathogenic organisms. They are intentionally used to disrupt a host's biological process. The main aim of these biological weapons is to kill or injure the host. The major targets of these weapons are living organisms such as humans, animals or vegetation. They also have the capacity of contaminating the non-living substances such as air, water and soil.

Some characteristics, which are considered before choosing an agent as a biological weapon are high toxicity, easy availability, cheap, easy transferability from one person to another, easy dispersion in aerosol form, and have no existing vaccine. It is difficult to find a dispersing medium for biological weapon; it can be done through aerosol, but this can be clogged while spraying. Dispersal through air is also difficult as it can be washed through rain or become ineffective when came in contact with UV rays. It can also be distributed by attaching it with explosive bombs, but with this method, microbes might probably explode as well.

### *Bioterrorism and Biological Warfare Agents*

Bioterrorism is considered as a criminal act against any unsuspecting civilians, which uses pathogenic biological agent, e.g., biological warfare agents. These biological warfare agents are generally colourless, odourless microorganisms (bacteria, viruses, fungi) or toxins, which are derived from microorganisms. It can disperse in air as aerosols or in edible items to infect mass population. They cannot be detected easily even after their release. Such attacks cause chaos and panic within a civilian population and terrorists can easily escape. The main purpose of these terrorists is to overwhelm emergency medical departments at local levels.

## 23.30.8 Bioleaching

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In the process of bioleaching, bacterial microorganisms are administered to extract precious metals, such as gold, from ore, in which it is embedded. The bacteria feed on nutrients in minerals, and separate the metal, which can be collected in a solution. It is a very eco-friendly method to extract the metals. Even if the concentration of the metal is very low and difficult to extract through smelting or roasting, bioleaching can easily extract the metals. Microorganisms act as a catalyst to speed up natural processes inside ore. Through oxidation (a chemical process), bacteria turn metal sulphide crystals into sulphates and pure metals.

### *Advantages of Bioleaching*

- **Economical:** Bioleaching is simple, cheaper and easy to operate and need few specialists for operating complex chemical plants. This technique is more profitable, due to the fact that necessary curbing of sulphur dioxide emissions at the time of smelting are costly.
- **Environmental:** As compared to the conventional extraction methods, it is more eco-friendly. For the company, there is occurrence of little damage to countryside due to the natural growth of bacteria; hence, the area around the mine and surrounding can be left without contamination. Bacteria are easily cultivated and recycle because they reproduce in the conditions of the mine.
- **Ore Concentration:** Bioleaching is useful in extracting metals from ores that have poor concentration of metal. It also replaces the extensive crushing and grinding process that make the process cost efficient and also reduce energy consumption.

### *Disadvantages of Bioleaching*

- **Economical:** It is a very slow process in comparison to smelting, thereby making it lesser profitable as well as the cash-flow delay.
- **Environmental:** Occasionally, the process produces toxic chemicals. Sulphuric acid and $H^+$ ions, so formed, can escape into the ground, thereby making the surface water acidic, causing damage to the environment. During acid mine drainage, heavy ions such as iron, zinc and arsenic are also leaked and cause 'Yellow Boy' pollution, when the pH of this solution rises.

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- For metals such as copper, bioleaching is not always economically viable or fast enough, even with its low cost.

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Table 23.1 List of biological organisms that may potentially be used as biological weapons

| Microbe | Natural Environment | Target Host | Mode of Contact | Diseases/Symp |
|---|---|---|---|---|
| *Bacillus anthracis* (Anthrax) | Soil | Humans, domestic animals | Open wounds, inhalation | Pulmonary anthrax septicemia, flu-like symptoms |
| Protein Toxin (RICIN) | Extracted from castor bean plants | Humans | Contaminated food or water, inhalation, injection | Severe abdominal pai watery and bloody dia vomiting, weakness, f cough, and pulmonary oedema |
| *Clostridium botulinum* (Botulism) | Soil | Humans | Contaminated food or water | Food poisoning |
| Smallpox (Varicella) | Eradicated from nature, now obtained from | Humans | Direct contact with bodily fluids or | Persistent fever, vomit rash on tongue and in mouth, rash and bumr |

## 23.30.9 Biomarker

**A biomarker is a naturally occurring substance, molecule, gene, etc.,** which is used as an indicator of a biological state, such as biological processes, pathogenic processes, or pharmacologic responses to a therapeutic intervention. It is used as a constructive tool in various scientific fields, which are of specific interest in the study of dose optimisation.

***Applications***

**In Medicine:** A biomarker can be a substance that is transferred into an organism to observe the function of the organ or other aspects of health; e.g., in the evaluation of heart muscle perfusion, a radioactive isotope, rubidium chloride is used.

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The detection of a particular substance in a body points towards a specific state of disease, like the presence of an antibody points towards an infection. By observing the change in the protein expression, scientists can correlate the severity off a disease and according to the progress, give treatment. Another application of biomarker is to indicate the presence of various environmental substances in epidemiology and toxicology.

In cell biology, a molecule acts as a biomarker for the detection and isolation of a particular cell type; for example, **to identify embryonic stem cells, the protein Oct-4 is used as a biomarker.** In genetics, a DNA sequence is identified as a genetic marker that causes disease or is associated with susceptibility to disease.

### 23.30.10 Bioinformatics

Bioinformatics is an inter-disciplinary approach, which includes application of computer science and information technology within the field of molecular biology. The term was given by **Pauline Hogeweg** in 1979. It is the development of databases, algorithms, computational and statistical observations for solving formal and practical problems that are faced during the managing and analysis of biological data. It is a mathematical and computing approach used to understand biological processes.

Commonly, bioinformatics process consists of the study and mapping of DNA and protein sequences, various DNA and protein sequences alignment for comparative study and screening and formation of 3D models of protein structures. The Human Genome Project has increased importance of Bioinformatics. The research will help develop and succeed in sequence alignment, prediction of protein structure, gene expression and protein – protein interaction, genome-wise relationship studies and other areas.

Bioinformatics differs significantly from biological computation and computational biology. Bio-engineering and biology is used by biological computation to build biological computers. On the other hand, bioinformatics uses computation to understand biology in a better manner. Bioinformatics undertakes the organising and analysing of basic biological data, whereas computational biology produces theoretical biological systems models.

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**TRANSCRIPTOME**

A gene is a specific DNA base-pair sequence. During a character (protein) synthesis, it transcribed into RNA (i.e., transcription) and then translates into protein synthesis (i.e., translation).

There are three types of RNA – messenger RNA (mRNA), transfer RNA (tRNA) and ribosomal RNA (rRNA). All RNAs play a significant role in producing proteins within the cytoplasm. At the time of this process, some DNA can be transformed into other types of RNA that are not coded protein but still may influence cell structure and regulating genes.

Analysis of the entire collection of RNA sequences in a cell is called transcriptome. Transcriptome is also an integral part of bioinformatics. Scientist can determine on and off of a gene to know its expression in living beings. Even in human beings, every cell has the same gene but expression in different cell varies. By the study of transcriptome, we can understand various normal and abnormal characters in human beings that can also reflect or contribute in any disease. Transcriptome data gives researchers a good place to start searching for a newly found gene's function.

National Human Genome Research Institute (NHGRI) in India has participated in two transcriptome-based projects that created various resources for researchers around the world:

1. Mammalian Gene Collection initiative
2. Mouse Transcriptome Project

The Mammalian Gene Collection initiative will setup a free, public library of human, mouse and rat mRNA sequences.

***Applications***

1. Storage of DNA sequences.
2. It helps track evolution of organisms by measuring changes in DNA.
3. Through sequence analysis, we are obtaining enough sequence data to map genes of complex diseases such as diabetes, infertility or Alzheimer's disease.
4. Helps in analysis of mutation in cancer.
5. Expression of genes determined by measuring mRNA levels.

## 23.30.11 Biosparging

Biosparging is a technology for fighting ground water pollution by using naturally occurring **microorganisms to biodegrade pollutants in soil layers above groundwater aquifers.** Biosparging is a groundwater

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remediation technique. Groundwater remediation enables extraction in which groundwater pollution is to be filtered. Contaminants found in groundwater cover the entire variety of physical, inorganic and organic chemical, bacteriological and radioactive parameters.

**In biosparging, air and nutrients are inserted into the soil layers for increasing the biological action of the naturally occurring microorganisms.** These microorganisms digest the pollutants as food and hence decrease their concentrations. Its usage is undertaken for reducing the concentrations of petroleum components, which dispersed in groundwater and the layers of soil.

However, it was recently discovered that the microorganisms used in biosparging can suffer from the high temperature caused by their own metabolic activities. This can slow down the cleanup of the groundwater and even lead to continuation of the spread of contamination. It has sparked a rethink on the way groundwater remediation systems are designed.

## QUESTIONS AND ANSWERS

### Objective Questions

1. **Aerial metagenomics' best refers to which one of the following situations?** **[UPSC (Pre) – 2023]**

   (a) Collecting DNA samples from air in a habitat at one go

   (b) Understanding the genetic makeup of avian species of a habitat

   (c) Using air-borne devices to collect blood samples from moving animals

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(d) Sending drones to inaccessible areas to collect plant and animal samples from land surfaces and water bodies.

**2. 'Microsatellite DNA' is used in the case of which one of the following?** **[UPSC (Pre) – 2023]**

(a) Studying the evolutionary relationships among various species of fauna.

(b) Stimulating 'stem cells' to transform into diverse functional tissues

(c) Promoting clonal propagation of horticultural plants.

(d) Assessing series of drug trial in a population

**3. Consider the following statements in respect of probiotics:** **[UPSC (Pre) – 2022]**

1. Probiotics are made of both bacteria and yeast.
2. The organisms in probiotics are found in food we ingest but they do not naturally occur in our gut.
3. Probiotics help in the digestion of milk sugars.

Which of the statements given above is/are correct?

(a) 1 only

(b) 2 only

(c) 1 and 3

(d) 2 and 3

**4. Consider the following statements: DNA Barcoding can be a tool to** **[UPSC (Pre) – 2022]**

1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

(a) 1 only

(b) 3 only

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(c) 1 and 2
(d) 2 and 3

**5. With reference to recent developments regarding 'Recombinant Vector Vaccines', Consider the following statements:** **[UPSC (Pre) – 2021]**

1. Genetic engineering is applied in the development of these vaccines.
2. Bacteria and viruses are used as vectors.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**6. In the context of hereditary diseases, consider the following statements:** **[UPSC (Pre) – 2021]**

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondrial replacement therapy either before or after in vitro fertilisation of egg.
2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**7. Bollgard I and Bollgard II technologies are mentioned in the context of** **[UPSC (Pre) – 2021]**

(a) clonal propagation of crop plants
(b) developing genetically modified crop plants
(c) production of plant growth substances

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(d) production of biofertilisers

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**8.Consider the following statements:** **[UPSC (Pre) – 2020]**

1. Genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent.
2. A person's genome can be edited before birth at the early embryonic stage.
3. Human induced pluripotent stem cells can be injected into the embryo of a pig.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 and 3 only
(c) 2 only
(d) 1, 2 and 3

**9. In the context of recent advances in human reproductive technology, 'Pronuclear Transfer' is used for** **[UPSC (Pre) – 2020]**

(a) fertilisation of egg in vitro by the donor sperm
(b) genetic modification of sperm producing cells
(c) development of stem cells into functional embryos
(d) prevention of mitochondrial diseases in offspring

**10. With reference to the recent developments in science, which one of the following statements is not correct?** **[UPSC (Pre) – 2019]**

(a) Functional chromosomes can be created by joining segments of DNA taken from cells of different species.
(b) Pieces of artificial functional DNA can be created in laboratories.
(c) A piece of DNA taken out from an animal cell can be made to replicate outside a living cell in a laboratory.
(d) Cells taken out from plants and animals can be made to undergo cell division in laboratory petri dishes.

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**11. 'RNA interference (RNAi)' technology has gained popularity in the last few years. Why?** **[UPSC (Pre) – 2019]**

1. It is used in developing gene silencing therapies.
2. It can be used in developing therapies for the treatment of cancer.
3. It can be used to develop hormone replacement therapies.
4. It can be used to produce crop plants that are resistant to viral pathogens.

Select the correct answer using the codes given below.

(a) 1, 2 and 4
(b) 2 and 3
(c) 1 and 3
(d) 1 and 4 only

**12. What is Cas9 protein that is often mentioned in news?** **[UPSC (Pre) – 2019]**

(a) A molecular scissors used in targeted gene editing
(b) A biosensor used in the accurate detection of pathogens in patients
(c) A gene that makes plants pest-resistant
(d) A herbicidal substance synthesised in genetically modified crops

**13. With reference to the Genetically Modified mustard (GM mustard) developed in India, consider the following statements:** **[UPSC (Pre) – 2018]**

1. GM mustard has the genes of a soil bacterium that give the plant the property of pest-resistance to a wide variety of pests.
2. GM mustard has the genes that allow the plant cross-pollination and hybridisation.
3. GM mustard has been developed jointly by the IARI and Punjab Agricultural University.

Which of the statements given above is/are correct?

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(a) 1 and 3
(b) Only 2
(c) 2 and 3
(d) 1, 2 and 3

**14. What is the application of Somatic Cell CJ Nuclear Transfer Technology?** **[UPSC (Pre) – 2017]**

(a) Production of bio-larvicides
(b) Manufacture of biodegradable plastics
(c) Reproductive cloning of animals
(d) Production of organisms free of diseases

**15. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future?** **[UPSC (Pre) – 2017]**

(i) Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
(ii) This technique helps in reducing the time required to develop new varieties of crop plants.
(iii) It can be used to decipher the host–pathogen relationships in crops.

Select the correct answer using the codes given below.

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (ii)
(d) All of these

**16. In the context of the developments in bioinformatics, the term transcriptome, sometimes seen in the news, refers to:**

(a) A range of enzymes used in genome editing **[UPSC (Pre) – 2016]**
(b) The full range of mRNA molecules expressed by an organism
(c) The description of the mechanism of gene expression
(d) A mechanism of genetic mutations taking place in cell

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**17. The Genetic Engineering Appraisal Committee is constituted under the** **[UPSC (Pre) – 2015]**

(a) Food Safety and Standards Act, 2006
(b) Geographical Indications of Goods (Registration and Protection) Act, 1999
(c) Environment (Protection) Act, 1986
(d) Wildlife (Protection) Act, 1972

**18. Which of the following statements is/are correct regarding vegetative propagation of plants?** **[UPSC (Pre) – 2014]**

(i) Vegetative propagation produces clonal population.
(ii) Vegetative propagation helps in eliminating the virus.
(iii) Vegetative propagation can be practiced most of the year.

Select the correct answer using the codes given below:

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**19. Consider the following techniques/phenomena**

(i) Budding and grafting in fruit plants. **[UPSC (Pre) – 2014]**
(ii) Cytoplasmic male sterility
(iii) Gene silencing

Which of the above is/are used to create transgenic crops?

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) None of these

**20. Recombinant DNA technology (Genetic Engineering) allows genes to be transferred** **[UPSC (Pre) – 2013]**

(i) across different species of plants
(ii) from animals to plants

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(iii)from microorganisms to higher organisms

Select the correct answer using the codes given below.

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**21. Mycorrhizal biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to:**

(i) Resist drought and increase absorptive area **[UPSC (Pre) – 2013]**
(ii) Tolerate extremes of pH
(iii)Resist disease infestation

Select the correct answer using the codes given below.

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**22. Consider the following kinds of organisms:** **[UPSC (Pre) – 2012]**

(i) Bacteria
(ii) Fungi
(iii)Flowering plants

Some species of which of the above kinds of organisms are employed as biopesticides?

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) All of these

**23. With reference to 'stem cells', frequently in the news, which of the following statements is/are correct?** **[UPSC (Pre) – 2012]**

(i) Stem cells can be derived from mammals only.

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(ii) Stem cells can be used for screening new drugs.
(iii) Stem cells can be used for medical therapies.

Select the correct answer using the codes given below:

(a) (i) and (ii)
(b) (ii) and (iii)
(c) Only (iii)
(d) All of these

**24. What are the reasons for the people's resistance to the introduction of Bt-brinjal in India?** **[UPSC (Pre) – 2012]**

(i) Bt-brinjal has been created by inserting a gene from a soil fungus into its genome.
(ii) The seeds of Bt-brinjal are terminator seeds and therefore, the farmers have to buy the seeds before every season from the seed companies.
(iii) There is an apprehension that the consumption of Bt-brinjal may have adverse impact on health.
(iv) There is some concern that the introduction of Bt-brinjal may have adverse effect on the biodiversity.

Select the correct answer using the codes given below:

(a) (i), (ii) and (iii)
(b) (ii) and (iii)
(c) (iii) and (iv)
(d) All of these

**25. At present, scientists can determine the arrangement or relative positions of genes or DNA sequences on a chromosome. How does this knowledge benefit us?** **[UPSC (Pre) – 2011]**

(i) It is possible to know the pedigree of livestock.
(ii) It is possible to understand the causes of all: human diseases.
(iii) It is possible to develop disease-resistant animal breeds.

Which of the statement(s) given above is/are correct?

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(a) (i) and (ii)
(b) Only (ii)
(c) (i) and (iii)
(d) All of these

**26. A genetically engineered form of brinjal, known as the Bt-brinjal, has been developed. The objective of this is** **[UPSC (Pre) – 2011]**

(a) to make it pest-resistant
(b) to improve its taste and nutritive qualities
(c) to make it drought-resistant
(d) to make its shelf-life longer

**27. Mon 863 is a variety of maize. It was in the news for the following reason:** **[UPSC (Pre) – 2011]**

(a) It is a genetically modified dwarf variety, which is resistant to drought.
(b) It is a genetically modified variety, which is pest-resistant.
(c) It is a genetically modified variety with ten-times higher protein content than regular maize crop.
(d) It is a genetically modified variety used exclusively for biofuel production.

**28. Which bacterial strain, developed from natural isolates by genetic manipulations, can be used for treating oil spills?** **[UPSC (Pre) – 2010]**

(a) Agrobacterium
(b) Clostridium
(c) Nitrosomonas
(d) Pseudomonas

**29. Genetically modified 'golden rice' has been engineered to meet human nutritional requirements. Which one of the following statements best qualifies golden rice?** **[UPSC (Pre) – 2010]**

(a) The grains have been fortified with genes to provide three times higher grain yield per acre than other high-yielding varieties.

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(b) Its grains contain pro-vitamin A, which upon ingestion, are converted to vitamin A in the human body.

(c) Its modified genes cause the synthesis of all the nine essential amino acids.

(d) Its modified genes cause the fortification of its grains with Vitamin D.

**30. Which one of the following pairs is not correctly matched?** **[UPSC (Pre) – 2006]**

| | *Recent Scientific Achievement* | *Country* |
|---|---|---|
| (a) | Creating the world's first mouse with a full human chromosome | United Kingdom |
| (b) | Cloning a human embryo for the first time | Germany |
| (c) | Guiding a spacecraft for collision with a comet | United States of America |
| (d) | Landing a spacecraft on an asteroid | Japan |

**31. The Genetic Engineering Approval Committee, whose permission is required for cultivation of any genetically modified crop such as Bt-Cotton in India, is under the Union Ministry of:** **[UPSC (Pre) – 2003]**

(a) Agriculture

(b) Environment and Forests

(c) Commerce and Industry

(d) Rural Development

**32. With reference to the latest developments in stem cell research, consider the following statements.** **[UPSC (Pre) – 2002]**

(i) The only source of human stem cells are the embryos at blastocyst stage.

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(ii) The stem cells can be derived without causing destruction to blastocysts.

(iii) The stem cells can regenerate themselves in vitro virtually forever.

(iv) Indian research centres also created a few cell lines, which can be developed into any types of tissues.

Which of these statements are correct?

(a) All of these

(b) (i), (ii) and (iii)

(c) (i), (ii) and (iv)

(d) (iii) and (iv)

**33.** ***Assertion:*** **Scientists can cut apart and paste together DNA molecules at will, regardless of the source of the molecules.**

***Reason:*** **DNA fragments can be manipulated using restriction endonucleases and DNA ligases.** **[UPSC (Pre) – 2001]**

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

**34. The American multinational company, Monsanto has produced an insect-resistant cotton variety that is undergoing field-trials in India. A toxin gene from which one of the following bacteria has been transferred to this transgenic cotton?** **[UPSC (Pre) – 2001]**

(a) Bacillus subtilis

(b) Bacillus thurigiensis

(c) Bacillus amyloliquifanciens

(d) Bacillus globilli

**35.** ***Assertion:*** **'DNA fingerprinting' has become a powerful tool to establish paternity and identity of criminals, in rape and assault cases.**

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*Reason:* **Trace evidences such as hairs, saliva arid dried semen are adequate for DNA.** **[UPSC (Pre) – 2000]**

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

**36. Insect-resistant cotton plants have been genetically engineered by inserting a gene from a/an:** **[UPSC (Pre) – 2000]**

(a) virus

(b) bacterium

(c) insect

(d) plant

**37. Which one of the following organisms can serve as a bio-fertiliser for ricecrop?** **[UPSC (Pre) – 2000]**

(a) Blue-green algae

(b) Rhizobium sp

(c) Mycorrhizal fungi

(d) Azotobacter sp

**38.** *Assertion:* **Dolly was the first cloned mammal.**

*Reason:* **Dolly was produced by in-vitro fertilisation.**

(a) Both A and R are true and R is the correct explanation of A. **[UPSC (Pre) – 1999]**

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

**39.** *Assertion:* **Insect-resistant transgenic cotton has been produced by inserting Bt gene.**

*Reason:* **The Bt gene is derived from a bacterium.**

(a) Both A and R are true and R is the correct explanation of A. **[UPSC (Pre) – 1999]**

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(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

**40. Indian farmers are unhappy over the introduction of 'Terminator Seed Technology' because the seeds produced by this technology are expected to:** **[UPSC (Pre) – 1999]**

(a) show poor germination
(b) form low-yielding plants, despite the high quality
(c) give rise to sexually sterile plants
(d) give rise to plants incapable of forming viable seeds

## Answer Key

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (a) | 2. (a) | 3. (c) | 4. (d) | 5. (c) | 6. (c) | 7. (b) | 8. (d) |
| 9. (d) | 10. (a) | 11. (a) | 12. (a) | 13. (b) | 14. (c) | 15. (d) | 16. (b) |
| 17. (c) | 18. (c) | 19. (b) | 20. (d) | 21. (d) | 22. (d) | 23. (b) | 24. (c) |
| 25. (c) | 26. (a) | 27. (b) | 28. (d) | 29. (b) | 30. (b) | 31. (b) | 32. (d) |
| 33. (a) | 34. (b) | 35. (a) | 36. (b) | 37. (a) | 38. (b) | 39. (b) | 40. (d) |

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UNIT-X Diseases and Various Immunisation Programmes

# CHAPTER 24
# RECENT HEALTH AND WELFARE PROGRAMMES INCLUDING DIFFERENT TYPES OF DISEASES

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24.11.12 Zika Virus

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India has a disproportionately high burden of cervical cancer, **contributing to about one-fourth of global cervical cancer deaths**.


24.11.13 MERS (Middle East Respiratory Syndrome)
24.11.14 NIPAH Virus
24.11.15 Kyasanur Forest Disease (KFD)
24.11.16 Monkey Pox
24.11.17 West Nile Virus
24.11.18 Adenovirus
24.11.19 Norovirus
24.11.20 Chandipura Virus
24.11.21 Disease X (Congo)
24.11.22 Neglected Tropical Diseases
24.11.23 Vaccine Drive to Fight Cervical Cancer (Cervavac)

**24.12 Superbug ISSUES**
**24.13 Drug-Resistant Tuberculosis (TB) – MDR, MDR+, XDR, XDR+**
**24.14 New Delhi Metallo-Beta-Lactamase (NDM-1)**
**24.15 Fixed Dose Combination (FDC) Drugs**
**24.16 Generic Drugs/Generic Medicine**


***Why are they cheaper?***

Since the manufacture of these generic drugs do not involve a repeat of the extensive clinical trials to prove their safety and efficacy, it costs less to develop them. Generic drugs are, therefore, cheaper. Because the compounds in the generic versions have the same molecular structure as the brand name version, **their quality is essentially the same.**

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24.16.1 Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP)

**24.17 Biosimilar Medicine**

**24.18 Biofilm**

**24.19 Phage Therapy**

**Phage therapy is a treatment approach that utilises bacteriophages, viruses that infect and kill specific bacteria, as an alternative to antibiotics.**

**24.20 Xenotransplantation**

**Part C: Different Immunisation Programmes In India**

**24.21 National Vector Borne Disease Control Programme (NVBDCP)**

24.21.1 National Anti-Malaria Programme

24.21.2 Japanese Encephalitis (JE) Control Programme

**24.22 Communicable Diseases Control Programme**

24.22.1 National AIDS Control Programme (NACP)

24.22.2 National Leprosy Eradication Programme (NLEP)

24.22.3 Revised National Tuberculosis Control Programme (RNTCP)

24.22.4 Pulse Polio Programme

**24.23 National Nutritional Programmes**

24.23.1 Mid-Day Meal Programme

**24.24 Others**

24.24.1 Universal Immunisation Programme (UIP)

24.24.2 National Rural Health Mission

24.24.3 National Nutrition Mission

24.24.4 Mission Indradhanush

24.24.5 Ayushman Bharat

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24.24.6 Poshan Abhiyan
24.24.7 'UMMID' Initiative
24.24.8 Pradhan Mantri Matru Vandana Yojana (PMMVY)

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# Part A: Conventional Diseases

## 24.1 Health, Diseases and Symptoms

Health denotes the state of complete physical, mental as well as social well-being. It is not only related to the physical and mental state of an organism but is closely linked to its environment and social well-being as well.

Disease refers to the condition when the proper functions of any plant, animal or human are disturbed and weakened. It is not simply because of the physical injuries and is generally expressed in the form of symptoms.

## 24.2 Different Types of Terminology Used on the Basis of Nature of Diseases

**Epidemic disease** is the disease that affects a community with a large number of people in it, for a given period of time.

**Endemic disease** is the disease that affects the same area or region regularly; it is fixed to particular area.

**Acute disease** is a quick, short-time course disease.

**Chronic disease** is a disease, which is slower, gradual and lasts for a longer course or duration.

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## 24.3 Congenital vs Acquired Diseases

**Congenital diseases** are innate diseases; these diseases are present from birth, which occur due to genetic variations. For example: colour blindness, etc.

**Acquired diseases** are post-featly, develop after birth.

## 24.4 Communicable vs Non-Communicable Diseases

**Communicable diseases (Infectious)** are transmissible by direct contact or indirect means with infected individuals or their discharge (by vector).

**Non-communicable diseases (Non-infectious)** are non-contagious and non-transmissible among people. These include organ or tissue malfunction.

## 24.5 Antigens vs Antibodies

***Antigens***

1. They represent large molecules of proteins or polysaccharides. In rare cases, lipids or nucleic acids can also act as antigens. Certain parts of microorganisms, others persons or animal tissues also constitute some of the antigens.
2. They are present on cell surface and lead to antibodies formation, specifically reacting with them. Antigenicity of large molecules depends upon the presence of 'Epitopes', on the surface.
3. **Haptens:** They are non-proteinaceous, acting as antigens; however, they do not induce antibody production.
4. MHC (Major Histocompatibility Complex) denotes a set of protein cell surface markers present on plasma membrane. The immune system

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uses it to identify 'self'. The antigen represents any substance that stimulates the immune system of the body.

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***Antibodies***

They are protein molecules called immunoglobulins (Ig) produced by lymphocytes. It is a molecule that specifically inactivates an antigen. Five major types of immunoglobulins (Igs) are: IgM, IgG, IgD, IgE and IgA, based on five types of heavy chains.

## 24.6 Fever

A fever, also called as a high fever or a high temperature, is not by itself an illness, but usually a symptom of an underlying condition, most often an infection.

A fever represents an inflammatory response that usually extends beyond the infection site and affects the entire body, leading to overall increase in body temperature. Usually, the inflammation of blood cells contributes to fever. The body temperature is normally regulated as well as maintained by the hypothalamus, that is an anatomical section of brain, functioning to maintain the homeostasis in the body.

Numerous conditions that can trigger fever include:

- Infections, that include the flu, pneumonia and common cold, and
- Some immunisations, like diphtheria or tetanus (in children)
- Teething (in infants)
- Certain inflammatory diseases, that include rheumatoid arthritis and Crohn's disease
- Blood clots
- Extreme sunburn
- Food poisoning
- Some medications, that include antibiotics

## 24.7 Communicable Diseases

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In these kinds of diseases, pathogens play the main role in transmitting diseases. So discussion on all pathogens is deemed necessary, for better understanding, pathogens are grouped as bacterial, viral, protozoan, helminthes, etc.

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## 24.7.1 Bacterial Diseases

### 24.7.1.1 Cholera

It is a highly communicable acute diarrheal disease caused by *Vibrio cholera,* through faecal–oral route by houseflies and it is treated by health education, proper sanitation of environment and water, drugs like tetracycline and chloramphenicol.

### 24.7.1.2 Pneumonia

Serious and inflammatory disease of lungs caused by *Streptococcus pneumonia* and *H. influenze,* characterised by rapid shallow breathing, chest pain and fever, cough. Important drugs for pneumonia are Penicillin, Streptomycin and Ampicillin.

***Pneumococcal 13-valent Conjugate Vaccine (PCV)***

- The government has launched a nationwide expansion of **Pneumococcal 13-valent Conjugate Vaccine (PCV)** under the **Universal Immunisation Programme (UIP)** as a part of **'Azadi ka Amrit Mahotsav'.**
- It was for the first time in the country that PCV would be available for universal use.
- **PCV13** protects against 13 types of bacteria that cause pneumococcal disease.

### 24.7.1.3 Typhoid

It is an acute enteric disease caused by *Salmonella typhi,* contracted through feco–oral route, which is found in the intestine of humans. Main symptoms are high fever, stool, body ache, etc. Antibiotics for typhoid are ampicillin and chloramphenicol.

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### 24.7.1.4 Leprosy

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It is also known as Hansen's disease and is caused by *Mycobacterium leprae,* which lives in the endothelial cells of RBC, affecting peripheral nerves and skin lesions and is also a chronic disease. Multi-Drug Therapy (MDT) is used for treating leprosy.

### 24.7.1.5 Tetanus

Serious bacterial diseases are caused by *Clostridium tetani,* which lives in intestine of horse, and is acquired through contaminated soil or wound. Main symptoms are lockjaw, muscle stiffness, rigidity, fever, etc.

**PNEUMOCOCCAL CONJUGATE VACCINES**

Pneumococcal is a bacterial disease, which infects lungs, blood and brain. Mostly it causes health issue to children of less than 5 years of age. Mostly risk of this disease is for 2-year-old children and old age people who are more than 65 years old. Those people have certain medical conditions and smokers are more susceptible to this disease. These diseases also include pneumonia, meningitis, sepsis, blood infection and ear infection. **Pneumococcal Conjugate Vaccine (PCV 13)** protects against 13 types of pneumococcal bacteria, whereas **Pneumococcal Polysaccharide Vaccine (PPSV 23)** protect against 23 types.

The growing resistance of *S. pneumonia* to commonly used antibiotics underlines the need for vaccines to be used to control the Pneumococcal disease. Antibiotic resistance happens when germs like bacteria and fungi develop the ability to defeat the drugs designed to kill them. The dependence on antibiotics that are not effective against drug-resistant bacteria can be reduced.

Most people who get a pneumococcal vaccine do not have any serious problems with it. These are usually mild and go away on their own within a few days, but serious reactions are possible.

### 24.7.1.6 Plague (Black Death Disease)

It is caused by *Yersiniapestis* (endoparasite of rat flea) and transmitted by fleas. Plague is primarily a disease of rats. It is of three types: (i) bubonic plague, (ii) pneumonic plague, (iii) septicemic plague.

1. **Bubonic plague** is the most common type caused when an infected rodent or flea bites. It is characterised by high fever, weakness and

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enlargement of lymph nodes.

2. **Pneumonic plague** is most lethal form and affects lungs.
3. **Septicemic plague** a life-threatening infection of the blood.

#### 24.7.1.7 Tuberculosis (TB)

***Mycobacterium tuberculosis*** causes this disease, affecting the lungs and causing pulmonary TB, but it also can affect any body parts. It is a chronic disease. TB affects animals also and it is transmitted through air by droplets, sputum. TB can be diagnosed based on radiological (-ray), biochemical (Montoux, PCR, etc.) and clinical signs, symptoms like cough with sputum, evening temperature, etc. TB can be treated by DOTS therapy in DOTS centre in RNTCP.

National Tuberculosis Control Programme (NTCP) since 1962 has been operating in the country. This did not bring good results. So, it was reviewed in 1992 by an expert committee and accordingly, the Revised National TB Control Programme (RNTCP 1), which is the Indian version or application of WHO's Directly Observed Treatment Short course (DOTS), was launched on 26 March 1997 and RNTCP II (2006).

#### 24.7.1.8 Pertussis/Whooping Cough

Pertussis is a disease of childhood and is caused by *Bordetella pertussis.* It affects respiratory passage with the characteristic 'whoop' (loud crowing inspiration) and fever. Whooping cough is spread through droplet infection and direct contact. Vaccine for pertussis is DPT for newborns.

#### 24.7.1.9 Diphtheria

It is caused by *Corneybacterium diphtheria* and antibiotics used are penicillin, erythromycin.

#### 24.7.1.10 Anthrax

It is caused by spore-forming *Bacillus anthracis.*

***Symptoms***

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1. **Cutaneous:** Itchy bump, which develops into painless necrotic ulcer.
2. **Inhalation:** Severe respiration problems and shock. It is dangerous.
3. **Intestinal:** Loss of appetite, nausea, fever, severe diarrhoea, abdominal pain, vomiting.

Table 24.1 Some diseases and their confirmatory immunological tests

| S. No. | Test | Disease |
|---|---|---|
| 1. | **Ames test** | Carcinogenicity |
| 2. | **Dick test** | Scarlet fever |
| 3. | **Mantoux test** | Tuberculosis |
| 4. | **Rose-Waaler test** | Rheumatoid fever |
| 5. | **Schick test** | Diphtheria |
| 6. | **Wassermann test** | Syphilis |
| 7. | **Widal test** | Typhoid |
| 8. | **Wayson stain test** | Plague |
| 9. | **ELISA (NS 1) Non-Structural Protein 1** | Dengue fever |
| 10. | **Lepromin test** | Leprosy |

## 24.7.2 Viral Diseases

### 24.7.2.1 Influenza (Also called flu)

Acute respiratory infection by influenza viruses *(Orthomyxoviridae)* type-A, type-B, and type-C. Both A and B-type viruses are responsible for epidemics of diseases throughout the world. Type-A has the highest antigen variation and comparatively lesser in type-B influenza, with no antigenic variability in type-C; there is no cross immunity between them. Major reservoir of influenza virus exists in animals and birds. It results in *coryza* (profound discharge of running rose), sneezing, sore throat, coughing,

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muscle pains, headache, chill and fever. Vaccines are available against influenza Type-A.

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#### 24.7.2.2 Mumps

It is caused by a ***Myxovirus,*** which effects human glands (parotid glands) and also the nervous system of human body. It has many symptoms like ear ache, stiffness in the affected side prior to the onset of swelling. Mumps may also affect the testicles, pancreas, CNS, ovaries, prostate, etc.; infection of testicles may cause sterility. In severe cases, there may be fever, headache and other constitutional symptoms, which may last 3–5 days. Vaccination is available as MMR (Measles, Mumps and Rubella).

#### 24.7.2.3 Measles/Rubeola

It is also caused by *myxovirus,* affecting respiratory tract and conjunctiva, which are transmitted by contact of fomite and droplet methods. In pre-eruptive stage (3–4 days), there is hoarseness, cough, sneezing, coryza, redness or watering of eyes (conjunctivitis), and blue-white Koplick's spots (with reddish base). Fourth day begins with itching, skin rash, rubeola starting from back of ears, reaching forehead, face (bloated and blotchy) and then body. The rashes begin to reduce on fifth or sixth day. Incubation period is 10 days from the onset of fever. Antibiotics and vitamin A are given to prevent complications. Vaccination (MMR of Mumps, Measles and Rubella) is available, which is given at the age of one. The infant carries passive immunity from the mother prior to this.

#### 24.7.2.4 General Measles (Rubella)

An acute childhood infection with low-grade fever, lymphadenopathy, which is affecting of the lymph nodes and a red and flat rash. Early pregnancy infection may amount to serious congenital defects. Immunoglobin are useful and vaccination is provided as MMR.

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#### 24.7.2.5 Small Pox

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Small pox has been eradicated globally. 'Edward Jenner' first developed vaccine against small pox.

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#### 24.7.2.6 Chicken Pox (Caused by Varicella Zoster)

Chickenpox is a fomite (that is clothes, furniture, etc.) transmitted, bursting/eruptive viral and contagious disease, which is common but rare in adults. Illness begins with slight fever, shivering and backache. There are then prodromal (symptom that starts before the onset of an illness) or transitory rashes. Uneasiness, aches and high fever accompany chicken pox. Itching and tendency to scratch is reduced by benzyl benzoate and calamine. There are no drugs and no vaccines for chicken pox. Antibiotics are given for life-long immunity.

#### 24.7.2.7 Poliomyelitis/Polio

It is caused by polio virus (have RNA nucleic acid) and transmitted through faeces, urine and nasal secretions affecting grey matter of the spinal cord, resulting in inability of the head to bend forward and stiffness of the neck. It also destroys motor nerve cells in the spinal cord. It is hence called infantile paralysis. Paralysis is accompanied by high fever, headache, chilliness and pain all over the body. Involvement of the laryngeal and pharyngeal muscles proves fatal. Oral vaccines are available (developed by Jonas Salk and Albert Sabin, 1940) to protect babies from polio attack. Public Pulse Polio Immunisation Programme is organised in India for eradicating polio.

#### 24.7.2.8 Rabies (Hydrophobia)

It is caused by Rabies virus/Lassa virus (RNA). It affects motor neurons of brains and is transmitted by the bites of carnivores. Rabies is contracted by humans through rabbit bite or saliva, dog bite, monkey, cats, etc. It leads to encephalitis, fear of water (hence called *hydrophobia*), high fever, severe headache, spasm of throat and chest, leading to death. Louis Pasteur developed vaccine against rabies.

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#### 24.7.2.9 Hepatitis

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A viral infection transmitted through faecal–oral route (Hepatitis A, E) or blood-born, affecting liver with symptoms like jaundice, abdominal pain, nausea and vomiting. Contamination of water and food may cause epidemic. Serum hepatitis (Hepatitis B) spreads by transfusion of contaminated blood and blood products. Personal cleanliness, boiled water, properly heated/cooked food are to be ensured during epidemic. Vaccine is available.

#### 24.7.2.10 Dengue Fever (Breakbone Fever)

It is an epidemic viral disease of humans caused by female tiger mosquito *Aedes aegypti,* during daytime and it is also acute tropical. Symptoms begin in 4–6 days, like sudden fever, headaches, joint and muscle pain, pain behind eye, vomiting, fatigue, nausea, skin rashes and mild bleeding (nose bleeding, bleeding in gums).

In severe cases, a person will suffer with haemorrhagic fever, a rare complication characterised by high fever. It damages lymph nodes and blood vessels in human body, bleeding from the nose and gums, swollen liver and also failure of the circulatory system. Due to progressive symptoms in dengue with massive bleeding, shock and death, it is also called as Dengue Shock Syndrome (DSS). Those people have weakened immune system and second offset of infection has maximum chances of haemorrhagic fever. People suffer with the loss of platelets in dengue. 'You and Me' campaign is related with Dengue.

Note: Female *Aedes aegypti* carries and transmits many diseases like Zika, dengue fever, chikungunya and yellow fever.

#### 24.7.2.11 Yellow Fever

Yellow fever is a haemorrhagic disease caused by arbovirus, infected by mosquito *Aedes aegypti.* Max Theilder, in 1951, got Nobel Prize for the development of vaccine for yellow fever.

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### 24.7.3 Protozoan Diseases

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Types of protozoan diseases are Malaria, Amoebiasis, kala-azar, Trypanosomiasis, etc.

#### 24.7.3.1 Malaria (Means Bad Air)

It is the most common protozoan diseases caused by the *Plasmodium sp. (belonging to class Sporozoa)* and transmitted by the bite of female *Anopheles* mosquito in human.

This disease is characterised by recurring rigours lasting 6–10 hours. There are three stages:

1. **Cold Stage:** Chill and shivering.
2. **Hot Stage:** Temperature rises to 100°F.
3. **Sweating Stage:** There is perspiration and gradual fall in temperature.

Malaria results in anaemia, toxaemia and splenomegaly (enlarged spleen).

***Measures Against Parasites of Malaria***

Originally, quinine derived from the bark of a Peruvian tree 'Cinchona' was used and hence, it served to be an old remedy to cure malaria.

These days synthetic drugs chloroquine phosphate and primaquine are used.

More effective than these is daraprim. It is slow in action, but in due course of time, kills the parasites both in the blood and the liver, and also in the mosquito that feeds on the blood of an infected person. Gambusia fish is also used in the protection of malaria because it eats up malaria larvae in ponds.

***Karnataka Sets 2027 as the Target to Become Malaria-Free***

- Karnataka has set a target to eliminate malaria by 2027, three years before the 2030 target set by the Union government.
- Karnataka has received national recognition and appreciation for its efforts to eliminate malaria in the past six years, between 2015 and 2021, as part of **the National Framework for Malaria Elimination in India (NFMEI) initiative.**

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- **China is the first country in the WHO Western Pacific Region** to be awarded a malaria-free certification **in more than three decades.**
- Other countries in the region that have achieved this status include Australia (1981), Singapore (1982) and Brunei Darussalam (1987).
- Globally, **43 countries and territories have been granted a malaria-free certification from WHO –** most recent being Cabo Verde (also known as Cape Verde) (2023).

***Mosquirix: The First Malaria Vaccine***

- **RTS.S/ASO1 (RTS.S),** trade name **Mosquirix**, was recently endorsed by **the World Health Organisation (WHO).**
- It is **the first and, to date, the only vaccine shown to have the capability of significantly reducing malaria,** and life-threatening severe malaria, in tests on young African children.

### 24.7.3.2 Kala Azar (Leishmaniasis)

It is caused by *Leishmania donovani* and is spread by sandfly *(Phlebotomus)* and characterised by enlarged spleen and liver with high fever. It is also known as dumdum fever.

### 24.7.3.3 Ciliary Dysentery (Balantidiasis)

It is caused by *Balantidium coli.* This disease is spread through cyst formation. Cysts pass out in the host's faeces. Infection occurs by ingesting cysts with food and drinks. *Balantidium coli* cause ulcers in the colon and invades mucous membrane by secreting an enzyme, hyaluronidase. This generally results in diarrhoea, but may also lead to cysts and can prevent human infection. Tetracycline or iodoquinol are effective treatments.

### 24.7.3.4 Giardiasis

This is caused by *Giardia intestinalis* (first parasitic protozoan known). This disease is transmitted by contamination of cyst with food and drinks.

### 24.7.3.5 Trypanosomiasis

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Trypanosomiasis is caused by different species of *Trypanosoma,* a flagellate protozoan. It is characterised by high fever, swelling of neck and arm, weakness, anaemia, muscle spasm, lethargy, unconsciousness and death.

### 24.7.3.6 Amoebiasis (Amoebic dysentery or Enteritis)

**Pathogen:** *Entamoeba histolytica*

**Vector:** Infection by contamination

**Symptoms:** *Entamoeba* secretes cytolysin that erodes the mucous membrane of intestine.

**Drug:** Metronidazole.

## 24.7.4 Helminthic Diseases

Helminthes are multicellular parasites, which cause many diseases. These consist of round-worms and flatworms. Helminthes diseases are transmitted through dirty hands, contaminated soil and food. Types of diseases are ascariasis, ancylostomiasis, filariasis, taeniasis, etc.

**Cysticercosis:** Infection by tapeworm can also occur by taking vegetables, which are not washed properly and water contaminated with tapeworm capsules. Thus, vegetarians can also get tapeworm infection. The capsules release embryos, which may reach the eyes or brain, develop into larvae, called cysticerci that encyst. The cysts so formed may cause blindness or epilepsy like symptoms. Another species, *Y. saginat,* called beef tapeworm, causes taeniasis in beef-taking humans. Its intermediate hosts are cattle.

Table 24.2 Diseases transmitted by Helminthic

| Name of the disease | Causative agent | Vector/transmitted by | Affected body part | Sym |
|---|---|---|---|---|
| Ascariasis | *Ascaris tumbricoides* (endoparasite) | Contamination of ascariasis-embryonated eggs with food and water | Intestine | Weak anaer to sha |

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| | | | | |
|---|---|---|---|---|
| | | (Vector: cockroach and vegetables). | | host's Toxin by the cause pain indige diarrh vomiti nause insom Diagn |

## 24.7.5 Sexually Transmitted Diseases (STD)

These diseases are sexually communicable diseases and caused by a wide range of bacterial agents, protozoan and fungal agents, viral agents and ectoparasites.

The former name of such diseases was venereal diseases (VD), which was later changed to Sexually Transmitted Diseases (STD), which indicates transformation from the past two decades.

Major sexually transmitted diseases are **syphilis** and **gonorrhoea.**

Refer to the List below for some STDs and their Pathogens.

**DISEASE**

**Bacterial**

- Syphilis
- Gonorrhoea
- Cancroid
- Vaginitis

**Viral**

- Herpes genitalis
- Condyloma acuminatum
- Molluscum contagiosum

**Protozoal**

- Trichosominiasis

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### 24.7.6 Diarrhoeal Diseases

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Diarrhoeal diseases are a group of infections of the intestinal tract. It is characterised by food poisoning, abdominal cramps, frequent passage of stool having blood and mucous, leading to dehydration.

#### *Bacterial Diarrhoea*

Bacteria causing diarrhoeal diseases are *E. coli, Shigella sp. Campylobacter* and *Salmonella.*

#### *Viral Diarrhoea*

- **Rotavirus** causes gastroenteritis with acute watery diarrhoea in infants and children.
- **Norwalk virus** causes gastroenteritis in all age groups.

#### *Protozoal Diarrhoea*

- ***Giardia intestinalis*** causes giardiasis.

ORS – ORS is Oral Rehydration Solution used for correcting dehydration caused by diarrhoea in all age groups. It contains 105 grams of glucose, 3.5 grams of NaCl (sodium chloride), 2.9 grams of KCl (potassium chloride) and 3.9 grams of trisodium citrate.

## 24.8 Non-Communicable Diseases

### 24.8.1 Diabetes Mellitus

It occurs due to either deficiency of insulin production or due to failure of cells to take up insulin from blood. It is characterised by **hyperglycemia** [It is a case when high glucose level is found in human blood (300–1200 mg/100 ml)]. There are many types of symptoms possible in diabetes mellitus (*Polyuria* – excessive urination due to increase in water content of urine, *Polydipsia* – excessive thirst, *Glycosuria* – glucose in urine, *Polyphagia* – excessive eating). Loss of weight, tiredness, dehydration, reduced healing power and blurred vision are some other prominent symptoms. Two types of diabetes are more prominent in human beings – Type-I and Type-II.

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(a) **Diabetes Mellitus Type-I (Insulin-Dependent Diabetes):** It is in a category of autoimmune diseases (are those when body destroys healthy cells, tissues, and organs in our body). In this type of diabetes, insulin synthesis does not increase during the consumption of sugar content. In the majority of cases, this type of diabetes appears before the patient is 40 years old. That is why this type of diabetes is also known as *Juvenile Diabetes* or *Childhood Diabetes.* About 15 per cent of all diabetes patients have Type-I.

Table 24.3 List of STDs

| | Name of disease | Causative agent | Symptoms | Control |
|---|---|---|---|---|
| a. | Syphilis | *Spirochaete, Treponemaallidum* | Ulcer on urogenital tract, lymph nodes, hair loss, swollen joints, heart disease and blindness, etc. | Antibiotics like penicillin, tetracycline |
| b. | Gonorrhoea | *Neisseria gonorrhoeae* | Discharge of pus or excessive vaginal secretion, leading to eye infection in children of gonorrhoea-afflicted mothers and arthritis. | Penicillin and ampicillin |

**DIABETES MELLITUS VS DIABETES INSIPIDUS**

**Diabetes Mellitus** is the problem of insulin hormone due to either destruction of Islet of Langerhans cells present in the pancreas or any other autoimmune cause. Here, there is increased level of sugar in blood. It can cause death when its level rises beyond a certain limit.

**Diabetes Insipidus** is the defect in secretion of vasopressin hormone (antidiuretic hormone), which is secreted from posterior pituitary gland present in hypothalamus. Its deficiency than the normal level increases dehydrolysis (removal of water through urine) even in low intake of water.

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Note: Caffeine consumption also decreases ADH level in human body, which induces dehydrolysis (more urination/ increased removal of water through urine).

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(b) **Diabetes Mellitus Type-II (Insulin-Independent Diabetes):** Two reasons behind Diabetes Mellitus Type-II: (1) Not enough insulin is being produced. (2) The insulin is not working properly – this is known as *insulin resistance* (The body produces insulin, but its insulin sensitivity is undermined and does not work as it should do – glucose in not entering the body's cells properly). Approximately 85 per cent of all diabetes patients have Type-II.

### 24.8.2 Cardiovascular Diseases

Hypertensive heart disease results in the damage of heart muscles and brain stroke (resulting in paralysis; loss of speech and other functions).

- **Arteriosclerosis:** Thickening of arteries and arterioles, resulting in blood pressure higher than normal.
- **Atherosclerosis:** It is the disease when arteries and arterioles become narrow because of fat deposition inside and over the tunica intima (inside blood vessel). It leads to high blood pressure, clot formation or *thrombosis.*

#### *Coronary Heart Disease*

- **Angina Pectoris:** Severe temporary heart pain, spreading especially to the left arm because of deficient oxygen supply or myocardial ischemia/reduced blood supply.
- **Heart Attack:** Formation of a clot (thrombus) in a narrowed coronary artery stops blood supply to the part of the heart muscle beyond the clot. The muscle cells of this part due to lack of oxygen and glucose. This condition is called 'heart attack' (coronary thrombosis or myocardial infarction) or MI.
- **Rheumatic Heart Disease:** The rheumatic heart disease is common in India under 20 years of age. In Rheumatic fever (due to *Streptococcal* infection), toxins are produced causing inflammation of different parts of the body, which also include endocardium (it is the inner most layer

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of tissues that lines the chambers of the heart) and joints. They give rise to murmur (heart sounds produced when blood flows across one of the heart valves), which are pathological (abnormal) heart sounds heard through a stethoscope.

**Prevention of Heart Disease:** Certain precautions can prevent heart diseases. These are as follows:

(i) Avoid strain by keeping cool in all circumstances.
(ii) Avoid saturated fats, smoking, drinking drugs especially after 35 years of age, to keep cholesterol level of the blood low.
(iii) Avoid becoming overweight and obese (bulky).
(iv) Take light exercise daily.
(v) Take preventive measures against infections of respiratory tract.
(vi) Consult a doctor in case of any pain in the chest.

**Stroke:** Stroke is cerebrovascular damage due to stopping of blood to a part caused by following reasons:

**NITRIC OXIDE SUPPLEMENTS BOOST OUR HEALTH AND PERFORMANCE**

Nitric Oxide Supplements Boost Our Health and Performance as it naturally synthesise in human body and do vasodilation (i.e. widening of blood vessels, allowing more blood to flow through them and lowering blood pressure).

It synthesise in inner lining of blood vessel i.e. Endothelial wall and Mitochondria also in certain condition.

Other health benefits includes:

1. Helps treat erectile dysfunction
2. May decrease muscle soreness
3. Lowers blood pressure
4. Boosts exercise performance
5. May help manage type 2 diabetes

(a) Clot/thrombus
(b) Breaking of an artery because of its wall weakening and hypertension
(c) Sustained contraction or spasm of an artery

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Such a stroke leads to paralysis, losing of consciousness, loss of hearing and even death.

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### 24.8.3 Arthritis

Common disease of old age, inflammation of joints and muscles resulting in pain and stiffness of the joints. This disease is of several types:

(a) **Osteoarthritis:** Wearing or breaking down of the cartilage that covers every end of the bones, causing great pain in the movement of joints and in future developed into bony spurs.

(b) **Rheumatoid Arthritis:** Erosion of joints due to unexpected attacking by the immune system of the body causing inflammation of the joints with great pain due to thickening of the tissue lining the joints; women are more often affected than men.

(c) **Gout:** This occurs in men, especially, in which sodium urate crystals and uric acid in excess form are gathered or deposited in the synovial joints, which causes arthritis with pain at night making it difficult to move.

**Osteoporosis:** Denotes a condition wherein an imbalance between the new bone formation and old bone reabsorption occurs. Either the body fails to form enough new bone, or there is too much old bone reabsorption, or even both can lead to it.

**Diagnosis:** A bone mineral density test (BMD test) uses X-rays for measuring the amount of minerals, i.e., calcium in the bones. This test greatly helps people who are at risk of osteoporosis, especially the women and older adults. Another name for the test is dual energy X-ray absorptiometry (DXA).

### 24.8.4 Kidney Stones

They are common urinary tract disorder that involves crystallised chemicals like uric acid, **calcium oxalate** and **calcium phosphate (Renal Calculi)**.

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**Treatment:** A medical procedure, which is used to treat certain types of kidney stones as well as stones in other organs, like the gallbladder or the liver, is termed as Lithotripsy.

This procedure uses sound waves (also known as high-energy shock waves) for breaking up large kidney stones into smaller pieces. Extracorporeal shock wave lithotripsy (ESWL) denotes the most common form of lithotripsy.

## 24.9 Occupational Diseases

Occupational diseases arise among those people who are exposed to air pollutants generated from the industries where they are working. Types of occupational disease are asbestosis, silicosis, byssinosis, bagassosis, anthracosis, etc.

**(i)** **Asbestosis** is characterised by pulmonary fibrosis leading to respiratory insufficiency and death, carcinoma of bronchus and gastrointestinal tract, etc.

**(ii)** **Silicosis** is a disease of the lungs caused due to inhalation of silica-containing dust.

**(iii)** **Byssinosis** is characterised by chronic cough, progressive dyspnea, ending in chronic bronchitis and emphysema.

**(iv)** **Bagassosis** is characterised by breathlessness, cough, haemoptysis and slight fever.

**(v)** **Anthracosis** is characterised with little ventilator impairment; progressive massive fibrosis causes severe respiratory disability and results in premature death.

Table 24.4 List of Occupational Diseases

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| Types of air pollutant | Dust | Disease |
|---|---|---|
| Inorganic | Coal dust<br>Silica<br>Asbestos<br>Iron | Anthraxosis<br>Silicosis<br>Asbestosis (lung cancer)<br>Siderosis |
| Organic | Cane fibre<br>Cotton dust<br>Tobacco<br>Hay/grain dust | Bagassosis<br>Byssinosis<br>Tobacossis<br>Farmer's lung |

All these diseases are caused by dust, called pneumoconiosis. Pneumoconiosis may gradually cripple a man by reducing his working capacity because of lung fibrosis and other complications.

Oxides of sulphur and its hydrates, $H_2SO_4$ in moderate concentration, cause suffocation and irritation in the upper respiratory tract and continuous exposure may cause lung diseases like chronic asthma, bronchitis, unconsciousness and even death.

Note: Carbon Monoxide Poisoning: A poisonous gas with no smell or taste, breathing in which can make you unwell, could even kill on being exposed to high levels of it, is called carbon monoxide (CO). It is produced on complete burning of fuels like gas, oil, coal and wood.

**How it affects:**

- On being breathed in, CO enters the bloodstream and thus mixes with haemoglobin (the pigment in RBCs, carrying oxygen around the body), to form carboxyhaemoglobin.
- And hence, it reduces the oxygen-carrying capacity of the blood and this lack of oxygen leads to body's cells and tissues to fail and ultimately die.

## 24.10 Vitamins

Funk (a Polish biochemist) is generally credited with being among the first to formulate the concept of vitamins, in 1912, which he called 'vital

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amines' or 'vitamines'. Vitamins are non-energy producing organic substances, which are vitally essential in traces for proper use of macronutrients in cell metabolism (because they play a role in energy transformations, usually acting as coenzyme in enzymatic system) and hence for proper growth and function (**Table 24.2**). They cannot be synthesised within the body, so must be acquired from exogenous source, i.e., diet. Vitamins are synthesised mostly by plants and certain intestinal bacteria. On the basis of their solubility in fats and water, vitamins are divided into two main categories: fat-soluble (A, D, E, K) and water-soluble vitamins (B, C).

### 24.10.1 Water-Soluble Vitamins

These vitamins help in energy production from fats and carbohydrates. Excess of these vitamins are excreted in the urine, which are generally non-toxic and there is a need for continuous supply of these vitamins in our diet. In addition, several other water-soluble substances are necessary as growth factor for some organisms but they occur in more than trace amounts. These groups include Inositol and choline. The deficiency of these vitamins greatly affects growing tissue like skin, the digestive tract, blood and the nervous system.

### 24.10.2 Fat-Soluble Vitamins

These vitamins are non-polar hydrophobic molecules, derivative of isoprene. They are digested and absorbed by fats to be transported in the blood. Most fat-soluble vitamins contain at least one –OH group (hydroxyl group), which results in ester formation with the fatty acids. Therefore, vitamins are soluble in organic solvents (like chloroform). These vitamins are toxic, if consumed in higher amounts. Vitamins A and D cause specific cells to decrease or increase certain genes' expression with their hormone actions. Only higher animals appear to require them from exogenous sources. An essential role of fat-soluble vitamins in plants and

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microorganisms has not been clearly established. They do not appear to serve as component of coenzyme.

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Note: **Hypervitaminosis** is the condition when there is excess of overall vitamins.

Table 24.5 Different types of vitamins, their chemical names, functions and deficiency problems in human body

| | Name | Sources | Functions | Effect of deficiency |
|---|---|---|---|---|
| | | **WATER-SOLUBLE VITAMINS** | | |
| 1. | B1 (Thiamine) | Rice bran, whole-wheat flour, oatmeal, eggs, yeast, meat, liver, etc. | Major component of coenzyme carboxylase required for carbohydrate and amino acid metabolism | Beri-beri: loss of appetite and vigour constipation, weak heartbeat, muscle atrophy, even paralysis. |
| 2. | B2 (Riboflavin) | Cheese, eggs, yeast, tomatoes, green vegetables, liver, meat, cereals, etc. | Combines with phosphoric acid to form coenzyme FAD essential for | Cheilosis (fissures a the corners of mout and lips), digestive disorders, burning sensations in skin a |

Note: Besides the vitamin-deficiency diseases, other important deficiency diseases involve that of proteins, namely Marasmus and Kwashiorkor, both being type of under-nutrition.

- **Marasmus:** Its occurrence is more often in young children and babies. The disease leads to weight loss and dehydration. Starvation is a form of this disorder.
- **Kwashiorkor:** People with severe protein deficiency may suffer from it. Children developing kwashiorkor are often older than those who develop marasmus. A diet that consists mainly of carbohydrates can lead to this condition.

## Part B: Popular and Recent Diseases with Various Issues

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## 24.11 Popular and Recent Diseases in News

### 24.11.1 Cancer

Hippocrates is a Greek physician and also called as 'Father of Medicine'. He has given the term Cancer, which means crab. Various agents are known to cause cancer, namely, genetics, tobacco consumption and radiation exposure.

**Symptoms:** Cancer does not have any specific symptoms. Many symptoms are possible, like: abnormal lumps, changes in your skin, persistent cough or hoarseness, unexplained weight loss, constant fatigue, persistent pain, indigestion or trouble swallowing, changes in bowel habits.

**Various Treatments are Adopted Against Cancer:** Stem-cell therapy; Drug treatment, which is also called Chemotherapy; Immunotherapy for improving immune system and vaccines, surgery; Radiation therapy using x-rays, gamma rays, bone-marrow transplantation, etc.

**Steps Taken by Government:** In 1975, initiation of the National Cancer Control Programme (NCCP) took place and later, in 1984–85 focussed on detection and prevention of cancer at early stage. In 2010, National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Disease and Stroke (NPCDCS) was launched by the government, which emphasised on three main sites of cancer: cervical, oral and breast. Later, during 2013–14, NPCDCS was brought under 'National Health Mission' for early detection of cancer, diagnosing and treatment and also prevention of cancer.

Twenty (20) State Cancer Institute (SCI) and Fifty (50) Tertiary Cancer Centres in different parts of India are to be assisted or given assistance by the government, under the NPCDCS Scheme 'Strengthening of tertiary care of cancer', ₹120 crore and ₹45 crore, respectively.

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In 2009, under the Rashtriya Arogya Nidhi, the 'Health Ministers Cancer Patient Fund (HMCPF)' was set up to provide funds for immediate financial assistance of up to ₹2 lakh to Below Poverty Line Cancer patients through Regional Cancer Centres, by providing them the fund.

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### 24.11.2 AIDS (Acquired Immuno-Deficiency Syndrome)

Scientists found out the source of HIV infection in humans from a type of chimpanzee in West Africa. Simian Immune Deficiency Virus (SIV) of chimpanzee was contracted and transferred to humans and in which mutation occurs, leading to HIV when it comes in contact with their infected blood. The virus slowly speeded across Africa and then later on to other parts of the world.

**Causative Agent:** HIV is caused by **lentivirus.** Lentivirus is from Retro-viridae family, having long incubation period.

**Symptoms:** Dry coughing or breath shortness, diarrhoea, pneumonia-like symptoms, fever and white spots, loss of vision and memory loss, severe headaches with neck stiffness, vomiting, nausea, lacking of coordination, depression and confusion, seizure or epilepsy.

**HIV** or **Human Immunodeficiency Virus** represents the virus that causes this disease, and leads to the initial stage of infection, while AIDS, that is Acquired Immuno-Deficiency Syndrome represents the set of symptoms or syndrome so developed in the later stage of this infection, when the immune system of the infected person becomes too weak.

***Government Measures***

- NACO (National AIDS Control Organisation) In 1986, Ministry of Health and Family Welfare constituted the National AIDS Committee. National AIDS Control Organisation (NACO) was also constituted to implement India's first National AIDS Control Programme, which was launched in 1992. NACO provides leadership to HIV/AIDS control programme in India through thirty-five (35) HIV/AIDS Prevention and Control Societies.

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**PROJECT SUNRISE**

For the prevention of AIDS in the eight north-eastern states, the government has launched a project called 'Project Sunrise' to create awareness about the disease in the North-East states. Although the North-East is less populated, it has, in comparison to other Indian states, some 1,00,000 people living with this condition.

### 24.11.3 HIV & AIDS (P&C) Act, 2017

A notification has been issued by The **Ministry of Health and Family Welfare** for bringing the ***Human Immunodeficiency Virus and Acquired Immune Deficiency Syndrome (Prevention and Control) Act, 2017*** in force from 10th September, 2018. This Act will safeguard the rights of people living with HIV and affected by HIV.

***Highlights of the Act***

1. **Provisions have been Made Related to the Role of Governments:** As per the Act, central and state governments **provide for anti-retroviral therapy (ART)** and management of opportunistic infections (infections that take advantage of weakness in the immune system and occur frequently). **It also forbids specific acts of discrimination by the state, or any other person, against HIV-positive people, or those living with such people.**
2. **Provisions Related to Discrimination:** It has laid down penal provisions for any discrimination practiced against a person with HIV/AIDS and breach of confidentiality. **The protection mandates its extension to the fields of healthcare services, educational services, employment, public facilities, property rights, holding public office and insurance.** It also provides for **confidentiality of HIV-related information** and has made it necessary to get informed consent for undertaking HIV tests, medical treatment and research.
3. **Provisions Related to Ombudsman: An ombudsman will be appointed by respective state government to enquire into complaints related to the violation of the Act and the provision of health care services.** A report shall be submitted to the state

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government every six months stating the number and nature of complaints received, the actions taken by the Ombudsman.

4. **Provisions Related to Supervision:** There are provisions related to supervision. A person with sufficient maturity level aged between 12 to 18 years who knows how to manage the affairs of his HIV or AIDS affected family shall be competent to act as a guardian of another sibling below 18 years of age. **The guardianship will be applied in matters relating to admission to educational establishments, operating bank accounts, managing property, care and treatment, amongst others.**
5. **Role of Courts: The Court must dispose off cases relating to HIV-positive persons on a priority basis.** While dealing with legal proceeding, if an HIV infected or affected person is a party, the court may pass orders that the proceedings be conducted (a) by hiding the identity of the person, (b) in camera, and (c) to restrain any person from publishing information that discloses the identity of the applicant. Medical expenses incurred by the applicant should be taken into account when any order is passed with regard to a maintenance application filed by an HIV infected or affected person.

***Background***

Around **21 lakh persons have been estimated to be living with HIV in India** and the percentage of patients receiving anti-retroviral therapy (ART) treatment currently stands at a mere 26.82% as against the global percentage of 41%, according to the 2015 Global Burden of Diseases (GBD).

### 24.11.4 Influenza

An influenza virus causes it. Three types of seasonal influenza viruses are present: (1) Influenza A; (2) Influenza B and (3) Influenza C. These viruses have different subtypes on the basis of various combinations and their surface proteins. In comparison to Influenza type A and Influenza type B,

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there is less case of occurrence for Type C Influenza. So the Influenza A and Influenza B viruses are included in seasonal influenza vaccines.

Haemogglutinin (H) and Neuraminidase (N) proteins cover the surface of Influenza A and Influenza B viruses (**Figure 24.1**).

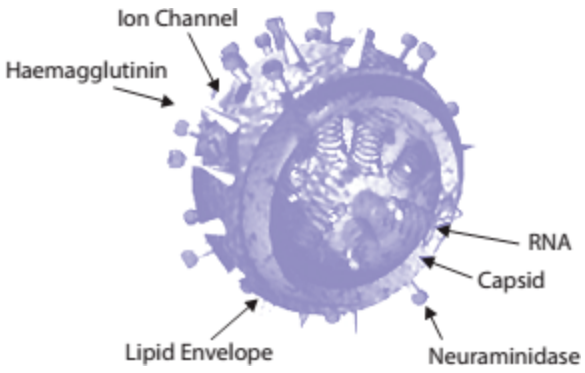


Figure 24.1 Structure of the influenza virion

In Influenza type A virus, there are 17 haemagglutinin types, ranging from H1 to H17 and 9 neuraminidase types, N1 to N9. Each influenza virus has at least one type of H (such as H1) and one type of N (such as N1).

The seasonal influenza A strains H1N1 and H3N2, currently circulating in humans, have contributed to lot more changes since their first introduction into humans.

Influenza type B strains circulation is not in animals, hence it cannot cause a pandemic. But, like influenza type A viruses, they continually change, so they are not immune to every strain. The other components of the flu vaccine are as follows:

H1N1 – (Spanish Flu, and Swine Flu)
H1N2 – (endemic in birds, human and pigs)
H2N2 – (Asian Flu) H3N2 – (Hong Kong Flu)
H5N1 – (Bird Flu)
H7N7 – (Zoonotic potential)
H7N2, H7N3
H9N2, H10N7

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**Symptoms:** Dry cough, headache, high fever, severe malaise, sore throat and runny nose, muscle and joint pain. Influenza can cause disease or illness or even death, especially in people at high risk (**Figure 24.2**).

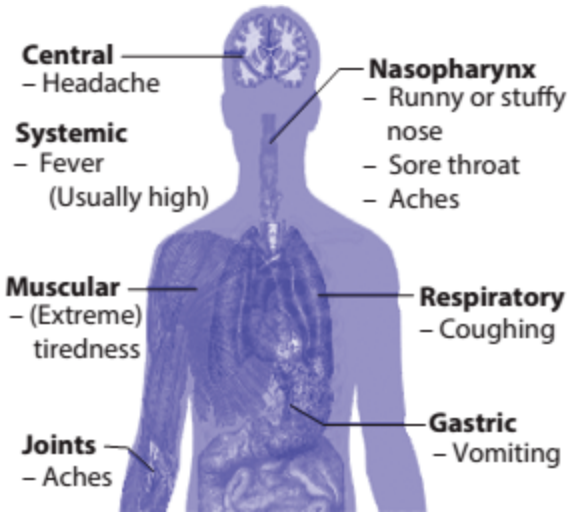


Figure 24.2 Symptoms of Influenza Disease

**Cure:** There are two classes of such medicines: Inhibitors of influenza neraminidase (zanamivir and oseltamivir), adamantanes (rimantadine and amantadine). Some influenza viruses have been reported to develop resistance to the antiviral medicines, thus, limiting the effectiveness of treatment.

***Differences between H1N1 Flu and Influenza Type A***

This H1N1 flu is a variety of Influenza type A. indicating the viral serotype; it is a type of shorthand for characteristics, which identify the virus to your immune system and enable the virus to enter your cells. H1N1 also has different strains of H1N1 flu.

## 24.11.5 Ebola

**History:** Ebola (named after the Ebola River in Zaire) first emerged in Sudan and Zaire in the year 1976. The first outbreak of Ebola (EBOLA-Sudan) was reported to infect over 284 people, with a mortality rate of 53

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per cent. Later, after a month, the second Ebola virus emerged from Yambuku, Zaire, Ebola-Zaire (EBOZ).

**Causative Agent:** Ebola virus is from filoviridae family and from the order mononegavirales. The causative agent of Ebola Hemorrhagic Fever (Ebola HF) is Ebola virus. The Ebola and Marburg viruses, from the filoviridae family, are among the most virulent pathogens in humans.

**Symptoms:** Diarrhoea, red eyes, vomiting, raised rash, cough, bleeding from the eye usually, chest pain, severe weight loss, bruising and internal bleeding.

**DIFFERENCES BETWEEN SWINE FLU AND BIRD FLU**

**Swine Flu**

It is a disease caused by H1N1 Virus. It infects people, pigs, birds and other animals. Swine flu is reported to normally infect pigs. Occasionally, the pigs transmit influenza viruses to people, mainly to hog farm workers and veterinarians. This flu also spreads quickly, affecting the respiratory system, but casualty rate is less as compared to Bird flu. Persons having chronic medical conditions and pregnant woman are vulnerable targets. Symptoms include diarrhoea, vomiting, coughing and fever.

**Bird Flu (Avian Influenza)**

It is a disease caused by H5N1 virus, carried by animals, especially birds, which rapidly spreads and causes sudden death of the one who is infected. The outbreak of avian influenza (H5N1) – started in poultry in South Korea in mid-December 2003, and affecting birds in many countries in Middle East, Africa, Europe and Asia. Bird flu spreads very slowly, but casualty rate is higher. Bird flu virus affects all human body systems. Symptoms include pneumonia, eye infections and other problems. Rural workers and children are victims who get infected with Bird flu.

**Treatment:** No specific treatment; Patient is to be managed in isolation facilities for preventing spreading of the infection. Intensive supportive care is required for very ill patients. Oral rehydration with solutions containing intravenous fluids or electrolytes is required as patients get dehydrated. Strict infection control measures by the health workers when caring for suspected patient.

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**Cure:** There are 2 licensed Ebola vaccines. Ervebo was licensed in November 2019 by the European Medicines Agency and prequalified by WHO. The United States Food and Drug Administration licensed the vaccine in December 2019.

**Important Preventions Measures for Travellers to be Observed:** Avoid travelling to affected areas, have good personal and environmental hygiene, use liquid soap or alcohol-based hand rub for cleaning hands before touching the nose, eyes and mouth, close contact with ill-persons, and avoid contact with blood or bodily fluids of patients, which includes items that may have come in contact with the infected person's blood or bodily fluids.

Contact with animals also needs to be avoided. Food should be thoroughly cooked before consumption; a close observation of the health conditions is necessary for 21 days after returning from an affected area. **The Government had advised Indians against non-essential travel to the four West African countries namely Guinea, Sierra Leone, Liberia, and Nigeria which were the epicentre of Ebola.**

### 24.11.6 Severe Acute Respiratory Syndrome (SARS)

**History:** This disease was initially transferred to humans through civets, mammals with catlike appearance. In 2002, SARS virus was firstly reported in Asia. As per the World Health Organization (WHO), July 2003, it causes death of many people worldwide due to its epidemic nature. This disease was reported in more than 30 countries in five continents.

**Causative Agent:** Severe Acute Respiratory Syndrome (SARS) is the term for a viral respiratory illness caused by a corona virus – the (SARS-CoV). Corona viruses are a group of viruses that have crown-like (corona) or halo-like appearance on viewing under a microscope and are a common cause of moderate-to-mild upper-respiratory illness in humans and causing severe diseases in animals like gastrointestinal, liver, respiratory and neurologic diseases.

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**Symptoms:** Fever, runny nose, headache, fatigue, muscle pain and ache, malaise (general discomfort feeling), decreasing appetite, diarrhoea, cough (dry), breath shortness, sore throat.

**Cure:** There is no cure for SARS but researchers are trying to find a vaccine. Treatment may include:

- Assistance with breathing by using a ventilator to deliver oxygen.
- The antibiotics to treat bacteria that cause pneumonia.
- The antiviral medications.
- High steroid doses for reducing swelling in the lungs.

### 24.11.7 Tuberculosis

TB is an infectious disease caused by **a bacterium *Mycobacterium tuberculosis.*** It typically affects the lungs (pulmonary TB) but can also affect other sites. The disease is spread when people who are sick with pulmonary TB expel bacteria into the air, for example by coughing. Diagnostic tests for TB disease include **Rapid molecular test, Sputum smear microscopy, Culture-based methods.** Without treatment, the mortality rate from TB is high.

#### *Some Facts on TB*

- TB remains the world's deadliest infectious killer. Each day, over 4000 people lose their lives to TB and close to 30,000 people fall ill with this preventable and curable disease.
- The **World Health Organization (WHO)** has launched a joint initiative **'Find. Treat. All. #EndTB'** with the Global Fund and Stop TB Partnership, with the aim of accelerating the TB response and ensuring access to care, in line with WHO's overall drive towards Universal Health Coverage.
- WHO publishes **a global TB report** every year since 1997.

#### *Initiatives in India*

1. TB treatment is free in India.
2. **Mass BCG vaccination** to prevent TB.

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3. In 1993, **the Revised National TB Control Programme (RNTCP)** was launched, offering free diagnosis and treatment for patients, rescuing them from otherwise sure death.
4. ₹12,000 crore fund to fight TB over the next 3 years.
5. **'TB Harega Desh Jeetega Campaign'** has been launched with an aim to eliminate tuberculosis from India.

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### *About 'TB Harega Desh Jeetega' Campaign*

- **Campaign aims to** improve and expand the reach of TB care services across the country by 2022.
- **Three Pillars of the Campaign –** Clinical approach, public health component and active community participation.
- **Supporting Aspects of the Campaign –** Patient support, private sector engagement, political and administrative commitment at all levels.
- **The government will ensure that** all patients, at private or public hospitals, receive free-of-cost and high-quality TB care.

**United Nations** has marked **2030 as a global target to eliminate TB worldwide** but the Government of **India has made 2025 as its target to eliminate tuberculosis from the country.**

### *World Tuberculosis Day*

**World Tuberculosis (TB) Day** is observed on **March 24** to raise public awareness about the devastating health, social and economic consequences of TB, and to step up efforts to end the global TB epidemic. The date marks **the day in 1882 when Dr Robert Koch announced that he had discovered the bacterium that causes TB,** which opened the way towards diagnosing and curing this disease. **The theme of World TB Day 2024** was ***'Yes! We can end TB!'*.**

### *Sustainable Development Goal (SDG)*

- The consolidated goal on health is **SDG 3.** One of these targets, **(Target 3.3),** explicitly mentions TB.
- **SDG 3 also includes a target (Target 3.8)** related to universal health coverage (UHC) in which TB is explicitly mentioned. This includes an indicator on the coverage of essential prevention, treatment and care interventions.

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### 24.11.8 Meningitis

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It is a rare infection that affect meninges layer which covers brain and spinal cord. Bacteria, virus and fungi are causing agents of meningitis. Bacterial meningitis is more contagious among people and can cause death. Viral meningitis is comparatively less severe and fungal meningitis is very rare form of meningitis.

Bacteria can be transmitted through coughing and sneezing from one person to the other. Bacteria travel through the bloodstream to the brain. Many bacteria are responsible for Meningitis: *Neisseria meningitidis, Streptococcus pneumoniae, Listeria monocytogenes, Haemophilus influenzae* type B. Most risky people to suffer from Meningitis: Meningitis is more dangerous for people with certain medical conditions, like absent or damaged spleen, immune system disorders, or chronic disease. Following are the most common symptoms of meningitis:

- Persistent and severe headache
- Stiffness of the neck and painful neck
- Vomiting
- Confusion and decrease in the level of consciousness
- Seizures

**Other symptoms of meningitis may include** Sluggishness, weakness and strange feelings (as tingling), muscle ache or weakness throughout the body. Eye becomes sensitive and eye pain from bright lights, skin rash and dizzy spells, etc.

***Antibiotics for Bacterial Meningitis***

**Branded Name:** Rocephin, Claforan

**Generic Name:** Ceftriaxone, Gentamicin, Vancomycin, Vancocin, Rifampin, Cefotaxime, Claforan, Penicillin G

**GLOBAL ROADMAP TO DEFEAT MENINGITIS BY 2030**

- The Global Roadmap to Defeat Meningitis by 2030 was recently released by **the World Health Organization (WHO)** and its partners.
- This is **the first-ever global strategy to defeat meningitis.**

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- It aims to eliminate epidemics of bacterial meningitis and to reduce deaths by 70% and halve the number of cases.

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## 24.11.9 Encephalitis

From the last few years, Japanese Encephalitis became a cause of many deaths of children and old-age people in eastern Uttar Pradesh. This disease is spread through contaminated water and carrier mosquito (female culex/Tiger Mosquito). Though there are no antiviral drugs developed till date but researches are underway to find the cure. Those people suffering with viral encephalitis are kept in fluids while monitoring for brain swelling. Japanese Encephalitis virus can infect the foetus and cause death.

## 24.11.10 Dengue

It is a viral disease spread due to bite of female Aedes mosquito. Dengue is also called as **break bone fever.**

**Symptoms of Dengue Fever:** Severe headaches, pain behind the eye, sudden high fever, nausea, severe muscle and pain, vomiting, skin rash, appearing three to four days after the onset of fever, mild bleeding (such as gums bleeding, easy bruising, nose bleed). In dengue infection, a person gets deficient in the human body with blood platelets.

Mostly younger children and old age people suffer due to this infection. Dengue haemorrhagic fever is a serious dengue, those appear in high fever, damage to blood vessels and lymph nodes, in severe cases have also bleeding from the gums and nose, liver will enlarge, and at last circulatory system stops to work. This is also called as dengue shock syndrome (DSS). This disease mostly occurs in people having weaker immune system.

***Treatment for Dengue Fever***

At present we don't have specific medicine for treating dengue infection. Pain relievers with acetaminophen can be used and medicines with aspirin as a constituent is to be avoided as it, could worsen bleeding. Resting, drinking plenty of fluids, and seeing a doctor is necessary.

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### 24.11.11 Chikungunya

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Chikungunya also often known as the 'monsoon disease' spread to people by the bite of an Aedes mosquito carrying the chikungunya virus.

**Symptoms of Chikungunya:** Overwhelming fatigue; inability to complete normal chores, weakness, high fever, pain/stiffness in the joints, headache, muscle pain, certain body parts, which are in pain may also swell up, sleeplessness, rashes in the body, nausea, ulcers in the mouth. It can be up to 12 days after being infected, that these symptoms begin to surface. There are times where infection occurs but no symptoms manifest. This situation/condition is known as silent chikungunya. People of any age group can suffer with this disease.

**Cure:** No cure for chikunguyna. Once contracted, the body develops immunity to it and will not suffer from it again. So it is important to be aware of the symptoms, to catch it early and get it treated so that we can recover fully and remain protected in the long run from contracting it again.

**Treatment:** Tulsi and carrots are recommended for effective recovery. Ayurvedic medicines that are made of several herbs, spices may be prescribed. Ayurvedic doctor can be visited for a check-up and individual treatment options.

***Taking Good Care of Our Health has a Big Role to Play in Preventing Chikungunya***

- Eating healthy food to build the immune system. Focussing on nuts, fruits, vegetables, seeds, fish, eggs, grain and lean meat and avoiding street food, fast food and greasy dishes.
- Regular exercise also helps makes the body stronger.
- Vitamins C, E help boost immunity. If daily requirement is not fulfilled through food, vitamin pills can be taken.
- Learning stress management, since stress has a debilitating effect on our immune system and hence the raises risk of catching infections.

### 24.11.12 Zika Virus

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During 1947, this disease was firstly reported in Uganda, in monkeys. But first, human case was detected in 1954 in Nigeria. Later, this disease spread in many continents, including Africa, South East Asia and Pacific Islands.

**Countries Affected:** Outbreaks of Zika virus disease have been recorded in Africa, the Americas, Asia and the Pacific.

**Cycle of Spread:** Aedes mosquitoes spread Zika virus worldwide.

**Symptoms:** Most virus carriers are without symptoms. It is a silent infection in a group of highly vulnerable individuals – pregnant women who, in turn, have a horrible outcome for their babies.

**Vaccines Developed so Far:** It was reported by US-Brazilian team of scientists, that two distinct vaccine candidates conferred powerful protection from the Zika infection when each vaccine was delivered by intra-muscular injection to mice. Scientists at Bharat Biotech International Ltd, Hyderabad, claim to have developed a Zika virus vaccine for humans and have patented it too. It is a big breakthrough by the Indian scientists to control the epidemic affecting the Americans. Trials are going on for the vaccine.

**WOLBACHIA METHOD**

Aedes mosquitoes are introduced with the bacterium in the Laboratory and then researchers release them into the wild.

It will compete with various viral diseases – dengue, zika, chikungunya, yellow fever, West Nile fever etc.

Wolbachia (wohl-bach-ee-uh) is a common type of bacteria found in insects. Approximately 6 in 10 of all types of insects, including butterflies, bees and beetles, around the world have Wolbachia.

## 24.11.13 MERS (Middle East Respiratory Syndrome)

MERS is a respiratory disease caused by virus in Middle East countries. It is a respiratory disease caused by coronavirus (MERS-CoV). This virus was firstly reported in Saudi Arabia in 2012. The symptoms may appear from

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normal cold to severe acute respiratory syndrome (SARS) level. This disease is transmitted from animals to humans (zoonotic). Many scientists believe that this disease originated in bats and is then transmitted to camels. Mostly, this disease transfers through human-to-human infections. Camels are the main reservoir host for MERS infection.

**Symptoms:** In mild infection: Fever, cough and shortness of breath are the normal symptoms. Sometimes, pneumonia is also found. Apart of these, gastrointestinal symptoms can also occur, like diarrhoea, etc. But in severe cases, it can lead to respiratory failure and cause death. Its mortality rate is approximately 36 per cent. Those people have weakened immune systems and suffer with chronic diseases like cancer, lung disease or diabetes; thus chance is more severe.

**Mode of Transmission:** Non-human to human transmission and human-to-human transmission.

**Prevention and Treatment:** It does not have any specific vaccine or treatment right now. Camel meat and camel-based products should be avoided. We should also try to make a distance with infected camel from our surroundings.

## 24.11.14 NIPAH Virus

According to WHO, the Nipah Virus infection is an emerging hooknoses, i.e., a disease transmitted from animals to humans. Nipah Virus belongs to a new genus termed *Henipavirus* (subfamily Paramyxovirinae). The fruit bats are natural host of the virus belonging to the family Pteropodidae. In 2004, many people were affected after the eating the date palm contaminated by infected fruit bats. Pigs can also act as intermediate hosts.

***When was it First Reported?***

It was first found in 1998 at Kampung Sungai Nipah village, Malaysia. The name of the virus has been taken from this village.

***What are the Symptoms in Humans?***

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- The Nipah infection has the similar symptoms like influenza: fever, muscle pain and respiratory problems. Disorientation can also be caused due to inflammation of the brain.
- Late arrival of Encephalitis can also occur. Sometimes a person can have an asymptomatic infection, and become a carrier of Nipah where it doesn't show any symptoms
- Currently, there are no vaccines available for both human and animals. The supportive treatment is being given to the people infected by Nipah in form of intensive supportive care.
- According to WHO, the symptoms of nausea, vomiting and convulsions associated with the disease can be reduced by **ribavarin.** Individuals infected required to be hospitalised and isolated. Special care should be taken to prevent human-to-human transmission. To detect the virus quickly and to initiate appropriate control measures.

### 24.11.15 Kyasanur Forest Disease (KFD)

KFD is caused by **the Kyasanur Forest Disease Virus (KFDV).** The virus was identified in 1957 when it was isolated from a sick monkey from **the Kyasanur Forest.** Since then, between 400 and 500 humans cases per year have been reported. **Hard ticks *(Haemaphysalis spinigera)* are the reservoir** of the KFD virus and once infected, remain so for life. Rodents, shrews and monkeys are common hosts for KFDV after being bitten by an infected tick. KFDV can cause **epizootics** with high fatality in primates.

#### *Transmission*

- ***Transmission to humans may occur after a tick bite or contact with an infected animal***, most importantly a sick or recently dead monkey.
- The disease as of now is stated to be transmitted through monkeys. Large animals such as goats, cows and sheep may become infected with KFD but play a limited role in the transmission of the disease.
- ***These animals provide the blood meals for ticks*** and it is possible for infected animals with viremia to infect other ticks, but transmission of KFDV to humans from these larger animals is extremely rare. Furthermore, **there is no evidence of disease transmission via the unpasteurised milk of any of these animals.**

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***Symptoms***

- After an ***incubation period of 3–8 days***, the symptoms of KFD begin suddenly with ***chills, fever and headache. Severe muscle pain with vomiting, gastrointestinal symptoms and bleeding problems may occur 3–4 days after initial symptom onset.***
- ***Patients may experience abnormally low blood pressure, and low platelet, red blood cell and white blood cell counts.***
- After 1–2 weeks of symptoms, some patients recover without complication. However, the ***illness is biphasic*** for a subset of patients (10–20%) who experience a second wave of symptoms at the beginning of the third week. These symptoms include fever and signs of neurological manifestations, such as severe headache, mental disturbances, tremors and vision deficits.

***Prevention***

- As per the doctors there is ***no specific treatment for KFD***, but early hospitalization and supportive therapy is important. ***Supportive therapy includes the maintenance of hydration and the usual precautions for patients with bleeding disorders.***
- A vaccine does exist for KFD and is used in endemic areas of India. Additional preventive measures include insect repellents and wearing protective clothing in areas where ticks are endemic.

## 24.11.16 Monkey Pox

- There has been a surge in reported cases of Monkeypox from several countries, notably in **Southeast Asia and the Western Pacific Region**.

- Monkeypox was named in 1970 after the virus responsible for the disease was initially found in captive monkeys in 1958, despite its

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limited association with these animals.

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- The virus likely persists in central African countries, primarily transmitted to humans through zoonosis from rodents in bush environments, where it has circulated for years.
- Monkeypox is a **rare viral infection similar to smallpox** although its symptoms are less severe.
- Monkeypox is a viral disease **primarily spread to the human population through zoonotic spillovers, with rodents and primates** serving as potential reservoirs.
- The **first human case** was reported in 1970 in the **Democratic Republic of Congo**.
- **Transmission:** Monkeypox can be **transmitted between humans through close contact and exposure to infected bodily fluids or lesions**. **Monkeypox spreads mainly through sexual transmission and close contact** — it is **not an airborne disease.**
- **Incubation Period:** The **incubation period of monkeypox typically ranges from 6 to 13 days** but can extend from 5 to 21 days.
- **Symptoms:** Common symptoms include **skin rash or mucosal lesions, fever, and swollen lymph nodes**, which may lead to various medical complications.
- **Vaccine:** Currently, there is **no effective vaccine** available for Monkeypox infection.

### 24.11.17 West Nile Virus

- West Nile Virus is an RNA virus transmitted through mosquito bites.

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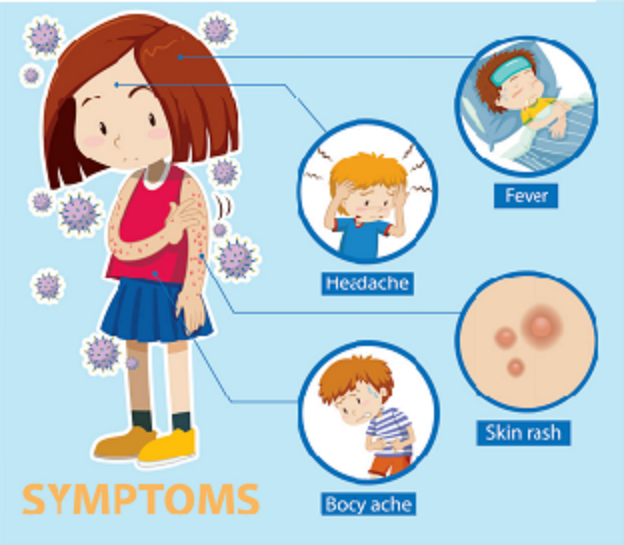


Figure 24.3 Monkeypox virus – Symptoms

- Belonging to the flavivirus family, it shares similarities with viruses causing diseases like St. Louis encephalitis, Japanese encephalitis, and yellow fever.
- The primary vectors for its spread are mosquitoes of the Culex species.

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- The virus spreads among humans and animals, including birds, which serve as its reservoir hosts.
- It was first isolated in a woman in the West Nile district of Uganda in 1937.
- It is commonly found in Africa, Europe, the Middle East, North America and West Asia.
- Older people, children and those with weakened immune systems are most at risk.

**Mosquitoes become infected when they feed on infected birds**, which circulate the virus in their blood for a few days. **The virus eventually gets into the mosquito's salivary glands**. During later blood meals (**when mosquitoes bite), the virus may be injected into humans and animals**, where it can multiply and possibly cause illness.

## 24.11.18 Adenovirus

Adenoviruses are a group of common viruses that infect the lining of your eyes, airways and lungs, intestines, urinary tract, and nervous system. They are common causes of fever, coughs, sore throats, diarrhoea, and pink eye. Infections happen in children more often than in adults, but anyone can get them. The infection usually cause only mild symptoms and get better on their own in a few days. But they can be more serious in people with weak immune systems, especially children.

### *How Adenoviruses Spread?*

These viruses are common in places with large groups of kids, such as day care centres, schools and summer camps. They are very contagious. They can spread when someone who is infected coughs or sneezes. Droplets containing the virus fly into the air and land on surfaces.

**Symptoms:** Each type of adenovirus can affect differently:

- Bronchitis: Cough, runny nose, fever, chills
- **Colds and other respiratory infections**: Stuffy and runny nose, cough, sore throat, and swollen glands
- **Ear infection**: Ear pain, irritability, fever

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- Pink eye **(conjunctivitis):** Red eyes, discharge from your eyes, tearing, feeling like there is something in your eye
- **Pneumonia**: Fever, cough, trouble breathing
- Stomach and **intestinal infections:** Diarrhoea, vomiting, headache, fever, stomach cramps
- **Swelling of the brain and spinal cord (meningitis** and **encephalitis):** Headache, fever, stiff neck, nausea, and vomiting (this is rare)
- **Urinary tract infections:** Burning and pain while urinating, frequent need to go to the washroom, blood in your urine

Call the doctor right away if your child has any of these more serious symptoms:

- Trouble breathing
- Swelling around their eyes
- Fever that does not go away after a few days
- Signs of dehydration, such as few tears or less wet diapers

## 24.11.19 Norovirus

A sharp increase in norovirus outbreaks was **reported in the United States in the winter of 2024**.

### *About Norovirus*

- It is a group of viruses that leads to **gastrointestinal illness.**
- It causes **inflammation of the lining of the stomach and intestines**, in addition to **severe vomiting and diarrhoea.**
- Norovirus is **resistant to many disinfectants** and **can heat up to 60°C.** Therefore, **merely steaming food or chlorinating water does not kill the virus**. The virus can **also survive many common hand sanitisers.**

### *Transmission*

- A person can be infected with different varieties of norovirus several times in their lives, but **developing immunity to one type doesn't provide protection against other varieties.**
- The virus can be **transferred from an infected to a healthy individual through contaminated surfaces or food.**
- The virus can primarily be **transmitted from oral-faecal.**

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- **Disease outbreaks typically occur aboard cruise ships, in nursing homes, dormitories and other closed spaces.**

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***Vulnerable Group***

The virus can affect people across all age groups but is known to cause serious symptoms in children, the elderly and people with comorbidities.

***Symptoms***

Diarrhoea, Vomiting, Stomach Ache, Gastrointestinal problems, etc., are the symptoms.

***Treatment***

**No vaccines are currently available to prevent norovirus.** Treatment for the infection focuses on relieving your symptoms.

***Status***

- There are 685 million cases annually, of which 200 million are detected among children younger than five years.
- **Nearly 50,000 children die every year due to diarrhoea caused by the virus.**
- These viruses are responsible for about 90% of viral gastroenteritis outbreaks and close to 50% of cases across the world.

## 24.11.20 Chandipura Virus

***Why in News?***

- Recently, many children have died of suspected Chandipura virus (CHPV) infection in Gujarat.

***About Chandipura Virus Infection***

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- It is a virus of the *Rhabdoviridae family*, which also includes other members such as the lyssavirus that causes rabies.

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Several species of sandflies like *Phlebotomine sandflies* and *Phlebotomus papatasi* and some mosquito species such as *Aedes aegypti* (which is also the vector for dengue) are considered vectors of CHPV.

- The virus resides in the salivary gland of these insects and can be transmitted to humans or other vertebrates like domestic animals through bites.
- The infection caused by the virus can then reach the central nervous system which can lead to encephalitis—inflammation of the active tissues of the brain.
- Disease progression can be as rapid as a **patient reporting high fever in the morning**, and **their kidneys or liver being affected by the evening.**

### *Symptoms*

- The CHPV infection presents initially with **flu-like symptoms** such as acute onset of fever, body ache and headache.
- It may then progress to **altered sensorium or seizures and encephalitis.**
- **Respiratory distress, bleeding tendencies or anaemia.**
- The infection often progresses rapidly after encephalitis, which may then lead to mortality within 24–48 hours of hospitalisation.
- **This infection has largely remained limited to children below 15 years.**

**Treatment:** The infection can only be **symptomatically managed** as currently there is no specific antiretroviral therapy or vaccine available for treatment.

### *Affected Regions in India*

- The CHPV infection was **first isolated in 1965 while investigating a dengue/chikungunya outbreak in Maharashtra.**
- However, one of the most significant outbreaks of the disease in India was seen in 2003–2004 in states such as **Maharashtra, northern Gujarat and Andhra Pradesh.**

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- The infection has **largely remained endemic to the central part of India**, where the population of CHPV **infection-spreading sandflies and mosquitoes is higher.**

## 24.11.21 Disease X (Congo)

A December 2024 outbreak in the Democratic Republic of Congo (DRC), causing over 400 deaths, raises concerns about Disease X.

***About Disease X***

- **Definition:** A **hypothetical term by WHO in 2018** for an **unknown pathogen capable of triggering a global pandemic**, highlighting the need for preparedness against emerging diseases.
- **Origins:** Emerged after the 2014–2016 West African Ebola epidemic.
- **Sources:** Can originate from viruses, bacteria, parasites, fungi or prions. Over 300 emerging diseases since 1940, 70% zoonotic.
- **Risk Factors:** High biodiversity regions, weak healthcare (e.g., Congo Basin), global connectivity, zoonotic spillovers, deforestation, agriculture, antimicrobial resistance, bioterrorism, lab accidents and climate change complicate predictions.

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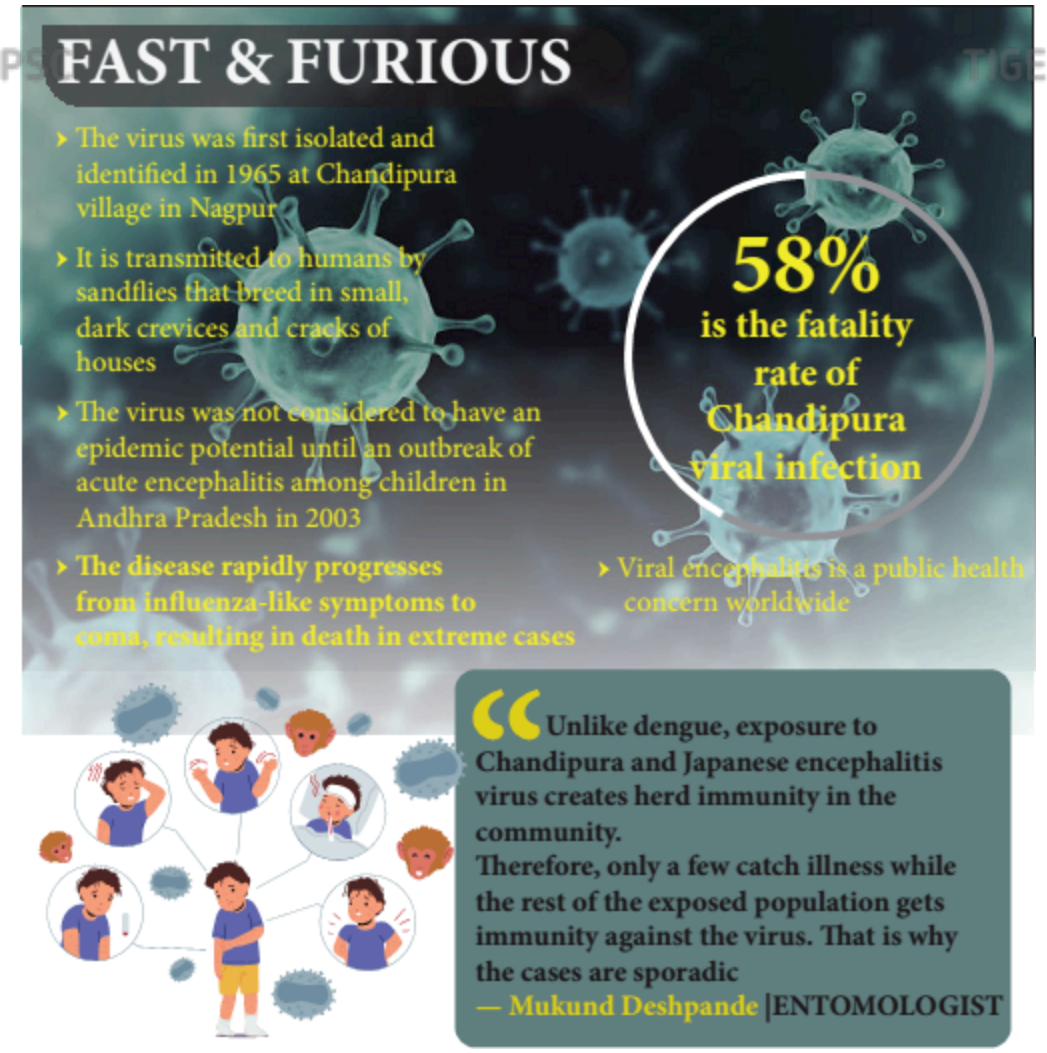


## 24.11.22 Neglected Tropical Diseases

***Why in News?***

- India is **progressing towards eliminating Neglected Tropical Diseases (NTDs)** through political support, community involvement and regional cooperation.

***About Neglected Tropical Diseases (NTDs)***

- **WHO Definition:** NTDs are a diverse **group of 20 conditions that are mainly prevalent in tropical areas**, where they mostly **affect impoverished communities** and disproportionately affect women and Children.
- The epidemiology of NTDs is complex and often **related to environmental conditions.**

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- They are **caused by variety of pathogens—viruses, bacteria, protozoa and parasitic worms (helminths).**
- They **are widespread where people in vulnerable situations** live, where water safety, sanitation and access to health care are inadequate or suboptimal.
- These diseases are called 'neglected' **because they have historically ranked very low on the global health agenda**, when they have appeared at all, receiving little attention and funding.
- **NTDs include Guinea worm, Chikungunya, Dengue, Kala Azar (*Visceral Leishmaniasis*) and Elephantiasis (*Lymphatic Filariasis*), among others. *India is home to about 12 NTDs*.**

***Which Are the Diseases Included in NTDs***

| | |
|---|---|
| • **Buruli ulcer**<br>• **Chagas uisease**<br>• **Dengue & Chikungunya**<br>• **Guinea-worm disease)**<br>• **Echinococcosis**<br>• **Foodborne Trematodiases**<br>• **Sleeping sickness**<br>• **Leishmaniasis**<br>• **Leprosy (Hansen's disease)**<br>• **Lymphatic filariasis**<br>• **Mycetoma** | • **Chromoblastomycosis and other deep mycoses**<br>• **Onchocerciasis (River Blindness)**<br>• **Podoconiosis**<br>• **Rabies**<br>• **Scabies**<br>• **Schistosomiasis**<br>• **Soil-transmitted helminthiases**<br>• **Snakebite envenoming**<br>• **Taeniasis/cysticercosis**<br>• **Trachoma**<br>• **Yaws and other endemic**<br>• **Treponematoses** |

- The epidemiology of **NTDs is complex**; some have animal and/or human reservoirs, many are vector-borne and most are associated with intricate life cycles. Preventing or eliminating NTDs is, therefore, challenging.
- It is estimated that NTDs affect more than 1 billion people, while the number of people requiring NTD interventions (both preventive and curative) is 1.5 billion.
- **Every year on 30th January, the Global community comes together to mark World NTDs Day** and renew its commitment to eliminating

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its burden worldwide.

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### *Treatment*

Antibiotics will not help adenovirus infections because these drugs only kill bacteria. Children often get over the illness on their own within a few days. Some infections, like pink eye or pneumonia, can last for a week or more.

Kids with a weak immune system may need treatment in the hospital to help them recover.

**ADENOVIRUSES VS RETROVIRUSES**

- Adenoviruses are double-stranded DNA genomes. Retroviruses are single-stranded RNA animal viruses that employ a double-stranded DNA intermediate for replication.
- Common cold is a symptom of adenovirus. The retrovirus known as human immunodeficiency virus (HIV) causes acquired immunodeficiency syndrome (AIDS) in humans.

## 24.11.23 Vaccine Drive to Fight Cervical Cancer (Cervavac)

- The **Indian Government plans to launch a three-phase vaccination campaign** against **Human Papillomavirus (HPV)** for girls aged 9–14.
- The **primary objective is to mitigate the risk of cervical cancer**, which is the **fourth most common type of cancer among women globally and the second most common in India.**

### *Vaccine Benefits*

- The HPV vaccine provides protection against HPV strains that cause **cervical cancer, as well as cancer of the anus, vagina, and oropharynx.**
- Additionally, the vaccine offers protection against HPV strains responsible for genital warts, offering comprehensive preventive benefits.

### *Indigenous HPV Vaccine – CERVAVAC*

- The Serum Institute of India launched an indigenous HPV vaccine called CERVAVAC in 2023.

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- This vaccine is likely to play a significant role in the government's vaccination campaign against HPV.
- It is said to be **effective against four strains of the virus - Type 6, Type 11, Type 16** and **Type 18**.
- CERVAVAC is based on **VLP (Virus-Like Particles),** similar to the **Hepatitis B** vaccination.

### *Cervical Cancer*

- **Cervical cancer develops in a woman's cervix** and is highly prevalent globally, ranking as the fourth most common cancer among women.
- India has a disproportionately high burden of cervical cancer**, contributing to about one-fourth of global cervical cancer deaths.**
- **Nearly all cervical cancer cases (99%) are linked to infection with high-risk HPV strains**, primarily transmitted through sexual contact.
- Effective primary prevention strategies, such as HPV vaccination, can significantly reduce the incidence of cervical cancer.

### *Strains and Oncogenic Potential*

- Persistent infections with high-risk HPV strains, particularly types 16 and 18, account for **approximately 85% of all cervical cancer cases** globally.
- At least **14 HPV types have been identified as oncogenic**, with types 16 and 18 being the most oncogenic and responsible for about 70% of cervical cancer cases.

### *Impact of Vaccination Campaign*

- The **three-phase HPV vaccination campaign aims to reduce the burden of cervical cancer in India by targeting girls aged 9–14,** a critical age group for preventive intervention.
- By providing comprehensive protection **against HPV-related cancers and genital warts**, the vaccination drive seeks to improve public health outcomes and reduce mortality associated with cervical cancer.

## 24.12 Superbug ISSUES

According to British dictionary, Superbug is defined as,

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*'An infective microorganism that has become resistant to antibiotics'.*

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Alexander Fleming has developed the world's first antibiotic Penicillin in 1928 from a fungi, *Penicillium notatum.* For this contribution, he got the Nobel Prize. But, the bacteria were already fighting back after many years due to resistance against these antibiotic. After some time, scientists also got a strain of *Staphylococcus aureus (S. aureus)* bacteria that became resistant against penicillin, due to genetic mutation.

Some bacteria can and have developed resistance to most of, if not all, the antibiotics that are being used against them. *Staphylococcus aureus is also another bacteria* that have become resistant to methicillin and many other antibiotics *(MRSA* – **methicillin resistant** *Staphylococcus aureus).* Recently, even MRSA strains have developed a resistance to vancomycin, that is a powerful antibiotic with frequent unpleasant side effects. Vancomycin was used for over 30 years as an antibiotic of last resort before resistance to it emerged.

## 24.13 Drug-Resistant Tuberculosis (TB) – MDR, MDR+, XDR, XDR+

There are different types of drug-resistant tuberculosis, namely:

### *MDR TB and XDR TB*

Rifampicin and isoniazid are the first level of medicine against TB bacteria. If these bacteria become resistant to this medicine, they are called as MDR (multi-drug resistant) TB level. XDR TB (extensively drug-resistant TB) is the next level of resistance, where bacteria become resistant to at least rifampicin and Isoniazid and additionally to fluoroquinolones, kanamycin or capreomycin andamikacin (these all are second line of tuberculosis drugs). Standard TB treatment is minimum for six months but in the case of MDR TB and XDR TB, they take two years or more. This treatment require more toxic and much more expensive drugs.

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**NEW ANTI-TUBERCULOSIS DRUG**

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US Food and Drug Administration (FDA) has approved a new drug **Pretomanid** for treating drug resistant tuberculosis – multidrug-resistant TB (MDR-TB) and extensively drug-resistant TB (XDR-TB).

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**Key Facts**

- **Pretomanid is only the third new anti-TB drug approved for use by FDA in more than 40 years.**
- Pretomanid will be part of the three-drug regimen for drug approval by the European Medicines Agency (EMA).
- **The duration of treatment for drug-resistant TB can be drastically cut** from 18 to 24 months to just six-nine months when pretomanid drug is used along with two already approved drugs – bedaquiline and linezolid.
- The all-oral, **three-drug regimen can also vastly improve the treatment success rate** and potentially decrease the number of deaths due to better adherence to treatment.

***'MDR(+) plus' and XDR(+) plus***

If first and second line of drug resistance case generate so it called as MDR+ case where patients not possible to be treated. If this case will reach up to totally drug resistant level so called as **'XDR (+) plus' TB.**

## 24.14 New Delhi Metallo-Beta-Lactamase (NDM-1)

NDM-1 is an enzyme, which makes bacteria resistant to a carbapeneme-family antibiotics. It is also a 'superbug' case. First case of NDM-1 came to limelight in December 2009 in New Delhi, where a patient was suffering from an infection caused by *Klebsiella pneumoniae.* Later on, it was detected in many nations – United Kingdom, Pakistan, United States, Canada and Japan. These all are Gram positive (+ve) bacteria like *Escherichia coli (E.coli)* and *Klebsiella pneumoniae,* got antibiotic resistance due to presence of NDM-1 gene those synthesis NDM-1 enzyme and can spread from one strain of the bacteria to another through horizontal gene transfer. This incidence was in news in India during 2010–2011.

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## 24.15 Fixed Dose Combination (FDC) Drugs

**An FDC is a blend of two or more active drug ingredients in a fixed ratio of doses.** According to US healthcare provider IMS Health, **almost half the drugs sold in India in 2014 were FDC, making it a world leader in combination drugs.**

With immediate effect **the manufacture, sale or distribution of 328 fixed dose combination (FDC) drugs for human use** has been banned by the Government.

The **health ministry's ban on FDCs includes painkillers, anti-diabetic, respiratory and gastrointestinal medicines, covering 6,000 brands.**

### *What Necessitated this?*

The expert panel probing the efficacy of 349 banned FDCs, after considering these drugs **'irrational',** cited safety issues and lack of therapeutic justification, and has given recommendation to continue the ban. There were findings that many FDCs were formulated without due attentiveness, **with dosing mismatches that could result in toxicity.**

### *Why are They Popular in India?*

- FDC is popular in India due to advantages such as **increased efficacy, better compliance, reduced cost and simpler logistics of distribution.**
- In treating infectious diseases particularly like **HIV, malaria and tuberculosis; FDC plays important role,** where giving multiple antimicrobial agents is the norm. FDCs are also useful for chronic conditions especially, when multiple disorders co-exist.

### *Is there a Flipside to FDCs?*

- Given that **much data is not available on drug–drug interaction and side-effects in FDC,** India's system for collecting data for problematic drug reactions is weak. **When multiple drugs from the same**

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**therapeutic group, like antibiotics, are clubbed together, drug resistance may be the result.**

- A lot of FDCs sold in India are unapproved because of lack of coordination between state and central regulators. **A study published in the journal of Public Library of Science (PLOS) in May, 2012 found that over 70% of non-steroidal anti-inflammatory drug (NSAID) combinations, which are used as painkillers, were being marketed in India without central government approval.**

**CENTRAL DRUGS STANDARD CONTROL ORGANISATION (CDSCO)**

The Central Drugs Standard Control Organization (CDSCO) under **Directorate General of Health Services, Ministry of Health & Family Welfare, Government of India is the National Regulatory Authority (NRA) of India. Under the Drugs and Cosmetics Act,** CDSCO have responsibility of approval of New Drugs, Conduct of Clinical Trials, laying down the standards for Drugs, control over the quality of imported Drugs in the country and coordination of the activities of State Drug Control Organizations by providing expert advice with a view of bring about the uniformity in the enforcement of the Drugs and Cosmetics Act.

Further CDSCO along with its state counterparts **is jointly responsible for grant of licences of certain specialised categories of critical drugs.**

## 24.16 Generic Drugs/Generic Medicine

Generic medicines are **unbranded medicines which are equally safe and having the same efficacy as that of branded medicines** in terms of their therapeutic value. The prices of generic medicines are much cheaper than their branded equivalent.

***Why are they Cheaper?***

- Since the manufacture of these generic drugs do not involve a repeat of the extensive clinical trials to prove their safety and efficacy, it costs less to develop them. Generic drugs are, therefore, cheaper.
- Because the compounds in the generic versions have the same molecular structure as the brand name version, **their quality is**

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**essentially the same.**

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***Why Aren't Generic Drugs More Popular?***

- **Lack of awareness about them.**
- Since they are cheap, **people who can afford branded drugs don't buy them believing them to be of inferior quality.** Chemists have to hand out exactly what's written on the prescription and most doctors except in government hospitals don't hand out generic drugs.
- Also, private doctors never hand out generic drugs because there are no kickbacks or incentives involved from pharma companies.
- The government or specifically the government's Department of Pharmaceuticals is also to blame for the lack of awareness.

***However, there are Three Fundamental Areas of Concern***

- **Efficacy of Indian-Made drugs:** Oftentimes, such drugs have been found to contain less than the required amount of active pharmaceutical ingredient (API), rendering them ineffective.
- **Lack of Data Integrity:** The poorly managed documentation practices of Indian generic firms featured as the primary criticism flagged by foreign regulatory authorities. The lack of reliable and complete data on the test results of specific drug batches, along with inconsistencies in the records presented, meant that inspection and verification of drug quality was extremely difficult.
- **Hygiene Standards of the Manufacturing Plants:** Individuals suffering from illness are especially susceptible to infections, and inspections of generic drug plants reveal pest infestations and dilapidated infrastructure.
- A proposal was received by the **Central Drugs Standard Control Organisation (CDSCO)** committee wherein **the Drugs Consultative Committee (DCC)** was apprised that registered medical practitioners can supply different categories of medicines including vaccines to their patients under the exemption provided, with certain conditions, under **Schedule K of the Drugs and Cosmetics Rules, 1945**. As of now there are no specified types of medicines which can be supplied by doctors to their patients.
- It is now proposed that **registered medical practitioners shall supply generic medicines only** and physicians samples shall be supplied free

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of cost.

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**JAN AUSHADHI SUGAM APP**

- **Union Ministry for Chemicals and Fertilisers** has launched a mobile application 'Jan Aushadhi Sugam'.
- The application **aims to enable people to search Jan Aushadhi generic medicines and the stores at the tip of their fingers.**
- It will also help analyse product comparison of Generic vs Branded medicine in form of MRP and overall savings.

### 24.16.1 Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP)

Pradhan Mantri Bhartiya Janaushadhi Pariyojana is **a campaign launched by the Department of Pharmaceuticals, Government of India, to provide quality medicines at affordable prices** to the masses through special kendras known as **Pradhan Mantri Bhartiya Jan Aushadhi Kendra.** This have been **set up to provide generic drugs,** which are available at lesser prices but are equivalent in quality and efficacy as expensive branded drugs. **Bureau of Pharma PSUs of India (BPPI)** is the implementing agency of PMBJP. BPPI (Bureau of Pharma Public Sector Undertakings of India) has been established under the Department of Pharmaceuticals, Govt. of India, with the support of all the CPSUs.

**PRADHAN MANTRI JAN-AUSHADHI YOJANA**

Initially **launched in 2008,** the scheme was **rechristened in 2015.** As on February 2024 there were 10, 454 PMBJKs in India.

***Salient Features of the Scheme***

- Ensure access to quality medicines.
- Extend coverage of quality generic medicines so as to reduce the out-of-pocket expenditure on medicines and thereby redefine the unit cost of treatment per person.

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- Create awareness about generic medicines through education and publicity so that quality is not synonymous with only high price.
- A public programme involving Government, PSUs, Private Sector, NGO, Societies, Cooperative Bodies and other Institutions.
- Create demand for generic medicines by improving access to better healthcare through low treatment cost and easy availability wherever needed in all therapeutic categories.

**RECENT AMENDMENTS IN DRUGS AND COSMETIC RULES, 1945**

Central Government has amended the Drugs Rules 1945 *vide* a Notification dated 28 December 2023 to revise the Schedule M to the said rules related to Good Manufacturing Practices and requirements of premises, plant and equipment for pharmaceutical products. [Press Release, PIB dated 9 February 2024].

## 24.17 Biosimilar Medicine

It is ***a biologic medical product*** that is almost ***an identical copy of an original product that is manufactured by a different company.*** They are ***officially approved versions of original 'innovator' products and can be manufactured when the original product's patent expires. Reference to the innovator product*** is an integral component of the approval.

### *Characteristics*

- Biological medicines ***contain active substances from a biological source***, such as living cells or organisms.
- Most biological medicines in current clinical use ***contain active substances made of proteins.***

### *Difference between Biosimilars and Generics*

- Biosimilars involve developing equivalent of ***biological entity***, while generics involve developing equivalent of a ***chemical entity*** – the Active Pharmaceutical Ingredient.
- **Cost:** Generic drugs are chemically identical to the original branded drug and, as such, cost significantly less because they don't require

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much testing. Because biosimilars are made from living organisms, though, and don't contain identical ingredients to their name-brand counterparts, they still require some testing. So, they cost more than generics, but less than the branded biologic.

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### *Need of the Hour*

- Biosimilars industry can act as ***a springboard for the pharma companies to innovate, excel and earn profit and the same needs to be promoted at the earliest.***
- We need ***to increase access through affordable pricing and some of the drugs need to be under price control.***
- Governments can support growth in production of complex generics and biosimilars by clarifying ***the regulatory framework*** for them, which is still evolving in many countries.

### *First Biosimilar Medicine – Trastuzumab*

WHO prequalifies ***first biosimilar medicine – trastuzumab,*** to increase worldwide access to life-saving breast cancer treatment. ***Trastuzumab – a monoclonal antibody*** – was included in ***the WHO Essential Medicines List*** in 2015 as an essential treatment for about 20% of breast cancers. Breast cancer is the most common form of cancer in women. 2.1 million women contracted breast cancer in 2018. 630,000 of them died from the disease, many because of late diagnosis and lack of access to affordable treatment.

## 24.18 Biofilm

An association of micro-organisms in which microbial cells adhere to each other on living or non-living surfaces within a self-produced matrix of extracellular polymeric substance is called biofilm.

- Its formation is a multi-step process starting with attachment to a surface then formation of micro-colony that leads to the formation of three-dimensional structure and finally ending with maturation followed by detachment.

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- During its formation many species of bacteria are able to communicate with one another through a specific mechanism called quorum sensing, which is a system of stimulus to co-ordinate different gene expressions.
- Bacterial biofilm is less accessible to antibiotics and the human immune system and thus poses a big threat to public health because of its involvement in a variety of infectious diseases.
- Biofilms may form on living or non-living surfaces and can be prevalent in natural, industrial and hospital settings.
- Biofilms can be present on the teeth of most animals as dental plaque, where they may cause tooth decay and gum disease.

A combination of several pathogens can synergistically interact to form biofilms in the food industry. In food-processing environments, bacteria are able to exist as multispecies biofilms, from where both spoilage and pathogenic bacteria can contaminate food. For instance, in the fishing industry, fresh fish products can suffer from biofilm formation by mixed pathogenic species (*Aeromonas hydrophila*, *L. monocytogenes*, *S. enterica*, or *Vibrio spp.*), which can imply significant health and economic issues. Synergistic interactions have been observed in a fresh-cut produce processing plant, where *E. coli* interacted with *Burkholderia caryophylli* and *Ralstonia insidiosa* to form mixed biofilms.

## 24.19 Phage Therapy

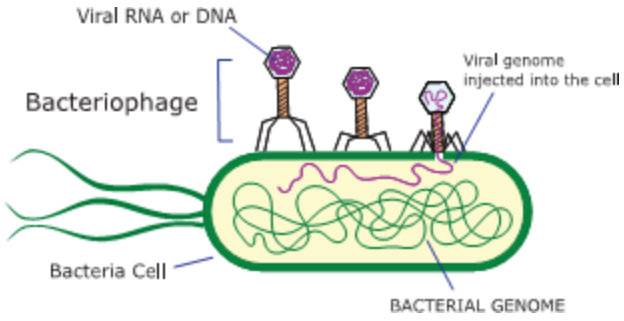


Figure 24.4 Phage Therapy

A study conducted by the **University of Exeter** shows that the public is aware of and accepts the use of **bacteria-killing viruses**, known as **phage**

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**therapy, as an alternative to antibiotics.**

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- **Phage therapy is a treatment approach that utilises bacteriophages, viruses that infect and kill specific bacteria, as an alternative to antibiotics.**
- **This method involves using these viruses to target and destroy bacterial infections, offering a promising solution for antibiotic-resistant infections.**

***Advantages of Phage Therapy***

- Phages are **highly specific in their action, targeting only the specific bacteria** they are programmed to attack.
- **Reduced Risk of Resistance:** This specificity potentially reduces the risk of antibiotic resistance, making phage therapy a valuable option for combating resistant infections.
- **Minimal Side Effects:** Patients undergoing phage therapy often experience minimal side effects compared to traditional antibiotic treatments.

***Bacteriophages***

- Bacteriophages are **viruses that infect bacteria** and **utilise them as hosts for their replication.**
- Bacteriophages are highly diverse and can target different types of bacteria, **offering a wide range of applications in phage therapy.**
- It was discovered by **Frederick Willian Twort in 1915** (Great Britain) and **Felix d'Herelle in 1917** (France).

## 24.20 Xenotransplantation

- **Definition:** According to the **US Food and Drug Administration (FDA),** 'Xenotransplantation is any procedure that involves the transplantation, implantation or infusion into a human recipient of either live cells, tissues or organs from a nonhuman animal source, or human body fluids, cells, tissues or organs that have had **ex vivo (outside living body)** contact with live nonhuman animal cells, tissues or organs'.

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- **Purpose:** The primary aim is to **address the shortage of human donor organs.** For instance, in the United States, nearly 90,000 people are waiting for a kidney transplant, with over 3,000 dying annually while still waiting.
- **Historical Context:** The practice dates back to the 1980s, with the heart being one of the first organs attempted for transplantation from animals to humans.
- **Procedure:** In xenotransplantation, the animal organ selected, such as a pig kidney, undergoes genetic modifications to improve its compatibility with the human body.
- This involves using gene-editing technologies like CRISPR-Cas9 to remove certain pig genes that produce sugars with antibodies our immune systems react to and adding human genes to improve the organ's compatibility.
- **Complications in Xenotransplantation:**
  - **Organ Rejection:** Preventing the human body from rejecting the pig organ is a significant challenge. Techniques such as embedding the pig's thymus gland with the kidney help reduce immune responses.
  - **Infection Risks:** The FDA highlights concerns about potential infections from both recognised and unknown infectious agents, which could spread to close contacts and the general population.
  - **Retroviruses:** There is a risk of cross-species infection by retroviruses, which may remain latent and cause diseases years after infection.
- **Xenotransplantation in India:** In 1997, a surgeon in Assam, performed a xenotransplantation procedure by transplanting a pig's heart into a human patient. Unfortunately, the patient passed away a week later, leading to legal repercussions.

**Why Are Pigs Often Used for Xenotransplantation:**

- **Historical Use:** Pig heart valves have been used in human surgeries for over 50 years.

## FIRST INDIGENOUS ANTIBIOTIC: NAFITHROMYCIN

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- In a ground-breaking step for **India's biotechnology sector**, **Union Minister formally launched the first indigenous antibiotic 'Nafithromycin' for resistant infections.**

**About Nafithromycin:**

- It is an antibiotic which has been **developed with the support of Biotechnology Industry Research Assistance Council (BIRAC).**
- It has been brought to market under the **trade name 'Miqnaf'.**
- It is the **country's first indigenously developed antibiotic aimed at tackling anti-microbial resistance (AMR).**
- This innovation is designed to treat community-acquired bacterial pneumonia (CABP).
- Nafithromycin's efficacy stands out as it targets both typical and atypical pathogens, **offering a potent solution where no new antibiotic in this class has been developed worldwide for over three decades**.
- The antibiotic has minimal gastrointestinal side effects, no significant drug interactions and remains unaffected by food, making it a versatile option for patients.
- It offers 10x efficacy with just three doses to combat drug-resistant pneumonia.
- **The development of Nafithromycin represents 14 years of dedicated research and an investment of ₹500 crores, with clinical trials spanning the United States Europe and India.**

- **Similarity to Humans:** Pigs have anatomical and physiological similarities to humans. Their widespread farming makes them a cost-effective and accessible source.
- **Size Matching:** Various pig breeds offer a range of organ sizes, which can be matched to the specific needs of human recipients.

# Part C: Different Immunisation Programmes In India

## 24.21 National Vector Borne Disease Control Programme (NVBDCP)

The National Vector Borne Disease Control Programme (NVBDCP) is a comprehensive programme for prevention and control of vector-borne

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diseases namely **Malaria, Filaria, Kala-azar, Japanese Encephalitis (JE), Dengue and Chikungunya**, which are covered under the overall umbrella of NRHM. Directorate of National Vector Borne Disease Control Programme (NVBDCP) is the central nodal agency for the prevention and control of vector-borne diseases.

### 24.21.1 National Anti-Malaria Programme

Malaria is a complex mosquito-borne infectious disease caused by a eukaryotic protist of the genus Plasmodium. It varies widely in epidemiology and clinical manifestation in different parts of the world. In India, Malaria is caused by *Plasmodium falciparum* or *Plasmodium vivax*, transmitted by nine major species of the female anopheles mosquitoes.

Malaria continues to pose a major public health threat in different parts of the country, particularly due to *Plasmodium falciparium*, as it is sometimes prone to complications and death, if not treated early. The major vector mosquito for rural malaria, viz., *Anopheles culicifacies*, is distributed all over the country and breeds in clean groundwater collections. Some of the vector species also breed in forest areas, mangroves, lagoons, etc., even in those with organic pollutants.

About 80 per cent of malaria burden is in Northeastern (NE) states, Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, Andhra Pradesh, Maharashtra, Gujarat, Rajasthan, West Bengal and Karnataka. The focus is on empowering grass-root workers in diagnosing and treating malaria cases even in remote and inaccessible areas by scaling-up the availability of bivalent Rapid Diagnostic Kits (RDK) and Artemisinin-based Combination Therapy (ACT). There is a need to give thrust for prevention/control of malaria (and other VBD also) in urban areas under the Urban Malaria Scheme, which is presently implemented in only 131 towns/cities. These efforts, coupled with integrated vector control strategies including distribution of Long Lasting Insecticide Treated Nets (LLIN) in endemic areas, will greatly reduce the malaria morbidity and mortality. Bi-valant Rapid Diagnostic Kit for improving diagnostic facilities for both types of malaria (*Plasmodium falciparum* and *Plasmodium vivax*) in difficult and inaccessible areas has been introduced in the programme. Performance-

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based incentive for ASHAs in endemic areas for Malaria and Kala-azar has been introduced in India.

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### 24.21.2 Japanese Encephalitis (JE) Control Programme

Japanese Encephalitis is a zoonotic disease, which is transmitted by vector mosquito, mainly belonging to *Culex vishnui* group. The transmission cycle is maintained in the nature by animal reservoirs of JE virus like pigs and water birds. Man is the dead-end host, i.e., JE is not transmitted from one infected person to other. Outbreaks are common in those areas where there is close interaction between animals/birds and human beings. The vectors of JE breed in large water bodies such as paddy fields. AES is emerging as a serious public health challenge. Given its complex etiology, medical complications and after-effects of illness, effort is on to develop a multi-pronged strategy, including safe water, sanitation, nutrition, community education, medical attention and rehabilitation to address the problem. The disease is endemic in 14 states; of which, Assam, Bihar, Haryana and Uttar Pradesh have been reporting outbreaks.

#### Jenvac

- In contrast to the Chinese vaccine, a trial using the **inactivated vaccine Jenvac**, developed by **Bharat Biotech in collaboration with the National Institute of Virology (NIV)**, Pune, using an Indian virus strain**, demonstrated superior protection, even with a single dose.**
- Seroprotection rates at the end of one year and two years were significantly higher for Jenvac compared to the Chinese vaccine, with 81.7% and 88.5% for Jenvac and 47.9% and 68.3% for the Chinese vaccine, respectively.
- **The trial also showed significantly higher levels of neutralising antibodies in children who received two doses of Jenvac compared to two doses of the Chinese vaccine.**

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# 24.22 Communicable Diseases Control Programme

## 24.22.1 National AIDS Control Programme (NACP)

Human Immunodeficiency Virus (HIV) is retro-virus. It is also known as human T-lymphotropic virus-III (HTLV-III), lymphadenopathy-associated virus (LAV) and AIDS-associated retrovirus (ARV). Transmission involves anal, vaginal or oral sex, blood transfusion, contaminated hypodermic needles, exchange between mother and baby during pregnancy, childbirth, breastfeeding or other exposure to one of the above bodily fluids.

***Objectives***

1. Suppressing the spread of HIV infection in India.
2. Strengthening India's capacity to respond towards HIV/AIDS on a long-term basis.

To reflect the extreme urgency with which HIV prevention and control must be pursued in India, the AIDS-II project of the National AIDS Control Programme is operational across all States and Union Territories. Moreover, a Centrally Sponsored Scheme with 100 per cent financial assistance from the Government of India direct the State AIDS Control Societies and certain selected Municipal Corporations/AIDS Control Societies.

Realising the potential threat due to HIV/AIDS, the world's AIDS day is celebrated on 1 December every year to spread the awareness about life-threatening diseases.

## 24.22.2 National Leprosy Eradication Programme (NLEP)

Leprosy is a chronic disease caused by a bacillus, *Mycobacterium leprae.* Official figures (WHO) show that more than 2,13,000 people, mainly in Asia and Africa, are infected, with approximately 2,49,000 new cases reported in 2008. *M. leprae* multiplies very slowly and the incubation period of the disease is about five years. Symptoms can take as long as 20 years to appear. Leprosy is not highly infectious. It is transmitted via

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droplets, from the nose and mouth, during close and frequent contacts with untreated cases. Untreated, leprosy can cause progressive and permanent damage to the skin, nerves, limbs and eyes. Early diagnosis and treatment with multi-drug therapy (MDT) remain the key elements in eliminating the disease as a public health concern.

***Objectives***

1. Early detection that can be achieved through active surveillance by the trained health workers;
2. Regular treatment of cases by provisions of Multi-Drug Therapy (MDT) at fixed incentres or a nearby village with moderate to low endemic areas/district;
3. Intensifying health education and public awareness campaigns for the removal of social stigma attached to the disease;
4. Appropriate medical rehabilitation and the leprosy ulcer care services.

### 24.22.3 Revised National Tuberculosis Control Programme (RNTCP)

In 1962, the National TB Control Programme was started with the aim to detect cases earliest and treat them. The District Tuberculosis Centre (DTC) and the Primary Health Institutions together assist in implementation of the programme, in the district. The state-level organisations support District Tuberculosis Programme (DTP) for the coordination and supervision of the programme. In 1993, the Revised National Tuberculosis Control Programme (RNTCP) that is based on the DOTS strategy began as a pilot project. It was launched as a national programme in 1997; however, rapid RNTCP expansion began in late 1998.

**NIKSHAY**, a web-based reporting for TB programme, has been another notable achievement that was initiated in 2012. It has enabled capture as well as transfer of individual patient data from country's remotest health institutions.

***Objectives***

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- Pursuing quality DOTS expansion and enhancement, by improving case finding and cure through an effective patient-centred approach so as to reach all patients, especially the poor.
- Addressing TB-HIV, MDR-TB and other challenges, through scaling up of TB-HIV joint activities, DOTS Plus, and other such relevant approaches.
- Contributing to health system strengthening, through collaboration with other health programmes and general services.
- Involving all health care providers-nongovernmental and private, public by scaling up approaches that are based on a public-private mix (PPM). This can ensure adherence to the international standards of TB care.
- Engaging people with TB, and affected communities to demand, and contribute towards effective care. This will involve scaling-up of the community TB care; communication and social mobilisation and creating demand through context-specific advocacy.
- Enabling and promoting research for the development of new drugs, vaccines and diagnostics. Operational Research will also be required to improve programme performance.

### 24.22.4 Pulse Polio Programme

Following World Health Assembly resolution in 1988, with the global initiative of eradication of polio in the year 1998, the Pulse Polio Immunisation Programme was launched in 1995, in India. Under this programme, children in the age group of 0–5 years are administered polio drops during the National and Sub-national immunisation rounds (in high risk areas) every year. As estimated, about 172 million children are immunised during each of the National Immunisation Day (NID).

On 24 February 2012, the WHO removed India from the list of the countries with active endemic wild poliovirus transmission.

***Objective***

- Achieving 100 per cent coverage under Oral Polio Vaccine. It aimed to immunise children through improved social mobilisation, plan mop-up operations in the areas where poliovirus has almost disappeared and thus maintain high level of morale among the public.

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Note: Two types of vaccine protecting against polio are inactivated poliovirus vaccine (IPV) and oral poliovirus vaccine (OPV). The IPV is given as an injection in the arm or leg, according to patient's age. The polio vaccine may be given simultaneously as other vaccines. Most people need to get polio vaccine during their early age. IPV is given to children in 4 doses at these ages: 2 months, 4 months, 6–18 months, and the booster dose at 4–6 years.

The first polio vaccine was the inactivated polio vaccine and was developed by Jonas Salk and came into use in 1955. While the oral polio vaccine was developed by Albert Sabin and in 1961 came into commercial use. In the US, OPV has not been used since 2000; however, it is still used in several parts of the world.

## 24.23 National Nutritional Programmes

### 24.23.1 Mid-Day Meal Programme

The Mid-day Meal Scheme involves provision of lunch at free of cost to school children on all working days. Its key objectives include: Protecting children from classroom hunger; Increasing school enrolment and attendance; Improved socialisation among children belonging to all castes; and Addressing malnutrition, and social empowerment through provision of employment to women.

The scheme has a long history, especially in Tamil Nadu and Gujarat, and has been expanded to all parts of India after a landmark direction by the Supreme Court of India on 28 November 2001.

Tamil Nadu was the first state to implement this programme.

***Objectives***

- Improvement in the nutritional status of children in classes I–VIII in Government, Local Body and Government-aided schools, as well as EGS and AIE centres.
- Encouraging poor children, who belong to disadvantaged sections, to attend school more regularly and help them in concentrating on classroom activities.
- Providing nutritional support to the children of primary stage in drought-affected areas.

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## 24.24 Others

### 24.24.1 Universal Immunisation Programme (UIP)

Universal Immunisation Programme is the name given to the vaccination programme, which was launched by the Government of India in 1985. In 1992, it became a part of Child Survival and Safe Motherhood Programme and is currently serving as one of the key areas under the National Rural Health Mission (NRHM) since 2005.

The programme consists of vaccination for 9 diseases: In 2007, addition of TB, Diphtheria, Pertussis/(whooping cough), Poliomyelitis, Diarrhoea, Japanese Encephalitis, Measles, Tetanus, Hepatitis B and Pneumonia. Hepatitis B was done to the UIP. Therefore, UIP has nine vaccine-preventable diseases under this programme.

In 2014, it was announced that four vaccines would be added to the programme, namely Rotavirus, Rubella and Japanese Encephalitis, as well as Injectable Polio Vaccine.

### 24.24.2 National Rural Health Mission

The National Rural Health Mission (NRHM) was launched by the Hon'ble Prime Minister on **12$^{th}$ April 2005**, to provide accessible, affordable and quality health care to the rural population, especially the vulnerable groups. The key features in order to achieve the goals of the Mission include making the public health delivery system fully functional and accountable to the community, human resources management, community involvement, decentralisation, rigorous monitoring & evaluation against standards, the convergence of health and related programmes from village level upwards, innovations and flexible financing and also interventions for improving the health indicators.

***Janani Suraksha Yojana***

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Janani Suraksha Yojana (JSY) is a safe motherhood intervention under the National Rural Health Mission (NHM), launched in 2005.

The scheme is under implementation in all states and Union Territories (UTs).

- Its objective is to reduce maternal and infant mortality among poor pregnant women.

***Cash Assistance for Institutional Delivery (in Rs)***

| Category | Rural Area | | Total | Urban Area | | Total |
|---|---|---|---|---|---|---|
| | Mother's package | ASHA's package* | | Mother's package | ASHA's package** | |
| **LPS** | 1400 | 600 | 2000 | 1000 | 400 | 1400 |
| **HPS** | 700 | 600 | 1300 | 600 | 400 | 1000 |

*ASHA package of ₹600 in rural areas include ₹300 for ANC component and ₹300 for facilitating institutional delivery.

**ASHA package of ₹400 in urban areas include ₹200 for ANC component and ₹200 for facilitating institutional delivery.

*Source:* nhm.gov.in (as accessed in July 2024)

- Its aim to promote institutional delivery among poor pregnant women.
- Its objective includes providing public health facilities to sick infants up to one year of age.

### 24.24.3 National Nutrition Mission

National Nutrition Mission was introduced by the Ministry of Women and Child Development, Government of India, to create awareness relating to malnutrition among pregnant women and lactating mothers through reduction of the incidence of anaemia among young children, adolescent girls and women.

In *Global Nutrition Report 2016,* India's progress was slow in addressing chronic malnutrition, manifest in stunting (low weight for age), wasting (low weight for height), micronutrient deficiencies and overweight. Even Bangladesh and Nepal are marginally better than India. As per the

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latest Global Nutrition Report 2022, the situation is no better. Barring the war-torn Afghanistan, India has performed worse than all the countries in the South Asian region in the Global Hunger Index 2022. It has ranked 107 out of 121 countries. India ranked 101 out of 116 countries in the Global Hunger Index (GHI) 2021.

The poor nutritional status of adolescent girls, combined with child marriage and multiple pregnancies even before becoming an adult is another reason that leads to 30 per cent of all children being born with low birth-weight.

### INTENSIFIED MISSION INDRADHANUSH 5.0

- **Intensified Mission Indradhanush (IMI) 5.0** is an initiative by the **Union Ministry of Health and Family Welfare** to **enhance routine immunisation** coverage in India.
- It **targets children up to 5 years of age and pregnant women** across all districts in the country.
- The **primary focus of IMI 5.0 is to improve Measles and Rubella vaccination coverage with the aim of eliminating both diseases by 2023 (ended on 14 October 2023)**.
- IMI 5.0 involves the **participation of Janpratinidhi** (people's representatives) and **social media influencers to raise awareness and encourage families to visit vaccination centres**.
- **Since its inception in 2014, 11 phases of Mission Indradhanush have been completed, with the ongoing 12th phase having vaccinated 5.06 crore children and 1.25 crore pregnant women**.
- IMI 5.0 is the fifth phase of Mission Indradhanush, **launched to achieve full immunisation coverage for all children and pregnant women.**
- Unlike previous phases targeting children up to 2 years old in selected districts, **IMI 5.0 covers all districts and includes children up to 5 years old.**
- The **campaign focuses on vaccines provided under the Universal Immunisation Programme (UIP)** such as **BCG, OPV, IPV, Pentavalent, Rotavirus, PCV, Measles-Rubella**, and others.
- It emphasises on improving Measles and Rubella vaccine coverage to prevent congenital anomalies and disabilities in children.
- IMI 5.0 uses the **U-WIN digital platform** for Routine Immunisation, facilitating online booking, tracking, and monitoring of immunisation services across all districts in the country.

**U-WIN: U-WIN is going to be the single source of information for immunisation services for the targeted population under the Universal Immunisation Programme.** The platform aims to register and provide vaccinations for each

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pregnant woman, document her delivery outcomes, register each newborn, administer birth doses, and manage all vaccination records.

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Under-nutrition (at both level of malnutrition and micronutrient deficiencies) among nearly 70 per cent of school-going children challenges the capacity for physical growth and cognitive development.

The past decade has seen a steady build-up of momentum around nutrition with the setting up of UN; the World Health Assembly adoption (in 2012) of the 2025 global targets for maternal, infant and development goals, which centre-stage the ending of all forms of malnutrition for all people by 2030, to name a few. India's progress clearly lags behind which is needed to eliminate malnutrition by 2030.

### 24.24.4 Mission Indradhanush

Mission Indradhanush was launched by the Ministry of Health and Family Welfare, Government of India, on 25 December 2014.

#### *Objective*

Mission Indradhanush aims to cover all those children under the age of 2 years by 2020, who are either unvaccinated, or are partially vaccinated against vaccine-preventable diseases. It will also cover all pregnant women against seven preventable diseases: **Diptheria, Hepatitis B, Whooping cough, Tetanus, Poliomyelitis, Tuberculosis** and **Measles.** In some states, vaccination against **Japanese Encephalitis and Haemophilus Influenza Type B are** also included. In 2016, **Rubella, Japanese Encephalitis, Injectable Polio Vaccine** and **Rota-virus** vaccine also become part of this mission. **Pneumonia** vaccine was also added to this mission in 2017.

#### *Intensified Mission Indradhanush (IMI)*

To further intensify the immunisation programme, the government launched the Intensified Mission Indradhanush (IMI) on October 8, 2017.

#### *Key Facts*

- Through this programme, Government of India aims to reach each and every child up to two years of age and all those pregnant women who

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have been left uncovered under the routine immunisation programme/UIP.

- Intensified Mission Indradhanush will cover low performing areas in the selected districts (high priority districts) and urban areas. Special attention will be given to unserved/low coverage pockets in sub-centre and urban slums with migratory population.

### 24.24.5 Ayushman Bharat

Government of India has introduced a **National Health Protection Scheme,** which will **cover over 10 crore poor vulnerable families (around 50 crore beneficiaries) providing coverage of up to 5 lakh (per family per year) for secondary and tertiary care hospitalisation. It will subsume the on-going centrally sponsored schemes – Rashtriya Swasthya Bima Yojana (RSBY)** and the **Senior Citizen Health Insurance Scheme (SCHIS). This scheme is named as Ayushman Bharat.**

***Highlights of the Scheme***

- **Coverage:** This scheme will have the benefit cover of ₹5 lakh per family per year. **There will be no cap on family size and age in the scheme** to ensure that no one is left out. It will **also include pre- and post-hospitalisation expenses.**
- **Target: More than 10 crore families belonging to poor and vulnerable population based on SECC database** will be the targeted beneficiaries. Benefits of the scheme are portable across the country and a **beneficiary covered under the scheme will be allowed to take cashless benefits from any public/private empanelled hospitals across the country.**
- **Role of State Governments:** State governments can expand AB-NHPM both horizontally and vertically. **States will be free to choose the modalities for implementation.** Implementation can be done through insurance company or directly through Trust/Society or a mixed model.
- **Council:** For guiding in policy directions and nurturing coordination between Centre and States, there is provision for setting up **Ayushman**

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**Bharat National Health Protection Mission Council (AB-NHPMC)** at apex level Chaired by Union Health and Family Welfare Minister.

Prime Minister Narendra Modi in 2018 rolled out the Centre's flagship scheme – **Pradhan Mantri Jan Arogya Abhiyaan, also known as Ayushman Bharat or the National Health Protection Mission (AB-NHPM).**

### *Significance of the Project*

- It is one of the **world's largest health scheme.**
- **Meant to help the poor and the economically deprived; the scheme will be available for 10.74 crore beneficiary families and about 50 crore Indian citizens.**
- According to health ministry officials, the **71st round of National Sample Survey Office (NSSO) says that 85.9% of rural households and 82% of urban households have no access to healthcare insurance/assurance.**

### *Ayushman Bharat Digital Mission*

- Launched in 2021 by the Prime Minister, the **Ayushman Bharat Digital Mission** will provide **a digital health ID** to the people who will hold their health records.
- The pilot project of Ayushman Bharat Digital Mission was announced on August 15, last year. Currently, the programme is being implemented in the pilot phase in six union territories (Chandigarh, Ladakh, Dadra and Nagar Haveli and Daman and Diu, Puducherry, Andaman and Nicobar Islands and Lakshadweep).

### *Features of the Mission*

- It is a digital health ecosystem under which **every Indian citizen will now have unique health IDs, digitised health records with identifiers for doctors and health facilities.**
- The scheme will come under **the Ayushman Bharat Pradhan Mantri Jan Arogya Yojana.**
- It **comprises six key building blocks –** HealthID, DigiDoctor, Health Facility Registry, Personal Health Records, e-Pharmacy and Telemedicine.

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- The **National Health Authority** has been given the mandate to design, build, roll-out and implement the mission in the country.
- **The core building block of the mission** is that the Health ID, DigiDoctor and Health Facility Registry shall be owned, operated and maintained by the Government of India.
- **Private stakeholders will have an equal opportunity** to integrate and create their own products for the market. The core activities and verifications, however, remain with the government.
- Under the Mission, **every Indian will get a Health ID card** that will store all medical details of the person including prescriptions, treatment, diagnostic reports and discharge summaries.
- **Health ID is a randomly generated 14-digit number** used for the purpose of uniquely identifying persons, authenticating them, and threading their health records (only with their informed consent) across multiple systems and stakeholders.
- The **citizens will be able to give their doctors and health providers one-time access to this data** during their visits to the hospital for consultation.

India eradicated many diseases in recent past – Small pox, Rinderpest, Polio, Yaws, Maternal and Neonatal Tetanus.

**Smallpox:** It is a viral disease eradicated in 1980.

**Rinderpest:** It affects cattle and ruminants. India became free from Rinderpest in October 2010.

**Polio:** In March 2014, South-East Asia Region was certified as polio-free. For the last time, it was recorded in West Bengal in 2011.

**Yaws:** It is a bacterial infection, which mostly affects tribal population. It targets skin, bones and joints and was eliminated from India in September 2006 but certified by WHO and UNICEF in July 2016.

**Maternal and Neonatal Tetanus (MNT):** This disease spreads in unhygienic circumstances where mothers and babies become vulnerable to MNT and die. In August 2015, this disease was completely eliminated.

***Need for this Mission***

- The mission aims to liberate citizens from the challenges of finding the right doctors, seeking appointment, payment of consultation fee, making several rounds of hospitals for prescription sheets among

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several others and will empower people to make an informed decision to avail the best possible healthcare.

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### 24.24.6 Poshan Abhiyan

POSHAN Abhiyan was launched by Prime Minister Shri Narendra Modi in Jhunjhunu, Rajasthan in March 2018. It targets to reduce level of under-nutrition and other related problems by ensuring convergence of various nutrition related schemes. It also targets stunting, under-nutrition, anaemia (among young children, women and adolescent girls) and low birth rate.

It will monitor and review implementation of all such schemes and utilise existing structural arrangements of line ministries wherever available. Its large component involves gradual scaling-up of interventions supported by on-going World Bank assisted Integrated Child Development Services (ICDS) Systems Strengthening and Nutrition Improvement Project (ISSNIP) to all districts in the country. Originally, the program was designed to span three years until March 2021, with a focus on addressing the prevalent issue of malnutrition in the nation. However, recognizing its effectiveness and potential, the government has consolidated multiple initiatives with similar goals, such as the Supplementary Nutrition Program and POSHAN Abhiyan, into a single comprehensive program known as Mission POSHAN 2.0. This integration aims to enhance operational efficiency, promote synergy, and adopt an integrated approach to delivering nutrition services.

These are key Nutrition strategies and interventions: IYCF (Infant and Young child feeding), Food and Nutrition, Immunisation, Institutional Delivery, WASH (Water, Sanitation and Hygiene), De-worming, ORS-Zinc, Food Fortification, Dietary Diversification, Adolescent Nutrition, Maternal Health and Nutrition, ECD (Early Childhood development)/ECCE (Early Childhood Care and Education), Convergence, ICT-RTM (Information and Communication. Technology enabled Real Time Monitoring) and Capacity Building etc.

#### *Anaemia Mukt Bharat*

The *Anaemia Mukt Bharat* – intensified Iron-plus Initiative aims to strengthen the existing mechanisms and foster newer strategies for tackling

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anaemia.

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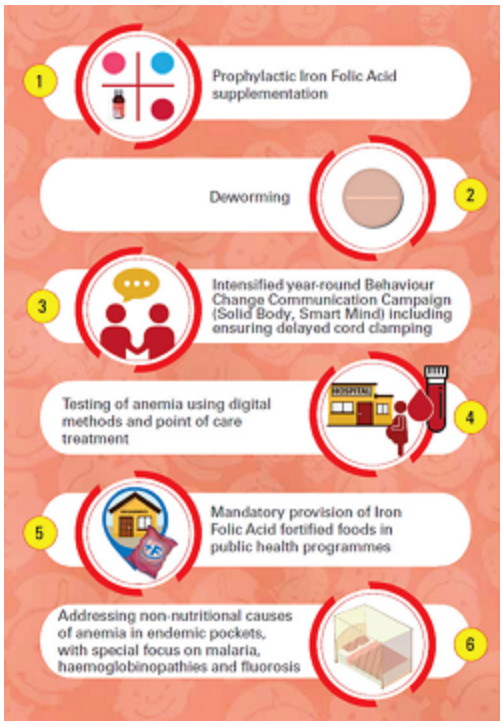
It focuses on six target beneficiary groups, through six interventions and six institutional mechanisms to achieve the envisaged target under the *Poshan Abhiyan*.

The Anaemia Mukt Bharat strategy has been designed to reduce prevalence of anaemia by 3 percentage points per year among children, adolescents and women in the reproductive age group (15–49 years), between the year 2018 and 2022.

The Anaemia Mukt Bharat strategy will be implemented in all villages, blocks, and districts of all the States/UTs of India through existing delivery platforms as envisaged in the National Iron Plus Initiative (NIPI) and Weekly Iron Folic Acid Supplementation (WIFS) programme.

It will focus on the following interventions:




## 24.24.7 'UMMID' Initiative

Government launches **'UMMID' initiative** to **tackle inherited genetic diseases of new born babies.**

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**UMMID initiative aims to** establish NIDAN **(National Inherited Diseases Administration)** Kendras to provide counselling, prenatal testing

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and diagnosis, management, and multidisciplinary care in Government Hospitals wherein the influx of patients is more.

- Produce skilled clinicians in Human Genetics.
- Undertake screening of pregnant women and new born babies for inherited genetic diseases in hospitals at aspirational districts.

#### *Need for and Significance of the Initiative*

- In India's urban areas, congenital malformations and genetic disorders are the third most common cause of mortality in newborns. With a very large population and high birth rate, and consanguineous marriage favoured in many communities, prevalence of genetic disorders is high in India.
- **Establishment of patient care services for genetic disorders** is the need of the time.

#### *UMMID (Unique Methods of Management and Treatment of Inherited Disorders) Initiative*

- **Department of Biotechnology** has started the UMMID Initiative which is designed on the concept of **'Prevention is better than Cure'.**
- UMMID aims to **create awareness about genetic disorders amongst clinicians and establish molecular diagnostics in hospitals** so that the fruits of development in medical genetics reach the patients in India.

### 24.24.8 Pradhan Mantri Matru Vandana Yojana (PMMVY)

***Pradhan Mantri Matru Vandana Yojana (PMMVY)*** is ***a maternity benefit*** rechristened from erstwhile ***Indira Gandhi Matritva Sahyog Yojana (IGMSY).*** The scheme is ***a conditional cash transfer scheme for pregnant and lactating women.*** It provides ***a partial wage compensation to women*** for wage-loss during childbirth and childcare and to provide conditions for safe delivery and good nutrition and feeding practices.

#### *Objectives*

1. Promoting appropriate practice, care and institutional service utilisation during pregnancy, delivery and lactation.
2. Encouraging the women to follow (optimal) nutrition and feeding practices, including early and exclusive breastfeeding for the first six

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months.

States/UTs and districts are awarded for best performance under PMMVY.

***Key Performers***

1. In the category of ***best performance since inception of the Scheme*** to States/UTs ***having population of more than 1 crore*** the ***first position was awarded to the state of Madhya Pradesh,*** followed by Andhra Pradesh and Haryana in the third position.
2. In the same category ***among States/UTs having population of less than 1 crore,*** Dadra & Nagar Haveli is in the first position. Himachal stood second and Chandigarh is at the third position.
3. ***In the District level awards for States/UTs with population of more than 1 crore,*** the ***first position went to Indore in Madhya Pradesh.*** In the same category for States/UTs with population under than 1 crore, the first position went to Serchhip in Mizoram.
4. Providing cash incentives for improved health and nutrition to pregnant and lactating mothers.

## QUESTIONS AND ANSWERS

### Objective Questions

**1. Which one of the following is synthesized in human body that dilates blood vessels and increase blood flow?[UPSC (Pre) – 2024]**

(a) Nitric Oxide

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(b) Nitrous Oxide
(c) Nitrogen Dioxide
(d) Nitrogen Pentoxide

**2. Consider the following statement in relation to Janani Suraksha Yojana:** **[UPSC (Pre) – 2023]**

1. It is a safe motherhood intervention of the State Health Department.
2. Its objective is to reduce maternal and neonatal mortality among poor pregnant women.
3. It aims to promote institutional delivery among poor pregnant women.
4. Its objective includes providing public health facilities to sick infants up to one year of age.

How many of the statements given above are correct?
(a) Only one
(b) Only two
(c) Only three
(d) Only four

**3. Consider the following statements in the context of intervention being undertaken under Anaemia Mukt Bharat Strategy:** **[UPSC (Pre) – 2023]**

1. It provides prophylactic calcium Supplements for pre-school Children, adolescents pregnant Women.
2. It runs a campaign for delayed cord Clamping at the time of child-birth.
3. It provides for periodic deworming to children and adolescents
4. It addresses non-nutritional causes of anaemia in endemic pockets with special focus on malaria, hemoglobinopathies and fluorosis.

How many of the statements given above are correct?
(a) Only one

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(b) Only two
(c) Only three
(d) All four

**4. 'Wolbachia method' is sometimes talked about with reference to which one of the following:** **[UPSC (Pre) – 2023]**

(a) Controlling the viral diseases spread by mosquitoes
(b) Converting crop residues into packing material
(c) Producing biodegradable plastics
(d) Producing biochar from thermochemical conversion of biomass

**5. Consider the following statements :** **[UPSC (Pre) – 2022]**

1. Biofilms can form on medical implants within human tissues.
2. Biofilms can form on food and food processing surfaces.
3. Biofilms can exhibit antibiotic resistance

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

**6. With reference to Ayushman Bharat Digital Mission, consider the following statements:** **[UPSC (Pre) – 2022]**

1. Private and public hospitals must adopt it.
2. As it aims to achieve universal health coverage, every citizen of India should be part of it ultimately.
3. It has seamless portability across the country.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

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**7. Consider the following statements:** **[UPSC (Pre) – 2021]**

1. Adenoviruses have single-stranded DNA genomes whereas retroviruses have double- stranded DNA genomes.
2. Common cold is sometimes caused by an adenovirus whereas AIDS is caused by a retrovirus.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**8. What is the importance of using Pneumococcal Conjugate Vaccines in India?** **[UPSC (Pre) – 2020]**

1. These vaccines are effective against pneumonia as well as meningitis and sepsis.
2. Dependence on antibiotics that are not effective against drug-resistant bacteria can be reduced.
3. These vaccines have no side effects and cause no allergic reactions.

Select the correct answer using the code given below:

(a) 1 only
(b) 1 and 2 only
(c) 3 only
(d) 1, 2 and 3

**9. Which one of the following statements is not correct?** **[UPSC (Pre) – 2019]**

(a) Hepatitis B virus is transmitted much like HIV.
(b) Hepatitis B, unlike Hepatitis C, does not have a vaccine.
(c) Globally, the number of people infected with Hepatitis B and C viruses are several times more than those infected with HIV.
(d) Some of those infected with Hepatitis Band C viruses do not show the symptoms for many years

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**10. Which of the following are the reasons for the occurrence of multi-drug resistance in microbial pathogens in India?** **[UPSC (Pre) – 2019]**

1. Genetic predisposition of some people
2. Taking incorrect doses of antibiotics to cure diseases
3. Using antibiotics in livestock farming
4. Multiple chronic diseases in some people

Select the correct answer using the code given below:

(a) 1 and 2
(b) 2 and 3 only
(c) 1, 3 and 4
(d) 2, 3 and 4

**11. Consider the following statements:** **[UPSC (Pre) – 2017]**

(i) In tropical regions, the same mosquito that transmits dengue also transmits zika virus.
(ii) Sexual transmission of zika virus is possible.

Which of the statements given above is/are correct?

(a) Only (i)
(b) Only (ii)
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

**12. Which of the following are the objectives of 'National Nutrition Mission'?** **[UPSC (Pre) – 2017]**

(i) To create awareness relating to malnutrition among pregnant women and lactating mothers.
(ii) To reduce the incidence of anaemia among young children, adolescent girls and women.
(iii) To promote the consumption of millets, coarse cereals and unpolished rice.
(iv) To promote the consumption of poultry eggs.

Select the correct answer using the code given below:

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(a) (i) and (ii)
(b) (i), (ii) and (iii)
(c) (i), (ii) and (iv)
(d) (iii) and (iv)

**13. 'Mission Indradhanush' launched by the Government of India pertains to** **[UPSC (Pre) – 2016]**

(a) Immunisation of children and pregnant women
(b) Construction of smart cities across the country
(c) India's own search for Earth-like planets in outer space
(d) New Educational Policy

**14. Among the following, which were frequently mentioned in the news for the outbreak of Ebola virus recently?** **[UPSC (Pre) – 2015]**

(a) Syria and Jordan
(b) Guinea, Sierra Leone and Liberia
(c) Philippines and Papua New Guinea
(d) Jamaica, Haiti and Surinam

**15. H1N1 virus is sometimes mentioned in the news with reference to which one of the following diseases?** **[UPSC (Pre) – 2015]**

(a) AIDS
(b) Bird flu
(c) Dengue
(d) Swine flu

**16. Consider the following pairs:** **[UPSC (Pre) – 2014]**

| *Vitamin* | *Deficiency disease* |
|---|---|
| (i) Vitamin C | Scurvy |
| (ii) Vitamin D | Rickets |
| (iii) Vitamin E | Night blindness |

Which of the pairs given above is/are correctly matched?
(a) (i) and (ii)

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(b) Only (iii)
(c) (i), (ii) and (ii)
(d) None of these

**17. Consider the following diseases:** **[UPSC (Pre) – 2014]**

(i) Diphtheria
(ii) Chickenpox
(iii)Smallpox

Which of the above diseases has/have been eradicated in India?

(a) (i) and (ii)
(b) Only (iii)
(c) (i), (ii) and (iii)
(d) None of these

**18. Widespread resistance of malaria parasite to drugs like chloroquinine has prompted attempts to develop a malarial vaccine to combat malaria. Why is it difficult to develop an effective malarial vaccine?** **[UPSC (Pre) – 2010]**

(a) Malaria is caused by several species of Plasmodium.
(b) Humans do not develop immunity to malaria during natural infection.
(c) Vaccines can be developed only against bacteria.
(d) Humans are only intermediate hosts and not the definitive hosts.

**19. Consider the following statements:** **[UPSC (Pre) – 2010]**

(i) Every individual in the population is equally susceptible host for Swine Flu.
(ii) Antibiotics have no role in the primary treatment of Swine Flu.
(iii)To prevent the future spread of swine flu in the epidemic area, the Swine (pigs) must all be clued.

Which of the statements given above is/are correct?

(a) (i) and (ii)
(b) Only (ii)

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(c) (ii) and (iii)
(d) All of these

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## Answer Key

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (a) | 2. (b) | 3. (c) | 4. (a) | 5. (d) | 6. (b) | 7. (b) | 8. (b) |
| 9. (b) | 10. (b) | 11. (c) | 12. (a) | 13. (a) | 14. (b) | 15. (d) | 16. (a) |
| 17. (b) | 18. (b) | 19. (b) | | | | | |

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# CHAPTER 25
# CORONA DIARY: COVID 19 – HISTORY AND UPDATES

## IN THIS CHAPTER...


**25.1 Coronavirus**

**25.2 What is a Public Health Emergency of International Concern**

**25.3 Schedule H1 Drug**

**25.4 WHO Declares Coronavirus a Pandemic**

**25.5 What is the 'Rapid Test'?**

**25.6 ACE 2 Receptors**

**25.7 Convalescent Plasma Therapy**

**25.8 B-Cells and T-Cells**


The most important function of B-cells is **humoral (antibody-driven) immunity** that is antibody production to fight bacteria and viruses.

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**25.9 Coronavirus Strain Isolated in India**

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25.11.4 Sputnik (Gamaleya)
**25.12 Monoclonal Antibodies Work Against COVID**
**25.13 mRNA Vaccine Technology**

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## 25.1 Coronavirus

It is ***a new virus identified by Chinese researchers.*** It was responsible for a new pneumonia-like illness that had swept Wuhan city of China since December 2019, leaving many ill and created panic.

### Key Facts

- Coronaviruses (CoV) are a large family of viruses (Coronaviridae family) that cause illness ranging from the common cold to more severe diseases such as Middle East Respiratory Syndrome (MERS-CoV) and Severe Acute Respiratory Syndrome (SARS-CoV).
- Coronaviruses are ***zoonotic,*** meaning they are transmitted between animals and people.
- Coronaviruses can affect mammals including pigs, cattle, cats, dogs, martens, camels, hedgehogs and some birds.

## 25.2 What is a Public Health Emergency of International Concern

- As per the WHO, it is 'an extraordinary event' that constitutes a 'public health risk to other States through the international spread of disease'

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and 'to potentially require a coordinated international response'. Previous emergencies include Ebola, Zika and H1N1.

- ***The responsibility of declaring an event as an emergency lies with the*** Director-General of the WHO and requires the convening of a committee of members.

## 25.3 Schedule H1 Drug

**The Drugs and Cosmetics Rules, 1945,** are the set of rules under **the Drugs and Cosmetics Act, 1940,** which has ***provisions for classification of drugs into different schedules and also guidelines for storage, sale and display.*** **Schedule H1** has been introduced through Gazette notification GSR 588 (E) dated 30-08-2013 **to check the indiscriminate use of antibiotics, anti-TB and some other drugs in the country.** The schedule contains certain **third and fourth generation antibiotics, certain habit-forming drugs and anti-TB drugs.**

## 25.4 WHO Declares Coronavirus a Pandemic

As fresh coronavirus cases continued to be reported from different countries, **the World Health Organization** finally declared the novel coronavirus **a 'pandemic'.** WHO said it was deeply concerned both by **the alarming levels of spread and severity of the disease and inaction.** Earlier, on 30 January, 2020 the WHO declared it as **a 'Public Health Emergency of International Concern'.**

**WHAT IS AN OUTBREAK, EPIDEMIC AND PANDEMIC**

1. An **outbreak** is understood to be a ***sudden rise in the cases of a disease*** at a particular place.
2. An **epidemic** is a ***large outbreak among a particular population or region.***

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3. A **pandemic** is ***'the worldwide spread of a new disease'.*** **There is no specific number of countries that a disease must touch for WHO to classify it as a pandemic.**

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## Symptoms of COVID-19

- COVID-19 is **similar to other respiratory illnesses and its symptoms include *fever, dry cough, sore throat and headache. There may also be aches and pains, fatigue and, in some cases, vomiting and diarrhoea.***
- While most cases are mild, some individuals may experience more severe symptoms such as ***shortness of breath and difficulty breathing, and may experience pneumonia in both lungs.*** Those with health issues or underlying conditions may also have a harder time recovering.
- ***It may take up to fourteen days after exposure for symptoms to appear.***

## 25.5 What is the 'Rapid Test'?

To tackle suspicions of whether the novel coronavirus has spread into the community and if not, then to prevent such a scenario, the **Kerala Government** announced **'rapid tests'** to **ensure speedy results within half-an-hour.**

- A rapid test is **conducted to determine whether there has been any kind of recent viral infection in a person's body.**
- When a pathogen enters a human body, specific antibodies are released as a response to the virus.
- ***A rapid test can detect the presence of such antibodies in blood, serum or plasma samples quickly, indicating a viral infection.*** Rapid testing is conducted usually **to check for community transmission of a virus during an epidemic.**
- According to the health department, it is a simple test that can be done with a person's blood sample and ***will give results within 10–30 minutes.*** It is also a low-cost test.

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## 25.6 ACE 2 Receptors

**The SARS-CoV-2** virus enters a human cell through 'ACE2 receptors', which are enzymes **that generate small proteins that are useful for the cell.**

In the search for treatment for COVID-19, many researchers focussed their attention on a specific protein that allows the virus to infect human cells. Called the angiotensin-converting enzyme 2, or ACE2 receptor, the protein provides the entry point for the coronavirus to hook into and infect a wide range of human cells.

ACE2 is a protein on the surface of many cell types. It is an enzyme that generates small proteins – by cutting up the larger protein angiotensinogen – which then go on to regulate functions in the cell.

Using the spike-like protein on its surface, the SARS-CoV-2 virus binds to ACE2–like a key being inserted into a lock–prior to entry and infection of cells. Hence, ACE2 acts as a cellular doorway – a receptor – for the virus that causes COVID-19.

## 25.7 Convalescent Plasma Therapy

**Plasma** is the liquid part of the blood. **Convalescent plasma,** extracted from the blood of patients recovering from an infection, is a source of antibodies against the infection. The therapy involves using their plasma to help others recover. **For COVID-19,** this has been one of the treatment options. The donor would have to be a documented case of COVID-19 and healthy for twenty-eight days since the last symptoms.

### *How Plasma Therapy Works?*

1. Blood is drawn from a person who has recovered from COVID-19 sickness.
2. The serum is separated and screened for virus-neutralising antibodies.

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3. **Convalescent serum,** that is the blood serum obtained from one who has recovered from an infectious disease and especially rich in antibodies for that pathogen, is then administered to a COVID-19 patient.
4. The sick acquires passive immunisation.

### *How Long the Antibodies will Remain in the Recipient?*

- After the antibody serum is given, it will stay in the recipient for **at least three to four days.** Various studies have confirmed that during this period, the sick person usually recovers.

### *Plasma Therapy Debate*

- The Indian Council of Medical Research (ICMR) has dropped the plasma therapy from the national guidelines as it found that convalescent plasma was ineffective in arresting COVID-19.
- In China and Netherlands too, the plasma therapy yielded very little or no results.

**INDIA'S FIRST PLASMA BANK**

- Established at the Institute of Liver and Biliary Sciences (ILBS), Delhi.
- Aim: To ease access to plasma that is being used as a trial to treat COVID-19 patients.

## 25.8 B-Cells and T-Cells

### *B-Cells*

- B stands for '**bone marrow**', in which these cells mature.
- B-cell function is threefold:
  1. Antigen presentation to other immune cells
  2. Cytokine secretion
  3. Antibody production

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- The most important function of B-cells is **humoral (antibody-driven) immunity** that is antibody production to fight bacteria and viruses.
- These antibodies are Y-shaped proteins that are specific to each pathogen and are able to lock onto the surface of an invading cell and mark it for destruction by other immune cells.
- When naive B-cells comes in contact with antigen that fits its receptor, it binds to it and takes it inside its membrane for processing which leads to formation of antigen-MHC complexes that T-cells bind to and release cytokines. These T-cell cytokines activate the B-cell.
- Inactivated B-lymphocytes also act as antigen-presenting cells (APCs) which attach to an antigen (foreign particle) and process it, producing membrane markers that act as a warning that a T-cell can understand.
- B-lymphocytes present these antigen-MHC complexes to T-cell receptors, causing T-cell activation.
- Where the non-specific immune system (the innate immune system) is concerned, this reaction produces the opposite effect–T-cell inactivation that is B-lymphocytes also help prevent auto-immune reactions.
- B-lymphocytes release cytokines (essential for cell-to-cell communication) which invites white blood cells in the form of phagocytes to the areas where B-cell antibodies have attached to antigens.

## *T-Cells*

- A type of white blood cell (lymphocyte) that is of key importance to the immune system.
- T-cells tailor the body's immune response to specific pathogens.
- They are like the soldiers who search out and destroy the targeted invaders.
- The 'T' stands for '**thymus**' – the organ in which these cells mature.
- **Types of T-cells:** There are two main types of T-cells:
  1. **Helper T-cells** stimulate B-cells to make antibodies and helps develop killer cells.
  2. **Killer T-cells** directly kill cells that have already been infected by a foreign invader.

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## *What do T-cells use?*

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T-cells use cytokines as messenger molecules to send chemical instructions to the rest of the immune system.

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Note: The **primary and secondary immune responses** are carried out with the help of two special types (T-cells) of lymphocytes present in our blood, i.e., B-lymphocytes (B-cells) and T-lymphocytes. The B-lymphocytes produce an army of proteins in response to pathogens into our blood to fight with them. These proteins are called antibodies. The T-cells themselves do not secrete antibodies but help B-cells produce them.

## 25.9 Coronavirus Strain Isolated in India

**India was only the fifth country** in the world besides Japan, Thailand, the United States and China **to have successfully isolated the COVID-19 virus strain,** helping it take the first step towards expediting the development of drugs, vaccines and rapid diagnostic kits in the country.

## 25.10 RT-PCR Test

The causative agent for COVID-19 is the **SARS-CoV-2 virus.** It is **an RNA virus that** means it infiltrates a healthy cell to multiply and survive. Thus, **the RT-PCR test is for the identification of SARS-CoV-2 RNA.** In this, **the RNA is converted to DNA through a process called 'reverse transcription' for detecting the virus.**

The SARS-CoV-2 RNA is **generally detectable in respiratory specimens** during the acute phase of infection.

1. For that **upper and lower respiratory specimens (such as nasal, nasopharyngeal) are collected.**
2. This **sample is treated** with several chemical solutions that remove substances, such as proteins and fats, and **extracts only the RNA present in the sample.**
3. Real-time RT-PCR setup usually **goes through 35 cycles,** which means that by the end of the process, **around 35 billion new copies of the**

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**sections of viral DNA are created** from each strand of the virus present in the sample.

4. As new copies of the viral DNA sections are built, **the marker labels attach to the DNA strands** and then release **a fluorescent dye,** which is measured by the machine's computer and presented in real-time on the screen. The computer tracks the amount of fluorescence in the sample after each cycle. **When the amount goes over a certain level of fluorescence, this confirms that the virus is present.**

The boon and bane of RT-PCR testing are in its **capacity to amplify even one viral gene segment in the sample to generate a detectable signal** – a positive test.

1. It is a boon because it **accurately detects the presence of virus.**
2. It is a bane because it is **prone to false-negative and false-positive results.**

### *False RT-PCR Reports*

- During sample preparation for testing, if even one gene segment falls into the tube from the laboratory environment, it will be amplified and the test will be positive – but, **false positive. A false negative PCR** means that a person with infection was missed by the test, but that is in the very nature of PCR. **The viral load is lower in the throat than in the nasopharynx.**
- Hence, throat swabs are falsely negative in 60% of tests and nasopharyngeal swabs in 30%, according to published studies.
- **An incorrectly taken nasal swab may miss the virus altogether and lead to a false-negative test.**

## 25.11 Important Vaccines Against Covid

### 25.11.1 Covaxin

- The first and the only indigenously developed vaccination against Covid-19 by **Bharat Biotech** in association with the **Indian Council of Medical Research (ICMR)** and the **National Institute of Virology (NIV).**

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- Covaxin is an **inactivated viral vaccine**. This vaccine is developed with Whole-Virion Inactivated Vero Cell-derived technology. They contain inactivated viruses, which cannot infect a person but still can teach the immune system to prepare a defence mechanism against the active virus.
- The **effectiveness of Covaxin is 81%** according to interim 3rd phase trial results.

### 25.11.2 Covishield

- Covishield has been developed by the **Oxford University** in collaboration with **AstraZeneca** and is being manufactured by the **Serum Institute of India (SII).**
- It has been prepared using the **viral vector platform.**
- A replication-deficient chimpanzee adenovirus (cold virus), ChAdOx1 which has been modified to enable it to carry the Covid-19 spike protein into the cells of humans.
- It is basically incapable of infecting the receiver but teaches the immune system to recognise the Covid-19 spike protein as a threat and work on building antibodies against it.
- The **effectiveness of the Covishield vaccine is nearly 90%** as per the global reports
- The interval between 1st and 2nd doses is 12–16 weeks.
- Efficacy after the second dose can vary from 70–90%.

### 25.11.3 Pfizer (Comirnaty/BioNTech)

- It is jointly developed by **Pfizer** and **BioNTech.**
- It uses **mRNA (messenger RNA) technology.**
- It **relies on synthetic genes** that can be generated and manufactured in weeks, and produced at a scale more rapidly than conventional vaccines.
- The vaccine works by introducing mRNA into the cells. This mRNA teaches cells to make a protein from the Covid-19 virus. The body detects this protein and triggers a robust immune response. This creates antibodies, as well as long-lasting immunity against SARS-CoV-2 infections.

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- After the body produces an immune response, it discards all of the vaccine ingredients, just as it would discard any substance that cells no longer need.

### 25.11.4 Sputnik (Gamaleya)

- It is an **adenovirus-based vaccine** created by the **Gamaleya National Centre of Epidemiology and Microbiology.**
- Sputnik V is a **combined vector vaccine**, which means it uses inactivated viruses called vectors.
- Unlike other types of vaccines, Sputnik V does not contain any form of the SARS-CoV-2 virus. Instead, it relies on a **modified version of two different adenoviruses – known as rAd26 and rAd5.**
- It has been modified to contain the SARS-CoV-2 spike protein to create immunity against Covid-19.
- When the vaccine enters the body, the adenovirus vectors within the vaccine enter the cells and recreate the spike protein found on the SARS-CoV-2 virus. As soon as the cell recognises this foreign spike protein, the immune system kicks in and begins to make antibodies against the virus.

## 25.12 Monoclonal Antibodies Work Against COVID

Immunity of human body works as a defence mechanism against pathogens (i.e., disease-causing agents, like virus, bacteria, etc.), those prompting the production of antibodies.

**The bodies of the majority of people who recover from COVID-19 produce antibodies to the SARS-CoV-2 virus. Scientists have found that these antibodies persist for at least 5–7 months after the infection.**

Scientists can also produce these antibodies (main-made protein) in a laboratory setting to be infused into the blood, i.e., **monoclonal antibodies (mAb).** mAb are identical copies of an antibody that targets one specific antigen. Scientists can make monoclonal antibodies by exposing the white blood cells to a particular antigen. They can then select a single white blood

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cell or clone and use this as the basis to produce many identical cells, making many identical copies of the monoclonal antibody. Monoclonal antibodies have been used earlier against Ebola and HIV disease and can be further use to kill various viral & bacterial pathogens. Each monoclonal antibody is specific to its matching antigen. In this way, against COVID-19, there are several authorised monoclonal antibody therapies applied. So, they stimulate various immunological function to kill antigens. They are used in a wide variety of application including the diagnosis, treatment of various diseases like cancer, autoimmune disorders and infections.

The SARS-CoV-2 virus's spike glycoprotein, which sits on its surface, functions to facilitate the virus's entry into the body's cells. Some SARS-CoV-2 antibodies bind to the spike protein and prevent the virus from entering the cell. Currently, all the monoclonal antibodies for COVID-19 for which the **FDA (Food and Drug Administration, USA)** have issued emergency use authorisation target the spike protein.

## 25.13 mRNA Vaccine Technology

Messenger RNA or mRNA technology works by teaching our cells to recognise and protect us against infectious diseases. One of the challenges with this new technology is that it must be kept cold to maintain stability during transport and storage.

### *mRNA Vaccines*

- mRNA vaccines **trick the body into producing some of the viral proteins itself.**
- They work by using **mRNA** which is **the molecule that essentially puts DNA instructions into action.**
- Inside a cell, mRNA is used as a template to build a protein.

### *How it Works?*

1. To produce an mRNA vaccine, scientists produce a synthetic version of the mRNA that a virus uses to build its infectious proteins.

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2. This mRNA is delivered into the human body, whose cells read it as instructions to build that viral protein, and therefore create some of the virus's molecules themselves.
3. These proteins are solitary, so they do not assemble to form a virus.
4. The immune system then detects these viral proteins and starts to produce a defensive response to them.

## Significance of mRNA Vaccines

- There are two parts to our immune system: **innate** (the defences we're born with) and **acquired** (which we develop as we come into contact with pathogens).
- **Classical vaccine molecules** usually only work with the acquired immune system and the innate immune system is activated by another ingredient called an **adjuvant.**
- Interestingly, **mRNA in vaccines could also trigger the innate immune system,** providing an extra layer of defence without the need to add adjuvants.

## Self-Replicating mRNA Covid-19 Vaccines

A California based pharmaceutical company has developed **ARCT-154, a self-amplifying mRNA vaccine** against Covid-19 infection.

**Benefits:** It offered 95% protection against severe Covid-19 and 55% against Covid infection.

## QUESTIONS AND ANSWERS

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## Objective Questions

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1. **With reference to monoclonal antibodies, often mentioned in news, consider the following statements:[UPSC (Pre) – 2025]**
   i. They are man-made proteins.
   ii. They stimulate immunological function due to their ability to bind to specific antigens.
   ii. They are used in treating viral infections like that of Nipah virus.

   Which of the statements given above are correct?
   (a) i and ii only
   (b) ii and iii only
   (c) i and iii only
   (d) i, ii and iii

2. **In the context of vaccines manufactured to prevent COVID-19 pandemic, consider the following statements: [UPSC (Pre) – 2022]**
   1. The Serum Institute of India produced COVID-19 vaccine named Covishield using mRNA platform.
   2. Sputnik V vaccine is manufactured using vector based platform.
   3. COVAXIN is an inactivated pathogen based vaccine.

   Which of the statements given above are correct ?
   (a) 1 and 2 only
   (b) 2 and 3 only
   (c) 1 and 3 only
   (d) 1, 2 and 3

3. **Which one of the following statements best describes the role of B cells and T cells in the human body?[UPSC (Pre) – 2022]**
   (a) They protect the body from environmental allergens.
   (b) They alleviate the body's pain and inflammation.
   (c) They act as immunosuppressants in the body.
   (d) They protect the body from the diseases caused by pathogens.

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**4. The term 'ACE2' is talked about in the context of: [UPSC (Pre) – 2021]**

(a) Genes introduced in the genetically modified plants.
(b) Development of India's own satellite navigation system.
(c) Radio collars for wildlife tracking.
(d) Spread of viral diseases.

**Answer Key**

| | | | |
|---|---|---|---|
| 1. (d) | 2. (b) | 3. (d) | 4. (d) |

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UNIT-XI Miscellaneous Topics

# CHAPTER 26
# MISCELLANEOUS TOPICS

## IN THIS CHAPTER...


**26.1 Heavy Water**
- 26.1.1 Various Uses of Heavy Water
- 26.1.2 Harmful Effects of Heavy Water
- 26.1.3 Indian Data regarding Heavy Water

**26.2 3D Printer, 4D Printing and Organ Printing**
- 26.2.1 3D Printing
- 26.2.2 4D Printing
- 26.2.3 Organ Printing/Bioprinting

**26.3 Narcoanalysis, Polygraphy Test and Brain Mapping**

**26.4 Kevlar: Bullet Proof Jackets**


A hydrogel is a 3D matrix of hydrophilic polymers capable of swelling in water and retaining significant amounts of water

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within its structure, upheld by chemical or physical cross-linking of polymer chains. The initial discovery of hydrogels dates back to 1960, credited to Wichterle and Lim.

**26.5 Agent Orange**

**26.6 Hydroponics, Aeroponics and Aquaponics**

**26.7 Biodigester Technology or Biotoilet**

**26.8 Drinking Water Purification/Water Purifier**

**26.9 Air Cleaners and Air Purifiers**

**26.10 Safety on Wheels**

26.10.1 Different types of Active Safety Features in Vehicle

**26.11 Wireless Transmission of Electricity, WiTricity and Wireless Charging**

**26.12 Cybernatic Organism (Cyborg)**

26.12.1 Neuralink

**26.13 Incandescent Bulb, CFL (Compact Fluorescent Lamps) and LED (Light-Emitting Diode)**

26.13.1 Incandescent Bulb

26.13.2 CFL (Compact Fluorescent Lamps)

26.13.3 Light-Emitting Diodes (LED)

**26.14 High-Speed Train: Bullet Train, Maglev (Magnetic Levitation) and Hyperloop**

26.14.1 Bullet Train

26.14.2 Maglev Technology

26.14.3 Hyperloop Technology

26.14.4 Vande Bharat (VB)/Train 18

**26.15 Indian Research Centre in Antarctica and Artic**

26.15.1 Antarctica Research Centre

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26.15.2 IndARC (India's Arctic Observatory)
**26.16 Television Technology: CRT, LCD, PLASMA, LED, OLED and QLED**
**26.17 Quantum Computing**
**26.18 Quantum Supremacy**
**26.19 Quantum Bits**
**26.20 Quantum Key Distribution (QKD)**
**26.21 Quantum Entanglement**
**26.22 Quantum Satellite**
**26.23 Quantum Teleportation**
**26.24 Difference between Classical Computer and Quantum Computer**
**26.25 Microwave Oven**
**26.26 Black Box in an Airplane**

All new cars sold in the European Union are now mandated to be equipped with sophisticated data recorders, named 'Event Data Recorder' (EDRs). Inspired by similar technology called "black boxes" used in airplanes, these devices promise to heighten accident investigation and road safety

**26.27 Carbon Fiber**
**26.28 KAVACH System**
**26.29 Rare Earth Elements**
**26.30 Accelerometer**
**26.31 Indian Standard: Electronic items, Food and Agriculture Related**
26.31.1 Bureau of Indian Standard (BIS)
26.31.2 Indian Standard Institutions (ISI)

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26.31.3 ECO-Mark Scheme
26.31.4 AGMARK
26.31.5 Food Safety and Standards Authority of India (FSSAI)

**26.32 Food and Health Related Issues**

26.32.1 MSG Maggie Issues
26.32.2 Bisphenol A (BPA)
26.32.3 Polished Rice
26.32.4 Antioxidants
26.32.5 Preservatives (Preserve to Reserve)
26.32.6 Aerated Drinks/Soft Drinks
26.32.7 Per- and Polyfluoroalkyl Substances (PFAS)
26.32.8 Fats: Types, Trans Fat and Omega 3
26.32.9 Gluten Intolerance and Celiac Disease
26.32.10 Milk Property and Milk Process
26.32.11 Alternative to Sugars
26.32.12 Anti-ageing
26.32.13 Fortified Rice
26.32.14 Triclosan
26.32.15 Polyethylene Terephthalate (PET)
26.32.16 Rice Bran Oil
26.32.17 Glycemic Index
26.32.18 A1 & A2 Milk
26.32.19 Weight Loss Drugs: Tirzepatide (Mounjaro) and Semaglutide (Wegovy)

## 26.1 Heavy Water

In 1932, Dr. H. C. Urey discovered heavy water and got the Nobel Prize. In heavy water, one of hydrogen's isotopes, deuterium is used. Heavy water is

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also popularly called as deuterium oxide ($D_2O$).

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### 26.1.1 Various Uses of Heavy Water

- Deuteration of drugs, altered metabolism of some drugs, which could lead to increased duration of pharmacological action and/or reduction of toxic manifestation.
- By deuterated drugs, therapeutic **efficiency is enhanced.**
- In the **preservation of several tissues/organs;** e.g., kidneys can be preserved without degradation or decay.
- **Deuteration of anticancer drugs decreases side effects**, while also retaining its efficacy.
- Heavy water can also be used in **nuclear magnetic resonance spectroscopy.** It can control blood pressure in humans.
- D2O is used as a **moderator** in certain nuclear reactors and it can also be used as cooling agent.
- **It increases the thermal stability of some vaccines** like oral polio vaccines or other macro-molecules, which had freezing temperature.

### 26.1.2 Harmful Effects of Heavy Water

- Its higher concentration in human body (approximately 50%) can cause cell dysfunctioning and death.
- It alters plant cell division process and hence plant growth is retarded.
- Its higher concentration (approximately 90%) severely affects aquatic life, i.e., kills fish, tadpoles, flatworms, drosophila etc.

**Differences between Normal water ($H_2O$) and Heavy water ($D_2O$):**

- $D_2O$ is more viscous than $H_2O$ due to high melting point and boiling point.
- $D_2O$ exhibits around 2600 times neutron-moderating ability as compared to $H_2O$.

### 26.1.3 Indian Data regarding Heavy Water

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India is the world's largest producer of heavy water. India has made its production self-reliant. India exports high-quality $D_2O$ to several countries like the United States and the Republic of Korea.

Heavy Water Board is a constituent unit, which is under the Department of Atomic Energy in the Government of India. The organisation is tasked with and is responsible for production of D2O that can be used both as a 'moderator' and a 'coolant' in nuclear reactors. But D2O is used as a good moderator and not as a coolant.

The research in D2O production was started by Bhabha Atomic Research Centre (BARC) in 1960s. The first heavy water reactor plant in India was established at Nangal, Punjab. Heavy Water Board is operating in seven heavy water plants around the country – Baroda in Gujarat, Tuticorin in Tamil Nadu, Kota in Rajasthan, Thal in Maharashtra, Hazira in Gujarat, Talcher in Odisha and Manuguru in Telangana.

Note: 1. **Global Heavy Water Market - 2023–2030:** Global Heavy Water Market reached US$ 61.3 million in 2022 and is expected to reach US$ 98.0 million by 2030, growing with a CAGR of 6.1% during the forecast period 2023–2030.

2. Argentina, the Soviet Union, the United States, Norway, Canada, Pakistan, France etc., are other countries of heavy water producer in the world.

## 26.2 3D Printer, 4D Printing and Organ Printing

### 26.2.1 3D Printing

**3D printing/additive manufacturing/rapid prototyping** is a process of making three-dimensional solid objects with the help of a digital file/format.

A 3D-object is printed by using additive processes or successive layering method. These layers are in the form of thinly sliced horizontal cross section of the eventual object.

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## Invention of 3D Printing

The first patent of 3D printing was given in 1986, when stereolithography apparatus (SLA) was used for object printing. This machine was first invented by Charles (Chuck) Hull in 1983. 3D printing becomes possible through three steps – Modelling, Printing and Finishing.

1. **Modelling:** Before going to print any object, a computer should have **computer-aided design (CAD)** or **animation modelling software**, which can store a digital format in computer.
2. **Printing:** After modelling, the next step is printing, where the liquid nature of cartridges used to dry in successive layers and form a complete object in a particular shape or geometric feature.
3. **Finishing:** In the case of any mismatch in object's shape or size, this object will again go in printing and then proper finishing becomes possible.

## Materials Used in 3D Printing

Many different materials can be used for 3D printing, such as glass-filled polyamide, stereolithography materials (epoxy resins), silver, titanium, steel, wax, photopolymers, ABS plastic, PLA, polyamide (nylon), polycarbonate etc.

## Uses of 3D Printing in Various Industries

Aerospace Industry, Architecture Industry, Automotive Industry, Commercial Products, Consumer Goods Industry, Consumer Electronics,

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Defence Industry, Dental Industry, Education Industry, Medical Industry, Moulding Industry etc.

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Note: **'Foodini'** (3D Printer for Food) is a 3D printer that uses edible ingredients.

### *Challenges Faced by 3D Printing*

Anything can be made by 3D printer, so criminals can misuse it to create illegal products. 3D printers can also create issues of Intellectual Property Rights for the 3D printer users.

## 26.2.2 4D Printing

- In **4D printing,** a **3D-printed** object will transform itself into another modification under the influence of some additional effects as temperature, light or other environmental stimuli.
- **3D-printed** objects maintain fixed shape like any plastic or metal parts but **4D-printed** objects can transform themselves over time.
- Different types of material can be used as an expandable element of smart materials like **hydrogel and elements with shape memory as polymeric** materials. **Hydrogel** is capable of absorbing a large amount of water and can be programmed to shrink or expand with changes in external environmental conditions.
- **Elements with Shape Memory:** It can be considered as smart materials because of their ability to return in its original shape from a deformed shape when stimuli is applied. The shape-memory materials are cheaper and have low density. Some of the elements under research are carbon fibres, smart textile materials, programmable wood etc.

### *Possible Applications of 4D Printing*

- **Medical Devices** – for stents placed into blood vessels. Drug capsules that release medicine.
- **Home Appliances** – control and adjust according to heat and humidity.
- **Clothes and Footwear**
  - ❑ ***Implants*** – for humans and animals made from biocompatible materials.
  - ❑ ***Soft Robots*** – can activate without reliance on an electric device.

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- **Smart Valves**
  - ❑ ***Sensors*** for infrastructure lines.

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**HYDROGEL**

A hydrogel is a 3D matrix of hydrophilic polymers capable of swelling in water and retaining significant amounts of water within its structure, upheld by chemical or physical cross-linking of polymer chains. The initial discovery of hydrogels dates back to 1960, credited to Wichterle and Lim.

Because of their soft, water-absorbent properties, hydrogels hold great promise for diverse applications in biomedical engineering, soft robotics and environmental studies.

- **It is used in controlled drug delivery in patients.**
- **In mobile air-conditioning systems, it can be used to absorb moisture from room to reduce humidity.**
- Hydrogels have received extensive attention as functional lubricants because of their **excellent anti-friction and anti-wear properties.**
- **Hydrogel in Food:** Hydrogels exhibit various applications in food packaging, nutrient enrichment, satiety and freshness detection.
- **Hydrogel in Agriculture:** Hydrogel works as water reservoirs around the root mass zones of the plant. It can trap irrigation and rainwater that can then be collected, stored and gradually released for crop requirements over prolonged durations.
- **Hydrogel in Medical Sector:** Medical dressing, Drug delivery, Cardiac repair, Neural tissue repair, Bone tissue repair, Spinal cord injury repair, Osteoarthritis cartilage damage repair, etc.

## 26.2.3 Organ Printing/Bioprinting

It is a mode of integrating biology to use 3D printing technology. It is a technique (3D printer) that is used for the development of artificial organs.

Many types of organs are developed in different shapes, **flat** – such as skin, **vascular** – such as blood vessels or **hollow** – such as the bladder.

Here, artificial organs are prepared for transplantation with the help of the recipient's own cells, so they have no chance of organ rejection.

### *Process of Organ Printing*

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In bioprinting, **bioink** is used as a cartridge, which is a mixture of specific cells like stem cells etc. These stem cell layering forms on biopaper, which is made up of collagen, water and hydrogel. All these layers will fuse in successive manner when biopaper will be removed or dry. The new bioink contains two different polymer components used in medical industry: **a natural polymer extracted from seaweed** and a **sacrificial synthetic polymer.**

### *Benefits of Bioprinting*

- No rejection of organs and DNA because own cells are used
- No need of immune-suppressant drugs that are required in normal organ transplant
- It will eliminate organ donation
- No need to wait for organ availability

## 26.3 Narcoanalysis, Polygraphy Test and Brain Mapping

This technique is used primarily in psychotherapy and in law enforcement for interrogation purposes. The term '**truth drug**' or '**truth serum**' refers to the barbiturate that is administered in narcoanalysis.

### *Concept*

In **the narcoanalysis test, the subject's inhibitions and mental defences are lowered by interfering with his/her nervous system with the use of drugs.**

**Narcoanalysis** is a type of **controlled administration** with the help of intravenous hypnotic medicine used in suspected person. **Hypnotic medicines are also called as truth drugs.**

**Polygraph** is also popular with the name of lie detector. It uses several instruments to check various activities in human body, such as blood pressure, pulse, respiration, breathing system and various other activities during any queries.

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**Brain mapping** is a neuroscience technique where brain wave frequency is analysed. Here, **a sensor is attached to head and connected with on-board computer screen. These sensors monitor brain-based electrical activity and the result appears in the form of certain waves due to stimulus.**

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**The narcoanalysis test is conducted by dissolving 3 grams of Sodium Pentothal or Sodium Amytal in 3000 millilitres of distilled water.**

**The drug suppresses the central nervous system, decreases blood pressure and then slows down the heart rate, thereby sending the subject into a hypnotic trance, resulting in a lack of inhibition.**

## *Criticisms*

The issue of adopting narcoanalysis test during interrogation in India has been widely debated. It raises serious ethical, legal and scientific questions. Some of the main points of criticism are:

1. **Violates Fundamental Rights: Right to silence and privacy violation.**
2. **Violates human rights and is a form of torture.**
3. **Unreliable and inadmissible in court.**
4. **Suspects who are used to drug abuse** may not experience the intended effect of the barbiturate and may still remain 'in control'.
5. A narcoanalysis test report may have validity but as it includes confessions made by a semiconscious person, it must not be totally held admissible in court.

## *In Favour*

1. The police must be permitted to use all available methods, so long as they do not violate any legal or ethical principles, since law enforcement is also important. Denying these methods can hamper the police in their investigations especially in cases such as terrorism where the defendant may use a variety of means to escape prosecution. Therefore, one has to take a balanced view.

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2. Polygraph and brain mapping are not intrusive techniques like narcoanalysis and must be permitted to be used by police in their investigations.

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### *P300 Test*

**When the brain recognises a sound of a person, it generates a special type of electric wave, which is called a P300**. Sensors are connected to the head of a person who is undergoing a P300 test and the subject is made to sit before a computer monitor. He/she is then shown certain images or made to hear certain sounds. The sensor monitors electrical activity in the brain and registers P300 waves, which are generated only if the subject has some connection with the stimulus, and in this case, it is pictures or sounds. The subject is not asked any question in P300 test. These tests are used by investigators to crosscheck their findings, and to determine if a suspect is telling the truth or make him/her reveal some facts pertaining to a case.

### *Status of Narco Test in India*

In a Supreme Court decision of 2010, it declared that narco-analysis, polygraph and brain mapping tests on an accused are illegal. However, the court allowed the use of similar techniques in criminal cases with the consent of the subject and under certain restrictions.

## 26.4 Kevlar: Bullet Proof Jackets

Kevlar is simply a super-strong plastic. There are literally hundreds of synthetic plastics made by **polymerisation (polymeric material)**, with all of them having different properties. Kevlar's amazing properties are partly due to its internal structure, how its molecules are naturally arranged in regular, parallel lines and partly due to the way it is made into fibres that are knitted tightly together.

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**POLYMERIC MATERIAL**

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It is a **large molecule or macromolecule** composed of millions of repeated linked units, each a relatively **light and simple molecule.**

**Polymeric materials** deal with the subject areas of **Material Science and Organic Chemistry.**

- The polymer is a substance made up of a **large number of molecules** that link together to form larger molecules.
- **Example of polymeric material –** an example of a synthetic polymer is plastic. An example of a natural polymer is rubber.
- Nylon 6.6 is a **synthetic polymer called a polyamide** because of the **characteristic monomers of amides in the backbone chain.** Two different molecules (**adipic acid and hexamethylenediamine**) are combined to create repeat units of 6 carbon atoms, thus the name nylon 6.6.

Super-strong Kevlar is known for its use in body armour (bullet-resistant vests). The following are some of Kevlar's properties:

- It is strong as well as light in weight.
- It does not melt like plastics.
- It can be used even at very low temperatures.
- High tensile strength, high toughness and chemical stability at high temperatures in aromatic polyamides.
- Kevlar is capable of preventing attacks from different chemicals. Though, long exposure to strong acids or bases will degrade it over a period of time.

Note: Kevlar is widely used as a friction material in the automotive industry and a combustion protection material in the aerospace industry.

## 26.5 Agent Orange

Agent Orange is the code name of a herbicide (plant-killing chemicals) developed for the US military, especially for use in tropical climates. 'Agent Orange' was named for the orange band that was used for marking of the drums it was stored in.

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The product was synthesised to deny an enemy cover and concealment in dense terrain by defoliating trees and shrubbery where the enemy can hide.

Agent Orange was tested in chemical warfare, under the name of **Operation Ranch Hand**, which the US military unleashed on **Vietnam during 1961–1971. Agent Orange was an equal proportion (50:50) of** two chemicals, viz., 2,4-dichlorophenoxyacetic acid (2, 4, D) and 2, 4, 5-trichlorophenoxyacetic acid (2, 4, 5, T).

The mixed product was further mixed with kerosene or diesel fuel and dispersed by aircraft, vehicle and hand spraying. An estimated 19 million gallons of Agent Orange were used in South Vietnam during the war. It had devastating effects on the health of the people, which is being even passed onto the children.

Note: The most widely used herbicide was Agent Orange, followed by Agent White. Other tactical herbicides that were used in Vietnam during the war include Agent Blue, Agent Purple, Agent Pink and Agent Green.

### *Agent Purple*

Agent Purple is reputed to have three times the dioxin levels of Agent Orange, 45 parts per million as opposed to 13 parts per million in Agent Orange.

### *Agent Blue*

Defined in medical literature as ‘an arsenic-based herbicide’ used to destroy broadleaf plants and trees (including grass, rice, bamboo, banana trees etc.), used especially on rice paddies during the Vietnam War.

## 26.6 HYDROPONICS, AEROPONICS AND AQUAPONICS

Refer **Infographic – Hydroponics vs. Aeroponics vs. Aquaponics.**

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## 26.7 BIODIGESTER TECHNOLOGY or BIOTOILET

The **Pitroda Committee** (for railway modernisation) and the **Kakodkar Committee** (for railway safety) had strongly recommended replacing the conventional open-discharge toilets with green toilets, with the aim of having cleaner, hygienic and safer railway ecosystem.

In recent years, bio-digester-based technology has been introduced with the help of **DRDO and Railway Ministry**. In this system, faecal waste will be stored in a close container and with the help of **bacterial inoculum**, organic waste will be converted into nutrient-rich liquid fertiliser and biogas with the use of the **anaerobic digestion** method. Many gases are generated in this process, like CH4, NH3, H2S etc. It can also be used for industrial or domestic purposes for managing waste and/or to release energy.

In 2015 budget, the Government introduced bio-toilets in Indian trains with the allocation of `1000 crore. This was further extended in the constitutive year's budget plan.

Note: 1. Sistema.bio eyes 20% growth in biodigester installations in India in 2024. 2. Celebrating one million biodigesters globally in January 2024.

### Bio-Toilets can be Installed in Many Places

Himalayan region, Glaciers, Railways coaches, Buses, on Highways, Rural areas household, Educational institutions, Pilgrimage sites, etc.

### *Advantages of Bio-Toilets*

- 100% sludge-free disposal of human waste or simply put, it eliminates the need for manual scavenging.
- Decomposition of solid waste to waste and biogas.
- Eco-friendly.
- Maintenance-free.
- Capable of functioning efficiently at sub-zero to –55°C.

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- Plays a critical role in preventable health care, since the bio-toilets eliminate disease-causing pathogens completely.

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### *Disadvantages of Bio-Toilets*

- Bio-toilets system is very sensitive. It has more than 100% of chocking probability, due to disposal of napkins, cigarette buds, gutka pouches, liquor bottles etc.
- It is difficult to procure spares.
- Apart from these, the Indian government has taken it as a challenge to make India open defecation-free under the **Swachh Bharat Abhiyan** and various state governments are also working positively in this direction.

## 26.8 DRINKING WATER PURIFICATION/WATER PURIFIER

It is the process of eliminating undesirable chemicals, biological contaminants, suspended solids and gases from contaminated water.

Purified water may diminish the concentration of particulate matter, including suspended particles, parasites, bacteria, algae, viruses, fungi, as well as minimising the amount of a range of dissolved and particulate material derived from the surfaces that come from runoff water due to rainfall.

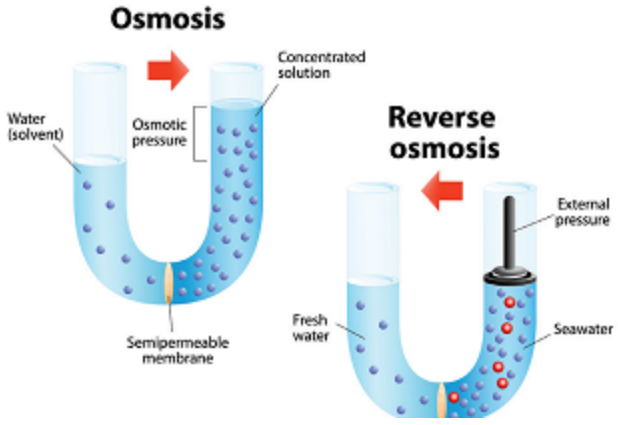


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Figure 26.1 Process of Reverse Osmosis

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There are different types of water filtration processes such as **reverse osmosis (RO)**, **ultrafiltration (UF)** and **ultraviolet** disinfection (UV). We should choose a water purification system depending on the quality of water.

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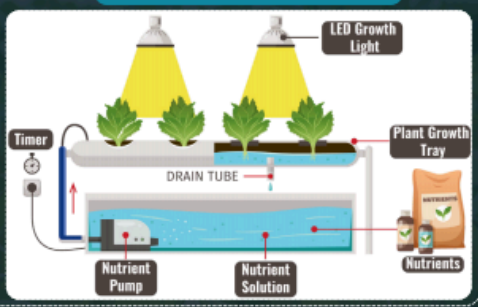
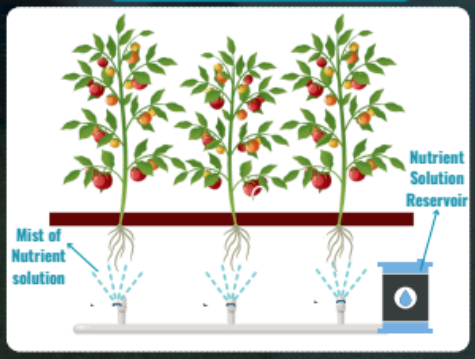
## HYDROPONICS

- **Definition:** Growing plants in a soil-free, nutrient-rich water solution.
- **Nutrient Delivery:** Direct to plant roots in a controlled environment.
- **Support Media:** Inert materials like perlite or coco coir.
- **System Types:** Nutrient Film Technique (NFT), Deep Water Culture (DWC), drip irrigation, and aeroponics.
- **Applications:** Suitable for indoor setups, greenhouses, or commercial operations.
- **Advantages of Hydroponics**
  - Precise control over environmental factors
  - Faster growth rates and increased yields
  - Efficient resource utilisation
  - Space efficiency
  - Year-round cultivation
  - Enhanced disease and pest control
  - Reduced environmental impact



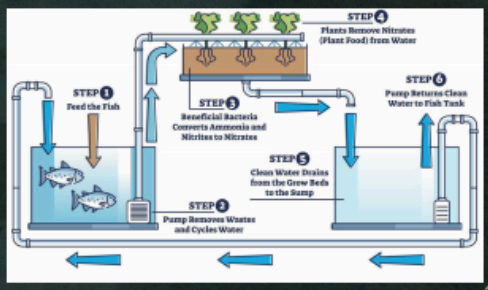

## AEROPONICS

- **Definition:** Plants grow in air or mist without a medium; nutrient-rich mist is sprayed directly onto roots.
- **Root Exposure:** Maximises oxygen exposure by eliminating growing medium, enhancing growth and nutrient absorption.
- **Nutrient Delivery:** Fine mist delivered to roots using misters or sprayers.
- **Space Efficiency:** Ideal for compact and vertical setups, suitable for urban farming and indoor gardening.
- **Resource Saving:** Efficient method that accelerates growth, increases yields, and eliminates soil use.
- **Advantages of Aeroponics**
  - Faster growth and higher yields
  - Efficient nutrient uptake
  - Water conservation
  - Space efficiency
  - Reduced disease and pest issues
  - Resource efficiency
  - Versatile and scalable




## AQUAPONICS

- **Definition:** Combines aquaculture (fish farming) and hydroponics (soilless plant cultivation) into a symbiotic system.
- **Nutrient Source:** Fish waste supplies nutrients for plants, which purify the water for fish.
- **Waste Utilisation:** Fish produce ammonia-rich waste; plants use this as nutrients, preventing toxic accumulation.
- **Nutrient Conversion:** Beneficial bacteria convert ammonia into nitrates, which plants absorb.
- **Water Purification:** Plants purify and oxygenate water, which cycles back to the fish.
- **Advantages of Aquaponics**
  - Sustainable and efficient
  - Dual harvest of fish and crops
  - Reduced environmental impact
  - Versatility and scalability
  - Educational and community benefits




## *Process of Reverse Osmosis*

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It is a simple water filtration mechanism. There are various components involved in a reverse osmosis system. The functions of the components that play an integral part in the RO system are as follows:

1. **Pre-Treatment:** The pre-treatment process maximises the efficiency of the treatment. Water passes through the sediment filters, carbon filters during the pre-treatment phase. **Sediment filters** help in removing silt whereas **carbon filters** help in removing the organic impurities present in water. These filters help remove dirt, chlorine content and other sediments present in water to protect the membranes from damage.
2. **Treatment with RO: Reverse osmosis is the process of osmosis in reverse.** Whereas Osmosis occurs naturally without energy required, to reverse the process of osmosis you need to apply energy to the more saline solution. A reverse osmosis membrane is a semi-permeable membrane that allows the passage of water molecules but not the majority of dissolved salts, organics, bacteria and pyrogens. However, you need to 'push' the water through the reverse osmosis membrane by applying pressure that is greater than the naturally occurring osmotic pressure in order to desalinate (demineralise or deionise) water in the process, allowing pure water through while holding back a majority of contaminants. Desalination is the process of removal of salt from seawater. Many water treatment plants now started using this reverse osmosis technique to deal with many water-related issues.

### *Ultraviolet Disinfection*

Ultraviolet disinfection usually involves a UV lamp. UV lamp provides pure and safe drinking water. The UV light used in this process is a high powered UV which is also known as UV-C or germicidal UV. UV-C rays penetrate into the body of pathogens to deactivate them. The UV rays modify the DNA of the pathogens in such a way that they cannot multiply. These rays have the ability to kill 99.9% of the microorganisms.

### *Ultrafiltration (UF)*

Ultrafiltration process uses hydrostatic pressure to force water against a semi-permeable membrane. This semi-permeable membrane helps in

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removing bacteria and different types of microorganisms and bad odour in the water. Ultrafiltration system helps in **removing colloids**, **bacteria**, **pathogens and other molecules that are larger than the size of the pores in the semi-permeable membrane**.

**Table 26.1** RO vs NF vs UF vs MF

| Feature | RO Purification | NF Purification | UF Purification | MF Purificatio |
|---|---|---|---|---|
| **Need for electricity** | Purification necessitates the use of electricity | Purification necessitates the use of electricity | Does not need electricity | Does not need electricity |
| **Extent of filtration** | Filters the water to remove all bacteria and viruses | All bacteria and viruses, as well as most natural organic matter and some natural minerals, including divalent ions that generate hard water, are killed | Filters the water to remove all bacteria and viruses | Filters out only bacteria |

Ultrafiltration is not very different from **reverse osmosis**, **microfiltration (MF) or nanofiltration (NF)**, except in terms of the size of the molecules it retains.

Note: **Ban on RO Water Purifier:** National Green Tribunal (NGT) in its Order dated the 20 May 2019 directed the Ministry of Environment, Forest and Climate Change (MoEF&CC) to come out with regulation on appropriate use of reverse osmosis based water purification system. Accordingly, MoEF&CC has published the Water Purification System (Regulation of Use) Rules, 2023. The rules outline guidelines for the proper management, storage, and utilisation of reject water and discarded elements generated by these systems. These rules were published on 10 November 2023, and shall come into force with effect from 10 November 2024. Bureau of Indian Standards (BIS) has also notified the "IS 16240: 2023 Reverse Osmosis based point of Use Water Treatment System for Drinking Purposes –

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Specification (First Revision)" on 16 March 2023. [Press Release, PIB dated 5 February 2024].

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**MEMBRANE BIOREACTORS**

Membrane bioreactors **combine membrane processes such as microfiltration or ultrafiltration** with the activated sludge process, a biological method for treating wastewater.

**Polymeric membranes**, notably polyvinylidene difluoride (PVDF), are widely employed in water and wastewater treatment due to their durability, chemical resistance, and mechanical strength, making them the **predominant choice in membrane bioreactors**.

These technologies are now **widely used for municipal and industrial wastewater treatment**.

## 26.9 Air Cleaners and Air Purifiers

### *Air Purifier*

- Dust, pollen, pet dander, mould spores and dust mite faeces can act as allergens, triggering allergies in sensitive people.
- Smoke particles and volatile organic compounds (VOCs) can pose a risk to health.

- Exposure to various components such as VOCs increases the symptoms of sick building syndrome (which can cause headaches, dizziness, nausea, and eye, ear and nose irritation).

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- In the 1950s, HEPA filters were commercialised, as main air-purifier filter.

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## Purifying Technologies

There are two types of air purifying technologies – Passive and Active.

1. Passive air purification units use air filters to remove pollutants. They are more efficient since all dust and particulate matter is permanently removed from the air and collected in the filters.
2. Active air purifiers use ionisation for cleaning the air and neutralise this pollution.

**Table 26.2** Technologies Developed for Air Purification

| Technologies | Pros | Cons |
|---|---|---|
| Pre-filters | • Captures large particles<br>• May be washable<br>• Extends life of HEPA filter | Requires maintenance |
| Activated carbon | • Absorbs odours, gases, and chemicals<br>• Leaves space, smelling fresh<br>• Removes tobacco smoke and pet odours | Heavy expensive |
| Ultraviolet light | • Neutralises microorganisms, like bacteria and viruses | |

## Pre-Filters

Captures the largest particles before they reach the more expensive HEPA filter.

- Most of the particles in air are large particles like dander and dust, not tiny ones like pollens and microbes.
- Even though HEPA filters have a large number of pleats to maximise their surface area, they can fill up quickly if larger particles are not pre-filtered out.

## Activated Carbon

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- Elements like activated carbon, zeolite or potassium permanganate are added to air purifier filters to increase efficiency to the filtering system.
- These elements work to absorb smoke, gases, chemicals and odours that are present in the air.
- Activated carbon neutralises odours and traps harmful chemicals and gases in its pores to provide relief from activities like second-hand smoke and harmful fumes produced from renovations.
- An effective filtering system that is also equipped with chemical adsorption materials can produce cleaner and fresher-smelling air.

### *Ultraviolet Light*

- Ultraviolet light (UV) technology is key to neutralising viruses and bacteria that accumulate on air purifier air filters.
- It is important to use the UV technology in combination with HEPA air filters and possibly activated carbon.
- Without a particulate filter system like a HEPA air filter, too many microorganisms may be hidden from the UV light, since it is not easily located on an air filter.

### *A HEPA filter is made up of a mat of randomly arranged fibres*

High efficiency particulate air (HEPA), originally called high-efficiency particulate absorber but also sometimes called high-efficiency particulate arresting or high-efficiency particulate arrestance is a type of air filter.

**Table 26.3** Types of Air Purifier

| Air Purifier | Pros | Cons |
|---|---|---|
| HEPA air purifiers | • Most effective technology for air purifiers<br>• Removes up to 99.97% of particles that pass through, like allergens, pollen, dust, dander and others that are 0.03 microns and larger in size | • Regular filter replacements |

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| | | |
|---|---|---|
| | • Used worldwide in medical cleanrooms | |
| Electrostatic air purifiers | • No air filter replacement necessary | • Removes only 30% of particles from the air<br>• Does not clean all of the air in an enclosed space<br>• Does not use HEPA air filters |
| Ozone | • Removes strong odours | • Extremely dangerous lung |

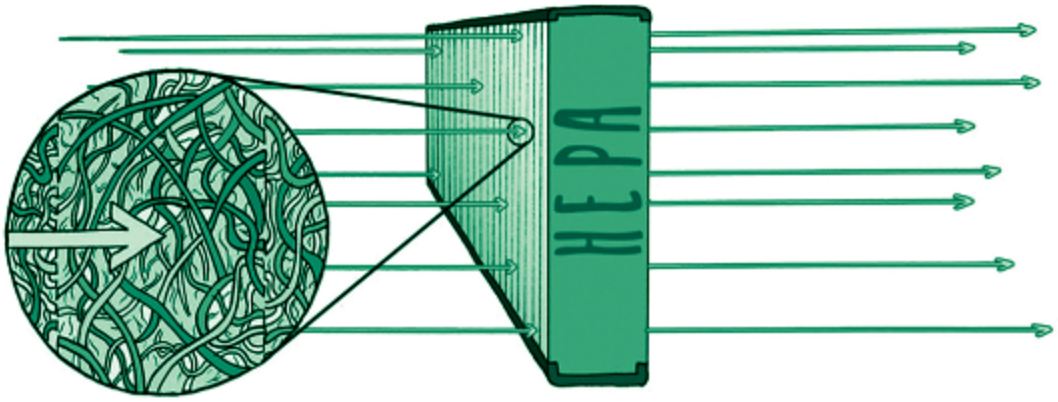


Figure 26.2 HEPA Filter

## 26.10 Safety on Wheels

As per Annual Report on 'Road Accidents in India-2022" published by the Ministry of Road Transport and Highways, a total of 4,61,312 road accidents have been reported by States and Union Territories (UTs) during the calendar year 2022, which claimed 1,68,491 lives and caused injuries to 4,43,366 persons. This marks an increase of 11.9% in accidents, 9.4% in fatalities, and 15.3% in injuries compared to the previous year. The United States of America has a distinguished record of the highest number of road accidents in the world.

There are two types of safety features in vehicles:

1. **Passive Safety Features:** **Airbags** are an example of 'passive safety features', which prevent or minimise injury to the passengers in a crash.

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**Principle of Airbag:** Generally **accelerometer** is required. It is responsible for detection of car crash/collision which results in the deployment of airbags almost **instantaneously [1/10 of second (30 milliseconds)].**

2. **Active Safety Features:** Safety features in a vehicle that help prevent a crash by providing the driver with better ways of controlling the vehicle and avoiding hazards are called 'active safety features'. Anti-lock breaking system, electronic stability control and automatic braking are some examples of 'active safety features'.

### 26.10.1 Different types of Active Safety Features in Vehicle

1. **Anti-Lock Braking System (ABS):** Skidding is the phenomenon of locking up the wheels of a vehicle when a sudden or extreme braking force is applied. When the front wheel(s) of a vehicle are locked, its ability to change direction is reduced.
2. **Autonomous Emergency Braking (AEB): Most AEB systems use radar, stereo camera and/or LiDAR (Light Detection and Ranging)-based technology to identify potential collision objects in front of the car.**
   If a potential collision is detected, generally AEB systems first try to alert the driver that an emergency action is needed. If still no action is taken and a collision is imminent, the system will then apply the brakes.
3. **Lane Departure Warning System:** Lane departure warning system warns the driver when the car is getting close to a lane marking. The type of warning can be different, depending on the system. While some systems give an audible signal, others produce a vibration in the steering wheel to simulate the feeling of the car running over a 'rumble strip'.
   **A camera is usually situated at the top of the windscreen behind the rear-view mirror.**
4. **Heads up Display:** HUD or heads up display is a **digital transparent display that is projected onto the windshield** (front mirror to save against air pressure) which takes their eyes on the road.
   This technology was initially developed especially for aircraft pilots, and for military aviation.

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5. **Hill Start Assist/Hill Control:** The hill start assist is a system that prevents the car from rolling down the slope while the driver **known as 'handbrake hill start',** in which the driver engages the handbrake of the car momentarily, while pulling away on an inclined surface.
6. **Electronic Stability Programme (ESP):** It monitors the wheel speed sensors, steering wheel angle, lateral acceleration, yaw rate etc., to **detect if the driver is losing control**. ESP employs in tandem, ABS, electronic brake force distribution (EBD), traction control system (TCS) and active yaw control (AYC) to **stabilise the vehicle and keep it on the road.**
7. **Pedestrian Airbag:** Volvo is the first company to introduce 'pedestrian airbag', a technology first used in their **Volvo V40**.
   The airbag is designed to protect pedestrians from accidental injuries at city speeds, from around 12 miles per hour to 31 miles per hour.
8. **Cruise Control:** Often, over long distances, driver's fatigue leads to vehicle crashes. Cruise control is a system that **automatically controls the speed of a vehicle on long stretches of highways with constant flow of traffic**. The purpose of cruise control is to accurately maintain a speed set by the driver, without any outside intervention by controlling the throttle–accelerator pedal
   linkage.

### ADAS: ADVANCED DRIVER-ASSISTANCE SYSTEMS

ADAS are electronic systems in a vehicle that use advanced technologies to assist the driver. They can include many active safety features, and often the terms 'ADAS' and 'active safety' are used interchangeably.

ADAS technology uses sensors, cameras and radar among other systems to monitor the environment around a vehicle. Based on this, ADAS provides drivers with real-time information and alerts to help them avoid collisions and other potential hazards. Autonomous application solutions are partitioned into various chips, called systems on a chip (SoCs). These chips connect sensors to actuators through interfaces and high-performance electronic controller units (ECUs).

Self-driving cars use a variety of these applications and technologies to gain 360-degree vision, both near (in the vehicle's immediate vicinity) and far.

Essential safety–critical ADAS applications include:

- Pedestrian detection/avoidance

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- Lane departure warning/correction
- Traffic sign recognition
- Automatic emergency braking
- Blind spot detection

These life-saving systems are key to the success of ADAS applications. They incorporate the latest interface standards and run multiple vision-based algorithms to support real-time multimedia, vision co-processing and sensor fusion subsystems.

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## 26.11 Wireless Transmission of Electricity, WiTricity and Wireless Charging

Wireless power transfer (WPT) has become very important nowadays, as portable technology has become a part of everyday life.

The **founder of AC electricity, 'Nikola Tesla', was the first to conduct experiments dealing with wireless charging** and accomplished a major breakthrough by transmitting 108 volts of high-frequency electric power over a distance of 25 miles to light 200 bulbs and run an electric motor.

In 1901, Tesla constructed the **Wardenclyffe Tower** to transfer electrical energy between America and Europe through the **Ionosphere**. The Tesla Tower was about 185-feet high, with a well about 100-feet deep, for use in experimenting with the transmission of electrical energy for power and lighting purposes through wireless medium.

The equipment cost was nearly $200,000**.** However, due to technological limitations (e.g., low system efficiency due to large-scale electric field), the idea has not been further developed and commercialised.

Later, during the 1920s and 1930s **magnetrons** were invented to convert electricity into microwaves, which enable wireless power transfer over long distances**.** However, there was no method to convert microwaves back to electricity. It was only in 1964 when W. C. Brown, regarded as the principal engineer of practical wireless charging, realised the conversion of microwaves to electricity through a **rectenna** or **a rectifying antenna (Witricity).**

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**Wireless power transfer (WPT),** using **magnetic resonance technology** that adopts the same basic theory that has already been developed for at least 30 years with the term inductive power transfer. At kilowatts power level, the transfer distance increases from several millimetres to several hundred millimetres with a grid-to-load efficiency above 90%. These advances make WPT very attractive for electric vehicle (EV) charging applications in both stationary and dynamic charging scenarios. The magnetic wireless power transfer over a distance of 6.5 feet can deliver 10 kilowatts of electric power with a coil-to-coil stationary efficiency of 97%.

## *Wireless Charging Technologies*

The development of wireless charging technologies is advancing towards two major directions. **'Radiative wireless charging' using radio frequency (RF) or microwaves is used for long-distance transmission of power.** Due to the safety issues raised by RF, exposed radiative wireless charging usually operates in a low-power region.

**'Non-radiative wireless charging'** (or coupling-based wireless charging) is **based on the coupling of the magnetic field between two coils within the distance of the coils'** dimension for energy.

It involves inductive coupling, **magnetic resonance** coupling and capacitive coupling. Due to safety implementation, non-radiative wireless charging has been widely used in our daily appliances.

**Inductive Charging:** Most of today's wireless chargers use 'inductive charging' with transmit and receiver coils in close proximity (only few centimetres). The action of an electrical transformer is the simplest instance of wireless energy transfer via induction. The transfer of energy takes place by electromagnetic coupling through a process known as mutual induction.

In resonance charging, the transfer efficiency and power transferred are a little lower compared to induction charging, but the transfer distance can be in metres.

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Most wireless charging devices these days follow the Qi (pronounced chee) standard, which has taken its time reaching ubiquity (the user experience has not always been great). The Wireless Power Consortium, which manages the charging protocol, announced the next-generation version called Qi2 in early 2023, and these days there are devices supporting it. It promises perfect alignment, with the potential for accessories to bridge the Android and iPhone divide.

The existing, non-magnetic wireless charging Extended Power Profile (EPP) has also been updated to comply with Qi2. This means that devices without magnets will be branded Qi and will still work with Qi2 chargers.

### Benefits of Qi2 over Qi

Greater efficiency, Faster charging, Wider compatibility for accessories, etc.

'**Radiate wireless power transfer'** is used to transfer large amounts of power over long distances through space, by means of microwaves. A **microwaves power transfer (MPT)** system converts direct current (DC) power to microwaves, transmits the microwave radiation to a receiver, which then converts the microwave radiation back to DC power.

## 26.12 CYBERNATIC ORGANISM (CYBORG)

The term 'Cyborg' was first coined in 1960 by Manfred Clynes and Nathan S. Kline to **describe man–machine systems** in which the control mechanisms of the human portion are modified externally by drugs or regulatory devices so that the being can live in an environment different from the normal one; for example, in an extra-terrestrial being.

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Cyborg, short for **'cybernetic organism',** is no longer an imaginary organism but an organism with organic and **biomechatronic body parts,** that is **partly human and partly machine**. They include people with cochlear implants, cardiac pacemakers, contact lenses or implants that make life easier for humans.

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## The Cyborgs

In 1997, **Philip Kennedy,** a scientist and physician, created the **world's first human cyborg** from Johnny Ray, a Vietnam veteran who suffered a stroke. Ray's body, as doctors called it, was 'locked in'. Kennedy embedded an implant he designed (and named 'neurotrophic electrode') near the part of Ray's brain so that Ray would be able to have some movement back in his body. The surgery went successfully, but in 2002, Johnny Ray died.

In 1998 Kevin Warwick, former Professor of Cybernetics at the University of Reading, initiated a series of pioneering experiments *involving the neurosurgical implantation of a device into the median nerves of his left arm in order to* **link his nervous system directly to a computer.**

In 2004, Neil Harbisson (**Fig. 26.3**)**,** an artist born with achromatopsia or extreme colour blindness, **permanently attached an antenna inside his skull.** He is the first human to be described as a Cyborg by any government.

## The Future

The practice of integrating technology with the human body is happening more often and beyond the medical field; perhaps, setting the stage for the next chapter in human evolution. Many researchers believe that we have already started down the path towards a Cyborg future; after all, many of us already depend on **bionic ears and eyes, insulin pump technology and prosthetics to help us survive.**

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Figure 26.3 Neil Harbisson with an Implanted Antenna

In 2016, the first Cyborg Olympics were **celebrated in Zurich Switzerland. 'Cybathlon' 2016 was the first Olympics for Cyborgs and the first worldwide and official celebration of Cyborg sports.** In this event, 16 teams of people with disabilities used technological developments to turn themselves into Cyborg athletes.

Figure 26.4 Doctor Octopus, a character from Spiderman movie, posing at the MCMExpo at the Excel Centre

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**SpaceX and Tesla's CEO, Elon Musk is backing a brain–computer interface venture called Neuralink. With Neuralink, Musk hopes to develop a micron-sized device to help link brains and machines, and treat brain injuries and their symptoms, such as strokes, paralysis and even memory loss in old age.**

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## 26.12.1 Neuralink

Neuralink, a pioneering American neurotechnology company founded by **Elon Musk in 2016**, has **successfully implanted a brain chip in its first human patient**.

This achievement marks a significant step forward in the field of **Brain-Computer Interfaces (BCIs)** and holds promise for revolutionising various aspects of healthcare and human-computer interaction.

### *Neuralink's Telepathy and Applications*

- Neuralink's first product, named **Telepathy**, is designed to **benefit individuals who have lost their limbs or have suffered traumatic injuries.**
- By implanting brain-computer interfaces, **Neuralink aims to enable individuals to control computers and other devices solely through their thoughts**.

### *Brain Implant Technology*

- **A brain implant, also known as a brain-computer interface (BCI),** is a sophisticated device that **establishes a direct connection between the brain and external technology.**

**BRAIN COMPUTER INTERFACE**

- **Neuralink's 'Blindsight'** is a **brain-computer interface (BCI)** implant designed to **restore vision** in individuals who are blind due to loss of both eyes or optic nerves.
- **Works only if the visual cortex is intact.**
- The device received **Breakthrough Device Designation** from the **US FDA**, which accelerates development of critical medical technologies.

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## What Is a Brain-Computer Interface (BCI)?

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- A BCI **detects brain signals**, processes them via a **computer** and sends commands to **external devices** without using muscles or nerves.
- A BCI is a computer-based system that acquires brain signals, analyses them and translates them into commands that are relayed to an output device to carry out a desired action.
- Key components:
  - ❑ **Sensors** to record brain activity (e.g. EEG or implants)
  - ❑ **Computer** to decode signals
  - ❑ **Output device** to perform the intended action
  - ❑ **Feedback loop** for learning and control

## Types of BCIs:

| Type | Description | Example |
|---|---|---|
| **Invasive** | Implanted in **brain tissue**; high accuracy | **Neuralink Implant** |
| **Non-Invasive** | Worn externally (e.g. on **scalp**) | EEG, fMRI (functional MRI) |
| **Partially Invasive** | Placed on brain surface under **skull** | ECoG |

## Electrocorticography (ECoG), a type of intracranial electroencephalography:

**Applications of BCIs:**

- **Medical:** Rehab for neurological conditions
- **Mental Wellness:** Monitor stress, emotions
- **Cognitive Enhancement:** Memory and decision support
- **Gaming/VR:** Hands-free immersive control

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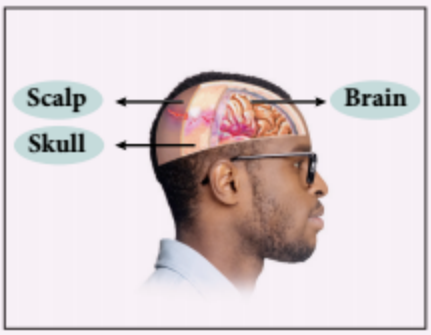


## Challenges:

- **Technical:** Interpreting complex brain signals, weak signal quality
- **Health Risks:** Invasive implants may harm brain tissues or cause infection
- **Privacy:** Risk of unauthorised brain data access
- **Security:** Potential for signal manipulation or 'brain hacking'
- **Legal and Ethical:** No strong laws on safety, consent or data protection; concerns about autonomy and 'cyborgisation'

- It operates by **translating signals from the brain into commands that can control various devices, such as computers, artificial limbs, or robots.**
- **This communication is facilitated by groups of electrodes strategically placed to capture brain activity and interpret its messages.**
- AI Powered chip.

## *Challenges and Concerns*

1. **Opposition from Animal Rights Groups:** Neuralink's testing on primates has drawn criticism from animal rights groups, raising **ethical concerns about the treatment of animals in scientific research.**
2. **Safety and Reliability:** Ensuring the safety and reliability of brain implants is crucial, as **glitches or malfunctions** could have serious consequences for patients. Rigorous testing and regulatory oversight are necessary to address these concerns.
3. **Privacy and Security:** Brain implants raise concerns about **data privacy and security**, as they involve the transmission and storage of sensitive neural information**. Safeguarding against unauthorised**

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**access and protecting individuals' privacy** is paramount in the development and deployment of brain-computer interfaces.

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## 26.13 Incandescent Bulb, CFL (Compact Fluorescent Lamps) and LED (Light-Emitting Diode)

### 26.13.1 Incandescent Bulb

It refers to the conversion of heat to light, which requires the filament inside an incandescent lamp to burn at a high temperature (350°F or 176°C). This conversion is very simple but the disadvantages are that only 5% of the total energy consumed by the lamp is used to generate light (95% is wasted as heat) and the lifetime is limited to about 2,000 hours.

### 26.13.2 CFL (Compact Fluorescent Lamps)

Fluorescence is the conversion of ultraviolet (UV) light to visible light. Electrons flow through the fluorescent lamp and collide with mercury atoms, causing photons of UV light to be released. The UV light is then converted into visible light, as it passes through the phosphor coating on the inside of the glass tube. This two-stage conversion process is much more efficient than incandescent lamp process, resulting in 25% of the total energy consumed used to generate light, lower lamp temperatures (40°C) and longer lifetime (10,000 hours).

#### *Environmental Impact*

CFLs contain toxic chemicals, unlike incandescent lamps. The mercury present in CFLs has been one of the most talked-about issues related to compact fluorescent light bulbs. Each contains about 4 milligrams of mercury, which is a cumulative poison. But because coal also contains mercury, which is released into the atmosphere when burned, this too is a source of mercury pollution. If we assume that all our electricity came from

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coal, then the amount of mercury pollution saved by switching to CFLs is about double that in the lamps themselves so their use could reduce the net mercury burden on the environment. CFLs are designed to operate within a specific temperature range. Temperatures below the range cause reduced output. There can also be problems with local pollution if they are not properly disposed.

### *CFLs Emit UV*

All fluorescent lamps are known to emit some UV. Typical fluorescent lamps, which include CFLs, which consumers would encounter, emit very low levels of UV. In order to measure any UV radiation from these lamps, very sensitive measuring equipment must be used.

Note: CFL emits both ultraviolet and visible light but LED emits only visible light, so LED does not have an adverse effect on human health.

## 26.13.3 Light-Emitting Diodes (LED)

LED lighting products produce light up to 90% more efficiently than incandescent light bulbs. An electrical current passes through a microchip, which illuminates the tiny light sources we call LEDs and the result is visible light.

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process.

**Table 26.4** Comparison among Incandescent Bulb, CFL and LED

| | Incandescent Light Bulbs | Compact Fluorescence Lamp (CFL) | LED |
|---|---|---|---|
| Lifespan (average) | 1,200 hours | 8,000 hours | 50,000 hours |

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| | | | |
|---|---|---|---|
| Watts of electricity used (equivalent to 60-watt bulb) | 60 watts | 13–15 watts | 6–8 watts |
| Kilowatts of electricity used | 3285 kWh/year | 767 kWh/year | 329 kWh/year |
| Turns on instantly | Yes | No, it takes time to warm up | Yes |
| Durability | Not very durable – glass or filament can break easily | Not very durable – glass can break easily | Very durable – LEDs can handle jarring and |

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LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction.

The colour of an LED is determined by the material used in the semiconducting element. The two primary materials used in LEDs are aluminium gallium indium phosphide alloys and indium gallium nitride alloys. Aluminium alloys are used to obtain red, orange and yellow light, and indium alloys are used to get green, blue and white light. Slight changes in the composition of these alloys change the colour of the emitted light.

## *Blue LED*

Exposure of blue light can cause permanent damage to the retina. Light-induced retinal damage can result from viewing either a very bright light for a short time (acute exposure) or a less bright light for a longer duration (development of age-related macular degeneration – AMD).

Blue LED light is most often used to treat acne. It may do this by reducing activity in the sebaceous glands, so they produce less of the oil that can plug the hair follicles, leading to acne. Blue light may also kill acne-causing bacteria known as Cutibacterium acnes.

In 2014, **Shuji Nakamura, Isamu Akasaki** and **Hiroshi Amano** were awarded the Nobel Prize in physics for their invention of blue LED, which made it possible to light **flat screen TVs, computer monitors, smart phones** and **tablets**. LED technology takes less energy and emits less heat, making it an ideal lighting source to save money and energy.

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## *Sodium Lamp vs LED lights*

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- Sodium lamps provide 360 degree light emitting. Whereas LED lights tend to direct light downward, rather than a full 360 degrees.
- Monochromatic light means that the light has a single wavelength. The LED light is not monochromatic because it is made up of different wavelengths, therefore is polychromatic. Often, a combination of wavelengths is used to create different colours.
- LEDs last longer than any light source commercially available in the market.

## 26.14 High-Speed Train: Bullet Train, Maglev (Magnetic Levitation) and Hyperloop

### 26.14.1 Bullet Train

A bullet train is not an official word used for high-speed railways, it is only the unofficial name of first high-speed train of Japan. A high-speed rail can be defined as a train that operates at a minimum speed of 200 km per hour on tracks built specially for high-speed travel.

On the basis of technology employed in the trains, it can be **classified into two categories**:

1. Tilt technology
2. Non-tilt technology

**Tilt Technology:** The trains based on this technology are known as tilting train. They have the capability to tilt their body towards the curve. This tilt allows the train to pass the curve at a higher speed. It may run at a speed of 200 km per hour for upgraded track and 250 km per hour or even faster for a new track. This technology is expensive and can generally function at larger passenger base regions.

**Non-Tilt Technology:** The trains based on this technology run on straight special tracks or tracks with large curve radius. Exclusive tracks are laid down which provide higher level of passenger comfort. Trains can run at

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250 km per hour or more speed on such tracks. This technology provides high efficiency of rail freight transport.

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### *Foundation Stone for Bullet Train in India*

India's first high-speed rail project (bullet train) project was jointly laid by Prime Minister Narendra Modi and his Japanese counterpart, Shinzo Abe at Sabarmati Station Ground in Ahmedabad in 2017. The 508 km-long, **Mumbai to Ahmedabad High Speed Rail (MAHSR) Project** would be built at a cost of 1.10 lakh crore.

**NEW UPDATES**

**India's Bullet train will run by August 2026: Ashwini Vaishnaw**

India's first bullet train is slated to run in August 2026 with a larger section planned for the year after, said Railways and Telecom Minister Ashwini Vaishnaw, noting that the project will give a boost to the economy, as many of the companies supplying to the project had started getting export orders **(14 March 2023).**

1. MAHSR project is a joint venture between **Indian Railways** and **Japan's Shinkansen Technology**.
2. The train will cover a distance of over 508 km from **Sabarmati (Gujarat) to Bandra (Mumbai, Maharashtra)** in around 2.58 hours at high speed of approximately 320 km per hour from the existing 7–8 hours. On the Ahmedabad–Mumbai route, **12 stations have been proposed**: Mumbai, Thane, Virar, Boisar, Vapi, Bilimora, Surat, Bharuch, Vadodara, Anand, Ahmedabad and Sabarmati (four stations in Maharashtra and eight stations in Gujarat).
3. As per plan, the entire corridor will be on elevated track, except 21 km, which will be underground; of which, 7 km will pass under the sea **(under water train in the world – the United States, Russia, Japan and China).**
4. Of the total project cost of 1,10,000 crore, 80% – 88,000 crore – will be provided by **Japan International Cooperation Agency (JICA)** as soft

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loan at an interest of 0.1% per annum. This loan will be repaid to Japan in 50 years, with 15 years grace period. The breakup of the cost has been finalised as **25% by Maharashtra, 25% by Gujarat and 50% by the Centre**. It has a proposed speed of approximately 320 km per hour with a train capacity of 900 to 1200 passengers.

5. **High Speed Rail Corporation of India Limited (HSRC):** It has been formed on the directions of the Ministry of Railways, Government of India, for the development and implementation of high-speed rail projects.

### 26.14.2 Maglev Technology

It is a new version of bullet trains or high-speed trains, based on magnetic levitation. They are levitated, propelled and guided by using magnets. The friction, energy consumption and the noise production are almost zero. This is a highly energy efficient and comfortable version of high-speed trains. Maglevs do not need maintenance frequently because wheels and the rails do not wear out. These trains uplift and move on track in friction-less condition, so speed can be easily increased. Attractive and repulsive forces are required to set the train in motion. Two types of Maglev trains are:

1. Electromagnetic propulsion
2. Electrodynamic suspension

Electrodynamic suspension train is now running in Japan with metal-based superconductor used. Metal at −278°C becomes superconductor and resistance becomes approximately zero. This bullet train is faster and its speed is around 601 km per hour.

**Table 26.5** Difference between Bullet Train and Maglev

| Maglev Train | Bullet Train |
|---|---|
| Does not need wheels | May or may not need wheels |
| Produces very low noise | Noise is a major issue |
| Highest energy efficiency | Is more efficient than conventional trains |

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| | |
|---|---|
| Capital cost is very high | Less expensive than Maglev |
| Technology available in a few countries | Available is most developed countries |

### 26.14.3 Hyperloop Technology

Hyperloop envisages pods or capsules travelling at high speeds through low-pressure tubes erected on columns or tunnelled underground. The system is fully autonomous and sealed, so no driver-related error is anticipated. In a sealed environment with almost no air resistance, the pods are expected to reach very high speeds.

#### *Mechanism of Hyperloop Train*

How to maintain good speed

1. **Decrease air density** in hyperloop train so air resistance become very less.
2. In front of the train, we will use an **air compressor fan,** which will reduce air pressure.
3. **Air bearing** is required to remove air outside.

#### *Hyperloop in India*

In 2018, the Maharashtra government signed an MoU with **British Virgin Group** to build the Mumbai–Pune hyperloop. The proposed hyperloop for Pune and Mumbai will link central Pune, Navi Mumbai International Airport and Mumbai in 25 minutes.

#### India's First Hyperloop to Connect Amaravati and Vijayawada

- **The US-based Hyperloop Transportation Technologies Inc. (HTT)** is gearing up to roll out India's first hyperloop project, connecting **Vijayawada and Andhra Pradesh's capital Amaravati.**
- Hyperloop, which uses a low-pressure tube to transport people at speeds as high as 1,200 km per hour, will cut down the nearly one-hour travel time for the Vijayawada–Amaravati ride to just around 5–6 minutes.

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- According to sources, the project, to be taken up on a public private partnership (PPP) mode, will cost around $200–250 million (`1,300–1,600 crore) and create over 2,500 jobs. Andhra Pradesh Economic Development Board (APEDB), along with the AP government, will assist HTT with all regulatory requirements, and develop and implement the hyperloop project in Amaravati.

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Hyperloop concept was firstly proposed by **Elon Musk, the CEO of SpaceX and Tesla**, in 2012. It should be noted that Musk is no longer directly involved in the development of this technology. Musk originally estimated that a route from **Los Angeles to San Francisco** would cost about $6 billion, or approximately $11.5 million per mile. But leaked documents from Hyperloop One (seen by Forbes) put that cost to $13 billion, or $121 million per mile.

Besides that, it still faces technological and regulatory hurdles, and several issues related to ownership of land that could slow down the prospects of setting up a hyperloop transportation system. Compressor fan and electromagnetic motor will further reduce the speed of train.

**HYPERLOOP IN WORLD**

1. **World's first hyperloop train** route will be built between Toronto and Montreal, Canada. (Note: Faster than passenger planes) **Toronto to Montreal** (640 km) will be covered in 39 minutes.
2. Los Angeles-based hyperloop: One has chosen two largest Canadian metropolitan areas for **world's first hyperloop commercial service**. **Los Angeles to San Francisco** would take just 50 minutes.
3. Other cities are also looking to set up hyperloop services. United Arab Emirates plans to build a hyperloop connection between **Abu Dhabi and Dubai**, two largest centres in terms of population in Emirates. These two cities are 150 km apart; hyperloop will cover this distance in 12 minutes and could be built as early as 2030.
4. A joint venture between Italian companies will aim to see the realisation of 'Hyper Transfer': the world's first Western operational commercial hyperloop system, which will travel between Venice-Mestre and Padua.

## 26.14.4 Vande Bharat (VB)/Train 18

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It is India's first indigenously built engineless semi-high speed train. Earlier, it was known by the name **Train 18**.

It runs between Delhi and Varanasi at a maximum speed of 160 kmph.

- The Train 18, later named **Vande Bharat Express**, was rolled out by the Integral Coach Factory, Chennai, under the Indian government's 'Make in India' initiative.
- The train was launched on **15 February 2019**.
- As of March 15, 2024, there are 51 operational Vande Bharat Express trains operating across 15 zones.
- Vande Bharat trains are self-propelled 'engineless' train sets.
- Its faster acceleration and deceleration results in reduced train travel time.
- Some of its passenger-friendly features include: European-style seats, diffused LED lighting, GPS-based infotainment systems, modular bio-toilets, fully sealed gangways for dust-free environment, centrally controlled entry/exit doors with sliding footsteps, divyang-friendly toilets and automatic sliding cabin doors.
- The Indian Railways plans to operate 60 New Vande Bharat trains by 2024.

**Table 26.6** Vande Bharat have different variants VB1.0, VB2.0, VB3.0 and VB4.0

| Features | VB1.0 | VB2.0 | VB3.0 | VB4.0 |
|---|---|---|---|---|
| **Exterior** | **Aerodynamic design** | **Aerodynamic design** | **Aerodynamic design** | **Aerodynamic design** |
| **Made up of** | **Stainless steal** | **Stainless steal** | Half of the VB3 train sets will have coaches made of steel, while the other half will have coaches made of aluminium | **Aluminium alloys** |

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| | | | | |
|---|---|---|---|---|
| | | | with articulated bogies. | |
| **Facility** | GPS-based passenger | • **Ditto**<br>• The VB2 is | • Ditto<br>• VB3 is | Improved passenger |

## 26.15 Indian Research Centre in Antarctica and Artic

### 26.15.1 Antarctica Research Centre

Antarctica is the fifth largest continent in the world, which is isolated, fully covered with snow and has the cleanest air in the world. The activities in Antarctica are regulated by Antarctica Treaty, 1959. It plays a significant role in all types of scientific researches because of its unique and untouched oceanic and climatic atmosphere.

Indian research stations in Antarctica are established under Indian Antarctic Programme. This programme was governed by National Centre for Antarctic and Ocean Research, Ministry of Earth Sciences and Government of India.

**Dakshin Gangotri:** In 1984, India established its first research station in Antarctica. It was located at approximately 2,500 km from the South Pole. It was later destroyed completely.

**Maitri:** It was the second permanent station of India in Antarctica, established in 1987. It was located on the rocky mountainous region called Schirmacher Oasis. India also built a freshwater lake around Maitri, known as Lake Priyadarshini.

**Bharti:** It was established in 2012. It is equipped with all the research facilities. Bharti is located at 3,000 km east of Maitri and was constructed using 134 recycled ships. It is a comfortable station and can accommodate up to 25 scientists.

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### 26.15.2 IndARC (India's Arctic Observatory)

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It is an Indian Arctic Observatory, which will be helpful to scientists in **understanding the Arctic climate and its influence on the Indian monsoon system.**

The deployment of IndARC, **the country's first underwater moored observatory in the Kongsfjorden fjord, half-way between Norway and the North Pole, represents a major milestone in India's scientific endeavours in the Arctic region.**

It was designed and developed by scientists from the Earth System Science Organisation (ESSO), National Centre for Antarctic and Ocean Research (NCAOR), National Institute of Ocean Technology (NIOT) and Indian National Centre for Ocean Information Services (INCOIS).

The observatory is anchored at a depth of 192 metres and has an array of 10 state-of-the-art oceanographic sensors, strategically positioned at various depths in the water. Certain sensors were programmed to collect real-time data on seawater temperature, salinity, ocean currents and other vital parameters of the fjord.

The Kongsfjorden is considered as a natural laboratory for studying the Arctic climate variability. Scientists predict that melting of the Arctic glaciers will trigger changes in weather patterns and ocean currents that could affect other parts of the world. It will be helpful to understand 'the interaction between the Arctic ice shelf and the deep sea and its influence on climate shift that requires detailed studies over an annual seasonal cycle'. One of the major constraints in such a study has been the difficulty in reaching the location to collect data during the harsh Arctic winter. The IndARC observatory is an attempt to overcome this lacuna.

## 26.16 Television Technology: CRT, LCD, PLASMA, LED, OLED and QLED

Television (TV) is a communication medium for transmitting and receiving moving images with or without accompanying sound. The word television

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came from **Greek words – tele and vision.** Tele means '**distant**' and vision means '**sight**'.

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There are different types of transmission modes in communication and information systems used. Transmission mode or communication mode is transferring of data between two devices. These are always in the direction of flow of information.

There are three types of transmission modes:

## (A) Simplex Mode

- Message sources are always transmitter and the destination source is receiver. In simplex mode, the direction of transmission is unchanging (from one antenna to the other only).
- A **radio station** and **television setups** are always a simplex channel because it always transmits the signal to its listeners and never allows them to transmit back.
- The advantage of simplex mode of transmission is that the entire bandwidth can be used because **data transmits only in one direction.**

## (B) Half-Duplex Channel

- It is a single physical channel in which the **direction may be reversed.**
- Here, **messages flow in two directions** as a half-duplex type, **but never at the same time.** At a single time, the transmission of data is done in only one direction.

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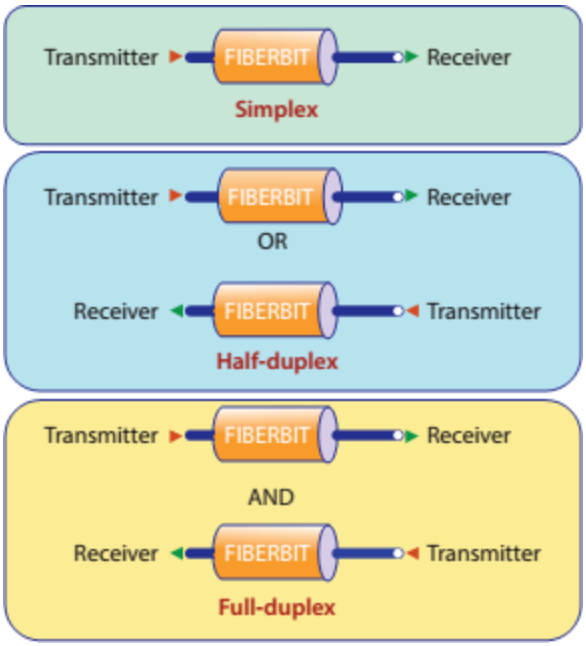


Figure 26.5 Different Types of Transmission Modes

- In a **telephone call**, we can observe this type of condition; we can see one party speaks while the other listens and when the other speaks, the first listens. When both speak simultaneously, it will result in a garbled sound that cannot be understood.
- The main problem of half-duplex mode of transmission is that since two channels are used, the bandwidth of the channel is decreased here.

### (C) Full-Duplex Channel

- A full-duplex channel is used for **transmitting simultaneous message exchange in both directions.**
- It has **two simplex channels**, one forward channel and other backward (reverse) channel, both linking at the same points.
- The main problem of the full-duplex mode of transmission is that we are using two channels simultaneously, so this **bandwidth** is also **decreased.**

There are different types of TV, depending upon the display device used. A display device is an output device for presentation of information in visual form. When the input information is supplied as an electrical signal, the display is called an electronic display.

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There are two main types of display devices:
(a) Cathode ray tube and (b) Flat-panel display.

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## (A) Cathode Ray Tube

The cathode ray tube (CRT) is a vacuum tube containing an electron gun (a source of electrons) and a fluorescent screen, with internal or external means to accelerate and deflect the electron beam, used to create images in the form of light emitted from the fluorescent screen. **Television has primary colours – red, blue and green.**

## (B) Flat-Panel Display

There are different technologies used under flat-panel display such as LCD, Plasma, LED, OLED.

### *I. Liquid Crystal Display (LCD)*

- LCD (liquid crystal display) television screens are made up of a thin layer of liquid crystals sandwiched between two glass plates. When an electric current is sent through the crystals, an array of tiny multi-coloured pixels light up to create a picture.
- LCD TV screens are thinner and lighter than plasma screens. They are the most used screens for computers, and are quickly gaining popularity as TV screens.
- LCD TV screens are anywhere from 1/4′′ to 4′′ thick and 2′′ to 65′′ wide.
- 'Twisted Nematic' liquid crystal substances are used in LCD television. These operate by twisting between 0 and 90 degrees, thus polarising the light as it passes through it. **4'-Propoxy-[1,1'-Biphenyl]-4carbonitrile (C16 H15 NO)** and **Trans-1-(bromomethyl)4-propylcyclohexane (C10 H19 Br)** liquid crystal materials are used in LCD TV. Another common chemical to be used in LCD screens is **MBBA**, or **N-(4-Methoxybenzylidene)-p-n-butylaniline.**

**COMPARISON WITH CRT AND LCD DISPLAY**

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- LCD is light weight (typically 1/5 of CRT)

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- In LCD, power consumption is lesser (typically ¼ of CRT)
- In LCD, there is no electromagnetic emission
- LCD are large screens (>20 inches) on desktops
- LCD are completely flat screen – no geometrical errors like CRT television
- LCD have crisp pictures – digital and uniform colours
- In LCD, fully digital signal processing is possible
- LCD has low luminance

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## *II. Plasma TV*

- They are called 'plasma' displays because the technology utilises small cells having electrically charged ionised gases.
- It is capable of producing deeper blacks, allowing a far superior contrast ratio. Plasma TV screens have a much higher resolution than CRT screens.
- In fact, the picture is so clear that it is almost like watching a scene through a window. Screen sizes range from 42′′ to 65′′ wide and are 3′′ to 4′′ thick.

## *III. LED TVs*

- LED is commonly known as '**backlight LCD television'**.
- In this technology, **light-emitting diodes use to backlight** the display device **instead of cathode fluorescent lights used in LCD television**.

**PLASMA VS LCD AND LED FEATURES**

- **Picture Quality:** Plasma TVs have better viewing angle – how far you can sit to one side of a TV screen before picture quality is affected**.**
- **Screen Life:** Plasma TVs have a screen life of about 30,000 to 60,00 hours, depending on the make and model, while LCD TVs have a screen life of 60,000 hours or more.
- **Plasma is a technology of large-size screen and is heaviest in all television (LCD and LED).** In mobile screen, plasma screen is not possible so we are using LCD and LED screen in mobile phone display.
- **Plasma TV consumes maximum power** in comparison to LCD and LED.

- The use of LED backlighting allows for a **thinner panel, lower power consumption, better heat dissipation, a brighter display** and **better**

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**contrast levels**.

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**DIFFERENCE BETWEEN LED AND LCD**

- LEDs produce images with greater dynamic contrast
- With edge-LED lighting can be extremely slim, some screens less than half an inch (0.92 cm) thick
- LED TVs do not heat-up and so have maximum screen life
- LEDs produce less environmental pollution on disposal
- LEDs are more expensive
- LEDs consume less power than LCD and Plasma
- LEDs have typically 20–30% lower power consumption
- LEDs are more reliable
- LEDs can allow a wider dimming range

## *IV. OLED (Organic Light-Emitting Diode)*

OLED (organic light-emitting diodes) is a flat light-emitting technology, made by **placing a series of organic (carbon based) thin films between two conductors.** OLED is a green technology that is very efficient and does not contain any bad metals.

The basic structure of an OLED is a **cathode (which injects electrons), an emissive layer** and an **anode (which removes electrons)**. More layers are used in modern OLED devices in order to make them more efficient, but the basic functionality remains the same.

### Benefits of OLEDs

- OLEDs have low power consumption and can be operated as a light source. OLEDs can produce white light and can provide the bulb of the future – low power (maybe even more efficient than CFL lamps) and thin area designs (and no bad metals).
- The materials are deposited by industrial coating processes, which are cheaper than the techniques required to make conventional LEDs.
- Because OLEDs emit light, they do not require a backlight and so are thinner and more efficient than LCD displays (which do require a white backlight).

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- OLEDs have a better picture quality – good colours, quick response rate and a wide viewing angle.
- They are inherently thin.
- They can be made on **flexible plastic substrates**, which will almost certainly lead to full colour flexible displays less than 1-millimetre thick.
- Multiple colours and all colours are possible.

### Disadvantages of OLEDS

- Today, it costs more to produce an OLED than it does to produce an LCD – although this should hopefully change in the future, as OLEDs have a potential to be even cheaper than LCDs because of their simple design.
- OLEDs have limited lifetime (like any display, really), that was quite a problem a few years ago but there has been constant progress, and today this is almost a non-issue.
- OLEDs can also be problematic in direct sunlight, because of their emissive nature. But companies are working to make it better.
- Certain materials in OLEDs are incredibly sensitive to moisture, which leads to short life, particularly on plastic substrates. This is the main reason for holding back flexible OLED manufacture.

## *Quantum LED (QLED)*

QLED stands for **quantum light-emitting diode.** It is like a regular LED TV; however, it uses tiny nanoparticles called **quantum dots** to super-charge its brightness and colour. The technology was introduced by Sony in 2013, but shortly after that, Samsung began selling its QLED TVs and established a licencing partnership with other manufacturers.

**Table 26.7** QLED vs OLED

| Parameter | QLED | OLED |
|---|---|---|
| Name | Quantum Light-emitting Diode | Organic Light-emitting Diode |

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| Technology | Transmissive | Emissive |
|---|---|---|
| Contrast and Black levels | Occurrence of Light Bleed | Perfect darkness |
| Brightness | Separate backlights | Own light |
| Colours | Excellent | Fantastic |
| Refresh rate, response time, and input lag | 2 to 8 millisecond - response time | Response time of 0.1 millisecond |
| Viewing angle | Centre | Up to 84 degrees |
| Screen Size | Higher range | Lower range |
| Lifespan | Excellent track record | Moderate |

## 26.17 Quantum Computing

Refer **Infographic – Quantum Computing**.

## 26.18 Quantum Supremacy

Refer **Infographic – Quantum Supremacy**.

## 26.19 Quantum Bits

Refer **Infographic – Quantum Bits**.

## 26.20 Quantum Key Distribution (QKD)

Refer **Infographic – Quantum Key Distribution**.

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## 26.21 Quantum Entanglement

Refer **Infographic – Quantum Entanglement**.

## 26.22 Quantum Satellite

Refer **Infographic – Quantum Satellite**.

## 26.23 Quantum Teleportation

Refer **Infographic – Quantum Teleportation.**

## 26.24 Difference between Classical Computer and Quantum Computer

Refer **Infographic – Classical Computers vs Quantum Computers**.

## 26.25 Microwave Oven

A microwave is much like the electromagnetic waves that zap through the air from TV and radio transmitters. It is an invisible up-and-down pattern of electricity and magnetism that races through the air at the speed of light (300,000 kilometres or 186,000 miles per second). Microwaves are effectively the shortest radio waves and the microwaves that cook food in ovens are just 12 centimetres (roughly 5 inches) long.

Despite their small size, microwaves carry a huge amount of energy. One drawback of microwaves is that they can **damage living cells and tissues**. This is why microwaves can be harmful to people and why

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microwave ovens are surrounded by strong metal boxes that do not allow the waves to escape.

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In normal operation, microwave ovens are perfectly safe. Even so, microwaves can be very dangerous, so never fool around with a microwave oven. Microwaves are also used in cellphones (mobile phones), where they carry your voice back and forth through the air and radar.

## History

Like many great inventions, microwave ovens were an accidental discovery. Back in the 1950s, American electrical engineer Percy Spencer (1894–1970) was carrying out some experiments with a magnetron at the Raytheon Manufacturing Company, where he worked. At that time, the main use for magnetrons was in radar, a way of using radio waves to help airplanes and ships find their way around in poor weather or darkness.

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# QUANTUM COMPUTING

## WHAT IS QUANTUM COMPUTING?

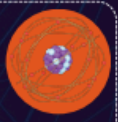
Quantum computing is an advanced computing paradigm that utilises principles from quantum physics to process and manipulate information.

Researchers and companies are actively working on developing and improving quantum hardware, algorithms, and error correction techniques to unlock the full potential of quantum computing.

Instead of classical bits, quantum computers use quantum bits, or qubits, which can represent multiple states simultaneously due to a property called superposition.

Quantum computers have the potential to solve complex problems more efficiently than classical computers.

Superposition allows quantum computers to perform parallel computations and explore different solutions simultaneously, potentially leading to exponential speedup in certain computational tasks.

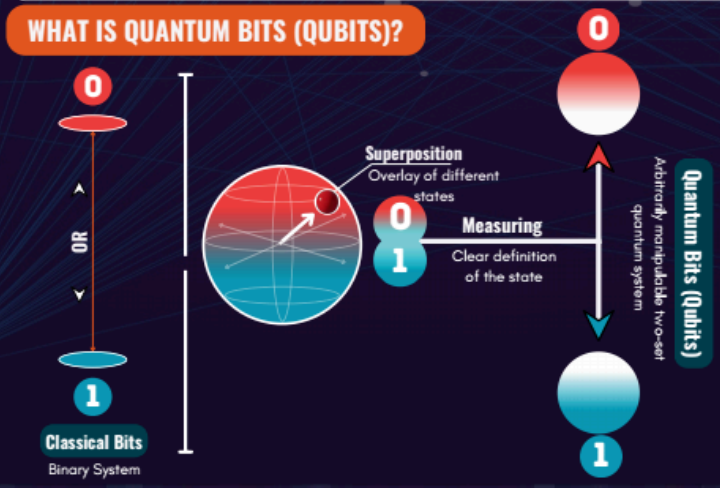


Quantum computing has the potential to revolutionise fields such as cryptography, optimisation, simulation, and machine learning.

## WHAT IS QUANTUM SUPREMACY?

- Quantum supremacy **refers to the state when a quantum computer can solve a specific problem that is practically infeasible for classical computers to solve within a reasonable amount of time.**
- It signifies **a milestone where quantum computers demonstrate their ability to outperform classical computers** in performing a particular computational task.
- Achieving quantum supremacy serves as an important proof-of-concept and showcases the potential power and capabilities of quantum computing.
- It also highlights the need for continued advancements in quantum hardware, algorithms, and error correction techniques to unlock the full potential of quantum computing in various domains.

## WHAT IS QUANTUM BITS (QUBITS)?




- **Definition:** A two-dimensional quantum-mechanical system, one of the simplest quantum systems.
- **Classical vs Quantum:**
  - **Classical bit:** Must be 0 or 1.
  - **Qubit:** Can be in a superposition of both 0 and 1 simultaneously.
- **Superposition:** Qubit can be a combination of 0 and 1 at the same time, a fundamental aspect of quantum mechanics and computing.
- **Computational Power:** The superposition ability of qubits provides unique computational power to quantum computers.
- **Creation:** Requires an object capable of achieving quantum superposition between two states.
- **Example:** An atomic nucleus can be used to create a qubit.

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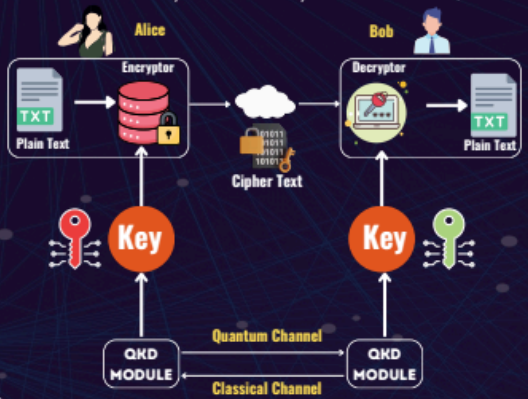
## WHAT IS QUANTUM KEY DISTRBUTION

- **Definition:** Cryptographic protocol using quantum mechanics to securely distribute encryption keys between two parties (Alice and Bob).
- **Goal:** Establish a secret key for secure communication.
- **Quantum Mechanics Principles:** Utilises uncertainty principle and no-cloning theorem.
- **Encoding:** Uses individual quantum particles (e.g., photons) to encode encryption key information.
- **Security:** Based on quantum mechanics, making it resistant to certain attacks; eavesdropping introduces detectable errors.
- **Application:** Secure key exchange for subsequent data encryption using classical cryptographic algorithms.
- **Benefits:** Potentially unbreakable key distribution method, enhancing security in finance, government and military sectors.

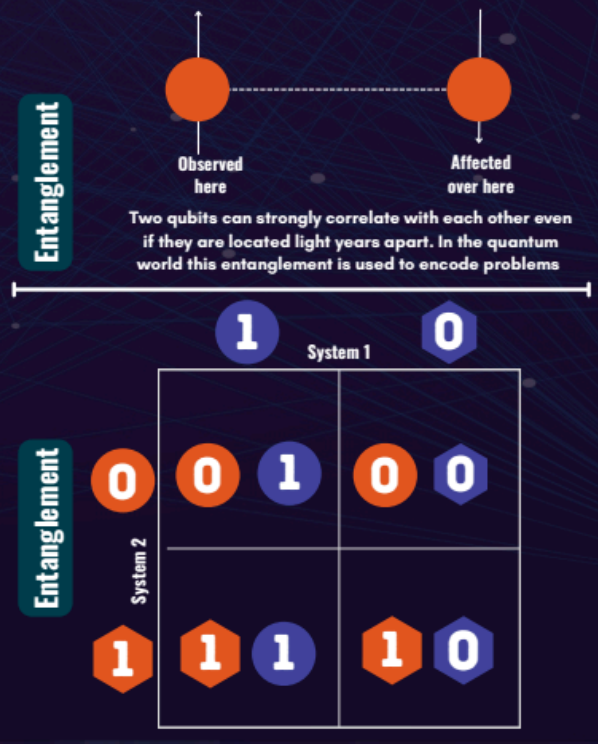



### PROCESS

- **Key Generation:**
  - Alice generates a stream of qubits, each representing a random bit (0 or 1) of the encryption key.
- **Quantum Transmission:**
  - Alice sends the qubits to Bob through a quantum channel (fiber-optic cable or free-space).
- **Measurement and Encoding:**
  - Bob receives and measures the qubits using randomly chosen compatible bases.
- **Key Exchange:**
  - Alice and Bob publicly share the bases used for each qubit, discarding those with different bases.
- **Error Correction:**
  - They perform error correction and privacy amplification to eliminate errors and potential information leakage, ensuring a final shared secret key.

## WHAT IS QUANTUM ENTANGLEMENT?

- **Definition:** Fundamental phenomenon where two or more particles become interconnected, regardless of distance.
- **Correlation:** Behavior of one entangled particle is instantly linked with the behavior of the other.
- **Combined System:** Entangled particles form a system with interdependent properties, losing individual identities.
- **Instantaneous Effect:** Measuring or changing one particle's state affects the other's state instantaneously.
- **Einstein's Description:** Referred to as "spooky action at a distance."
- **State Superposition:** Particles can exist in a superposition of states (0 and 1 simultaneously), unlike classical binary states.

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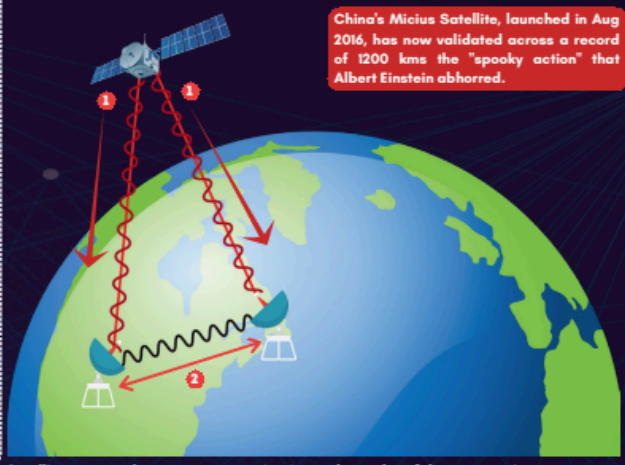
## WHAT IS QUANTUM SATELLITE?

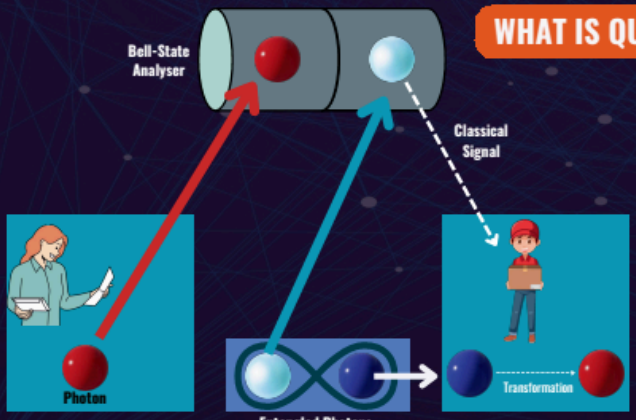
- A quantum satellite, also known as a quantum communication satellite, is a **specialised satellite designed to enable quantum communication and quantum information processing between ground-based stations.** It leverages the unique properties of quantum mechanics to enable secure communication and other quantum-enabled applications.
- **Purpose:** Enables quantum communication and information processing between ground stations.
- **Entangled Photon Distribution:** Distributes entangled photon pairs between ground stations.
- **Quantum Communication Protocols:** Uses entangled photons for protocols like quantum key distribution (QKD).
- **Secure Communication:** Establishes cryptographic keys using entangled photons.
- **Long-Distance Transmission:** Allows long-distance photon transmission with minimal signal loss or interference.
- **Global Quantum Networks:** Potential to create secure, global quantum communication networks.
- **Technological Advancements:** Active research to enhance capabilities and explore new applications.

**China's Micius Satellite, launched in Aug 2016, has now validated across a record of 1200 kms the "spooky action" that Albert Einstein abhorred.**




**Satellite-to-ground quantum communication and samples of the quantum Internet.**
1. **A satellite produces an entangled pair of photons and sends one photon to each station. The probability that each photon arrives at the destination.**
2. **A link between the two ground stations is formed when both photons arrive.**

## WHAT IS QUANTUM TELEPORTATION?




- Quantum teleportation is a remarkable concept in quantum physics that involves the **transfer of quantum information from one location to another without physically moving the particle itself.** It allows for the instantaneous transfer of the quantum state of a particle from a sender (Alice) to a receiver (Bob), even if they are separated by a large distance.
- Here's a breakdown of quantum teleportation:
  a. **Initial Entanglement:** Alice and Bob establish an entangled pair of particles (Bell pair) with strong correlation.
  b. **Preparing the Quantum State:** Alice combines the unknown quantum state with her entangled particle.
  c. **Measurement and Communication:** Alice measures both particles and obtains classical information.
  d. **Sending Classical Information:** Alice sends the measurement results to Bob via a classical communication channel.
  e. **State Reconstruction:** Bob uses the received information to perform operations on his entangled particle, replicating Alice's original quantum state.

- **No Physical Transport:** Involves replicating a quantum state at a different location without moving the actual particle.
- **Principles:** Relies on entanglement and transfer of classical information.
- **Applications:** Crucial for quantum computing, secure communication, and quantum networking.
- **Significance:** Highlights the power of quantum entanglement and the transfer of quantum information, demonstrating the counterintuitive nature of quantum mechanics.

### Majorana 1 (New Quantum Chip)

- **Microsoft** has announced the arrival of **Majorana 1**, a **new quantum chip** that produces more reliable and scalable qubits than its predecessors. Qubits are the building blocks of quantum computers.
- The tech giant considers Majorana 1 to be a major milestone in the road to the future of useful quantum computing.
- Microsoft CEO Satya Nadella said that the company will be able to build quantum computers for commercial use between 2027 and 2029.

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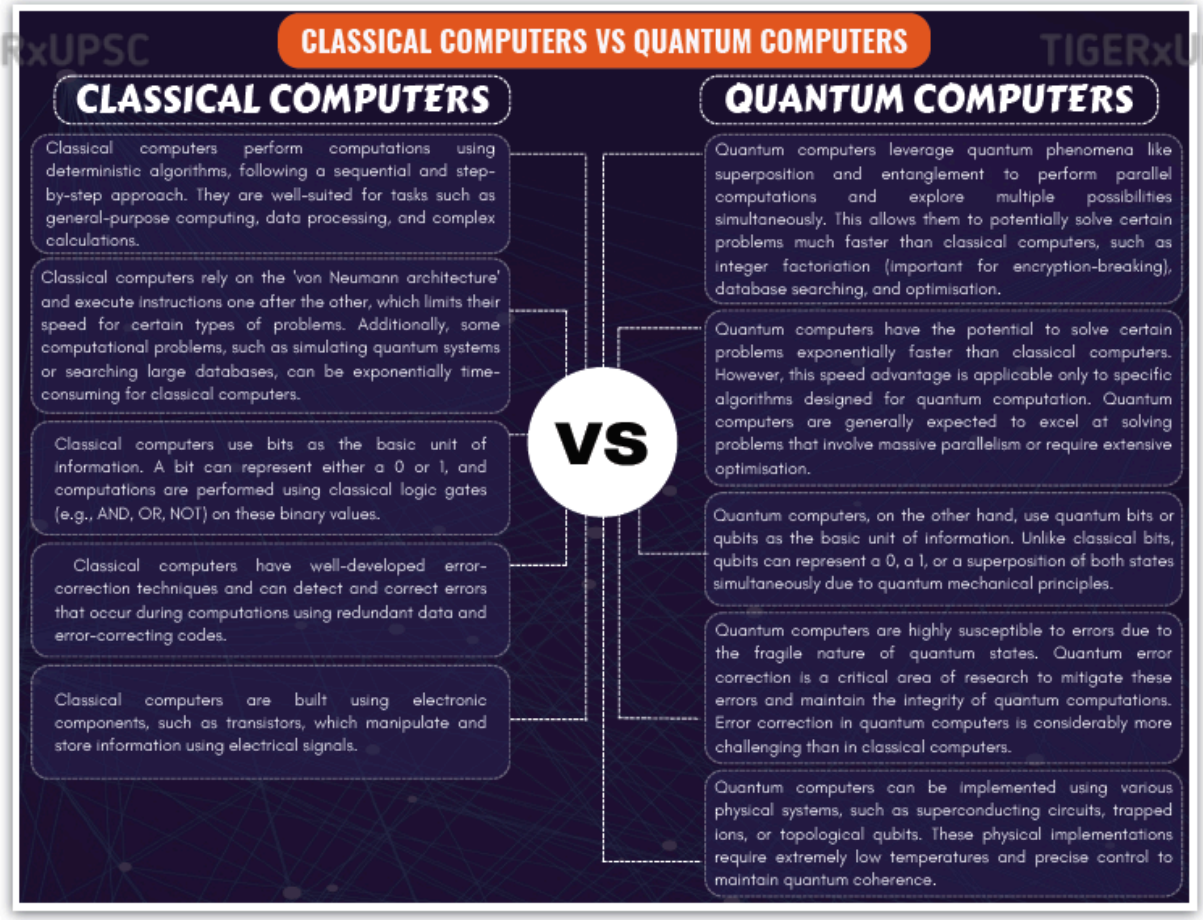


## *Working Mechanism of Microwave Oven*

- Inside the strong metal box, there is a microwave generator called a **magnetron**. When you start cooking, the magnetron takes electricity from the power outlet and converts it into high-powered radio waves.
- The magnetron blasts these waves into the food compartment through a channel called a waveguide.
- The food sits on a turntable, spinning slowly around so the microwaves cook it evenly.
- The microwaves bounce back and forth off the reflective metal walls of the food compartment, just like light bounces off a mirror. When the microwaves reach the food itself, they do not simply bounce off. Just as radio waves can pass straight through the walls of your house, microwaves penetrate inside the food. As they travel through it, they make the molecules inside it vibrate more quickly. Vibrating molecules have heat, so, the faster the molecules vibrate, the hotter the food

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becomes. Thus, the microwaves pass their energy onto the molecules in the food, rapidly heating it up.

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### *Health Hazards*

Microwave radiation can heat body tissue in the same way as it heats food. Its higher exposure can cause a painful burn. Its higher level on eye lens causes cataracts. Microwave ovens always used low level of microwaves, within the region of non-ionising radiation. The effects of long-term exposure of microwaves on human body are still uncertain. It is best to stay away (an arm's length) in reducing its exposure.

## 26.26 Black Box in an Airplane

Any commercial aeroplane or corporate jet is required to be equipped with a **cockpit voice recorder** and a **flight data recorder**. It is these two items of separate equipment which we commonly refer to as a '**black box**.' While they do nothing to help the plane when it is in the air, both these pieces of equipment are vitally important should the plane crash, as they help crash investigators find out what happened just before the crash.

To help locate the cockpit voice recorder and a flight data recorder in the aftermath of a plane crash that occurs at sea, each recorder has a device fitted to it known as an **underwater locator beacon (ULB)**. The device is activated as soon as the recorder comes into contact with water and it can

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transmit from a depth as deep as 14,000 feet. Also, to help investigators find them; a black box is not actually black at all, but bright orange.

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**EUROPEAN CARS MANDATED TO ADOPT AIRPLANE-INSPIRED 'BLACK BOXES' BY JULY 2024**

All new cars sold in the European Union are now mandated to be equipped with sophisticated data recorders, named 'Event Data Recorder' (EDRs). Inspired by similar technology called 'black boxes' used in airplanes, these devices promise to heighten accident investigation and road safety.

These EDRs are small devices hidden inside a car›s electronics and gets activated automatically when the airbags or seat belt pretensioners are engaged.

## 26.27 Carbon Fiber

- Carbon fiber, also known as graphite fibre, is a lightweight and high-strength material composed of carbon atoms arranged in a crystalline structure.
- It is a type of fiber-reinforced polymer (FRP) where the reinforcing material is carbon fiber.
- The production of carbon fiber involves several steps. It begins with the precursor material, usually a polymer, such as polyacrylonitrile (PAN), rayon or pitch. The precursor undergoes a series of treatments, including oxidation and carbonisation, to remove non-carbon elements and convert the material into pure carbon fibers.

### *Characteristics and Applications of Carbon Fibers*

- Physical strength, specific toughness, light weight aerospace, road and marine transport, sporting goods.
- While recycling carbon fiber is more challenging than recycling other materials, it is possible through various methods like mechanical and chemical recycling.
- High dimensional stability, low coefficient of thermal expansion and low abrasion missiles, aircraft brakes, aerospace antenna and support

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structure, large telescopes, optical benches, waveguides for stable high-frequency (GHz) precision measurement frames.

- Good vibration damping, strength and toughness of audio equipment, loudspeakers for Hi-fi equipment, pickup arms, robot arms.
- Electrical conductivity automobile hoods, novel tooling, casings and bases for electronic equipments, EMI and RF shielding, brushes.
- Biological inertness and x-ray permeability, medical applications in prostheses, surgery and x-ray equipment, implants, tendon/ligament repair.
- Fatigue resistance, self-lubrication, high damping Textile machinery, gentra engineering.
- Chemical inertness, high corrosion resistance chemical industry; nuclear field; valves, seals, and pump components in process plants.
- Electromagnetic properties, large generator retaining rings, radiological equipment.

### *Applications of Carbon Fibre:*

- **Aerospace and Aviation:** Carbon fiber composites are extensively used in the aerospace industry to reduce weight while maintaining structural strength.
- **Automotive:** Carbon fiber is used in high-performance vehicles to reduce weight and improve fuel efficiency.
- **Sports and Recreation:** Carbon fiber is widely used in sporting goods and equipment, such as tennis rackets, golf clubs, bicycle frames, and helmets. Its high strength and lightweight properties enhance performance and durability.
- **Wind Energy:** Carbon fiber composites are used in wind turbine blades to provide strength and flexibility while minimising weight.
- **Civil Engineering and Construction:** Carbon fiber is employed in structural reinforcements, such as beams, columns and bridge decks to enhance strength and durability. It is also used in seismic retrofitting to reinforce buildings against earthquakes.

## 26.28 KAVACH System

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## Kavach System—India's Indigenous Train Protection System

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Kavach is India's own indigenously developed **Automatic Train Protection (ATP) system.** It has been developed by the **Research Design and Standards Organisation (RDSO),** which is the research wing of Indian Railways, in partnership with Indian industry vendors like Medha Servo Drives Pvt. Ltd., HBL Power Systems Ltd. & Kernex Microsystems.

Kavach, formerly known as **Train Collision Avoidance System (TCAS),** is India's answer to enhancing railway safety. It is designed to prevent collisions by automatically controlling train speed if required. It continuously monitors the train's speed and applies brakes if the driver fails to respond to warnings, preventing scenarios like **over speeding, passing a signal at danger (SPAD),** or **head-on collisions** (if both trains are equipped with Kavach).

### *National Rail Plan (NRP) 2030*

The NRP 2030 document aims to develop infrastructure by 2030 to cater to the projected traffic growth up to 2050. Key focus areas include capacity augmentation on saturated routes, reducing transit times, improving safety, and increasing the modal share of freight transport by rail.

### *How it Works*

The Kavach system is a sophisticated safety system. While it does use trackside transponders (which are a type of RFID technology) fitted on the track, the statement is potentially misleading or incomplete. These transponders are installed at various locations along the track, including station limits, gradient changes, curves, and at specific intervals, not just exclusively "in station section" as implied. The complete Kavach system involves trackside radio towers, locomotive units, and station interfaces communicating with each other.

### *Key Features and Applications*

- Prevents over-speeding and signal violations
- Automatically halts trains if another train is detected on the same track

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- Transmits signal information to the locomotive continuously—especially useful in low visibility (e.g. fog)
- Ensures safe operation in inclement (difficult) weather
- Enhances driver assistance and collision avoidance

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## 26.29 Rare Earth Elements

Rare earth elements (REEs) are a group of 17 metallic elements, comprising the 15 lanthanides (atomic numbers 57–71) along with scandium (Sc) and yttrium (Y). These elements are not particularly rare in terms of abundance in the Earth's crust, but they are rarely found concentrated in economically viable deposits. They are crucial for many modern technologies, including electronics, renewable energy and defense.

### Here is a More Detailed Breakdown

#### *What are They?*

- **Lanthanides:** These are the 15 elements from lanthanum (La) to lutetium (Lu) in the periodic table.
- **Scandium (Sc) and Yttrium (Y):** These are transition metals that share similar chemical properties with the lanthanides and are often found in the same ore deposits.
- **Light vs. Heavy:** REEs are sometimes categorised as light (La-Sm) or heavy (Eu-Lu), based on their atomic weight and ionic radii.

#### *Why are They Important?*

- **High-Tech Applications:** REEs are essential components in a wide range of technologies, including:
  - ❑ **Permanent Magnets:** Used in electric vehicles, wind turbines and electronics.
  - ❑ **Catalysts:** Used in catalytic converters in cars and other industrial processes.
  - ❑ **Batteries:** Used in rechargeable batteries for various applications.

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- ❑ **Electronics:** Used in smartphones, computers and other digital devices.
- ❑ **Lighting:** Used in energy-efficient lighting and display technologies.
- ❑ **Medical Imaging:** Used in MRI and other medical imaging techniques.

- **Renewable Energy:** REEs play a crucial role in the development of renewable energy technologies like wind turbines and electric vehicles.
- **National Security:** Many REEs are vital for military applications, making them strategically important.

### *Challenges and Concerns*

- **Concentration:** REEs are widely dispersed in the Earth's crust and are not typically found in high concentrations.
- **Extraction and Processing:** Extracting and processing REEs can be complex, costly and environmentally challenging.
- **Geopolitical Concerns:** China currently dominates the production and processing of REEs, raising concerns about supply chain vulnerability.
- **Sustainability:** There is growing concern about the environmental impact of REE mining and processing, and the need for more sustainable practices.

**Note:** China has the largest fraction (37 percent), followed by Brazil and Vietnam (18 percent each), Russia (15 percent) and the remaining countries (12 percent).

## 26.30 Accelerometer

Refer **Infographic – Accelerometer.**

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# 26.31 Indian Standard: Electronic items, Food and Agriculture Related

## 26.31.1 Bureau of Indian Standard (BIS)

The Bureau of Indian Standards (BIS) is the national standards body of India under the **Ministry of Consumer Affairs, Food and Public Distribution,** Government of India. It was established by the **Bureau of Indian Standards Act, 1986,** and came to effect from 23 December 1986. The BIS is engaged in the activities of standardisation, conformity assessment and quality assurance of goods, articles, processes, systems and services.

The BIS certification scheme is basically **voluntary** in nature. However, for a number of products compliance to Indian Standards is made compulsory by the Central Government under various considerations viz. **public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security.**

The BIS is the National Standards Body of India (NSB). It has **three major activities:**

1. **Standard formulation**
2. **Certification Activities:** This is the primary activity of BIS with a majority of its workforce engaged in certification activities. It includes factory visits, examining product at the workplace, checking factory laboratories for availability of proper testing equipment, handling new

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licence applications, renewing/rejecting old ones, enforcement activities etc.

3. **Laboratories:** This activity includes receiving samples sent by certification officer, applying proper tests, preparing reports and also (kind of) giving a verdict on whether the product meets the corresponding standard.

### 26.31.2 Indian Standard Institutions (ISI)

The ISI mark is a standards-compliance mark for **industrial products in India** since 1950. The mark certifies that a product conforms to an Indian standard (IS) developed by the Bureau of Indian Standards (BIS), the national standards body of India.

On 1 January 1978, it was renamed to the Bureau of the Indian Standards. The ISI mark is mandatory for a certain products to be sold in India, such as many of the electrical appliances like switches, electric motors, wiring cables, heaters, kitchen appliances etc., and other products like Portland Cement, LPG valves, LPG cylinders, automotive tyres etc. In the case of most other products, ISI marks are optional. Counterfeiting ISI marks is a punishable offence by the law, but enforcement is uncommon.

### 26.31.3 ECO-Mark Scheme

ECO Mark Scheme was instituted by the Government of India for labelling of environment-friendly products. The scheme is being administered by the Bureau of Indian Standards.

The scheme first saw its application in 1991. It is voluntary in nature and helps in identifying eco-friendly products in the market. The eco-mark scheme falls under the ambit of Ministry of Environment, Forest and Climate Change.

The scheme covers various product categories like soaps and detergents, paints, food items, lubricating oils, packing/packaging materials, architectural paints and powder coatings, batteries, electrical and electronic goods, food additives, wood substitutes, cosmetics, aerosols and propellants, plastic products, textiles, fire-extinguisher, leather and coir and coir products.

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The presence of ECO logo along with ISI mark on a product indicates that the product meets certain environmental criteria along with the quality requirements as specified in the relevant Indian Standard.

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For implementation of the scheme, BIS is responsible for the following functions:

- Renewal, suspension and cancellation of the licence.
- Undertaking inspections and taking samples for analysis of any material or substance in relation to which the ECO mark has been used.

To operate the scheme, BIS has included additional requirements for ECO mark in the concerned Indian Standards. The terms and conditions governing operation of the licence shall be as per the Bureau of Indian Standards Act, Rules and the Regulations.

### 26.31.4 AGMARK

AGMARK is a certification mark for agricultural produce, assuring that they conform to a grade standard notified by Directorate of Marketing and Inspection (DMI), Department of Agriculture, Cooperation and Farmers Welfare, **Ministry of Agriculture and Farmers Welfare** under Agricultural Produce (Grading Marking) Act, 1937.

These standards differentiate between quality and 2–3 grades are prescribed for each commodity. Till date, grade standards for 22 agricultural commodities have been notified. These include **fruits, vegetables, cereals, pulses, oilseeds, vegetable oils, ghee, spices, honey, creamery butter, wheat, atta, besan** etc.

While framing the standards, the existing standards in The Food Safety and Standards Act, 2006, **Codex Alimentarius Commission**, International Organization for Standardization, etc. are considered.

### 26.31.5 Food Safety and Standards Authority of India (FSSAI)

Food Safety and Standards Authority of India (FSSAI) is an **autonomous body** established under **the Ministry of Health and Family Welfare**, Government of India.

It has been established under the Food Safety and Standards Act, 2006 which is a consolidating statute related to food safety and regulation in India.

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**CODEX ALIMENTARIUS COMMISSION (CAC)**

The Codex Alimentarius Commission or CAC is the body responsible for all matters regarding the implementation of the Joint FAO/WHO Food Standards Programme.

Membership of the Commission is open to all Member Nations and Associate Members of FAO and WHO which are interested in international food standards.

The Commission meets in regular session once a year alternating between Geneva and Rome.

The programme of work of the Commission is funded through the regular budgets of WHO and FAO with all work subject to approval of the two governing bodies of the parent organisations.

FSSAI is responsible for **protecting and promoting public health** through the regulation and supervision of food safety.

The FSSAI has its headquarters at **New Delhi**. The authority also has **6 regional offices** located in Delhi, Guwahati, Mumbai, Kolkata, Cochin and Chennai. A total of **14 referral laboratories** notified by FSSAI, 72 State/UT laboratories located throughout India and 112 laboratories are National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited private laboratories notified by FSSAI.

## 26.32 Food and Health Related Issues

### 26.32.1 MSG Maggie Issues

Maggi is an international brand of seasonings, instant noodles and soups that was established in Switzerland. Nestle acquired the brand in 1947.

Maggi noodles are highly popular in both India and Malaysia. Nestle has 39% market share in Malaysia, where Maggi is synonymous with instant noodles, and 90% market share in India prior to a **nationwide ban by the Food Safety and Standards Authority of India.** The ban reduced the market share to 53% in India. In June 2015, quality tests conducted in India found high amounts of lead in Maggi Noodles.

**Food Safety Regulators at Barabanki, Uttar Pradesh**, in May 2015, reported that samples of Maggi Noodles had **unexpectedly high levels of**

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**monosodium glutamate**, which is **around 17 times the permissible limit of lead.** This report led to multiple market withdrawals and investigations in India and beyond.

The Supreme Court of India has lifted a stay on criminal proceedings against Nestle over the alleged presence of lead and MSG or monosodium glutamate in popular noodles.

### *What's the Issue?*

- The apex court had allowed the case against Nestle to reopen after the Swiss food giant had admitted the presence of toxic ingredients in its noodles during a court hearing.
- Nestle's lawyers have admitted that toxic elements like lead and MSG are found in its sample tests conducted by **CFTRI (Central Food Technological Research Institute), Mysuru**. However, the company has argued that the lead was within the permissible limit.

## Effects of Lead and MSG on Health

### *(A) Lead*

- According to the **World Health Organization (WHO),** lead is a cumulative toxicant that affects multiple body systems, including your liver and kidneys in particular.
- It is especially **harmful to infant and pregnant women** as it can cause damage to the **central nervous system in children.**
- Since lead is a heavy metal it can get stored in the teeth and bones, where it accumulates over time. In pregnant women, prolonged exposure to lead can result in **miscarriage, prematurity, low birth weight** etc.

### *(B) Monosodium Glutamate*

- MSG, short form for monosodium glutamate, is an amino acid found in your body and most foods. It is a flavour enhancer commonly added to Chinese food, canned vegetables, soups and processed meats.
- When consumed in excess, it can cause several health issues, including a headache, nausea, sweating, weakness, chest pain, tingling or burning

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in the face as well as neck and other areas. It is claimed that MSG can cause asthma and even brain damage.

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**NATIONAL CONSUMER DISPUTES REDRESSAL COMMISSION (NCDRC)**

- The National Consumer Disputes Redressal Commission (NCDRC), India is a quasi-judicial commission in India which was set up in 1988 under the Consumer Protection Act of 1986. The 1986 Act later got repealed by a new Act of 2019.
- The commission is headed by a sitting or retired judge of the Supreme Court of India.

### Statutory Provisions

- Section 58 of Consumer Protection Act, 2019, provides that the National Commission shall have jurisdiction to entertain a complaint valued more than ten crore and also have Appellate and Revisional jurisdiction from the orders of State Commissions or the Central Authority as the case may be.
- Section 67 of Consumer Protection Act, 2019, provides that any person aggrieved by an order of National Commission may prefer an Appeal against such order to Supreme Court of India within a period of 30 days from the date of such order.

- MSG may cause adverse symptoms in some people, including a condition called Chinese restaurant syndrome or MSG symptom complex. However, there is no definitive evidence that MSG is solely responsible for causing these symptoms.

## 26.32.2 Bisphenol A (BPA)

Bisphenol A is used in the production of **polycarbonate plastics** and **epoxy resins**.

**Polycarbonate plastics** have many applications, some food and drink packaging; for example, water and in front bottles, compact discs, impact safety equipment and medical devices etc. **Epoxy resins** are used as lacquers to coat metal products such as food cans, bottles tops and water supply pipes.

Bisphenol A is found in **hard plastic bottle**. BPA exposure leads to wide variety of health issues, such as **obesity and diabetes**. **Vitamin D**

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**deficiency, cancer, insomnia, arthritis, heart diseases** and **disturbances of hormones** in human body are other problems in human body.

### *Exposure of BPA in Human Body*

The main source of exposure to BPA for most people is through diet, while air, dust and water are other possible sources of exposure. BPA in food and beverages accounts for the majority of daily human exposure.

Bisphenol A can leach into food from the protective internal epoxy resin coatings of canned foods and from consumer products such as polycarbonates, table ware, food-storage containers and water bottles. BPA can also be **found in breast milk**.

### *Prevent Exposures to BPA*

- Do not put polycarbonate plastic food containers in microwaves. Polycarbonate is strong and durable but over time, it may break down from overuse at high temperatures.
- Reduced use of canned food.
- Use bottles that are BPA-free.
- It is better to opt for glass, porcelain or stainless steel containers, particularly for hot food or liquids.
- Plastic containers have recycle codes on the bottom. Some, but not all, plastics that are marked with recycle codes 3 or 7 may be made with BPA.

## 26.32.3 Polished Rice

The removal of outer bran, husk and germ layer of the rice grain is through the process known as **milling**; after that, rice grain goes through the process of **polishing**, which gives a shiny, whitish appearance to the seeds.

These processes **remove the vital nutrients** from the grain, which is very essential for health. Bran contains many important nutrients, **approximately 80% of the minerals, fibre and essential oils** come from the grains.

Many studies conclude that the fibres prevent many diseases such as **gastrointestinal diseases and heart disease**. Rice bran oil **decreases**

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**serum cholesterol**, which is the potent factor in heart diseases. The germ is loaded with **vitamins, minerals, unsaturated fats** etc.

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## 26.32.4 Antioxidants

It is a substance that **inhibits/delays the oxidation process** of other molecules. Oxidation is a chemical process in which oxygen is pooled with other molecules and **produces free radicals that can damage the cells**.

Antioxidants, present naturally in our body and also added to food as a supplement, provide various benefits, and avoid diseases. Some common antioxidants include enzymes and other substances like some Vitamins, β carotene etc. Most common vitamins used as antioxidants are A, C and E. Vitamins D, K and folic acid also increase antioxidant property.

### *Benefits of Antioxidants*

- **Repairs Damaged Cells**: Many plants and animals produce and store large amounts of anti-oxidants. It helps in reducing the cell damage process and also has the ability to repair damaged cells.
- **Preventing Cancer:** National Cancer Institute performed many experiments on animals and observed 'antioxidants help in preventing free radicals' damage that is related to cancer.
- Antioxidants help in **reducing the process of ageing** and may even **lead to longevity**.

## 26.32.5 Preservatives (Preserve to Reserve)

Preservatives help in maintaining the nutritional viability and extension of duration of food from getting spoiled through the action of microorganisms, specifically bacteria, moulds and yeasts.

Freezing, boiling, smoking, salting, pasteurising, dehydrating and pickling are some traditional natural ways of preserving food. 'Natural food preservatives' like salt, sugar alcohol, vinegar are the preservatives for making pickles, jams and juices. Coffee and soup powder dehydrated and freeze in dry form for preservation. Citrus acid and ascorbic acid work on enzymes and disrupt metabolism and help in preservation.

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'**Chemical preservatives'** are most effective for longer shelf-life. Some examples are benzoates (sodium benzoate, benzoic acid), nitrates (sodium nitrite), sulphites (sulphur dioxide), sorbates (sodium sorbate, potassium sorbate), vitamin C, BHA (butylated hydroxyanisole) and bacterial growth inhibitors like sodium nitrite, sulphur dioxides and benzoic acid. Ethanol is also a chemical preservative in food and wine.

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There are **three types of artificial preservatives** of food:

- **Antimicrobial agents** (benzoates, sodium benzoate, sorbates, nitrites)
- **Antioxidants** (sulphites, vitamin E, vitamin C and butylated hydroxytoluene (BHT))
- **Chelating agent** (disodium ethylene diamine tetra acetic acid, polyphosphates and citric acid)

**Natural Preservatives** include rosemary (Gulmehandi – *Rosmarinus officinalis*) and oregano (ajwain leaves etc.) extract, hops, salt, sugar, vinegar, alcohol, diatomaceous earth and castor oil. Hops are **the flowers (also called seed cones or strobiles) of the hop plant** Humulus lupulus, a member of the Cannabaceae family of flowering plants. Diatomaceous earth is also called 'Kieselguhr'. Diatomaceous earth is a common alternate name, but the term is more appropriate for unconsolidated or less lithified rock of the same origin.

| Disease/Problems | Causes |
|---|---|
| **Asthma** | • Sodas contain sodium benzoate, which is used as a preservative to control microbial growth.<br>• Sodium preservatives get mixed with the diet and reduce the content of potassium in our body.<br>• This increases the chances of Asthma, and eczema.<br>• Survey facts: Every day, 370 people die of asthma resulting from these beverage products. |
| **Dissolves Tooth Enamel** | • Soft drinks contain high-reactive solvents, which when mixed with saliva, or stomach's digestive acids become acidic in nature.<br>• This starts effecting immediately, in the next second of consumption. |

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### 26.32.6 Aerated Drinks/Soft Drinks

A soft drink contains **carbonated water, a sweetener and a natural or artificial flavouring**.

The sweetener may be sugar, high-fructose corn syrup, aspartame (neotame, acesulfame and sucralose are sweeteners being used sometimes), fruit juice, sugar substitutes (in the case of diet drinks), or some combination of these.

Acids like phosphoric acid in colas and citric acid in citrus-flavoured drinks are added to soft drinks for extra bite and feel. Carbonated water (carbon dioxide dissolved in water under pressure) is also mildly acidic (chemically carbonic acid, $H2CO3$).

Caffeine is added as a stimulant, but it has a bitter taste that is a component in many soft drinks. Orange soda often contains **glyceryl abietate** and **brominated vegetable oil**. These help keep fatty flavours suspended in the liquid (density balancers and emulsifiers). Gums and modified food starches are also used for this purpose.

**Sodium benzoate** is used as a **broad spectrum antimicrobial**, inhibiting bacteria, moulds and yeasts. The high acid content of the soft drink is necessary for the preservative action.

**Sodium citrate** buffers the acids, so the pH stays low (acidic). It also emulsifies any fats or fat-soluble compounds in the flavourings, keeping them in solution.

**Potassium sorbate** is added to inhibit yeasts and fungi.

**Ascorbic acid (vitamin C)** is used as an antioxidant. In colas, the colour comes from caramel colouring (burnt sugar).

**Red 40** and **other colours** are used in fruit-flavoured drinks such as orange soda.

### 26.32.7 Per- and Polyfluoroalkyl Substances (PFAS)

Per- and polyfluoroalkyl substances (PFAS) are a large, complex group of synthetic chemicals that have been used in consumer products around the world since the 1950s. They are ingredients in various everyday products.

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For example, PFAS are used to keep food from sticking to packaging or cookware, make clothes and carpets resistant to stains and create firefighting foam that is more effective.

PFAS molecules have a chain of linked carbon and fluorine atoms. Because the carbon-fluorine bond is one of the strongest, these chemicals do not degrade easily in the environment.

### *How PFAS exposures happen to human being?*

PFAS are used in the aerospace, automotive, construction, and electronics industries. Over time, PFAS may leak into the soil, water, and air.

People are most likely exposed to these chemicals by consuming PFAS-contaminated water or food, using products made with PFAS, or breathing air containing PFAS. Because PFAS break down slowly, if at all, people and animals are repeatedly exposed to them, and blood levels of some PFAS can build up over time.

Multiple health effects associated with PFAS exposure have been identified and are supported by different scientific studies.

**Concerns about the public health impact of PFAS have arisen because of the following reasons:**

- **Widespread Occurrence:** Studies find PFAS in the blood and urine of people, and scientists want to know if they cause health problems.
- **Numerous Exposures:** PFAS are used in hundreds of products globally, with many opportunities for human exposure.
- **Growing Numbers:** PFAS are a group of nearly 15,000 synthetic chemicals, according to a chemicals database (CompTox) maintained by the U.S. Environmental Protection Agency.
- **Persistent:** PFAS remain in the environment for an unknown amount of time.
- **Bioaccumulation:** People may encounter different PFAS chemicals in various ways. Over time, people may take in more of the chemicals than they excrete, a process that leads to bioaccumulation in bodies.

Because there are many types of PFAS chemicals, which often occur in complex mixtures and in various everyday products, researchers face challenges in studying them. More research is needed to fully understand all sources of exposure and if and how they may cause health problems.

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## 26.32.8 Fats: Types, Trans Fat and Omega 3

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Fats are nutrients in food that the body uses to build nerve tissue (including the brain and nerves) and hormones. Human body also uses fat as fuel. Unused fats are stored by the body in fat cells. Fats are required to human body for absorption of some vitamins (vitamins A, D, E and K are fat-soluble, meaning they can only be absorbed if there is fat in a person's diet).

There are different levels of importance to human beings:

- Fats are the building blocks of hormones.
- Fats are necessary for insulating all nervous system tissues in the body.
- Fats help people feel full, so they are less likely to eat as much.
- Fat is a great source of energy but has twice the amount of calories, compared with the same amount of carbohydrates or protein.
- For example, 1 gram of fat provides 9 calories, whereas 1 gram of both carbohydrates and protein provide 4 grams each.
- Desserts and snacks (including potato chips, chocolate, cakes, doughnuts, pastries and cookies) are a significant source of fat.
- Kids also get fat from whole-milk products and high-fat meats, such as bacon, hot dogs and non-lean red meat.

**Types of Fats:** There are two types of fats on the basis of chemical compounds and its structure.

1. **Unsaturated Fats:** Found in plant foods and fish, these fats are seen as neutral or even beneficial to heart health. The types of unsaturated fats are:
   - ❑ **Monounsaturated,** found in avocados and olives, peanuts and canola oils.
   - ❑ **Polyunsaturated**, found in most vegetable oils. **Omega-3 fatty acids are a type of polyunsaturated fat found in oily fish like Tuna and Salmon.**
2. **Saturated Fats:** Saturated fats are found in meat and other animal products, such as butter, shortening, lard, cheese and milk (except skim or non-fat). Saturated fats are also found in palm and coconut oils, which are often used in commercial baked goods. Eating too much saturated fat can raise blood cholesterol levels and increases the risk of heart disease.

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## 1. The Fat Family Tree

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The family of fat is highly complex, so to make it less confusing, picture it as a family tree.

At the top, there are two different families of fat – **saturated fat** and **unsaturated fat**.

Saturated fat like butter is packed with hydrogen atoms, making it solid at room temperature. Unsaturated fat (like olive oil) contains fewer hydrogen atoms, so it is liquid at room temperature.

The family of unsaturated fat includes two children: **monounsaturated fat** and **polyunsaturated fat.** In the polyunsaturated fat family, you will find omega-6 fatty acids and omega-3 fatty acids, and it is the omega-3 family that has been making headlines in the nutrition world.

## 2. Omega 3 and its Types

There are three types of fatty acids that are collectively called omega 3: **ALA (alpha-linolenic acid), EPA (eicosapentaenoic)** and **DHA (docosahexaenoic acid).**

### *Mega Health Benefits*

- Extensive research has indicated that omega-3 fats reduce inflammation, helping to prevent inflammatory diseases like **heart disease** and **arthritis.** In addition to warding off inflammation, omega 3s are also **essential to the brain**, which impact behaviour and cognitive function, and are especially necessary during **foetal development**. According to the University of Maryland Medical Centre (UMM), omega 3s may also improve artery health by helping to **reduce plaque build-up and blood clots in arteries** that lead to the brain.
- Improve cholesterol with the lowering tri-glycerides and elevating HDL (good cholesterol) levels. These benefits come primarily from DHA and EPA. Improve joint health by reducing joint tenderness and stiffness associated with arthritis and osteoarthritis. Improve bone health by positively impacting the body's calcium levels, reducing the incidence of bone loss.
- Improve mental health by helping to insulate nerve cells in the brain, allowing these nerve cells to better communicate with one another.

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People who are deficient in omega 3s may suffer from depression, bipolar disorder, schizophrenia and eating disorders.

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- Improve skin health by alleviating of symptoms related to skin disorders like acne and psoriasis.
- Improve bowel health by reducing inflammation of the bowels, helping alleviate symptoms of Crohn's disease and ulcerative colitis.
- Improve lung health by reducing inflammation in diseases like asthma. Improve menstrual health by reducing the pain associated with PMS (pre-menstrual syndrome) and menstruation.
- Help to **prevent cancer – colon, breast and prostate** cancers have all been correlated with low intakes of omega 3s.

## *Sources of Omega 3*

The three different types of omega 3 are found in specific types of foods.

ALA is found primarily in foods of plant origin, but it is also found in other foods such as beef. The richest source of ALA is flaxseed, but it is also found in hempseed, canola oil, soybeans, soybean oil, pumpkin seeds, pumpkin seed oil, linseeds, walnuts and walnut oil.

Once ingested, the body converts ALA into EPA and DHA, allowing it to be more readily used by the body. However, this conversion is not very efficient. That is why experts recommend including EPA and DHA sources in your diet as well.

## *Facts Corner*

**DHA** is found in seafood, algae and cold-water fish such as salmon, sardines and albacore tuna.

Note: Fish oil supplements and vegetarian DHA supplements (containing algae) are also available in liquid and capsule form, but always consult your health care provider before taking any supplements. Only use fish oil supplements that have been certified to be free of heavy metal contaminants like mercury.

**EPA** is found in many of the same foods as DHA, which includes cold-water fish such as salmon, and sardines, as well as cod liver, herring, mackerel and halibut.

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Note: Fish oil and vegetarian algae supplements are also good sources of EPA, but always consult your health care provider before taking any supplements. Only use fish oil supplements that have been certified to be free of heavy metal contaminants like mercury.

**Enriched eggs** containing all three types of omega-3 fatty acids are readily available these days. These eggs are enriched by adding flaxseed or algae to the hen's diet so that they produce eggs that are rich in healthy fats. According to the Flax Council, omega-3-enriched eggs provide almost half of the recommended daily level of ALA and one-quarter of the recommended daily level of EPA and DHA – the same amount that can be found in 3 ounces of fish.

### 3. Good and Bad Cholesterol

- Cholesterol is a waxy, fat-like material, which is found in various parts of the body. It has two sources: the liver produces it, and we consume it in animal products, particularly meat and dairy products.
- There are two main types of cholesterol – HDL and LDL.
- **LDL cholesterol** is more likely to clog blood vessels because it carries the cholesterol away from the liver into the bloodstream, where it can stick to the blood vessels.
- **HDL (high-density lipoprotein) cholesterol,** on the other hand, carries the cholesterol back to the liver where it is broken down.
- LDL cholesterol is the bad kind, so call it 'lousy' cholesterol – 'l' for lousy. The HDL is the good cholesterol, so remember it as 'healthy' cholesterol – 'h' for healthy.
- Cholesterol floats around in your blood and can get into the walls of the blood vessels and stay there.
- If you have too much cholesterol in your bloodstream, it can collect in the blood vessel walls, causing these 'pipes' to become narrower. This can clog the blood vessels.
- If the clogging gets worse over many years, it can cause damage to important body parts, like the **heart (heart attack)** and **brain (stroke).**

### 4. Trans Fat

Trans fat is created by a process called **hydrogenation,** which adds hydrogen molecules to highly unsaturated (liquid) oil, such as corn oil,

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vegetable oil, or soybean oil, etc. After hydrogenation, the oil is called 'partially hydrogenated' when listed on the package's ingredients list, and it contains trans fats.

### *Why do Manufacturers Use Hydrogenated Oils?*

Earlier, manufacturers predominately used animal fats such as lard, beef tallow and butter when making baked and fried foods. Later, when scientists discovered that these saturated fats contributed to heart disease and 'bad' (LDL) cholesterol levels, food companies started to look for alternatives to these saturated fats.

Since hydrogenation makes oils more stable and solid at room temperature, this improves the baking characteristics of the liquid oil as well as the taste and texture of the end product. Partially hydrogenated oil provided a good alternative when it came to taste, texture and stability, and manufacturers started the use of these oils instead of animal fats. Years later, scientists discovered that both saturated fats and trans fats increase the risk for heart disease.

### *Which Products Contain Partially Hydrogenated Oil and Trans Fat?*

Food products that contain trans fat are vegetable shortenings, harder stick **margarines, crackers, candies, cookies, snack, baking mixes and icing foods, fried foods, doughnuts, pastries and store-bought baked goods.** You may think that trans fat primarily comes from margarine, but margarine accounts for less than 20% of the trans fat in the average American's diet. Some meats and dairy products naturally contain small amounts of trans fat.

Note: Saturated fats, trans fats and cholesterol tend to raise 'bad' (LDL) cholesterol levels in the blood, which in turn increases the risk for heart disease.

**KIDS AND FAT INTAKE**

Although some people may think it is wise to try to cut fat altogether, it is crucial for fat to stay a part of a child's diet.

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- For young kids, especially, fat and cholesterol play important roles in brain development.
- Ages 2 and under – fat should not be restricted.
- Ages 1 to 3 years – 30 to 35% of calories coming from fat.
- Ages 4 to 18 years – 25 to 35% of calories.

## 5. Food Safety and Standards Authority of India (FSSAI) Campaign

The Food Safety and Standards Authority of India (FSSAI) had launched a new mass media campaign called "Heart Attack Rewind", the 30 second public service announcement (PSA) – the first mass media campaign of its kind, in order to create awareness about trans fats and eliminate them in India by 2023.

### *Trans Fats*

FSSAI launches 'Trans-Fat Free' logo. This marked an important milestone in the movement against trans fats and also provided a momentum to accelerate the 'Eat Right India' movement of FSSAI.

### *Harmful Effects*

- TFAs pose a higher risk of heart disease than saturated fats. While saturated fats raise total cholesterol levels, TFAs not only raise total cholesterol levels but also reduce the good cholesterol (HDL), which helps to protect us against heart disease. Trans-fats consumption increases the risk of developing heart disease and stroke.

**FSSAI LAUNCHES AWARENESS CAMPAIGN ON FOOD SAFETY ASPECTS AT PROMINENT MARKETS OF DELHI**

Food Business Operators sensitized about importance of organic farming, raw material testing, ill effects of pesticide residues, artificial ripening and wax coating using unapproved chemicals

The Food Safety and Standards Authority of India (FSSAI) has launched an extensive awareness and sensitization program targeting major markets in the national capital, in collaboration with the Food Safety Department, Delhi.

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Kicking off from the popular Khan Market and INA Market of Delhi on 8th April, 2024, the awareness campaign will focus particularly on the detection and mitigation of pesticide residue and contaminants in food products. [Press Release, PIB dated 8 April 2024]

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- It is also associated with a higher risk of developing obesity, type 2 diabetes, heart disease, metabolic syndrome, insulin resistance, infertility, certain types of cancers and can also lead to compromised foetal development causing harm to the yet-to-be born baby.

### *Why they are being Increasingly Used?*

- TFA containing oils can be preserved longer; they give the food the desired shape and texture and can easily substitute 'pure ghee'. These are comparatively far lower in cost and thus add to profit/saving.

**INITIATIVE BY KERALA**

Kerala has launched an initiative to enforce dietary guidelines, involving the reduction of trans-fatty acids (TFAs), salt and sugar in commercially available foods in the State. The initiative, with technical support from the World Bank, WHO and the Food Safety and Standards Authority of India (FSSAI), is launched as unhealthy diet is pushing up metabolic syndrome and premature deaths due to non-communicable diseases (NCDs) among Keralites.

## 26.32.9 Gluten Intolerance and Celiac Disease

In some people, mostly in children, gluten, a **type of protein triggers an autoimmune reaction** where the body produces antibodies that attack its own tissues.

A **mixture of gliadin and glutenin** causes inflammation and damage to the lining of the small intestine and results in Celiac disease or CD, which also goes by names like **gluten enteropathy, gluten sensitivity** and **idiopathic spruce**.

Besides **wheat**, the other major cereals containing this protein are **barley, rye** and to a lesser extent, **oats**.

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Doctors have begun to call it an **'impending epidemic'**.

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It is believed that the new strains of wheat that are cultivated nowadays have a highly antigenic gluten, unlike the traditional ancient varieties, which may have caused the rising incidence of the disease.

In milder forms of celiac disease, the **microvilli of the absorbing epithelial cells on the villi of the intestine are destroyed,** thereby decreasing the absorptive surface area as much as twofold. In more severe forms, the villi themselves become blunted or disappear altogether, further reducing its absorptive area.

More common in children, may include **abdominal bloating and pain, chronic diarrhoea, vomiting, constipation, pale foul smelling or fatty stools** and **weight loss.**

Besides, improper absorption of nutrients also causes malnutrition, hampering normal growth and development, leading to other problems like delayed growth, short stature, delayed puberty and enamel defects of the permanent teeth. Irritability is also a common feature in children. On the other hand, digestive symptoms are less prominent in adults who develop unexplained iron deficiency, **anaemia, fatigue, bone-joint pains, arthritis, bone loss, depression, anxiety, seizure and infertility**. Over time, they also get complications like **malnutrition, liver disease** and **cancer of the intestine.** The disease can be confirmed by **duodenal biopsy**.

The only treatment for celiac disease available at present is a strictly gluten-free diet, which requires avoiding wheat, barley, rye, oat and their derivatives, while depending completely on rice or corns as the staple food.

### 26.32.10 Milk Property and Milk Process

The milk goes through different processes, which can be called as **'dairy process'**. It is an industrial process in which milk components are separated and then re-constituted to produce desired milk products. From milking to consumption, milk may undergo a lot of dairy processes depending upon the product desired.

Let us look at some of these processes.

**Cooling:** Microorganism growth is prevented by immediately chilling the milk to 5°C or below. At this temperature, milk will retain its freshness for

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three to four days.

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**Standardisation:** If milk has to be sold as buffalo milk, the minimum fat per cent should be standardised to 6% for most of the states in India. Milk may be standardised to full-cream milk, toned milk, double-toned milk etc.

- **Full-cream** milk should contain minimum 6.0% fat and 9.0% **solids not fat (SNF),** while standardised milk should contain minimum 4.5% fat and 8.5% SNF.
- **Toned milk** and **double-toned milk** should contain minimum 3.0% fat and 8.5% SNF and minimum 1.5% fat and 9.0% SNF, respectively.

**Solids Not Fat (SNF):** A substance in milk other than water and butterfat are termed as solids not fat (SNF). It is a residual component left after the complete evaporation of water from milk.

**Pasteurisation:** Pasteurisation eliminates all pathogens and most of the spoilage type of microorganisms. For this purpose, milk is heated to such a time–temperature combination that ensures destruction of even the most heat-resistant pathogen. Mycobacterium (TB germ) is the most heat-resistant pathogen and it is taken as the index for deciding pasteurisation temperature.

Pasteurisation may be done by two methods: **Low-temperature long time (LTLT)** or **high-temperature short time (HTST)**. Nowadays, the HTST method is mostly used. After pasteurisation, the milk must be immediately cooled down to 5°C or below so that the surviving microorganisms do not multiply further.

**Cream Separation:** In cream separation process, fat is removed from the milk by passing it through a cream separator in which it is rotated at a very high speed (around 8000 revolutions per minute) in thin films. Milk cream or fat, being lighter (specific gravity 0.9), remains in the centre while the skimmed milk (skim milk contains the entire constituent except fat), being heavier (specific gravity 1.036), goes towards the periphery in the cream separator. The separated cream and milk are collected from different outlets provided in the cream separator.

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**Homogenisation:** The fat in milk is present in shape of fat globules that are of different sizes. The size may vary from 0.1 to 22 microns and generally it is from 2 to 8 microns. Homogenisation is the process wherein fat globules are broken down mechanically to a **size of about one micron.** Homogenisation is generally used in reconstitution of milk and fat-rich products like **ice cream, condensed milk, sterilised flavoured milk** etc. Homogenised milk does not stick to the container and gives a viscous appearance and rich mouth taste.

**Sterilisation:** Some bacterial spores may not get destroyed even by boiling of milk. They are destroyed through sterilisation that is the process of heating milk to 120°C for 30 seconds or 150°C for 2–3 seconds so that a shelf-life of 6 months of milk at room temperature is ensured.

## *Adulteration in Milk*

The most common adulteration in milk is water. This is added to increase the volume of milk during loose milk distribution.

The second category of adulterants are those, which are added to mask the adulteration of water in milk. These are added to increase the SNF level. The number of such adulterants is very large. The most common are **starch, matlodextrins, sugar, salt, urea** and a lot more.

In addition, traders may also add other chemicals to milk. When milk is transported to farther places for distribution, the bacteria in milk deteriorate it producing lactic acid. This lactic acid destabilises the proteins in milk and when such milk is heated, it gets coagulated. To prevent such curdling, traders add alkaline neutralisers like **caustic soda** and **sodium bicarbonate** so that the developed acidity is neutralised and milk does not curdle on heating. Another class of additives to milk may be preservatives. These are substances that have bacteriostatic or bactericidal effects and prevent the growth of microorganisms. This enhances the shelf-life of milk. But addition of these preservatives is illegal and such substances are harmful for human health. The most common preservatives are hydrogen peroxide and formalin.

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## 26.32.11 Alternative to Sugars

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Alternative sugars are those, which provide **sweet taste but very low or minimum calories**.

These artificial sweeteners are sometimes called as **non-nutritive sweeteners** because they do not provide energy.

There are different types of alternative sweeteners:

(A) **Artificial sweeteners:** These are synthetic sweeteners, like **Aspartame, Saccharin, Neo-tame, Alitmate, Acesulfame-k, Sucralose, Tagatose, Alitmate and cyclamate.**

(B) **Sugar replacers**

(C) **Herbal alternative: Stevia** (also popular with the name of '**Sugar Tulsi**')

### (A) Some important artificial sweeteners

#### *Aspartame*

- Aspartame contains two amino acids – **Aspartic acid** and **Phenyl alanine** with **methyl group**. It is 200 times sweeter than normal sugar.
- A person who is suffering with Phenylketonurics (PKU) should avoid aspartame because it contains aspartame. It is also dangerous for children. The accumulation of phenylalanine and its by-product is toxic for the nervous system and causes irreversible brain damage.
- During metabolism of Aspartame in human body, both amino acids are absorbed in human body as a source of energy but methyl group get metabolised and become methanol, formaldehyde and other toxic substances.
- Some researchers said that consumption of Aspartame also causes various other problems, like Fibromyalgia syndrome, multiple sclerosis symptoms, dizziness, headaches, menstrual problems etc.

#### *Saccharin*

- Saccharin, as an artificial sweetener, was banned earlier but now it got licence for human consumption by FDA (Food and Drug Administration), the US. Researchers said that saccharin with smoke is responsible for bladder cancer.

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- It's made in a laboratory by oxidising the chemicals **o-toluene sulphonamide** or **phthalic anhydride**. It looks like white, crystalline powder.
- It is 200 to 700 times sweeter than table sugar (sucrose) and it does not contain any calories.

Note:

1. Sucralose, Acesulfame-K and Saccharin are not metabolised in the body but Aspartame digest as a protein in human body (4 kcalories per gram), but its energy contribution is very less.
2. The Food Safety and Standards Authority of India (FSSAI) has approved six artificial sweeteners, namely **saccharin sodium, aspartame, acesulfame potassium, sucralose, neotame** and **isomaltulose** for use in food.

## (B) Sugar Replacers

- Sugar replacers also provide calories, but fewer than their carbohydrate cousins, sugars. Sugar replacers yield some amount of energy, so they are also called as **nutritive sweeteners**.
- Sugar alcohols can be used as sugar replacers; they occur naturally in fruits and vegetables. They absorb very slowly in human body than conventional sugar.
- Sugar alcohol also has side effects like gas, abdominal discomfort and diarrhoea.
- There are various examples of sugar alcohol, those we are using in many food items like chewing-gum (Orbit), jams, jellies, cookies etc.: Erythritol, Isomalt, Lactitol, Maltitol, Mannitol, Sorbitol, Xylitol.

## (C) Herbal Alternative

- **Stevia** is a **natural sweetener** and **sugar substitute** derived from the leaves of the plant species Stevia rebaudiana, native to **Paraguay** and **Brazil.**
- The active compounds are **steviol glycosides**, which have about 50 to 300 times the sweetness of sugar. Stevia is heat-stable, pH-stable and not fermentable. The human body does not metabolise the glycosides in stevia, so it contains zero calories as a non-nutritive sweetener.
- Stevia is also popular with many names like **Meethi Tulsi** or **Meethi Patti** in Hindi and **Madhu Patra** in Sanskrit.

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Some studies observed that Stevia had an 'inhibitory effect on bacterial communication' those contribute to an imbalance of good and bad bacteria in the gut, which can result in gastrointestinal symptoms like diarrhoea, gas and nausea.

Due to more duration of intake of Stevia trigger certain proteins those are responsible for inflammation in both the liver and kidneys, which over time could damage the organs.

Note: It has mixed opinion by researchers. In the United States, people are using it as a dietary supplement but not used as a food additive.

## 26.32.12 Anti-ageing

Anti-ageing is defined as to slowing, preventing or reversing the ageing process in human body. **Ayerst Laboratories** (a pharmaceutical company in Montreal) isolated a compound from a rare strain of bacteria Streptomyces hygroscopicus as anti-ageing compound. There are many reasons identified behind anti-ageing like Rapamycin drug, Calorie restriction, Resveratrol (red wine compound) etc.

### (A) Rapamycin

- In 2009, a large National Institute of Health-funded study established that rapamycin and its derivative helped mice live longer. Male mice on rapamycin lived 9% longer. The female's lifespan was extended by 14%.
- Rapamycin acts on what is called the **mechanistic Target of Rapamycin (mTOR)** complex, a set of genes that acts as the cell growth regulating kinase. When mTOR is blocked, it can push cells into a life-extending survival mode.
- Giving such a drug to older patients, whose immune systems are often already diminished, would make them vulnerable to life-threating infections, defeating the purpose.

### (B) Calorie Restriction

- The most well-known strategy to extend an animal's lifespan is **calorie restriction (CR)** or **dietary restriction (DR)**. Discovered about 70

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years ago, CR involves reducing the calorie intake of an animal by 30–40% compared to its normal calorie intake.

- It is thought to lower metabolic rate and cause the body to generate fewer damaging free radicals.
- CR delays or prevents age-related diseases and extends lifespan in species ranging from yeast to primates. Evidence that CR retards ageing and extends median and a maximal lifespan was first presented in the 1930s.
- But radical dieting for calorie restriction is not a reasonable idea to increase lifespan among humans, for two reasons; firstly, because CR has shown to affect fertility of the organism in many cases; secondly, because the person would most likely die of starvation, if not of ageing.

### (C) Resveratrol (Red Wine – Extending Life)

- **Resveratrol** is a phytoalexin polyphenol that is found naturally in the grape skins and is present at low concentrations in red wine, has shown diverse health benefits ranging from reducing the risk for heart disease and increasing human lifespan.

#### THE LONG-LIVING JEWS

There is a historically isolated and culturally insular population from Eastern and Central Europe, known as the **Ashkenazi Jews**. It has been observed that these Jews unusually tend to live longer and suffer from less cardiovascular problems compared to other populations. The secret of this interesting group of people could help us pinpoint the longevity genes responsible for ageing.

- Researchers at Yeshiva University's Albert Einstein School of Medicine examined 477 Ashkenazi Jews, looking for factors that may have contributed to their longevity. They observed that the Ashkenazi Jews who have crossed the age of 95, are more likely to possess a genetic variant that down-regulates insulin- like growth factor 1 (IGF-1). Studies suggest that such mutations could be responsible for delaying old age.
- Disruption of IGF-1 has caused an increase in the lifespan of mice by 30–40%.
- Some scientists are of the opinion that Sir2 which regulates the low metabolic state of calorie restriction may be part of the IGF-1 pathway.
- The complex interlinked dynamics of calorie restriction, Sirtuins, IGF1 and many other 'longevity genes' presents itself like a puzzle before scientists, that needs to be solved if any serious progress can be made in fighting ageing.

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- **Resveratrol-Sir 2:** Resveratrol acts as an activator of **Silent information regulator 2 (Sir2)** proteins, or **sirtuins**. When an extra copy of the SIR 2 gene was added to the yeast **S. ceravisiae**, the formation of extra chromosomal rDNA circles was repressed which led to a 30% extension in the cell's lifespan. Mammals possess seven types of sirtuins, SIR T1 to 7.

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## 26.32.13 Fortified Rice

Fortification is **the practice of deliberately increasing the content of an essential micronutrient,** i.e., vitamins and minerals (including trace elements) in a food, in order to improve the nutritional quality of the food supply and provide a public health benefit with minimal risk to health.

### *Rice Fortification*

- Rice is the world's most important staple food. An estimated 2 billion people eat rice every day, forming the mainstay of diets across large part of Asia and Africa.
- Regular milled rice is low in micronutrients and serves primarily as a source of carbohydrate only. The fortification of rice is a major opportunity to improve nutrition.
- Fortified rice contains vitamin A, vitamin B1, vitamin B12, folic acid, iron and zinc.

### *Food Fortification in India*

- **Food Safety and Standards Authority of India (FSSAI)** has formulated a comprehensive regulation on fortification of foods namely **'Food Safety and Standards (Fortification of Foods) Regulations, 2016'.** These regulations set the standards for food fortification and encourage the production, manufacture, distribution, sale and consumption of fortified foods.
- The regulations also provide for specific role of FSSAI in promotion for food fortification and to make fortification mandatory. This sets the premise for the national summit on fortification of food.

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### *Benefits of Fortification*

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- If consumed on a regular and frequent basis, fortified foods will maintain body stores of nutrients more efficiently and more effectively than will intermittent supplements.
- Fortified foods are also better at lowering the risk of the multiple deficiencies that can result from seasonal deficits in the food supply or a poor quality diet.
- Fortification can be an excellent way of increasing the content of vitamins in breast milk and thus reducing the need for supplementation in postpartum women and infants.
- Fortification of widely distributed and widely consumed foods has the potential to improve the nutritional status of a large proportion of the population, both poor and wealthy.
- Fortification is often more cost-effective than other strategies, especially if the technology already exists and if an appropriate food distribution system is in place.

To tackle the menace of malnutrition, NITI Aayog seeks creation of roadmap by Department of Food and Public Distribution for taking **the Rice Fortification Pilot Scheme** Pan India.

### 26.32.14 Triclosan

Triclosan is a common ingredient found in many **grooming products, including facial washes, toothpaste and deodorants.** However, it is mainly found in **antibacterial hand washes, soaps, gels and household cleaners.**

Triclosan is an ingredient added to many consumer products intended **to reduce or prevent bacterial contamination.**

Some short-term animal studies have shown that exposure to high doses of triclosan is associated with a decrease in the levels of some thyroid hormones. Other studies have raised the possibility that exposure to triclosan contributes to **making bacteria resistant to antibiotics**. One is a study investigating the potential of developing **skin cancer after a long-term exposure** to triclosan in animals.

In 1997, FDA reviewed extensive effectiveness data on triclosan in Colgate toothpaste. The evidence showed that triclosan in that product was effective in preventing gingivitis.

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### *Regulation on Triclosan under FDA (Food and Drug Administration)*

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1. Triclosan is **an ingredient added to many consumer products intended to reduce or prevent bacterial contamination.**
2. **The antibacterial compound triclosan has been linked to numerous human health problems**. Exposures come mainly by absorption through the skin or through the lining of the mouth. These exposures have resulted in contact dermatitis, or skin irritation, and an increase in allergic reactions, especially in children.
3. Triclosan **functions as an antimicrobial agent by impairing the production of bacterial lipids.** More specifically, it blocks the active site of a bacterial enzyme known as enoyl-acyl carrier protein reductase. Since humans lack this enzyme, triclosan has been generally accepted as harmless to human health.
4. **Effects of triclosan:** The use of triclosan is linked to the following effects on human health: **Abnormal endocrine system/thyroid hormone signalling. Weakening of immune system.** Children exposed to antibacterial products at an early age have an increased chance of developing allergies, asthma and eczema.
5. **Colgate remove triclosan:** Researchers examined the antibacterial compound triclosan, which federal authorities banned in over-the-counter products like antibacterial soaps and gels in 2016 after **manufacturers were unable to prove it was safe for long-term use or more effective at preventing infection spread than soap and water.**
6. **Is triclosan a carcinogen?** Triclosan, found in antibacterial soap and other products, causes cancer in mice. Long-term exposure to triclosan – an antibacterial ingredient found in some soap, toothpaste and other common household products – **may lead to liver cancer in mice**, according to a new study.

Note: India lacks any such regulation so far on the use of Triclosan-based products.

## 26.32.15 Polyethylene Terephthalate (PET)

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Polyethylene terephthalate is the chemical name of **polyester**. It is a clear, strong and lightweight plastic which is widely used for **packaging of foods and beverages, usually convenience-sized soft drinks, juices and water.**

**They are often used in durable-press blends with other fibres such as rayon, wool, cotton and reinforcing the inherent properties.**

PET plastic is the **recyclable plastic** and **BPA free**. But still PET is not safe and creates various health risks such as stunted growth, reproduction issues, low energy levels, body balance issues and inability to process stress. **Maharashtra government has announced the ban on sale of alcohol in PET bottles because alcoholic beverages have carbonated material.**

PET an inert material is **widely accepted by health authorities also as a safe plastic**, which is one of the reasons it is so commonly used in food and beverage packageing. Similar to glass **it is hygienic and generally resistant to attack by bacteria and other microorganisms.**

PET made material is 60–70% degraded within 6 months but not completely biodegradable as a **non-biodegradable polymer.**

**PET is known to leach antimony trioxide and phthalates. Both of these are dangerous to health.** While antimony may contribute to cancer development, skin problems, menstrual and pregnancy issues, phthalates are endocrine disruptors. So the use of PET has mixed declarations by the corporate sector and scientists.

## 26.32.16 Rice Bran Oil

**Department of Food and Public Distribution** launched National Agricultural Cooperative Marketing Federation of India Ltd. **(NAFED)** Fortified Rice Bran Oil. NAFED comes under the Ministry of Agriculture and deals with marketing cooperatives for agricultural produce in India.

It is popularly known as **heart oil** and also health food in western world. Rice bran oil is derived from rice. Rice has several layers; the outermost layer is called bran. Bran is a brown coloured, hard and oily layer between the rice kernel and the outer husk. The rice bran oil comes from the bran.

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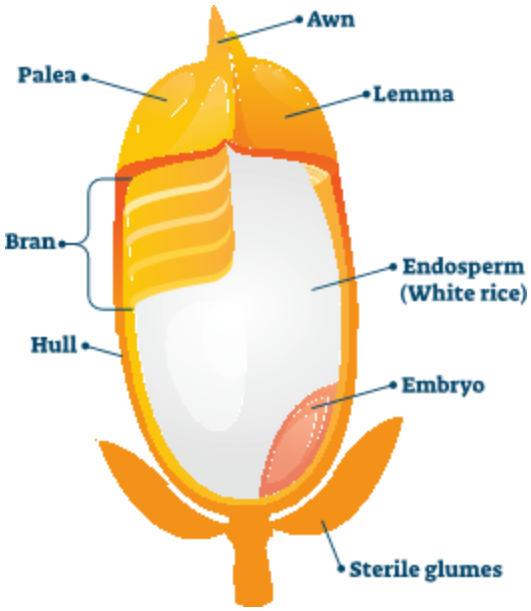


Figure 26.6 Composition of Rice Grain

Rice bran oil is not sticky, thus edible and suitable for cooking. Rice bran oil is a natural source of a powerful antioxidant, **Oryzanol** (studies claim that Oryzanol assists in **reducing cardiovascular diseases**). Rice bran oil is nutritionally rich and has mild flavour.

It has a high-smoking point of 232°C (450°F) which makes it useful for high heat cooking like stir frying, deep frying etc. This means the oil retains its nutrition even at high temperatures. Rice bran oil can also be used for nourishing, cleansing and brightening skin.

**COMPOSITION OF RICE BRAN OIL**

One table spoon of rice bran oil contains

- Calories: 120
- Protein: 0 grams
- Fat: 14 grams
- Carbohydrates: 0 grams
- Fibre: 0 grams
- Sugar: 0 grams

Rice bran oil is a good source of

- Vitamin E

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- Vitamin K

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## Health Benefits of Rice Bran Oil

- Helps improve cholesterol, blood sugar levels and blood pressure.
- Helps maintain a healthy heart: **The striking balance of monounsaturated and polyunsaturated fats in rice bran oil makes it healthy for the heart.**
- Helps reduce post-menopause symptoms.
- Enhances vitamin E levels: **It also acts as an immunity booster and reduces the risk of cancer due to the high amount of vitamin E it contains.**
- Excellent supplement to regulate thyroid functioning.
- Helps decrease calcium absorption in kidneys and prevents stone formation.
- This oil is recommended by The American Heart Association and the World Health Organization (WHO) as one of the best substitutes for other edible oils.

## NAFED Fortified Rice Bran Oil

- Rice bran oil from NAFED will be fortified and it will contain additional nutrients and vitamins and will be marketed by NAFED.
- According to the FSSAI, fortified oil can help a person fulfil 25–30% of the recommended dietary intake for vitamins A and D.
- NAFED fortified rice bran oil will be available at all NAFED stores and also on various online platforms.

## Significance

- This initiative by NAFED will significantly reduce the country's consumption dependence on imported edible oils in future.
- It will provide easy access to NAFED branded high quality rice bran oil, which will also give a boost to the indigenous oil manufacturing industries.
- It will further give an impetus to the Prime Minister's Aatmanirbhar Bharat initiative.

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### *Side-Effects of Rice Bran Oil*

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- Increasing the amount of bran in the diet can cause unpredictable bowel movements, intestinal gas and stomach discomfort.
- People having gastrointestinal conditions or digestive tract problem such as intestinal ulcers, adhesions, conditions that cause narrowing or blockage of the digestive tract, slow digestion, or other stomach or intestinal disorders are not recommended to use rice bran oil.
- The fibre in rice bran can block your digestive tract causing indigestion.

## 26.32.17 Glycemic Index

- **Definition:** GI is a measure of how quickly a carbohydrate-containing food raises blood sugar levels after it is consumed.
- The **index ranks** the carbohydrate-rich foods on a scale of **0 to 100 based on their ability to raise blood sugar levels as compared to pure glucose** (which has a GI of 100).

- **Proposed by:** Prof. David Jenkins of the University of Toronto in 1981.
- **Factors that Determine GI**: The **GI of a food is determined by various –**

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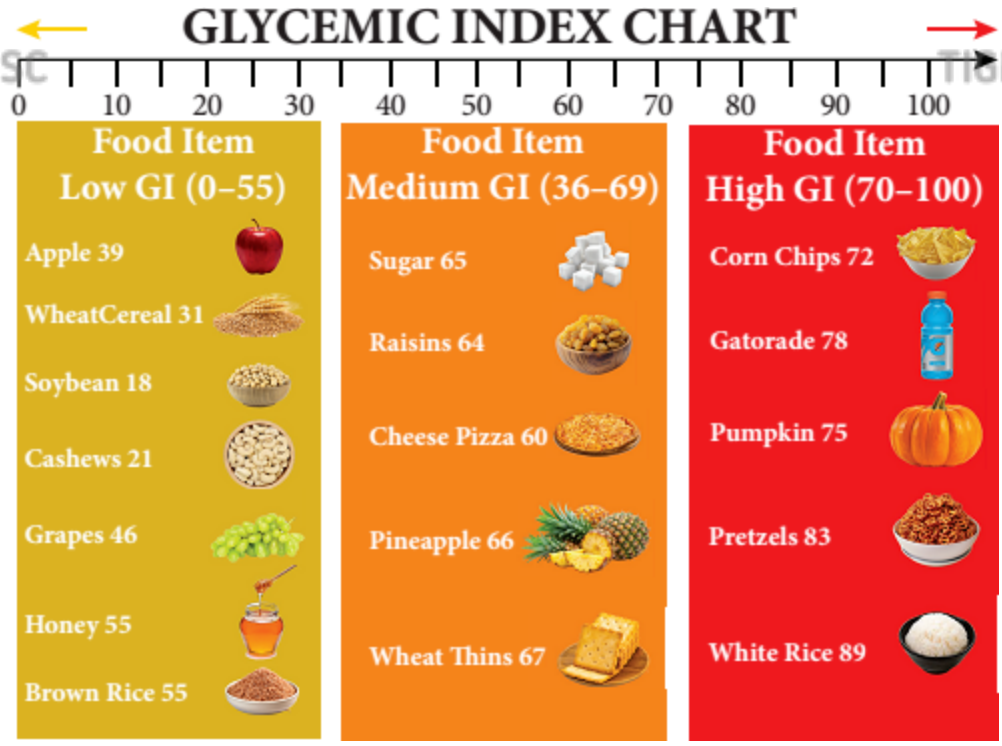


- ❑ **Internal factors** like amylose, lipids, protein, phytic acid, dietary fibre, resistant Starch etc. and
- ❑ **External factors** like cooking, processing, retro-gradation, soaking and germination.

## *About Glycemic Load (GL)*

- **Glycemic Load (GL):** It uses GI and the amount of total Carbohydrates in a serving of a specific food to estimate how quickly and how much blood sugar will rise after its consumption.
- **Calculation:** The GL of a food is obtained by multiplying the quality of carbohydrate in a given food (GI) by the amount of carbohydrate in a serving.

## *Link between GI and Human Health*

- ***Complications for People with Diabetes:*** High GI foods cause rapid fluctuations in blood sugar levels, which may strain the body's ability to produce insulin or utilise it effectively.
- ***Issues to Cardio Vascular Health:*** Diets rich in GI lead to weight gain, higher triglyceride levels and blood pressure in the body thereby

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making the individual susceptible to cardiovascular complications in the long run.

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## 26.32.18 A1 & A2 Milk

### *Basis of the Classification*

- A1 and A2 are genetic variants of Beta (β)-casein protein. Casein (make 80 % of milk protein) is one of the two types of protein found in milk. The other one is Whey.
  - ❑ Differentiation in both is linked to difference in structure of amino acid sequence.

  - ❑ Also, **A1 evolved from A2 through natural mutation.**
- Regular milk contains both A1 and A2 beta-casein, while A2 milk is unique in that it contains only the A2 variant.
  - ❑ Studies of the National Bureau of Animal Genetic Resources (NBAGR) have confirmed that indigenous cows and buffaloes produce A2 milk.

### *About Proteins*

- These are large molecules made of amino acids. It is of two types:
  - ❑ **Essential Amino Acids:** Cannot be made by the body. As a result, they must come from food.
  - ❑ **Nonessential Amino Acids:** Can be synthesised in the body
- Proteins are the main structural constituents of the cells and tissues of the body. Muscles and organs are largely made of proteins

**Food Safety and Standards Authority of India (FSSAI)** withdraw its direction to **Food Business Operators (FBOs) to not market their milk**

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**and milk products in the name of A1 and A2.**

- Earlier, FSSAI stated that standards of milk as specified in Food Safety and Standards (Food Product Standards and Food Additives) Regulations, 2011, do not mention/ recognise any differentiation of milk on the basis of A1 and A2 types.
- Food Safety and Standards (Food Product Standards and Food Additives) Regulations, 2011, has been enacted under the Food Safety and Standards Act 2006.

**BCM-7, or beta-casomorphin-7, is an opioid peptide released during the digestion of A1 beta-casein in bovine milk and have several health concerns.**

| A1 Milk | A2 Milk |
|---|---|
| Produced by A1 cow | Produced by A2 cow |
| People with milk intolerance can't digest it (Al milk is not easy to digest) | It is easy and light to digest |
| Entails A1 beta-casein protein | Entails A2 beta-casein protein |
| Obtained from the international cow breeds found in countries like Northern Europe, Australia, and the United States | Obtained from the desi Indian cow breeds like Gir, Sahiwal, and Tharparkar |
| Decreases the immunity of the body | Boosts the immunity power |
| Yields BCM-7 | Does not produce BCM-7 |
| Causes gas, bloating, stomach upsets, and ulcers | Strengthens the digestive system and helpful in treating acidity |
| As it's not immunity-supportive, it's not fit for Covid patients | It's a healthy and immunity-friendly treat for Covid patients |

## 26.32.19 Weight Loss Drugs: Tirzepatide (Mounjaro) and Semaglutide (Wegovy)

An **expert committee of India's drug regulator** gave the **green light to the drug tirzepatide** and India could soon allow 'game-changing' weight-loss drug Tirzepatide.

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## About Tirzepatide and Semaglutide

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- These two weight loss drugs have been **approved by the FDA** for chronic weight management in adults.
- **Semaglutide and tirzepatide are polypeptides,** small proteins that boost the levels of naturally occurring hormones in the body, including that of **glucagon-like-peptide 1 (GLP-1),** which control weight through the brain and digestive tract.

## Functioning

- **Higher GLP-1 levels, released in the gut**, spark a reaction by stimulating neurons that alter gut function, **leading to a sense of fullness.**
- This process also taps into a brain mechanism that lights up neural pathways, triggering the sensation of satiety—the feeling of being satisfied and having had enough to eat.
- They **also help manage glucose levels**, making them an effective treatment for diabetes.
- **Semaglutide only targets GLP-1 receptors.**
- On the other hand, **tirzepatide also boosts a second hormone: glucose-dependent insulinotropic polypeptide (GIP).**
- The GIP also regulates weight through receptors in brain and fat cells.
- These drugs can be **prescribed to those who are obese (with a body mass index of over 30), or overweight (with a BMI between 27 and 30),** and **have at least one other health condition related to their weight (such as high blood pressure, high cholesterol or type 2 diabetes).**

## Side Effects

- Nausea
- Diarrhoea
- Vomiting
- Constipation
- Abdominal Pain,
- Indigestion
- Injection-Site Reactions

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- Fatigue
- Allergic Reactions
- Belching
- Hair loss
- Heartburn

## QUESTIONS AND ANSWERS

### Objective Questions

**1. Consider the following statements:** **[UPSC (Pre) – 2025]**

i. It is expected that Majorana 1 chip will enable quantum computing.

ii. Majorana 1 chip has been introduced by Amazon Web Services (AWS).

iii. Deep learning is a subset of machine leaning.

Which of the statements given above are correct?

(a) i and ii only
(b) ii and iii only
(c) I and iii only
(d) i, ii and iii

**2. Consider the following statements:** **[UPSC (Pre) – 2025]**

i. Indian Railways have prepared a National Rail Plan (NRP) to create a 'future ready' railway system by 2028.

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ii. 'Kavach' is an Automatic Train Protection system developed in collaboration with Germany.
iii. The 'Kavach' system consists of RFID tags fitted on the track in station sections.

Which of the above statements given above are not correct?
(a) i and ii only
(b) ii and iii only
(c) i and iii only
(d) i, ii and iii

**3. Consider the following statements:** **[UPSC (Pre) – 2025]**

**Statement I:** Some rare earth elements are used in the manufacturing of flat television screen and computer monitors.

**Statement II:** Some rare earth elements have phosphorescence properties.

Which one the following is correct in respect of the above statements?
(a) Both Statement I and Statement II are correct and Statement II explains Statement I.
(b) Both Statement I and Statement II are correct and Statement II does not explains Statement I.
(c) Statement I is correct but Statement II is not correct.
(d) Statement I is not correct but Statement II is correct.

**4. In which of the following are hydrogels used?** **[UPSC (Pre) – 2024]**
1. Controlled drug delivery in patients
2. Mobile air-conditioning systems
3. Preparation of industrial lubricants

Select the correct answer using the code given below:
(a) 1 only
(b) 1 and 2 only

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(c) 2 and 3 only
(d) 1, 2 and 3

5. **"Membrane Bioreactors" are often discussed in the context of: [UPSC (Pre) – 2024]**
   (a) Assisted reproductive technologies
   (b) Drug delivery nanotechnologies
   (c) Vaccine production technologies
   (d) Waste water treatment technologies

6. **With reference to perfluoroalkyl and polyfluoroalkyl substances (PFAS) that are used in making many consumer products, consider the following statements:[UPSC (Pre) – 2024]**
   1. PFAS are found to be widespread in drinking water, food and food packaging materials.
   2. PFAS are not easily degraded in the environment.
   3. Persistent exposure to PFAS can lead to bioaccumulation in animal bodies.

   Which of the statements given above are correct ?
   (a) 1 and 2 only
   (b) 2 and 3 only
   (c) 1 and 3 only
   (d) 1, 2 and 3

7. **Consider the following action:[UPSC (Pre) – 2023]**
   1. Detection of car crash/collision which results in the deployment of airbags almost instantaneously.
   2. Detection of accidental free fall of a laptop towards the ground which results in the immediate turning off of the hard drive.
   3. Detection of the tilt of the smart phone which results in the rotation of display between portrait and landscape mode.

   In how many of the above action is the function of accelerometer required?

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(a) Only 1
(b) Only 2
(c) Only 3
(d) None of these

**8. Consider the following statements:** **[UPSC (Pre) – 2023]**

1. Carbon fibres are used in the Manufacture of components used In automobiles and aircrafts.
2. Carbon fibres once used cannot be recycled.

Which of the statements given above is/are correct?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**9. With reference to polyethylene terephthalate, the use of which is so widespread in our daily lives, consider the following statements:** **[UPSC (Pre) – 2022]**

1. Its fibres can be blended with wool and cotton fibres to reinforce their properties.
2. Containers made of it can be used to store any alcoholic beverage.
3. Bottles made of it can be recycled into other products.
4. Articles made of it can be easily disposed of by incineration without causing greenhouse gas emissions.

Which of the statements given above are correct?
(a) 1 and 3
(b) 2 and 4
(c) 1 and 4
(d) 2 and 3

**10. Bisphenol A (BPA), a cause of concern, is a structural/key component in the manufacture of which of the following kinds of**

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**plastics?**

(a) Low density polyethylene **[UPSC (Pre) – 2021]**
(b) Polycarbonate
(c) Polyethylene terephthalate
(d) Polyvinyl chloride

**11. 'Triclosan', considered harmful when exposed to high levels for a long time, is most likely present in which of the following? [UPSC (Pre) – 2021]**

(a) Food preservatives
(b) Fruit-ripening substances
(c) Reused plastic containers
(d) Toileteries

**12. With reference to street-lighting, how do sodium lamps differ from LED lamps? [UPSC (Pre) – 2021]**

1. Sodium lamps produce light in 360 degrees but it is not so in the case of LED lamps.
2. As street-lights, sodium lamps have longer life span than LED lamps.
3. The spectrum of visible light from sodium lamps is almost monochromatic while LED lamps offer significant colour advantages in street-lighting.

Select the correct answer using the code given below.

(a) 3 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

**13. In India, the use of carbofuran, methyl parathion, phorate and triazophos is viewed with apprehension. These chemicals are used as**

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(a) Pesticides in agriculture **[UPSC (Pre) – 2019]**
(b) Preservatives in processed foods
(c) Fruit-ripening agents
(d) Moisturising agents in cosmetics

**14. ‘3D Printing’ has applications in which of the following?[UPSC (Pre) – 2018]**

1. Preparing of confectionery items
2. Manufacture of bionic ears
3. Automotive industry
4. Reconstructive surgeries
5. Data processing technologies

Select the correct answer using the code given below:
(a) 1, 3 and 4
(b) 2, 3 and 5
(c) 1 and 4
(d) All of these

**15. Consider the following pairs:**
**[UPSC (Pre) – 2018] – (Inter-disciplinary)**

| | *Terms sometimes seen in news* | *Context/Topic* |
|---|---|---|
| 1. | Belle II experiment | Artificial intelligence |
| 2. | Blockchain Cryptocurrency | Digital/technology |
| 3. | CRISPR Cas 9 | Particle Physics |

Which of the pairs given above is/are correctly matched?

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(a) 1 and 3
(b) Only 2
(c) 2 and 3
(d) 1, 2 and 3

**16. Consider the following pairs:[UPSC (Pre) – 2017]**

| | *Commonly used/ consumed matter* | *Unwanted or controversial chemicals* |
|---|---|---|
| (i) | Lipstick | Lead |
| (ii) | Soft drinks | Brominated vegetable oil |
| (iii) | Chinese fast food | Monosodium Glutamate |

Which of the pairs given above is/are correctly matched:
(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii) and (iii)

**17. Organic Light-Emitting Diodes (OLEDs) are used to create digital display in many devices. What are the advantages of OLED display over liquid crystal display?[UPSC (Pre) – 2017]**

(i) OLED displays can be fabricated on flexible plastic substrates.
(ii) Roll-up-displays embedded in clothing can be made using OLEDs.
(iii) Transparent displays are possible using OLEDs.

Select the correct answer given below:

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(a) (i) and (iii)
(b) Only (ii)
(c) (i), (ii) and (iii)
(d) None of these

**18.** **Consider the following statements:** **[UPSC (Pre) – 2017]**

(i) The standard mark of Bureau of Indian Standards (BIS) is mandatory for automotive tyres and tubes.
(ii) AGMARK is a quality certification mark issued by the Food and Agriculture Organisation (FAO).

Which of the statements given above is/are correct?

(a) Only (i)
(b) Only (ii)
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

**19.** **Why does the Government of India promote the use of 'Neem-coated Urea' in agriculture?** **[UPSC (Pre) – 2016]**

(a) Release of Neem oil in the soil increases nitrogen fixation by the soil micro-organisms.
(b) Neem coating slows down the rate of dissolution of urea in the soil.
(c) Nitrous oxide, which is a greenhouse gas, is not at all released into atmosphere by crop fields.
(d) It is a combination of a weedicide and a fertiliser for particular crops.

**20.** **The term 'Ind ARC' sometimes seen in the news, is the name of** **[UPSC (Pre) – 2015]**

(a) An indigenously developed radar system inducted into Indian Defence

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(b) India's satellite to provide services to the countries of Indian Ocean Rim
(c) A scientific establishment set up by India in Antarctic region
(d) India's underwater observatory to scientifically study the Arctic region

**21. With reference to bio-toilets used by the Indian Railways, consider the following statements: [UPSC (Pre) – 2015]**

(i) The decomposition of human waste in the bio-toilets is initiated by fungal inoculums.
(ii) Ammonia and water vapour are the only end products in this decomposition, which are released into the atmosphere.

Which of the statements given above is/are correct?
(a) Only (i)
(b) Only (ii)
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

**22. What is the role of ultraviolet (UV) radiation in the water purification systems? [UPSC (Pre) – 2012]**

(i) It inactivates/kills harmful microorganisms in water.
(ii) It removes all the undesirable odours from the water.
(iii) It quickens the sedimentation of solid particles, removes turbidity and improves the clarity of water.

Which of the statements given above is/are correct?
(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii) and (iii)

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**23. A company marketing food products advertises that its items do not contain trans fats. What does this campaign signify to the customers? [UPSC (Pre) – 2011]**

(i) The food products are not made out of hydrogenated oils.
(ii) The food products are not made out of animal fats/oils.
(iii) The oils used are not likely to damage the cardiovascular health of the consumers.

Which of the statement(s) given above is/are correct?

(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii) and (iii)

**24. Aspartame is an artificial sweetener sold in the market. It consists of amino acids and provides calories like other amino acids. Yet, it is used as a calorie sweetening agent in food items. What is basis of this use? [UPSC (Pre) – 2011]**

(a) Aspartame is as sweet as table sugar, but unlike table sugar, it is not readily oxidised in human body due to lack of requisite enzymes.
(b) When aspartame is used in food processing, the taste remains, but it becomes resistant to oxidation.
(c) Aspartame is as sweet as sugar, but after ingestion into the body, it is converted into metabolites that yield no calories.
(d) Aspartame is several times sweeter than table sugar, hence food items made with small quantities of aspartame yield fewer calories on oxidation.

**25. What is the difference between a CFL and an LED lamp? [UPSC (Pre) – 2011]**

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(i) To produce light, a CFL uses mercury vapour and phosphor while an LED lamp uses semiconductor material.
(ii) The average life span of a CFL is much longer than that of an LED lamp.
(iii) A CFL is less energy-efficient as compared to an LED lamp.

Which of the statement(s) given above is/are correct?
(a) Only (i)
(b) (ii) and (iii)
(c) (i) and (iii)
(d) (i), (ii) and (iii)

**26. A new optical disc format known as the Blu-ray Disc (BD) is becoming popular. In what way is it different from the traditional DVD?** **[UPSC (Pre) – 2011]**

(i) DVD supports Standard Definition video while BD supports High Definition video.
(ii) Compared to a DVD, the BD format has several times more storage capacity.
(iii) Thickness of BD is 2.4 mm while that of DVD is 1.2 mm.

Which of the statement(s) given above is/are correct?
(a) Only (i)
(b) (i) and (ii)
(c) (ii) and (iii)
(d) (i), (ii) and (iii)

**27. Regular intake of fresh fruits and vegetables is recommended in the diet since they are a good source of antioxidants. How do antioxidants help a person maintain health and promote longevity?** **[UPSC (Pre) – 2011]**

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(a) They activate the enzymes necessary for vitamin synthesis in the body and help prevent vitamin deficiency.

(b) They prevent excessive oxidation of carbohydrates, fats and proteins in the body and help avoid unnecessary wastage of energy.

(c) They neutralise the free radicals produced in the body during metabolism.

(d) They activate certain genes in the cells of the body and help delay the ageing process.

**28. With reference to the treatment of cancerous tumours, a tool called cyber-knife has been making the news. In this context, which one of the following statements is correct?[UPSC (Pre) – 2011]**

(a) It is a robotic image-guided system

(b) It delivers an extremely precise dose of radiation

(c) It has the capability of achieving sub-millimetre accuracy

(d) It can map the spread of tumour in the body

## Answer Key

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. (a) | 2. (d) | 3. (a) | 4. (d) | 5. (d) | 6. (d) | 7. (c) | 8. (b) |
| 9. (a) | 10. (b) | 11. (d) | 12. (c) | 13. (a) | 14. (d) | 15. (b) | 16. (d) |
| 17. (c) | 18. (a) | 19. (b) | 20. (d) | 21. (d) | 22. (a) | 23. (c) | 24. (d) |
| 25. (c) | 26. (b) | 27. (c) | 28. (d) | | | | |

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# PART - B UPSC Mains Answers — New Pattern — Science and Technology (2022–2024)

PART B

2024 MAINS ANSWERS

2023 MAINS ANSWERS

2022 MAINS ANSWERS

**2021–2013 Main Answer is available on the above QR code**

## 2024 Mains Answers

1. Explain the role of millets for ensuring health & nutritional security in India.(10 Marks/150 words)

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**Ans:** The UN and FAO declared 2023 as the **International Year of Millets** following India's proposal. This step underscores the need to understand & adopt higher nutritional value & climate resilience of millets compared to the more widely consumed cereals.

- Millets are coarse cereals that provide **protein**, **dietary fibre**, **iron** and **calcium** along with carbohydrates, making them a much **nutrient-dense** option for the malnourished population of India.
- As per research, millets are also known to be a good defence against diabetes by **lowering blood sugar spikes** after consumption. In a country like India which is called the '**diabetic capital'** of the world, millets could be helpful in managing the health of a huge number of affected patients as an **alternative** to rice or wheat.
- Millets use **70% less water** than rice, grow in half the time of wheat and need **40% less energy** in processing, making them a sustainable crop as compared to the water-intensive wheat and rice.
- Millets can also **withstand extreme heat & drought** conditions, and hence are labelled as hardy crops that are **adaptable** to climate change and global warming.
- Millets, once cultivated widely, lost significance due to the Green Revolution's focus on rice & wheat reducing millets as a fodder crop.
- However, with **government support**, millets are making a comeback. Their hardy nature empowers small farmers to adapt with changing climatic conditions, promote biodiversity & even achieve sustainable development.
- Even after **75 years of independence**, malnutrition continues to plague India. In this regard, the beneficial properties of millets can help in achieving **nutritional security** among the affected population.
- With proper awareness & **government support**, millets can become a potent tool to address **health & nutritional concerns** in India.

**2. What is the present world scenario of intellectual property rights with respect to life materials? Although India is second**

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**in the world to file patents, still only a few have been commercialised. Explain the reasons behind this low commercialisation. (10 Marks / 150 words)**

**Ans:** Life materials refer to biological resources like plants, animals, micro-organisms, genetic materials & biotechnological products. The existence of most life materials in nature and their ability to manipulate them makes this realm of intellectual property rights (IPR) very complex.

In this regard, international agreements like TRIPS & biological organisations like Convention on Biodiversity (CBD) have set certain regulations but their acceptance differs with various countries based on various parameters:

1. **Biotechnological Innovations:** Patents on such life materials, while necessary for fields of healthcare, raise questions on the ethical implications from a moral & legal perspective, for example, CRISPR–CAS 9 technology is patented in the USA but restricted in Europe due to the potential of gene editing on a wide range of materials or species.

2. **Protection of Traditional Knowledge:** There is a possibility of companies creating monopolies by patenting life materials. Hence, India invented the Traditional Knowledge Digital Library (TKDL) to block patents based on traditional practices at the international level.

3. **IP Disputes:** Ethical concerns regarding the application of patents have led to international legal disputes which lead to restrictions on usage of such technologies in the approved countries.

6. **Restriction on Patenting Naturally Occurring Substances:** Most countries agree on the prohibition of patents for materials or changes which occur as a result of natural processes. For example, revocation of BRCA1 & two patents from Myriad Genetics by the US Supreme Court.

In the context of IPR, India stands at the second position globally in terms of filing patents. However, commercialisation stands at only **0.4%** in

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2022–23. This suggests a disconnect between the purpose & application of patents in India:

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1. **Academic Patents:** Most patents filed from academic institutions do not have the capacity for commercial translation. They are more focused on research innovation over practical usage of the same.
2. **Market Barrier:** A high amount of investment is required to make most patents commercially viable & this provides no guarantee of returns. Hence, most patent holders are sceptical to commercialise their patents.
3. **Defensive Patents:** A lot of patents are filed to block competitors rather than for commercialisation.
4. **Legal Disputes:** Inadequate enforcement of IPR & prolonged legal battles discourage investment towards commercialisation.

While these challenges persist, **regulatory reforms**, **government funding** and better collaboration between academia & industry can pave the way for higher **commercialisation** of patents as well as boost the **economy through innovative products**.

**3. Describe the context & salient features of the Digital Personal Data Protection Act 2023. (10 Marks / 150 words)**

**Ans:** The **Digital Personal Data Protection Act (DPDP Act) 2023** traces its origin to the **KS Puttaswamy vs Union of India** case, where the Supreme Court of India recognised the **right to privacy** as a **fundamental right** under Article 21 of the Indian Constitution.

Consequently, the Justice BN Srikrishna committee set up by MeitY proposed the initial draft bill in 2018. After a series of changes & public consultations, **DPDP became an Act in 2023**.

**Aims of the Act:** DPDP Act aims to regulate the **processing of personal data** while ensuring:

- Individual's **right to protect** their personal data
- Need to process data for **lawful purposes**

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***Salient Features of the DPDP Act***

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1. **It defines**
   - Obligations of data fiduciaries for data processing
   - Rights & duties of data principals (i.e., owners of data)
   - Financial penalties for breach of obligations & rights
2. **It is based on seven principles:**
   - Consent for lawful use of personal data
   - Purpose limitation & right to be forgotten
   - Data minimisation
   - Data accuracy
   - Storage limitation
   - Reasonable security safeguards
   - Principle of accountability
3. It provides for the **establishment of Data Protection Board of India (DPBI)** to monitor compliance, impose penalties and hear grievances. Any further appeals will be handled by the **Telecom Disputes Settlement & Appellate Tribunal (TDSAT)**.
4. It also provides **exemptions** where the primary obligations & rights won't apply, including for matters of **security** or **sovereignty of the state**, **prevention of offences**, **enforcing legal rights** & **other specified purposes**

While the DPDP Act has **certain limitations** like inadequate safeguards, **violation of RTI** and **complicated grievance redressal**, it is the **first step** to achieve **regulatory protection over personal digital data**.

Further **amendments**, **clarifications** and **real-life use cases** can highlight the limitations or strengths of the Act for effective enforcement & protection of individual's data in the **public realm**.

**4. What are asteroids? How real is the threat of them causing extinction of life? What strategies have been developed to prevent such a catastrophe? (15 Marks / 250 words)**

**Ans:** Asteroids are rocky, metallic remnants left over from the early formation of the solar system ~4.6 billion years ago. They are **smaller than**

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**planets** & usually found in the **inner solar system**, concentrated in a belt around Mars & Jupiter—also called the **asteroid belt**.

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These asteroids in the belt follow a stable elliptical orbit usually. However, in the event of a collision with other objects, an asteroid may deviate from its path or escape the gravitational pull of Jupiter, potentially altering its trajectory.

In such a situation, if an asteroid is directed towards Earth, a potential collision can cause **widespread damage** ranging from localised impacts to global catastrophe—if the asteroid **burns in the Earth's atmosphere** (bolide event) or impacts the surface.

***Some Notable Events or Cases***

1. November 2013 **Chelyabinsk meteor**
2. Mass extinction—**Anamoly Hypothesis**
3. **Tsunamis** or **firestorms**
4. **Impact winter** (similar to nuclear winter)

While the **probability of a major collision is low**, the **increasing planetary defense mechanisms** to protect the human population from a potentially hazardous / **catastrophic asteroid impact** is significant.

***Some of The Collision Strategies Include***

1. **HAMMER:** Hypervelocity Asteroid Mitigation Mission for Emergency Response—This aims to explore the use of **nuclear explosion** in order to destroy the approaching asteroid.
2. **DART:** Double Asteroid Redirection Test—This technology aims to deflect an asteroid to a different trajectory using the principle of **momentum transfer** where a spacecraft at high velocity impacts the asteroid.
3. **Gravity Tractor:** It is a theoretical spacecraft which would be used to deflect the asteroid into a non-colliding orbit by utilising **gravitational pull** between the 2 objects.
4. **Focused Solar Energy:** This theory proposes **deflecting an asteroid** by focusing solar energy from a nearby spacecraft onto its surface to create thrust from **resulting vaporisation** of the material.

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While many methods have been proposed, their efficacy would be proved only in the event of a **successful deflection** of an asteroid. So far, only the **DART technology** has been successfully tested against the **Didymos** & **Dimorphos pair** of asteroids. Hence, more methods can prove to be an effective deterrent against **potential asteroid collision**.

**5. Social media and encrypted messaging services pose a serious security challenge. What measures have been adopted at various levels to address the security implications of social media? Also suggest any other remedies to address the problem.**

**(15 Marks / 250 words)**

**Ans:** While social media & encrypted messaging services provide a reliable platform for users to connect & share information without risk of it leaking to the public, the high-level encryption systems have also proved to be a **security nightmare** in terms of **tracking miscreants and criminals** -

1. Spread of **misinformation** rapidly
2. Social media platforms can promote **radicalisation** & **extremism**
3. End-to-end encryption makes it **difficult** for law enforcement to **tackle** or **monitor** criminal activities
4. **Propaganda** can be used to control **election** & **military narratives**

However, to tackle such threats, the government & law enforcement agencies have undertaken various measures such as—

1. **IT Rules 2021**
   - Mandates social media platforms to **trace first originator** of misleading posts
   - Social media platforms are directed to **take down** controversial or misleading content
2. **DPDP Act 2023**
   - Ensures **data localisation** within Indian borders
   - Provides **exemptions** to government to regulate data in case of **national security**
3. **Collaborative Action:** Government works with platforms like **WhatsApp** and **X** to curb fake news, hate speech & **traceability of**

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**crimes** when required by law.

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4. **Cybercrime Monitoring:** Creation of **Indian Cybercrime Coordination Centre (I4C)** and replication of the cyberdome model to combat crimes like **child pornography**, **data breach**, etc.

**While the problems continue to plague the system, suitable remedies can help tackle the issue in an efficient manner:**

- Balanced encryption laws to ensure both **individual privacy** and **lawful access** by law enforcement agencies
- **Digital literacy campaigns** for both minors & adults
- **AI-based content monitoring** to target radicalism or misinformation
- Govt. **fact-checking agencies** to stop the spread of fake news
- **International collaboration** to tackle cross-border cybercrimes

India continues to battle security issues propagated by **extremist elements** and **propaganda spread** from criminal elements across the borders.

To control this effectively and safeguard the **sovereignty of India**, the government is taking bold decisions to assert its position against **cyberterrorism & digital fraud**.

A **comprehensive strategy** combining the balance of **privacy & security**, **technology adoption**, suitable **restrictions on freedom of speech** & **legal opinion** is required for **national security**.

**6. What is the technology being employed for electronic toll collection on highways? What are its advantages and limitations? What are the proposed changes that make this process seamless? Should this transition carry any political keywords?**

**Ans:** Electronic toll collection is **vehicle traffic** **(10 Marks / 150 words)**

**alternative to toll booths** which uses **radio signals** for detection & corresponding payment using an electronic payment system.

Under the **NETC programme** (National Electronic Toll Collection), NPCI has conceptualised & developed the **FASTag** technology for payment on any toll plaza across the country. Salient features of FASTag include:

- It employs **Radio Frequency Identification (RFID)** for detection of moving vehicles.

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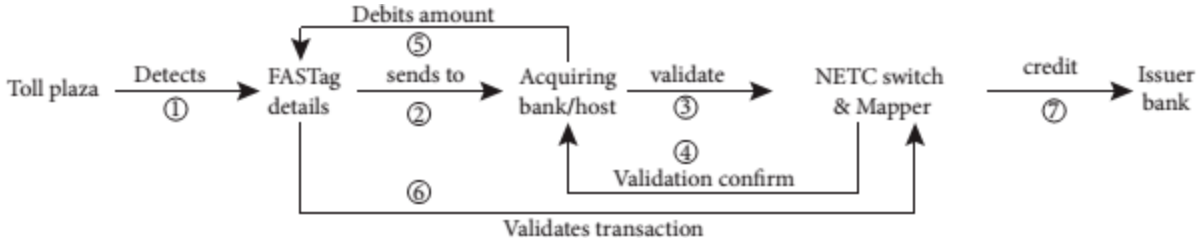
- **Interoperability** across banks allows for deduction from FASTag prepaid **wallets** linked with the particular FASTag sticker.
- **VAHAN integration** to be integrated with vehicle **registration data** for authentication & enforcement.




This technology has a number of advantages ranging from

- **Reduced human interface** & transparent revenue collection
- **Minimisation of toll plaza congestion**
- **Environmentally beneficial** by cutting down fuel emissions & consumption during idling.

1. **Cashless payment** ensures user convenience.
2. **Interoperability** across banks promotes flexibility for consumers.
3. Integration with **VAHAN** ensures **compliance** & ease of **enforcement**.
4. Shift from unskilled labour collecting toll to **skilled employment** in technology development driving growth of **human capital**.

However these advantages also can be overshadowed by performance issues like:

- **Digital literacy** can be challenging for elderly people
- **Technical glitches** like RFID scanner malfunction can lead to missed tags & double collections
- **Delays** in tag activation can cause issues in emergencies
- **Penalty** for non-FASTag vehicles can cause concerns during emergencies.

To enhance this technology, several changes have been proposed which if implemented can further increase **efficiency** of toll collection. Alongside, the **concerns** also need to be addressed during implementation & changes.

1. **GPS-based tolling system** and **Automatic Number Plate Recognition (ANPR)** system could eliminate need for physical toll

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booths & ensure uniform toll collection. However, AI-based data collection raises concerns of **data misuse**, **data breaches**, & **DDoS attacks**.

2. **One Nation One FASTag** scheme implementation can lead to uniformity of the toll collection & **interoperability** across all toll plazas in India. However, a **centralised system** can make it **vulnerable** to widespread technical collapse or cyber-attacks **threatening national security**.

Adopting any new technology is a **balancing act** between safety and efficiency. FASTag has revolutionised highway tolling in India & further developments should ensure **inclusion** & **data protection** norms.

## 2023 Mains Answers

**1. How does e-technology help farmers in production and marketing of agricultural produce? Explain. (10 Marks / 150 words)**

**Ans:** **e-Technology** has revolutionised many industries including agriculture, which is a vital sector in India in terms of food security, sustainable development and reduction of poverty. With the increase in population, food demand has scaled rapidly and e-technology has thoroughly helped in meeting the needs through enhanced **productivity** and effective **monitoring** of agricultural products.

***Role of E-Technology in Agricultural Production***

1. **Precision Farming Techniques:** Using data, sensors & drones to optimise irrigation, fertiliser & pesticide application on crops
2. **Determining Soil Quality:** through satellite & sensor data to analyse crop types & agricultural input suitable for soil.
3. **Agricultural Inputs & Machinery:** can be procured from online applications at competitive prices & good quality.

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4. **Distance Education:** Farmers can leverage govt. initiatives like Kisan Suvidha, Pusa Krishi, Agri apps to gain insights to improve the quality of their produce.

***Role of E-Technology in Agricultural Marketing:***

1. **E-commerce platforms** like **e-NAM** enable farmers to sell their produce at will without being dependent on APMCs to market their produce.
2. **Market intelligence platforms** like **AGMARKNET portals** can help farmers understand **demand** & current **prices** to avail maximum **profit** from their produce.
3. **Corporate initiatives** like **ITC's e-Choupal** can help farmers sell their produce **directly to companies** at **competitive rates**.

The **growth of food demand** requires **effective supply** of food to markets with **least wastage** while maintaining **standard qualities**. Thus, **e-technology** can be **leveraged** in transforming the **production** & **supply landscape** of agriculture for benefit of all.

**2. Introduce the concept of (AI) Artificial Intelligence. How does AI help in clinical diagnosis? Do you perceive any threat to the privacy of the individuals in the use of AI in healthcare?**

**Ans:** **Artificial intelligence** is a **technology** **(10 Marks / 150 Words)** that mimics the working of a human brain in reference to its **performance**, **accuracy** & **reliability**. It does so using the concept of **neural networks (Deep Learning)**, that is the flow of signals through neurons in order to **eliminate information** to & fro from the brain.

The **ability of AI to take decisions based on data** including **recognising patterns**, **analysing images** and **understanding human commands** can enable it to **potentially revolutionise healthcare**—

1. **Medical Imaging:** AI in MRI & CT can enable deep learning of every organ as well as aid doctors to detect diseases based on generated images, potentially reducing human oversight.
2. **Analysis of Electronic Health Records (EHRs):** AI can help identify patients with high risk of chronic diseases, improvement in **preventive**

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**care** and reduce need for emergency interventions through continuous monitoring.

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3. **Addressing the Healthcare Crisis:** AI can aid accessibility of doctors by reducing their mundane tasks like scheduling, form filling, report writing, managing administrative workload and enabling doctors to focus on **patient care**.
4. **Reduce expenditure**—Due to its **efficiency** & ability to handle a large amount of complex tasks, AI can help **reduce expenditure** as well as **improve health outcomes simultaneously**.

However, such convenience often comes with **challenges**. In context of AI, **data privacy remains a significant concern**:

- Health data of individuals can be used by**insurance companies** to **manipulate claims**.
- **Data breaches** of sensitive health data into wrong hands can be used as tools for **biological warfare**.
- **Unsuspecting individuals** can be scammed by **private healthcare providers** to avail **unnecessary treatments** by manipulating individual data.
- If health data is made public to **employers**, it can result in **workplace discrimination** based on **personal health**.

Hence, a **fine balance** requires to be achieved while using AI in healthcare ensuring that the **cons are always outweighed by the benefits** of technology **integration** for better service & **safety of personal data**.

**3. Describe several ways in which microorganisms can help in meeting the current fuel shortage. (10 Marks / 150 words)**

**Ans:** Just like in the environment, microorganisms play a very important role even in the production of fuel. Through various chemical reactions, microbes have the capacity to meet fuel shortage by providing alternatives to the traditional fossil fuel.

***Some of The Methods Include***

1. **Biofuel Production:** These are fuels produced from living organisms where microbes play a role in converting the raw material to fuel.

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2. **Bioethanol:** Produced using carbohydrate-rich edible plants like sugarcane & rotting microbes like yeast aid the fermentation process to produce bioethanol, which can be blended with petrol.
3. **Biodiesel:** It is usually produced from lipids & food sources where the algae helps in transesterification reaction to produce biodiesel.
4. **Biogas Production:** Biogas can be produced through anaerobic respiration process—dumps & sewage in presence of microbial consortium to produce biogas which is a combination of $CH_4$ + $CO_2$ + $NH_3$ etc.
5. **Microbial Fuel Cell:** It functions as a hydrogen fuel cell; it uses organic waste along with anaerobic bacteria & a metallic catalyst to generate H+ (protons). This in turn produces an ionic load which can be used as electricity (fuel directly converted to electrical energy).

**'Thus by mimicking the natural processes of anaerobic respiration and microbial decomposition, fuels can be generated in order to meet the rising energy needs of the country'.**

**4. The use of unmanned aerial vehicles (UAVs) by our adversaries across the borders to ferry arms/ammunition, drugs, etc. is a serious threat to internal security. Comment on the measures being taken to tackle this threat. (10 Marks / 50 words)**

**Ans:** UAVs are aircraft piloted by **remote control** or programmed to perform a particular task using onboard computers. As a consequence, they can be used by adversaries on a small scale to **smuggle arms, drugs** as well as **create explosions**.

To combat such threats, drone operations are governed by **regulations set under the DGCA—Directorate General of Civil Aviation**. Some measures include:

1. **Mandatory registration** of all drones (>250 gm) for all purposes over the **Digital Sky Platform**. Nano drones (<250 gm) also require registration +
licensing if used for commercial purposes.

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2. **No permission, no take-off:** operators need to comply with NPNT (No Permission, No Take-off) rule by obtaining **permission** for each flight on Digital Sky.
3. **No-fly zones:** certain areas like airports, military installations, government offices and other sensitive locations are off-limits.
4. **Altitude & visual line of sight:** Unless specifically exempted, drones cannot be flown over **400 ft** and drone operators **should always keep drones in line of sight**.
5. **Daylight operations:** Unless prior permission is obtained, drones are restricted for **daytime flying only**.
6. Other measures like **import restrictions** and **penalties for non-compliance** of **Drone Rules 2021** help manage the menace of certain illegal drones to a great extent.

However, due to the advancement of **stealth technology** & **explosive drones**, **detection** & **containment of damage** becomes challenging. In this regard, **development of advanced radar systems** & **drone interceptors** are the need of the hour. India's **S-400 air defence system** and **Barak-8** have provided the much-needed security in this regard.

**5. The adoption of electric vehicles is rapidly growing worldwide. How do electric vehicles contribute to reducing carbon emissions and what are the key benefits they offer compared to traditional combustion engine vehicles? (15 Marks / 250 words)**

**Ans:** There is a global shift towards adoption of eco-friendly **technologies** with a view to curbing carbon emissions, which is the primary cause for global warming. Among clean technologies, **electric vehicles (EVs)** have become very common due to their **efficiency** and lower **lifecycle emissions**.

EVs based on the **technology used** can reduce **carbon emission** in numerous ways

- **No tailpipe emissions**, leading to significant reduction in urban air pollution and heat islands.
- **Energy efficiency**—leads to lower demand of electricity & less heat generation caused by fossil.

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- If the electricity grid is powered by **renewable sources**, emissions can be curbed to **near-zero levels**. However, even in coal-dominated grids, EVs produce less total $CO_2$ as compared to fossil fuel vehicles.
- Possibility of **battery reuse & recycling** can compensate for the emissions released during **battery manufacturing**.

For India, road transport contributes ~12% of transport emissions, which can be reduced using EVs. The growth of EV sector, with also will be helpful to India to meet its **net-zero targets by 2070** as well as boost the economy by **reducing dependence on crude oil imports**.

In addition, EVs also provide several benefits over IC engine (ICE) based vehicles

1. **No air Pollution:** ICEs release toxic pollutants like NOx, $SO_2$ & CO which are significant contributors to air pollution. But EVs have no tailpipe emissions & curb air pollution entirely.
2. **No Noise Pollution:** ICE vehicles, especially diesel, contribute to a lot of urban noise pollution. In comparison EVs are virtually noiseless.
3. **Green Fuel:** If electricity is sourced from clean sources, EVs leave a minimal carbon footprint. But ICE vehicles will always use fossil fuel sources leading to high carbon emission.
4. **Low Maintenance:** As EVs have fewer moving parts compared to ICE vehicles, low maintenance is required thereby curbing cost.

While initial cost of adoption can be high for EVs compared to ICE (Internal Combustion Engine), government subsidies like **FAME-II**, **PM-eBus schemes** and **state-level incentives** like lower **GST**, no **registration charges** can enable faster adoption of EVs.

India continues to develop the required infrastructure including **charging stations**, **maintenance depots and battery replacement** for EVs in order to reduce carbon emissions & lower environmental impact due to ICE vehicles.

**6. What is the main task of India's third moon mission which could not be achieved in its earlier mission? List the countries that have achieved this task. Introduce the subsystems in the spacecraft**

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**launched & explain the role of 'Virtual Launch Control Centre' in Bengaluru? (15 Marks / 250 words)**

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**Ans:** India's third moon mission **Chandrayaan—3** was launched in 2023 from the **Satish Dhawan Space Centre (SDSC)**, Sriharikota aboard **LVM-3M4 (GSLV-MK3 M4)** and carried **3 modules**—propulsion, lander (**Vikram**) & rover (**Pragyaan**).

Along with orbiter, the **lander & rover** were also part of **Chandrayaan–2**, which encountered an **unsuccessful landing** on the moon's surface. Hence, these two missions were carried over to **Chandrayaan–3**, & successfully landed on **23rd August 2023** at the **Shiv Shakti Point**.

Among all countries who have achieved soft landing on the moon—**Russia, USA & China**. **India is the first and only country** to successfully achieve soft-landing at the **lunar South Pole (69.3°S)**. Due to this commendable achievement, Chandrayaan–3 received the **World Space Award** by **International Aeronautical Federation (2024)** & **23rd August** is celebrated as **National Space Day**.

***Subsystems in Chandrayaan–3***

1. **Propulsion module** | 1 payload | **SHAPE**:

   **SHAPE** stands for Spectro-polarimetry of Habitable Planet Earth. Its role was to find similar exoplanets to earth where habitation is possible using a form of passive remote sensing.

2. **Vikram Lander** | 4 payloads | **RAMBHA, ChaSTE, ILSA, LRA**
   - **RAMBHA:** Radio Anatomy of Moon Bound Hypersensitive Ionosphere or Atmosphere, i.e., its role was to determine the characteristics of surface plasma in the lunar near-surface atmosphere.
   - **ChaSTE:** Chandrayaan's Surface Thermo-physical Experiment, i.e., it studied the thermal properties of the lunar surface near the south pole.
   - **ILSA:** Instrument for Lunar Seismic Activity by recording ground acceleration at landing site.

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- **LRA:** LASER Retroreflector Array to track the lander's position.

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3. **Pragyaan Rover** | 2 payloads | **LIBS, APXS** (around landing site)
    - **LIBS**—Laser Induced Breakdown Spectroscopy to study the chemical composition of lunar soil through vaporising plasma using laser light.
    - **APXS**—Alpha Particle X-ray Spectrometer to analyse elemental composition of lunar soil using X-ray measurement of waves emitted after bombarding the sample with α-particles.

All these subsystems combined together studied the lunar surface to understand moon's mineralogy & determine the possibility of habitable conditions on the moon & surrounding exoplanets.

**The Virtual Launch Control Centre (VLCC)** at the Vikram Sarabhai Space Centre (VSSC) in Bengaluru plays a crucial role in ensuring the success of ISRO's missions by enabling remote monitoring and control of launch operations. It acts as a centralised hub for real-time data analysis, anomaly detection and coordination among various teams and systems involved in the launch.

## 2022 Mains Answers

**1. Each year a large amount of plant material, cellulose, is deposited on the surface of planet Earth. What are the natural processes this cellulose undergoes before yielding $CO_2$, water or other end products? (10 Marks / 150 words)**

**Ans:** Cellulose is an organic material, the most abundant on Earth. It is found in the cell walls of plant stems and cells. This cellulose when deposited in the atmosphere undergoes a variety of processes:

1. **Physical Process**
    - Involves the use of **weathering** aided by **heat, snow** and **rain** to **weaken** the plant matter.

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- **Swelling** occurs in cellulose fibres which can now be easily decomposed faster.

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2. **Chemical Process**
   - The physically broken-down cellulose fibres can be further subjected to:
     a. **Hydrolysis:** water in presence breaks cellulose into **simpler sugars**
     b. **Microbial degradation:** carried out in **aerobic/anaerobic conditions** where cellulolytic microbes break down cellulose to **simple sugars** like **glucose**
3. **End Products**
   - The broken-down sugar & leftover cellulose are finally broken down by **fungi** into **$CO_2$ & $H_2O$**.
   - (Some cellulose might remain partially decomposed to form a fertile layer of **humus on soil**.)

While natural decomposition of cellulose yields $CO_2$ & $H_2O$, they can be artificially processed in anaerobic conditions in presence of bacteria to convert them into simple sugars & further into biofuels. Understanding the impact of cellulose decomposition is crucial for managing global warming induced climate change & reducing carbon emissions.

**2. Do you think India will meet 50% of its energy needs from renewable energy by 2030? Justify your answer. How will the shift of available power from fossil fuels to renewables help achieve the above objective? Explain. (15 Marks / 250 words)**

**Ans:** India's geographical location & availability of a number of water bodies give it an edge when it comes to switching to renewable energy. The advantage driven by its **optimum implementation** of clean energy initiatives has led to significant progress on this.

As per Central Electricity Authority (CEA), renewables already account for **46.3%** (203.16 GW) of renewable energy out of the total electricity generation capacity of India in 2021. This keeps India on the

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track to both achieving 50% and 500 GW of energy from renewable sources by 2030.

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***Renewable energy (220 GW) [as of Mid-2025]***

- Solar energy—105.65 GW
- Wind power—51.29 GW
- SHEP (Small hydroelectric power)—5.1 GW
- Biomass & others—Remaining

While fossil fuels covered 85% of generation & 75% of electricity needs, the increasing installed capacity of renewables contributing to a majority of the added renewable energy electricity generation reduces the carbon footprint & achieves the target of 50% by 2030. But at the same time, schemes like PM Kusum, Rooftop Solar, Solar Parks, PM Surya Ghar Yojana (Har Ghar Bijli Yojana) & other effective wind energy projects etc. can prove to be instrumental.

Apart from the initiatives which provide a **structural shift**, the real drivers of the change lie in a **behavioral change** propelled by India's fossil fuel **subsidy reform**: As per Asia-Pacific Climate Report 2024, released by ADB (Asian Development Bank), '**India cut its fossil fuel subsidies by 85% between 2013 & 2023'**

This reform was aided by a '**remove, target & shift' strategy** (P18) which included gradual phasing out

1. **Gradual phasing out** of petrol & diesel subsidies (2010–14)
2. **Tax hikes** on fossil fuels for fiscal support to renewable projects
3. **Excise duty hikes** on fossil fuels to boost tax revenue (2014–17)
4. **New subsidies for renewable energy adoption** through various schemes—
   - **Solar Energy:** PM-KUSUM, PM–Surya Ghar, NSM (National Solar Mission)
   - **National Green Hydrogen Mission**
   - **Green Energy Corridors**
   - **PM E-drive scheme**

These steps have been successful in bringing about a **slow but monumental shift** towards renewable energy. Both at individual &

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commercial level, the subsidies have **attracted consumers** to shift to renewable energy sources, with **solar energy being the most widely adopted**.

**3. Launched on 25th December 2021, James Webb Space Telescope has been much in news since then. What are its unique features which make it superior to its predecessor space telescope? What are the key goals of this mission? What potential benefits does it hold for the human race? (15 Marks / 250 words)**

**Ans:** James Webb Space Telescope (JWST), launched on 25th December 2021, is a joint project of NASA, ESA and Canadian Space Agency. Designed to be a successor of the Hubble Telescope, it is equipped with a multitude of advanced instruments & placed at L2 halo orbit of the sun. It has a number of unique features like—

1. **Orbit:** JWST is placed at L2 instead of orbiting the earth like Hubble. As a result, it is shielded from light interferences by the moon, sun and Earth.
2. **Light Mirror:** The bigger mirror of Ø 6.5m compared to 2.4m mirror diameter of Hubble allows for greater light to be captured enabling study of older & far-off galaxies.
3. **Strength:** It is 100 times more powerful than Hubble enabling it to observe images upto 250 million years from the Big Bang while Hubble can observe upto 400 million years only.
   technologically (JWST—body galaxies, HST—toddler galaxies)
4. **Spectrum:** The usage of infrared light in JWST compared to UV light used in Hubble allows JWST to observe clearer & more detailed images of objects in the universe. e.g., Einstein ring

With such **stronger capabilities**, the JWST aims to:

1. **Investigate** a broad range of fields of astronomy & cosmology.
2. Understand the **origins** of the universe & our solar system.
3. Look out for **life** in far-away planets.
4. Look out for **properties** of various **cosmic bodies** like galaxies, & faraway exoplanets.

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5. Study properties of black holes, dark matter, dark energy & associated phenomena like gravitational waves, lensing & **entwining**.

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The detailed study of this wide range of phenomena can have a number of **benefits** including:

- Understanding if the **Standard Model of cosmology** is **applicable** when studying the origin & composition of the universe.
- Getting closer to understanding the **Big Bang Theory**, considered as the **starting point** of our universe.
- Understanding the exact **conditions** responsible for **life** & replicating them possibly in the future on other viable planets.
- Understanding **impact** of gravitational waves, solar flares on the electronic systems of Earth & their **capability of interfering** with EM waves.

The JWST is a unique tool that has led to **notable results** like the presence of carbon on exoplanets (K2–18b) and brightness of galaxies being **disproportionate** to size which have **challenged** existing assumptions & **furthered space research** rapidly in the last few years.

**4. What is the basic principle behind vaccine development? How do vaccines work? What approaches were adopted by the Indian vaccine manufacturers to produce COVID-19 vaccines?**

**Ans:** The basic principle of vaccination lies in eliciting **immunological response** of the body. Vaccines contain a harmless element of the infectious agent which activates the **acquired immunity** of the disease. **(15 Marks / 250 words)**

***How do Vaccines Work?***

**Step 1:** A vaccine containing a weakened virus or dead virus (i.e. the protein causing infection is killed) is **injected** into the host.

(Illustration of injected dead virus and host—'no antibody')

**Step 2:** The antigen (or DNA/RNA segment of antigen) elicits an immune response from the body & **new antibodies** are produced—**Primary Response**

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(Illustration of dead virus and immune response)

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- new antibodies produced
- memory cells produced
- binds to antigen to neutralise it

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**Step 3:** These new antibodies bind to the antigen and neutralise the virus and protect the body from infection. **Immunologically**, memory cells are created so that if the same microbe attacks the body again, it produces **secondary antibodies** if a real pathogen enters the host in the future—**Secondary Response**

During COVID-19, similar principles were used to produce vaccines to combat the SARS-CoV-2 virus based on available **technology** and **resources**.

- **Recombinant DNA Technology:** A viral vector vaccine using adenovirus vector was used to produce **Covishield** by Serum Institute of India. This was the most widely available vaccine with an efficacy of ~80% after the second dose.
- **Inactivated Vaccine: Covaxin** is a whole inactivated SARS-CoV-2 virus-based vaccine developed by Bharat Biotech with ICMR—National Institute of Virology.
- **Nucleic Acid vaccine: ZyCoV-D** was a DNA vaccine (3rd generation) developed by Zydus Lifesciences. It was the first plasmid-DNA vaccine to receive **emergency authorisation for human use** in India. It received approval from DCGI for usage only in severe restricted cases.
- **Herd Immunity:** This approach is used by all manufacturers cumulatively to increase efficacy of their vaccines. When a lot of people in the **community** are vaccinated, the pathogen gradually stops circulating as it encounters mostly immune systems & dies off after it is unable to locate a host for a certain period.

**Thus**, a collective approach coupled with **government support** to push vaccination to a mission-mode during COVID-19 helped the country combat one of the deadliest pandemics in history in a time-bound and efficient manner.

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